MINISTRY OF EDUCATION AND TRAINING

FPT UNIVERSITY

Capstone Project Document

E-Library - Intelligent Library System

SP25SE126	
Group Members	Lê Xuân Phước - SE171957 Nguyễn Vũ Quang Huy - SE171850 Trần Trương Văn - SE170128 Đoàn Viết Thanh - SE171867
Supervisor	Nguyễn Thị Cẩm Hương
Ext Supervisor	
Capstone Project code	GSP25SE87

– Ho Chi Minh city, April 2025 –

TABLE OF CONTENTS

Acknowledgement **25**
Definition and Acronyms **25**
I. Project Introduction **26**
1. Overview 26
1.1 Project Information 26
1.2 Project Team 26
2. Product Background 26
3. Existing Systems 27
3.1 Koha 27
3.2 BiblioCommons 27
4. Business Opportunity 27
5. Software Product Vision 28
6. Project Scope & Limitations 28
6.1 Major Features 28
6.2 Limitations & Exclusions 33
II. Project Management Plan **35**
1. Overview 35
1.1 Scope & Estimation 35
1.2 Project Objectives 37
1.3 Project Risks 38
2. Management Approach 40
2.1 Project Process 40
2.2 Quality Management 40
2.3 Training Plan 41
3. Project Deliverables 41
4. Responsibility Assignments 42
5. Project Communications 43
5.1 Communication Plan 43
5.2 External Interface 44
6. Configuration Management 44
6.1 Document Management 44
6.2 Source Code Management 44
6.3 Tools & Infrastructures 45
III. Software Requirement Specification **45**
1. Product Overview 45
2. User Requirements 46
2.1 Actors 46
2.2 Use Cases 48
2.2.1 Diagram(s) 48
2.2.2 Descriptions 50
3. Functional Requirements 60
3.1 System Functional Overview 60
3.1.1 Screens Flow 60

3.1.2 Screen Descriptions 62
3.1.3 Screen Authorization 68
3.1.4 Non-Screen Functions 72
3.1.5 Entity Relationship Diagram 72
3.2 Applications for Guests and Users 78
3.2.1 Login Page 78
3.2.2 Registration Page 79
3.2.3 Email Verification Page (without account) 80
3.2.4 Reset password 81
3.2.5 Home Page 82
3.2.6 Library Item Detail 84
3.2.7 Search the Library item 88
3.2.8 Predict the upload image 91
3.2.9 Locate the Library item 95
3.2.10 Borrow the digital resources 100
3.2.11 Borrow the Library item remotely 104
3.2.12 Create the borrow request 106
3.3 Website for Guests and Users 108
3.3.1 Patron’s profile 108
3.3.2 Patron’s library card 110
3.3.3 Patron’s borrow request list 113
3.3.4 Patron’s borrow request detail 116
3.3.5 Patron’s borrow record list 120
3.3.6 Patron’s borrowed record detail 122
3.3.7 Patron’s borrow digital list 124
3.3.8 Patron’s borrow digital detail 126
3.3.9 Patron’s reservation list 128
3.3.10 Patron’s reservation detail 131
3.3.11 Patron’s transaction list 133
3.3.12 Patron’s transaction detail 136
3.3.13 Patron’s security setting 138
3.3.14 Patron’s notification list 140
3.3.15 Patron’s notification detail 142
3.3.16 Recommended for patrons 143
3.4 Website for Librarian and Admin 144
3.4.1 Dashboard management 144
3.4.2 Library item management 148
3.4.3 Create library item 152
3.4.4 Library item detail management 155
3.4.5 Authors management 165
3.4.6 Categories management 172
3.4.7 Packages management 176
3.4.8 Conditions management 179
3.4.9 Patrons management 181

3.4.10 Patron detail management 185
3.4.11 Create new patron 188
3.4.12 Library cards management 190
3.4.13 Library card detail management 193
3.4.14 Notifications management 195
3.4.15 Fines management 197
3.4.16 Suppliers management 199
3.4.17 Trackings management 203
3.4.18 Create Tracking by Import Excel 205
3.4.19 Create Tracking manually 208
3.4.20 Tracking detail 211
3.4.21 User management 215
3.4.22 Employee management 221
3.4.23 Borrow Records Management 227
3.4.24 Borrow Record Detail Management 230
3.4.25 Crete Borrow Record 231
3.4.26 Return Library Items 234
3.4.27 Borrow Requests Management 239
3.4.28 Borrow Request Detail Management 240
3.4.29 Borrow Digitals Management 241
3.4.30 Borrow Digital Detail Management 243
3.4.31 Reservations Management 244
3.4.32 Reservation Detail Management 245
3.4.33 Roles Management 246
3.4.34 Permissions Management 249
3.4.35 Transaction Management 251
3.4.36 Train Sessions 253
3.4.37 System Configuration 255
4. Non-Functional Requirements 258
4.1 External Interfaces 258
4.2 Quality Attributes 259
5. Requirement Appendix 261
5.1 Business Rules 261
5.2 Application Messages List 267
IV. Software Design Description 270
1. System Design 270
1.1 System Architecture 270
1.2 Package Diagram 271
1.2.1. Backend Package Diagram 271
1.2.2. Web Application Package Diagram 272
1.2.3. Mobile Application Package Diagram 273
2. Database Design 273
3. Detailed Design 279
3.1 Sign In 279

3.1.1 Class Diagram 279
3.1.2 Sequence Diagram 280
3.1.3 Class Specification 281
3.2 Sign Up 281
3.2.1 Class Diagram 281
3.2.2 Sequence Diagram 282
3.2.3 Class Specification 282
3.3. Manage Users 283
3.3.1. Create User 283
3.3.2. Get All Users 285
3.3.3. Update User 286
3.3.4. Delete User 288
3.4. Manage Employees 290
3.4.1. Create Employee 290
3.4.2. Get All Employee 292
3.4.3. Update Employee 294
3.4.4. Delete Employee 295
3.5 Manage Role Permissions 298
3.5.1 Create Role with Default Permission 298
3.5.2. Get Feature Permission 301
3.5.3. Update Permission 303
3.6. Manage Categories 305
3.6.1. Create Category 305
3.6.2. Get All Category 307
3.6.3. Update Category 308
3.6.4. Delete Category 309
3.7. Manage Authors 311
3.7.1. Create Author 311
3.7.3. Update Author 314
3.7.4. Delete Author 315
3.8. Manage Suppliers 317
3.8.1. Create Supplier 317
3.8.2. Get All Supplier 320
3.8.3. Update Supplier 321
3.8.4. Delete Supplier 322
3.9. Manage Warehouse Trackings 324
3.9.1. Create Warehouse Tracking 324
3.9.2. Get All Warehouse Trackings 326
3.9.3. Update Warehouse Tracking 327
3.9.4. Update Warehouse Tracking Status 328
3.9.5. Delete Warehouse Tracking 330
3.10. Manage Warehouse Tracking Details 331
3.10.1. Add to Warehouse Tracking 331
3.10.2. Update Barcode Range Registration 333

3.11. Manage Library Items 335
3.11.1. Create Library Item 335
3.11.2. Get All Library Items 337
3.11.3. Update Library Item 338
3.11.4. Update Library Item Status 339
3.11.5. Update Shelf Location 342
3.11.6. Delete Library Item 343
3.12. Manage Library Item Instances 346
3.12.1. Update In Shelf 346
3.12.2. Update Out of Shelf 348
3.12.3. Mark as Lost 349
3.12.4. Mark Lost as Found 351
3.12.5. Get by Barcode 353
3.13. Manage Library Item Conditions 355
3.13.1. Create Item Condition 355
3.13.2. Update Item Condition 356
3.13.3. Delete Item Condition 358
3.13.4. Get All Item Conditions 359
3.14. Manage Library Resources 361
3.14.1. Upload Resource To Cloud 361
3.14.2. Update Resource In Cloud 362
3.14.3. Delete Resource From Cloud 364
3.14.4. Add Resource To Library Item 366
3.14.5. Get Preview of Resource 368
3.14.6. Get Borrowed Resource 370
3.15. Manage Library Location 372
3.15.1. Get All Shelves 372
3.15.2. Get Map Shelf Detail by ID 373
3.15.3. Get Map by Floor ID 375
3.16. Manage Training Session 377
3.16.1. Get Untrained Items Group 377
3.16.2. Train Item Group 379
3.16.3. Get All Training Session With Detail 382
3.17. Manage Library Cards 383
3.17.1. Register Library Card 383
3.17.2. Confirm Card Creation 386
3.17.3. Get All Library Card 387
3.18. Transactions 389
3.18.1. Create Transaction 389
3.18.2. Create Fine Payment 391
3.18.3. Pay for Transaction 393
3.19. Manage Fine Policies 395
3.19.1. Create Fine Policy 395
3.19.2. Delete Fine Policy 397

3.19.3. Get Fine Policy 398
3.19.4. Update Fine Policy 399
3.20. Borrow Requests 401
3.20.1. Create Borrow Request 401
3.20.2. Cancel Specific Item 403
3.20.3. Cancel Specific Digital Resource 405
3.20.4. Cancel Specific Reservation 407
3.20.5. Cancel Borrow Request 409
3.20.6. Get All Borrow Request 411
3.21. Borrow Records 412
3.21.1. Create Borrow Record 412
3.21.2. Extend Borrow Record 415
3.21.3. Process Return 417
3.21.4. Get All Borrow Record 419
3.22. Manage Reservations 420
3.22.1. Get Assignable Item Instances by Reservation ID 420
3.22.2. Assign Specific Item Instance 423
3.22.3. Confirm Apply Label 425
3.23. Manage Digital Borrows 427
3.23.1. Pay For Digital Borrow 427
3.23.2. Get Digital Borrow 429
3.24. Manage Notification 430
3.24.1. Create Notification 430
3.24.2. Get All Notification 431
3.24.3. Update Notification 433
V. Software Testing Documentation 435
1. Scope of Testing 435
2. Test Strategy 444
2.1 Testing Types 444
2.2 Test Levels 444
2.3 Supporting Tools 445
3. Test Plan 445
3.1 Human Resources 445
3.2 Test Environment 445
3.3 Test Milestones 446
4. Test Cases 446
VI. Release Package & User Guides 447
1. Deliverable Package 447
2. Installation Guides 448
2.1 System Requirements 448
2.1.1 Hardware requirements 448
2.1.2 Software requirements 449
2.2 Installation Instruction 449
2.2.1 Front-end 449

2.2.2 Mobile 453
2.2.3 Back-end 460
3. User Manual 465
3.1 Overview 465
3.2 Workflow 1 466
3.3 Workflow 2 475
3.4 Workflow 3 488
3.5 Workflow 4 498

LIST OF TABLES

Definition and Acronyms **25**
Table 1.1 : List of definition and acronyms word 25
I. Project Introduction **26**
Table 1.2 : List of member in project team 26
II. Project Management Plan **35**
Table 2.1.1 : Project Management Plan 35
Table 2.1.2 : Project Objectives 37
Table 2.1.3: Project Risks 38
Table 2.2.3 : Training Plan 41
Table 2.3: Project Deliverables 41
Table 2.4 : Responsibility Assignments 43
Table 2.5.1 : Communication plan 43
Table 2.5.2 : External Interface 44
Table 2.6.3 : Tools & Infrastructure 45
III. Software Requirement Specification **45**
Table 3.2.1 : List of actors 46
Table 3.2.2.2: Uses case description 50
Table 3.1.2.a : Website Screen Descriptions 62
Table 3.1.2.b : Mobile Screen Flow 67
Table 3.1.3.a : Website Authorization 68
Table 3.1.3.b : Mobile Authorization 71
Table 3.1.4: Non-Screen Functions 72
Table 3.1.5 : Entity Description 74
Table 5.1 : Business Rules 261
Table 5.2 : Application Messages List 267
IV. Software Design Description **270**
Table 4.2 : Table Descriptions 275
Table 3.1.3.a: Authentication Controller Description Table 281
Table 3.1.3.b: Authentication Service Description Table 281
Table 3.2.3.a: Authentication Controller Description Table 282
Table 3.2.3.b: Authentication Service Description Table 283
Table 3.3.1.3.a: User Controller Description Table 284
Table 3.3.1.3.b: User Service Description Table 284

Table 3.3.2.3.a: User Controller Description Table 286
Table 3.3.2.3.b: User Service Description Table 286
Table 3.3.3.3.a: User Controller Description Table 288
Table 3.3.2.3.b: User Service Description Table 288
Table 3.3.4.3.a: User Controller Description Table 289
Table 3.3.4.3.b: User Service Description Table 290
Table 3.4.1.3.a: Employee Controller Description Table 292
Table 3.4.1.3.b: Employee Service Description Table 292
Table 3.4.2.3.a: Employee Controller Description Table 294
Table 3.4.2.3.b: Employee Service Description Table 294
Table 3.4.3.3.a: Employee Controller Description Table 295
Table 3.4.3.3.b: Employee Service Description Table 295
Table 3.4.4.3.a: Employee Controller Description Table 297
Table 3.4.4.3.b: Employee Controller Description Table 298
Table 3.4.4.3.b: Employee Service Description Table 298
Table 3.5.1.3.a: Role Permission Controller Description Table 300
Table 3.5.1.3.b: Role Permission Service Description Table 300
Table 3.5.2.3.a: Role Permission Controller Description Table 302
Table 3.5.2.3.b: Role Permission Service Description Table 302
Table 3.5.3.3.a: Role Permission Controller Description Table 304
Table 3.5.3.3.b: Role Permission Service Description Table 304
Table 3.6.1.3.a: Category Controller Description Table 306
Table 3.6.1.3.b:Category Service Description Table 306
Table 3.6.2.3.a:Category Controller Description Table 308
Table 3.6.2.3.b:Category Service Description Table 308
Table 3.6.3.3.a: Category Controller Description Table 309
Table 3.6.3.3.b: Category Service Description Table 309
Table 3.6.4.3.a: Category Controller Description Table 310
Table 3.6.4.3.b: Category Service Description Table 311
Table 3.7.1.3.AuthorController Description Table 312
Table 3.7.1.3.b:Author Service Description Table 312
Table 3.7.2.3.a Author Controller Description Table 314
Table 3.7.2.3.b Author Service Description Table 314
Table 3.7.3.1.a: Author Controller Description Table 315
Table 3.7.3.2.b:Author Service Description Table 315
Table 3.7.4.3.a: Author Controller Description Table 317
Table 3.7.4.3.b: Author Service Description Table 317
Table 3.8.1.3 Supplier Controller Description Table 319
Table 3.8.1.3.b Supplier Service Description Table 319
Table 3.8.2.3.a Supplier Controller Description Table 321
Table 3.8.2.3.b Supplier Service Description Table 321
Table 3.8.3.3.a: Supplier Controller Description Table 322
Table 3.8.3.3.b:Supplier Service Description Table 322
Table 3.8.4.3.a: Supplier Controller Description Table 323

Table 3.8.4.3.b: Supplier Service Description Table 324
Table 3.9.1.3.a: Warehouse Tracking Controller Description Table 325
Table 3.9.1.3.b: Warehouse Tracking Service Description Table 326
Table 3.9.2.3.a: Warehouse Tracking Controller Description Table 327
Table 3.9.2.3.b: Warehouse Tracking Service Description Table 327
Table 3.9.3.3.a: Warehouse Tracking Controller Description Table 328
Table 3.9.3.3.b: Warehouse Tracking Service Description Table 328
Table 3.9.4.3.a: Warehouse Tracking Controller Description Table 329
Table 3.9.4.3.b: Warehouse Tracking Service Description Table 329
Table 3.9.5.3.a: Warehouse Tracking Controller Description Table 330
Table 3.9.5.3.b: Warehouse Tracking Service Description Table 330
Table 3.10.1.3.a: Warehouse Tracking Detail Controller Description Table 332
Table 3.10.1.3.b: Warehouse Tracking Detail Service Description Table 333
Table 3.10.2.3.a: Warehouse Tracking Detail Controller Description Table 334
Table 3.10.2.3.b: Warehouse Tracking Detail Service Description Table 334
Table 3.11.1.3.a: Library Item Controller Description Table 336
Table 3.11.1.3.b: Library Item Service Description Table 336
Table 3.11.2.3.a: Library Item Controller Description Table 338
Table 3.11.2.3.b: Library Item Service Description Table 338
Table 3.11.3.3.a: Library Item Controller Description Table 339
Table 3.11.3.3.b: Library Item Service Description Table 339
Table 3.11.4.3.a: Library Item Controller Description Table 341
Table 3.11.4.3.b: Library Item Service Description Table 341
Table 3.11.5.3.a: Library Item Controller Description Table 343
Table 3.11.5.3.b: Library Item Service Description Table 343
Table 3.11.6.3.a: Library Item Controller Description Table 345
Table 3.11.6.3.b: Library Item Service Description Table 346
Table 3.12.1.3.a: Library Item Instance Controller Description Table 347
Table 3.12.1.3.b: Library Item Instance Service Description Table 347
Table 3.12.2.3.a: Library Item Instance Controller Description Table 349
Table 3.12.2.3.b: Library Item Instance Service Description Table 349
Table 3.12.3.3.a: Library Item Instance Controller Description Table 351
Table 3.12.3.3.b: Library Item Instance Service Description Table 351
Table 3.12.4.3.a: Library Item Instance Controller Description Table 353
Table 3.12.4.3.b: Library Item Instance Service Description Table 353
Table 3.12.5.3.a: Library Item Instance Controller Description Table 355
Table 3.12.5.3.b: Library Item Instance Service Description Table 355
Table 3.13.1.3.a: Library Item Condition Controller Description Table 356
Table 3.13.1.3.b: Library Item Condition Service Description Table 356
Table 3.13.2.3.a: Library Item Condition Controller Description Table 358
Table 3.13.2.3.b: Library Item Condition Service Description Table 358
Table 3.13.3.3.a: Library Item Condition Controller Description Table 359
Table 3.13.3.3.b: Library Item Condition Service Description Table 359
Table 3.13.4.3.a: Library Item Condition Controller Description Table 360

Table 3.13.4.3.b: Library Item Condition Service Description Table 361
Table 3.14.1.3.a: Resource Controller Description Table 362
Table 3.14.1.3.b: Cloud Service Description Table 362
Table 3.14.2.3.a: Resource Controller Description Table 363
Table 3.14.2.3.b: Cloud Service Description Table 364
Table 3.14.4.3.a: Library Resource Controller Description Table 367
Table 3.14.4.3.b: Library Resource Service Description Table 368
Table 3.14.5.3.a: Library Resource Controller Description Table 369
Table 3.14.5.3.b: Library Resource Service Description Table 369
Table 3.15.6.3.a: Library Resource Controller Description Table 371
Table 3.15.6.3.b: Library Resource Service Description Table 371
Table 3.15.1.3.a: Library Shelf Controller Description Table 373
Table 3.15.1.3.b: Library Shelf Service Description Table 373
Table 3.15.2.3.a: Library Shelf Controller Description Table 374
Table 3.15.2.3.b: Library Shelf Service Description Table 375
Table 3.15.3.3.a: Library Shelf Controller Description Table 376
Table 3.15.3.3.b: Library Shelf Service Description Table 377
Table 3.16.1.3.a: Custom Vision Controller Description Table 379
Table 3.16.1.3.b: AI ClassificationService Description Table 379
Table 3.16.2.3.a: Custom Vision Controller Description Table 381
Table 3.16.2.3.b: AI ClassificationService Description Table 381
Table 3.16.3.3.a: Custom Vision Controller Description Table 383
Table 3.16.3.3.b: AI ClassificationService Description Table 383
Table 3.17.1.3.a: Library Card Controller Description Table 385
Table 3.17.1.3.b: Library Card Description Table 385
Table 3.17.2.3.a: Library Card Controller Description Table 387
Table 3.17.2.3.b: Library Card Description Table 387
Table 3.17.3.3.a: Library Card Controller Description Table 388
Table 3.17.3.3.b: Library Card Description Table 389
Table 3.18.1.3.a: Transaction Controller Description Table 391
Table 3.18.1.3.b: Transaction Service Description Table 391
Table 3.18.2.3.a: Transaction Controller Description Table 393
Table 3.18.2.3.b: Transaction Service Description Table 393
Table 3.18.3.3.a: Transaction Controller Description Table 395
Table 3.18.3.3.b: Transaction Service Description Table 396
Table 3.19.1.3.a: Fine Policy Controller Description Table 397
Table 3.19.1.3.b: Fine Policy Service Description Table 397
Table 3.19.2.3.a: Fine Policy Controller Description Table 398
Table 3.19.2.3.b: Fine Policy Service Description Table 398
Table 3.16.3.3.a: Fine Policy Controller Description Table 400
Table 3.16.3.3.b: Fine Policy Service Description Table 400
Table 3.19.4.3.a: Fine Policy Controller Description Table 401
Table 3.19.4.3.b:Fine Policy Service Description Table 402
Table 3.20.1.3.a: Borrow Request Controller Description Table 403

Table 3.20.1.3.b: Borrow Request Service Description Table 403
Table 3.20.2.3.a: Borrow Request Controller Description Table 405
Table 3.20.2.3.b: Borrow Request Service Description Table 405
Table 3.20.3.3.a: Borrow Request Controller Description Table 407
Table 3.20.3.3.b: Borrow Request Service Description Table 407
Table 3.20.4.3.a: Borrow Request Controller Description Table 409
Table 3.20.4.3.b: Borrow Request Service Description Table 409
Table 3.20.5.3.a: Borrow Request Controller Description Table 411
Table 3.20.5.3.b: Borrow Request Service Description Table 411
Table 3.20.6.3.a: Borrow Request Controller Description Table 413
Table 3.20.6.3.b: Borrow Request Service Description Table 413
Table 3.21.6.3.a: Borrow Record Controller Description Table 415
Table 3.21.6.3.b: Borrow Record Service Description Table 415
Table 3.21.2.3.a: Borrow Record Controller Description Table 417
Table 3.21.2.3.b: Borrow Record Service Description Table 417
Table 3.21.3.3.a: Borrow Record Controller Description Table 419
Table 3.21.3.3.b: Borrow Record Service Description Table 420
Table 3.21.4.3.a: Borrow Record Controller Description Table 421
Table 3.21.4.3.b: Borrow Record Service Description Table 421
Table 3.22.1.3.a: Reservation Controller Description Table 423
Table 3.22.1.3.b: Reservation Service Description Table 423
Table 3.22.2.3.a: Reservation Controller Description Table 425
Table 3.22.2.3.b: Reservation Service Description Table 425
Table 3.22.3.3.a: Reservation Controller Description Table 427
Table 3.22.3.3.b: Reservation Service Description Table 427
Table 3.23.1.3.a: Payment Controller Description Table 428
Table 3.23.1.3.b: Digital Borrow Service Description Table 428
Table 3.23.2.3.a: Digital Borrow Controller Description Table 429
Table 3.23.2.3.b: Digital Borrow Service Description Table 430
Table 3.24.1.3.a: Notification Controller Description Table 431
Table 3.24.1.3.b: Notification Service Description Table 431
Table 3.24.2.3.a: Notification Controller Description Table 432
Table 3.24.2.3.b: Notification Service Description Table 433
Table 3.24.3.3.a: Notification Controller Description Table 434
Table 3.24.3.3.b: Notification Service Description Table 434
V. Software Testing Documentation 435
Table 5.1 : Scope of Testing 435
Table 5.2.1 : Testing Types 444
Table 5.2.2 : Test Levels 444
Table 5.2.3: Supporting Tools 445
Table 5.3.1 Human Resources 445
Table 5.3.2 : Test Environment 445
Table 5.3.3 : Test Milestones 446
VI. Release Package & User Guides 447

Table 6.2.1.1 : Hardware requirements 451
Table 6.2.1.2 : Software requirements 452

LIST OF FIGURES

Definition and Acronyms **25**
I. Project Introduction **26**
Figure 1.6.1 : Major features 33
II. Project Management Plan **35**
Figure 2.2.1 : Project Process 40
III. Software Requirement Specification **45**
Figure 3.1: E-Library - Intelligent Library System 46
Figure 3.2.2.1: Uses case diagram 49
Figure 3.1.1.a : Website Screen Flow 62
Figure 3.1.1.b : Mobile Screen Flow 62
Figure 3.1.5.1: Conceptual ERD 73
Figure 3.1.5.2: Logical ERD 74
Figure 3.2.1: Login Page 79
Figure 3.2.2: Registration Page 80
Figure 3.2.3: Email Verification Page 81
Figure 3.2.4: Reset Password Page 82
Figure 3.2.5.1: Home Page 83
Figure 3.2.5.2: Home Page 84
Figure 3.2.6.1: Library Item Detail 86
Figure 3.2.6.2: Library Item Detail 86
Figure 3.2.6.3: Library Item Detail 87
Figure 3.2.6.4: Library Item Detail 87
Figure 3.2.6.5: Library Item Detail 88
Figure 3.2.7.1: Search the Library item 89
Figure 3.2.7.2: Search the Library item 89
Figure 3.2.7.3: Search the Library item 90
Figure 3.2.7.4: Search the Library item 90
Figure 3.2.7.5: Search the Library item 91
Figure 3.2.8.1: Predict the upload image 92
Figure 3.2.8.2: Predict the upload image 93
Figure 3.2.8.3: Predict the upload image 93
Figure 3.2.8.4: Predict the upload image 94
Figure 3.2.8.5: Predict the upload image 94
Figure 3.2.8.6: Predict the upload image 94
Figure 3.2.8.7: Predict the upload image 95
Figure 3.2.9.1: Locate the Library item 96
Figure 3.2.9.2: Locate the Library item 96
Figure 3.2.9.3: Locate the Library item 97
Figure 3.2.9.4: Locate the Library item 97

Figure 3.2.9.5: Locate the Library item 98
Figure 3.2.9.6: Locate the Library item 98
Figure 3.2.9.7: Locate the Library item 99
Figure 3.2.9.8: Locate the Library item 99
Figure 3.2.9.9: Locate the Library item 100
Figure 3.2.9.10: Locate the Library item 100
Figure 3.2.10.1: Borrow the digital resources 102
Figure 3.2.10.2: Borrow the digital resources 102
Figure 3.2.10.3: Borrow the digital resources 103
Figure 3.2.10.4: Borrow the digital resources 103
Figure 3.2.10.5: Borrow the digital resources 104
Figure 3.2.10.6: Borrow the digital resources 104
Figure 3.2.11.1: Borrow the Library item remotely 105
Figure 3.2.11.2: Borrow the Library item remotely 106
Figure 3.2.11.3: Borrow the Library item remotely 106
Figure 3.2.12.1: Create the borrow request 108
Figure 3.2.12.2: Create the borrow request 108
Figure 3.3.1.1: Patron’s profile 109
Figure 3.3.1.2: Patron’s profile 110
Figure 3.3.2.1: Patron’s library card 111
Figure 3.3.2.2: Patron’s library card 111
Figure 3.3.2.3: Patron’s library card 112
Figure 3.3.2.4: Patron’s library card 112
Figure 3.3.2.5: Patron’s library card 113
Figure 3.3.3.1: Patron’s request list 114
Figure 3.3.3.2: Patron’s request list 115
Figure 3.3.3.3: Patron’s request list 115
Figure 3.3.3.4: Patron’s request list 116
Figure 3.3.4.1: Patron’s borrow request detail 117
Figure 3.3.4.2: Patron’s borrow request detail 117
Figure 3.3.4.2: Patron’s borrow request detail 118
Figure 3.3.4.3: Patron’s borrow request detail 118
Figure 3.3.4.4: Patron’s borrow request detail 119
Figure 3.3.4.5: Patron’s borrow request detail 119
Figure 3.3.4.6: Patron’s borrow request detail 120
Figure 3.3.5.1: Patron’s borrow record list 121
Figure 3.3.5.2: Patron’s borrow record list 122
Figure 3.3.5.3 Patron’s borrow record list 122
Figure 3.3.5.4: Patron’s borrow record list 122
Figure 3.3.6.1: Patron’s borrow record detail 123
Figure 3.3.6.2: Patron’s borrow record detail 124
Figure 3.3.7.1: Patron’s borrow record detail 123
Figure 3.3.6.3: Patron’s borrow digital list 125
Figure 3.3.7.2: Patron’s borrow digital list 125

Figure 3.3.7.3: Patron’s borrow digital list 126
Figure 3.3.7.4: Patron’s borrow digital list 126
Figure 3.3.8.1: Patron’s borrow digital detail 127
Figure 3.3.8.2: Patron’s borrow digital detail 127
Figure 3.3.8.3: Patron’s borrow digital detail 128
Figure 3.3.8.3: Patron’s borrow digital detail 128
Figure 3.3.9.1: Patron’s reservation list 130
Figure 3.3.9.2: Patron’s reservation list 130
Figure 3.3.9.3: Patron’s reservation list 130
Figure 3.3.9.4: Patron’s reservation list 131
Figure 3.3.10.1: Patron’s reservation detail 132
Figure 3.3.10.2: Patron’s reservation detail 132
Figure 3.3.10.3: Patron’s reservation detail 133
Figure 3.3.10.4: Patron’s reservation detail 133
Figure 3.3.11.1: Patron’s transaction list 135
Figure 3.3.11.2: Patron’s transaction list 135
Figure 3.3.11.3: Patron’s transaction list 135
Figure 3.3.11.4: Patron’s transaction list 136
Figure 3.3.11.5: Patron’s transaction list 136
Figure 3.3.12.1: Patron’s transaction detail 137
Figure 3.3.12.2: Patron’s transaction detail 137
Figure 3.3.12.3: Patron’s transaction detail 138
Figure 3.3.12.4: Patron’s transaction detail 138
Figure 3.3.13.1: Patron’s security setting 139
Figure 3.3.13.2: Patron’s security setting 140
Figure 3.3.13.3: Patron’s security setting 140
Figure 3.3.14.1: Patron’s notification list 141
Figure 3.3.14.2: Patron’s notification list 142
Figure 3.3.14.3: Patron’s notification list 142
Figure 3.3.15.1: Patron’s notification detail 143
Figure 3.3.15.2: Patron’s notification detail 143
Figure 3.3.16.1: Patron’s notification detail 144
Figure 3.4.1.1: Dashboard management 146
Figure 3.4.1.2: Dashboard management 146
Figure 3.4.1.3: Dashboard management 147
Figure 3.4.1.4: Dashboard management 147
Figure 3.4.1.5: Dashboard management 148
Figure 3.4.1.6: Dashboard management 148
Figure 3.4.2.1: Library item management 150
Figure 3.4.2.2: Library item management 151
Figure 3.4.2.3: Library item management 151
Figure 3.4.2.4: Library item management 152
Figure 3.4.3.1: Create Library item 153
Figure 3.4.3.2: Create Library item 154

Figure 3.4.3.3: Create Library item 154
Figure 3.4.3.4: Create Library item 155
Figure 3.4.3.5: Create Library item 155
Figure 3.4.4.1: Library item detail management 160
Figure 3.4.4.2: Library item detail management 160
Figure 3.4.4.3: Library item detail management 160
Figure 3.4.4.4: Library item detail management 161
Figure 3.4.4.5: Library item detail management 161
Figure 3.4.4.6: Library item detail management 162
Figure 3.4.4.7: Library item detail management 162
Figure 3.4.4.8: Library item detail management 163
Figure 3.4.4.9: Library item detail management 163
Figure 3.4.4.10: Library item detail management 164
Figure 3.4.4.11: Library item detail management 164
Figure 3.4.4.12: Library item detail management 165
Figure 3.4.4.13: Library item detail management 165
Figure 3.4.5.1: Authors management 168
Figure 3.4.5.2: Authors management 168
Figure 3.4.5.3: Authors management 169
Figure 3.4.5.4: Authors management 169
Figure 3.4.5.5: Authors management 169
Figure 3.4.5.6: Authors management 170
Figure 3.4.5.7: Authors management 170
Figure 3.4.5.8: Authors management 170
Figure 3.4.5.9: Authors management 171
Figure 3.4.5.10: Authors management 171
Figure 3.4.5.11: Authors management 172
Figure 3.4.6.1: Categories management 173
Figure 3.4.6.2: Categories management 174
Figure 3.4.6.3: Categories management 174
Figure 3.4.6.4: Categories management 175
Figure 3.4.6.5: Categories management 175
Figure 3.4.6.6: Categories management 176
Figure 3.4.6.7: Categories management 176
Figure 3.4.7.1: Packages management 177
Figure 3.4.7.2: Packages management 177
Figure 3.4.7.3: Packages management 178
Figure 3.4.7.4: Packages management 178
Figure 3.4.7.5: Packages management 179
Figure 3.4.8.1: Conditions management 180
Figure 3.4.8.2: Conditions management 180
Figure 3.4.8.3: Conditions management 180
Figure 3.4.8.4: Conditions management 181
Figure 3.4.8.5: Conditions management 181

Figure 3.4.9.1: Patrons management 184
Figure 3.4.9.2: Patrons management 184
Figure 3.4.9.3: Patrons management 185
Figure 3.4.9.4: Patrons management 185
Figure 3.4.10.1: Patrons detail management 188
Figure 3.4.10.: Patrons detail management 188
Figure 3.4.11.1: Create new patron 190
Figure 3.4.11.2: Create new patron 190
Figure 3.4.12.1: Library cards management 192
Figure 3.4.12.2: Library cards management 193
Figure 3.4.12.3: Library cards management 193
Figure 3.4.13.1: Library cards detail management 195
Figure 3.4.14.1: Notifications management 196
Figure 3.4.14.2: Notifications management 197
Figure 3.4.15.1: Fines management 199
Figure 3.4.15.2: Fines management 199
Figure 3.4.16.1: Suppliers management 201
Figure 3.4.16.2: Suppliers management 201
Figure 3.4.16.2: Suppliers management 201
Figure 3.4.16.3: Suppliers management 202
Figure 3.4.16.4: Suppliers management 202
Figure 3.4.16.5: Suppliers management 203
Figure 3.4.17.1: Trackings management 204
Figure 3.4.17.2: Trackings management 204
Figure 3.4.17.3: Trackings management 205
Figure 3.4.18.1: Create Trackings by Import Excel 208
Figure 3.4.19.1: Create Trackings manually 210
Figure 3.4.19.2: Create Trackings manually 210
Figure 3.4.19.3: Create Trackings manually 211
Figure 3.4.19.4: Create Trackings manually 211
Figure 3.4.20.1: Tracking detail 213
Figure 3.4.20.2: Tracking detail 213
Figure 3.4.20.3: Tracking detail 214
Figure 3.4.20.4: Tracking detail 214
Figure 3.4.20.5: Tracking detail 215
Figure 3.4.21.1: User management 218
Figure 3.4.21.2: User management 219
Figure 3.4.21.3: User management 219
Figure 3.4.21.4: User management 220
Figure 3.4.21.5: User management 220
Figure 3.4.21.6: User management 221
Figure 3.4.21.7: User management 221
Figure 3.4.21.8: User management 221
Figure 3.4.22.1: Employee management 224

Figure 3.4.22.2: Employee management 225
Figure 3.4.22.3: Employee management 225
Figure 3.4.22.4: Employee management 226
Figure 3.4.22.5: Employee management 226
Figure 3.4.22.6: Employee management 227
Figure 3.4.22.7: Employee management 227
Figure 3.4.22.8: Employee management 227
Figure 3.4.23.1: Borrow records management 229
Figure 3.4.23.2: Borrow records management 229
Figure 3.4.23.3: Borrow records management 229
Figure 3.4.24.1: Borrow records detail 231
Figure 3.4.25.1: Create borrow records 233
Figure 3.4.25.2: Create borrow records 234
Figure 3.4.26.1: Return Library items 236
Figure 3.4.26.2: Return Library items 236
Figure 3.4.26.3: Return Library items 237
Figure 3.4.26.4: Return Library items 237
Figure 3.4.26.5: Return Library items 238
Figure 3.4.26.6: Return Library items 238
Figure 3.4.26.7: Return Library items 239
Figure 3.4.27.1: Borrow request management 240
Figure 3.4.27.2: Borrow request management 240
Figure 3.4.28.1: Borrow request detail management 241
Figure 3.4.29.1: Borrow digitals management 242
Figure 3.4.29.2: Borrow digitals management 243
Figure 3.4.30.1: Borrow digital detail management 244
Figure 3.4.31.1: Reservations management 245
Figure 3.4.31.2: Reservations management 245
Figure 3.4.32.1: Reservation detail management 246
Figure 3.4.33.1: Roles management 248
Figure 3.4.33.2: Roles management 248
Figure 3.4.33.3: Roles management 249
Figure 3.4.33.4: Roles management 249
Figure 3.4.33.5: Roles management 249
Figure 3.4.33.6: Roles management 249
Figure 3.4.34.1: Permission management 251
Figure 3.4.34.2: Permission management 251
Figure 3.4.34.3: Permission management 251
Figure 3.4.35.1: Transaction management 252
Figure 3.4.35.2: Transaction management 252
Figure 3.4.35.3: Transaction management 253
Figure 3.4.36.1: Train sessions 254
Figure 3.4.36.2: Train sessions 254
Figure 3.4.36.3: Train sessions 255

Figure 3.4.36.4: Train sessions 255
Figure 3.4.37.1: System configuration 257
Figure 3.4.37.2: System configuration 257
Figure 3.4.37.3: System configuration 258
Figure 3.4.37.4: System configuration 258
IV. Software Design Description 270
Figure 4.1.1.1 System Architecture 270
Figure 4.1.2.1 Backend Package Diagram 272
Figure 4.1.2.2 Web Application Package Diagram 273
Figure 4.1.2.3 Mobile Application Package Diagram 273
Figure 4.2.1 : Conceptual ERD 274
Figure 4.2.2 : Logical ERD 275
Figure 3.1.1 Class Diagram 280
Figure 3.1.2 Sequence Diagram 281
Figure 3.2.1 Class Diagram 282
Figure 3.2.2 Sequence Diagram 282
Figure 3.3.1.1 Class Diagram 283
Figure 3.3.1.2 Sequence Diagram 284
Figure 3.3.2.1 Class Diagram 285
Figure 3.3.2.2 Sequence Diagram 286
Figure 3.3.3.1 Class Diagram 287
Figure 3.3.3.2 Sequence Diagram 287
Figure 3.3.4.1 Class Diagram 289
Figure 3.3.4.2 Sequence Diagram 289
Figure 3.4.1.1 Class Diagram 291
Figure 3.4.1.2 Sequence Diagram 292
Figure 3.4.2.1 Class Diagram 293
Figure 3.4.2.2 Sequence Diagram 293
Figure 3.4.3.1 Class Diagram 294
Figure 3.4.3.2 Sequence Diagram 295
Figure 3.4.4.1 Class Diagram 296
Figure 3.4.4.2 Sequence Diagram 297
Figure 3.5.1.1 Class Diagram 299
Figure 3.5.1.2 Sequence Diagram 300
Figure 3.5.2.1 Class Diagram 301
Figure 3.5.2.2 Sequence Diagram 302
Figure 3.5.3.1 Class Diagram 303
Figure 3.5.3.2 Sequence Diagram 304
Figure 3.6.1.1 Class Diagram 305
Figure 3.6.1.2 Sequence Diagram 306
Figure 3.6.2.1 Class Diagram 307
Figure 3.6.2.2 Sequence Diagram 308
Figure 3.6.3.1 Class Diagram 308
Figure 3.6.3.1 Class Diagram 309

Figure 3.6.4.1 Class Diagram 310
Figure 3.6.4.2 Sequence Diagram 310
Figure 3.7.1.1 Class Diagram 311
Figure 3.7.1.2 Sequence Diagram 312
Figure 3.7.2.1 Class Diagram 313
Figure 3.7.2.2 Sequence Diagram 313
Figure 3.7.3.1 Class Diagram 314
Figure 3.7.3.2 Sequence Diagram 315
Figure 3.7.4.1 Class Diagram 316
Figure 3.7.4.2 Sequence Diagram 316
Figure 3.8.1.1 Class Diagram 318
Figure 3.8.1.2 Sequence Diagram 319
Figure 3.8.2.1 Class Diagram 320
Figure 3.8.2.2 Sequence Diagram 321
Figure 3.8.3.1 Class Diagram 322
Figure 3.8.3.2 Sequence Diagram 322
Figure 3.8.4.1 Class Diagram 323
Figure 3.8.4.2 Sequence Diagram 323
Figure 3.9.1.1 Class Diagram 325
Figure 3.9.1.2 Sequence Diagram 325
Figure 3.9.2.1 Class Diagram 326
Figure 3.9.2.2 Sequence Diagram 327
Figure 3.9.3.1 Class Diagram 327
Figure 3.9.3.2 Sequence Diagram 327
Figure 3.9.4.1 Class Diagram 328
Figure 3.9.4.2 Sequence Diagram 329
Figure 3.9.5.1 Class Diagram 330
Figure 3.9.5.2 Sequence Diagram 330
Figure 3.10.1.1 Class Diagram 331
Figure 3.10.1.2 Sequence Diagram 332
Figure 3.10.2.1 Class Diagram 333
Figure 3.10.2.2 Sequence Diagram 334
Figure 3.11.1.1 Class Diagram 335
Figure 3.11.1.2 Sequence Diagram 336
Figure 3.11.2.1 Class Diagram 337
Figure 3.11.2.2 Sequence Diagram 337
Figure 3.11.3.1 Class Diagram 338
Figure 3.11.3.2 Sequence Diagram 339
Figure 3.11.4.1 Class Diagram 340
Figure 3.11.4.2 Sequence Diagram 341
Figure 3.11.5.1 Class Diagram 342
Figure 3.11.5.2 Sequence Diagram 342
Figure 3.11.6.1 Class Diagram 344
Figure 3.11.6.2 Sequence Diagram 345

Figure 3.12.1.1 Class Diagram 347
Figure 3.12.1.2 Sequence Diagram 347
Figure 3.12.2.1 Class Diagram 348
Figure 3.12.2.2 Sequence Diagram 349
Figure 3.12.3.1 Class Diagram 350
Figure 3.12.3.2 Sequence Diagram 351
Figure 3.12.4.1 Class Diagram 352
Figure 3.12.4.2 Sequence Diagram 353
Figure 3.12.5.1 Class Diagram 354
Figure 3.12.5.2 Sequence Diagram 354
Figure 3.13.1.1 Class Diagram 355
Figure 3.13.1.2 Sequence Diagram 356
Figure 3.13.2.1 Class Diagram 357
Figure 3.13.2.2 Sequence Diagram 357
Figure 3.13.3.1 Class Diagram 358
Figure 3.13.3.2 Sequence Diagram 359
Figure 3.13.4.1 Class Diagram 360
Figure 3.13.4.2 Sequence Diagram 360
Figure 3.14.1.1 Class Diagram 361
Figure 3.14.1.2 Sequence Diagram 362
Figure 3.14.2.1 Class Diagram 363
Figure 3.14.2.2 Sequence Diagram 363
Figure 3.14.3.1 Class Diagram 364
Figure 3.14.3.2 Sequence Diagram 365
Figure 3.14.4.1 Class Diagram 366
Figure 3.14.4.2 Sequence Diagram 367
Figure 3.14.5.1 Class Diagram 368
Figure 3.14.5.2 Sequence Diagram 369
Figure 3.14.6.1 Class Diagram 370
Figure 3.14.6.2 Sequence Diagram 371
Figure 3.15.1.1 Class Diagram 372
Figure 3.15.1.2 Sequence Diagram 373
Figure 3.15.2.1 Class Diagram 374
Figure 3.15.2.2 Sequence Diagram 374
Figure 3.15.3.1 Class Diagram 376
Figure 3.15.3.2 Sequence Diagram 376
Figure 3.16.1.1 Class Diagram 378
Figure 3.16.1.2 Sequence Diagram 379
Figure 3.16.2.1 Class Diagram 380
Figure 3.16.2.2 Sequence Diagram 381
Figure 3.16.3.1 Class Diagram 382
Figure 3.16.3.2 Sequence Diagram 382
Figure 3.17.1.1 Class Diagram 384
Figure 3.17.1.2 Sequence Diagram 385

Figure 3.17.2.1 Class Diagram 386
Figure 3.17.2.2 Sequence Diagram 387
Figure 3.17.3.1 Class Diagram 388
Figure 3.17.3.2 Sequence Diagram 388
Figure 3.18.1.1 Class Diagram 389
Figure 3.18.1.2 Sequence Diagram 390
Figure 3.18.2.1 Class Diagram 391
Figure 3.18.2.2 Sequence Diagram 392
Figure 3.18.3.1 Class Diagram 393
Figure 3.18.3.2 Sequence Diagram 394
Figure 3.19.1.1 Class Diagram 395
Figure 3.19.1.2 Sequence Diagram 396
Figure 3.19.2.1 Class Diagram 397
Figure 3.19.2.2 Sequence Diagram 397
Figure 3.19.3.1 Class Diagram 398
Figure 3.19.3.2 Sequence Diagram 399
Figure 3.19.4.1 Class Diagram 400
Figure 3.19.4.2 Sequence Diagram 400
Figure 3.20.1.1 Class Diagram 402
Figure 3.20.1.2 Sequence Diagram 402
Figure 3.20.2.1 Class Diagram 403
Figure 3.20.2.2 Sequence Diagram 404
Figure 3.20.3.1 Class Diagram 405
Figure 3.20.3.2 Sequence Diagram 406
Figure 3.20.4.1 Class Diagram 407
Figure 3.20.4.2 Sequence Diagram 408
Figure 3.20.5.1 Class Diagram 409
Figure 3.20.5.2 Sequence Diagram 410
Figure 3.20.6.1 Class Diagram 411
Figure 3.20.6.2 Sequence Diagram 412
Figure 3.21.6.1 Class Diagram 413
Figure 3.21.6.2 Sequence Diagram 414
Figure 3.21.2.1 Class Diagram 415
Figure 3.21.2.2 Sequence Diagram 416
Figure 3.21.3.1 Class Diagram 417
Figure 3.21.3.2 Sequence Diagram 418
Figure 3.21.4.1 Class Diagram 420
Figure 3.21.4.2 Sequence Diagram 420
Figure 3.22.1.1 Class Diagram 421
Figure 3.22.1.2 Sequence Diagram 422
Figure 3.22.2.1 Class Diagram 423
Figure 3.22.2.2 Sequence Diagram 424
Figure 3.22.3.1 Class Diagram 425
Figure 3.22.3.2 Sequence Diagram 426

Figure 3.23.1.1 Class Diagram 427
Figure 3.23.1.2 Sequence Diagram 428
Figure 3.23.2.1 Class Diagram 429
Figure 3.23.2.2 Sequence Diagram 429
Figure 3.24.1.1 Class Diagram 431
Figure 3.24.1.2 Sequence Diagram 431
Figure 3.24.2.1 Class Diagram 432
Figure 3.24.2.2 Sequence Diagram 432
Figure 3.24.3.1 Class Diagram 433
Figure 3.24.3.2 Sequence Diagram 434
V. Software Testing Documentation 435
Figure 5.4.1 : Unit test case 446
Figure 5.4.2 : Test Statistic 447
VI. Release Package & User Guides 447
Figure 6.2.2.1.1: Front-end 450
Figure 6.2.2.1.2: Front-end 450
Figure 6.2.2.1.3: Front-end 450
Figure 6.2.2.1.4: Front-end 451
Figure 6.2.2.1.5: Front-end 451
Figure 6.2.2.1.6: Front-end 452
Figure 6.2.2.1.7: Front-end 452
Figure 6.2.2.1.8: Front-end 453
Figure 6.2.2.2.1: Mobile 454
Figure 6.2.2.2.2: Mobile 454
Figure 6.2.2.2.3: Mobile 455
Figure 6.2.2.2.4: Mobile 455
Figure 6.2.2.2.5: Mobile 456
Figure 6.2.2.2.6: Mobile 456
Figure 6.2.2.2.7: Mobile 457
Figure 6.2.2.2.8: Mobile 457
Figure 6.2.2.2.9: Mobile 457
Figure 6.2.2.2.10: Mobile 458
Figure 6.2.2.2.11: Mobile 458
Figure 6.2.2.2.12: Mobile 459
Figure 6.2.2.2.13: Mobile 459
Figure 6.2.2.2.14: Mobile 460
Figure 6.2.2.2.15: Mobile 460
Figure 6.2.2.3.1: Project Folder 461
Figure 6.2.2.3.2: Open cmd screen 462
Figure 6.2.2.3.3: Run the command screen 464
Figure 6.2.2.3.4 : API screen 464
Figure 6.3.2.1: Login screen 466
Figure 6.3.2.2: Create warehouse tracking screen 466
Figure 6.3.2.3: Login screen 467

Figure 6.3.2.4: Warehouse tracking detail screen 467
Figure 6.3.2.5: Catalog screen 468
Figure 6.3.2.6: Catalog screen 468
Figure 6.3.2.7: Catalog screen 469
Figure 6.3.2.8: Catalog screen 469
Figure 6.3.2.9: Printer screen 470
Figure 6.3.2.10: Catalog screen 471
Figure 6.3.2.11: Catalog screen 471
Figure 6.3.2.12: Library item management detail screen 472
Figure 6.3.2.13: Library item management detail screen 472
Figure 6.3.2.14: Change instance’s status screen (Mobile app) 473
Figure 6.3.2.15: Library item management detail screen 473
Figure 6.3.2.16: Train AI screen 474
Figure 6.3.2.17: Train session detail screen 474
Figure 6.3.2.18: Create notification screen 475
Figure 6.3.3.1: Login screen 475
Figure 6.3.3.2: Create patron screen 476
Figure 6.3.3.3: Patron’s Gmail Inbox 476
Figure 6.3.3.4: Login screen 477
Figure 6.3.3.5: Search screen 477
Figure 6.3.3.6: Search results screen 478
Figure 6.3.3.7: AI prediction screen 479
Figure 6.3.3.8: AI prediction screen 479
Figure 6.3.3.9: AI prediction screen 480
Figure 6.3.3.10: Library item detail screen 481
Figure 6.3.3.11: Library item detail screen 481
Figure 6.3.3.12: Library item detail screen 482
Figure 6.3.3.13: Borrow list screen 482
Figure 6.3.3.14: Borrow request detail screen 483
Figure 6.3.3.15: Borrow request detail screen 483
Figure 6.3.3.16: Borrow digitals screen 484
Figure 6.3.3.17: Borrow resource detail screen (ebook) 484
Figure 6.3.3.18: Borrow resource detail screen (audiobook) 485
Figure 6.3.3.19: Borrow request detail screen (Mobile App) 485
Figure 6.3.3.20: Map screen (Mobile App) 486
Figure 6.3.3.21: Login screen 486
Figure 6.3.3.22: Create borrow record screen 487
Figure 6.3.3.23: Create borrow record screen 487
Figure 6.3.4.1: Login screen 488
Figure 6.3.4.2: Return library items screen 488
Figure 6.3.4.3: Return library items screen 489
Figure 6.3.4.4: Return library items screen 489
Figure 6.3.4.5: Return library items screen 490
Figure 6.3.4.6: Return library items screen 491

Figure 6.3.4.7: Return library items screen 491
Figure 6.3.4.8: Return library items screen 491
Figure 6.3.4.9: Assign reservations screen 492
Figure 6.3.4.10: Assign reservations screen 492
Figure 6.3.4.11: Assign reservations screen 493
Figure 6.3.4.12: Printer screen 493
Figure 6.3.4.13: Assign reservations screen 494
Figure 6.3.4.14: Login screen 494
Figure 6.3.4.15: Borrow record detail screen 495
Figure 6.3.4.16: Borrow record detail screen 495
Figure 6.3.4.17: Recommend for you screen 496
Figure 6.3.4.18: Library item detail screen 496
Figure 6.3.4.19: Library item detail screen 497
Figure 6.3.4.20: Recommend for you screen 497
Figure 6.3.5.1: Dashboard screen 498
Figure 6.3.5.2: Dashboard screen 498
Figure 6.3.5.3: Dashboard screen 499
Figure 6.3.5.4: Dashboard screen 499
Figure 6.3.5.5: Dashboard screen 500
Figure 6.3.5.6: Dashboard screen 500
Figure 6.3.5.7: Dashboard screen 500
Figure 6.3.5.8: Dashboard screen 500
Figure 6.3.5.9: Dashboard screen 501
Figure 6.3.5.10: Create supplement request screen 501
Figure 6.3.5.11: Create supplement request screen 502
Figure 6.3.5.12: Supplement request detail screen 502
Figure 6.3.5.13: Supplement request detail screen 503
Figure 6.3.5.14: Supplement request detail screen 503
Figure 6.3.5.15: Login screen 504
Figure 6.3.5.16: Warehouse tracking detail screen 504
Figure 6.3.5.17: View PDF screen 505

Acknowledgement

We would like to express our sincere gratitude to our supervisor, Ms. Nguyen Thi Cam Huong, for her insightful advice and exceptional guidance throughout our capstone project. Her clear explanations and patient mentoring helped us navigate each stage of our work, from defining our research questions to refining our final deliverables. Ms. Huong's support encouraged us to think critically and remain focused on our goals.

Our thanks also go to the library staff at FPT University for generously sharing their knowledge of library operations and business rules. Their detailed explanations of classification systems, circulation procedures, and digital cataloguing allowed our team to model real-world processes accurately and complete our project in a timely manner.

Additionally, we are particularly grateful to Mr. Nguyen, Mr. Tai, Mr. Sang, and Mr. Tri for taking the time to review our work and provide thoughtful feedback. Their careful reading, constructive comments, and practical suggestions greatly improved the clarity and quality of our project. We appreciate their dedication and the effort they invested in helping us reach our objectives.

Finally, we extend our deepest appreciation to our team members for their collaborative spirit and late-night discussions that drove this project forward, and to our families for their unwavering encouragement and understanding. Their combined support made this project possible.

Definition and Acronyms

Table 1.1 : List of definition and acronyms words

Acronym	Definition
AWS	Amazon Web Services
BA	Business Analysis
BR	Business Rule
ERD	Entity Relationship Diagram
GUI	Graphical User Interface
PM	Project Manager
SDD	Software Design Description
SPMP	Software Project Management Plan
SRS	Software Requirement Specification
UAT	User Acceptance Test
UC	Use Case

API	Application Program Interface

I. Project Introduction

1. Overview

1.1 Project Information

- Project name: E-Library - Intelligent Library System
- Project code: ELS
- Group name: SP25SE126
- Software type: Web and Mobile App

1.2 Project Team

Table 1.2 : List of member in project team

Full Name	Role	Email	Mobile
Nguyễn Thị Cẩm Hương	Lecturer	Huongntc2@fe.edu.vn	
Lê Xuân Phước	Leader	Phuoclxse171957@fpt.edu.vn	0777155780
Nguyễn Vũ Quang Huy	Member	Huynvqse171850@fpt.edu.vn	0826794531
Trần Trương Văn	Member	Vanttse170128@fpt.edu.vn	0933131464
Đoàn Viết Thanh	Member	Thanhdvse171867@fpt.edu.vn	0364745067

2. Product Background

The rapid evolution of digital technology and changing patron habits have exposed significant limitations in traditional library systems. Users face limited search options and spend a long time finding books in large libraries. It is also hard to borrow or reserve books without going to the library. Additionally, valuable user data remains underutilized, hindering the ability to offer personalized book suggestions or know which books are most popular. This makes it difficult to add new books that patrons might like.

The **E-Library - Intelligent Library System (ELS)** has been developed to address these challenges. The system integrates advanced search engine technology with AI-driven book prediction and recommendation, including the capability to search by capturing a book cover image. It further enhances the user experience with a 2D indoor navigation feature that guides users to the exact location of their desired books. Remote borrowing and reservation functions expand access, while user reading habits, reading history and behavioral data drive personal recommendations. With clear statistical insights, this system transforms the library experience by ensuring efficiency, accuracy, and heightened user engagement.

3. Existing Systems

3.1 Koha

Koha is an open-source integrated library system designed for modern libraries around the world, and it is widely used in Vietnam. It offers key functions such as cataloging, circulation, acquisitions, and patron management - the essential features of any e-library system. Librarians use Koha to manage collections efficiently and simplify daily operations, while users benefit from easy access to complete catalog information. Although Koha does not include built-in AI recommendations or indoor navigation, it provides a strong foundation for adding advanced features like improved search options, remote borrowing, and image-based book lookup. These enhancements can transform a traditional library into a fully interactive e-library.

3.2 BiblioCommons

BiblioCommons serves as an intelligent discovery layer that enhances existing library systems, turning standard catalogs into dynamic, user-friendly portals. By adding social reviews, personalized recommendations, and real-time popularity metrics, it offers an engaging experience similar to modern e-library platforms. Users receive customized suggestions and a more interactive search experience, while library staff can use behavioral data to refine service delivery and resource management. Like many leading e-library systems that combine technology with traditional collections, BiblioCommons connects conventional library management with innovative digital experiences, making the discovery process more intuitive and engaging.

4. Business Opportunity

The E-Library - Intelligent Library System (ELS) is a modern solution built to update and improve traditional libraries. It uses digital technology and AI to solve common problems and make finding books easier. ELS offers features like an advanced search engine (even letting users search by taking a picture of a book cover), 2D indoor navigation to guide users to the right section, remote borrowing and reservations, and personalized book recommendations based on user data.

Market Opportunity:

- Modern Technology: Many libraries still use old systems that make searching for books slow and confusing. ELS uses new digital tools and AI to make searching faster and simpler.
- Better User Experience: Finding books in a big library can be time-consuming. ELS helps by providing clear directions and reducing the time spent searching.
- Remote Services: Traditional libraries require you to visit in person to borrow or reserve books. ELS allows users to do this online, meeting the growing demand for digital access.
- Personalized Recommendations: Most libraries do not use user data to suggest books. With ELS, libraries can offer recommendations that match each patron’s interests, making the experience more personal.

5. Software Product Vision

For library users who seek a modern, efficient, and personalized experience, the E-Library - Intelligent Library System (ELS) is a web and mobile-based solution that transforms how books are discovered, borrowed, and recommended. ELS leverages advanced search technology, AI-driven recommendations, and 2D indoor navigation to simplify book location and access, while offering remote borrowing and reservation features. Unlike traditional library systems, ELS creates a data-informed system that meets the diverse needs of today's patrons and enhances library operations.

6. Project Scope & Limitations

6.1 Major Features

- **FE-01:** Authentication
    - Sign In With Email/Password
    - Sign In With Google
    - Sign Out
    - Sign Up
    - Verify Email
    - Forgot Password
    - Change Password
    - Enable MFA (Multi-Factors-Authentication)
    - MFA Backup
- **FE-02:** Role/Permission Management
    - Get Roles
    - Get User Permissions
    - Create Role
    - Update Role
    - Delete Role
    - Change User's Role
    - Change Role Access Level
- **FE-03:** User Management
    - Get Users
    - Create User
    - Get User Detail
    - Update User Status
    - Update User
    - Soft Delete User
    - Soft Delete Multiple Users
    - Undo Soft Delete User
    - Undo Soft Delete Multiple Users
    - Delete User
    - Delete Multiple Users
    - Import/Export Users

- **FE-04:** Employee Management
    - Get Employees
    - Create Employee
    - Update Employee
    - Soft Delete Employee
    - Soft Delete Multiple Employees
    - Undo Soft Delete Employee
    - Undo Soft Delete Multiple Employees
    - Delete Employee
    - Delete Multiple Employees
    - Import/Export Employees
- **FE-05:** Author Management
    - Get Authors
    - Create Author
    - Update Author
    - Soft Delete
    - Soft Delete Multiple Authors
    - Undo Soft Delete Author
    - Undo Soft Delete Multiple Authors
    - Delete Author
    - Delete Multiple Authors
    - Import/Export Authors
- **FE-06:** Category Management
    - Get Categories
    - Create Category
    - Update Category
    - Delete Category
    - Import/Export Categories
- **FE-07:** Library Item Management
    - Get Library Items
    - Get Library Item by ID
    - Create Library Item
    - Train AI Image Detecting
    - Train AI Image Detecting for Multiple Library Items
    - Group Library Item with Ubiquitous Features
    - Check OCR Item Cover Image Information
    - Update Library Item
    - Update Library Item Status (Draft/Public)
    - Update Library Item Shelf Location
    - Soft Delete Library Item
    - Soft Delete Multiple Library Items
    - Undo Soft Delete Library Item

    - Undo Soft Delete Multiple Library Items
    - Delete Library Item
    - Delete Multiple Library Items
    - Add Author to Library Item
    - Add Multiple Authors to Library Item
    - Delete Author from Library Item
    - Delete Multiple Authors from Library Item
    - Import/Export Library Items
- **FE-08:** Library Item Instance Management
    - Get Library Item Instances
    - Add Multiple Library Item Instances
    - Update Library Item Instance
    - Update Multiple Library Item Instances
    - Soft Delete Library item Instance
    - Soft Delete Multiple Library Item Instances
    - Undo Soft Delete Library Item Instance
    - Undo Soft Delete Multiple Library Item Instances
    - Delete Library Item Instance
    - Delete Multiple Library Item Instances
- **FE-09:** Library Item Condition Management
    - Get Library Item Conditions
    - Get Library Item Condition by ID
    - Create Library Item Condition
    - Update Library Item Condition
    - Delete Library Item Condition
- **FE-10:** Library Resource Management
    - Get Resources
    - Get Resource by ID
    - Add Multiple Resources
    - Update Resource
    - Soft Delete Resource
    - Soft Delete Multiple Resources
    - Undo Soft Delete Resource
    - Undo Soft Delete Multiple Resources
    - Delete Resource
    - Delete Multiple Resources
- **FE-11:** Library Card Management
    - Get Library Cards
    - Get Library Card by ID
    - Check Allow to Extend Library Card
    - Update Library Card
    - Archive Library Card

    - Confirm Library Card
    - Reject Library Card
    - Extend Library Card
    - Extend Library Card Borrow Amount
    - Suspend Library Card
    - Unsuspended Library Card
    - Delete Library Card
- **FE-12:** Library Card Holder Management
    - Get Library Card Holders
    - Get Library Card Holder by ID
    - Get Borrow Request by User ID
    - Get Borrow Record by User ID
    - Get Digital Borrow by User ID
    - Get Transaction by User ID
    - Get All Privacy Notifications by User ID
    - Create Library Card Holder (both User & Library Card)
    - Add Library Card
    - Update Library Card Holder
- **FE-13:** Library Location Management
    - Get Map Floors
    - Get Map by Floor ID
    - Get Shelves
    - Get Shelf by ID with Floor, Zone, Section
    - Get Shelf Units Summary Detail
- **FE-14:** Borrow Record Management
    - Get Borrow Records
    - Get Borrow Record by ID
    - Process Borrow Request
    - In-Library Checkout
    - Process Return
    - Extend Expiration Date
- **FE-15:** Borrow Request Management
    - Create Borrow Request
    - Add Items to Borrow Request
    - Cancel Specific Item
    - Cancel Specific Reservation
    - Cancel Specific Digital
- **FE-16:** Reservation Management
    - Get Reservations
    - Get Reservation by ID
    - Get Assignable by ID
    - Check Assignable

    - Assign Item Instances
    - Confirm Apply Label
    - Get Applied Label
    - Extend Pickup Date
    - Re-apply Label
- **FE-17:** Notification Management
    - Get All Notifications by User ID
    - Send Public/Private Notification
    - Get Privacy Notifications by User Id
    - Mark as Read
    - Mark All as Read
- **FE-18:** Transaction Management
    - Get Transactions
    - Get Transaction by ID
    - Get Transactions by User ID
    - Get Transaction Histories by User ID
    - Create Transaction for Extend/Register Library Card, Fine, Digital Borrow
    - Cancel Payment
- **FE-19**: Fine Management
    - Get Fine Policies
    - Get Fine Policy by ID
    - Delete Fine Policy
    - Update Fine Policy
- **FE-20:** Warehouse Tracking Management
    - Get Warehouse Trackings
    - Get Warehouse Tracking Detail
    - Create Stock-In Warehouse Tracking
    - Create Stock-Out Warehouse Tracking
    - Create Transfer Warehouse Tracking
    - Add Warehouse Tracking Detail
    - Update Warehouse Tracking
    - Update Warehouse Tracking Detail
    - Delete Warehouse Tracking
    - Delete Warehouse Tracking Detail
    - Import/Export Warehouse Tracking
- **FE-21:** System Configuration Management
    - Config App Settings
    - Config AI Settings
    - Config Borrow Settings
    - Config Payment Settings
- **FE-22:** Dashboard
    - Get Dashboard Overview

- Get Dashboard Circulation
- Get Dashboard Digital Resource
- Get Dashboard Financial and Transaction
- Get All Overdue Borrows
- Get All latest Borrows
- Get All Assignable Reservations
- Get All Top Circulation Items

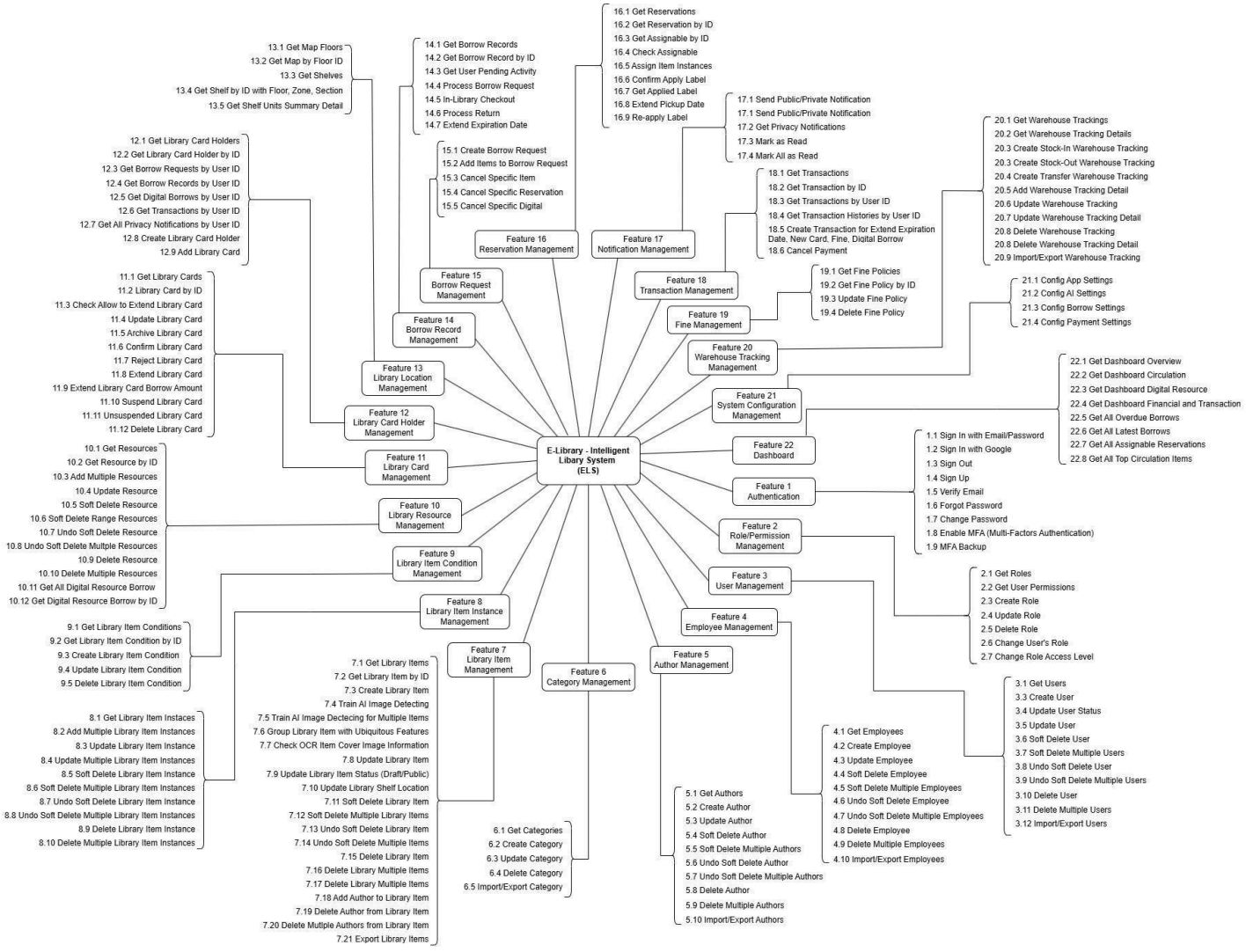


Figure 1.6.1 : Major features

6.2 Limitations & Exclusions

- **LI-1:** The 2D navigation feature is somewhat inconvenient for librarians because they have to import a manually created map beforehand, instead of being able to customize the map directly on the web.
- **LI-2**: The effectiveness of the voice recognition feature depends on the user, as there may be background noise during voice recording.
- **LI-3**: The OCR feature and smart recommendations through images may produce slow results because the outcome depends on the level of AI training.

- **LI-4**: The system does not completely prevent users from exposing books, taking pictures of system resources.
- **LI-5:** The AI training part is difficult because of capacity limitations, for example only 5000 training images uploaded, 50 tags, 20 iterations.
- **LI-6:** Elastic search has the risk of synchronizing with the database. In addition, if elastic search suddenly fails or loses connection, the filter part of the system will have major problems such as affecting users searching for books.
- **LI-7**: Lack of resource connectivity between libraries. The system is not applicable for use by multiple libraries/library branches at the same time.
- **LI-8**: The system performance may degrade when a large number of users access it concurrently, especially during peak hours.
- **LI-9**: The system lacks multilingual support. Currently, it only supports the two primary system languages (Vietnamese and English) and does not cater to users who speak other languages.
- **LI-10:** The integration of Artificial Intelligence in the system remains vague and lacks clear, practical implementations. Current AI features are underdeveloped and do not significantly enhance the user experience or management capabilities.
- **LI-11:** The library item recommendation mechanism is currently subjective and based primarily on basic user interactions. It does not incorporate advanced behavioral analysis or personalized algorithms, leading to less relevant suggestions for users.
- **LI-12:** The user interface has not yet been fully optimized in terms of usability and accessibility, potentially affecting user satisfaction and efficiency in navigation.
- **LI-13:** Both the web and mobile applications require a stable internet connection to function, limiting access for users in low-connectivity areas.
- **LI-14:** The system supports only a few payment options, which may not cater to all users' preferences or local payment practices.
- **LI-15:** The mobile application does not fully mirror the functionalities and data of the web application, leading to a fragmented user experience.
- **LI-16:** The current system does not support efficient inventory tracking or stock-out processes, posing challenges in material control and logistical operations.
- **LI-17:** The management of book information is not fully optimized. Categorization options are limited, and metadata management lacks flexibility, making it difficult to maintain an organized and diverse collection.
- **LI-18:** Penalty fees for damaged or lost items are currently determined by the librarians' evaluation, which may lead to inconsistency and a lack of transparency.
- **LI-19:** The system lacks a clear mechanism to handle reserved items once all copies are checked out. Reserved documents are not separated from general shelves, which causes confusion among users when they see reserved items still accessible on public shelves.
- **LI-20:** The current system has limited capacity in handling a large number of concurrent users or data-intensive operations. This may lead to reduced performance or downtime under heavy load, making it unsuitable for large-scale deployment without further optimization.

- **EX-1:** The system is restricted to library spaces only. It does not offer navigation outside the library, such as navigating to other campus buildings or providing campus-wide route guidance.
- **EX-2:** The system does not provide a way to login via social platforms such as Facebook, Instagram, Github, etc.
- **EX-3:** Custom map editing or design is not available through the frontend interface. Librarians must use third-party tools to generate maps and upload them manually.
- **EX-4:** The mobile application does not support AI features such as image-based book recommendations, which is currently only available on the web platform.

- **EX-5:** The system excludes the use of authentication methods (e.g., fingerprint or facial recognition) for login or verification due to device compatibility and privacy concerns.
- **EX-6:** The system does not include a real-time chat feature for direct communication with librarians or support staff.
- **EX-7:** The application does not provide offline functionality. Users cannot access core features such as book searching, reservation, or navigation without an internet connection.
- **EX-8:** The system does not include dynamic pricing or fine calculation features based on item popularity, return history, or demand fluctuations.
- **EX-9:** The system does not support auto-scaling or cloud-based deployment configurations out-of-the-box. Any scaling or load balancing setup must be manually configured and maintained by system administrators.

II. Project Management Plan

1. Overview

1.1 Scope & Estimation

Table 2.1.1 : Project Management Plan

#	WBS Item	Complexity	Est. Effort (man-days)
1	**Initiating**		
1.1	Collect Requirements	Medium	14
1.2	Deliver Report 1 – Project Introduction	Simple	7
1.3	Deliver Report 2 – Project Management Plan	Simple	7
1.4	Deliver Report 3 – System Requirement Specification	Medium	15
2	**Design**		
2.1	Design user interface prototypes	Complex	10
2.2	Deliver Report 4 – Software Design Description	Medium	18
3	**Training**		
3.1	Learn backend, frontend, and mobile frameworks	Medium	5
3.2	Study 2D indoor navigation systems (e.g., Woosmap)	Complex	7
3.2	Research AI-powered features (e.g., OCR, image-based book recommendations)	Complex	7
3.3	Research cloud service integration (e.g., Cloudinary)	Medium	5

3.4	Research payment gateway solutions (e.g., PayOS)	Medium	5
4	**Implementation (Core Features)**		
4.1	Web Application (Librarian)		
4.1.1	Warehouse tracking (stock-in, stock-out, stocktaking)	Medium	3
4.1.2	Cataloging library items (author, title, publisher, unique registration, etc.)	Medium	5
4.1.3	Image-based AI training	Medium	5
4.1.4	Digital resource management (eBooks, audiobooks)	Simple	2
4.1.5	Item condition history tracking	Simple	2
4.1.6	Audit log monitoring	Simple	2
4.1.7	Library patron management	Medium	5
4.1.8	Borrow/return operations	Medium	3
4.1.9	Shelf, zone, and section organization	Medium	3
4.1.10	Report and statistics generation	Medium	5
4.2	Web, Mobile Application (User)		
4.2.1	View library items	Simple	2
4.2.2	Utilize advanced search (e.g., OPAC, Elasticsearch, auto-complete)	Medium	3
4.2.3	Register for a library card	Simple	2
4.2.4	Borrow, return and reserve both physical and digital resources	Medium	5
4.2.5	Access transaction history for borrowings, returns, and reservations	Simple	2
4.2.6	Receive notifications	Simple	2
4.2.7	Leverage 2D indoor navigation to quickly locate library items	Complex	7
4.3	Web Application (Admin)		
4.3.1	Manage user roles and permissions	Medium	5

4.3.2	Configure system settings	Medium	5
4.3.3	Manage payment transactions	Simple	3
4.3.4	Oversee audit logs and system activities	Simple	3
5	**Integration**		
5.1	Integrate cloud services	Medium	3
5.2	Integrate payment gateway solutions	Medium	3
5.3	Integrate an AI-powered recommendation system	Complex	7
5.4	Implement a CI/CD pipeline using GitHub Actions	Complex	7
6	**Testing**		
6.1	Unit tests for backend services	Complex	12
6.2	Unit tests for frontend services	Medium	8
6.3	Integration tests	Complex	12
6.4	Delivery Report 5 - Software Test Documentation	Complex	18
7	**Deployment**		
7.1	Configure CD for the backend using Docker and Azure Container Registry	Medium	7
7.2	Deploy the frontend application on AWS Amplify	Medium	4
7.3	Final deployment and launch	Medium	7
7.4	Set up CI/CD with GitHub action and expo	Medium	7
8	**Closing**		
8.1	Deliver Report 6 – Software User Guide	Medium	7
8.2	Deliver Report 7 – Final Project Report	Simple	14

Total Estimated Effort: 290 (man-days)

1.2 Project Objectives

Table 2.1.2 : Project Objectives

#	Testing Stage	No. of Defects	% of Defect	Notes
1	Reviewing	10	10	
2	Unit Test	15	20	
3	Integration Test	45	55	

4	System Test	20	15	
	Total	100	100	

1.3 Project Risks

Table 2.1.3: Project Risks

No.	Risk Description	Impact	Possibility	Response Plans
1	Out of scope change in requirements	High	High	- Create a deliverable plan with clear objectives and requirements. - Make frequent communication with teammates and Mentor.
2	There is a team member who is quite busy with work and militia duty	Low	High	Take advantage of late nights to code.
3	Members of the team get sick	High	Low	When they recover, assign new tasks and move existing ones to the proper location.
4	Delay in timeline task	High	High	Ensure that each team member contributes the same amount of effort to the task.
5	Insufficient Testing	High	High	Create a thorough schedule that incorporates testing at every stage of the development process.
6	Failures of Payment Systems	High	Low	To guarantee redundancy, integrate several payment gateways, and keep an eye on transaction logs in real time for problems.
7	Unreliable User Interface	Medium	Medium	Iterative UI improvements are made through frequent usability testing and feedback loops with both customers and employees.
8	Lost or damaged data	High	Medium	Perform regular backups, use data recovery mechanisms.
9	Personal information leakage	High	Low	Use data encryption, access control, API protection.

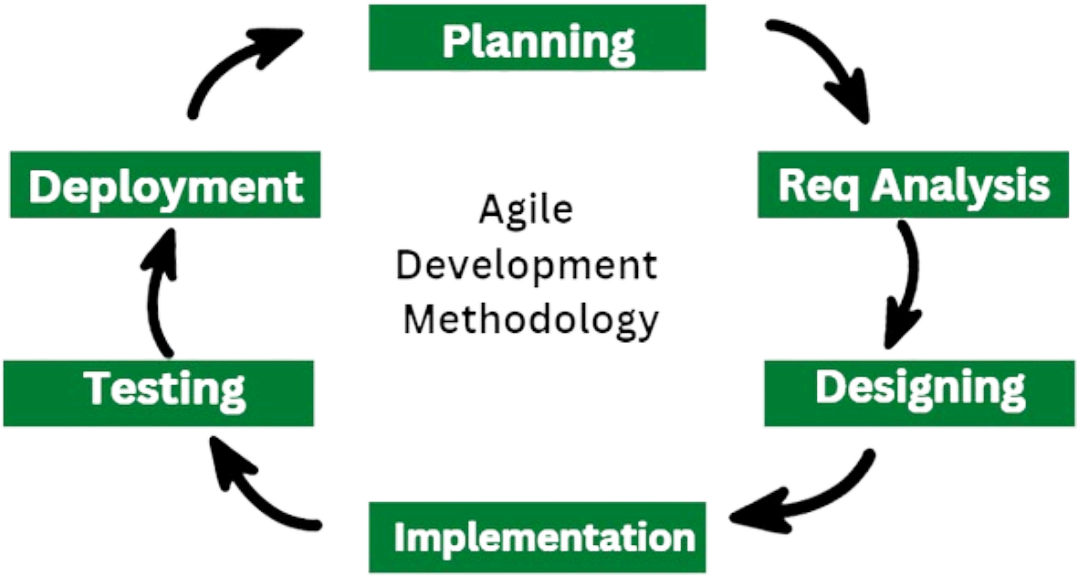
10	System errors cause service interruptions	High	Low	System monitoring, have a plan to quickly fix the problem.
11	Security attacks (SQL Injection, XSS, Brute Force,...)	High	High	Update security regularly, use a firewall, check data input.
12	Account security unsafe	High	High	Two-factor authentication (2FA), check for unusual login activity.
13	System overload when many users access at the same time	High	Medium	Optimize performance, expand infrastructure according to demand..
14	Abuse of admin's authority	High	High	Restrict rights by role, log data changes.
15	Bugs or unfriendly features	Medium	High	Test carefully before deployment, collect feedback from users.
16	Late book borrowing due to system errors	High	Low	Provide a complaint mechanism, quick customer support.
17	Cannot change the structure and properties of each POI directly on the app (ELS)	High	High	Go through the library's Console UI that supports navigation maps.
18	Response from AI features like OCR, recommendation, voice to text maybe wrong and delay	Medium	Medium	Improve AI accuracy by training models on diverse datasets, enabling user feedback, and incorporating manual validation mechanisms.
19	System updates expired and suspended time wrong (background service work improperly)	Medium	Medium	The system will aggregate failed tasks and retry them within a specific timeframe. If the retries continue to fail beyond a certain threshold, the system will notify the admin.
20	The email service provider failed to send the announcement email, resulting in the failure to remind	High	Medium	The system records failed attempts. A background service will aggregate them at the end of the day and retry sending emails.

	users to borrow books			

2. Management Approach

2.1 Project Process

Figure 2.2.1 : Project Process



Reference: https://www.geeksforgeeks.org/what-is-agile-methodology

This project applied the Agile model to ensure the flexibility, adaptability to changes. It is also separated into 2-week-long sprints and includes following stages for each sprint.

- **Planning** – The initial phase where requirements are gathered, goals are defined, and the overall roadmap for development is created.
- **Requirement Analysis** – Gather and define the project requirements
- **Designing** – Creating wireframes, prototypes, and system architecture to guide the development process.
- **Implementation** – Writing and implementing code to build the software solution.
- **Testing** – Conducting testing (unit, integration, and user testing) to ensure quality and functionality.
- **Deployment** – Releasing the product to users, either in a staging or production environment.

2.2 Quality Management

To ensure the quality of the ROMS, the following practices will be followed:

- **Code Review:** Code must be reviewed by at least one other team member before merging into the main branch.
- **API convention:** Clear route naming and standard response structures.
- **Unit Testing:** Ensure individual components work as expected.
- **Integration Testing:** Ensure seamless interaction between client and server.
- **System Testing:** Full system validation to ensure stability.

2.3 Training Plan

Table 2.2.3 : Training Plan

Training Area	Participants	When, Duration	Waiver Criteria
Figma design	Le Xuan Phuoc, Doan Viet Thanh	Week 1 - 6 days	Mandatory
Integrate real time with SignalR	Nguyen Vu Quang Huy, Tran Truong Van	Week 1 - 6 days	Mandatory
Integrate OCR, Voice to Text, AI Classification	Nguyen Vu Quang Huy, Doan Viet Thanh, Tran Truong Van	Week 2 - 14 day	Mandatory
Integrate payment gateway PayOS	Le Xuan Phuoc,Nguyen Vu Quang Huy	Week 4 - 6 days	Mandatory
Integrate Elastic Search	Le Xuan Phuoc, Tran Truong Van	Week 4 - 6 days	Mandatory
Integrate Face Detection, Object Detection AI	Le Xuan Phuoc, Nguyen Vu Quang Huy	Week 6 - 6 days	Mandatory
Create library map and locate items	Tran Truong Van, Le Xuan Phuoc	Week 7 -14 days	Mandatory
Integrate book recommendation based on user’s favorite	Le Xuan Phuoc	Week 7 -14 days	Mandatory

3. Project Deliverables

Table 2.3: Project Deliverables

No	Deliverable	Due Date	Deliverable Scope

1	Project Plan Document, SRS Document	Week 4	This deliverable includes a clear project plan that outlines the objectives, timelines, and resources required. The System Requirements Specification (SRS) will detail both functional and non-functional requirements, along with user profiles and business rules.
2	Design Document & UI Prototypes	Week 6	Describes the system's architecture, component interactions, and database schema. Also includes user interface prototypes that emphasize intuitive and accessible design.
3	Core Features Implementation (Phase 1)	Week 8	Implements fundamental modules such as authentication, account management, and basic library functionalities (cataloging, warehouse tracking). Introduces initial AI components and indoor navigation features.
4	Extended Features Implementation (Phase 2)	Week 10	Adds advanced functionalities: AI-based image recognition, digital resource management, user module enhancements (e.g., borrow/return/reserve), and admin tools (roles, payment transactions). Ensures deeper integration of learned frameworks.
5	Integration & Testing	Week 12	Integrates cloud services, payment gateways, and AI recommendations. Conducts unit, integration, and system tests.
6	Deployment	Week 14	Configures continuous deployment using Docker and Azure Container Registry for the backend, deploys the frontend to AWS Amplify, and finalizes system launch. Also includes setting up CI/CD pipelines.
7	Final Documentation & Closure	Week 15	Delivers the Software User Guide, Final Project Report, and completes project closure activities. This includes handing over all documentation.

4. Responsibility Assignments

D~Do; R~Review; S~Support; I~Informed; <blank>- Omitted

Table 2.4 : Responsibility Assignments

Responsibility	PhuocLX SE171957	HuyNVQ SE171850	VanTT SE170128	ThanhDV SE171867	Nguyễn Thị Cẩm Hương
Project Planning & Tracking	D	R	R	D	I
Prepare Project Introduction Document	D	S	S	D	I
Prepare SRS Document (Overview Part)	R	S	D	D	I
Prepare SRS Document (User Requirements)	D	D	S	S	I
Create and Prioritize Backlog	D	S	I	I	I
Sprint Planning	D	S	S	R	I
Daily Stand-ups	D	I	I	I	I
Development of Features	R	D	S	D	I
Testing & Quality Assurance	R	D	S	D	I
Sprint Review	I	D	S	I	I
Release Planning	D	S	I	I	I

5. Project Communications

5.1 Communication Plan

Table 2.5.1 : Communication plan

Communication Item	Who/ Target	Purpose	When, Frequency	Type, Tool, Method(s)
Working with advisor	Supervisor and team members	• Review documentation • Demonstrate features	1 time per week	Google Meet

		• Evaluate progress and result		
Working in team	Team members	• Raise opinions, problems • Ask for help from other members	many times per day	Messenger, Discord,Google Meet
Working in team	Team members	• Report the working status to the leader	1 time per week	Google Meet

5.2 External Interface

Table 2.5.2 : External Interface

Function	Contact Person (name, position)	Contact address (email, telephone)	Responsibility
Supervisor	Ms. Nguyễn Thị Cẩm Hương	huongntc2@fe.edu.vn	- Provide document template - Give instructions to the project team - Review deliverables - Supervise project status

6. Configuration Management

6.1 Document Management

- Use Google Docs, Google Sheets to save documents, system messages, issue logs and for real-time editing.
- Google docs: Summary links
- Private github organization: Github repository

6.2 Source Code Management

- Our team uses GitHub to manage our source code efficiently. The version control system enables team members to track commits, collaborate seamlessly, and merge code effectively.
- Follow Git management best practices:
    - Commit frequently with meaningful messages.
    - Review changes, format, lint, and check the build before committing
    - Use branches for new features.
    - Establish and follow an agreed-upon workflow.
    - Review and resolve conflicts before merging.

6.3 Tools & Infrastructures

Table 2.6.3 : Tools & Infrastructure

Category	Tools / Infrastructure
Technology	NextJS (front end), React Native (mobile app), .NET Core (back end), Elasticsearch (search engine)
Database	SQL Server
IDEs/Editors	Visual Studio Code, Visual Studio 2022, Rider
Diagramming	Visual Diagram, Draw.io
Documentation	Microsoft Office, Google Docs/Sheets/Slides
Version Control	GitHub (source code), Google Docs (documents)
Deployment server	Azure Container Registry, AWS Amplify
Project management	Jira
Design Tool	Figma, V0

III. Software Requirement Specification

1. Product Overview

The E-Library—Intelligent **Library System (ELS)** is an advanced library management solution designed specifically for large, multi-floor libraries. Utilizing 2D indoor mapping technology, ELS helps users navigate expansive library areas effortlessly, allowing them to find their desired books with ease. The system incorporates advanced search engine technology with AI-driven book prediction and recommendations, allowing users to search using a captured image of a book cover.

ELS aims to modernize traditional library functions by incorporating mobile and web applications that simplify the processes of borrowing, returning, and reserving books, which helps to minimize wait times. Additionally, a powerful data analytics dashboard provides library staff with valuable insights into inventory management, book popularity, and user engagement. This approach overcomes the challenges of traditional systems, making the library experience more efficient, user-friendly, and informed by data.

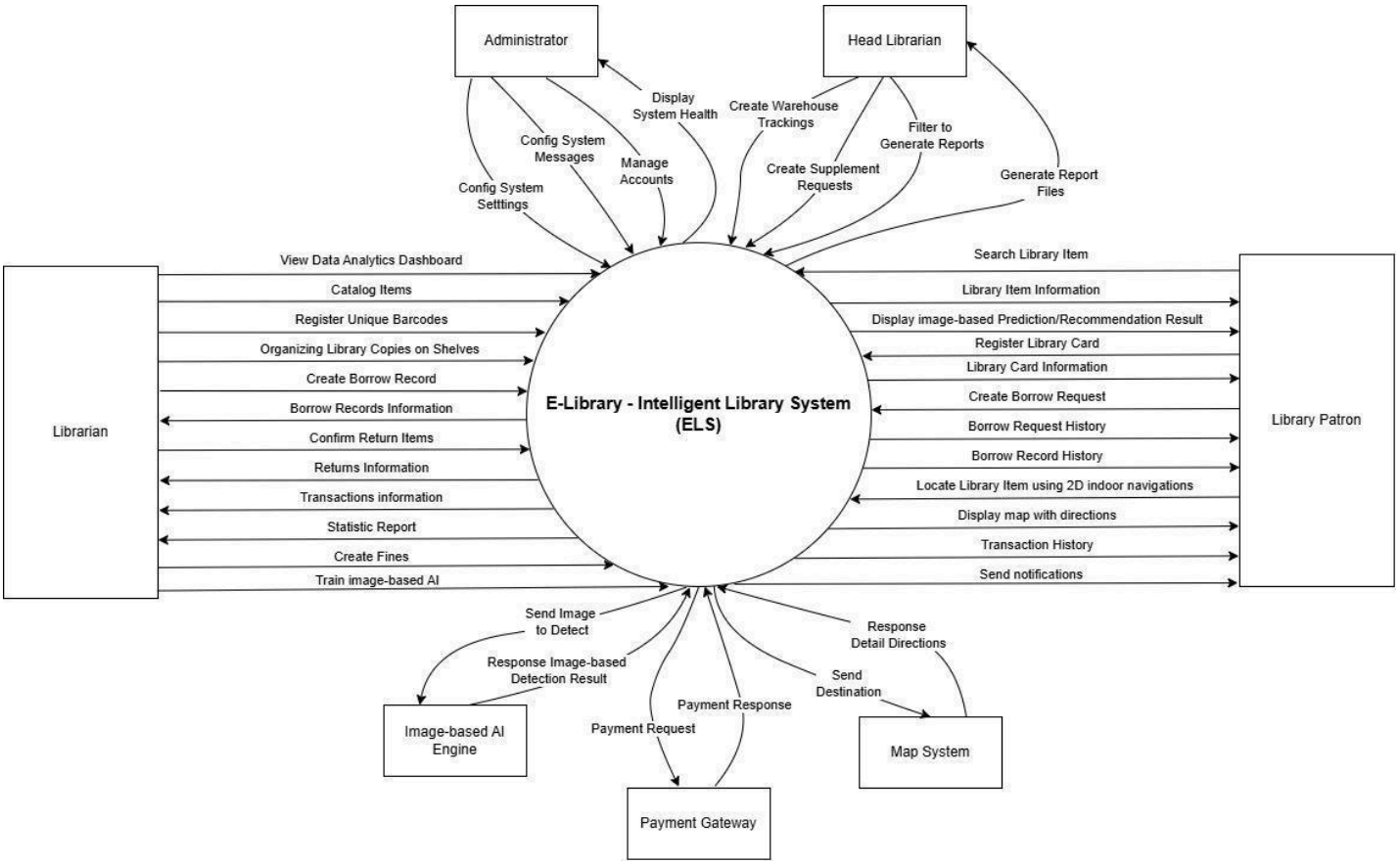


Figure 3.1: E-Library - Intelligent Library System

2. User Requirements

2.1 Actors

Table 3.2.1 : List of actors

#	Actor	Description
1	Library Patron	As a user, I want: 1. An easy-to-use search feature that supports multiple query methods including advanced text search and image-based (book cover) recognition, so that I can find my desired reading material without wasting time. 2. Integrated 2D indoor maps that help me navigate multi-floor libraries and find the exact location of my selected books, so that I can save time and avoid confusion in large libraries. 3. Personalized book suggestions based on my reading history, preferences, and behavioral data, so that I can discover new books and authors that match my interests.

		4. The ability to borrow or reserve books remotely via mobile or web applications, so that I don't need to wait in line or visit the library in person. 5. As a user, I want to review my borrowing and payment history in a dashboard, so that I can track my activity and manage my reading list effectively. 6. A system automatically sends notifications or reminders when my borrowed books are about to be due or when my reservation is about to expire, so that I can avoid late fees and better manage my borrowed items. 7. Access to eBooks or audiobooks through an interactive, user-friendly interface, so that I can enjoy reading or listening on various devices without physically visiting the library. 8. Rate books, leave comments, and read reviews from other library users, so that I can share my opinions and discover quality books more easily.
2	Librarian	As a librarian, I need: 1. Catalog detail metadata and assign unique identifiers (e.g., barcodes) for each new item, so that I can organize and retrieve resources efficiently. 2. Process item check-ins, check-outs, and returns quickly, so that I can minimize wait times for users and keep availability information accurate. 3. View user details and borrowing history, so that I can efficiently deal with overdue or lost book returns. 4. Upload or validate images of book covers, so that the AI-based recommendation and detection module becomes more accurate and helpful for users. 5. Detailed reports on borrowing rates, overdue items, and user activity, so that I can gain insights into library usage and improve resource planning.
3	Head Librarian	As a head librarian, I want: 1. To create bulk book data and track inventory in real time, so that I can maintain accurate, up-to-date records of all library items. 2. To create and process requests for additional copies or versions of popular or highly demanded items, so that high-usage resources meet user needs. 3. To filter and generate comprehensive reports for both stock-in entries and supplement requests, so that I can track all items added to the

		inventory and identify high-demand items that require additional purchases.
4	Administrator	As an administrator, I want: 1. To set and manage overall library policies (borrowing limits, fines, membership tiers), so that the system aligns with library rules and operational standards. 2. To create and assign roles with different access levels, so that system usage remains secure and well-structured. 3. To enforce data protection measures (e.g. multi-factor authentication), so that user information and library records remain safe. 4. Detailed audit logs of librarian actions for transparency and troubleshooting.

2.2 Use Cases

2.2.1 Diagram(s)

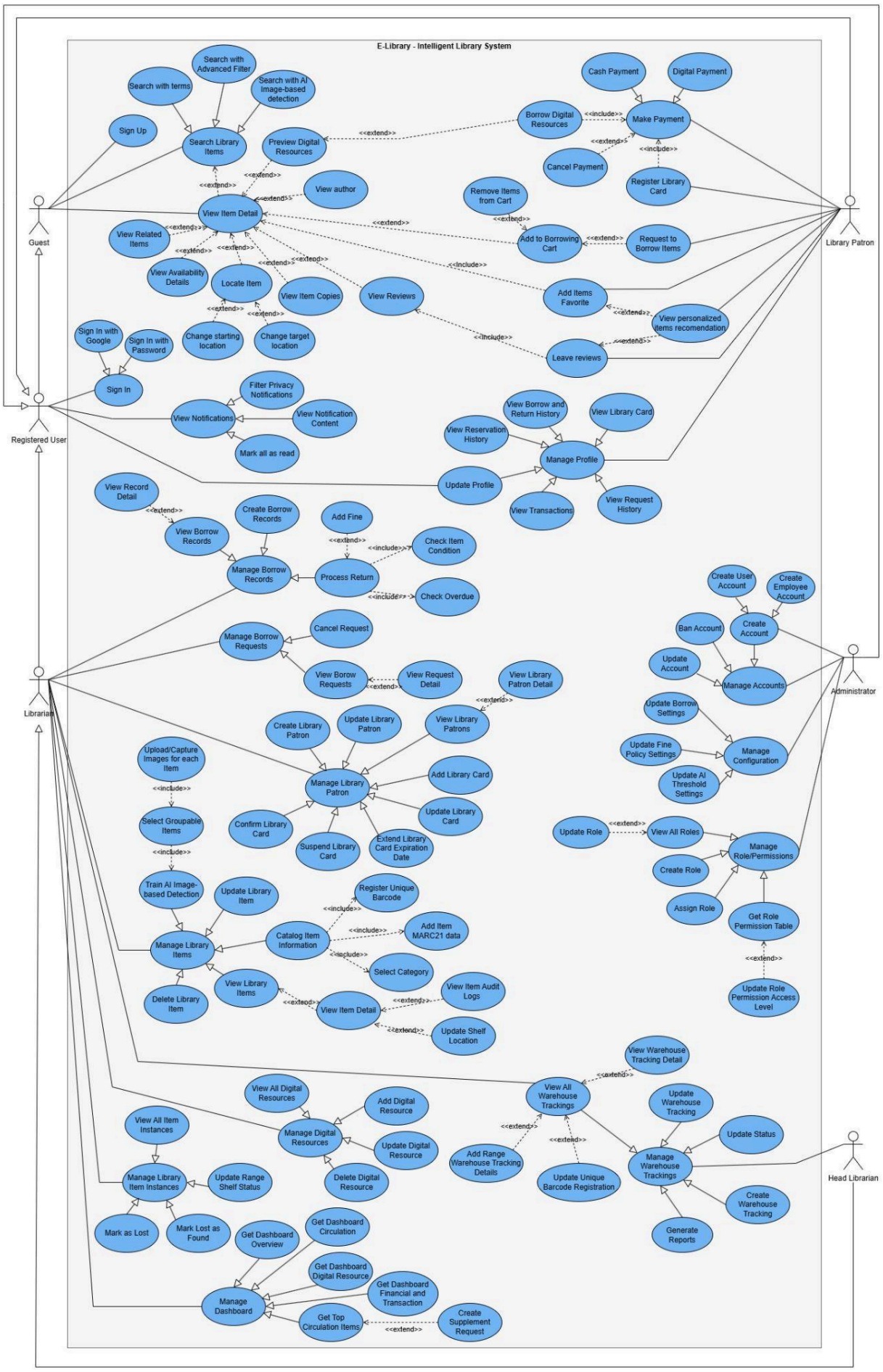

E-Library - Intelligent Library System
Guest
Registered User
Library Patron
Librarian
Administrator
Head Librarian
Search with Advanced Filter
Search with terms
Search with AI Image-based detection
Sign Up
Search Library Items
Preview Digital Resources
View author
View Item Detail
View Related Items
View Availability Details
Locate Item
View Item Copies
View Reviews
Change starting location
Change target location
Sign In with Google
Sign In with Password
Sign In
Filter Privacy Notifications
View Notifications
View Notification Content
Mark all as read
Cash Payment
Digital Payment
Borrow Digital Resources
Make Payment
Cancel Payment
Register Library Card
Remove Items from Cart
Add to Borrowing Cart
Request to Borrow Items
Add Items Favorite
View personalized items recomendation
Leave reviews
View Borrow and Return History
View Library Card
View Reservation History
Manage Profile
Update Profile
View Transactions
View Request History
View Record Detail
View Borrow Records
Create Borrow Records
Add Fine
Check Item Condition
Manage Borrow Records
Process Return
Check Overdue
Cancel Request
Manage Borrow Requests
View Borow Requests
View Request Detail
View Library Patron Detail
Create Library Patron
Update Library Patron
View Library Patrons
Manage Library Patron
Add Library Card
Update Library Card
Confirm Library Card
Suspend Library Card
Extend Library Card Expiration Date
Upload/Capture Images for each Item
Select Groupable Items
Train AI Image-based Detection
Update Library Item
Register Unique Barcode
Catalog Item Information
Add Item MARC21 data
Select Category
Manage Library Items
View Library Items
Delete Library Item
View Item Audit Logs
View Item Detail
Update Shelf Location
Create User Account
Create Employee Account
Ban Account
Create Account
Update Account
Manage Accounts
Update Borrow Settings
Update Fine Policy Settings
Update AI Threshold Settings
Manage Configuration
Update Role
View All Roles
Create Role
Assign Role
Manage Role/Permissions
Get Role Permission Table
Update Role Permission Access Level
View Warehouse Tracking Detail
View All Warehouse Trackings
Add Range Warehouse Tracking Details
Update Unique Barcode Registration
Update Warehouse Tracking
Update Status
Manage Warehouse Trackings
Create Warehouse Tracking
Generate Reports
View All Digital Resources
Add Digital Resource
Manage Digital Resources
Update Digital Resource
Delete Digital Resource
View All Item Instances
Manage Library Item Instances
Update Range Shelf Status
Mark as Lost
Mark Lost as Found
Get Dashboard Overview
Get Dashboard Circulation
Get Dashboard Digital Resource
Get Dashboard Financial and Transaction
Manage Dashboard
Get Top Circulation Items
Create Supplement Request
<<include>>
<<extend>>

Figure 3.2.2.1: Uses case diagram

2.2.2 Descriptions

Table 3.2.2.2: Uses case description

ID	Use Case	Actors	Use Case Description
1	Sign Up	Guest	Guest provides personal details (name, email, password) to register for a new account.
2	Sign In	Registered User	Registered user chooses a preferred authentication method and enters the necessary credentials.
3	Sign In with Password	Registered User	Registered user logs in by entering their email or username along with a password.
4	Sign In with Google	Registered User	Registered user selects to sign in using their Google account.
5	Sign Out	Registered User	Registered user signs out to end their session and the application no longer recognizes them as authenticated.
6	Search Library Items	Guest	Guest accesses the search interface to search for items within the library catalog.
7	Search with Terms	Guest	Guest enters keywords or phrases into the search field to search for a desired item.
8	Search with Advanced Filter	Guest	Guest refines their search by applying various filters such as publication date, category, or author.
9	Search with AI-based detection	Guest	Guest uses an AI-driven search feature by providing an image or AI input to predict items for uploaded image or recommend for the items that share the

			same genre, author, or topical terms.
10	View Item Detail	Guest	Guest can view library items with comprehensive details including the title, author, summary, and additional metadata.
11	Preview Digital Resources	Guest	Guest can preview a digital resource with a sample or a few pages from an eBook or digital content, enabling them to assess the material before deciding to borrow or purchase it.
12	View Author	Guest	Guest can access the author's profile, presenting a biography and a list of other works by the author.
13	View Reviews	Guest	Guest can read reviews and ratings submitted by other users for a specific item.
14	View Availability Details	Guest	Guest can view the availability of an item, indicating whether it is on the shelf, checked out, or on hold.
15	View Item Copies	Guest	Guest can view the number of copies available for an item.
16	Locate Item	Guest	Guest can locate an item position, including the specific library floor, branch, section and shelf location.
17	Change Starting Location	Guest	Guest can update their current position on the map.
18	Change Target Location	Guest	Guest can update their target location to pick up library items.

19	View Related Items	Guest	Guest can view suggestion items that share subjects, or metadata with the item currently being viewed.
20	Leave Reviews	Library Patron	Library patron able to submit feedback by writing a review and providing a rating for an item.
21	Update Profile	Registered User	Registered user modifies specific profile fields such as name, address, or phone number.
22	View Request History	Library Patron	Library patron can view all request history, including details of items and pick-up expiration date.
23	View Borrow / Return History	Library Patron	Library patron can view all borrowed records and returned items with a detailed display of dates, item titles, and any associated fees or fines.
24	View Reservation History	Library Patron	Library patron can view their reserved item history, which includes the earliest and latest available dates, the reserved item, and the pick-up expiration date.
25	View Transactions	Library Patron	Library patron views all payment transactions they have made, including dates, amounts, and payment methods.
26	Register Library Card	Library Patron	Library patron may register a new library card if none exists.
27	View Library Card	Library Patron	Library patron views their library card details, such as card number, expiration date, total borrow amount once, and extension histories.

28	Add to Borrowing Cart	Library Patron	Library patron adds physical library items to a temporary cart, allowing the user to request to borrow many items at the same time.
29	Remove Items from Cart	Library Patron	Library patron reviews items in their borrowing cart and removes any no longer desired.
30	Request to Borrow Items	Library Patron	Library patron finalizes a borrowing request by confirming items in the cart.
31	Borrow Digital Resources	Library Patron	Library patron can borrow digital resources that are not already recorded in their borrowing history.
32	Make Payment	Library Patron	Library patron can create payment for digital resource borrow, library card registration or fines.
33	Cash Payment	Library Patron	Library patron selects a cash-based payment method for their payment session.
34	Digital Payment	Library Patron	Library patron selects a digital payment method for their payment session.
35	Cancel Payment	Library Patron	Library patron decides to cancel the ongoing payment process before completion.
36	View personalized items recommendation	Library Patron	Library patron can view suggestions based on their borrowing history, preferences, or popular items.
37	Add Items Favorite	Library Patron	Library patron can mark certain items as favorites.

38	View Notifications	Library Patron	Library patron view list of recent notifications about new arrival items, due dates or fines.
39	View Notification Content	Registered User	Registered user selects a specific notification to read its details.
40	Filter Privacy Notifications	Registered User	Registered user applies filters to privacy messages.
41	Mark All as Read	Registered User	Registered user marks all unread notifications as read.
42	View Borrow Requests	Librarian	Librarian can view a list of items currently requested by patrons.
43	View Request Detail	Librarian	Librarian can view detailed borrow requests that include requested items and pick-up expiration dates.
44	Cancel Request	Librarian	Librarian can cancel a patron's request if the item is reported lost or cancellation required by the patron.
45	View Borrow Records	Librarian	Librarian can view a list of items currently borrowed by patrons.
46	View Record Detail	Librarian	Librarian can view detailed borrow records that include borrowed items, borrow and due dates, borrow request and reservation details, overdue status, fines, and extension requests.
47	Create Borrow Records	Librarian	Librarian creates a borrow record by processing approved borrow requests and directly checking out items, recording

			borrow and due dates along with patron information.
48	Process Return	Librarian	Librarian completes each return transaction, checks for possible fines, and updates return condition.
49	Add Fines	Librarian	Librarian apply a fee to a patron's account due to late returns or item damage, specifying the amount and reason for the charge.
50	Create Library Patron	Librarian	Librarian add a new patron by entering personal information, contact details to register the individual as a library user.
51	Update Library Patron	Librarian	Librarian modify existing patron information, such as correcting contact details, or updating any other relevant data in the patron's record.
52	View Library Patrons	Librarian	Librarian can view a list of all registered patrons.
53	Add Library Card	Librarian	Librarian can issue a new library card to a patron.
54	Update Library Card	Librarian	Librarian can modify library card information, which can include updating card information, expanding the total borrow amount once, marking a lost card as inactive or archiving the card if lost.
55	Extend Library Card Expiration Date	Librarian	Librarian lengthen the validity period of a patron's library card.

56	Suspend Library Card	Librarian	Librarian can suspend a patron's library card due to policy violations or other reasons, preventing checkouts until the suspension is over.
57	Confirm Library Card	Librarian	Librarian can verify library card information for the patron, who registers the card online.
58	View Library Patron Detail	Librarian	Librarian can view information about a specific patron, including borrowing history, reservation history, fines, and their payment transactions.
59	Catalog Item Information	Librarian	Librarian can apply MARC21 fields (e.g., title, author, publication data) to ongoing items for standardized cataloging and assign a unique barcode to each physical item for efficient checkouts and returns.
60	Train AI Image-based Detection	Librarian	Librarian can select groupable items, captures or uploads at least five images per item within each group, and processes AI-based image recognition training.
61	View Library Items	Librarian	Librarian can view a list of existing items in the system.
62	View Item Detail	Librarian	Librarian can view details of a specific item (including metadata, availability, instances, digital resources and shelf location).
63	Update Library Item	Librarian	Librarian can modify the existing details of an item (e.g., title, author, classification number, cutter number, etc.).

64	Delete Library Item	Librarian	Librarian can remove an item from the catalog when it is no longer in circulation (due to damage or loss).
65	Update Shelf Location	Librarian	Librarian can change an item's physical placement in the library.
66	View All Digital Resources	Librarian	Librarian can view a list of digital resources (e-books and audiobooks) along with their borrowing histories.
67	Add Digital Resource	Librarian	Librarian can add a new digital resource (such as an ebook or audiobook file) to a specific item, supplying relevant metadata (title, price, total borrow days).
68	Update Digital Resource	Librarian	Librarian can modify information about a digital resource (e.g., title, price, total borrow days, or updating file links).
69	Delete Digital Resource	Librarian	Librarian can remove a digital resource that is no longer valid or required.
70	View Item Instances	Librarian	Librarian can view a detailed list of all instances for a specific item, including unique identifiers, condition, and availability status.
71	Update Range Shelf Status	Librarian	Librarian can update range shelf status for item's instances (e.g., available, requested, borrowed, reserved, or lost).
72	Update Range Condition	Librarian	Librarian can update the condition of multiple item instances (e.g.,good, worn, damage, lost).

73	View Item Audit Logs	Librarian	Librarian can view an item's record of past changes, including updates, relocations, or borrowing activities.
74	View All Warehouse Trackings	Head Librarian, Librarian	Head librarian and librarian can view a list of warehouse tracking records that detail stock-in and stock-out events.
75	View Warehouse Tracking Details	Head Librarian, Librarian	Head librarian and librarian can view detailed warehouse tracking records, including the item title, total quantity, ISBN, and stock transaction type.
76	Create Warehouse Tracking	Head Librarian	Head librarian can create new warehouse records for multiple stock-in and stock out items.
77	Add Range Warehouse Tracking Details	Head Librarian	Head librarian can manually add details or scan for each warehouse tracking record, including the item title, total quantity, ISBN and stock transaction type.
78	Update Warehouse Tracking	Head Librarian	Head librarian can modify details of an existing warehouse record.
79	Update Unique Barcode Registration	Head Librarian, Librarian	Head librarian and librarian can update and register a unique barcode range for each tracking detail item based on their total stock in amount.
80	Update Status	Head Librarian	Head librarian modifies an item's warehouse status (e.g., draft, completed, cancelled)
81	Create Account	Administrator	Administrator creates a new account (user or employee) in the system.

82	Ban Account	Administrator	Administrator restricts or suspends a user or employee account, preventing further logins or activities due to policy violations.
83	Update Account	Administrator	Administrator modifies account information, such as changing personal details.
84	Update Fine Policy Settings	Administrator	Administrator modifies the rules governing library fines, defining fee amounts, and other fine-related parameters.
85	Update Borrow Settings	Administrator	Administrator configures borrowing policies, such as maximum checkout limits, and borrow durations.
86	Update AI Threshold Settings	Administrator	Administrator configures AI-related parameters, for detection accuracy thresholds for image-based searches or recommendation features.
87	View System Performance	Administrator	Administrator views system performance (e.g., response times, resource usage).
88	Manage Role/Permissions	Administrator	Administrator organizes roles and associated permissions, determining which actions each role can perform within the system.
89	View All Roles	Administrator	Administrator can view a list of existing roles in the system.
90	Create Role	Administrator	Administrator creates new roles with all system permissions restricted by default.

91	Assign Role	Administrator	Administrator assigns a specific role to a user or employee.
92	Get Role Permission Table	Administrator	Administrator can view details of which actions each role is allowed to perform.
93	Update Role Permission Access Level	Administrator	Administrator can modify a user's permission for a specific role with a specific feature within the system.

3. Functional Requirements

3.1 System Functional Overview

3.1.1 Screens Flow

a) Website

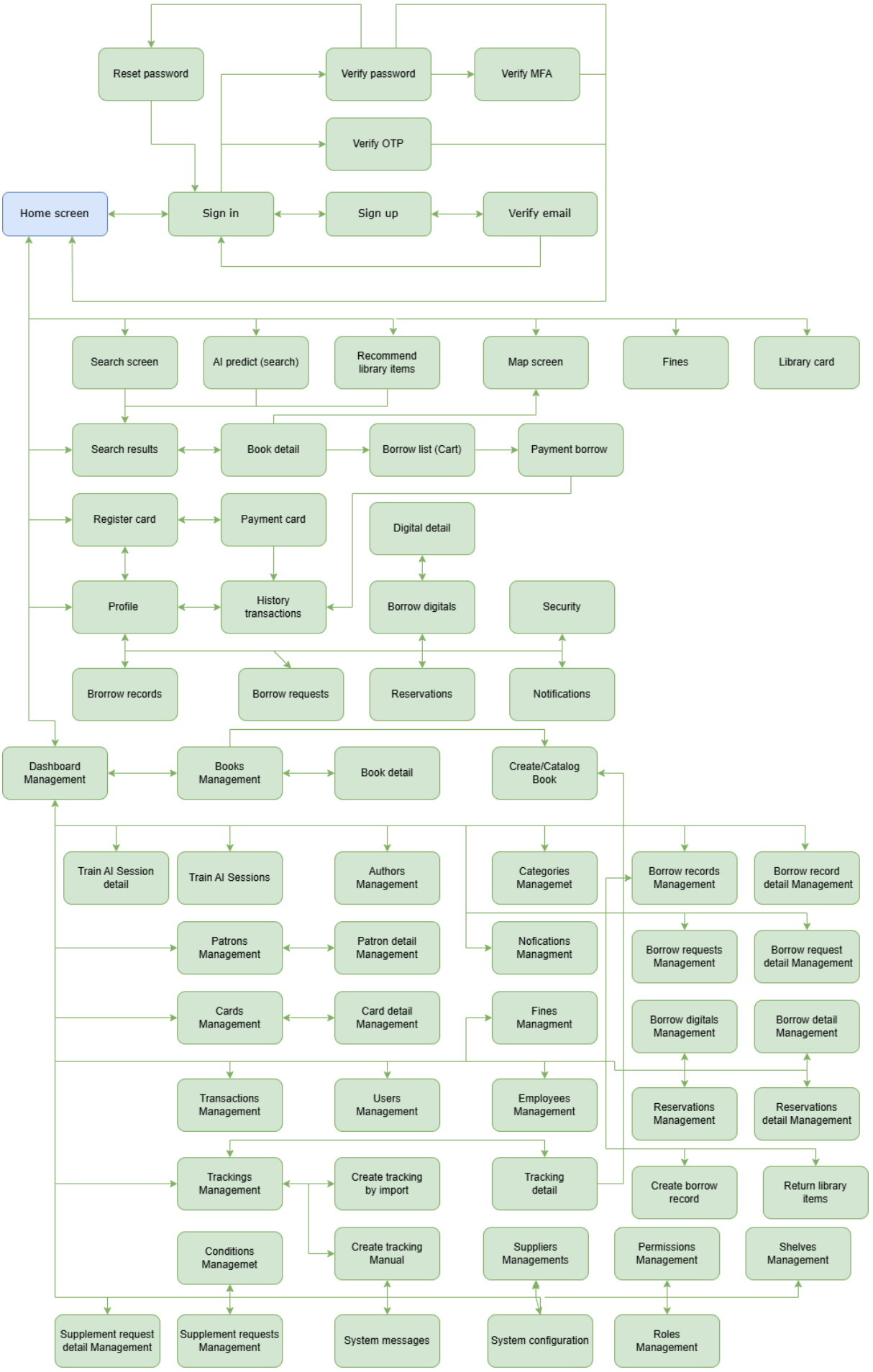

Reset password
Verify password
Verify MFA
Verify OTP
Home screen
Sign in
Sign up
Verify email
Search screen
AI predict (search)
Recommend library items
Map screen
Fines
Library card
Search results
Book detail
Borrow list (Cart)
Payment borrow
Register card
Payment card
Digital detail
Profile
History transactions
Borrow digitals
Security
Brorrow records
Borrow requests
Reservations
Notifications
Dashboard Management
Books Management
Book detail
Create/Catalog Book
Train AI Session detail
Train AI Sessions
Authors Management
Categories Managemet
Borrow records Management
Borrow record detail Management
Patrons Management
Patron detail Management
Nofications Managment
Borrow requests Management
Borrow request detail Management
Cards Management
Card detail Management
Fines Managment
Borrow digitals Management
Borrow detail Management
Transactions Management
Users Management
Employees Management
Reservations Management
Reservations detail Management
Trackings Management
Create tracking by import
Tracking detail
Create borrow record
Return library items
Conditions Managemet
Create tracking Manual
Suppliers Managements
Permissions Management
Shelves Management
Supplement request detail Management
Supplement requests Management
System messages
System configuration
Roles Management

Figure 3.1.1.a : Website Screen Flow

b) Mobile

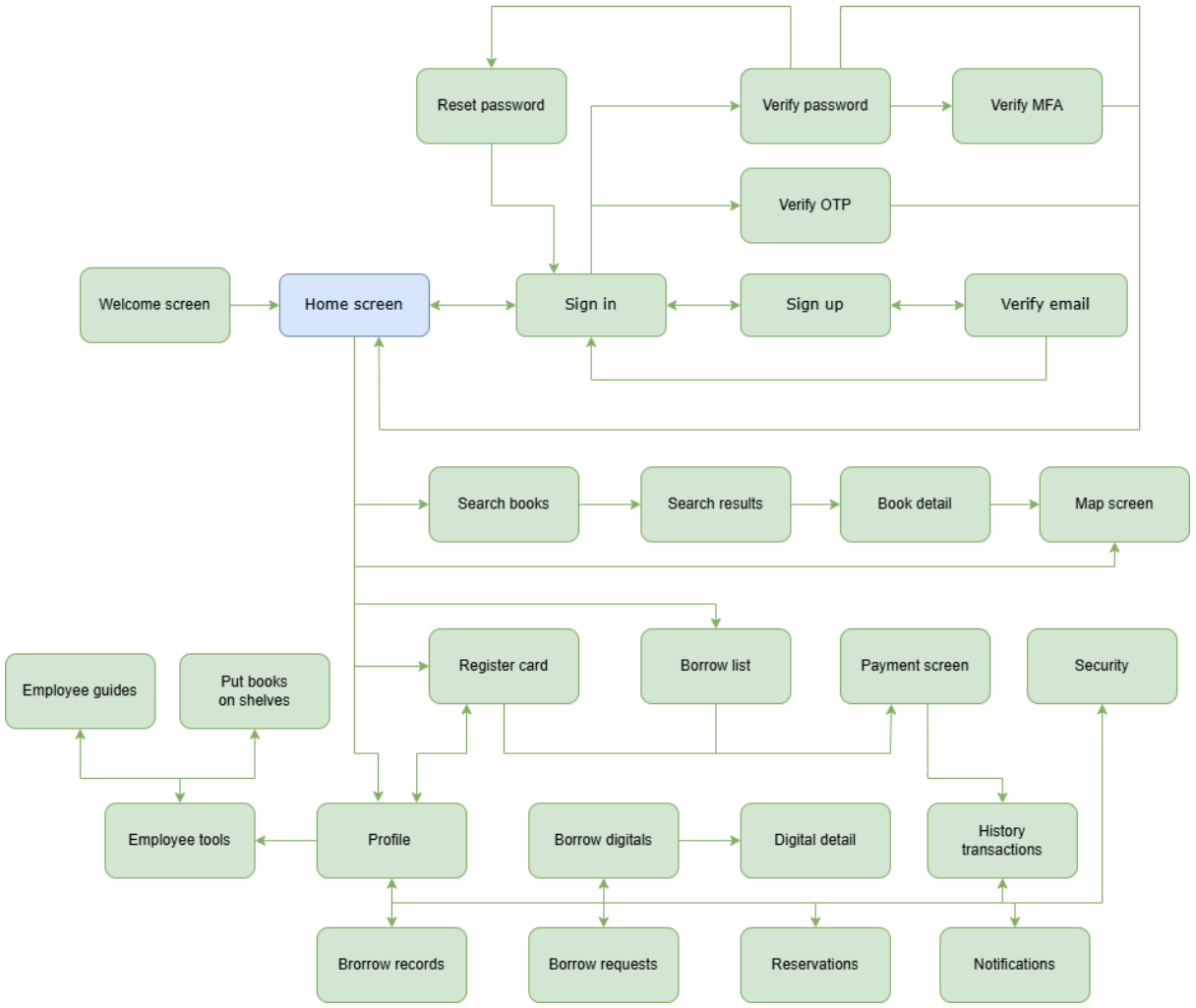


Figure 3.1.1.b : Mobile Screen Flow

3.1.2 Screen Descriptions

a) Website

Table 3.1.2.a : Website Screen Descriptions

#	Feature	Screen	Description
1	Home	Home screen	The initial screen that displays main navigation options and features for the users.
2	Authentication	Sign in	Allow users to sign in using Google One Tap authentication or enter their email to proceed with the login process. Alternatively, they can click "Sign Up" if they do not have an account.

3	Authentication	Verify password	Users need to enter their password to complete the login process or will be redirected to the MFA verification screen if two-factor authentication is enabled.
4	Authentication	Verify OTP	Users need to enter the OTP sent via email to complete the login process or will be redirected to the MFA verification screen if two-factor authentication is enabled.
5	Authentication	Verify MFA	Users need to enter the MFA code from an authenticator app or similar application to complete the login process.
6	Authentication	Sign up	Users can sign up using Google One Tap authentication or by entering all the required information to complete the registration.
7	Authentication	Verify email	Users need to enter the code sent via email to complete the registration process.
8	Authentication	Reset password	Users need to enter the verification code sent via email before they can reset their password.
9	Search	Search screen	Provide library patrons with three types of search options: quick search, basic search, and advanced search, allowing them to easily find library items within the system.
10	Search	Search results	Display all relevant results after a search.
11	Search	AI predict	Allow library patrons to upload an image of a book and then provide corresponding search results.
12	Search	Recommend library items	Provide book recommendations based on interactions of library patrons.
13	Indoor navigation	Map Screen	Allow library patrons to navigate the library's internal map and provide directions to specific locations.
14	Library items	Library item detail	Allow library patrons to view detailed information about a library item.
15	Borrow	Borrow list	Allow library patrons to view the items they have added to their list before deciding to borrow them.
16	Borrow	Payment borrow	Allow library patrons to make payments for digital materials.

17	Library cards	Register card	Allow library patrons to register for a library card.
18	Library cards	Payment card	Allow library patrons to make a payment for their library card to complete the registration process.
19	Library patrons	Profile	Allow library patrons to view and edit their personal information.
20	Library patrons	History transactions	Allow library patrons to view their transaction history.
21	Library patrons	Borrow requests	Allow library patrons to view their borrow requests.
22	Library patrons	Borrow records	Allow library patrons to view their borrowed records.
23	Library patrons	Borrow digital	Allow library patrons to view the digital materials they have borrowed.
24	Library patrons	Digital detail	Allow library patrons to read and/or listen to digital materials.
25	Library patrons	Reservations	Allow library patrons to view their reservations.
26	Library patron	Notifications	Allow library patrons to view notifications sent to them.
27	Library patron	Security	Allow library patrons to change their password, enable MFA, or generate new MFA backup codes.
28	Report management	Dashboard management	Allow librarians to view an overview of system statistics.
29	Library items management	Library items management	Allow librarians to view, search, import, and export library items.
30	Library items management	Library item detail management	Allow librarians to view detailed information about a library item, edit its details, and update its status.
31	Library items management	Train AI	Display groups of untrained library items and allows selecting one or multiple groups for training.
32	Authors management	Authors management	Allow librarians to view/search authors, create, edit, and delete authors.
33	Categories management	Categories management	Allow librarians to view/search categories, create, edit, and delete categories.

34	Notifications management	Notifications management	Allow librarians to view/search notifications, create notifications
35	Fines management	Fines management	Allow librarians to view/search fines, create, edit, and delete fines.
36	Patrons management	Patrons management	Allow librarians to view, search, import, and export patrons.
37	Patrons management	Patron detail	Allows librarians to edit or delete patron information. They can also add, modify, approve, reject, delete, extend, suspend, or archive a patron's library card.
38	Library cards management	Library card management	Allow librarians to view, search, import, and export library cards.
39	Library cards management	Library cards detail	Allow librarians to modify, approve, reject, delete, extend, suspend, or archive a patron's library card.
40	Transactions management	Transactions management	Allow librarians to view, search transactions.
41	Users management	Users management	Allows librarians to view/search, import, and export users. They can also add, edit user information, delete users, and modify user roles.
42	Employees management	Employees management	Allows librarians to view/search, import, and export users. They can also add, edit employee information, delete employees, and modify employee roles.
43	Suppliers management	Suppliers management	Allow librarians to view/search suppliers and create, edit, and delete suppliers.
44	Conditions management	Conditions management	Allow librarians to view/search conditions, create, edit, and delete conditions.
45	Trackings management	Trackings management	Allow librarians to view, search trackings.
46	Trackings management	Create tracking by import	Allows librarians to create trackings by uploading an Excel file.
47	Trackings management	Create tracking manual	Allows librarians to create trackings by manually entering information for each library item.
48	Trackings management	Tracking detail	Allows librarians to view, edit, and delete tracking. They can also catalog, modify, confirm glue barcodes to each library item in the tracking and export the tracking file.

49	Supplement requests Management	Supplement requests Management	Allow librarians to view, search supplement requests.
50	Supplement requests Management	Supplement request detail	Allows librarians to view, edit, and delete supplement requests. They can export the supplement request file.
51	Borrow Management	Borrow records Management	Allow librarians to view, search the borrowed records.
52	Borrow Management	Borrow record Detail Management	Allow librarians to view detailed borrowing records.
53	Borrow Management	Create borrow record	Allow librarians to scan patron's cards and library items to create a borrow record.
54	Borrow Management	Return library items	Allow librarians to scan patron's cards, library items, assign fines to return library items.
55	Borrow Management	Borrow requests Management	Allow librarians to view, search the borrow requests.
56	Borrow Management	Borrow request Detail Management	Allow librarians to view detailed borrow requests.
57	Borrow Management	Borrow digitals Management	Allow librarians to view, search the borrowed digitals.
58	Borrow Management	Borrow digital Detail Management	Allow librarians to view details of digital borrows.
59	Borrow Management	Reservations Management	Allow librarians to view, search the reservations.
60	Borrow Management	Reservation Detail Management	Allow librarians to view detailed reservations. They can also assign an item to the reservation and extend the pickup day.
61	Shelves Management	Shelves Management	Allow librarians to view floors, zones, sections, shelves in indoor maps.
62	Roles Management	Roles Management	Allows admin to view, add, and edit system roles.
63	Permissions Management	Permissions Management	Allows admin to view and edit permissions for each role.

64	System	System messages	Allows admin to view system messages.
65	System	System configurations	Allows users to view and modify system settings.

b) Mobile

Table 3.1.2.b : Mobile Screen Flow

#	Feature	Screen	Description
1	Home	Home	Displays the list of library items based on the available category.
2	Authentication	Sign in	Displays the login form allowing to fill out email or login by Google.
3	Authentication	Verify password	Allow users to enter their password to complete the login process or will be redirected to the MFA verification screen if two-factor authentication is enabled.
4	Authentication	Verify OTP	Allow users to enter the OTP sent via email to complete the login process or will be redirected to the MFA verification screen if two-factor authentication is enabled.
5	Authentication	Verify MFA	Allow users to enter the MFA code from an authenticator app or similar application to complete the login process.
6	Authentication	Sign up	Displays the registration form to fill out email, firstname, lastname, password, and confirmed password.
7	Authentication	Verify email	Allow users to enter the code sent via email to complete the registration process.
8	Authentication	Reset password	Allow users to enter the verification code sent via email before they can reset their password.
9	Search	Select search	Displays 3 search navigation buttons: search page, search by AI prediction, and search by AI recommendation.
10	Search	Search page	Allow library patrons to search with quick search, basic search, or advanced search.

11	Search	Search page result	The system navigates users to the result search page.
12	Search	AI prediction	Allow library patrons to search by uploaded images from their image gallery.
13	Search	AI prediction result	Displays the AI prediction result with 4 tabs: Ocr result, ocr detail, ocr detection, and the other matches
14	Library item	Library item detail	Displays library item details, including cover image, library item, and author information.
15	Show Map	Map	Displays the shelf information and the directions of specific library items.
16	QR	QR scanning	Allow to scan with a camera.
17	Profile	Profile	Displays personal information like email, first name, last name, phone number, address, gender, avatar.
18	Library cards	Register card	Allow library patrons to register for a library card.
19	Library cards	Payment card	Allow library patrons to make a payment for their library card to complete the registration process.
20	Borrow request	Borrow request	Displays the details of the borrow request including the list of library items, reserved items, and resources.

3.1.3 Screen Authorization

a) Website

Table 3.1.3.a : Website Authorization

Screen	Guest	Patrons	Librarian	Head Librarian	Admin
Home screen	x	x	x	x	x
Sign in		x	x	x	x
Verify password		x	x	x	x
Verify OTP		x	x	x	x
Verify MFA		x	x	x	x
Sign up	x	x	x	x	x
Verify email	x				
Reset password		x	x	x	x

Screen	Guest	Patrons	Librarian	Head Librarian	Admin
Search screen	x	x	x	x	x
Search results	x	x	x	x	x
AI predict	x	x	x	x	x
Recommend library items	x	x			
Map Screen	x	x	x	x	x
Library item detail	x	x	x	x	x
Borrow list		x			
Payment borrow		x			
Register card		x			
Payment card		x			
Profile		x	x	x	x
History transactions		x			
Borrow requests		x			
Borrow records		x			
Borrow digital		x			
Digital detail		x			
Reservations		x			
Notifications		x	x	x	x
Security		x	x	x	x
Dashboard management			x	x	x
Library items management			x	x	x
Library item detail management			x	x	x
Train AI			x	x	x
Authors management			x	x	x
Categories management			x	x	x
Notifications management			x	x	
Fines management			x	x	
Patrons management			x	x	x
Patron detail			x	x	x
Library card management			x	x	x

Screen	Guest	Patrons	Librarian	Head Librarian	Admin
Library cards detail			x	x	x
Transactions management			x	x	
Users management					x
Employees management					x
Suppliers management			x	x	
Conditions management			x	x	x
Trackings management			x	x	
Create tracking by import			x	x	
Create tracking manual			x	x	
Tracking detail			x	x	
Supplement requests Management			x	x	
Supplement request detail			x	x	
Borrow records Management			x	x	
Borrow record Detail Management			x	x	
Create borrow record			x	x	
Return library items			x	x	
Borrow requests Management			x	x	
Borrow request Detail Management			x	x	
Borrow Digital Management			x	x	
Borrow digital Detail Management			x	x	
Reservations Management			x	x	
Reservation Detail Management			x	x	

Screen	Guest	Patrons	Librarian	Head Librarian	Admin
Shelf Management			x	x	x
Permissions Management					x
Roles Management					x
System message					x
System configuration					x

b) Mobile

Table 3.1.3.b : Mobile Authorization

Screen	Guest	Patrons	Librarian	Head Librarian	Admin
Home	x	x	x	x	x
Sign in		x	x	x	x
Verify password		x	x	x	x
Verify OTP		x	x	x	x
Verify MFA		x	x	x	x
Sign up	x	x	x	x	x
Verify email	x				
Reset password		x	x	x	x
Select search	x	x	x	x	x
Search page	x	x	x	x	x
Search page result	x	x	x	x	x
AI prediction	x	x	x	x	x
AI prediction result	x	x	x	x	x
Library item detail	x	x	x	x	x
Map	x	x	x	x	x

Screen	Guest	Patrons	Librarian	Head Librarian	Admin
QR scanning			x		
Profile		x	x	x	x
Register card		x			

3.1.4 Non-Screen Functions

Table 3.1.4: Non-Screen Functions

#	Feature	System Function	Description
1	Authentication and Authorization	JWT Token Generation	Generates JSON Web Tokens (JWT) for secure authentication and authorization, allowing users to access protected resources.
2	Background Job Processing	Job Scheduling	Schedules background jobs for tasks such as sending notifications, deactivating contracts, and handling other asynchronous processes.
3	Notification Service	Email Notifications	Sends email notifications for important events or updates.
4	Security	Data Encryption	Encrypts sensitive payment history and account money-related data.

3.1.5 Entity Relationship Diagram

Conceptual ERD

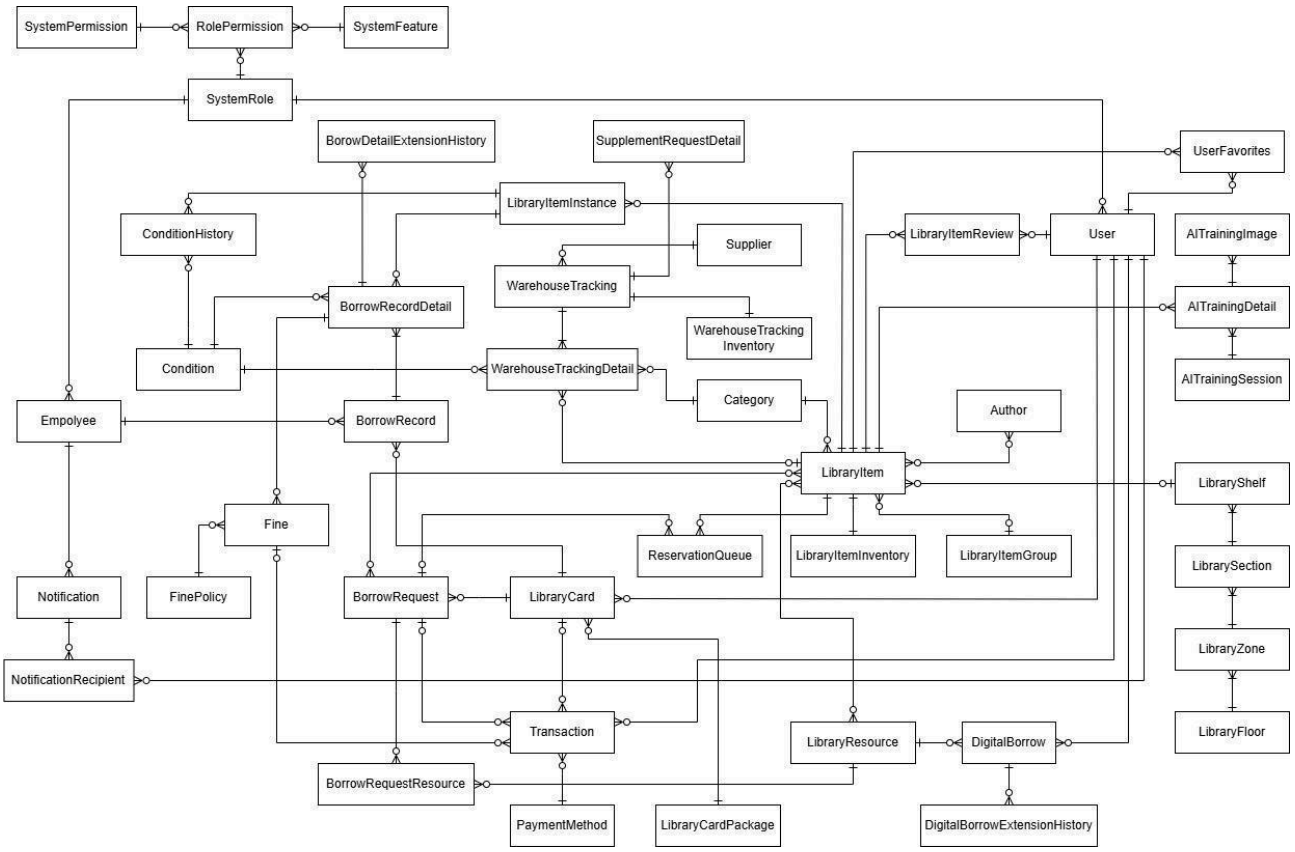


Figure 3.1.5.1: Conceptual ERD

Logical ERD

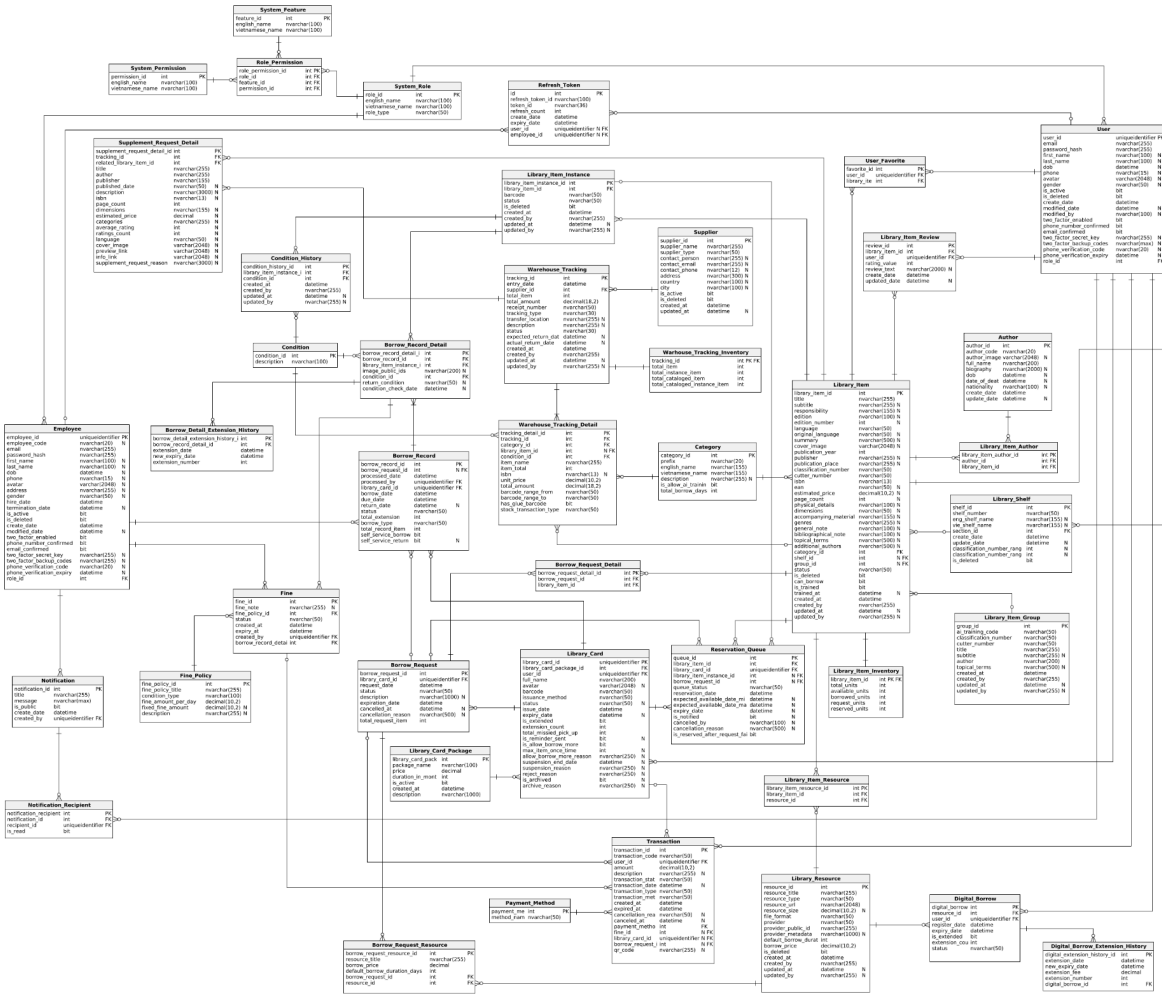


Figure 3.1.5.2: Logical ERD

Entities Description

Table 3.1.5 : Entity Description

#	Entity	Description
1	AI_Training_Session	Contains AI training session information. - Primary keys: training_session_id - Foreign keys:
2	AI_Training_Detail	Store a detailed AI training session for a specific library item. - Primary keys: training_detail_id - Foreign keys: training_session_id, library_item_id
3	AI_Training_Image	Stores images URL for AI training details. - Primary keys: training_image_id - Foreign keys: training_detail_id
4	Author	Represents an author with biographical details. - Primary keys: author_id - Foreign keys:

5	Borrow_Request	Manages a borrow request submitted by a library patron. - Primary keys: borrow_request_id - Foreign keys: library_card_id
6	Borrow_Request_Detail	Details the items requested in a borrowing request. - Primary keys: borrow_request_detail_id - Foreign keys: borrow_request_id, library_item_id, library_item_instance_id
7	Borrow_Request_Resource	Stores additional resource details. - Primary keys: borrow_request_resource_id - Foreign keys: borrow_request_id, resource_id, transaction_id
8	Borrow_Record	Stores borrow history with details for a library patron. - Primary keys: borrow_record_id - Foreign keys: borrow_request_id, library_card_id, proceesed_by
9	Borrow_Record_Detail	Stores details for items borrowed in a record. - Primary keys: borrow_record_detail_id - Foreign keys: borrow_record_id, library_item_instance_id, condition_id, return_condition_id, processed_return_by
10	Borrow_Detail_Extension_ History	Stores extension information on a borrowed record detail. - Primary keys: borrow_detail_extension_history_id - Foreign keys: borrow_record_detail_id
11	Category	Defines categories for library items. - Primary keys: category_id - Foreign keys:
12	Digital_Borrow	Tracks digital borrow transactions for resources by library patrons. - Primary keys: digital_borrow_id - Foreign keys: resource_id, user_id
13	Digital_Borrow_Extension_ History	Tracks the extension history for a digital borrowing record. - Primary keys: digital_extension_history_id - Foreign keys: digital_borrow_id
14	Employee	Represents library staff with employment details. - Primary keys: employee_id - Foreign keys: role_id
15	Fine	Records fines imposed for overdue, damaged, or lost returned items. - Primary keys: fine_id - Foreign keys: borrow_record_detail_id, fine_policy_id, createdBy
16	Fine_Policy	Stores fine policy configurations. - Primary keys: fine_policy_id

		- Foreign keys:
17	Library_Card	Contains data for library membership cards. - Primary keys: library_card_id - Foreign keys: library_card_package_id, user_id
18	Library_Card_Package	Defines membership packages. - Primary keys: library_card_package_id - Foreign keys:
19	Library_Floor	Represents library floors. - Primary keys: floor_id - Foreign keys:
20	Library_Zone	Defines zones within the library associated with a floor. - Primary keys: zone_id - Foreign keys: floor_id
21	Library_Section	Represents a section within the library. - Primary keys: section_id - Foreign keys: zone_id
22	Library_Shelf	Represents a shelf in a library section. - Primary keys: shelf_id - Foreign keys: section_id
23	Library_Item	Catalogs library items with bibliographical, physical, and cataloguing information. - Primary keys: library_item_id - Foreign keys: category_id, shelf_id, group_id
24	Library_Item_Instance	Represents a specific physical copy of a library item. - Primary keys: library_item_instance_id - Foreign keys: library_item_id
25	Library_Item_Author	Represents works for specific authors. - Primary keys: library_item_author_id - Foreign keys: library_item_id, author_id
26	Library_Item_Inventory	Stores real-time counts for library items. - Primary keys: library_item_id - Foreign keys: library_item_id
27	Library_Item_Resource	Links a library item to additional digital resources. - Primary keys: library_item_resource_id - Foreign keys: library_item_id, resource_id
28	Library_Item_Condition	Defines standard conditions for library items. - Primary keys: condition_id - Foreign keys:
29	Library_Item_Condition_History	Store history changes for the condition of a specific library item instance. - Primary keys: condition_history_id - Foreign keys: library_item_instance_id, condition_id
30	Library_Item_Group	Groups library items for cataloguing or AI training purposes. - Primary keys: group_id

		- Foreign keys:
31	Library_Item_Review	Contains reviews and ratings for library items. - Primary keys: review_id - Foreign keys: library_item_id, user_id
32	Library_Resource	Stores metadata about digital resources. - Primary keys: resource_id - Foreign keys:
33	Notification	Contains system notifications. - Primary keys: notification_id - Foreign keys: created_by
34	Notification_Recipient	Maps notifications to recipients and tracks the read status for each notification. - Primary keys: notification_recipient_id - Foreign keys: notification_id, recipient_id
35	Payment_Method	Lists available payment methods - Primary keys: payment_method_id - Foreign keys:
36	Reservation_Queue	Tracks reservations placed on library items. - Primary keys: queue_id - Foreign keys: library_item_id, library_item_instance_id, library_card_id, borrow_request_id
37	Refresh_Token	Manages refresh tokens for user or employee session management. - Primary keys: id - Foreign keys: user_id, employee_id
38	Role_Permission	Associates system roles with specific features and permissions. - Primary keys: role_permission_id - Foreign keys: role_id, feature_id, permission_id
39	System_Feature	Lists system features. - Primary keys: feature_id - Foreign keys:
40	System_Permission	Defines permission levels. - Primary keys: permission_id - Foreign keys:
41	System_Role	Represents roles within the system. - Primary keys: role_id - Foreign keys:
42	Supplier	Contains supplier details, including contact information and address. - Primary keys: supplier_id - Foreign keys:
43	Supplement_Request_Detail	Stores supplementary bibliographic information for items requested to be added to the library. - Primary keys: supplement_request_detail_id

		- Foreign keys: related_library_item_id, tracking_id
44	Warehouse_Tracking	Stores records for warehouse operations. - Primary keys: tracking_id - Foreign keys: supplier_id
45	Warehouse_Tracking_Detail	Provides detailed tracking data per library item in the warehouse. - Primary keys: tracking_detail_id - Foreign keys: tracking_id, library_item_id, category_id, condition_id
46	Warehouse_Tracking_Inventory	Summarizes warehouse inventory counts related to a tracking record. - Primary keys: tracking_id - Foreign keys: tracking_id
47	Transaction	Manages financial and operational transactions that occur within the system. - Primary keys: transaction_id - Foreign keys: user_id, fine_id, resource_id, library_card_package_id, payment_method_id
48	User	Stores user account information. - Primary keys: user_id - Foreign keys: role_id, library_card_id
49	User_Favorite	Stores a user's favorite library items. - Primary keys: favorite_id - Foreign keys: user_id, library_item_id

3.2 Applications for Guests and Users

3.2.1 Login Page

- **Function Description:** The login page allows users to log in using a previously created account or sign in with Google. This page also allows users to navigate to the Register page or the Home page.
- **Function Trigger:**

  When the user is not logged in, the Login page will display the Login form:

  - **Log in to the web application via Google:**
    - **Button/Navigation:** There is a Google login button with the content "Continue with Google."
    - **Trigger:** When clicked, it shows the Google Login Form.
    - **Function:** Users can log in to their Google account.
  - **Verify Email:**
    - **Button/Navigation:** There is an email input in the login form.
    - **Trigger:** After the user enters their email address, they click the button with the text "Continue."

        - **Function:** The system checks whether the email exists. If the email already exists and is valid, the user is redirected to the password entry page. If the email does not exist, an error is displayed prompting the user to enter a different email.

    - **Navigate to the home page:**
        - **Button/Navigation:** There is a “Home Page” button at the right of the login form.
        - **Trigger:** When clicked, it navigates users to the home page.
        - **Function:** The web application displays the whole content on the home page.
- **Business Rules:** BR-42, BR-69

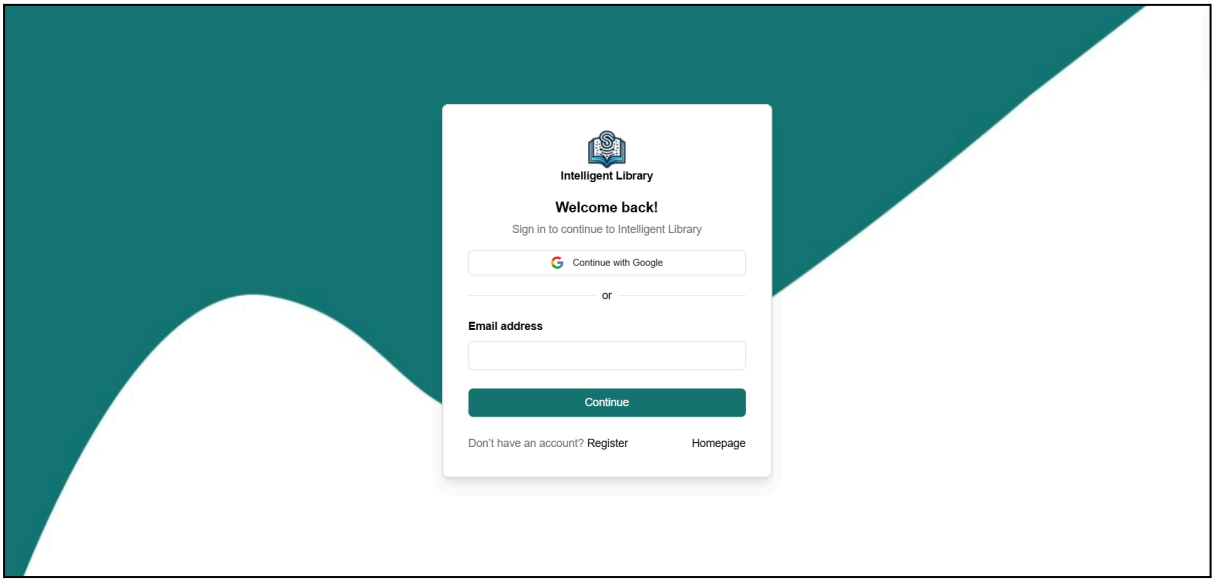


Figure 3.2.1: Login Page

3.2.2 Registration Page

- **Function Description:** The registration page allows users to register a new account by filling out the account registration form. This page also allows users to log in via the user’s Google account, navigate to the home page, or come back to the login page.
- **Function Trigger:**

    When the user has not had an account before, the registration page will display the registration form:

    - **Log in to the web application via Google:**
        - **Button/Navigation:** There is a Google login button with the content “Continue with Google."
        - **Trigger:** When clicked, it shows the Google Login Form.
        - **Function:** Users can log in to their Google account.

    - **Register a new account:**

- **Button/Navigation:** There is a registration form in the center of the screen.
- **Trigger:** User has to fill out all of the required fields, like first name, last name, email address, password, and confirmed password.
- **Function:** The system checks whether the email exists. If the email already exists and is valid, the user is redirected to the password entry page. If the email does not exist, an error is displayed prompting the user to enter a different email.

❖ **Navigate to the home page:**

- **Button/Navigation:** There is a “Homepage” button at the right of the login form.
- **Trigger:** When clicked, it navigates users to the home page.
- **Function:** The web application displays the whole content on the home page.

❖ **Navigate to the login page:**

- **Button/Navigation:** There is a “Sign in” button at the bottom of the registration form.
- **Trigger:** When clicked, it navigates users to the login page.
- **Function:** The web application displays the whole content on the login page.

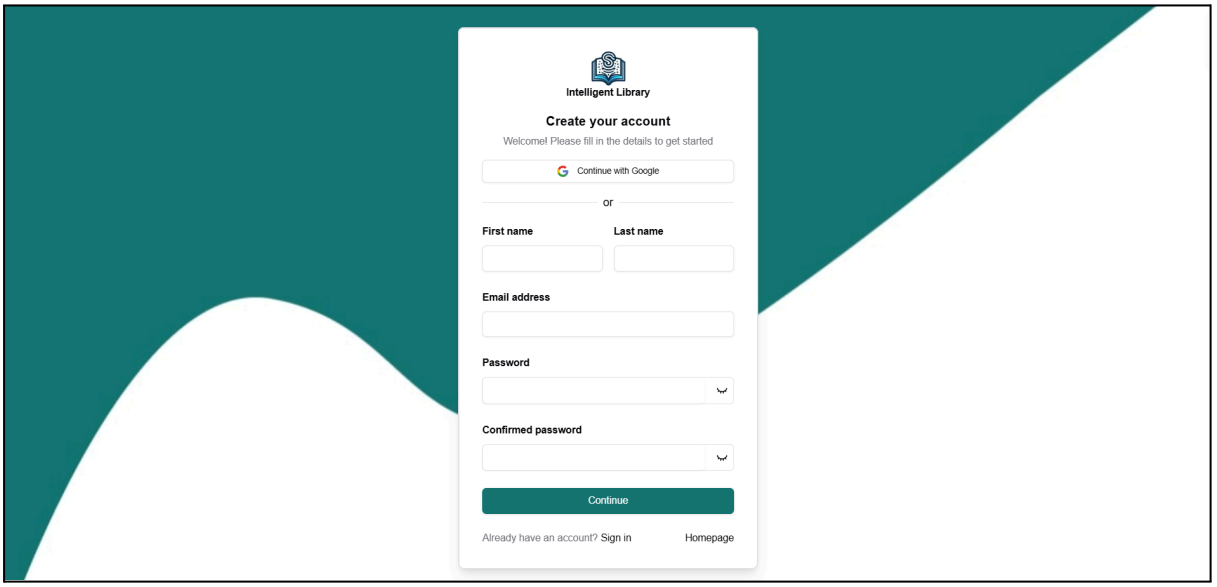


Figure 3.2.2: Registration Page

3.2.3 Email Verification Page (without account)

- **Function Description:** The user must type the OTP code that was sent to the user's email to verify the code.
- **Function Trigger:**
  When the user has not had an account before, the registration page will display the registration form:

❖ **Enter the OTP code:**
- **Button/Navigation:** There is a OTP input in the center of the Email
- **Trigger:** User types the OTP code received from the user's email, then clicks the "Continue" button.
- **Function:** The system verifies the OTP code.

❖ **Resend the OTP code via email to the user:**
- **Button/Navigation:** There is a "Resend" button below OTP input.
- **Trigger:** When clicked, the user receives a new OTP.
- **Function:** The system resends a new OTP code via email to the user.

❖ **Navigate to the login page:**
- **Button/Navigation:** There is a "Back to login" button at the bottom of the email verification form.
- **Trigger:** When clicked, it navigates users to the login page.
- **Function:** The web application displays the whole content on the login page.

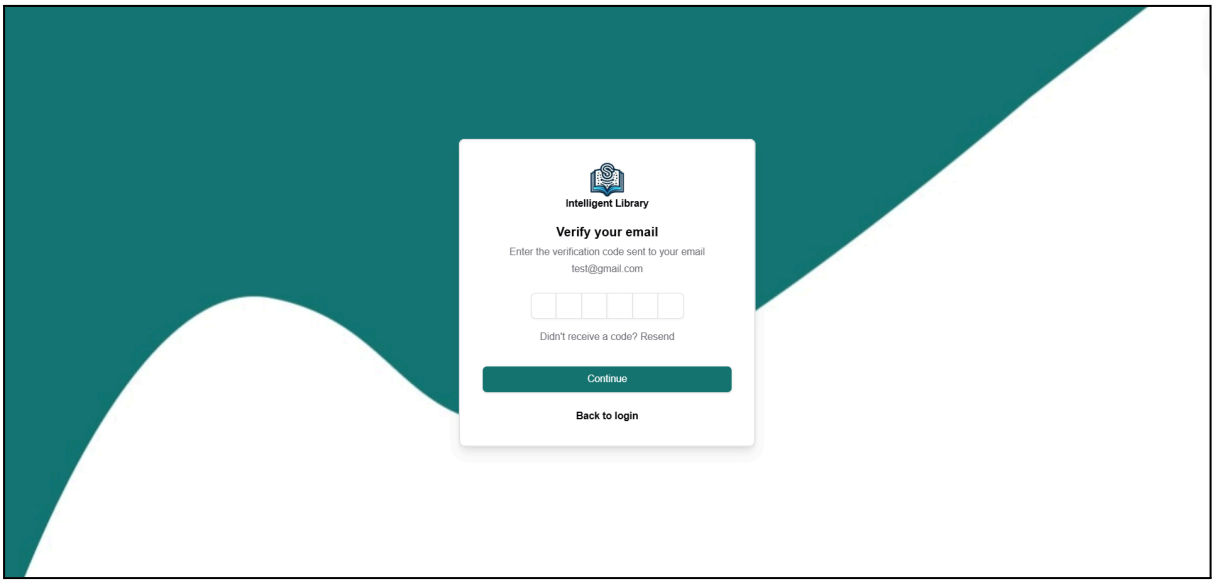


Figure 3.2.3: Email Verification Page

3.2.4 Reset password

- **Function Description:** The user types an OTP value that was sent to the user's email to reset the password.
- **Function Trigger:**

User receives a new OTP code, which was sent to the user's email

❖ **Enter the OTP code:**
- **Button/Navigation:** There is a OTP input in the center of the Email
- **Trigger:** User types the OTP code received from the user's email, then clicks the "Continue" button.
- **Function:** The system verifies the OTP code.

❖ **Resend the OTP code via email to the user:**

- **Button/Navigation:** There is a “Resend” button below OTP input.
- **Trigger:** When clicked, the user receives a new OTP.
- **Function:** The system resends a new OTP code via email to the user.

❖ **Navigate to the login page:**

- **Button/Navigation:** There is a “Back to login” button at the bottom of the email verification form.
- **Trigger:** When clicked, it navigates users to the login page.
- **Function:** The web application displays the whole content on the login page.

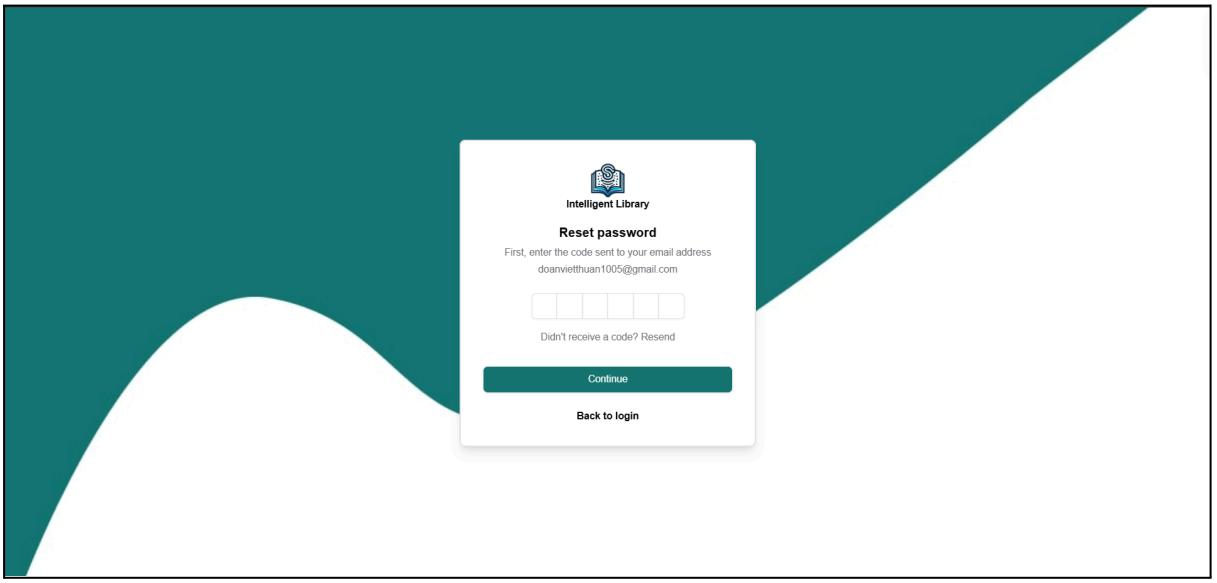


Figure 3.2.4: Reset Password Page

3.2.5 Home Page

- **Function Description:** The home page allows customers (both logged-in users and guests) to view books based on available book genres from the system, displays a list of recently accessed books, searches for books (voice search, filtering books by fields), multilingual conversion (Vietnamese-English), can switch between light and dark themes, allows logging in and registering for a library card (if not available yet).
- **Function Trigger:**

When the user is not logged in, the home page will display basic information about the library and include the following functions:

❖ **Log in to the web application:**

- **Button/Navigation:** There is a login button at the bottom of the sidebar.
- **Trigger:** When clicked, it navigates to the authentication page.
- **Function:** Users can log in or register an account.

❖ **Switch to multi-language:**

- **Button/Navigation:** There is a multi-language button on the home page.
- **Trigger:** When clicked, it allows users to select the Vietnamese or English language.

        - **Function:** All website content is converted between Vietnamese and English depending on user choice.
    - ❖ **Filter library items:**
        - **Button/Navigation:** There is a filter button on the header of the web application.
        - **Trigger:** When clicked, it shows the filter section.
        - **Function:** Displays three tabs like quick search, basic search, and advanced search.
    - ❖ **Search in voice:**
        - **Button/Navigation:** There is a mic icon on the header of the web.
        - **Trigger:** When clicked, it allows users to record their voice.
        - **Function:** Displays library items based on users' record of voice.
    - ❖ **Register the library card:**
        - **Button/Navigation:** There is a "Register library card" button.
        - **Trigger:** When clicked, it navigates the user to the Library Card page in profile management (in case the user logged in and the user hasn't registered a personal library card).
        - **Function:** Once logged in, users can register the personal library card to borrow library resources.

- **Business Rules:** BR-01, BR-17, BR-38, BR-40, BR-79

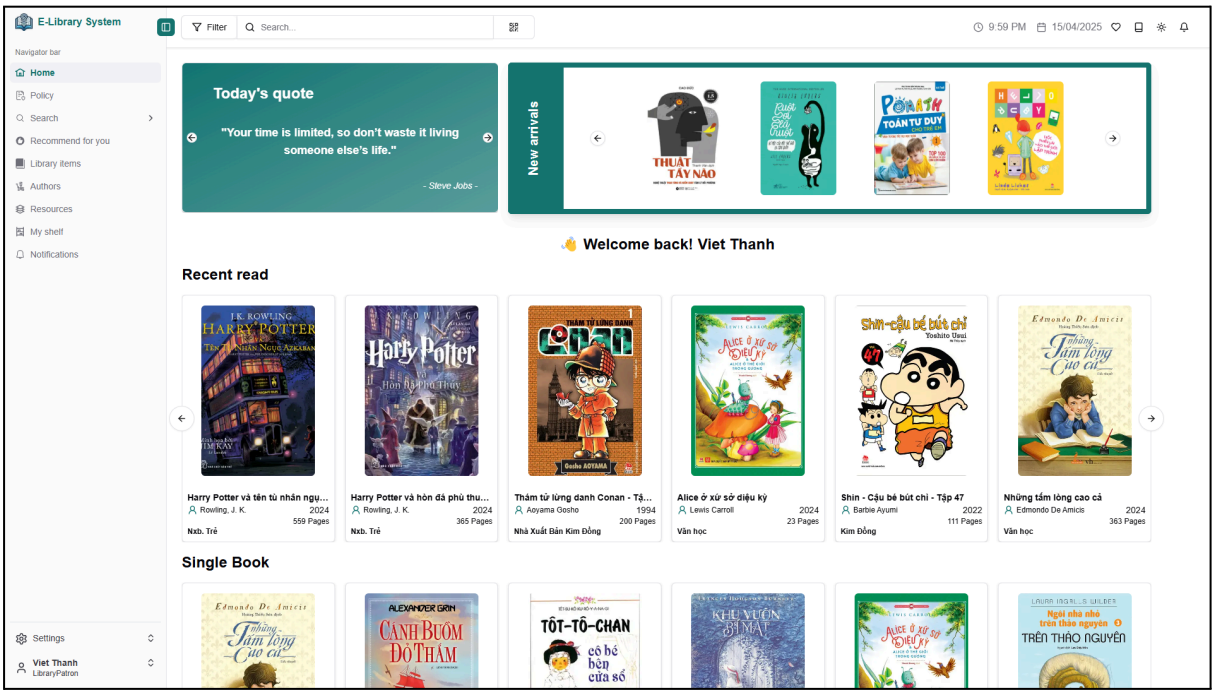


Figure 3.2.5.1: Home Page

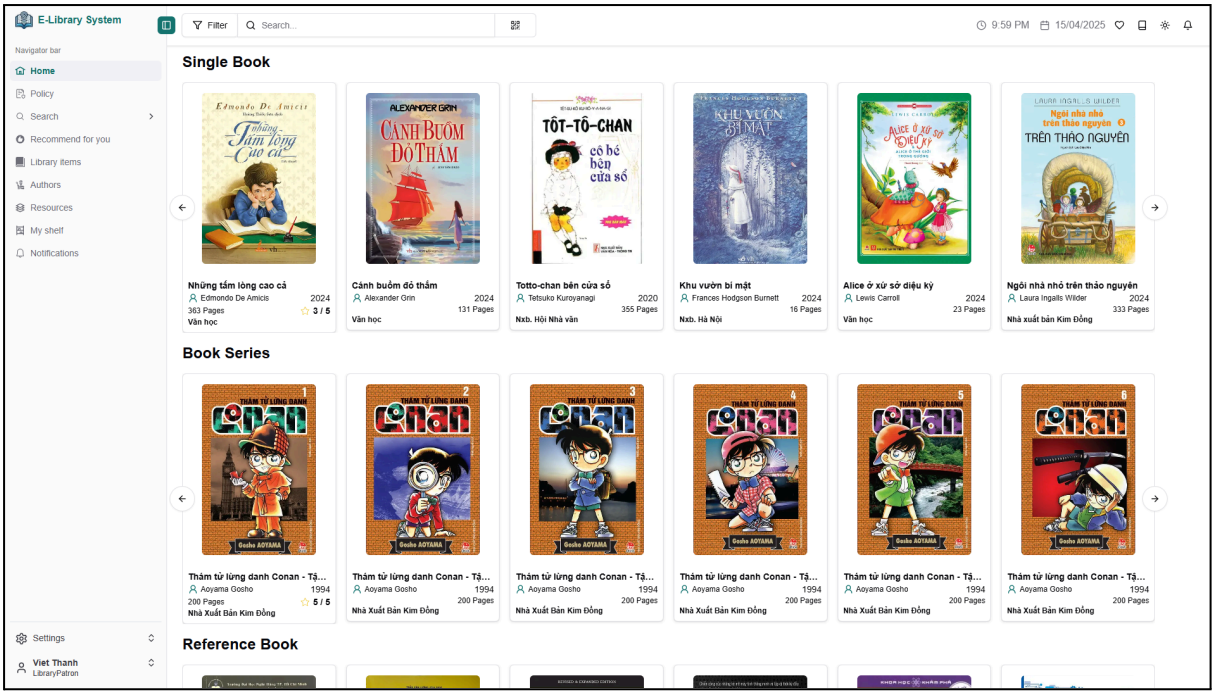


Figure 3.2.5.2: Home Page

3.2.6 Library Item Detail

- **Function Description:** The users can view library item details, including cover image, library item, and author information. Furthermore, the user can also locate the position, borrow, view resources of the library item.
- **Function Trigger:**

  - **Locate the library item:**
    - **Button/Navigation:** There is a “Locate” button in the book preview section
    - **Trigger:** The users click the “Locate” button to find the exact location of the library item.
    - **Function:** The system will display the location map of the library item directly on the phone along with the directions.

  - **Add the library item to the user’s favorite list:**
    - **Button/Navigation:** There is an “Add to favorite list” button in the book information section
    - **Trigger:** Users click the “Add to favorite list” button to add the library items to their favorite list
    - **Function:** The system saves users’ favorite lists of library items on their local web.

  - **Add the library item to the user’s borrow list:**
    - **Button/Navigation:** There is a “Borrow” button in the book information section.
    - **Trigger:** Users click the “Borrow” button to add the library items to their borrow list

        - **Function:** The system saves users' borrow lists of library items on their local web.

    - **View the resources of the library item:**
        - **Button/Navigation:** The library item information section has some resource buttons.
        - **Trigger:** When clicked, the web opens the resources of the library item.
        - **Function:** The system navigates users to open the library item resources (if users paid for the resources). Otherwise, the system navigates users to the library card registration page to pay the personal library card.

    - **View the overview of the library item:**
        - **Button/Navigation:** There is the "Overview" tab below the library item information.
        - **Trigger:** When clicked, the web opens the resources of the library item.
        - **Function:** The system displays the library item information like publication year, publisher, language, page count, published location, category, cutter number, dimensions, responsibility, genre, topic terms, general notes, classification number, bookshelf ID, and ISBN code.

    - **View the editions of the library item:**
        - **Button/Navigation:** There is the "View edition" tab below the library item information.
        - **Trigger:** When clicked, the web opens the editions of the library item.
        - **Function:** The system displays the editions of the chosen library item (title, authors, language information), including the pagination and search bar. The users can click each of the editions to view more details.

    - **View the instances of the library item:**
        - **Button/Navigation:** There is the "Instances" tab below the library item information.
        - **Trigger:** When clicked, the web opens the resources of the library item
        - **Function:** The system displays the instances (code, barcode, and status) of the chosen library item, including the pagination and search bar.

    - **View the reviews of the library item:**
        - **Button/Navigation:** There is the "Review" tab below the library item information.
        - **Trigger:** When clicked, the web opens the resources of the library item
        - **Function:** The system displays the review list of the chosen library item.

    - **View the related items of the library item:**
        - **Button/Navigation:** There is the "Related items" tab below the library item information.
        - **Trigger:** When clicked, the web opens the resources of the library item
        - **Function:** The system displays the related item of the chosen library item, including the pagination and search bar. The users can click the "Detail" button to view more information.

- **Business Rules:** BR-26, BR-60

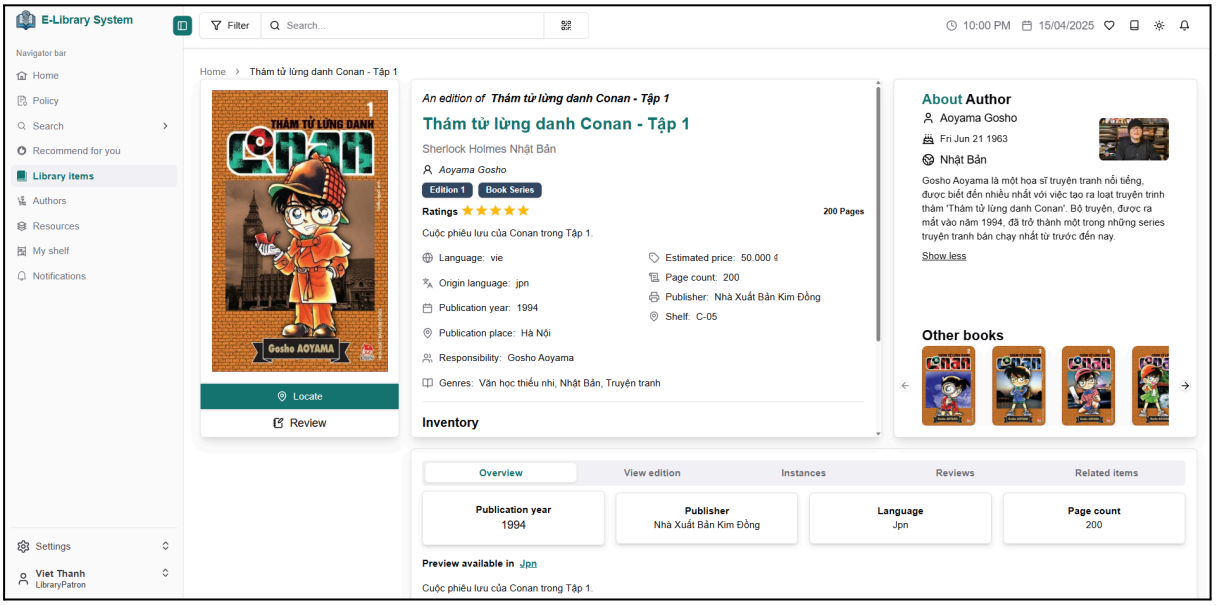


Figure 3.2.6.1: Library Item Detail

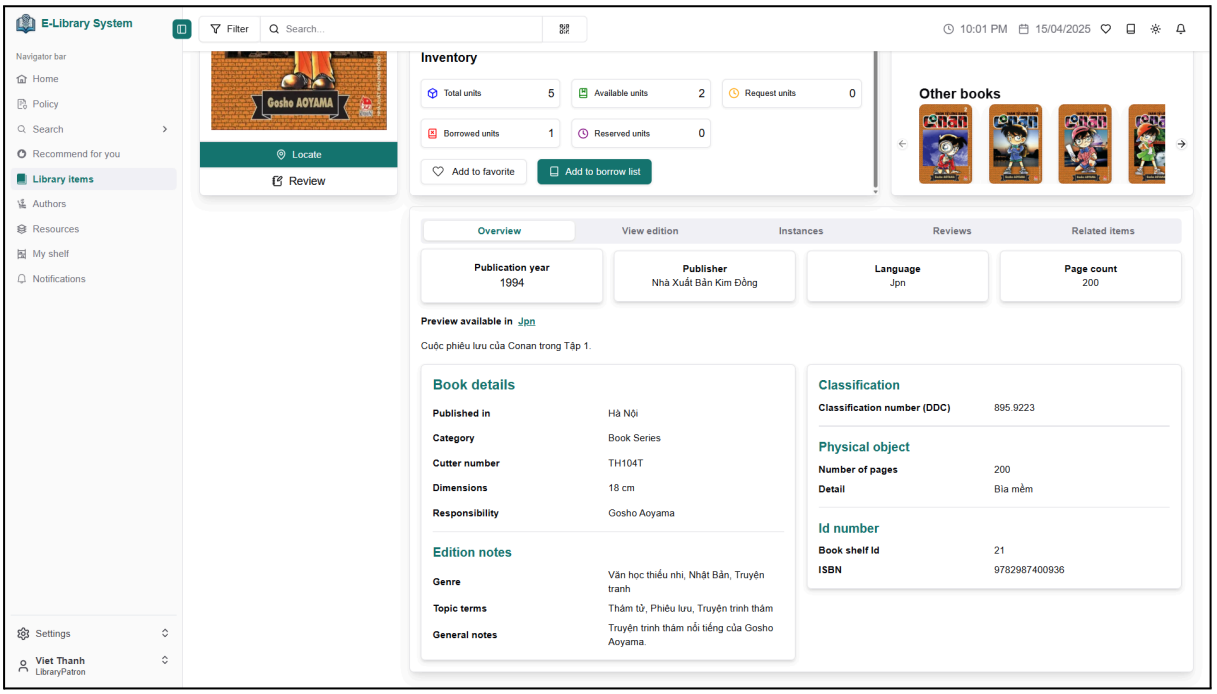


Figure 3.2.6.2: Library Item Detail

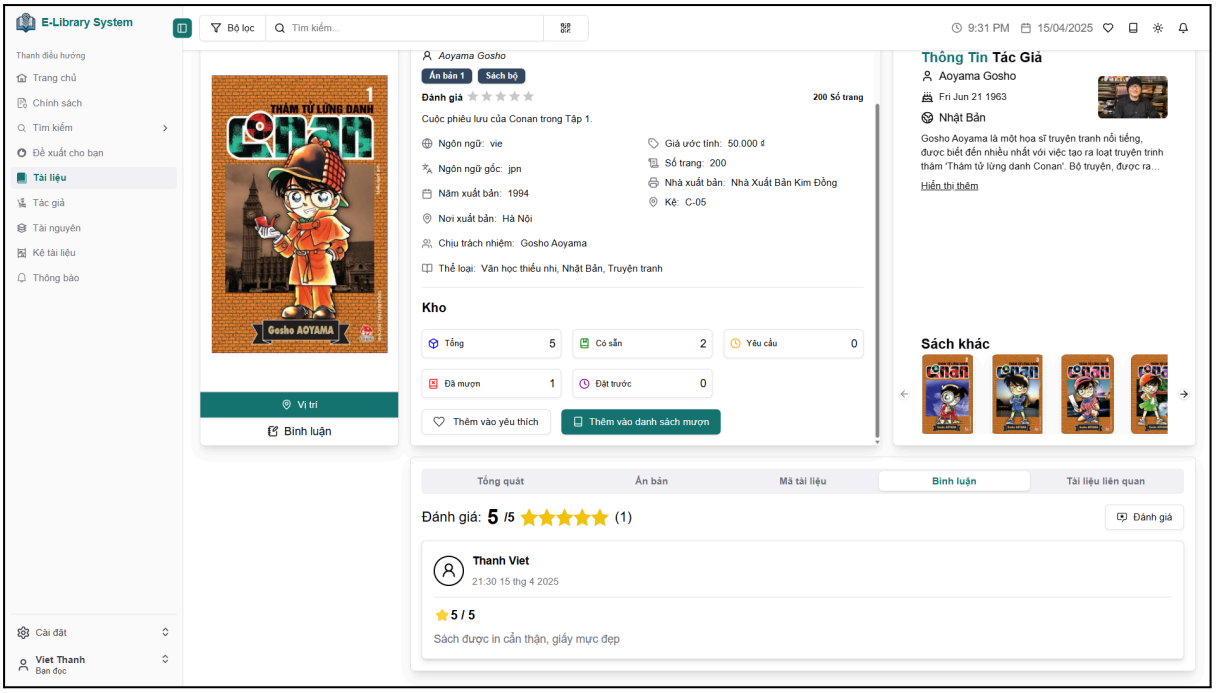


Figure 3.2.6.3: Library Item Detail

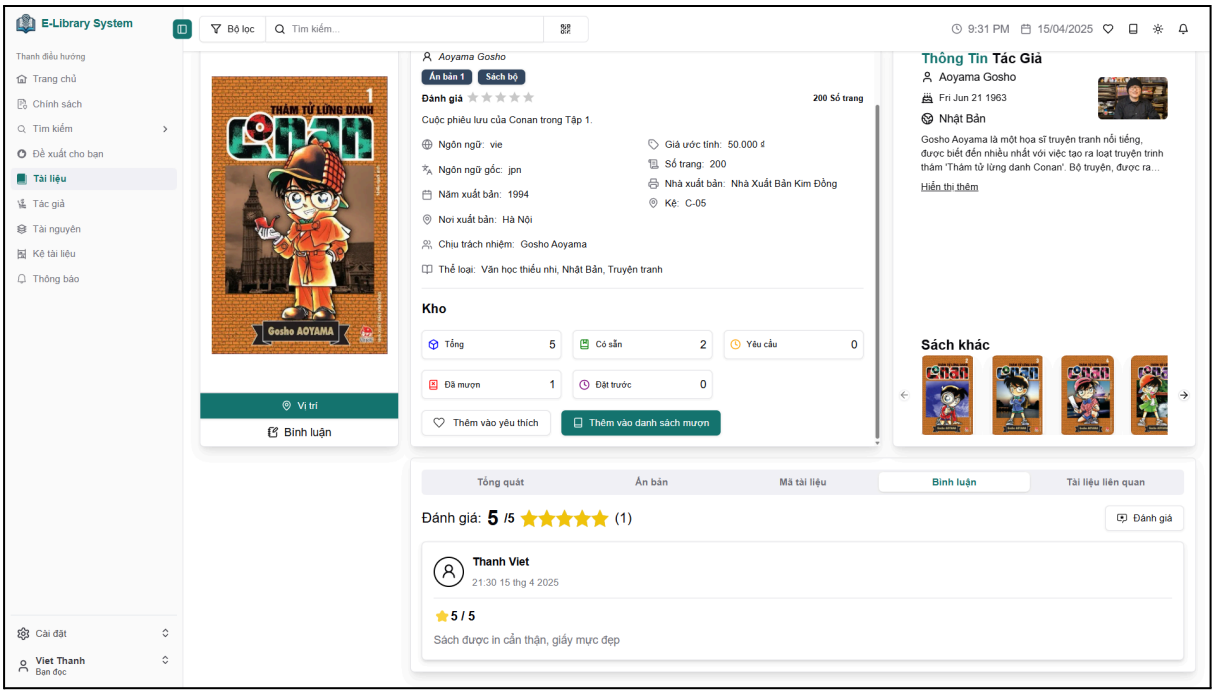


Figure 3.2.6.4: Library Item Detail

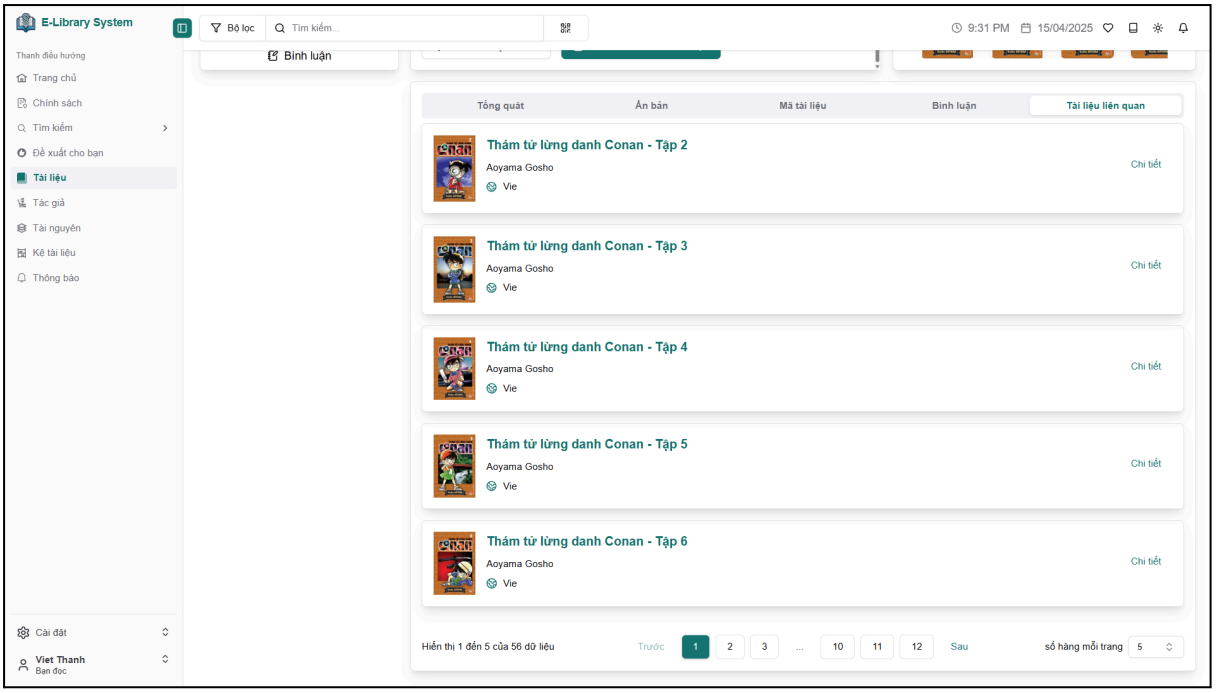


Figure 3.2.6.5: Library Item Detail

3.2.7 Search the Library item

- **Function Description:** The users can search the library items they want by typing the keyword on the search bar. Moreover, they can use some search features (quick search, basic search and advanced search).
- **Function Trigger:**

    - ❖ **Search the library item:**
        - **Button/Navigation:** There is a search input on the search bar, and 3 searching tabs on the search page.
        - **Trigger:** The users type any keyword related to the library items they want.
        - **Function:** The system will display the filtered library items.

- **Business Rules:** BR-32, BR-70, BR-78

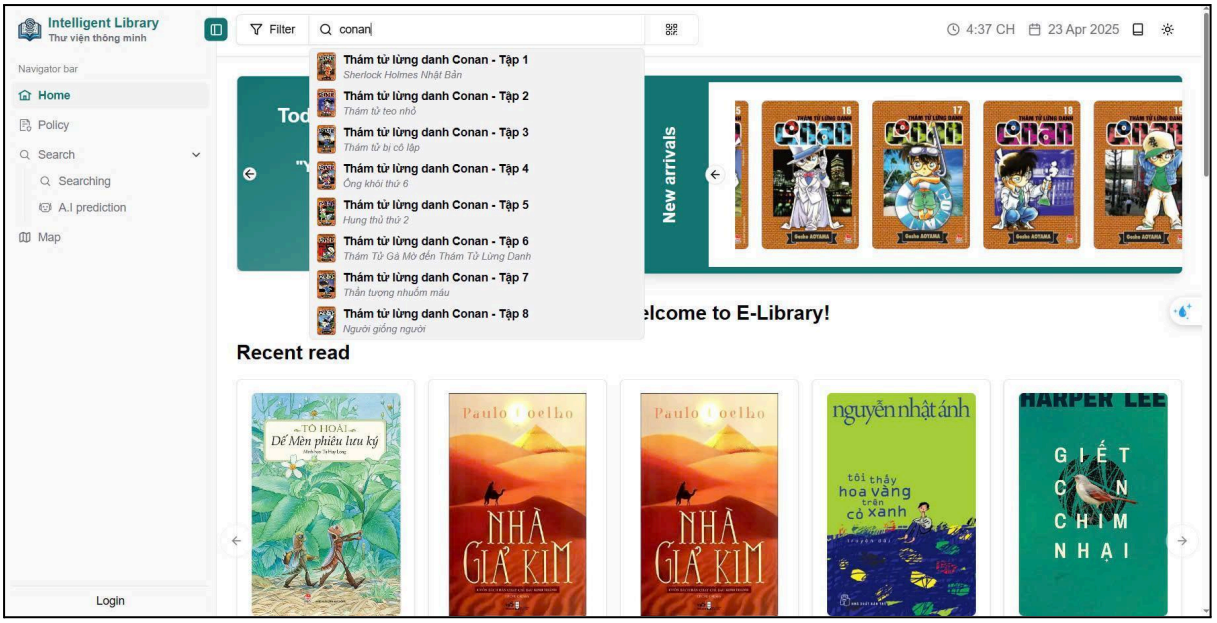


Figure 3.2.7.1: Search the Library item

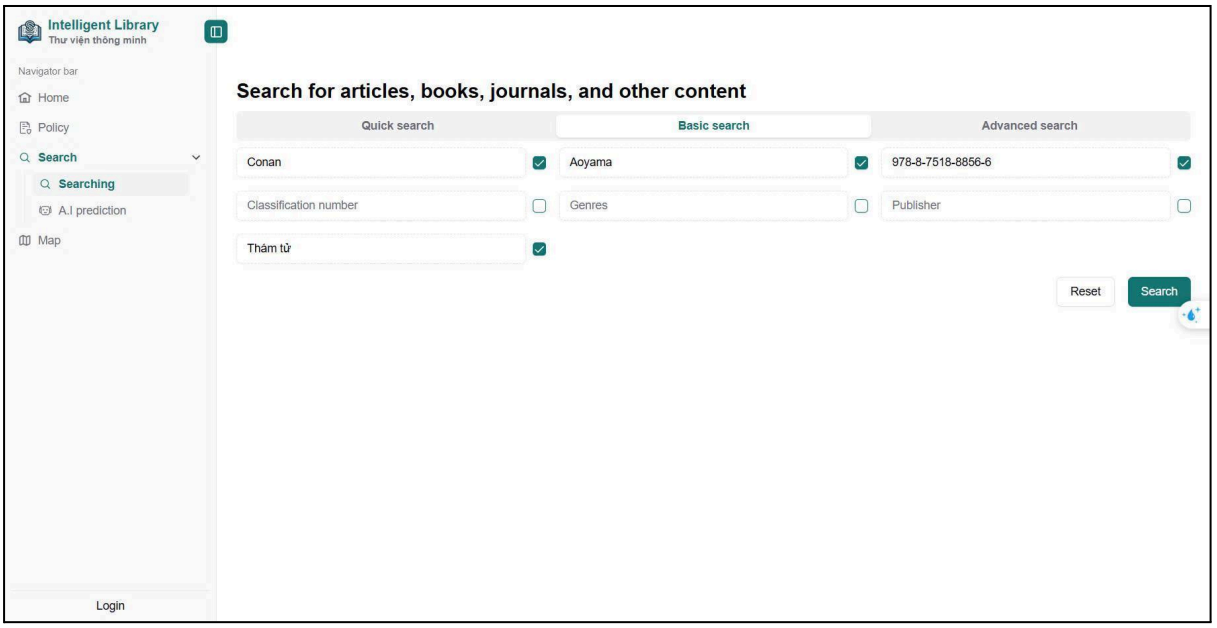


Figure 3.2.7.2: Search the Library item

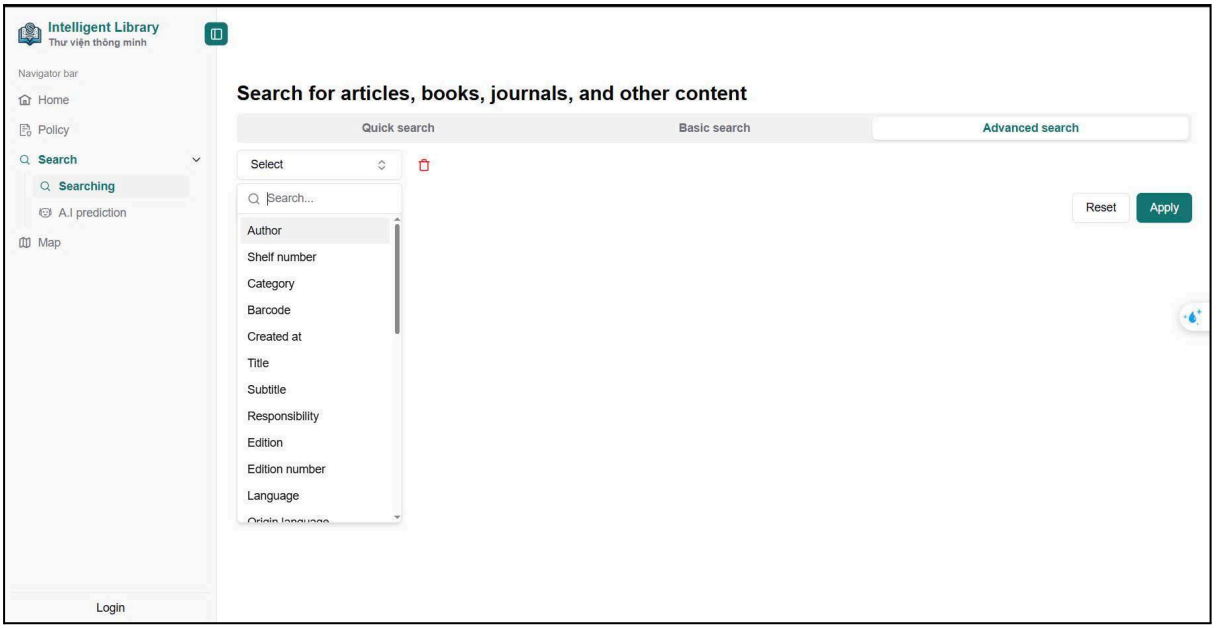


Figure 3.2.7.3: Search the Library item

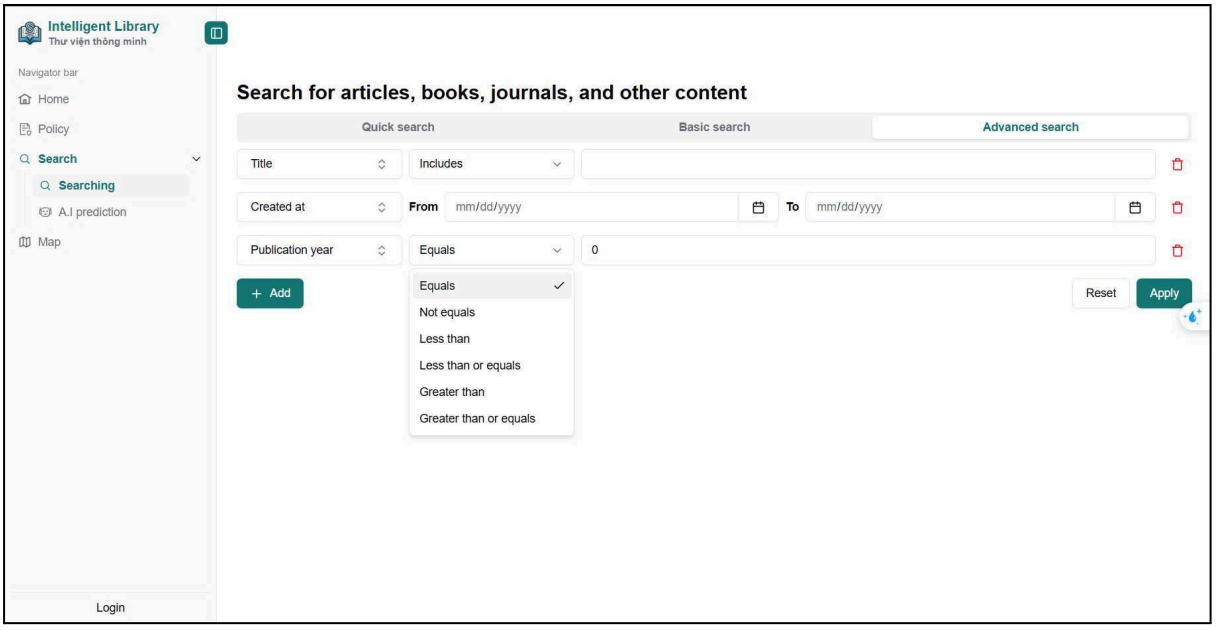


Figure 3.2.7.4: Search the Library item

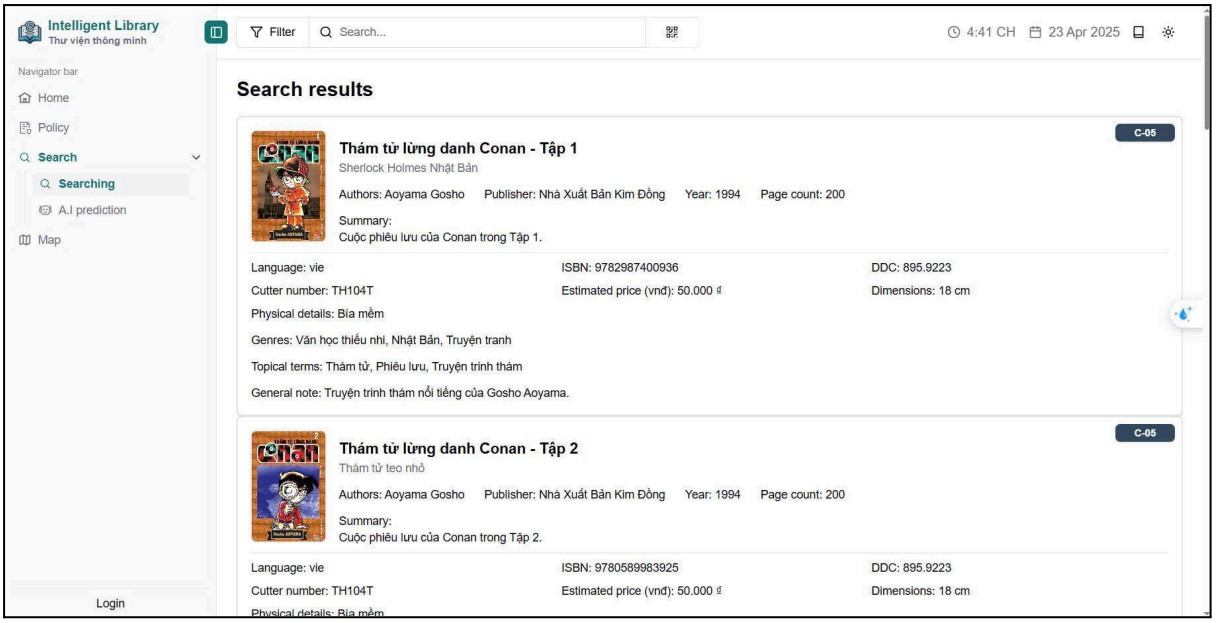


Figure 3.2.7.5: Search the Library item

3.2.8 Predict the upload image

- **Function Description:** The image prediction feature allows users to upload images from their devices, after which the system will analyze, evaluate, and predict to provide results that best match the uploaded image.
- **Function Trigger:**

  - **Choose the image from the user's device:**
    - **Button/Navigation:** There is an image drop zone in the center of the page.
    - **Trigger:** The user clicks the image drop zone or drag and drop the specific image from the user's device to upload the image.
    - **Function:** The system will show the uploaded image in the image drop zone.

  - **Clear the uploaded image:**
    - **Button/Navigation:** There is a "Clear" button at the top right page.
    - **Trigger:** The user clicks the "Clear" button.
    - **Function:** The page will clear the uploaded image, then allow the user to upload the other new image.

  - **Process the uploaded image:**
    - **Button/Navigation:** There is a "Process" button at the top right page, which is on the right of the "Clear" button.
    - **Trigger:** The user clicks the "Process" button.
    - **Function:** The page will predict the uploaded image, then navigate the user to the predicted result page.

  - **View the result of the uploaded image prediction:**

        - **Button/Navigation:** There is a “Result” tab trigger on the header of the tab list.
        - **Trigger:** The user clicks the “Result” tab trigger.
        - **Function:** The page will display the matchest result related to the upload image. Furthermore, the page shows the overview information of the predicted image, and displays the AI-predict section on the right side including some prediction percentage like title, publisher, authors, threshold value, match overview, and the result status.

    - **View the OCR detail of the uploaded image prediction:**
        - **Button/Navigation:** There is a “OCR detail” tab trigger on the header of the tab list.
        - **Trigger:** The user clicks the “OCR detail” tab trigger.
        - **Function:** The page will display the OCR detail including three main parts
            + The result section shows the value of match percentage and overall threshold
            + OCR text shows some match phrase including the assumption values of title, author, publisher
            + The comparison table presents some evidence proving the result section based on match phrase, fuzziness, threshold, match overall, and match percentage for title, author, and publisher.

    - **View the other matches of the uploaded image prediction:**
        - **Button/Navigation:** There is a “Other matches” tab trigger on the header of the tab list.
        - **Trigger:** The user clicks the “Other matches” tab trigger.
        - **Function:** The page will show the list of other matches. Each matched item displays the library item detail, image, and the AI-predict on the right side of it, which shows the percentage values of title, publisher, author, threshold value, match overall, and status.

- **Business Rules:** BR-33, BR-80, BR-91

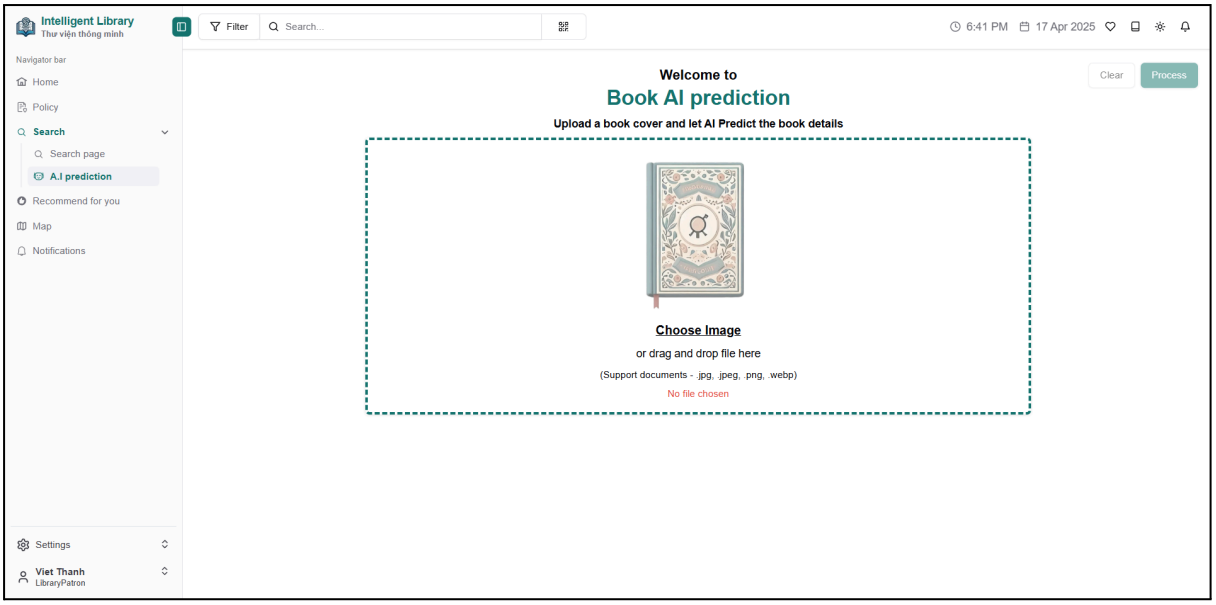


Figure 3.2.8.1: Predict the upload image

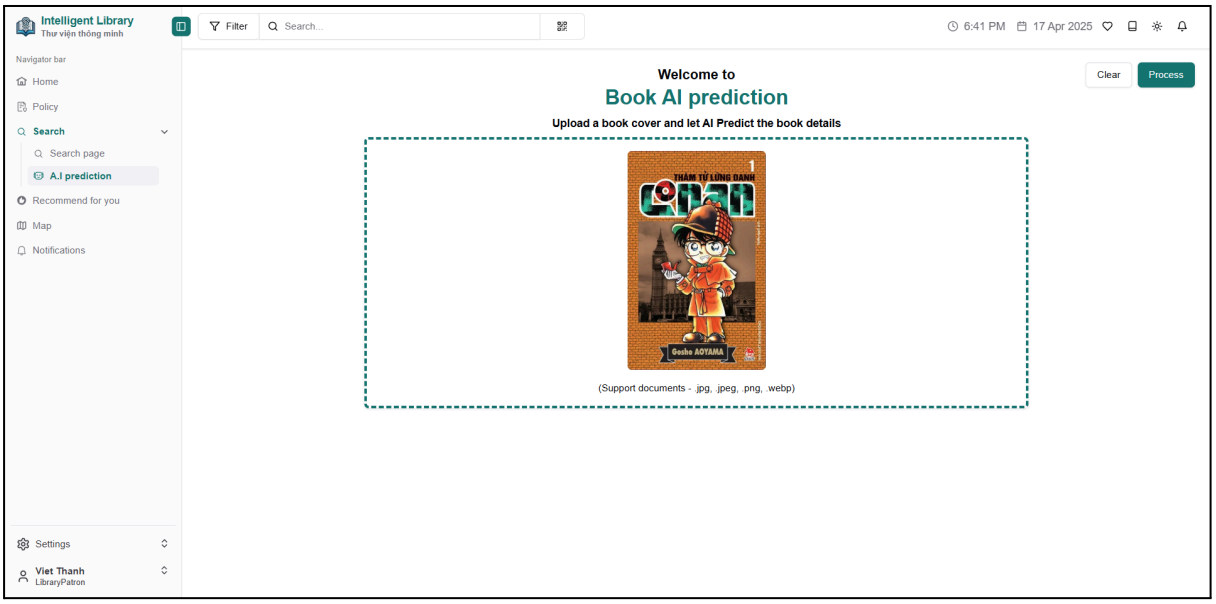


Figure 3.2.8.2: Predict the upload image

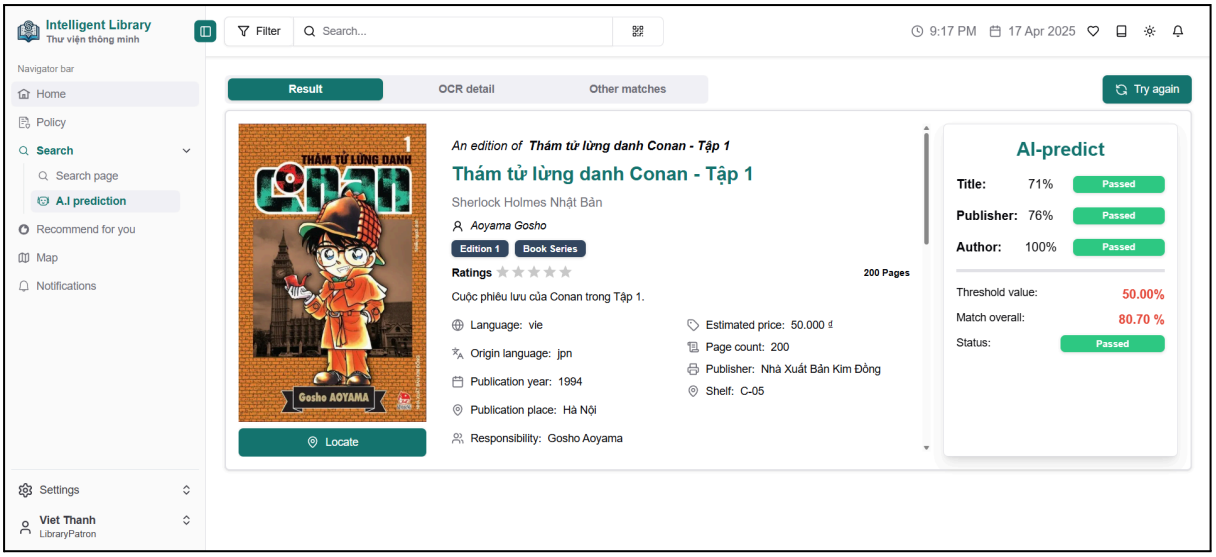


Figure 3.2.8.3: Predict the upload image

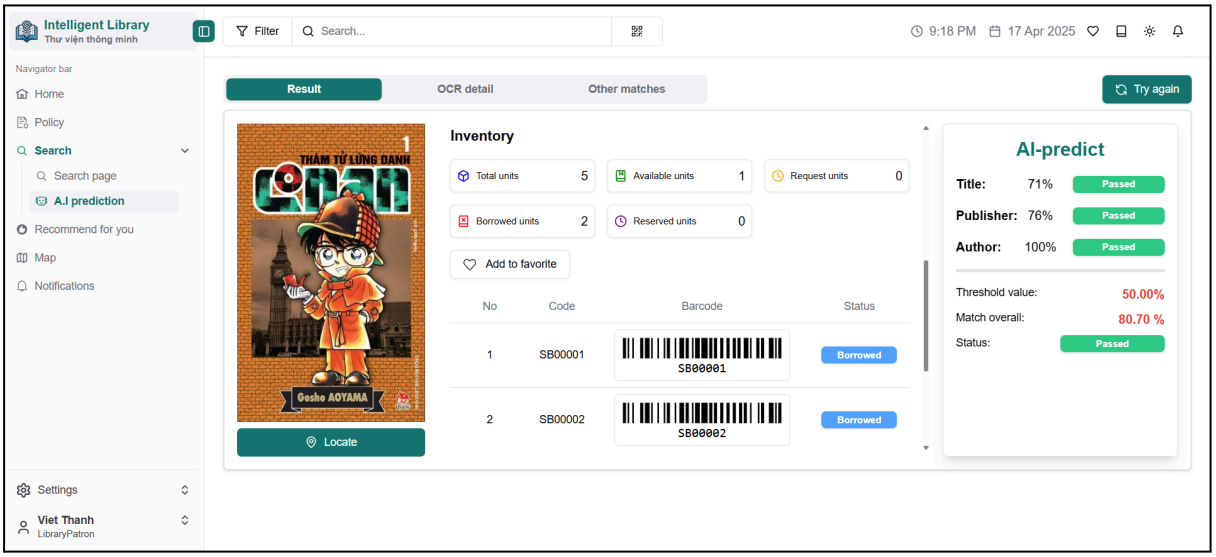


Figure 3.2.8.4: Predict the upload image

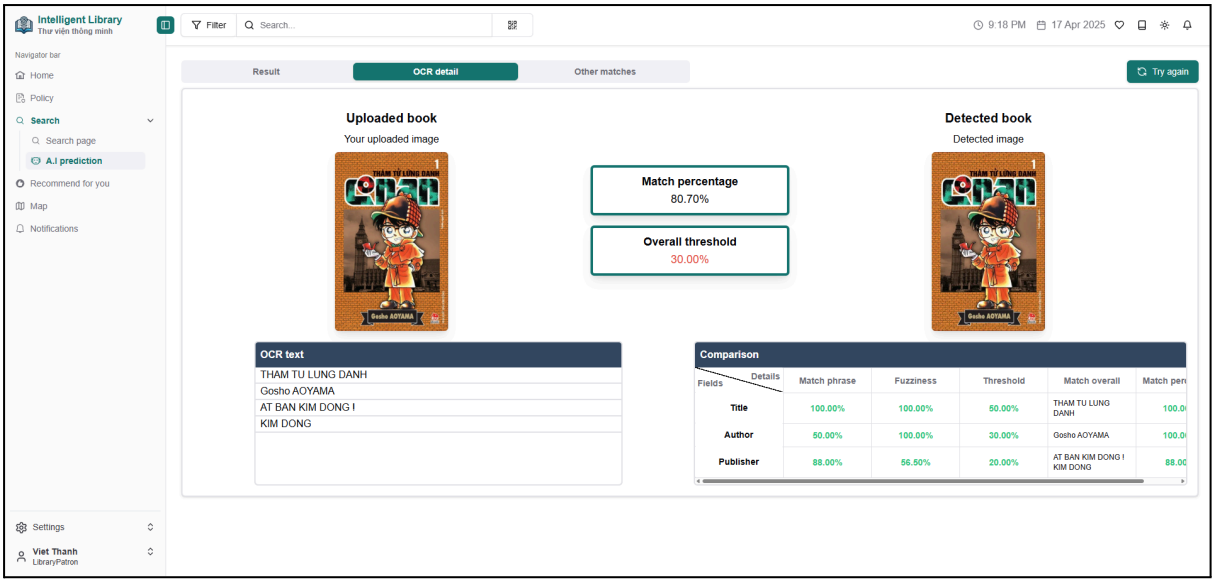


Figure 3.2.8.5: Predict the upload image

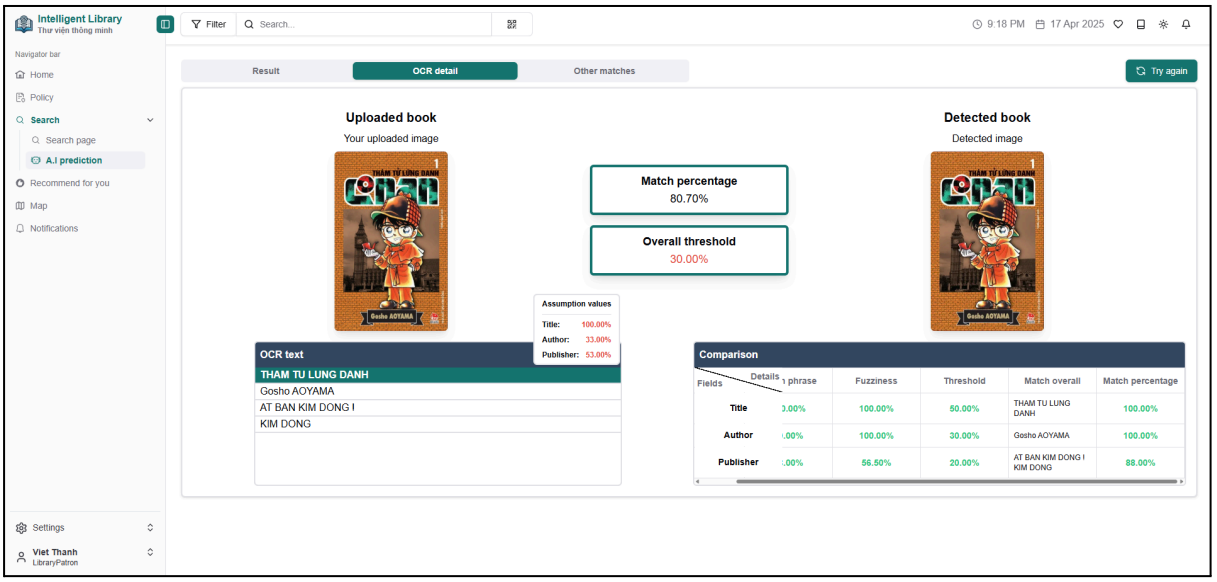


Figure 3.2.8.6: Predict the upload image

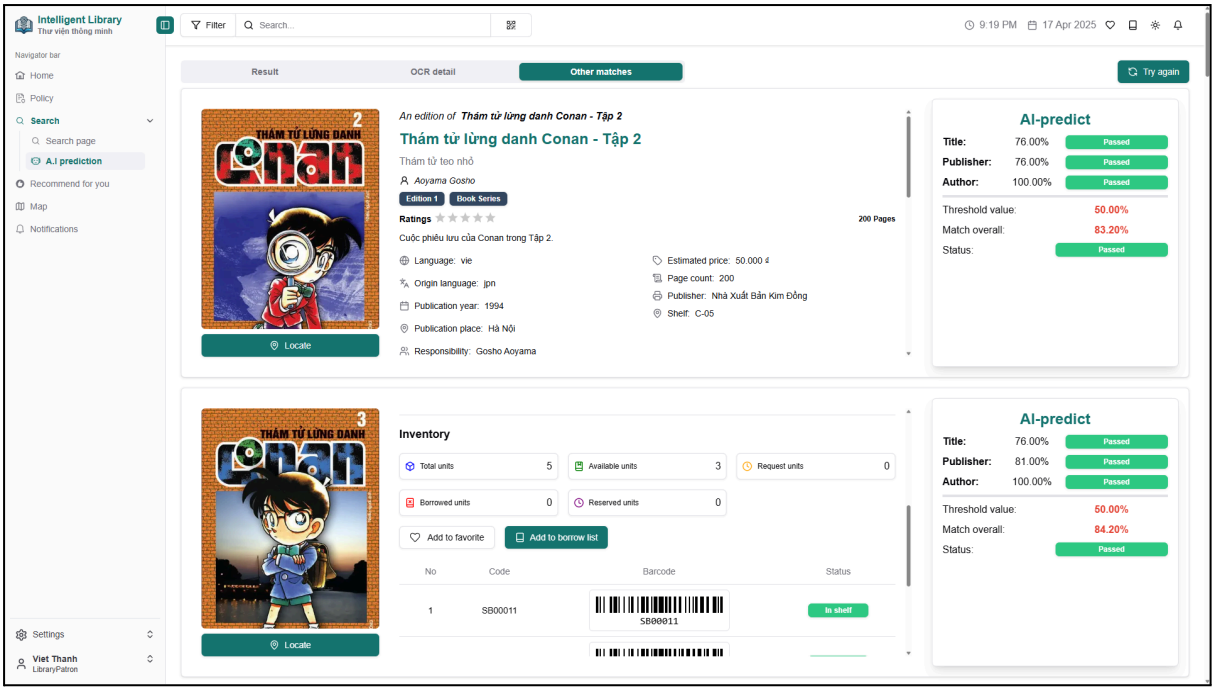


Figure 3.2.8.7: Predict the upload image

3.2.9 Locate the Library item

- **Function Description:** To facilitate searching for books directly at the library, the directions feature on the library map will reduce search time, as well as indicate the exact location of the library items.
- **Function Trigger:**

  ❖ **Open the location dialog of the library item:**
  - **Button/Navigation:** There is a “Locate” button in the book preview section, which is under the library item Image.
  - **Trigger:** The users click the “Locate” button to open the overview position of the library item.
  - **Function:** The page will open the position dialog showing some information like the floor, shelf, section, and zone.

  ❖ **Cancel the location dialog of the library item:**
  - **Button/Navigation:** There is a “Cancel” button in the position of the library item dialog, which is at the right bottom of the dialog.
  - **Trigger:** The users click the “Cancel” button.
  - **Function:** The page will close the position dialog of the library item.

  ❖ **Locate the location of the library item:**
  - **Button/Navigation:** There is a “Locate” button in the shelf location dialog.
  - **Trigger:** The users click the “Locate” button to open the map of the library item.
  - **Function:** The page will show the real map built before.

❖ **Search the direction of the library item:**

- **Button/Navigation:** There are two inputs for typing a origin location and typing a destination location.
- **Trigger:** User can type the address of the start location and the end location.
- **Function:** The page will show the map directly to indicate the exact shelf location of the library item.

- **Business Rules:** BR-03

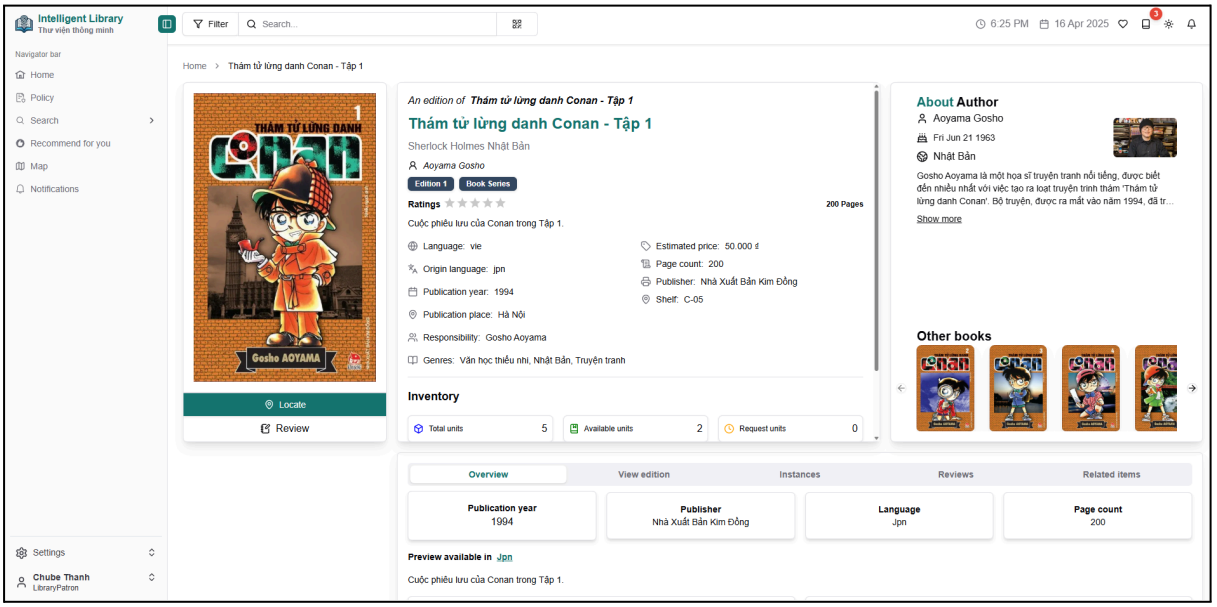


Figure 3.2.9.1: Locate the Library item

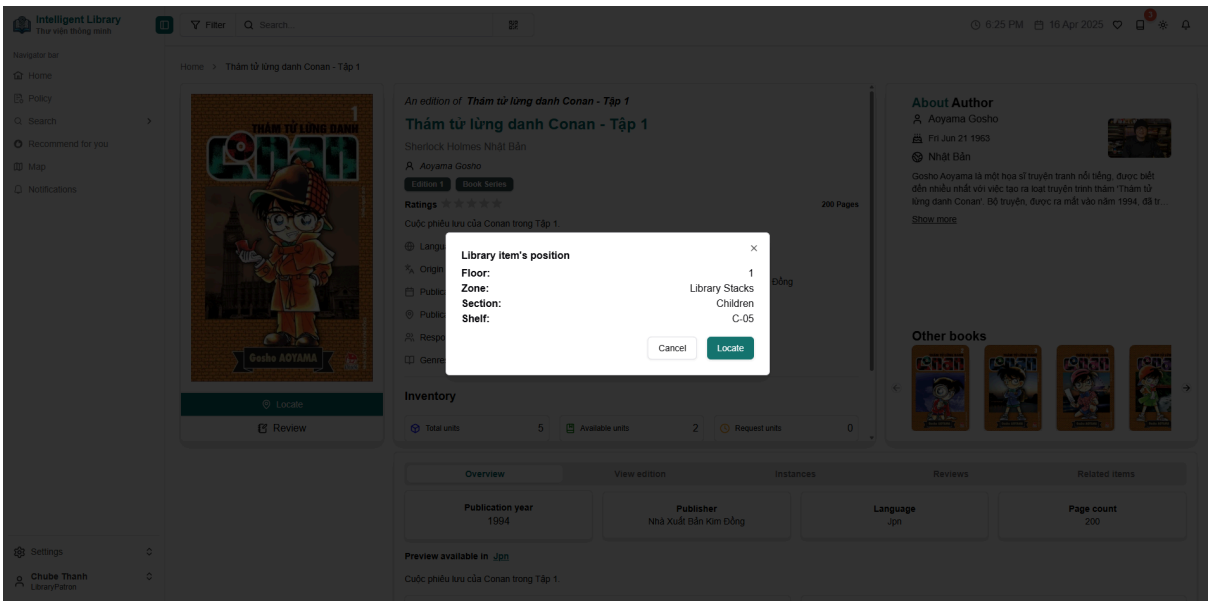


Figure 3.2.9.2: Locate the Library item

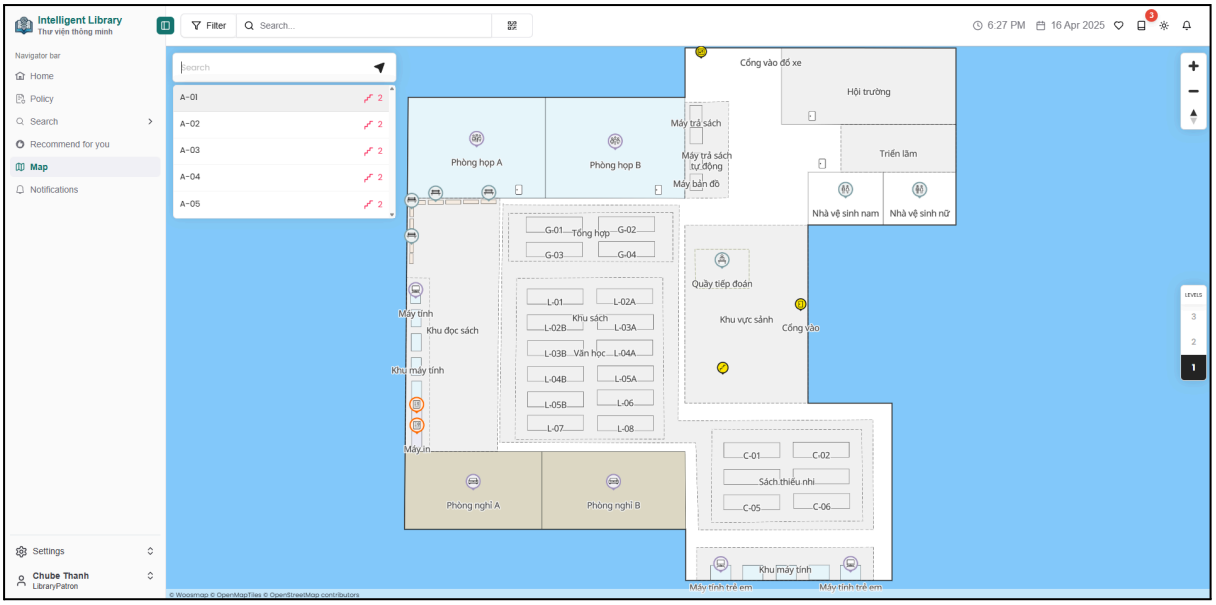


Figure 3.2.9.3: Locate the Library item

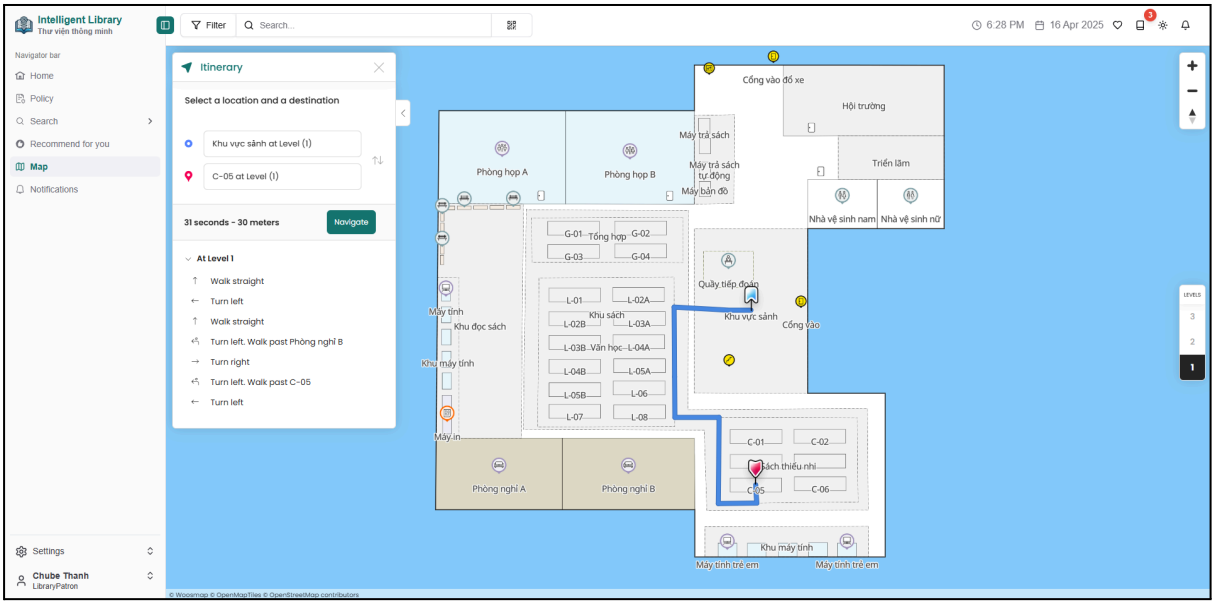


Figure 3.2.9.4: Locate the Library item

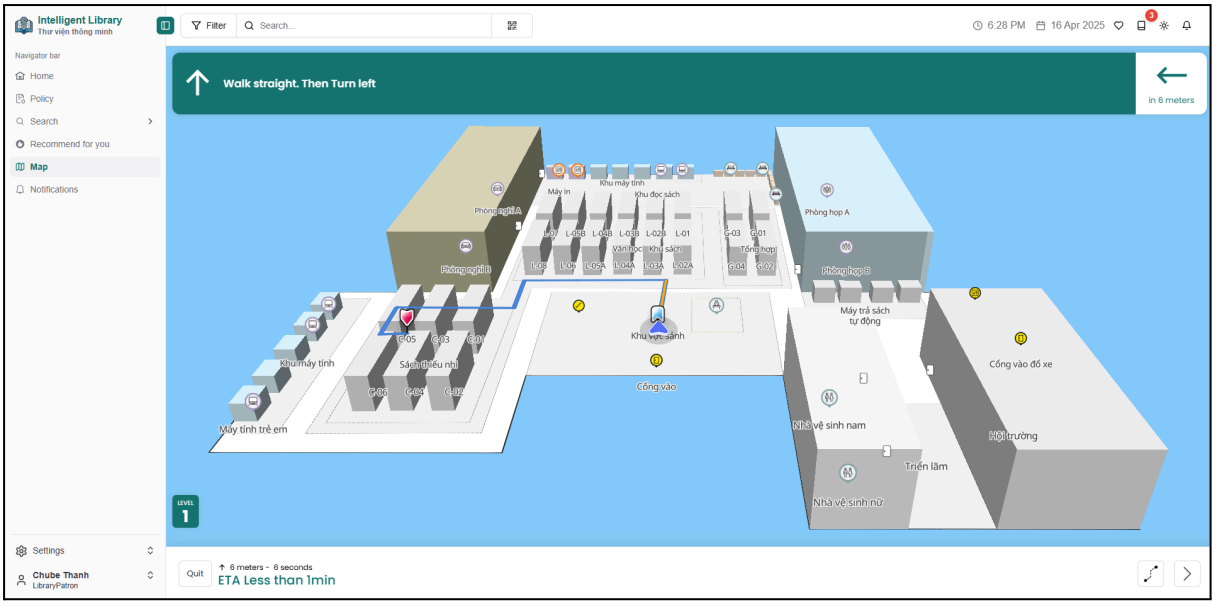


Figure 3.2.9.5: Locate the Library item

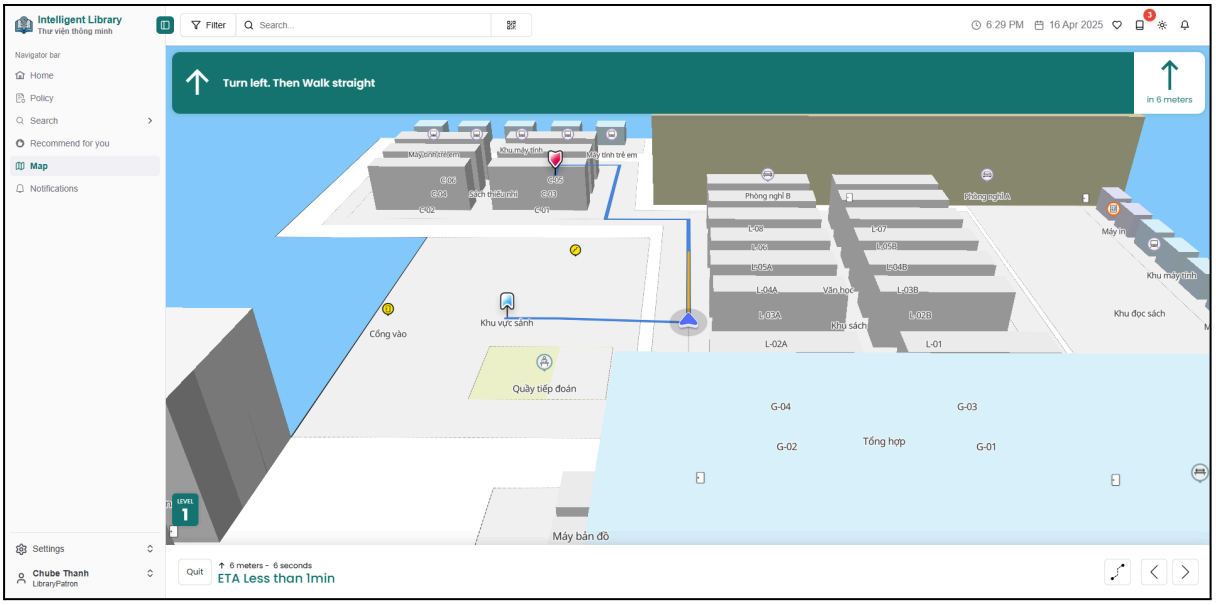


Figure 3.2.9.6: Locate the Library item

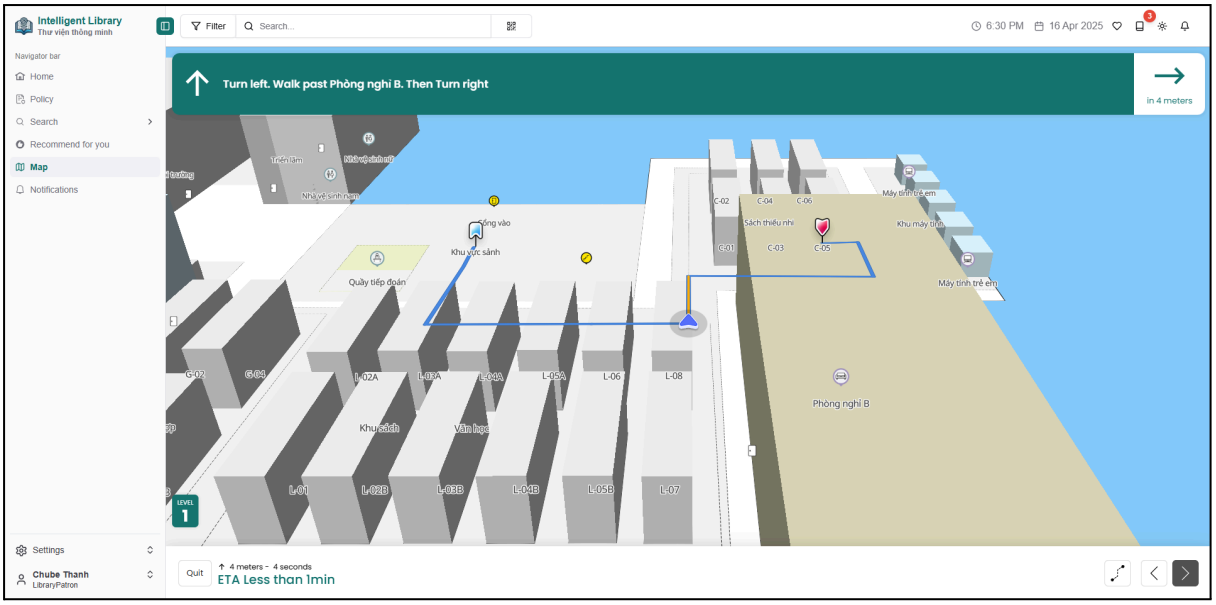


Figure 3.2.9.7: Locate the Library item

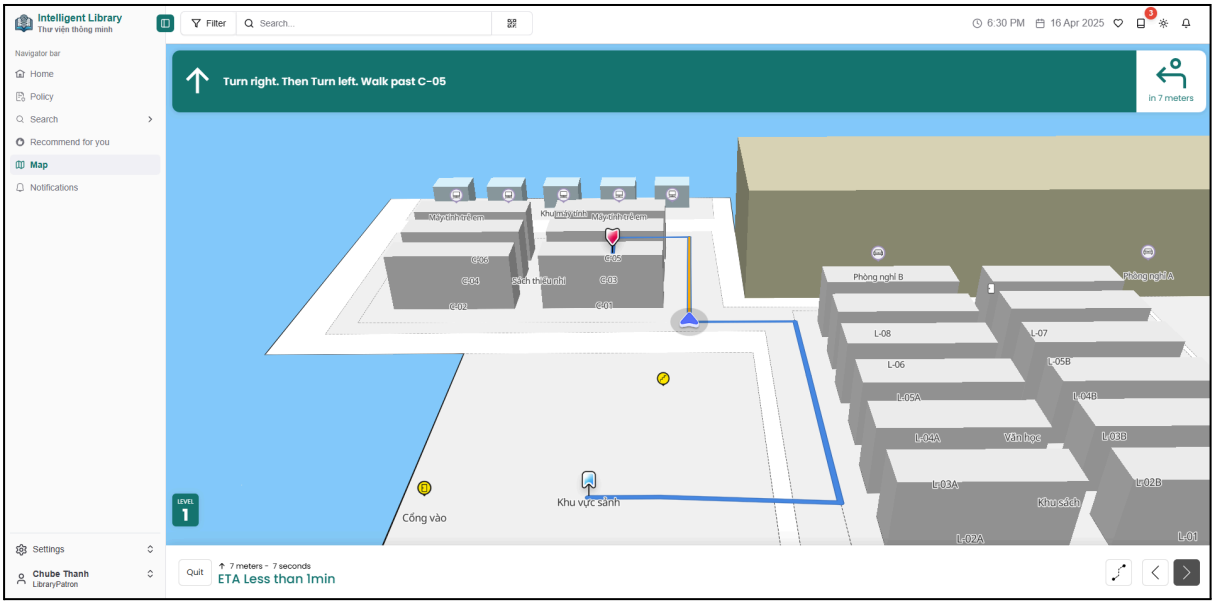


Figure 3.2.9.8: Locate the Library item

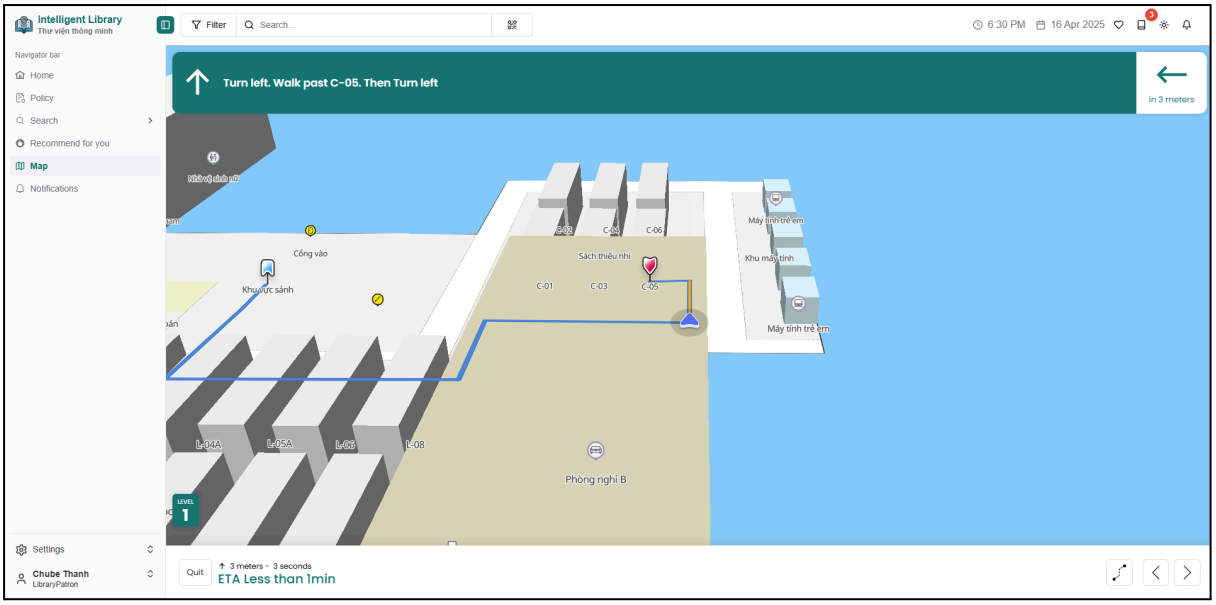


Figure 3.2.9.9: Locate the Library item

Figure 3.2.9.10: Locate the Library item

3.2.10 Borrow the digital resources

- **Function Description:** The library patrons can have a look at the resource list, watch the preview of the digital library resource, and start to borrow the resources they like.
- **Function Trigger:**

  - ❖ **View the list of the digital resources:**
    - **Button/Navigation:** There is a “List digital resources” button in the library item information section.

        - **Trigger:** Patron clicks the “List digital resources” button to show the list of the available resources belonging to the library item inside the digital resources dialog.
        - **Function:** The system displays the digital resources dialog, showing the list of the available resources belonging to the library item based on the resource type, like “Ebook” or “Audiobook.”

    - **View the preview of the digital resources:**
        - **Button/Navigation:** There is a “Preview” button in the digital resources dialog.
        - **Trigger:** Patron clicks the “Preview” button to view the ebook or audiobook resources.
        - **Function:** The system navigates the users to view the digital resources preview page that shows some the beginning page of ebook.

    - **Show the digital resources confirmation dialog:**
        - **Button/Navigation:** There is a “Borrow” button in the digital resources dialog.
        - **Trigger:** Patron clicks the “Borrow” button to borrow the digital resources they want.
        - **Function:** The system displays the form of borrowing the digital resource. Furthermore, it shows some information about the digital resources, like title, resource type, borrowing duration, and borrowing price.

    - **Show the borrowing digital resources confirmation:**
        - **Button/Navigation:** There is a “Borrow” button in the borrowing digital resources confirmation dialog.
        - **Trigger:** Patron clicks the “Borrow” button to confirm the digital resources they want.
        - **Function:** The system displays the real-time countdown QR payment. After the library patrons paid successfully, the system redirected them to the library item detail page.

    - **Cancel the borrowing digital resources confirmation:**
        - **Button/Navigation:** There is a “Cancel” button in the borrowing digital resources confirmation dialog.
        - **Trigger:** Patron clicks the “Cancel” button to cancel the chosen digital resources they want.
        - **Function:** The system cancels the library patron’s payment, and closes the dialog.

    - **Pay the borrowing digital resources:**
        - **Button/Navigation:** There is a “Borrow” button on the borrowing digital resources confirmation dialog.
        - **Trigger:** When the patron clicks the “Borrow” button, the page shows the payment confirmation dialog, then the patron can choose the payment method to pay the resource before scanning the payment QR.
        - **Function:** The patron can access the whole content of the paid resource.

- **Business Rules:** BR-10

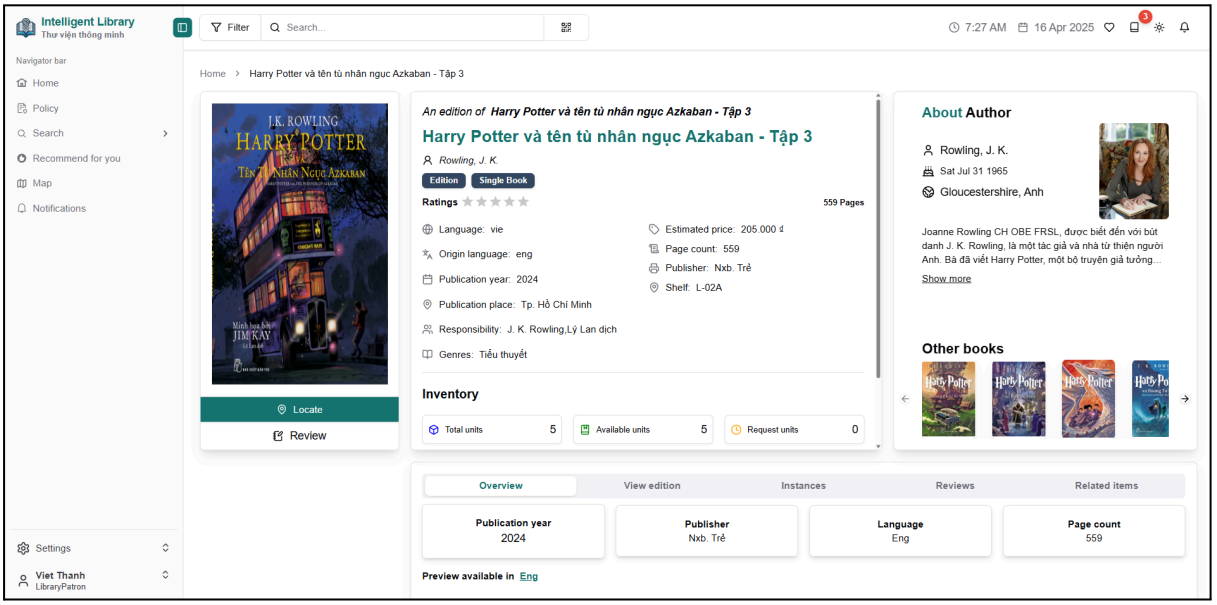


Figure 3.2.10.1: Borrow the digital resources

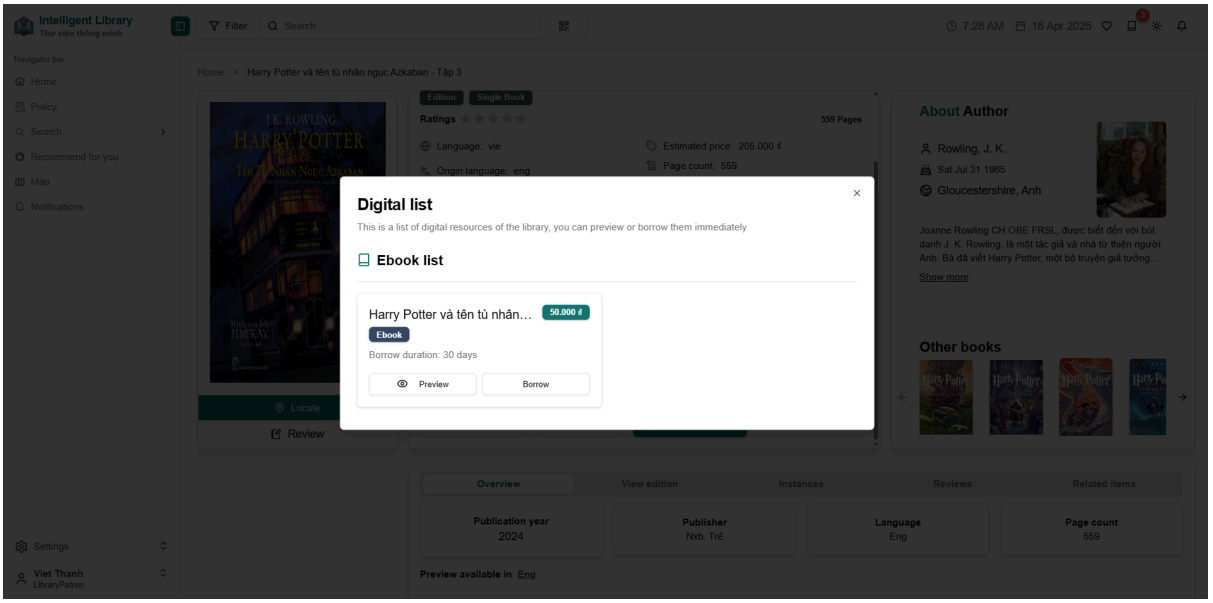


Figure 3.2.10.2: Borrow the digital resources

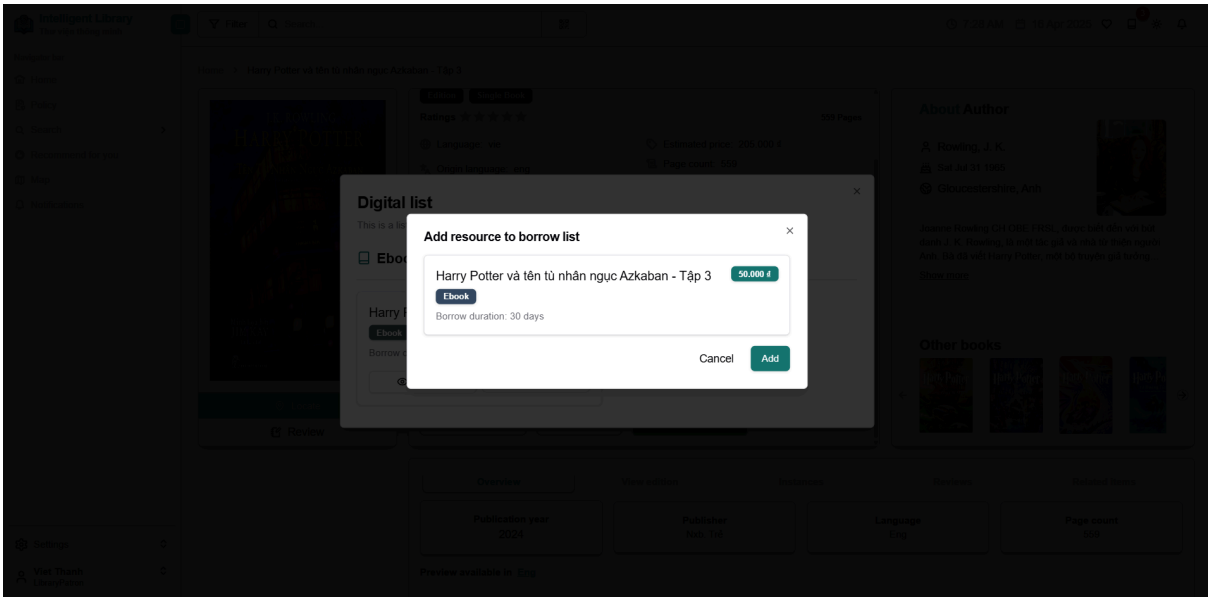


Figure 3.2.10.3: Borrow the digital resources

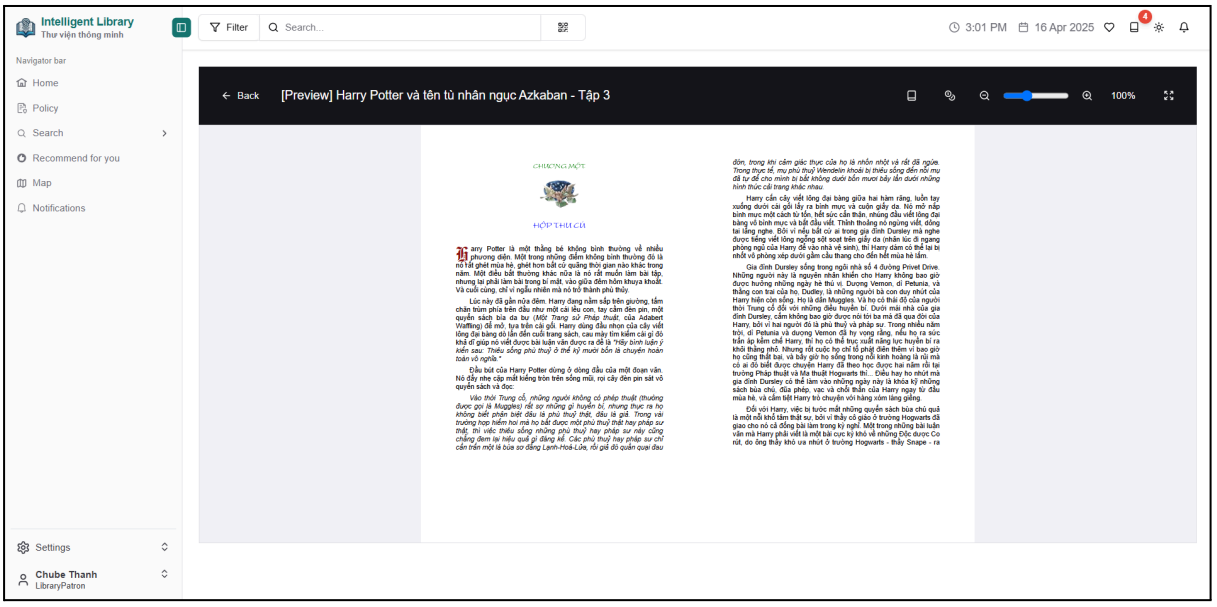


Figure 3.2.10.4: Borrow the digital resources

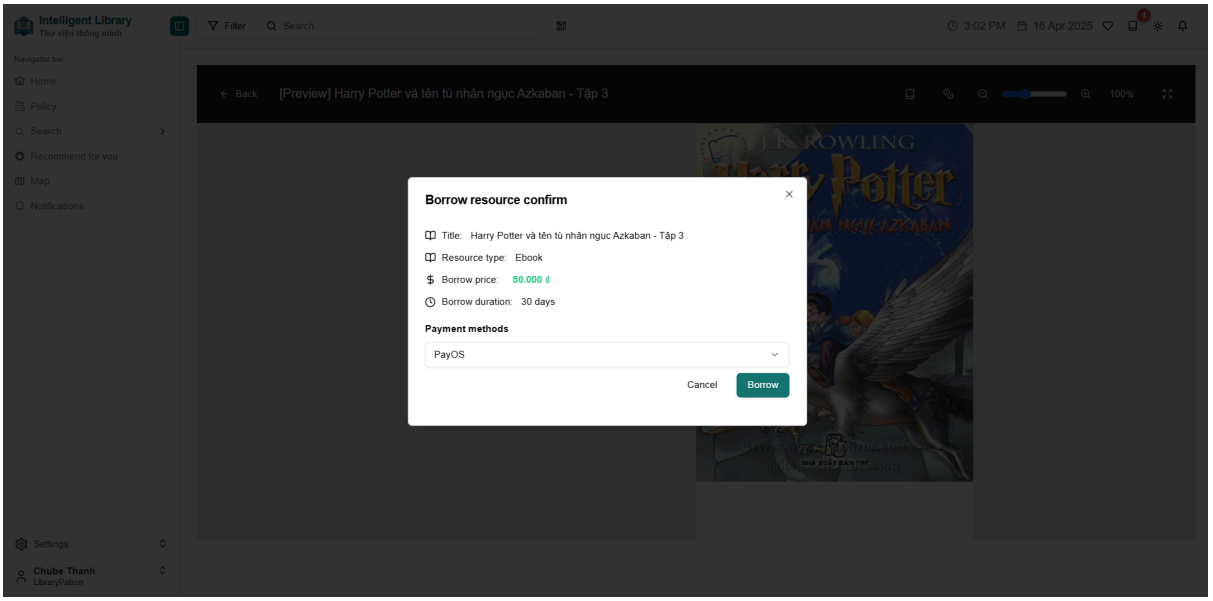


Figure 3.2.10.5: Borrow the digital resources

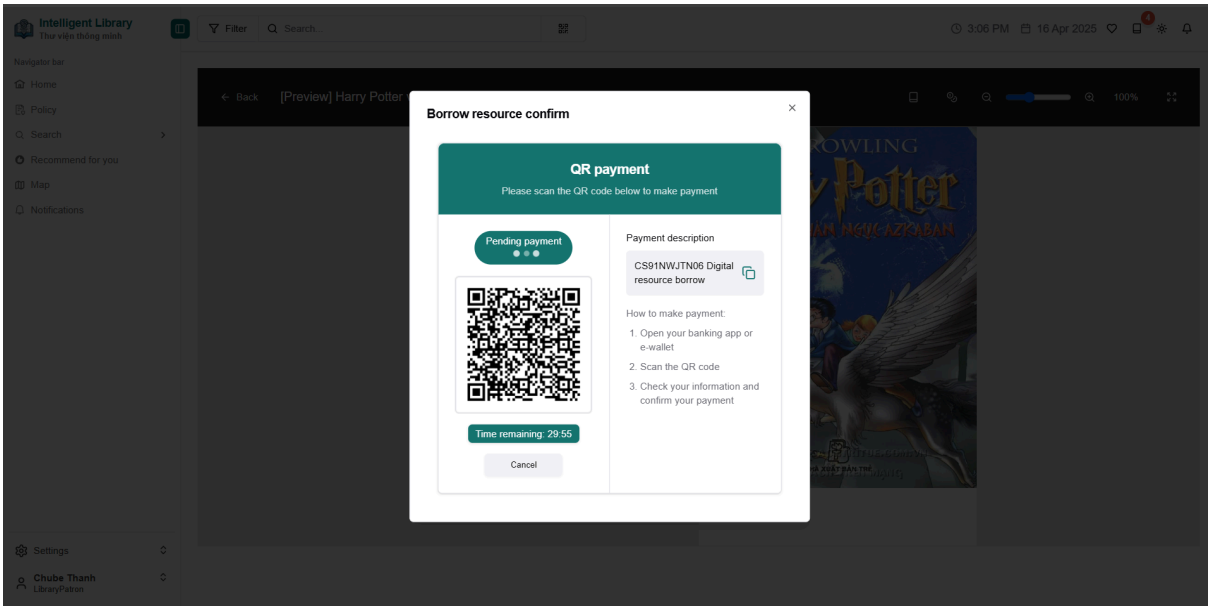


Figure 3.2.10.6: Borrow the digital resources

3.2.11 Borrow the Library item remotely

- **Function Description:** The library patrons can have a look at the resource list, watch the preview of the digital library resource, and start to borrow the resources they want.
- **Function Trigger:**
  - ❖ **Add library item to the borrow list:**
    - **Button/Navigation:** There is a “Add to borrow list” button in the library item information section.
    - **Trigger:** When clicked, the page shows the borrowed additional confirmation dialog.

- **Function:** The page adds the library item to the personal borrow list.

❖ **View the borrow list:**
- **Button/Navigation:** There is a “Borrow list” button in the bottom right corner of the web application.
- **Trigger:** Patron clicks the “Borrow list” to open the right sidebar-sheet showing the list of the library items added to the temporary borrow list.
- **Function:** The page displays the personal borrow list of the library items, and the resources.

❖ **Remove the item in the borrow list:**
- **Button/Navigation:** There is a “Trash” icon in the library item card.
- **Trigger:** Patron clicks the “Trash” icon to remove the library item from the borrow list.
- **Function:** The system removes the library item from the borrowing list.

❖ **Remove all the items in the borrow list:**
- **Button/Navigation:** There is a “Remove all” button in the bottom of the right sheet sidebar.
- **Trigger:** Patron clicks the “Remove all” button to clear the borrowing list.
- **Function:** The system clears all the library items in the borrow list.

❖ **Navigate to the borrow list:**
- **Button/Navigation:** There is a “Borrow all” button in the end of the right sidebar.
- **Trigger:** Patron clicks the “Borrow all” button.
- **Function:** The page navigates the users to the borrow page.

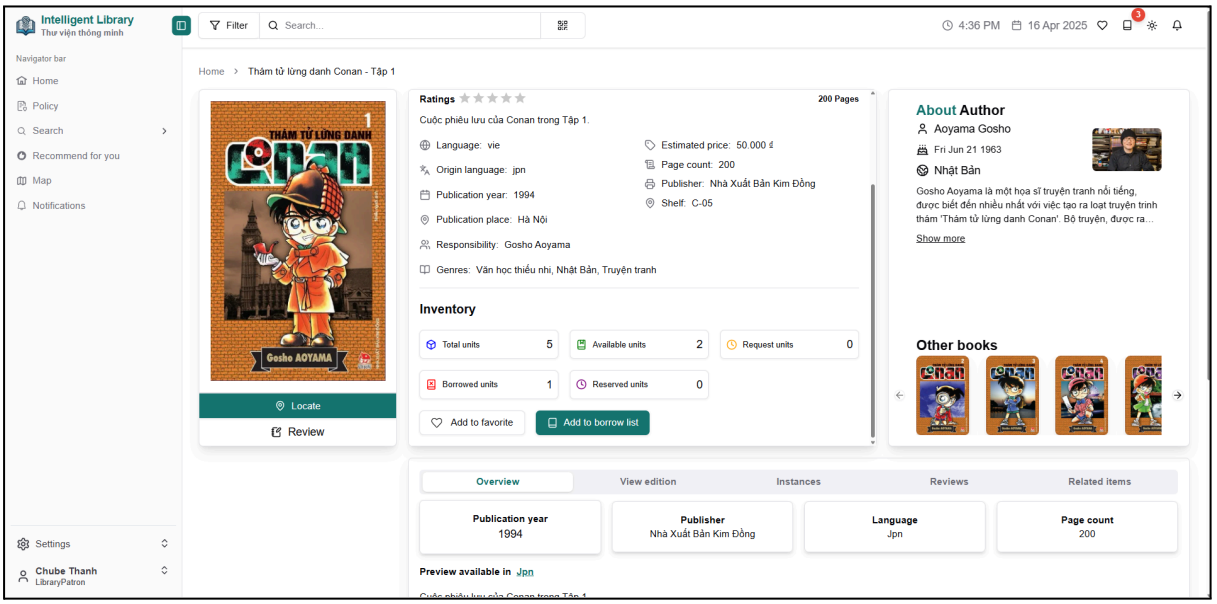


Figure 3.2.11.1: Borrow the Library item remotely

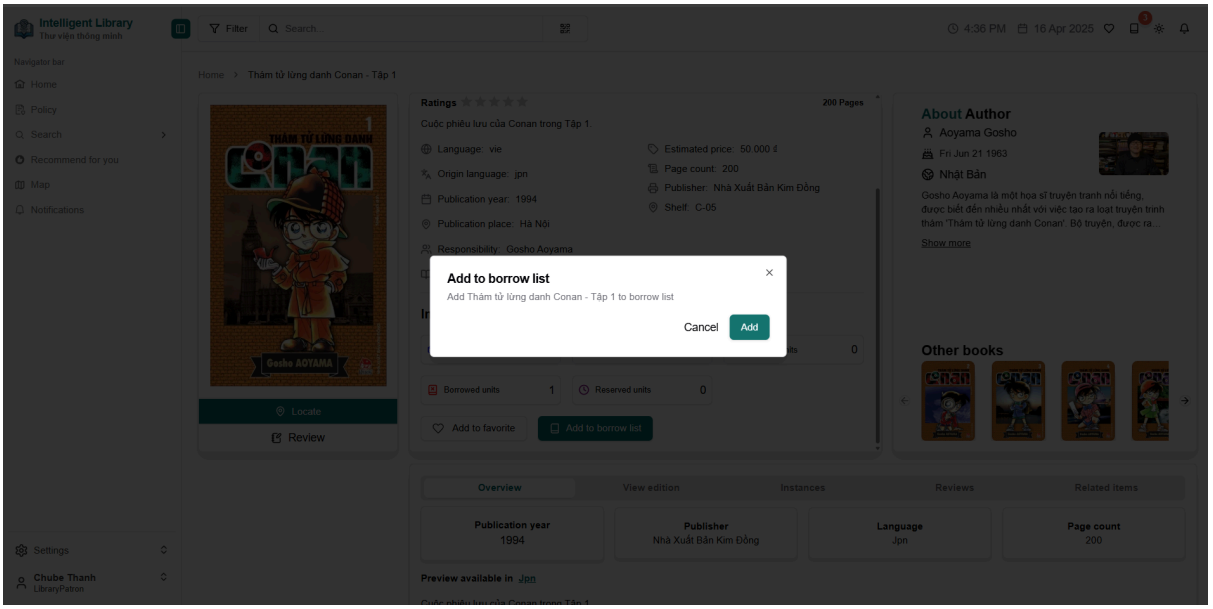


Figure 3.2.11.2: Borrow the Library item remotely

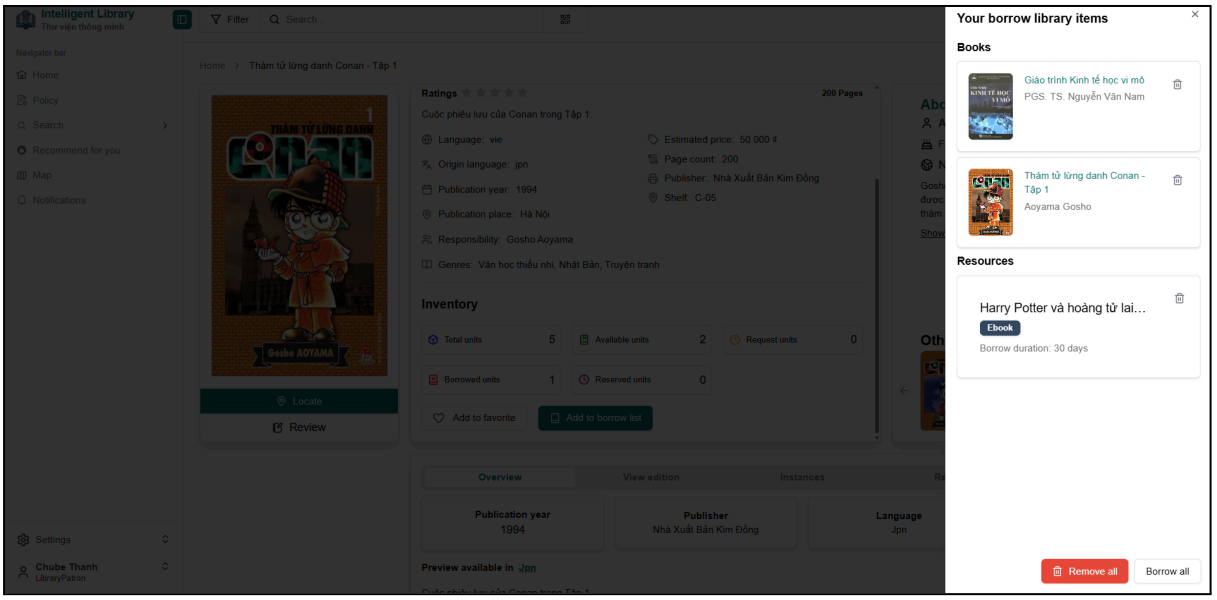


Figure 3.2.11.3: Borrow the Library item remotely

3.2.12 Create the borrow request

- **Function Description:** The user moves to the borrow list page, then selects and checks the availability of library items or library resources before creating the borrow request.
- **Function Trigger:**
  - ❖ **Select all the library items and digital resources:**
    - ■ **Button/Navigation:** There is a “Select all” button at the top of the borrow list.
    - ■ **Trigger:** When the user checks the “Select all” button, all the library items, and digital resources are selected.

        - **Function:** The page adds all the library items, and digital resources to the availability borrow request dialog.
    - **Check to borrow the specific library items or digital resources:**
        - **Button/Navigation:** There are the “Select borrow” buttons in each of the library item cards or digital resource cards.
        - **Trigger:** When the user checks the “Select borrow” button .
        - **Function:** The page adds the specific item to the availability borrow request dialog.
    - **Cancel the specific checked library items or digital resources:**
        - **Button/Navigation:** There is a “Trash” icon in each of the library item cards or digital resource cards.
        - **Trigger:** When the user checks the “Trash” icon.
        - **Function:** The page removes the specific item the availability borrow request dialog.
    - **Open the availability dialog of the borrow request:**
        - **Button/Navigation:** There is a “Borrow” button on the top of the borrow list page.
        - **Trigger:** When clicked, the page opens the availability borrow request dialog.
        - **Function:** The page checks the availability of each item which is selected to borrow.
    - **Cancel the availability dialog of the borrow request:**
        - **Button/Navigation:** There is a “Cancel” button at the end of the borrow request dialog.
        - **Trigger:** The user clicks a “Cancel” button to cancel the borrow request.
        - **Function:** The page clears and closes the dialog of the borrow request.
    - **Create the borrow request:**
        - **Button/Navigation:** There is a “Continue” button at the end of the borrow request dialog, which is next to the “Cancel” button.
        - **Trigger:** When clicked, the page navigates the users to the.
        - **Function:** The system creates a borrow request based on the selected library items and digital resources, then it navigates the patron to the profile page.

- **Business Rules:** BR-09**,** BR-15, BR-18, BR-19, BR-21, BR-39, BR-81, BR-86

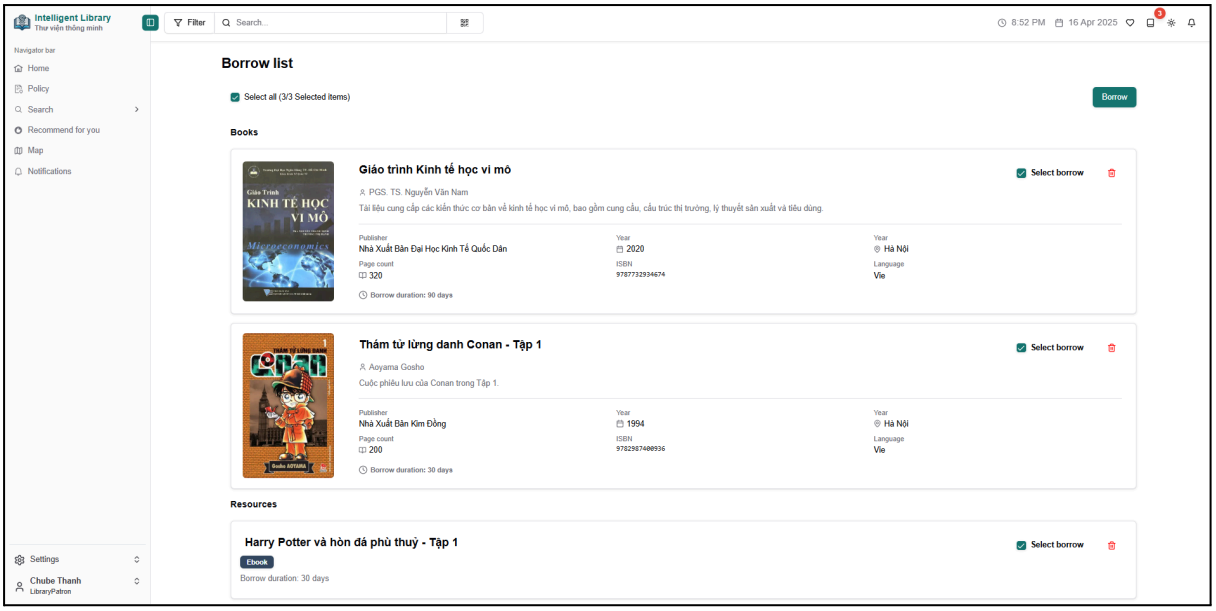


Figure 3.2.12.1: Create the borrow request

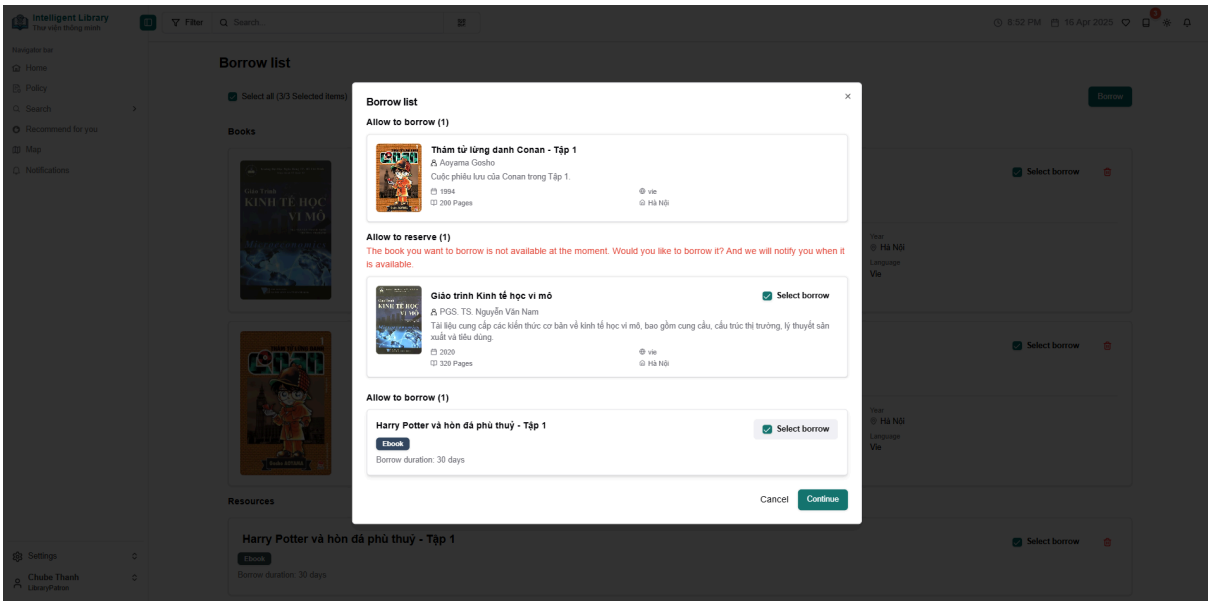


Figure 3.2.12.2: Create the borrow request

3.3 Website for Guests and Users

3.3.1 Patron's profile

- **Function Description:** The library patron profile page allows the library patrons to view and update their personal information.
- **Function Trigger:**
  - ❖ **View the user's profile:**
    - ■ **Button/Navigation:** There is a "Account" button located in the user dropdown menu on the sidebar.

- **Trigger:** The user clicks the “Account” button to show the personal profile.
- **Function:** The page displays the user’s personal details, including full name, phone number, date of birth, gender, email, and address. Furthermore, the user’s profile picture is shown with an option to change it.

❖ **Reset the user’s profile:**

- **Button/Navigation:** There is a “Reset” button at the bottom right corner of the profile form.
- **Trigger:** The user clicks the “Reset” button to reset all the values in the profile form.
- **Function:** The page resets the user’s personal details, including full name, phone number, date of birth, gender, email, and address to default values.

❖ **Update the user’s profile:**

- **Button/Navigation:** There is a “Update” button at the bottom right corner of the profile form.
- **Trigger:** The user clicks the “Update” button to update the personal profile.
- **Function:** The system saves the user’s personal details, including full name, phone number, date of birth, gender, email, and address.

- **Business Rules:** BR-05

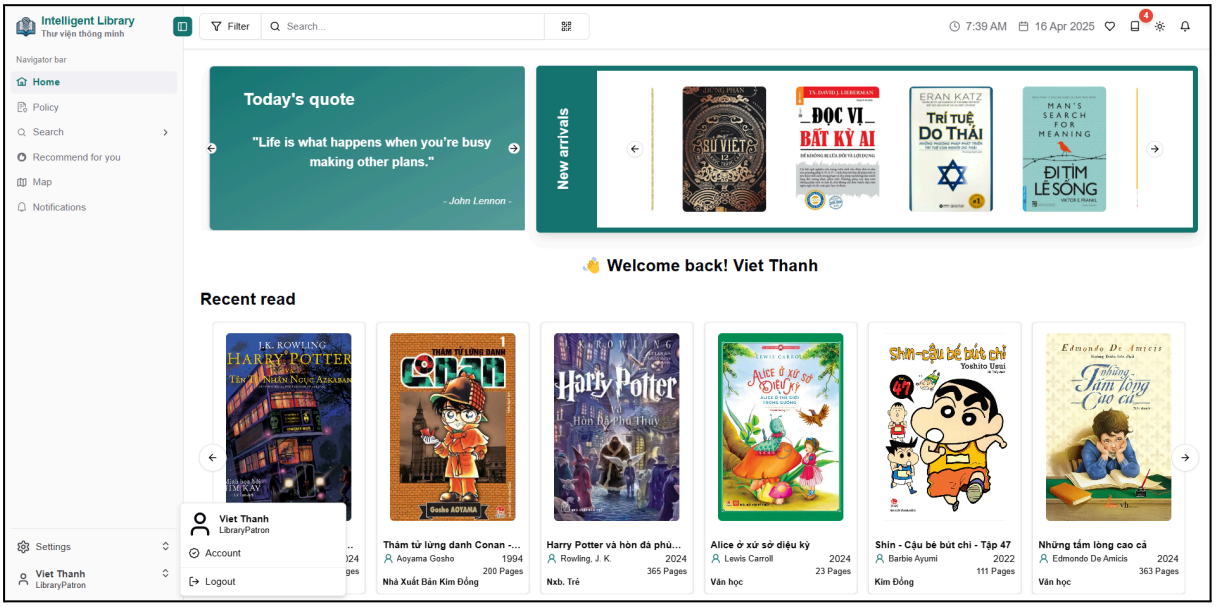


Figure 3.3.1.1: Patron’s profile

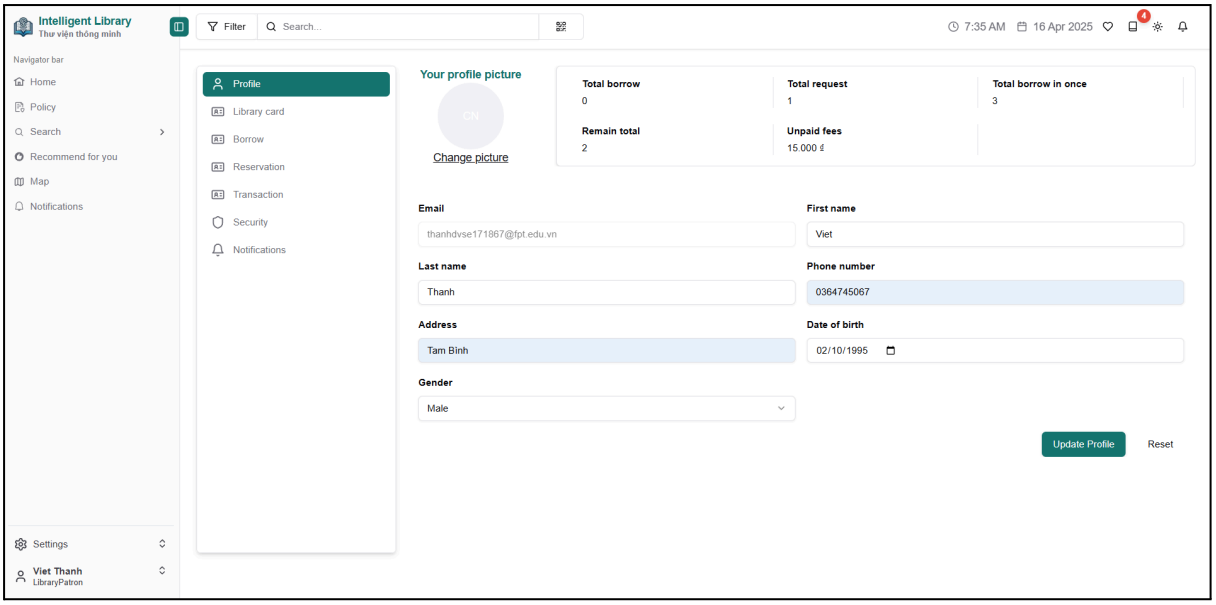


Figure 3.3.1.2: Patron's profile

3.3.2 Patron's library card

- **Function Description:** In the case, the patron registered the personal library card, the page shows a member card of a patron including some information like card name, email, birthday, address, issue date, expiry date, and patron's barcode. Furthermore, it also shows additional information about the library card like insurance method, allow borrowing more, max items at once, total missed pickups, reminder sent, extended, extension count, and expiry date. Otherwise, the patron has to register the personal library card.
- **Function Trigger:**
  - ❖ **View the user's profile:**
    - ■ **Button/Navigation:** There is a "Library card" button on the profile tab.
    - ■ **Trigger:** The user clicks the "Library card" button to navigate to the personal library card page.
    - ■ **Function:** The page shows name, email, birthday, address, issue date, expiry date, and patron's barcode. Moreover, it also shows additional information about the library card like insurance method, allow borrowing more, max items at once, total missed pickups, reminder sent, extended, extension count, and expiry date. Otherwise, the patron has to register the personal library card.
- **Business Rules:** BR-07, BR-16, BR-50

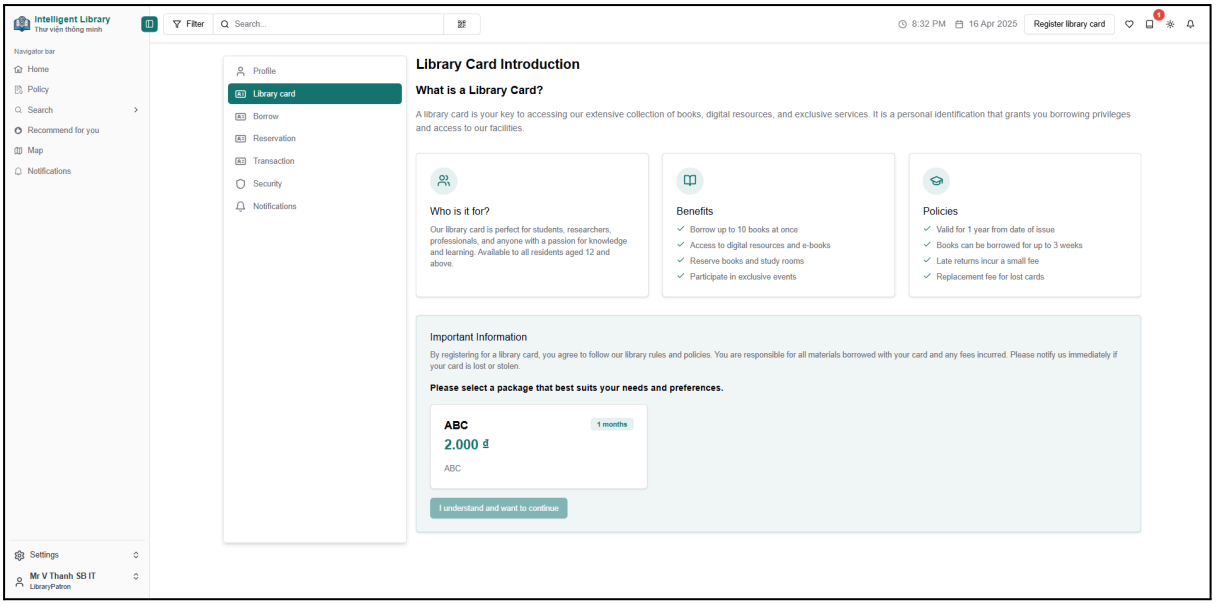


Figure 3.3.2.1: Patron's library card

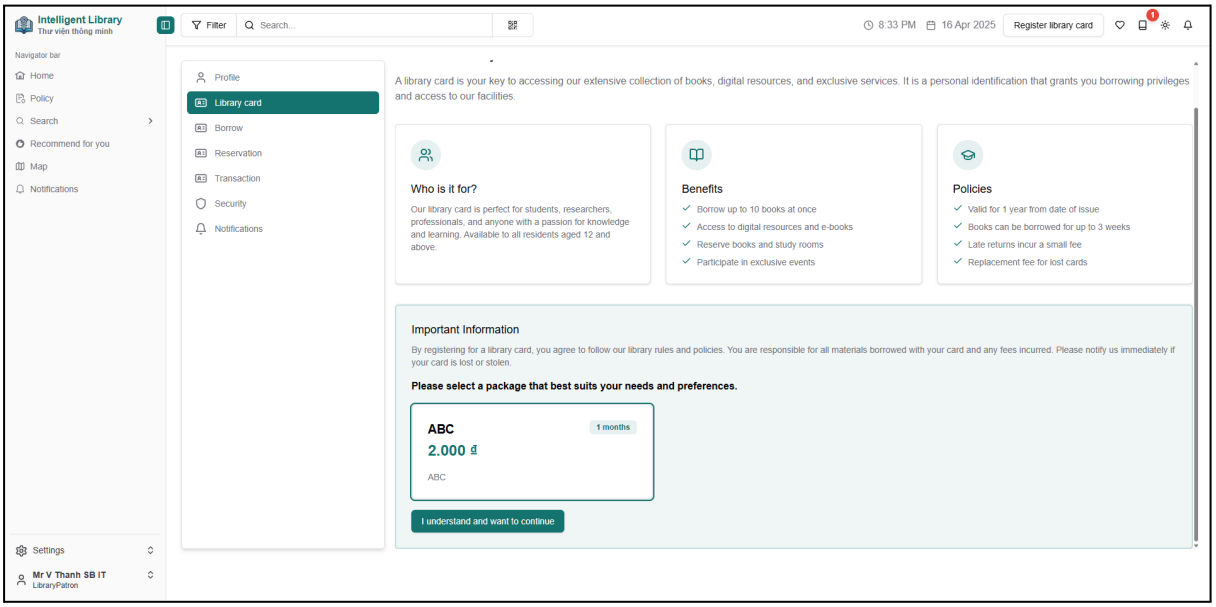


Figure 3.3.2.2: Patron's library card

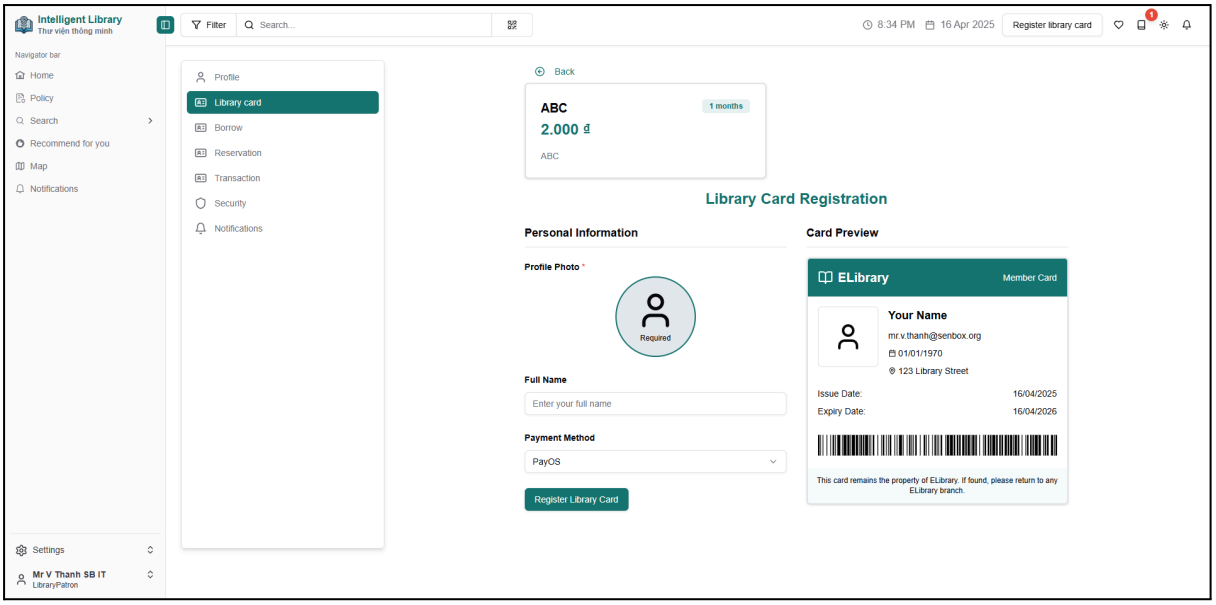


Figure 3.3.2.3: Patron's library card

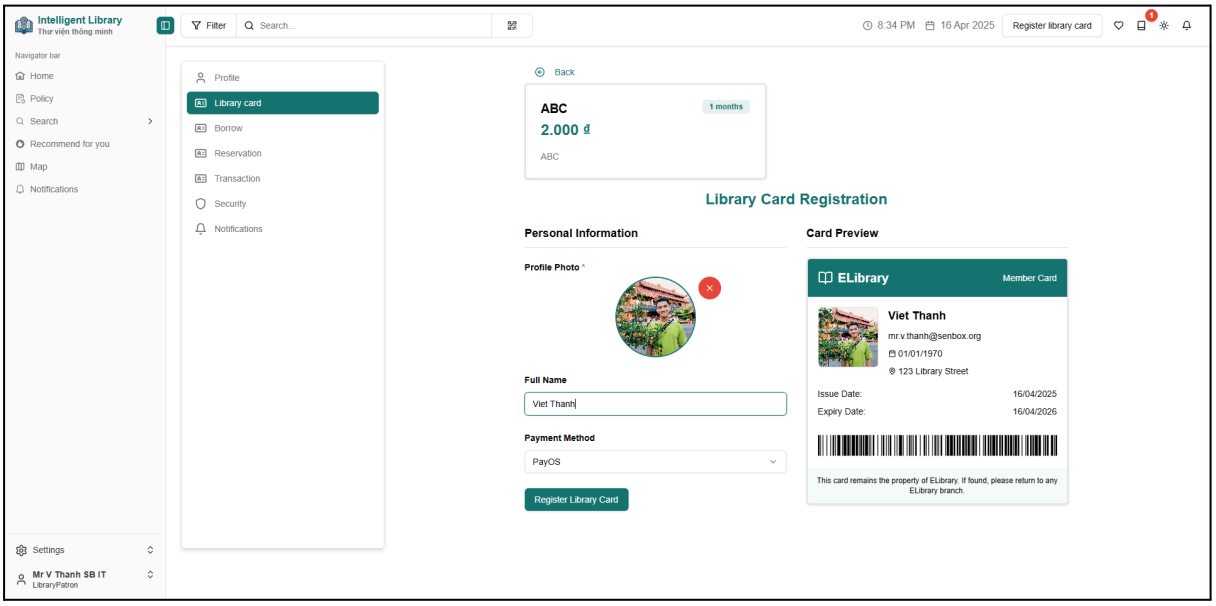


Figure 3.3.2.4: Patron's library card

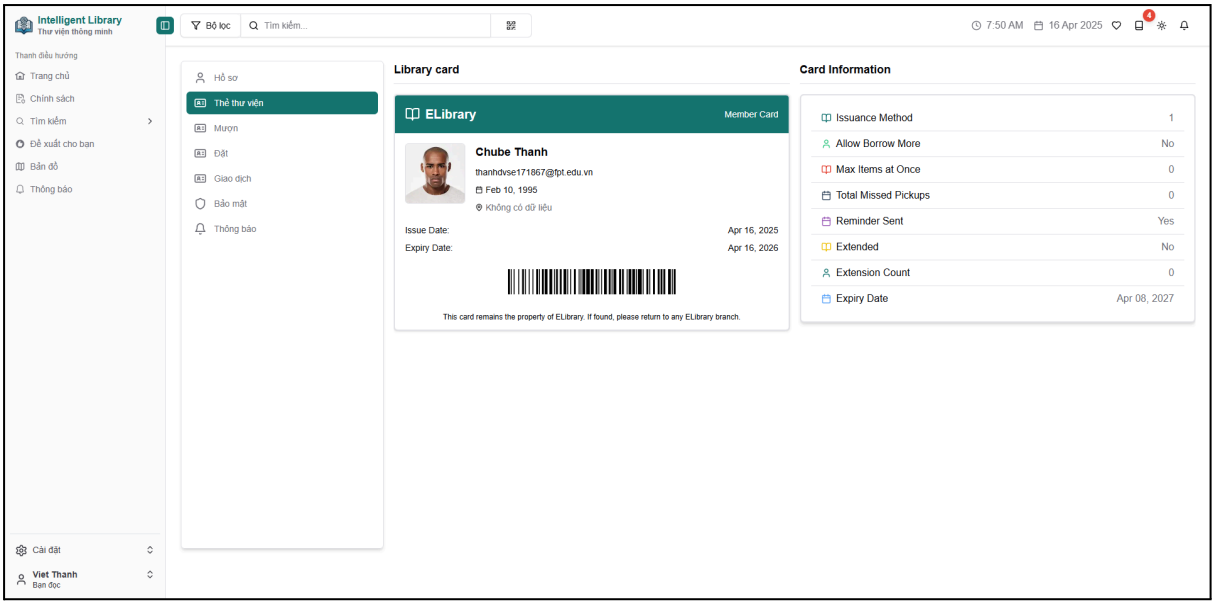


Figure 3.3.2.5: Patron's library card

3.3.3 Patron's borrow request list

- **Function Description:** The page shows the list of the borrow requests.
- **Function Trigger:**
  - **View the list of borrow requests:**
    - **Button/Navigation:** There is a "Borrow request" tab in the borrow page.
    - **Trigger:** The user clicks the "Borrow request" tab to see the list of the borrowed request.
    - **Function:** The page shows the borrow request list information including request date, expiration date, total request items, status (created, expired, borrowed, cancelled)
  - **Search borrow request:**
    - **Button/Navigation:** There is a search bar on the top of the borrow request list.
    - **Trigger:** The user types the keyword of the borrow request.
    - **Function:** The system displays the list of borrow requests based on the entered keyword.
  - **Limit the number of borrow requests:**
    - **Button/Navigation:** There is a row per page button on the bottom of the borrow request list.
    - **Trigger:** The user selects the value of the row per page.
    - **Function:** The number of notifications is shown by the selected value of the row per page (5, 10, 30, 50, 100).
  - **View the detail of borrow requests:**
    - **Button/Navigation:** There is a row per page button on the bottom of the borrow request list.
    - **Trigger:** The user selects the value of the row per page.
    - **Function:** The number of notifications is shown by the selected value of the row per page (5, 10, 30, 50, 100).
  - **Filter borrow requests:**

- **Button/Navigation:** There is a filter button which is on the top of the borrow request list, and next to the search bar.
- **Trigger:** The user clicks the filter button, the page displays the filter dialog. Then, the user can filter the borrow request with status, request date, expiration.
- **Function:** The page allows the patron to filter the borrow requests based on status (created, expired, borrowed, cancelled), request date, expiration date, cancelled date, and advanced search

❖ **Control filter borrow requests dialog:**

- **Button/Navigation:** There are 3 “Cancel”, “Reset”, and “Apply” control buttons of the filter borrow request dialog at the end of the dialog.
- **Trigger:** The user can click each control button to cancel, reset, and apply the filter borrow request.
- **Function:**
  + When the user clicks the “Cancel” button, the value of the filter borrow request dialog will be clear and the page will close the dialog.
  + When the user clicks the “Reset” button, the value of filter borrow request dialog will be clear.
  + When the user clicks the “Apply” button, the page will show the list of borrow requests based on the entered value of filter borrow request dialog.

- **Business Rules:** BR-83, BR-84, BR-85

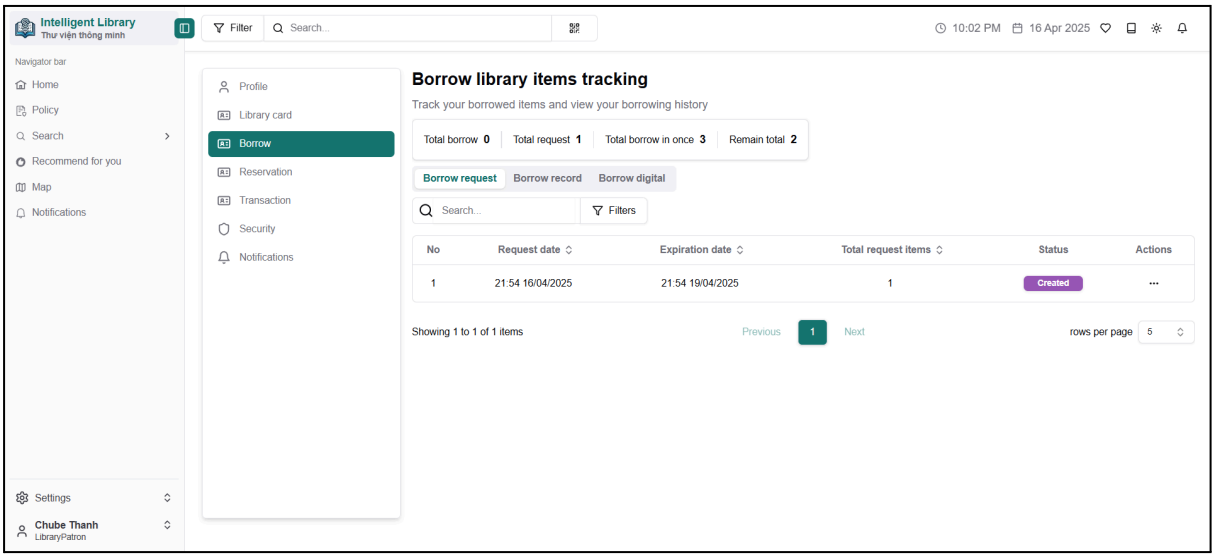


Figure 3.3.3.1: Patron’s request list

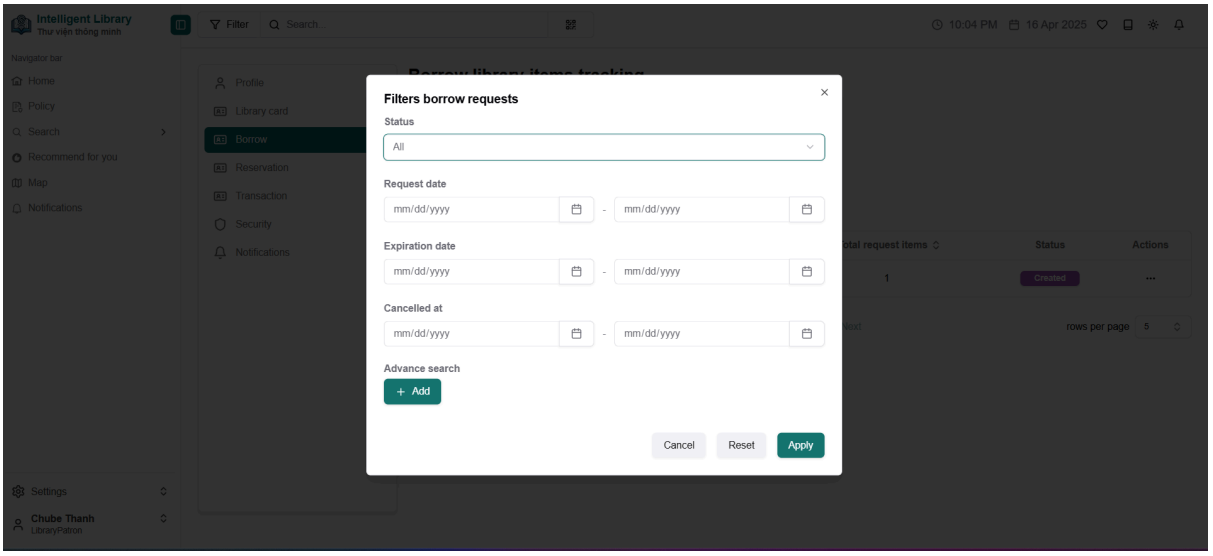


Figure 3.3.3.2: Patron’s request list

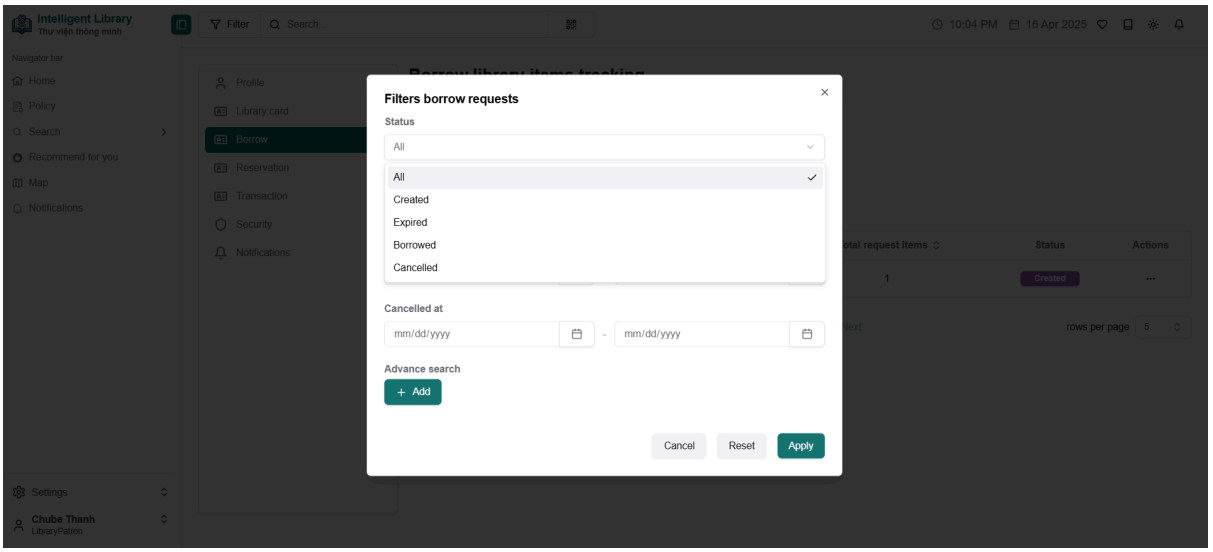


Figure 3.3.3.3: Patron’s request list

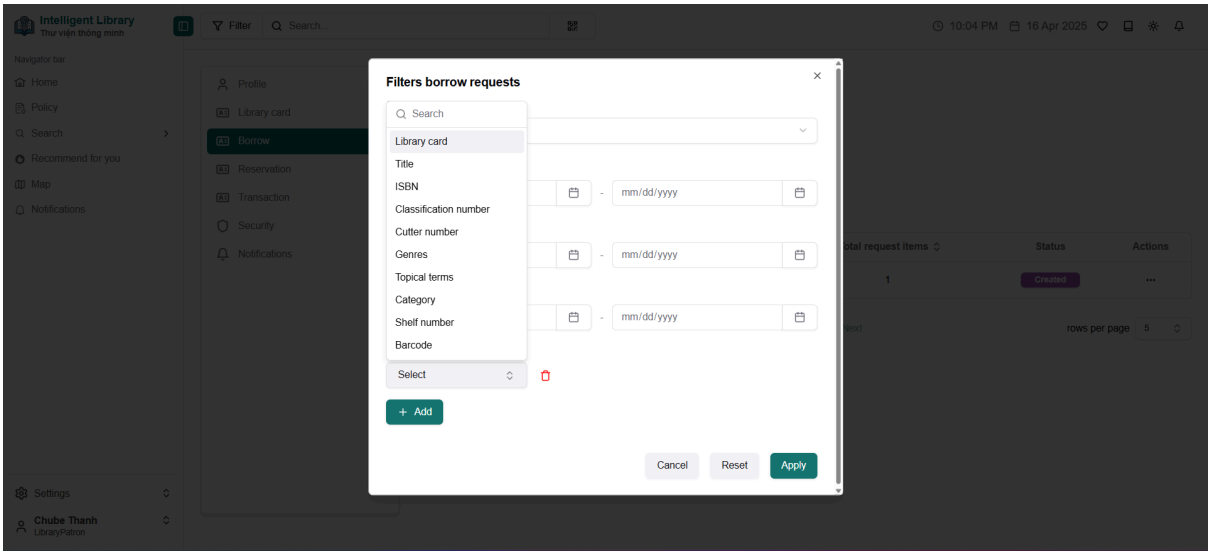


Figure 3.3.3.4: Patron's request list

3.3.4 Patron's borrow request detail

- **Function Description:** The page shows the details of the borrowed request including the library item, reserved item, and the digital resource.
- **Function Trigger:**
  - ❖ **Navigate to the borrow request detail:**
    - ■ **Button/Navigation:** There is a "View detail" button on the action cell of each borrow request.
    - ■ **Trigger:** The patron clicks the "View detail" button.
    - ■ **Function:** The page will navigate the patron to the borrow request detail page showing the list of library items, reserved items and digital resources.
  - ❖ **Cancel the item in the borrow request detail:**
    - ■ **Button/Navigation:** There is a "Cancel" button at the end of the Cancel dialog.
    - ■ **Trigger:** The patron clicks the "Cancel" button.
    - ■ **Function:** The page cancel and close the dialog
  - ❖ **Submit cancel the item in the borrow request detail:**
    - ■ **Button/Navigation:** There is a "Submit" button at the end of the Cancel dialog, which is next to the "Cancel" button.
    - ■ **Trigger:** The patron clicks the "Submit" button.
    - ■ **Function:** The page will remove the indicated item in the borrow request.

- **Business Rules:** BR-20, BR-36, BR-63

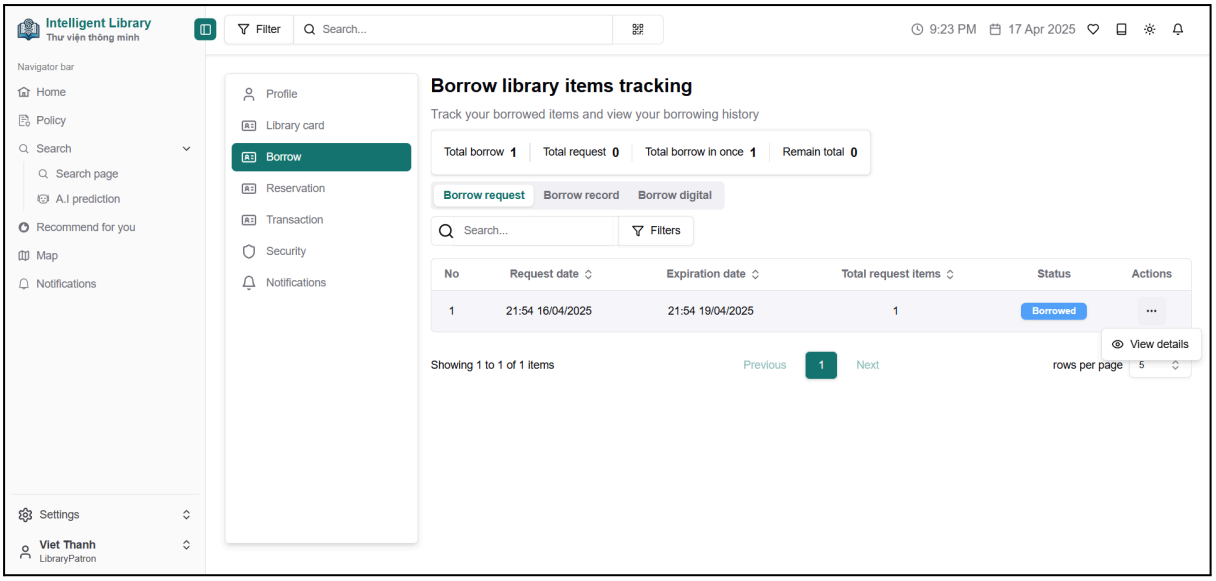


Figure 3.3.4.1: Patron's borrow request detail

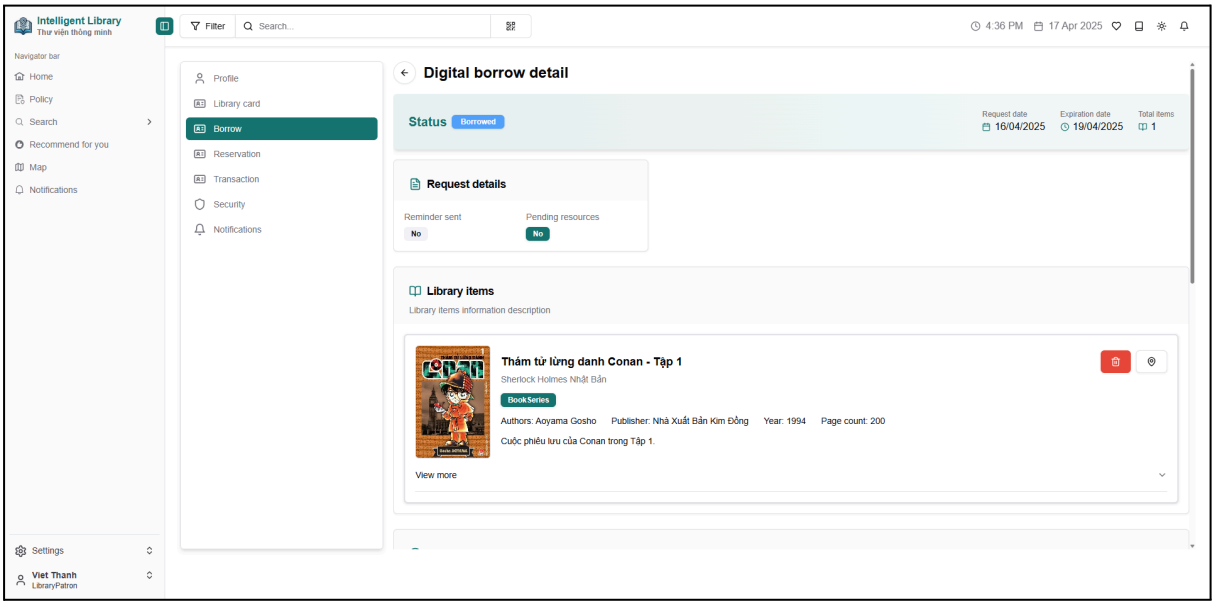


Figure 3.3.4.2: Patron's borrow request detail

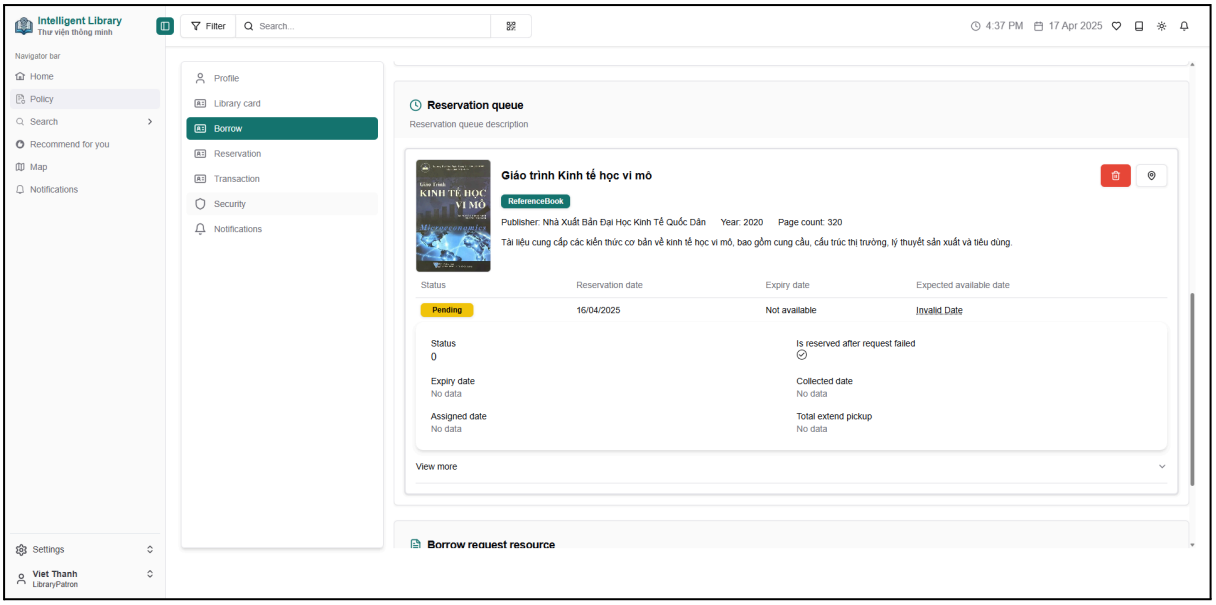


Figure 3.3.4.2: Patron's borrow request detail

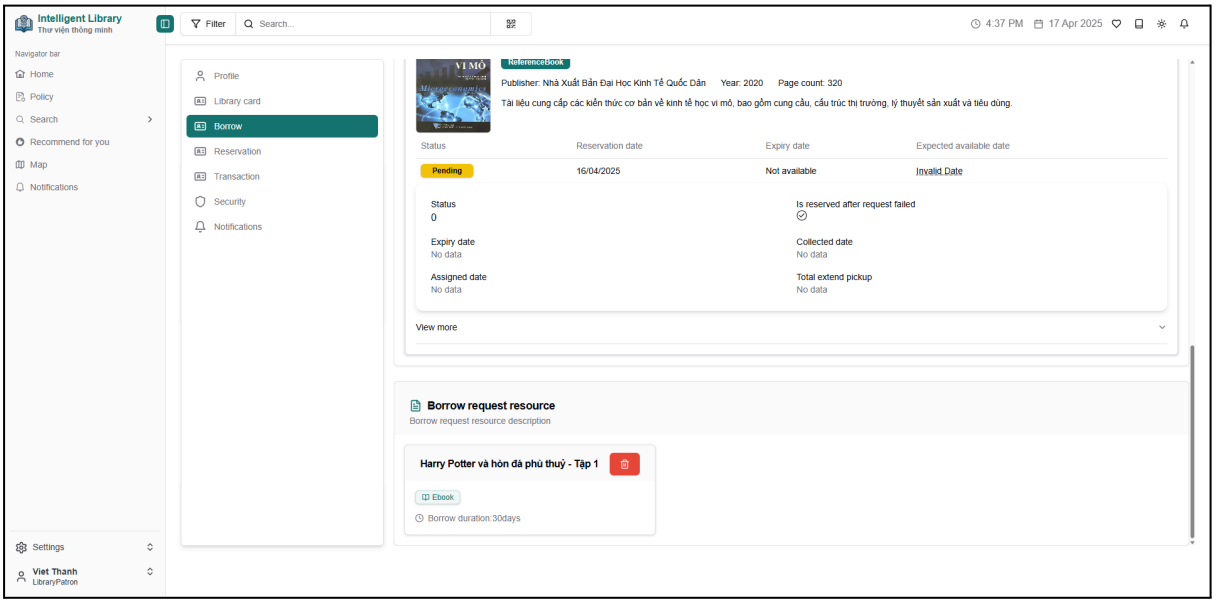


Figure 3.3.4.3: Patron's borrow request detail

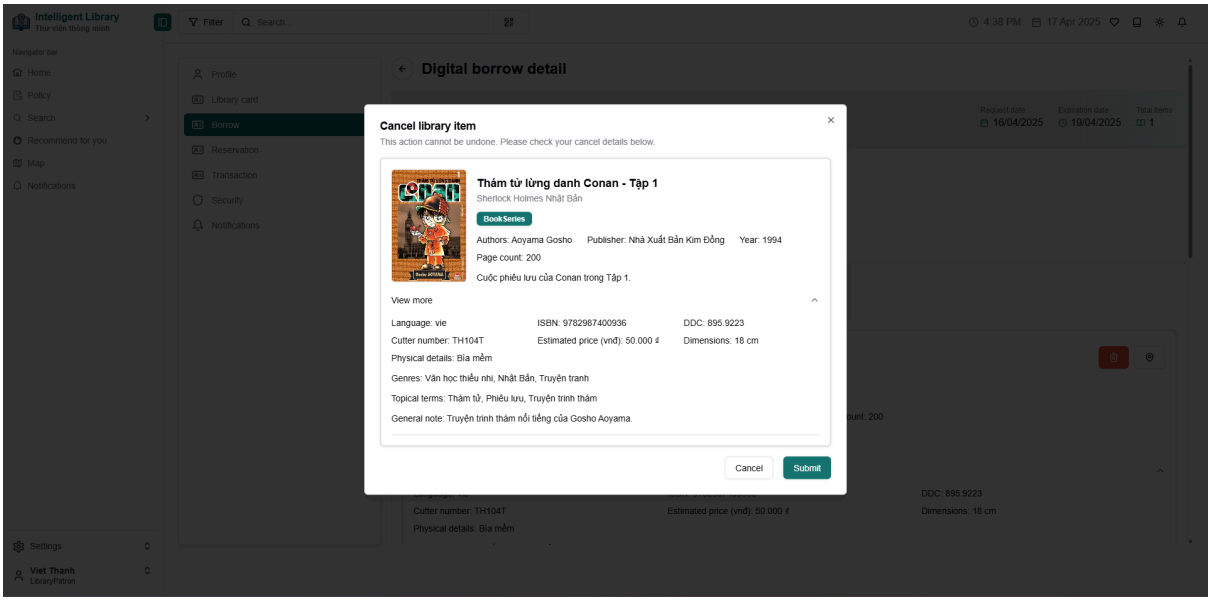


Figure 3.3.4.4: Patron's borrow request detail

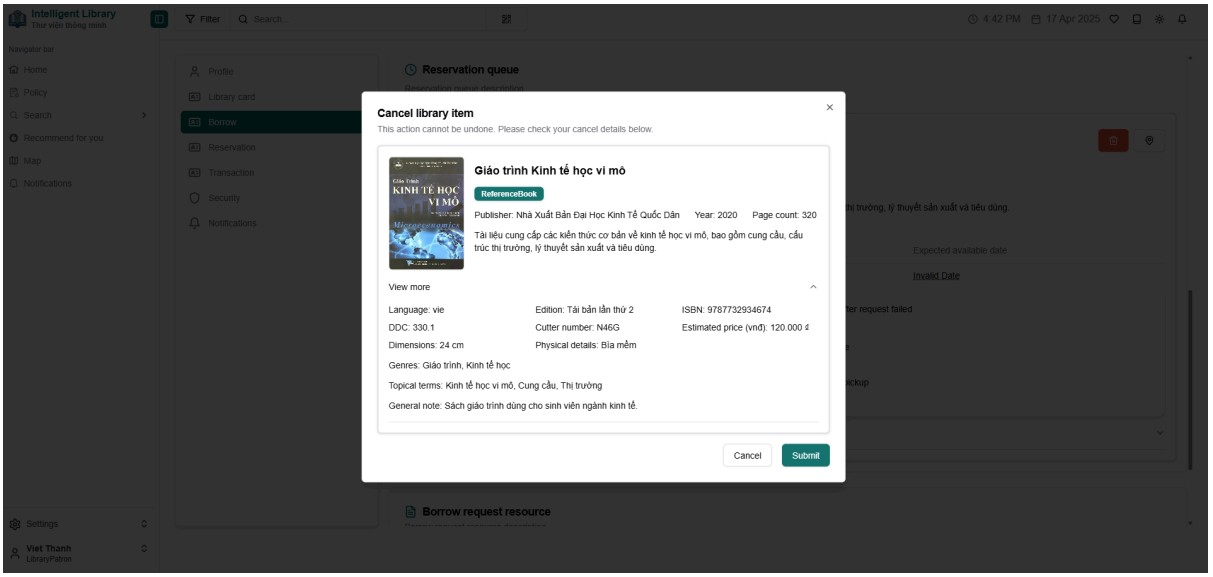


Figure 3.3.4.5: Patron's borrow request detail

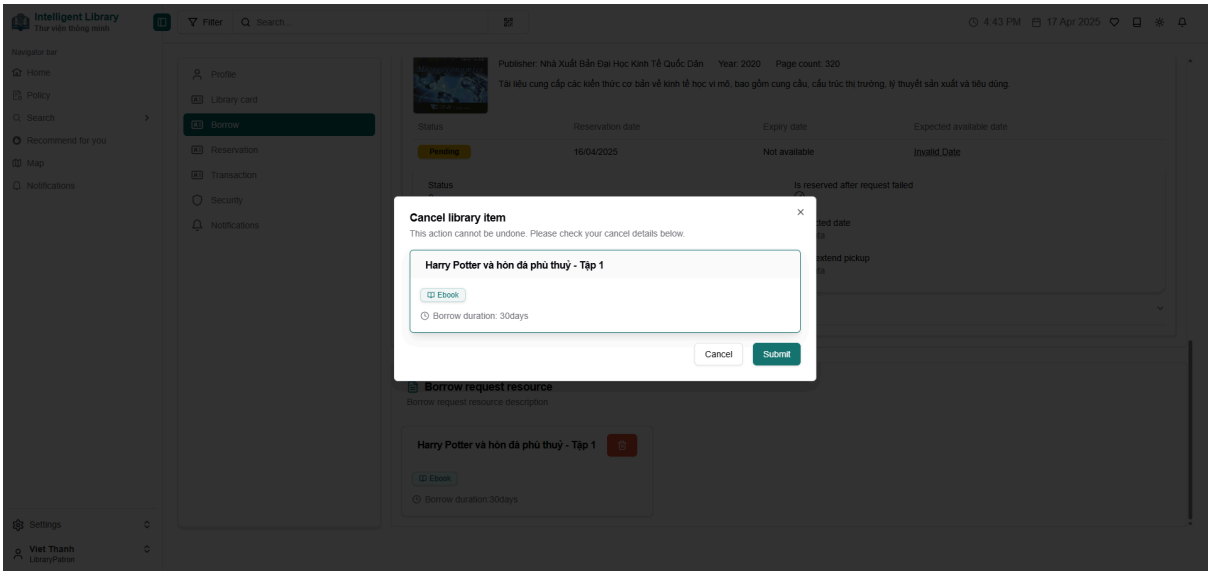


Figure 3.3.4.6: Patron's borrow request detail

3.3.5 Patron's borrow record list

- **Function Description:** The borrowed record page shows the list of patrons' borrowed records. The page also shows some information about borrow date, borrow type (borrowing, returned, overdue, lost), total record items, fine to payment, and process by.
- **Function Trigger:**
  - **Search borrow records:**
    - **Button/Navigation:** There is a search bar on the top of the borrowed record list.
    - **Trigger:** The user types the keyword of the borrowed record.
    - **Function:** The system displays the list of borrowed records based on the entered keyword.
  - **Limit the number of borrowed records:**
    - **Button/Navigation:** There is a row per page button on the bottom of the borrowed records list.
    - **Trigger:** The user selects the value of the row per page.
    - **Function:** The number of borrowed records is shown by the selected value of the row per page (5, 10, 30, 50, 100).
  - **Filter borrowed records:**
    - **Button/Navigation:** There is a filter button which is on the top of the borrowed record list, and next to the search bar.
    - **Trigger:** The user clicks the filter button, the page displays the filter dialog. Then, the user can filter the borrowed records with borrow type (borrowing, returned, overdue, lost), borrow date, and advanced search (library card, title, ISBN, Classification number, cutter number, genres, topical terms, category, shelf number, barcode)
    - **Function:** The system displays the list of borrowed records based on the entered keyword.
  - **Control filter borrowed records dialog:**

- **Button/Navigation:** There are 3 “Cancel”, “Reset”, and “Apply” control buttons of the filter borrowed record dialog at the end of the dialog.
- **Trigger:** The user can click each control button to cancel, reset, and apply the filter borrowed records.
- **Function:**
  + When the user clicks the “Cancel” button, the value of the filter borrowed record dialog will be clear and the page will close the dialog.
  + When the user clicks the “Reset” button, the value of filter borrowed record dialog will be clear.
  + When the user clicks the “Apply” button, the page will show the list of borrowed records based on the entered value of filter borrowed record dialog.

❖ **Business Rules:** BR-43, BR-53, BR-82

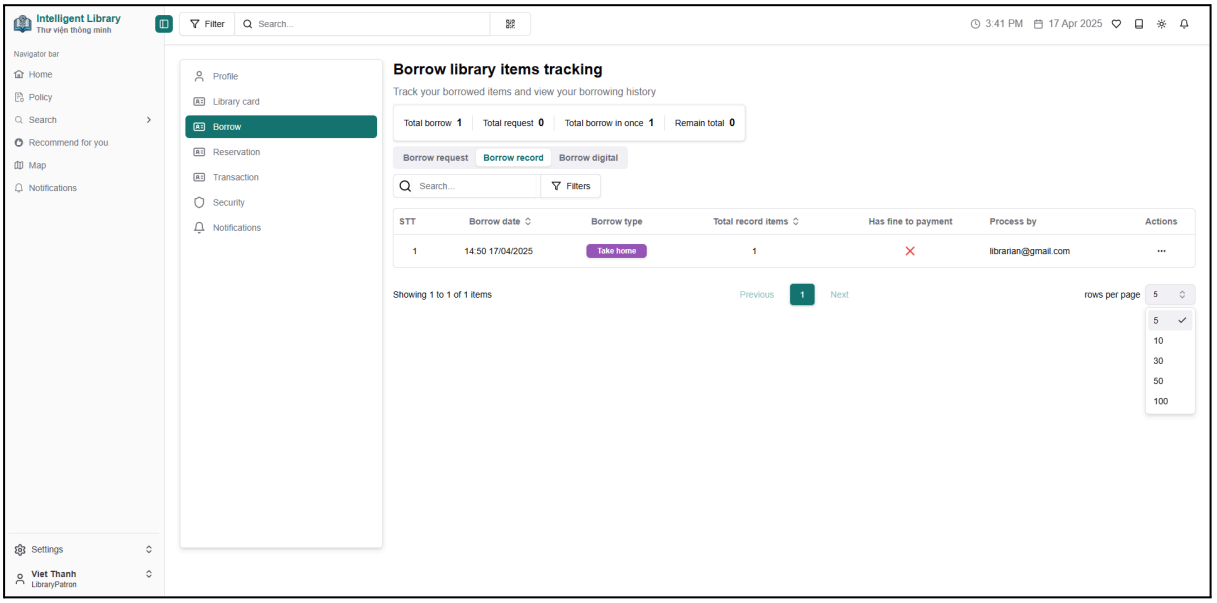


Figure 3.3.5.1: Patron’s borrow record list

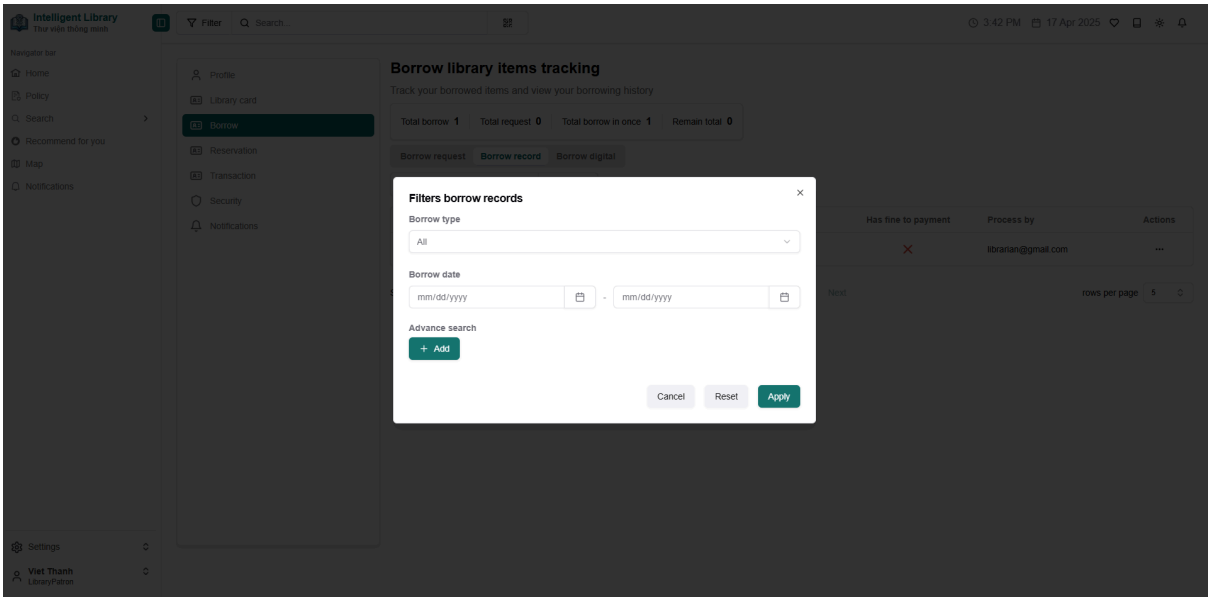


Figure 3.3.5.2: Patron’s borrow record list

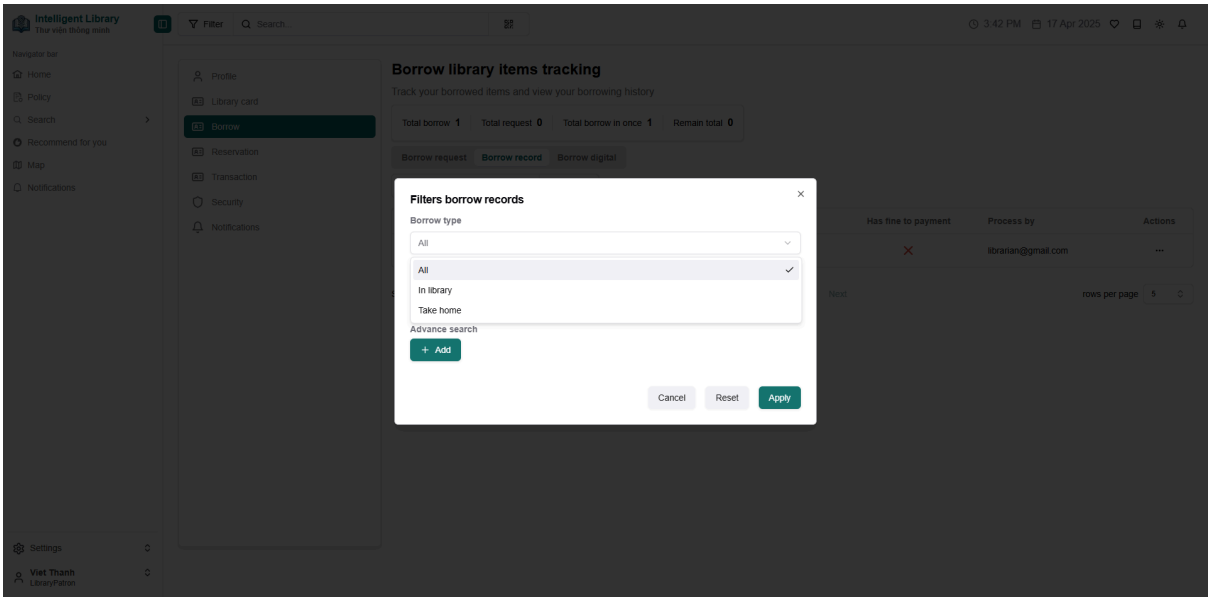


Figure 3.3.5.3 Patron’s borrow record list

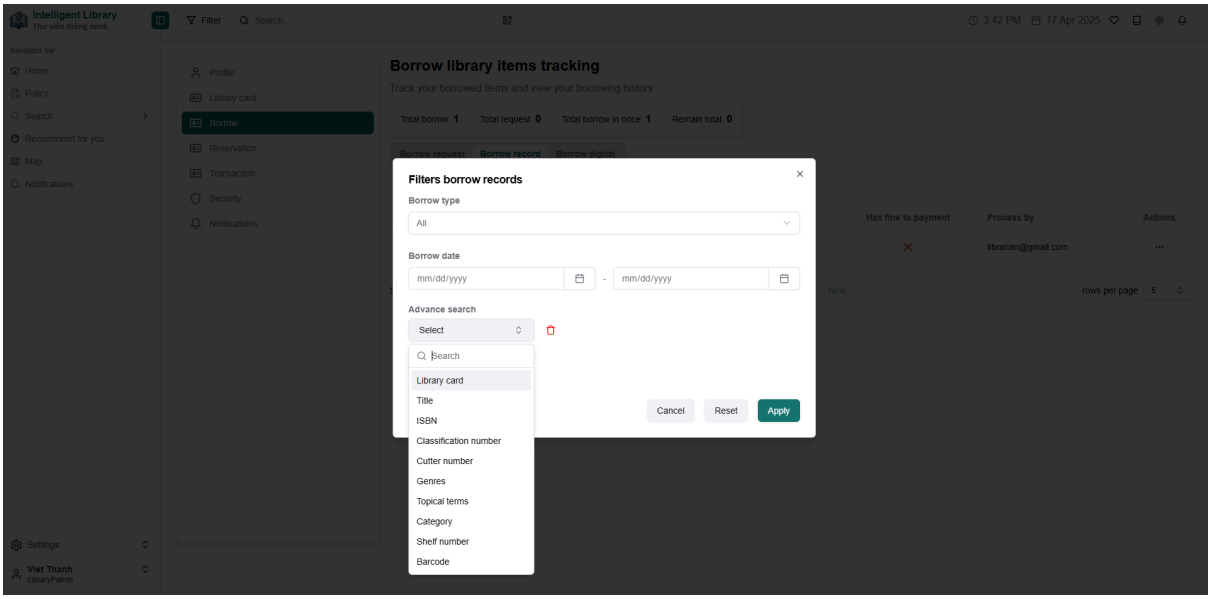


Figure 3.3.5.4: Patron's borrow record list

3.3.6 Patron's borrowed record detail

- **Function Description:** The page shows the details of the borrowed record.
- **Function Trigger:**
  - ❖ **Navigate to the borrow record detail:**
    - ■ **Button/Navigation:** There is a "View detail" button on the action cell of each borrowed record.
    - ■ **Trigger:** The patron clicks the "View detail" button.
    - ■ **Function:** The page will navigate the patron to the borrowed record detail page that show:
      - + Borrow information: borrow date, borrow type (take home, and in library), total record items, has fine to payment, and process by
      - + Borrow record details: library item, unique barcode, due date, return date, status (borrowing, returned, overdue, lost), total extension, condition, return condition, is reminder sent, and has fines.

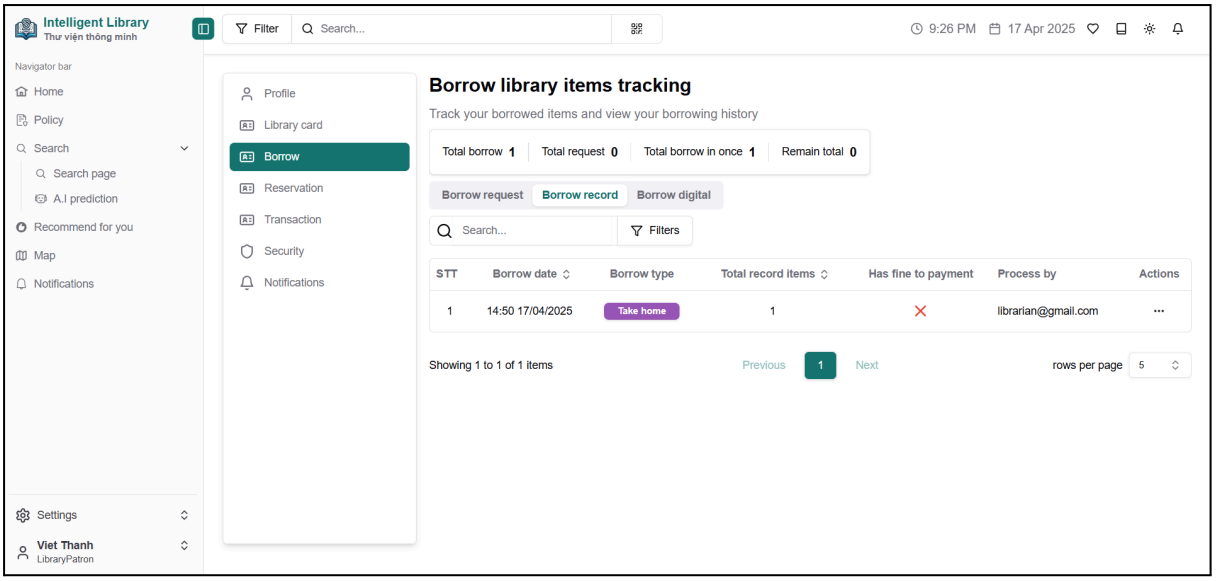


Figure 3.3.6.1: Patron's borrow record detail

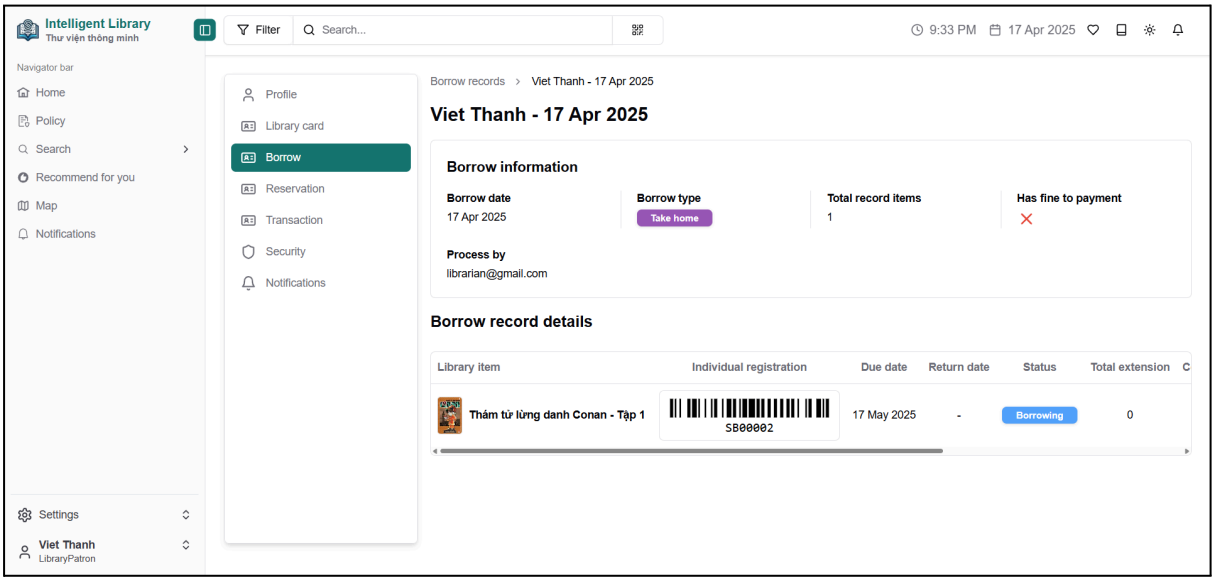


Figure 3.3.6.2: Patron's borrow record detail

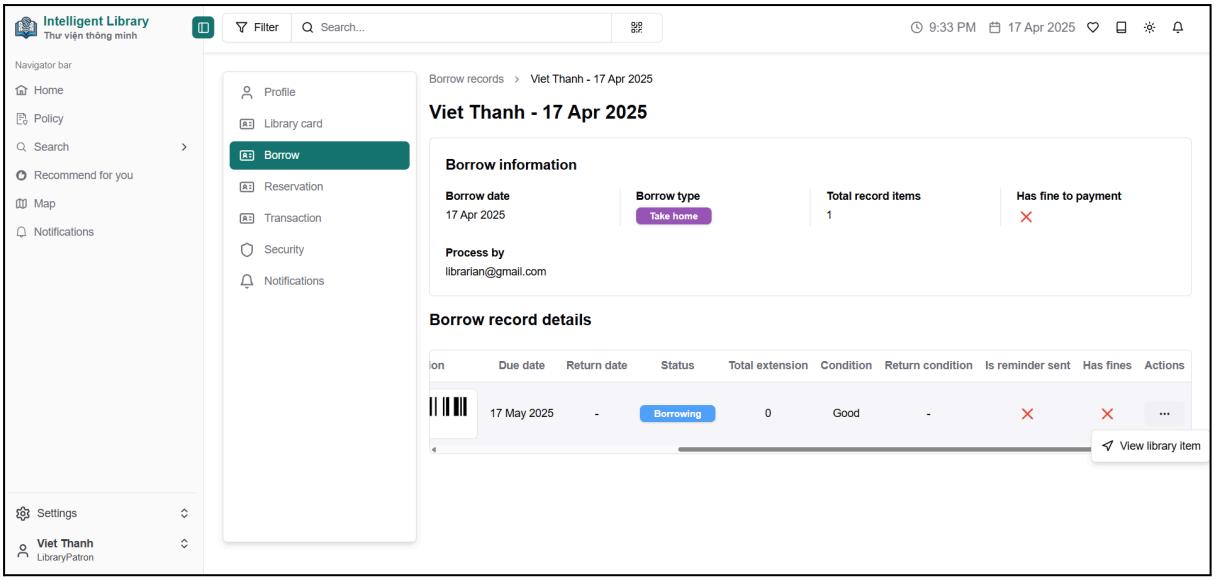


Figure 3.3.6.3: Patron's borrow record detail

3.3.7 Patron's borrow digital list

- **Function Description:** The borrowed digital page shows the list of patron's borrowed digitals. The page also shows some information about title, type (AudioBook, Ebook), registration date, expiration date, borrow price, is extended, extension count, and status (active, expired, cancelled, preparing).
- **Function Trigger:**
  - **Search borrowed digitals:**
    - **Button/Navigation:** There is a search bar on the top of the borrowed digital list.
    - **Trigger:** The user types the keyword of the borrowed digital.
    - **Function:** The system displays the list of borrowed digitals based on the entered keyword.
  - **Limit the number of borrowed digitals:**
    - **Button/Navigation:** There is a row per page button on the bottom of the borrowed digital list.
    - **Trigger:** The user selects the value of the row per page.
    - **Function:** The number of borrowed digitals shown by the selected value of the row per page (5, 10, 30, 50, 100).
  - **Filter borrowed digitals:**
    - **Button/Navigation:** There is a filter button which is on the top of the borrowed digital list, and next to the search bar.
    - **Trigger:** The user clicks the filter button, the page displays the filter dialog. Then, the user can filter the borrowed digitals with status (all, active, expired, cancelled, preparing), extended (all, yes, no), borrow date, expiry date, and advanced search (resource type, resource size, default borrow duration days, and borrow price).
    - **Function:** The system displays the list of borrowed digitals based on the entered keyword.
  - **Control filter borrowed digitals dialog:**

- **Button/Navigation:** There are 3 “Cancel”, “Reset”, and “Apply” control buttons of the filter borrowed digital dialog at the end of the dialog.
- **Trigger:** The user can click each control button to cancel, reset, and apply the filter borrowed digital.
- **Function:**
  + When the user clicks the “Cancel” button, the value of the filter borrowed digital dialog will be clear and the page will close the dialog.
  + When the user clicks the “Reset” button, the value of the filter borrowed digital dialog will be clear.
  + When the user clicks the “Apply” button, the page will show the list of borrowed digitals based on the entered value of filter borrowed digital dialog.

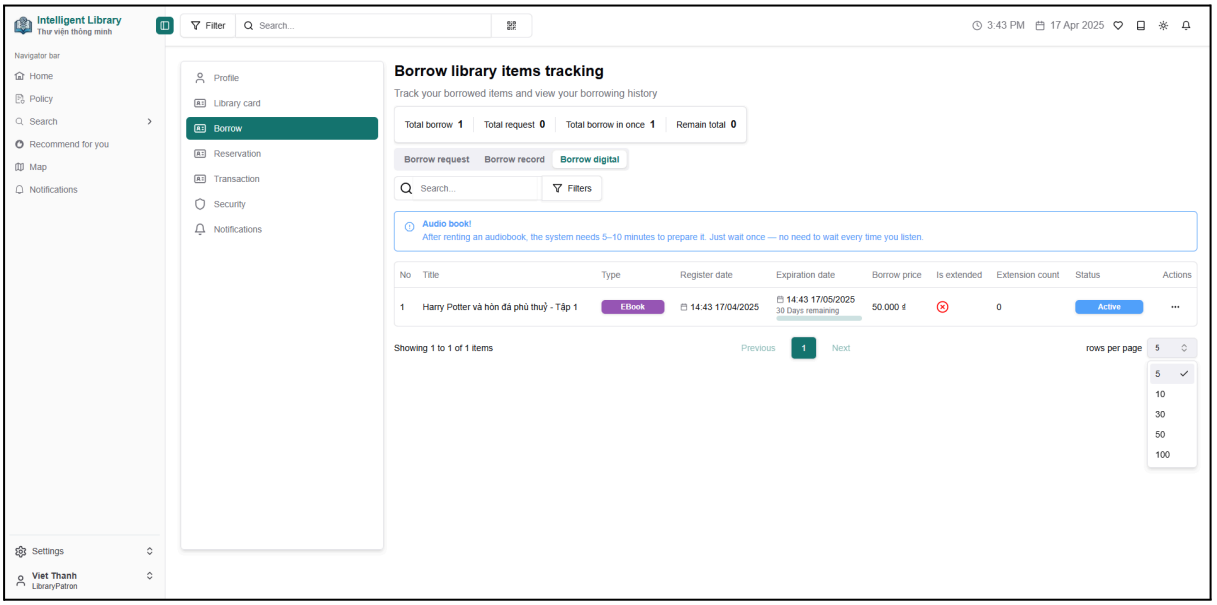


Figure 3.3.7.1: Patron’s borrow digital list

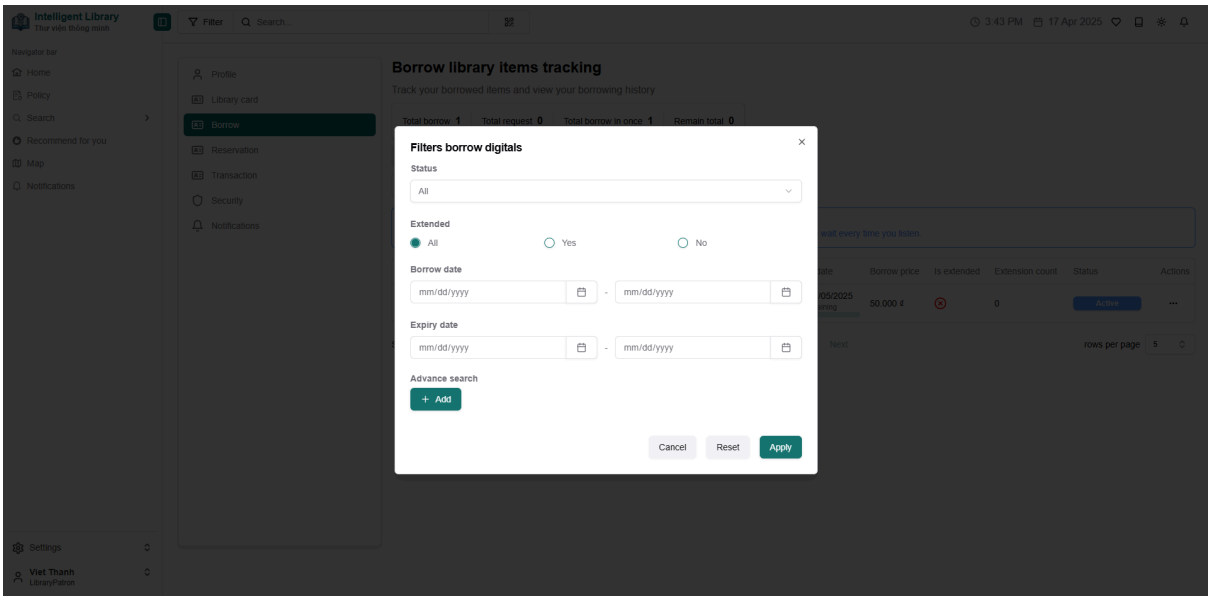


Figure 3.3.7.2: Patron’s borrow digital list

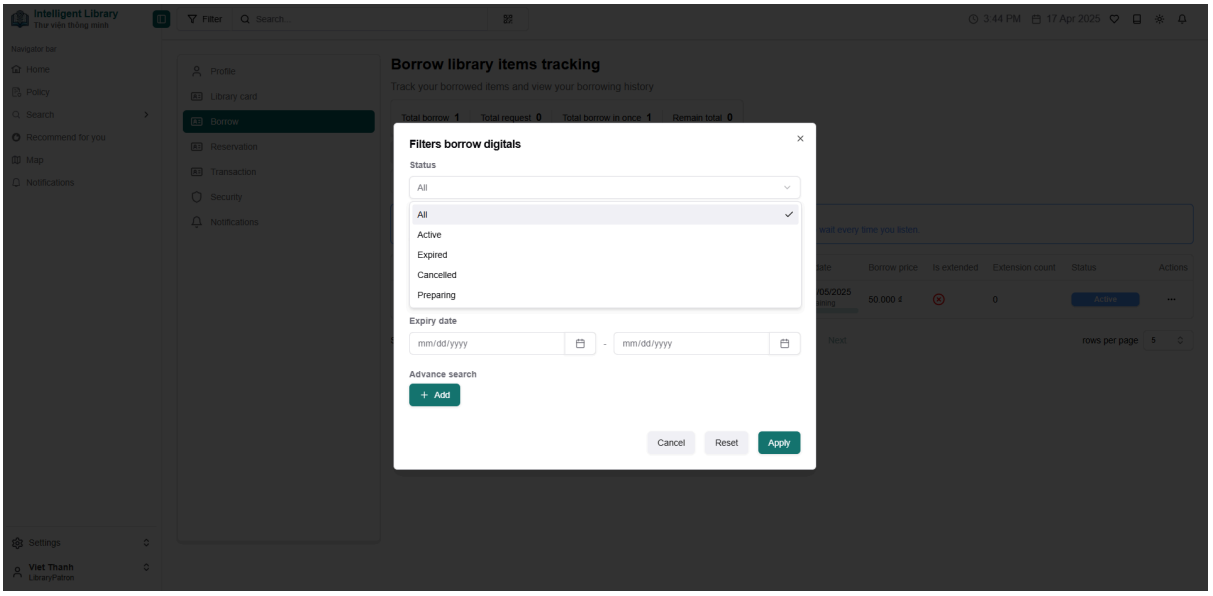


Figure 3.3.7.3: Patron’s borrow digital list

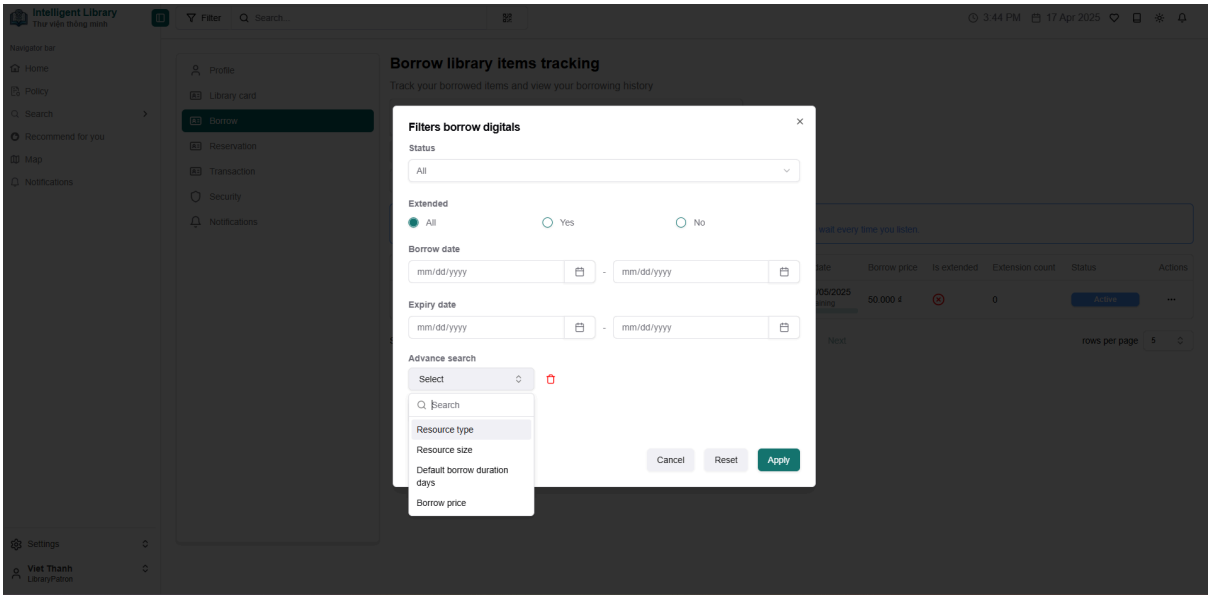


Figure 3.3.7.4: Patron’s borrow digital list

3.3.8 Patron’s borrow digital detail

- **Function Description:** The page shows the details of the borrowed digital.
- **Function Trigger:**
  - ❖ **Navigate to the borrow digital detail:**
    - **Button/Navigation:** There is a “View detail” button on the action cell of each borrowed digital.
    - **Trigger:** The patron clicks the “View detail” button.
    - **Function:** The page will navigate the patron to the borrowed digital detail page.

❖ **Navigate to the borrow resource:**

- **Button/Navigation:** There is a “View resource” button on the action cell of each borrowed digital.
- **Trigger:** The patron clicks the “View resource” button.
- **Function:** The page will navigate the patron to the resource embedded with watermark in order to enhance the security.

- **Business Rules:** BR-22, BR-23, BR-25

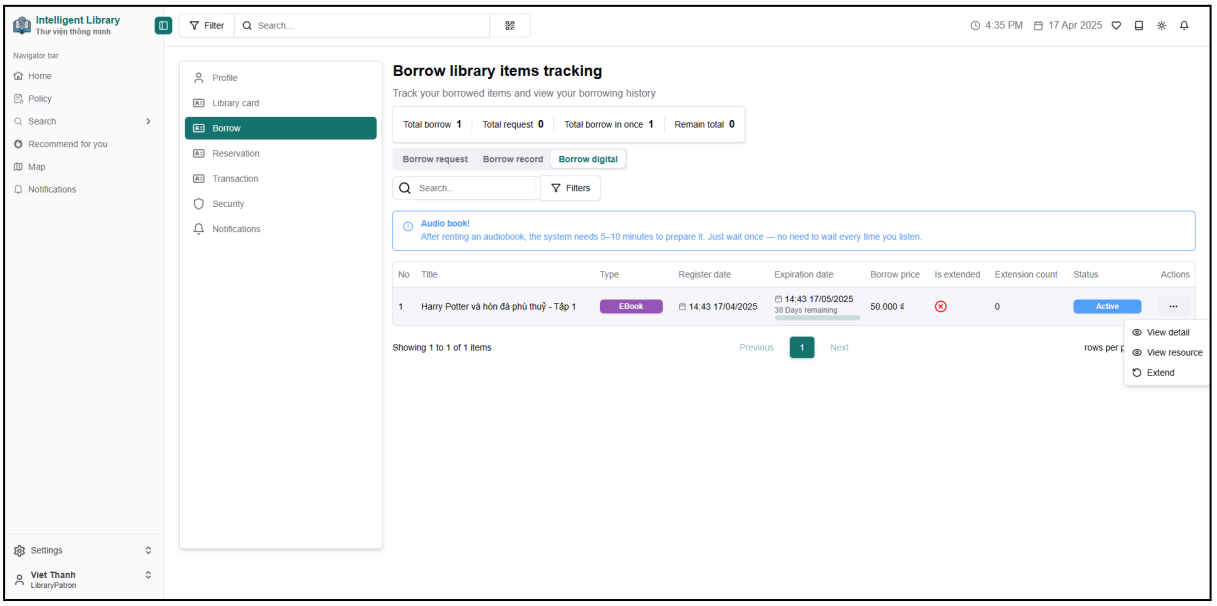


Figure 3.3.8.1: Patron’s borrow digital detail

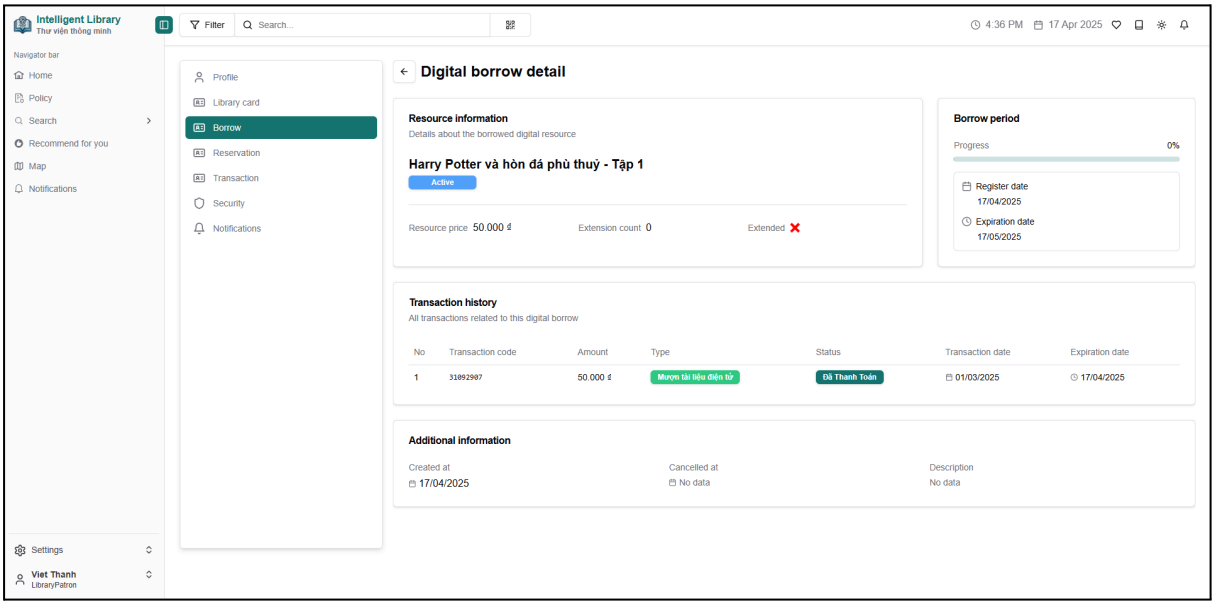


Figure 3.3.8.2: Patron’s borrow digital detail

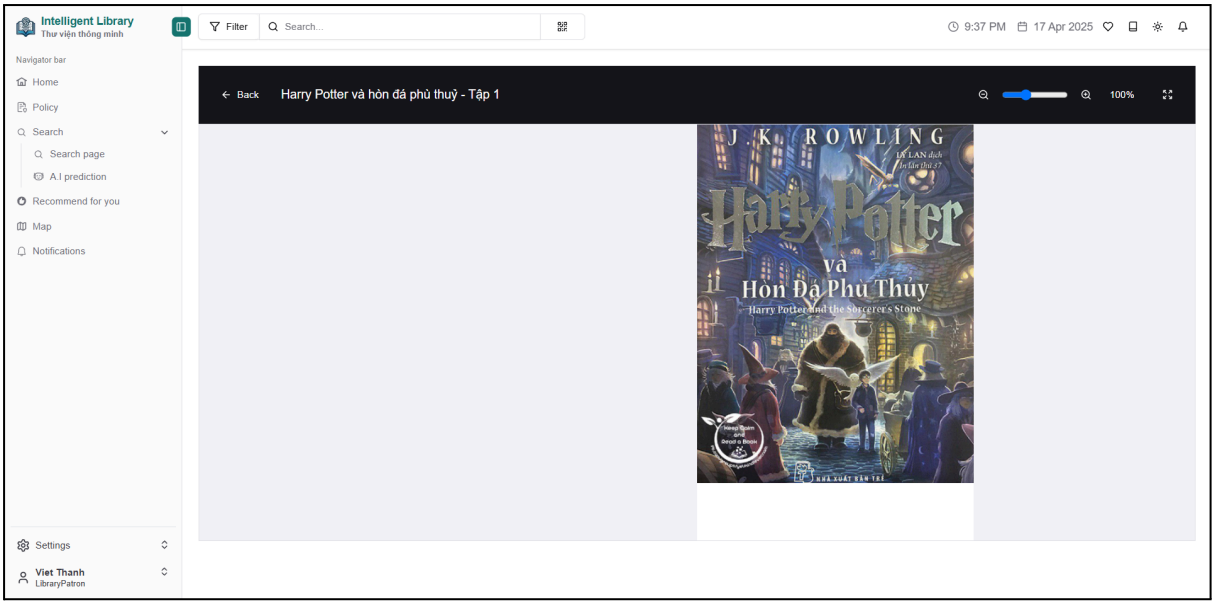


Figure 3.3.8.3: Patron's borrow digital detail

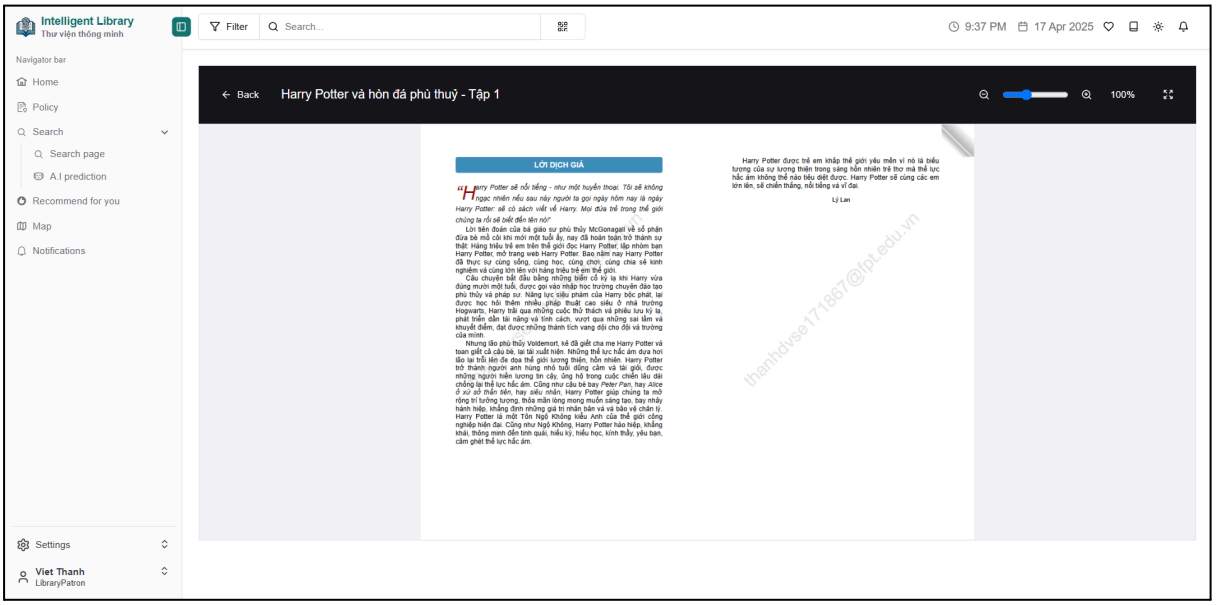


Figure 3.3.8.3: Patron's borrow digital detail

3.3.9 Patron's reservation list

- **Function Description:** The reservation page shows the list of patron's reservations. The page also shows some information about the reserved library item, reservation code, status (pending, assigned, collected, expired, and cancelled), and reservation date.
- **Function Trigger:**
  - **Search reservations:**
    - **Button/Navigation:** There is a search bar on the top of the reservation list.
    - **Trigger:** The user types the keyword of the reservation.

- **Function:** The system displays the list of reservations based on the entered keyword.

❖ **Limit the number of reservations:**

- **Button/Navigation:** There is a row per page button on the bottom of the reservation list.
- **Trigger:** The user selects the value of the row per page.
- **Function:** The number of reservations is shown by the selected value of the row per page (5, 10, 30, 50, 100).

❖ **Filter reservations:**

- **Button/Navigation:** There is a filter button which is on the top of the reservation list, and next to the search bar.
- **Trigger:** The user clicks the filter button, the page displays the filter dialog. Then, the user can filter the reservations with some fields like applied label, notified, reserved after request failed, status (pending, assigned, collected, expired, and cancelled), reservation date, expiry date, assigned date, collected date, expected date min, and expected date max.
- **Function:** The system displays the list of reservations based on the entered keyword.

❖ **Control filter reservations dialog:**

- **Button/Navigation:** There are 3 “Cancel”, “Reset”, and “Apply” control buttons of the filter reservation dialog at the end of the dialog.
- **Trigger:** The user can click each control button to cancel, reset, and apply the filter reservation.
- **Function:**
  + When the user clicks the “Cancel” button, the value of the filter reservation dialog will be clear and the page will close the dialog.
  + When the user clicks the “Reset” button, the value of filter reservation dialog will be clear.
  + When the user clicks the “Apply” button, the page will show the list of reservations based on the entered value of filter reservation dialog.

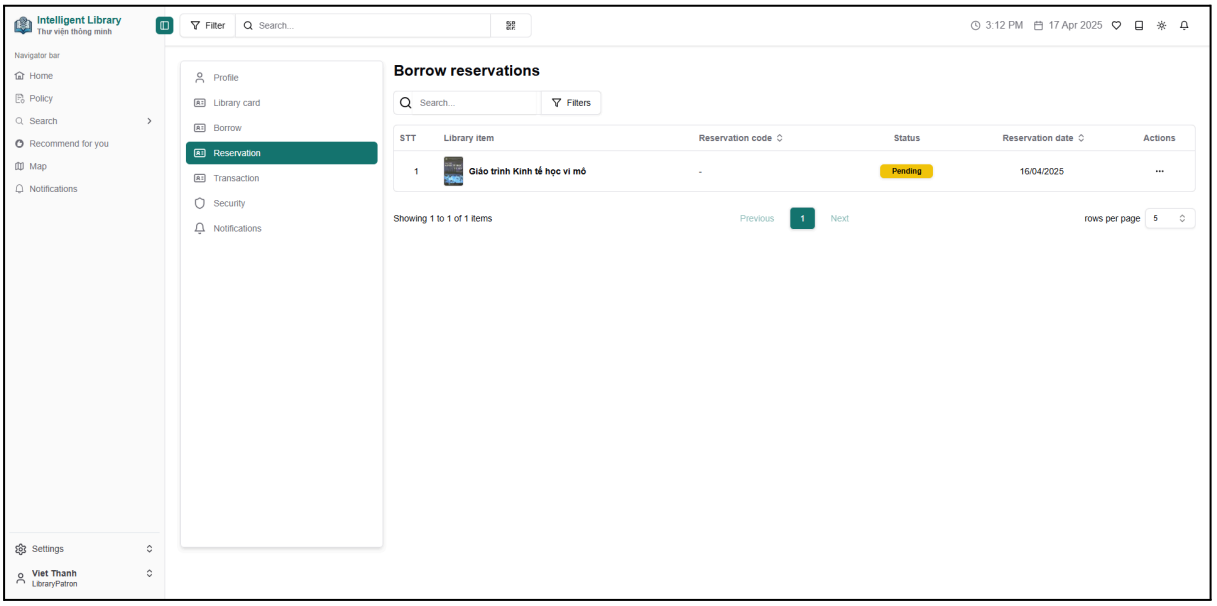


Figure 3.3.9.1: Patron’s reservation list

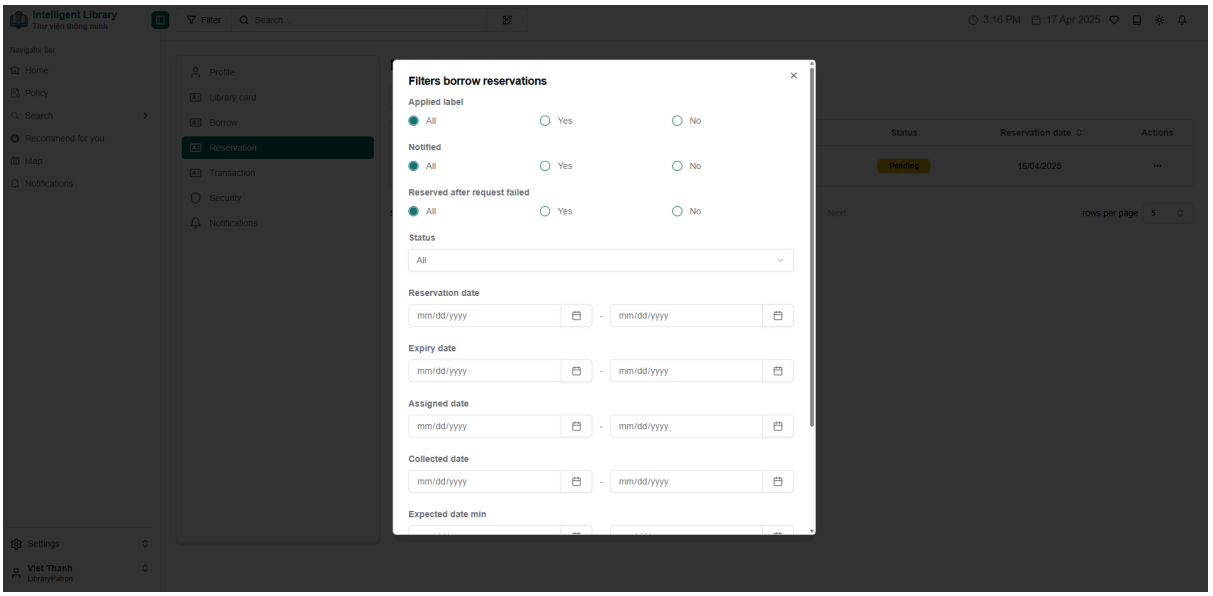


Figure 3.3.9.2: Patron’s reservation list

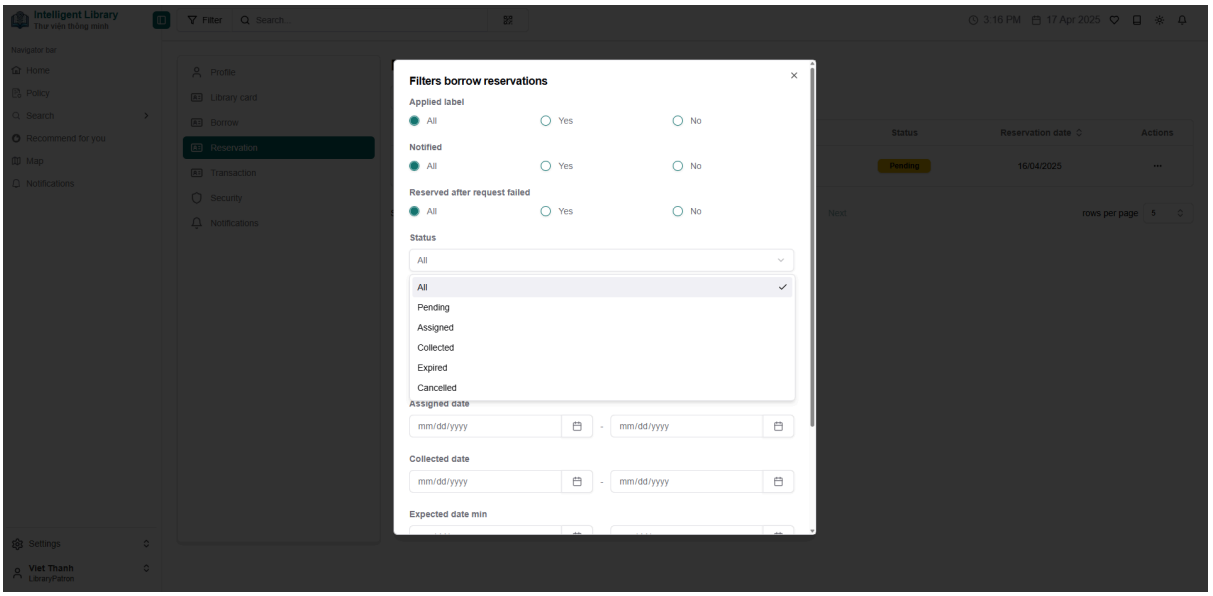


Figure 3.3.9.3: Patron’s reservation list

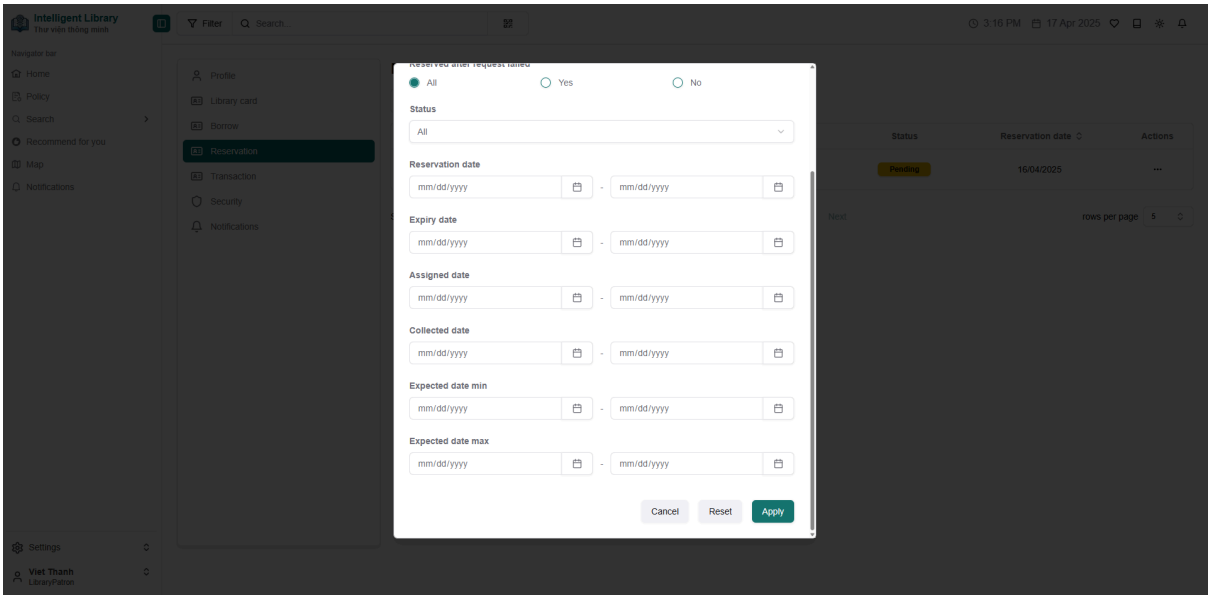


Figure 3.3.9.4: Patron's reservation list

3.3.10 Patron's reservation detail

- **Function Description:** The page shows the details of the reservation including two main sections like reservation information, and library item information.
- **Function Trigger:**
  - ❖ **Navigate to the reservation page:**
    - ■ **Button/Navigation:** There is a "View detail" button on the action cell of each reservation in the reservation list page.
    - ■ **Trigger:** The patron clicks the "View detail" button.
    - ■ **Function:** The page will navigate the patron to the reservation detail page that shows:
      - + Reservation information: reservation code, status, assignable, applied label, reservation date, expected date, assigned date, collected date, expiry date, total extended pickup, notified, cancelled by, cellation reason, reserved after request failed, reserved item, and assigned barcode.
      - + Library item: title, author, publisher, publish year, page count, summary, language, edition, ISBN, DDC, cutter number, price, genres, topical terms, and general note.

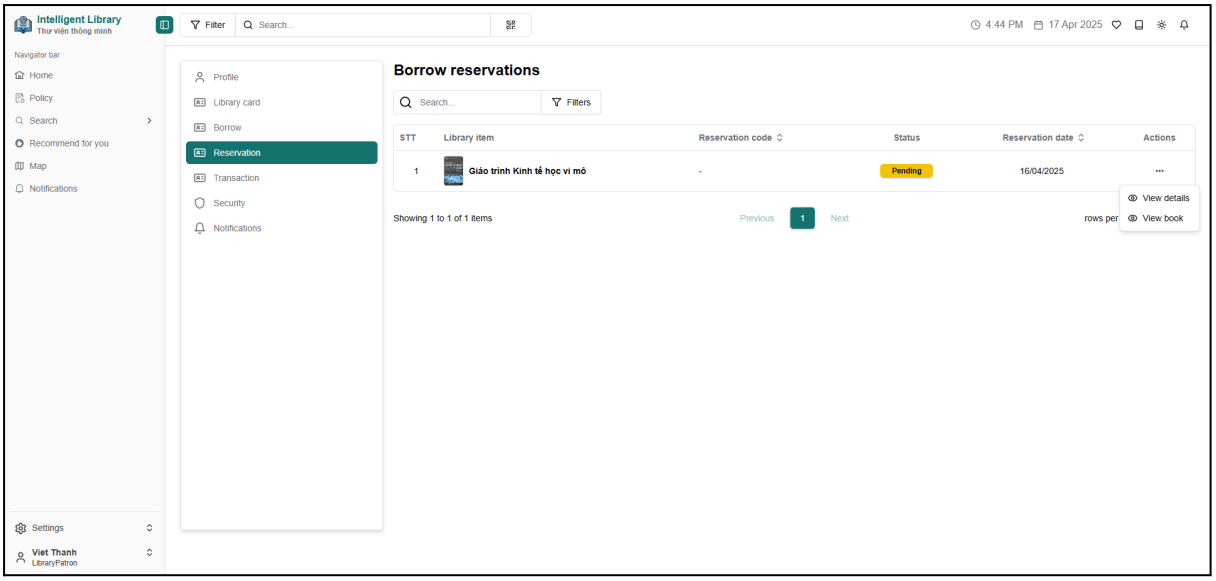


Figure 3.3.10.1: Patron's reservation detail

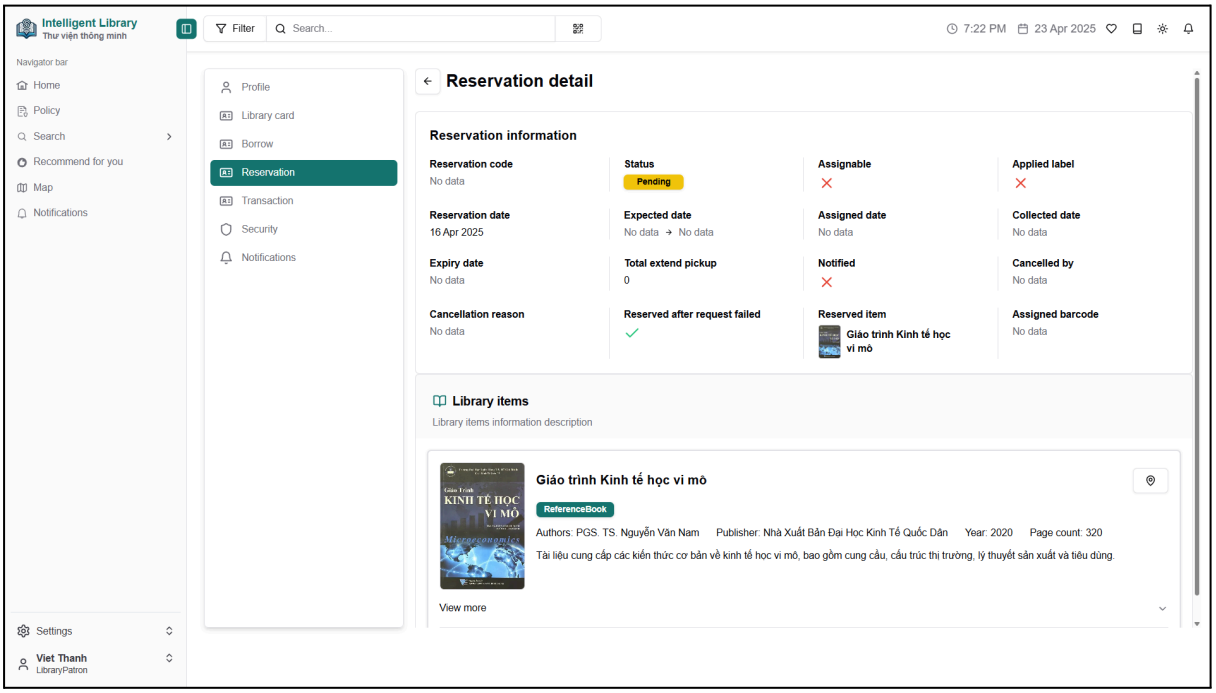


Figure 3.3.10.2: Patron's reservation detail

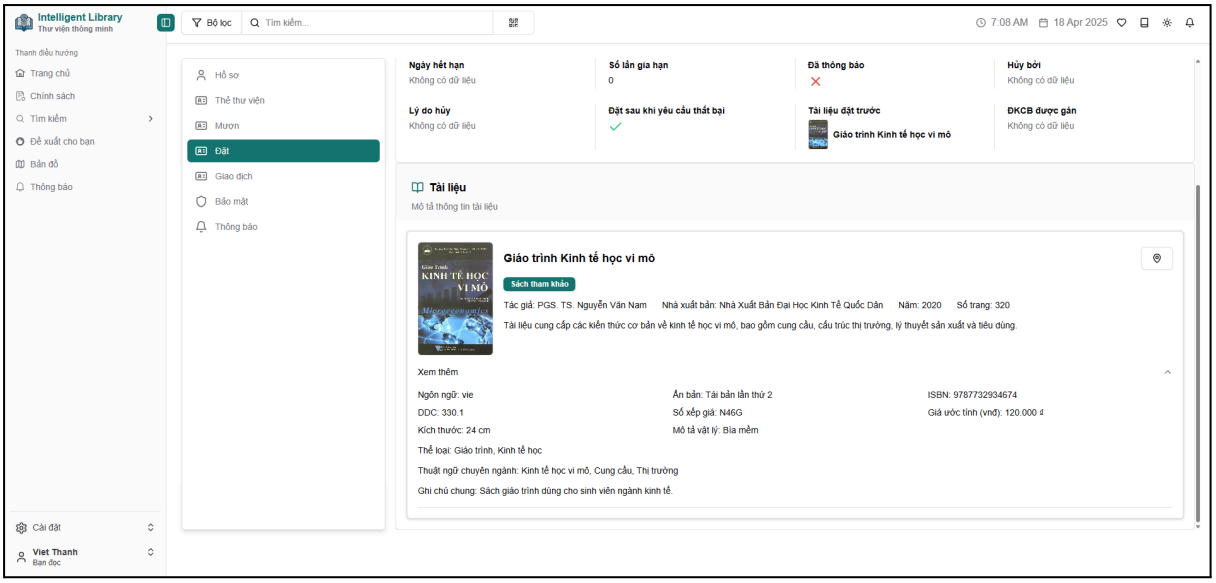


Figure 3.3.10.3: Patron's reservation detail

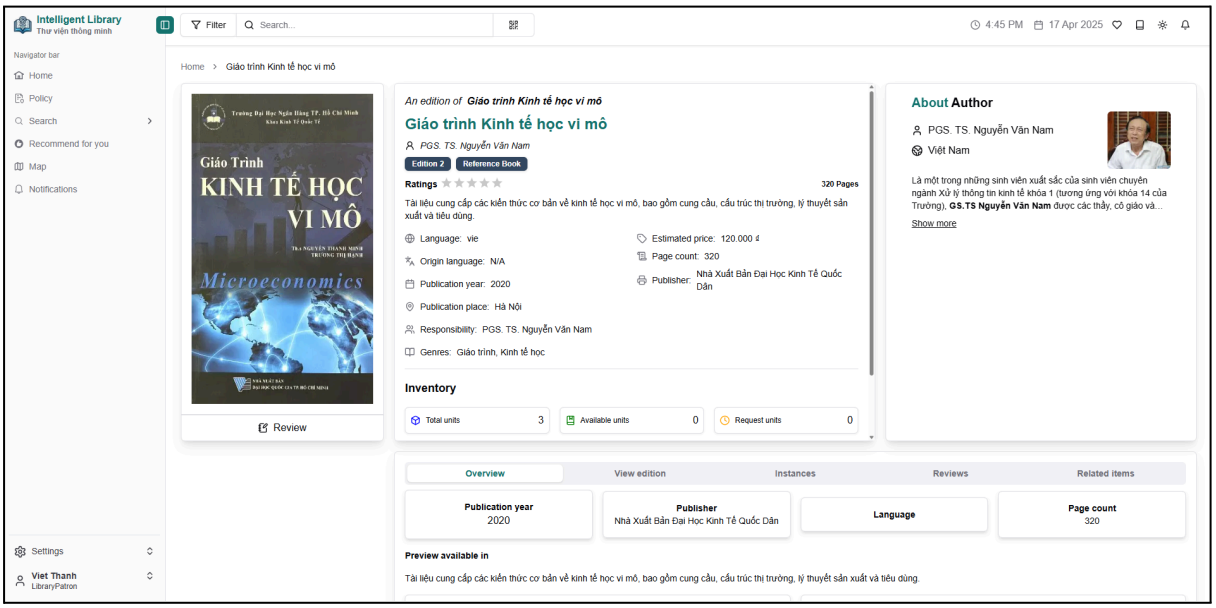


Figure 3.3.10.4: Patron's reservation detail

3.3.11 Patron's transaction list

- **Function Description:** The transaction page shows the list of patron's transactions. The page also shows some information about transaction code, amount, description, status (pending, expired, cancelled, paid), transaction type (fine, digital borrow, library card registration, library card extension, and digital extension), transaction date, expired at, created at, cancelled at, and cancelled reason.
- **Function Trigger:**
  - **Search transactions:**
    - **Button/Navigation:** There is a search bar on the top of the transaction list.

- **Trigger:** The user types the keyword of the transaction.
- **Function:** The system displays the list of transactions based on the entered keyword.

❖ **Limit the number of transactions:**

- **Button/Navigation:** There is a row per page button on the bottom of the transaction list.
- **Trigger:** The user selects the value of the row per page.
- **Function:** The number of transactions is shown by the selected value of the row per page (5, 10, 30, 50, 100).

❖ **Filter transactions:**

- **Button/Navigation:** There is a filter button which is on the top of the transaction list, and next to the search bar.
- **Trigger:** The user clicks the filter button, the page displays the filter dialog. Then, the user can filter the transactions with status (pending, expired, cancelled, paid), transaction type (fine, digital borrow, library card registration, library card extension, and digital extension), transaction date, transaction method (cash, digital), amount, transaction date, cancelled at, and cancelled reason
- **Function:** The system displays the list of transactions based on the entered keyword.

❖ **Control filter transactions dialog:**

- **Button/Navigation:** There are 3 “Cancel”, “Reset”, and “Apply” control buttons of the filter transaction dialog at the end of the dialog.
- **Trigger:** The user can click each control button to cancel, reset, and apply the filter transaction
- **Function:**
  + When the user clicks the “Cancel” button, the value of the filter transaction dialog will be clear and the page will close the dialog.
  + When the user clicks the “Reset” button, the value of filter transaction dialog will be clear.
  + When the user clicks the “Apply” button, the page will show the list of transactions based on the entered value of filter transaction dialog.

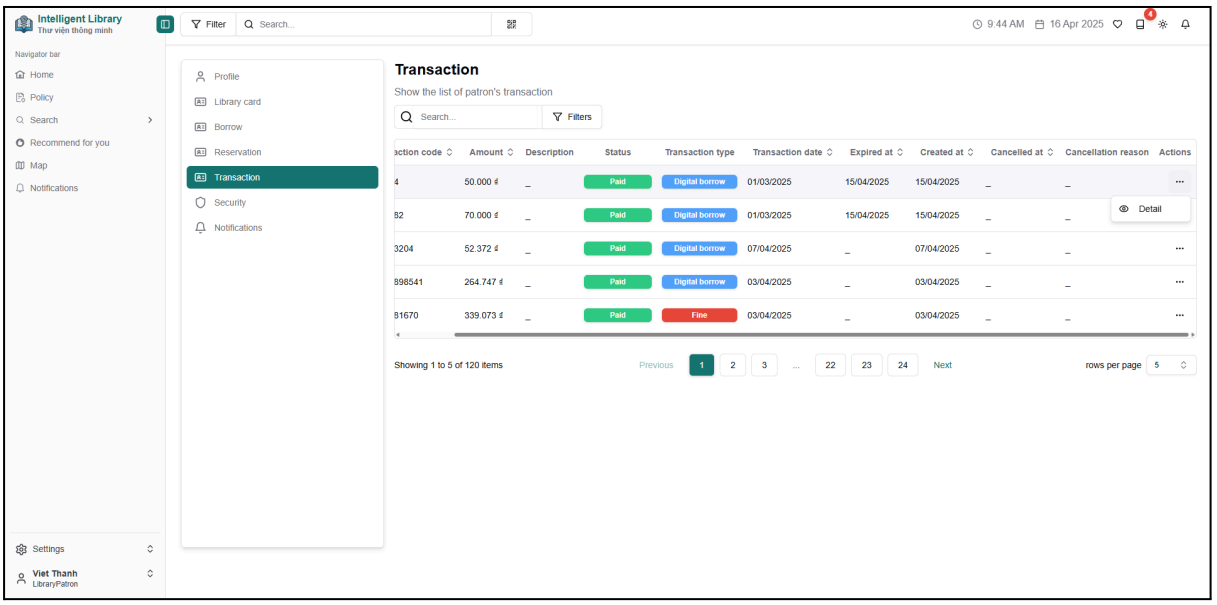


Figure 3.3.11.1: Patron’s transaction list

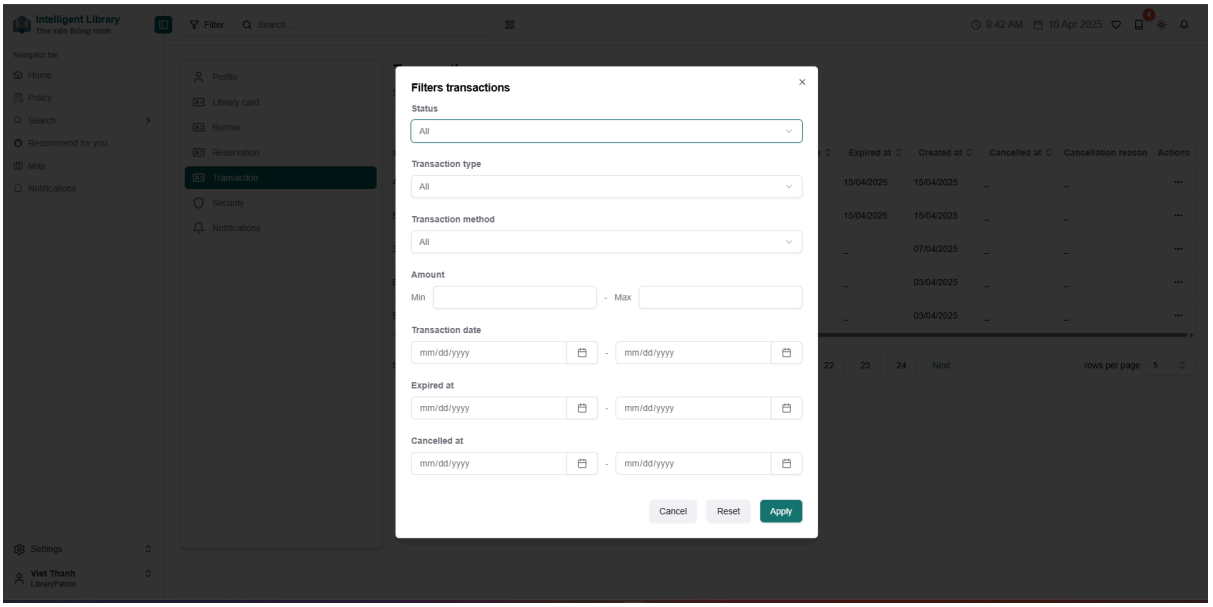


Figure 3.3.11.2: Patron’s transaction list

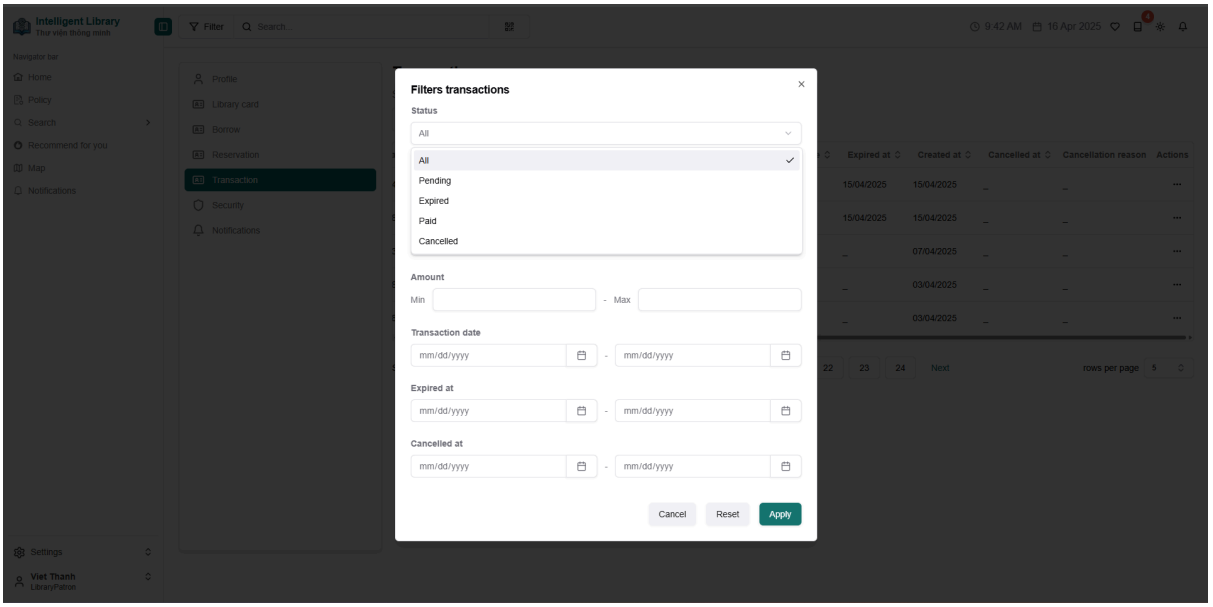


Figure 3.3.11.3: Patron’s transaction list

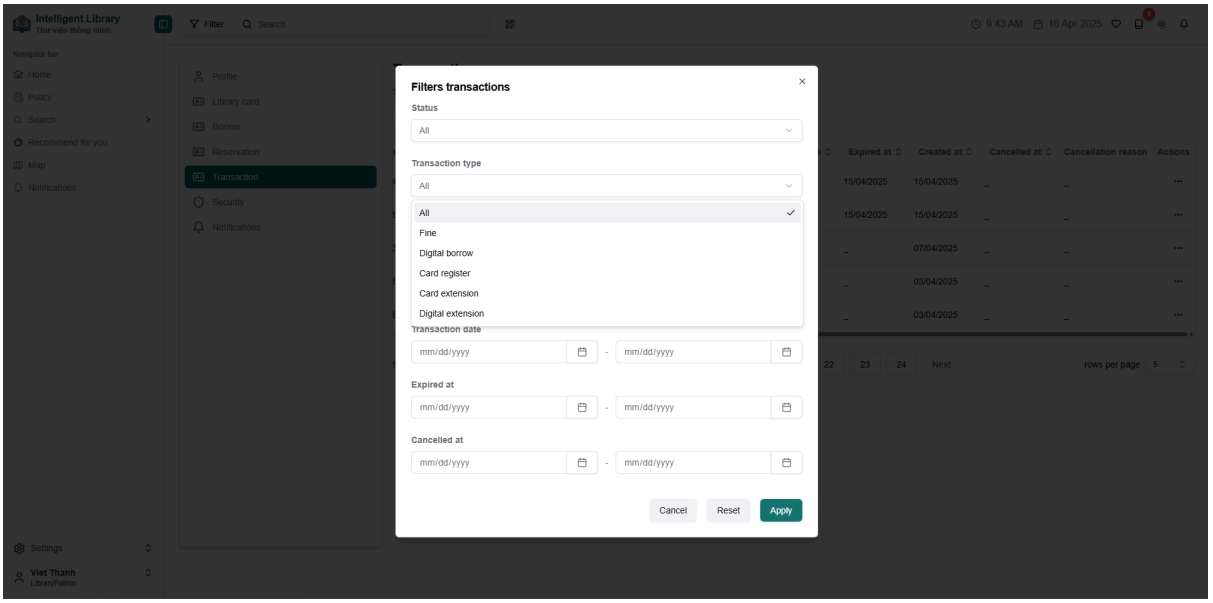


Figure 3.3.11.4: Patron's transaction list

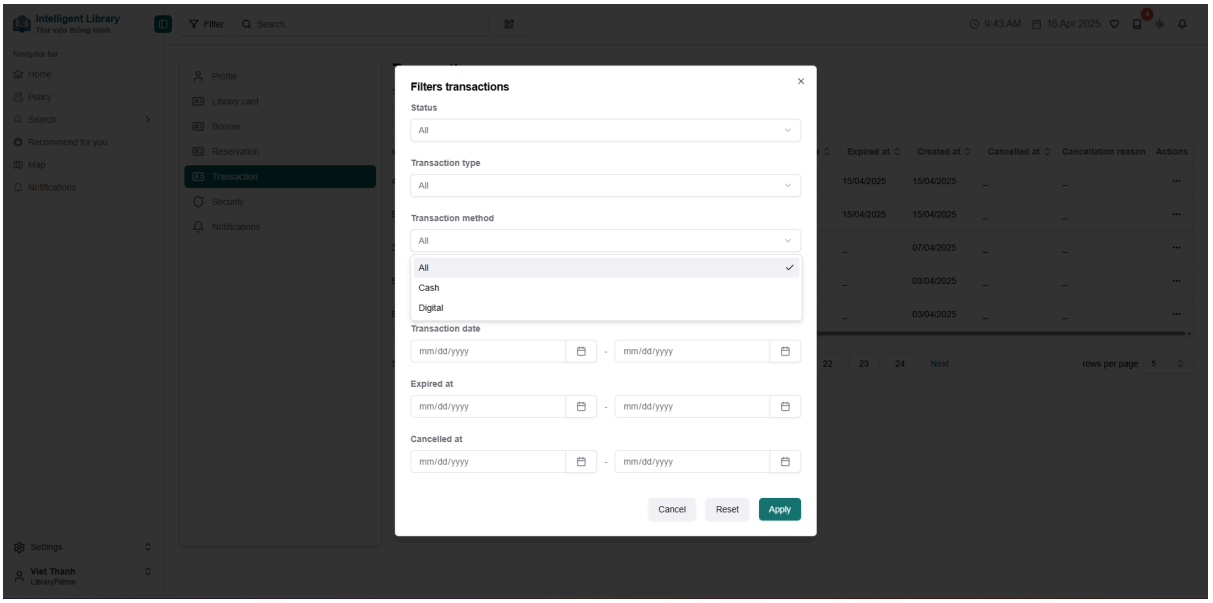


Figure 3.3.11.5: Patron's transaction list

3.3.12 Patron's transaction detail

- **Function Description:** The page shows the details of the patron's transaction.
- **Function Trigger:**
  - ❖ **Navigate to the reservation page:**
    - **Button/Navigation:** There is a "View detail" button on the action cell of each transaction in the transaction list page.
    - **Trigger:** The patron clicks the "View detail" button.
    - **Function:** The page will navigate the patron to the reservation detail page that shows:

+ Reservation information: reservation code, status, assignable, applied label, reservation date, expected date, assigned date, collected date, expiry date, total extended pickup, notified, cancelled by, cellation reason, reserved after request failed, reserved item, and assigned barcode.
+ Library item: title, author, publisher, publish year, page count, summary, language, edition, ISBN, DDC, cutter number, price, genres, topical terms, and general note.

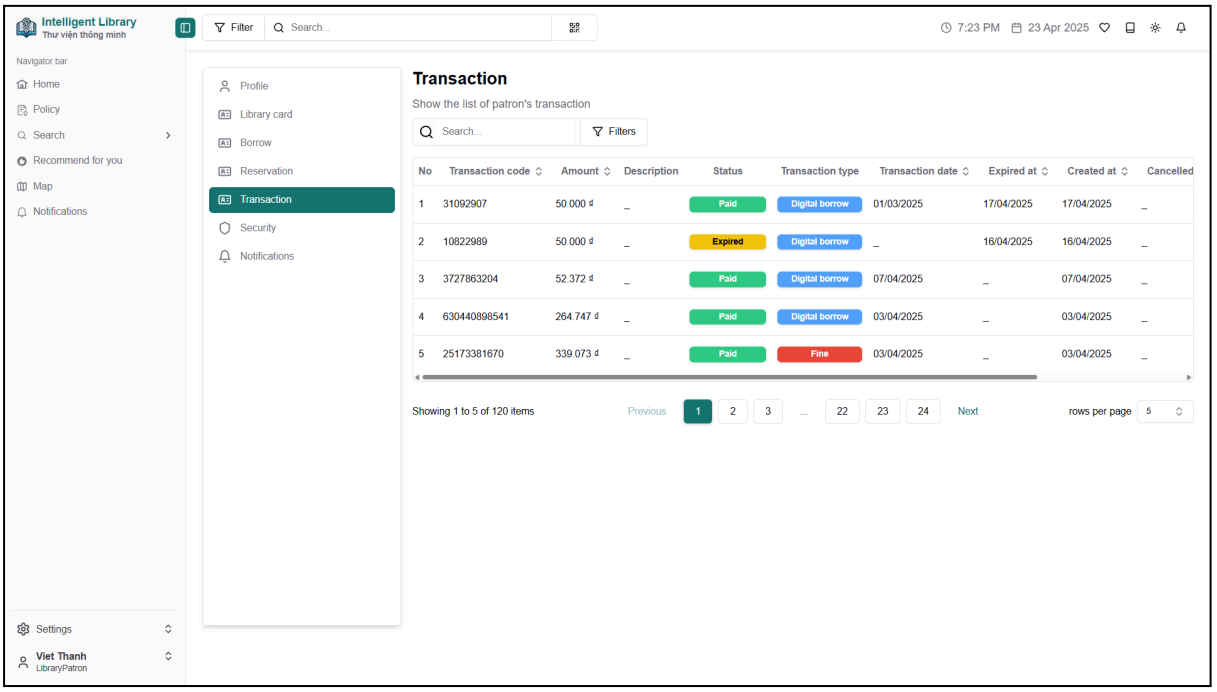


Figure 3.3.12.1: Patron's transaction detail

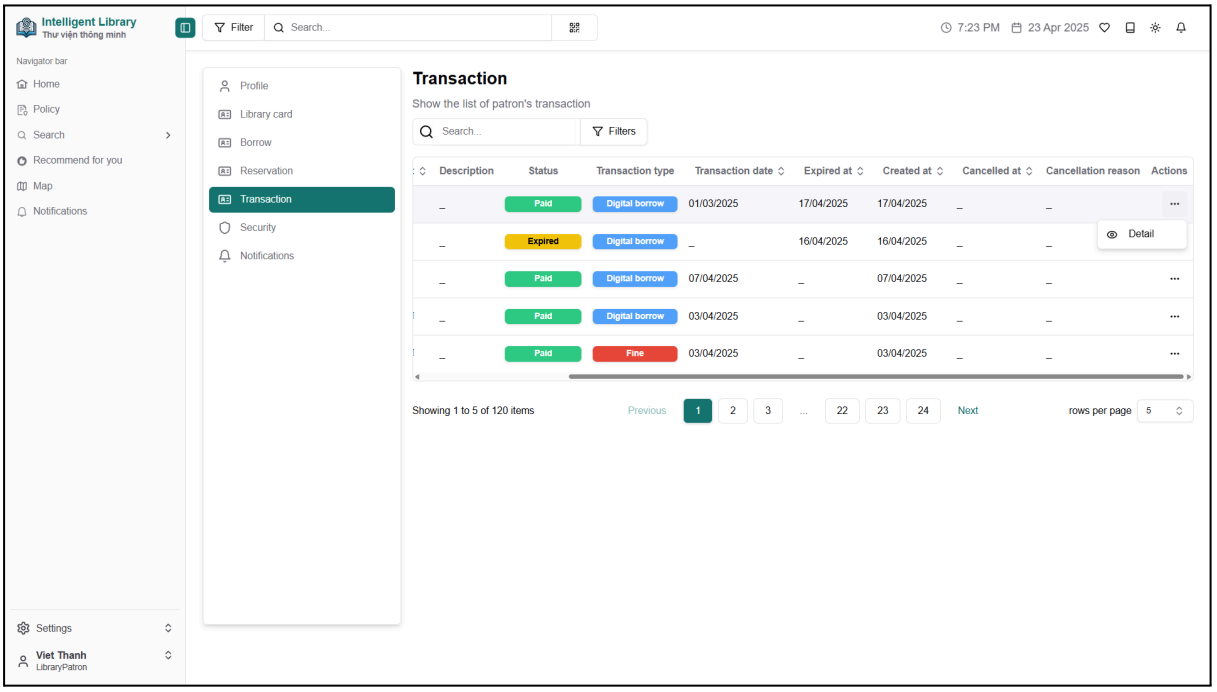


Figure 3.3.12.2: Patron's transaction detail

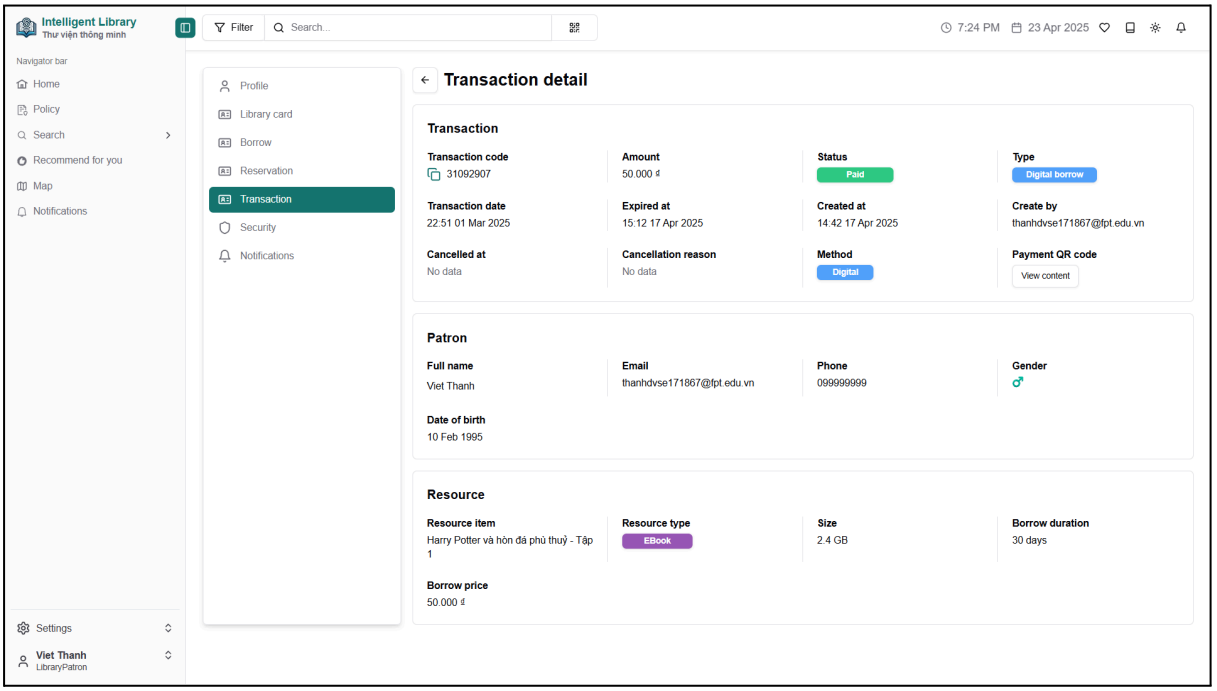


Figure 3.3.12.3: Patron's transaction detail

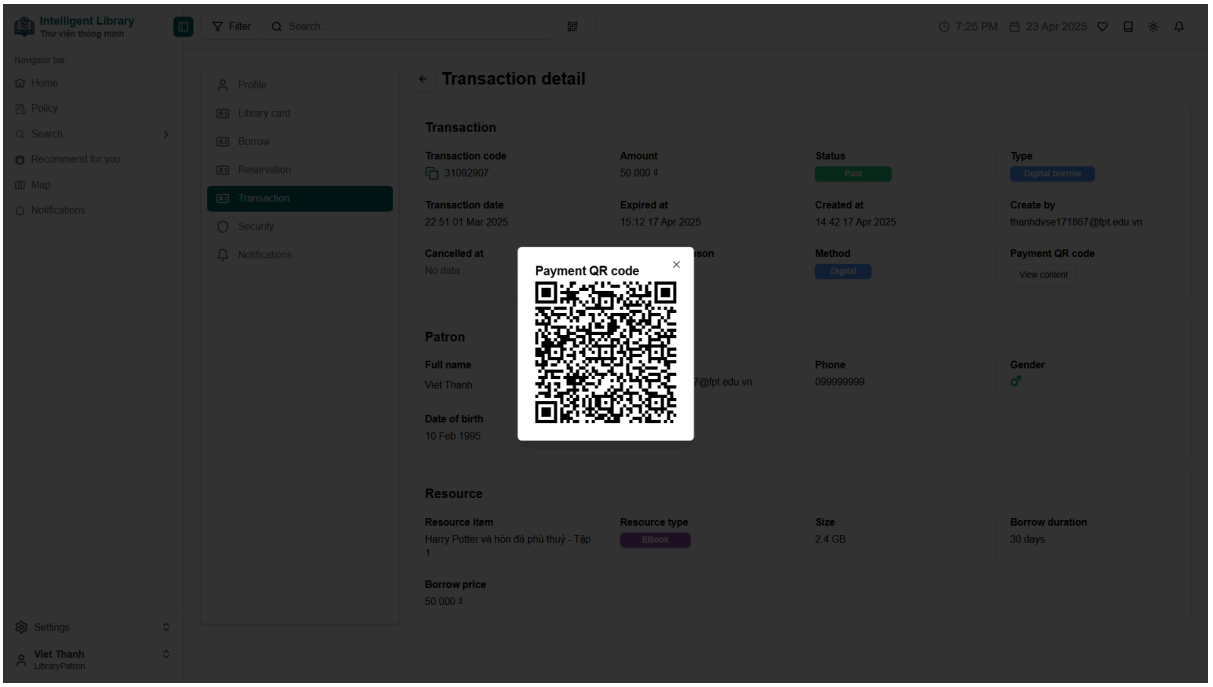


Figure 3.3.12.4: Patron's transaction detail

3.3.13 Patron's security setting

- **Function Description:** The page shows the patron's security setting, which can activate the Multi-factor authentication, and update the password.
- **Function Trigger:**
  - ❖ **Open the Update password form:**
    - ■ **Button/Navigation:** There is a "Update password" button next to the password.
    - ■ **Trigger:** The patron clicks the "Update password" button to open the Reset password form.
    - ■ **Function:** The page will open the update password form.
  - ❖ **Update new password:**
    - ■ **Button/Navigation:** There are two inputs for New password and Confirmed password, Cancel button and Continue button.
    - ■ **Trigger:** The patron types 2 input entries, and makes sure that the confirmed password matches with the new password, then clicks the Continue button to update the password.
    - ■ **Function:** The patrons have to type the new password and confirmed password to reset their password, then click the "Continue" button.
  - ❖ **Enable the Multi-factor Authentication (MFA):**
    - ■ **Button/Navigation:** There is a "Enable MFA" button.
    - ■ **Trigger:** The patron clicks the "Enable MFA" button to open MFA form and types the MFA codes.
    - ■ **Function:** The patron types the MFA codes to activate the Multi-factor Authentication feature to improve the account security.

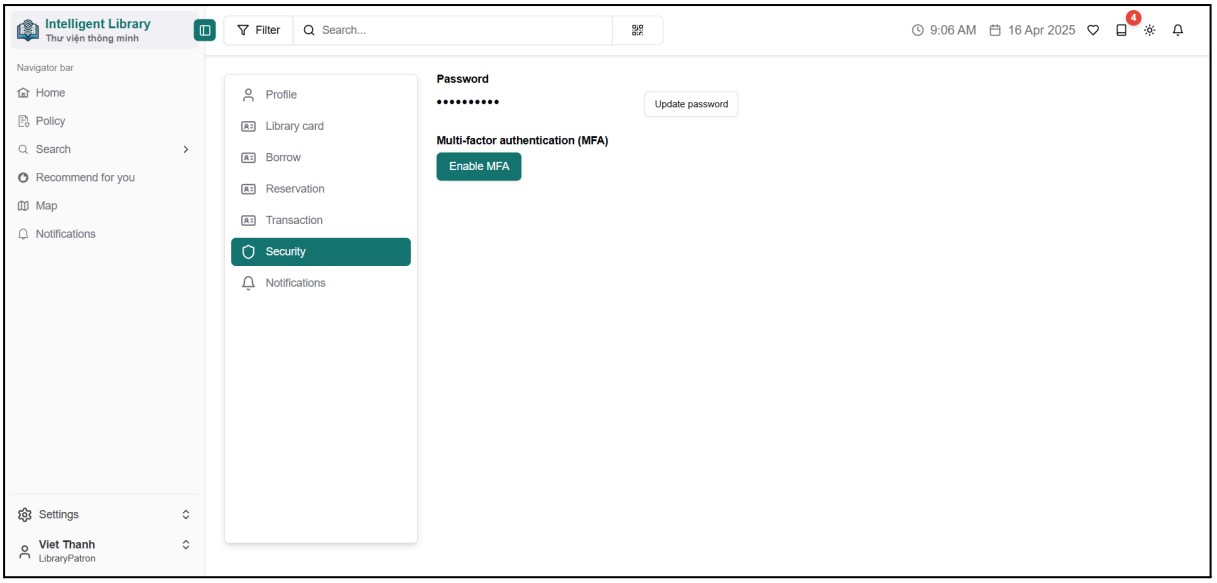


Figure 3.3.13.1: Patron's security setting

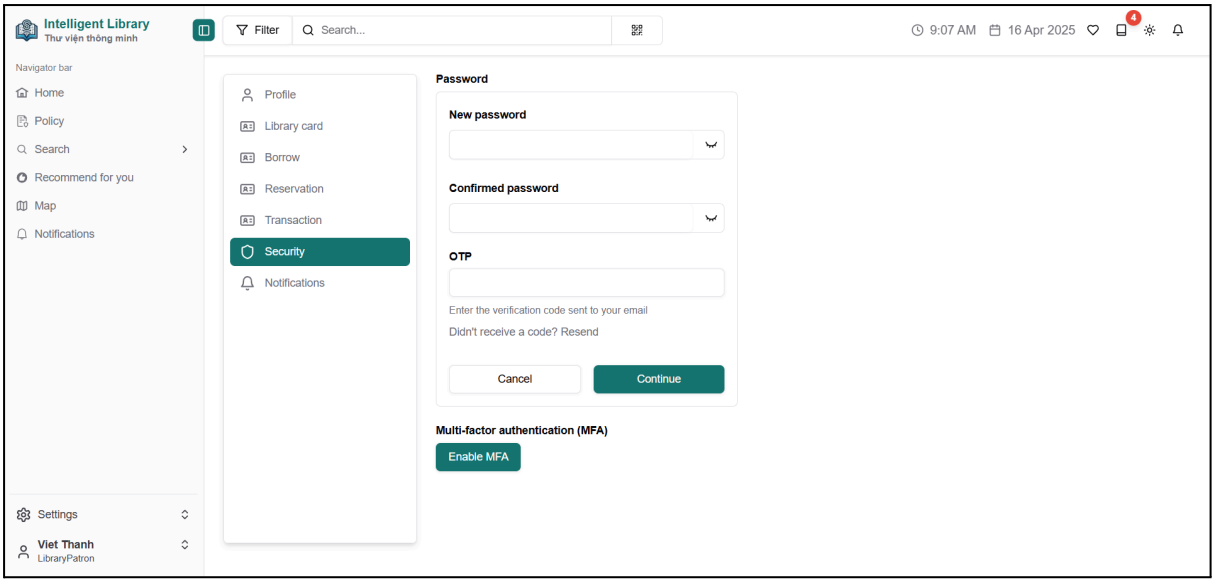


Figure 3.3.13.2: Patron's security setting

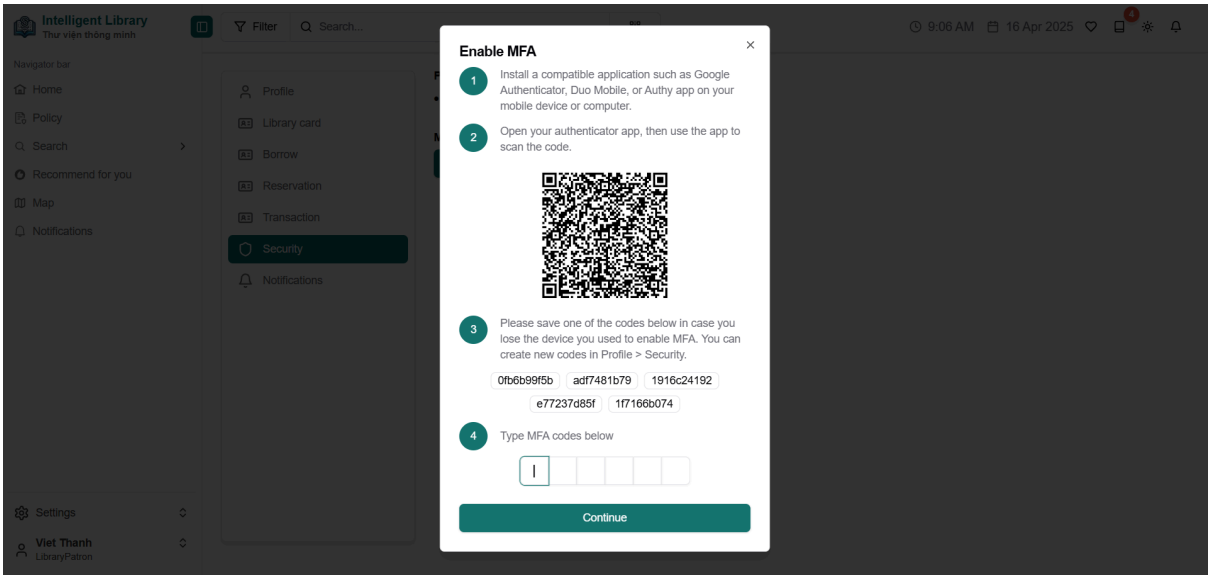


Figure 3.3.13.3: Patron's security setting

3.3.14 Patron's notification list

- **Function Description:** The notification page shows the list of notifications related to the library item news. In addition, users can filter or view details of the notification by clicking the notification.
- **Function Trigger:**
  - **Search notifications:**
    - **Button/Navigation:** There is a search bar on the top of the notification list.
    - **Trigger:** The user types the keyword of the notification.
    - **Function:** The system displays the list of notifications based on the entered keyword.

- **Limit the number of notifications:**
  - **Button/Navigation:** There is a row per page button on the bottom of the notification list.
  - **Trigger:** The user selects the value of the row per page.
  - **Function:** The number of notifications is shown by the selected value of the row per page.
- **Filter notifications:**
  - **Button/Navigation:** There is a filter button which is on the top of the notification list, and next to the search bar.
  - **Trigger:** The user clicks the filter button, the page displays the filter dialog. Then, the user can filter the notifications with notification type (all, event, reminder, and notice), and create date fields.
  - **Function:** The system displays the list of notifications based on the entered keyword.
- **Control filter notifications dialog:**
  - **Button/Navigation:** There are 3 “Cancel”, “Reset”, and “Apply” control buttons of the filter notification dialog at the end of the dialog.
  - **Trigger:** The user can click each control button to cancel, reset, and apply the filter notification.
  - **Function:**
    + When the user clicks the “Cancel” button, the value of the filter notification dialog will be clear and the page will close the dialog.
    + When the user clicks the “Reset” button, the value of filter notification dialog will be clear.
    + When the user clicks the “Apply” button, the page will show the list of notifications based on the entered value of filter notification dialog.
- **Business Rules:** BR-55

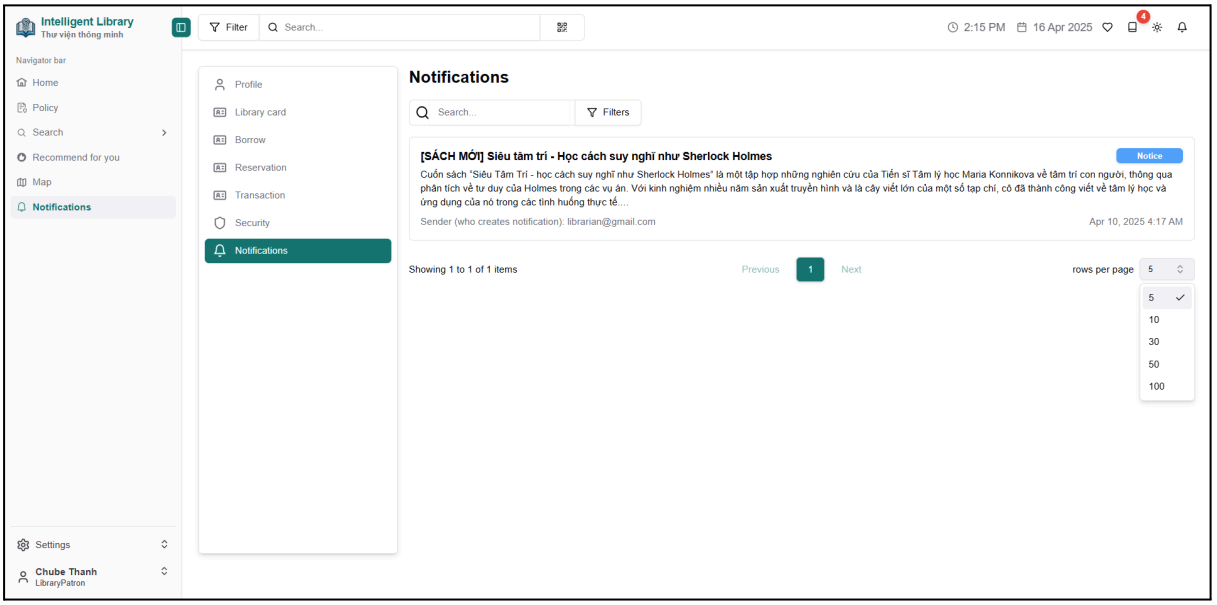


Figure 3.3.14.1: Patron’s notification list

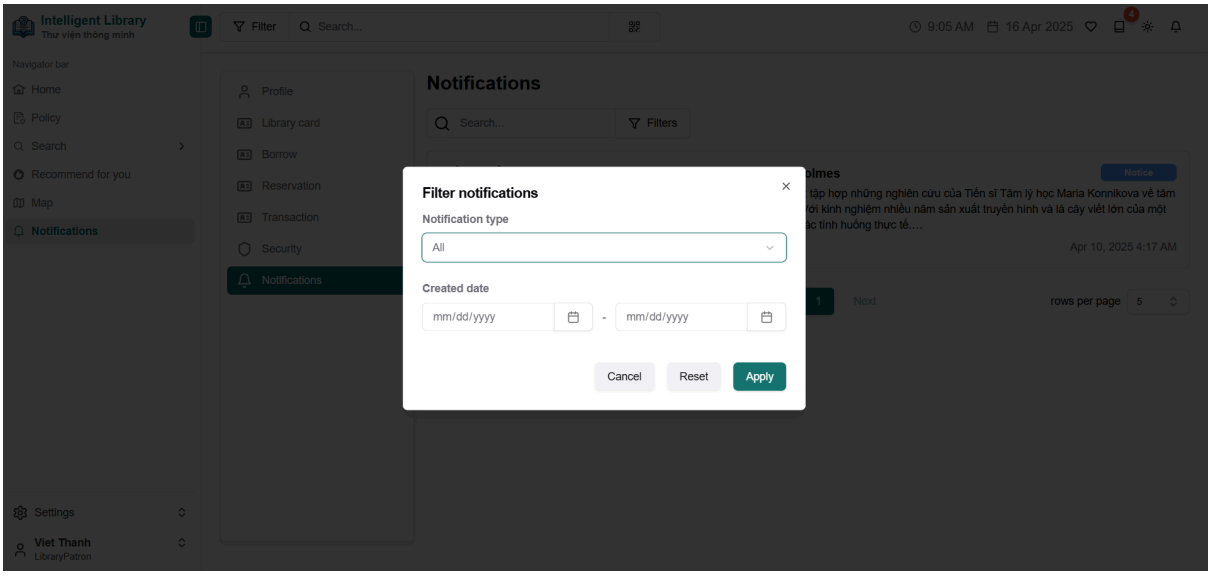


Figure 3.3.14.2: Patron's notification list

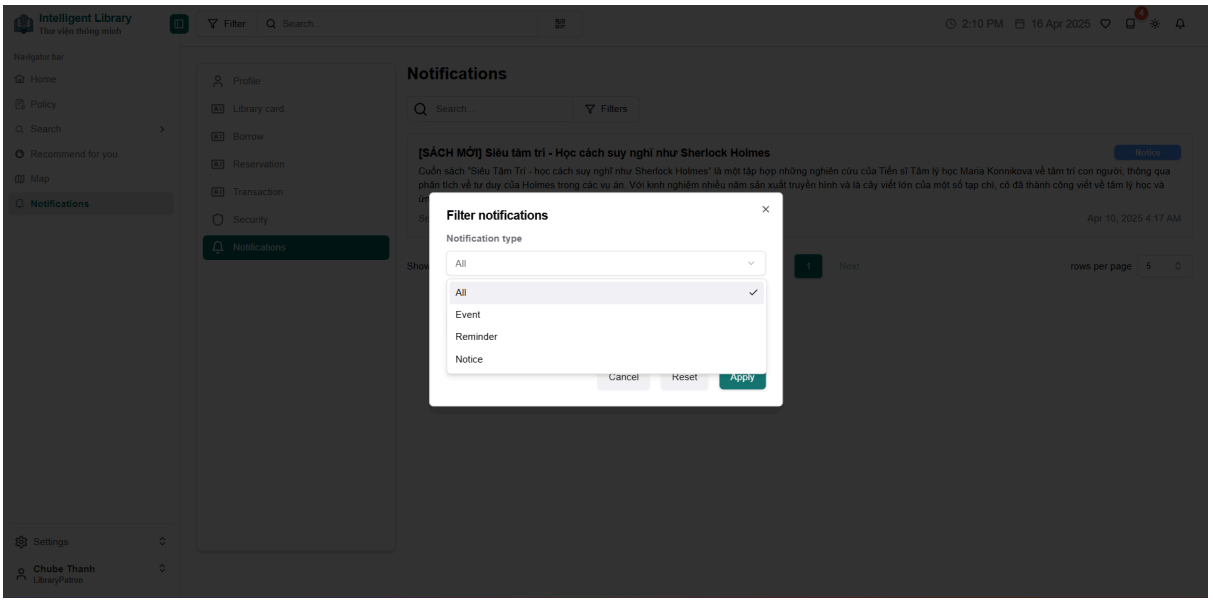


Figure 3.3.14.3: Patron's notification list

3.3.15 Patron's notification detail

- **Function Description:** The notification detail page shows more details of the specific notification.
- **Function Trigger:**
  - **Navigate to the notification detail page:**
    - **Button/Navigation:** There are the notification cards in the notification list.
    - **Trigger:** The user clicks the notification card to move to the notification detail page.
    - **Function:** The page shows the title, sender information, badge of notification type, created at, and the notification content.

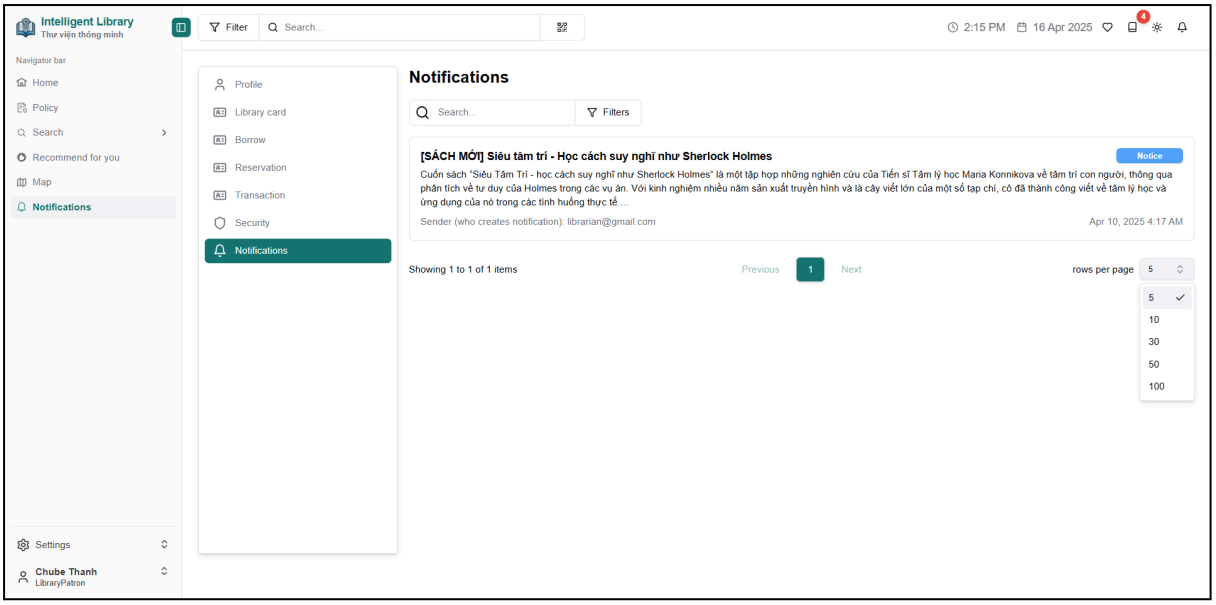


Figure 3.3.15.1: Patron's notification detail

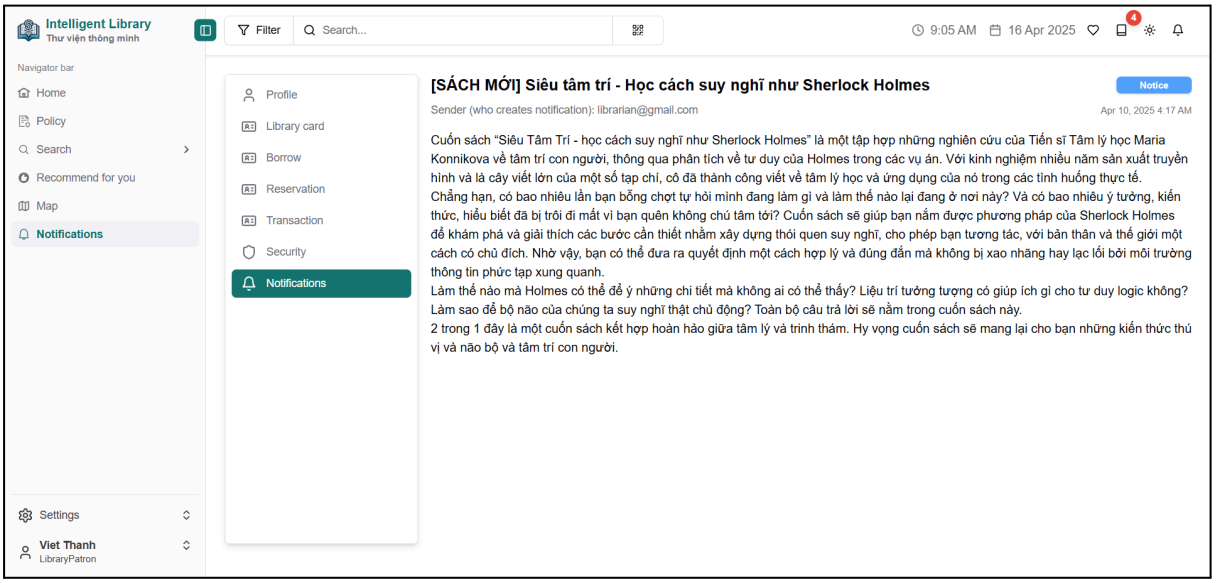


Figure 3.3.15.2: Patron's notification detail

3.3.16 Recommended for patrons

- **Function Description:** The recommendation page suggests the list of suitable library items for the patrons.
- **Function Trigger:**
  - **Navigate to the recommendation page:**
    - **Button/Navigation:** There is a "Recommended for you" button on the sidebar.
    - **Trigger:** The user clicks the "Recommended for you" button to navigate the recommendation page.
    - **Function:** The page shows the list of suggested library items.

❖ **Business Rules:** BR-44, BR-54, BR-90, BR-92

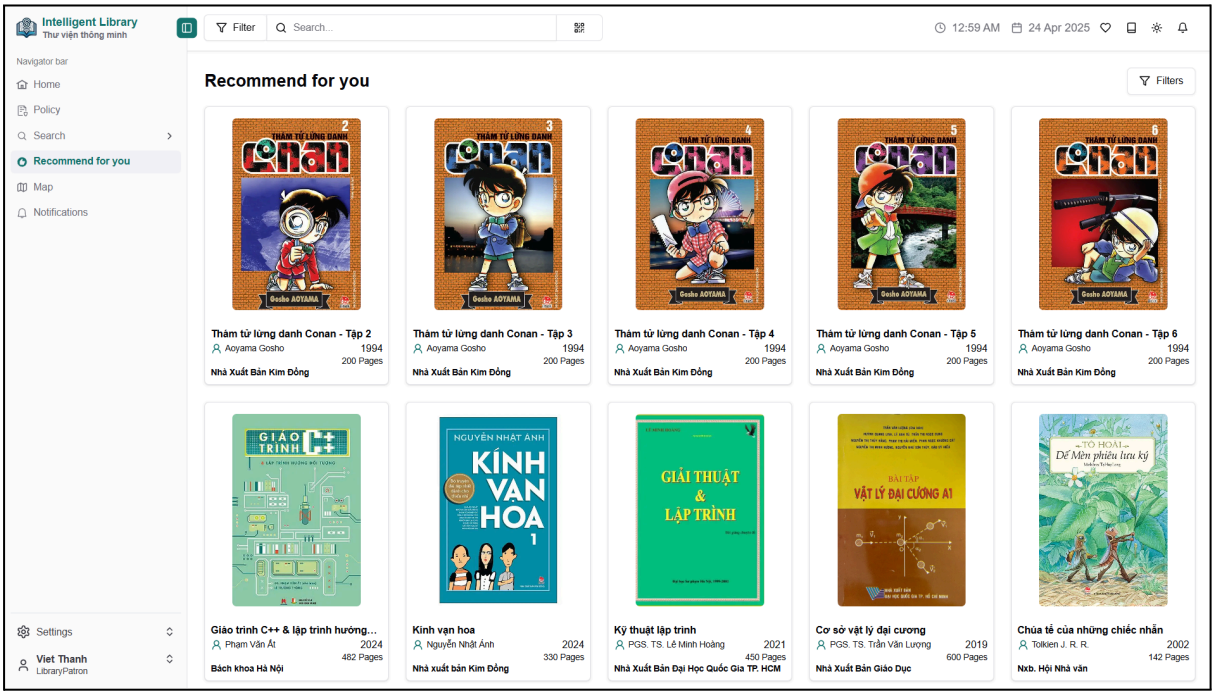


Figure 3.3.16.1: Patron's notification detail

3.4 Website for Librarian and Admin

3.4.1 Dashboard management

- **Function Description:** This screen provides a comprehensive overview of library operations, displaying key metrics like total patrons, employees, library items, and stock percentages. It includes sections for borrow/return trends, circulation activity, patron analysis, financial transactions, overdue borrows, and available vs. need units, using charts, tables..
- **Function Trigger:**

  ❖ **View Overview Metrics:**
  - **Button/Navigation:** The "Overview" section displays stats like total patrons, employees, and library items.
  - **Trigger:** Metrics are automatically displayed upon accessing the page.
  - **Function:** The system provides a quick snapshot of key library statistics for administrative oversight.

  ❖ **Analyze Borrow/Return Trends:**
  - **Button/Navigation:** The "Borrow Trends" and "Return Trends" sections show line graphs of borrowing and returning activity over time.
  - **Trigger:** Graphs are automatically rendered on page load.
  - **Function:** The system visualizes borrowing and returning patterns, helping administrators identify peak usage periods.

  ❖ **Review Borrow/Return Activity:**
  - **Button/Navigation:** The "Borrow/Return Activity Analysis" section shows metrics like total requests, failed borrows, and borrow failed rates.
  - **Trigger:** Metrics are automatically displayed on page load.

    - **Function:** The system summarizes circulation activity, highlighting success and failure rates.
- **Track Borrow/Return Trends:**
    - **Button/Navigation:** The "Borrow Trends" and "Return Trends" sections display line graphs of borrowing and returning activity over time.
    - **Trigger:** Graphs are automatically rendered on page load.
    - **Function:** The system visualizes borrowing and returning patterns, helping identify peak usage periods.
- **Examine Circulation Details:**
    - **Button/Navigation:** The "Borrow Failed vs Overdue Count" and "Borrow Failed vs Overdue Rates" sections show bar charts comparing failed and overdue counts/rates by item type.
    - **Trigger:** Charts are automatically displayed on page load.
    - **Function:** The system highlights circulation issues, showing which item types have higher failure or overdue rates.
- **Analyze Digital Resources:**
    - **Button/Navigation:** The "Digital Resources Analysis" section includes metrics like total digital resources and a table of top borrowed digital items with details.
    - **Trigger:** Data is automatically populated on page load with pagination controls.
    - **Function:** The system tracks digital resource usage, identifying popular items and borrowing patterns.
- **Monitor Financial Transactions:**
    - **Button/Navigation:** The "Financial & Transaction Analysis" section shows a line graph of revenue trends and a table of latest transactions with pagination.
    - **Trigger:** Graph and table are automatically displayed on page load.
    - **Function:** The system tracks financial activity, including fines and payments, for budget oversight.
- **Track Overdue Borrows and Reservations:**
    - **Button/Navigation:** The "Overdue Borrow" and "Assignable Reserves" sections display lists of overdue items and reservations.
    - **Trigger:** Lists are automatically shown on page load.
    - **Function:** The system identifies overdue items and pending reservations for follow-up.
- **Review Latest Borrows:**
    - **Button/Navigation:** The "Latest Borrows" section lists recent borrowing activity with details like library item, patron, due date, and status.
    - **Trigger:** The list is automatically populated on page load with pagination.
    - **Function:** The system shows current borrowing activity, allowing administrators to monitor active loans.
- **Assess Available vs Need Units:**
    - **Button/Navigation:** The "Available vs Need Units Analysis" section includes a bar chart and a table of top circulation items with borrow counts and rates
    - **Trigger:** Chart and table are automatically displayed on page load with a "Create Supplement Request" button.
    - **Function:** The system highlights resource gaps, assisting in inventory planning and acquisition.

- **Business Rules:** BR-49, BR-95, BR-99, BR-100, BR-101

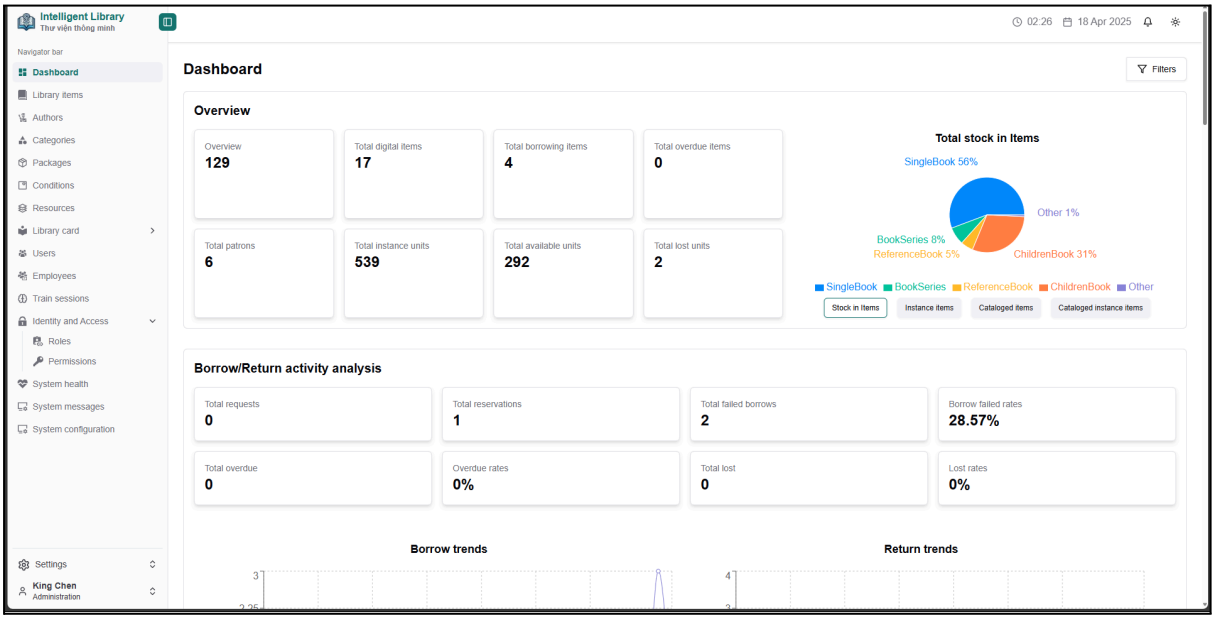


Figure 3.4.1.1: Dashboard management

Figure 3.4.1.2: Dashboard management

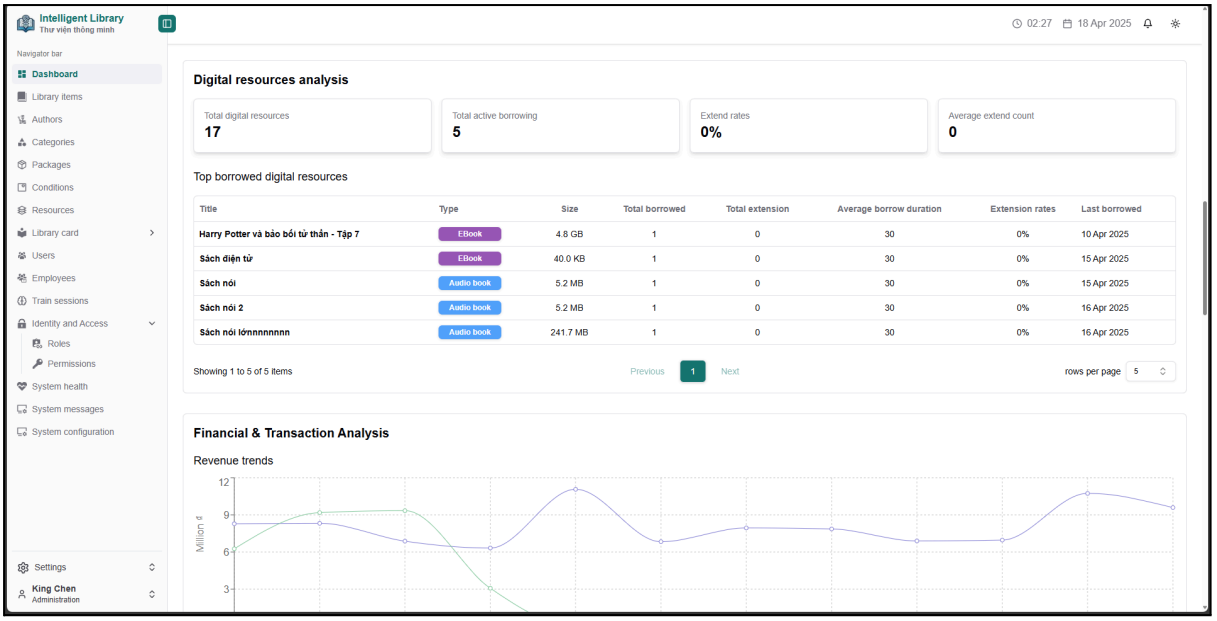


Figure 3.4.1.3: Dashboard management

Figure 3.4.1.4: Dashboard management

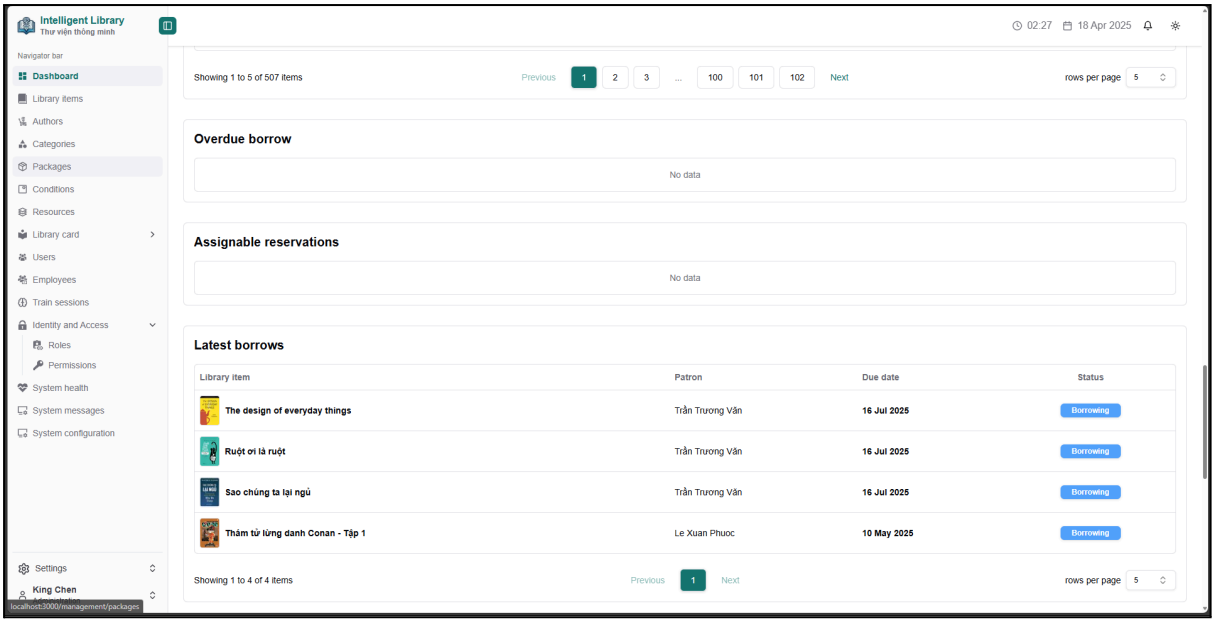


Figure 3.4.1.5: Dashboard management

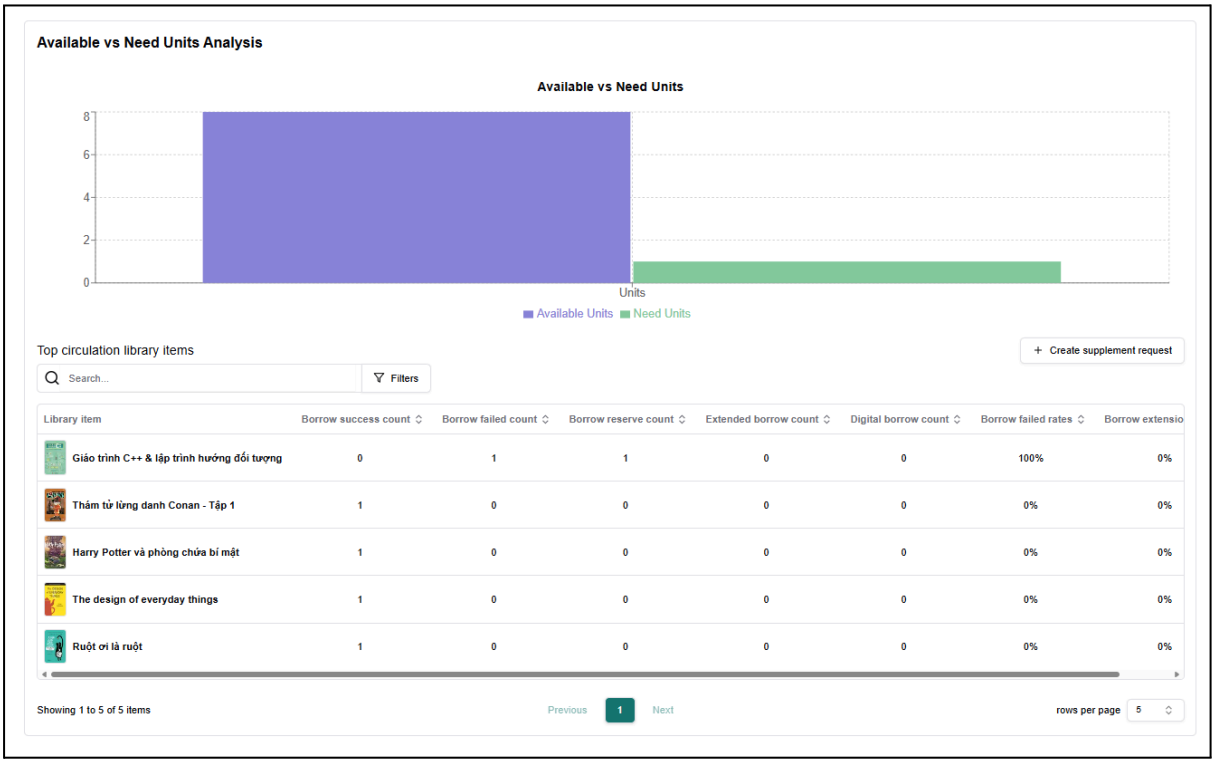


Figure 3.4.1.6: Dashboard management

3.4.2 Library item management

- **Function Description:** This screen allows **librarians** to view, search, and manage library items efficiently. It provides various search methods, data import/export functionalities, and actions for handling library items.
- **Function Trigger:**

- **Search library items:**
  - **Button/Navigation:** There is a search input on the top of the screen.
  - **Trigger:** The librarian enters the keyword that they want to search for and then presses "Enter".
  - **Function:** The system will filter out the corresponding search results.

- **Filter library items:**
  - **Button/Navigation:** There is a "Filter" button right next to the search input.
  - **Trigger:** When clicked, a dialog box will appear, allowing the librarian to choose between two tabs: Basic Search and Advanced Search. The librarian can enter the necessary inputs or select options from dropdown menus to refine the search. Finally, clicking "Search" will execute the query.
  - **Function:** The system will filter out the corresponding filter results.

- **Custom display:**
  - **Button/Navigation:** There is a “Columns” button on the top of the screen.
  - **Trigger:** When clicked, the system displays a dialog allowing the librarian to choose which columns to show or hide. Once done, they can press "Apply" to confirm the changes.
  - **Function:** The system will hide/show the columns that the librarian has previously selected.

- **Sort a column:**
  - **Button/Navigation:** The table headers with up/down icons indicate sortable columns.
  - **Trigger:** When clicked, the action will proceed.
  - **Function:** Clicking the header toggles the sort order between ascending and descending. The icon changes accordingly (up = ascending, down = descending).
- **Import library items:**
  - **Button/Navigation:** There is an "Import" button on the top of the screen.
  - **Trigger:** When clicked, the system presents a dialog where the librarian can upload an Excel file and fill in the required fields. Once completed, they can press "Import" to upload the file.
  - **Function:** New library items will be added to the system.
- **Export library items:**
  - **Button/Navigation:** There is a "Export" button on the top of the screen.
  - **Trigger:** Librarians can press this button to perform the data export process.
  - **Function:** The system will download an Excel file containing the library items’ data to the librarian’s computer.

- **Navigate to the create library item page:**
  - **Button/Navigation:** There is a “Add library item” button at the top of the screen.
  - **Trigger:** When clicked, it navigates users to the **create library item** page.

- **Function:** The web application will navigate to and display the Create Library Item page.

❖ **Action with library items:**
- **Button/Navigation:** A “Hành động” button at the top of the screen will be clickable when librarians select at least a row on the table.
- **Trigger:** When clicked, the system will display options such as "Move to Trash", "Restore", "Permanently Delete"
- **Function:** The system will perform the selected action (Move to Trash, Restore, or Permanently Delete) on the chosen library items. For sensitive actions, a confirmation dialog may appear before proceeding.

❖ **Pagination**
- **Button/Navigation:** Simple numbered buttons, "Next" and "Previous" links, at the bottom of the table.
- **Trigger:** Librarian clicks on any of the pagination controls.
- **Function:** Changes the page of the table, allowing the librarians to navigate through multiple entries.

❖ **Adjust page size**
- **Button/Navigation:** A number selection dropdown at the bottom of the screen
- **Trigger:** When clicked, different options will be displayed so that the librarian can choose the page size they want.
- **Function:** The system will display more or less rows per page of the table.

❖ **View different types of library items**
- **Button/Navigation:** There are two different tabs that the librarian can choose from located above the table.
- **Trigger:** When clicked, the system will filter and display the corresponding library items.
- **Function:**
  - ➢ **"Active" Tab**: Displays active library items where isDeleted = false.
  - ➢ **"Deleted" Tab**: Shows deleted library items where isDeleted = true.

- **Business Rules:** BR-06, BR-29, BR-41, BR-47, BR-61, BR-64

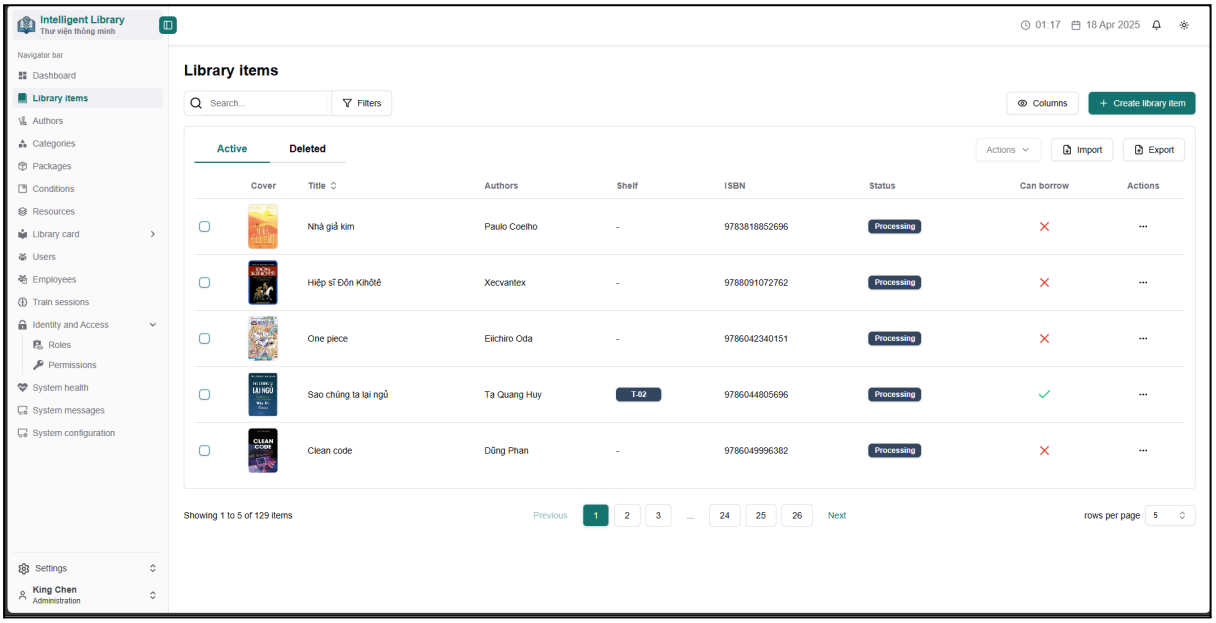


Figure 3.4.2.1: Library item management

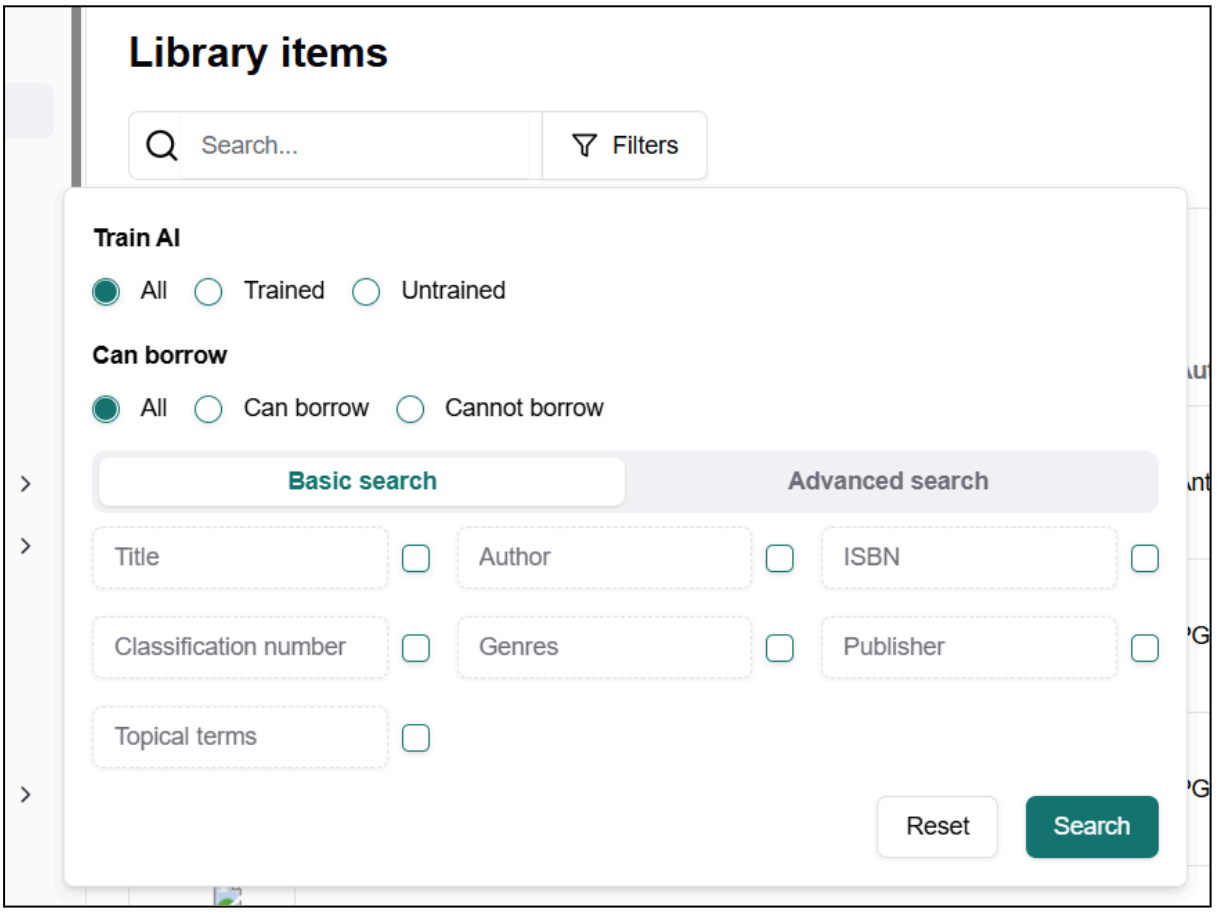


Figure 3.4.2.2: Library item management

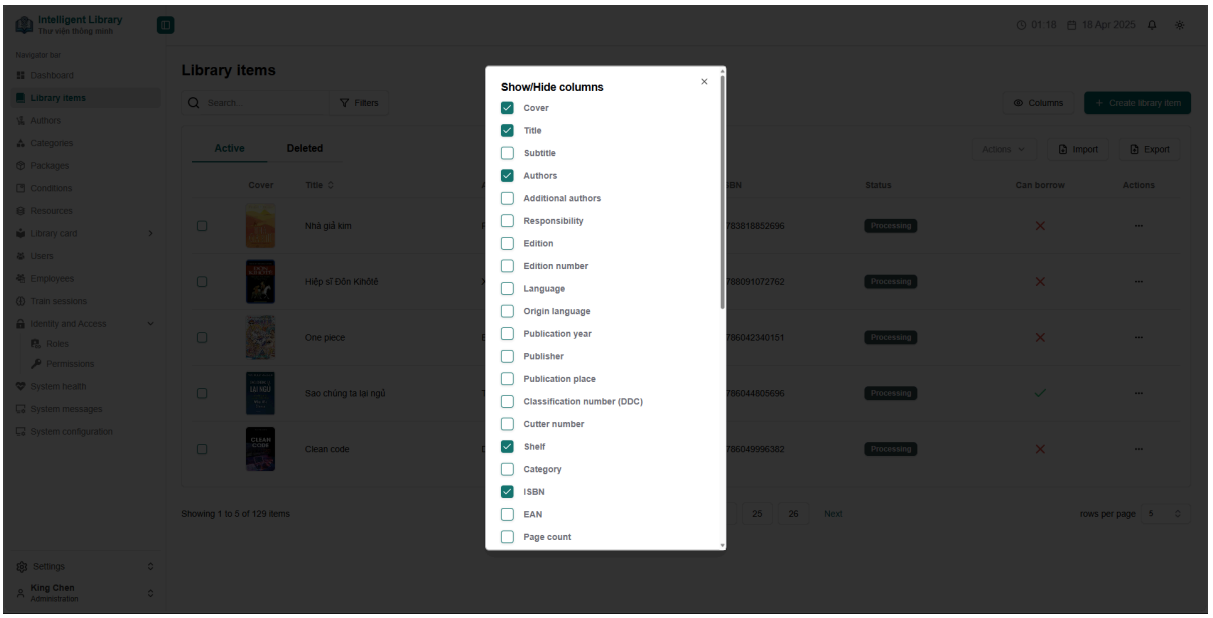


Figure 3.4.2.3: Library item management

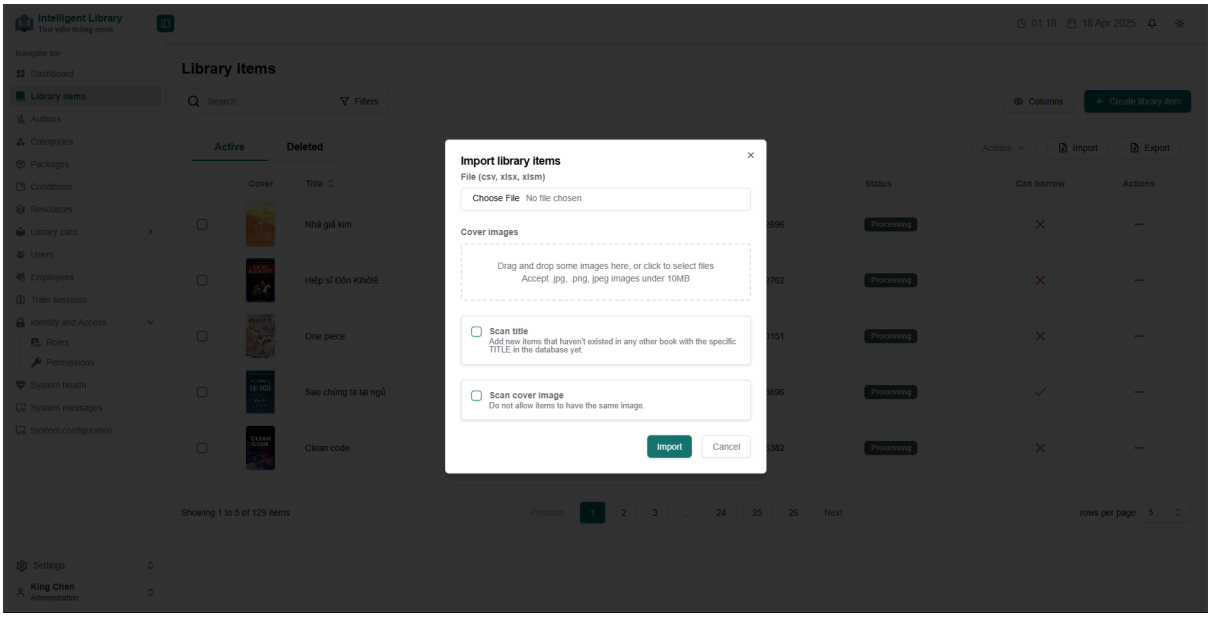


Figure 3.4.2.4: Library item management

3.4.3 Create library item

- **Function Description:** This screen allows librarians to catalog new library items through a multi-step process across: Category, Catalog, Glue Unique Barcodes, and Resources. It supports selecting categories, entering catalog details manually or via MARC21, generating barcodes for physical copies, and optionally adding digital resources like eBooks or audiobooks.
- **Function Trigger:**

    ❖ **Select Category**
    - **Button/Navigation:** In the "Category" tab, librarians choose the material type with a prefix.
    - **Trigger:** Selecting a category from the list and proceeding to the next step.
    - **Function:** The system assigns the selected category and prefix to the new library item, determining its classification and usage rules.

    ❖ **Catalog Item Details**
    - **Button/Navigation:** In the "Catalog" tab, librarians can manually enter details or use the "Copy Catalog" button to input a MARC21 record.
    - **Trigger:** Clicking "Copy Catalog," entering the MARC21 data, and clicking "Continue," or manually filling the form.
    - **Function:** The system populates the form with details extracted from the MARC21 record or saves the manually entered data for the library item.

    ❖ **Generate Barcodes for Copies**
    - **Button/Navigation:** In the "Unique Barcodes" tab, the system auto-generates book copy codes and barcodes with a "Print" button.
    - **Trigger:** No input required; barcodes are displayed automatically with an option to print.

- **Function:** The system creates unique barcodes for each copy, allowing librarians to print and affix them to physical items for tracking.

❖ **Add Digital Resources (Optional)**

- **Button/Navigation:** In the "Resources" tab, librarians can add eBooks or audiobooks by entering details and uploading files via drag-and-drop
- **Trigger:** Clicking "More eBook" or "More Audio" and uploading a file or filling in details.
- **Function:** The system attaches digital resources to the library item, making them available for borrowing if added.

❖ **Navigate and Save**

- **Button/Navigation:** "Back" and "Continue" buttons are available at the bottom of each tab.
- **Trigger:** Clicking "Continue" to move to the next tab or "Back" to return to the previous tab.
- **Function:** The system saves progress and guides the librarian through the cataloging process, ensuring all steps are completed before finalizing the new library item.

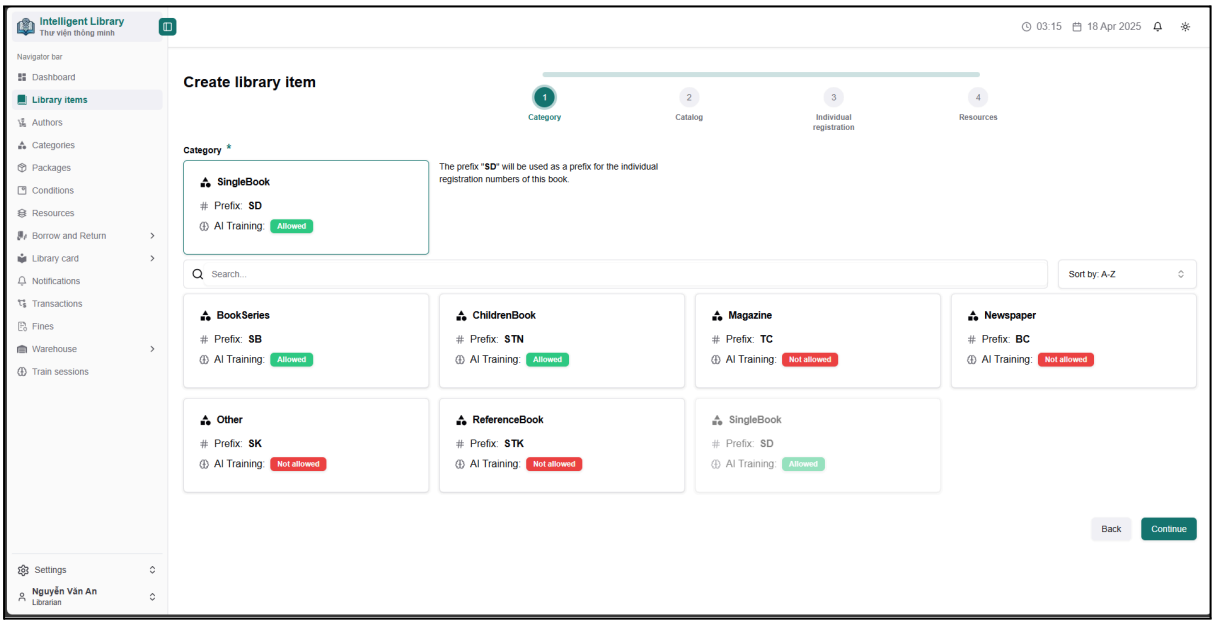


Figure 3.4.3.1: Create Library item

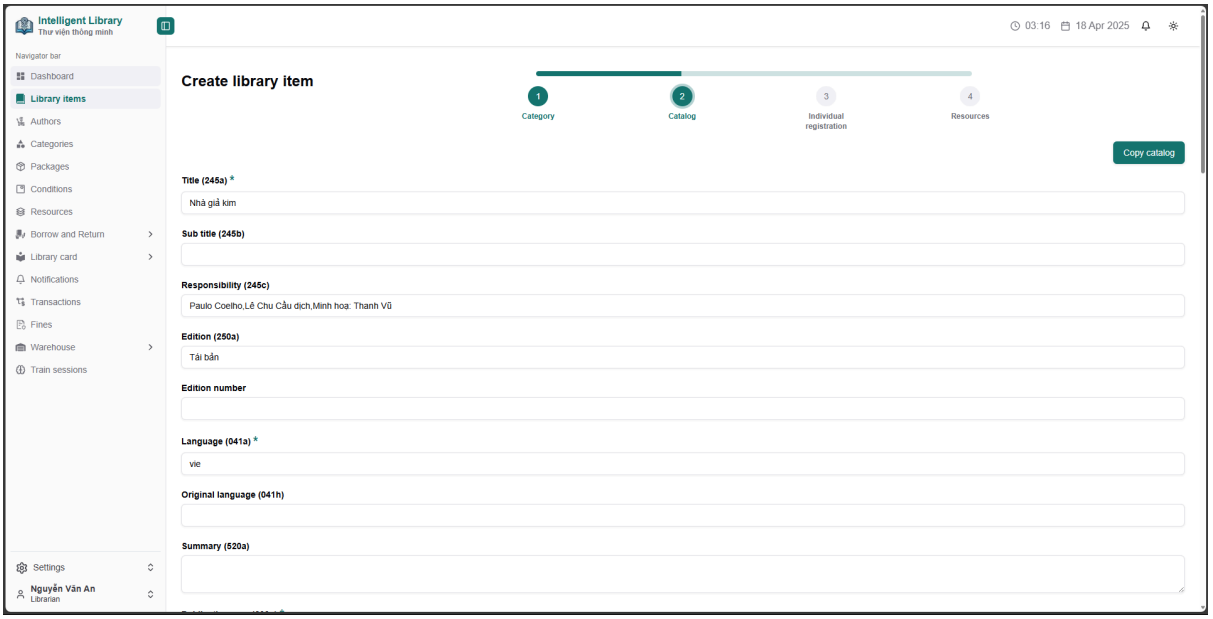


Figure 3.4.3.2: Create Library item

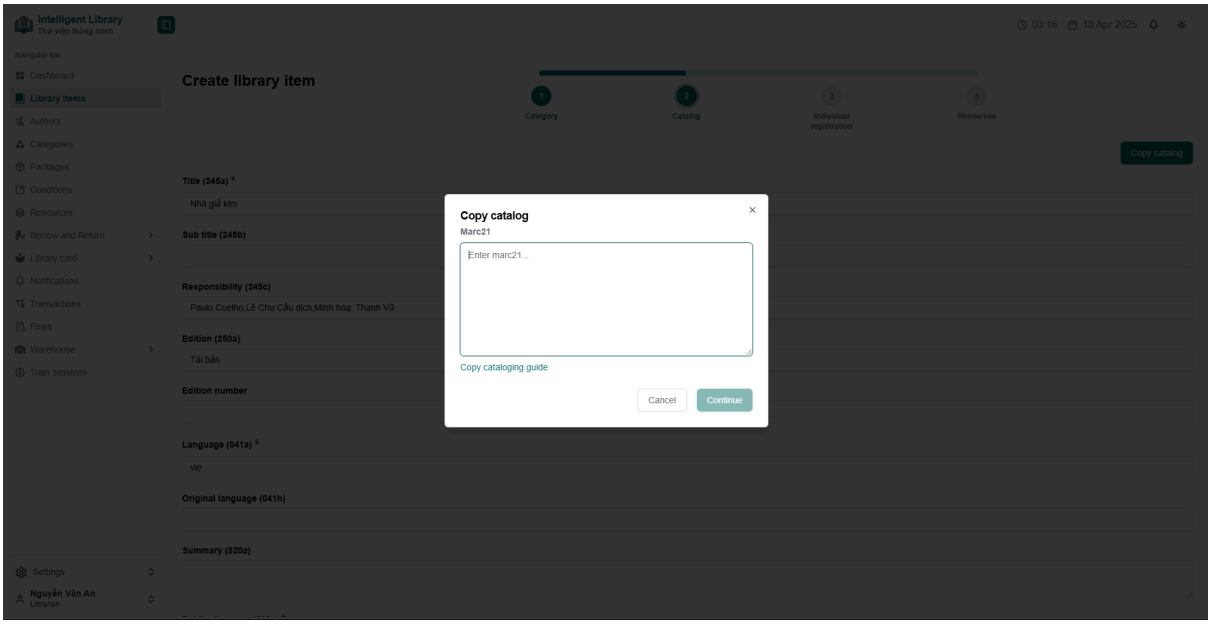


Figure 3.4.3.3: Create Library item

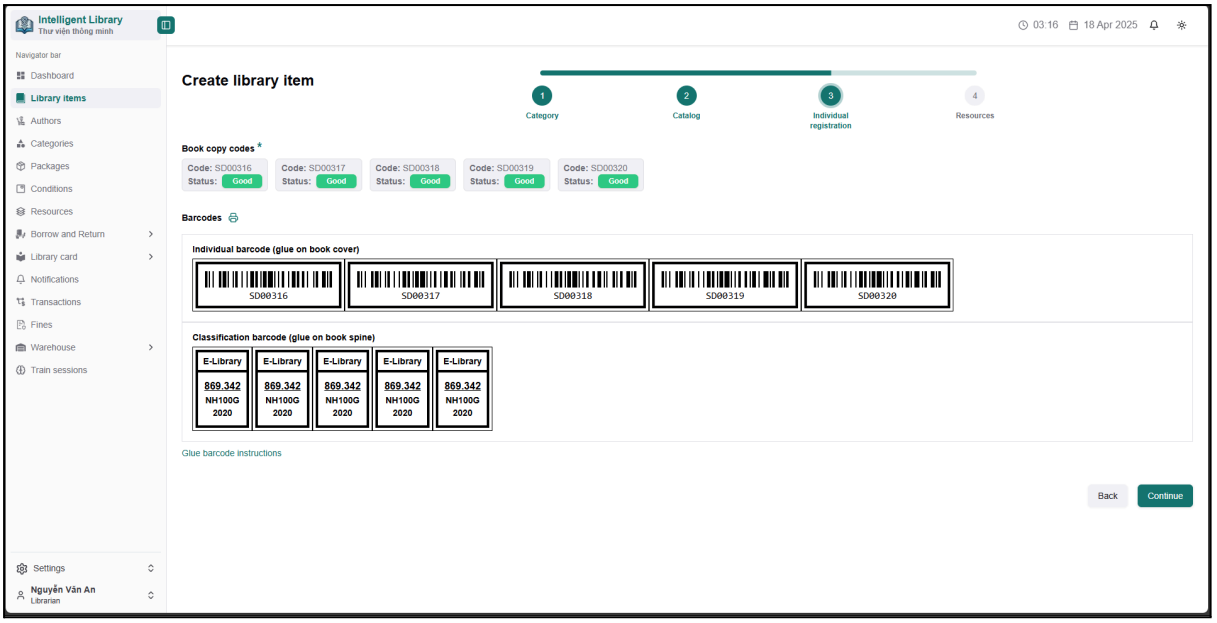


Figure 3.4.3.4: Create Library item

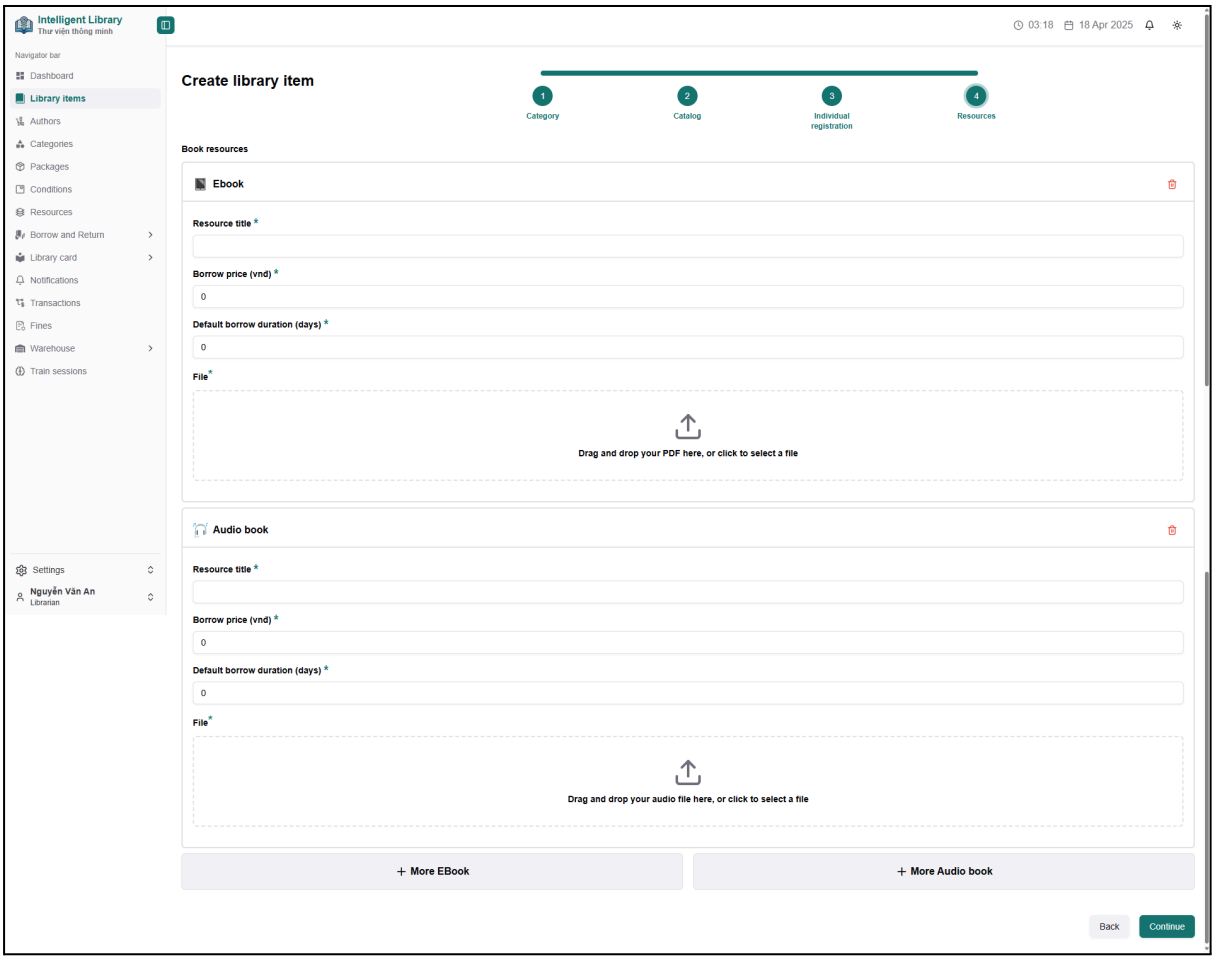


Figure 3.4.3.5: Create Library item

3.4.4 Library item detail management

- **Function Description:** This screen allows the librarian to view detailed library item information, edit general information, edit bookshelf, move library item to the trash, restore library item, permanently delete library item, and change the library item's status to **published** or **draft**. It also allows managing authors, digital resources, physical copies, and the library item's edit history.
- **Function Trigger:**

  - **View Detailed Information of Library Items**
    - **Button/Navigation:** A row of tabs is displayed at the top of the screen, immediately below the library item's title. The librarian can select one of the available tabs (including: Basic Info, Edition Info, Classification, Physical Details, Content Info, Inventory, Train AI, Metadata).
    - **Trigger:** When the librarian clicks on any tab, the system will initiate the corresponding process.
    - **Function:** The system will display the detailed information of the library item based on the selected tab.

  - **Edit Library Item Information**
    - **Button/Navigation:** At the top-right corner of the screen, there is a dropdown menu labeled "Actions." When clicked, the librarian will see the option "Edit Information."
    - **Trigger:** Upon selecting the "Edit Information" option, the system will display a dialog. The librarian can then edit the details of the library item and click "Continue" to proceed.
    - **Function:** The system will update the information of the library item accordingly.

  - **Edit shelf location**
    - **Button/Navigation:** At the top-right corner of the screen, there is a dropdown menu labeled "Actions." When clicked, the librarian will see the option "Edit Shelf."
    - **Trigger:** Upon selecting the "Edit Shelf" option, the system will display a dialog. The librarian can then search for and select an appropriate shelf for the library item.
    - **Function:** The system will update the shelf location of the library item accordingly.

  - **Toggle Publish/Draft Status of Library Item**
    - **Button/Navigation:** At the top-right corner of the screen, there is a dropdown menu labeled "Actions." When clicked, the librarian will see the option "Publish Item" if the library item is currently in draft status, or the option "Change to Draft" if the library item is currently in published status.
    - **Trigger:** Upon selecting either option, the system will initiate the corresponding process.
    - **Function:** The system will update the status of the library item to the selected option (either published or draft).

  - **Move Library Item to Trash**

- **Button/Navigation:** At the top-right corner of the screen, there is a dropdown menu labeled "Actions." When clicked, the librarian will see the option "Move to Trash" if the library item has the field isDeleted=false.
- **Trigger:** Upon selecting the "Move to Trash" option, the system will initiate the process.
- **Function:** The system will update the field isDeleted to true for the library item.

❖ **Restore Library Item**

- **Button/Navigation:** At the top-right corner of the screen, there is a dropdown menu labeled "Actions." When clicked, the librarian will see the option "Restore" if the library item has the field isDeleted=true.
- **Trigger:** Upon selecting the "Restore" option, the system will display a dialog prompting the user to confirm the action of restoring the library item. If the user confirms, the system will proceed with the action.
- **Function:** The system will update the field isDeleted to false for the library item.

❖ **Delete Library Item**

- **Button/Navigation:** At the top-right corner of the screen, there is a dropdown menu labeled "Actions." When clicked, the librarian will see the option "Delete" if the library item has the field isDeleted=true.
- **Trigger:** Upon selecting the "Delete" option, the system will display a dialog prompting the librarian to confirm the action of deleting the library item. If the librarian confirms, the system will proceed with the action.
- **Function:** The system will remove the library item from the database.

❖ **Add Authors to Library Item**

- **Button/Navigation:** At the bottom of the screen, there is a row of tabs for entities related to the library item, including the "Authors" tab. When the "Authors" tab is selected, a button labeled "Add Authors" will appear at the bottom-right corner of the tab row.
- **Trigger:** Upon clicking the "Add Authors" button, the system will display a dialog. The librarian can then select the authors to add and click "Save" to confirm.
- **Function:** The system will add the selected authors to the library item.

❖ **Remove Authors from Library Item**

- **Button/Navigation:** At the bottom of the screen, there is a row of tabs for entities related to the library item, including the "Authors" tab. When the "Authors" tab is selected, the authors are displayed in a table format. The librarian can select one or multiple rows by clicking the checkbox in each row.
- **Trigger:** Once at least one row is selected, the system will display a "Delete" button next to the "Add Authors" button. When the librarian clicks the "Delete" button, the process will be initiated.
- **Function:** The system will remove the selected authors from the library item.

- **Create Resource for Library Item**
  - **Button/Navigation:** At the bottom of the screen, there is a row of tabs for entities related to the library item, including the "Resources" tab. When the "Resources" tab is selected, a button labeled "Create Resource" will appear at the bottom-right corner of the tab row.
  - **Trigger:** Upon clicking the "Create Resource" button, the system will display a dialog. The librarian can fill all required fields and click "Save" to confirm.
  - **Function:** The system will create new resources for the library item.

- **Remove Resources from Library Item**
  - **Button/Navigation:** At the bottom of the screen, there is a row of tabs for entities related to the library item, including the "Resources" tab. When the "Resources" tab is selected, the resources are displayed in a table format. The librarian can select one or multiple rows by clicking the checkbox in each row.
  - **Trigger:** Once at least one row is selected, the system will display a "Delete" button next to the "Create Resource" button. When the librarian clicks the "Delete" button, the process will be initiated.
  - **Function:** The system will remove the selected resource from the library item.

- **Put Physical Copies on Shelf**
  - **Button/Navigation:** At the bottom of the screen, there is a row of tabs for entities related to the library item, including the "Copies" tab. When the "Copies" tab is selected, the physical copies are displayed in a table format. The librarian can select one or multiple copies by clicking the checkbox in each row. Once at least one copy is selected, a dropdown menu labeled "Actions" will appear at the top-right corner of the table, containing the option "Put on Shelf."
  - **Trigger:** Upon selecting the "Put on Shelf" option from the "Actions" dropdown, the system will initiate the process.
  - **Function:** The system will update the status of the selected physical copies to "In Shelf," indicating that they are now placed on the shelf.

- **Remove Physical Copies from Shelf**
  - **Button/Navigation:** At the bottom of the screen, there is a row of tabs for entities related to the library item, including the "Copies" tab. When the "Copies" tab is selected, the physical copies are displayed in a table format. The librarian can select one or multiple copies by clicking the checkbox in each row. Once at least one copy is selected, a dropdown menu labeled "Actions" will appear at the top-right corner of the table, containing the option "Remove on Shelf."
  - **Trigger:** Upon selecting the "Remove on Shelf" option from the "Actions" dropdown, the system will initiate the process.
  - **Function:** The system will update the status of the selected physical copies to "Out Of Shelf," indicating that they are no longer on the shelf.

- **Move Physical Copies to Trash**

- **Button/Navigation:** At the bottom of the screen, there is a row of tabs for entities related to the library item, including the "Copies" tab. Within the "Copies" tab, there are two sub-tabs: "Active" and "Deleted." When the "Active" sub-tab is selected, the physical copies are displayed in a table format. The librarian can select one or multiple copies by clicking the checkbox in each row. Once at least one copy is selected, a dropdown menu labeled "Actions" will appear at the top-right corner of the table, containing the option "Move to Trash."
- **Trigger:** Upon selecting the "Move to Trash" option from the "Actions" dropdown, the system will initiate the process.
- **Function:** The system will update the selected physical copies by moving them to the "Deleted" sub-tab, indicating that they have been moved to the trash.

❖ **Restore Physical Copies**

- **Button/Navigation:**At the bottom of the screen, there is a row of tabs for entities related to the library item, including the "Copies" tab. Within the "Copies" tab, there are two sub-tabs: "Active" and "Deleted." When the "Deleted" sub-tab is selected, the physical copies that have been moved to the trash are displayed in a table format. The librarian can select one or multiple copies by clicking the checkbox in each row. Once at least one copy is selected, a dropdown menu labeled "Actions" will appear at the top-right corner of the table, containing the option "Restore."
- **Trigger:** Upon selecting the "Restore" option from the "Actions" dropdown, the system will initiate the process.
- **Function:** The system will update the selected physical copies by moving them back to the "Active" sub-tab, indicating that they have been restored.

❖ **Delete Physical Copies**

- **Button/Navigation:** At the bottom of the screen, there is a row of tabs for entities related to the library item, including the "Copies" tab. Within the "Copies" tab, there are two sub-tabs: "Active" and "Deleted." When the "Deleted" sub-tab is selected, the physical copies that have been moved to the trash are displayed in a table format. The librarian can select one or multiple copies by clicking the checkbox in each row. Once at least one copy is selected, a dropdown menu labeled "Actions" will appear at the top-right corner of the table, containing the option "Delete."
- **Trigger:** Upon selecting the "Delete" option from the "Actions" dropdown, the system will initiate the process.
- **Function:** The system will permanently remove the selected physical copies from the database, deleting them entirely from the "Deleted" sub-tab.

❖ **View History of Changes to Library Item**

- **Button/Navigation:** At the bottom of the screen, there is a row of tabs for entities related to the library item, including the "Histories" tab. When the "Histories" tab is selected, a table displays the history of changes

made to the library item, with columns such as "When," "Who," "Type," and a "View Details" button for each entry.

- **Trigger:** When the librarian clicks on the "View Details" button for a specific history entry, the system will display a dialog showing detailed changes.
- **Function:** The system will present a detailed view of the changes made to the library item, including the field modified, the old value, and the new value, in a structured format (e.g., JSON table). The dialog also includes options to view additional details if needed.

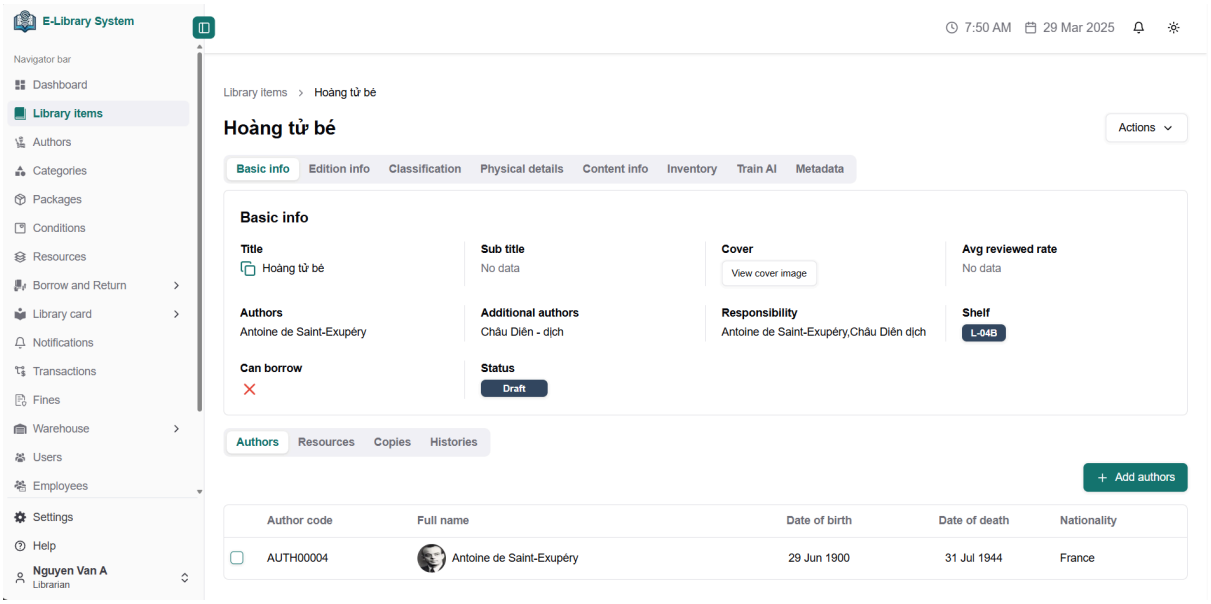


Figure 3.4.4.1: Library item detail management

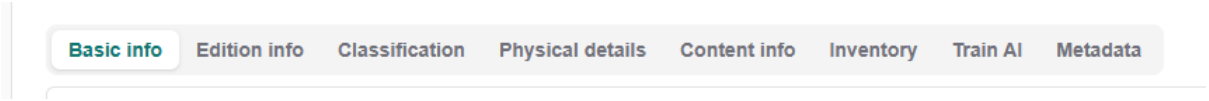


Figure 3.4.4.2: Library item detail management

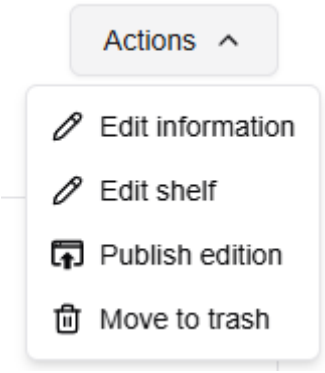


Figure 3.4.4.3: Library item detail management

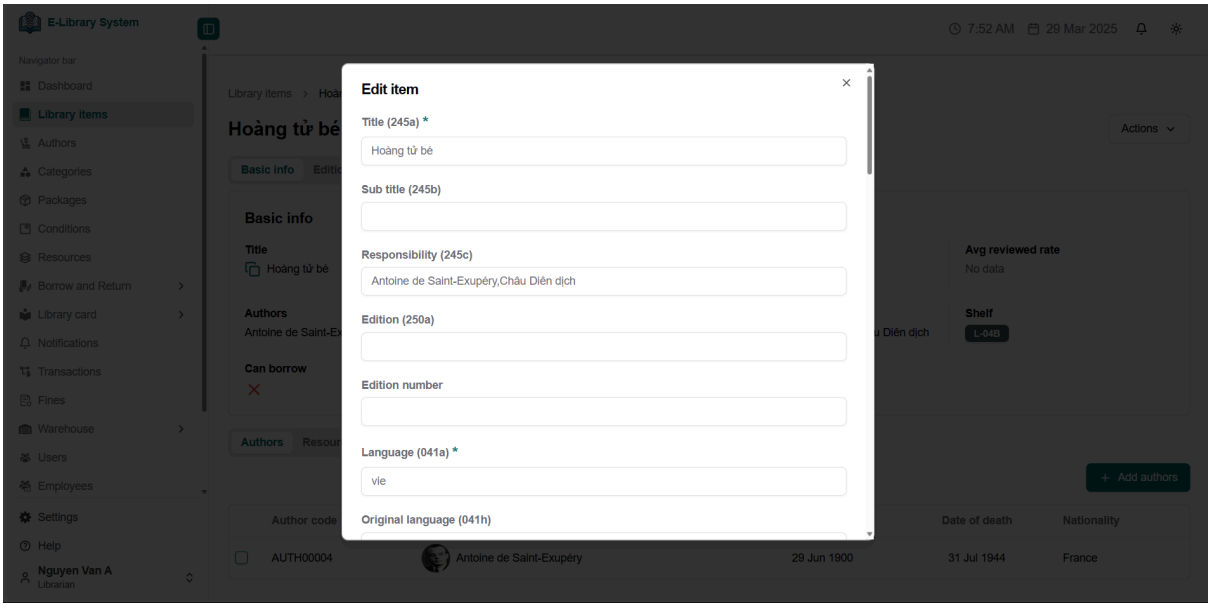


Figure 3.4.4.4: Library item detail management

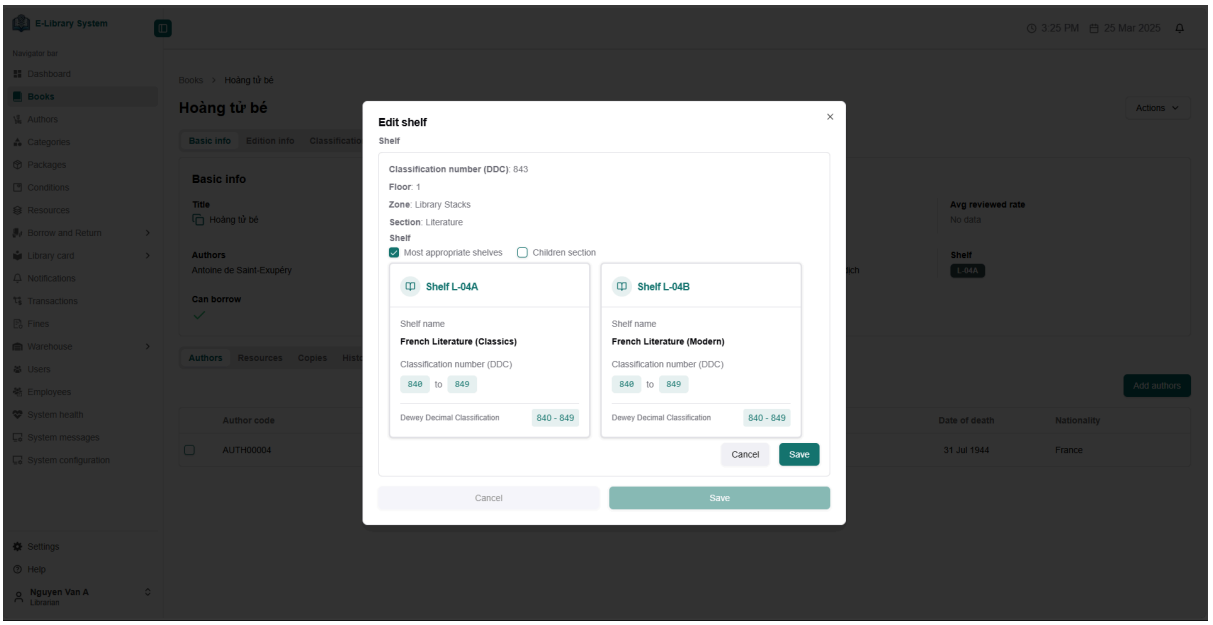


Figure 3.4.4.5: Library item detail management

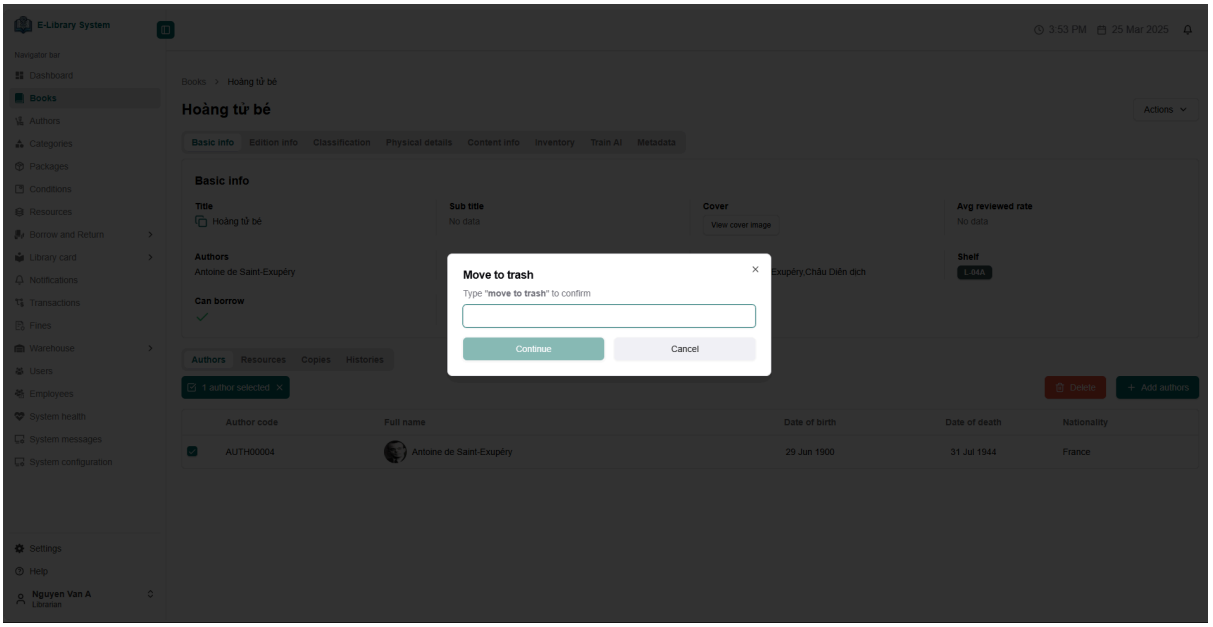


Figure 3.4.4.6: Library item detail management

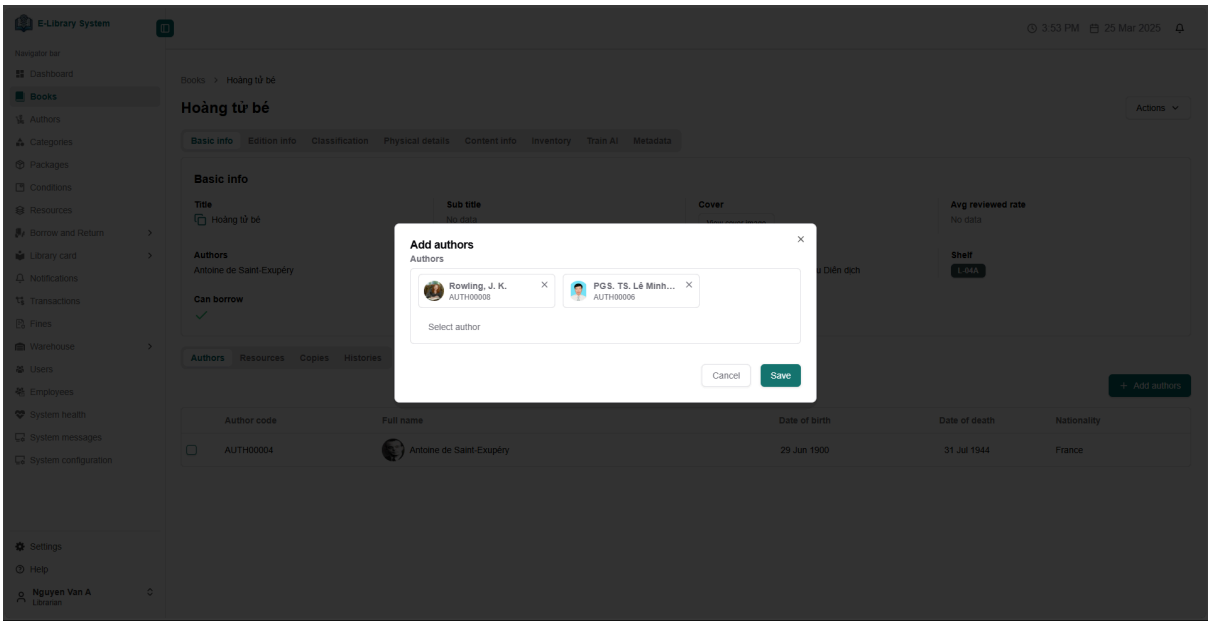


Figure 3.4.4.7: Library item detail management

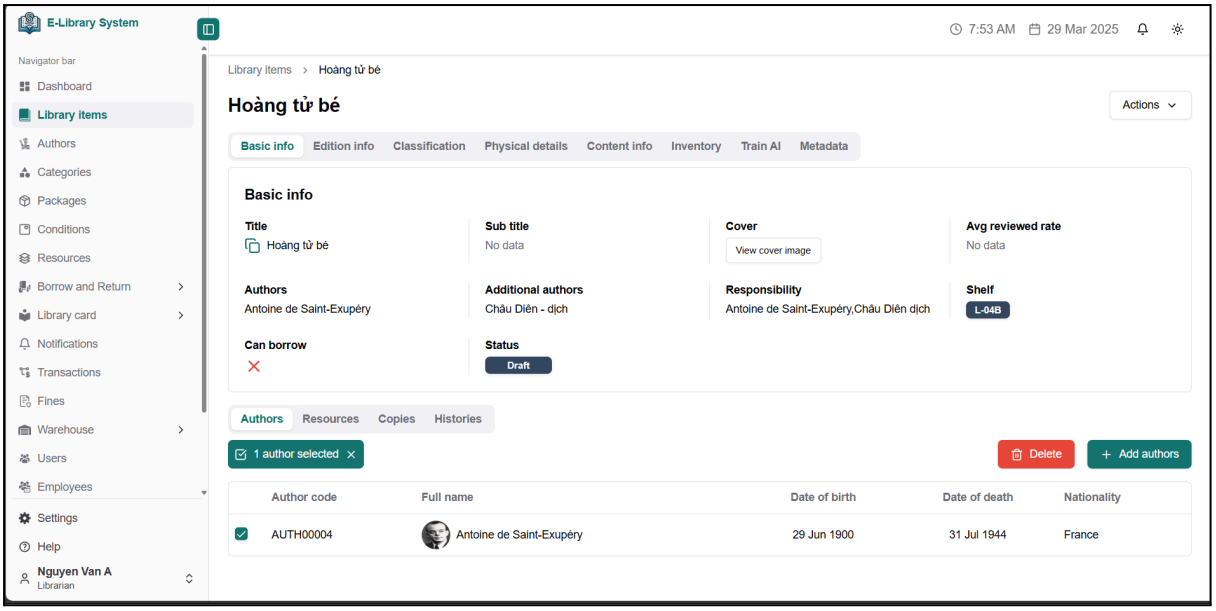


Figure 3.4.4.8: Library item detail management

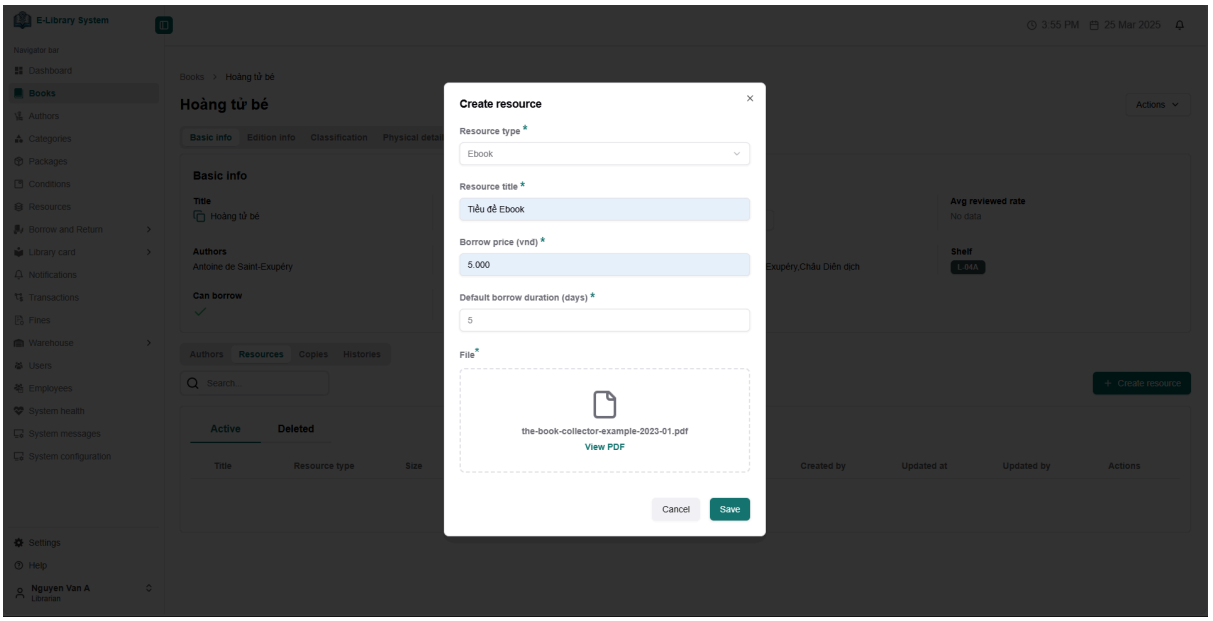


Figure 3.4.4.9: Library item detail management

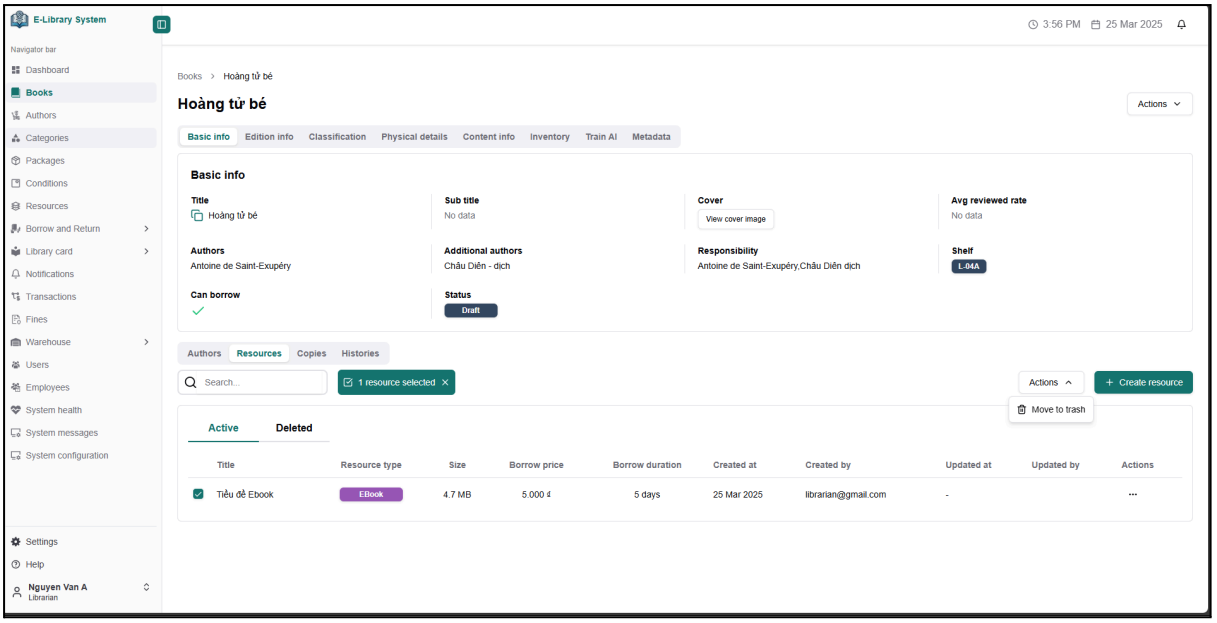


Figure 3.4.4.10: Library item detail management

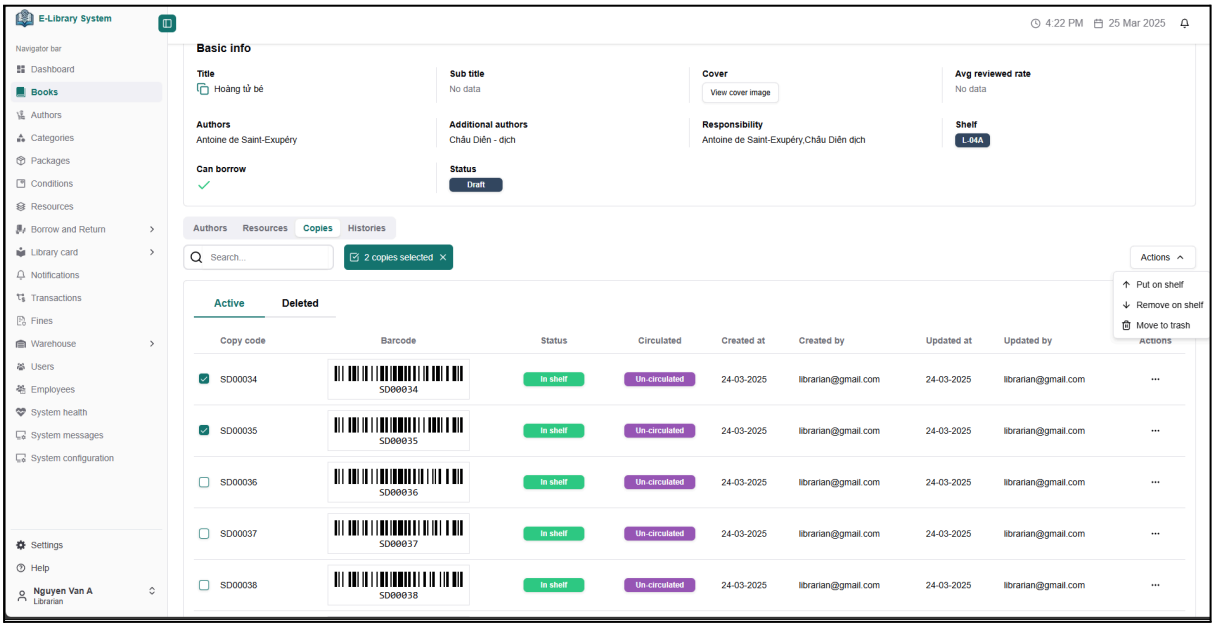


Figure 3.4.4.11: Library item detail management

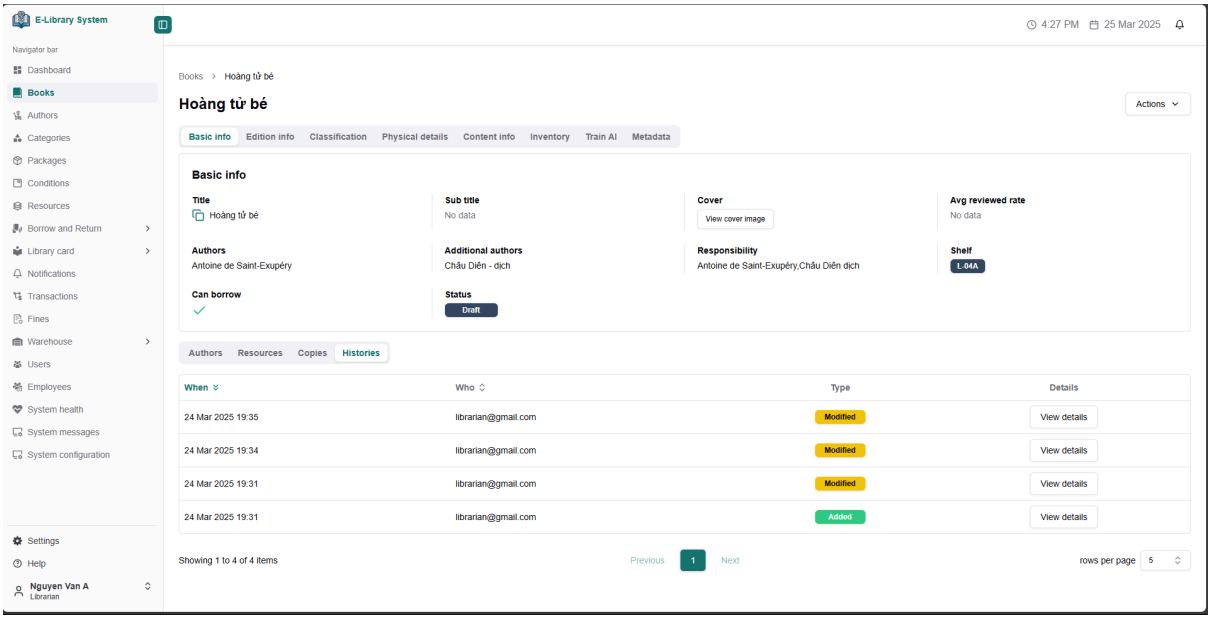


Figure 3.4.4.12: Library item detail management

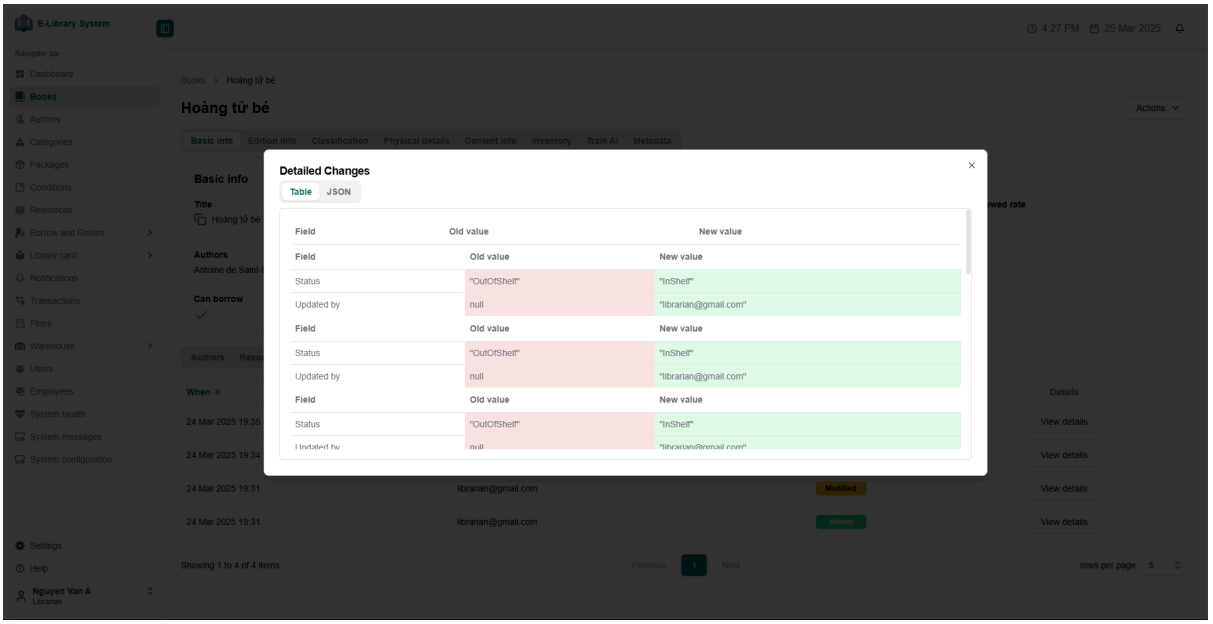


Figure 3.4.4.13: Library item detail management

3.4.5 Authors management

- **Function Description:** This screen allows **librarians** to view, search, and create authors. It provides various filters methods, data import/export functionalities, and actions for handling authors.
- **Function Trigger:**

    - ❖ **Search authors:**
        - **Button/Navigation:** There is a search input on the top of the screen.
        - **Trigger:** The librarian enters the keyword they want to search for and then presses "Enter".

- **Function:** The system will filter out the corresponding search results.

❖ **Filter authors:**
- **Button/Navigation:** There is a "Filter" button right next to the search input.
- **Trigger:** When clicked, a dialog will appear. Librarians can then enter the necessary inputs or select options from dialog to refine the search. Finally, clicking "Apply" will execute the query.
- **Function:** The system will filter out the corresponding filter results.

❖ **Import authors:**
- **Button/Navigation:** There is an “Import” button on the top of the screen.
- **Trigger:** When clicked, the system presents a dialog where the librarian can upload an Excel file and fill in the required fields. the librarian can press 'Import' to upload the file.
- **Function:** New authors will be added to the system.

❖ **Export authors:**
- **Button/Navigation:** There is an “Export” button on the top of the screen.
- **Trigger:** Librarians can press this button to perform the data export process.
- **Function:** The system will download to the user's computer an excel file containing data of the authors.

❖ **Create author:**
- **Button/Navigation:** There is a “Create author” button at the top of the screen.
- **Trigger:** When clicked, the system will display a dialog, the librarian needs to fill in the required fields then press the “Continue” button.
- **Function:** The system will create a new author with the provided information.

❖ **Sort a column:**
- **Button/Navigation:** The table headers with up/down icons indicate sortable columns.
- **Trigger:** When clicked, the action will proceed.
- **Function:** Clicking the header toggles the sort order between ascending and descending. The icon changes accordingly (up = ascending, down = descending).

❖ **Edit an author:**
- **Button/Navigation:** Click on the "..." button in the last column of each row in the table. A dropdown will appear with the "Edit author" option.
- **Trigger:** When clicked, the system will display a dialog allowing the librarian to edit the author's information, then click "Save" to confirm.
- **Function:** The system will update the author’s information.

❖ **Move to trash an author:**
- **Button/Navigation:** At the “Active” tab of the table. Click on the "..." button in the last column of each row in the table. A dropdown will appear with the Move to trash" option.

- **Trigger:** When clicked, a dialog will appear, asking the librarian to confirm moving the author to the trash. After confirming, the librarian clicks the "Continue" button.
- **Function:** The system will update the **isDeleted** field of the selected author to **true**.

❖ **Restore an author:**
- **Button/Navigation:** At the “Deleted” tab of the table. Click on the "..." button in the last column of each row in the table. A dropdown will appear with the “Restore" option.
- **Trigger:** When clicked, the action will proceed.
- **Function:** The system will update the **isDeleted** field of the selected author to **false**.

❖ **Delete permanently an author:**
- **Button/Navigation:** At the “Deleted” tab of the table. Click on the "..." button in the last column of each row in the table. A dropdown will appear with the **"Delete permanently"** option.
- **Trigger:** When clicked, a dialog will appear, asking the librarian to confirm to delete permanently the author. After confirming, the librarian clicks the "Continue" button.
- **Function:** The system will permanently delete the author from the database.

❖ **Action with authors:**
- **Button/Navigation:** A “Hành động” button at the top of the screen will be clickable when librarians select at least a row on the table.
- **Trigger:** When clicked, the system will display options such as "Move to Trash", "Restore", "Permanently Delete"
- **Function:** The system performs the selected action on the chosen authors.

❖ **Pagination**
- **Button/Navigation:** Simple numbered buttons, "Next" and "Previous" links, at the bottom of the table.
- **Trigger:** Librarian clicks on any of the pagination controls.
- **Function:** Changes the page of the table, allowing the librarians to navigate through multiple entries.

❖ **Adjust page size**
- **Button/Navigation:** A number selection dropdown at the bottom of the screen
- **Trigger:** When clicked, different options will be displayed so that the librarian can choose the page size they want.
- **Function:** The system will display more or less rows per page of the table.

❖ **View different types of authors**
- **Button/Navigation:** There are two different tabs that the librarian can choose from located above the table.
- **Trigger:** When clicked, the system will filter the corresponding authors.
- **Function:**
  - **"Active" Tab**: Displays active authors where isDeleted = false.

➢ **"Deleted" Tab**: Shows deleted authors where isDeleted = true.

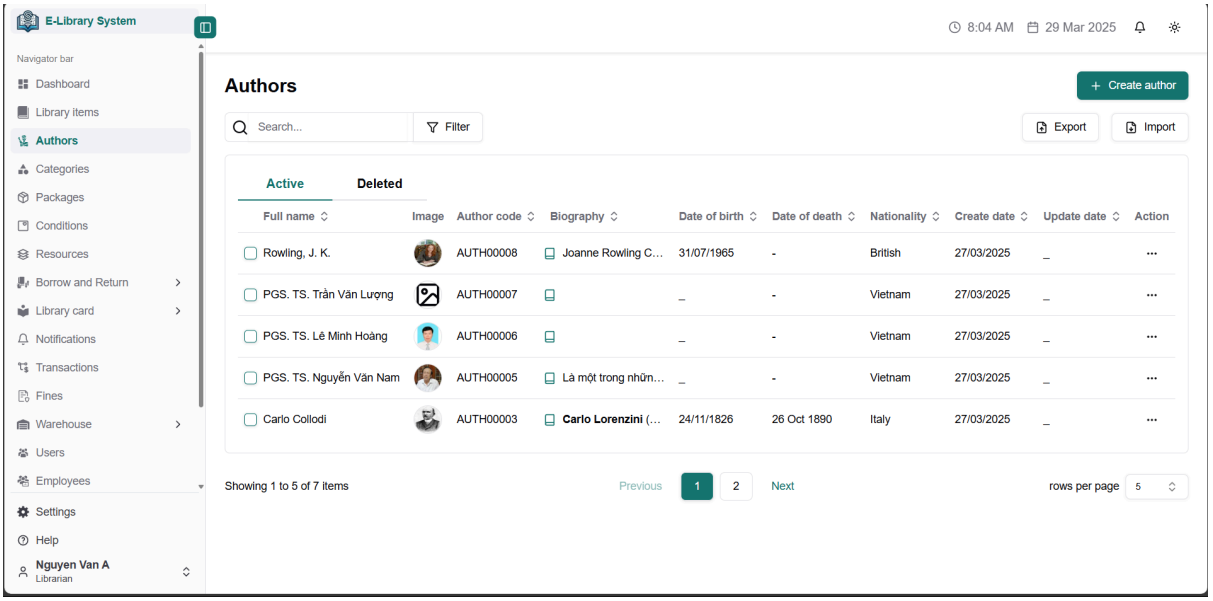


Figure 3.4.5.1: Authors management

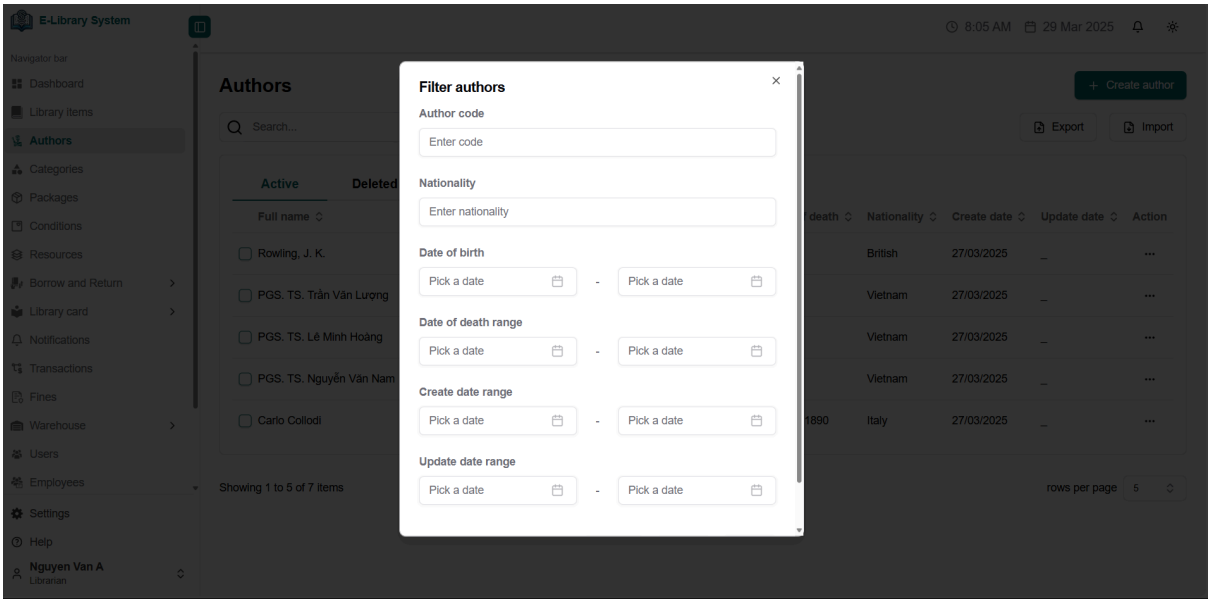


Figure 3.4.5.2: Authors management

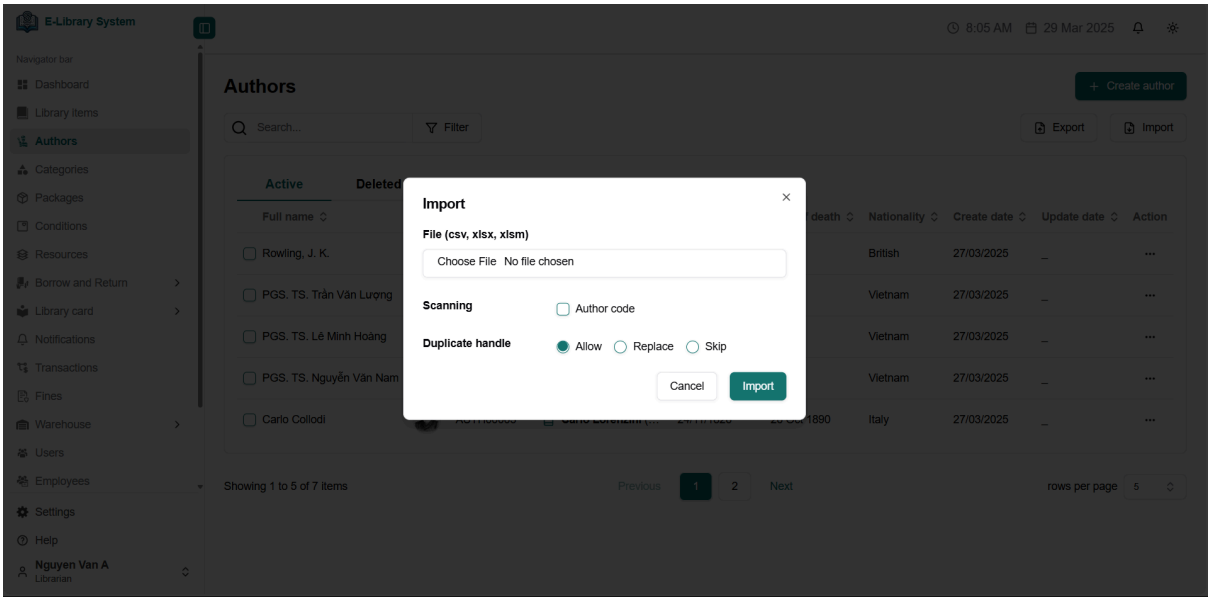


Figure 3.4.5.3: Authors management

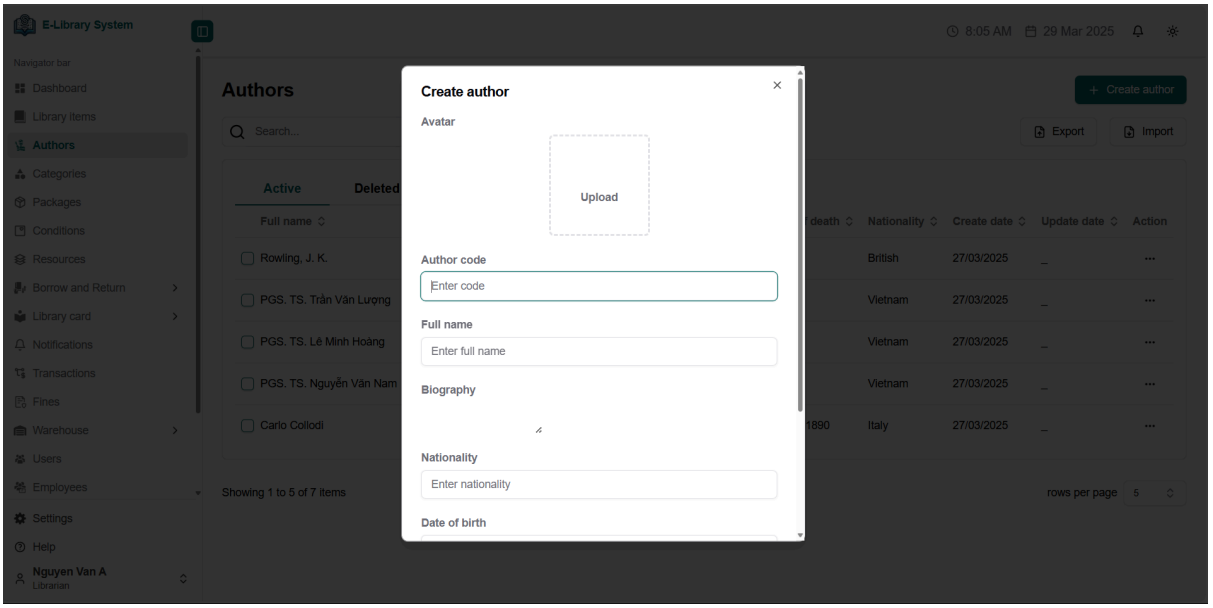


Figure 3.4.5.4: Authors management

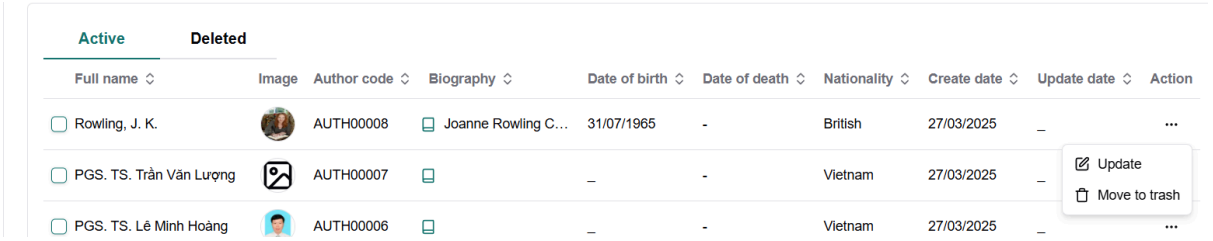


Figure 3.4.5.5: Authors management

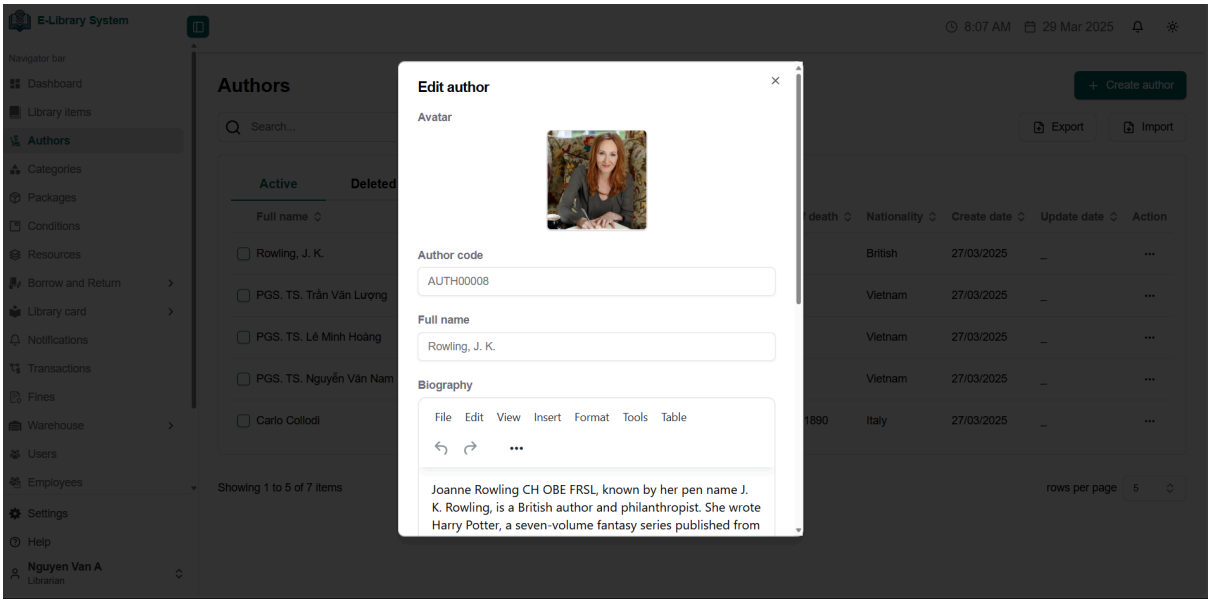


Figure 3.4.5.6: Authors management

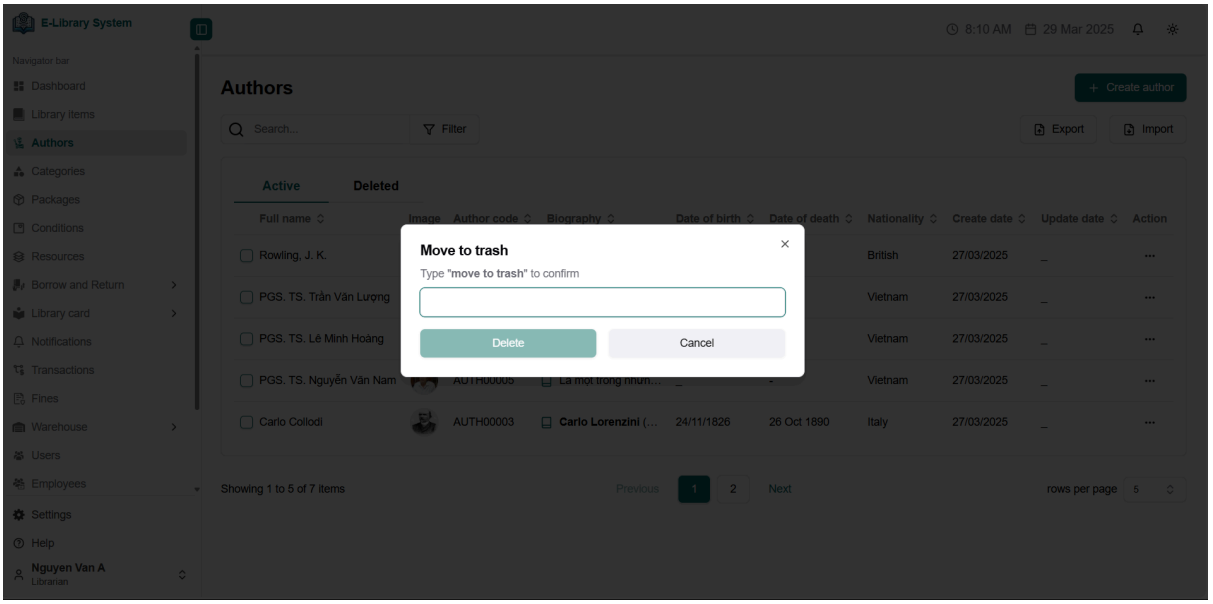


Figure 3.4.5.7: Authors management

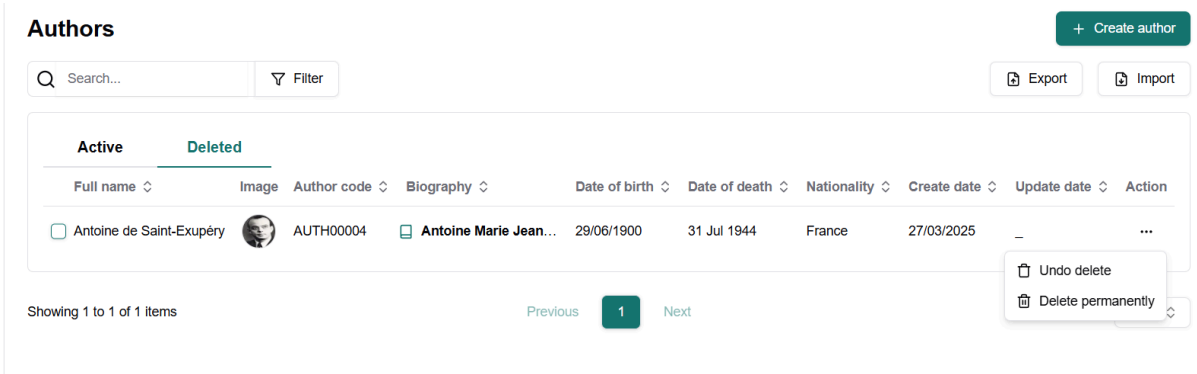


Figure 3.4.5.8: Authors management

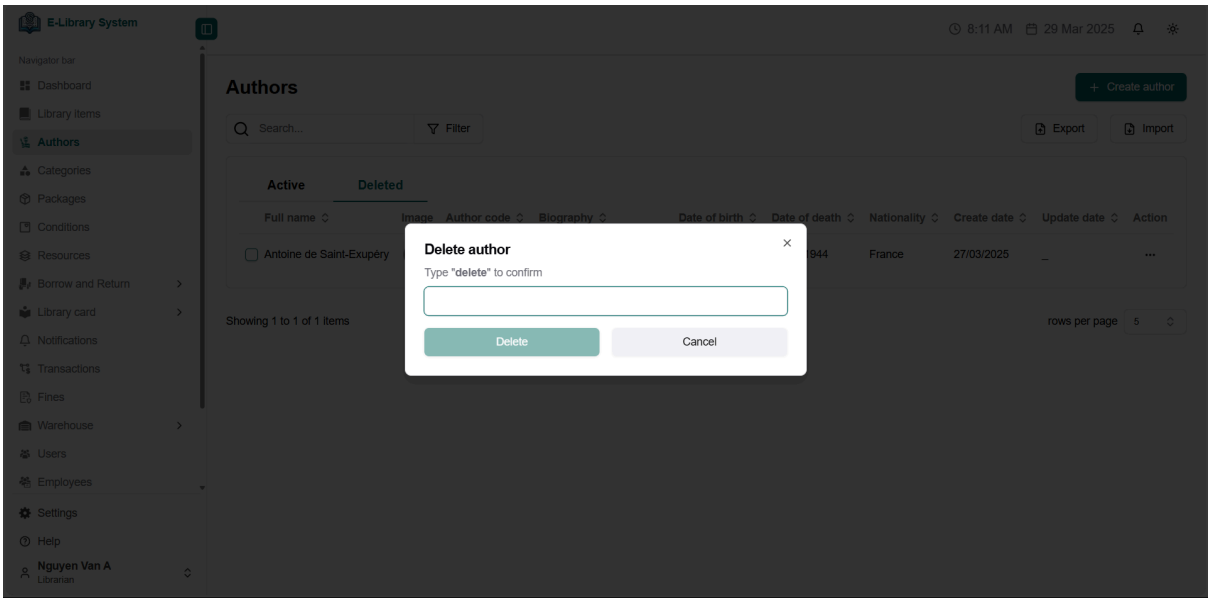


Figure 3.4.5.9: Authors management

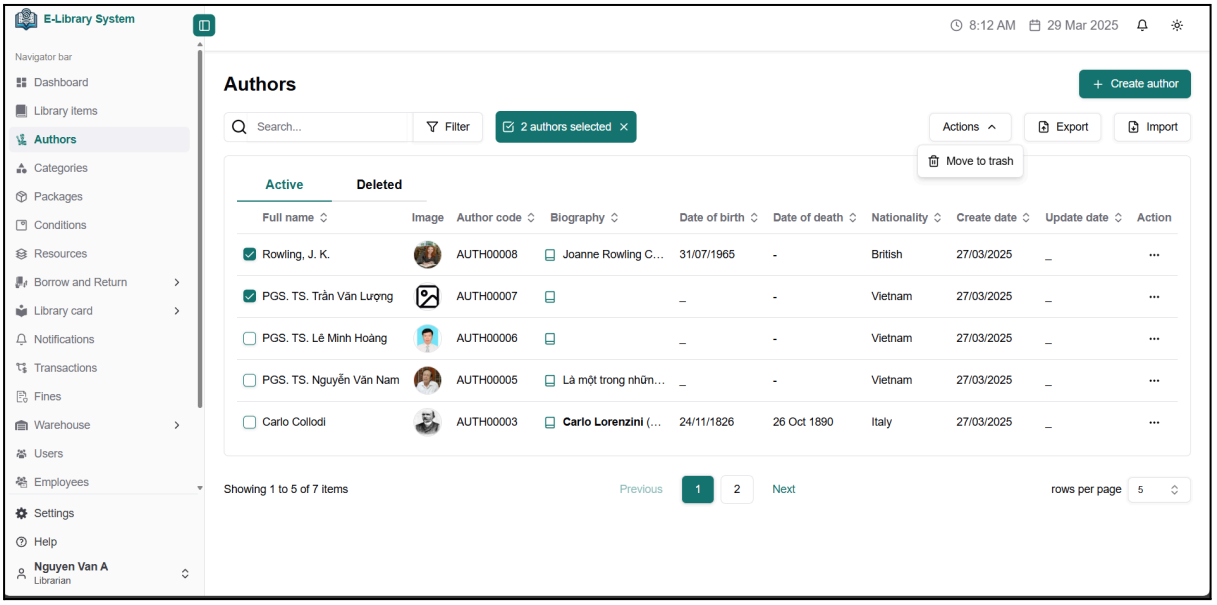


Figure 3.4.5.10: Authors management

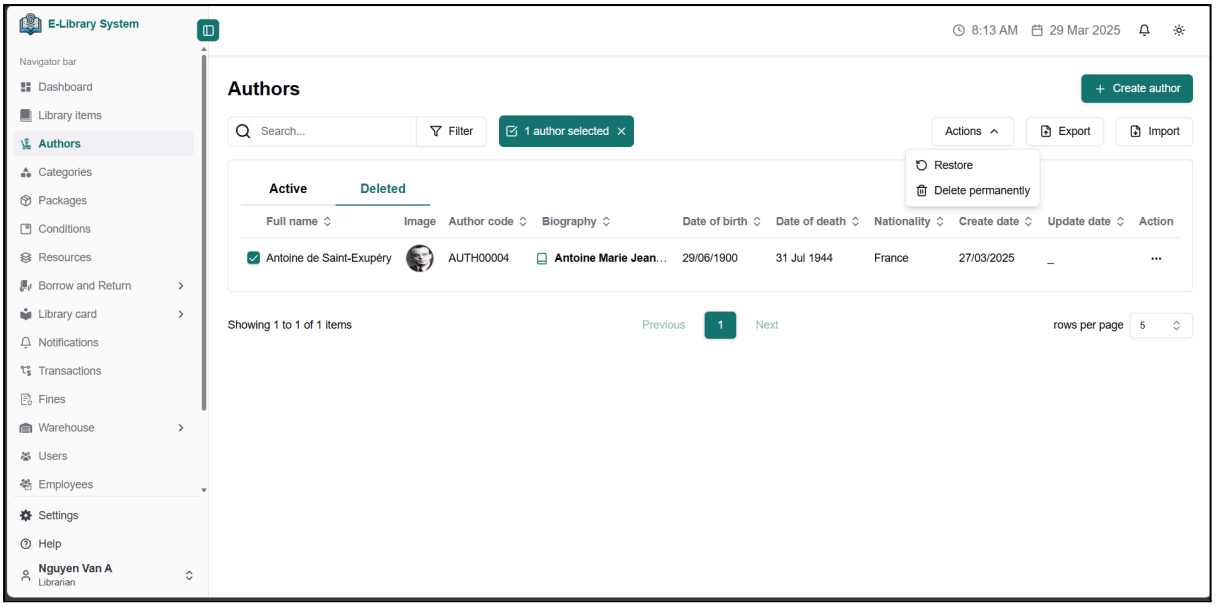


Figure 3.4.5.11: Authors management

3.4.6 Categories management

- **Function Description:** This screen allows **librarians** to view, search, and import categories. It also provides create, edit or delete each category.
- **Function Trigger:**

  - ❖ **Search categories:**
    - **Button/Navigation:** There is a search input on the top of the screen.
    - **Trigger:** The librarian enters the keyword they want to search for and then presses "Enter".
    - **Function:** The system will filter out the corresponding search results.

  - ❖ **Sort categories:**
    - **Button/Navigation:** There is a search sort dropdown to the right of the search input.
    - **Trigger:** When clicked, the system will display a dropdown with two options: **"A-Z"** and **"Z-A"**. The librarian can select one to execute the sorting.
    - **Function:** The system will sort the categories based on the option selected by the librarian.
  - ❖ **Import categories:**
    - **Button/Navigation:** There is an “Import” button on the top of the screen.
    - **Trigger:** When clicked, the system presents a dialog where the librarian can upload an Excel file and fill in the required fields. Once completed, they can press "Import" to proceed
    - **Function:** New categories will be added to the system.

  - ❖ **Create a category:**
    - **Button/Navigation:** There is a “Create category” button at the top of the screen.

- **Trigger:** When clicked, the system will display a dialog, the librarian needs to fill in the required fields then press the “Continue” button.
- **Function:** The system will create a new category.

❖ **Edit a category:**

- **Button/Navigation:** The librarian can right-click on a **category card**, and a context menu will appear with the **"Edit"** option.
- **Trigger:** When clicked, the system will display a dialog allowing the librarian to edit the category's information, then click "Save" to confirm.
- **Function:** The system will update the category’s information.

❖ **Delete a category:**

- **Button/Navigation:** The librarian can right-click on a **category card**, and a context menu will appear with the **"Delete"** option.
- **Trigger:** When clicked, a dialog will appear, asking the librarian to confirm to delete the category. After confirming, the librarian clicks the "Continue" button.
- **Function:** The system will delete the category from the database.

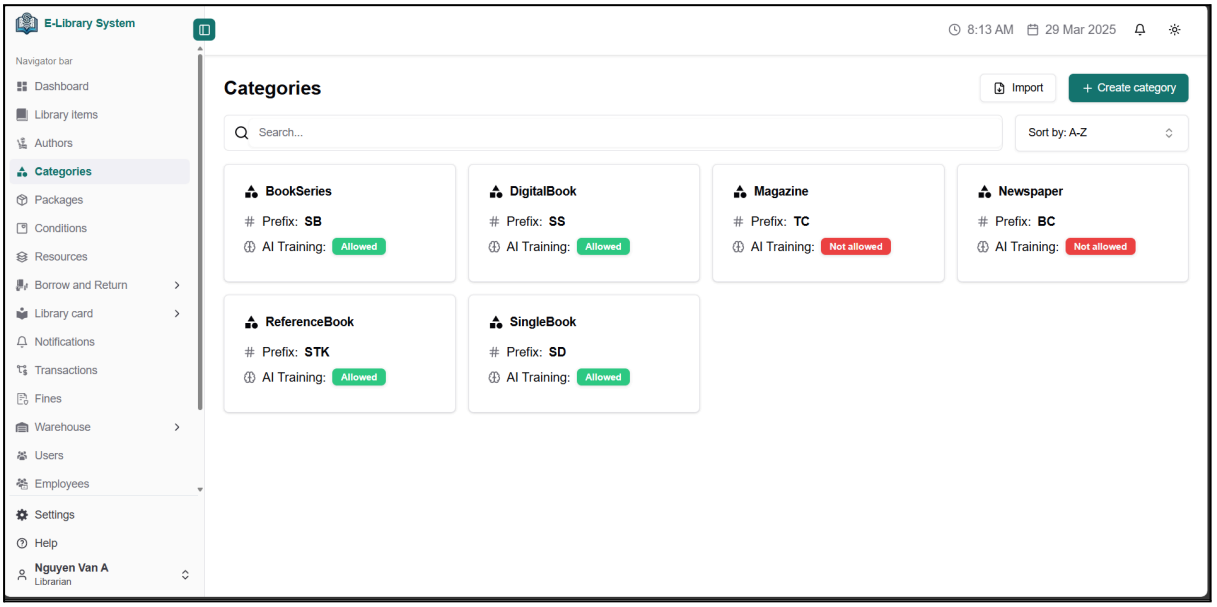


Figure 3.4.6.1: Categories management

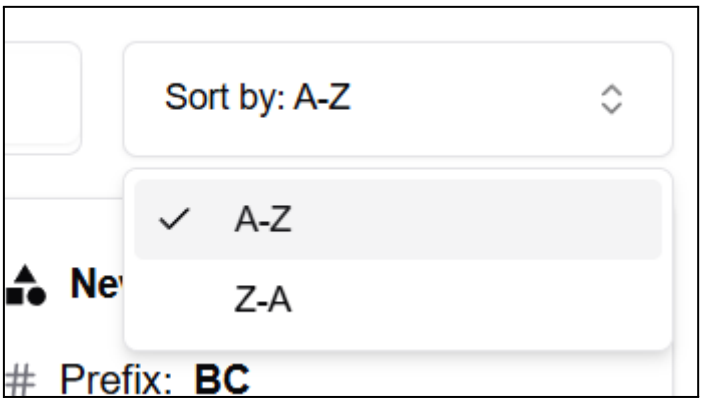


Figure 3.4.6.2: Categories management

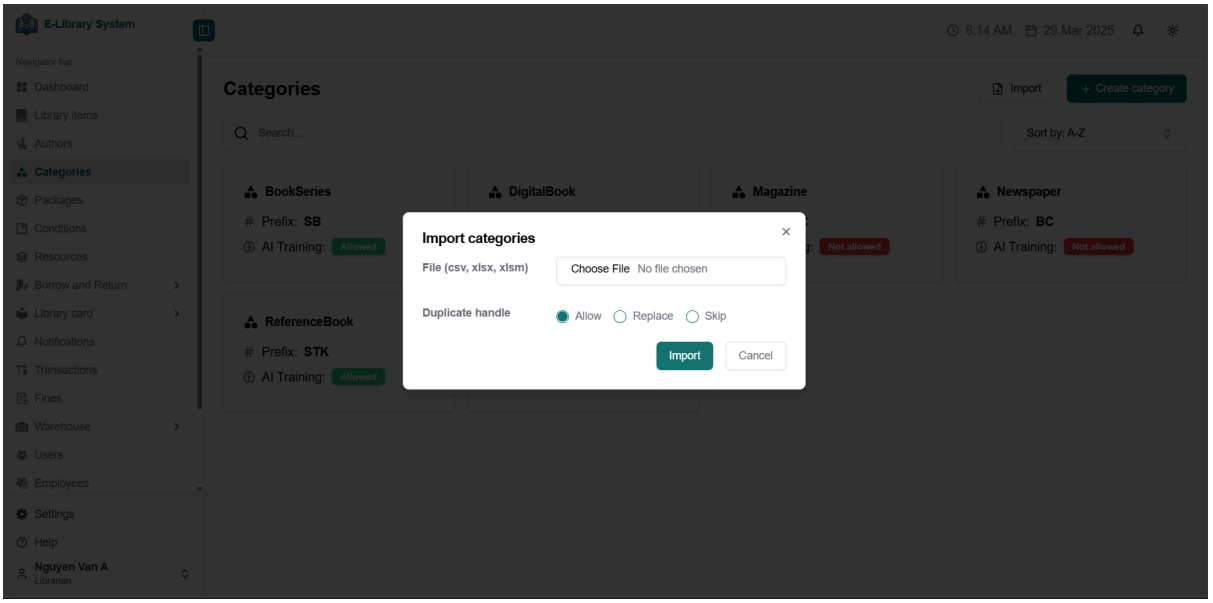


Figure 3.4.6.3: Categories management

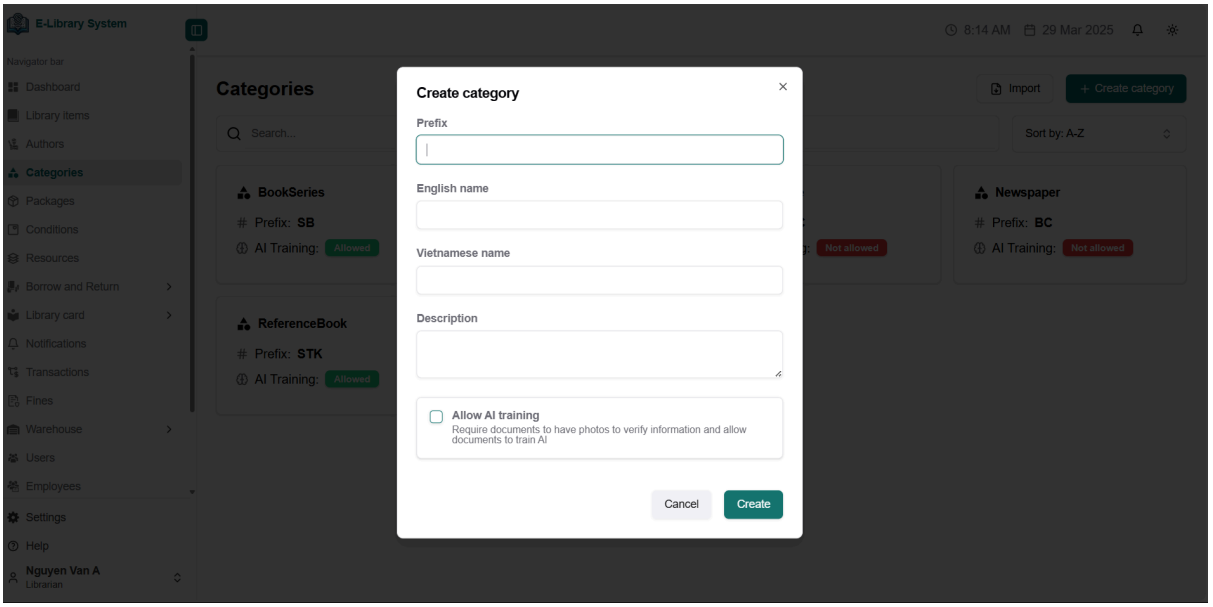


Figure 3.4.6.4: Categories management

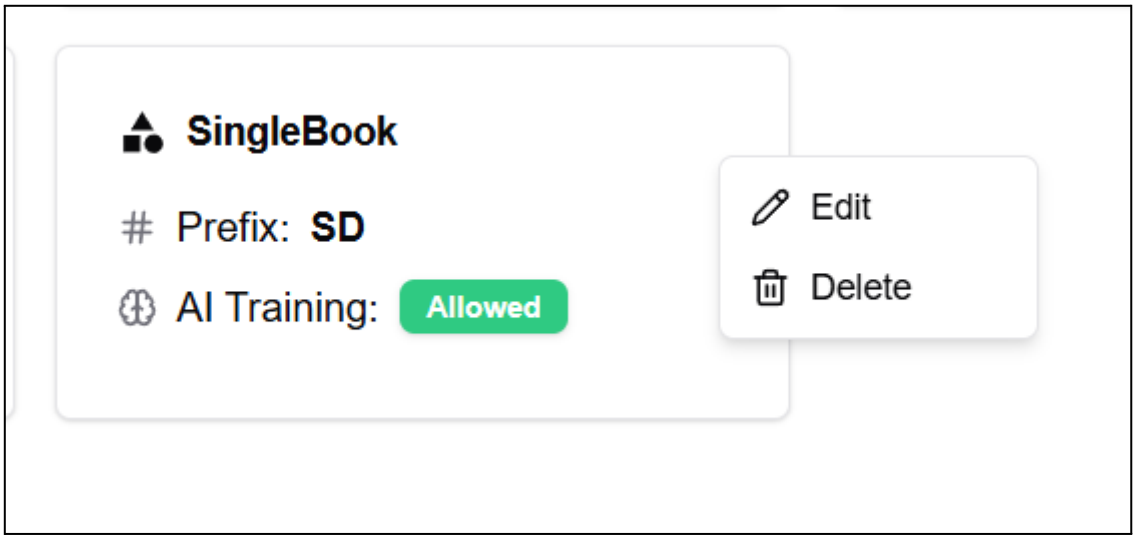


Figure 3.4.6.5: Categories management

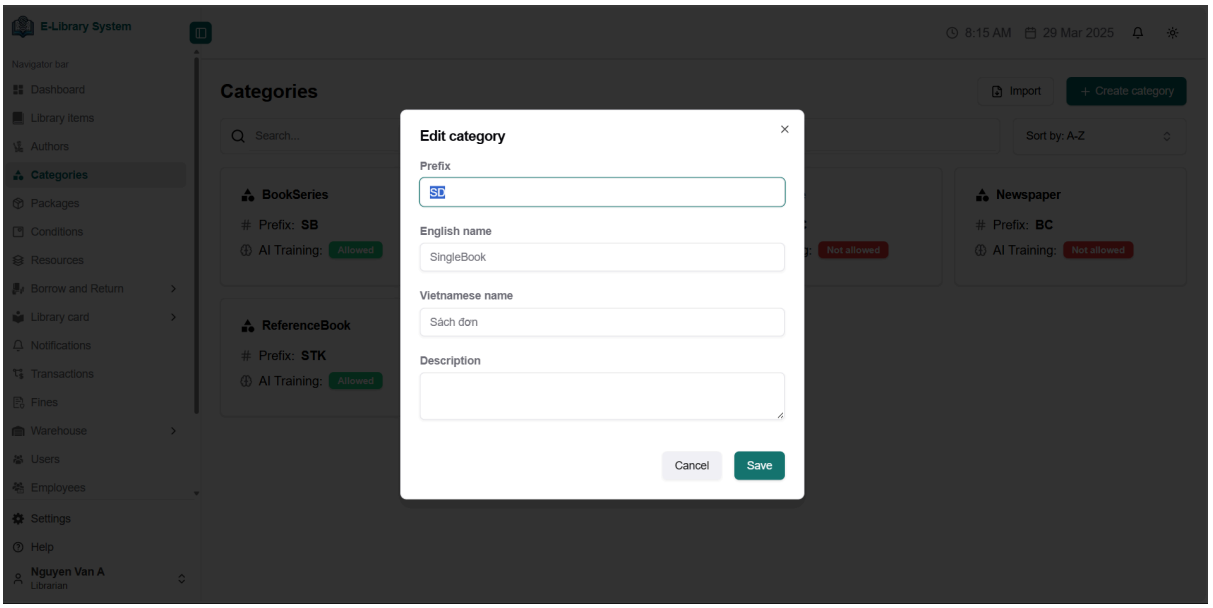


Figure 3.4.6.6: Categories management

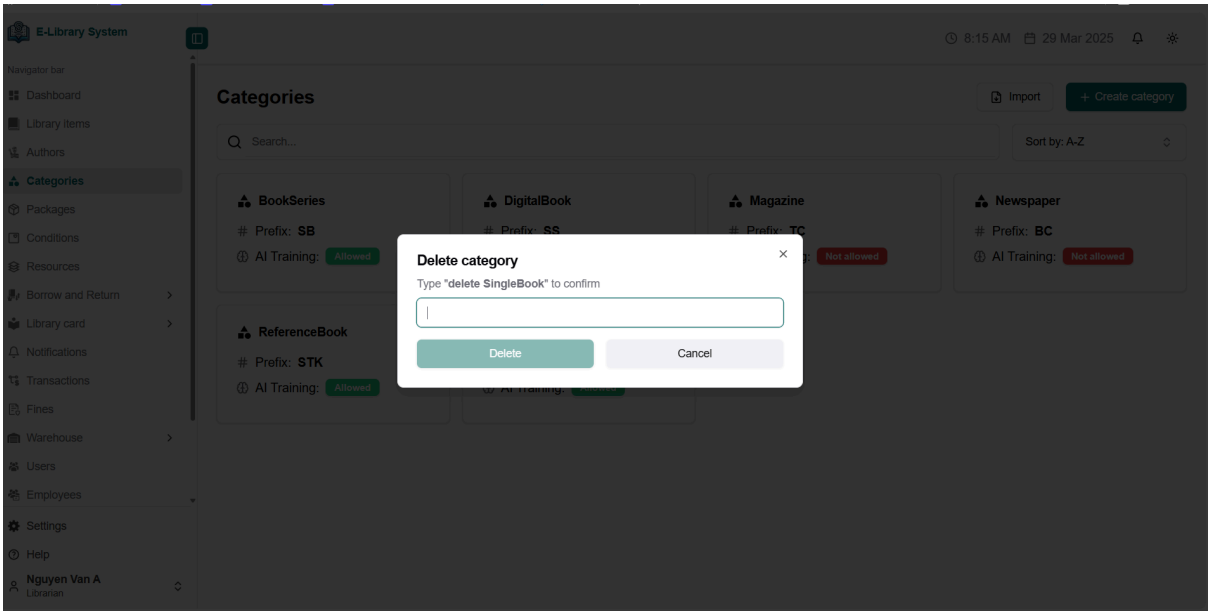


Figure 3.4.6.7: Categories management

3.4.7 Packages management

- **Function Description:** This screen allows **librarians** to view packages. It also provides create, edit or delete each package.
- **Function Trigger:**

  - **Create a package:**
    - **Button/Navigation:** There is a “Create package” button at the top of the screen.
    - **Trigger:** When clicked, the system will display a dialog, the librarian needs to fill in the required fields then press the “Continue” button.
    - **Function:** The system will create a new package.

  - **Edit a package:**
    - **Button/Navigation:** The librarian can right-click on a **package card**, and a context menu will appear with the **"Edit"** option.
    - **Trigger:** When clicked, the system will display a dialog allowing the librarian to edit the package’s information, then click "Save" to confirm.
    - **Function:** The system will update the package’s information.

  - **Delete a package:**
    - **Button/Navigation:** The librarian can right-click on a **package card**, and a context menu will appear with the **"Delete"** option.
    - **Trigger:** When clicked, a dialog will appear, asking the librarian to confirm to delete the package. After confirming, the librarian clicks the "Continue" button.
    - **Function:** The system will delete the package from the database.

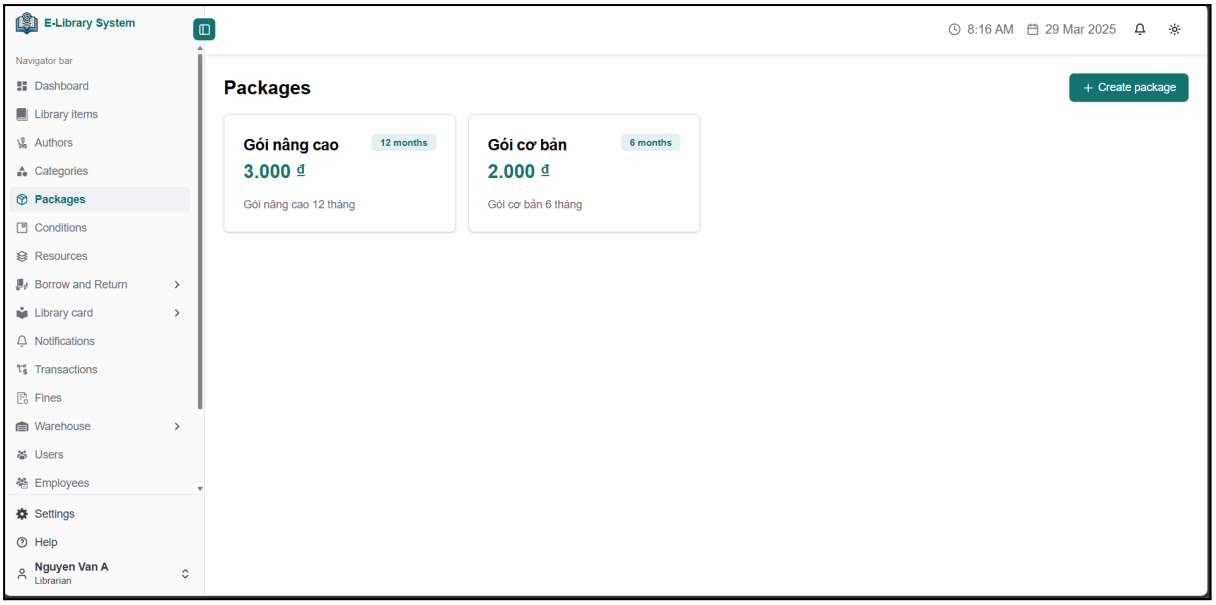


Figure 3.4.7.1: Packages management

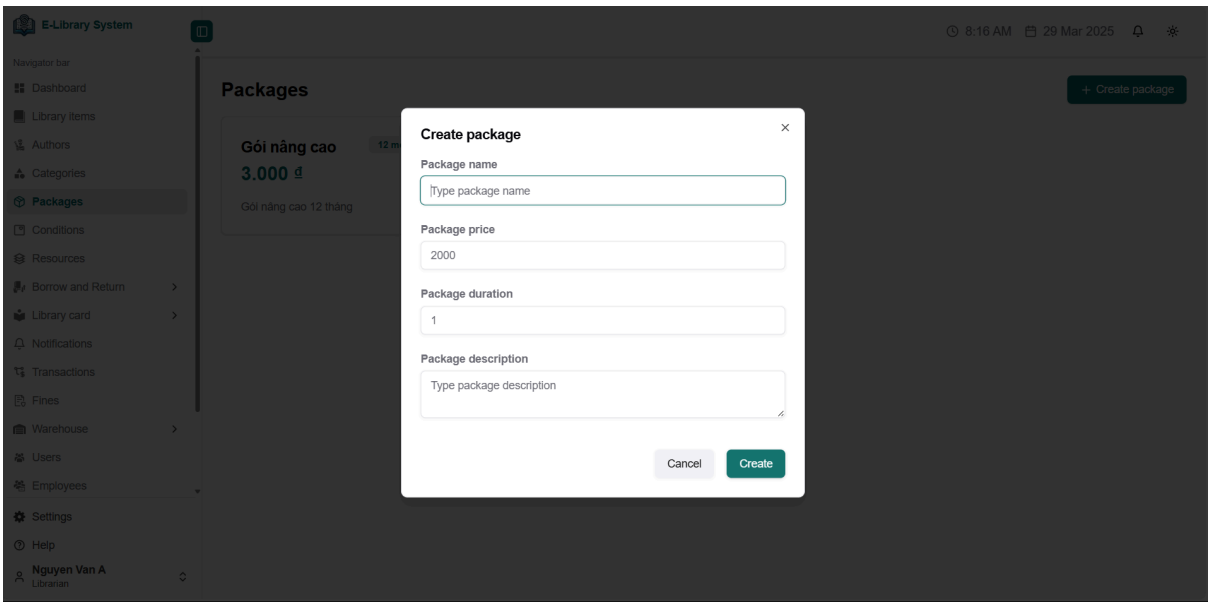


Figure 3.4.7.2: Packages management

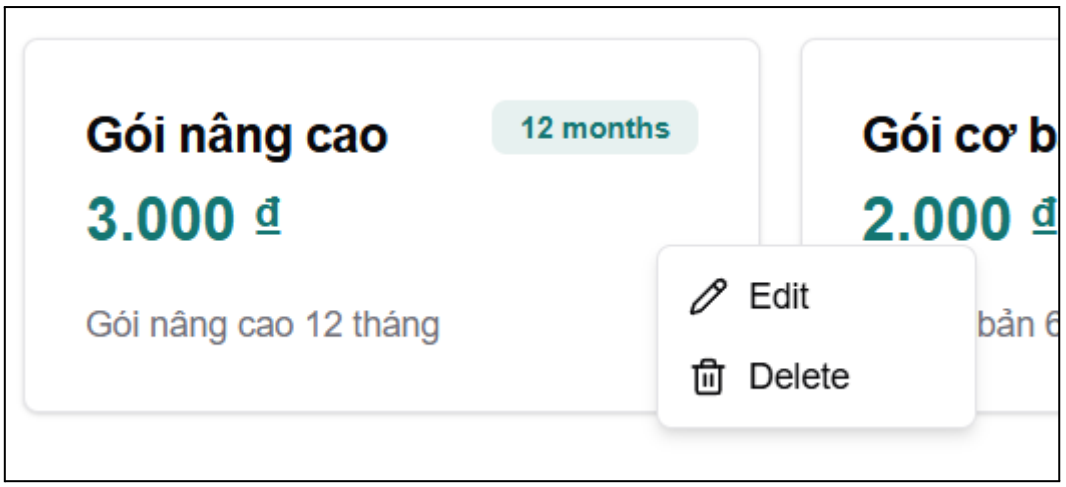


Figure 3.4.7.3: Packages management

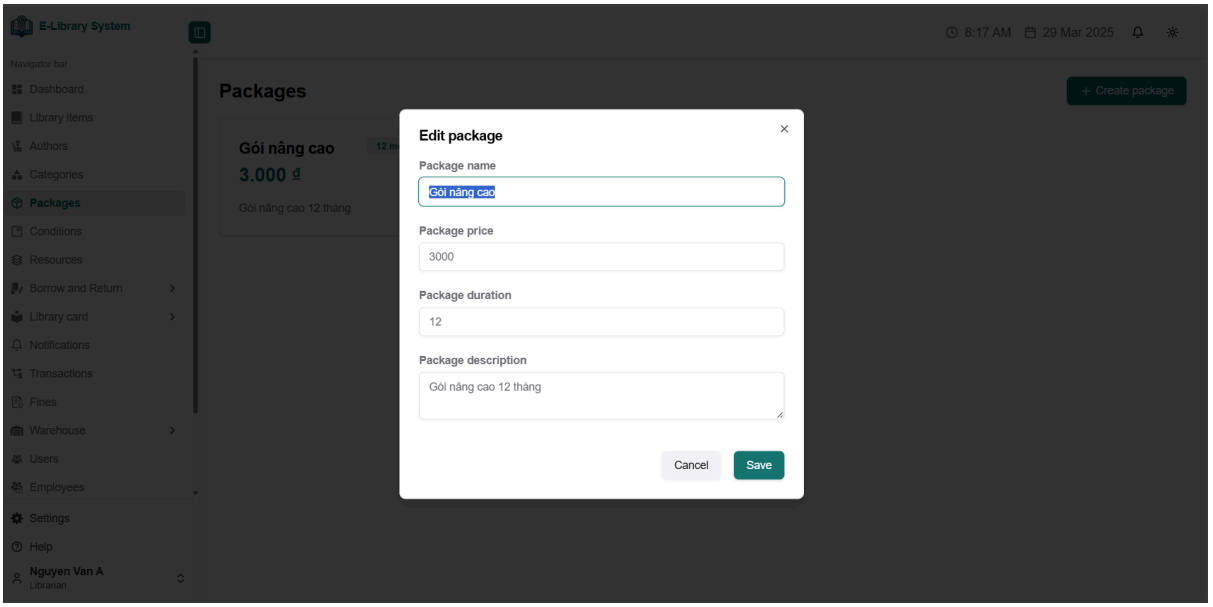


Figure 3.4.7.4: Packages management

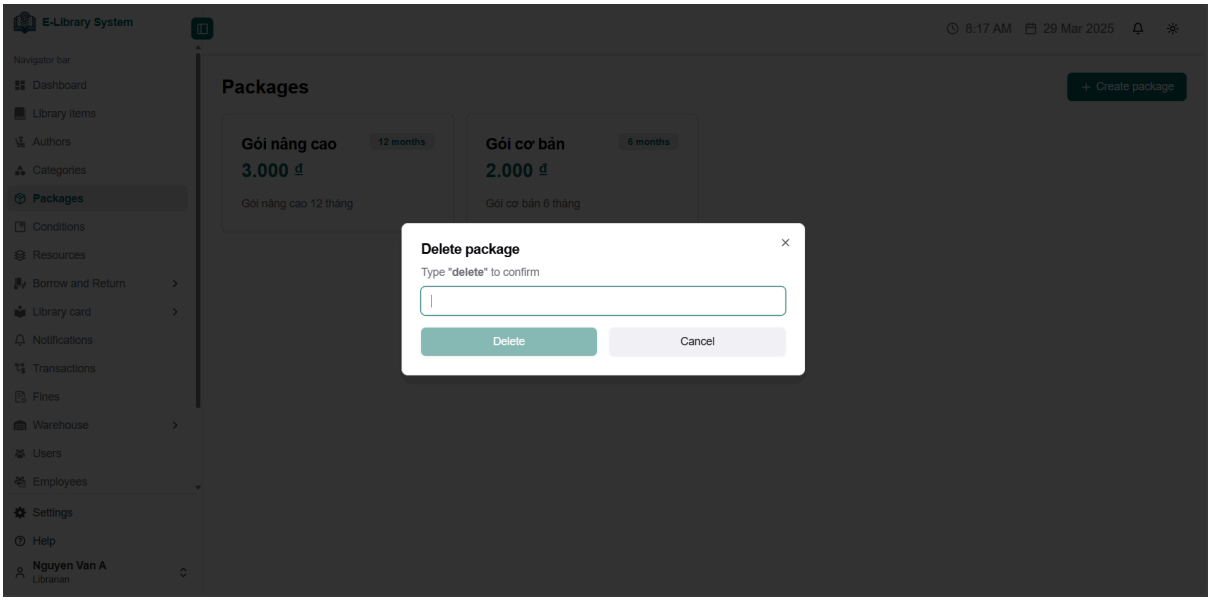


Figure 3.4.7.5: Packages management

3.4.8 Conditions management

- **Function Description:** This screen allows **librarians** to view conditions. It also provides create, edit or delete each condition.
- **Function Trigger:**

    - **Create a condition:**
        - **Button/Navigation:** There is a “Create condition” button at the top of the screen.
        - **Trigger:** When clicked, the system will display a dialog, the librarian needs to fill in the required fields then press the “Continue” button.
        - **Function:** The system will create a new condition.

    - **Edit a condition:**
        - **Button/Navigation:** The librarian can right-click on a **condition card**, and a context menu will appear with the **"Edit"** option.
        - **Trigger:** When clicked, the system will display a dialog allowing the librarian to edit the condition’s information, then click "Save" to confirm.
        - **Function:** The system will update the condition’s information.

    - **Delete a condition:**
        - **Button/Navigation:** The librarian can right-click on a **condition card**, and a context menu will appear with the **"Delete"** option.
        - **Trigger:** When clicked, a dialog will appear, asking the librarian to confirm to delete the condition. After confirming, the librarian clicks the "Continue" button.
        - **Function:** The system will delete the condition from the database.

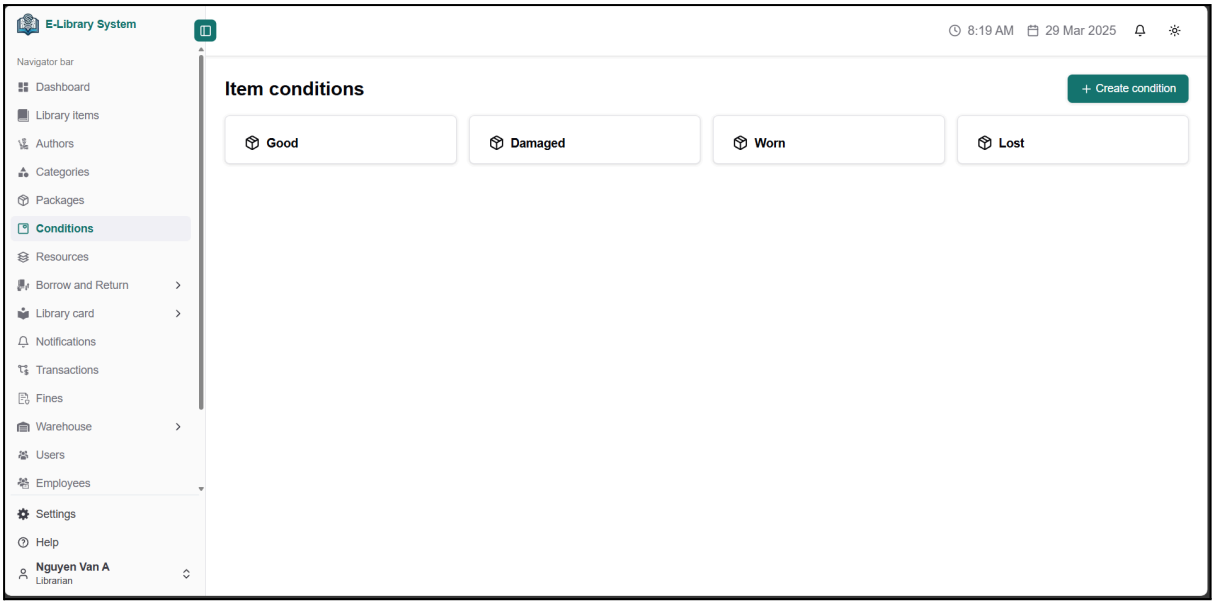


Figure 3.4.8.1: Conditions management

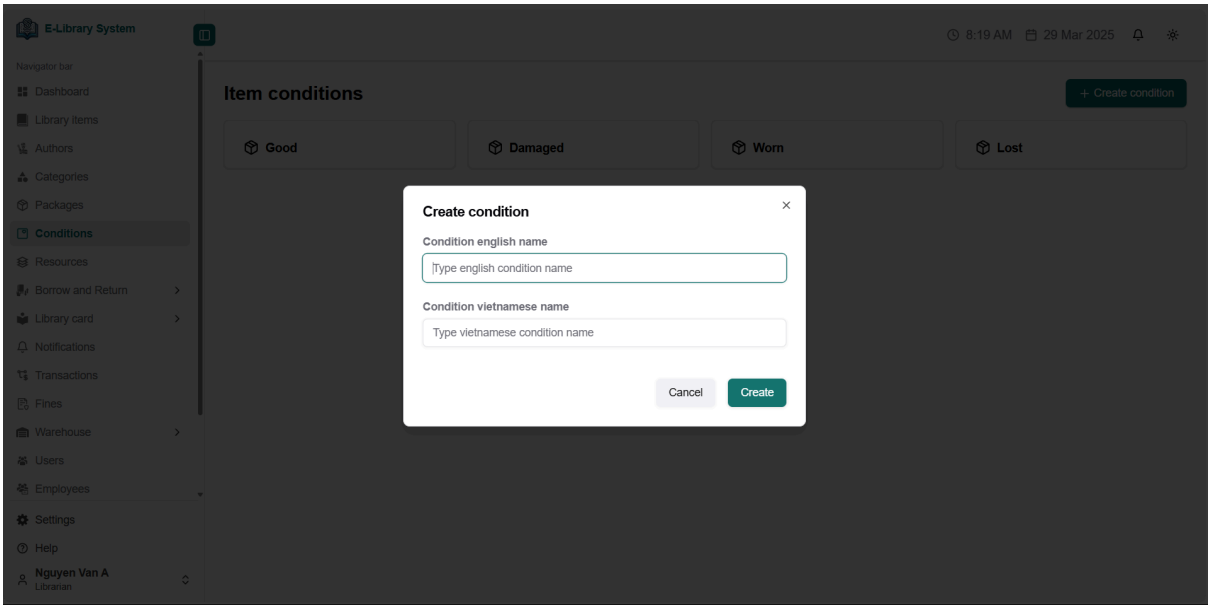


Figure 3.4.8.2: Conditions management

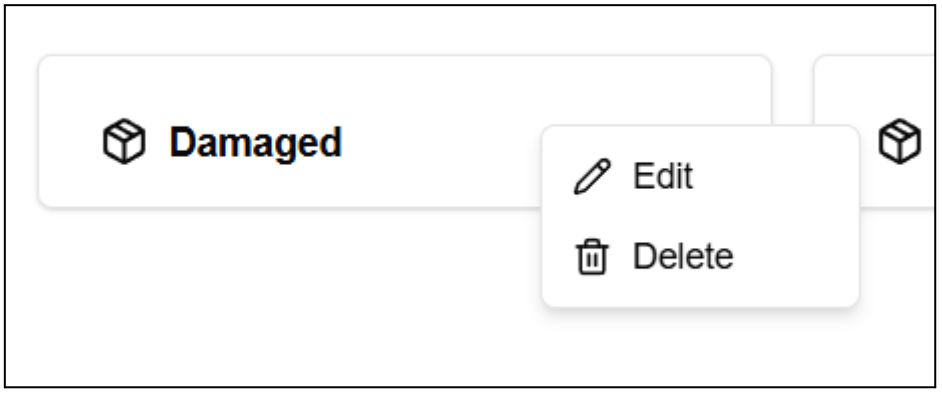


Figure 3.4.8.3: Conditions management

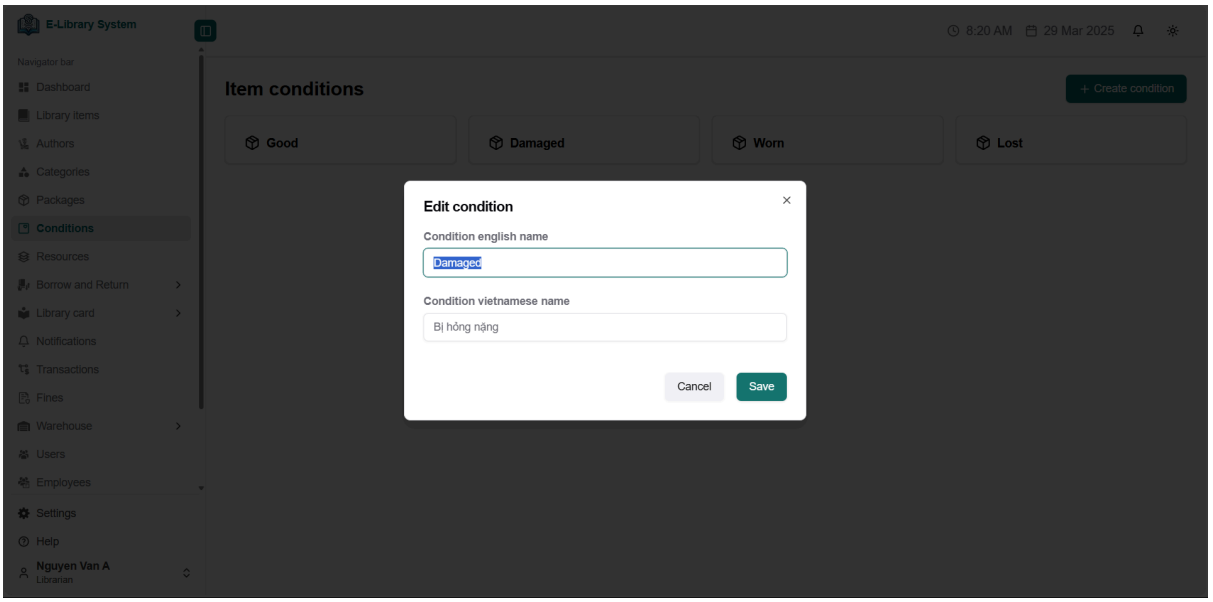


Figure 3.4.8.4: Conditions management

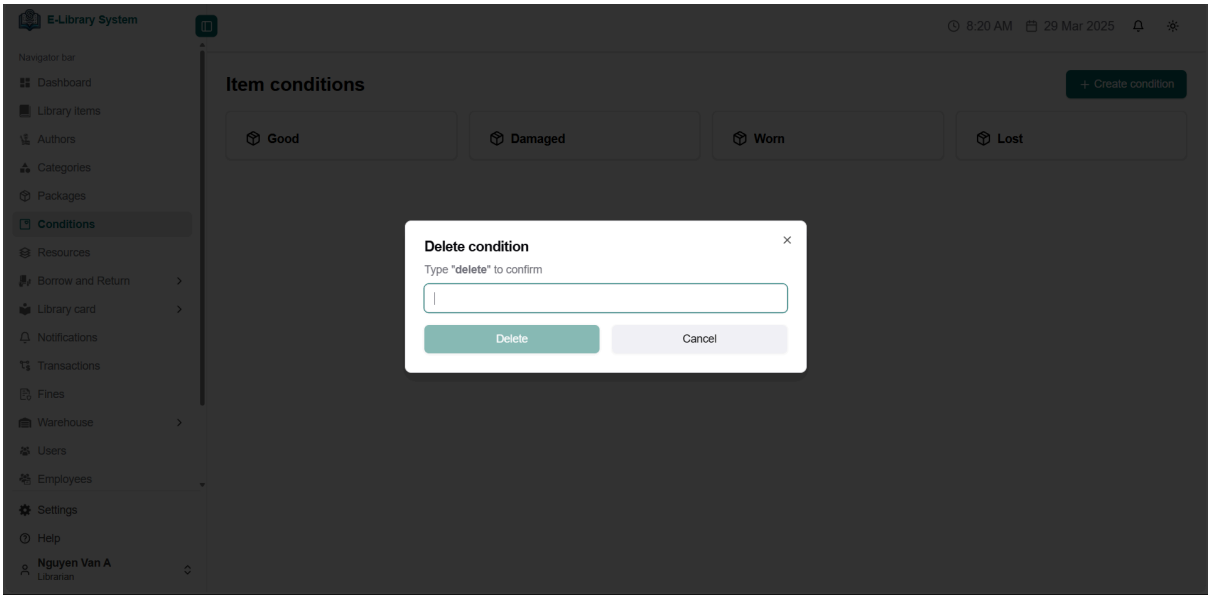


Figure 3.4.8.5: Conditions management

3.4.9 Patrons management

- **Function Description:** This screen allows **librarians** to view, search, and manage patrons efficiently. It provides various search methods, data import/export functionalities, and actions for handling patrons.
- **Function Trigger:**

    - **Search patrons:**
        - **Button/Navigation:** There is a search input on the top of the screen.

- **Trigger:** The librarian enters the keyword they want to search for and then presses "Enter".
- **Function:** The system will filter out the corresponding search results.

❖ **Filter patrons:**
- **Button/Navigation:** There is a "Filter" button right next to the search input.
- **Trigger:** When clicked, a dialog will appear. Librarians can then enter the necessary inputs or select options from dialog to refine the search. Finally, clicking "Apply" will execute the query.
- **Function:** The system will filter out the corresponding filter results.

❖ **Custom display:**
- **Button/Navigation:** There is a "Columns" button on the top of the screen.
- **Trigger:** When clicked, the system displays a dialog where the librarian can choose to hide or show specific fields, then click "Apply" to confirm.
- **Function:** The system will hide/show the columns that the librarian has previously selected.

❖ **Sort a column:**
- **Button/Navigation:** The table headers with up/down icons indicate sortable columns.
- **Trigger:** When clicked, the action will proceed.
- **Function:** Clicking the header toggles the sort order between ascending and descending. The icon changes accordingly (up = ascending, down = descending).

❖ **Import patrons:**
- **Button/Navigation:** There is a “Import” button on the top of the screen.
- **Trigger:** When clicked, the system presents a dialog where the librarian can upload an Excel file and fill in the required fields of the form. Once completed, they can press "Continue" to proceed
- **Function:** New patrons will be added to the system.

❖ **Export patrons:**
- **Button/Navigation:** There is a “Export” button on the top of the screen.
- **Trigger:** Librarians can press this button to perform the data export process.
- **Function:** The system will download an Excel file containing data of the patrons to the librarian’s computer.

❖ **Navigate to the create patron page:**
- **Button/Navigation:** There is a “Create patron” button at the top of the screen.
- **Trigger:** When clicked, it navigates users to the **create patron** page.
- **Function:** The system navigates to and displays the "Create Patron" page.

❖ **Action with patrons:**
- **Button/Navigation:** A “Actions” button at the top of the screen will be clickable when librarians select at least a row on the table.
- **Trigger:** When clicked, the system will display options such as "Move to Trash", "Restore", "Permanently Delete"

- **Function:** The system will take action on the selected patrons based on the chosen option.

❖ **Pagination**
- **Button/Navigation:** Simple numbered buttons, "Next" and "Previous" links, at the bottom of the table.
- **Trigger:** Librarian clicks on any of the pagination controls.
- **Function:** Changes the page of the table, allowing the librarians to navigate through multiple entries.

❖ **Adjust page size**
- **Button/Navigation:** A number selection dropdown at the bottom of the screen
- **Trigger:** When clicked, different options will be displayed so that the librarian can choose the page size they want.
- **Function:** The system will display more or less rows per page of the table.

❖ **View different types of patrons**
- **Button/Navigation:** There are two different tabs that the librarian can choose from located above the table.
- **Trigger:** When clicked, the system will filter the corresponding patrons.
- **Function:**
  - ➢ **"Active" Tab**: Displays active patrons where isDeleted = false.
  - ➢ **"Deleted" Tab**: Shows deleted patrons where isDeleted = true.

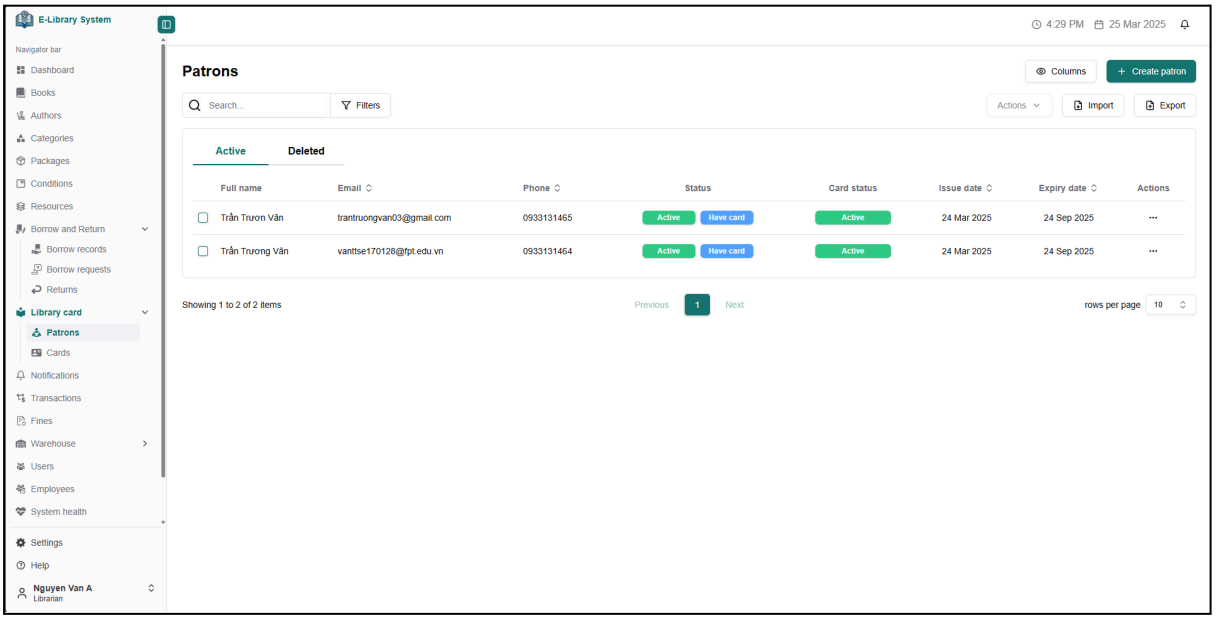


Figure 3.4.9.1: Patrons management

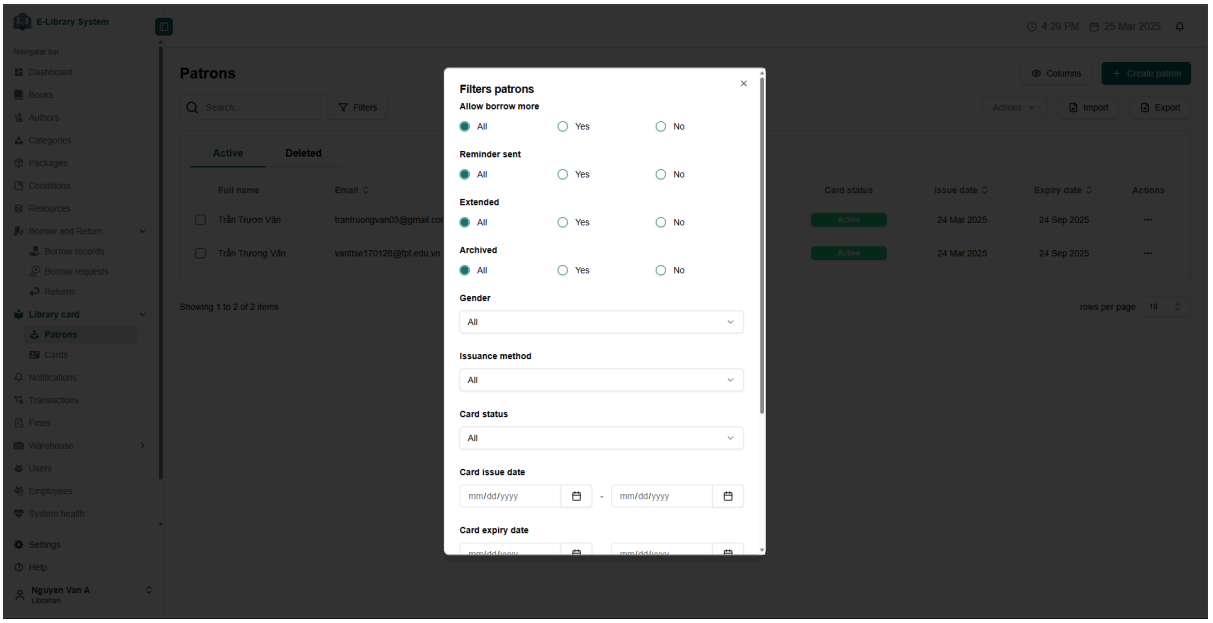


Figure 3.4.9.2: Patrons management

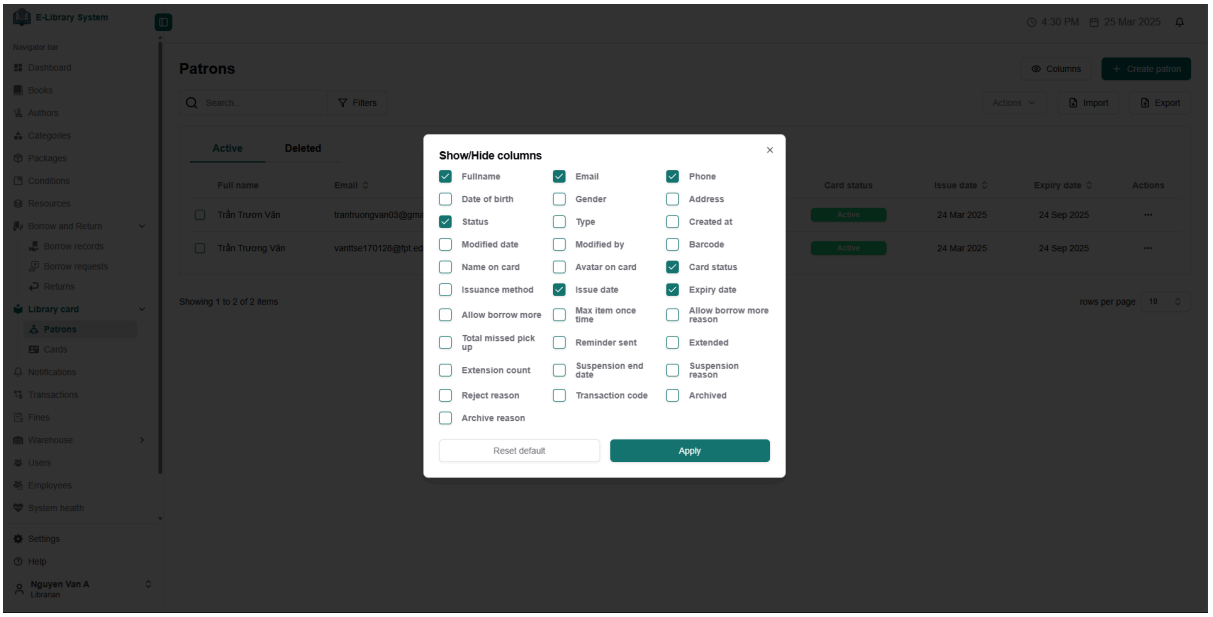


Figure 3.4.9.3: Patrons management

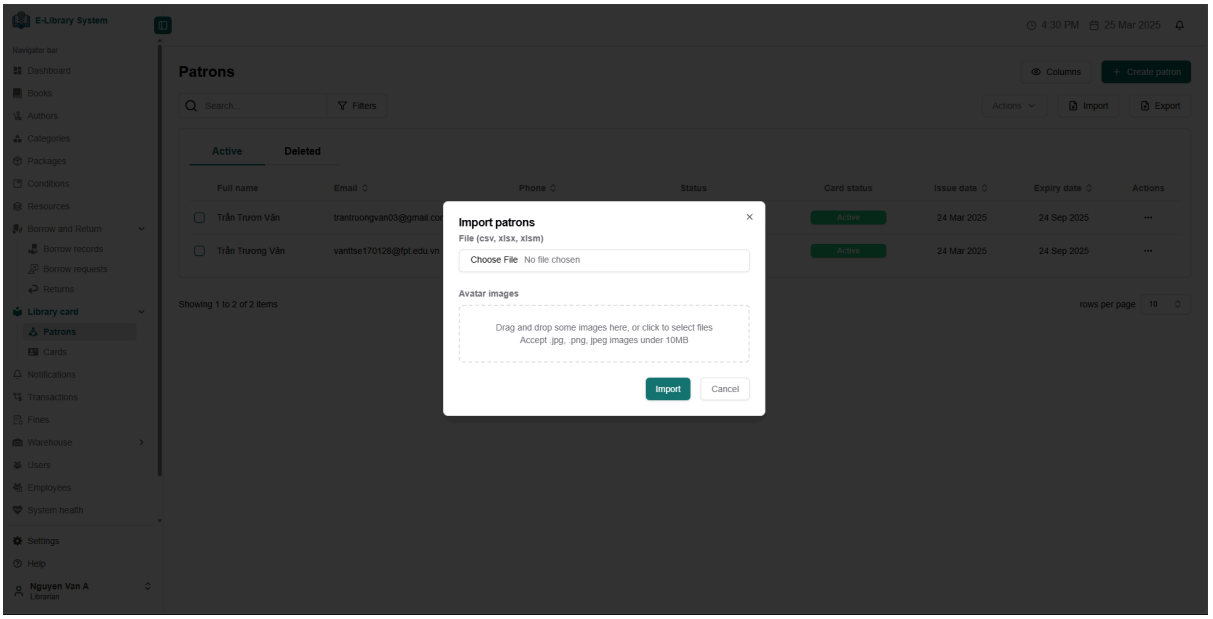


Figure 3.4.9.4: Patrons management

3.4.10 Patron detail management

- **Function Description:** This screen provides a comprehensive view of a patron's information. It displays key personal details such as the patron's name, email, phone, etc. along with the creation and modification dates. The page also includes a "Card Information" section, showing the library card's details, including the cardholder's name, barcode, issuance method, status, and borrowing limits. Librarians can manage the patron and their card through the "Actions" dropdowns, which offer options like editing information, moving to trash, confirming, suspending, uns suspending, archiving, or deleting the card. This page ensures librarians can efficiently oversee and update patron records and their associated library cards.

- **Function Trigger:**

  - **Edit Patron Information**
    - **Button/Navigation:** At the top-right corner of the screen, there is a dropdown menu labeled "Actions." When clicked, the librarian will see the option "Edit Information."
    - **Trigger:** Upon selecting the "Edit Information" option, the system will display a dialog where the librarian can modify the patron's details.
    - **Function:** The system will update the patron's information based on the changes made in the dialog.

  - **Move Patron to Trash**
    - **Button/Navigation:** At the top-right corner of the screen, there is a dropdown menu labeled "Actions." When clicked, the librarian will see the option "Move to Trash" if the patron is not already in the trash (i.e., not deleted)

- **Trigger:** Upon selecting the "Move to Trash" option, the system will initiate the process.
- **Function:** The system will mark the patron as deleted, effectively moving them to the trash.

❖ **Restore Patron**
- **Button/Navigation:** At the top-right corner of the screen, there is a dropdown menu labeled "Actions." When clicked, the librarian will see the option "Restore" if the patron has been previously moved to the trash
- **Trigger:** Upon selecting the "Restore" option, the system will initiate the process.
- **Function:** The system will update the patron's status by removing the deleted mark, effectively restoring them to an active state.

❖ **Delete Patron**
- **Button/Navigation:**At the top-right corner of the screen, there is a dropdown menu labeled "Actions." When clicked, the librarian will see the option "Delete" if the patron is currently in the trash.
- **Trigger:** Upon selecting the "Delete" option, the system will initiate the process.
- **Function:** The system will permanently delete the patron from the system.

❖ **Edit Library Card Information**
- **Button/Navigation:** In the "Card Information" section, at the top-right corner, there is a dropdown menu labeled "Actions." When clicked, the librarian will see the option "Edit Information."
- **Trigger:** Upon selecting the "Edit Information" option, the system will display a dialog where the librarian can modify the library card details.
- **Function:** The system will update the library card information based on the changes made in the dialog.

❖ **Confirm Card**
- **Button/Navigation:** In the "Card Information" section, at the top-right corner, there is a dropdown menu labeled "Actions." When clicked, the librarian will see the option "Confirm Card" if the library card is in a "Pending" state.
- **Trigger:** Upon selecting the "Confirm Card" option, the system will initiate the process.
- **Function:** The system will update the library card's status to "Active," indicating that the librarian has approved the card for the patron, enabling them to use it for borrowing library items.

❖ **Extend Borrow Amount**
- **Button/Navigation:** In the "Card Information" section, at the top-right corner, there is a dropdown menu labeled "Actions." When clicked, the librarian will see the option "Extend Borrow Amount."
- **Trigger:** Upon selecting the "Extend Borrow Amount" option, the system will initiate the process.

- **Function:** The system will increase the borrowing limit of the patron as specified by the librarian, allowing them to borrow more items simultaneously.

❖ **Suspend Card**
- **Button/Navigation:** In the "Card Information" section, at the top-right corner, there is a dropdown menu labeled "Actions." When clicked, the librarian will see the option "Suspend Card."
- **Trigger:**Upon selecting the "Suspend Card" option, the system will display a dialog where the librarian can specify the suspension details, such as the reason and duration.
- **Function:** The system will update the library card's status to "Suspended," preventing the patron from borrowing items until the suspension is lifted, and will record the suspension reason and end date.

❖ **Unsuspend Card**
- **Button/Navigation:** In the "Card Information" section, at the top-right corner, there is a dropdown menu labeled "Actions." When clicked, the librarian will see the option "Unsuspend Card" if the library card is currently in a suspended state (e.g., due to a previous suspension).
- **Trigger:** Upon selecting the "Unsuspend Card" option, the system will initiate the process.
- **Function:** The system will update the library card's status to "Active," indicating that the librarian has approved the card for the patron, enabling them to use it for borrowing library items.

❖ **Archive Card**
- **Button/Navigation:** In the "Card Information" section, at the top-right corner, there is a dropdown menu labeled "Actions." When clicked, the librarian will see the option "Archived Card."
- **Trigger:** Upon selecting the "Archived Card" option, the system will initiate the process.
- **Function:** The system will mark the library card as archived, removing it from the patron's active use while retaining its record in the database for historical purposes. This is typically used when the physical card is lost, ensuring the card is no longer usable but its data is preserved.

❖ **Delete Card**
- **Button/Navigation:**In the "Card Information" section, at the top-right corner, there is a dropdown menu labeled "Actions." When clicked, the librarian will see the option "Delete Card."
- **Trigger:** Upon selecting the "Delete Card" option, the system will initiate the process.
- **Function:** The system will permanently delete the library card from the patron's record.

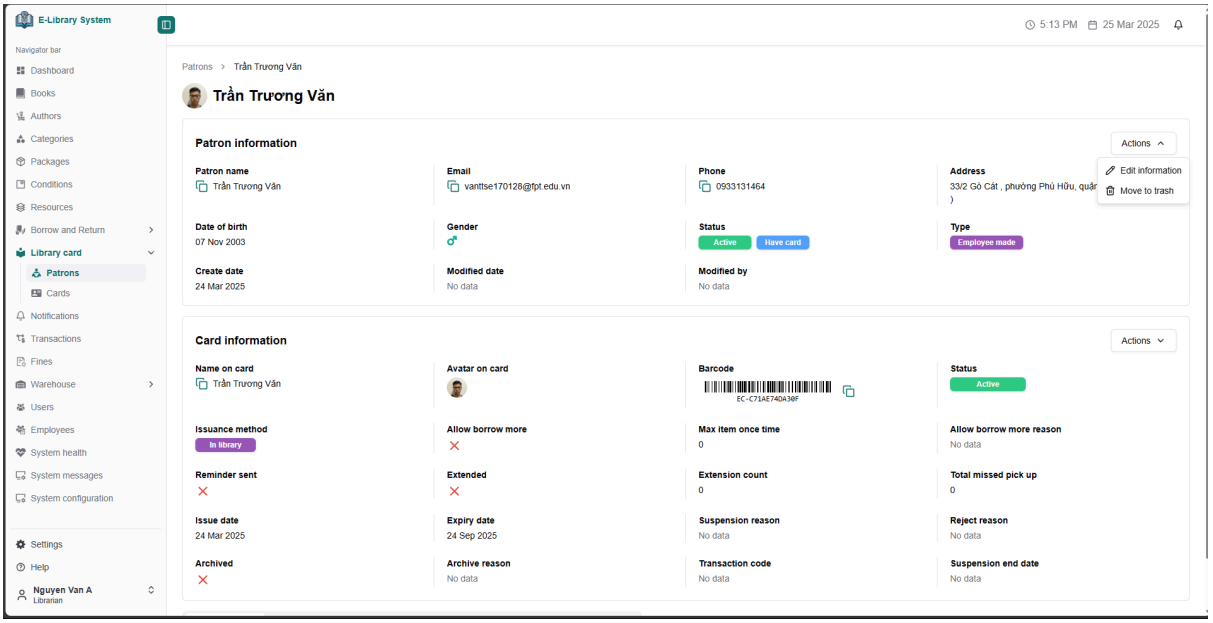


Figure 3.4.10.1: Patrons detail management

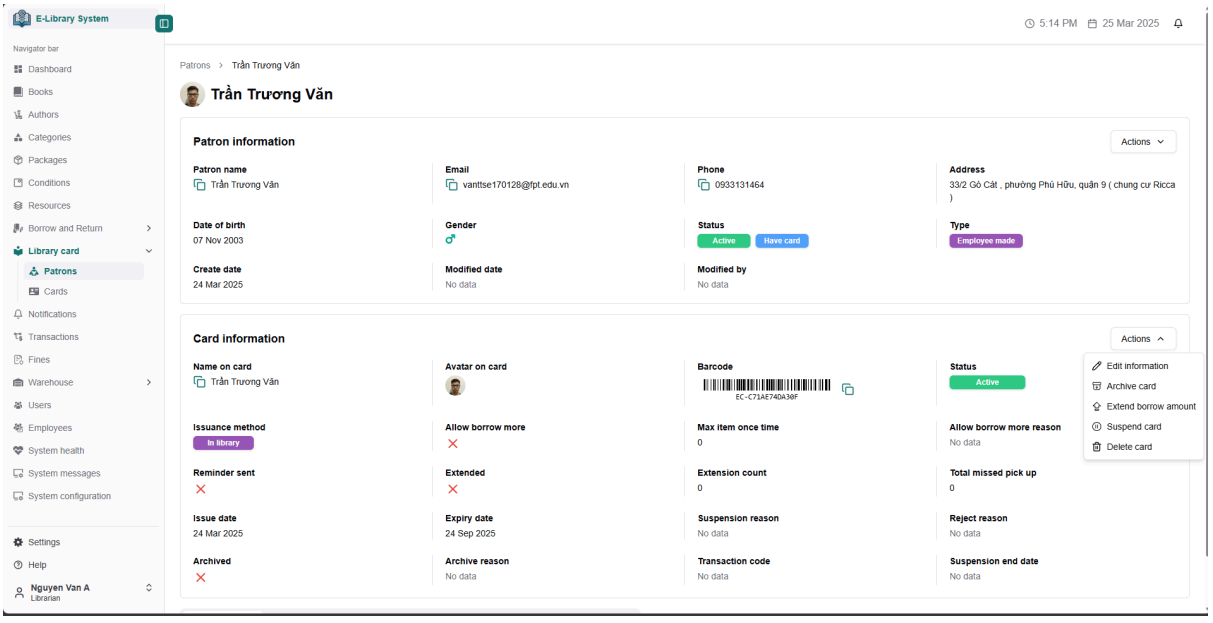


Figure 3.4.10.: Patrons detail management

3.4.11 Create new patron

- **Function Description:** This screen allows **librarians** to view, search, and manage library cards efficiently. It provides various search and filter methods
- **Function Trigger:**

    - **Select Package for Patron:**
        - **Button/Navigation:** On the "Create Patron" page, a drop down labeled "Package" is available at the top.
        - **Trigger:** Clicking the dropdown displays a list of available packages.

- **Function:** The system assigns the selected package to the patron, determining the subscription cost and duration.

❖ **Enter Patron Information**
- **Button/Navigation:** The form includes required fields for "Email," "First Name," "Last Name," "Phone," "Address," and "Date of Birth."
- **Trigger:** The librarian inputs the patron's details into the respective fields.
- **Function:** The system records the patron's personal information, ensuring all required fields are filled before proceeding.

❖ **Upload and Validate Avatar**
- **Button/Navigation:** An "Avatar" section allows uploading an image (under 10MB) with a drag-and-drop or click-to-select option.
- **Trigger:** After uploading an image, the system checks for a face and displays the result (e.g., "Failed" if no face is detected).
- **Function:** The system validates the uploaded avatar, ensuring it contains a face for use on the patron's library card, and displays a preview of the image.

❖ **Select Transaction Method**
- **Button/Navigation:** A "Transaction Method" section offers two radio buttons: "Cash" and "Digital."
- **Trigger:** Selecting a method activates the corresponding workflow.
- **Function:** The system records the chosen payment method for the patron's subscription. If "Cash" is selected, a warning appears to confirm payment; if "Digital" is selected, it proceeds to the QR payment step.

❖ **Confirm Cash Payment**
- **Button/Navigation:** When "Cash" is selected, a warning message appears with a checkbox: "I acknowledge that the patron has paid in cash."
- **Trigger:** The librarian checks the box to confirm the cash payment.
- **Function:** The system allows the creation process to proceed only after the cash payment is acknowledged, ensuring payment verification.

❖ **Initiate QR Payment for Digital Transaction**
- **Button/Navigation:** When "Digital" is selected, clicking "Continue" opens a "QR Payment" dialog.
- **Trigger:** The dialog displays the package details, a payment description, and a QR code with a 30-minutes timer.
- **Function:** The system generates a QR code for the patron to scan and make the payment via a banking app, with instructions to confirm the payment within the time limit.

❖ **Cancel Patron Creation**
- **Button/Navigation:** A "Cancel" button is available at the bottom-right of the form or QR payment dialog.
- **Trigger:** Clicking "Cancel" exits the process.
- **Function:** If canceled before clicking "Continue," the system discards all entered data without creating a patron. If canceled in the QR payment

dialog (after clicking "Continue" with "Digital" selected), the patron remains in an "Unpaid" state, and no further action is taken.

- **Continue Patron Creation (Digital Payment)**
  - **Button/Navigation:** A "Continue" button is available at the bottom-right of the form.
  - **Trigger:** Clicking "Continue" after filling required fields and selecting "Digital" as the transaction method submits the form and opens the QR payment.
  - **Function:** The system creates a new patron record in an "Unpaid" state with the provided details and package. The QR payment dialog then displays a QR code for the patron to scan and complete the payment, updating the status to "Paid" upon successful payment confirmation.

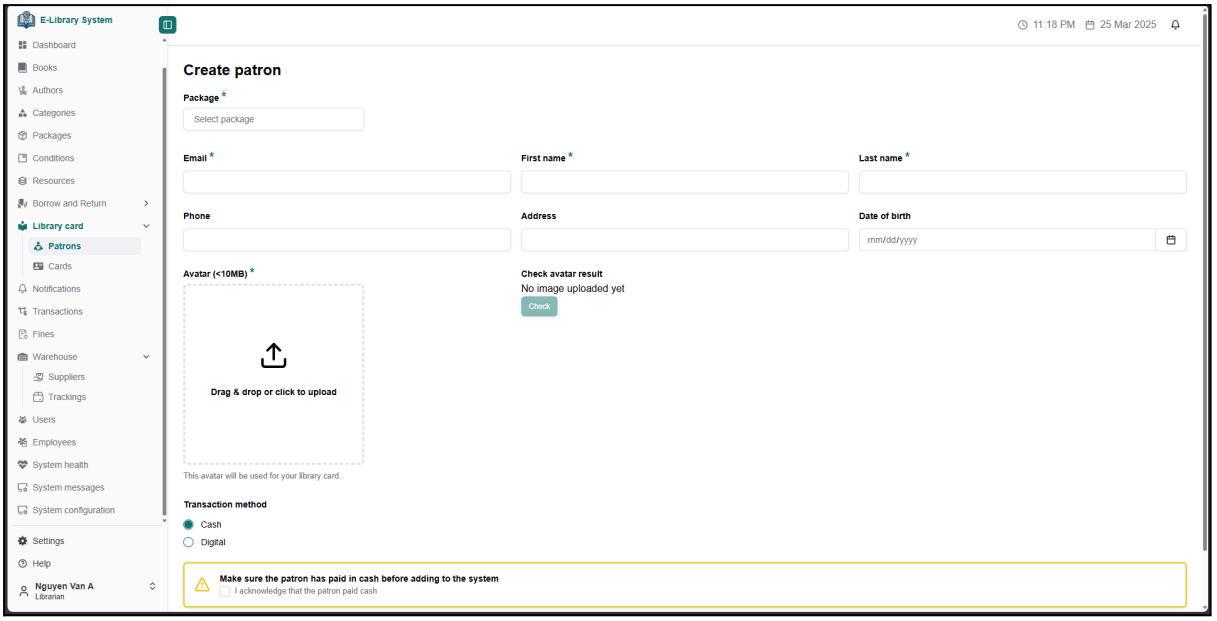


Figure 3.4.11.1: Create new patron

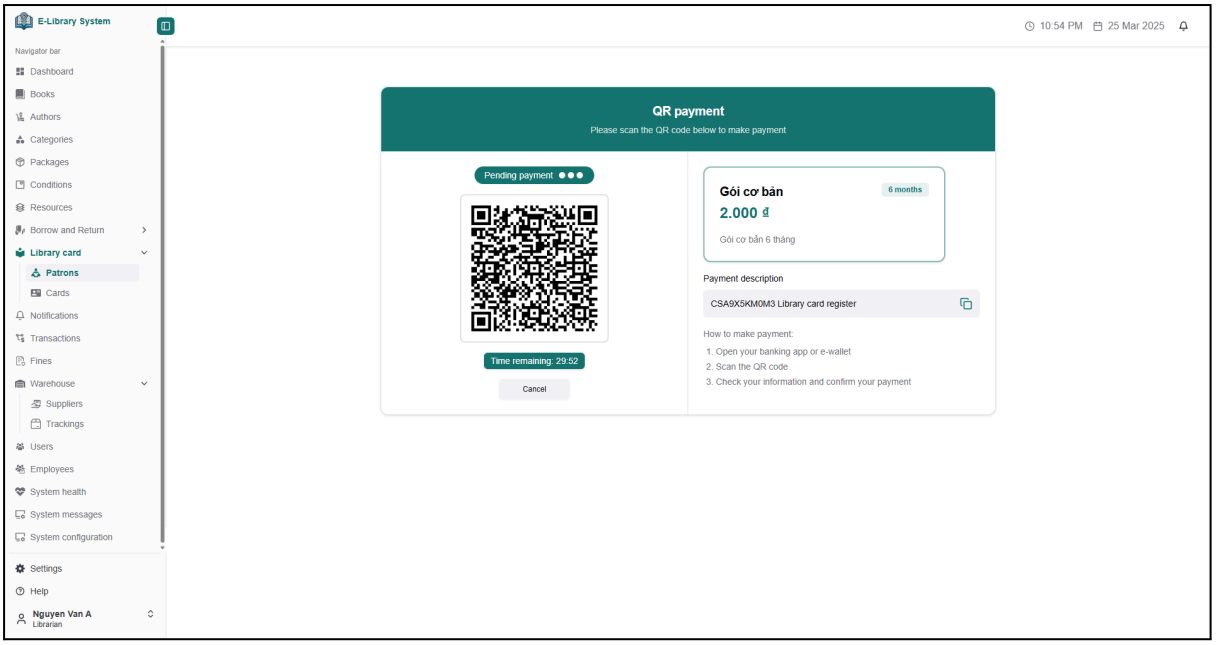


Figure 3.4.11.2: Create new patron

3.4.12 Library cards management

- **Function Description:** This screen allows **librarians** to view, search, and manage library cards efficiently. It provides various search and filter methods
- **Function Trigger:**

    - **Search cards:**
        - **Button/Navigation:** There is a search input on the top of the screen.
        - **Trigger:** The librarian enters the keyword they want to search for and then presses "Enter".
        - **Function:** The system will filter out the corresponding search results.
    - **Filter cards:**
        - **Button/Navigation:** There is a "Filter" button right next to the search input.
        - **Trigger:** When clicked, a dialog will appear. Librarians can then enter the necessary inputs or select options from dialog to refine the search. Finally, clicking "Apply" will execute the query.
        - **Function:** The system will filter out the corresponding filter results.
    - **Custom display:**
        - **Button/Navigation:** There is a “Columns” button on the top of the screen.
        - **Trigger:** When clicked, the system displays a dialog so that the librarian can choose to hide or show fields and then press "Enter"
        - **Function:** The system will hide/show the columns that the librarian has previously selected.
    - **Sort a column:**
        - **Button/Navigation:** The table headers with up/down icons indicate sortable columns.
        - **Trigger:** When clicked, the action will proceed.
        - **Function:** Clicking the header toggles the sort order between ascending and descending. The icon changes accordingly (up = ascending, down = descending).

    - **Navigate to the create patron page:**
        - **Button/Navigation:** There is a “Create card” button at the top of the screen.
        - **Trigger:** When clicked, it navigates users to the **create patron** page.
        - **Function:** The web application displays the whole content on the **create patron** page.
    - **Pagination**
        - **Button/Navigation:** Simple numbered buttons, "Next" and "Previous" links, at the bottom of the table.
        - **Trigger:** Librarian clicks on any of the pagination controls.
        - **Function:** Changes the page of the table, allowing the librarians to navigate through multiple entries.
    - **Adjust page size**
        - **Button/Navigation:** A number selection dropdown at the bottom of the screen

- **Trigger:** When clicked, different options will be displayed so that the librarian can choose the page size they want.
- **Function:** The system will display more or less rows per page of the table.

❖ **View different types of cards**

- **Button/Navigation:** There are two different tabs that the librarian can choose from located above the table.
- **Trigger:** When clicked, the system will filter the corresponding patrons.
- **Function:**
  - ➢ **"Active" Tab**: Displays active cards where isArchived= false.
  - ➢ **"Archived" Tab**: Shows deleted cards where isArchived= true.

- **Business Rules:** BR-08, BR-24

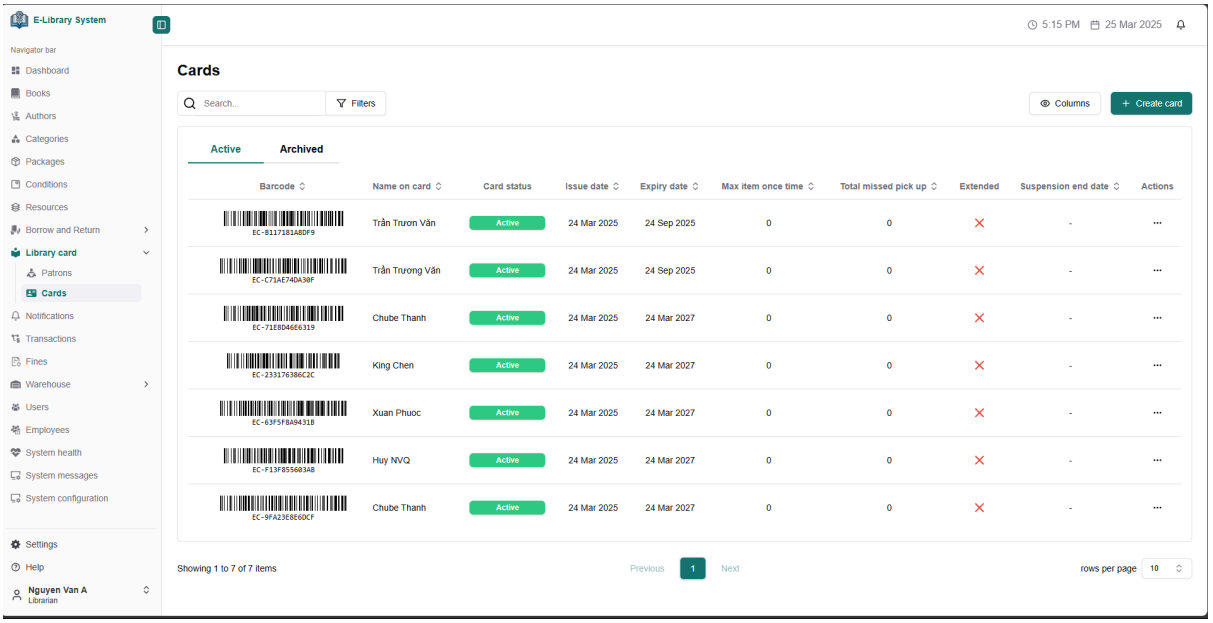


Figure 3.4.12.1: Library cards management

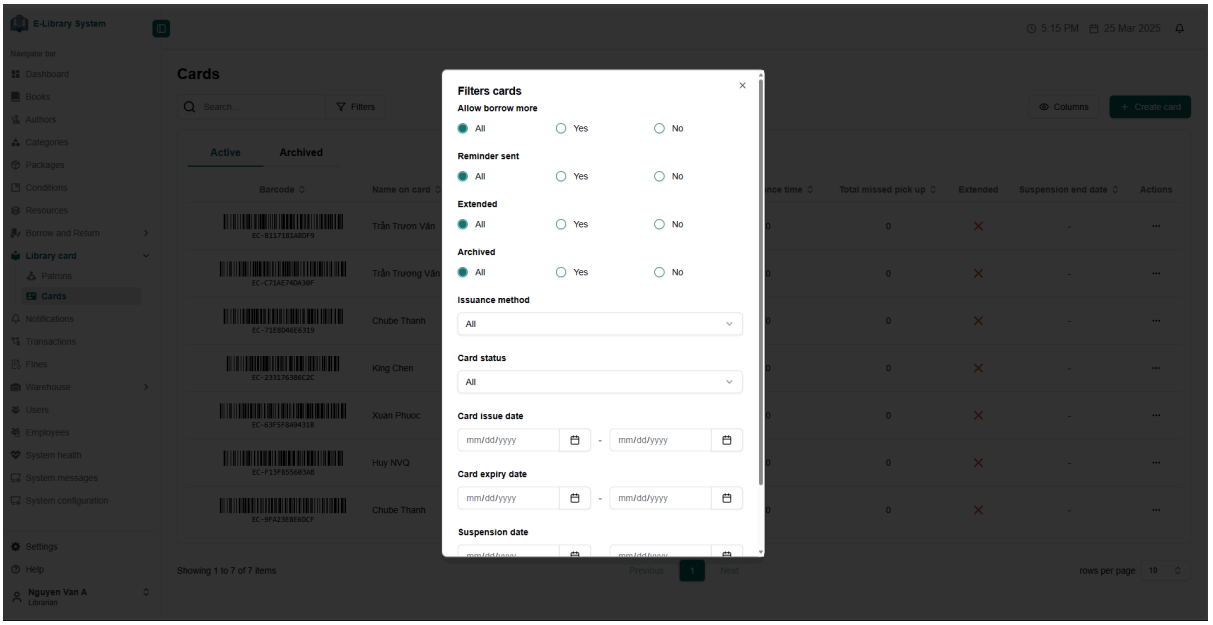


Figure 3.4.12.2: Library cards management

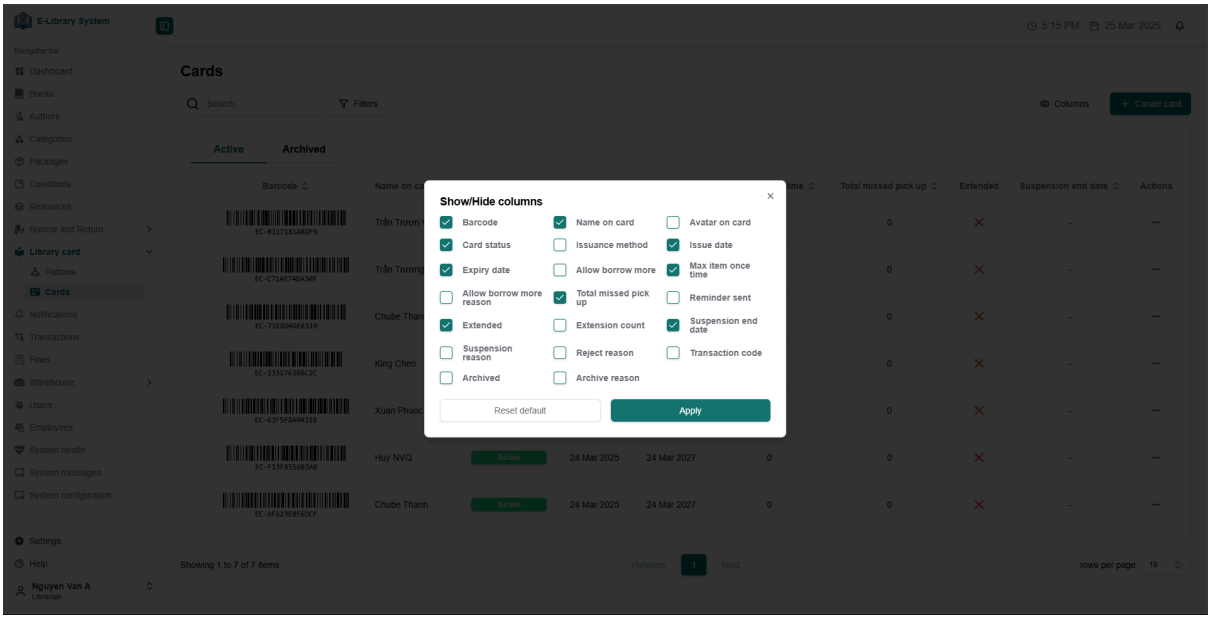


Figure 3.4.12.3: Library cards management

3.4.13 Library card detail management

- **Function Description:** This screen provides a "Card Information" section, showing the library card's details, including the cardholder's name, barcode, issuance method, status, and borrowing limits. Librarians can manage the patron and their card through the "Actions" dropdowns, which offer options like editing information, moving to trash, confirming, suspending, uns suspending, archiving, or deleting the card. This page ensures librarians can efficiently oversee and update patron records and their associated library cards.

- **Function Trigger:**

  - **Edit Library Card Information**
    - **Button/Navigation:** In the "Card Information" section, at the top-right corner, there is a dropdown menu labeled "Actions." When clicked, the librarian will see the option "Edit Information."
    - **Trigger:** Upon selecting the "Edit Information" option, the system will display a dialog where the librarian can modify the library card details.
    - **Function:** The system will update the library card information based on the changes made in the dialog.

  - **Confirm Card**
    - **Button/Navigation:** In the "Card Information" section, at the top-right corner, there is a dropdown menu labeled "Actions." When clicked, the librarian will see the option "Confirm Card" if the library card is in a "Pending" state.
    - **Trigger:** Upon selecting the "Confirm Card" option, the system will initiate the process.

- **Function:** The system will update the library card’s status to "Active," indicating that the librarian has approved the card for the patron, enabling them to use it for borrowing library items.

❖ **Extend Borrow Amount**
- **Button/Navigation:** In the "Card Information" section, at the top-right corner, there is a dropdown menu labeled "Actions." When clicked, the librarian will see the option "Extend Borrow Amount."
- **Trigger:** Upon selecting the "Unsuspend Card" option, the system will initiate the process.
- **Function:** The system will update the library card’s status by removing the suspension, setting it back to "Active," and clearing any associated suspension details (e.g., suspension reason and end date), allowing the patron to resume borrowing.

❖ **Suspend Card**
- **Button/Navigation:** In the "Card Information" section, at the top-right corner, there is a dropdown menu labeled "Actions." When clicked, the librarian will see the option "Suspend Card."
- **Trigger:**Upon selecting the "Suspend Card" option, the system will display a dialog where the librarian can specify the suspension details, such as the reason and duration.
- **Function:** The system will update the library card’s status to "Suspended," preventing the patron from borrowing items until the suspension is lifted, and will record the suspension reason and end date.

❖ **Unsuspend Card**
- **Button/Navigation:** In the "Card Information" section, at the top-right corner, there is a dropdown menu labeled "Actions." When clicked, the librarian will see the option "Unsuspend Card" if the library card is currently in a suspended state (e.g., due to a previous suspension).
- **Trigger:** Upon selecting the "Confirm Card" option, the system will initiate the process.
- **Function:** The system will update the library card’s status to "Active," indicating that the librarian has approved the card for the patron, enabling them to use it for borrowing library items.

❖ **Archived Card**
- **Button/Navigation:** In the "Card Information" section, at the top-right corner, there is a dropdown menu labeled "Actions." When clicked, the librarian will see the option "Archived Card."
- **Trigger:** Upon selecting the "Archived Card" option, the system will initiate the process.
- **Function:** The system will mark the library card as archived, removing it from the patron’s active use while retaining its record in the database for historical purposes. This is typically used when the physical card is lost, ensuring the card is no longer usable but its data is preserved.

❖ **Delete Card**

- **Button/Navigation:**In the "Card Information" section, at the top-right corner, there is a dropdown menu labeled "Actions." When clicked, the librarian will see the option "Delete Card."
- **Trigger:** Upon selecting the "Delete Card" option, the system will initiate the process.
- **Function:** The system will permanently delete the library card from the patron's record, removing all associated borrowing privileges.

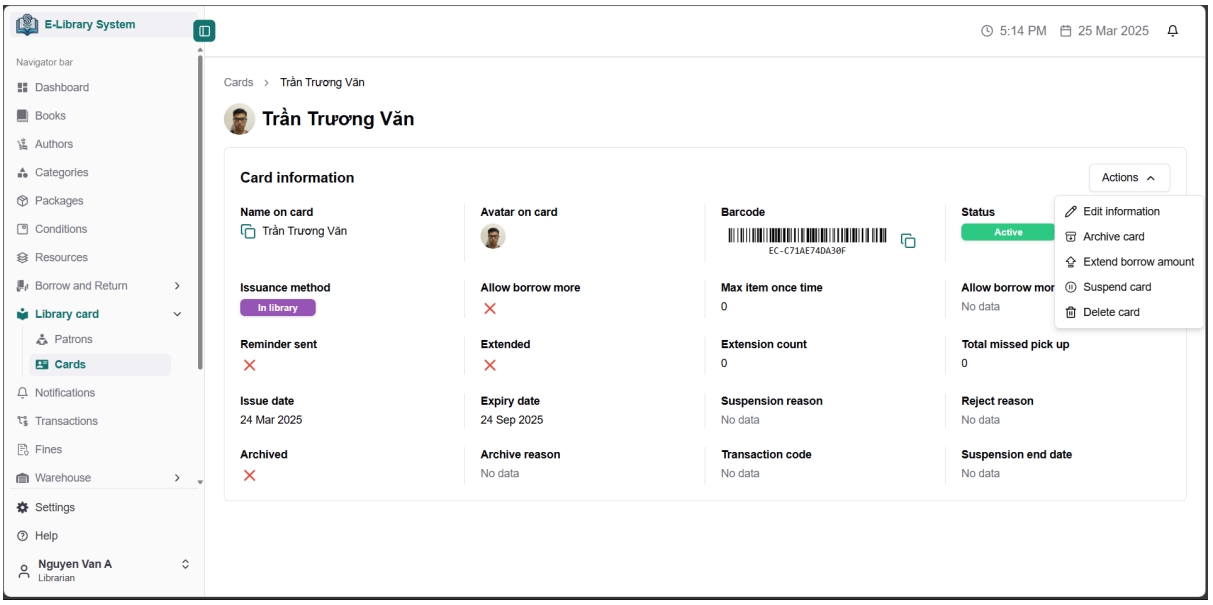


Figure 3.4.13.1: Library cards detail management

3.4.14 Notifications management

- **Function Description:** This screen allows **librarians** to view, search, and manage notifications efficiently. It provides various search and filter methods.
- **Function Trigger:**

❖ **Search notifications:**
- **Button/Navigation:** There is a search input on the top of the screen.
- **Trigger:** The librarian enters the keyword they want to search for and then presses "Enter".
- **Function:** The system will filter out the corresponding search results.

❖ **Filter notifications:**
- **Button/Navigation:** There is a "Filter" button right next to the search input.
- **Trigger:** When clicked, a dialog will appear. Librarians can then enter the necessary inputs or select options from dialog to refine the search. Finally, clicking "Apply" will execute the query.
- **Function:** The system will filter out the corresponding filter results.

❖ **Sort a column:**
- **Button/Navigation:** The table headers with up/down icons indicate sortable columns.

- **Trigger:** When clicked, the action will proceed.
- **Function:** Clicking the header toggles the sort order between ascending and descending. The icon changes accordingly (up = ascending, down = descending).

❖ **Create a notification:**
- **Button/Navigation:** There is a “Create notification” button at the top of the screen.
- **Trigger:** When clicked, the system will display a dialog, the librarian needs to fill in the required fields then press the “Create” button.
- **Function:** The system will create a new notification.

❖ **Pagination**
- **Button/Navigation:** Simple numbered buttons, "Next" and "Previous" links, at the bottom of the table.
- **Trigger:** Librarian clicks on any of the pagination controls.
- **Function:** Changes the page of the table, allowing the librarians to navigate through multiple entries.

❖ **Adjust page size**
- **Button/Navigation:** A number selection dropdown at the bottom of the screen
- **Trigger:** When clicked, different options will be displayed so that the librarian can choose the page size they want.
- **Function:** The system will display more or less rows per page of the table.

- **Business Rules:** BR-68

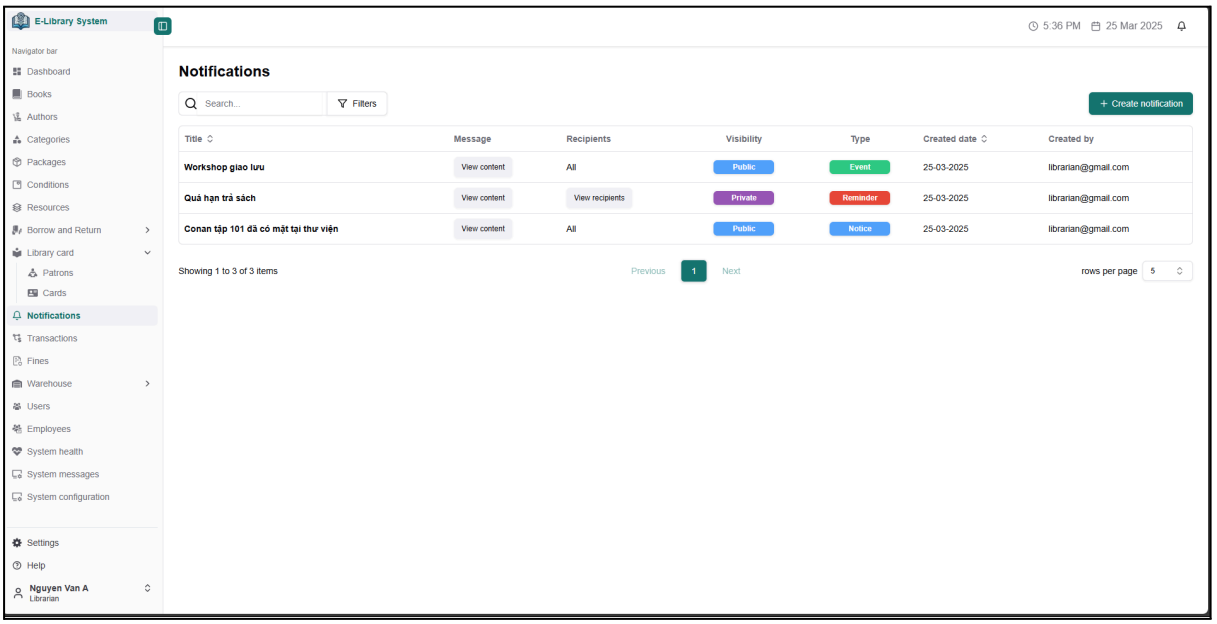


Figure 3.4.14.1: Notifications management

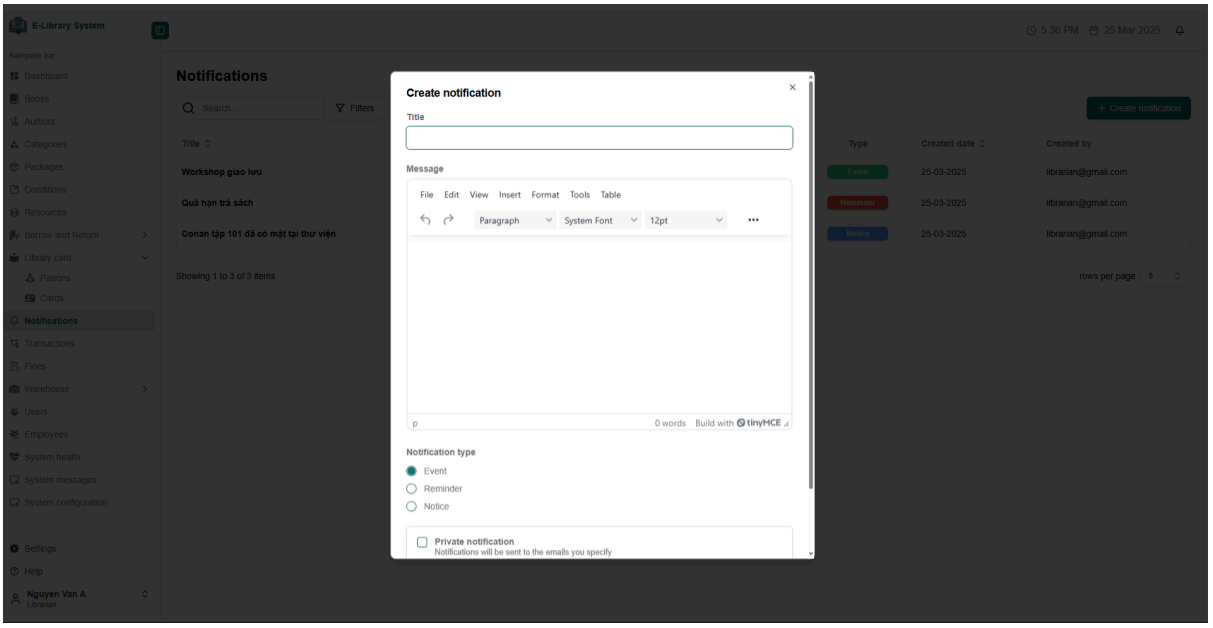


Figure 3.4.14.2: Notifications management

3.4.15 Fines management

- **Function Description:** This screen allows **librarians** to view, search, and manage fines efficiently. It provides various search and filter methods.
- **Function Trigger:**

  - **Search fines:**
    - **Button/Navigation:** There is a search input on the top of the screen.
    - **Trigger:** The librarian enters the keyword they want to search for and then presses "Enter".
    - **Function:** The system will filter out the corresponding search results.

  - **Sort a column:**
    - **Button/Navigation:** The table headers with up/down icons indicate sortable columns.
    - **Trigger:** When clicked, the action will proceed.
    - **Function:** Clicking the header toggles the sort order between ascending and descending. The icon changes accordingly (up = ascending, down = descending).
  - **Import fines:**
    - **Button/Navigation:** There is an “Import” button on the top of the screen.
    - **Trigger:** When clicked, the system presents a dialog where the librarian can upload an Excel file and fill in the required fields. Once completed, they can press "Continue" to proceed
    - **Function:** New fines will be added to the system.

  - **Create a fine:**
    - **Button/Navigation:** There is a “Create fine” button at the top of the screen.

        - **Trigger:** When clicked, the system will display a dialog, the librarian needs to fill in the required fields then press the “Create” button.
        - **Function:** The system will create a new fine.

    - **Edit fine:**
        - **Button/Navigation:** Each fine row has an "Actions" dropdown on the right side. When clicked, the librarian will see the option "Edit."
        - **Trigger:** Upon selecting the "Edit" option, the system will display a dialog pre-filled with the current details of the fine, allowing the librarian to modify fields such as the title, condition type, fixed fine amount, or fine amount per day.
        - **Function:** The system will update the fine’s details based on the changes made in the dialog and reflect the updated information in the fines table.

    - **Delete fine:**
        - **Button/Navigation:** Each fine row has an "Actions" dropdown on the right side. When clicked, the librarian will see the option "Delete."
        - **Trigger:** Upon selecting the "Delete" option, the system will initiate the process.
        - **Function:** The system will permanently remove the selected fine from the fines list, deleting its record from the database.

    - **Delete Multiple Fines:**
        - **Button/Navigation:** The librarian can select multiple fines by clicking the checkbox in each row. Once at least one fine is selected, a "Delete" button will appear at the top-right corner of the table, next to the "Import" and "Create fine” buttons
        - **Trigger:** Upon clicking the "Delete" button, the system will initiate the process
        - **Function:** The system will permanently remove the selected fines from the fines list, deleting their records from the database.

    - **Pagination**
        - **Button/Navigation:** Simple numbered buttons, "Next" and "Previous" links, at the bottom of the table.
        - **Trigger:** Librarian clicks on any of the pagination controls.
        - **Function:** Changes the page of the table, allowing the librarians to navigate through multiple entries.
    - **Adjust page size**
        - **Button/Navigation:** A number selection dropdown at the bottom of the screen
        - **Trigger:** When clicked, different options will be displayed so that the librarian can choose the page size they want.
        - **Function:** The system will display more or less rows per page of the table.
- **Business Rules:** BR-59

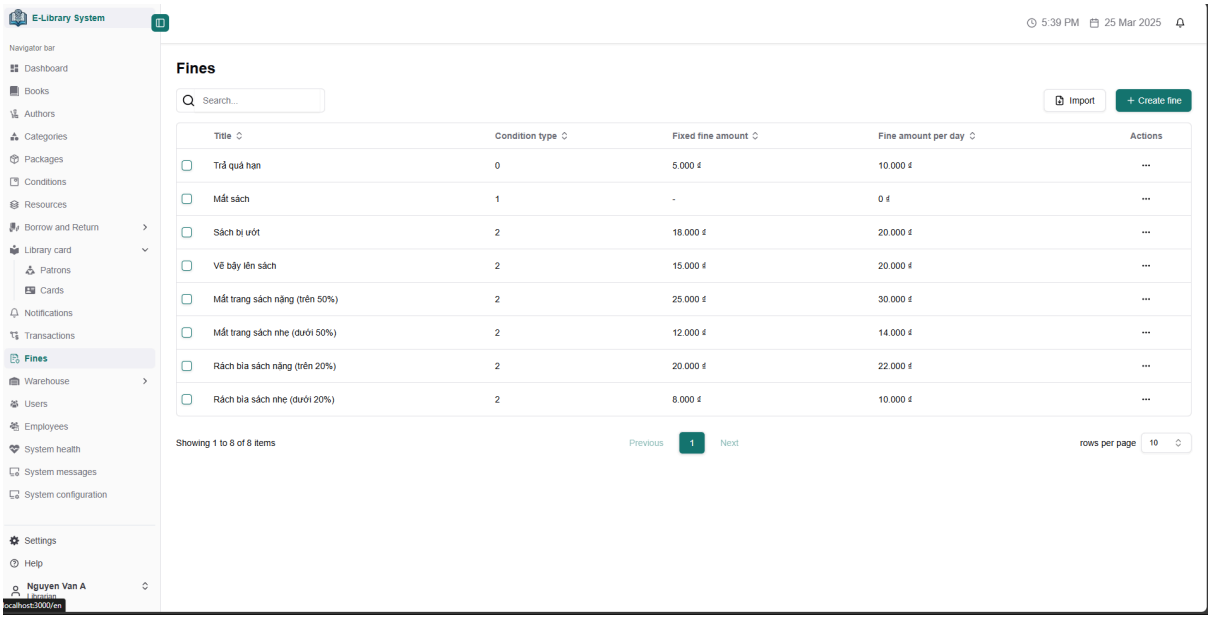


Figure 3.4.15.1: Fines management

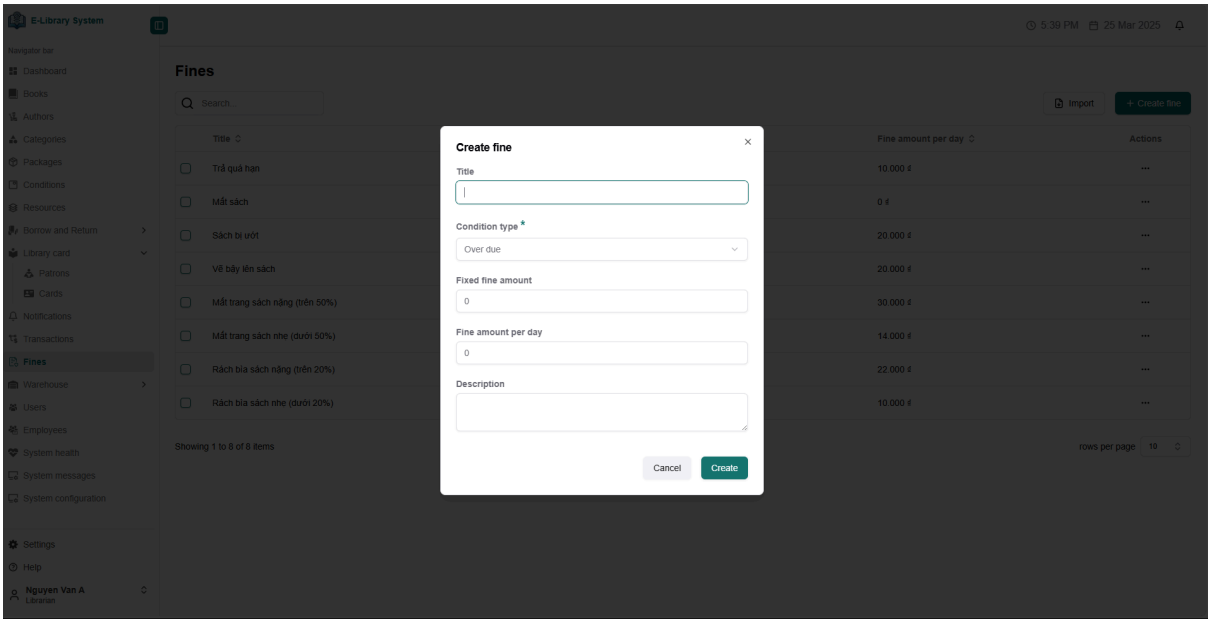


Figure 3.4.15.2: Fines management

3.4.16 Suppliers management

- **Function Description:** This screen allows **librarians** to view, search, and import/export suppliers. It also provides create, edit or delete each supplier.
- **Function Trigger:**

    - **Search suppliers:**
        - **Button/Navigation:** There is a search input on the top of the screen.

- **Trigger:** The librarian enters the keyword they wants to search for and then presses "Enter".
- **Function:** The system will filter out the corresponding search results.

❖ **Sort suppliers:**
- **Button/Navigation:** There is a search sort dropdown next to the right the search input.
- **Trigger:** When clicked, the system will display a dropdown with two options: **"A-Z"** and **"Z-A"**. The librarian can select one to execute the sorting.
- **Function:** The system will sort the suppliers based on the option selected by the librarian.

❖ **Import suppliers:**
- **Button/Navigation:** There is an “Import” button on the top of the screen.
- **Trigger:** When clicked, the system presents a dialog where the librarian can upload an Excel file and fill in the required fields. Once completed, they can press "Import" to proceed
- **Function:** New categories will be added to the system.

❖ **Export suppliers:**
- **Button/Navigation:** There is a "Export" button on the top of the screen.
- **Trigger:** Librarians can press this button to perform the data export process.
- **Function:** The system will download to the user's computer an excel file containing data of the suppliers.

❖ **Create a supplier:**
- **Button/Navigation:** There is a “Create supplier” button at the top of the screen.
- **Trigger:** When clicked, the system will display a dialog, the librarian needs to fill in the required fields then press the “Create” button.
- **Function:** The system will create a new supplier.

❖ **Edit a supplier:**
- **Button/Navigation:** The librarian can right-click on a **supplier card**, and a context menu will appear with the **"Edit"** option.
- **Trigger:** When clicked, the system will display a dialog allowing the librarian to edit the supplier’s information, then click "Save" to confirm.
- **Function:** The system will update the supplier’s information.

❖ **Delete a supplier:**
- **Button/Navigation:** The librarian can right-click on a **supplier card**, and a context menu will appear with the **"Delete"** option.
- **Trigger:** When clicked, a dialog will appear, asking the librarian to confirm to delete the supplier. After confirming, the librarian clicks the "Continue" button.
- **Function:** The system will delete the supplier from the database.

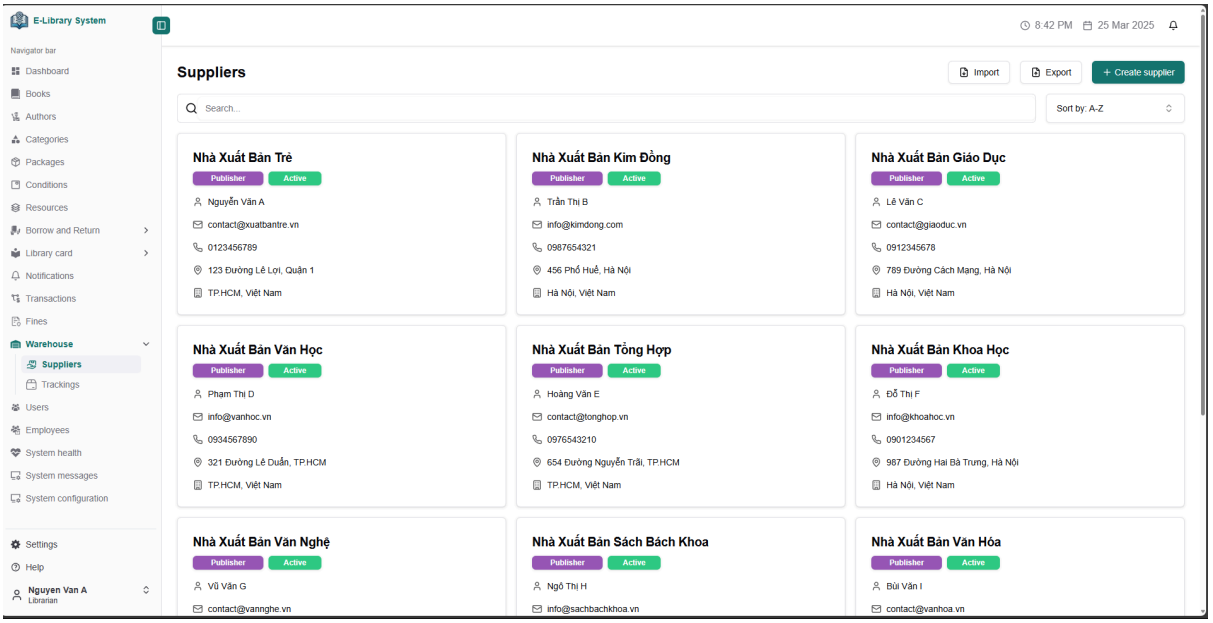


Figure 3.4.16.1: Suppliers management

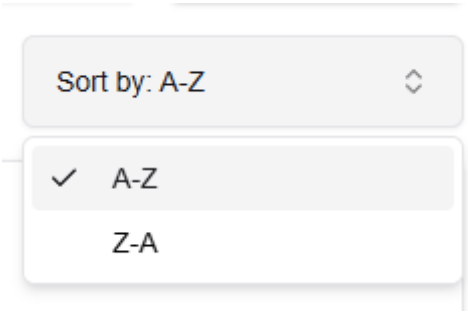


Figure 3.4.16.2: Suppliers management

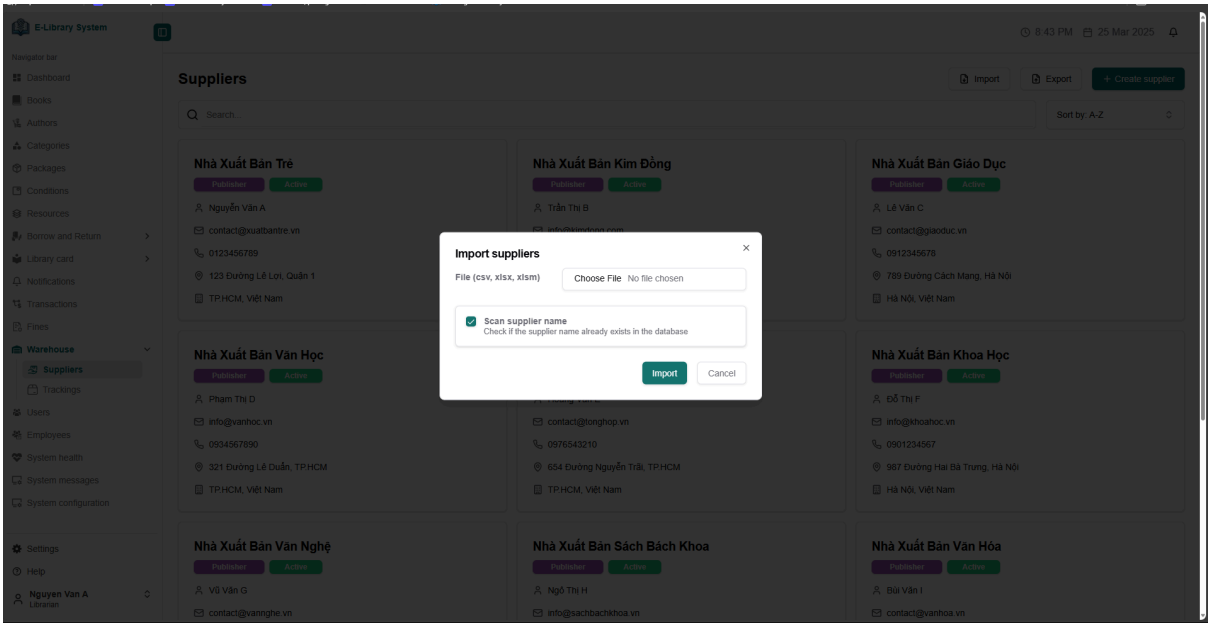


Figure 3.4.16.2: Suppliers management

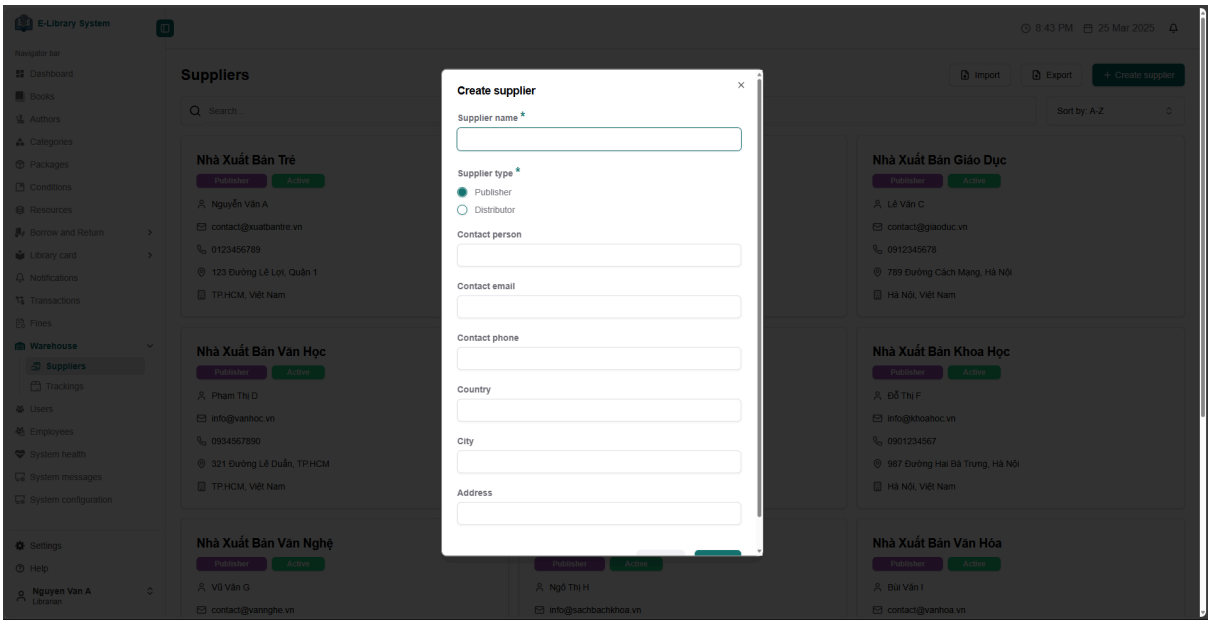


Figure 3.4.16.3: Suppliers management

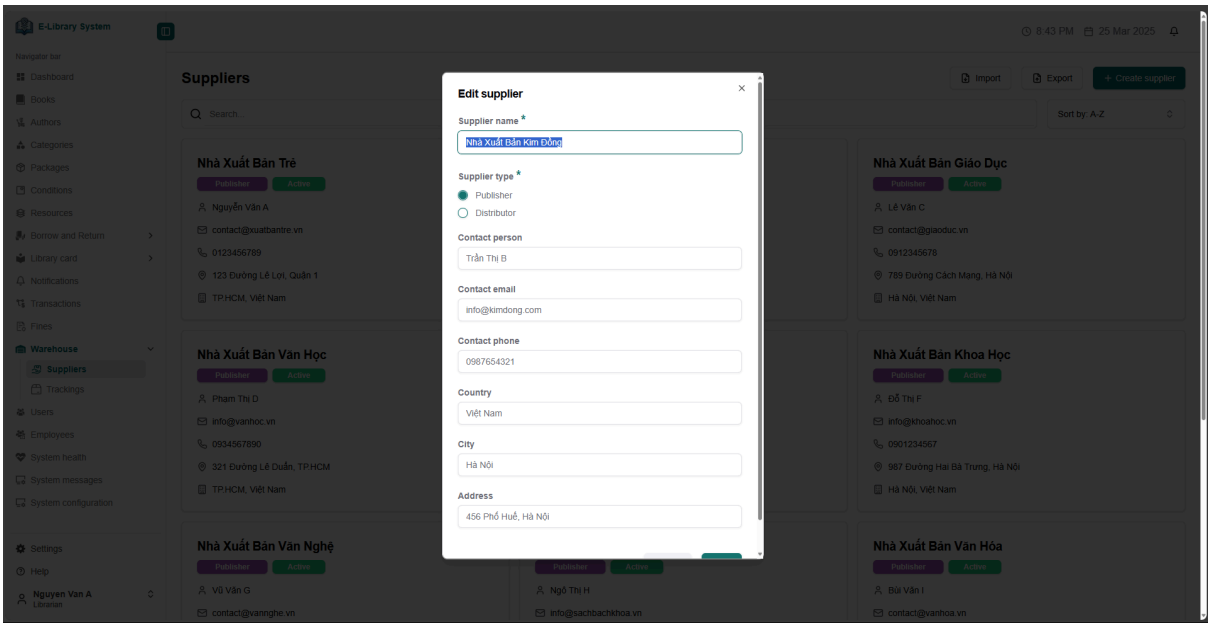


Figure 3.4.16.4: Suppliers management

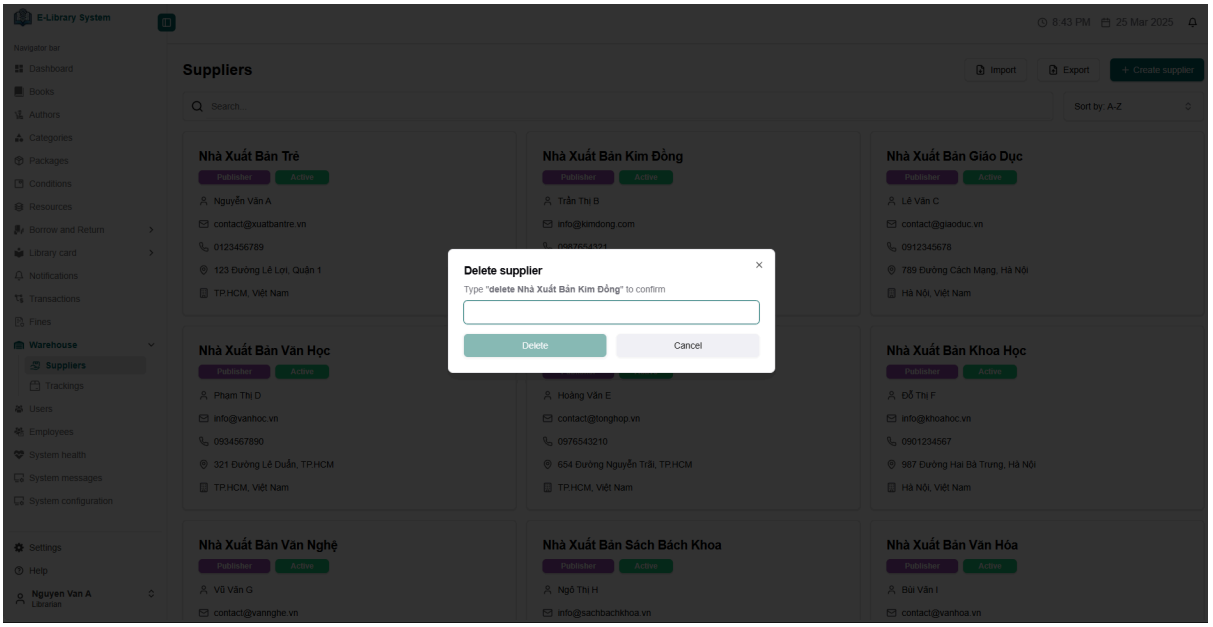


Figure 3.4.16.5: Suppliers management

3.4.17 Trackings management

- **Function Description:** This screen allows **librarians** to view, search, and manage trackins efficiently. It provides various search and filter methods

- **Function Trigger:**

  - **Search trackings:**
    - **Button/Navigation:** There is a search input on the top of the screen.
    - **Trigger:** The librarian enters the keyword they want to search for and then presses "Enter".
    - **Function:** The system will filter out the corresponding search results.
  - **Sort a column:**
    - **Button/Navigation:** The table headers with up/down icons indicate sortable columns.
    - **Trigger:** When clicked, the action will proceed.
    - **Function:** Clicking the header toggles the sort order between ascending and descending. The icon changes accordingly (up = ascending, down = descending).

  - **Navigate to Create Tracking by Import Excel Page**
    - **Button/Navigation:** On the "Warehouse Tracking" page, a button labeled "Create Tracking" is located at the top-right corner of the screen. When clicked, a dialog will appear, offering the option "Import Excel."
    - **Trigger:** Upon selecting the "Import Excel" option, the system will initiate the process.
    - **Function:** The system will navigate to the "Create Tracking by Import Excel" page and display its interface, allowing the librarian to proceed with importing tracking data via an Excel file.

❖ **Navigate to Create Tracking Manually Page**

- **Button/Navigation:** On the "Warehouse Tracking" page, a button labeled "Create Tracking" is located at the top-right corner of the screen. When clicked, a dialog will appear, offering the option "Manual."
- **Trigger:** Upon selecting the “Manual" option, the system will initiate the process.
- **Function:** The system will navigate to the "Create Tracking Manually" page and display its interface, allowing the librarian to manually input tracking details for the warehouse.

- **Business Rules:** BR-71

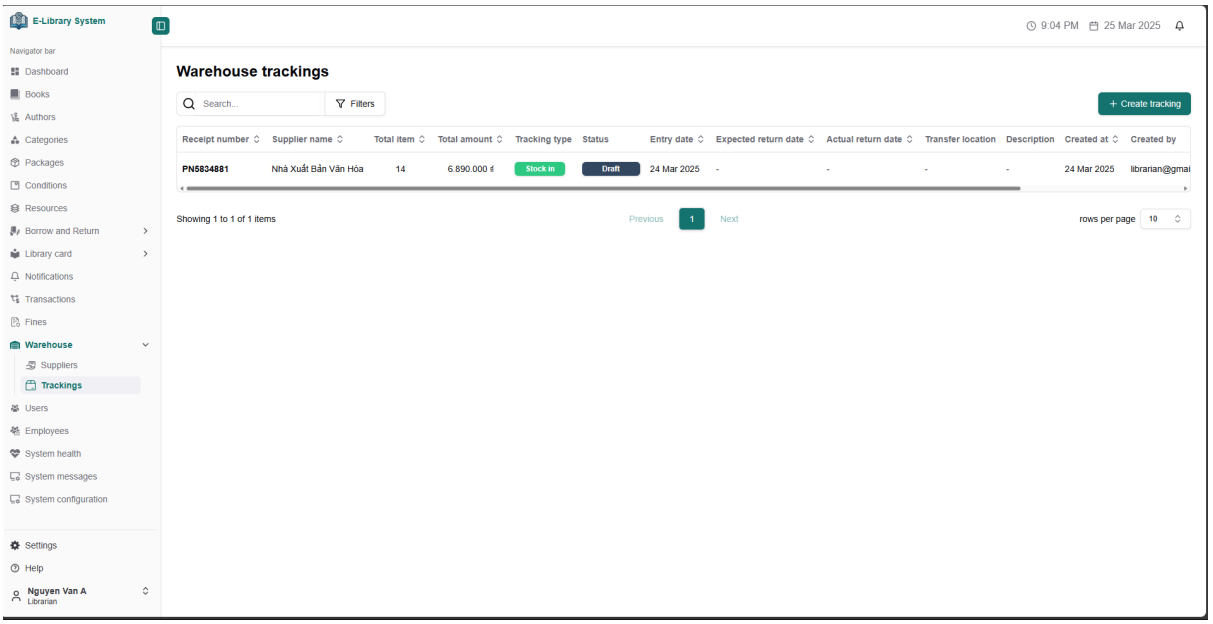


Figure 3.4.17.1: Trackings management

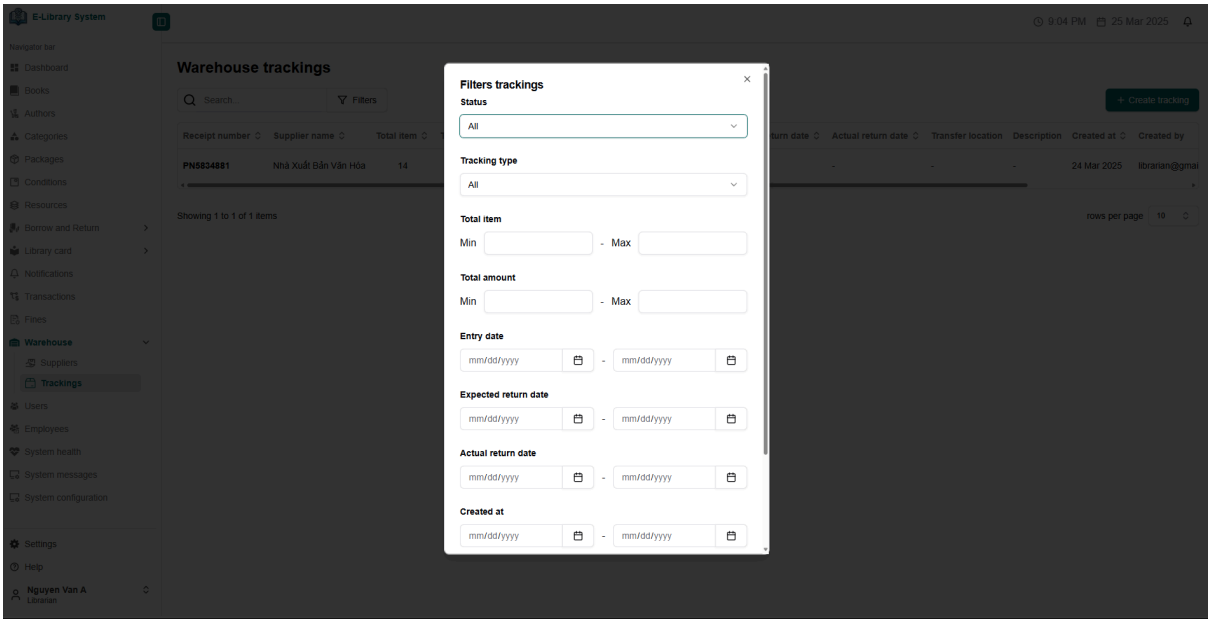


Figure 3.4.17.2: Trackings management

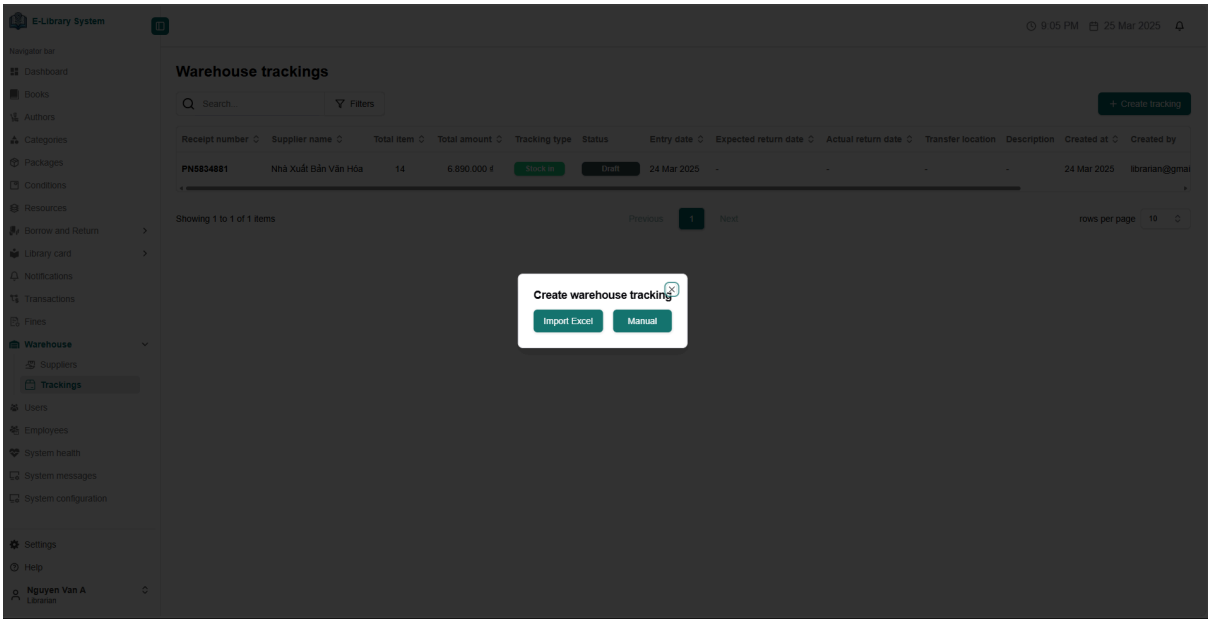


Figure 3.4.17.3: Trackings management

3.4.18 Create Tracking by Import Excel

- **Function Description:** The "Create Tracking by Import Excel" page allows librarians to create warehouse tracking entries by uploading an Excel file (CSV, XLSX, XLSM). Users can add cover images, select a supplier, specify total items, amount, entry date, tracking type, and description. Scanning options ensure data integrity by checking for duplicates. Required fields are marked, and librarians can create or cancel the entry.

- **Function Trigger:**

  - **Upload Excel File for Tracking:**
    - **Button/Navigation:** On the "Create Tracking by Import Excel" page, there is a section labeled "File (csv, xlsx, xlsm)" with a "Choose File" button.
    - **Trigger:** When the librarian clicks the "Choose File" button, a file explorer dialog will open, allowing them to select an Excel file
    - **Function:** The system will upload the selected Excel file, which contains the tracking data, and prepare it for processing once the "Create" button is clicked.

  - **Upload Cover Images for Tracking**
    - **Button/Navigation:** On the "Create Tracking by Import Excel" page, there is a section labeled "Cover Images" with a drag-and-drop area or a "Click to Select Files" option.
    - **Trigger:** When the librarian drags and drops image files (JPG, PNG, JPEG under 10MB) into the area or clicks to select files, the system will upload the images.
    - **Function:** The system will attach the uploaded cover images to the tracking entry, associating them with the items specified in the Excel file.

- **Select Supplier for Tracking**
  - **Button/Navigation:** On the "Create Tracking by Import Excel" page, there is a dropdown field labeled "Supplier" marked as required.
  - **Trigger:** When the librarian clicks the dropdown, a list of available suppliers will be displayed, and they can select one.
  - **Function:** The system will associate the selected supplier with the tracking entry, linking the warehouse tracking to the chosen supplier.

- **Specify Total Items for Tracking**
  - **Button/Navigation:** On the "Create Tracking by Import Excel" page, there is a field labeled "Total Item" marked as required.
  - **Trigger:** The librarian can manually input the total number of items in the tracking entry.
  - **Function:** The system will record the total number of items specified, ensuring it aligns with the data in the uploaded Excel file during validation.

- **Specify Total Amount for Tracking**
  - **Button/Navigation:** On the "Create Tracking by Import Excel" page, there is a field labeled "Total Amount" marked as required.
  - **Trigger:** The librarian can manually input the total monetary amount for the tracking entry.
  - **Function:** The system will record the total amount specified, associating it with the tracking entry for financial tracking purposes.

- **Specify Entry Date for Tracking:**
  - **Button/Navigation:** On the "Create Tracking by Import Excel" page, there is a field labeled "Total Amount" marked as required.
  - **Trigger:** The librarian can manually input the total monetary amount for the tracking entry.
  - **Function:** The system will record the total amount specified, associating it with the tracking entry for financial tracking purposes.
- **Select Tracking Type**
  - **Button/Navigation:** On the "Create Tracking by Import Excel" page, there is a section labeled "Tracking Type" marked as required, with radio button options: "Stock In," "Stock Out," and "Transfer."
  - **Trigger:** The librarian can select one of the radio buttons to specify the type of tracking.
  - **Function:** The system will record the selected tracking type, determining whether the entry is for stock coming in, going out, or being transferred to another location.
- **Add Description for Tracking**
  - **Button/Navigation:** On the "Create Tracking by Import Excel" page, there is a text area labeled "Description."
  - **Trigger:** The librarian can type a description in the text area to provide additional details about the tracking entry.
  - **Function:** The system will save the description as part of the tracking entry, providing context or notes for future reference.
- **Configure Scanning Options for Tracking**

- **Button/Navigation: On the "Create Tracking by Import Excel" page, there are three checkbox options under scanning settings: "Scan Item," "Scan Title," and "Scan Cover Image."**
  - **Scan Item: "Check the name of the item should not be duplicated."**
  - **Scan Title: "Add new books that haven't existed in any other book with the specific TITLE in the database yet."**
  - **Scan Cover Image: "Do not allow books to have the same image."**
- **Trigger:** The librarian can check or uncheck these options to enable or disable specific validation rules during the import process.
- **Function:** The system will apply the selected scanning rules when processing the Excel file, ensuring data integrity by checking for duplicates in item names, titles, or cover images as specified.

❖ **Create Tracking Entry**
- **Button/Navigation:** On the "Create Tracking by Import Excel" page, there is a "Create" button at the bottom-right corner of the form.
- **Trigger:** When the librarian clicks the "Create" button after filling in the required fields and uploading the Excel file, the system will initiate the process.
- **Function:** The system will process the uploaded Excel file, apply the selected scanning options, and create a new warehouse tracking entry with the provided details (supplier, total items, total amount, entry date, tracking type, description, and cover images).

❖ **Cancel Tracking Creation**
- **Button/Navigation:** On the "Create Tracking by Import Excel" page, there is a "Cancel" button at the bottom-right corner of the form, next to the "Create" button.
- **Trigger:** When the librarian clicks the "Cancel" button, the system will initiate the process.
- **Function:** The system will discard all entered data and navigate back to the "Warehouse Tracking" page without creating a new tracking entry.

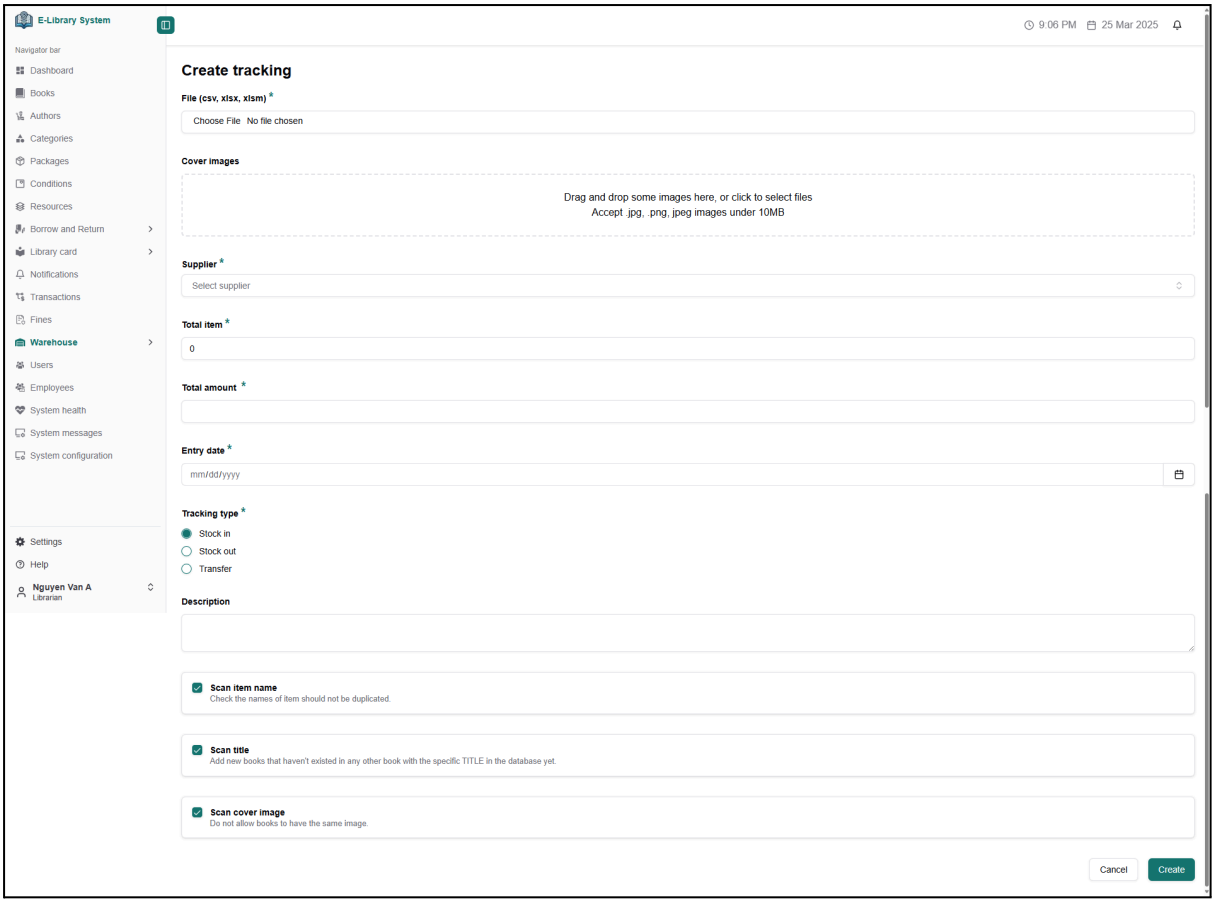


Figure 3.4.18.1: Create Trackings by Import Excel

3.4.19 Create Tracking manually

- **Function Description:** The "Create Tracking Manually" page enables librarians to manually create warehouse tracking entries by selecting a supplier, entry date, and total items/amount. It supports "New" (create new items) or "Additional" (select existing items) transactions, with a "Fast Input" option for pasting Excel data. Item details like ISBN and pricing are entered in a table. Librarians can create or cancel the entry.

- **Function Trigger:**

  - **Select Supplier and Tracking Details**
    - **Button/Navigation:** On the "Create Tracking" page, librarians can select a supplier from a dropdown, specify the entry date, and input the total items and amount (auto-calculated).
    - **Trigger:** Upon selecting or entering the required fields, the system updates the form.
    - **Function:** The system records the supplier, entry date, total items, total amount, and description for the tracking entry.

  - **Fast Input via Excel Data**
    - **Button/Navigation:** A "Fast Input" button in the "Tracking Details" section opens a dialog for pasting Excel data.

- **Trigger:** Clicking "Fast Input," pasting data, and confirming adds the data to the tracking table.
- **Function:** The system parses the pasted Excel data to quickly populate the tracking details table with item information.

❖ **Fast Input via Excel Data**
- **Button/Navigation:** A "Fast Input" button in the "Tracking Details" section opens a dialog for pasting Excel data.
- **Trigger:** Clicking "Fast Input," pasting data, and confirming adds the data to the tracking table.
- **Function:** The system parses the pasted Excel data to quickly populate the tracking details table with item information.

❖ **Add a new row to tracking table**
- **Button/Navigation:** In the "Tracking Details" section, librarians choose between "New item" (for new items) or "Additional item" (for existing items) via buttons.
- **Trigger:** Selecting "New item" allows manual item creation; selecting "Additional" item requires choosing an existing library item from a dropdown.
- **Function:** The system determines whether to create a new item or add to an existing one based on the selected transaction type.

❖ **Catalog new library item**
- **Button/Navigation:** In each “New item” row, librarians can click "Catalog" to input the item’s title and optionally catalog full details (e.g., ISBN, publisher, year).
- **Trigger:** Clicking "Catalog" opens a form to enter item details.
- **Function:** The system creates a new library item with the provided details and associates it with the tracking detail entry.

❖ **Add Existing Item**
- **Button/Navigation:** In each “Additional item” row, librarians choose an existing item from the "Select Library Item" dropdown.
- **Trigger:** Selecting an item populates its details (e.g., title, ISBN, category).
- **Function:** The system links the existing item to the tracking entry for additional stock.

❖ **Create Tracking Entry**
- **Button/Navigation:** A "Create" button at the bottom-right of the form finalizes the tracking entry.
- **Trigger:** Clicking "Create" submits the form after filling required fields.
- **Function:** The system creates a new warehouse tracking entry with all provided details and items.

❖ **Cancel Tracking Creation:**
- **Button/Navigation:** A "Cancel" button at the bottom-right of the form discards the entry.
- **Trigger:** Clicking "Cancel" exits the form.

- **Function:** The system discards all entered data and returns to the "Warehouse Tracking" page.

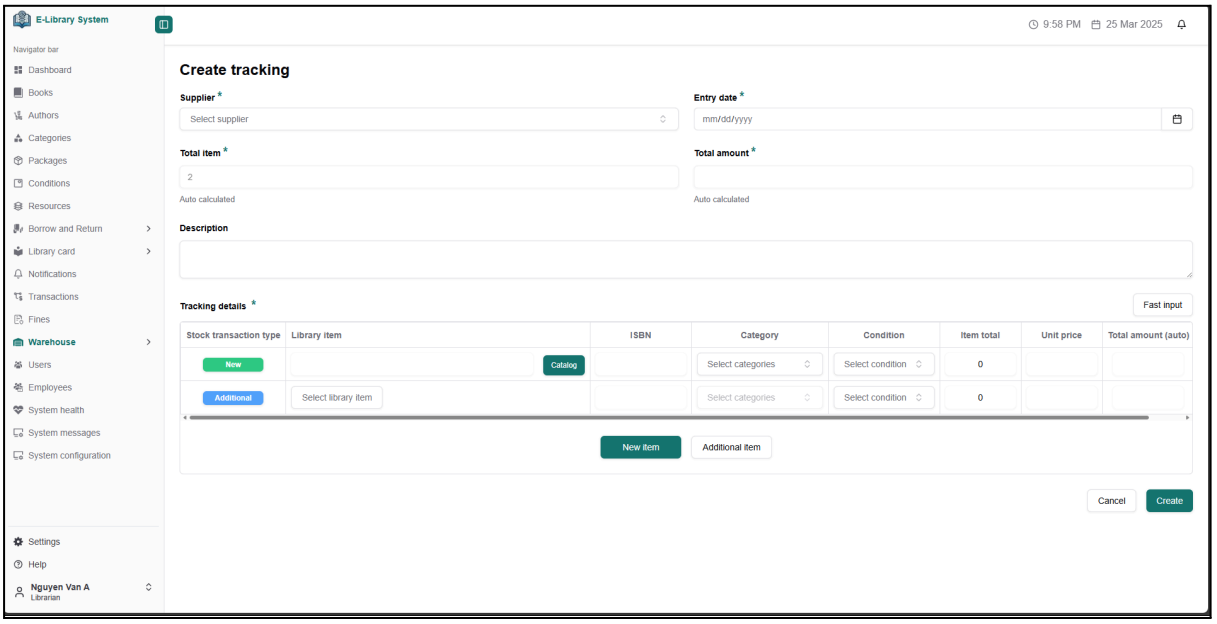


Figure 3.4.19.1: Create Trackings manually

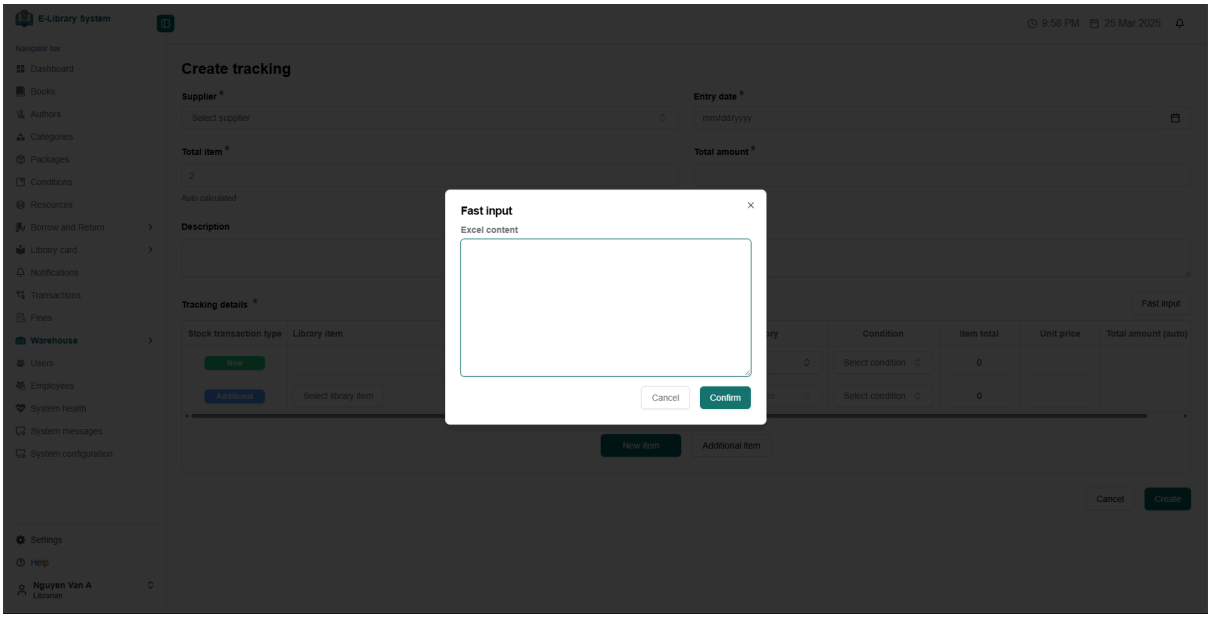


Figure 3.4.19.2: Create Trackings manually

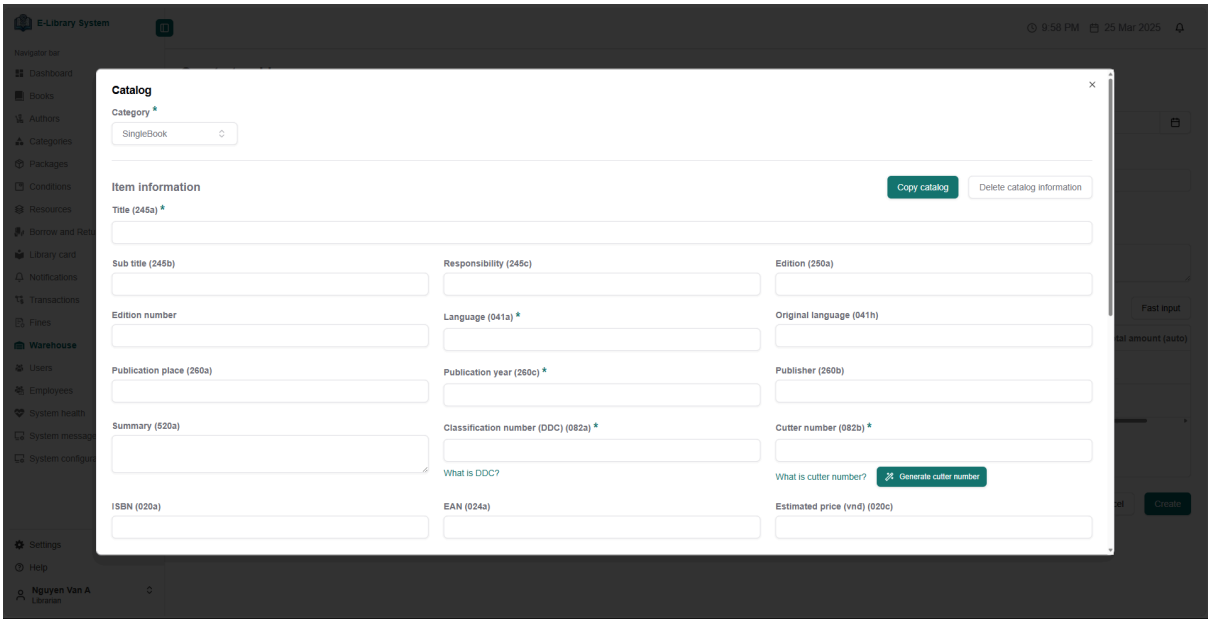


Figure 3.4.19.3: Create Trackings manually

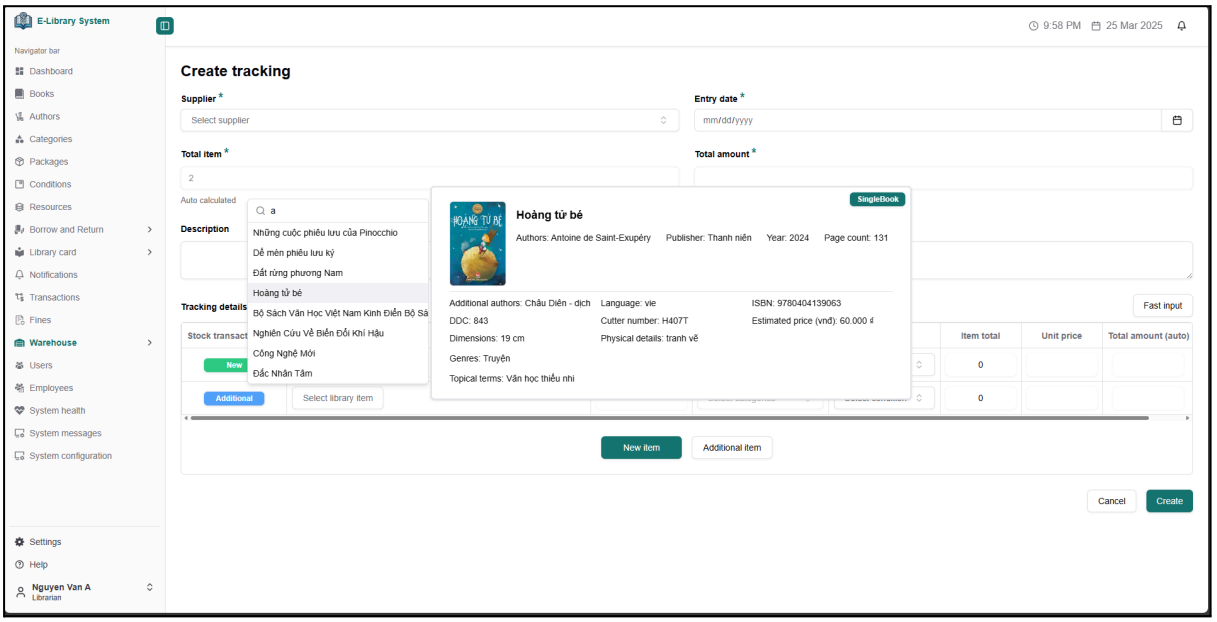


Figure 3.4.19.4: Create Trackings manually

3.4.20 Tracking detail

- **Function Description:** The "Tracking Details" page displays a specific warehouse tracking entry with details like receipt number, supplier, total items, amount, and status. It includes statistics on library items, instances, and cataloged items. The "Tracking Details" table lists items with their ISBN, category, barcode range, total items, unit price, amount, glue barcode status, condition, and creation/update

information. Librarians can edit/delete the tracking entry, add/import new details, view item details, catalog items, or confirm barcode gluing.

- **Function Trigger:**

  - **Edit Tracking Information**
    - **Button/Navigation:** On the "Tracking Details" page, an "Actions" dropdown at the top-right offers the "Edit" option.
    - **Trigger:** Selecting "Edit" opens a form with the tracking details.
    - **Function:** The system updates the tracking entry (e.g., supplier, total amount, entry date) based on the changes made.

  - **Delete Tracking Entry**
    - **Button/Navigation:** OIn the "Actions" dropdown at the top-right, the "Delete" option is available.
    - **Trigger:** Selecting "Delete" initiates the process.
    - **Function:** The system permanently removes the tracking entry from the database.

  - **View Library Item Details**
    - **Button/Navigation:** In the "Tracking Details" table, each row has an "Actions" dropdown with a "View Library Item" option.
    - **Trigger:** Selecting "View Library Item" navigates to the item's details page.
    - **Function:** The system displays the full details of the selected library item (e.g., title, ISBN, category).

  - **Edit Tracking Detail**
    - **Button/Navigation:** In the "Tracking Details" table, each row has an "Actions" dropdown with an "Edit" option.
    - **Trigger:** Selecting "Edit" opens a form with the item's details.
    - **Function:** The system updates the specific tracking detail (e.g., total items, unit price, condition) based on the changes made.

  - **Delete Tracking Detail**
    - **Button/Navigation:**In the "Tracking Details" table, each row has an "Actions" dropdown with a "Delete" option.
    - **Trigger:** Selecting "Delete" initiates the process.
    - **Function:** The system removes the specific tracking detail (item) from the tracking entry.

  - **Navigate to Catalog Page (Create Library Item Page)**
    - **Button/Navigation:** In the "Tracking Details" table, each row has a "Catalog" button if the item is uncataloged.
    - **Trigger:** Upon selecting the "Catalog" option, the system will initiate the process.
    - **Function:** The system will navigate to the "Create Library Item" page and display its interface, allowing the librarian to catalog.

  - **Glue Barcode**

- **Button/Navigation:** In the "Tracking Details" table on the "Tracking Details" page, each row has a "Glue Barcode" button under the "Has Glue Barcode" column if the barcode has not yet been glued.
- **Trigger:** Clicking the "Glue Barcode" button opens a dialog displaying unique barcodes and classification barcodes for the item, along with a "Confirm Has Glued" button.
- **Function:** The system allows the librarian to print the barcodes for gluing onto the library item. Upon clicking "Confirm Has Glued," the system updates the "Has Glue Barcode" status to "Yes" (checkmark), confirming that the barcodes have been glued to the item.

❖ **Import Tracking Details**

- **Button/Navigation:** An "Import" button at the top-right of the "Tracking Details" table is available.
- **Trigger:** Clicking "Import" opens a dialog to upload an Excel file.
- **Function:** The system processes the uploaded file to add multiple tracking details (items) to the tracking entry at once.

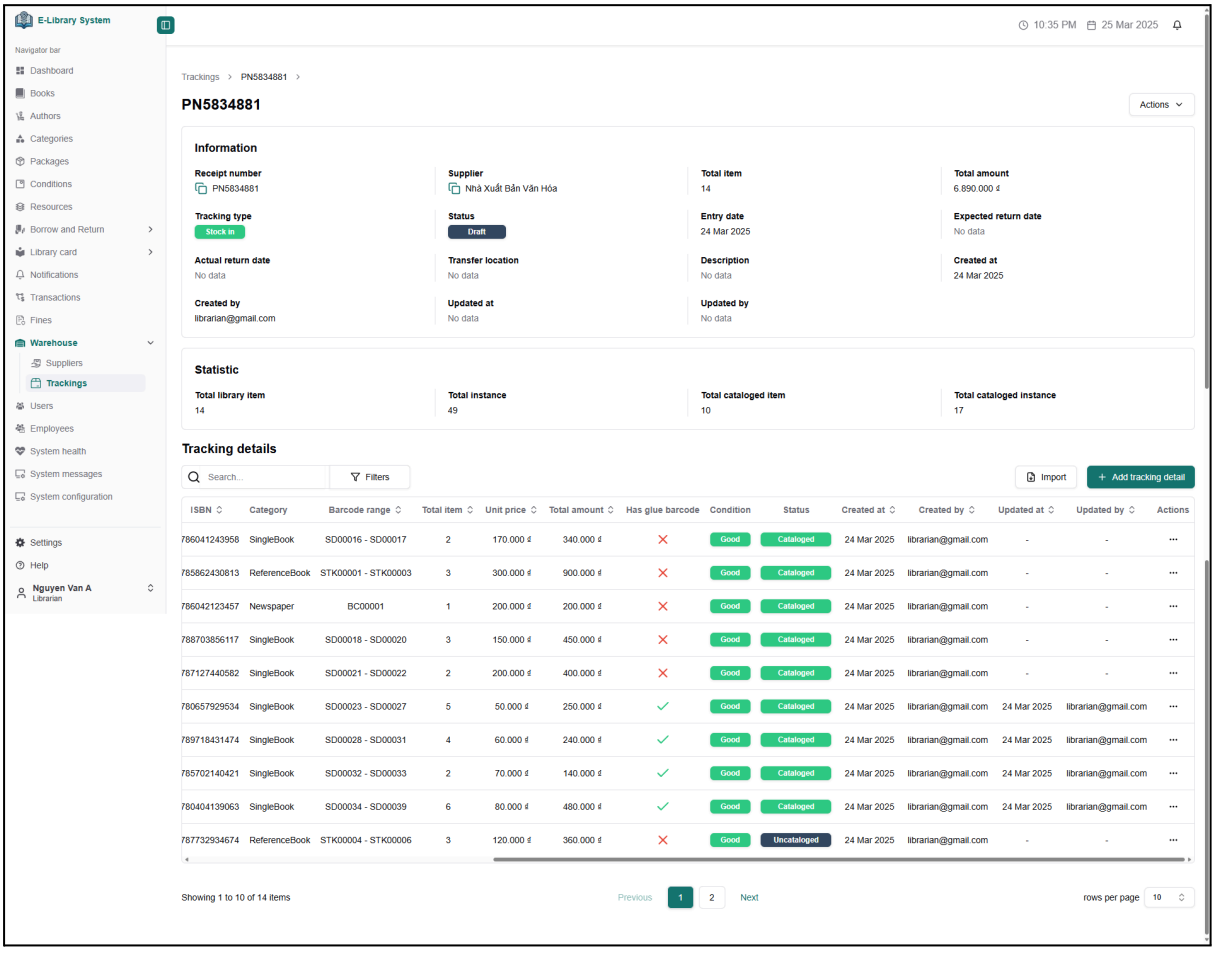


Figure 3.4.20.1: Tracking detail

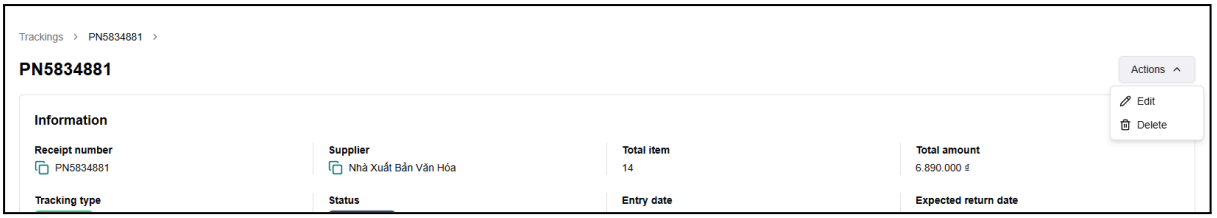


Figure 3.4.20.2: Tracking detail

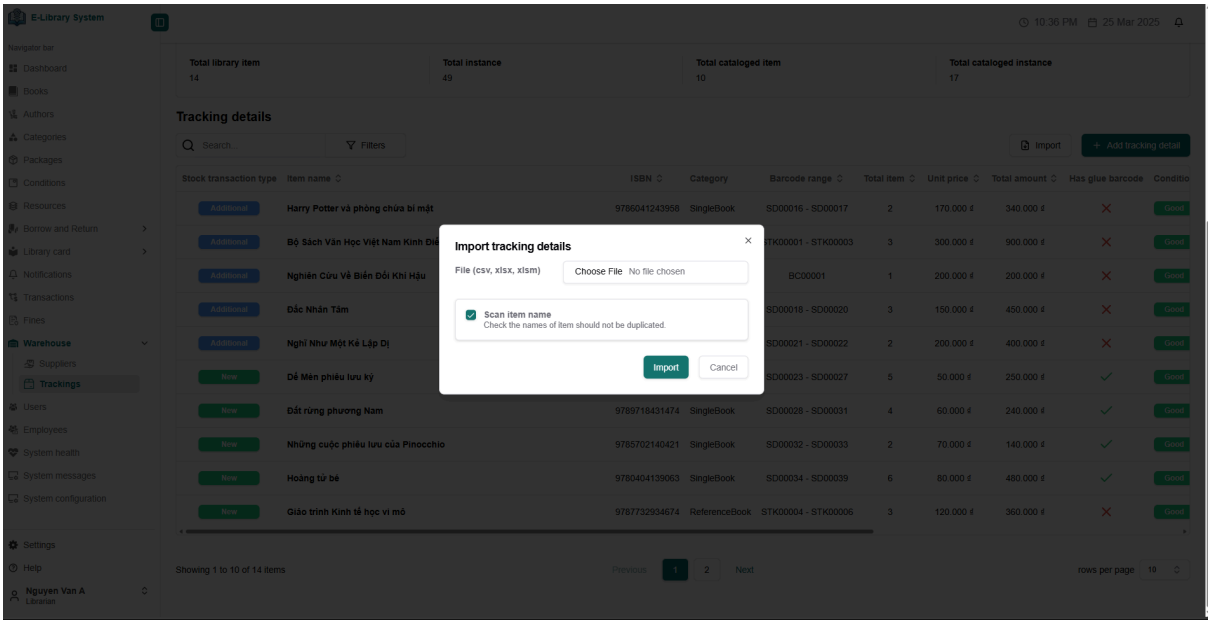


Figure 3.4.20.3: Tracking detail

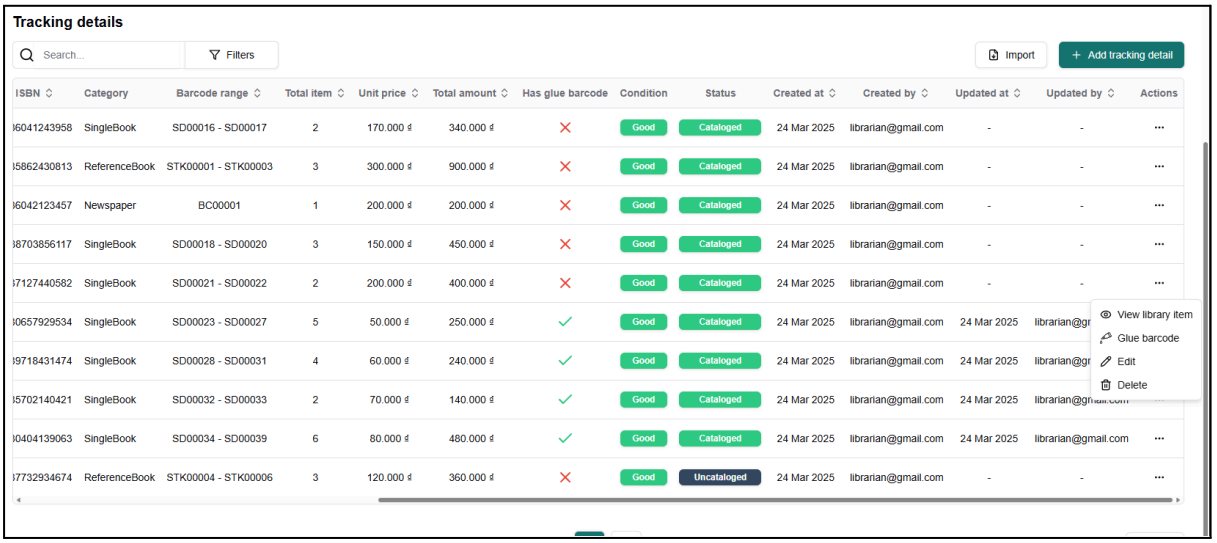


Figure 3.4.20.4: Tracking detail

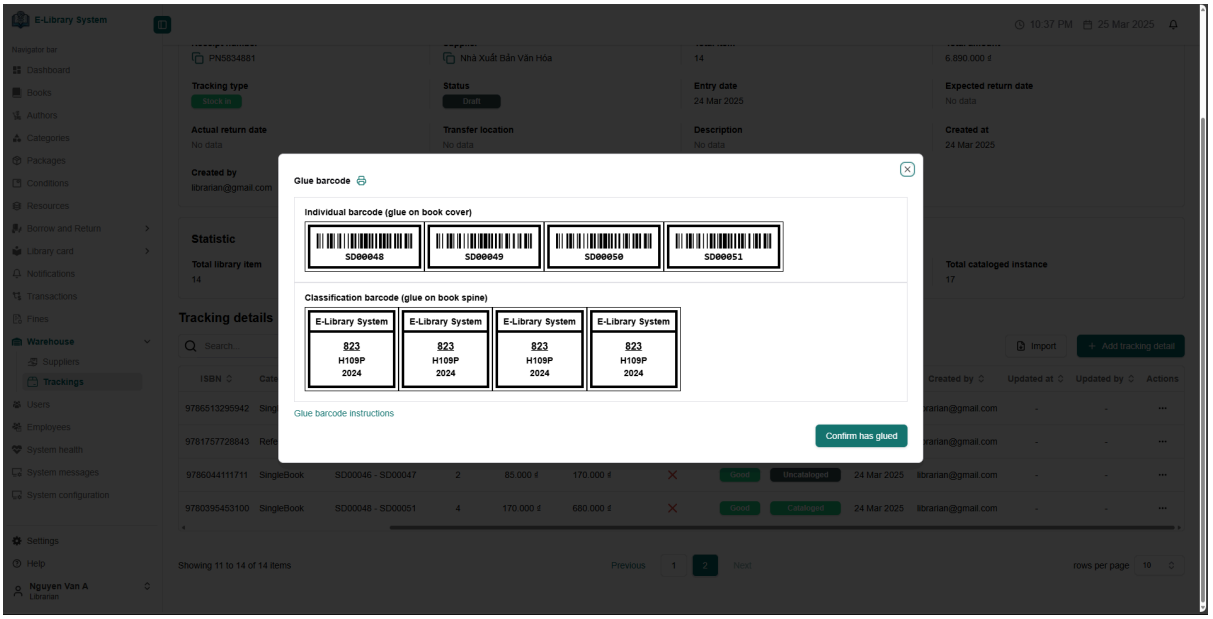


Figure 3.4.20.5: Tracking detail

3.4.21 User management

- **Function Description:** This screen allows admin to view, search, and create users. It provides various filters methods, data import/export functionalities, and actions for handling users.
- **Function Trigger:**

  - ❖ **Search users:**
    - **Button/Navigation:** There is a search input on the top of the screen.
    - **Trigger:** The admins enter the keyword they want to search for and then presses "Enter".
    - **Function:** The system will filter out the corresponding search results.
  - ❖ **Filter users:**
    - **Button/Navigation:** There is a "Filter" button right next to the search input.
    - **Trigger:** When clicked, a dialog will appear. Admins can then enter the necessary inputs or select options from dialog to refine the search. Finally, clicking "Apply" will execute the query.
    - **Function:** The system will filter out the corresponding filter results.
  - ❖ **Import users:**
    - **Button/Navigation:** There is an “Import” button on the top of the screen.
    - **Trigger:** When clicked, the system presents a dialog where the admin can upload an Excel file and fill in the required fields. The admin can press 'Import' to upload the file.
    - **Function:** New users will be added to the system.

  - ❖ **Export users:**
    - **Button/Navigation:** There is an “Export” button on the top of the screen.
    - **Trigger:** Admins can press this button to perform the data export process.

- **Function:** The system will download to the user's computer an excel file containing data of the users.

❖ **Create user:**
- **Button/Navigation:** There is a “Create user” button at the top of the screen.
- **Trigger:** When clicked, the system will display a dialog, the admin needs to fill in the required fields then press the “Continue” button.
- **Function:** The system will create a new author with the provided information.

❖ **Sort a column:**
- **Button/Navigation:** The table headers with up/down icons indicate sortable columns.
- **Trigger:** When clicked, the action will proceed.
- **Function:** Clicking the header toggles the sort order between ascending and descending. The icon changes accordingly (up = ascending, down = descending).

❖ **Edit a user:**
- **Button/Navigation:** Click on the "..." button in the last column of each row in the table. A dropdown will appear with the "Edit user" option.
- **Trigger:** When clicked, the system will display a dialog allowing the librarian to edit the user's information, then click "Save" to confirm.
- **Function:** The system will update the user’s information.

❖ **Move to trash a user:**
- **Button/Navigation:** At the “Active” tab of the table. Click on the "..." button in the last column of each row in the table. A dropdown will appear with the Move to trash" option.
- **Trigger:** When clicked, a dialog will appear, asking the admin to confirm moving the user to the trash. After confirming, the librarian clicks the "Continue" button.
- **Function:** The system will update the **isDeleted** field of the selected user to **true**.

❖ **Restore a user:**
- **Button/Navigation:** At the “Deleted” tab of the table. Click on the "..." button in the last column of each row in the table. A dropdown will appear with the “Restore" option.
- **Trigger:** When clicked, the action will proceed.
- **Function:** The system will update the **isDeleted** field of the selected user to **false**.

❖ **Delete permanently a user:**
- **Button/Navigation:** At the “Deleted” tab of the table. Click on the "..." button in the last column of each row in the table. A dropdown will appear with the **"Delete permanently"** option.

- **Trigger:** When clicked, a dialog will appear, asking the admin to confirm to delete permanently the user. After confirming, the admin clicks the "Continue" button.
- **Function:** The system will permanently delete the user from the database.

❖ **Action with users:**
- **Button/Navigation:** A “Actions” button at the top of the screen will be clickable when admins select at least a row on the table.
- **Trigger:** When clicked, the system will display options such as "Move to Trash", "Restore", "Permanently Delete"
- **Function:** The system performs the selected action on the chosen users.

❖ **Pagination**
- **Button/Navigation:** Simple numbered buttons, "Next" and "Previous" links, at the bottom of the table.
- **Trigger:** Admin clicks on any of the pagination controls.
- **Function:** Changes the page of the table, allowing the admins to navigate through multiple entries.

❖ **Adjust page size**
- **Button/Navigation:** A number selection dropdown at the bottom of the screen
- **Trigger:** When clicked, different options will be displayed so that the admin can choose the page size they want.
- **Function:** The system will display more or less rows per page of the table.

❖ **View different types of users**
- **Button/Navigation:** There are two different tabs that the admin can choose from located above the table.
- **Trigger:** When clicked, the system will filter the corresponding users.
- **Function:**
  - **"Active" Tab**: Displays active users where isDeleted = false.
  - **"Deleted" Tab**: Shows deleted users where isDeleted = true.

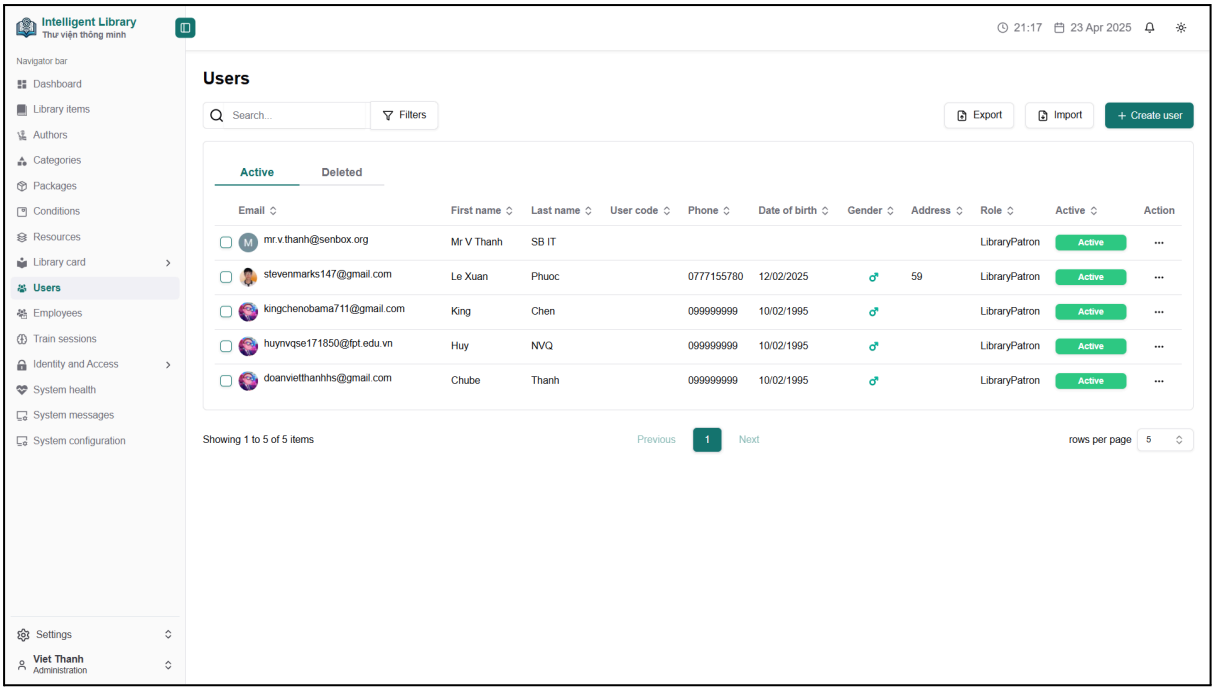


Figure 3.4.21.1: User management

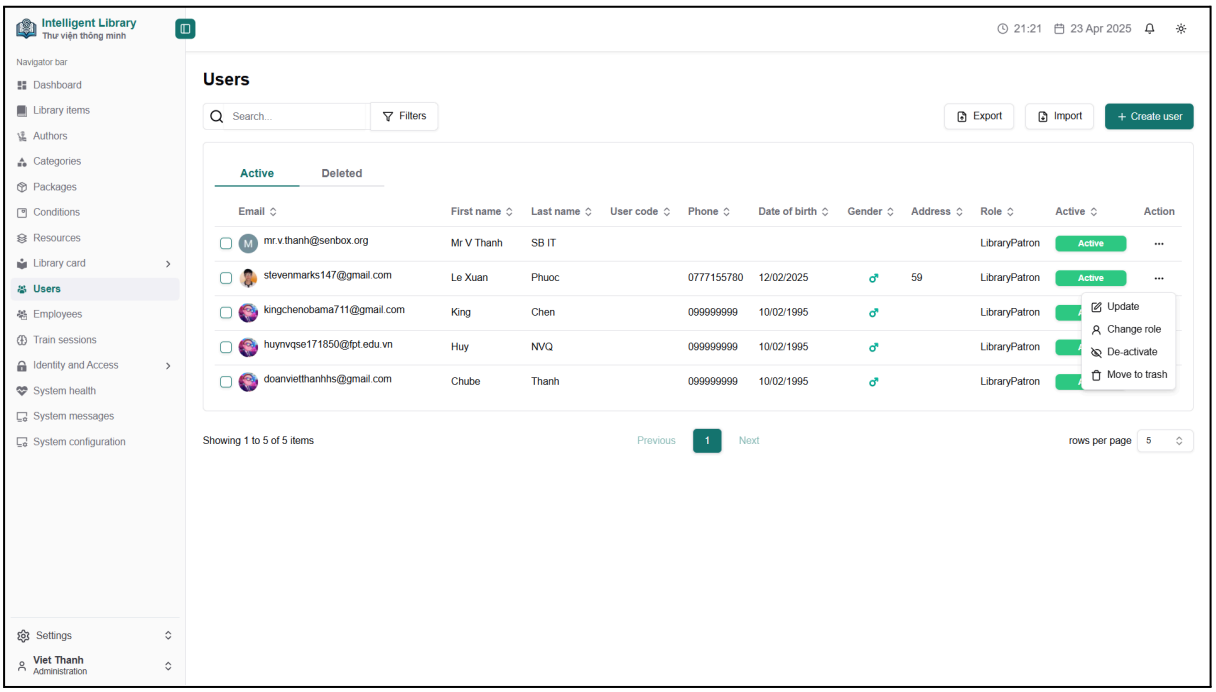


Figure 3.4.21.2: User management

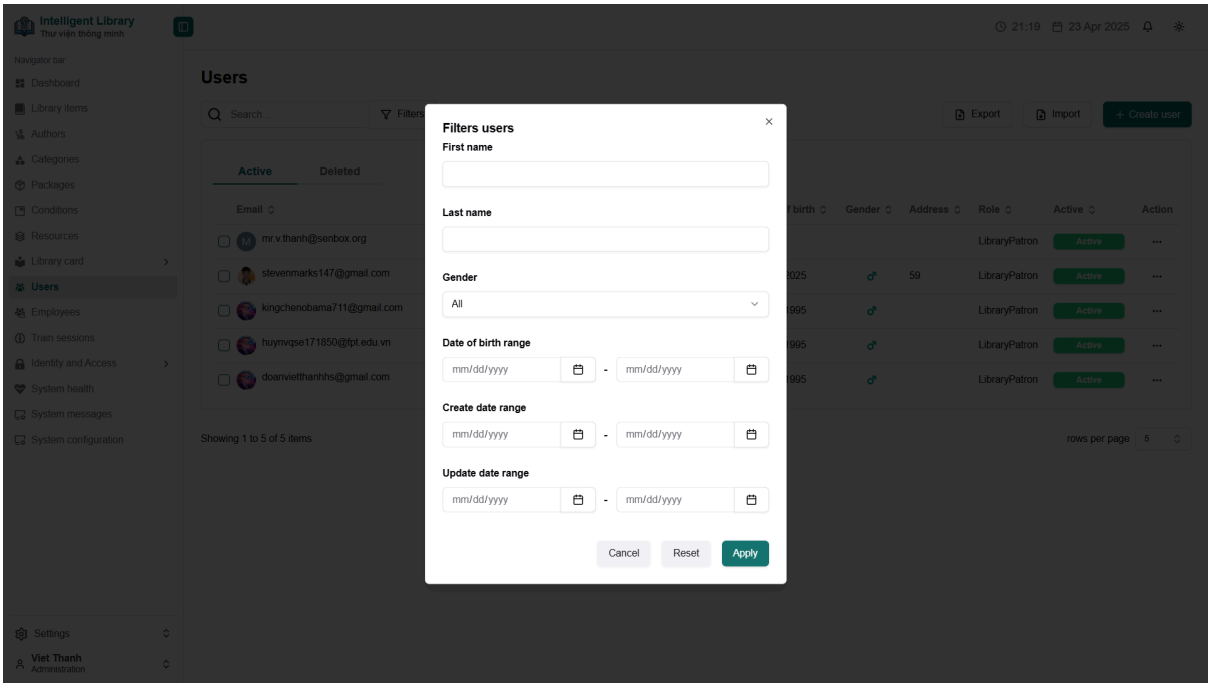


Figure 3.4.21.3: User management

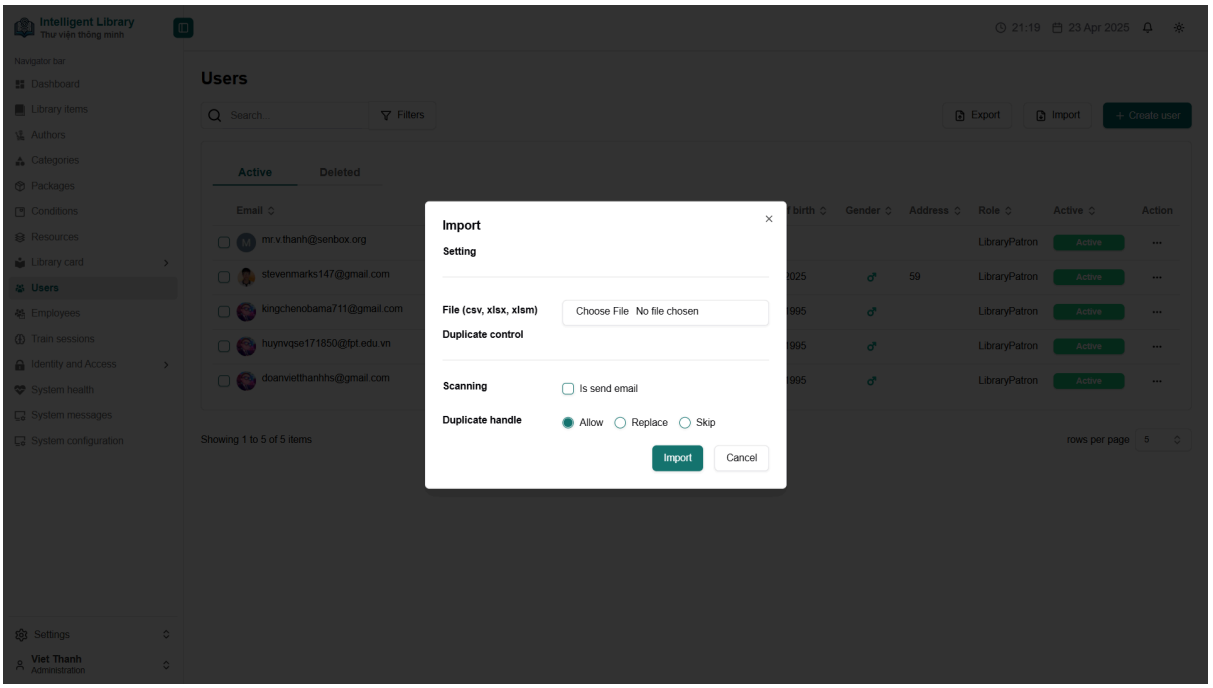


Figure 3.4.21.4: User management

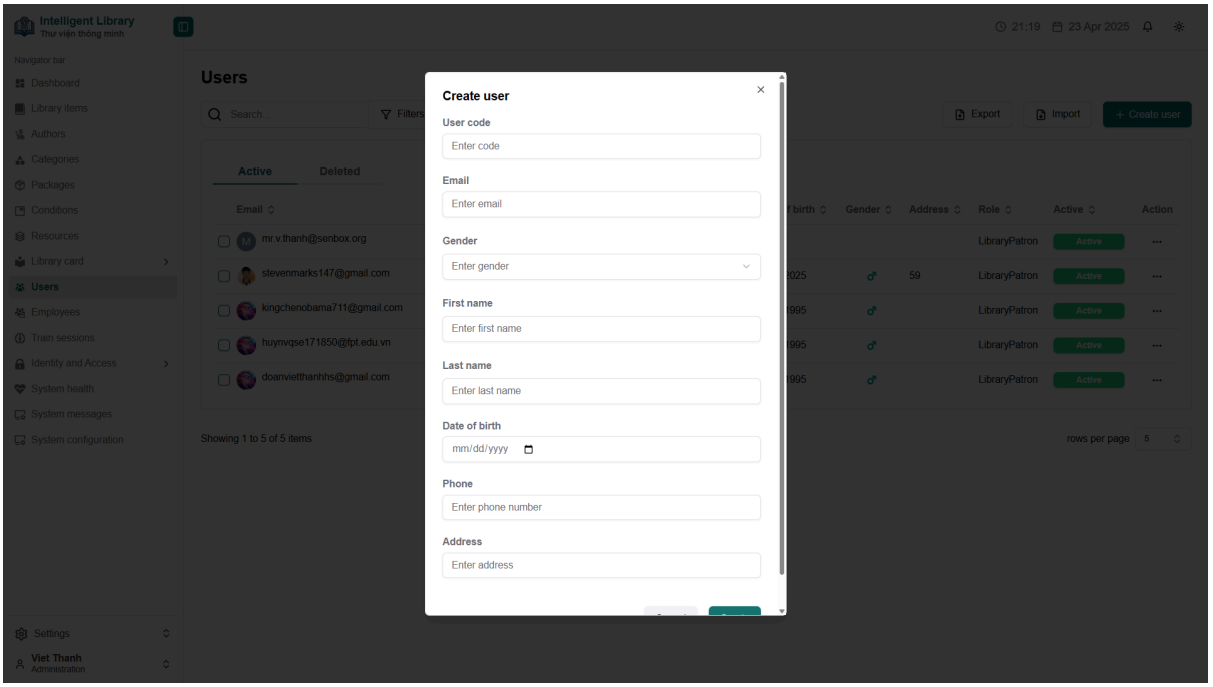


Figure 3.4.21.5: User management

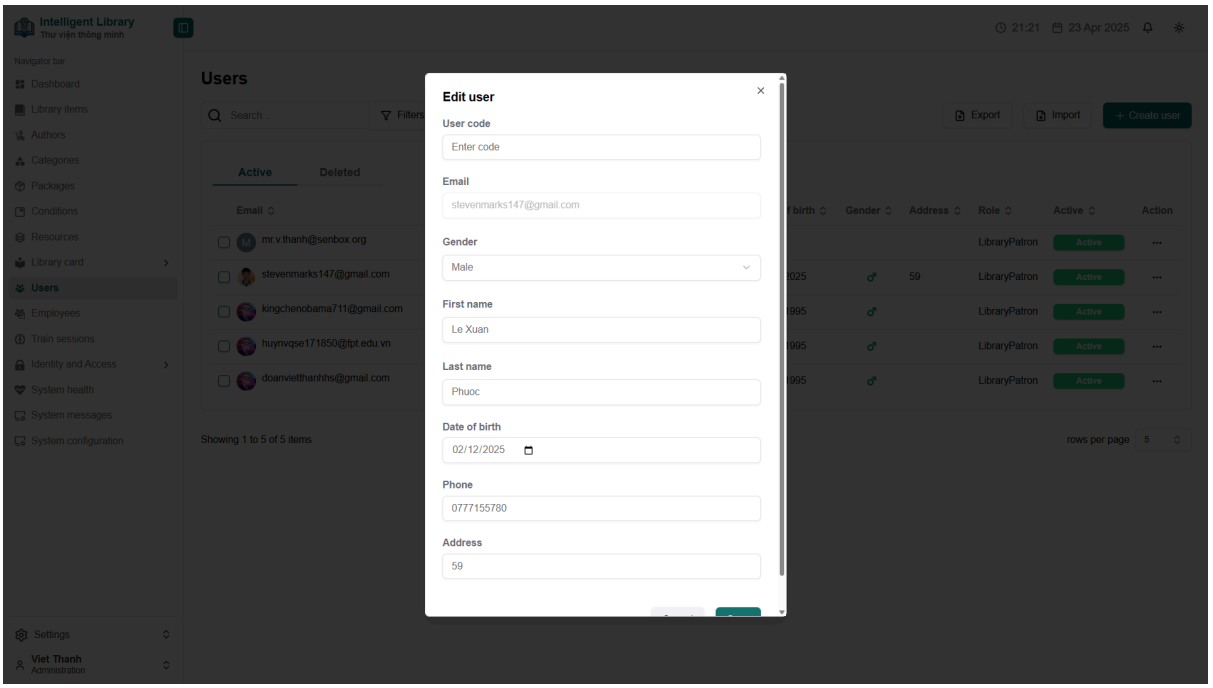


Figure 3.4.21.6: User management

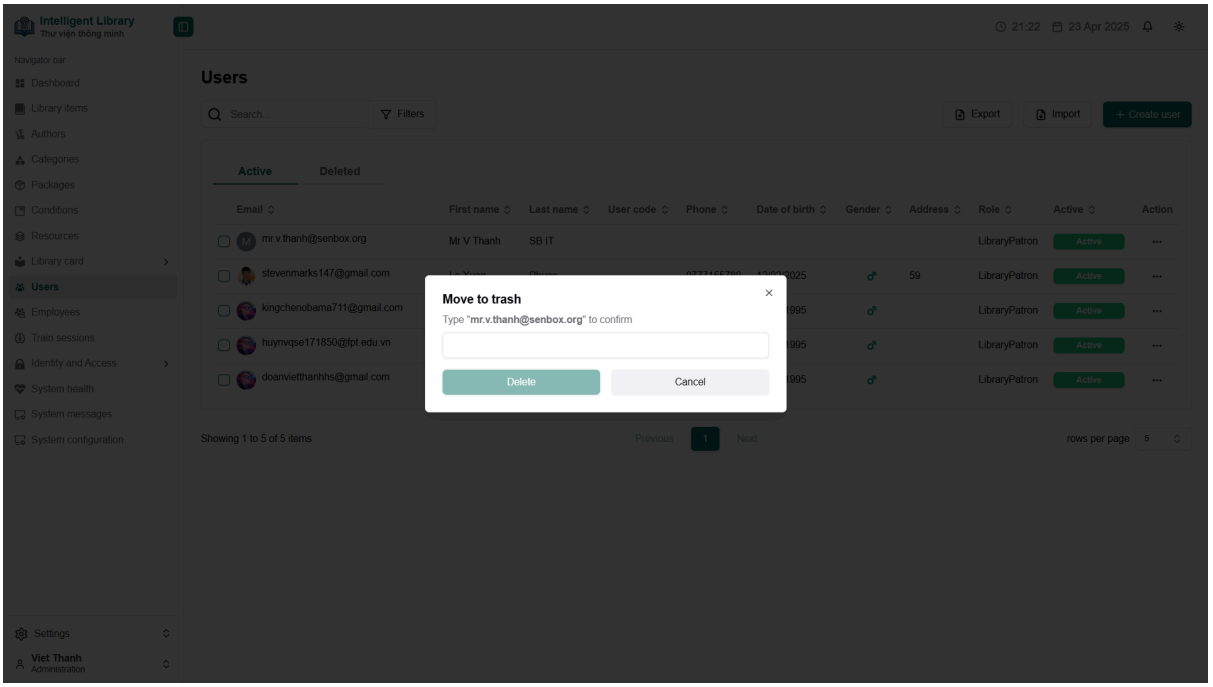


Figure 3.4.21.7: User management

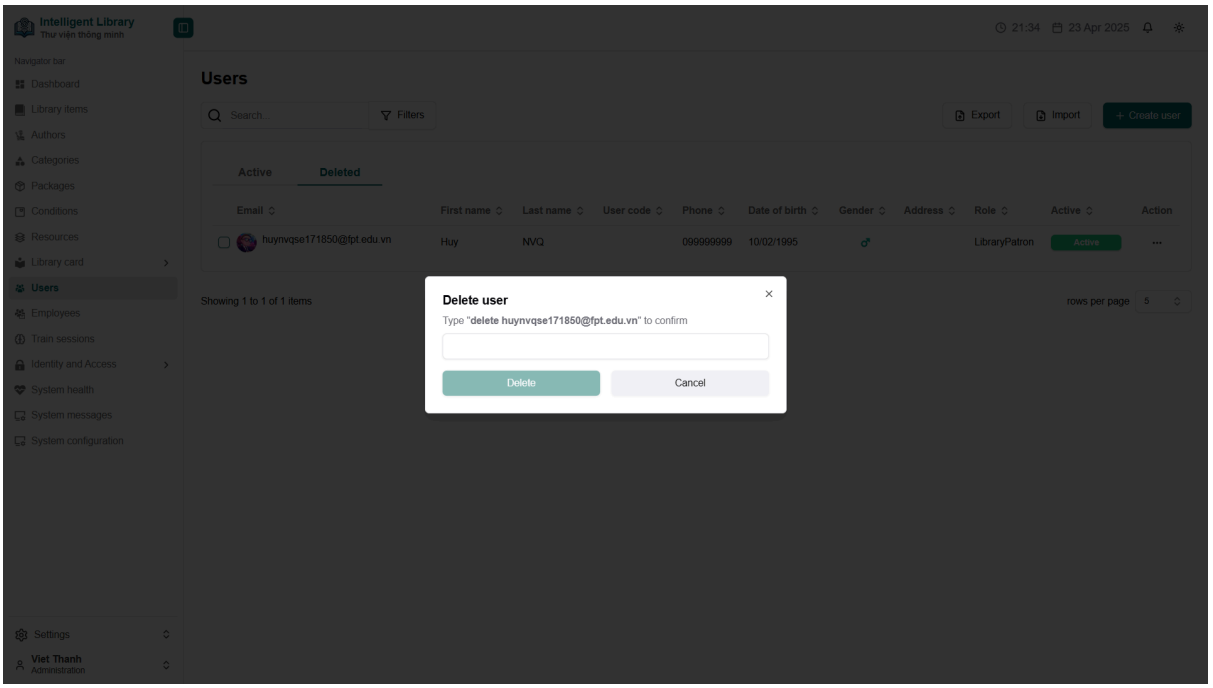


Figure 3.4.21.8: User management

3.4.22 Employee management

- **Function Description:** This screen allows admin to view, search, and create employees. It provides various filters methods, data import/export functionalities, and actions for handling employees.
- **Function Trigger:**

- **Search employees:**
  - **Button/Navigation:** There is a search input on the top of the screen.
  - **Trigger:** The admins enter the keyword they want to search for and then presses "Enter".
  - **Function:** The system will filter out the corresponding search results.
- **Filter employees:**
  - **Button/Navigation:** There is a "Filter" button right next to the search input.
  - **Trigger:** When clicked, a dialog will appear. Admins can then enter the necessary inputs or select options from dialog to refine the search. Finally, clicking "Apply" will execute the query.
  - **Function:** The system will filter out the corresponding filter results.
- **Import employees:**
  - **Button/Navigation:** There is an “Import” button on the top of the screen.
  - **Trigger:** When clicked, the system presents a dialog where the admin can upload an Excel file and fill in the required fields. The admin can press 'Import' to upload the file.
  - **Function:** New employees will be added to the system.

- **Export employees:**
  - **Button/Navigation:** There is an “Export” button on the top of the screen.
  - **Trigger:** Admins can press this button to perform the data export process.
  - **Function:** The system will download to the user's computer an excel file containing data of the employees.

- **Create employee:**
  - **Button/Navigation:** There is a “Create employee” button at the top of the screen.
  - **Trigger:** When clicked, the system will display a dialog, the admin needs to fill in the required fields then press the “Continue” button.
  - **Function:** The system will create a new author with the provided information.

- **Sort a column:**
  - **Button/Navigation:** The table headers with up/down icons indicate sortable columns.
  - **Trigger:** When clicked, the action will proceed.
  - **Function:** Clicking the header toggles the sort order between ascending and descending. The icon changes accordingly (up = ascending, down = descending).

- **Edit a employee:**
  - **Button/Navigation:** Click on the "..." button in the last column of each row in the table. A dropdown will appear with the "Edit employee" option.
  - **Trigger:** When clicked, the system will display a dialog allowing the librarian to edit the employee's information, then click "Save" to confirm.
  - **Function:** The system will update the employee’s information.

- **Move to trash a employee:**

- **Button/Navigation:** At the “Active” tab of the table. Click on the "..." button in the last column of each row in the table. A dropdown will appear with the Move to trash" option.
- **Trigger:** When clicked, a dialog will appear, asking the admin to confirm moving the employee to the trash. After confirming, the librarian clicks the "Continue" button.
- **Function:** The system will update the **isDeleted** field of the selected employee to **true**.

❖ **Restore an employee:**
- **Button/Navigation:** At the “Deleted” tab of the table. Click on the "..." button in the last column of each row in the table. A dropdown will appear with the “Restore" option.
- **Trigger:** When clicked, the action will proceed.
- **Function:** The system will update the **isDeleted** field of the selected employee to **false**.

❖ **Delete permanently an employee:**
- **Button/Navigation:** At the “Deleted” tab of the table. Click on the "..." button in the last column of each row in the table. A dropdown will appear with the **"Delete permanently"** option.
- **Trigger:** When clicked, a dialog will appear, asking the admin to confirm to delete permanently the employee. After confirming, the admin clicks the "Continue" button.
- **Function:** The system will permanently delete the employee from the database.

❖ **Action with employees:**
- **Button/Navigation:** A “Actions” button at the top of the screen will be clickable when admins select at least a row on the table.
- **Trigger:** When clicked, the system will display options such as "Move to Trash", "Restore", "Permanently Delete"
- **Function:** The system performs the selected action on the chosen employees.

❖ **Pagination**
- **Button/Navigation:** Simple numbered buttons, "Next" and "Previous" links, at the bottom of the table.
- **Trigger:** Admin clicks on any of the pagination controls.
- **Function:** Changes the page of the table, allowing the admins to navigate through multiple entries.

❖ **Adjust page size**
- **Button/Navigation:** A number selection dropdown at the bottom of the screen
- **Trigger:** When clicked, different options will be displayed so that the admin can choose the page size they want.
- **Function:** The system will display more or less rows per page of the table.

❖ **View different types of employees**
- **Button/Navigation:** There are two different tabs that the admin can choose from located above the table.

- **Trigger:** When clicked, the system will filter the corresponding employees.
- **Function:**
  - **"Active" Tab**: Displays active employees where isDeleted = false.
  - **"Deleted" Tab**: Shows deleted employees where isDeleted = true.

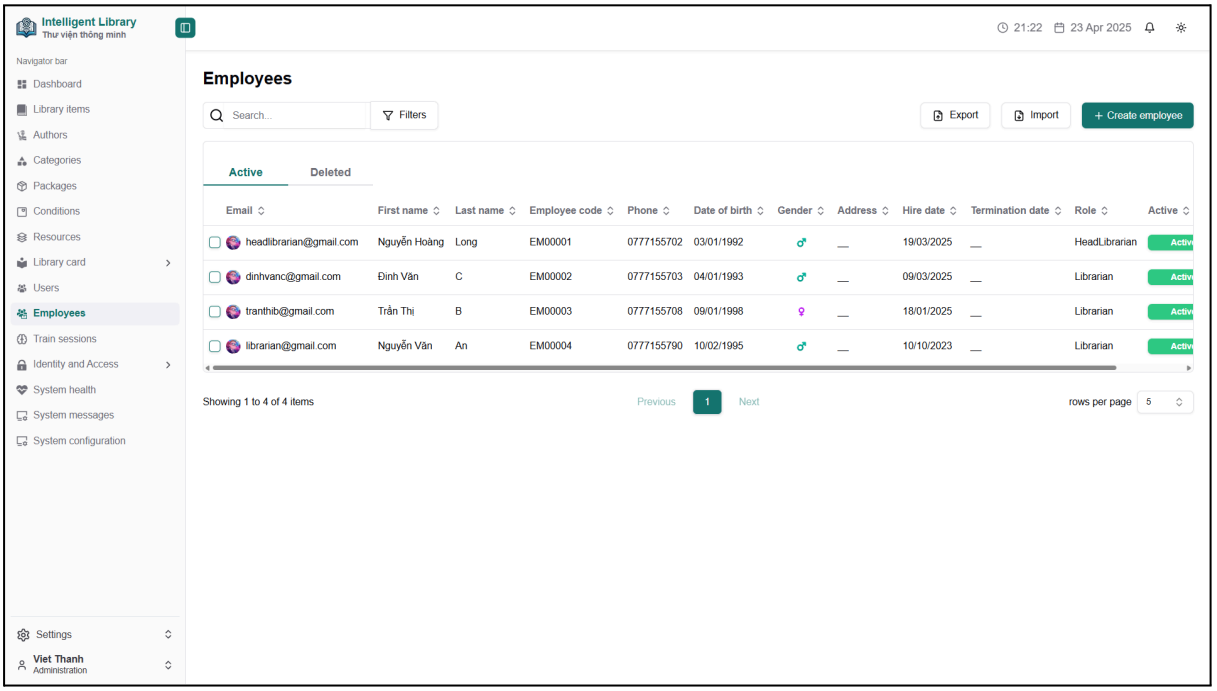


Figure 3.4.22.1: Employee management

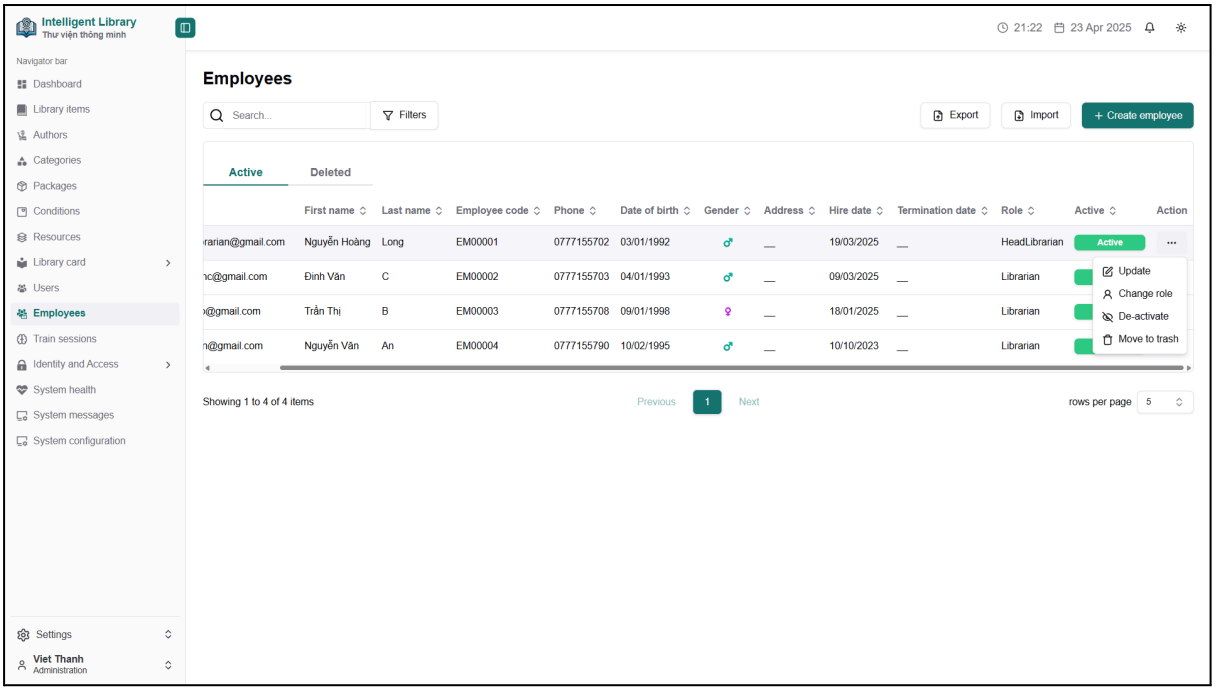


Figure 3.4.22.2: Employee management

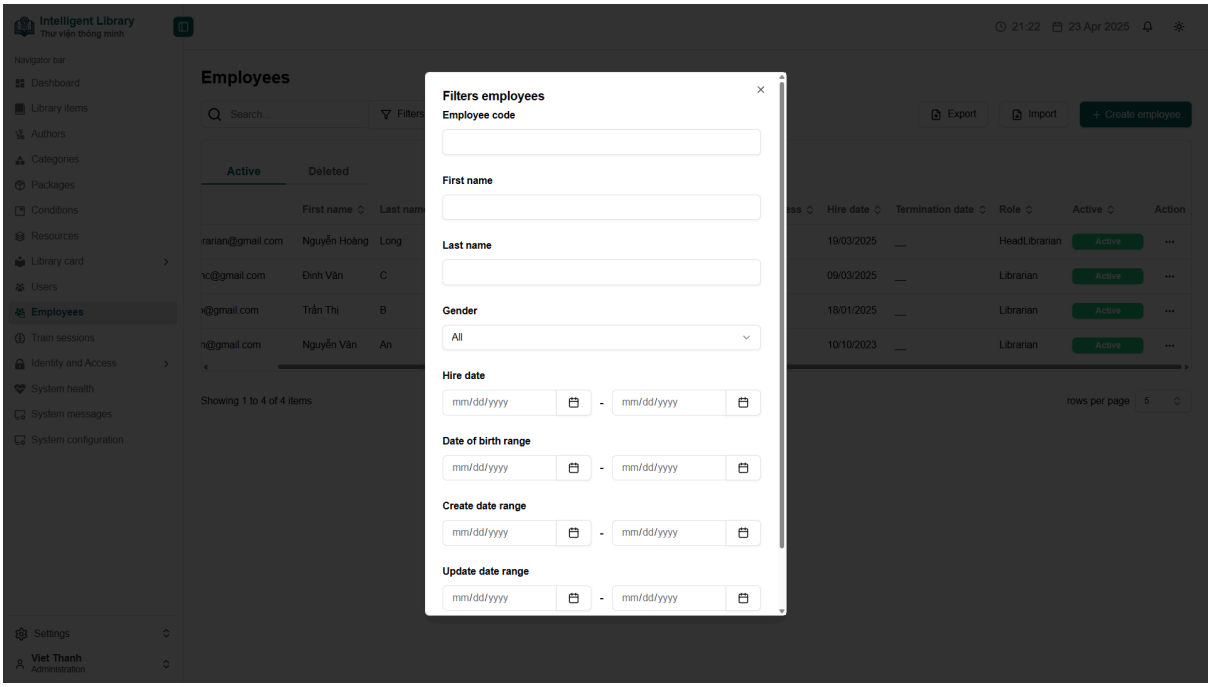


Figure 3.4.22.3: Employee management

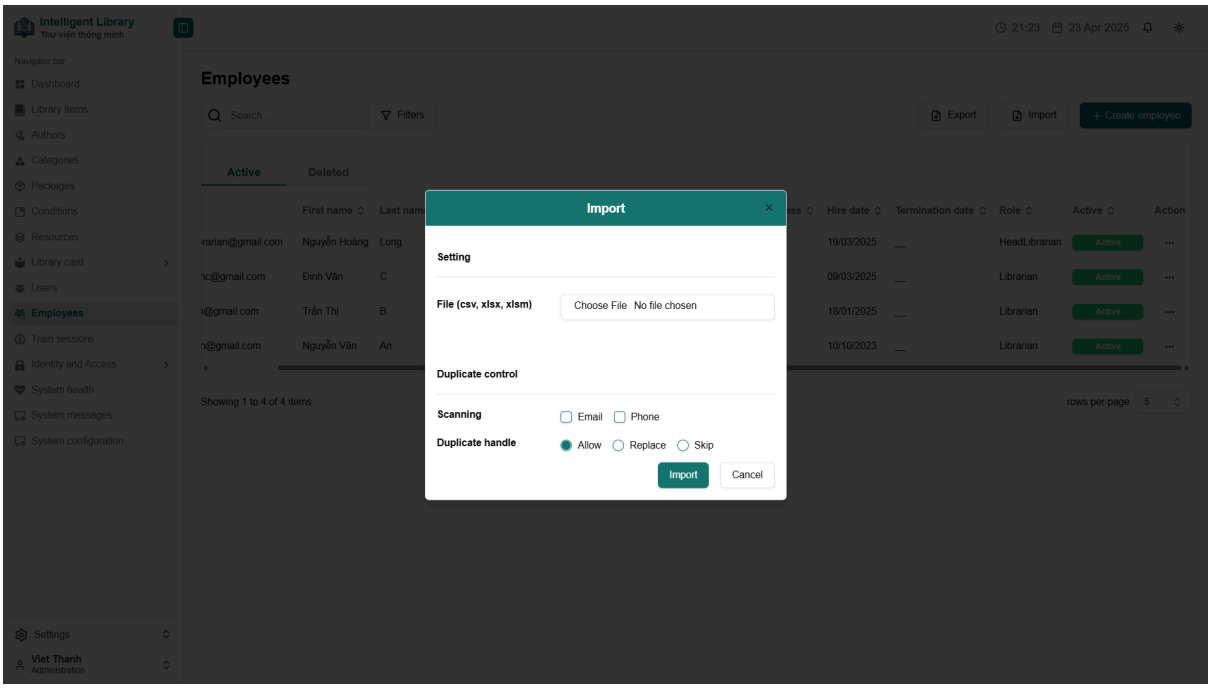


Figure 3.4.22.4: Employee management

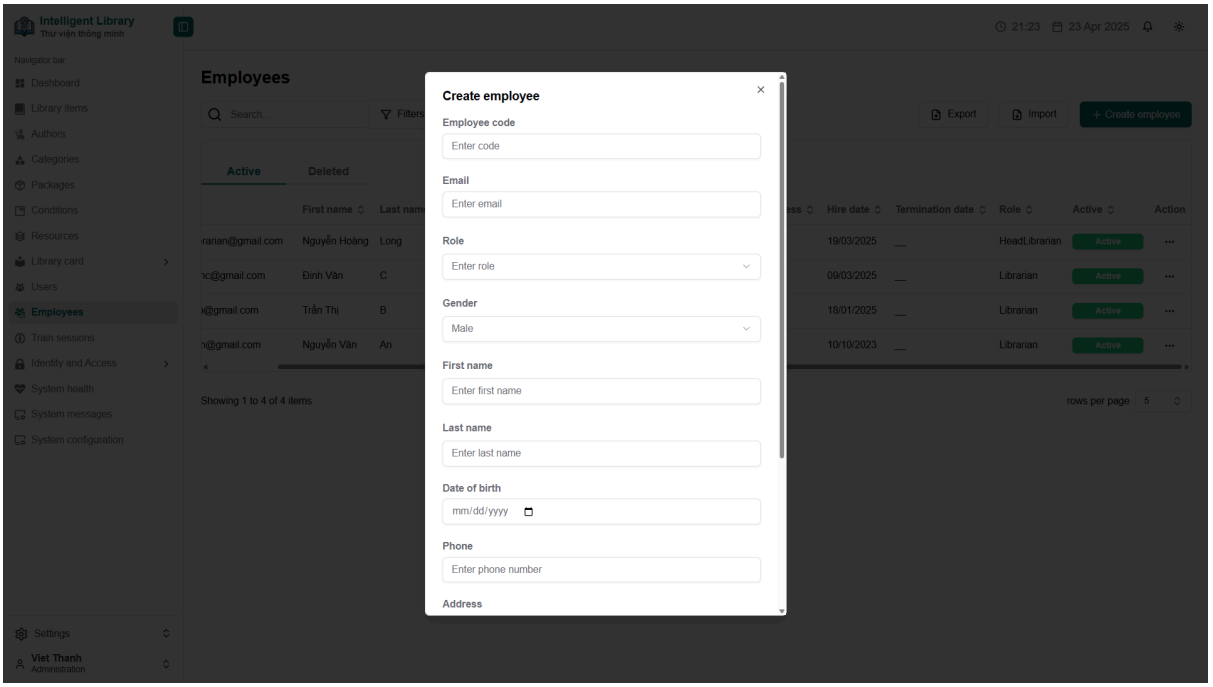


Figure 3.4.22.5: Employee management

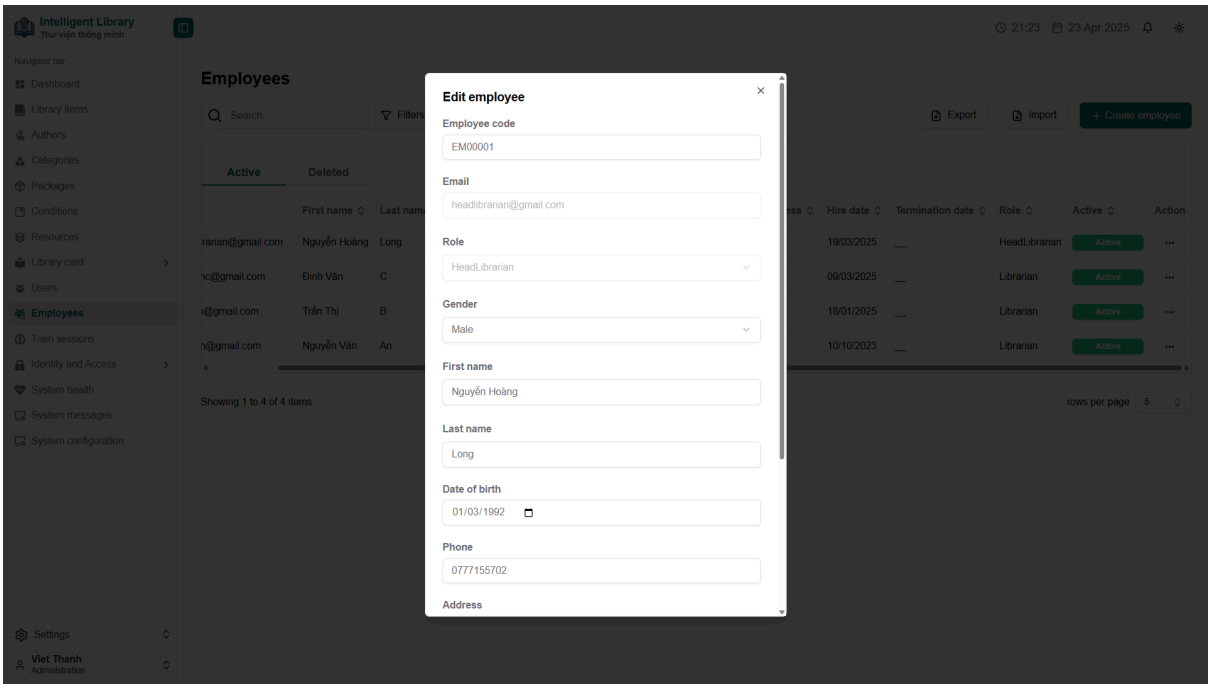


Figure 3.4.22.6: Employee management

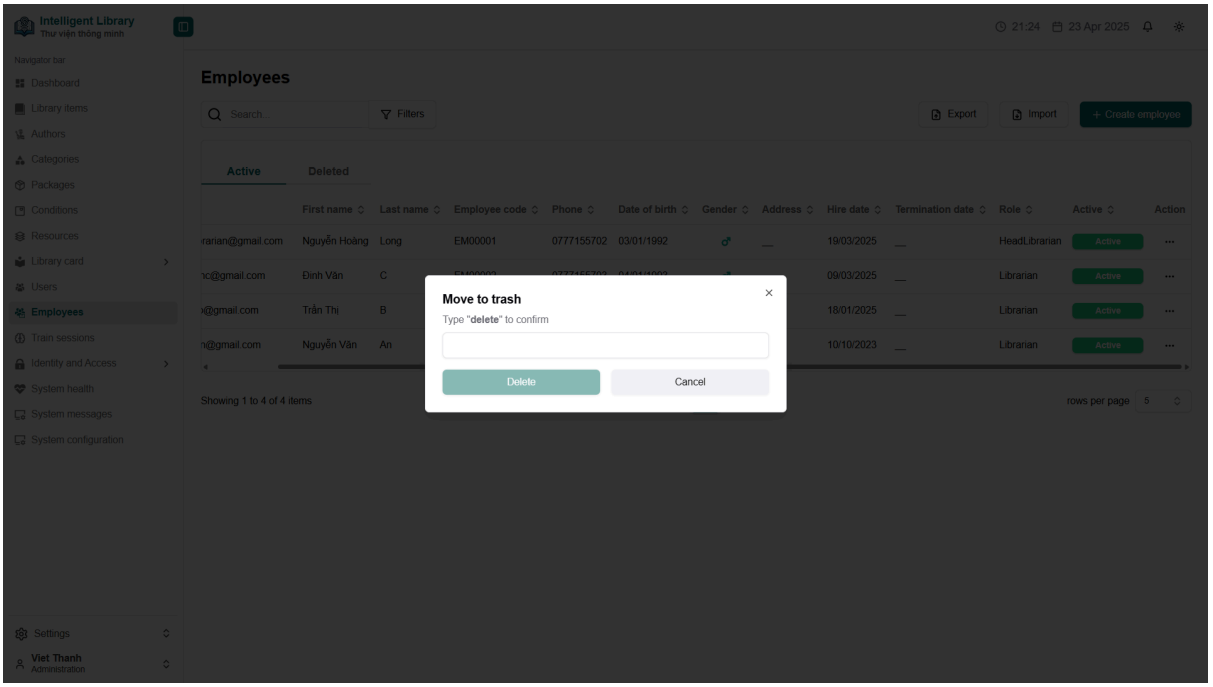


Figure 3.4.22.7: Employee management

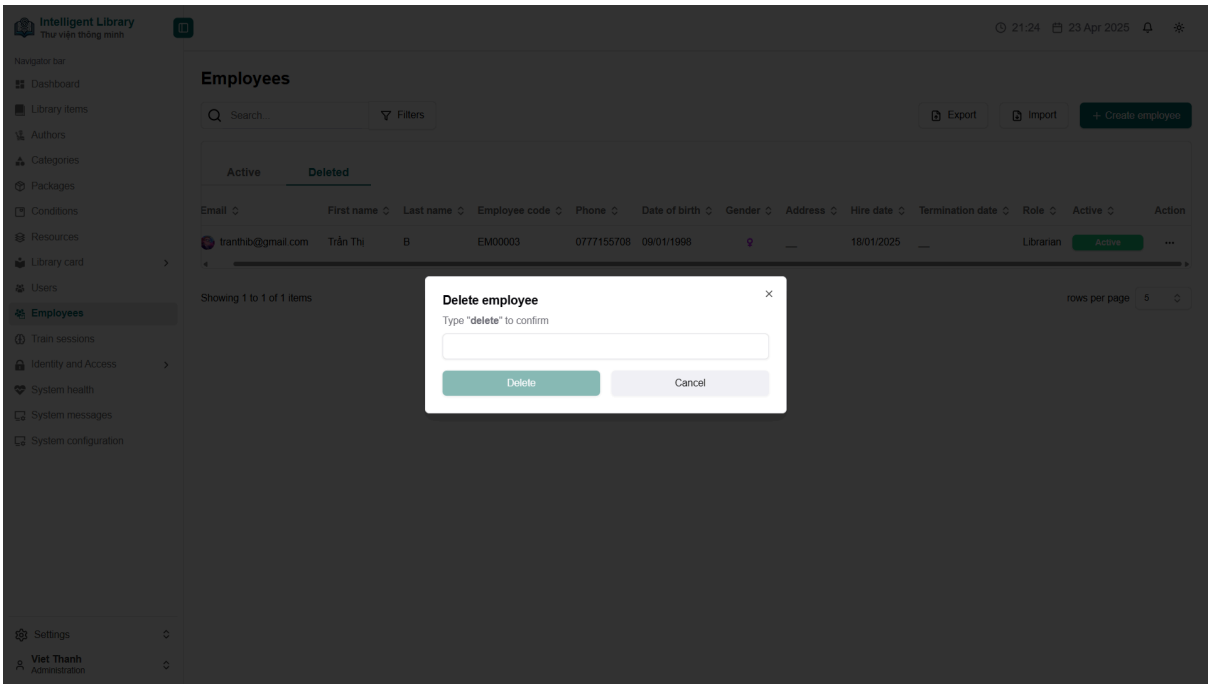


Figure 3.4.22.8: Employee management

3.4.23 Borrow Records Management

- **Function Description:** This screen allows **librarians** to view, search, and manage borrow records efficiently. It provides various search and filter methods.

- **Function Trigger:**

  - **Search borrow records:**
    - **Button/Navigation:** There is a search input on the top of the screen.
    - **Trigger:** The librarian enters the keyword they want to search for and then presses "Enter".
    - **Function:** The system will filter out the corresponding search results.
  - **Filter borrow records:**
    - **Button/Navigation:** There is a "Filter" button right next to the search input.
    - **Trigger:** When clicked, a dialog will appear. Librarians can then enter the necessary inputs or select options from dialog to refine the search. Finally, clicking "Apply" will execute the query.
    - **Function:** The system will filter out the corresponding filter results.
  - **Sort a column:**
    - **Button/Navigation:** The table headers with up/down icons indicate sortable columns.
    - **Trigger:** When clicked, the action will proceed.
    - **Function:** Clicking the header toggles the sort order between ascending and descending. The icon changes accordingly (up = ascending, down = descending).

  - **Navigate to Create Borrow Record Page**
    - **Button/Navigation:** On the "Borrow Records" page, a button labeled "Create borrow record" is located at the top-right corner of the screen. **Trigger:** When clicked, it navigates users to the **Create Borrow Record** page.
    - **Function:** The system will navigate to the "**Create Borrow Record**" page and display its interface, allow the librarian to scan the card and the items patron wish to borrow

  - **Navigate to Return Library Items Page**
    - **Button/Navigation:** On the "Borrow Records" page, a button labeled Return Items" is located at the top-right corner of the screen.
    - **Trigger:** When clicked, it navigates users to the **Return Library Items** page.
    - **Function:** The system will navigate to the **Return Library Items** page and display its interface, allow the librarian to scan the card and the items patron wish to return
  - **Pagination**
    - **Button/Navigation:** Simple numbered buttons, "Next" and "Previous" links, at the bottom of the table.
    - **Trigger:** Librarian clicks on any of the pagination controls.
    - **Function:** Changes the page of the table, allowing the librarians to navigate through multiple entries.
  - **Adjust page size**
    - **Button/Navigation:** A number selection dropdown at the bottom of the screen
    - **Trigger:** When clicked, different options will be displayed so that the librarian can choose the page size they want.

- **Function:** The system will display more or less rows per page of the table.

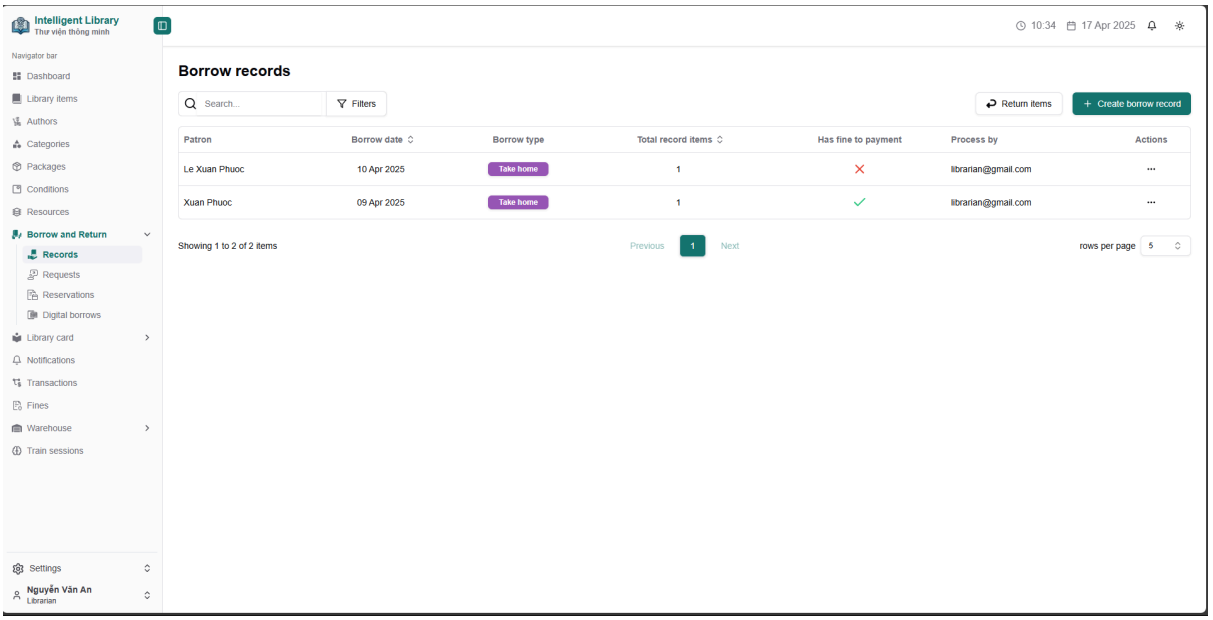


Figure 3.4.23.1: Borrow records management

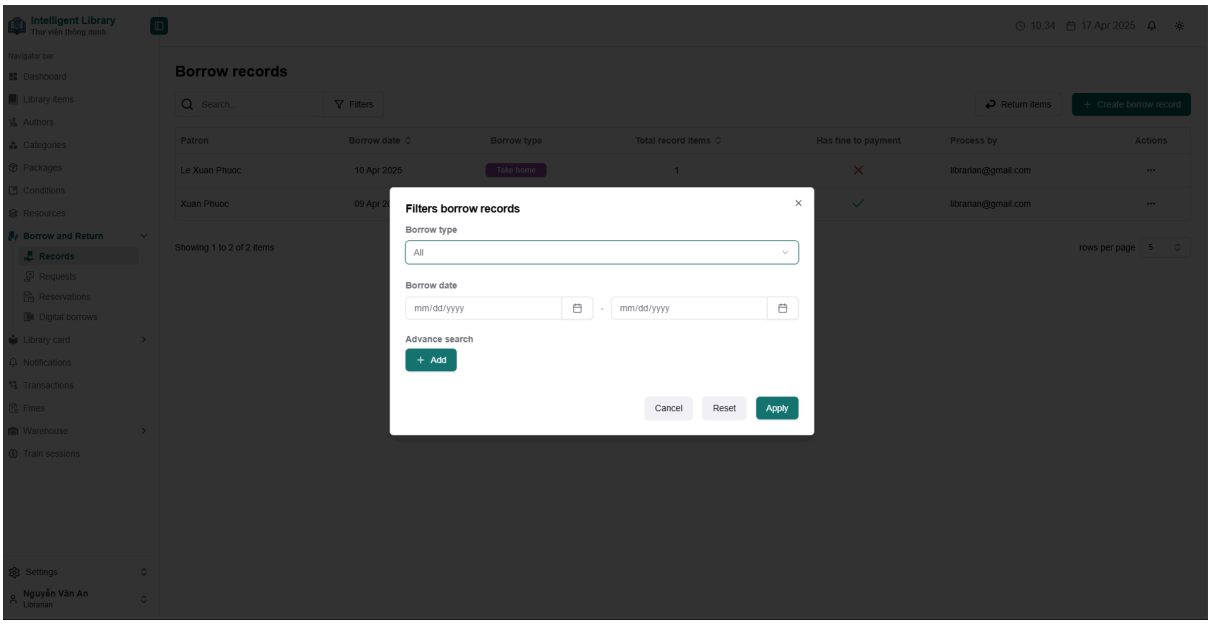


Figure 3.4.23.2: Borrow records management

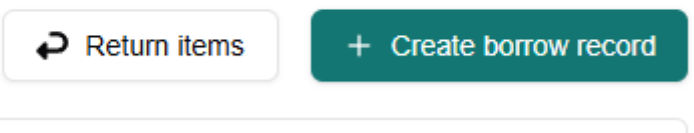


Figure 3.4.23.3: Borrow records management

3.4.24 Borrow Record Detail Management

- **Function Description:** The screen offers a read-only view of a patron's borrowing transaction, showing borrow record details (date, type, total items, payment status), patron information (name, card barcode, permissions), borrow progress, and borrowed item specifics (title, barcode, condition, due date).

- **Function Trigger:**

  - **View Borrow Information:**
    - **Button/Navigation:** The "Borrow Information" section displays details like borrow date, type, total items, and payment status.
    - **Trigger:** Information is automatically displayed upon accessing the page.
    - **Function:** The system shows the borrowing transaction's metadata, such as "Borrow Date: 10 Apr 2025," "Borrow Type: Take home," and "Has Fine Payment: No."
  - **View Patron Information:**
    - **Button/Navigation:** The "Patron Information" section includes fields like name, avatar, card barcode, and borrowing permissions.
    - **Trigger:** Details are automatically displayed upon accessing the page.
    - **Function:** The system presents the patron's details, such as "Name on Card: Le Xuan Phuoc," card barcode, and permissions like "Allow Borrow: Yes."
  - **View Borrow Progress:**
    - **Button/Navigation:** The "Progress" section shows a timeline of phases (e.g., "Request Phase" and "Borrowing Phase").
    - **Trigger:** The timeline is automatically displayed upon accessing the page.
    - **Function:** The system displays the transaction's progress, including timestamps like "Request Phase on 04:37 10/04/2025" and the number of items requested or borrowed.

  - **View Borrowed Item Details:**
    - **Button/Navigation:** The "Borrowing Phase" section lists borrowed items with details like title, barcode, initial condition, due date, and return status.
    - **Trigger:** Item details are automatically displayed upon accessing the page.
    - **Function:** The system shows specifics of each borrowed item, such as "Title: Thám tử lừng danh Conan - Tập 1," "Initial Condition: Good," and "Due Date: 04:44 10/05/2025."
- **Business Rules:** BR-56, BR-57

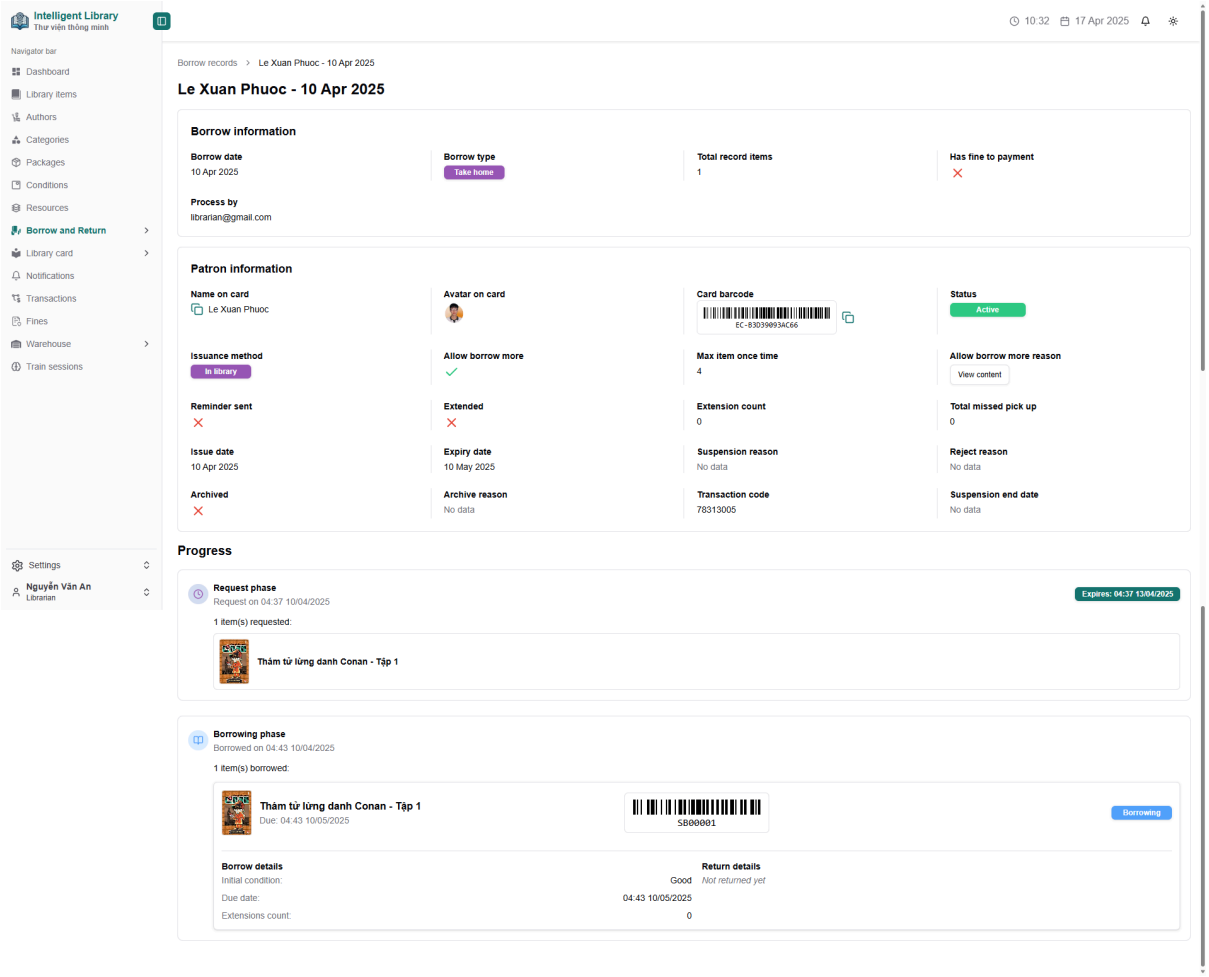


Figure 3.4.24.1: Borrow records detail

3.4.25 Crete Borrow Record

- **Function Description:** This screen allows librarians to initiate a borrow record by scanning a patron's library card and the unique barcodes of library items. It displays patron details, activity summary. The screen lists borrowed request items and assigned reservation items, allowing the librarian to scan barcodes to add items to the borrowed record.
- **Function Trigger:**

  - ❖ **Scan Patron's Library Card:**
    - **Button/Navigation:**A "Scan Patron's Card" section with a barcode icon is displayed initially.
    - **Trigger:** The librarian uses a barcode scanner to scan the patron's library card barcode.
    - **Function:** The system retrieves and displays the patron's details and their borrowing activity summary upon successful scanning.
  - ❖ **Select Borrow Type:**
    - **Button/Navigation:** A "Borrow Type" section offers two radio buttons: "Take Home" and "In Library."
    - **Trigger:**The librarian selects the appropriate borrow type.

        - **Function:** The system sets the borrowing mode, determining whether items are borrowed for home use or in-library use, affecting due dates and policies.
    - **View Borrow Request Items:**
        - **Button/Navigation:** The "Borrow Request Items" section lists items with details like title, authors, and borrow duration
        - **Trigger:** Items are automatically displayed if the patron has pending borrow requests.
        - **Function:** The system shows requested items that the patron intends to borrow, aiding the librarian in item selection.
    - **View Assigned Reservation Items:**
        - **Button/Navigation:** The "Assigned Reservation Items" section lists items with details like title, authors, and a barcode field.
        - **Trigger:** Items are automatically displayed if the patron has reservations with assigned items.
        - **Function:** The system displays reserved items, indicating their status for the librarian to process.
    - **Scan Unique Barcodes of Items:**
        - **Button/Navigation:** Each item in the “"Borrow Request Item” and "Assigned Reservation Items" section has a barcode field with a "Scanned" label once scanned.
        - **Trigger:** The librarian uses a barcode scanner to scan the unique barcode of each library item.
        - **Function:** The system updates the item’s status to "Scanned," linking the item to the borrow record and confirming it as part of the borrow record.
    - **Cancel Borrow Record Creation:**
        - **Button/Navigation:** A "Cancel" button is available at the bottom-right of the page.
        - **Trigger:** Clicking the "Cancel" button.
        - **Function:** The system discards the borrow record creation process and returns to the previous page without saving any changes.
    - **Continue Borrow Record Creation:**
        - **Button/Navigation:** A "Continue" button is available at the bottom-right of the page.
        - **Trigger:** Clicking the "Continue" button after scanning the patron’s card and all item barcodes.
        - **Function:** The system finalizes the borrow record, linking the scanned items to the patron, and navigates to the next step.

- **Business Rules:** BR-30, BR-51, BR-72, BR-73

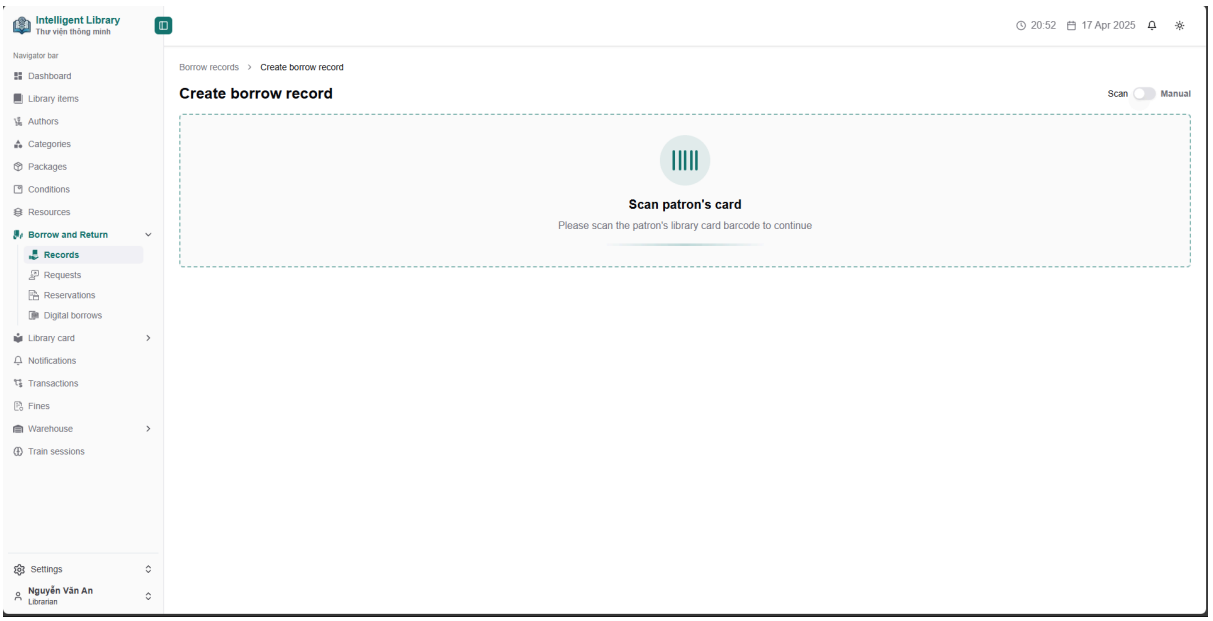


Figure 3.4.25.1: Create borrow records

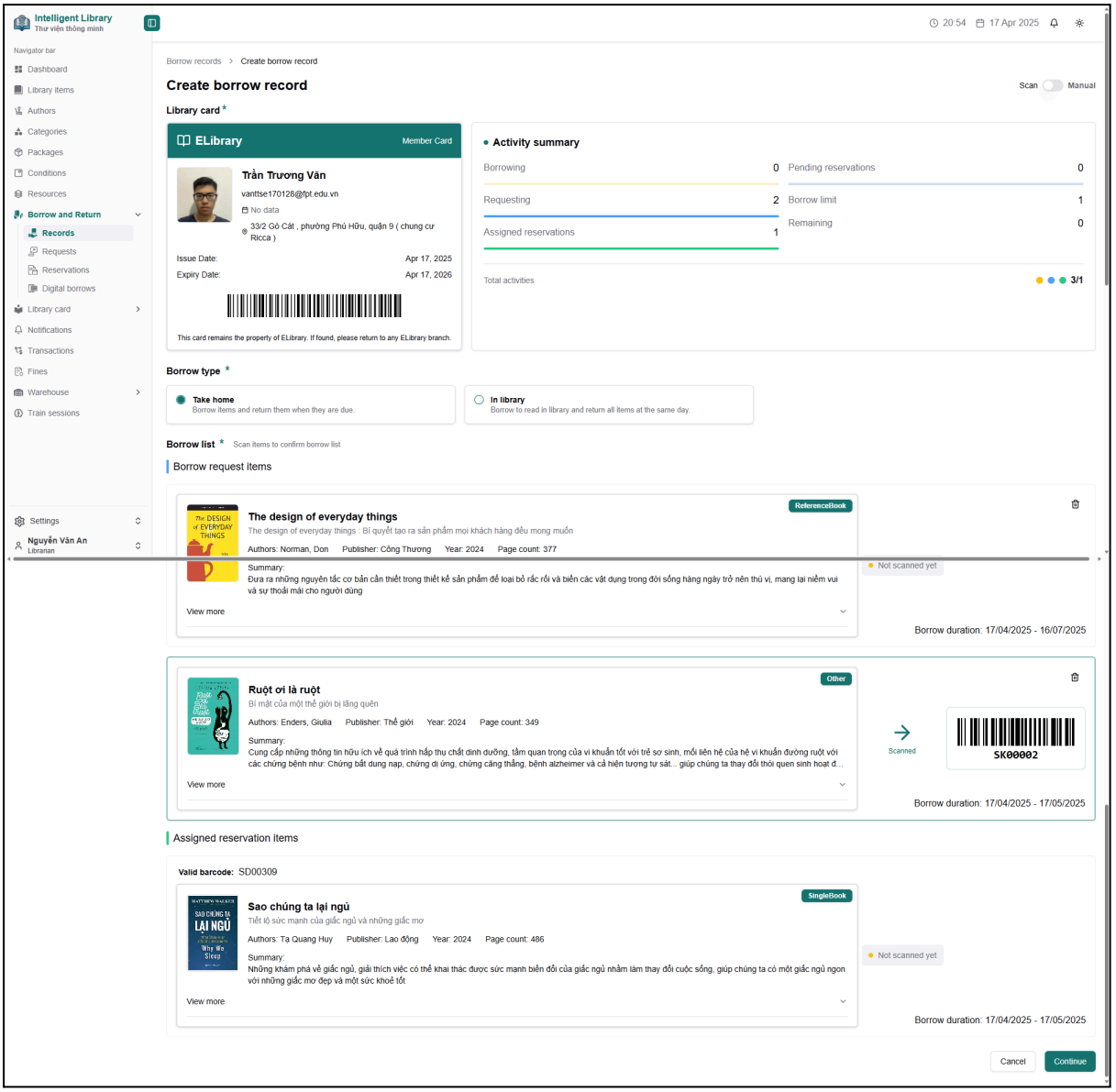


Figure 3.4.25.2: Create borrow records

3.4.26 Return Library Items

- **Function Description:** This screen allows librarians to process the return of borrowed items by scanning the patron's library card and the unique barcodes of the items. After scanning the patron's card, their details and borrowing activity summary are displayed. Once an item is scanned, its details appear with fields to select the return condition and an option to add fines if applicable, ensuring accurate tracking of returns and penalties.
- **Function Trigger:**

  - **Scan Patron's Library Card:**
    - **Button/Navigation:** A "Scan Patron's Card" section with a barcode icon is displayed initially/.
    - **Trigger:** The librarian uses a barcode scanner to scan the patron's library card barcode.

- **Function:** The system retrieves and displays the patron’s details and their borrowing activity summary.

❖ **Scan Unique Barcode of Items:**
- **Button/Navigation:** After scanning the patron’s card, a section lists borrowed items as their barcodes are scanned.
- **Trigger:** The librarian uses a barcode scanner to scan the unique barcode of each returned library item.
- **Function:**The system identifies the item, marks it as "Scanned," and displays its details along with options to select return condition and add fines.

❖ **Select Return Condition:**
- **Button/Navigation:** A "Return Condition" dropdown appears for each scanned item with options.
- **Trigger:** The librarian selects the condition from the dropdown after scanning an item.
- **Function:**The system records the item’s return condition, which may affect fines or item status.

❖ **Add Fines:**
- **Button/Navigation:** A "Fines" section with an " Add Fine" button appears for each scanned item, and clicking it opens a dialog to select fine types.
- **Trigger:** The librarian clicks "+ Add Fine" and selects a fine type with predefined fine amounts.
- **Function:**The system applies the selected fine to the patron’s record, displaying the fine amount per day and total, ensuring accurate penalty tracking.

❖ **Cancel Return Process:**
- **Button/Navigation:** A "Cancel" button is available at the bottom-right of the page.
- **Trigger:** Clicking the "Cancel" button.
- **Function:**The system discards the return process and returns to the previous page without saving any changes.

❖ **Continue Return Process:**
- **Button/Navigation:** A "Continue" button is available at the bottom-right of the page.
- **Trigger:** Clicking the "Continue" button after scanning items and setting conditions/fines.
- **Function:** The system finalizes the return process, updates the patron’s borrow record, and applies any fines.

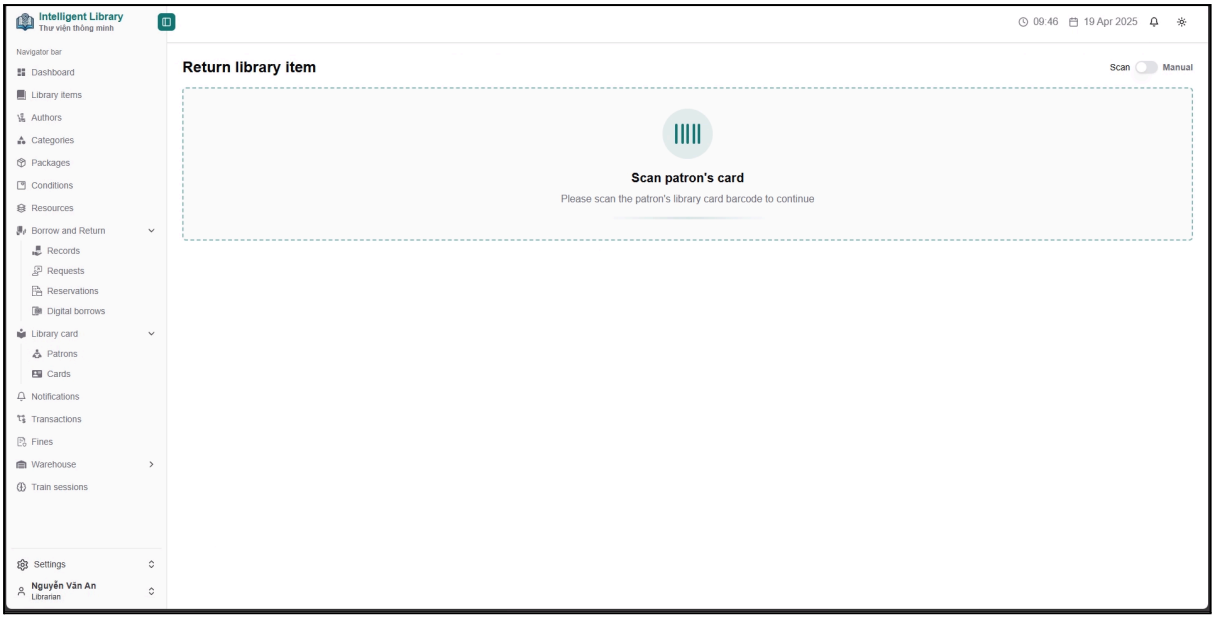


Figure 3.4.26.1: Return Library items

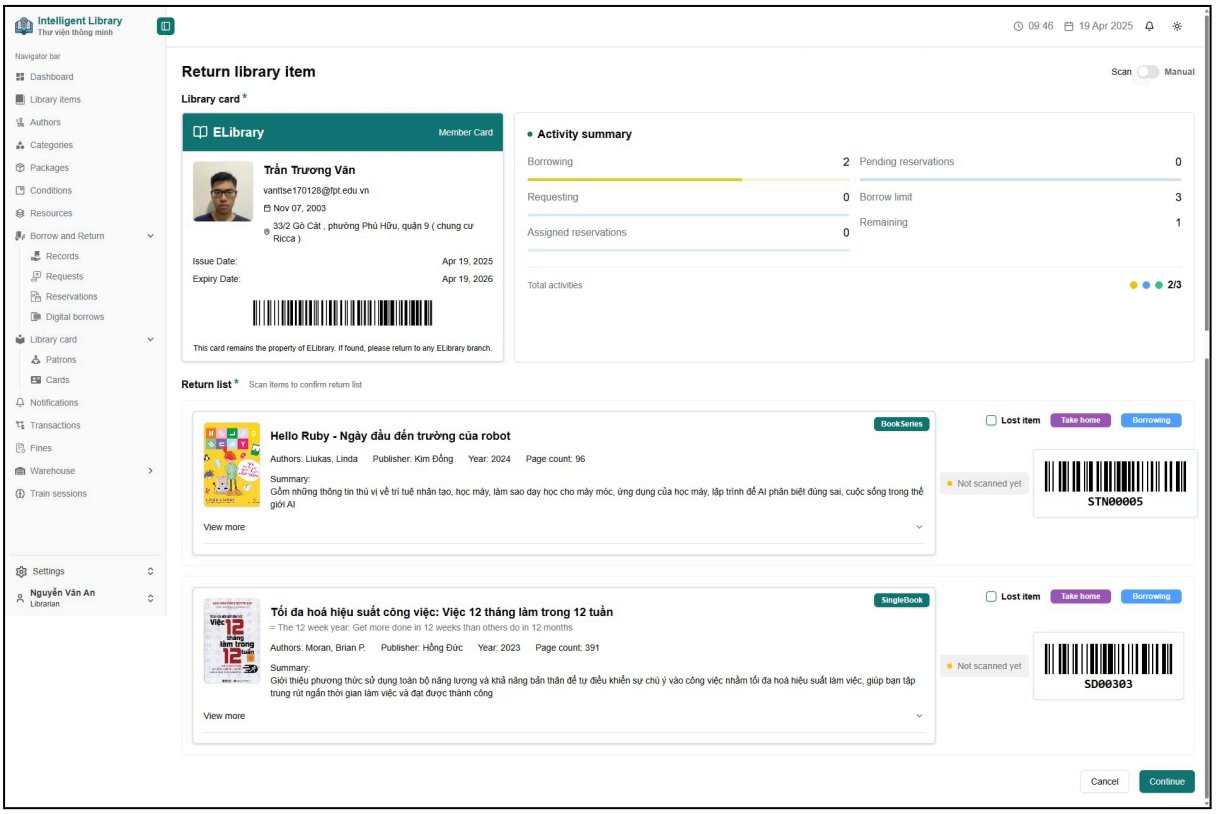


Figure 3.4.26.2: Return Library items

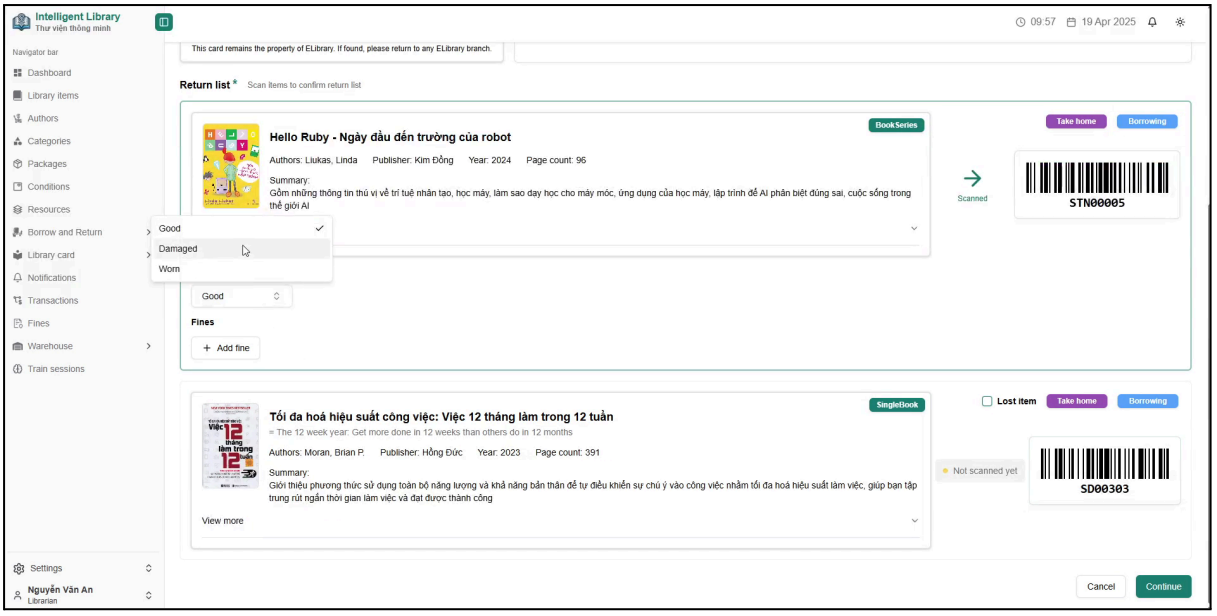


Figure 3.4.26.3: Return Library items

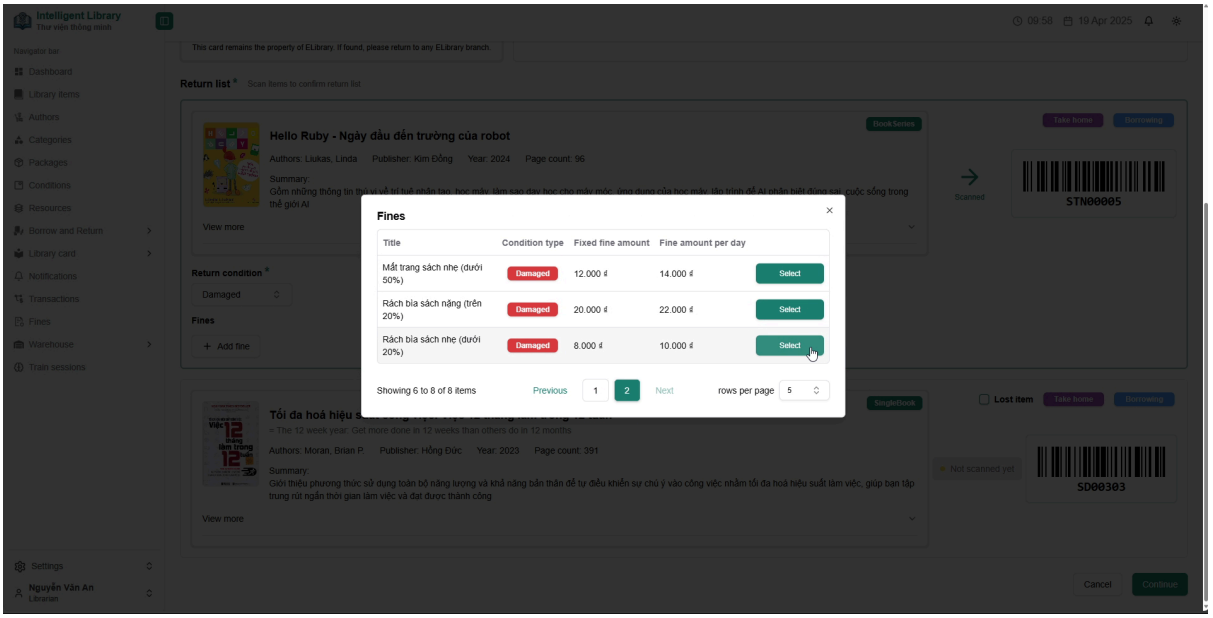


Figure 3.4.26.4: Return Library items

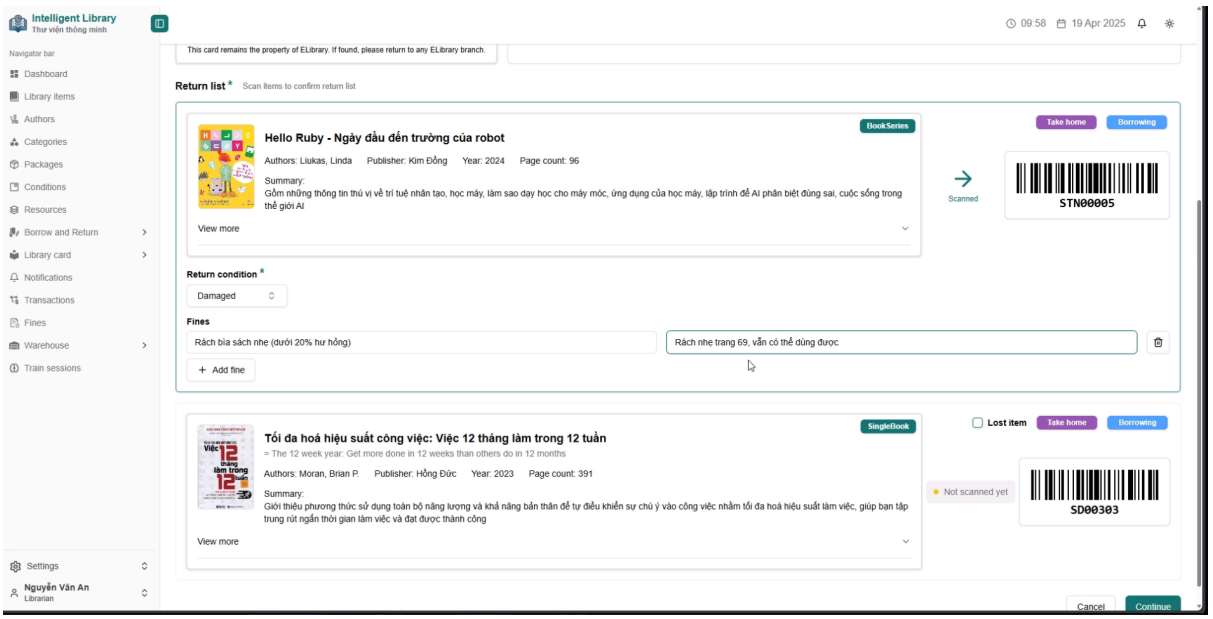


Figure 3.4.26.5: Return Library items

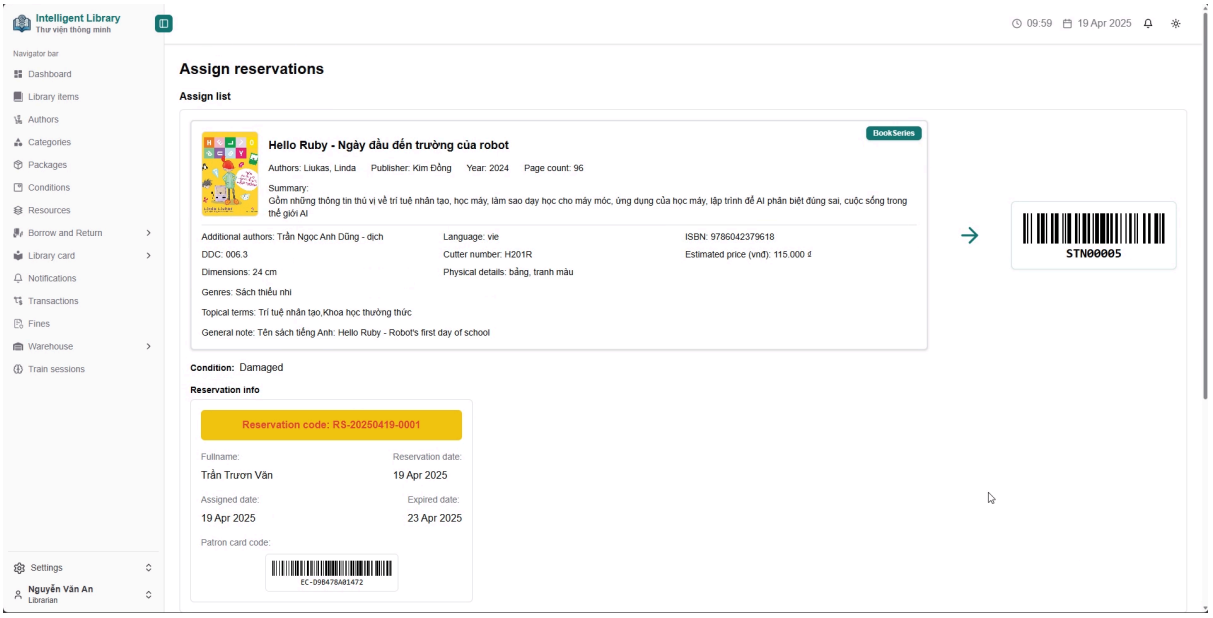


Figure 3.4.26.6: Return Library items

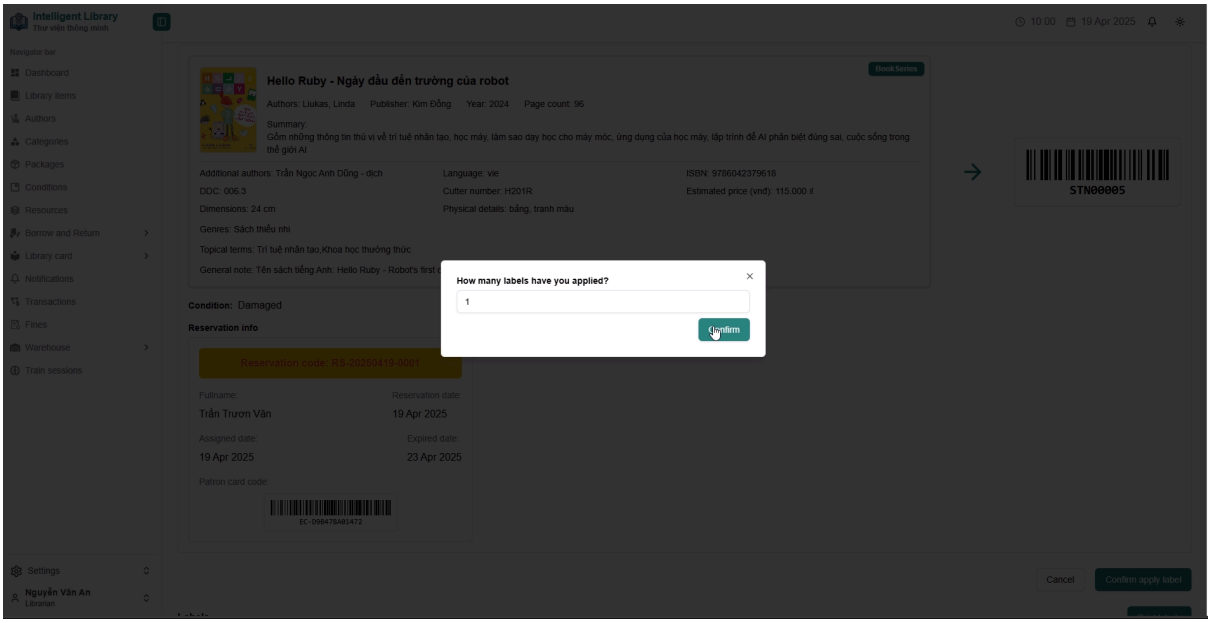


Figure 3.4.26.7: Return Library items

3.4.27 Borrow Requests Management

- **Function Description:** This screen allows **librarians** to view, search, and manage borrow requests efficiently. It provides various search and filter methods.
- **Function Trigger:**

  - **Search borrow requests:**
    - **Button/Navigation:** There is a search input on the top of the screen.
    - **Trigger:** The librarian enters the keyword they want to search for and then presses "Enter".
    - **Function:** The system will filter out the corresponding search results.
  - **Filter borrow requests:**
    - **Button/Navigation:** There is a "Filter" button right next to the search input.
    - **Trigger:** When clicked, a dialog will appear. Librarians can then enter the necessary inputs or select options from dialog to refine the search. Finally, clicking "Apply" will execute the query.
    - **Function:** The system will filter out the corresponding filter results.
  - **Sort a column:**
    - **Button/Navigation:** The table headers with up/down icons indicate sortable columns.
    - **Trigger:** When clicked, the action will proceed.
    - **Function:** Clicking the header toggles the sort order between ascending and descending. The icon changes accordingly (up = ascending, down = descending).
  - **Pagination**
    - **Button/Navigation:** Simple numbered buttons, "Next" and "Previous" links, at the bottom of the table.
    - **Trigger:** Librarian clicks on any of the pagination controls.
    - **Function:** Changes the page of the table, allowing the librarians to navigate through multiple entries.

❖ **Adjust page size**

- **Button/Navigation:** A number selection dropdown at the bottom of the screen
- **Trigger:** When clicked, different options will be displayed so that the librarian can choose the page size they want.
- **Function:** The system will display more or less rows per page of the table.

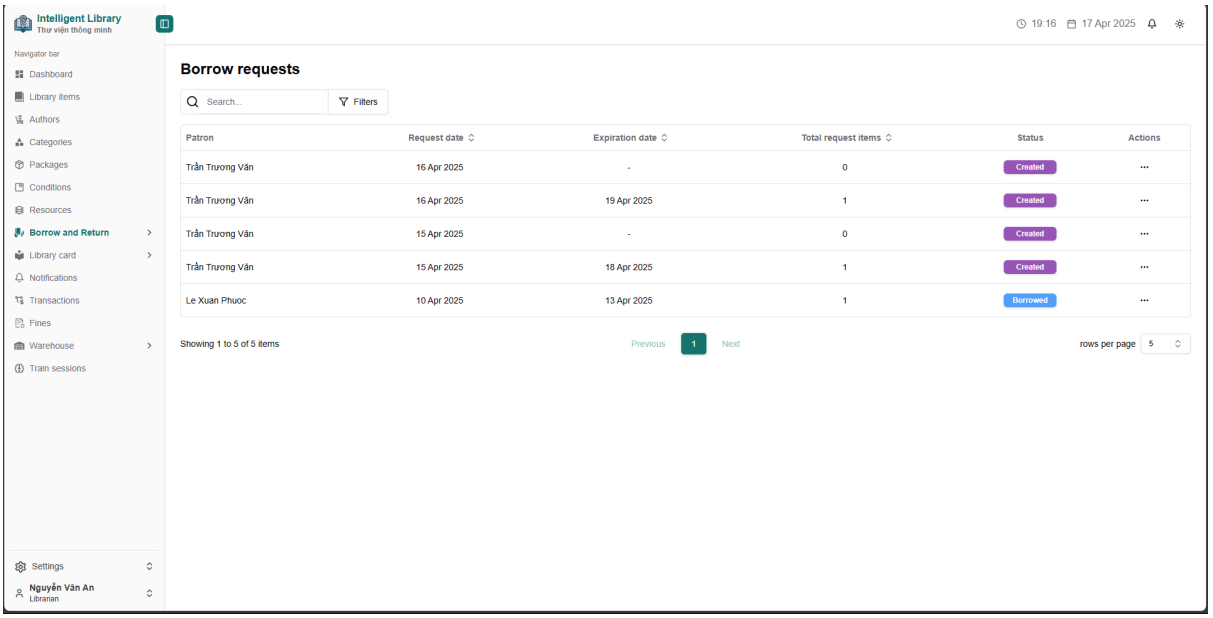


Figure 3.4.27.1: Borrow request management

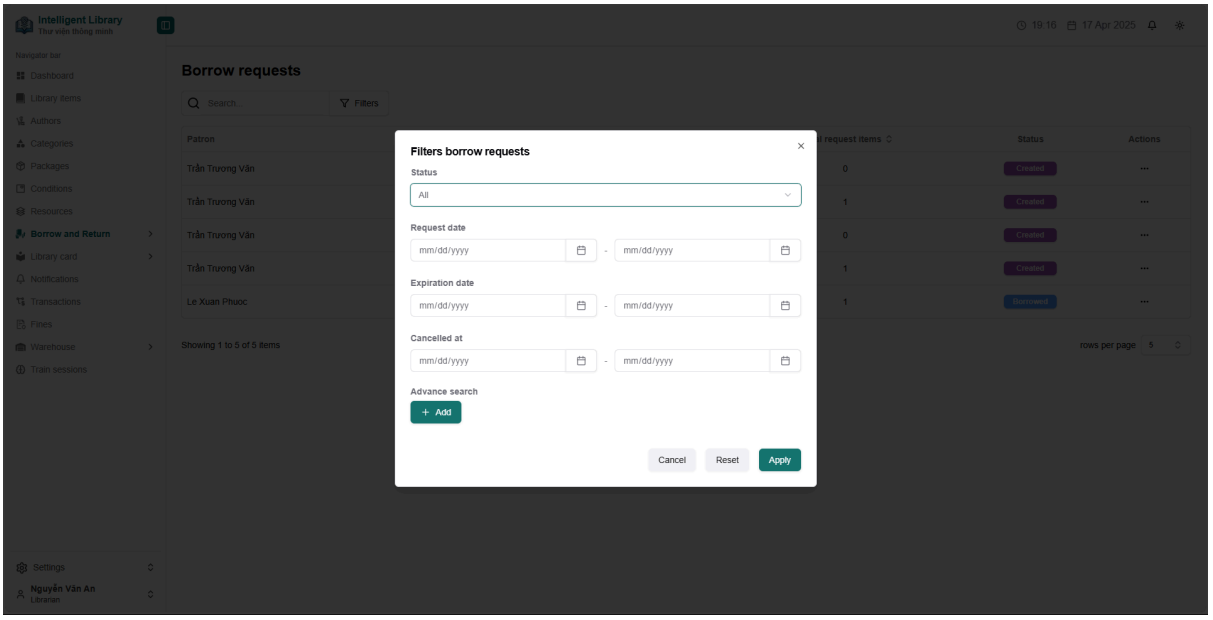


Figure 3.4.27.2: Borrow request management

3.4.28 Borrow Request Detail Management

- **Function Description:** This screen displays a read-only view of a patron's borrow request, showing request details, patron information like name and card barcode, and a list of requested items. It is solely for viewing request details.
- **Function Trigger:**

  - **View Request Information:**
    - **Button/Navigation:** The "Request Information" section shows details like request date, expiration date, total items, and status.
    - **Trigger:** Information is automatically displayed upon accessing the page.
    - **Function:** The system presents the borrow request's metadata
  - **View Patron Information:**
    - **Button/Navigation:** The "Patron Information" section displays fields like name, avatar, card barcode, and borrowing permissions.
    - **Trigger:** Details are automatically displayed upon accessing the page
    - **Function:** The system shows patron details, including "name on card", "card barcode", and permissions like "allow borrow more", etc.
  - **View Requested Items:**
    - **Button/Navigation:** The "Request Items" section includes a table with columns for "Library Item," "Authors," "Shelf Number," and "ISBN."
    - **Trigger:** The table is automatically displayed upon accessing the page.
    - **Function:** The system lists requested library items with their metadata.

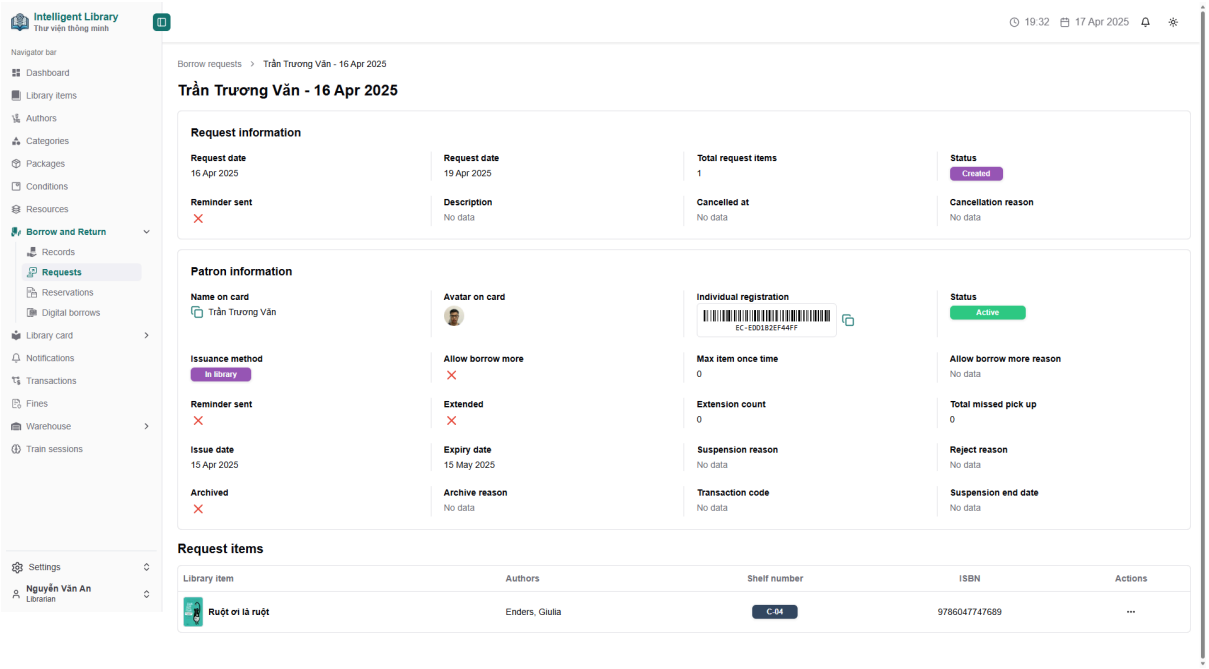


Figure 3.4.28.1: Borrow request detail management

3.4.29 Borrow Digitals Management

- **Function Description:** This screen allows **librarians** to view, search, and manage borrow digitals efficiently. It provides various search and filter methods.
- **Function Trigger:**

  - **Search borrow digitals:**
    - **Button/Navigation:** There is a search input on the top of the screen.

- **Trigger:** The librarian enters the keyword they want to search for and then presses "Enter".
- **Function:** The system will filter out the corresponding search results.

❖ **Filter borrow digitals:**

- **Button/Navigation:** There is a "Filter" button right next to the search input.
- **Trigger:** When clicked, a dialog will appear. Librarians can then enter the necessary inputs or select options from dialog to refine the search. Finally, clicking "Apply" will execute the query.
- **Function:** The system will filter out the corresponding filter results.

❖ **Sort a column:**

- **Button/Navigation:** The table headers with up/down icons indicate sortable columns.
- **Trigger:** When clicked, the action will proceed.
- **Function:** Clicking the header toggles the sort order between ascending and descending. The icon changes accordingly (up = ascending, down = descending).

❖ **Pagination**

- **Button/Navigation:** Simple numbered buttons, "Next" and "Previous" links, at the bottom of the table.
- **Trigger:** Librarian clicks on any of the pagination controls.
- **Function:** Changes the page of the table, allowing the librarians to navigate through multiple entries.

❖ **Adjust page size**

- **Button/Navigation:** A number selection dropdown at the bottom of the screen
- **Trigger:** When clicked, different options will be displayed so that the librarian can choose the page size they want.
- **Function:** The system will display more or less rows per page of the table.

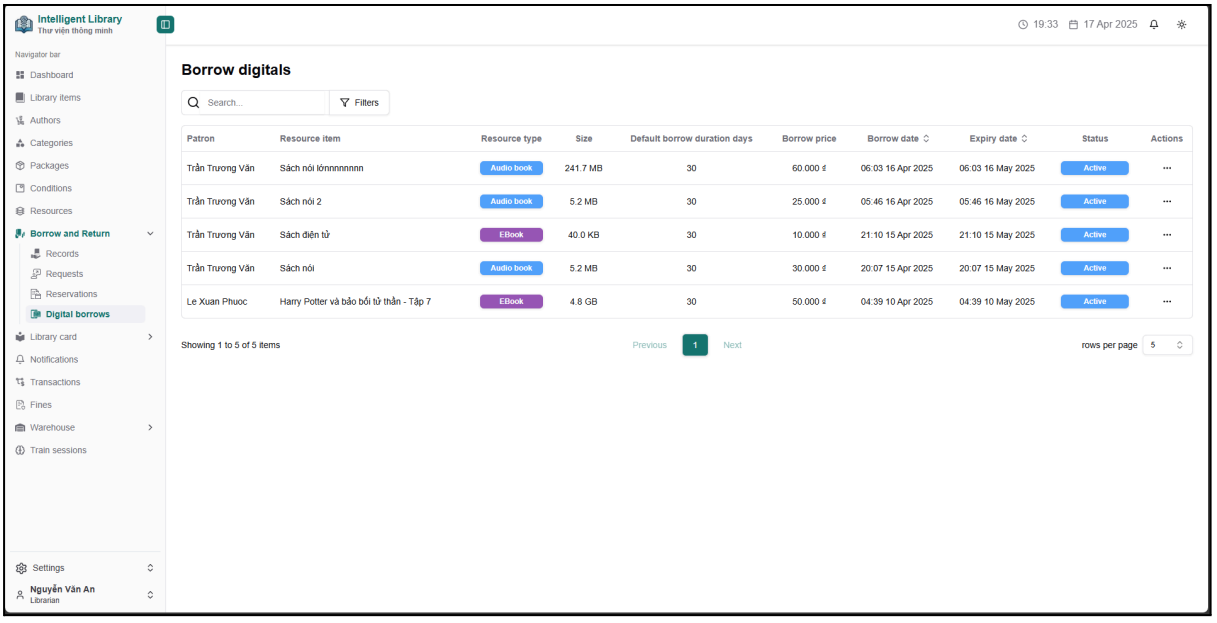


Figure 3.4.29.1: Borrow digitals management

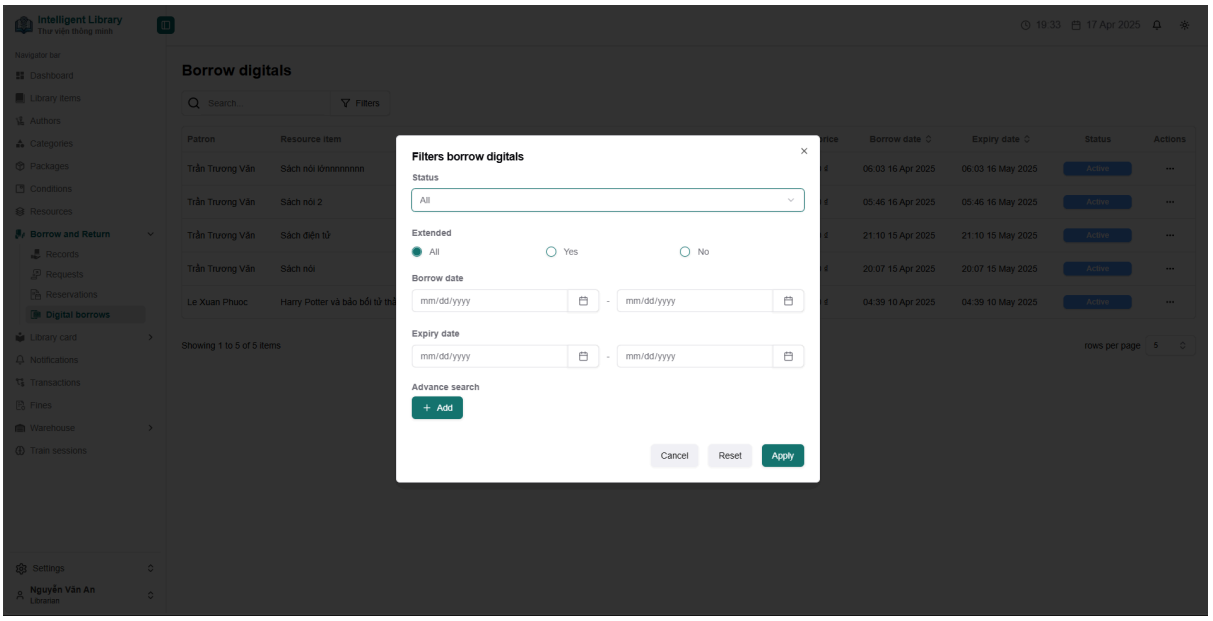


Figure 3.4.29.2: Borrow digitals management

3.4.30 Borrow Digital Detail Management

- **Function Description:** This screen displays a digital borrowing transaction for a patron, including borrow date, expiry date, and status. It shows the resource and transaction details like payment status and a QR code.
- **Function Trigger:**

  - **View Borrow Information:**
    - **Button/Navigation:** The "Borrow Information" section shows details like borrow date, expiry date, extended status, and extension count.
    - **Trigger:** Information is automatically displayed upon accessing the page.
    - **Function:** The system presents the digital borrow's metadata.
  - **View Resource Details:**
    - **Button/Navigation:** The "Resource" section displays fields like resource item, type, size, borrow price, and duration.
    - **Trigger:**Details are automatically displayed upon accessing the page
    - **Function:** The system shows details of the borrowed digital resource.
  - **View Transaction Information:**
    - **Button/Navigation:** The "Transaction" section includes fields like transaction code, amount, status, type, method, and a "View QR Code" button.
    - **Trigger:** Information is automatically displayed, with the QR code accessible via the "View QR Code" button.
    - **Function:** The system provides transaction details

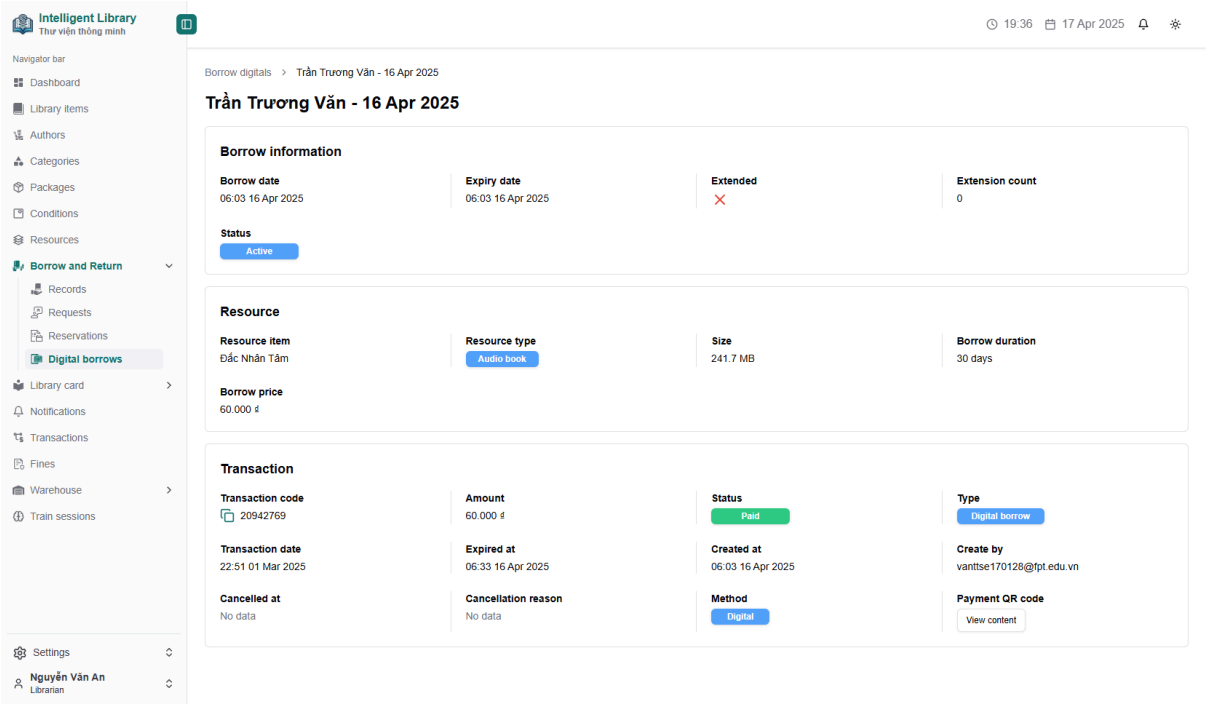


Figure 3.4.30.1: Borrow digital detail management

3.4.31 Reservations Management

- **Function Description:** This screen allows **librarians** to view, search, and manage reservations efficiently. It provides various search and filter methods.
- **Function Trigger:**

  - **Search reservations:**
    - **Button/Navigation:** There is a search input on the top of the screen.
    - **Trigger:** The librarian enters the keyword they want to search for and then presses "Enter".
    - **Function:** The system will filter out the corresponding search results.
  - **Filter reservations:**
    - **Button/Navigation:** There is a "Filter" button right next to the search input.
    - **Trigger:** When clicked, a dialog will appear. Librarians can then enter the necessary inputs or select options from dialog to refine the search. Finally, clicking "Apply" will execute the query.
    - **Function:** The system will filter out the corresponding filter results.
  - **Sort a column:**
    - **Button/Navigation:** The table headers with up/down icons indicate sortable columns.
    - **Trigger:** When clicked, the action will proceed.
    - **Function:** Clicking the header toggles the sort order between ascending and descending. The icon changes accordingly (up = ascending, down = descending).
  - **Pagination**
    - **Button/Navigation:** Simple numbered buttons, "Next" and "Previous" links, at the bottom of the table.

- **Trigger:** Librarian clicks on any of the pagination controls.
- **Function:** Changes the page of the table, allowing the librarians to navigate through multiple entries.

❖ **Adjust page size**

- **Button/Navigation:** A number selection dropdown at the bottom of the screen
- **Trigger:** When clicked, different options will be displayed so that the librarian can choose the page size they want.
- **Function:** The system will display more or less rows per page of the table.

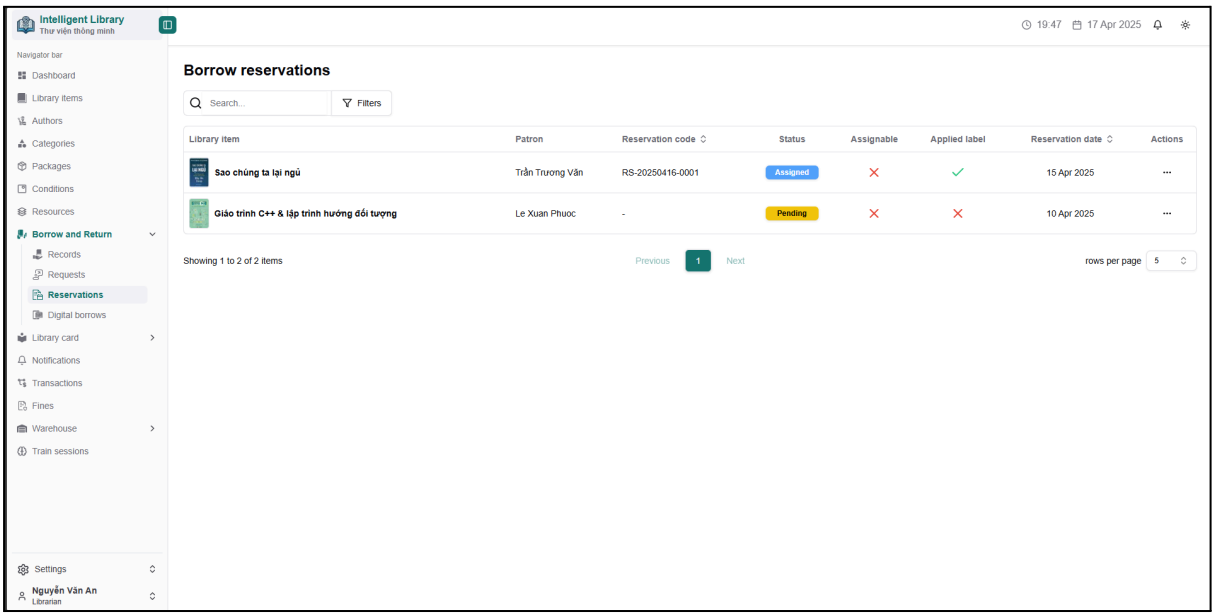


Figure 3.4.31.1: Reservations management

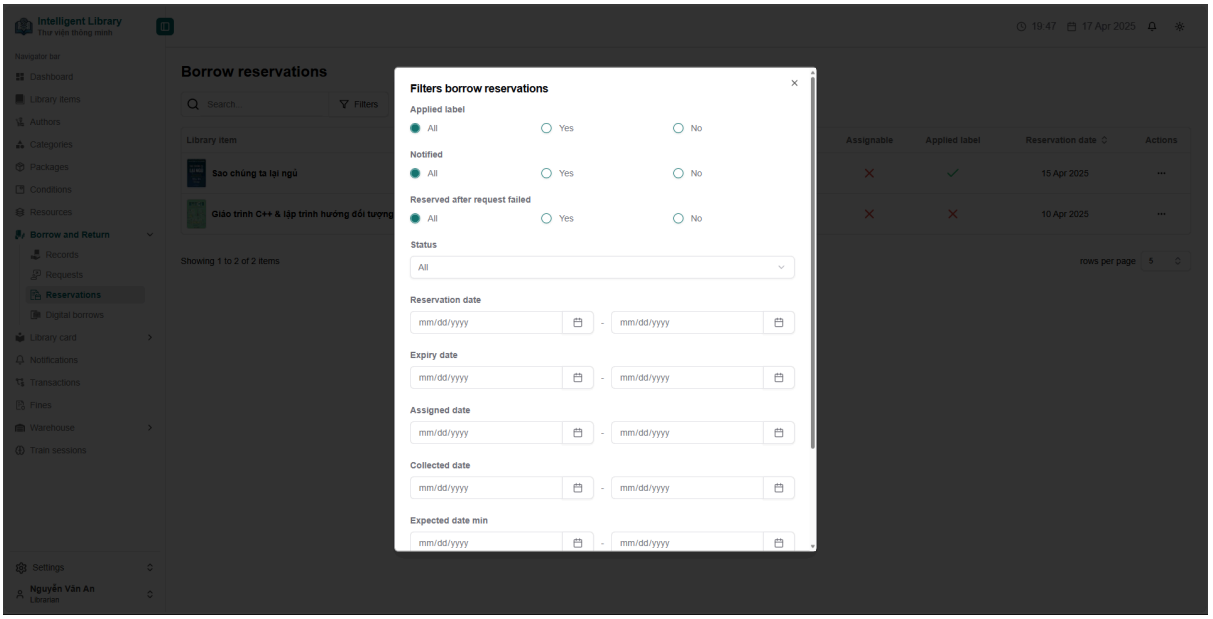


Figure 3.4.31.2: Reservations management

3.4.32 Reservation Detail Management

- **Function Description:** This screen allows **librarians** to view, search, and manage reservations efficiently. It provides various search and filter methods.
- **Function Trigger:**
    - ❖ **View Reservation Information:**
        - **Button/Navigation:** The "Reservation Information" section displays details like reservation date, expiry date, status, and reserved item.
        - **Trigger:** Information is automatically displayed upon accessing the page.
        - **Function:** The system shows the reservation's metadata.
    - ❖ **View Patron Information::**
        - **Button/Navigation:** The "Patron Information" section includes fields like name, avatar, card barcode, and borrowing permissions.
        - **Trigger:** Details are automatically displayed upon accessing the page.
        - **Function:** The system presents patron details.
    - ❖ **Assign Item to Reservation:**
        - **Button/Navigation:** An "Actions" dropdown at the top-right contains an "Assign Item" option.
        - **Trigger:** Clicking "Assign Item" in the dropdown.
        - **Trigger:** Details are automatically displayed upon accessing the page.
        - **Function:** The system allows the librarian to assign a specific library item instance to the reservation.
- **Business Rules:** BR-58

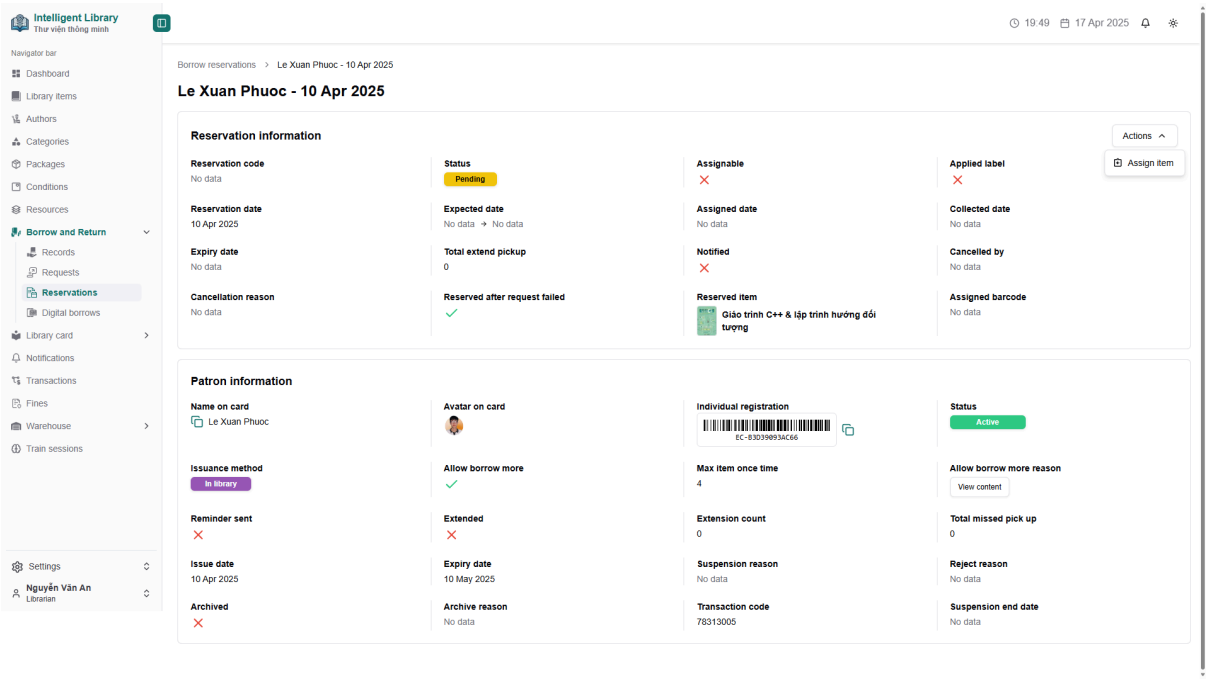


Figure 3.4.32.1: Reservation detail management

3.4.33 Roles Management

- **Function Description:** This screen displays a list of user roles within the library system, each with a description of their permissions. It allows administrators to search, sort, create, edit, and delete roles.

- **Function Trigger:**

  - **View Roles List:**
    - **Button/Navigation:** The page displays a grid of roles with their names and descriptions..
    - **Trigger:** The list is automatically displayed upon accessing the page.
    - **Function:** The system shows all defined roles.
  - **Search Roles:**
    - **Button/Navigation:** A search bar is located above the roles grid.
    - **Trigger:** Entering keywords into the search bar.
    - **Function:** The system filters the roles list based on the entered keywords, allowing administrators to quickly find specific roles.
  - **Sort Roles:**
    - **Button/Navigation:** A "Sort by: A-Z" dropdown is available above the roles grid, with options "A-Z" and "Z-A.".
    - **Trigger:** Selecting a sorting option from the dropdown.
    - **Function:** The system reorders the roles list alphabetically, either from A to Z or Z to A, based on the selected option.
  - **Create New Role:**
    - **Button/Navigation:** A "Create Role" button is located at the top-right of the page.
    - **Trigger:** Clicking the "Create Role" button
    - **Function:** The system opens a form to define a new role, allowing administrators to specify the role name, and role type, then save it to the list.
  - **Edit Role:**
    - **Button/Navigation:** Each role has an "Edit" option in a dropdown menu.
    - **Trigger:** Clicking "Edit" in the dropdown opens an "Edit Role" dialog with fields for English and Vietnamese names.
    - **Function:** The system allows administrators to modify the role’s details and save the changes.
  - **Delete Role:**
    - **Button/Navigation:** Each role has a "Delete" option in a dropdown menu.
    - **Trigger:** Clicking "Delete" opens a "Delete Role" confirmation dialog, requiring the user to type the role name to confirm.
    - **Function:** The system removes the role from the list upon confirmation, ensuring accidental deletions are prevented through the confirmation step.

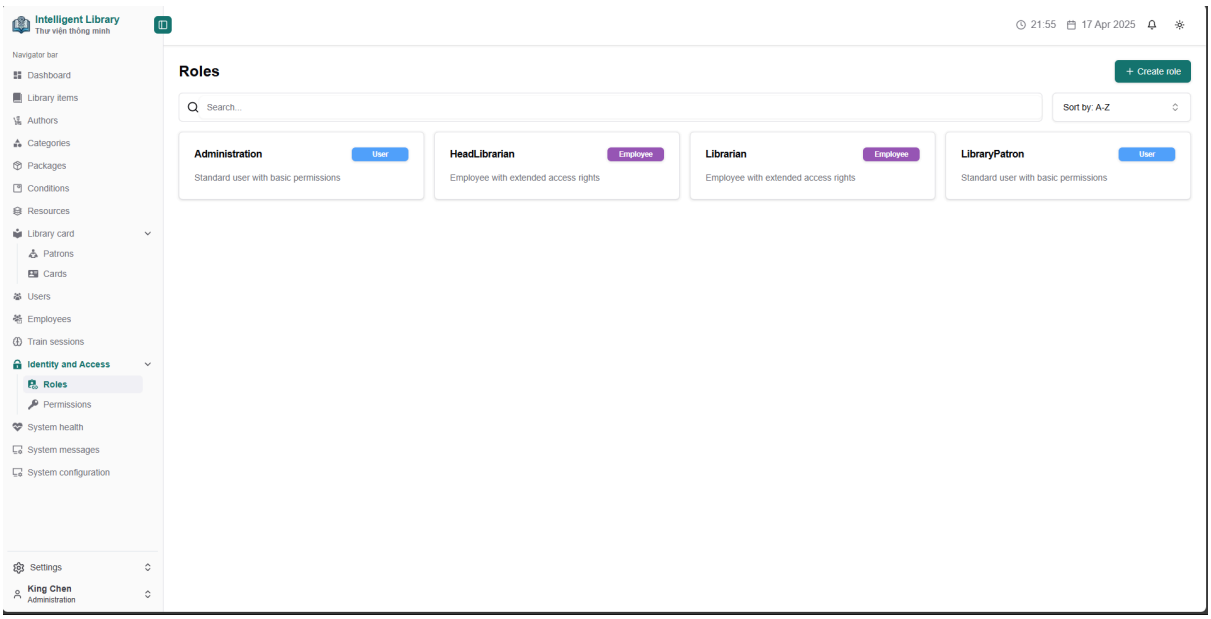


Figure 3.4.33.1: Roles management

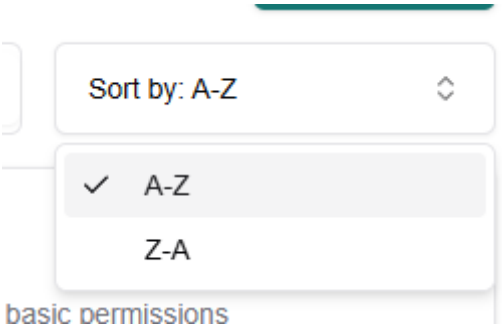


Figure 3.4.33.2: Roles management

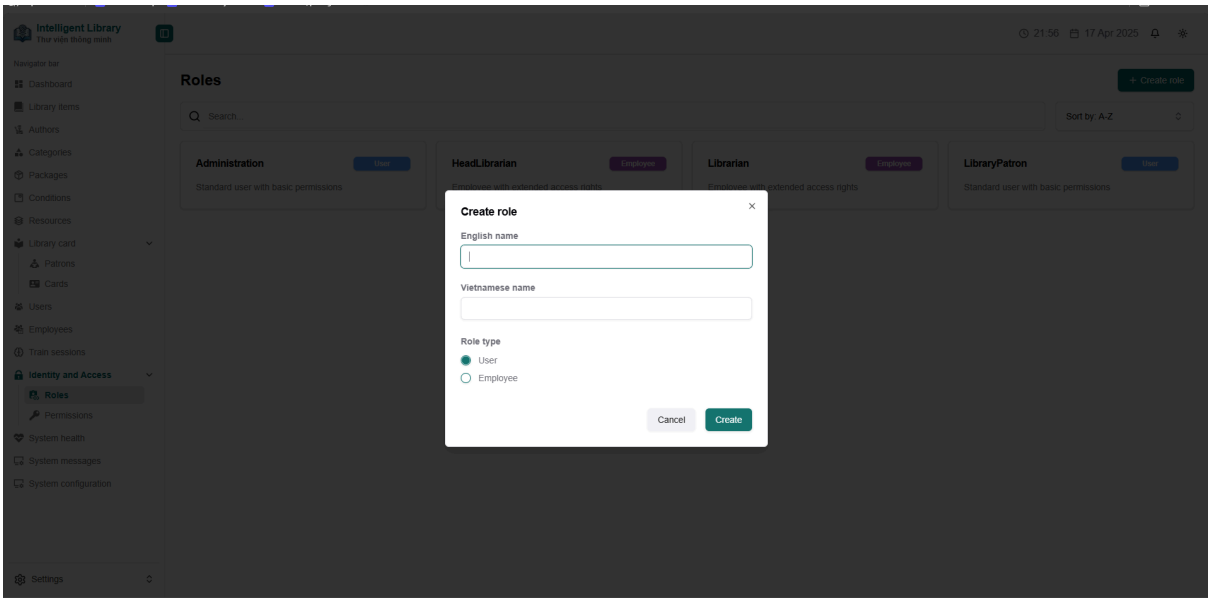


Figure 3.4.33.3: Roles management

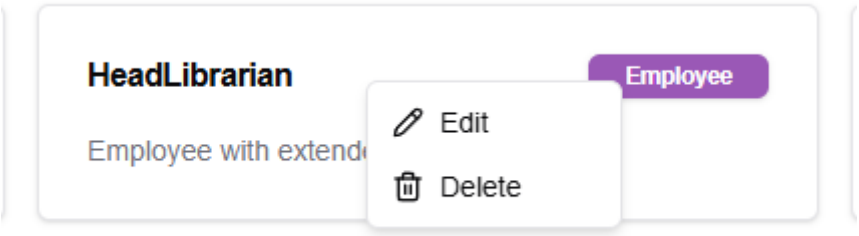


Figure 3.4.33.4: Roles management

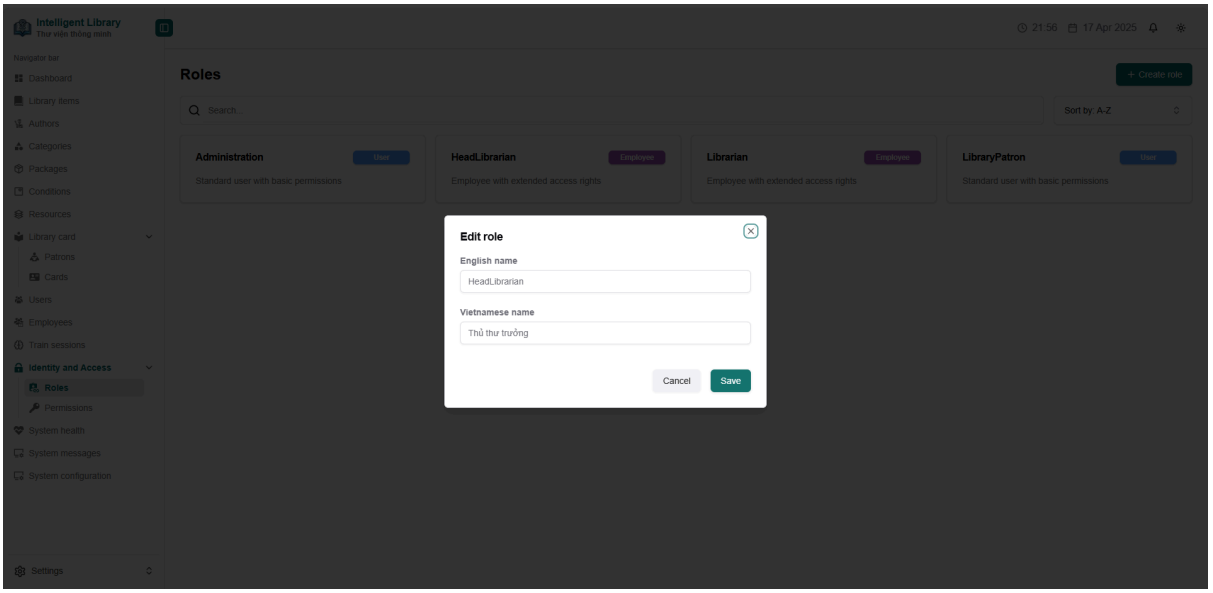


Figure 3.4.33.5: Roles management

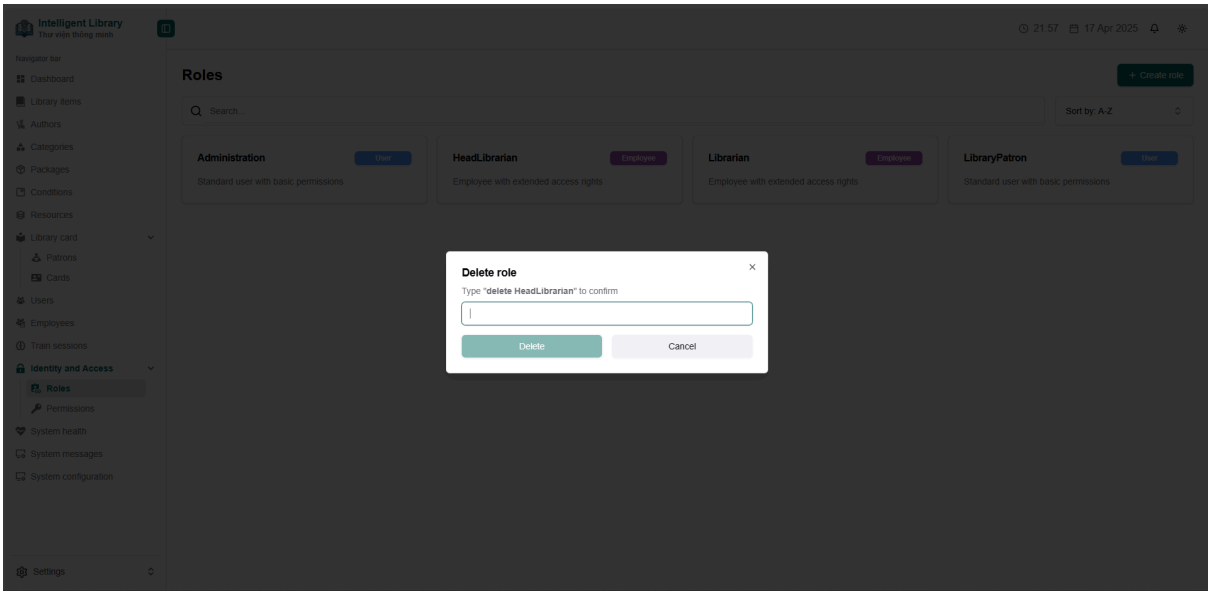


Figure 3.4.33.6: Roles management

3.4.34 Permissions Management

- **Function Description:** This screen displays a table mapping roles to features with their access levels. It allows administrators to toggle the table orientation and modify access levels for each role per feature, ensuring precise control over permissions.
- **Function Trigger:**

  - **View Permissions Table:**
    - **Button/Navigation:** The page displays a table with roles as columns and features as rows, showing access levels
    - **Trigger:** The table is automatically displayed upon accessing the page.
    - **Function:** The system provides an overview of permissions, showing which roles have access to specific features within the library system.
  - **Toggle Table Orientation:**
    - **Button/Navigation:** A "Role Horizontal" dropdown at the top-right offers options "Role Horizontal" and "Role Vertical."
    - **Trigger:** Selecting an option from the dropdown.
    - **Function:** The system switches the table orientation, either displaying roles as columns (horizontal) or rows (vertical), improving readability based on user preference.
  - **Modify Access Levels:**
    - **Button/Navigation:** Each cell in the table shows the access level with a dropdown menu
    - **Trigger:** Clicking on an access level opens the dropdown to select a new level.
    - **Function:** The system updates the permission for the selected role and feature, allowing administrators to grant or restrict access.

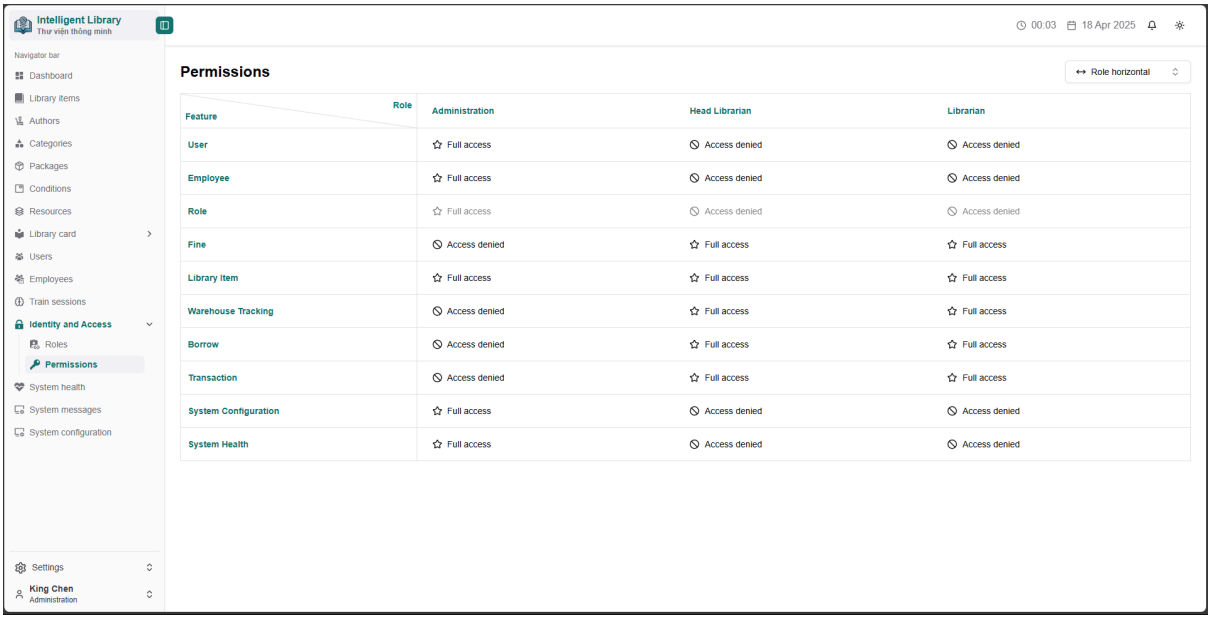


Figure 3.4.34.1: Permission management

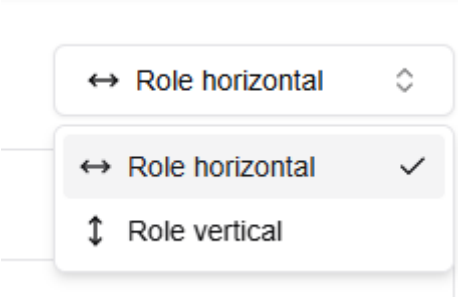


Figure 3.4.34.2: Permission management

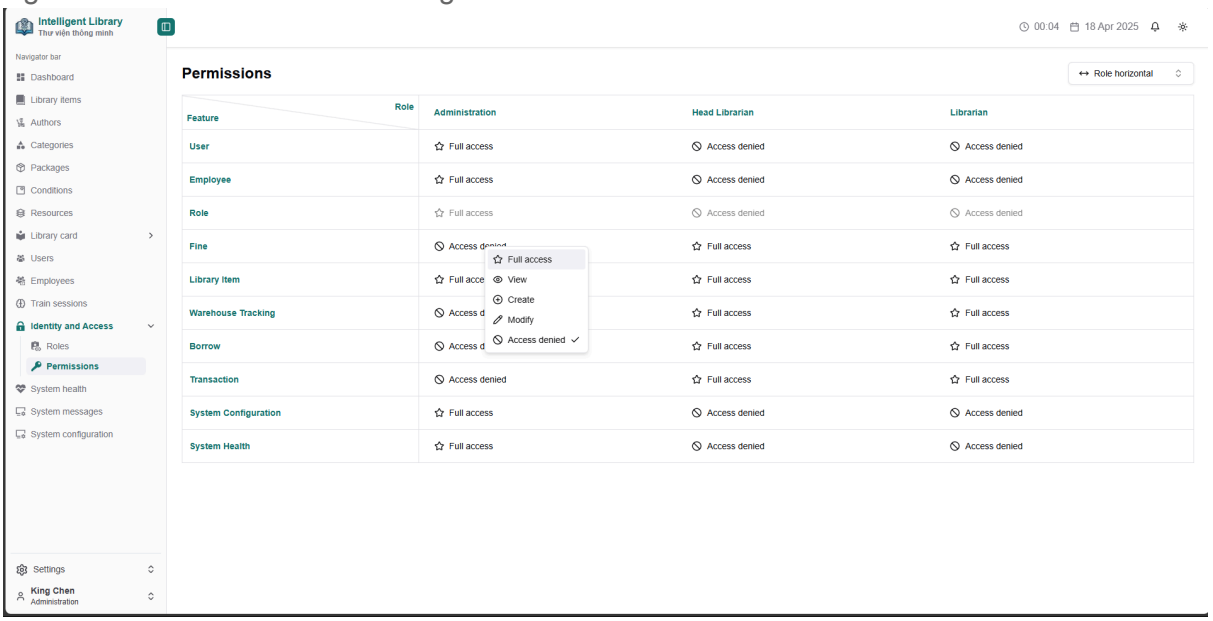


Figure 3.4.34.3: Permission management

3.4.35 Transaction Management

- **Function Description:** Transaction Management page shows the list of transactions within the library system. It allows administrators to search, sort, and filter the transactions.
- **Function Trigger:**

  - **Filter the transactions:**
    - **Button/Navigation:** There is a “Filter" button next to the search input.
    - **Trigger:** The administrators click the “Filter” button to show the filter dialog.
    - **Function:** The system filters the transactions based on the amount, status (pending, expired, paid, cancelled), transaction type (fine, digital borrow, digital extension, card register, card extension), transaction method (cash, digital), transaction date, expiry time, and cancelled time.

  - **View the transaction detail:**
    - **Button/Navigation:** There is a “Filter" button next to the search input.
    - **Trigger:** The administrators click the “Filter” button to show the filter dialog.

- **Function:** The system filters the transactions based on the amount, status (pending, expired, paid, cancelled), transaction type (fine, digital borrow, digital extension, card register, card extension), transaction method (cash, digital), transaction date, expiry time, and cancelled time.

- **Business Rules:** BR-45

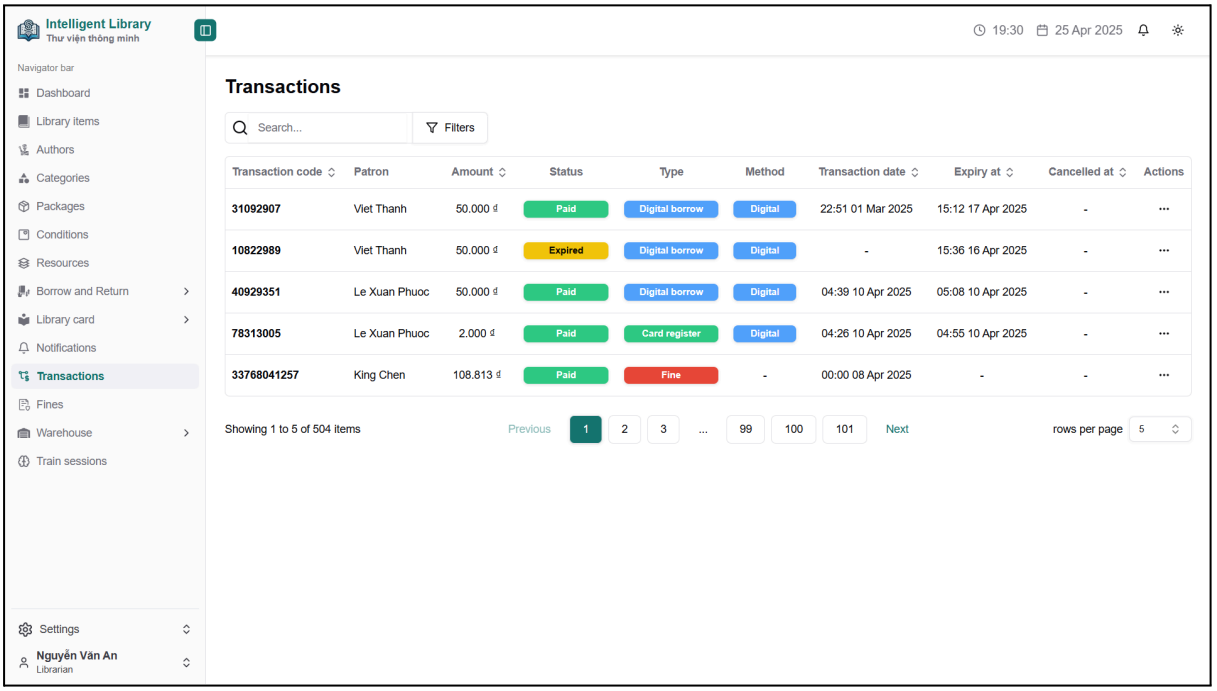


Figure 3.4.35.1: Transaction management

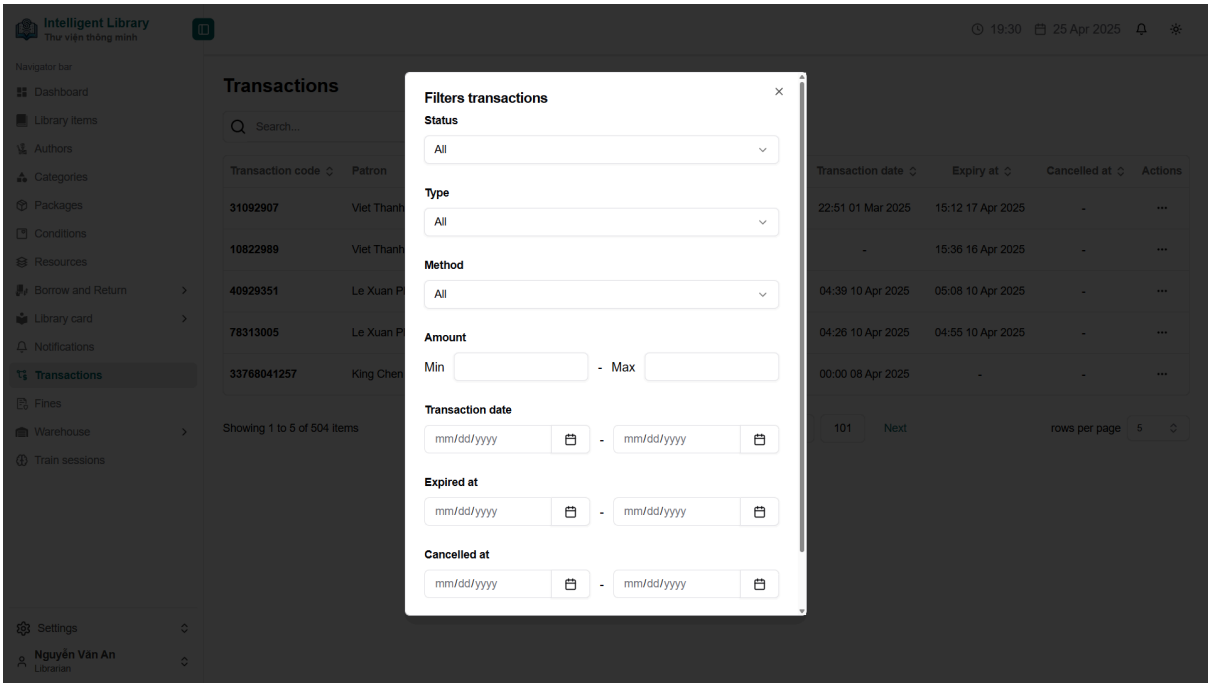


Figure 3.4.35.2: Transaction management

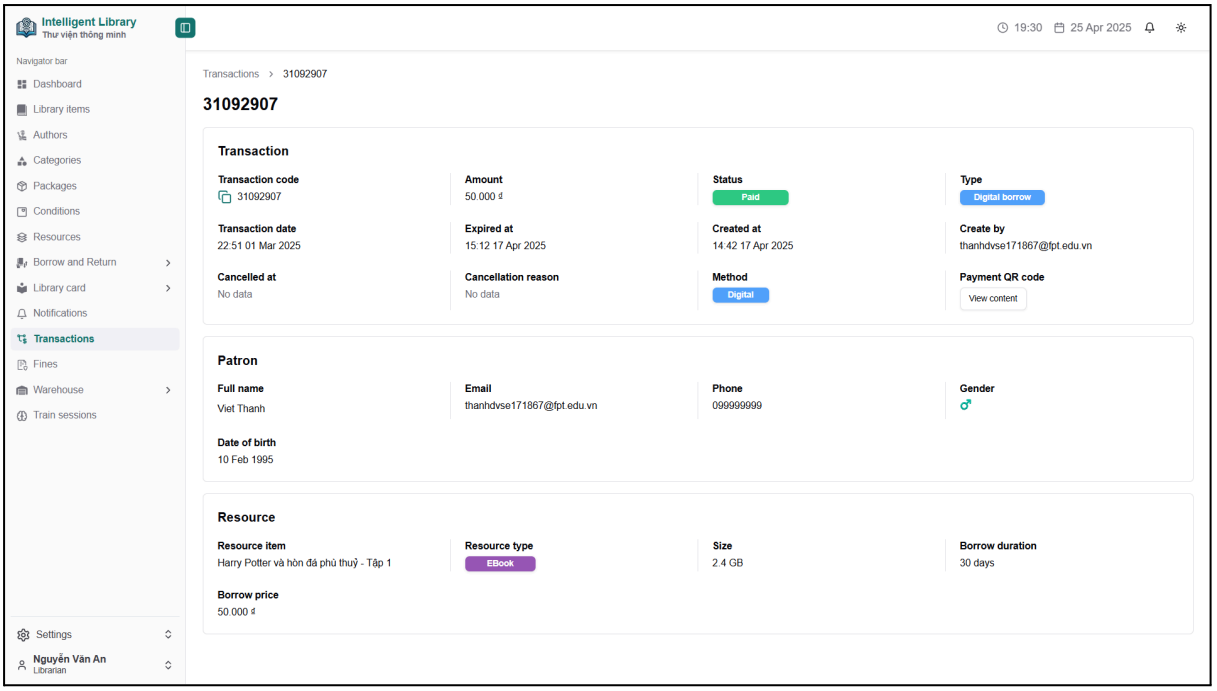


Figure 3.4.35.3: Transaction management

3.4.36 Train Sessions

- **Function Description:** The screen displays the list of trained library items. It allows administrators to search, sort, and filter the trained library items.
- **Function Trigger:**

  - **Navigate to Train AI page:**
    - **Button/Navigation:** There is a “Train AI” button on the top-right corner of the Train sessions page.
    - **Trigger:** The administrator clicks the “Train AI” button.
    - **Function:** The system navigates the administrator to the Train AI page.

  - **Drag and drop the library item image:**
    - **Button/Navigation:** There is a drag and drop zone.
    - **Trigger:** The administrator clicks the drag and drop zone to upload the image they want to train for the library items.
    - **Function:** The administrators upload the library item images to train AI .
- **Function Trigger:** BR-04, BR-46, BR-64, BR-65

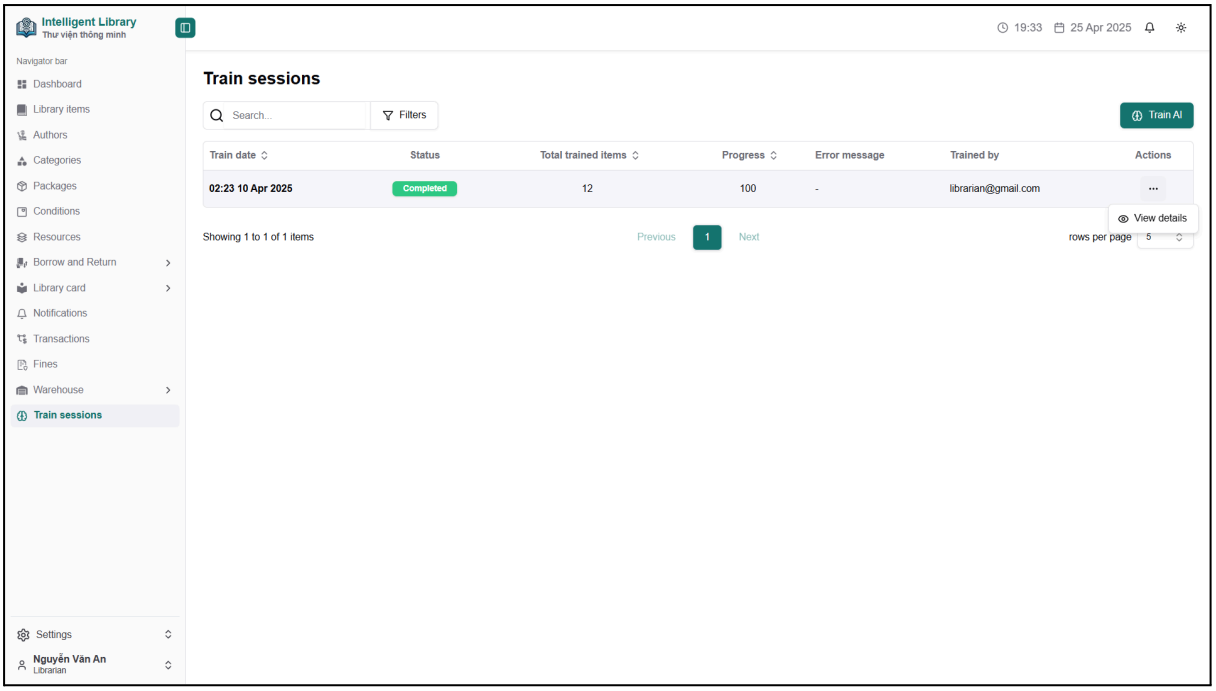


Figure 3.4.36.1: Train sessions

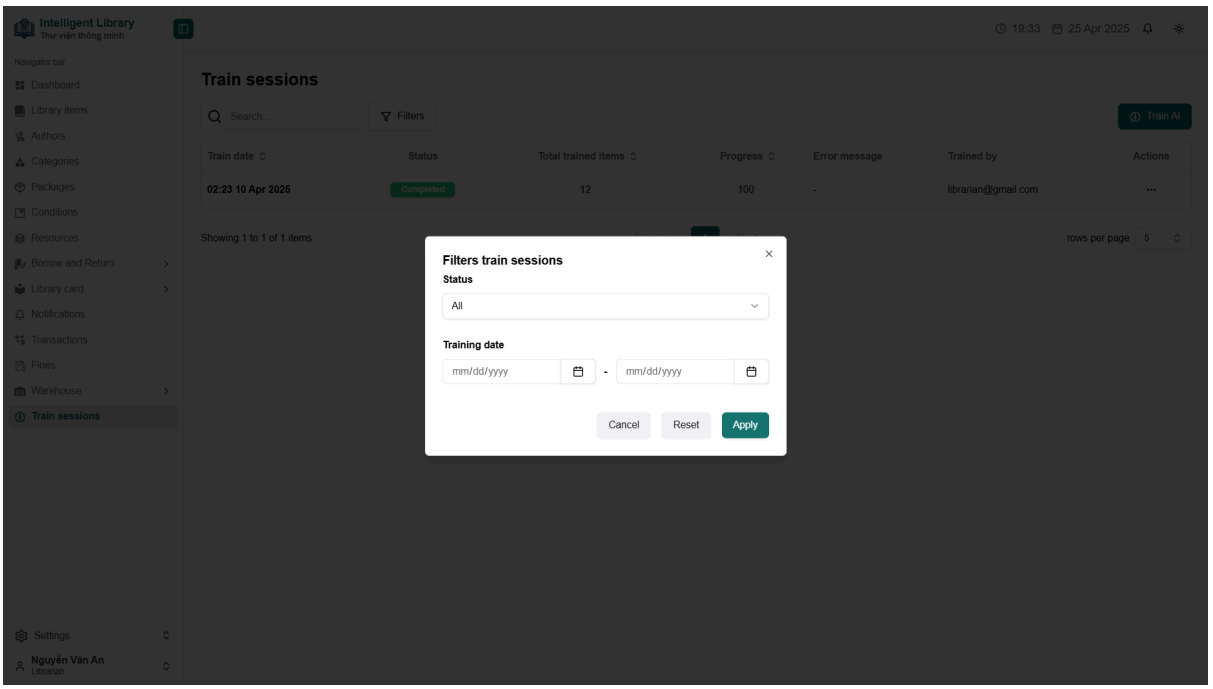


Figure 3.4.36.2: Train sessions

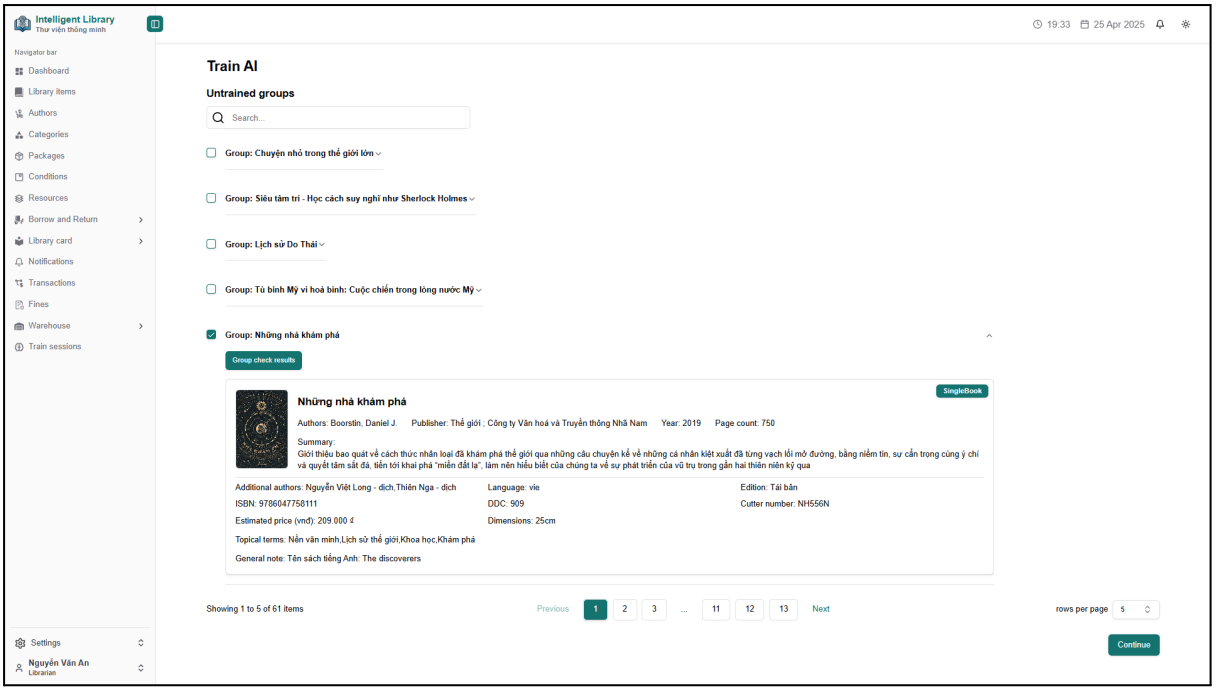


Figure 3.4.36.3: Train sessions

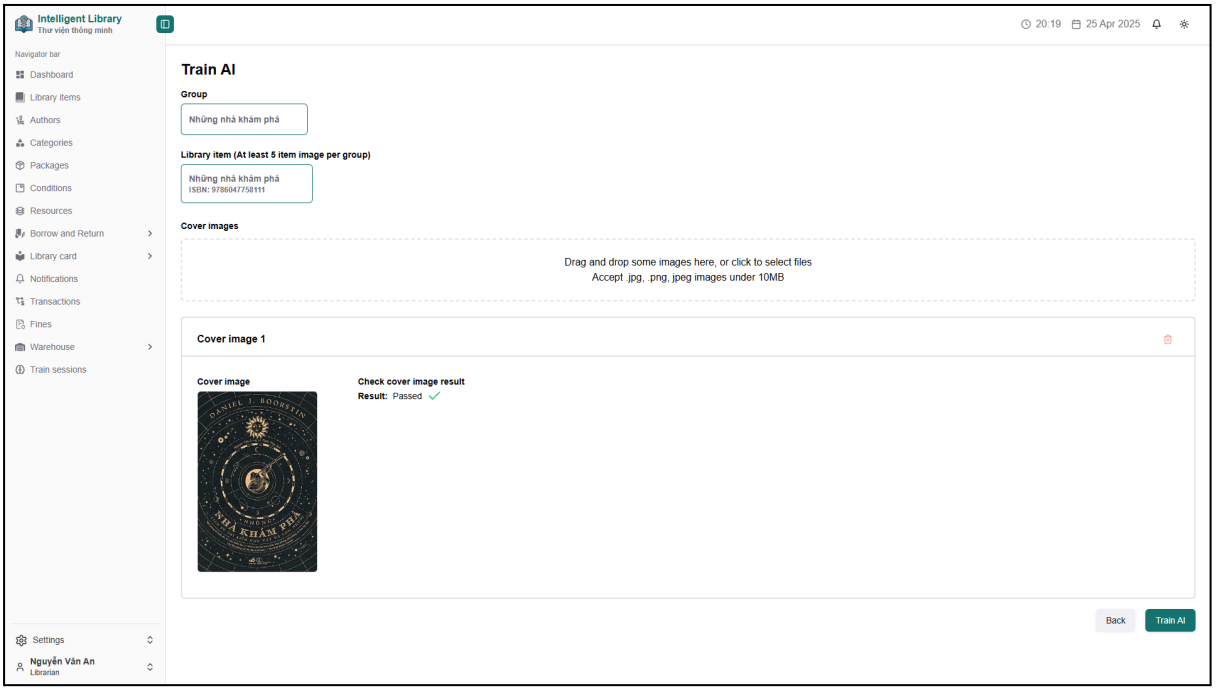


Figure 3.4.36.4: Train sessions

3.4.37 System Configuration

- **Function Description:** This screen allows administrators to manage library system settings across multiple tabs: Ads Script, App Settings, Circulation Settings, and Digital Borrow. It includes configurable fields like ad scripts in different languages, app details, circulation rules, and digital borrowing settings.
- **Function Trigger:**

- ❖ **Manage Ads Script Settings:**
  - **Button/Navigation:** The "Ads Script" tab contains fields for ad scripts in English and Vietnamese.
  - **Trigger:** Editing the text fields in the respective language sections.
  - **Function:** The system updates the advertisement scripts displayed to users based on the entered text.

- ❖ **Configure App Settings:**
  - **Button/Navigation:** The "App Settings" tab includes fields, library barcode prefix, library location, and page size, etc.
  - **Trigger:** Editing the fields
  - **Function:** The system updates general application settings, affecting library operations and display configurations.

- ❖ **Adjust Circulation Settings:**
  - **Button/Navigation:** The "Circulation Settings" tab contains fields like borrow limits, extension periods, and fine expiration days.
  - **Trigger:** Modifying the numeric fields.
  - **Function:** The system updates borrowing rules, ensuring compliance with library policies for loans and returns.

- ❖ **Set Digital Borrow Settings:**
  - **Button/Navigation:** The "Digital Borrow" tab includes fields like minimum minutes to add ads and page load settings.
  - **Trigger:** Editing the fields
  - **Function:** he system adjusts digital borrowing configurations, controlling ad frequency and page load behavior for digital resources.

- ❖ **Save Configuration Changes:**
  - **Button/Navigation:** A "Save Changes" button is available at the bottom of each tab.
  - **Trigger:** Clicking the "Save Changes" button after editing fields.
  - **Function:** The system saves all modified settings, applying them across the library system for consistent operation.

- **Business Rules:** BR-34, BR-35, BR-48, BR-96

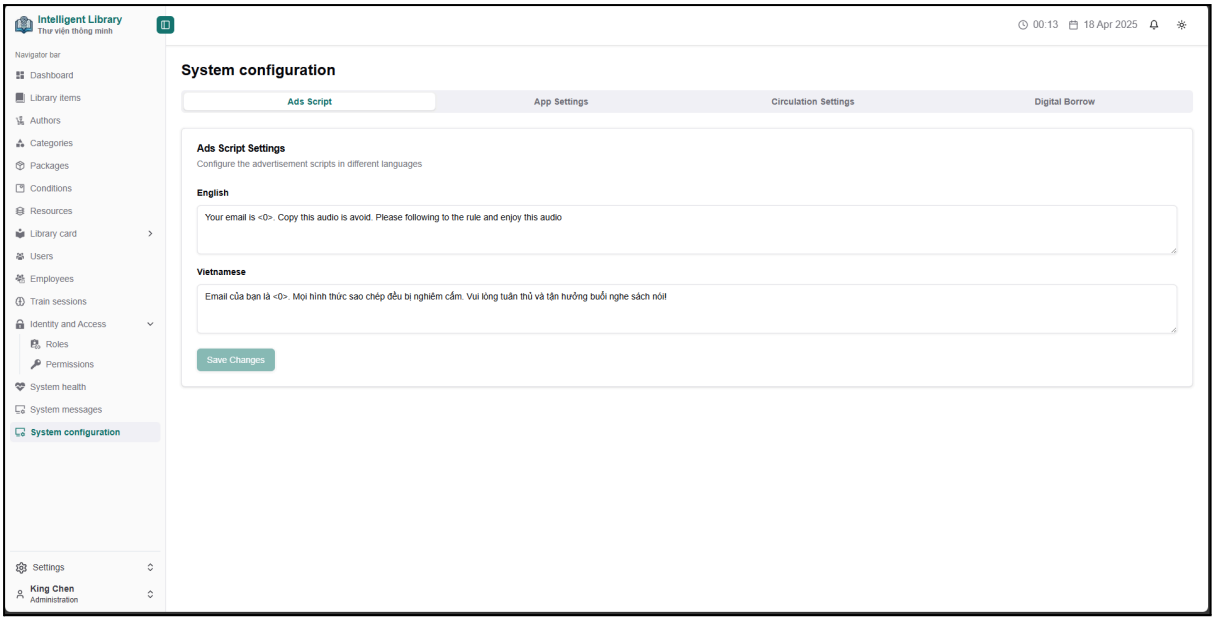


Figure 3.4.37.1: System configuration

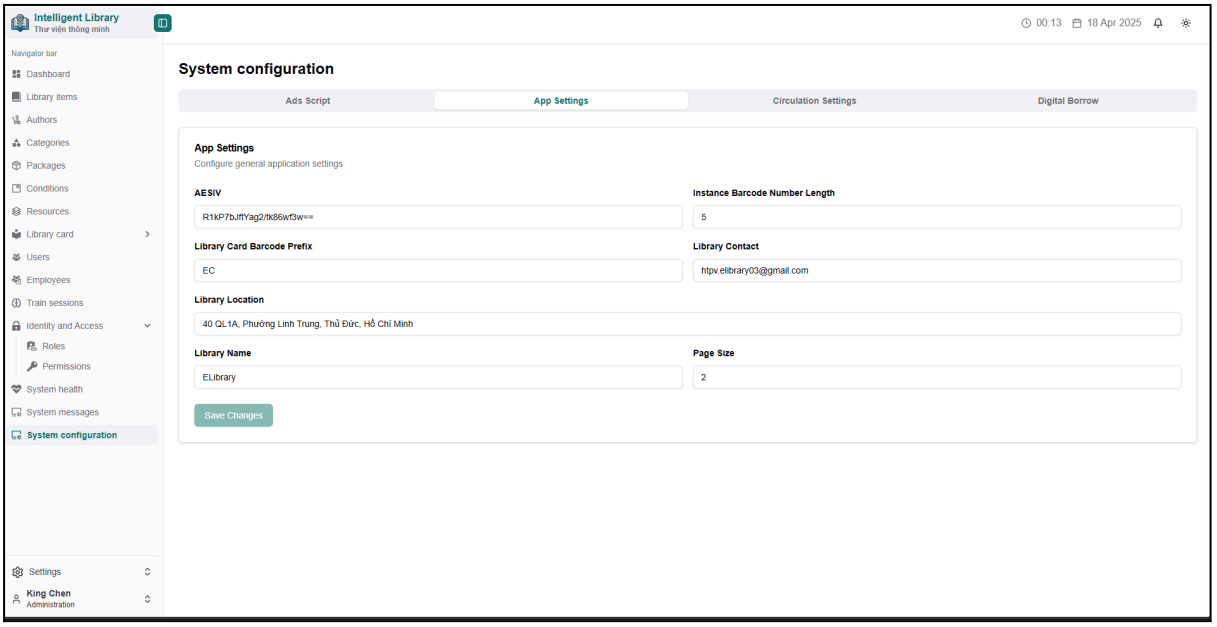


Figure 3.4.37.2: System configuration

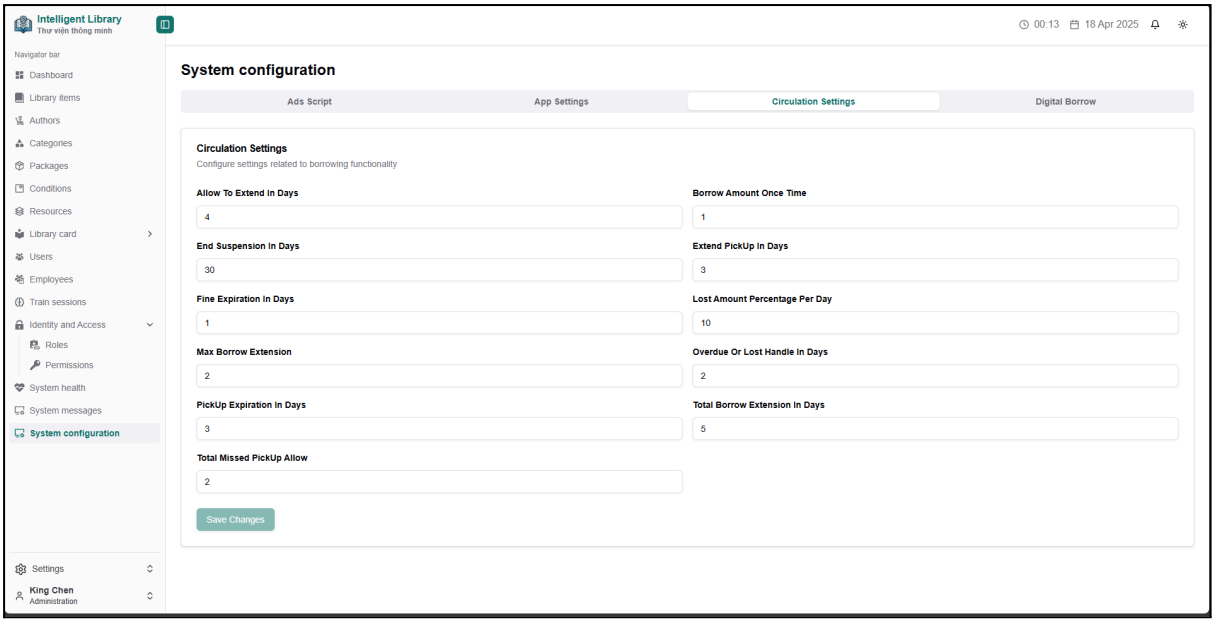


Figure 3.4.37.3: System configuration

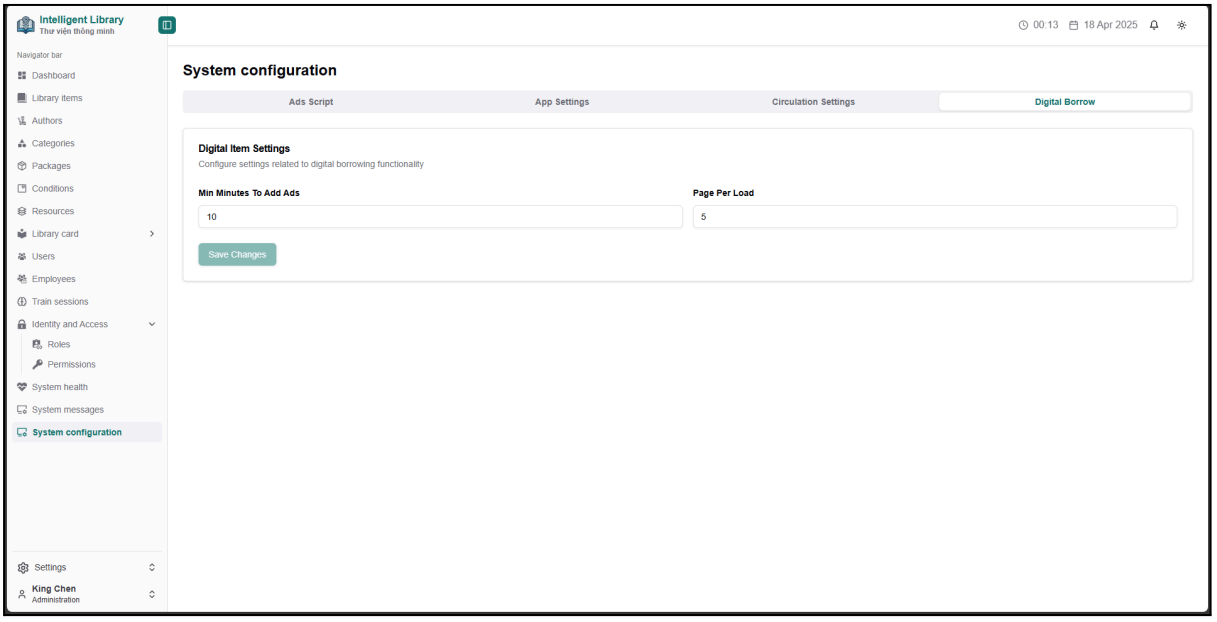


Figure 3.4.37.4: System configuration

4. Non-Functional Requirements

4.1 External Interfaces

This section defines how ELS will interact with users and other system elements:

- **Web Browsers:**
    - The web application must be accessible across multiple browsers and operating systems, ensuring consistent functionality.
    - Interfaces should include remote borrowing/reservation features and display the 2D indoor navigation map effectively.

- **Mobile Devices:**
    - The ELS mobile application must integrate with smartphone hardware (such as cameras) to support image-based book searches.
- **Library Infrastructure Systems:**
    - ELS will connect with existing library databases to manage inventory, user data, and book reservation/borrowing operations.
- **External Systems**
    - **Payment Gateways (PayOS):** The system will integrate with external payment gateways (e.g., PayOS) to handle any payment-related operations. Moreover, all communications with payment systems must be secured and adhere to standard security protocols (e.g., HTTPS, TLS).
    - **Mapping Services Integration (Woosmap):** The system will incorporate integration with external mapping services such as Woosmap, allowing their UI components to be embedded directly within the ELS for improved indoor navigation.

4.2 Quality Attributes

This section ensures the ELS is intuitive and easy to learn for various user profiles:

4.2.1 Usability

- **Training Time:**
    - **Normal Users (Patrons):** A new user can become productive with basic search, navigation, and borrowing functions within 30 minutes of initial use.
    - **Power Users (Library Staff):** Library staff should be able to become proficient in managing tasks (e.g., book inventory management, data analytics dashboard) within 2-3 days of hands-on usage. This proficiency should be supported by an easy-to-follow training manual and visual tutorials.
- **Familiarity and Standards:**
    - The system’s interface design will adhere to common usability standards such as Microsoft's GUI standards and modern UX principles. This approach will emphasize ease of use, clarity, and intuitive workflows, ensuring that both new and experienced users can navigate the system with confidence.
- **Task Efficiency:**
    - **Book Search and Navigation:** Patrons should be able to locate a desired book using text or image-based search and receive directions via the 2D indoor navigation feature within 2–3 minutes on average.
    - **Remote Borrowing/Reservation:** The process for reserving or borrowing a book remotely should be streamlined so that users can complete the transaction in under 3 minutes.

4.2.2 Reliability

- **Availability:** ELS should be available for at least 98% of its operational hours, ensuring that remote borrowing, reservations, and indoor navigation functions are accessible when needed.
- **Mean Time Between Failures (MTBF):** The system is expected to operate continuously for a minimum of 12 months on average before a major failure occurs, reflecting a robust design.
- **Mean Time To Repair (MTTR):**
    - For critical issues - such as failures affecting core functionality (e.g., the advanced search, indoor navigation, or remote reservation functions) - the system must be restored within 3-4 hours.
    - Non-critical issues should be resolved within 24 hours.
- **Accuracy**:
    - The accuracy of key functions - such as image-based search results, and indoor navigation mapping - must meet a 98% precision target, ensuring that users receive highly relevant and correct information in real time.
- **Bug and Defect Rate:**
    - The acceptable limit is set at no more than 5 bugs per 1,000 lines of code (KLOC) for critical functions (e.g., process remote transactions, or correctly display navigation).

4.2.3 Performance

- **Response Time:**
    - Indoor navigation map updates must render within 2–3 seconds.
    - Text-based and AI image-based searches should return results within 30 seconds on average.
    - Remote borrowing and reservation transactions should complete within 3 seconds to ensure a smooth user experience.
- **Throughput**:
    - The system should support up to 500 concurrent users across both web and mobile platforms without any noticeable degradation.
    - During peak usage periods, ELS should be capable of handling at least 50 transactions per second, where a transaction may include a search query, a navigation update, or a remote borrowing/reservation action.
- **Capacity**:
    - **User Accounts:** ELS will be designed to support up to 50,000 registered patron accounts.
    - **Daily Transactions:** The system should efficiently manage up to 5,000 daily transactions, including searches, navigation requests and remote borrowing/reservation operations.

- **Resource Utilization:**
    - **Memory Usage:** The application should not exceed 80% of available system memory during normal operations.
    - **Disk Usage:** The system must use disk space efficiently, and maintain logs to prevent excessive consumption of disk space.
    - **Communication:** Bandwidth utilization should remain controlled, ensuring that the system's network load does not exceed 75% of total capacity during peak usage.

5. Requirement Appendix

5.1 Business Rules

Table 5.1 : Business Rules

ID	Rule Definition
BR-01	The recent read section shows a list of 10 books accessed by the user recently.
BR-02	The system will track cases where the user's borrow request fails as a basis for future book purchases.
BR-03	The system needs an available map to help the user track the location of library items.
BR-04	Each book in Training Book AI needs 5 pictures.
BR-05	Library patrons can modify their personal information.
BR-06	Books will be arranged on the shelf according to the DDC (Dewey Decimal Classification) system.
BR-07	Users must register for a library card via a subscription package from the system before borrowing any resources.
BR-08	Users must wait for the librarian to approve the card.
BR-09	If a guest wants to borrow audiobooks or ebooks, they must log in with a user account before proceeding with borrowing the resources.
BR-10	The ebook version will be displayed as a flipbook to enhance the user experience, with a watermark containing user information at the bottom of each page to improve security and reduce ebook piracy.
BR-11	The audiobook version will include an additional MP3 file to strengthen copyright protection and security of the resource.

BR-12	If a user wants to borrow more than three items, they must provide a reason when filling out the form for the librarian's approval.
BR-13	Users can add library items to the wish list after requesting a borrow fail.
BR-14	Materials can only be renewed when there are 4 days left before the return deadline.
BR-15	Renewals cannot be done twice consecutively, even if there is a reservation for the item (this means that even if someone has reserved the item, it can still be renewed once).
BR-16	Overdue materials cannot be extended.
BR-17	Users can change languages between Vietnamese and English.
BR-18	If users have not paid a fine, they will not be able to borrow the book.
BR-19	If the user borrows late twice, their library card will be suspended for 1 month.
BR-20	If a user successfully places an order to borrow a book but does not pick it up on time, the book will be prevented from borrowing or reserving for 1 month.
BR-21	Library patrons must pay to borrow digital resources.
BR-22	Library patrons have a specific time to access their paid digital resource.
BR-23	Library patrons could repay to extend their digital borrowing.
BR-24	When a library card is lost, the library patron must tell the librarian to deactivate and archive that lost card.
BR-25	If there is a leaked resource, caught library patrons would be banned.
BR-26	Library patrons can rate and leave comments on books after borrowing them.
BR-27	Library patrons can report damaged or incorrect information about books to the library.
BR-28	If a book is lost, library patrons must compensate the library based on the value of the item.
BR-29	Borrowed items will be locked if they are overdue and not renewed or returned.

BR-30	All items will be checked before being borrowed to ensure they are complete and in good condition.
BR-31	If returned items do not have good condition status, library patrons have to pay a fee based on their condition.
BR-32	The system will provide advanced search tools for users.
BR-33	Library patrons could find the relative book through the cover image of another book.
BR-34	Admin could have an overview of the system health.
BR-35	Admin could change the number of books shown on the website.
BR-36	Every successful transaction of borrowing a digital book would be sent to the library patron’s email.
BR-37	Library patrons also use the application on their phones.
BR-38	All users could choose a light or dark background.
BR-39	Library patrons could order books if those books are not available.
BR-40	Users could choose multi-factor authentication.
BR-41	Librarians could relocate books to other bookshelves.
BR-42	Admin must login by multi factor authentication.
BR-43	Library patrons are able to check their borrowed history.
BR-44	Library patrons will see recommended books that are related to their favorite books.
BR-45	The system will automatically change the transaction status if library patrons did not pay on time.
BR-46	Librarians could track the training process.
BR-47	Admin could change the information of the library.
BR-48	Admin could configure for borrowing duration shorter or longer.

BR-49	Librarians could see statistics about the income and best favorite book from the user.
BR-50	Library patrons could purchase a library card using cash or online payment methods like PayOS.
BR-51	Librarians could scan the barcode on the book to check the status of the book.
BR-52	A book's availability status (available or on hold) must be updated in real-time after every transaction.
BR-53	When library patrons check out digital resources, the patrons can view the record of the time of borrowing and expiration.
BR-54	The system will track library patron preferences to recommend books based on past borrowing habits.
BR-55	Library patrons will be notified when a book they reserved becomes available for pick-up.
BR-56	All digital books will expire automatically after the borrowing period ends and will be inaccessible.
BR-57	The system will automatically generate and email the overdue notices to library patrons.
BR-58	The system will allow library patrons to cancel their reservation request for a book if they no longer need it.
BR-59	If a library patron returns a book after the due date, a fine will be calculated automatically based on the number of days late.
BR-60	The library will allow library patrons to borrow physical books only during the working hours of the library.
BR-61	A book’s availability for borrowing will be checked before placing an order to ensure it is in stock.
BR-62	Library patrons can report bugs or errors in the system through a designated support page.
BR-63	Library patrons can cancel or modify their borrowing request for a book if it has not yet been processed.
BR-64	Librarians could not train more when one training session is in progress.
BR-65	Librarians must give the right images of the book to train.
BR-66	Library patrons could see matched statistics of the AI prediction.
BR-67	Librarians must give book details based on MARC-21 metadata.

BR-68	Librarians could send private notifications for a group of library patrons.
BR-69	Library patrons are allowed to sign in with a Google account.
BR-70	The system will automatically suggest possible search terms if the entered keywords are unclear or misspelled.
BR-71	Librarians can create, update, and delete catalog entries for books, including assigning new editions or media formats (e.g., digital audiobook).
BR-72	Administrators must approve and verify all librarian and staff account creations before granting access to sensitive areas of the system.
BR-73	The system will automatically log users out after 30 minutes of inactivity to prevent unauthorized access.
BR-74	Librarians must scan the barcode on each borrowed or returned item to update its availability status in real time.
BR-75	Warehouse tracking records must be updated when physical items are added or removed from the system to maintain accurate stock information.
BR-76	All physical items are required to have a unique barcode that can be scanned for inventory management purposes.
BR-77	The library will send email reminders about upcoming events that a user has registered for, including detailed schedules and any requirements.
BR-78	Users can search library items by filling out the search bar with 3 options of search features like quick search, basic search, and advanced search.
BR-79	Users can search the library by recording their voice with 2 types of languages: Vietnamese and English. The result of voice search is based on the user's pronunciation.
BR-80	Users can use AI prediction to hint at the library items they want.
BR-81	To create a borrow request, the library patrons must make sure that the borrowed list doesn't include their already borrowed library items, or already reserved library items, or already requested library items.
BR-82	The library patrons also can reserve library items that are in an unavailable status.
BR-83	The library patrons can observe their borrow history on their profile page.
BR-84	The borrow history page displayed three tabs: borrow request, borrow record, and borrow digital showing borrow history.
BR-85	Only the library patrons can borrow and reserve books. Guests may search and view limited information but cannot borrow or reserve.

BR-86	The library patrons may not request to borrow a book if all available copies are already borrowed, they would reserve for it instead.
BR-87	Borrow requests and assigned item reservations are valid for a set period (e.g., 3 days). If the patron does not pick up the book within the timeframe, the book becomes available for other patrons.
BR-88	Overdue fines are automatically calculated based on the overdue period and the daily fine rate set by the library's policy.
BR-89	Books returns are processed in real time. Once a book is returned, any associated overdue fines are added to the library patron's account.
BR-90	The book recommendation feature only displays books that match a user's interests based on relevant metadata (title, author, genres, topical terms, etc.) and usage data (borrowed, reserved history, favorites list, etc.).
BR-91	Image-based searches (via captured book cover) must adhere to accuracy thresholds, returning a list of potential matches sorted by relevance.
BR-92	Personalized recommendations for guest users may receive popular or trending books.
BR-93	The inventory records must be accurate when librarians update book status (available, checked out, reserved, lost, etc.).
BR-94	All fine payments or fees are handled through a secure payment gateway (e.g., PayOS).
BR-95	The system's analytics dashboard provides insights (e.g., popular books, top borrow failed books, borrow or return trends, etc).
BR-96	The system allows librarians to filter for a specific timeframe to retrieve, calculate and aggregate the data in the dashboard.
BR-97	Newly borrowed items must be displayed real-time in the dashboard.
BR-98	The system displays a list of items that are overdue, including borrower details, due dates for process impose fines in the dashboard.
BR-99	The system shows financial insights related to library operations, such as fines collected, library card registers, digital borrows, or other transactions in the dashboard.
BR-100	The system displays all reservation requests that need to be handled in the dashboard.
BR-101	The system can identify the library books with the highest circulation rates (e.g., reservation, borrow failed rates) to assist the librarian in making decisions about whether to acquire additional copies or new books related to popular items.
BR-102	Librarians can generate reports (e.g., stock-in, supplement requests) for specific library operations timeline.

5.2 Application Messages List

Table 5.2 : Application Messages List

No	Message Code	Message Type	Context	Content
1	AIService.Success0002	Toast message	AI training	Train thành công
2	AIService.Success0004	Toast message	AI prediction	Dự đoán thành công
3	AIService.Warning0001	In red, below image preview	Image validation	Dữ liệu không trùng khớp với ảnh: <0>
4	AIService.Warning0003	In red, below image preview	Book detection	Không xác định quyển sách nào trong ảnh
5	Auth.Success0002	Toast message	Login	Đăng nhập thành công
6	Auth.Warning0007	In red, under login form	Login error	Sai tài khoản hoặc mật khẩu
7	Auth.Warning0005	In red, under OTP field	OTP validation	Mã OTP không đúng, vui lòng gửi lại
8	Auth.Warning0006	In red, under email field	Email validation	Email đã tồn tại
9	Auth.Warning0008	Toast message	Email not verified	Tài khoản chưa xác thực email
10	Auth.Warning0010	In red, under OTP login	2FA required	Yêu cầu xác thực 2 lớp
11	Borrow.Fail0001	Toast message	Borrow registration	Có lỗi xảy ra, đăng ký mượn tài liệu thất bại
12	Borrow.Fail0005	Toast message	Borrow extension	Gia hạn mượn tài liệu điện tử thất bại vì <0>
13	Borrow.Success0001	Toast message	Borrow success	Tổng <0> tài liệu đã được đăng ký mượn thành công
14	Borrow.Warning0002	Toast message	Book availability check	Số lượng tài liệu đã được mượn hết

15	Borrow.Warning0003	Toast message	Duplicate borrow validation	Không được mượn trùng tài liệu. Mỗi lần mượn chỉ được lấy một bản
16	Borrow.Warning0005	Toast message	Max borrow limit	Tối đa <0> tài liệu cùng một lúc. Hiện tại, bạn chỉ có thể mượn thêm <1>
17	Cloud.Fail0001	Toast message	Image upload	Tải hình thất bại
18	Cloud.Success0001	Toast message	Image upload	Tải hình thành công
19	Cloud.Warning0001	In red, near image viewer	Image not found	Không tìm thấy hình
20	File.Warning0001	In red, near file input	File upload error	Tệp đã tải lên không được hỗ trợ
21	Fine.Fail0001	Toast message	Fine payment	Thanh toán phí phạt thất bại vì <0>
22	Fine.Success0001	Toast message	Fine payment	Thanh toán <0> phí phạt thành công
23	Fine.Warning0001	Toast message	Unpaid fine warning	Bạn còn <0> phí phạt chưa thanh toán. Vui lòng thành toán để tiếp tục
24	LibraryCard.Fail0001	Toast message	Library card registration	Đăng ký thẻ thư viện thất bại
25	LibraryCard.Success0002	Toast message	Library card registration	Đăng ký thẻ thư viện thành công
26	LibraryCard.Warning0001	Toast message	Activation required	Thẻ thư viện chưa được kích hoạt. Vui lòng liên hệ thư viện
27	LibraryCard.Warning0002	Toast message	Card expired	Thẻ thư viện đã hết hạn. Vui lòng gia hạn
28	LibraryCard.Warning0004	Toast message	Card required for service	Bạn cần có thẻ thư viện để sử dụng dịch vụ này
29	LibraryItem.Fail0003	Toast message	Shelf action	Xếp kệ cho tài liệu thất bại
30	LibraryItem.Success0001	Toast message	Shelf action	Tài liệu đã được xếp kệ thành công

31	LibraryItem.Warning0007	In red, under ISBN field	ISBN duplication	Mã ISBN <0> đã tồn tại
32	LibraryItem.Warning0005	In red, under ĐKCB field	ĐKCB duplication	Số đăng ký cá biệt <0> đã tồn tại
33	LibraryItem.Warning0024	Toast message	Pending request check	Hiện đang có <0> yêu cầu đặt mượn chưa được xử lí
34	Notification.Fail 0001	Toast message	Notification sending	Gửi thông báo thất bại
35	Notification.Success 0001	Toast message	Notification sending	Gửi thông báo thành công
36	Reservation.Fail0001	Toast message	Reservation status update	Bản sao đã được bạn đọc đặt mượn, nhưng không cập nhật được trạng thái
37	Reservation.Warning0001	Toast message	No copy found	Không tìm thấy bản sao phù hợp để xử lý
38	Employee.Warning0001	In red, under employee ID input	Duplicate ID	Mã nhân viên đã tồn tại
39	Author.Warning0001	In red, under author ID input	Duplicate author ID	Mã tác giả đã tồn tại
40	Auth.Fail0001	Toast message	Password change	Thay đổi mật khẩu thất bại
41	Auth.Success0006	Toast message	Password change	Đổi mật khẩu thành công
42	Auth.Success0009	Toast message	Email verification	Xác thực email thành công, vui lòng xem email để đổi lại mật khẩu
43	Auth.Warning0013	In red, under feature access	Login required	Bạn cần đăng nhập để sử dụng chức năng này
44	Borrow.Warning0020	In red, under extend panel	Extension limit warning	Không thể gia hạn vì có bạn đọc khác đặt mượn
45	Borrow.Warning0025	In red, under return form	Fine required on late return	Vui lòng thêm phí phạt cho tài liệu trả trễ hạn

46	LibraryItem.Warning0011	In red, under copy status	Cannot change status	Không thể thay đổi trạng thái bản sao. <0>
47	LibraryItem.Warning0020	In red, under shelf panel	Cannot shelve item	Không thể xếp kệ cho tài liệu vì <0>
48	LibraryItem.Warning0026	In red, under shelf panel	Insufficient shelf stock	Không thể hạ kệ cho tài liệu <0> vì số lượng trên kệ không đủ
49	Cloud.Fail0002	Toast message	Video upload	Tải video thất bại
50	Cloud.Success0002	Toast message	Video upload	Tải video thành công

IV. Software Design Description

1. System Design

1.1 System Architecture

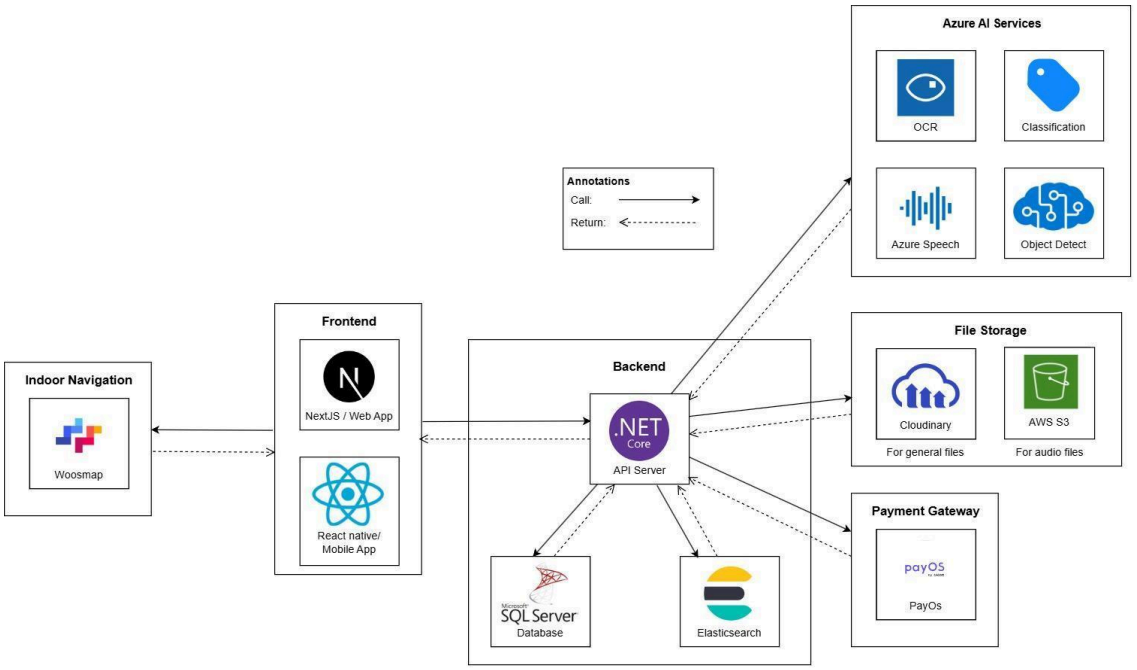


Figure 4.1.1.1 System Architecture

1.2 Package Diagram

1.2.1. Backend Package Diagram

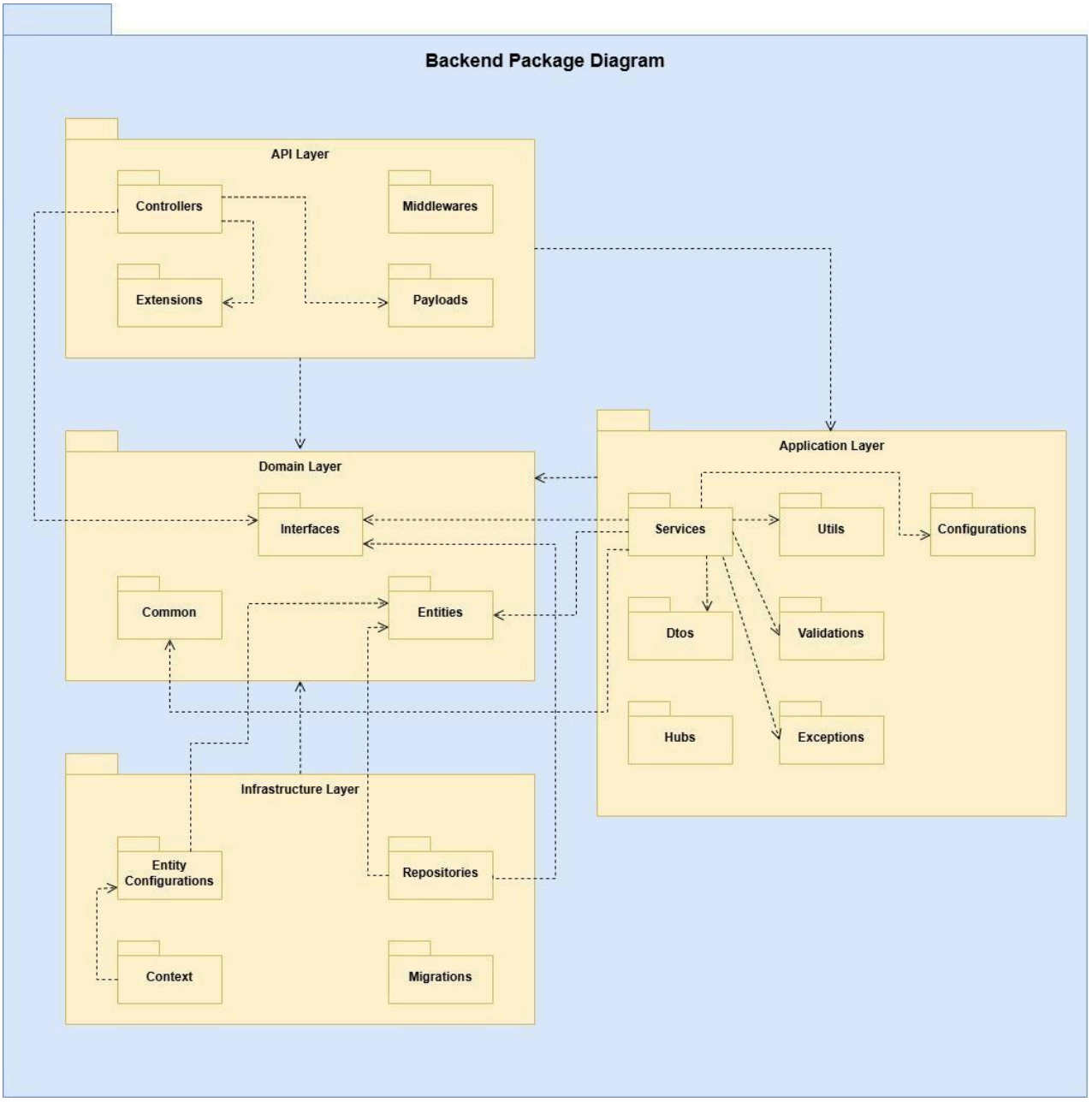


Figure 4.1.2.1 Backend Package Diagram

1.2.2. Web Application Package Diagram

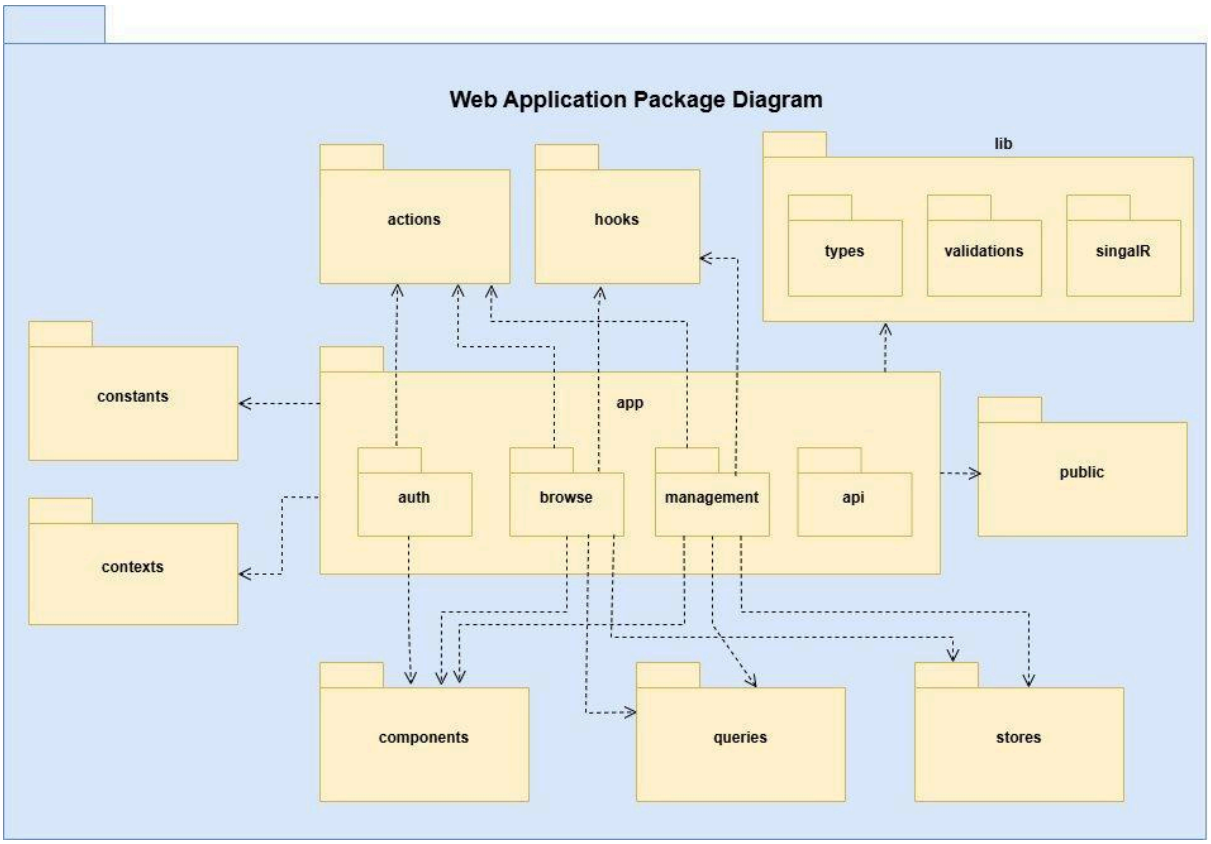


Figure 4.1.2.2 Web Application Package Diagram

1.2.3. Mobile Application Package Diagram

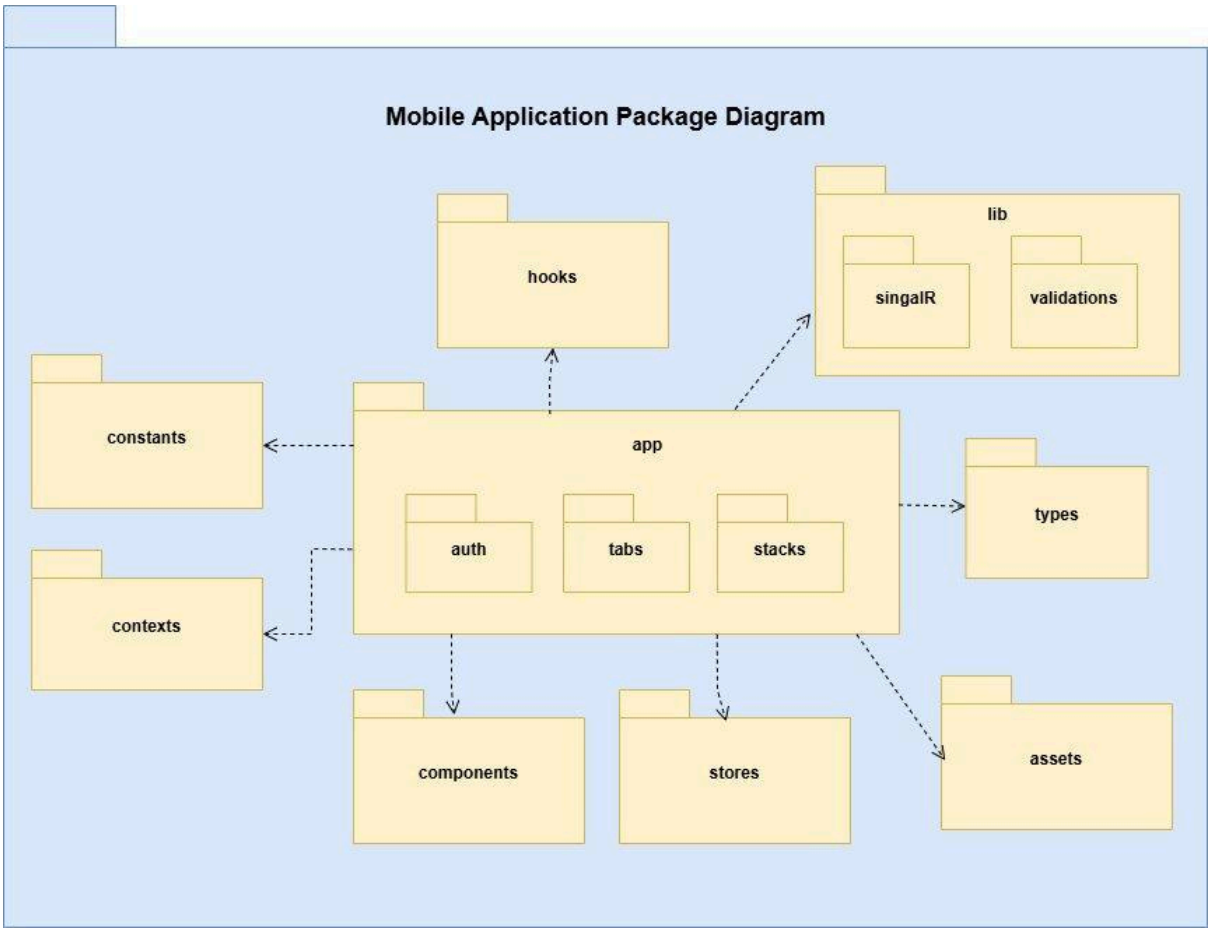


Figure 4.1.2.3 Mobile Application Package Diagram

2. Database Design

Conceptual ERD

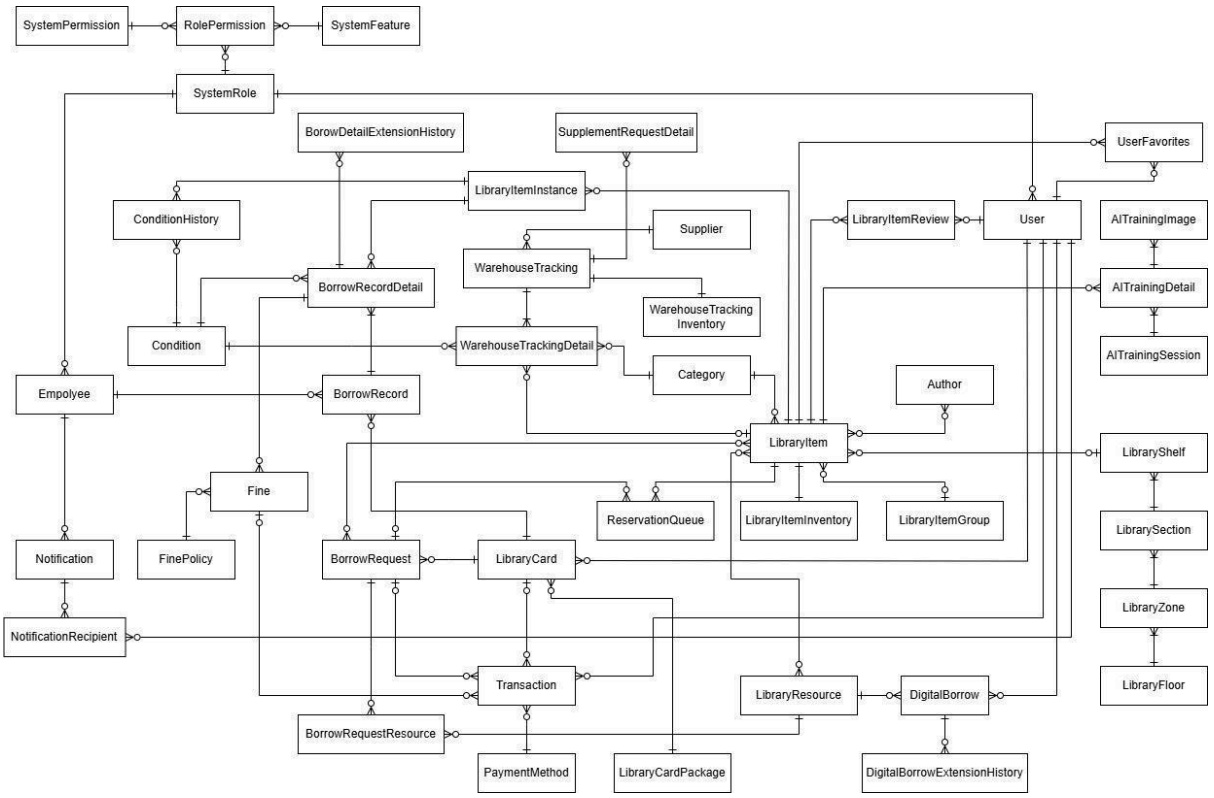


Figure 4.2.1 : Conceptual ERD

Logical ERD

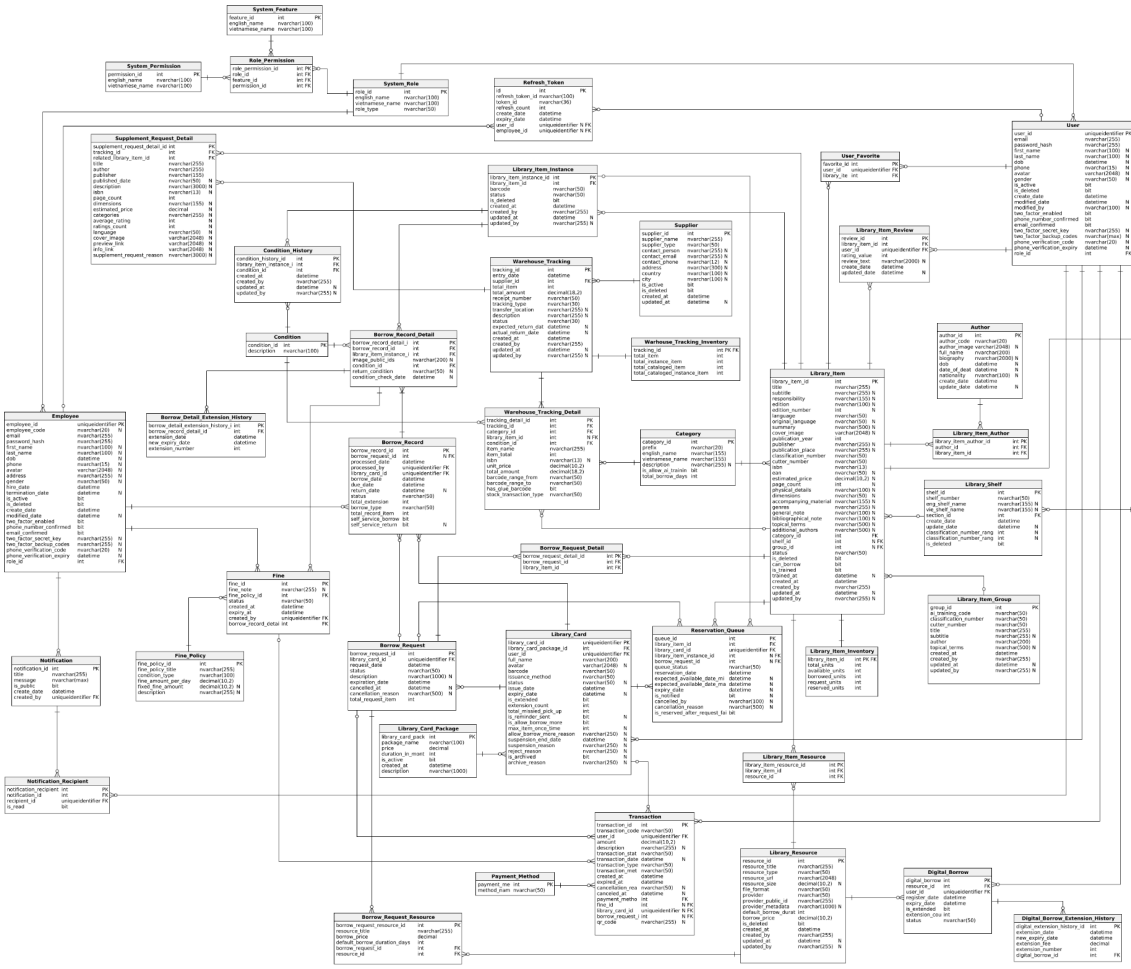


Figure 4.2.2 : Logical ERD

Entities Description

Table 4.2 : Table Descriptions

#	Entity	Description
1	AI_Training_Session	Contains AI training session information. - Primary keys: training_session_id - Foreign keys:
2	AI_Training_Detail	Store a detailed AI training session for a specific library item. - Primary keys: training_detail_id - Foreign keys: training_session_id, library_item_id
3	AI_Training_Image	Stores images URL for AI training details. - Primary keys: training_image_id - Foreign keys: training_detail_id
4	Author	Represents an author with biographical details. - Primary keys: author_id - Foreign keys:

5	Borrow_Request	Manages a borrow request submitted by a library patron. - Primary keys: borrow_request_id - Foreign keys: library_card_id
6	Borrow_Request_Detail	Details the items requested in a borrowing request. - Primary keys: borrow_request_detail_id - Foreign keys: borrow_request_id, library_item_id, library_item_instance_id
7	Borrow_Request_Resource	Stores additional resource details. - Primary keys: borrow_request_resource_id - Foreign keys: borrow_request_id, resource_id, transaction_id
8	Borrow_Record	Stores borrow history with details for a library patron. - Primary keys: borrow_record_id - Foreign keys: borrow_request_id, library_card_id, proceesed_by
9	Borrow_Record_Detail	Stores details for items borrowed in a record. - Primary keys: borrow_record_detail_id - Foreign keys: borrow_record_id, library_item_instance_id, condition_id, return_condition_id, processed_return_by
10	Borrow_Detail_Extension_ History	Stores extension information on a borrowed record detail. - Primary keys: borrow_detail_extension_history_id - Foreign keys: borrow_record_detail_id
11	Category	Defines categories for library items. - Primary keys: category_id - Foreign keys:
12	Digital_Borrow	Tracks digital borrow transactions for resources by library patrons. - Primary keys: digital_borrow_id - Foreign keys: resource_id, user_id
13	Digital_Borrow_Extension_ History	Tracks the extension history for a digital borrowing record. - Primary keys: digital_extension_history_id - Foreign keys: digital_borrow_id
14	Employee	Represents library staff with employment details. - Primary keys: employee_id - Foreign keys: role_id
15	Fine	Records fines imposed for overdue, damaged, or lost returned items. - Primary keys: fine_id - Foreign keys: borrow_record_detail_id, fine_policy_id, createdBy
16	Fine_Policy	Stores fine policy configurations. - Primary keys: fine_policy_id

		- Foreign keys:
17	Library_Card	Contains data for library membership cards. - Primary keys: library_card_id - Foreign keys: library_card_package_id, user_id
18	Library_Card_Package	Defines membership packages. - Primary keys: library_card_package_id - Foreign keys:
19	Library_Floor	Represents library floors. - Primary keys: floor_id - Foreign keys:
20	Library_Zone	Defines zones within the library associated with a floor. - Primary keys: zone_id - Foreign keys: floor_id
21	Library_Section	Represents a section within the library. - Primary keys: section_id - Foreign keys: zone_id
22	Library_Shelf	Represents a shelf in a library section. - Primary keys: shelf_id - Foreign keys: section_id
23	Library_Item	Catalogs library items with bibliographical, physical, and cataloguing information. - Primary keys: library_item_id - Foreign keys: category_id, shelf_id, group_id
24	Library_Item_Instance	Represents a specific physical copy of a library item. - Primary keys: library_item_instance_id - Foreign keys: library_item_id
25	Library_Item_Author	Represents works for specific authors. - Primary keys: library_item_author_id - Foreign keys: library_item_id, author_id
26	Library_Item_Inventory	Stores real-time counts for library items. - Primary keys: library_item_id - Foreign keys: library_item_id
27	Library_Item_Resource	Links a library item to additional digital resources. - Primary keys: library_item_resource_id - Foreign keys: library_item_id, resource_id
28	Library_Item_Condition	Defines standard conditions for library items. - Primary keys: condition_id - Foreign keys:
29	Library_Item_Condition_History	Store history changes for the condition of a specific library item instance. - Primary keys: condition_history_id - Foreign keys: library_item_instance_id, condition_id
30	Library_Item_Group	Groups library items for cataloguing or AI training purposes. - Primary keys: group_id

		- Foreign keys:
31	Library_Item_Review	Contains reviews and ratings for library items. - Primary keys: review_id - Foreign keys: library_item_id, user_id
32	Library_Resource	Stores metadata about digital resources. - Primary keys: resource_id - Foreign keys:
33	Notification	Contains system notifications. - Primary keys: notification_id - Foreign keys: created_by
34	Notification_Recipient	Maps notifications to recipients and tracks the read status for each notification. - Primary keys: notification_recipient_id - Foreign keys: notification_id, recipient_id
35	Payment_Method	Lists available payment methods - Primary keys: payment_method_id - Foreign keys:
36	Reservation_Queue	Tracks reservations placed on library items. - Primary keys: queue_id - Foreign keys: library_item_id, library_item_instance_id, library_card_id, borrow_request_id
37	Refresh_Token	Manages refresh tokens for user or employee session management. - Primary keys: id - Foreign keys: user_id, employee_id
38	Role_Permission	Associates system roles with specific features and permissions. - Primary keys: role_permission_id - Foreign keys: role_id, feature_id, permission_id
39	System_Feature	Lists system features. - Primary keys: feature_id - Foreign keys:
40	System_Permission	Defines permission levels. - Primary keys: permission_id - Foreign keys:
41	System_Role	Represents roles within the system. - Primary keys: role_id - Foreign keys:
42	Supplier	Contains supplier details, including contact information and address. - Primary keys: supplier_id - Foreign keys:
43	Supplement_Request_Detail	Stores supplementary bibliographic information for items requested to be added to the library. - Primary keys: supplement_request_detail_id

		- Foreign keys: related_library_item_id, tracking_id
44	Warehouse_Tracking	Stores records for warehouse operations. - Primary keys: tracking_id - Foreign keys: supplier_id
45	Warehouse_Tracking_Detail	Provides detailed tracking data per library item in the warehouse. - Primary keys: tracking_detail_id - Foreign keys: tracking_id, library_item_id, category_id, condition_id
46	Warehouse_Tracking_Inventory	Summarizes warehouse inventory counts related to a tracking record. - Primary keys: tracking_id - Foreign keys: tracking_id
47	Transaction	Manages financial and operational transactions that occur within the system. - Primary keys: transaction_id - Foreign keys: user_id, fine_id, resource_id, library_card_package_id, payment_method_id
48	User	Stores user account information. - Primary keys: user_id - Foreign keys: role_id, library_card_id
49	User_Favorite	Stores a user's favorite library items. - Primary keys: favorite_id - Foreign keys: user_id, library_item_id

3. Detailed Design

3.1 Sign In

3.1.1 Class Diagram

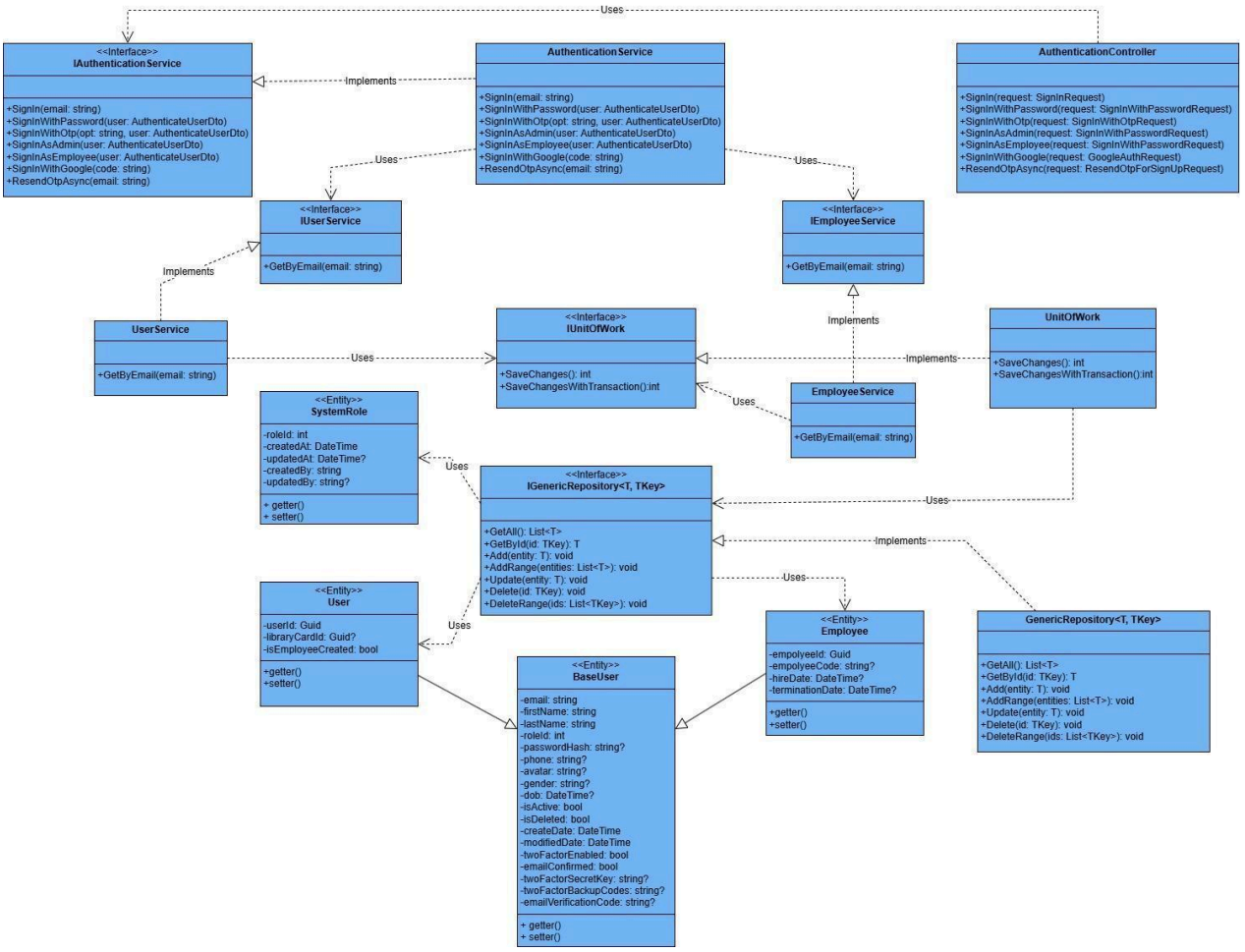


Figure 3.1.1 Class Diagram

3.1.2 Sequence Diagram

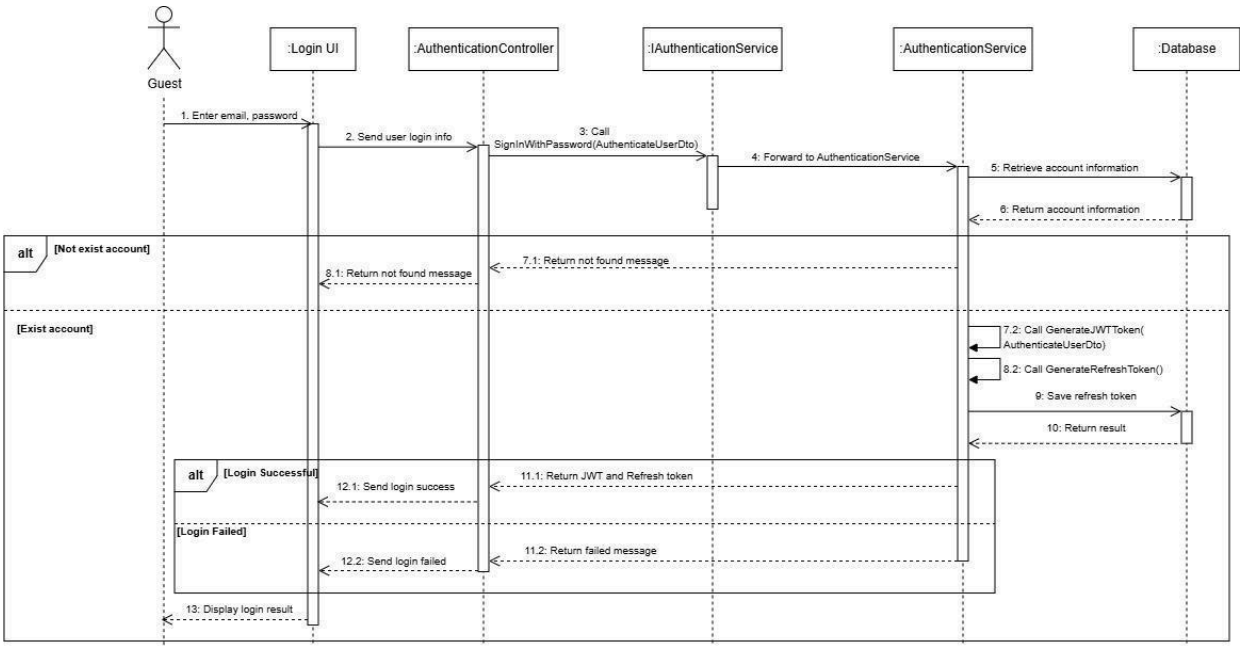


Figure 3.1.2 Sequence Diagram

3.1.3 Class Specification

3.1.3.a Authentication Controller

Table 3.1.3.a: Authentication Controller Description Table

No	Method	Description
1	SignIn(SignInRequest)	Receives user credentials (email and password) via an HTTP request and initiates the login process.

3.1.3.b Authentication Service

Table 3.1.3.b: Authentication Service Description Table

No	Method	Description
1	SignInWithPassword(AuthenticateUserDto authenticateUserDto)	Validates the provided email and password to authenticate the user and grant access.
2	GenerateJWTToken(AuthenticateUserDto authenticateUserDto)	Creates a JWT token to authorize access to protected endpoints.
3	GenerateRefreshToken()	Creates a refresh token, allowing to request a new access token without re-entering credentials.

3.2 Sign Up

3.2.1 Class Diagram

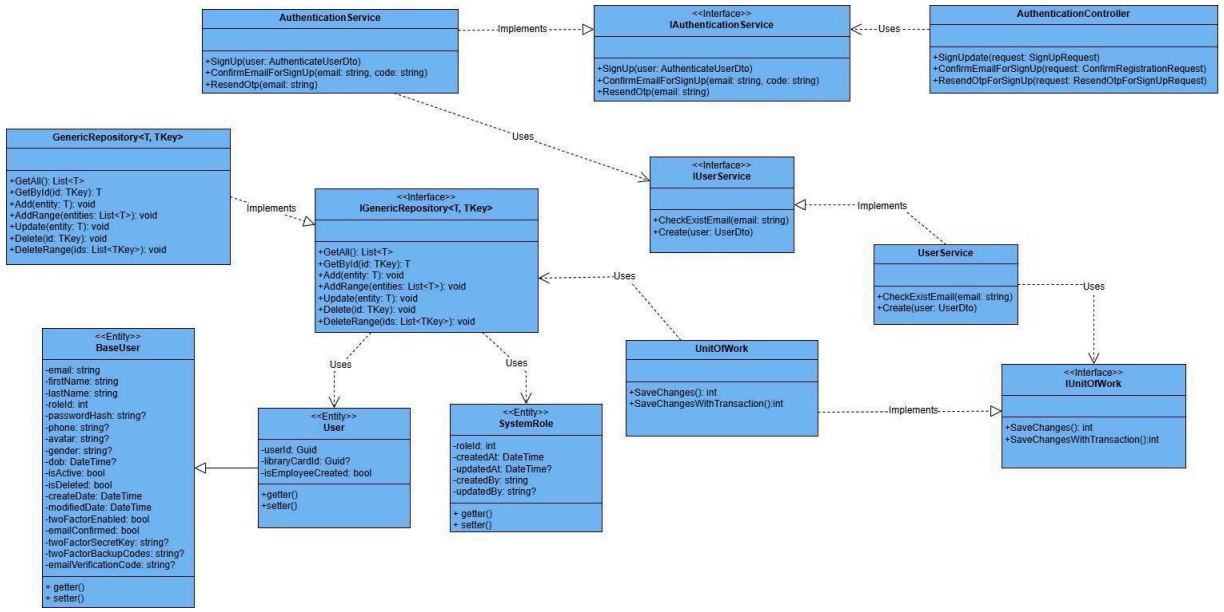


Figure 3.2.1 Class Diagram

3.2.2 Sequence Diagram

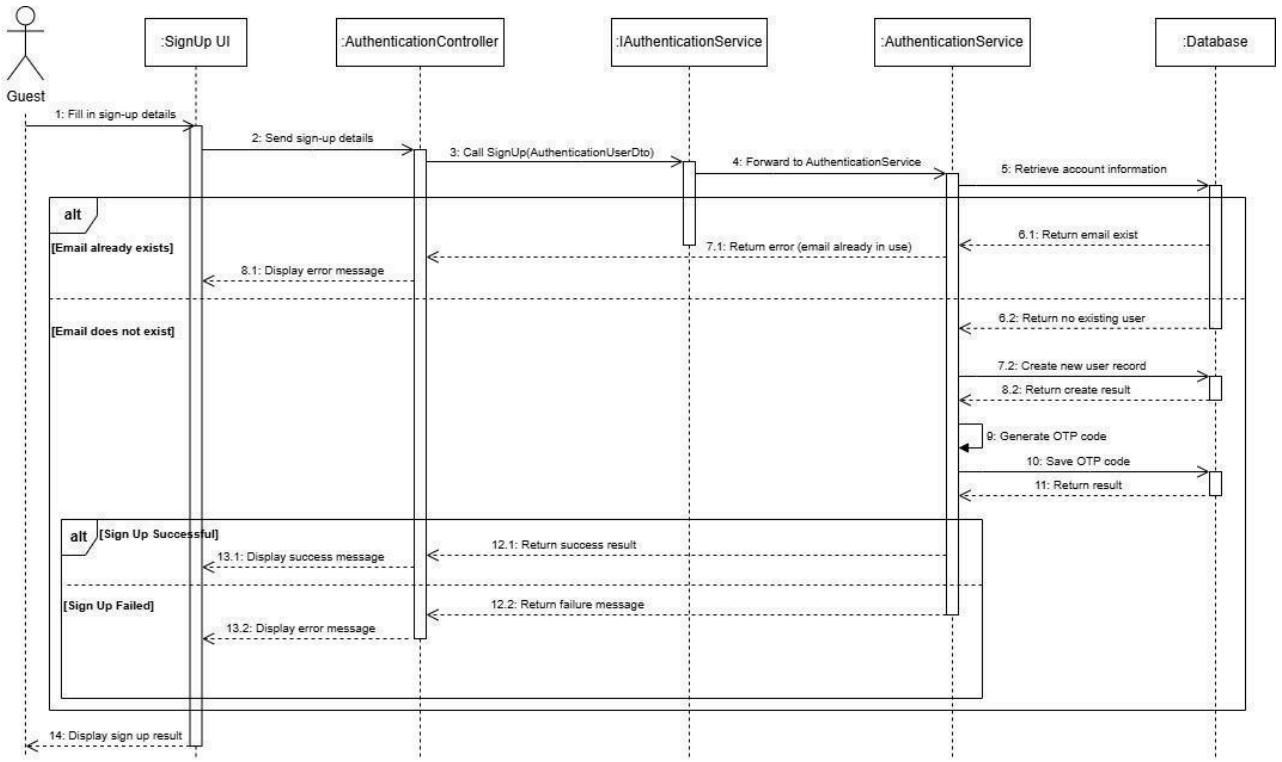


Figure 3.2.2 Sequence Diagram

3.2.3 Class Specification

3.2.3.a Authentication Controller

Table 3.2.3.a: Authentication Controller Description Table

No	Method	Description
1	SignUp(SignUpRequest signUpRequest)	Receives user registration details (e.g., email, password) and forwards them to the service layer to create a new account.
2	ConfirmEmailForSignUp(ConfirmRequest confirmRequest)	Verifies the user's email address by matching a provided confirmation code.
3	ResendOtp(ConfirmOtpRequest confirmRequest)	Resends the one-time code (OTP) for email verification.

3.2.3.b Authentication Service

Table 3.2.3.b: Authentication Service Description Table

No	Method	Description
1	SignUp(AuthenticateUserDto authenticateUserDto)	Registers a new user by validating and storing the provided credentials (e.g., email, password).
2	ConfirmEmailForSignUp(string email, string code)	Verifies the user's email address by matching a provided confirmation code.
3	ResendOtp(string email)	Resends the one-time code (OTP) for email verification.

3.3. Manage Users

3.3.1. Create User

3.3.1.1 Class Diagram

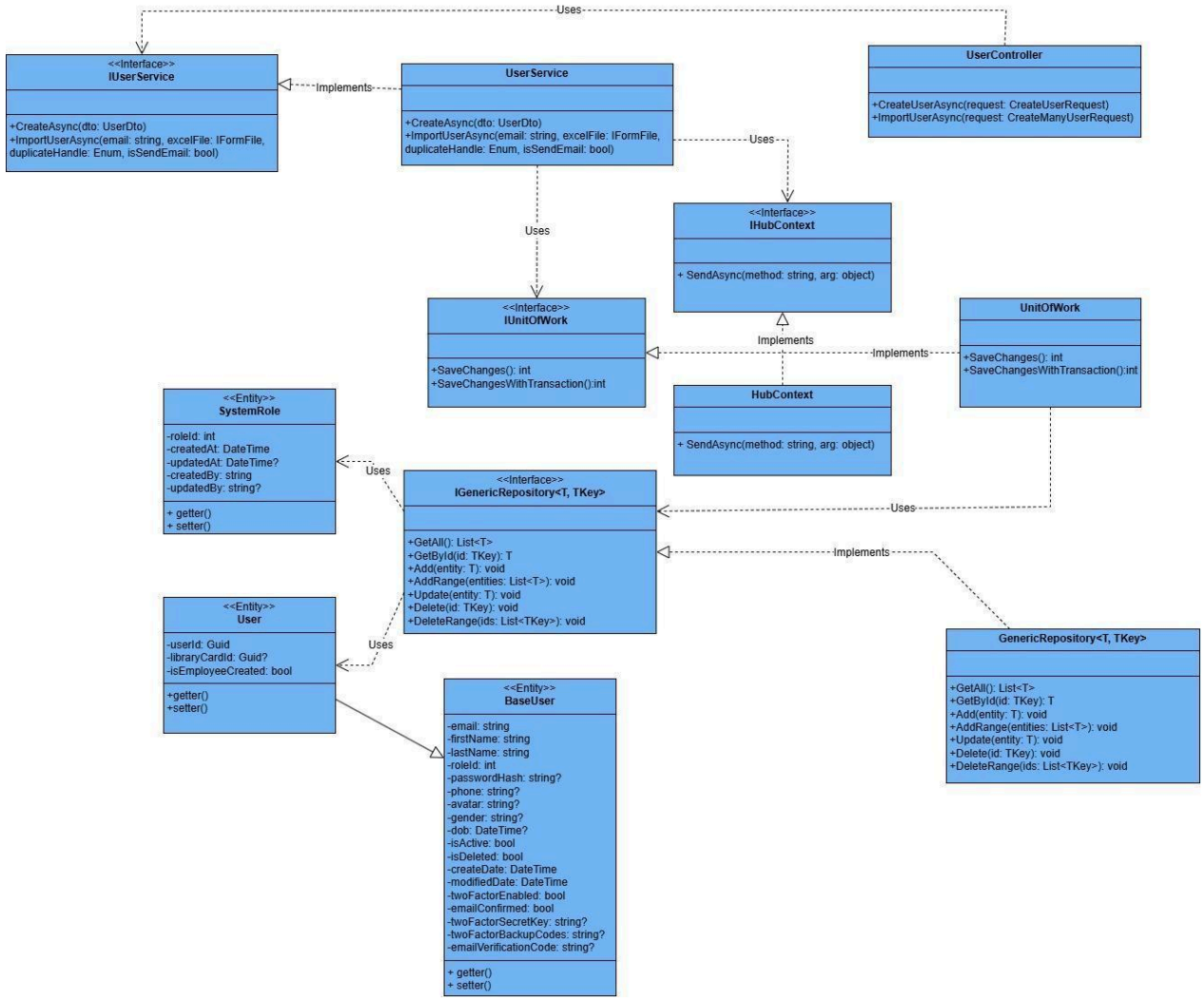


Figure 3.3.1.1 Class Diagram

3.3.1.2 Sequence Diagram

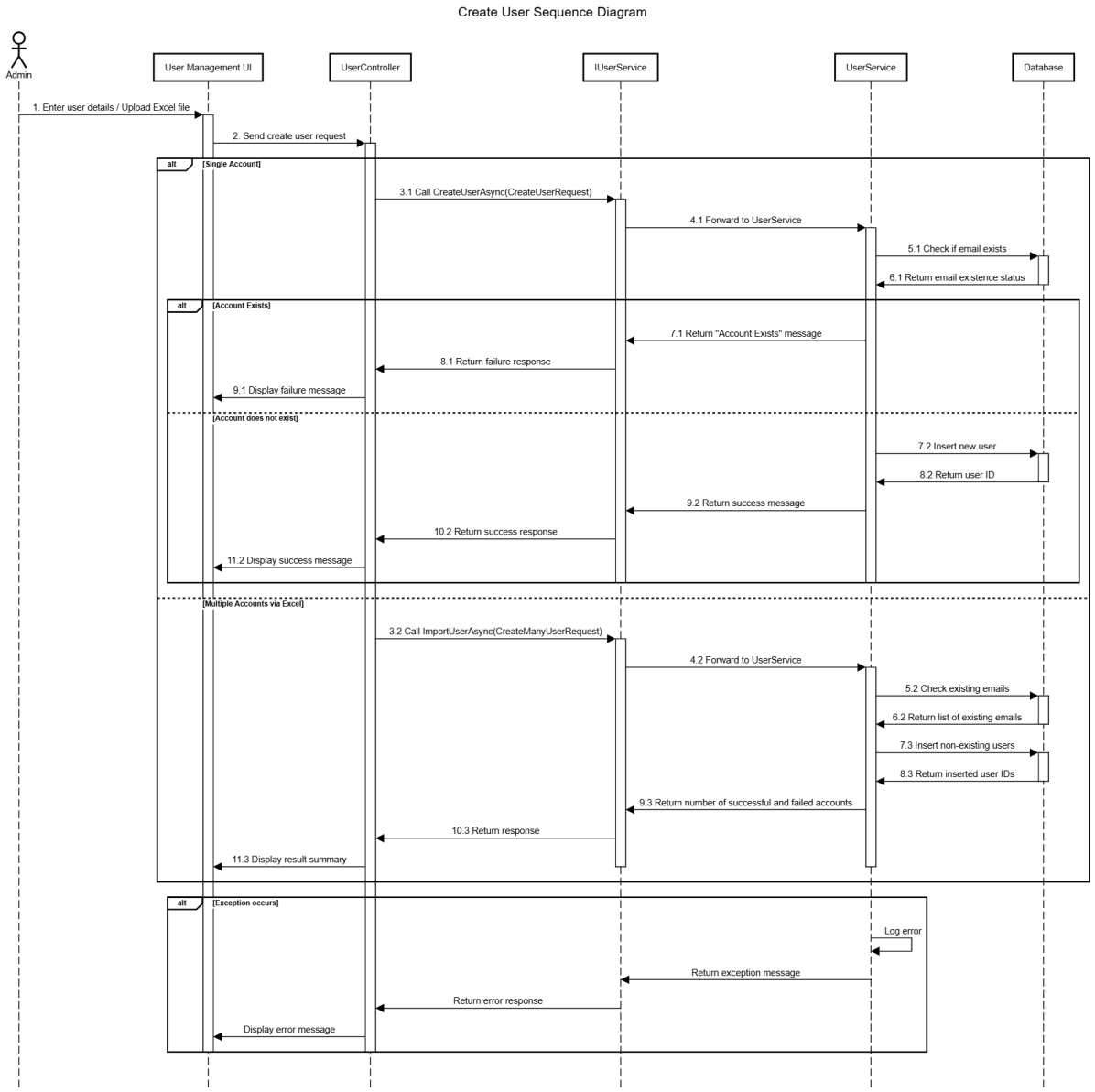


Figure 3.3.1.2 Sequence Diagram

3.3.1.3 Class Specification

3.3.1.3.a User Controller

Table 3.3.1.3.a: User Controller Description Table

No	Method	Description
1	CreateUserAsync(CreateUserRequest createUserRequest)	Fill the form with account information (email, username, DoB,etc) and forward to the service layer to create a new account.
2	ImportUserAsync(CreateManyUserRequest createManyUserRequest)	Create many accounts at the same time via excel file which include account information as create single account.

3.3.1.3.b User Service

Table 3.3.1.3.b: User Service Description Table

No	Method	Description
1	CreateUserAsync(UserDto userDto)	Create a new account.
2	ImportUserAsync(string email,IFormFile excelFile,DuplicateHandle duplicateHandle,bool isSendEmail);	Create many accounts at the same time via excel file.

3.3.2. Get All Users

3.3.2.1 Class Diagram

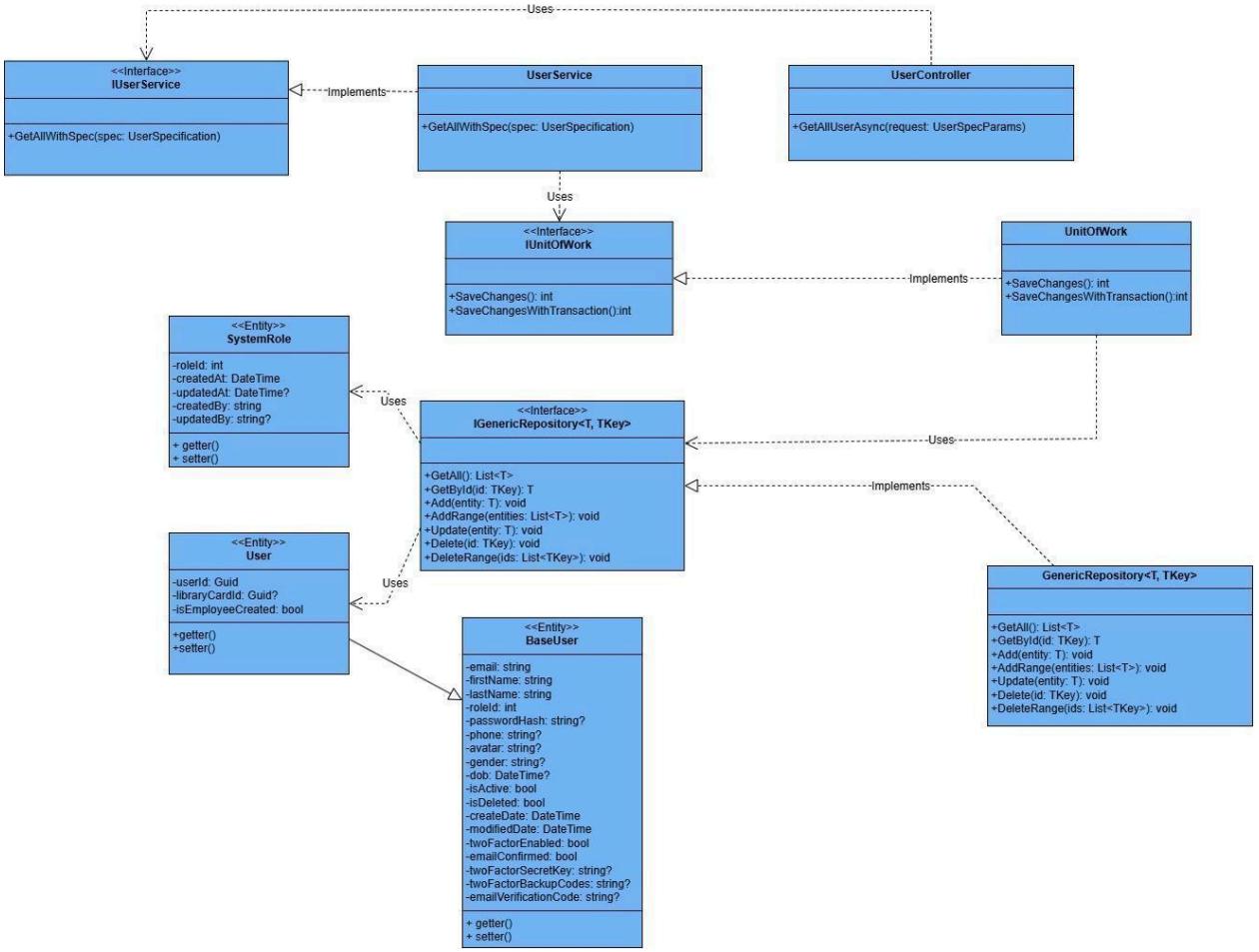


Figure 3.3.2.1 Class Diagram

3.3.2.2 Sequence Diagram

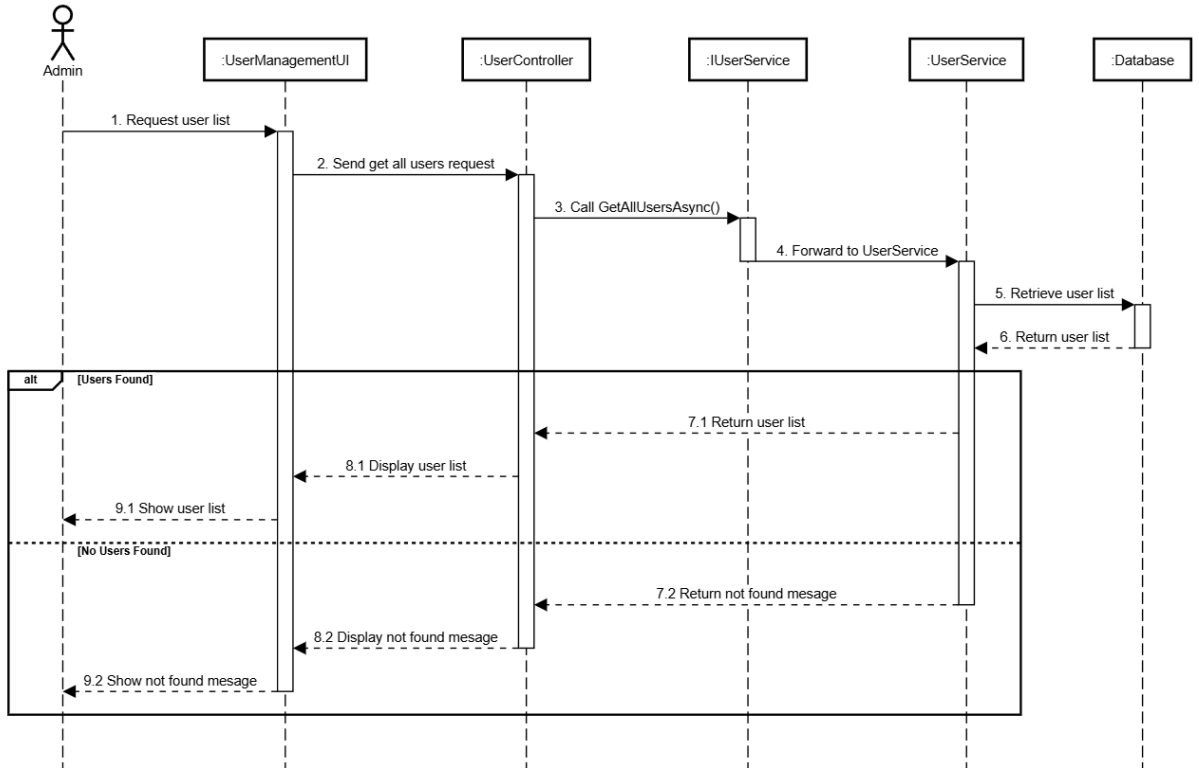


Figure 3.3.2.2 Sequence Diagram

3.3.2.3 Class Specification

3.3.2.3.a User Controller

Table 3.3.2.3.a: User Controller Description Table

No	Method	Description
1	GetAllUsersAsync(UserSpecParams specParams)	Get all users in the system and support searching and filtering.

3.3.2.3.b User Service

Table 3.3.2.3.b: User Service Description Table

No	Method	Description
1	GetAllWithSpec(UserSpecification spec)	Get all users that suit conditions. If there is no condition, get all users joined in the system (except admin account).

3.3.3. Update User

3.3.3.1 Class Diagram

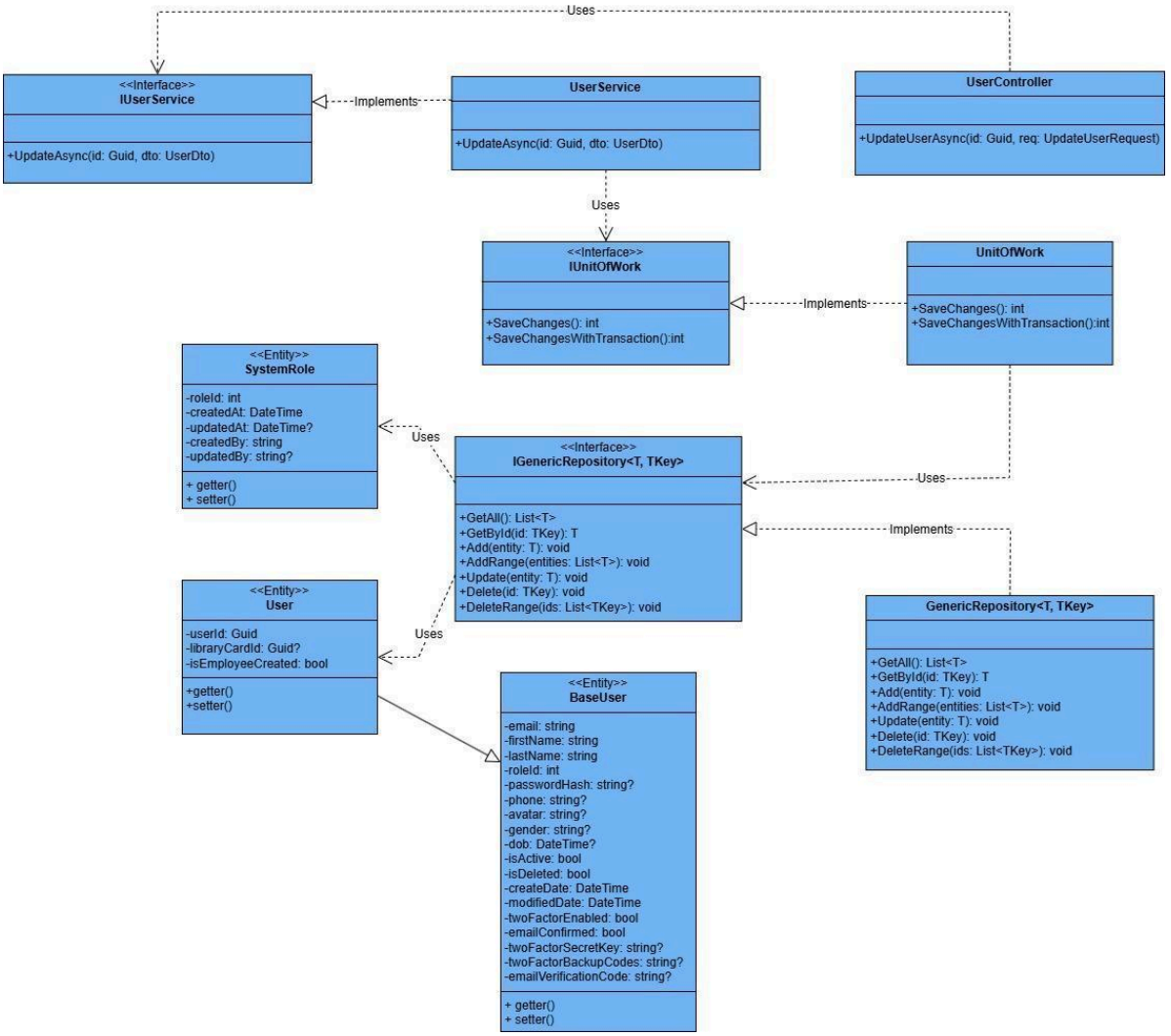


Figure 3.3.3.1 Class Diagram

3.3.3.2 Sequence Diagram

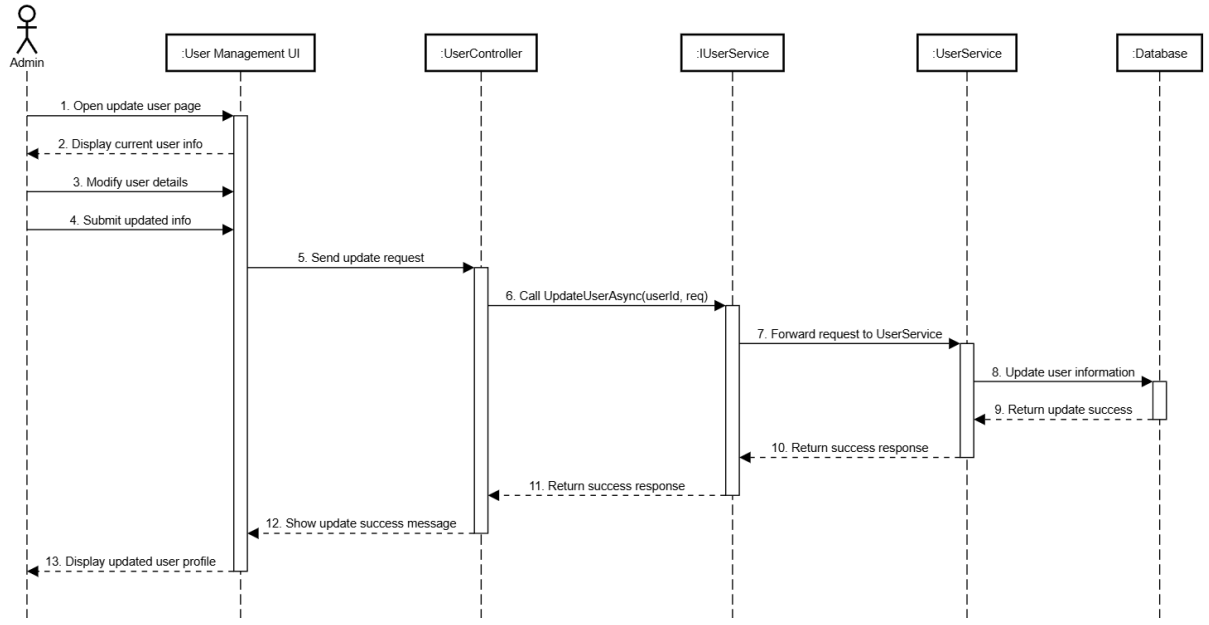


Figure 3.3.3.2 Sequence Diagram

3.3.3.3.a User Controller

Table 3.3.3.3.a: User Controller Description Table

No	Method	Description
1	UpdateUserAsync(Guid userId, UpdateUserRequest updateRequest)	Update user information that displays in profile.

3.3.3.3.b User Service

Table 3.3.2.3.b: User Service Description Table

No	Method	Description
1	UpdateAsync(Guid userId, UpdateUserRequest updateRequest)	Update user information that displays in profile.

3.3.4. Delete User

3.3.4.1 Class Diagram

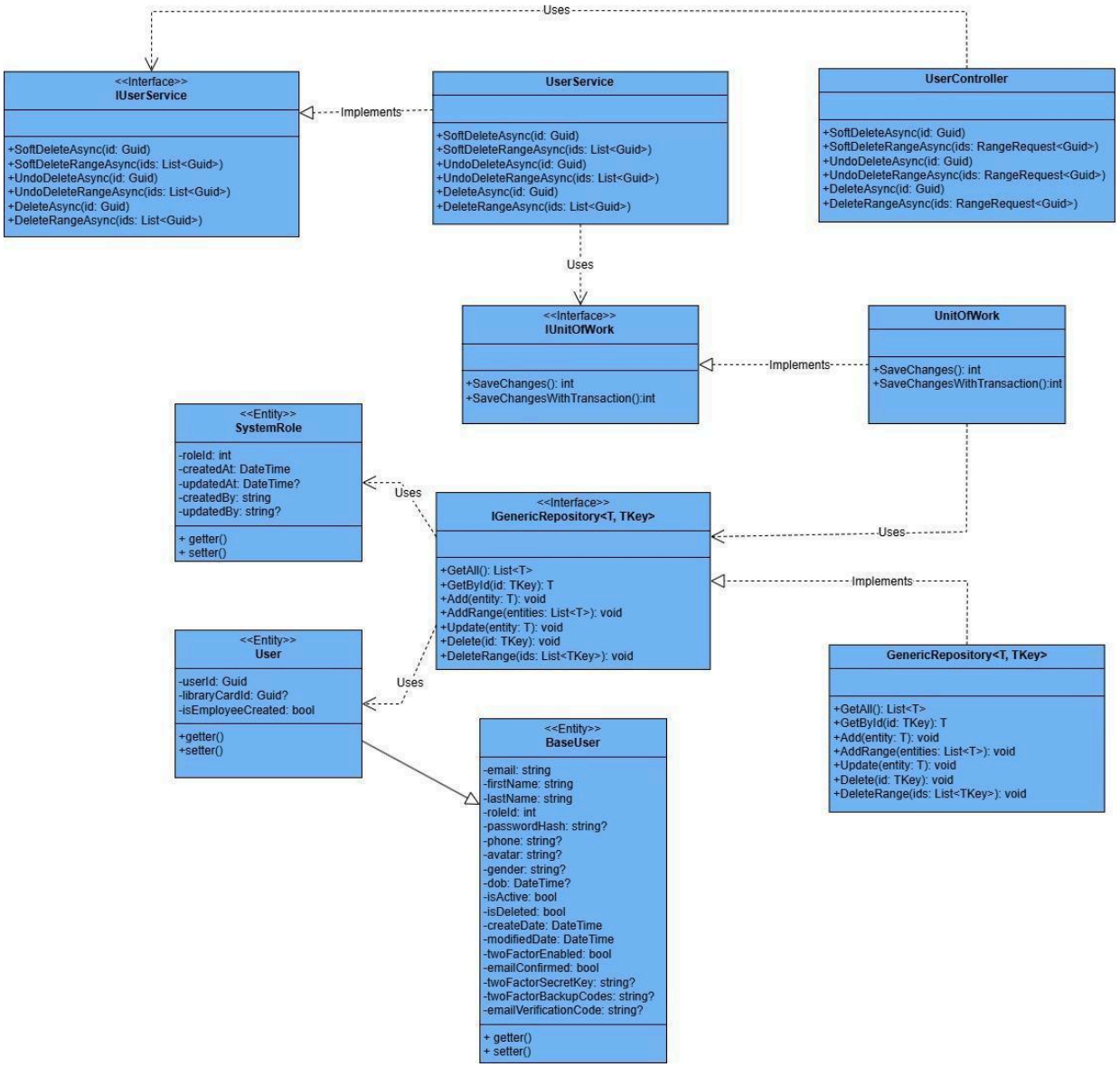


Figure 3.3.4.1 Class Diagram

3.3.4.2 Sequence Diagram

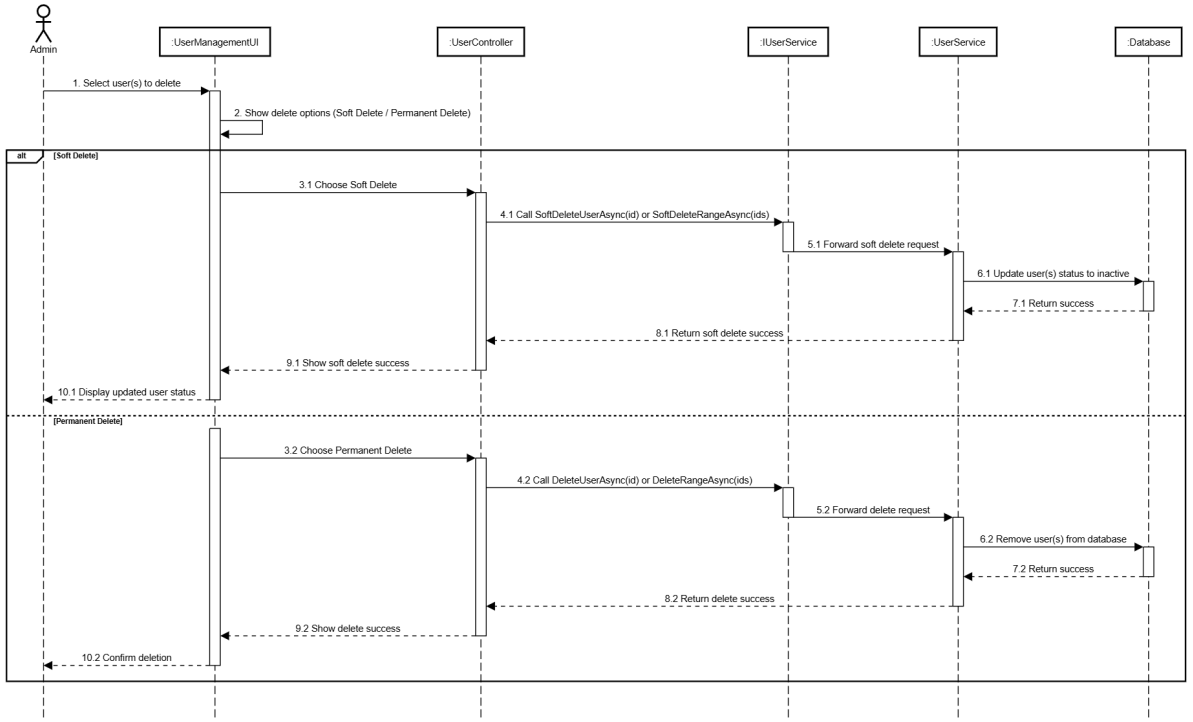


Figure 3.3.4.2 Sequence Diagram

3.3.4.3 Class Specification

3.3.4.3.a User Controller

Table 3.3.4.3.a: User Controller Description Table

No	Method	Description
1	SoftDeleteUserAsync(Guid userId)	Deactivate account. The account is still in the database but can not log in.
2	SoftDeleteRangeUserAsync(RangeRequest req)	Deactivate many accounts at the time. Those accounts can not log in to the system.
3	UndoDeleteUserAsync(Guid userId)	Activate account again. The account could be used like every normal account.
4	UndoDeleteRangeUserAsync(RangeRequest req)	Activate many accounts at the time. Those accounts could be used normally.
5	DeleteUserAsync(Guid userId)	Remove account from the database. Admin would not see this account in user management any more.
6	DeleteRangeUserAsync(RangeRequest req)	Remove accounts from the database. Those accounts no longer exist in the system.

3.3.4.3.b User Service

Table 3.3.4.3.b: User Service Description Table

No	Method	Description

1	SoftDeleteAsync(Guid userId)	Retrieve an account with a given userId to deactivate that account.
2	SoftDeleteRangeAsync(List<Guid> ids)	Retrieve a list of accounts that match with the input and deactivate them.
3	UndoDeleteAsync(Guid userId)	Retrieve an account that was inactive then activate it again.
4	UndoDeleteRangeAsync(List<Guid> ids)	Activate many accounts at the time. Those accounts could be used normally.
5	DeleteAsync(Guid userId)	Remove account from the database. Admin would not see this account in user management any more.
6	DeleteRangeAsync(List<Guid> ids)	Remove accounts from the database. Those accounts no longer exist in the system.

3.4. Manage Employees

3.4.1. Create Employee

3.4.1.1 Class Diagram

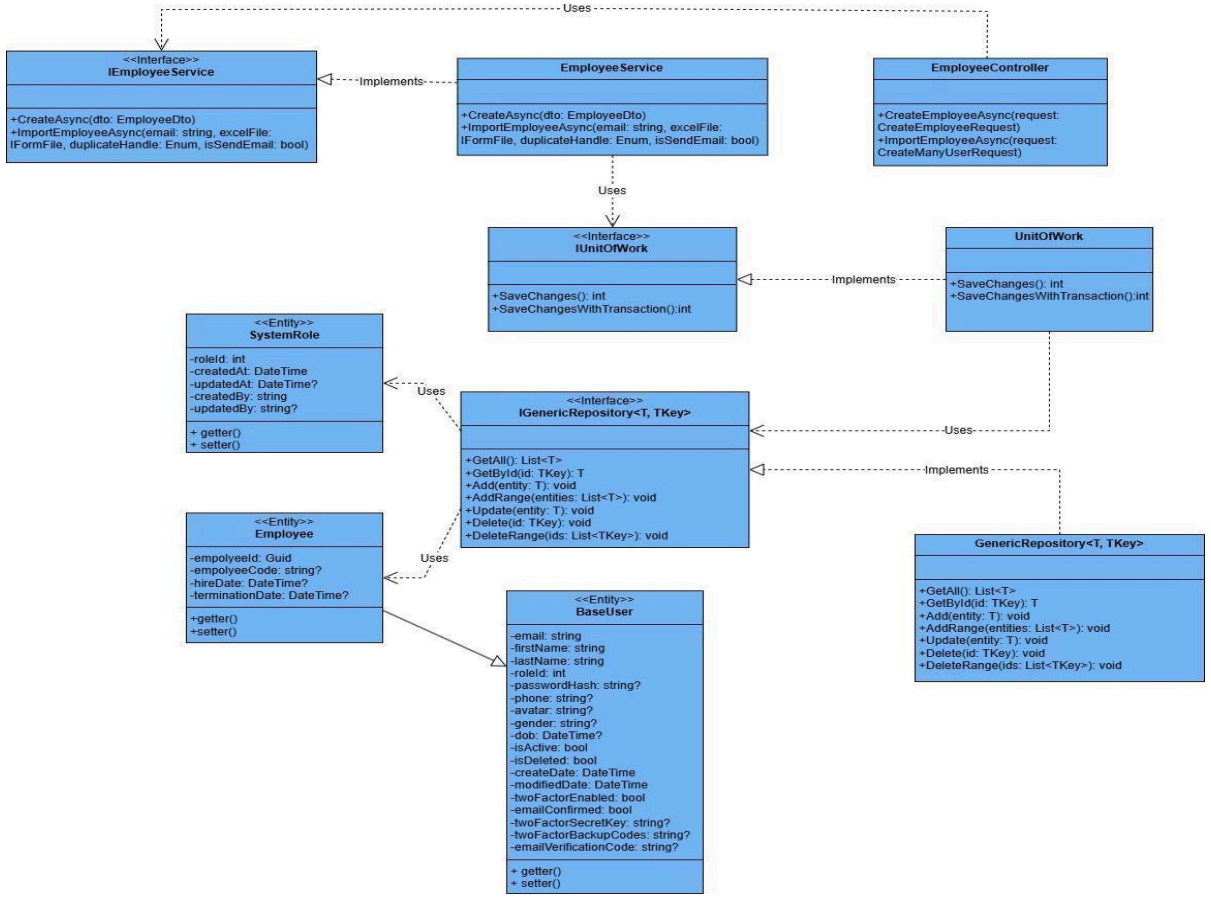


Figure 3.4.1.1 Class Diagram

3.4.1.2 Sequence Diagram

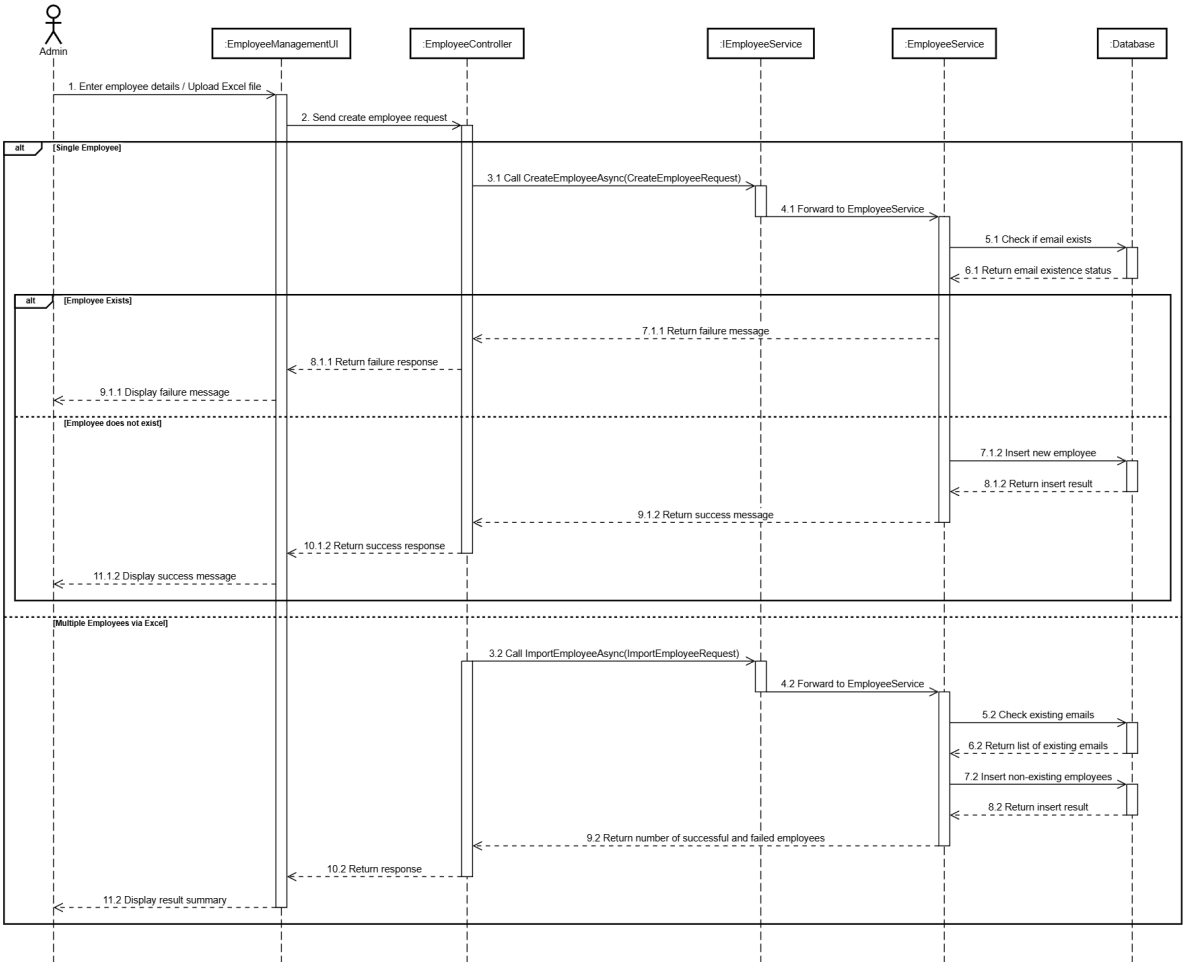


Figure 3.4.1.2 Sequence Diagram

3.4.1.3 Class Specification

3.4.1.3.a Employee Controller

Table 3.4.1.3.a: Employee Controller Description Table

No	Method	Description
1	CreateEmployeeAsync(CreateEmployee Request createEmployeeRequest)	Fill the form with account information (email, employee code, hire date, DoB,etc) and forward to the service layer to create a new account for the employee.
2	ImportEmployeeAsync(CreateManyEmpl oyeeRequest createManyRequest)	Create many accounts with roles as employees at the same time via excel file which include account information as create single account.

3.4.1.3.b Employee Service

Table 3.4.1.3.b: Employee Service Description Table

No	Method	Description
1	CreateEmployeeAsync(EmployeeDto employeeDto)	Create a new employee account.
2	ImportEmployeeAsync(string email,IFormFile excelFile,DuplicateHandle duplicateHandle,bool isSendEmail);	Create many accounts at the same time via excel file where each row contains employee details.

3.4.2. Get All Employee

3.4.2.1 Class Diagram

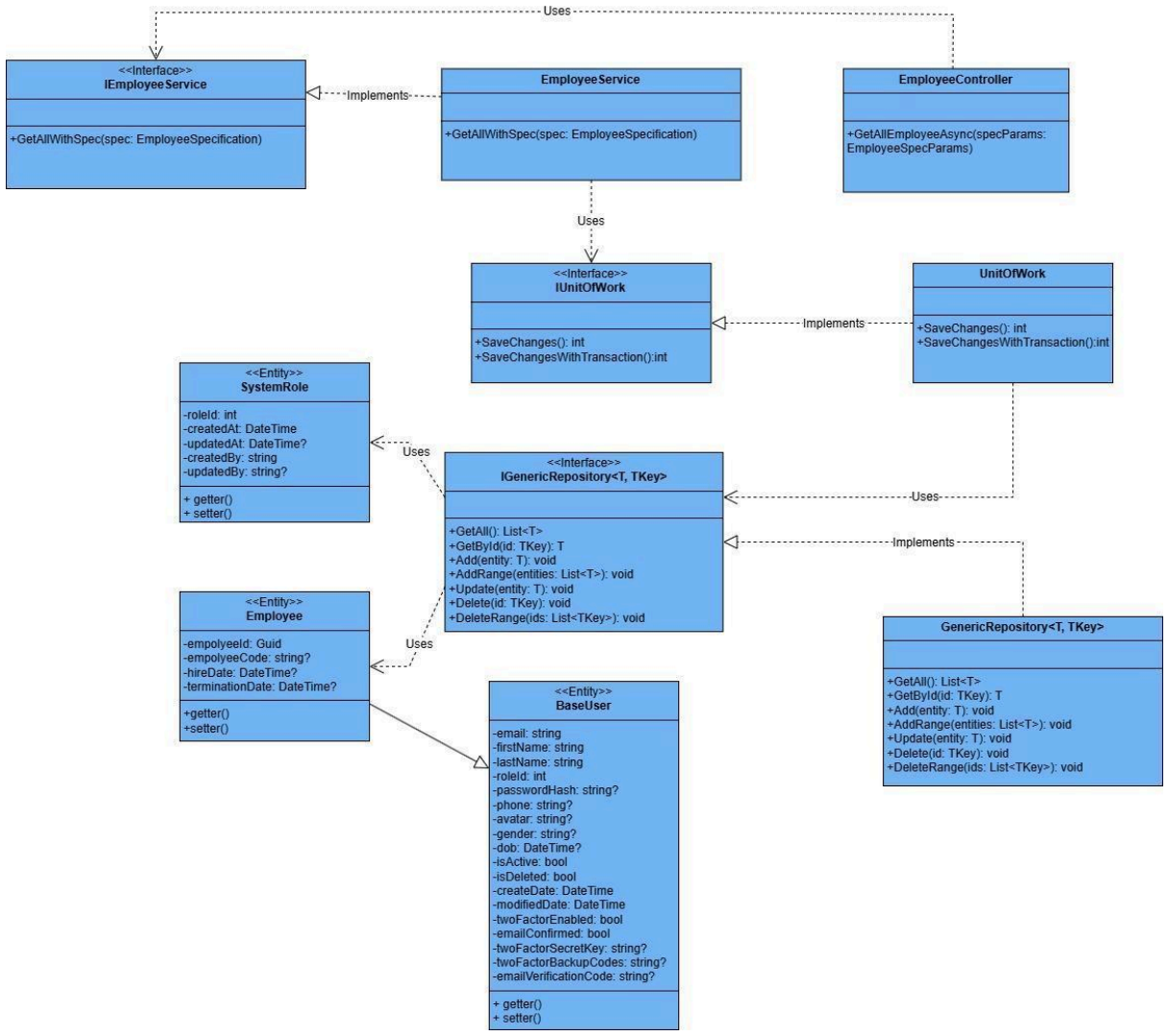


Figure 3.4.2.1 Class Diagram

3.4.2.2 Sequence Diagram

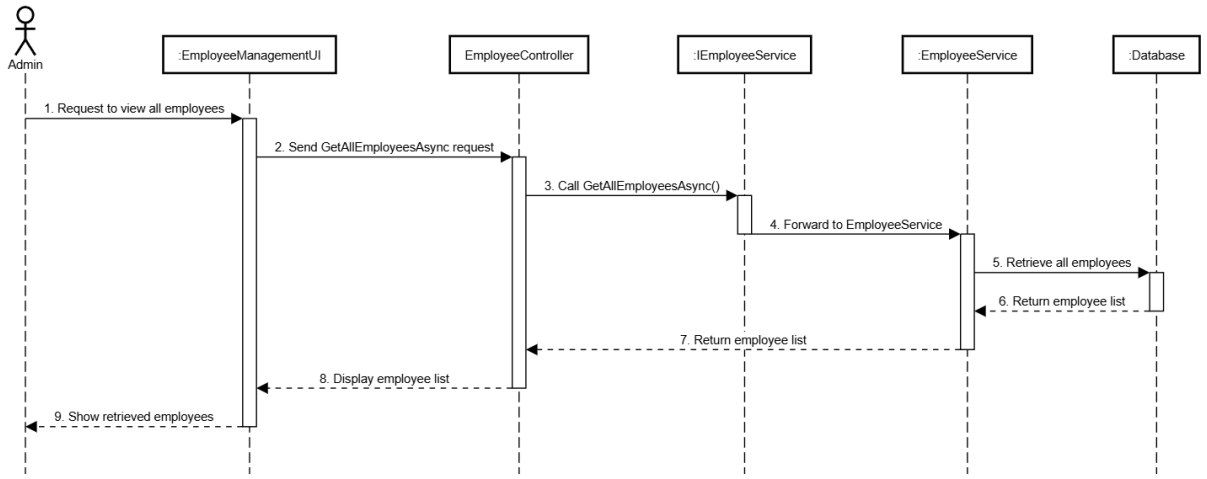


Figure 3.4.2.2 Sequence Diagram

3.4.2.3 Class Specification

3.4.2.3.a Employee Controller

Table 3.4.2.3.a: Employee Controller Description Table

No	Method	Description
1	GetAllEmployeeAsync(EmployeeSpecParams specParams)	Get all employee details in the system and support searching and filtering.

3.4.2.3.b Employee Service

Table 3.4.2.3.b: Employee Service Description Table

No	Method	Description
1	GetAllWithSpec(EmployeeSpecification spec)	Get all employees that suit conditions. If there is no condition, get all employee accounts in the system.

3.4.3. Update Employee

3.4.3.1 Class Diagram

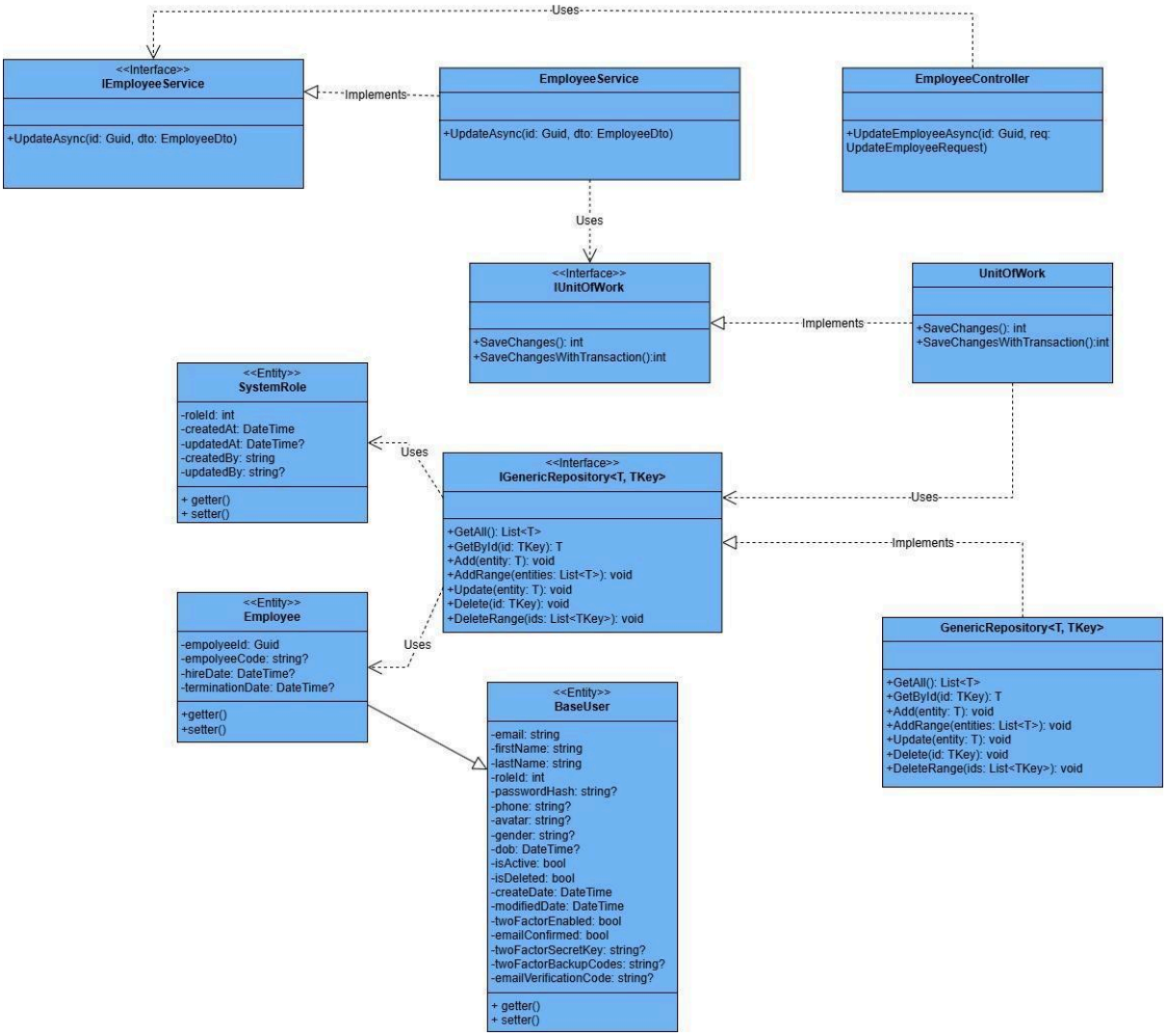


Figure 3.4.3.1 Class Diagram

3.4.3.2 Sequence Diagram

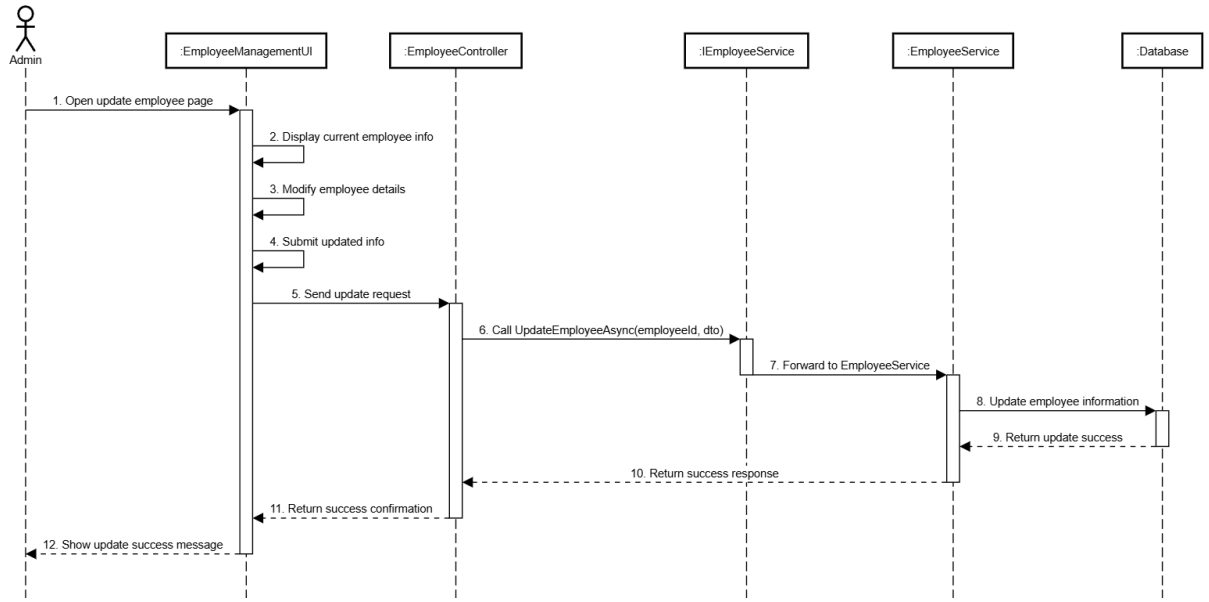


Figure 3.4.3.2 Sequence Diagram

3.4.3.3 Class Specification

3.4.3.3.a Employee Controller

Table 3.4.3.3.a: Employee Controller Description Table

No	Method	Description
1	UpdateEmployeeAsync(Guid employeeId, UpdateEmployeeRequest updateRequest)	Update employee information that display in profile.

3.4.3.3.b Employee Service

Table 3.4.3.3.b: Employee Service Description Table

No	Method	Description
1	UpdateAsync(Guid employeeId, EmployeeDto employeeDto)	Update employee information that displays in profile.

3.4.4. Delete Employee

3.4.4.1 Class Diagram

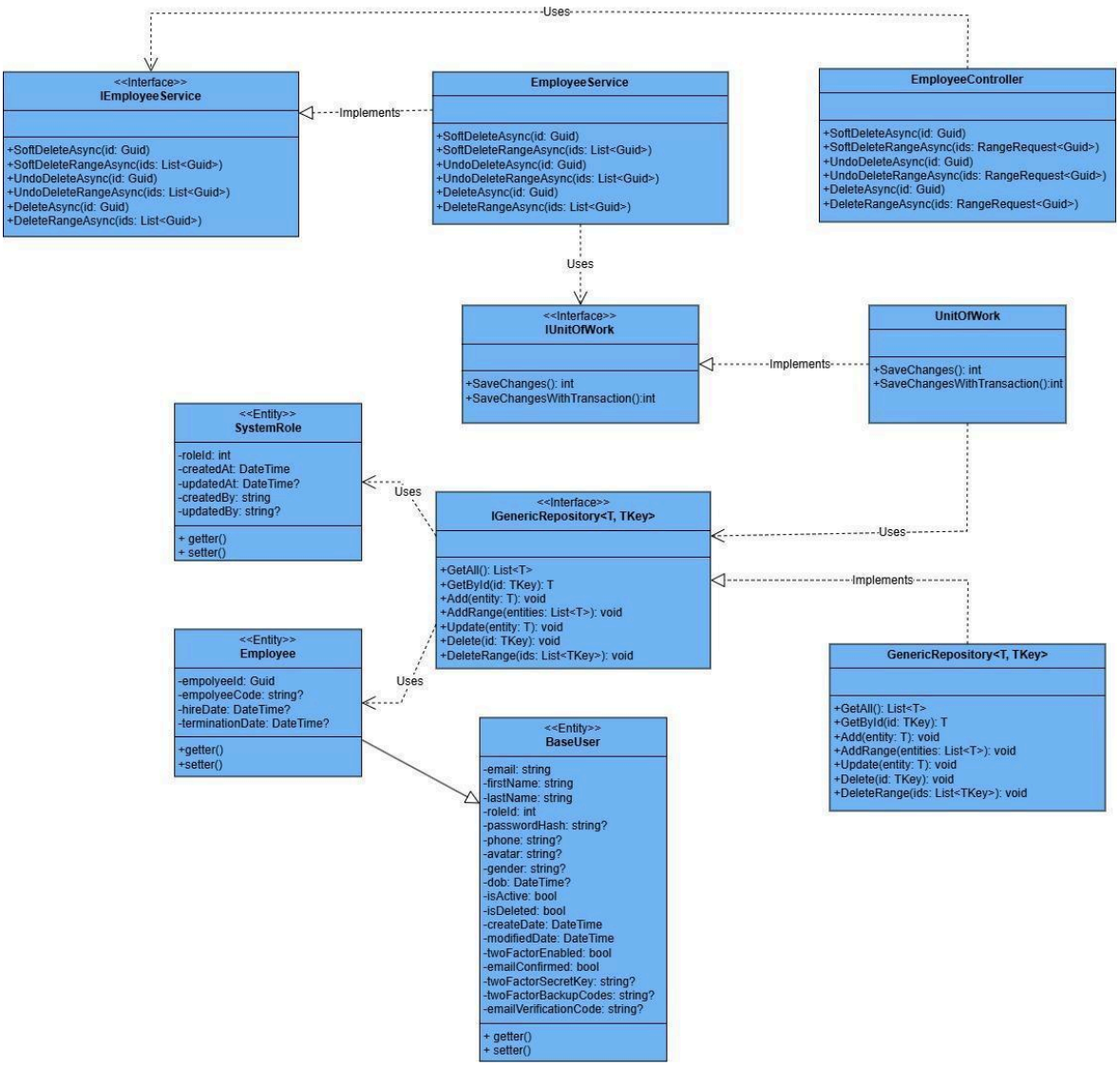


Figure 3.4.4.1 Class Diagram

3.4.4.2 Sequence Diagram

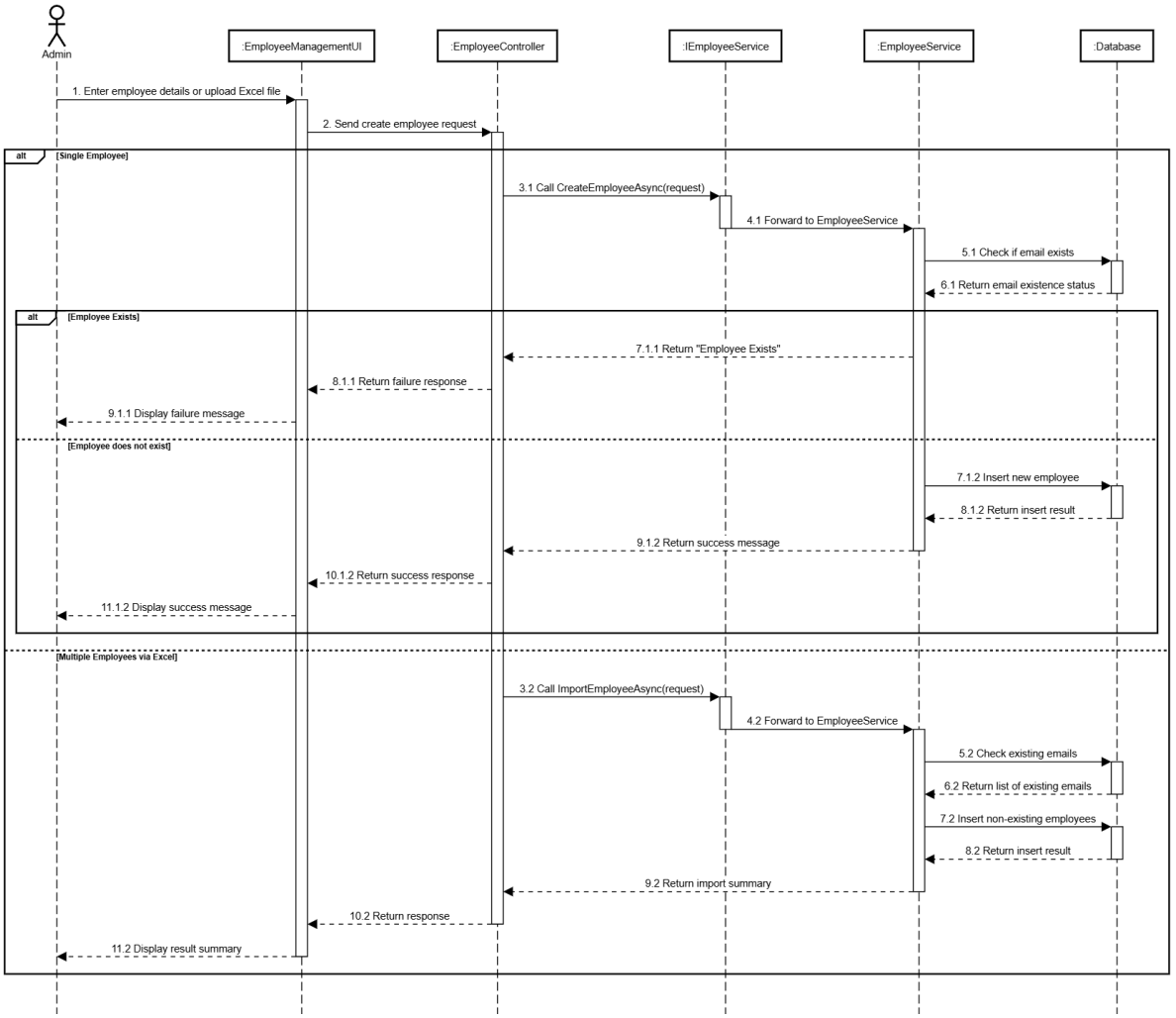


Figure 3.4.4.2 Sequence Diagram

3.4.4.3 Class Specification

3.4.4.3.a Employee Controller

Table 3.4.4.3.a: Employee Controller Description Table

No	Method	Description
1	SoftDeleteEmployeeAsync(Guid employeeId)	Deactivate employee account. The account is still in the database but can not log in.
2	SoftDeleteRangeEmployeeAsync(RangeRequest req)	Deactivate many employee accounts at the time. Those accounts can not log in to the system.
3	UndoDeleteEmployeeAsync(Guid employeeId)	Activate employee's account again. The account could be used like every normal account.
4	UndoDeleteRangeEmployeeAsync(RangeRequest req)	Activate many employee accounts at the time. Those account could be used normally.

5	DeleteEmployeeAsync(Guid employeeId)	Remove employee's account from the database. Admin would not see this account in user management any more.
6	DeleteRangeEmployeeAsync(RangeRequest req)	Remove employee accounts from the database. Those accounts no longer exist in the system.

3.4.4.3.b Employee Service

Table 3.4.4.3.b: Employee Controller Description Table

No	Method	Description
1	SoftDeleteAsync(Guid employeeId)	Retrieve employee account with given employeeId to deactivate that account
2	SoftDeleteRangeAsync(List<Guid> employeeIds)	Retrieve a list of employee accounts that match with the input and deactivate them.
3	UndoDeleteAsync(Guid employeeId)	Retrieve an employee account that was inactive then activate it again.
4	UndoDeleteRangeAsync(List<Guid> employeeIds)	Activate many employee accounts at the time. Those accounts could be used normally/
5	DeleteAsync(Guid employeeId)	Remove account from the database. Admin would not see this employee account in user management any more.
6	DeleteRangeAsync(List<Guid> employeeIds)	Remove accounts from the database. Those employee accounts no longer exist in the system.

Table 3.4.4.3.b: Employee Service Description Table

3.5 Manage Role Permissions

3.5.1 Create Role with Default Permission

3.5.1.1 Class Diagram

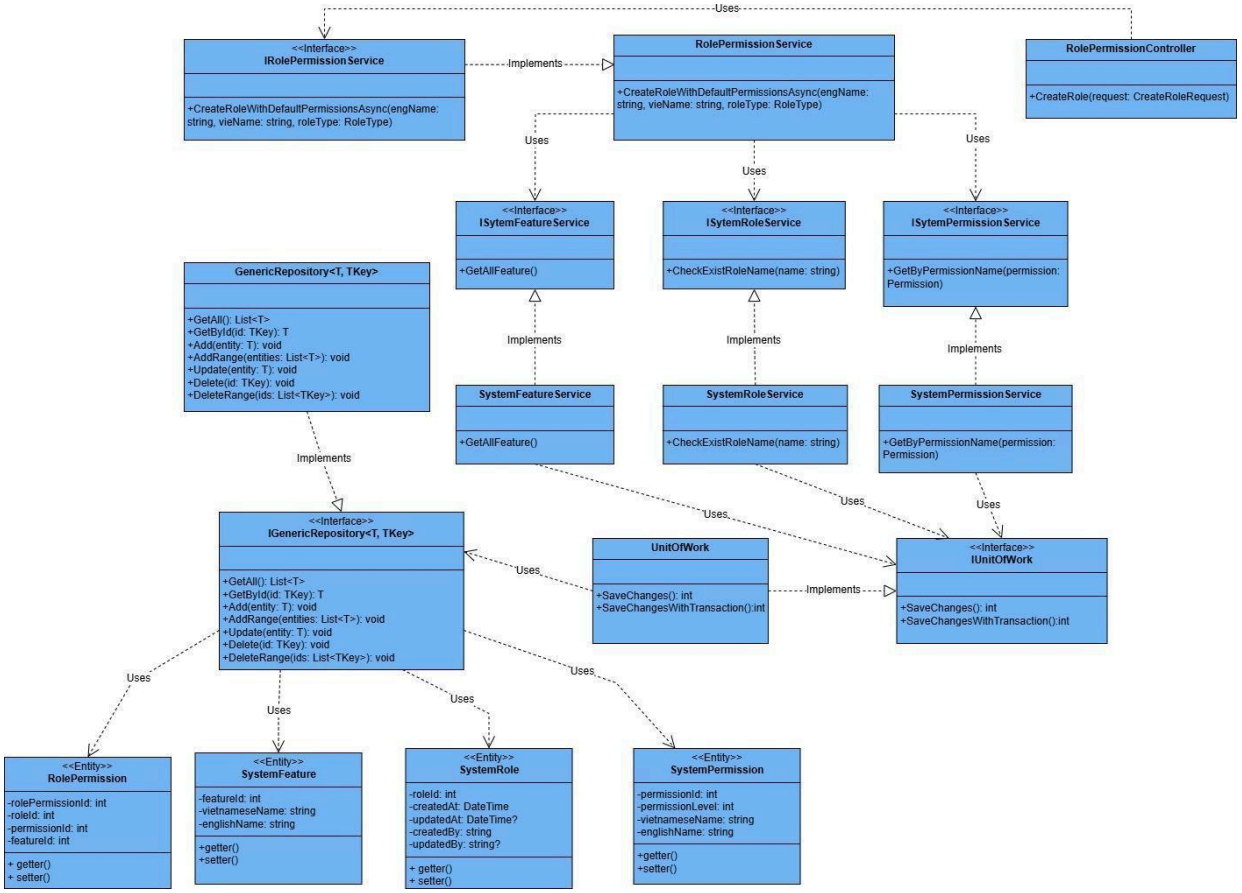


Figure 3.5.1.1 Class Diagram

3.5.1.2 Sequence Diagram

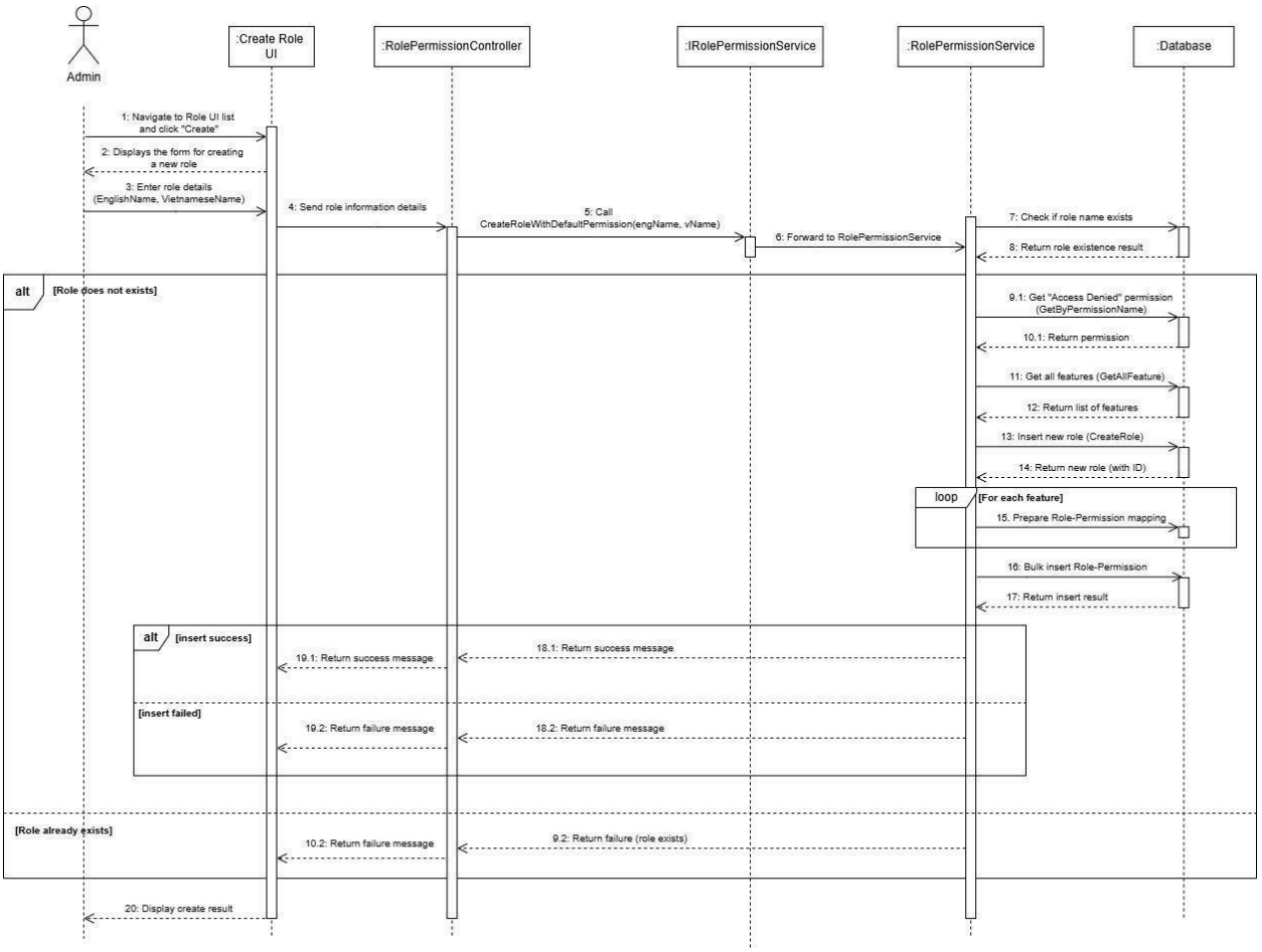


Figure 3.5.1.2 Sequence Diagram

3.5.1.3 Class Specification

3.5.1.3.a Role Permission Controller

Table 3.5.1.3.a: Role Permission Controller Description Table

No	Method	Description
1	CreateRole(CreateRoleRequest request)	Receives role details (e.g., English and Vietnamese names) and forwards them to the service layer to create a new role.

3.5.1.3.b Role Permission Service

Table 3.5.1.3.b: Role Permission Service Description Table

No	Method	Description
1	CreateRoleWithDefaultPermission(string engName, string vieName)	Creates a new role with default permissions. Checks if the role already exists. if not, retrieves the “Access Denied” permission, fetches all system features,

		inserts the new role into the database, and assigns defaults.
2	GetAllFeature()	Retrieves all available system features that can be assigned to roles.
3	GetByPermissionName(PermissionEnum permission)	Retrieves permission details based on the provided permission name (e.g., “Access Denied”).

3.5.2. Get Feature Permission

3.5.2.1 Class Diagram

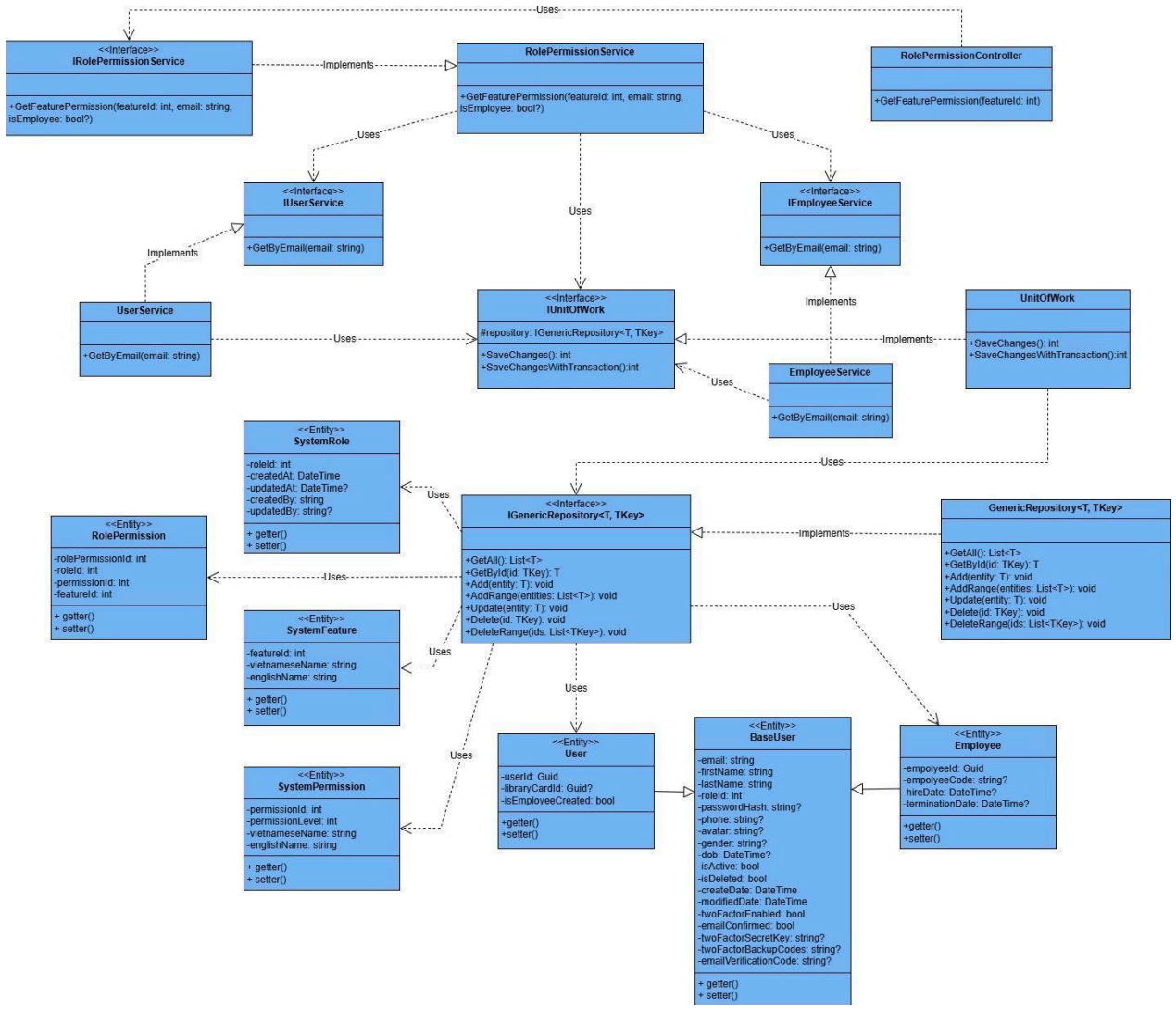


Figure 3.5.2.1 Class Diagram

3.5.2.2 Sequence Diagram

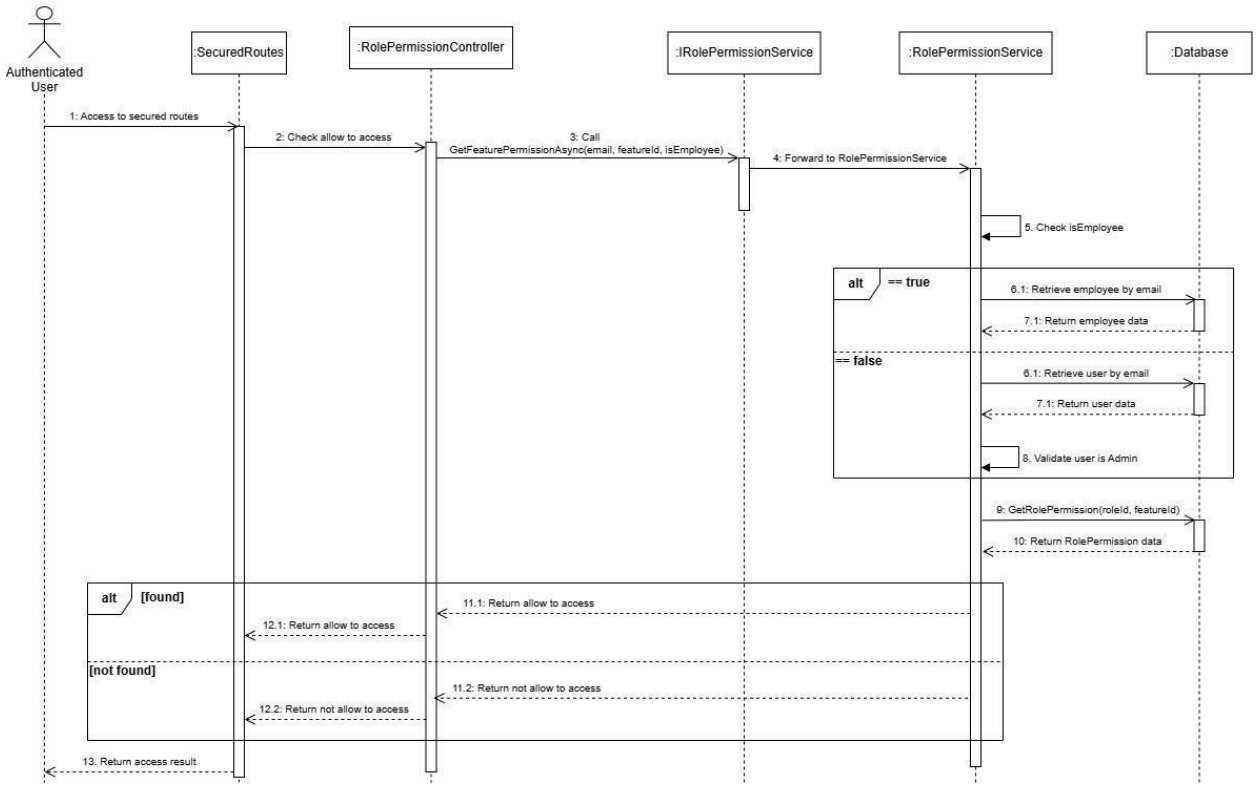


Figure 3.5.2.2 Sequence Diagram

3.5.2.3 Class Specification

3.5.2.3.a Role Permission Controller

Table 3.5.2.3.a: Role Permission Controller Description Table

No	Method	Description
1	GetFeaturePermission(int featureId)	Extracts the user's email and user type from the request context, then forwards to the service layer to check whether the user (employee or admin) has permission for the specified feature.

3.5.2.3.b Role Permission Service

Table 3.5.2.3.b: Role Permission Service Description Table

No	Method	Description
1	GetFeaturePermission(int featureId, string email, bool? isEmployee)	Verifies the user's role (employee or admin) using the provided email and user type, then retrieves the matching role-permission record for the specified feature.

2	GetRolePermission(int roleId, int featureId)	Get permission data for the specified role and feature to access permission level.

3.5.3. Update Permission

3.5.3.1 Class Diagram

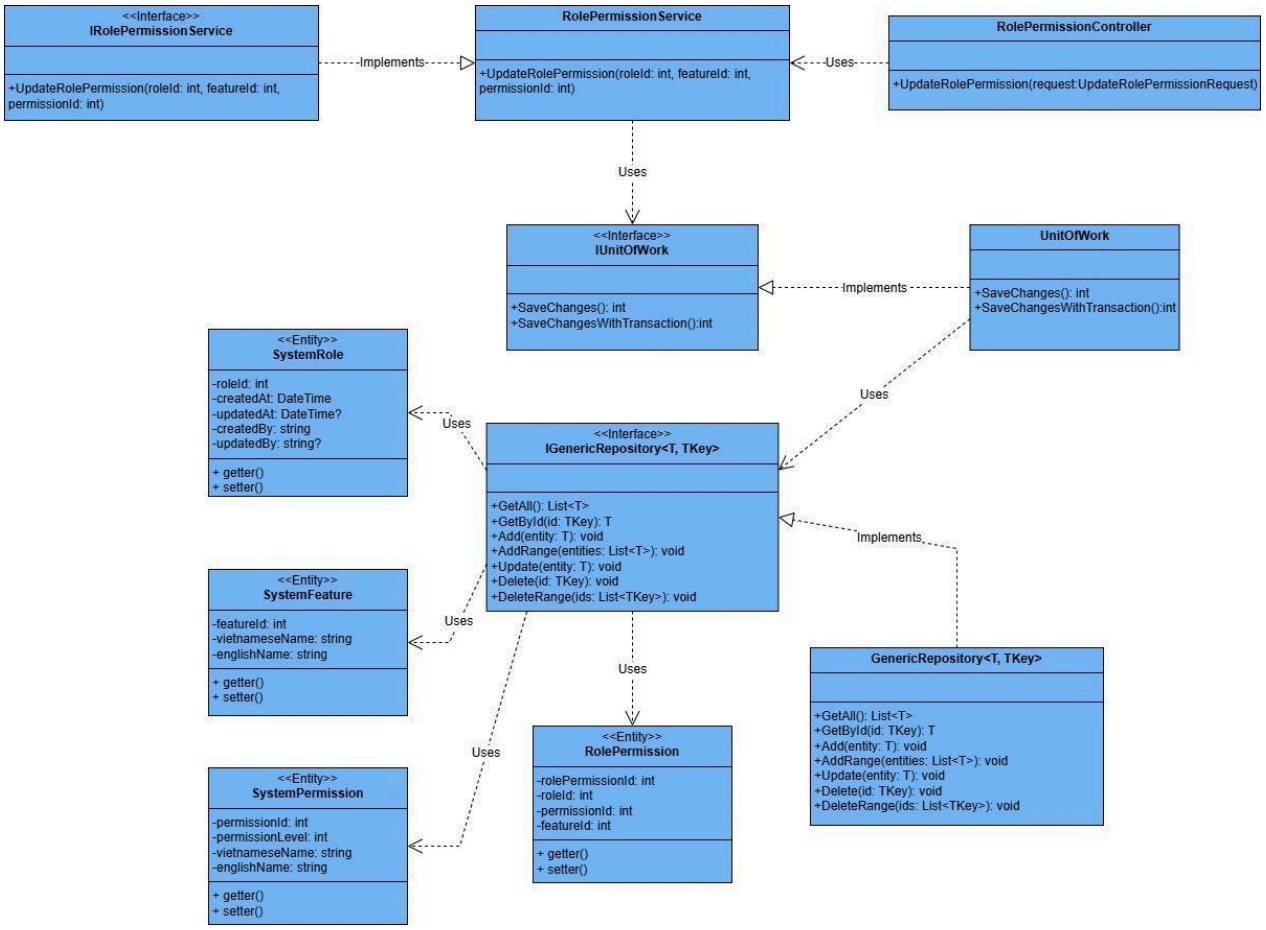


Figure 3.5.3.1 Class Diagram

3.5.3.2 Sequence Diagram

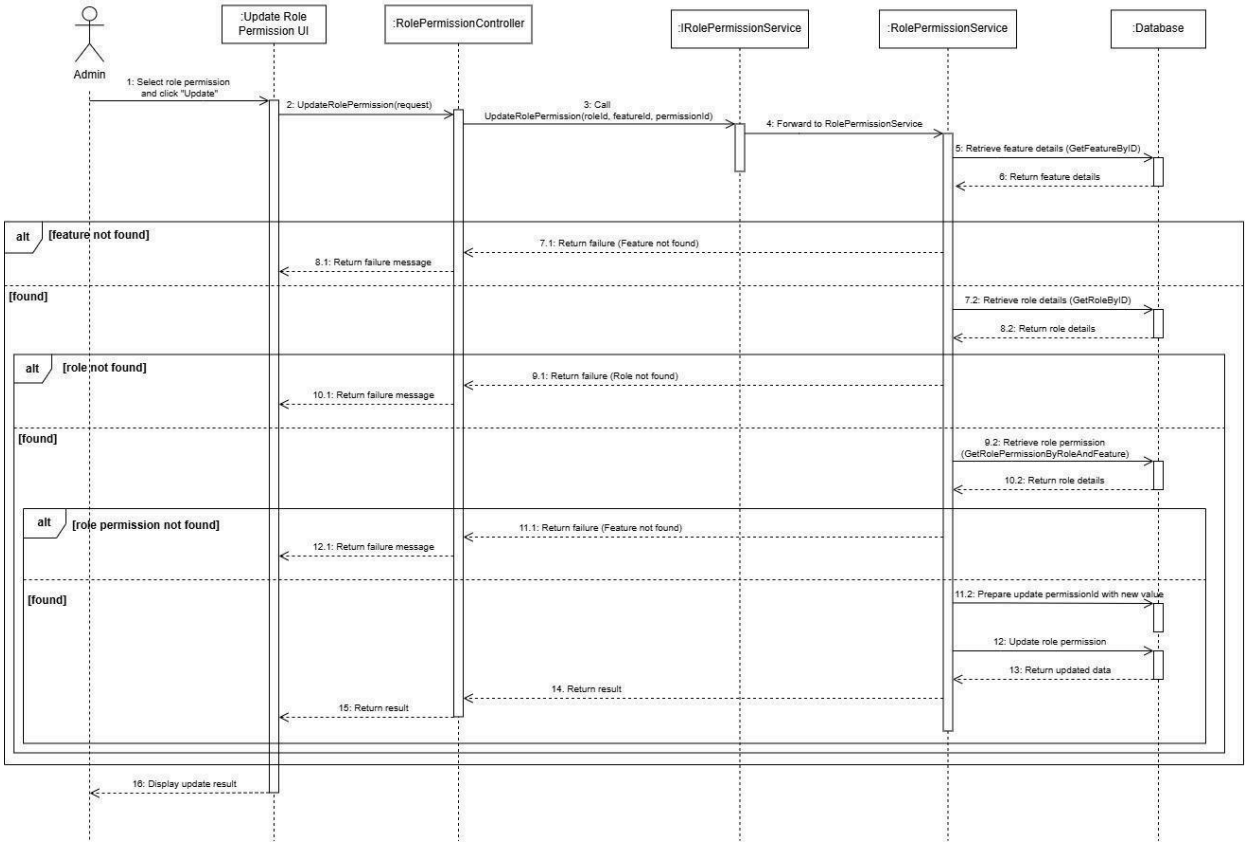


Figure 3.5.3.2 Sequence Diagram

3.5.3.3 Class Specification

3.5.3.3.a Role Permission Controller

Table 3.5.3.3.a: Role Permission Controller Description Table

No	Method	Description
1	UpdateRolePermission(UpdateRolePermissionRequest request)	Receives the updated role-permission information (role ID, feature ID, and new permission ID) and forwards them to the service layer to update role permission.

3.5.3.3.b Role Permission Service

Table 3.5.3.3.b: Role Permission Service Description Table

No	Method	Description
1	UpdateRolePermission(int roleId, int featureId, int permissionId)	After confirming the existence of the specified role, feature, and permission, this method updates the role-permission mapping to assign the new permission to the role for the given feature.

2	GetFeatureById(int featureId)	Retrieves the feature entity from the database using the provided feature ID.
3	GetRoleById(int roleId)	Retrieves the role entity from the database using the provided feature ID.
4	GetPermissionById(int permissionId)	Retrieves the permission entity from the database using the provided feature ID.

3.6. Manage Categories

3.6.1. Create Category

3.6.1.1 Class Diagram

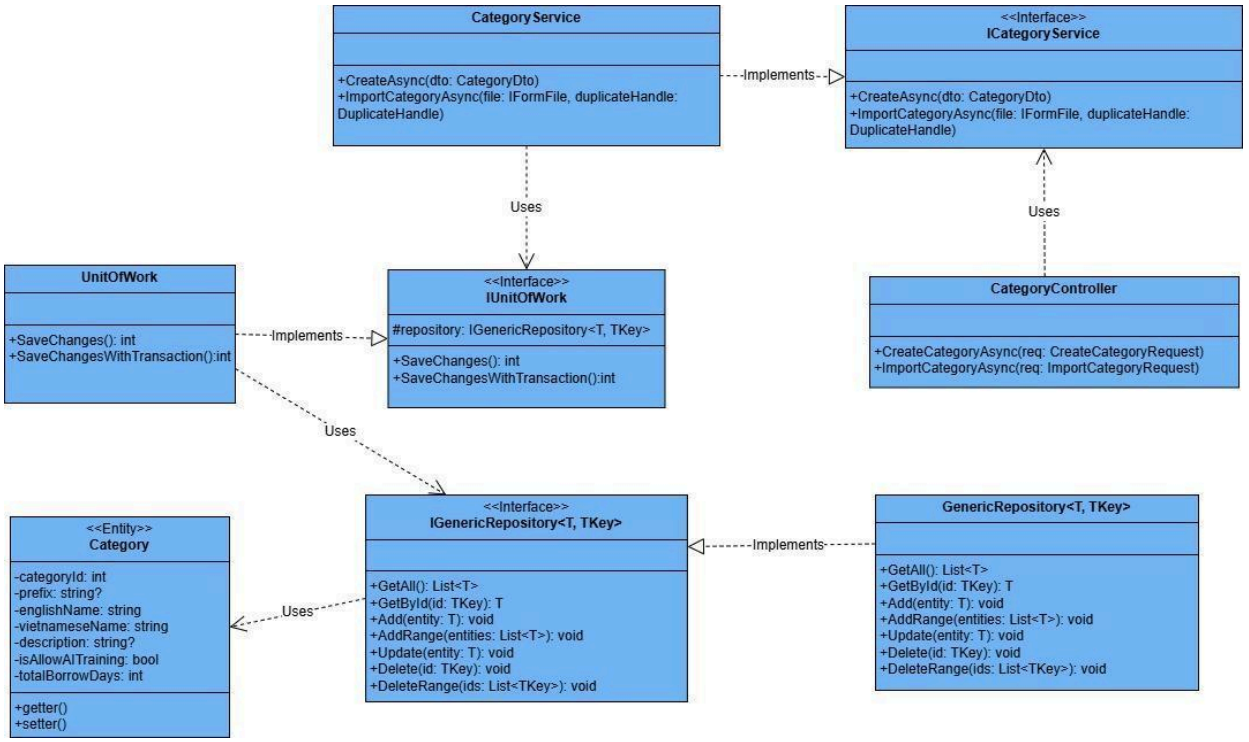


Figure 3.6.1.1 Class Diagram

3.6.1.2 Sequence Diagram

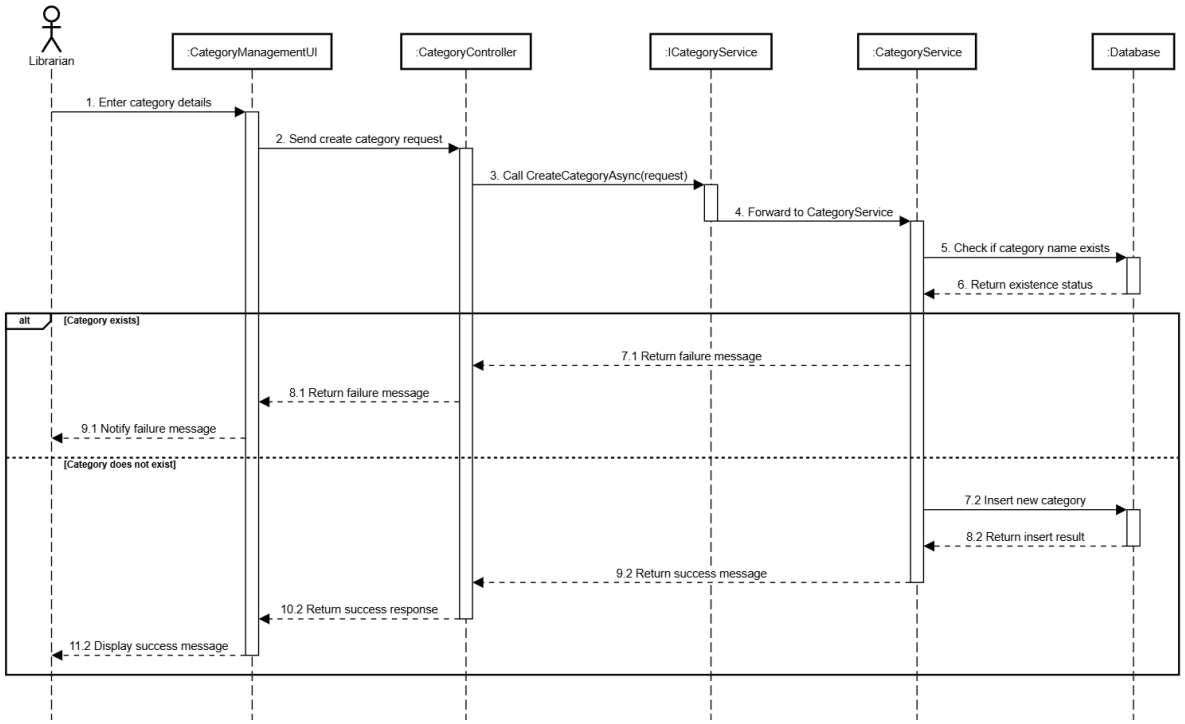


Figure 3.6.1.2 Sequence Diagram

3.6.1.3 Class Specification

3.6.1.3.a Category Controller

Table 3.6.1.3.a: Category Controller Description Table

No	Method	Description
1	CreateCategoryAsync(CreateCategoryRequest req)	Fill the form with category information and forward to the service layer to create a new book category.
2	ImportCategoryAsync(ImportBookCategoryRequest req)	Create many categories at the same time via excel file which include account information to create a single category.

3.6.1.3.Category Service

Table 3.6.1.3.b:Category Service Description Table

No	Method	Description
1	CreateCategoryAsync(CreateCategoryRequest req)	Create many categories at the same time via excel file.

2	ImportCategoryAsync(IFormFile file,DuplicateHandle duplicateHandle)	Create new categories by reading excel files.

3.6.2. Get All Category

3.6.2.1 Class Diagram

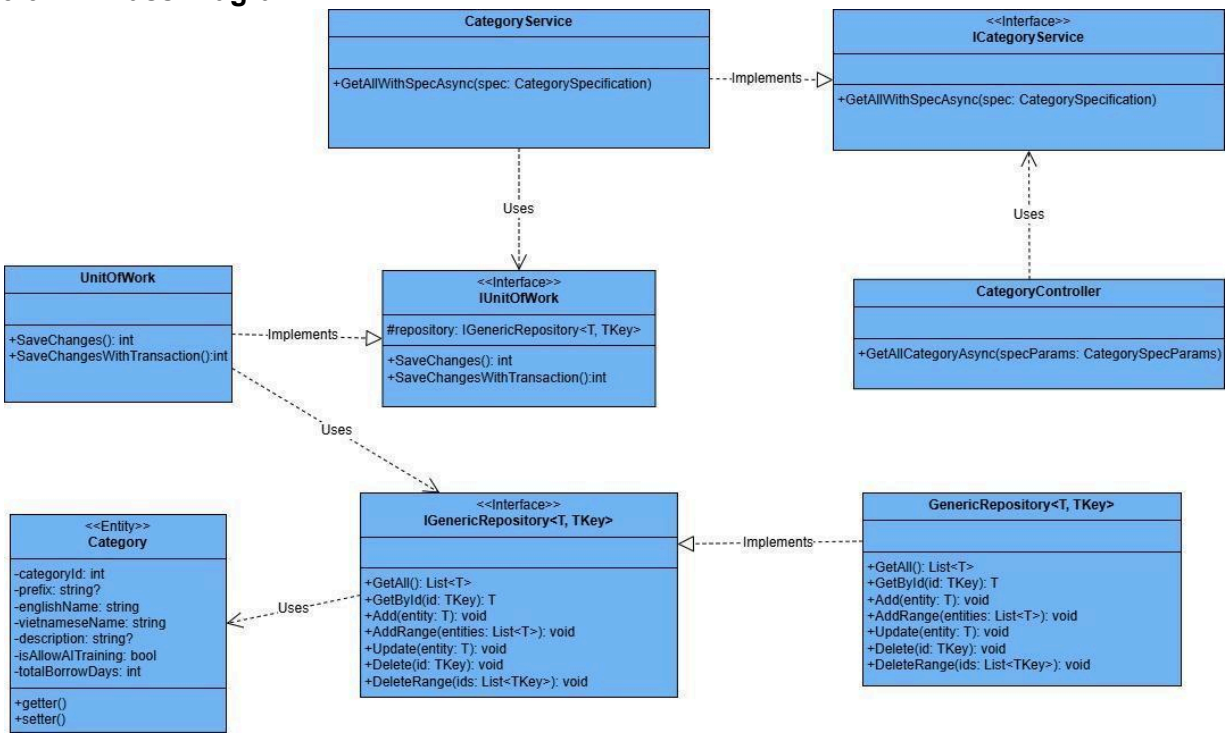


Figure 3.6.2.1 Class Diagram

3.6.2.2 Sequence Diagram

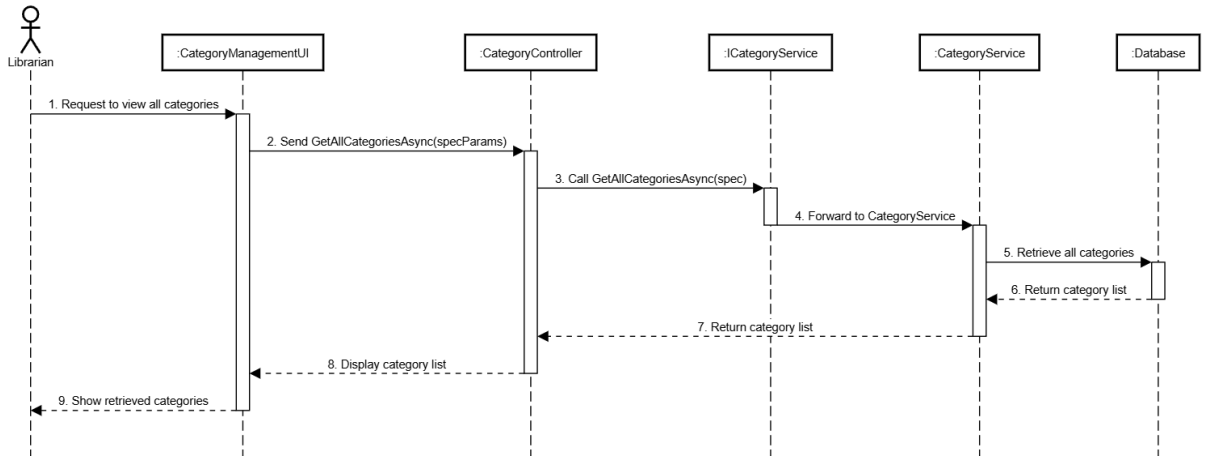


Figure 3.6.2.2 Sequence Diagram

3.6.2.3 Class Specification

3.6.2.3.a Category Controller

Table 3.6.2.3.a:Category Controller Description Table

No	Method	Description
1	GetAllCategoriesAsync(CategorySpecParams specParams)	Get all categories in the system and support searching and filtering.

3.6.2.3.b Category Service

Table 3.6.2.3.b:Category Service Description Table

No	Method	Description
1	GetAllWithSpec(CategorySpecification spec)	Get and filter all public categories in the database.

3.6.3. Update Category

3.6.3.1 Class Diagram

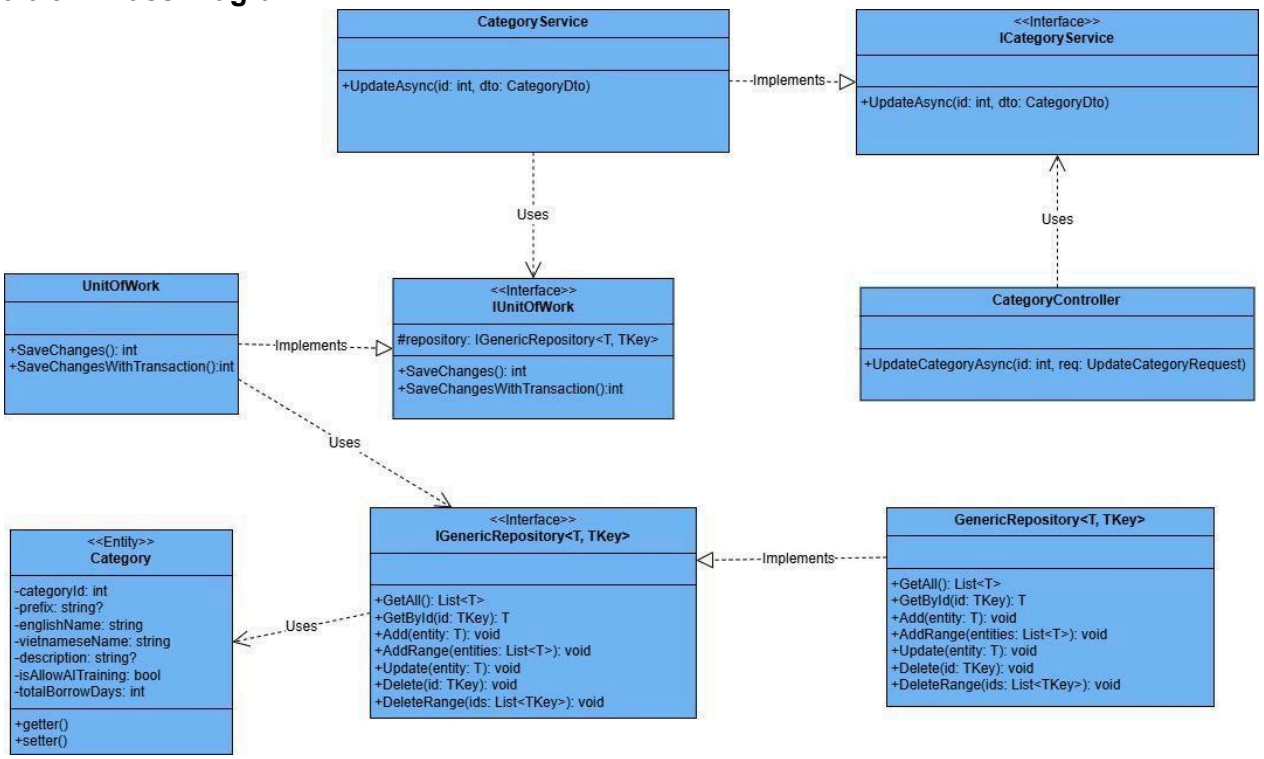


Figure 3.6.3.1 Class Diagram

3.6.3.2 Sequence Diagram

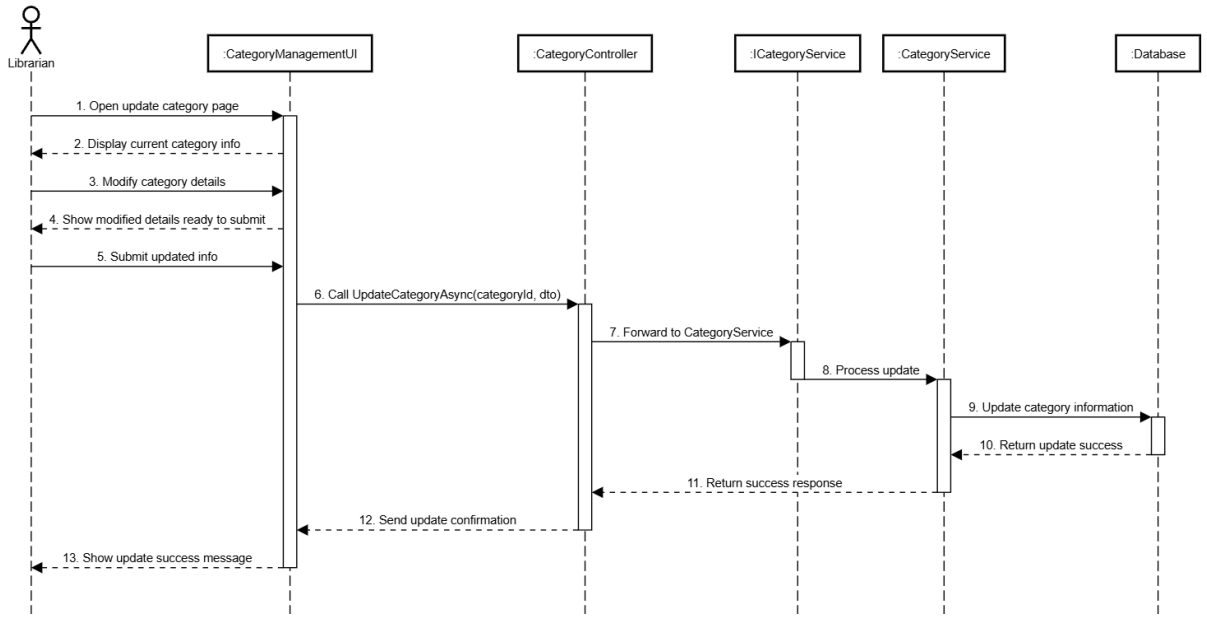


Figure 3.6.3.1 Class Diagram

3.6.3.3 Class Specification

3.6.3.3.a Category Controller

Table 3.6.3.3.a: Category Controller Description Table

No	Method	Description
1	UpdateCategoryAsync(int CategoryId, UpdateCategoryRequest updateRequest)	Update Category information that displays in category management UI.

3.6.3.3.b Category Service

Table 3.6.3.3.b: Category Service Description Table

No	Method	Description
1	UpdateAsync(Guid CategoryId, UpdateCategoryRequest updateRequest)	Validate and update the changed information of the category.

3.6.4. Delete Category

3.6.4.1 Class Diagram

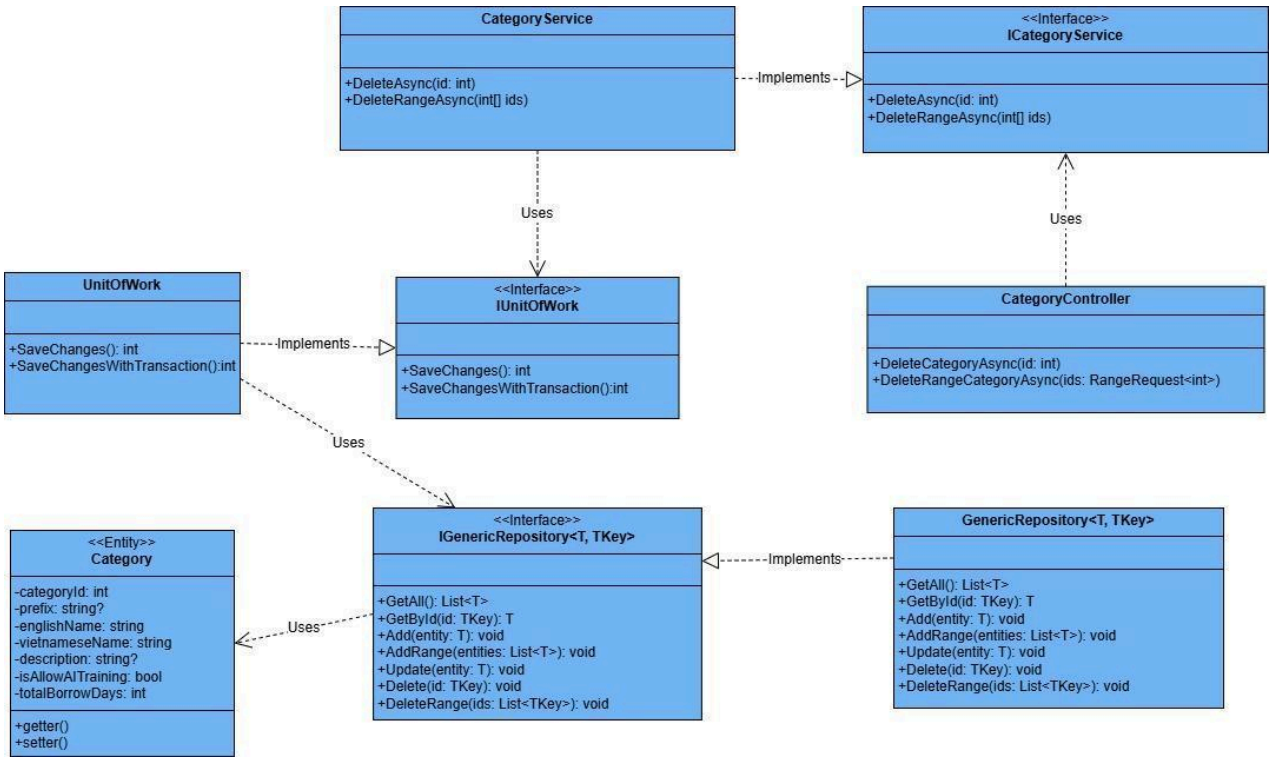


Figure 3.6.4.1 Class Diagram

3.6.4.2 Sequence Diagram

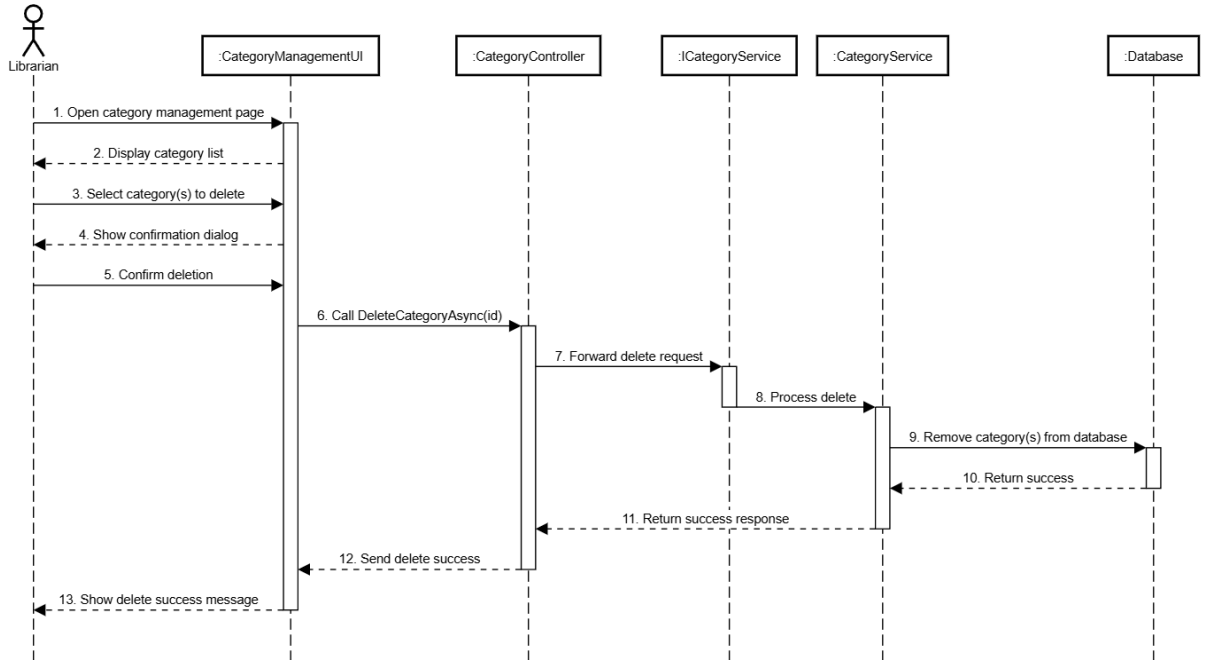


Figure 3.6.4.2 Sequence Diagram

3.6.4.3 Class Specification

3.6.4.3.a Category Controller

Table 3.6.4.3.a: Category Controller Description Table

No	Method	Description

1	DeleteCategoryAsync(int categoryId)	Choose a category that is not used any more to delete.
2	DeleteAsync(int categoryId)	Choose many categories that are not used to delete.

3.6.4.3.b Category Service

Table 3.6.4.3.b: Category Service Description Table

No	Method	Description
1	DeleteAsync(int CategoryId)	Remove category and it could not be recovered after deletion.
2	DeleteRangeAsync(int[] req)	Remove categories from the database and could not be recovered after deletion.

3.7. Manage Authors

3.7.1. Create Author

3.7.1.1 Class Diagram

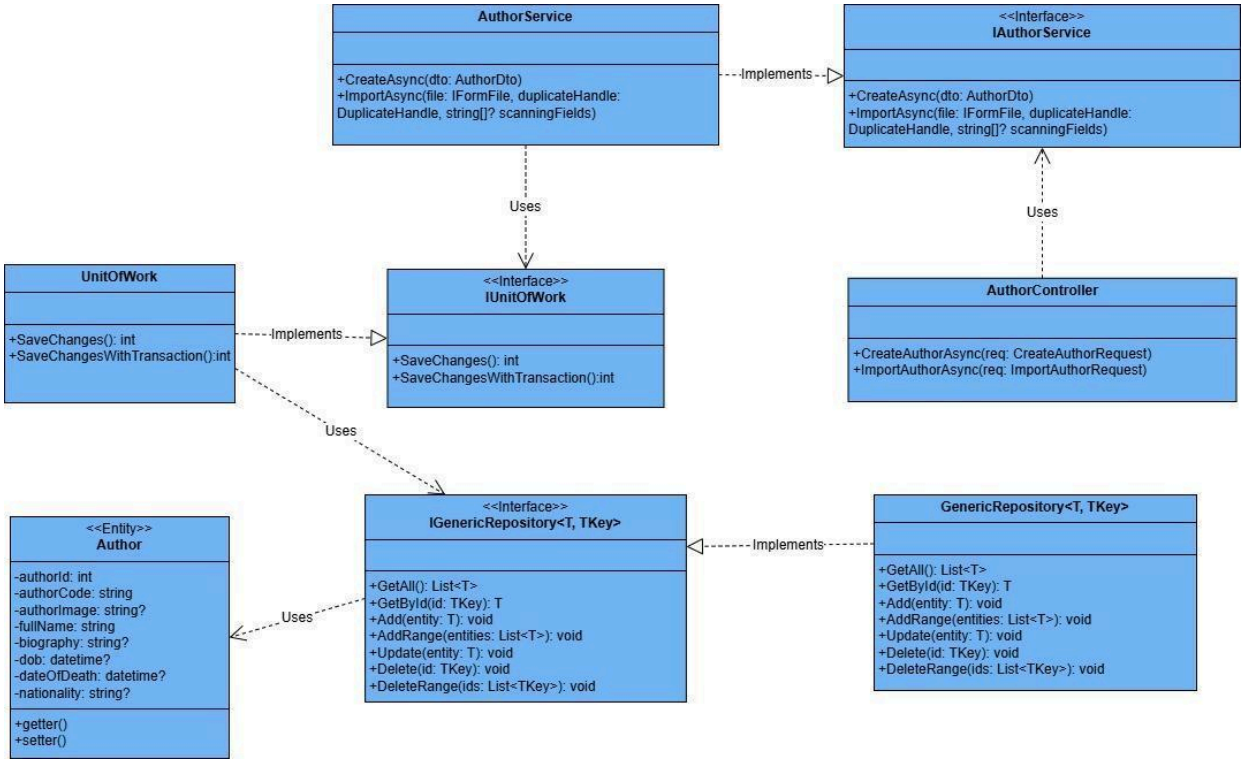


Figure 3.7.1.1 Class Diagram

3.7.1.2 Sequence Diagram

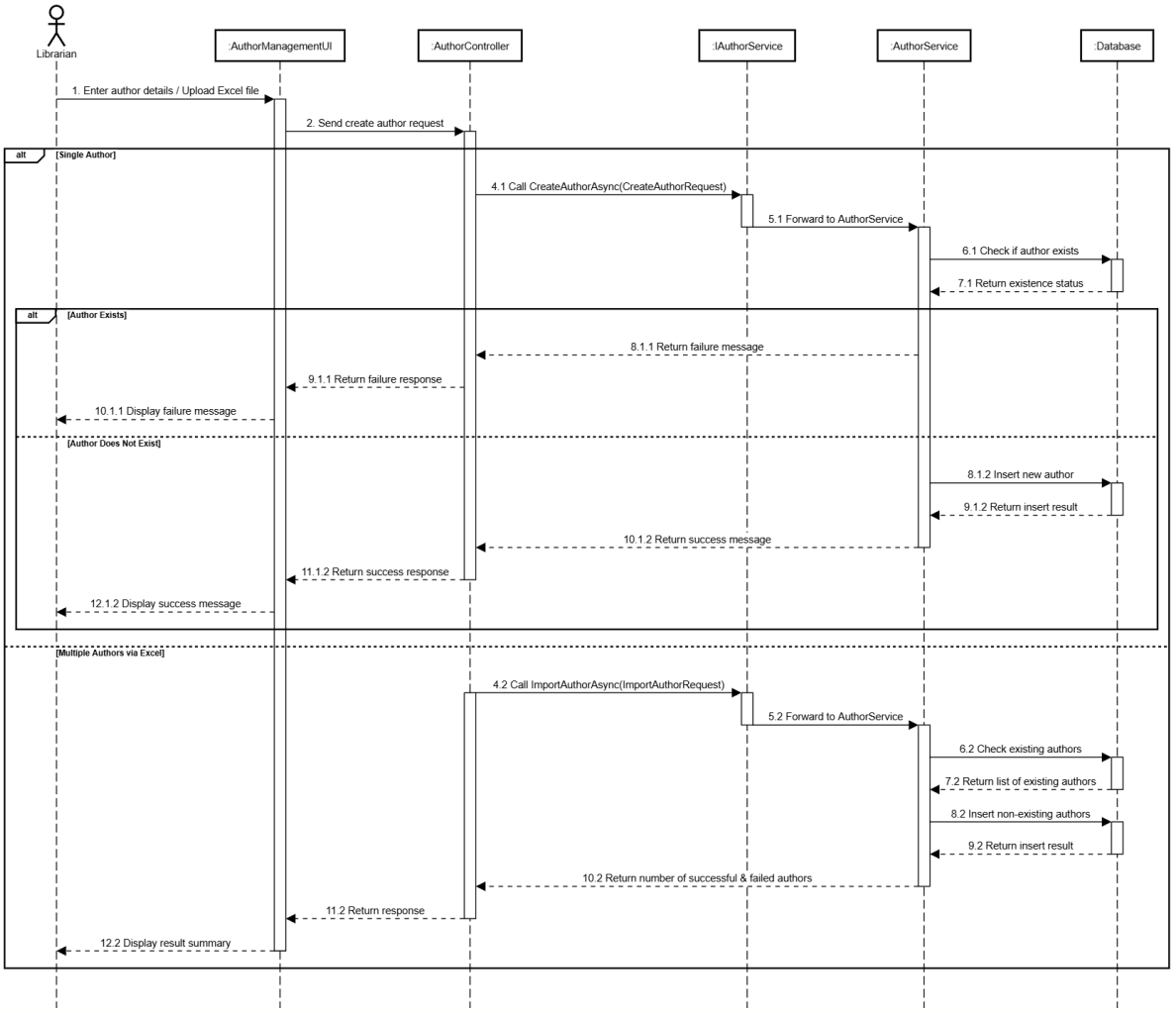


Figure 3.7.1.2 Sequence Diagram

3.7.1.3 Class Specification

3.7.1.3.a AuthorController

Table 3.7.1.3.AuthorController Description Table

No	Method	Description
1	CreateAuthorAsync(CreateAuthorRequest req)	Create a new author with required information.
2	ImportAuthorAsync(ImportAuthorRequest req)	Create authors by importing excel files that contain author’s information.

3.7.1.3.b Author Service

Table 3.7.1.3.b:Author Service Description Table

No	Method	Description
1	CreateAuthorAsync(AuthorDto req)	Create a single author with given information.
2	ImportAuthorAsync(IFormFile file,DuplicateHandle duplicateHandle,string[]? ScanningField)	Handle create multi authors via excel file at the same time.

3.7.2.1 Class Diagram

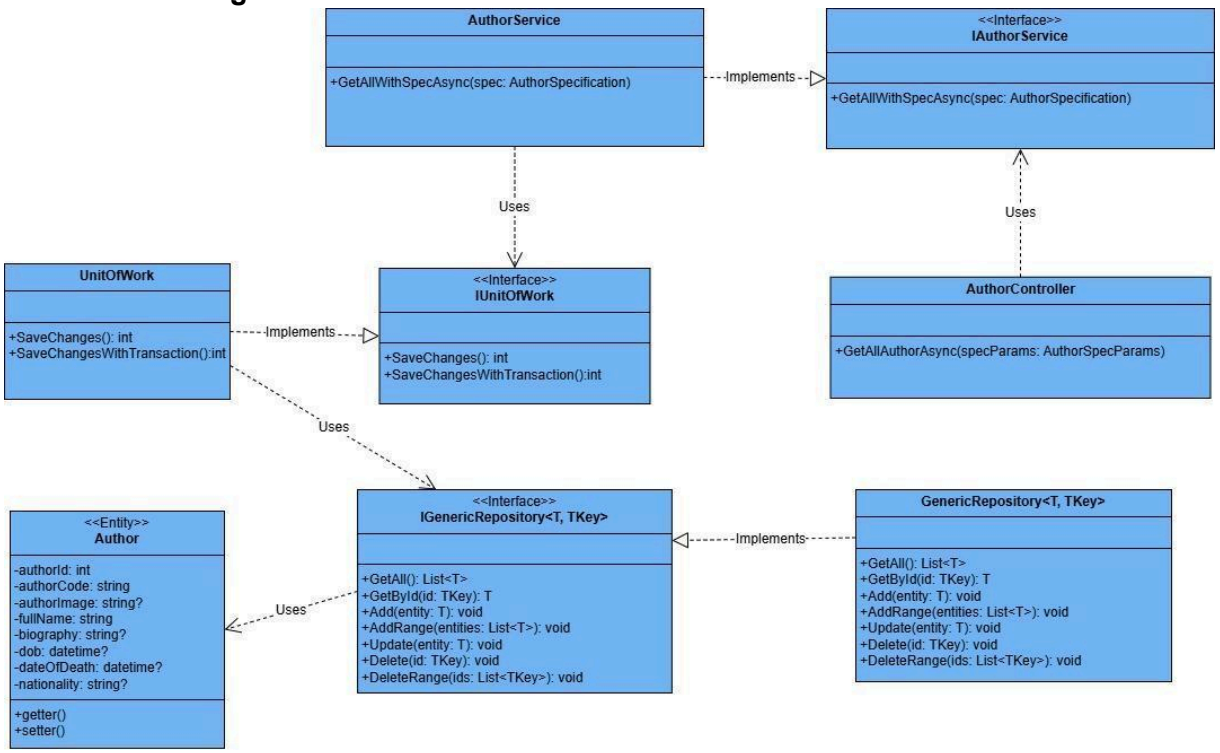


Figure 3.7.2.1 Class Diagram

3.7.2.2 Sequence Diagram

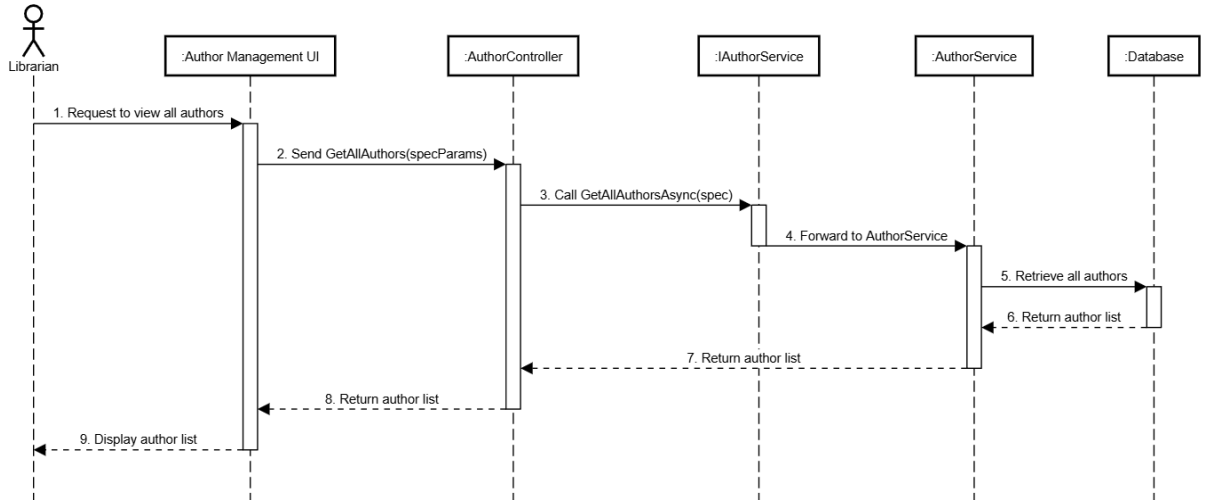


Figure 3.7.2.2 Sequence Diagram

3.7.2.3 Class Specification

3.7.2.3.a Author Controller

Table 3.7.2.3.a Author Controller Description Table

No	Method	Description
1	GetAllAuthorFromPublic(AuthorSpeciParams authorSpecParams)	Get all authors in the system and support searching and filtering.

3.7.2.3.b Author Service

Table 3.7.2.3.b Author Service Description Table

No	Method	Description
1	GetAllWithSpec(AuthorSpecParam spec,bool tracked)	Get and filter all public authors in the database.

3.7.3. Update Author

3.7.3.1 Class Diagram

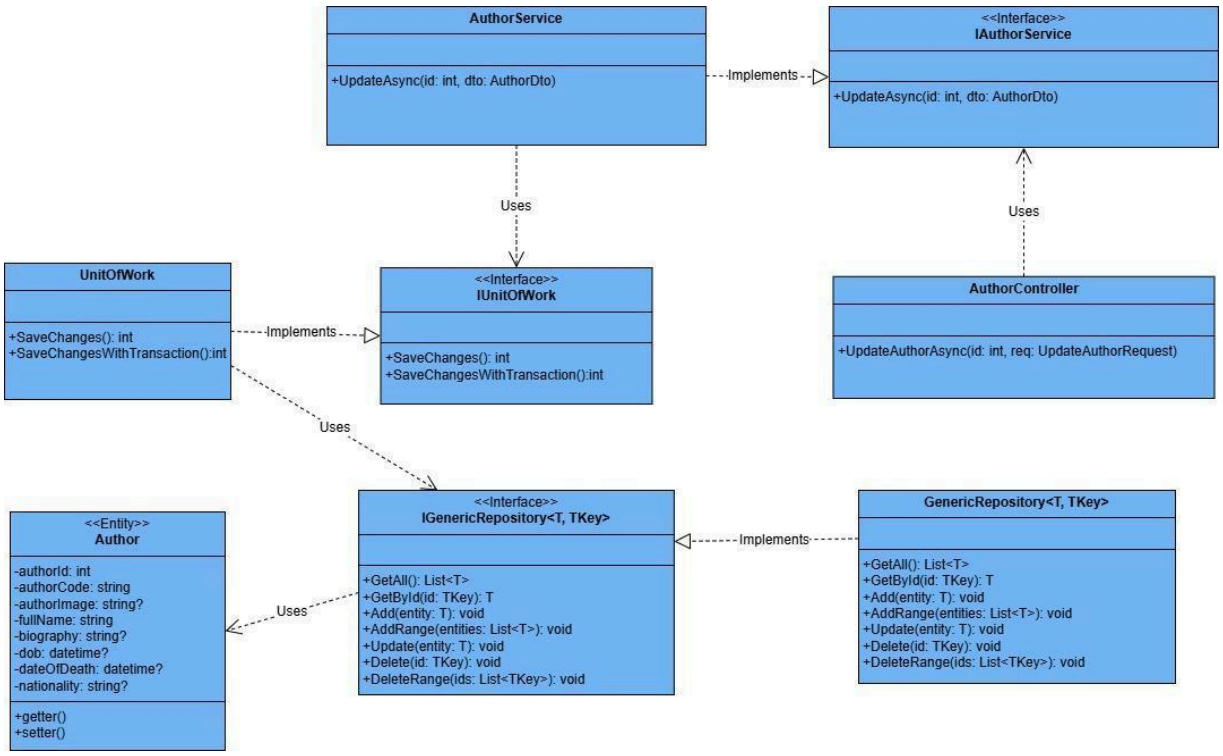


Figure 3.7.3.1 Class Diagram

3.7.3.2 Sequence Diagram

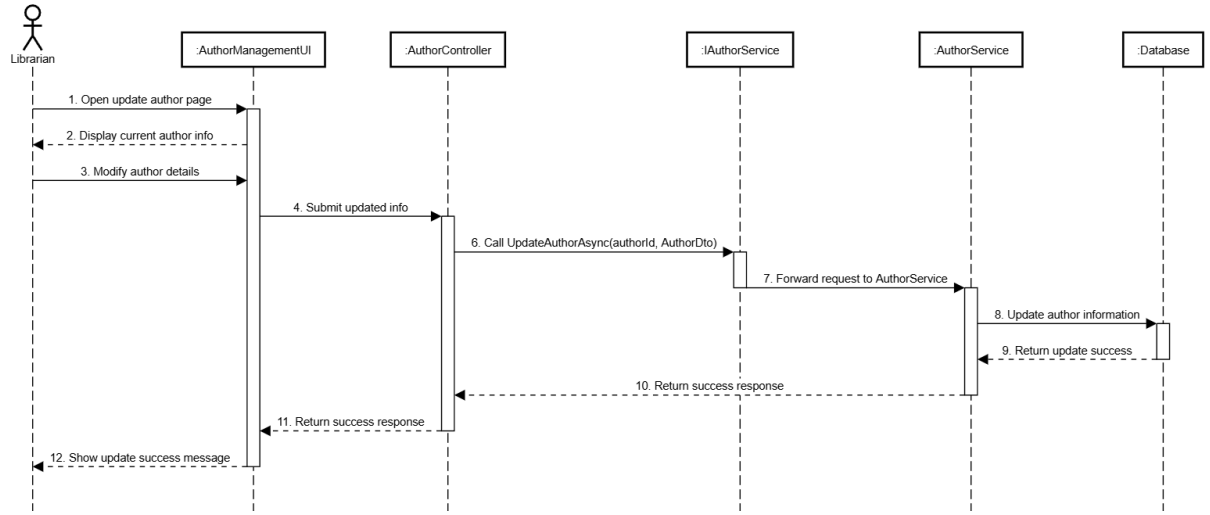


Figure 3.7.3.2 Sequence Diagram

3.7.3 Class Specification

3.7.3.1.a Author Controller

Table 3.7.3.1.a: Author Controller Description Table

No	Method	Description
1	UpdateAuthorAsync(int AuthorId, UpdateAuthorRequest updateRequest)	Update Author information that displays in the Author management UI.

3.7.3.2.b Author Service

Table 3.7.3.2.b:Author Service Description Table

No	Method	Description
1	UpdateAsync(int AuthorId, UpdateAuthorRequest updateRequest)	Validate and update the changed information of the author.

3.7.4. Delete Author

3.7.4.1 Class Diagram

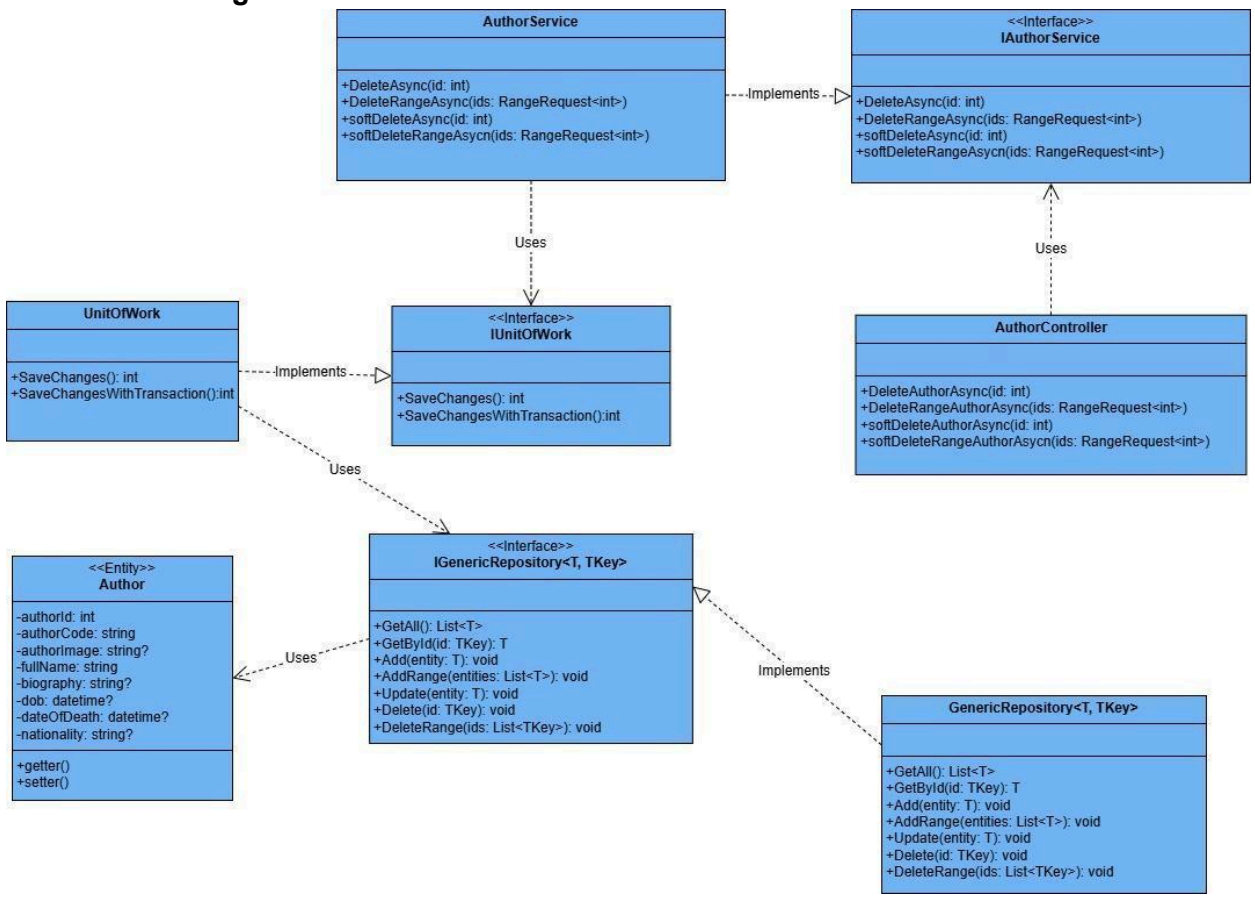


Figure 3.7.4.1 Class Diagram

3.7.4.2 Sequence Diagram

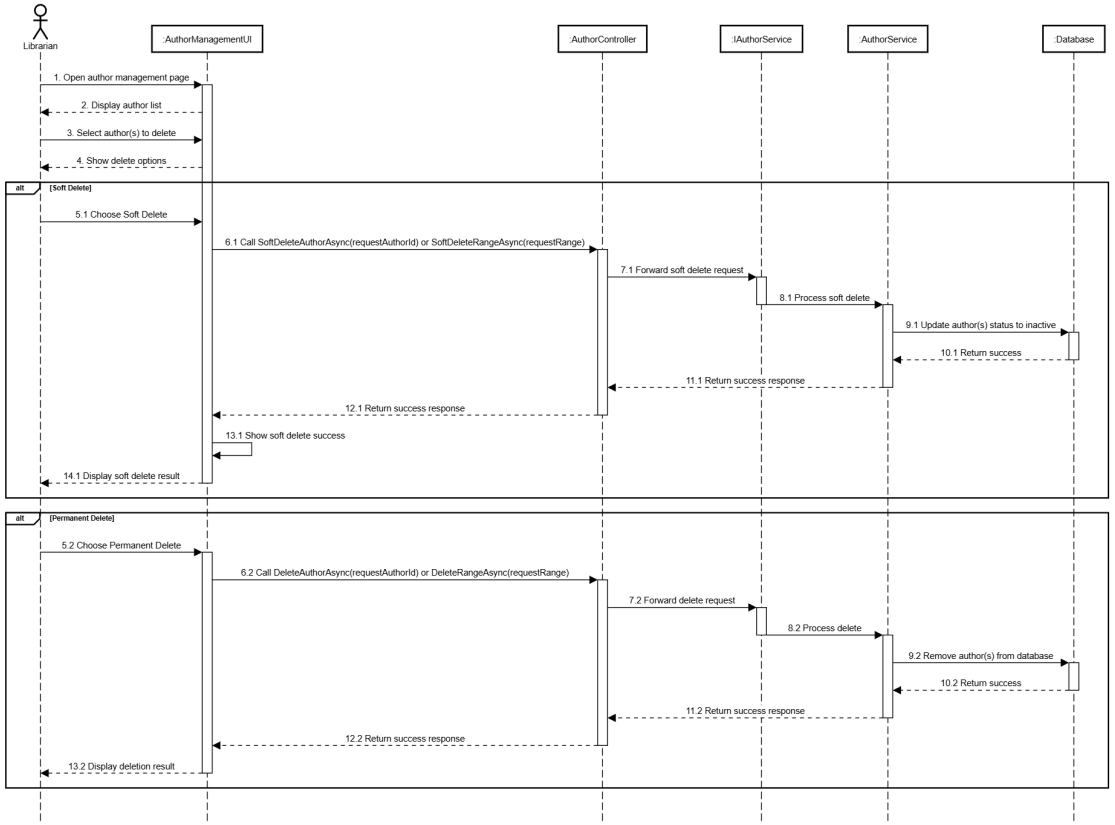


Figure 3.7.4.2 Sequence Diagram

3.7.4.3 Class Specification

3.7.4.3.a Author Controller

Table 3.7.4.3.a: Author Controller Description Table

No	Method	Description
1	DeleteAuthorAsync(int AuthorId)	Choose an Author that is not used any more to delete.
2	DeleteRangeAuthorAsync(RangeRequest <int> req)	Choose many categories that are not used to delete.

3.7.4.3.b Author Service

Table 3.7.4.3.b: Author Service Description Table

No	Method	Description
1	DeleteAsync(int AuthorId)	Remove Author and it could not be recovered after deletion.

2	DeleteRangeAsync(int[] req)	Remove categories from the database and could not be recovered after deleted.

3.8. Manage Suppliers

3.8.1. Create Supplier

3.8.1.1 Class Diagram

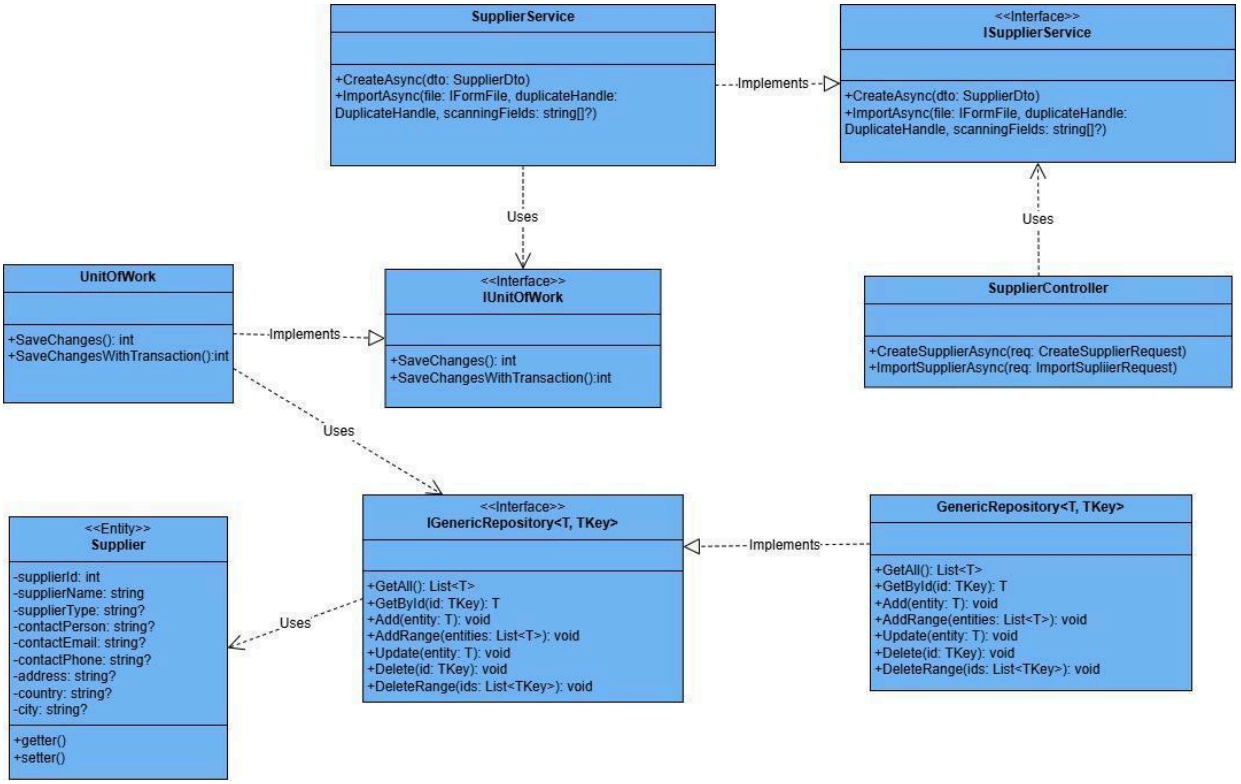


Figure 3.8.1.1 Class Diagram

3.8.1.2 Sequence Diagram

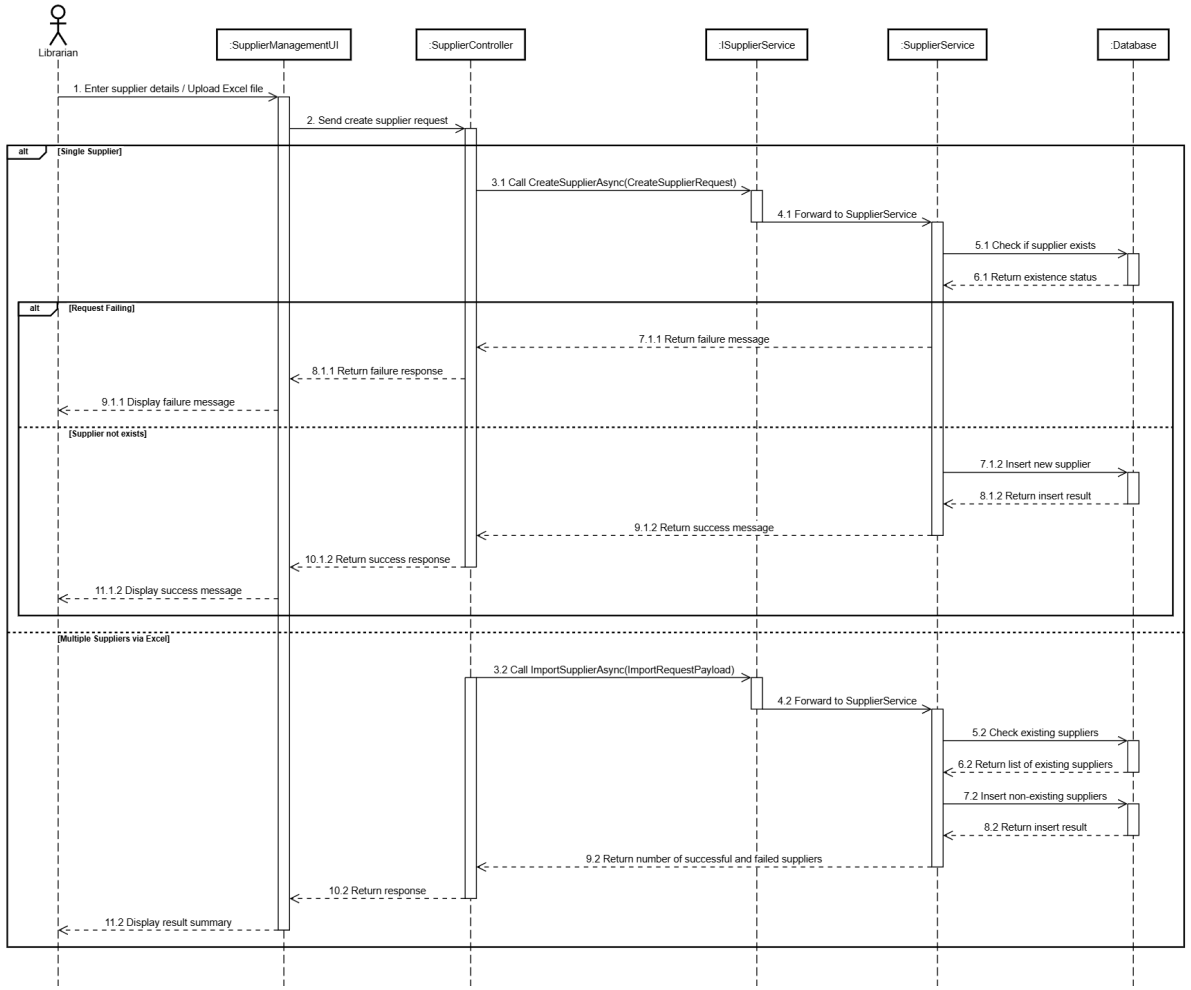


Figure 3.8.1.2 Sequence Diagram

3.8.1.3 Class Specification

3.8.1.3.Supplier Controller

Table 3.8.1.3 Supplier Controller Description Table

No	Method	Description
1	CreateSupplierAsync(CreateSupplierRequest req)	Create a new Supplier with required information.
2	ImportSupplierAsync(ImportSupplierRequest req)	Create Suppliers by importing an excel file that contains Supplier's information.

3.8.1.3.Supplier Service

Table 3.8.1.3.b Supplier Service Description Table

No	Method	Description
1	CreatSupplierAsync(SupplierDto req)	Create a single Supplier with given information.
2	ImportSupplierAsync(IFormFile file,DuplicateHandle duplicateHandle,string[]? ScanningField)	Handle create multi Suppliers via excel file at the same time.

3.8.2. Get All Supplier

3.8.2.1 Class Diagram

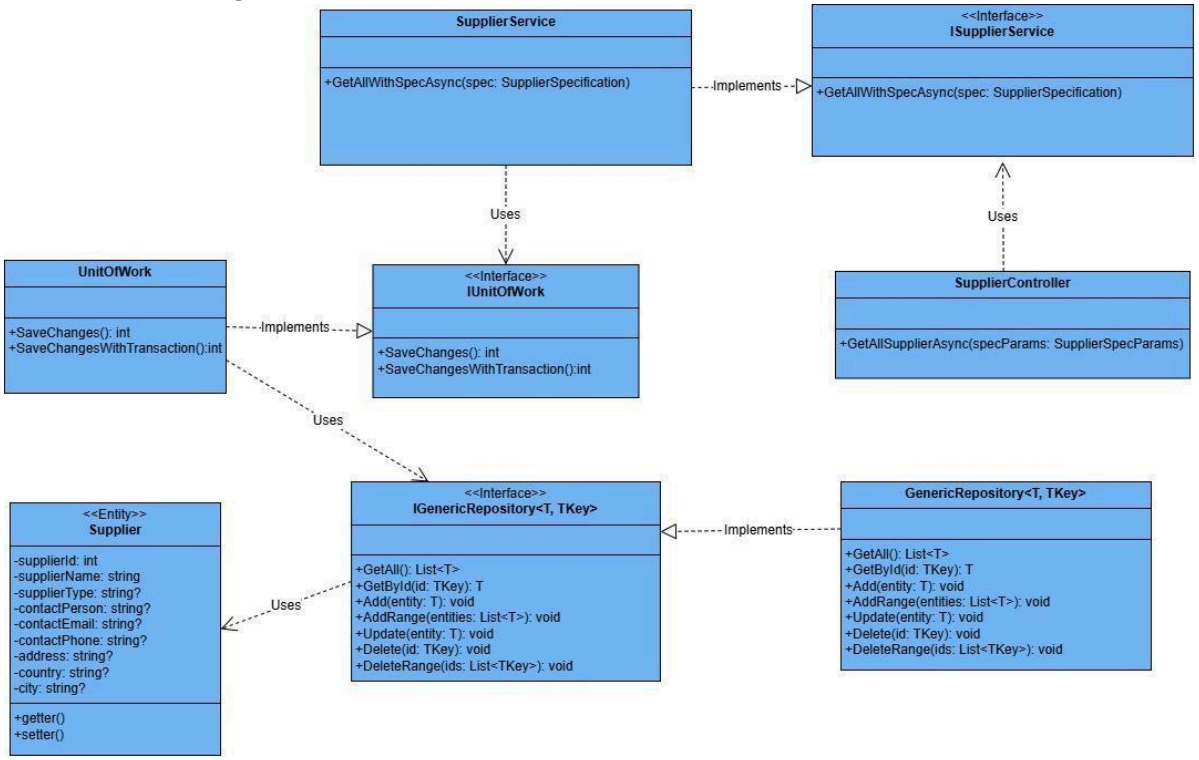


Figure 3.8.2.1 Class Diagram

3.8.2.2 Sequence Diagram

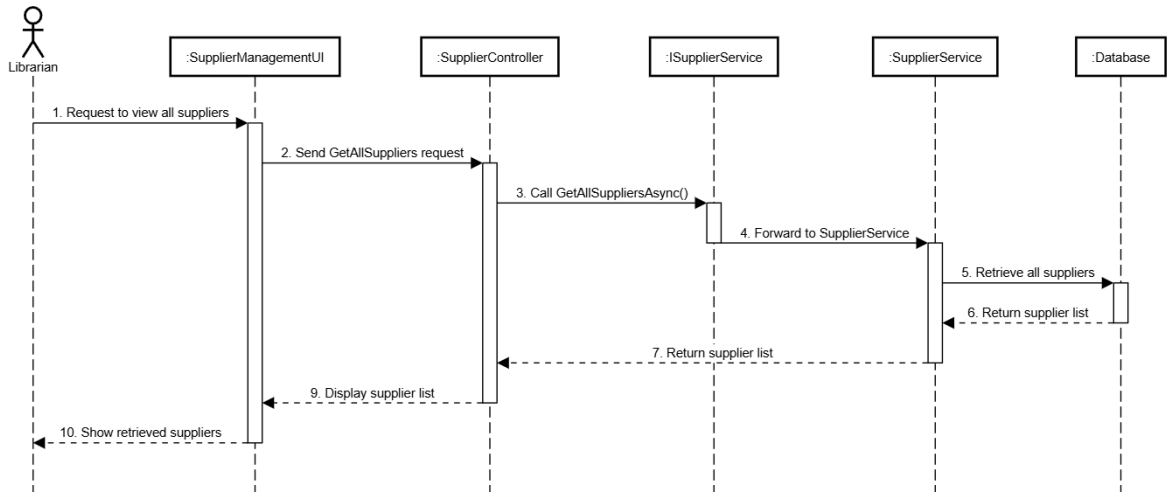


Figure 3.8.2.2 Sequence Diagram

3.8.2.3 Class Specification

3.8.2.3.a Supplier Controller

Table 3.8.2.3.a Supplier Controller Description Table

No	Method	Description
1	GetAllSupplierFromPublic(SupplierSpeciParams SupplierSpecParams)	Get all Suppliers in the system and support searching and filtering.

3.8.2.3.b Supplier Service

Table 3.8.2.3.b Supplier Service Description Table

No	Method	Description
1	GetAllWithSpec(SupplierSpecification spec)	Get and filter all public Suppliers in the database.

3.8.3. Update Supplier

3.8.3.1 Class Diagram

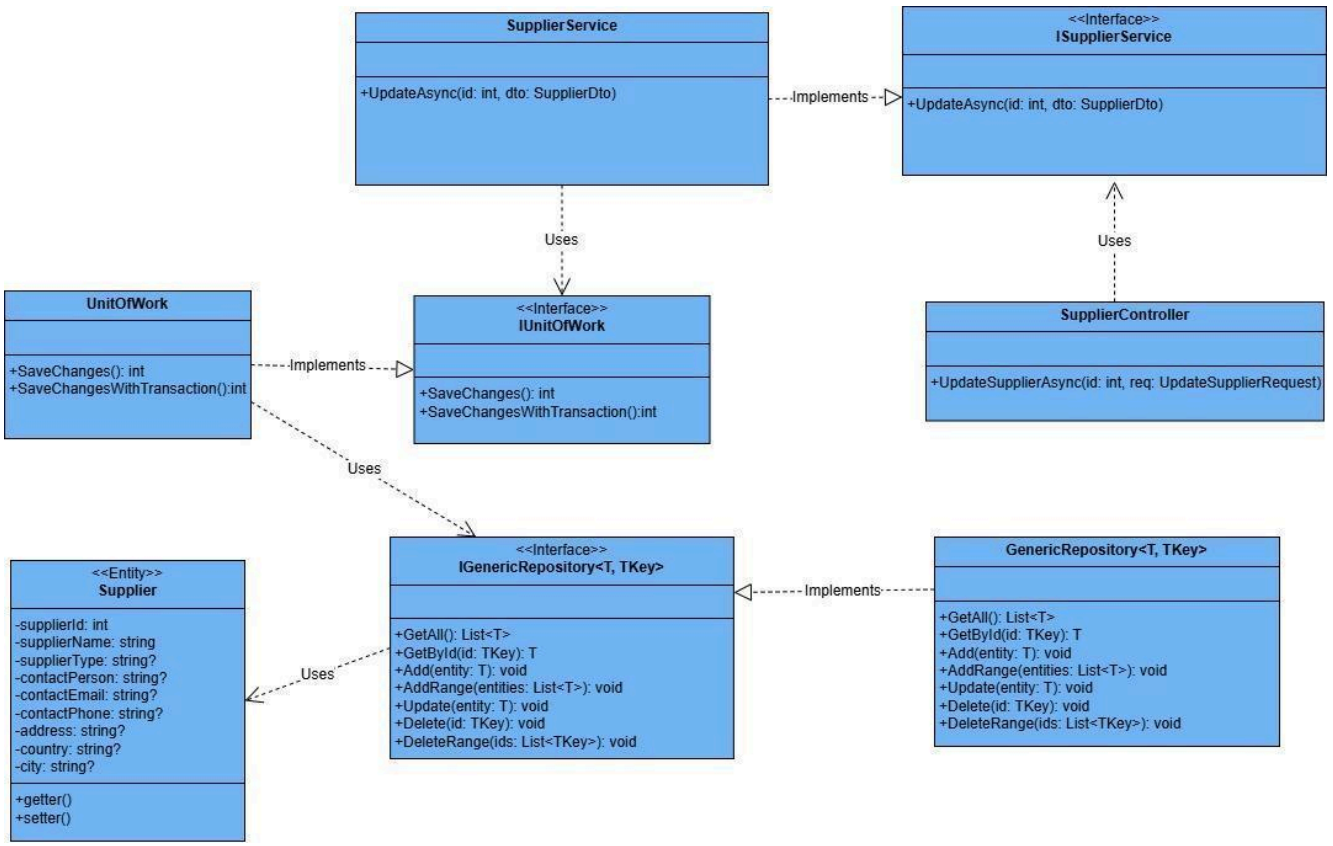


Figure 3.8.3.1 Class Diagram

3.8.3.2 Sequence Diagram

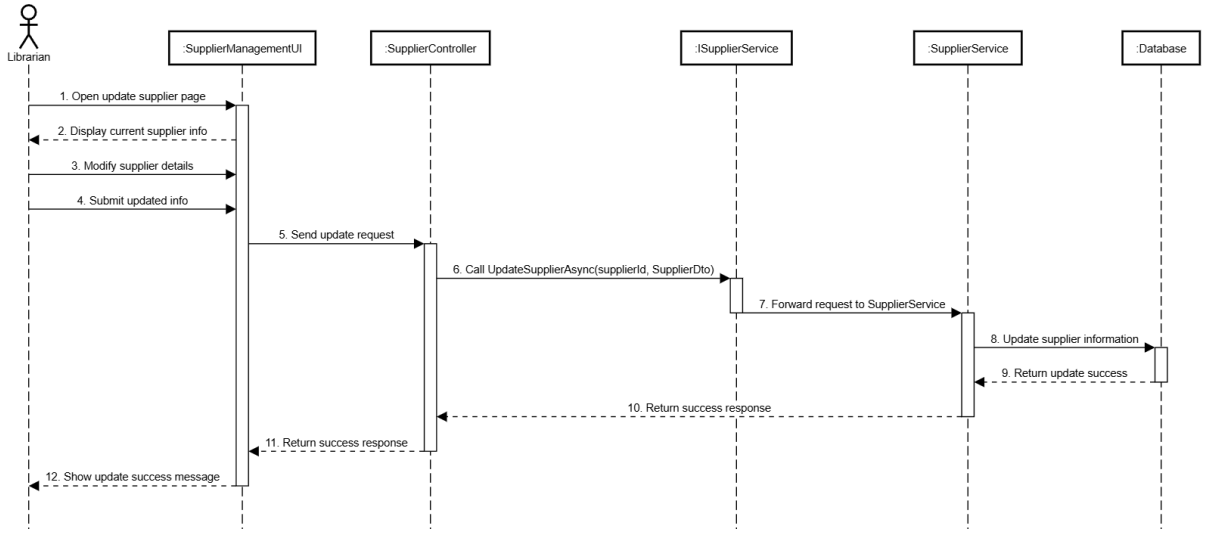


Figure 3.8.3.2 Sequence Diagram

3.8.3.3 Class Specification

3.8.3.3.a Supplier Controller

Table 3.8.3.3.a: Supplier Controller Description Table

No	Method	Description

1	UpdateSupplierAsync(int SupplierId, UpdateSupplierRequest updateRequest)	Update Supplier information that displays in Supplier management UI.

3.8.3.3.b Supplier Service

Table 3.8.3.3.b:Supplier Service Description Table

No	Method	Description
1	UpdateAsync(int SupplierId, SupplierDto supplierDto)	Validate and update the changed information of Supplier.

3.8.4. Delete Supplier

3.8.4.1 Class Diagram

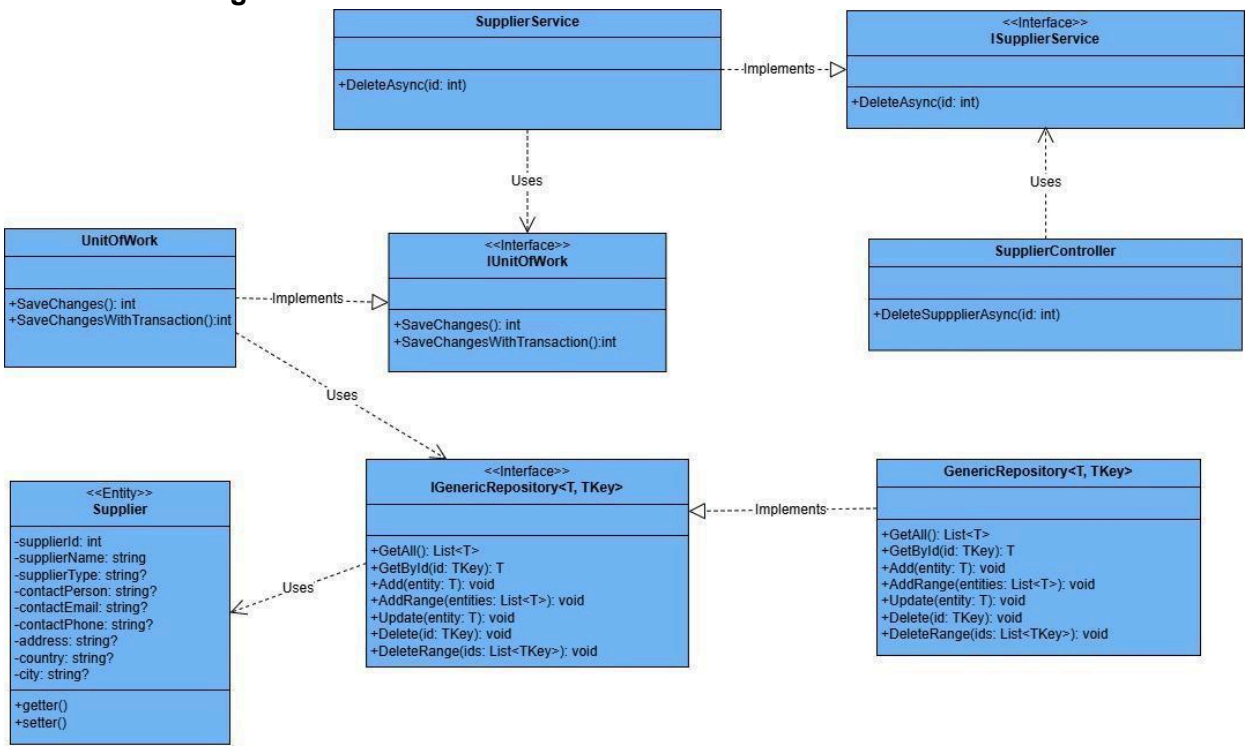


Figure 3.8.4.1 Class Diagram

3.8.4.2 Sequence Diagram

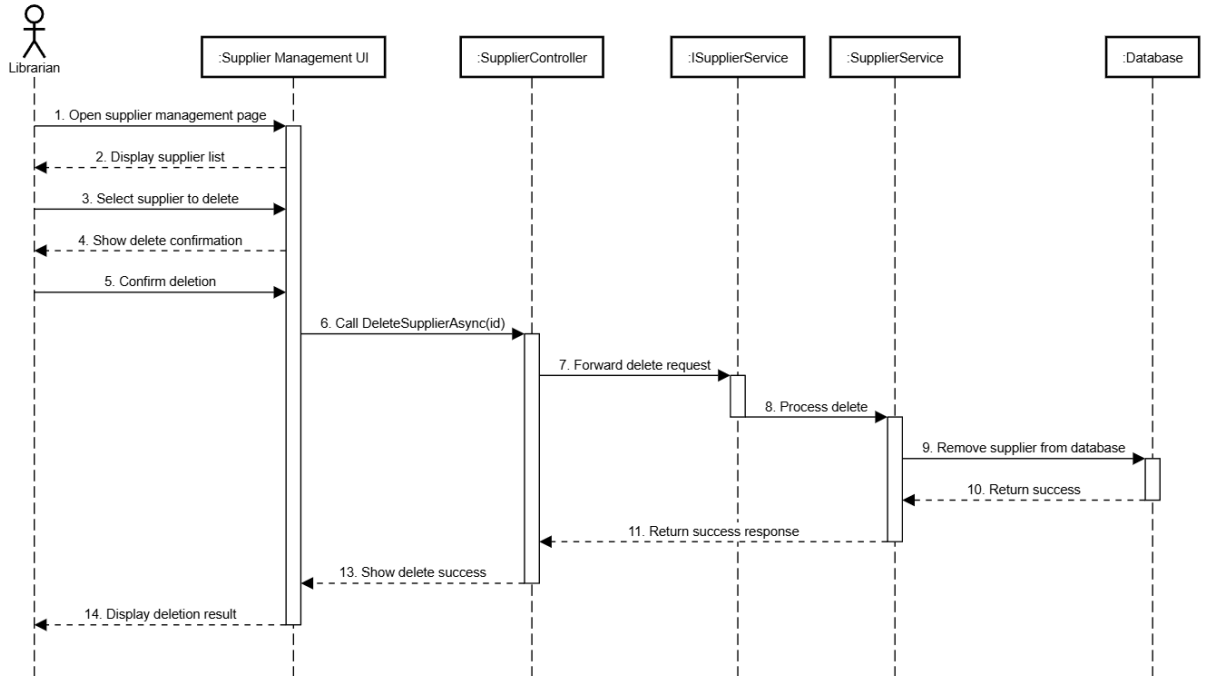


Figure 3.8.4.2 Sequence Diagram

3.8.4.3 Class Specification

3.8.4.3.a Supplier Controller

Table 3.8.4.3.a: Supplier Controller Description Table

No	Method	Description
1	DeleteSupplierAsync(int SupplierId)	Choose a Supplier that is not used any more to delete.
2	DeleteRangeSupplierAsync(RangeRequest <int> req)	Choose many suppliers that are not used to delete.

3.8.4.3.b Supplier Service

Table 3.8.4.3.b: Supplier Service Description Table

No	Method	Description
1	DeleteAsync(int SupplierId)	Remove Supplier and it could not be recovered after deletion.
2	DeleteRangeAsync(int[] req)	Remove suppliers from the database and could not be recovered after deleted

3.9. Manage Warehouse Trackings

3.9.1. Create Warehouse Tracking

3.9.1.1 Class Diagram

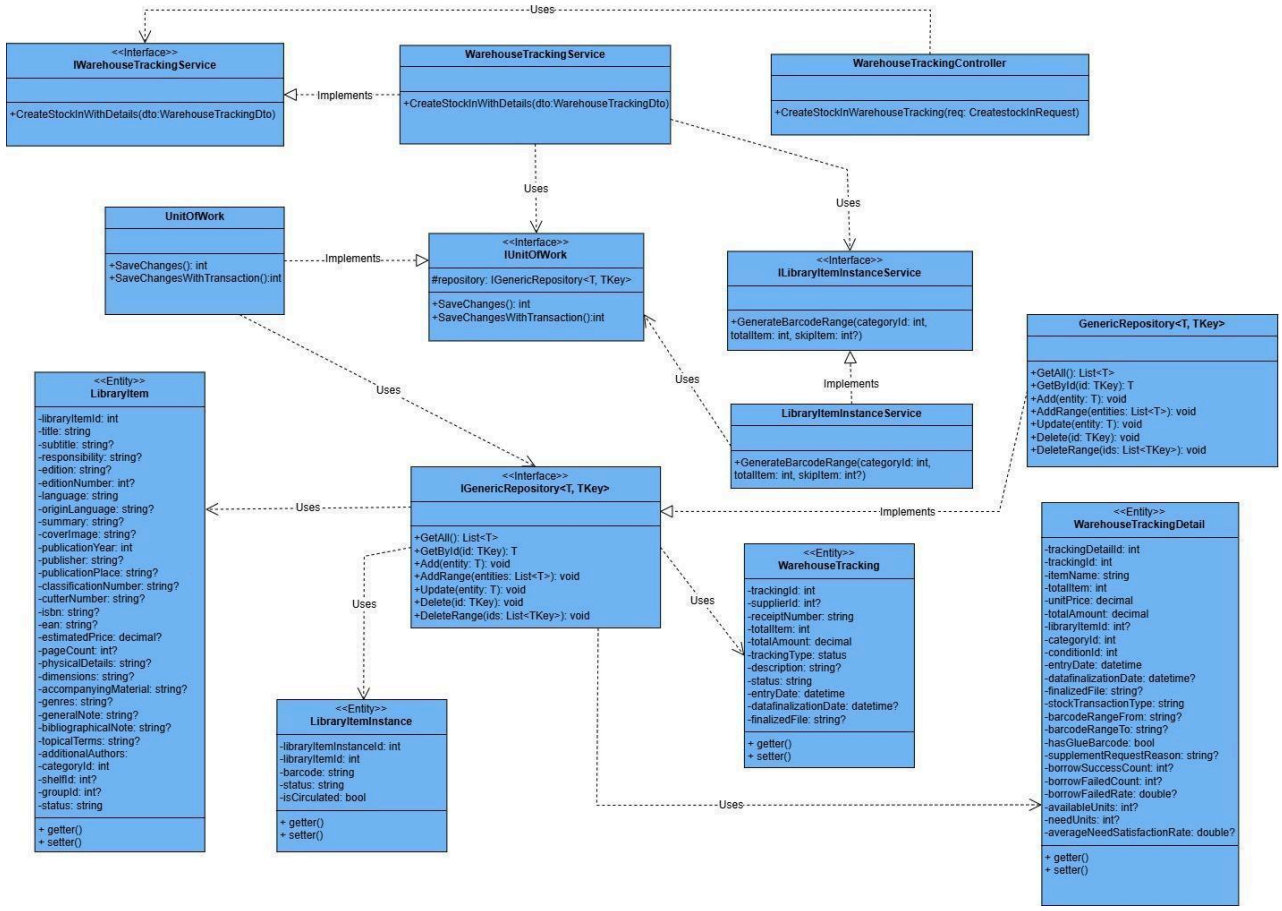


Figure 3.9.1.1 Class Diagram

3.9.1.2 Sequence Diagram

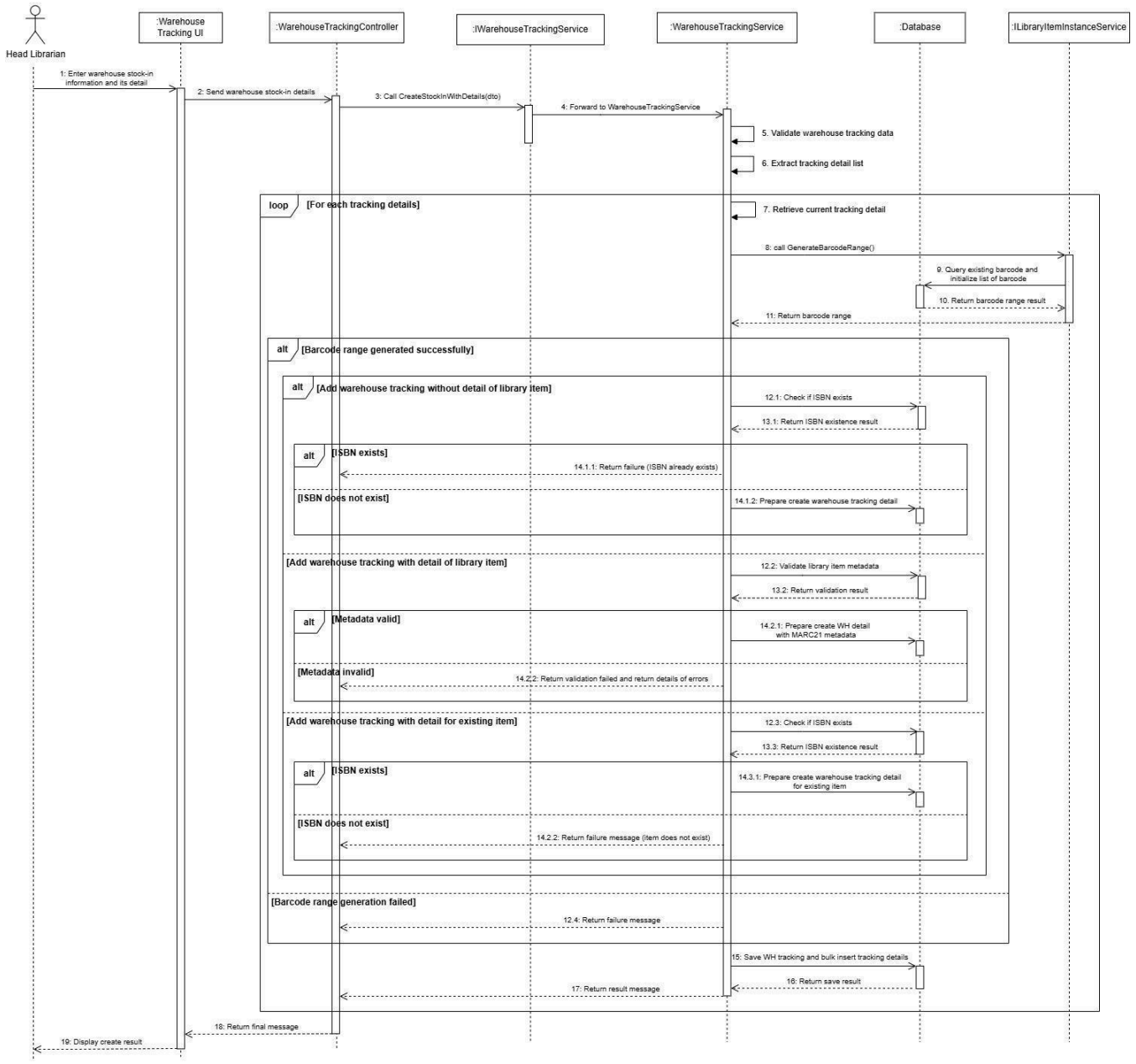


Figure 3.9.1.2 Sequence Diagram

3.9.1.3 Class Specification

3.9.1.3.a Warehouse Tracking Controller

Table 3.9.1.3.a: Warehouse Tracking Controller Description Table

No	Method	Description
1	CreateStockInWithDetails(CreateStockIn Request req)	Receives request containing bulk stock-in details for library items, and invokes the service to process create new stock-in entries.

3.9.1.3.b Warehouse Tracking Service

Table 3.9.1.3.b: Warehouse Tracking Service Description Table

No	Method	Description
1	CreateStockInWithDetails(WarehouseTrackingDto)	Validates and processes the provided bulk stock-in data for library items, applies business rules (e.g. checking item existence, quantities, etc.).

3.9.2. Get All Warehouse Trackings

3.9.2.1 Class Diagram

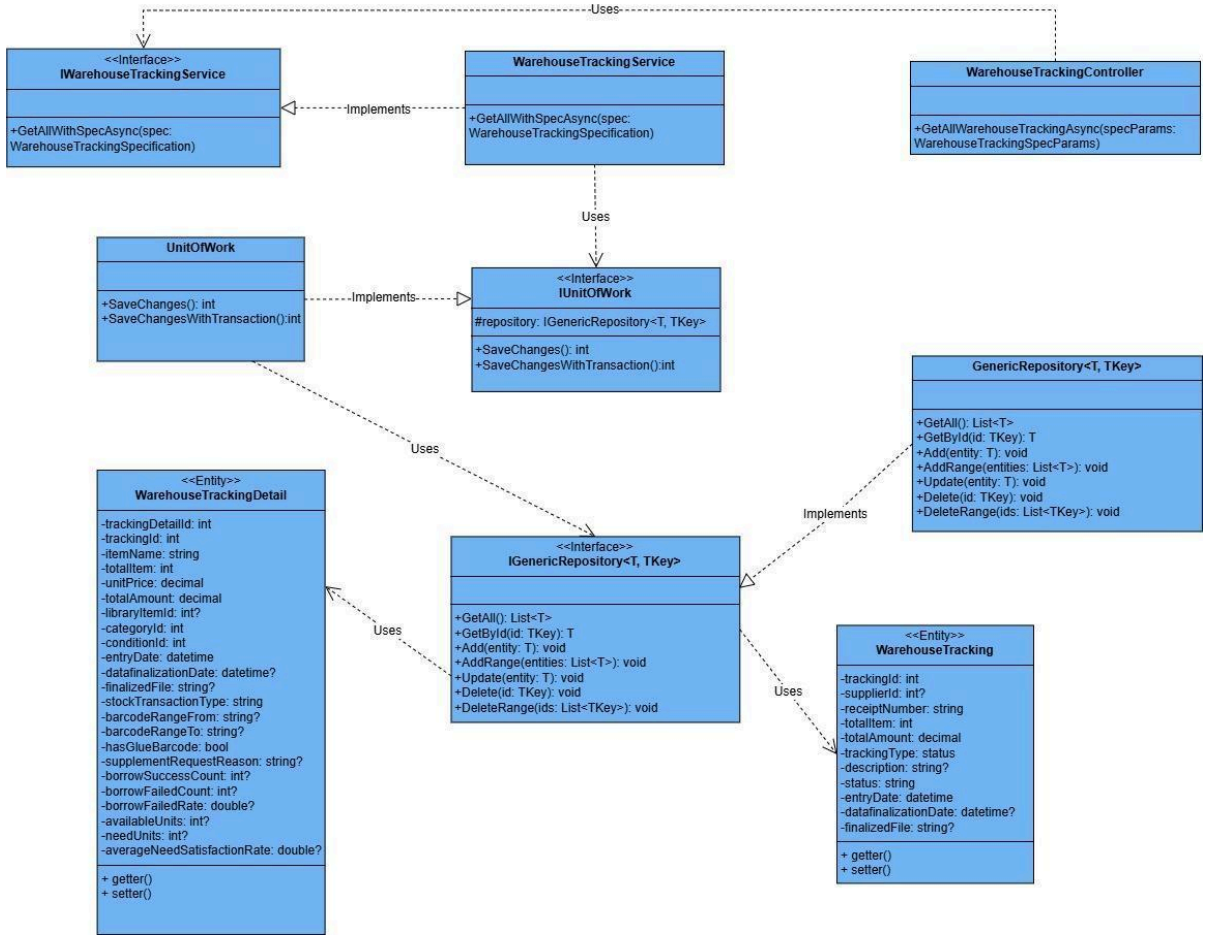


Figure 3.9.2.1 Class Diagram

3.9.2.2 Sequence Diagram

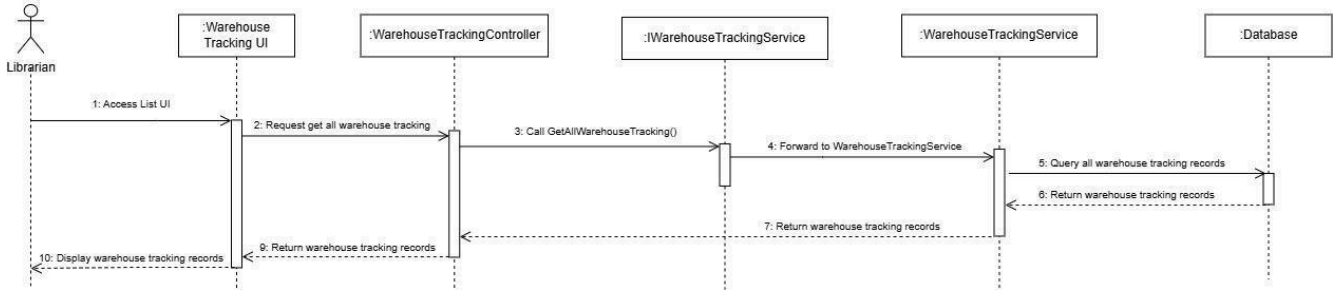


Figure 3.9.2.2 Sequence Diagram

3.9.2.3 Class Specification

3.9.2.3.a Warehouse Tracking Controller

Table 3.9.2.3.a: Warehouse Tracking Controller Description Table

No	Method	Description
1	GetAllWarehouseTracking(WarehouseTrackingSpecParam params)	Receives request with filtering, sorting and pagination parameters.

3.9.2.3.b Warehouse Tracking Service

Table 3.9.2.3.b: Warehouse Tracking Service Description Table

No	Method	Description
1	GetAllWithSpec(WarehouseTrackingSpecification spec)	Applies the given parameters to retrieve all matching warehouse tracking entities.

3.9.3. Update Warehouse Tracking

3.9.3.1 Class Diagram

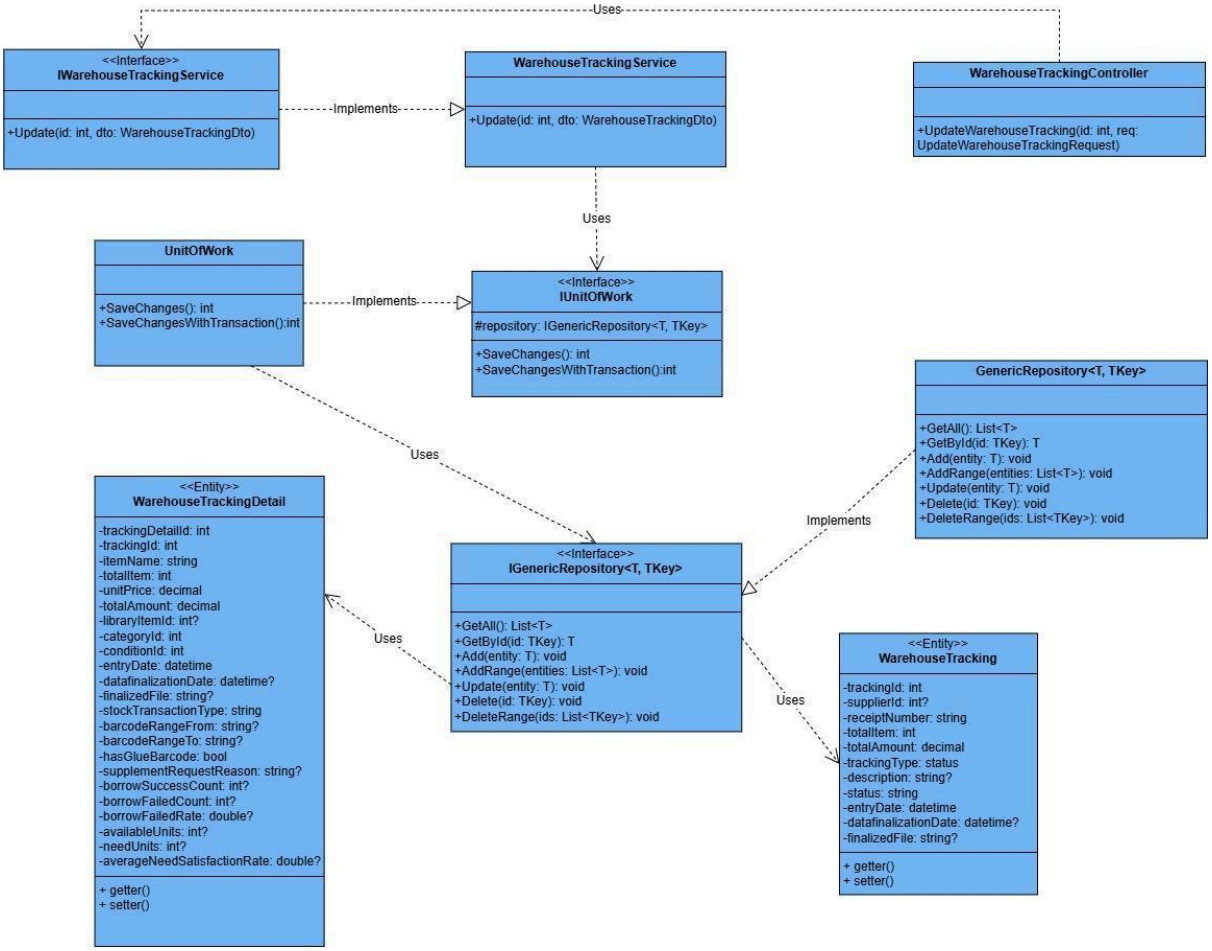


Figure 3.9.3.1 Class Diagram

3.9.3.2 Sequence Diagram

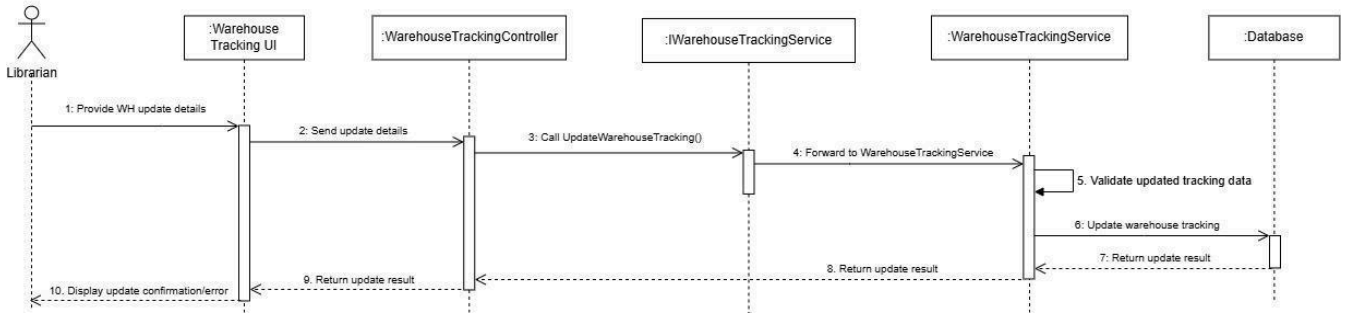


Figure 3.9.3.2 Sequence Diagram

3.9.3.3 Class Specification

3.9.3.3.a Warehouse Tracking Controller

Table 3.9.3.3.a: Warehouse Tracking Controller Description Table

No	Method	Description
1	UpdateWarehouseTracking(int trackingId, UpdateWarehouseTrackingRequest request)	Retrieves warehouse tracking modification request with the tracking ID and updated warehouse tracking payload.

3.9.3.3.b Warehouse Tracking Service

Table 3.9.3.3.b: Warehouse Tracking Service Description Table

No	Method	Description
1	Update(int trackingId, WarehouseTrackingDto whTrackingdto)	Validates that the specified tracking exists, applies provided update fields and save the changes to the database.

3.9.4. Update Warehouse Tracking Status

3.9.4.1 Class Diagram

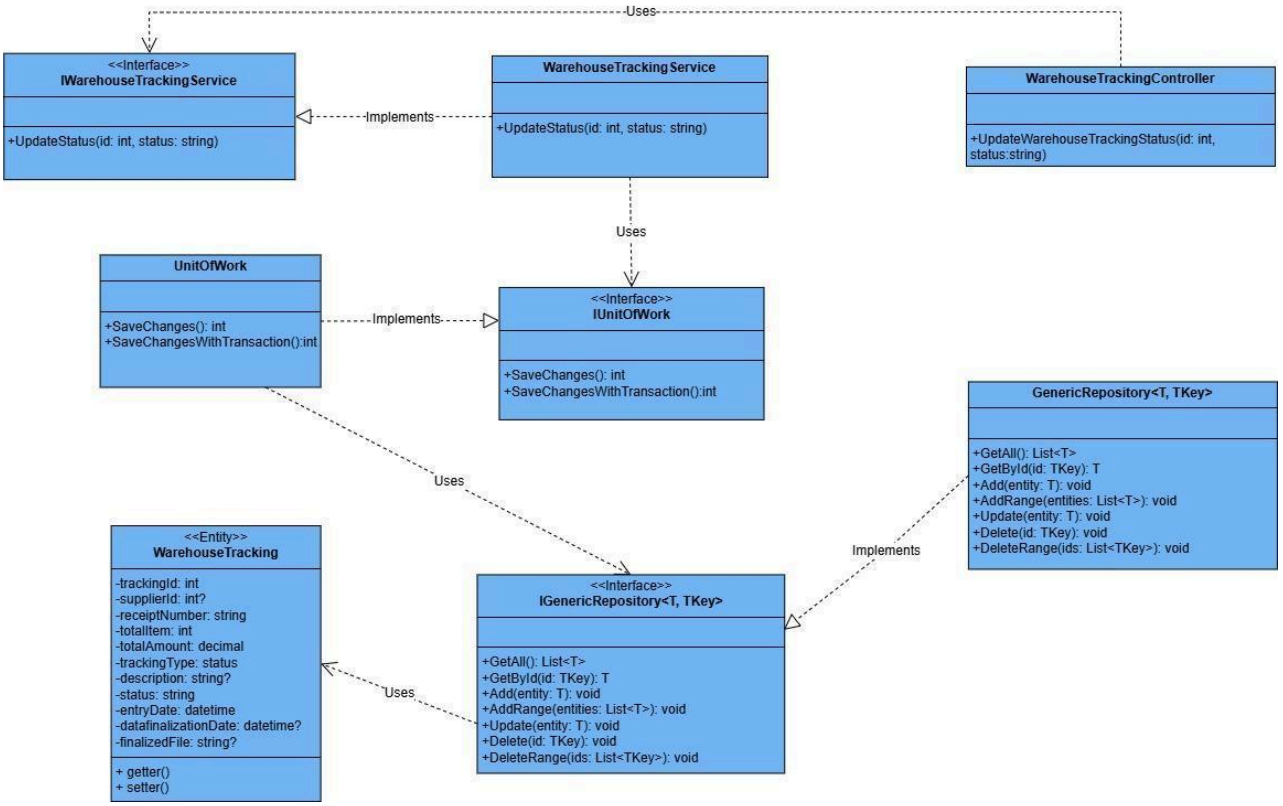


Figure 3.9.4.1 Class Diagram

3.9.4.2 Sequence Diagram

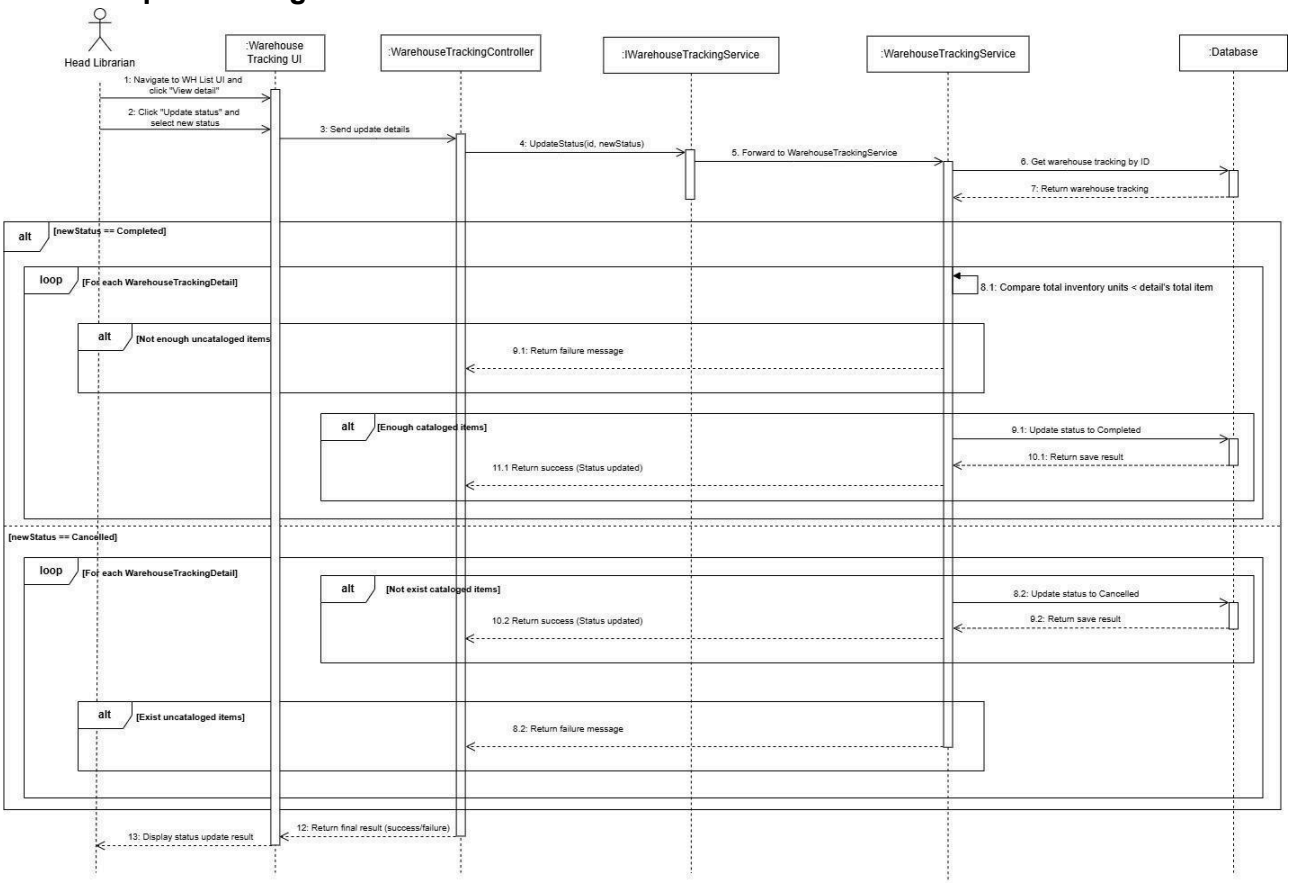


Figure 3.9.4.2 Sequence Diagram

3.9.4.3 Class Specification

3.9.4.3.a Warehouse Tracking Controller

Table 3.9.4.3.a: Warehouse Tracking Controller Description Table

No	Method	Description
1	UpdateWarehouseTrackingStatus(int trackingId, WarehouseTrackingStatus status)	Receives update status request containing the tracking ID and new status.

3.9.4.3.b Warehouse Tracking Service

Table 3.9.4.3.b: Warehouse Tracking Service Description Table

No	Method	Description
1	UpdateStatus(int trackingId, WarehouseTrackingStatus status)	Verifies that the tracking with the given ID exists, and applies the new status.

3.9.5. Delete Warehouse Tracking

3.9.5.1 Class Diagram

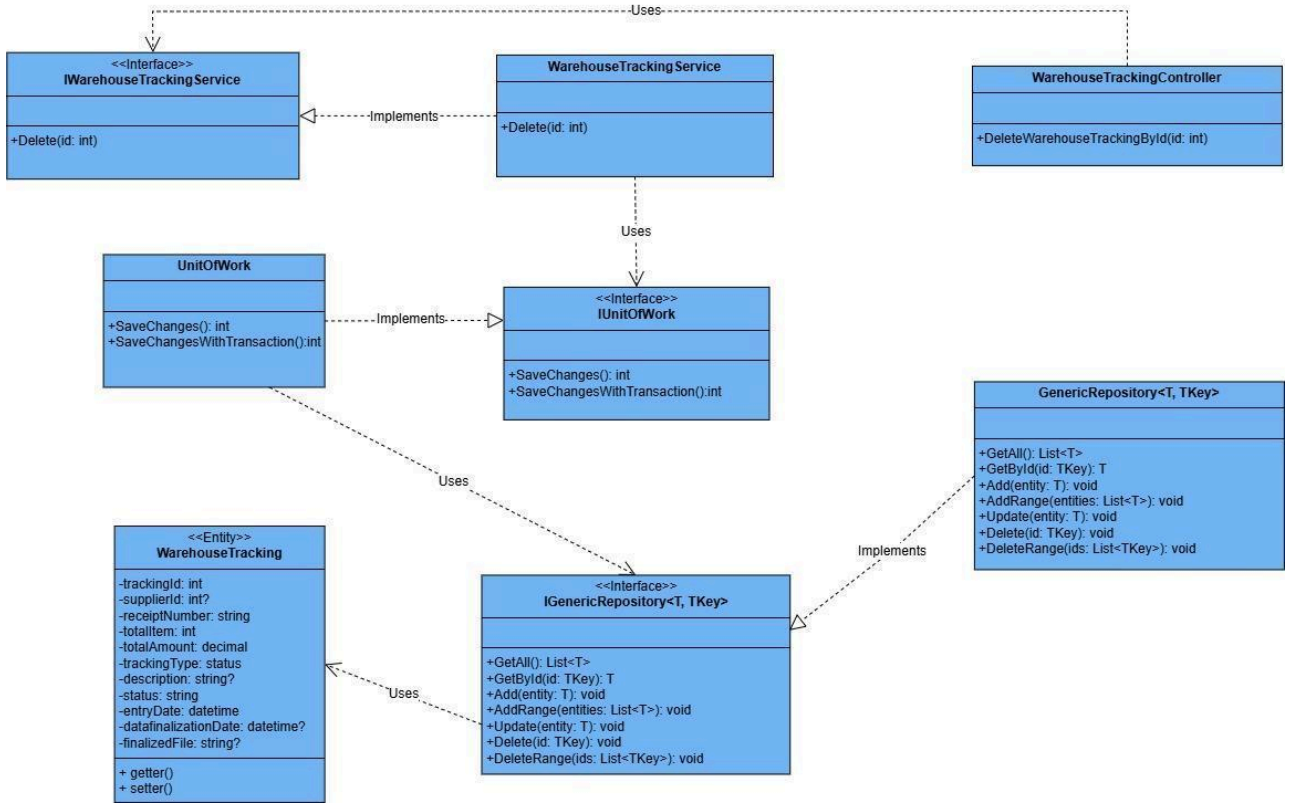


Figure 3.9.5.1 Class Diagram

3.9.5.2 Sequence Diagram

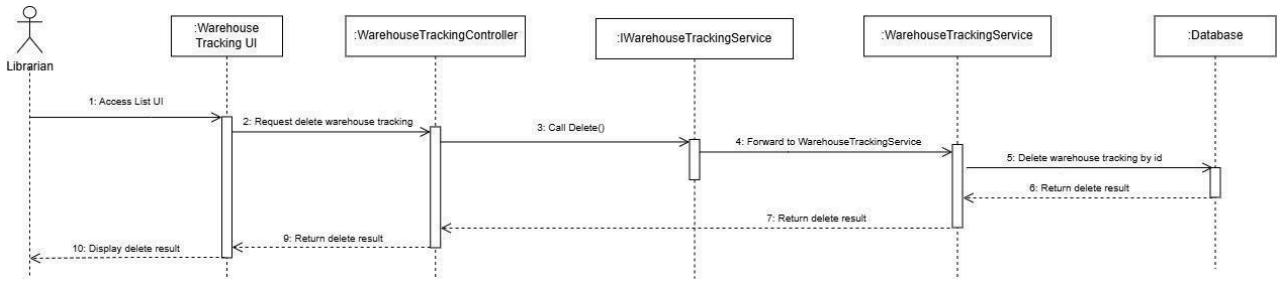


Figure 3.9.5.2 Sequence Diagram

3.9.5.3 Class Specification

3.9.5.3.a Warehouse Tracking Controller

Table 3.9.5.3.a: Warehouse Tracking Controller Description Table

No	Method	Description
1	DeleteWarehouseTracking(int trackingId)	Receives deletion request with the specified tracking ID.

3.9.5.3.b Warehouse Tracking Service

Table 3.9.5.3.b: Warehouse Tracking Service Description Table

No	Method	Description
1	Delete(int trackingId)	Verifies the existence of the tracking with tracking ID, deletes it from the database.

3.10. Manage Warehouse Tracking Details

3.10.1. Add to Warehouse Tracking

3.10.1.1 Class Diagram

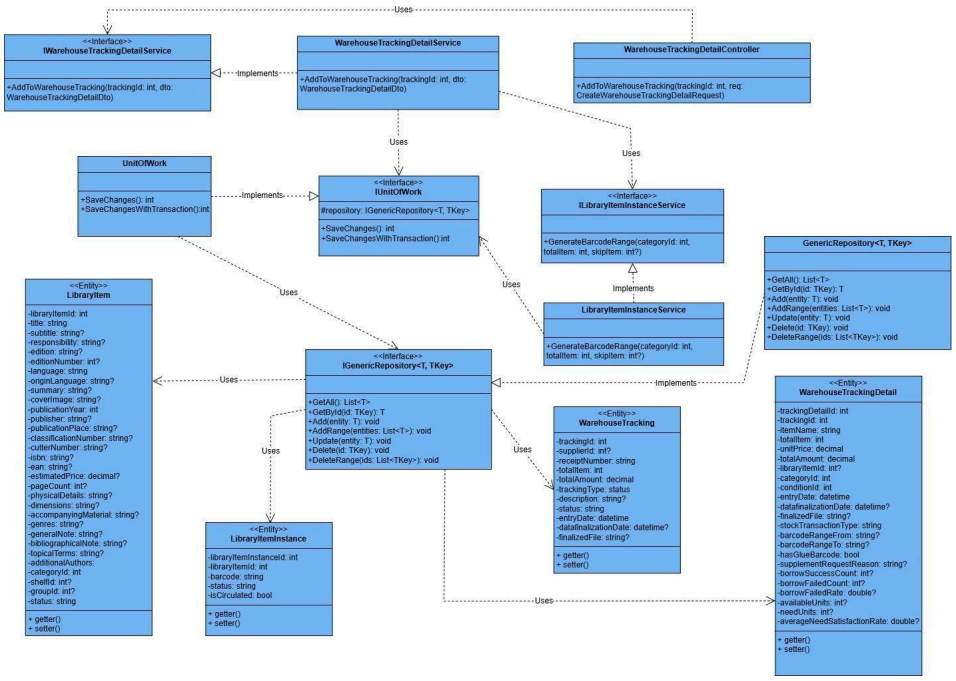


Figure 3.10.1.1 Class Diagram

3.10.1.2 Sequence Diagram

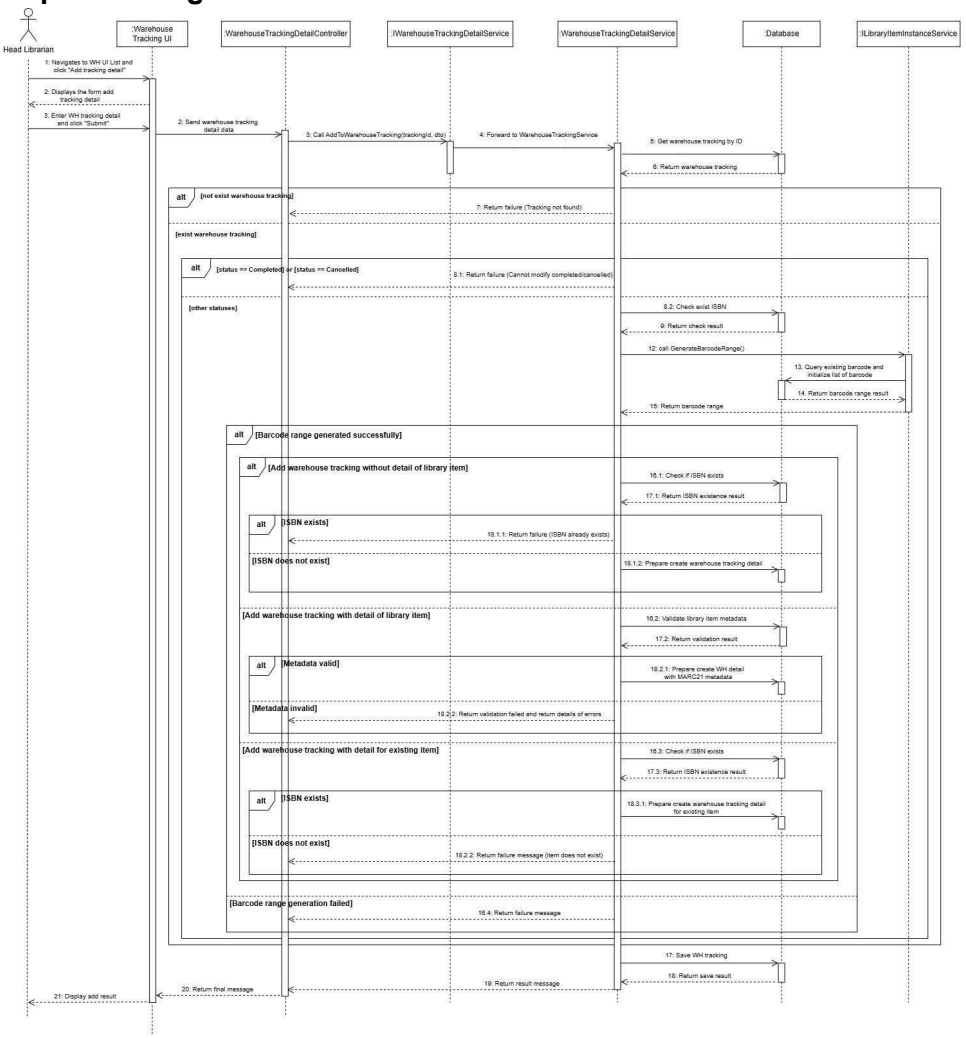


Figure 3.10.1.2 Sequence Diagram

3.10.1.3 Class Specification

3.10.1.3.a Warehouse Tracking Detail Controller

Table 3.10.1.3.a: Warehouse Tracking Detail Controller Description Table

No	Method	Description
1	AddToWarehouseTracking(int trackingId, CreateWarehouseTrackingDetailRequest req)	Receives request with tracking ID and payload containing new tracking detail entries.

3.10.1.3.b Warehouse Tracking Detail Service

Table 3.10.1.3.b: Warehouse Tracking Detail Service Description Table

No	Method	Description
1	AddToWarehouseTracking(int trackingId, WarehouseTrackingDetailDto whTrackingDetailDto)	Verifies that the warehouse tracking with the given ID exists, validates each detail in DTO and performs create and save to the database.

3.10.2. Update Barcode Range Registration

3.10.2.1 Class Diagram

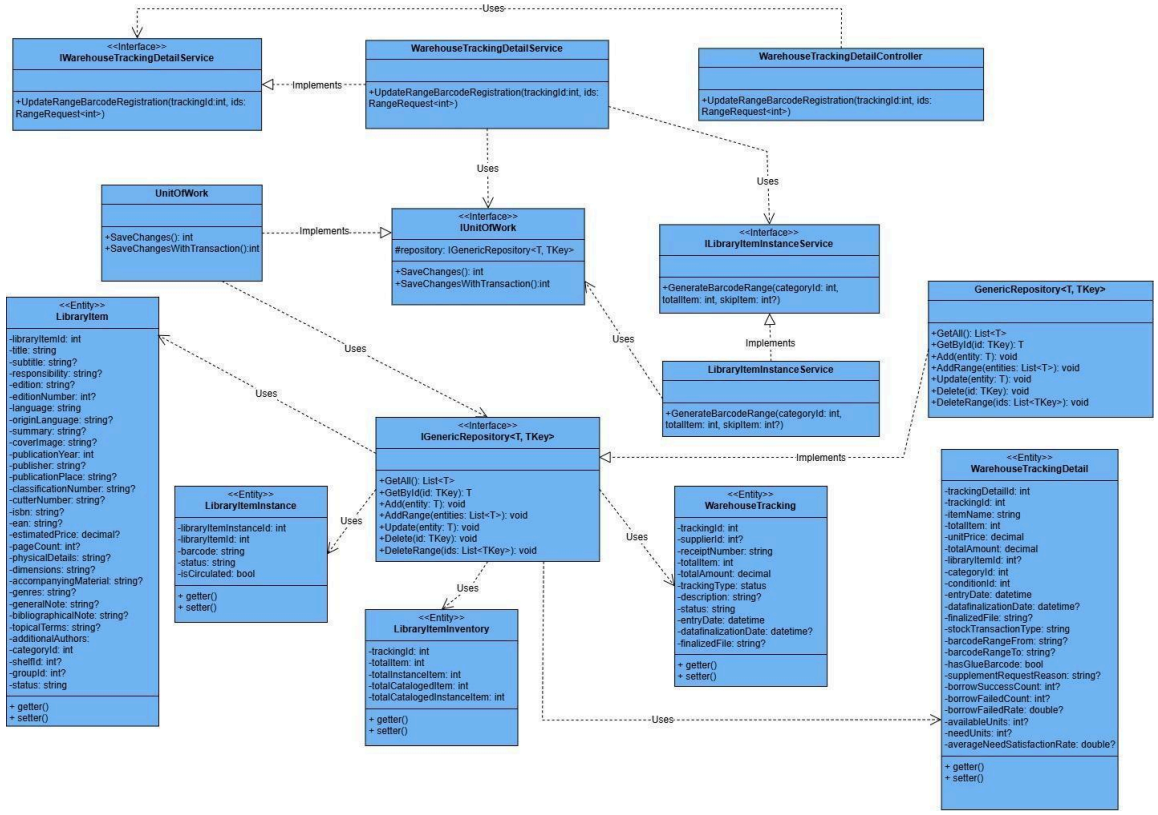


Figure 3.10.2.1 Class Diagram

3.10.2.2 Sequence Diagram

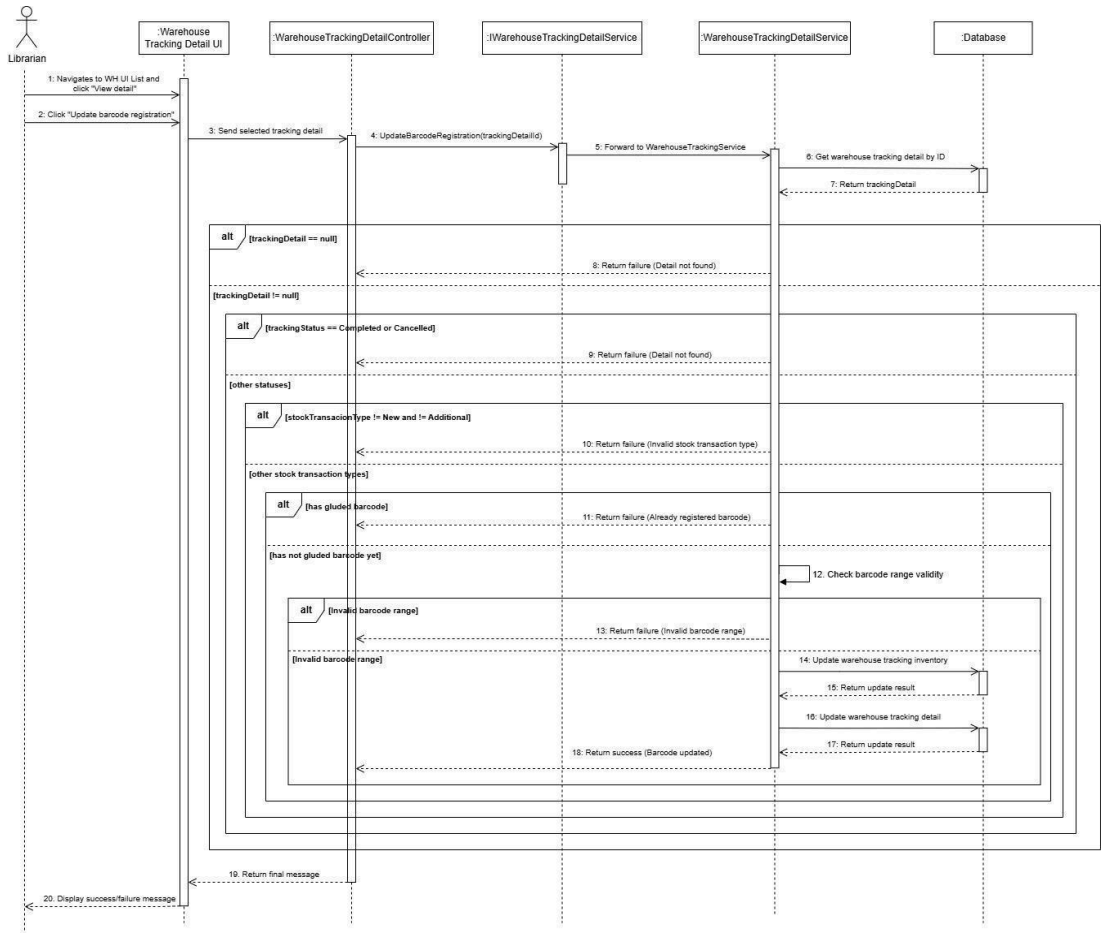


Figure 3.10.2.2 Sequence Diagram

3.10.2.3 Class Specification

3.10.2.3.a Warehouse Tracking Detail Controller

Table 3.10.2.3.a: Warehouse Tracking Detail Controller Description Table

No	Method	Description
1	UpdateRangeBarcodeRegistration(int trackingDetailId)	Receives request with the specified tracking detail ID, invokes the service to update range the barcode registration status.

3.10.2.3.b Warehouse Tracking Detail Service

Table 3.10.2.3.b: Warehouse Tracking Detail Service Description Table

No	Method	Description
1	UpdateRangeBarcodeRegistration(int trackingDetailId)	Verifies that the given warehouse tracking exists, updates its tracking details' barcode registration status.

3.11. Manage Library Items

3.11.1. Create Library Item

3.11.1.1 Class Diagram

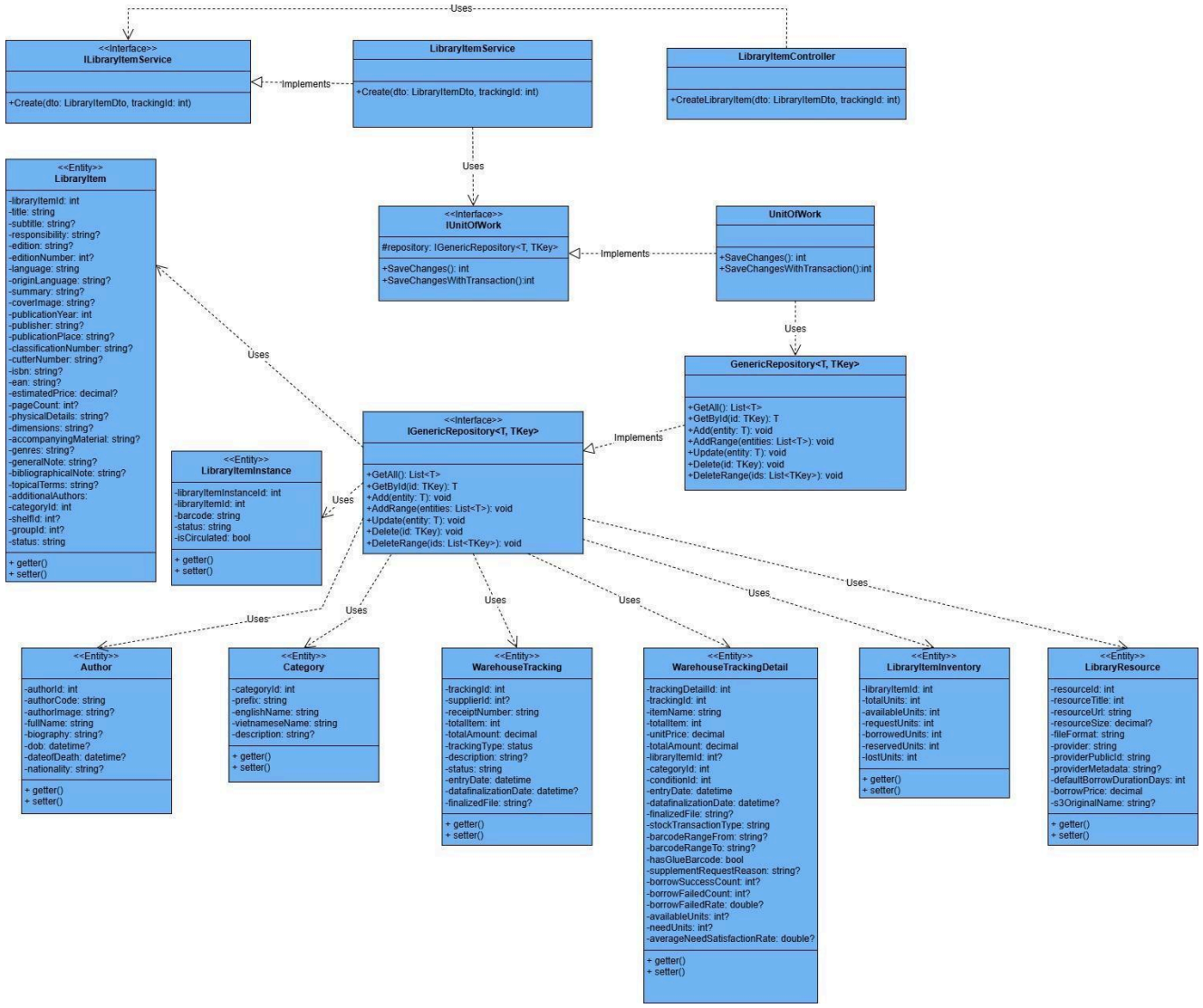


Figure 3.11.1.1 Class Diagram

3.11.1.2 Sequence Diagram

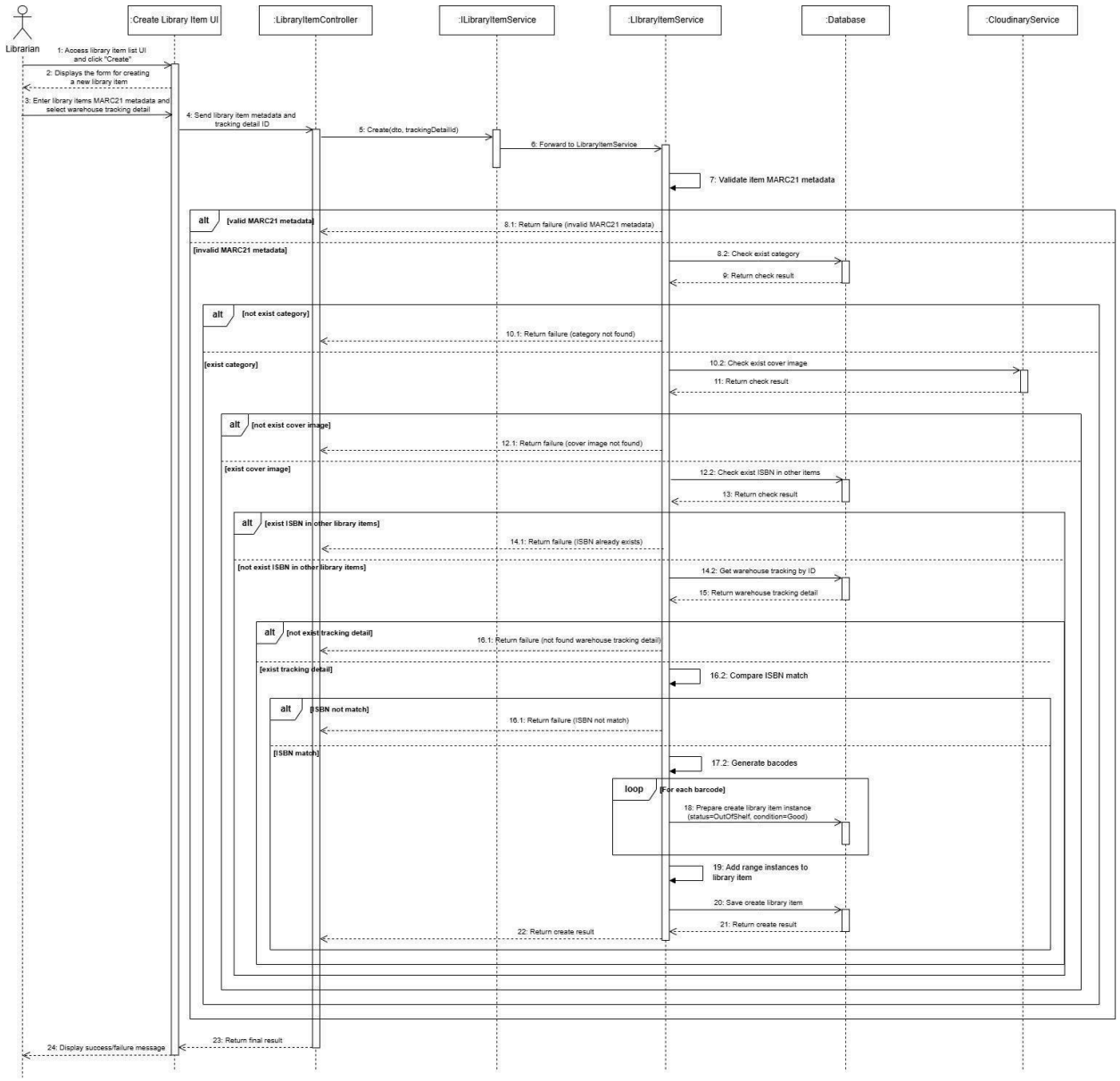


Figure 3.11.1.2 Sequence Diagram

3.11.1.3 Class Specification

3.11.1.3.a Library Item Controller

Table 3.11.1.3.a: Library Item Controller Description Table

No	Method	Description
1	CreateLibraryItem(CreateLibraryItemRequest req, int trackingDetailId)	Receives request containing the new library item payload and specified tracking detail ID, invokes the services to create new library item.

3.11.1.3.b Library Item Service

Table 3.11.1.3.b: Library Item Service Description Table

No	Method	Description

1	Create(LibraryItemDto libItemDto, int trackingDetailId)	Validates provided library item DTO and check whether the tracking detail exists, save the new library item linked to that tracking detail in the database.

3.11.2. Get All Library Items

3.11.2.1 Class Diagram

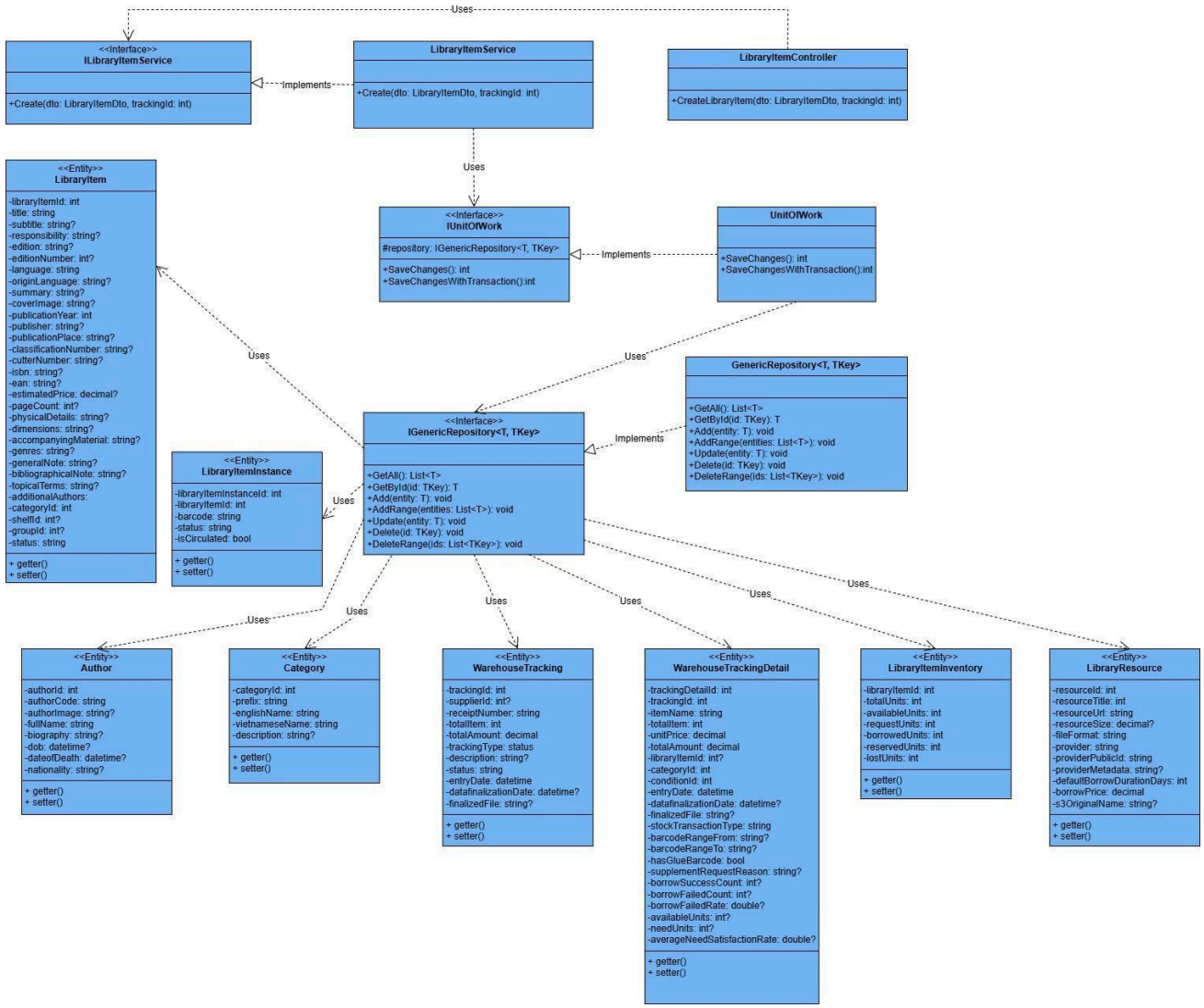


Figure 3.11.2.1 Class Diagram

3.11.2.2 Sequence Diagram

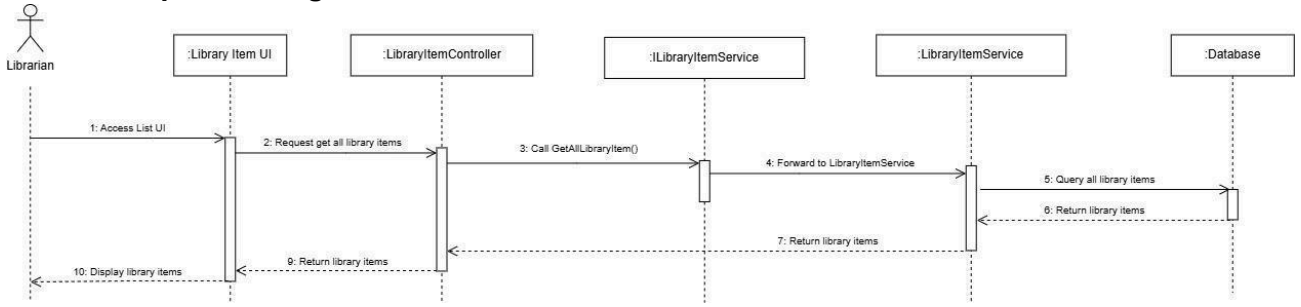


Figure 3.11.2.2 Sequence Diagram

3.11.2.3 Class Specification

3.11.2.3.a Library Item Controller

Table 3.11.2.3.a: Library Item Controller Description Table

No	Method	Description
1	GetAllLibraryItem(LibraryItemSpecParams specParams)	Receives the request with filtering, sorting, and pagination parameters, invokes the services to fetch all matching items.

3.11.2.3.b Library Item Service

Table 3.11.2.3.b: Library Item Service Description Table

No	Method	Description
1	GetAllWithSpec(LibraryItemSpecification spec)	Applies the given parameters to retrieve all matching library items.

3.11.3. Update Library Item

3.11.3.1 Class Diagram

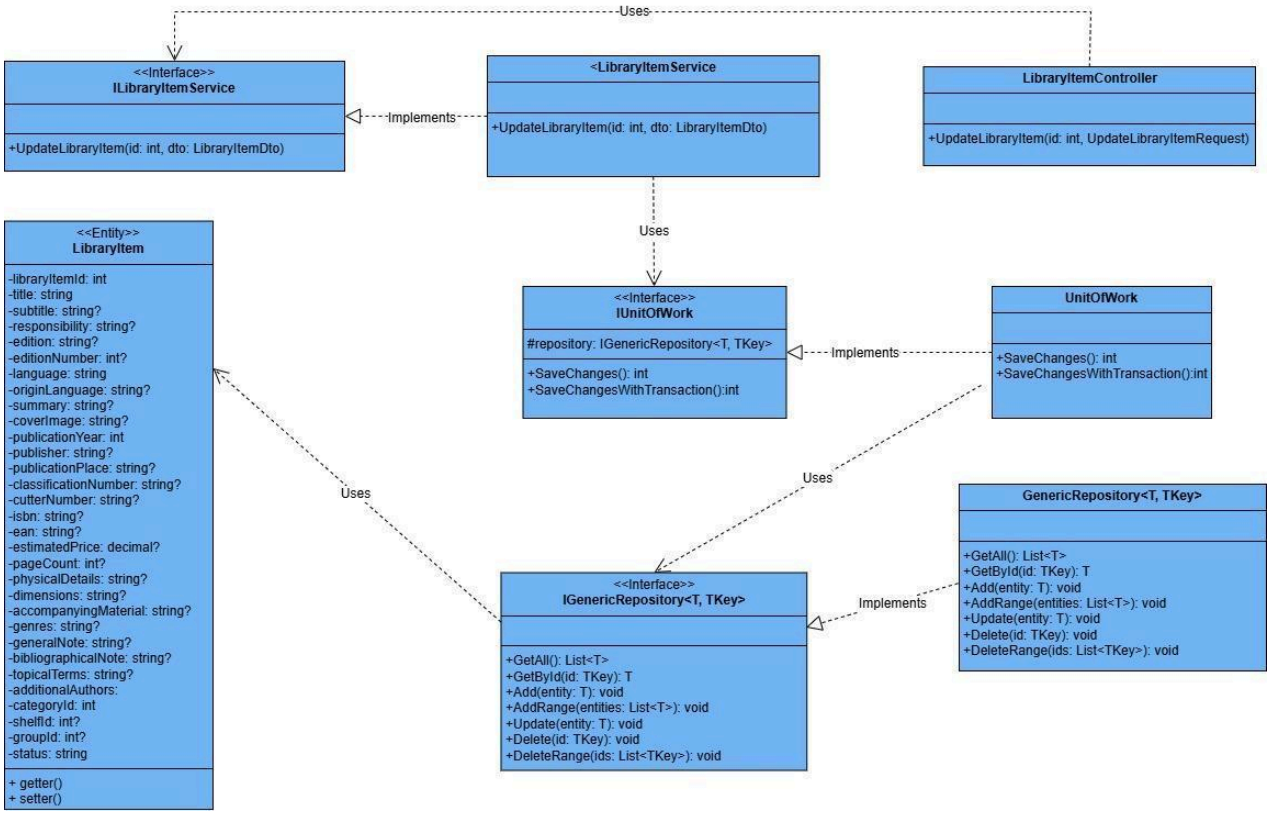


Figure 3.11.3.1 Class Diagram

3.11.3.2 Sequence Diagram

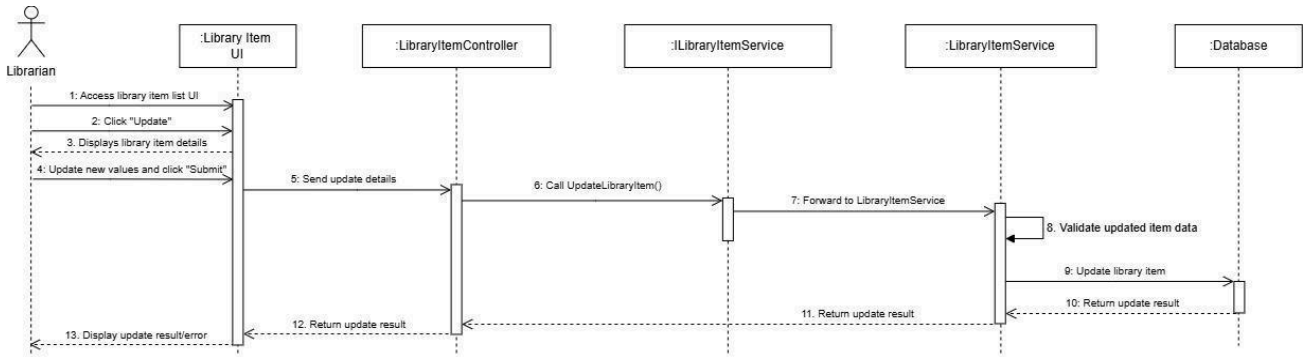


Figure 3.11.3.2 Sequence Diagram

3.11.3.3 Class Specification

3.11.3.3.a Library Item Controller

Table 3.11.3.3.a: Library Item Controller Description Table

No	Method	Description
1	UpdateLibraryItem(int libraryItemId, UpdateLibraryItemRequest req)	Receives the request with the specified libraryItemId and update fields, calls the service to apply modification.

3.11.3.3.b Library Item Service

Table 3.11.3.3.b: Library Item Service Description Table

No	Method	Description
1	UpdateLibraryItem(int libraryItemId, LibraryItemDto libItemDto)	Checks that a library item with the given libraryItemId exists, applies the updated values and save changes to the database.

3.11.4. Update Library Item Status

3.11.4.1 Class Diagram

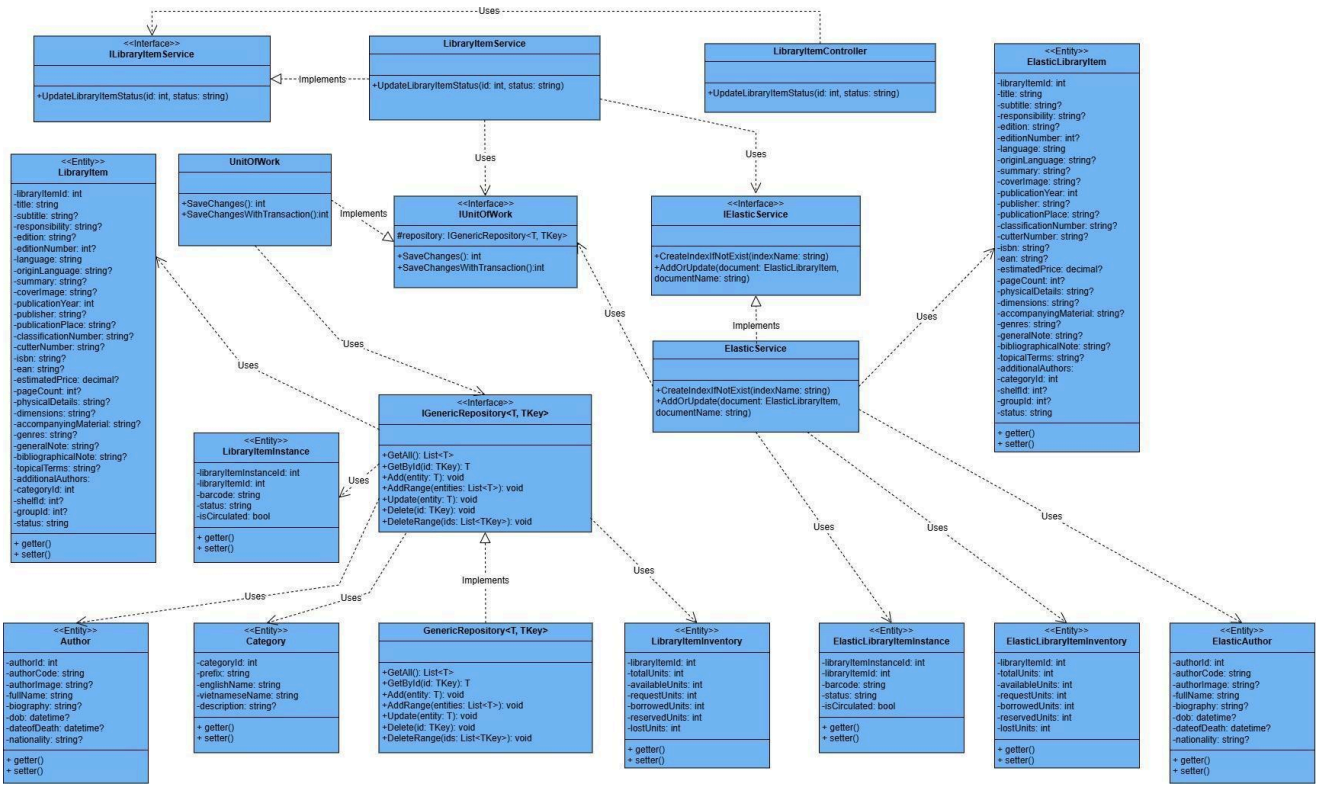


Figure 3.11.4.1 Class Diagram

3.11.4.2 Sequence Diagram

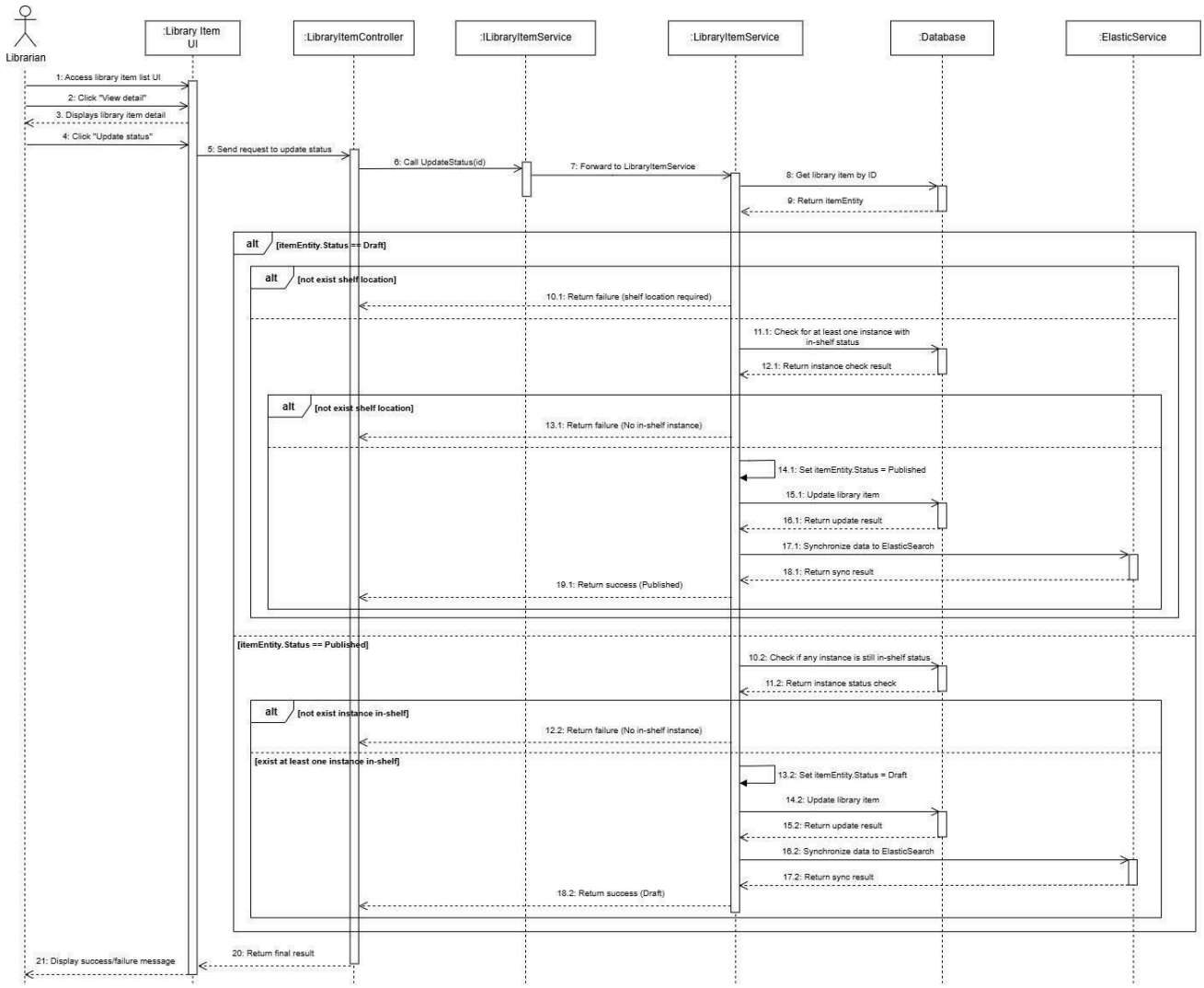


Figure 3.11.4.2 Sequence Diagram

3.11.4.3 Class Specification

3.11.4.3.a Library Item Controller

Table 3.11.4.3.a: Library Item Controller Description Table

No	Method	Description
1	UpdateLibraryItemStatus(int libraryItemId, LibraryItemStatus status)	Receives request with the library item ID and new status, invokes the service to update only the status field of the specified library item.

3.11.4.3.b Library Item Service

Table 3.11.4.3.b: Library Item Service Description Table

No	Method	Description
1	UpdateStatus(int libraryItemId, LibraryItemStatus status)	Verifies that the library item with given library item ID exists, applies the new status and save changes to the database.

3.11.5. Update Shelf Location

3.11.5.1 Class Diagram

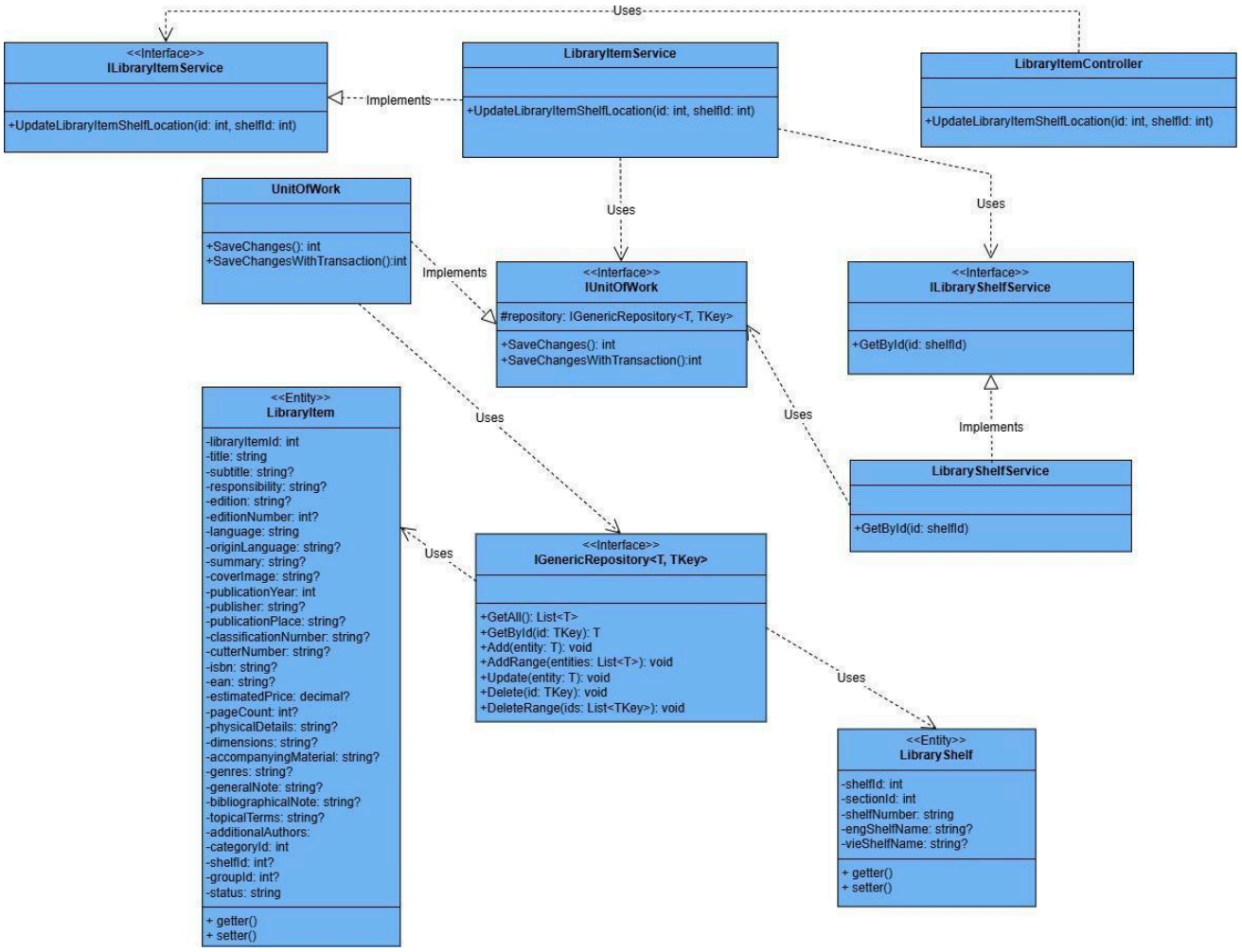


Figure 3.11.5.1 Class Diagram

3.11.5.2 Sequence Diagram

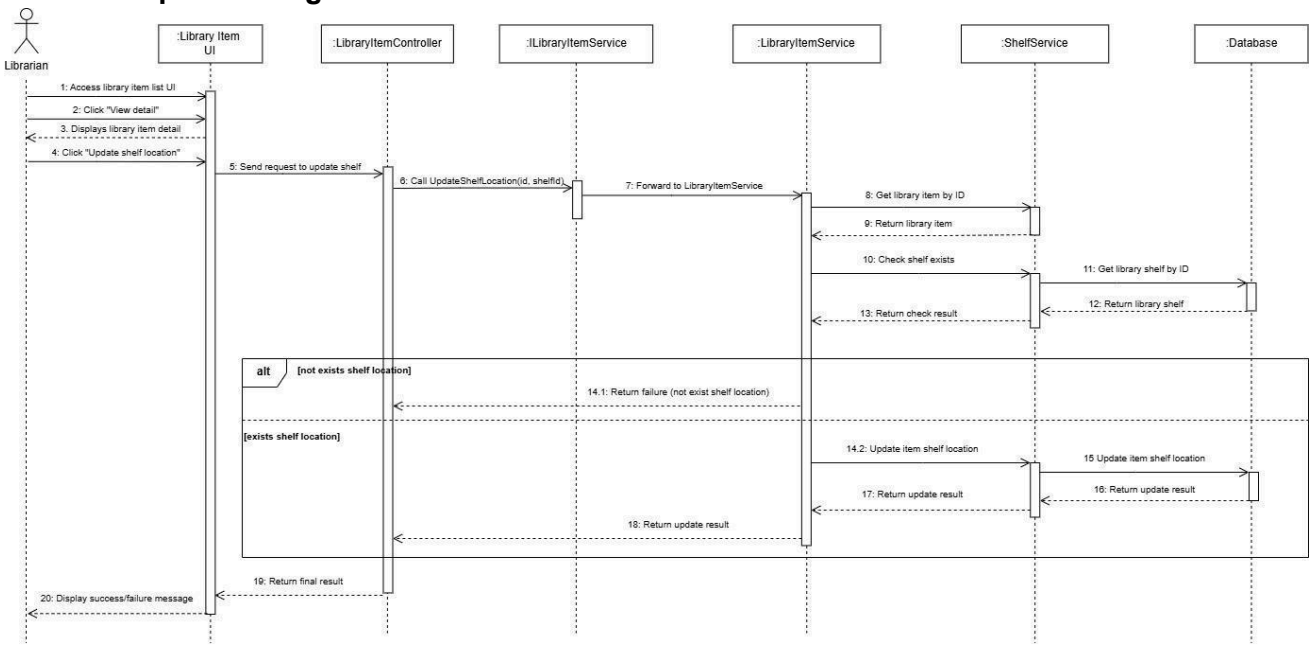


Figure 3.11.5.2 Sequence Diagram

3.11.5.3 Class Specification

3.11.5.3.a Library Item Controller

Table 3.11.5.3.a: Library Item Controller Description Table

No	Method	Description
1	UpdateLibraryItemShelfLocation(int libraryItemId, int shelfId)	Receives request with the library item ID and new shelf ID, invokes the service to update only the shelf location field of the specified item.

3.11.5.3.b Library Item Service

Table 3.11.5.3.b: Library Item Service Description Table

No	Method	Description
1	UpdateShelfLocation(int libraryItemId, int shelfId)	Verifies that the library item within the given library item ID and shelf ID exists, updates the shelf location property on the library item entity and save changes to the database.

3.11.6. Delete Library Item

3.11.6.1 Class Diagram

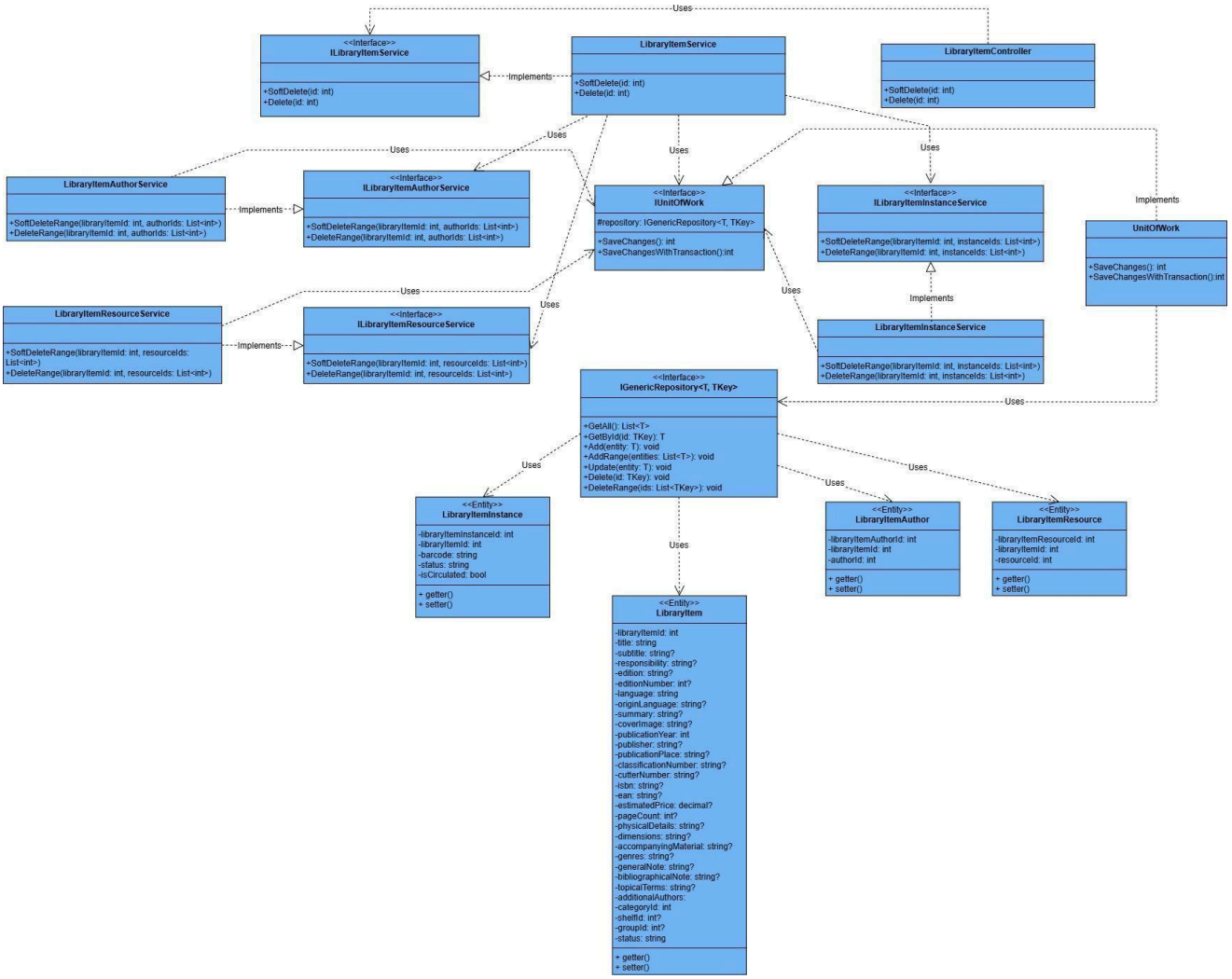


Figure 3.11.6.1 Class Diagram

3.11.6.2 Sequence Diagram

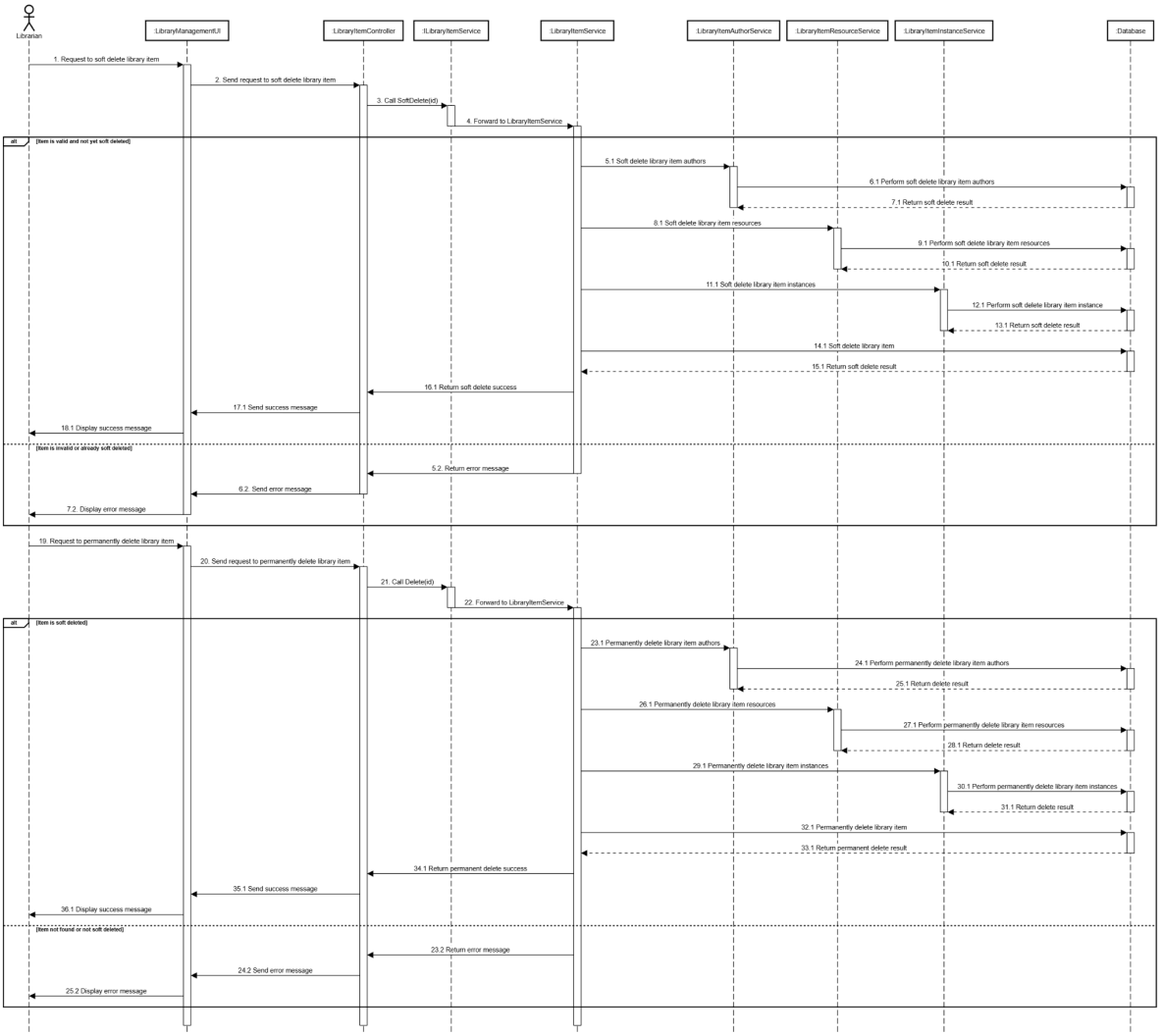


Figure 3.11.6.2 Sequence Diagram

3.11.6.3 Class Specification

3.11.6.3.a Library Item Controller

Table 3.11.6.3.a: Library Item Controller Description Table

No	Method	Description
1	SoftDeleteLibraryItem(int libraryItemId)	Receives request with the library item ID, invokes service to mark the specified library item as delete (soft delete).
2	DeleteLibraryItem(int libraryItemId)	Receives request with the library item ID, invokes service to permanently remove the specified library item.

3.11.6.3.b Library Item Service

Table 3.11.6.3.b: Library Item Service Description Table

No	Method	Description

1	SoftDelete(int libraryItemId)	Verifies that a library item with the given library item ID exists, sets the soft delete flag to true.
2	Delete(int libraryItemId)	Check that the library item with the given library item ID exists, permanently delete from the database.

3.12. Manage Library Item Instances

3.12.1. Update In Shelf

3.12.1.1 Class Diagram

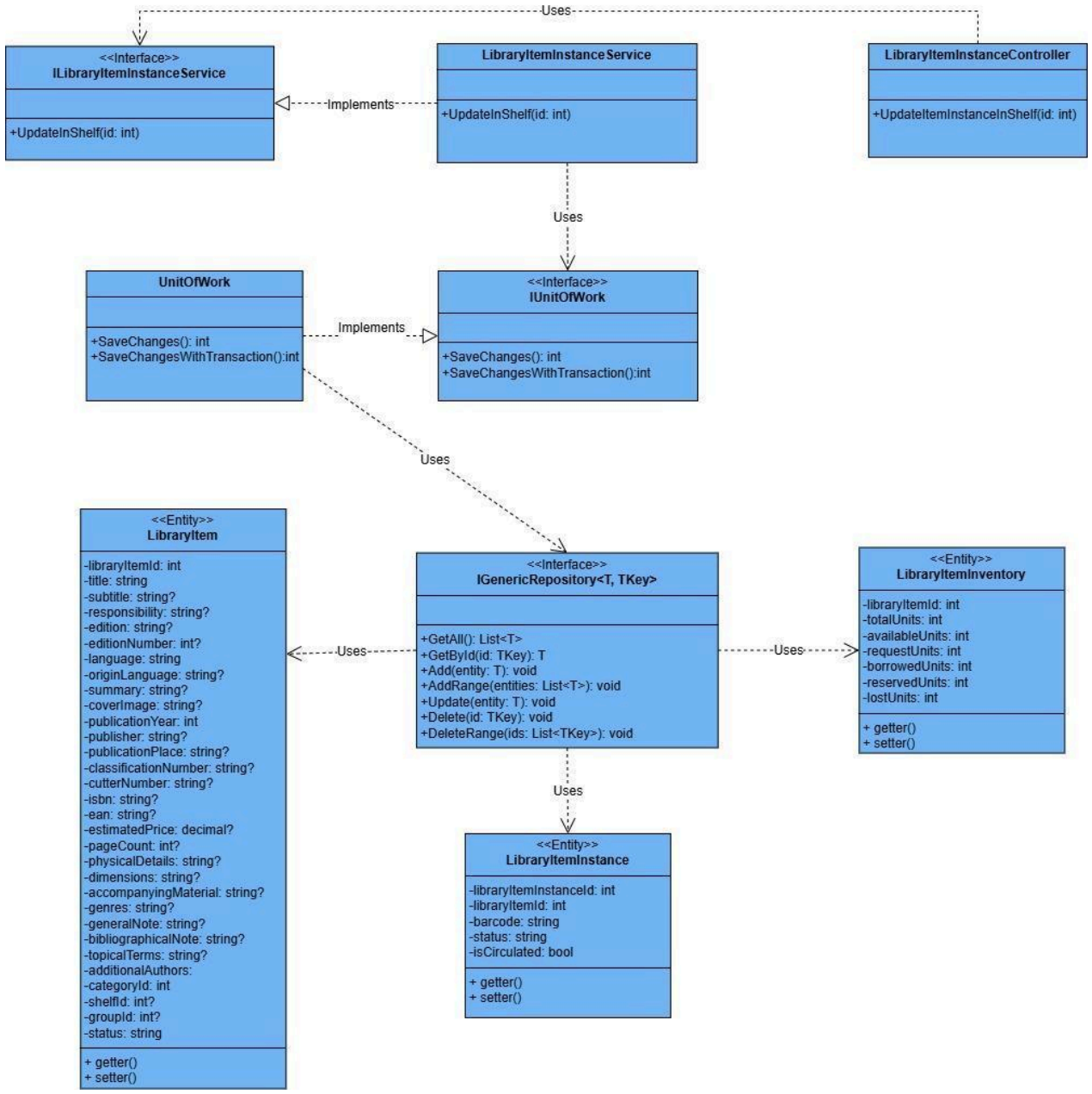


Figure 3.12.1.1 Class Diagram

3.12.1.2 Sequence Diagram

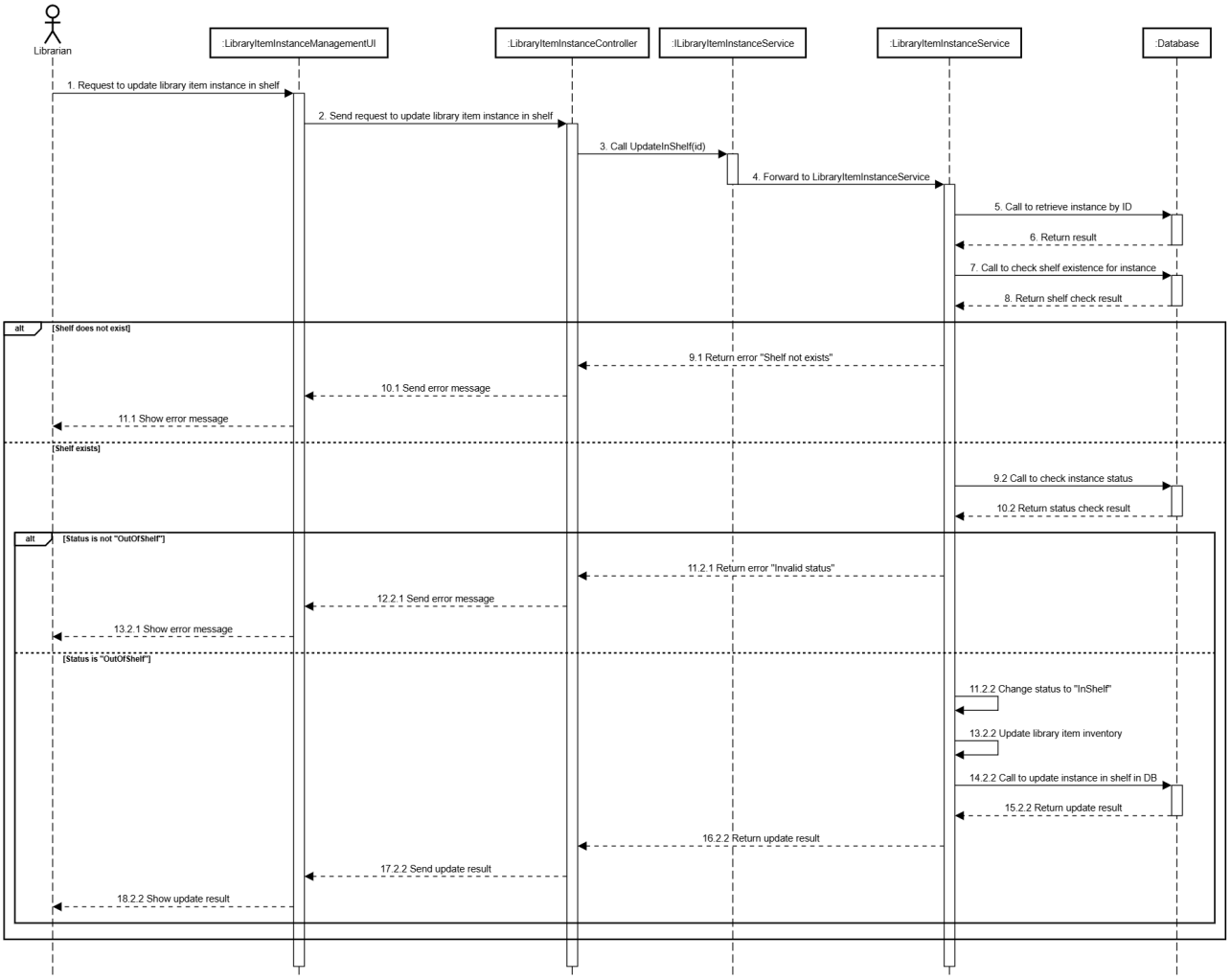


Figure 3.12.1.2 Sequence Diagram

3.12.1.3 Class Specification

3.12.1.3.a Library Item Instance Controller

Table 3.12.1.3.a: Library Item Instance Controller Description Table

No	Method	Description
1	UpdateItemInstanceInShelf(int libraryItemInstanceId)	Receives request with the library item instance ID, invokes the service to mark the specified item instance as in-shelf.

3.12.1.3.b Library Item Instance Service

Table 3.12.1.3.b: Library Item Instance Service Description Table

No	Method	Description
1	UpdateInShelf(int libraryItemInstanceId)	Verifies that a library item instance ID exists, updates its in-shelf.

3.12.2. Update Out of Shelf

3.12.2.1 Class Diagram

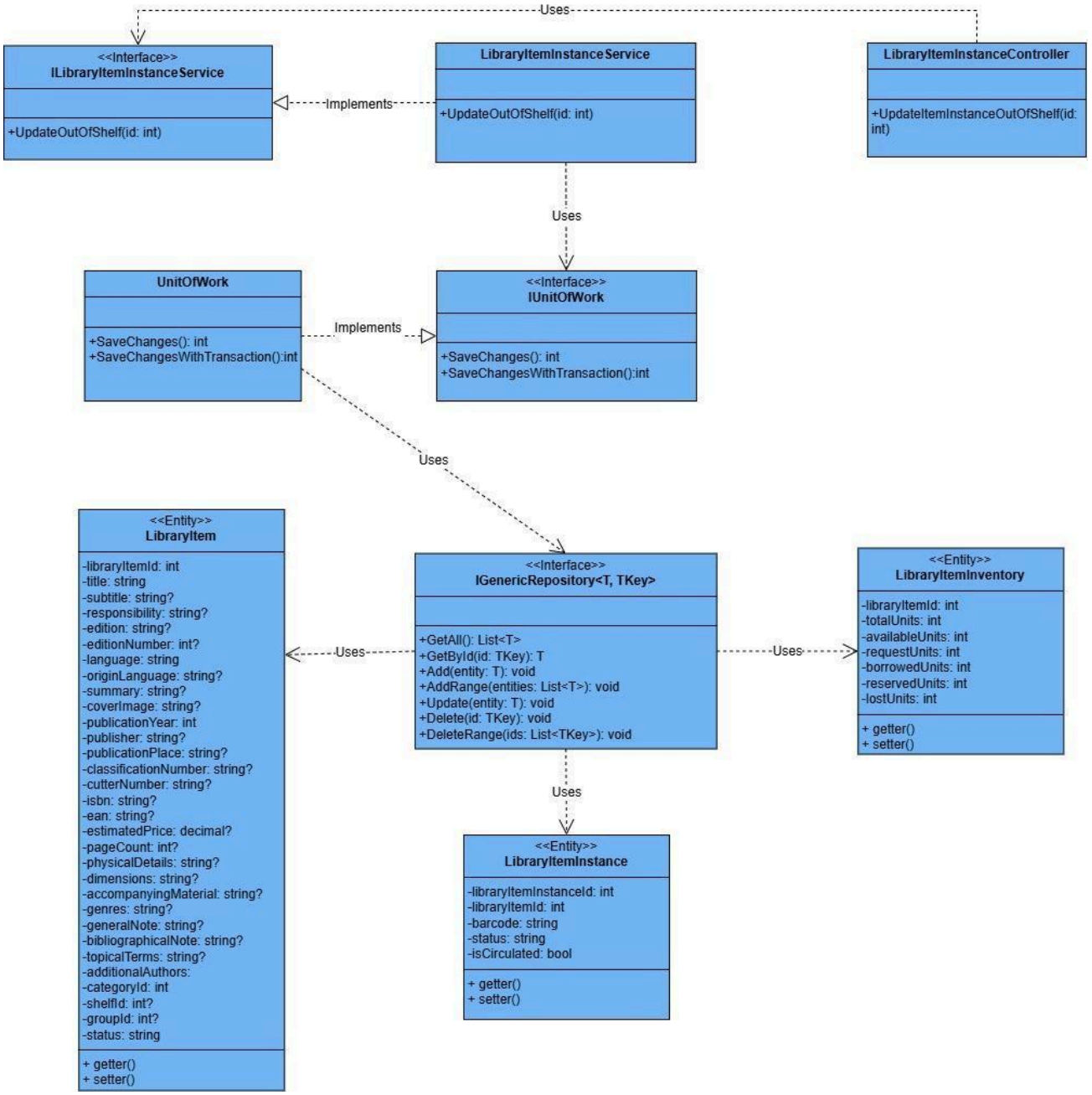


Figure 3.12.2.1 Class Diagram

3.12.2.2 Sequence Diagram

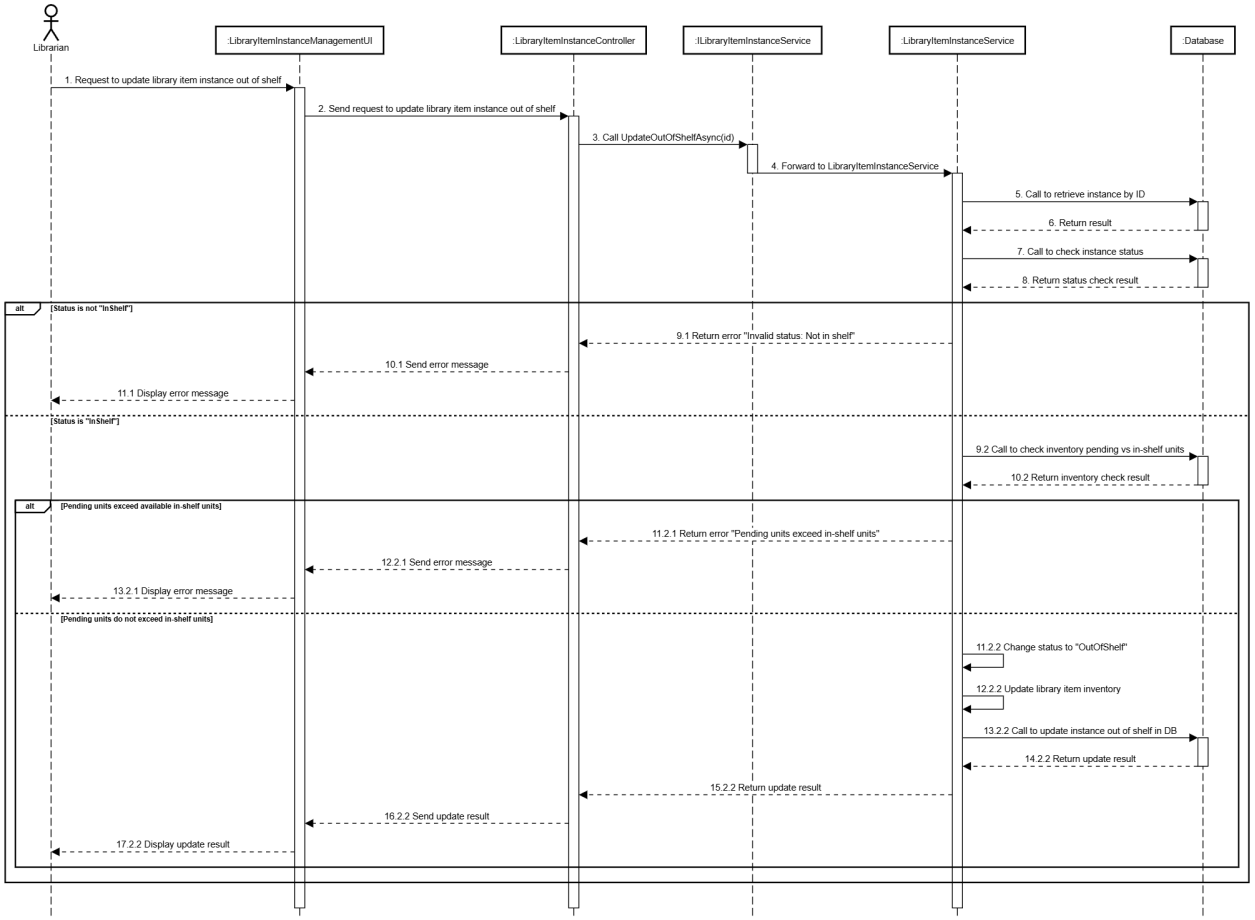


Figure 3.12.2.2 Sequence Diagram

3.12.2.3 Class Specification

3.12.2.3.a Library Item Instance Controller

Table 3.12.2.3.a: Library Item Instance Controller Description Table

No	Method	Description
1	UpdateItemInstanceOutOfShelf(int libraryItemInstanceId)	Receives request with the library item instance ID, invokes the service to mark the specified item instance as out-of-shelf.

3.12.2.3.b Library Item Instance Service

Table 3.12.2.3.b: Library Item Instance Service Description Table

No	Method	Description
1	UpdateOutOfShelf(int libraryItemInstanceId)	Verifies that a library item instance ID exists, updates its out-of-shelf.

3.12.3. Mark as Lost

3.12.3.1 Class Diagram

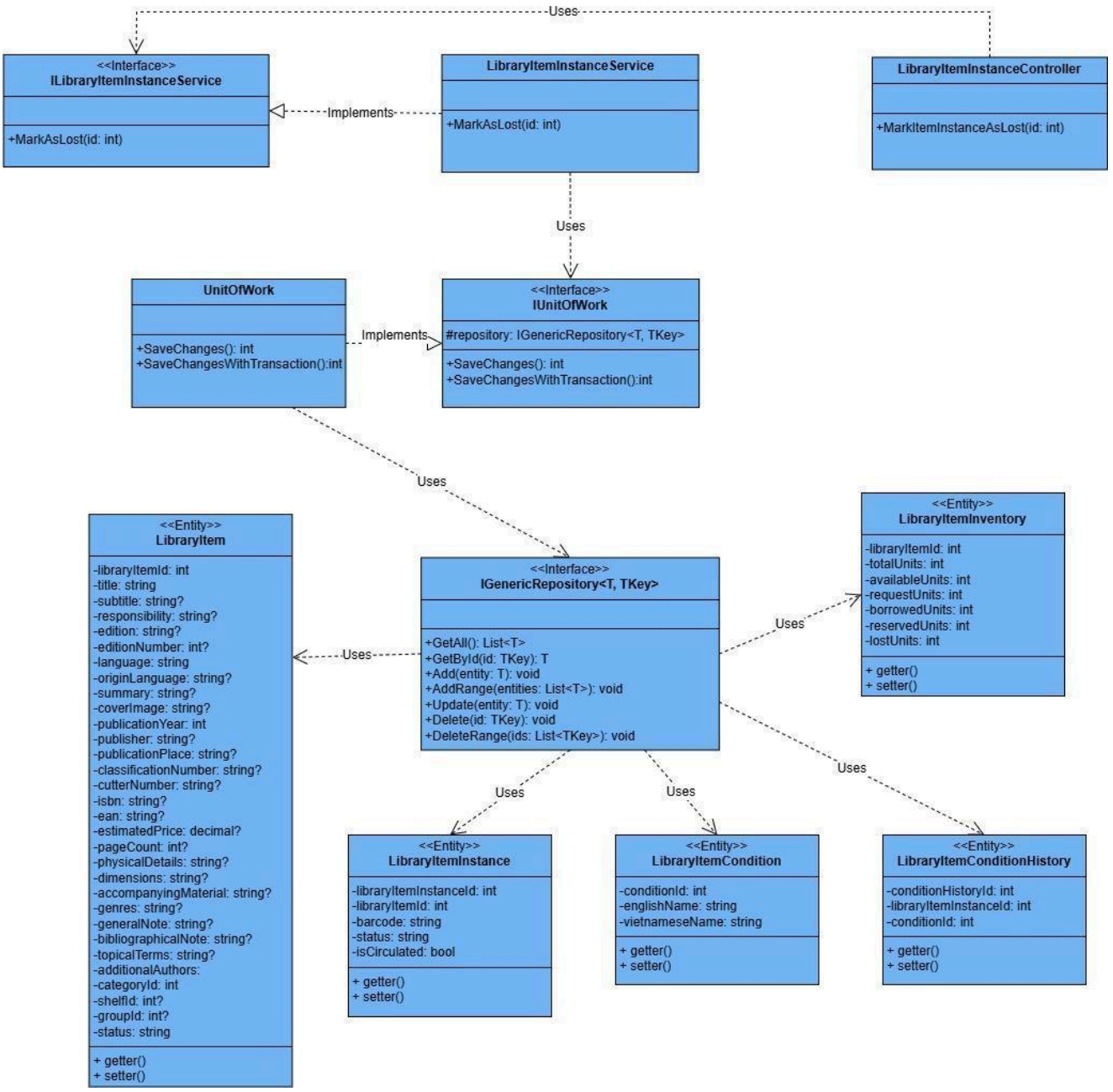


Figure 3.12.3.1 Class Diagram

3.12.3.2 Sequence Diagram

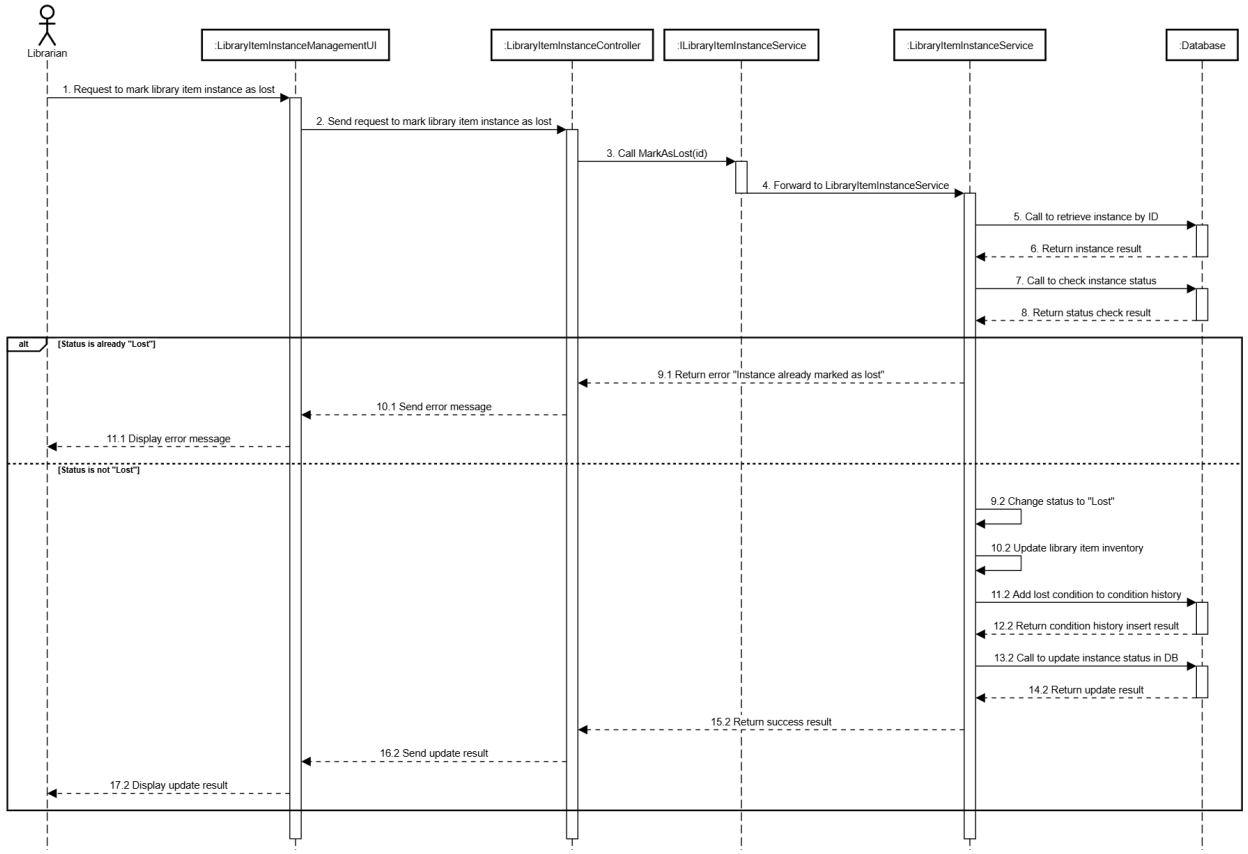


Figure 3.12.3.2 Sequence Diagram

3.12.3.3 Class Specification

3.12.3.3.a Library Item Instance Controller

Table 3.12.3.3.a: Library Item Instance Controller Description Table

No	Method	Description
1	MarkItemInstanceAsLost(int libraryItemInstanceId)	Receives request with the library item instance ID, invokes the service to mark the specified item as lost.

3.12.3.3.b Library Item Instance Service

Table 3.12.3.3.b: Library Item Instance Service Description Table

No	Method	Description
1	MarkAsLost(int libraryItemInstanceId)	Verifies that a library item instance with the given ID exists, sets its status to lost.

3.12.4. Mark Lost as Found

3.12.4.1 Class Diagram

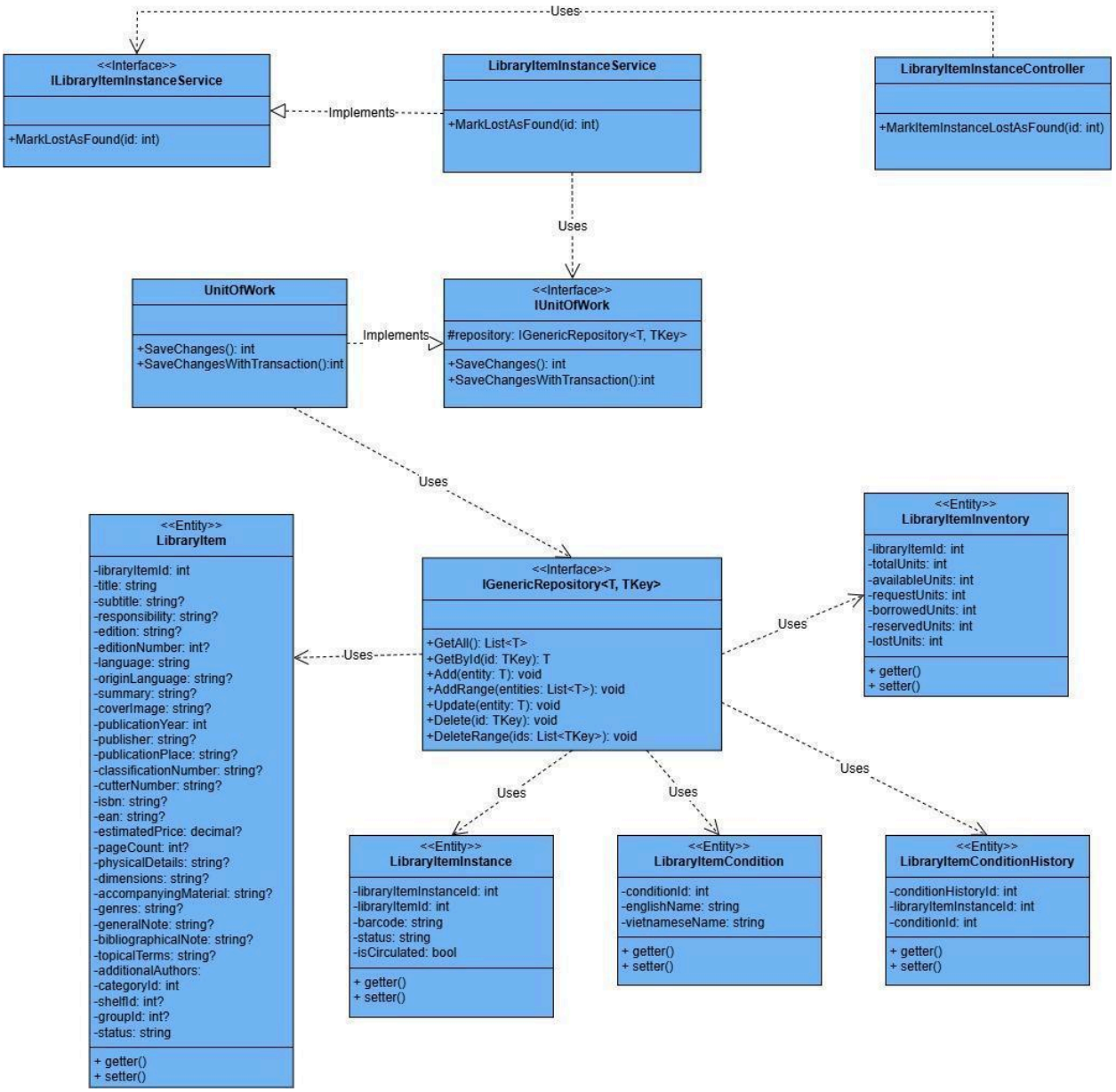


Figure 3.12.4.1 Class Diagram

3.12.4.2 Sequence Diagram

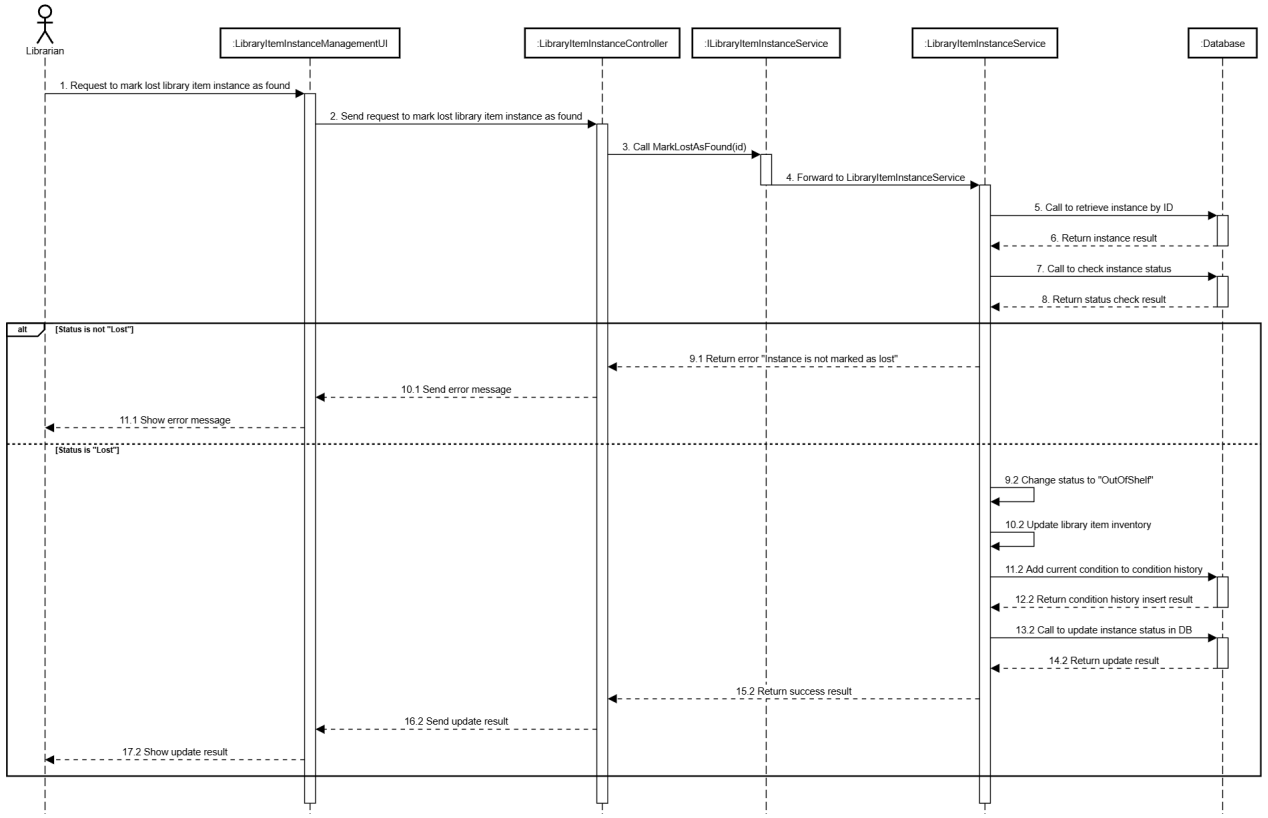


Figure 3.12.4.2 Sequence Diagram

3.12.4.3 Class Specification

3.12.4.3.a Library Item Instance Controller

Table 3.12.4.3.a: Library Item Instance Controller Description Table

No	Method	Description
1	MarkItemInstanceLostAsFound(int libraryItemInstanceId)	Receives request with the library item instance ID, invokes the service to mark the specified item as found (out-of-shelf).

3.12.4.3.b Library Item Instance Service

Table 3.12.4.3.b: Library Item Instance Service Description Table

No	Method	Description
2	MarkLostAsFound(int libraryItemInstanceId)	Verifies that a library item instance with the given ID exists, sets its status to found (out-of-shelf).

3.12.5. Get by Barcode

3.12.5.1 Class Diagram

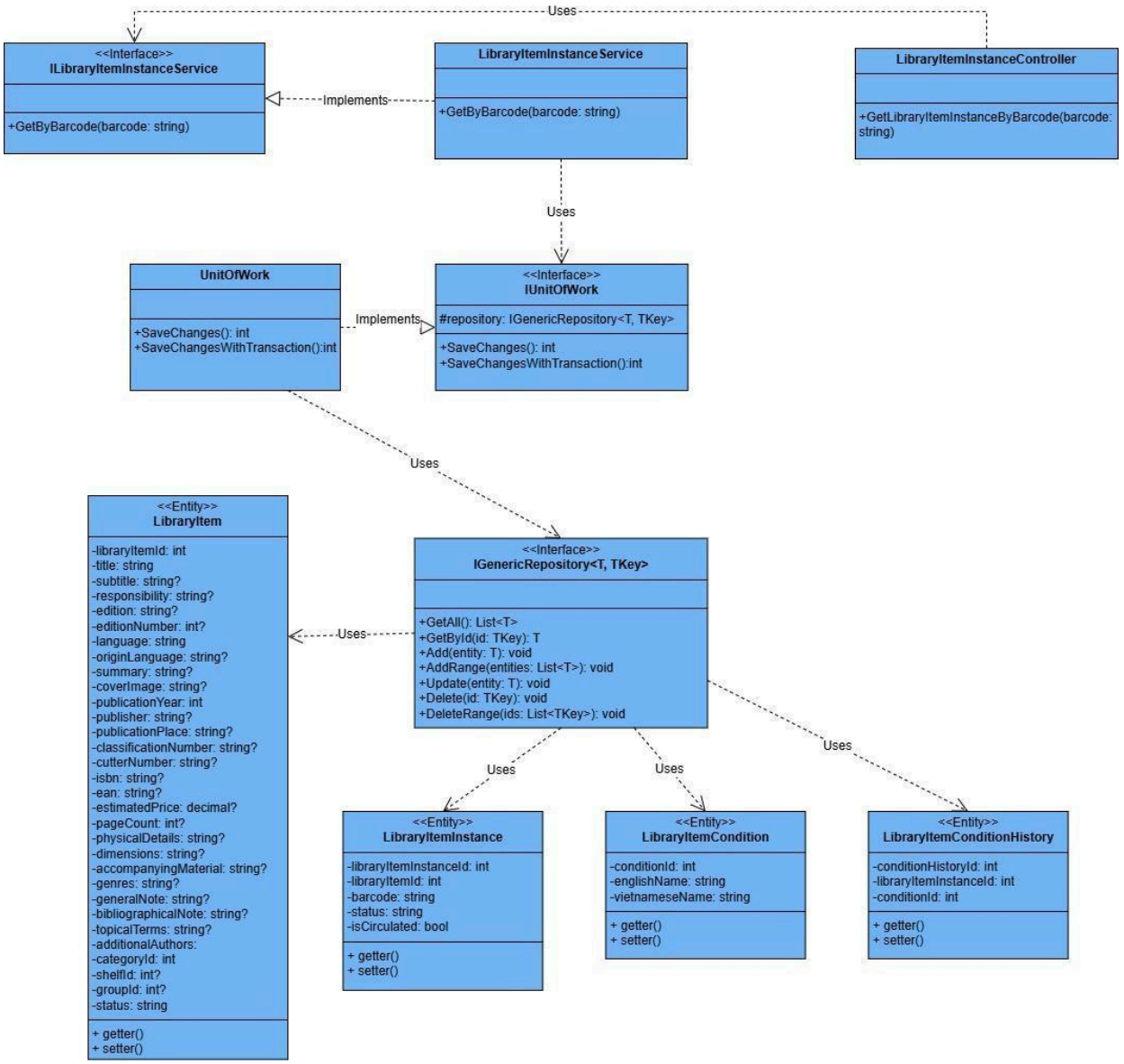


Figure 3.12.5.1 Class Diagram

3.12.5.2 Sequence Diagram

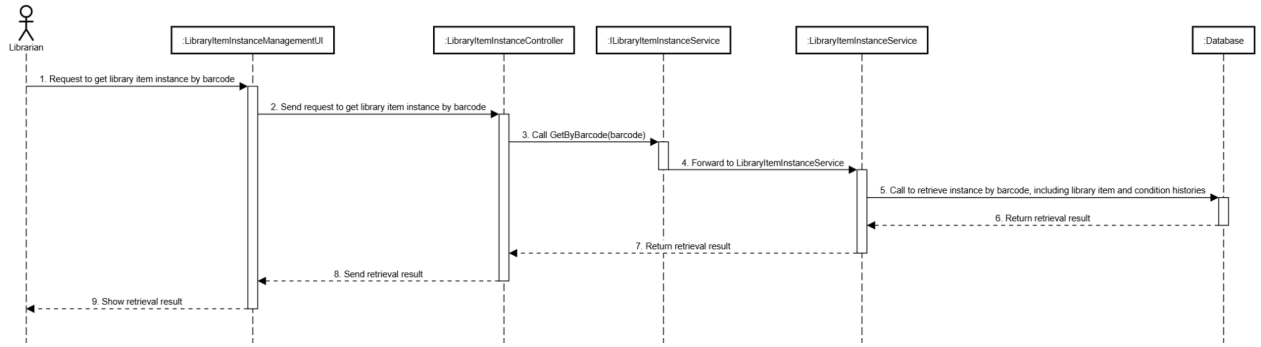


Figure 3.12.5.2 Sequence Diagram

3.12.5.3 Class Specification

3.12.5.3.a Library Item Instance Controller

Table 3.12.5.3.a: Library Item Instance Controller Description Table

No	Method	Description
1	GetByBarcode(string barcode)	Receives a request with a barcode, invokes the service to look up the library item instance matching that barcode.

3.12.5.3.b Library Item Instance Service

Table 3.12.5.3.b: Library Item Instance Service Description Table

No	Method	Description
2	GetByBarcode(string barcode)	Retrieves a library item instance with the given barcode.

3.13. Manage Library Item Conditions

3.13.1. Create Item Condition

3.13.1.1 Class Diagram

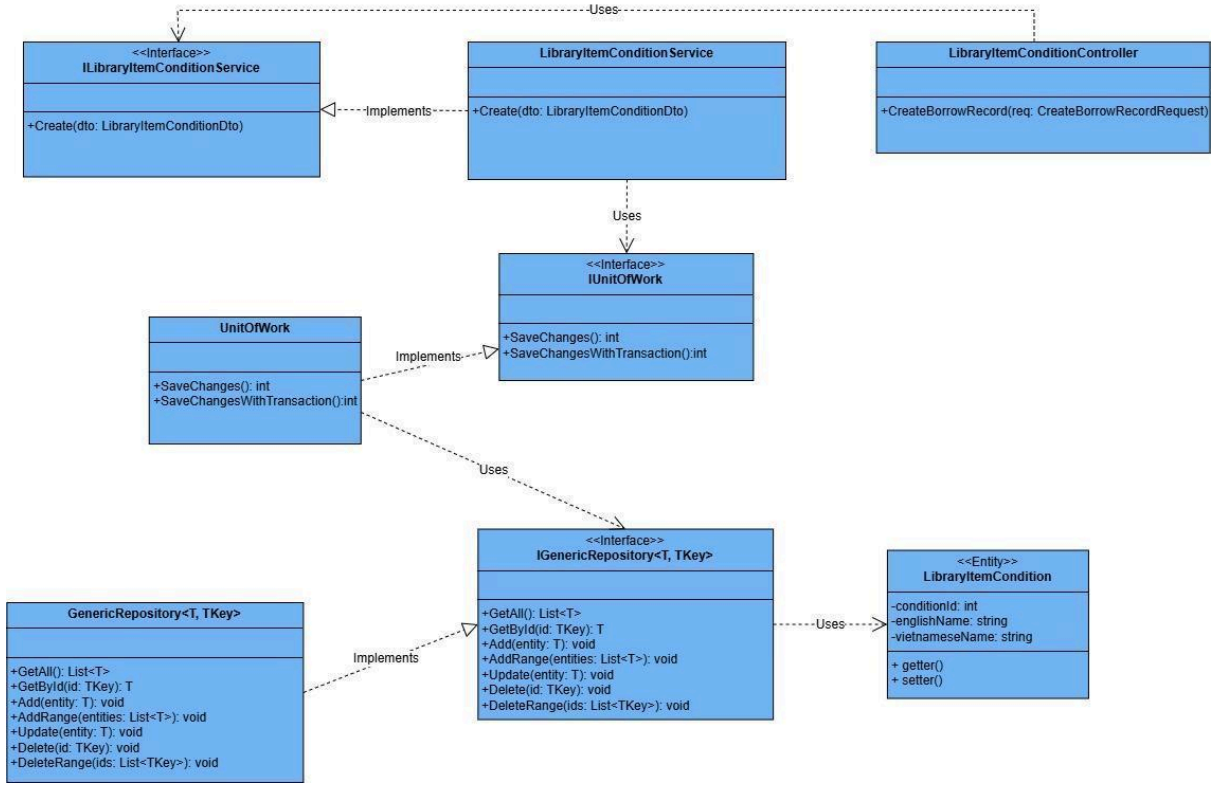


Figure 3.13.1.1 Class Diagram

3.13.1.2 Sequence Diagram

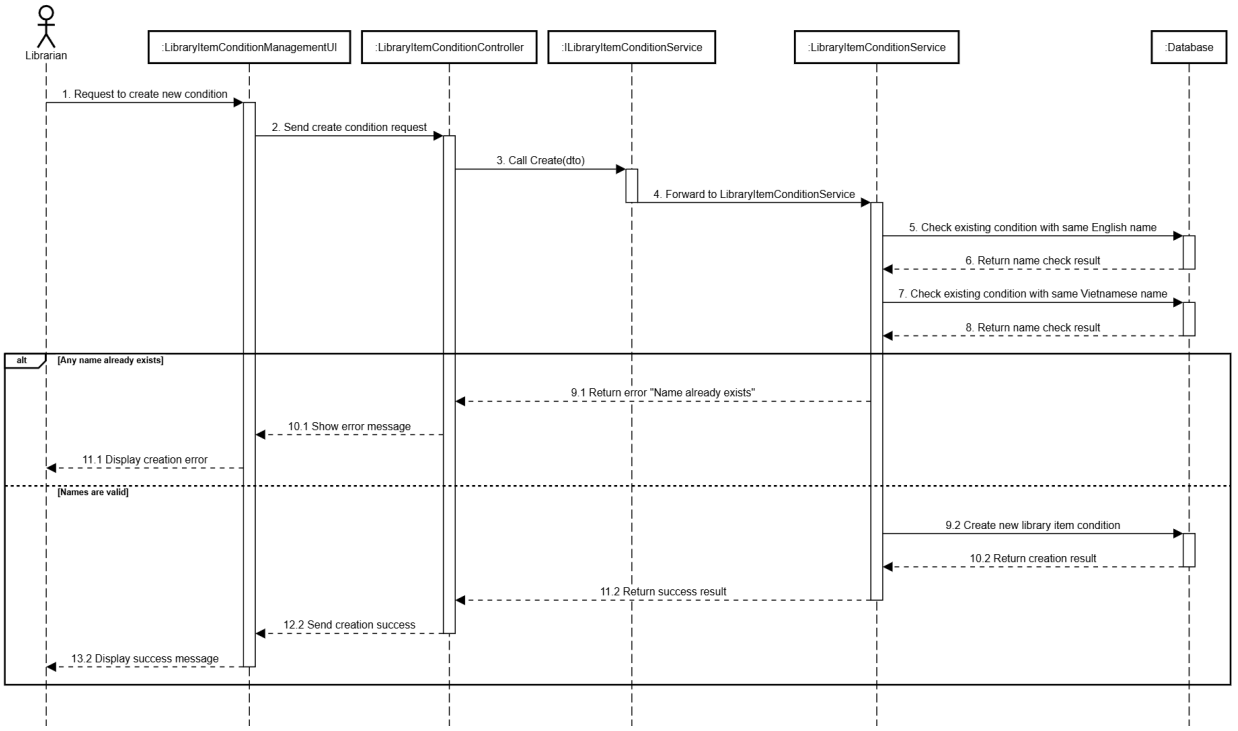


Figure 3.13.1.2 Sequence Diagram

3.13.1.3 Class Specification

3.13.1.3.a Library Item Condition Controller

Table 3.13.1.3.a: Library Item Condition Controller Description Table

No	Method	Description
1	CreateLibraryItemCondition(CreateLibraryItemConditionRequest req)	Receives request containing the new library item condition payload, invokes the service to perform create new library item condition.

3.13.1.3.b Library Item Condition Service

Table 3.13.1.3.b: Library Item Condition Service Description Table

No	Method	Description
2	Create(LibraryItemConditionDto itemConditionDto)	Validate the provided library item condition, and create a new record to the database.

3.13.2. Update Item Condition

3.13.2.1 Class Diagram

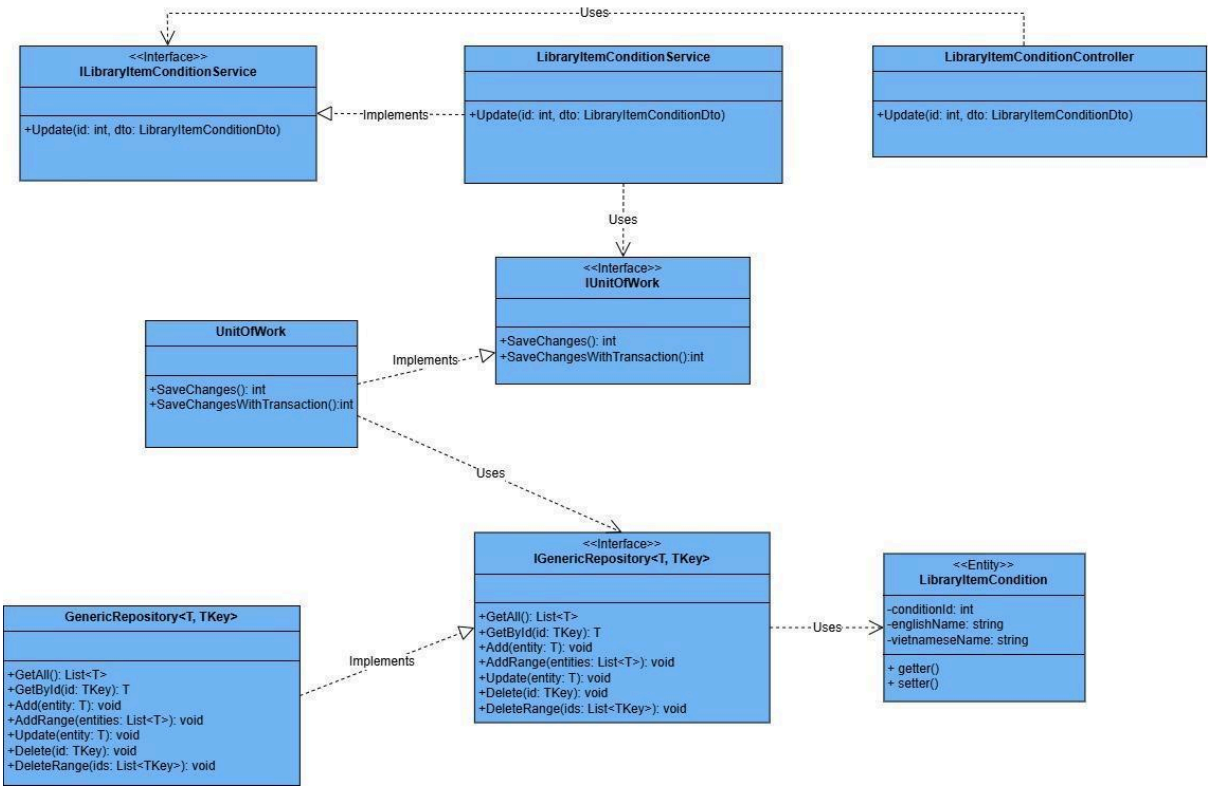


Figure 3.13.2.1 Class Diagram

3.13.2.2 Sequence Diagram

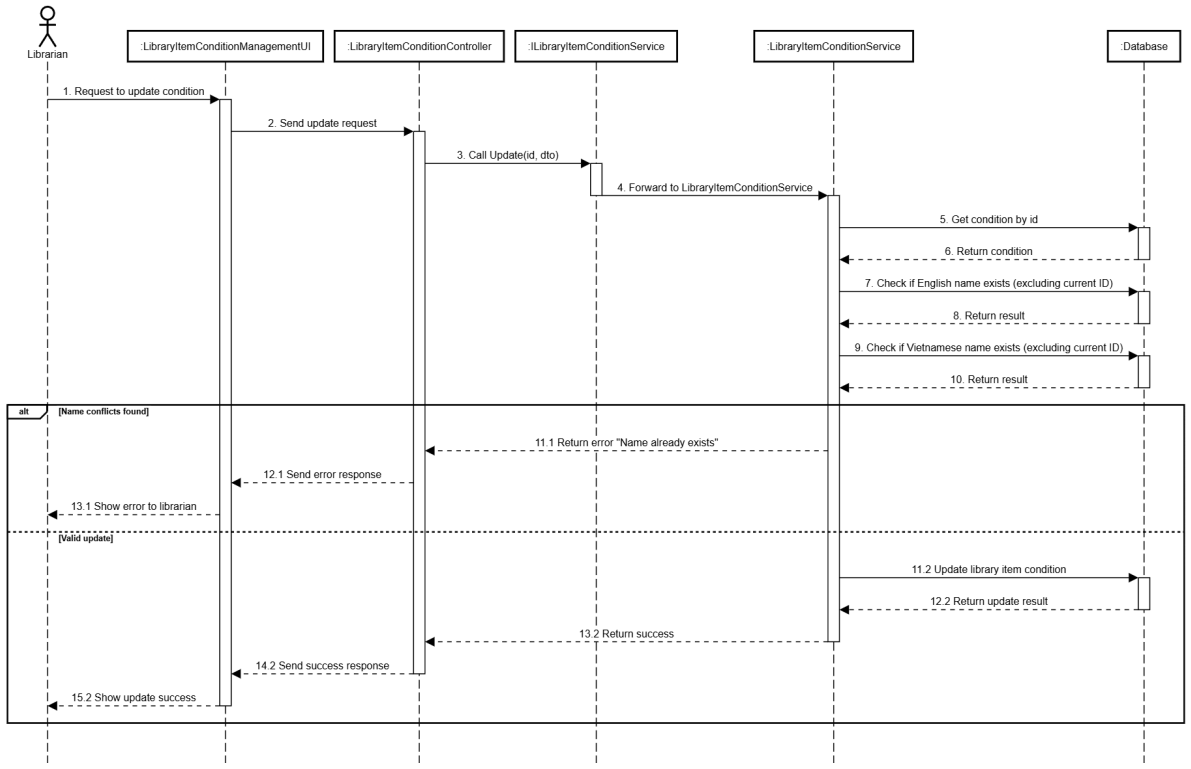


Figure 3.13.2.2 Sequence Diagram

3.13.2.3 Class Specification

3.13.2.3.a Library Item Condition Controller

Table 3.13.2.3.a: Library Item Condition Controller Description Table

No	Method	Description
1	UpdateLibraryItemCondition(int conditionId, UpdateLibraryItemConditionRequest req)	Receives request with the specific condition ID and the updated details, calls the service to apply and persist changes.

3.13.2.3.b Library Item Condition Service

Table 3.13.2.3.b: Library Item Condition Service Description Table

No	Method	Description
2	Update(int conditionId, LibraryItemConditionDto conditionDto)	Verifies that a condition with the given condition ID exists, update fields based on the provided fields and save changes to the database.

3.13.3. Delete Item Condition

3.13.3.1 Class Diagram

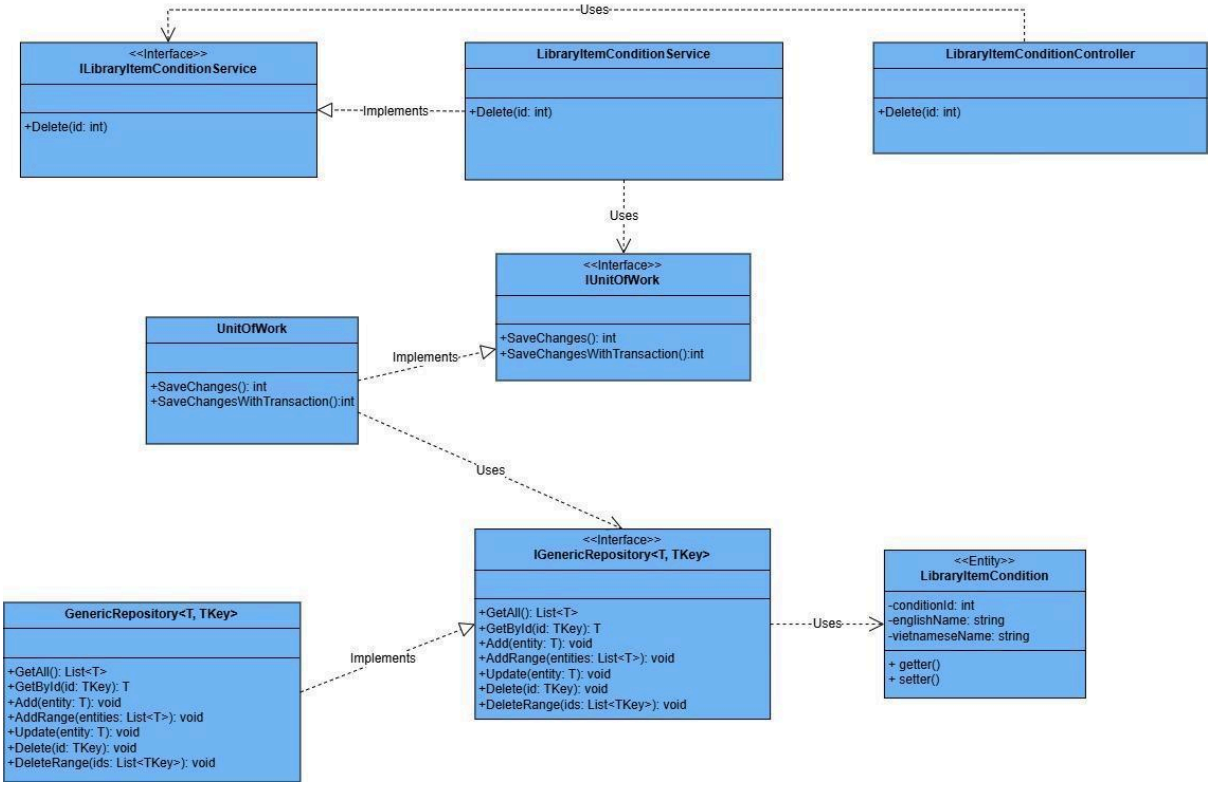


Figure 3.13.3.1 Class Diagram

3.13.3.2 Sequence Diagram

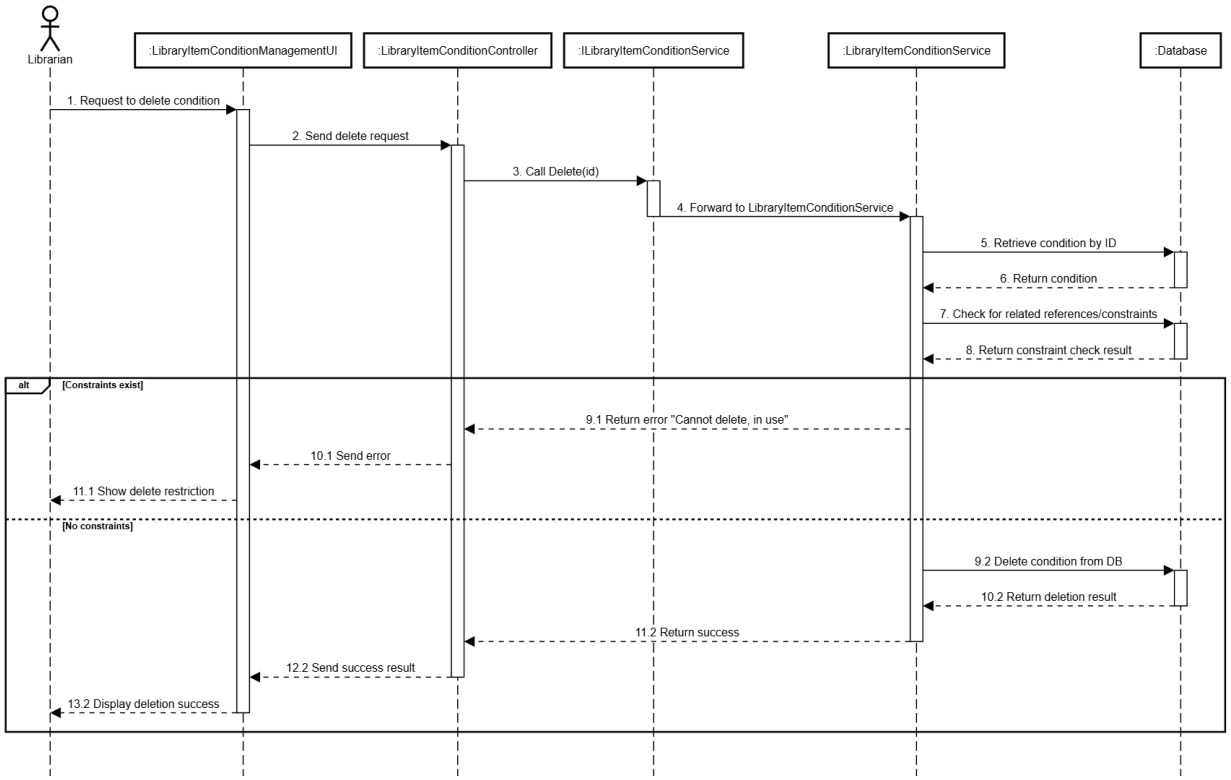


Figure 3.13.3.2 Sequence Diagram

3.13.3.3 Class Specification

3.13.3.3.a Library Item Condition Controller

Table 3.13.3.3.a: Library Item Condition Controller Description Table

No	Method	Description
1	DeleteLibraryItemCondition(int conditionId)	Receives the specified condition ID, calls the service to remove the specified library item condition.

3.13.3.3.b Library Item Condition Service

Table 3.13.3.3.b: Library Item Condition Service Description Table

No	Method	Description
1	Delete(int conditionId)	Check that the library item condition exists, permanently remove it from the database.

3.13.4. Get All Item Conditions

3.13.4.1 Class Diagram

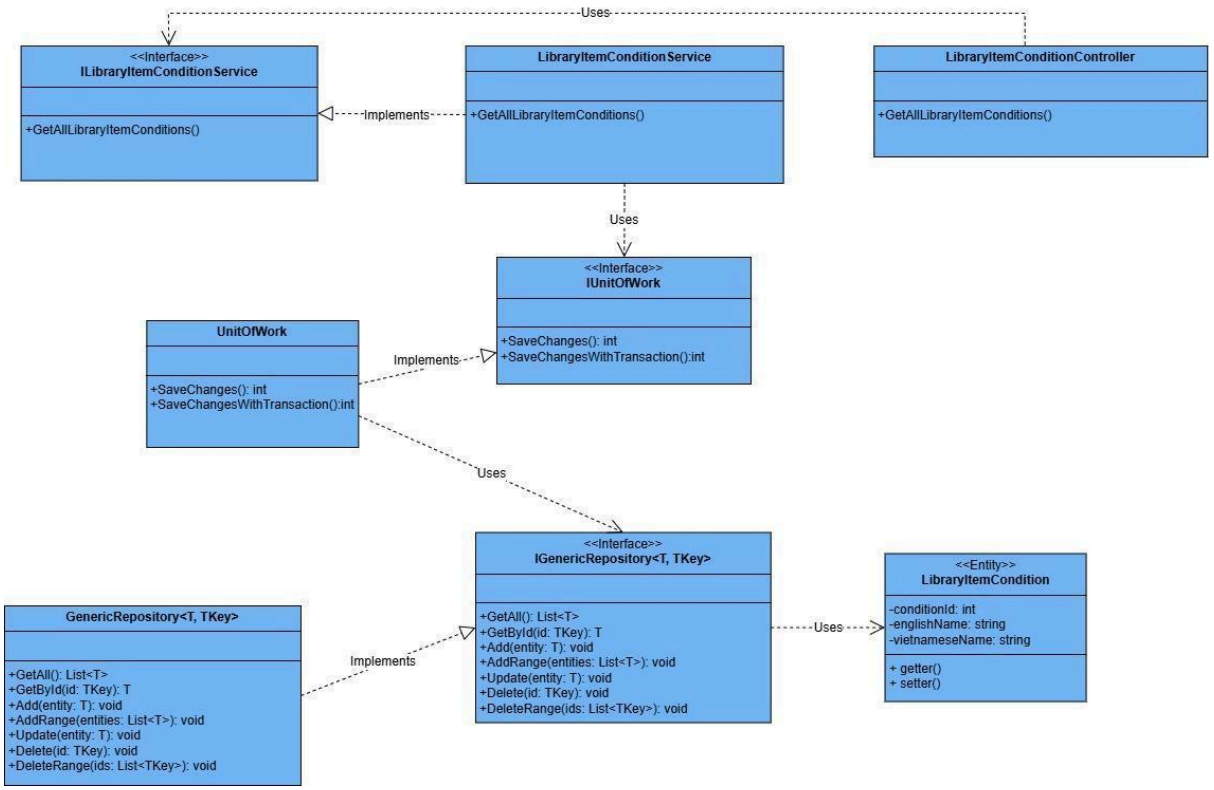


Figure 3.13.4.1 Class Diagram

3.13.4.2 Sequence Diagram

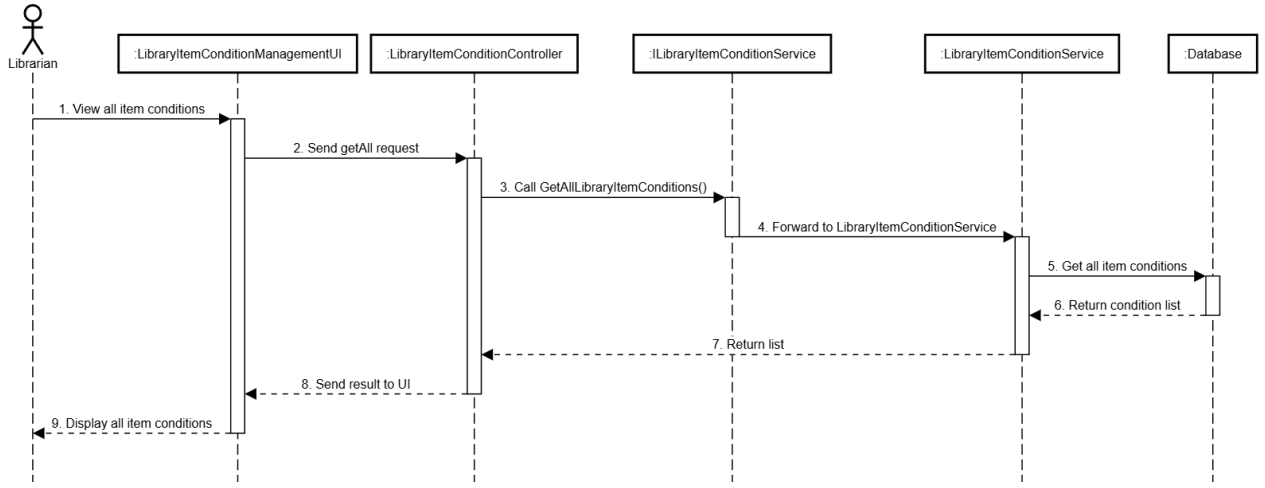


Figure 3.13.4.2 Sequence Diagram

3.13.4.3 Class Specification

3.13.4.3.a Library Item Condition Controller

Table 3.13.4.3.a: Library Item Condition Controller Description Table

No	Method	Description
1	GetAllLibraryItemConditions()	Receives request with no parameters, invokes the service to retrieve all existing library item conditions.

3.13.4.3.b Library Item Condition Service

Table 3.13.4.3.b: Library Item Condition Service Description Table

No	Method	Description
1	GetAll()	Retrieve all exist library item conditions.

3.14. Manage Library Resources

3.14.1. Upload Resource To Cloud

3.14.1.1 Class Diagram

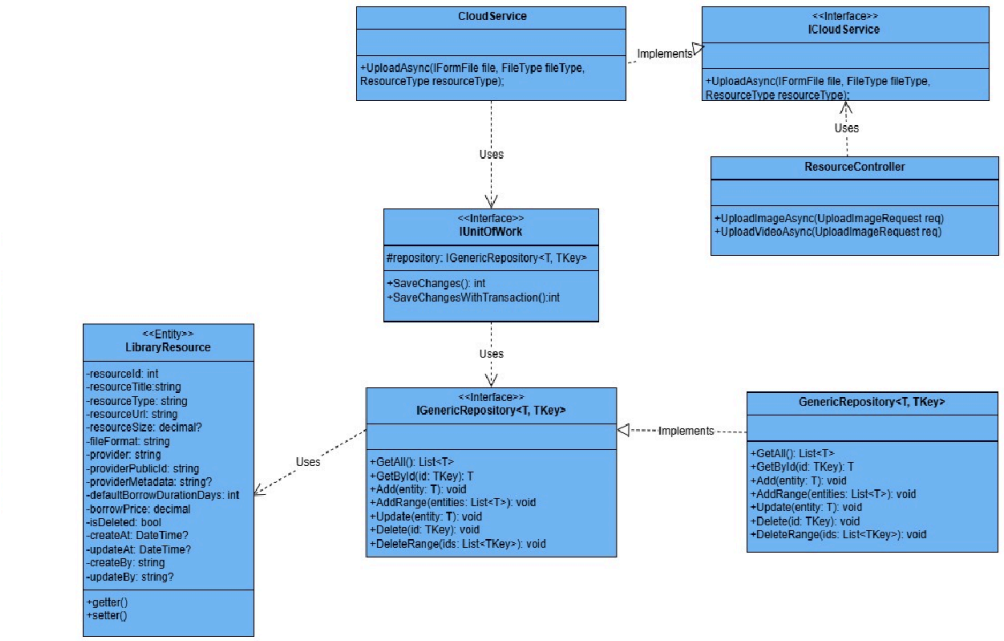


Figure 3.14.1.1 Class Diagram

3.14.1.2 Sequence Diagram

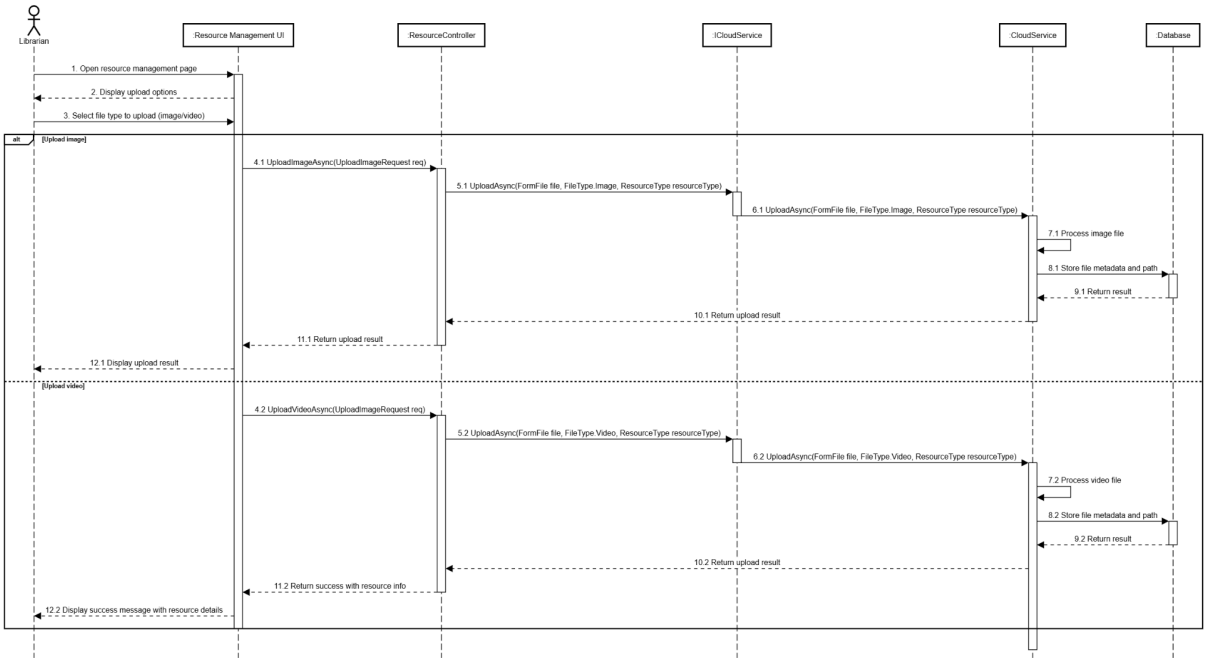


Figure 3.14.1.2 Sequence Diagram

3.14.1.3 Class Specification

3.14.1.3.a Resource Controller

Table 3.14.1.3.a: Resource Controller Description Table

No	Method	Description
1	UploadVideoAsync(UploadVideoRequest req)	Upload video from librarian's computer to cloud storage
2	UploadImageAsync(UploadImageRequest req)	Upload cover image, user image, PDF file to cloud storage.

3.14.1.3.b Cloud Service

Table 3.14.1.3.b: Cloud Service Description Table

No	Method	Description
1	UploadAsync(IFormFile file, FileType fileType, ResourceType resourceType)	Upload file from librarian's computer to cloud storage

3.14.2. Update Resource In Cloud

3.14.2.1 Class Diagram

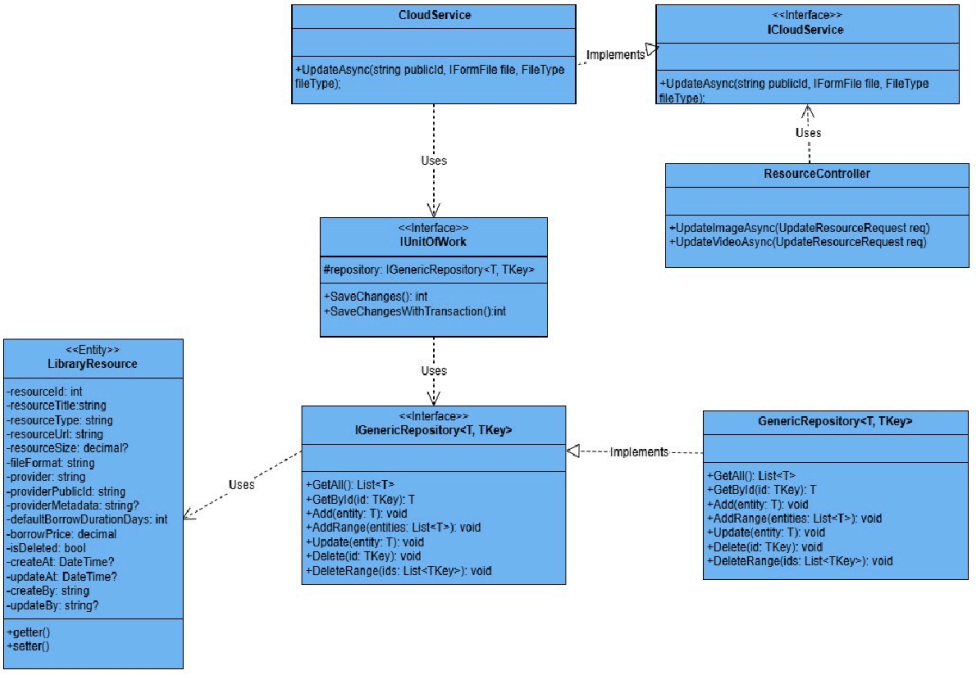


Figure 3.14.2.1 Class Diagram

3.14.2.2 Sequence Diagram

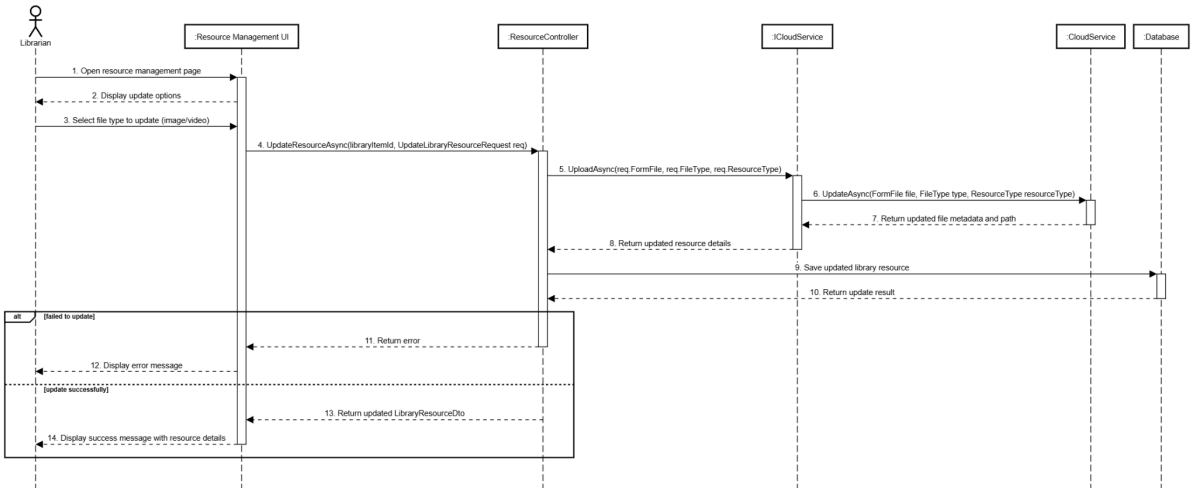


Figure 3.14.2.2 Sequence Diagram

3.14.2.3 Class Specification

3.14.2.3.a Resource Controller

Table 3.14.2.3.a: Resource Controller Description Table

No	Method	Description

1	UpdateVideo(UpdateVideoRequest req)	Replace video that existed in cloud storage with another.
2	UpdateImage(UpdateImageRequest req)	Replace cover image, user image, PDF file in cloud storage.

3.14.2.3.b Cloud Service

Table 3.14.2.3.b: Cloud Service Description Table

No	Method	Description
1	UpdateAsync(IFormFile file, FileType fileType, ResourceType resourceType)	Replace file in cloud storage with the new one

3.14.3. Delete Resource From Cloud

3.14.3.1 Class Diagram

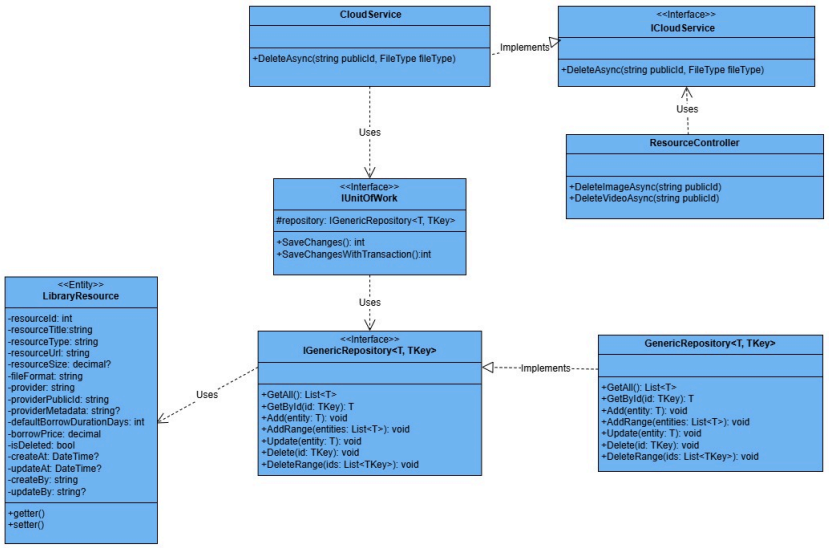


Figure 3.14.3.1 Class Diagram

3.14.3.2 Sequence Diagram

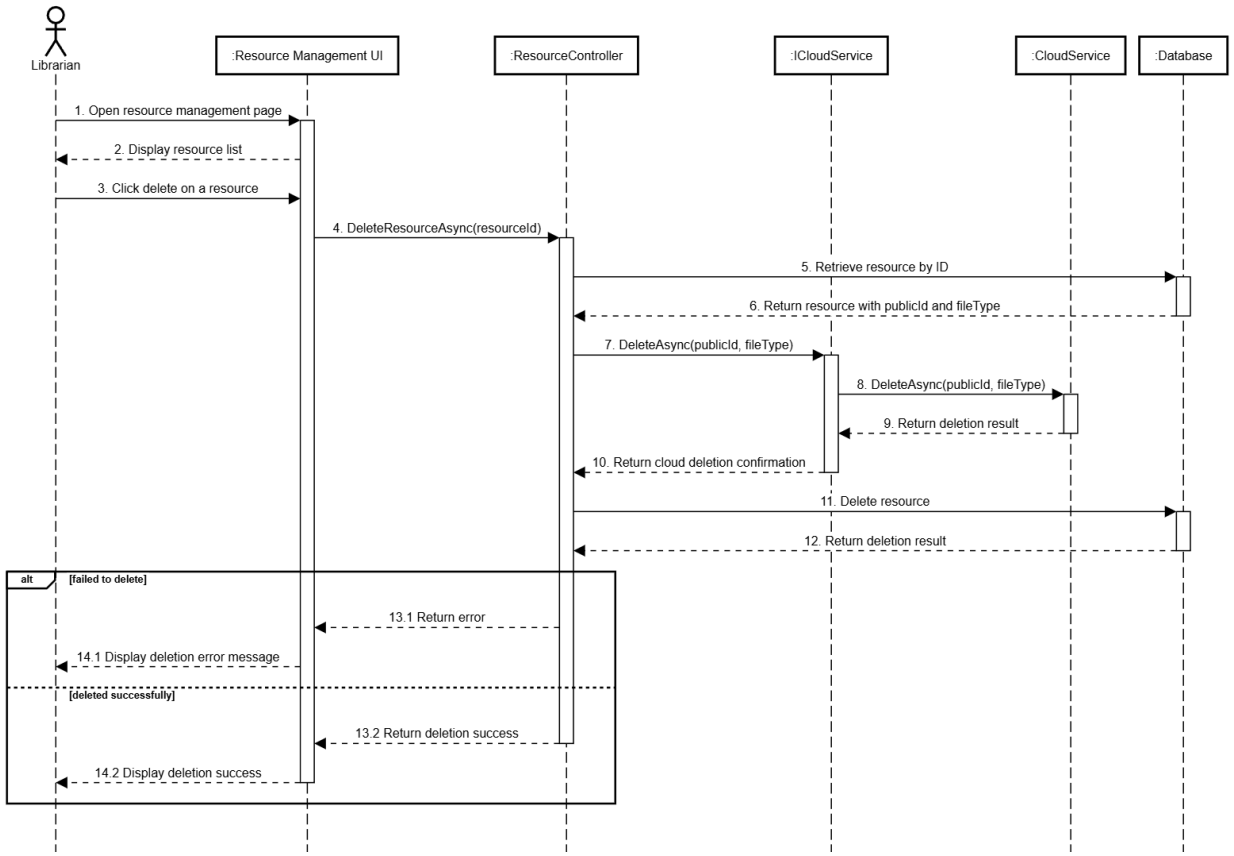


Figure 3.14.3.2 Sequence Diagram

3.14.3.3 Class Specification

3.14.3.3.a Resource Controller

Table 3.14.3.3.a: Resource Controller Description Table

No	Method	Description
1	DeleteVideoAsync(string publicId)	Delete video that existed in cloud storage with another.
2	DeleteImageAsync(string publicId)	Delete cover image, user image, PDF file in cloud storage.

3.14.3.3.b Cloud Service

Table 3.14.3.3.b: Cloud Service Description Table

No	Method	Description

1	DeleteAsync(string publicId, FileType fileType)	Delete file in cloud storage with the new one

3.14.4. Add Resource To Library Item

3.14.4.1 Class Diagram

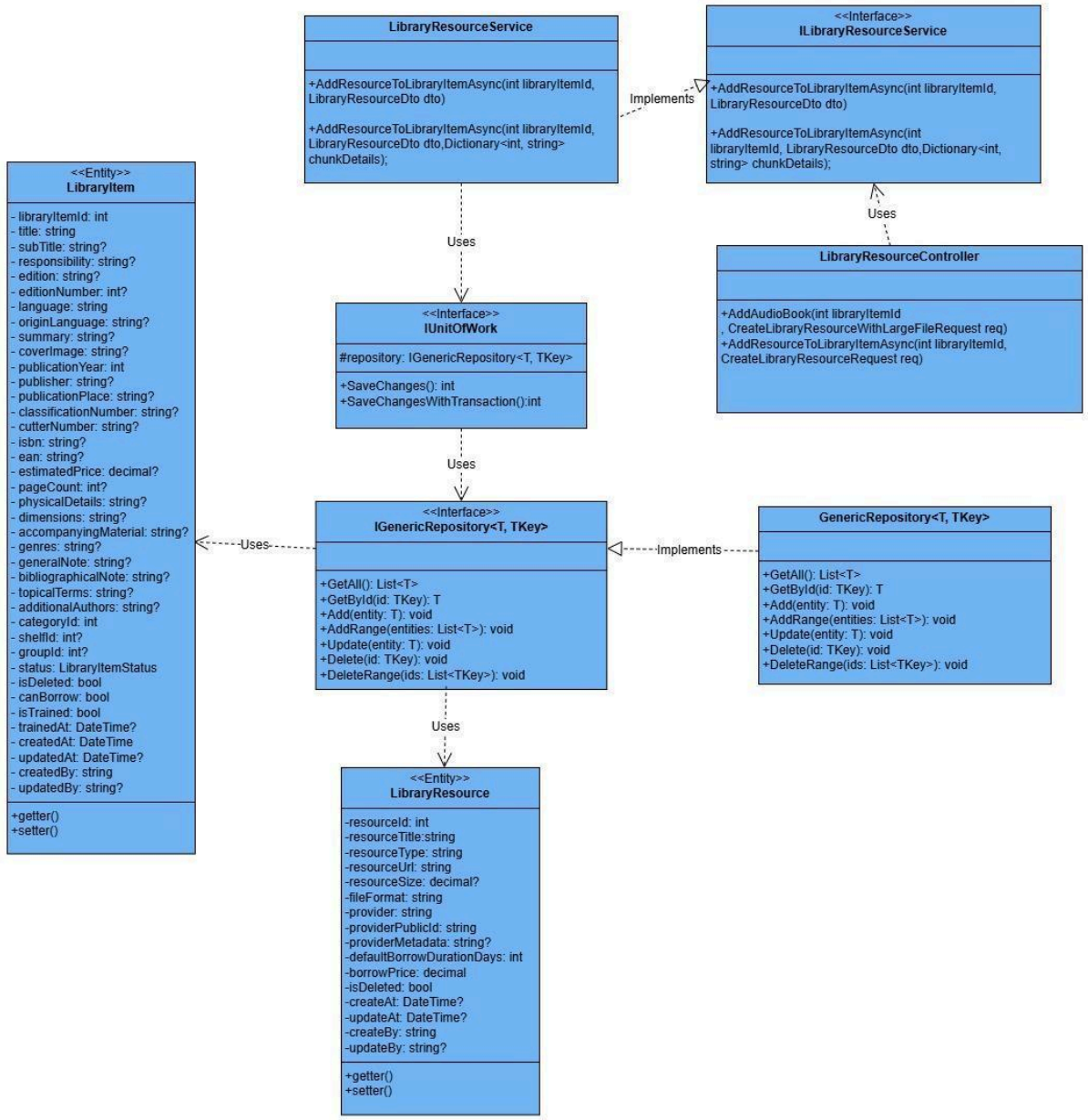


Figure 3.14.4.1 Class Diagram

3.14.4.2 Sequence Diagram

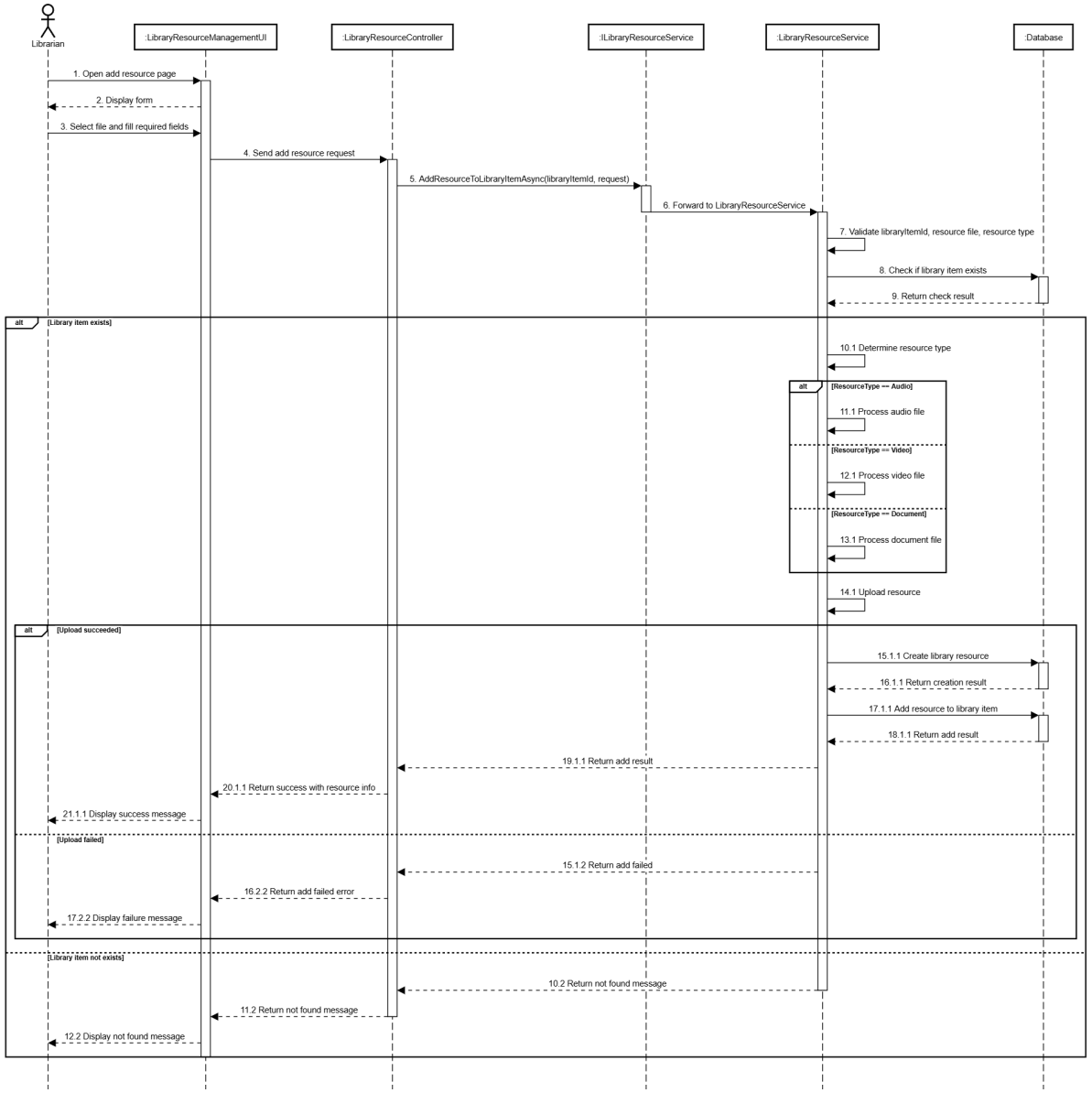


Figure 3.14.4.2 Sequence Diagram

3.14.4.3 Class Specification

3.14.4.3.a Library Resource Controller

Table 3.14.4.3.a: Library Resource Controller Description Table

No	Method	Description
1	AddAudioBook(int libraryItemId,CreateLibraryResourceWithLargeFileRequest req)	Add an audio file to a book that has an audio version for the user.

2	AddResourceToLibraryItemAsync(int libraryItemId,CreateLibraryResourceRequest req)	Add a PDF file to a book that hasebook version for the user.

3.14.4.3.b Library Resource Service

Table 3.14.4.3.b: Library Resource Service Description Table

No	Method	Description
1	AddResourceToLibraryItemAsync(int libraryId, LibraryResourceDto libraryResourceDto)	Add a PDF file to a book that has an ebook version for users.
2	AddResourceToLibraryItemAsync(int libraryId, Dictionary<int,string>chunkDetails)	Add a large audio file to a book that has audio version for user

3.14.5. Get Preview of Resource

3.14.5.1 Class Diagram

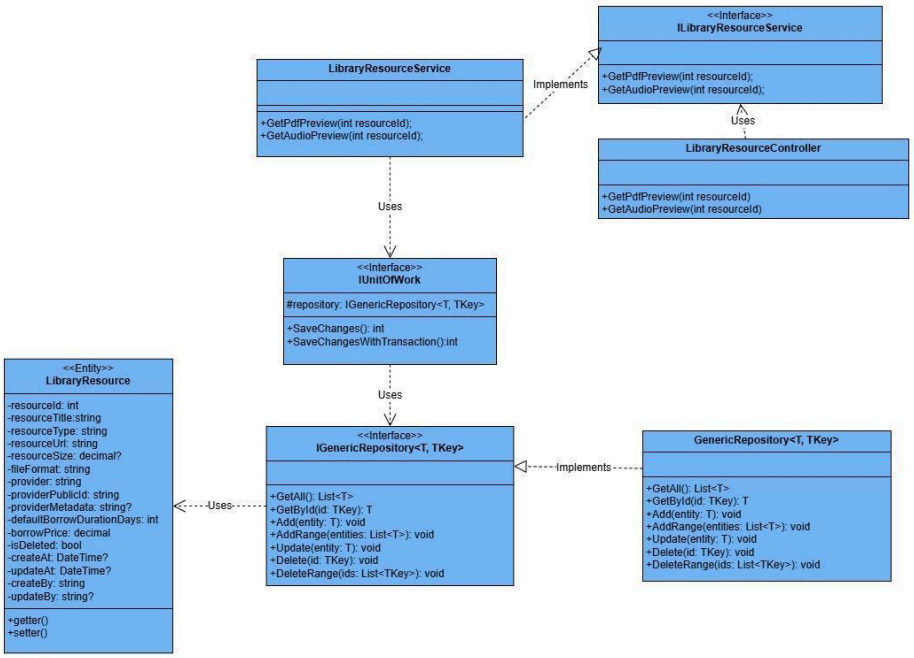


Figure 3.14.5.1 Class Diagram

3.14.5.2 Sequence Diagram

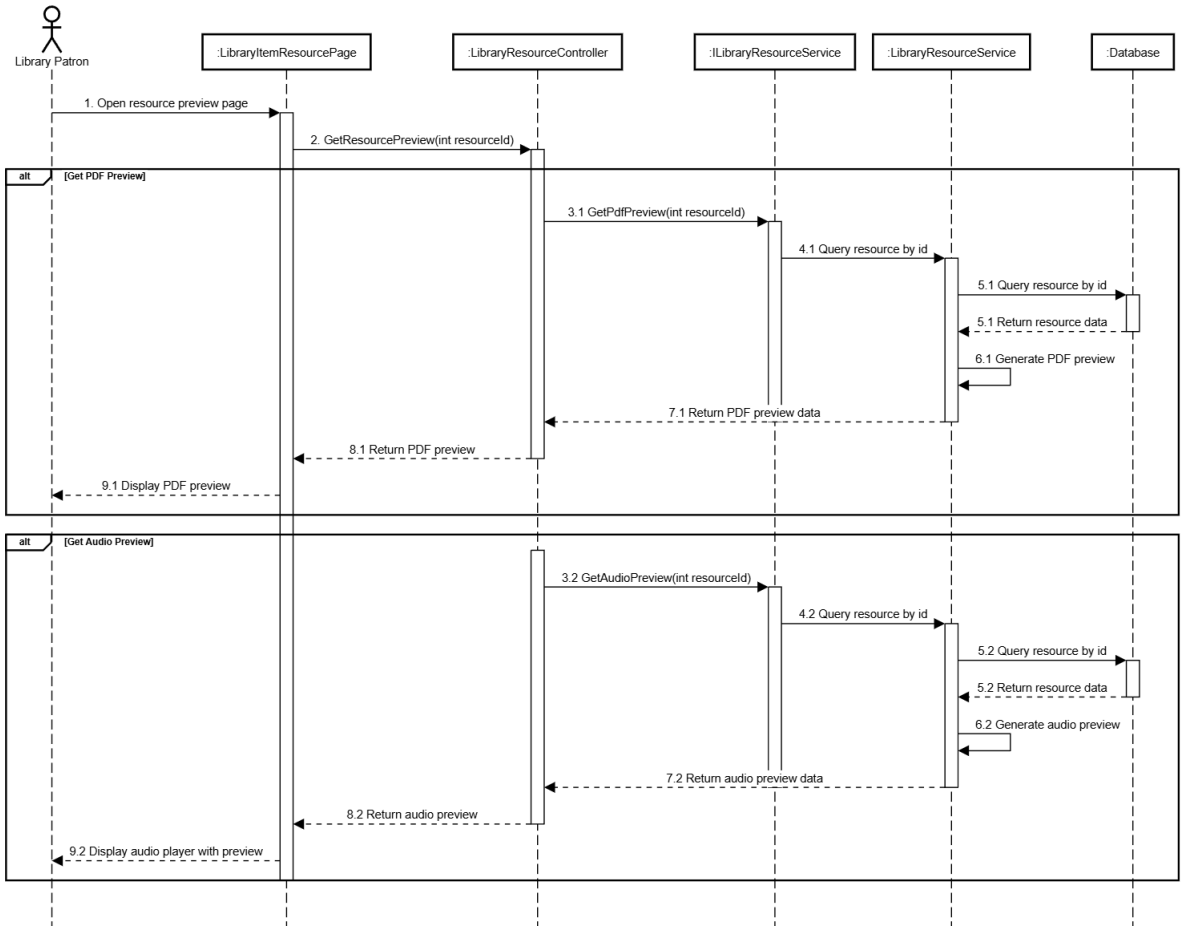


Figure 3.14.5.2 Sequence Diagram

3.14.5.3 Class Specification

3.14.5.3.a Library Resource Controller

Table 3.14.5.3.a: Library Resource Controller Description Table

No	Method	Description
1	GetPdfPreview(int resourceId)	Get 5 first pages for user to have an overview about a book
2	GetAudioPreview(int resourceId)	Get 5-minute part of the whole audio file for user to preview

3.14.5.3.b Library Resource Service

Table 3.14.5.3.b: Library Resource Service Description Table

No	Method	Description
1	GetPdfPreview(int resourceId)	Get 5 first pages without watermark for user to have an overview about a book
2	GetAudioPreview(int resourceId)	Get 5-minute part without watermark of the whole audio file for user to preview

3.14.6. Get Borrowed Resource

3.14.6.1 Class Diagram

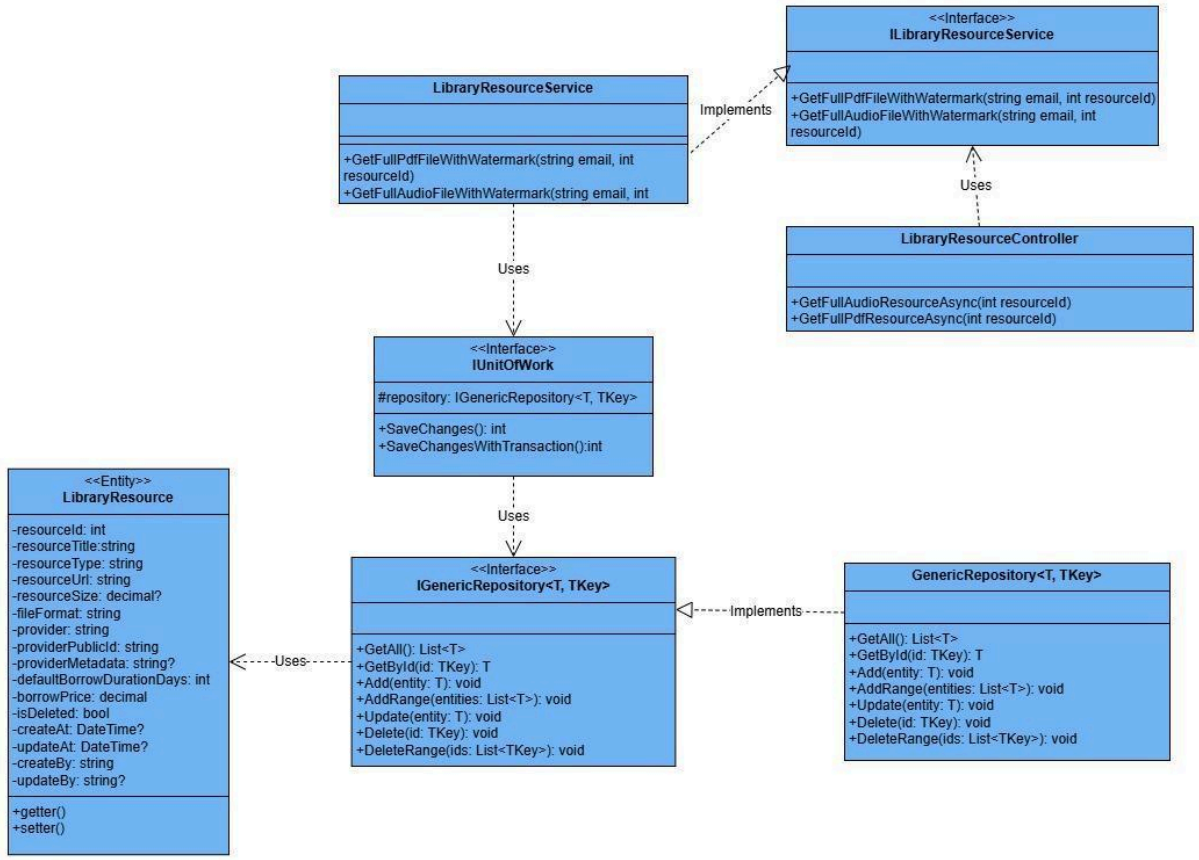


Figure 3.14.6.1 Class Diagram

3.14.6.2 Sequence Diagram

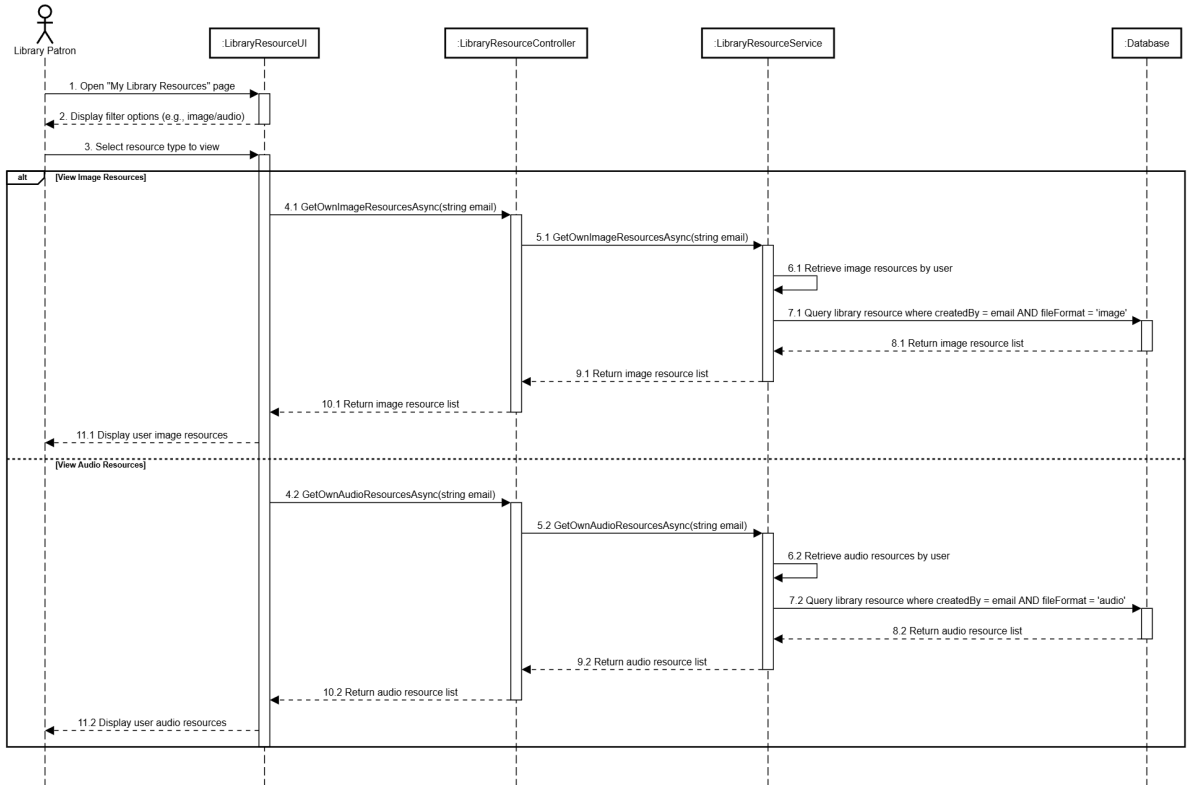


Figure 3.14.6.2 Sequence Diagram

3.15.6.3 Class Specification

3.15.6.3.a Library Resource Controller

Table 3.15.6.3.a: Library Resource Controller Description Table

No	Method	Description
1	GetFullPdfFileWithWatermark(int resourceId)	Get a PDF file with a watermark in every single page with the text of the user email.
2	GetFullAudioFileWithWatermark(int resourceId)	Get the audio version of the book with a reminding policy voice.

3.15.6.3.b Library Resource Service

Table 3.15.6.3.b: Library Resource Service Description Table

No	Method	Description

1	GetFullPdfFileWithWatermark(string email, int resourceId)	Add watermark to the original file and return the full version of the watermarked PDF file to the user.
2	GetFullAudioFileWithWatermark(string email, int resourceId)	Add voice reminder with specified email to original audio and return full version of watermarked audio file to user

3.15. Manage Library Location

3.15.1. Get All Shelves

3.15.1.1 Class Diagram

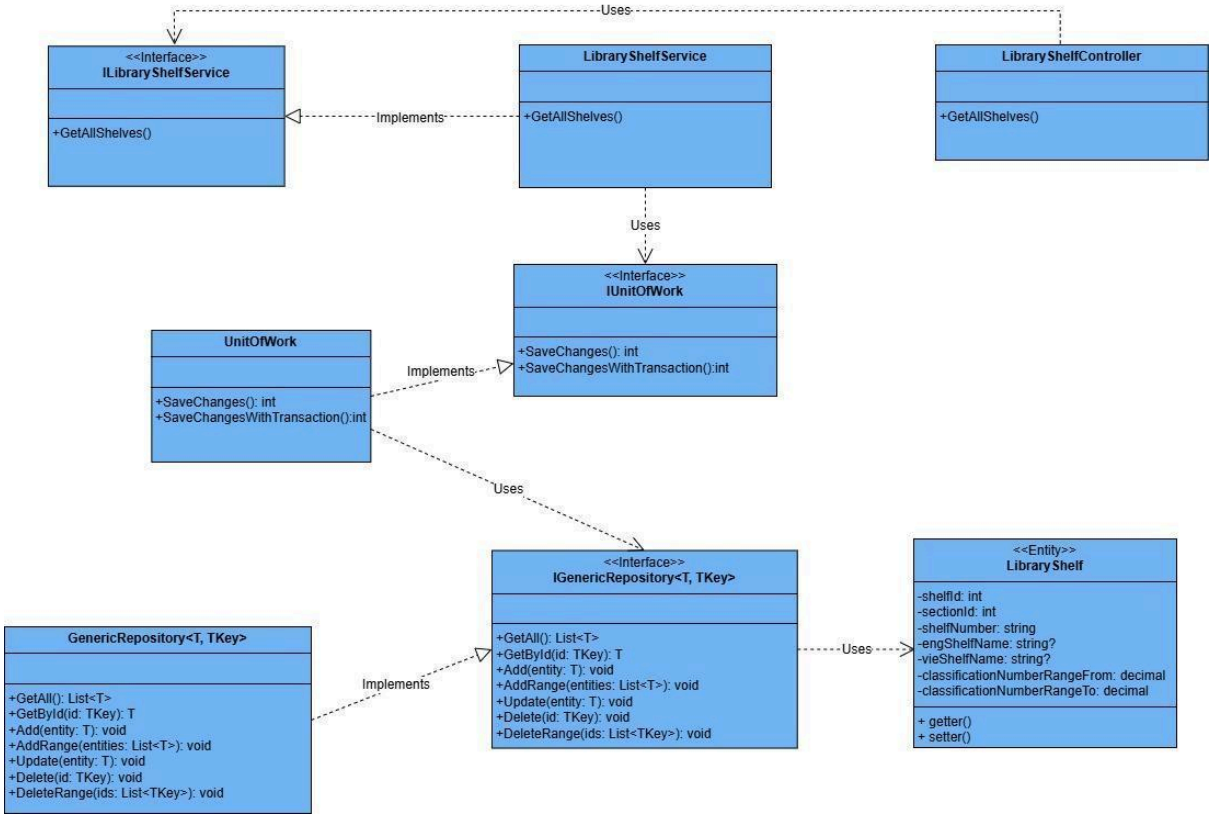


Figure 3.15.1.1 Class Diagram

3.15.1.2 Sequence Diagram

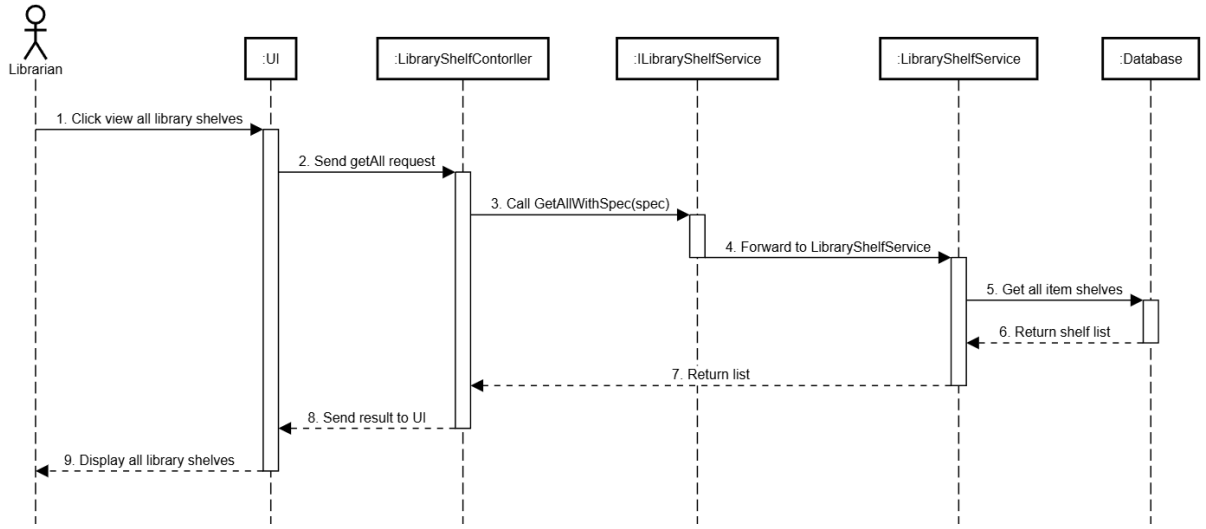


Figure 3.15.1.2 Sequence Diagram

3.15.1.3 Class Specification

3.15.1.3.a Library Shelf Controller

Table 3.15.1.3.a: Library Shelf Controller Description Table

No	Method	Description
1	GetAllLibraryShelves(LibraryShelfSpecParams specParams)	Request with filtering, sorting, and pagination, invokes service to retrieve all existing shelves match.

3.15.1.3.b Library Shelf Service

Table 3.15.1.3.b: Library Shelf Service Description Table

No	Method	Description
1	GetAllWithSpec(LibraryShelfSpecification spec)	Applies filtering, sorting, and paging logic to retrieve all library shelves match.

3.15.2. Get Map Shelf Detail by ID

3.15.2.1 Class Diagram

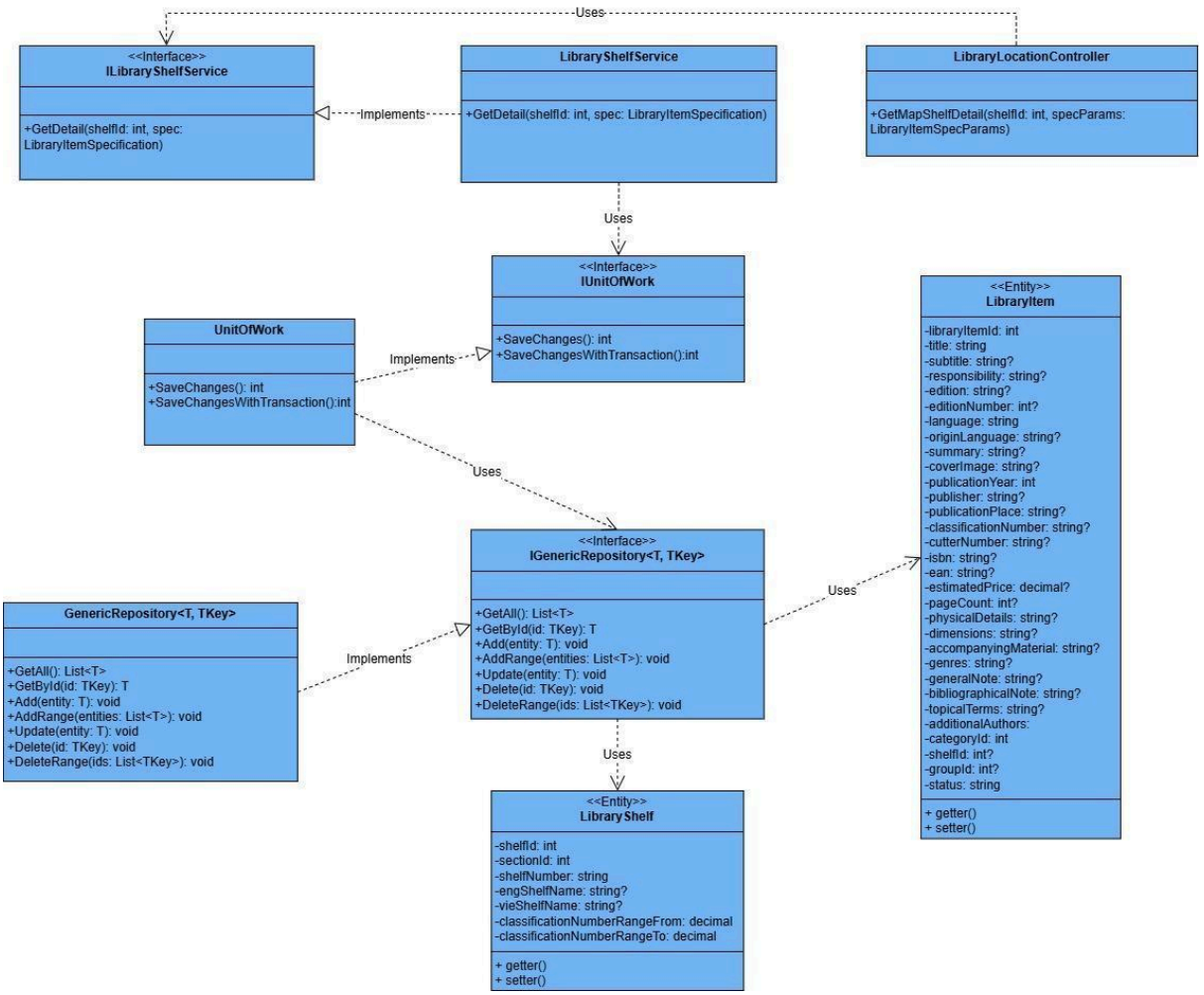


Figure 3.15.2.1 Class Diagram

3.15.2.2 Sequence Diagram

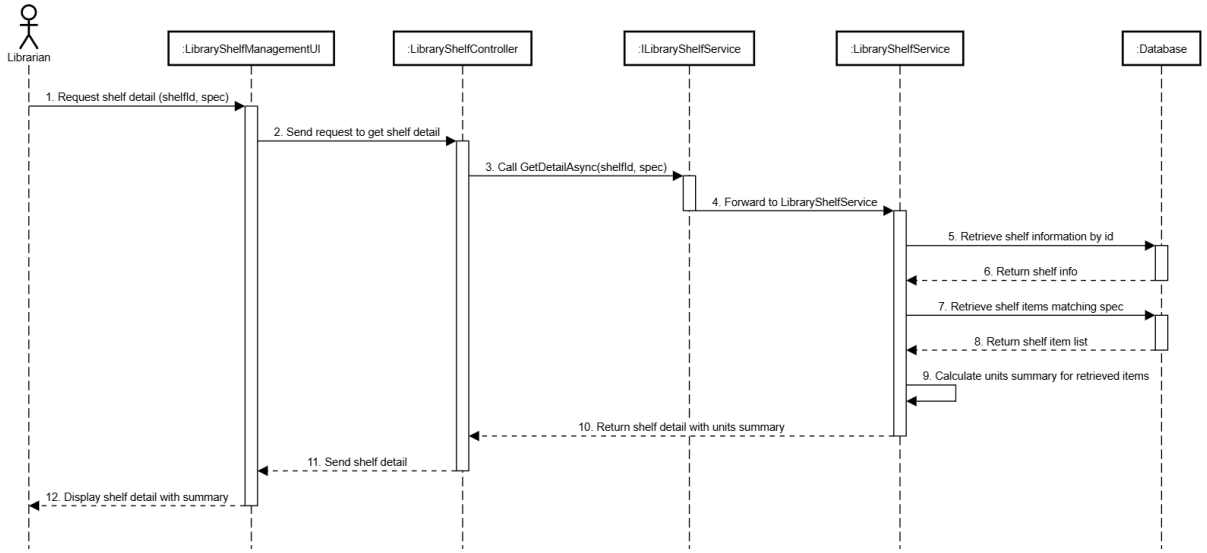


Figure 3.15.2.2 Sequence Diagram

3.15.2.3 Class Specification

3.15.2.3.a Library Shelf Controller

Table 3.15.2.3.a: Library Shelf Controller Description Table

No	Method	Description
1	GetMapShelfDetail(int shelfId, LibraryItemSpecParams specParams)	Receives request to retrieve detailed information about a specific library shelf, including the library items mapped to it.

3.15.2.3.b Library Shelf Service

Table 3.15.2.3.b: Library Shelf Service Description Table

No	Method	Description
1	GetMapShelfDetail(int shelfId, LibraryItemSpecification spec)	Retrieves the shelf that matches the provided shelf ID, applying filtering and sorting criteria from provided specification to include library items.

3.15.3. Get Map by Floor ID

3.15.3.1 Class Diagram

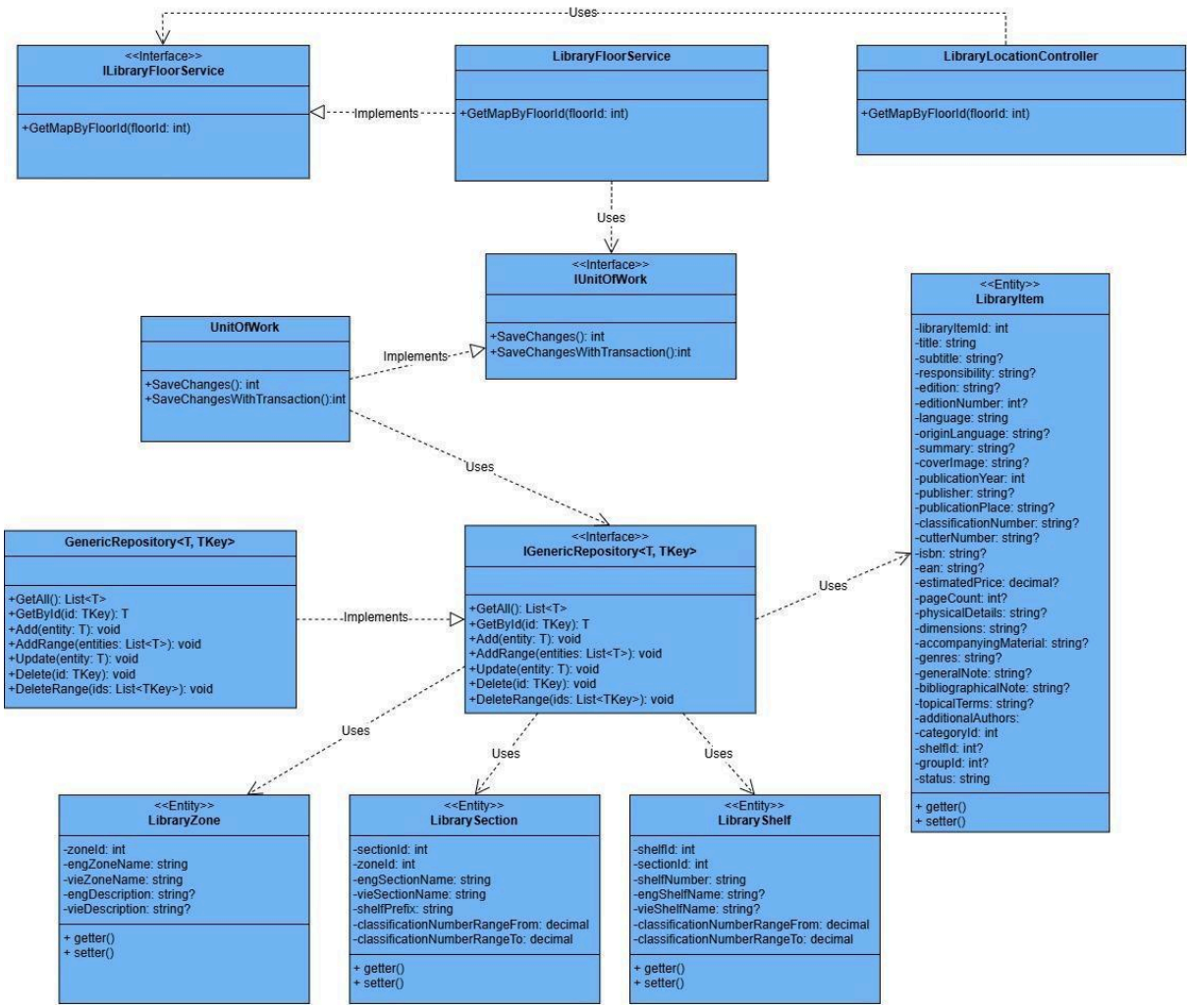


Figure 3.15.3.1 Class Diagram

3.15.3.2 Sequence Diagram

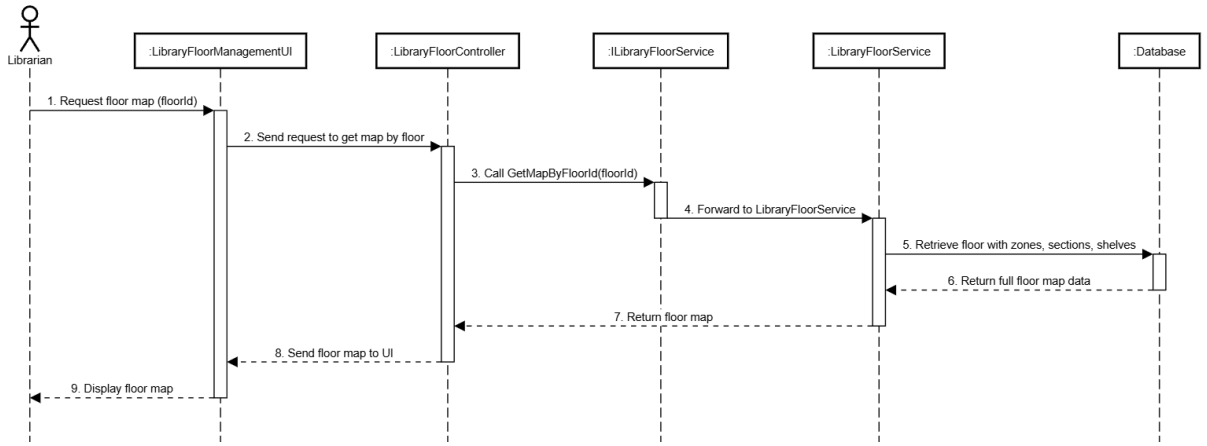


Figure 3.15.3.2 Sequence Diagram

3.15.3.3 Class Specification

3.15.3.3.a Library Shelf Controller

Table 3.15.3.3.a: Library Shelf Controller Description Table

No	Method	Description
1	GetMapByFloorId(int floorId)	Receives request to retrieve a map view of all library shelves located on a specific floor.

3.15.3.3.b Library Shelf Service

Table 3.15.3.3.b: Library Shelf Service Description Table

No	Method	Description
1	GetMapByFloorId(int floorId)	Retrieves structural and mapping data for all library shelves assigned to the given floor ID, including zones and sections.

3.16. Manage Training Session

3.16.1. Get Untrained Items Group

3.16.1.1 Class Diagram

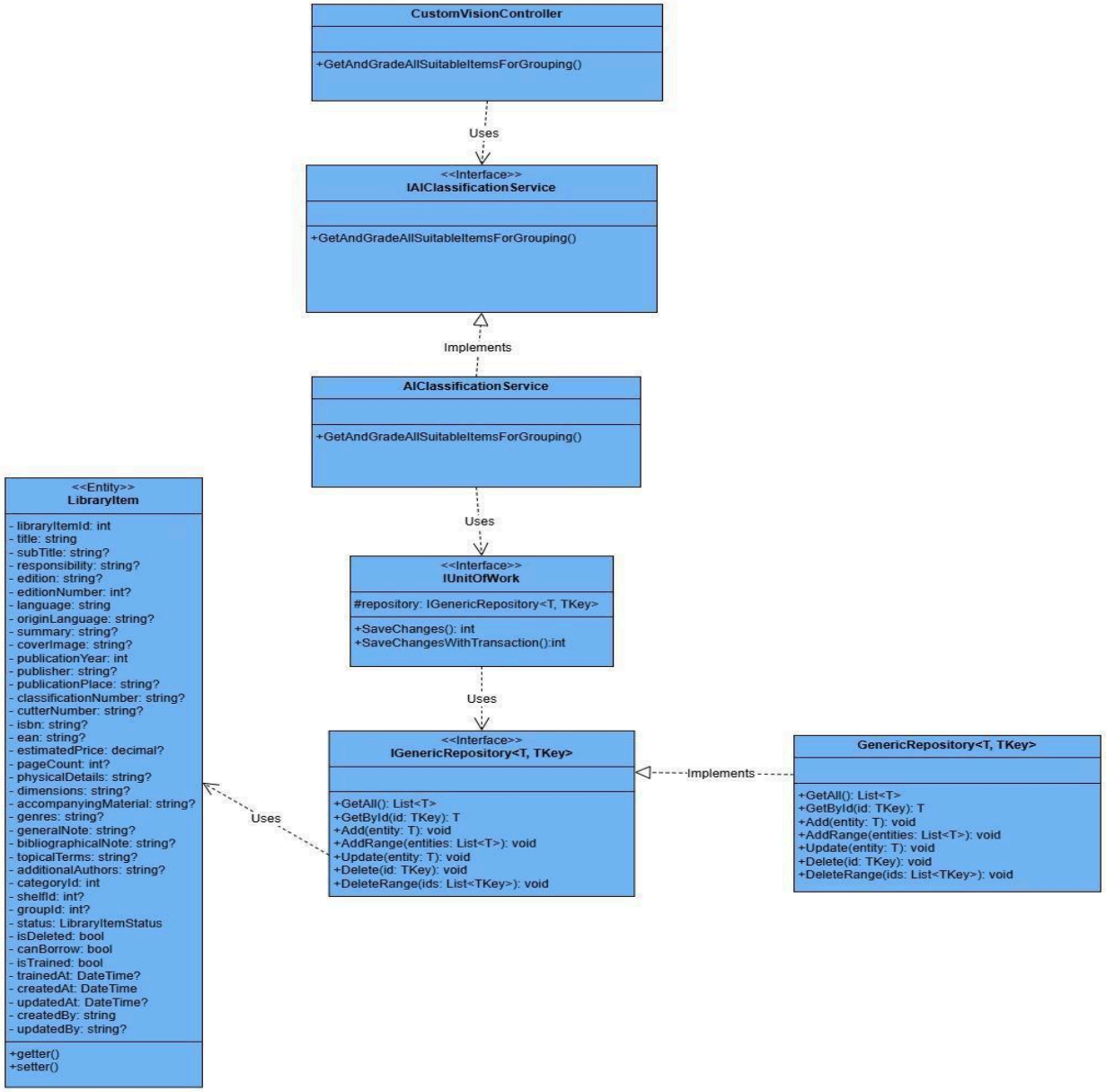


Figure 3.16.1.1 Class Diagram

3.16.1.2 Sequence Diagram

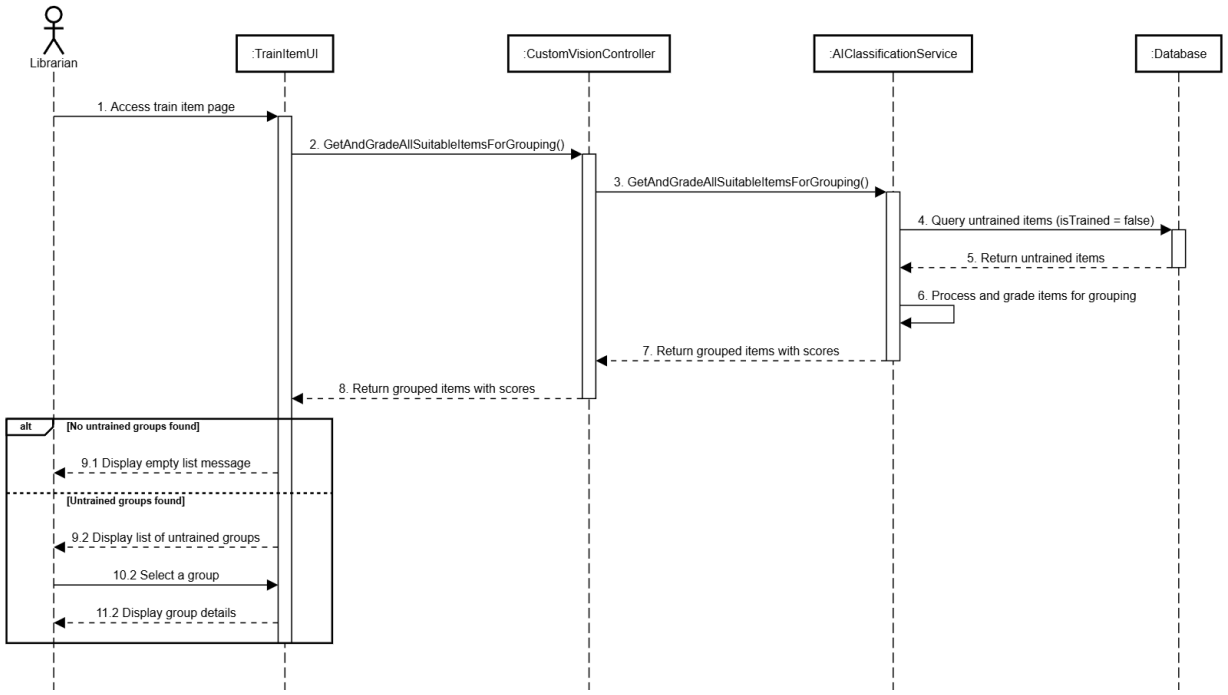


Figure 3.16.1.2 Sequence Diagram

3.16.1.3 Class Specification

3.16.1.3.a Custom Vision Controller

Table 3.16.1.3.a: Custom Vision Controller Description Table

No	Method	Description
1	GetAndGradeAllSuitableGroupForItemForGrouping()	Get group include books that have relationship to train for AI

3.16.1.3.b AIClassification Service

Table 3.16.1.3.b: AI ClassificationService Description Table

No	Method	Description
1	GetAndGradeAllSuitableGroupForItemForGrouping()	Get group include books that have relationship to train for AI

3.16.2. Train Item Group

3.16.2.1 Class Diagram

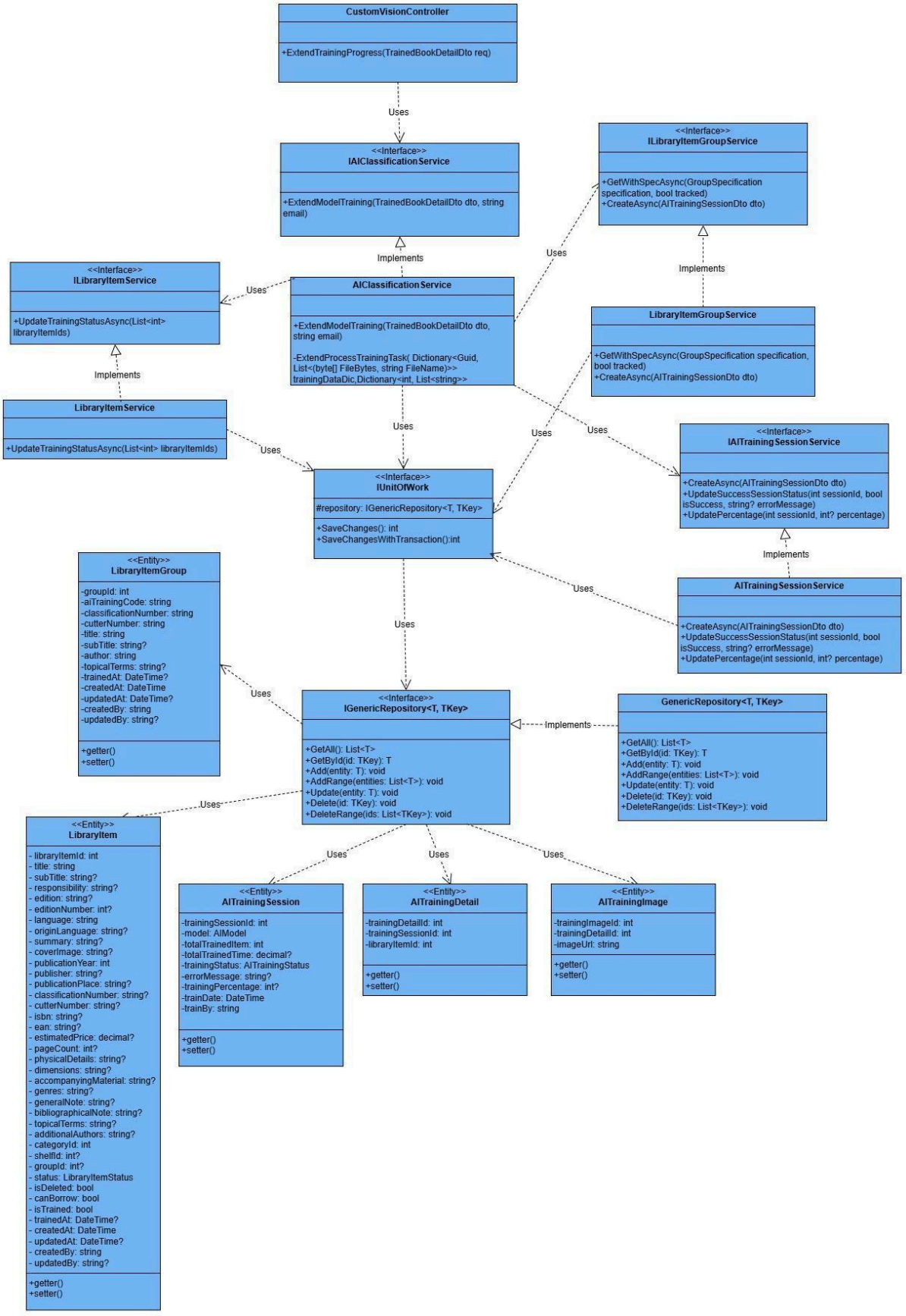


Figure 3.16.2.1 Class Diagram

3.16.2.2 Sequence Diagram

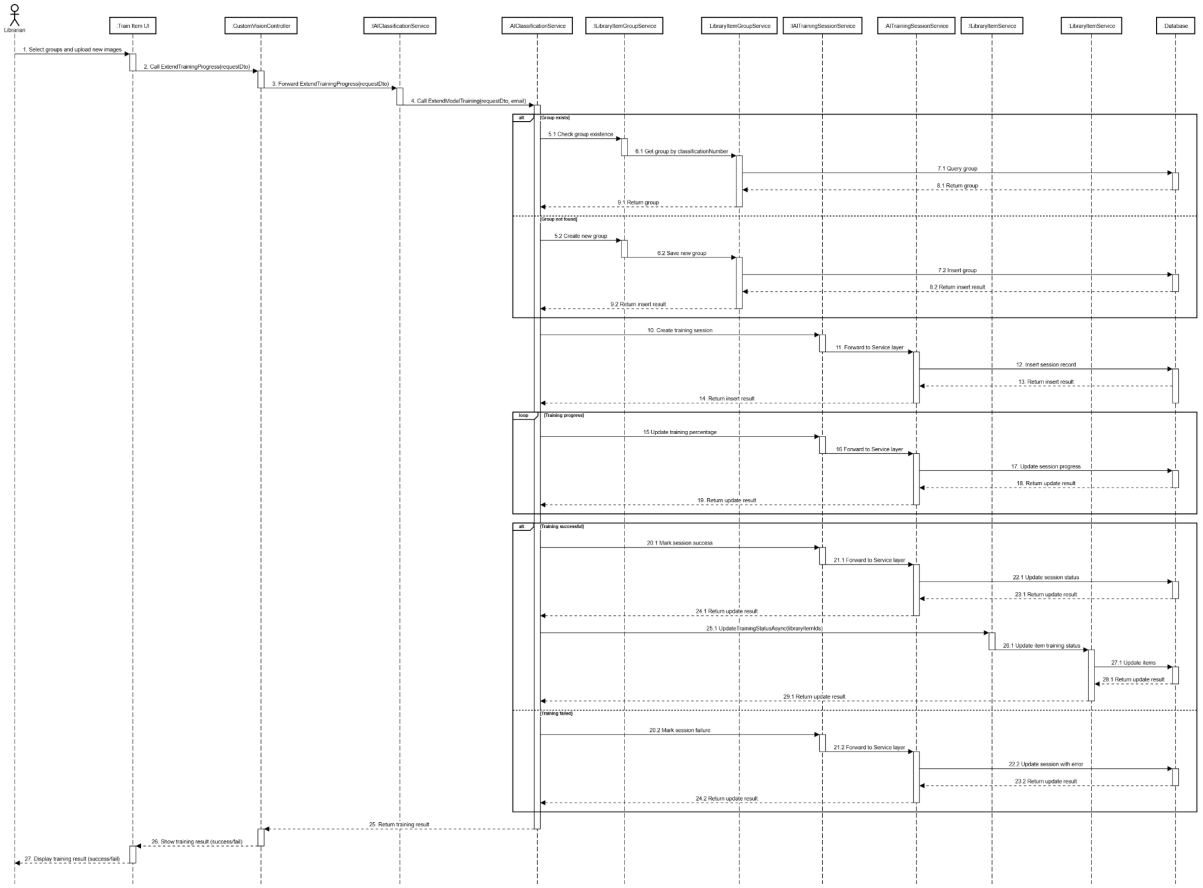


Figure 3.16.2.2 Sequence Diagram

3.16.2.3 Class Specification

3.16.2.3.a Custom Vision Controller

Table 3.16.2.3.a: Custom Vision Controller Description Table

No	Method	Description
1	ExtendTrainingProgress(TrainedBookDetailedDto req)	Training AI for recognising book cover images.

3.16.2.3.b AIClassification Service

Table 3.16.2.3.b: AI ClassificationService Description Table

No	Method	Description
1	ExtendModelTraining(TrainedBookDetailedDto detailDto, string email)	Training AI and creating training history that include information about the training process.

3.16.3. Get All Training Session With Detail

3.16.3.1 Class Diagram

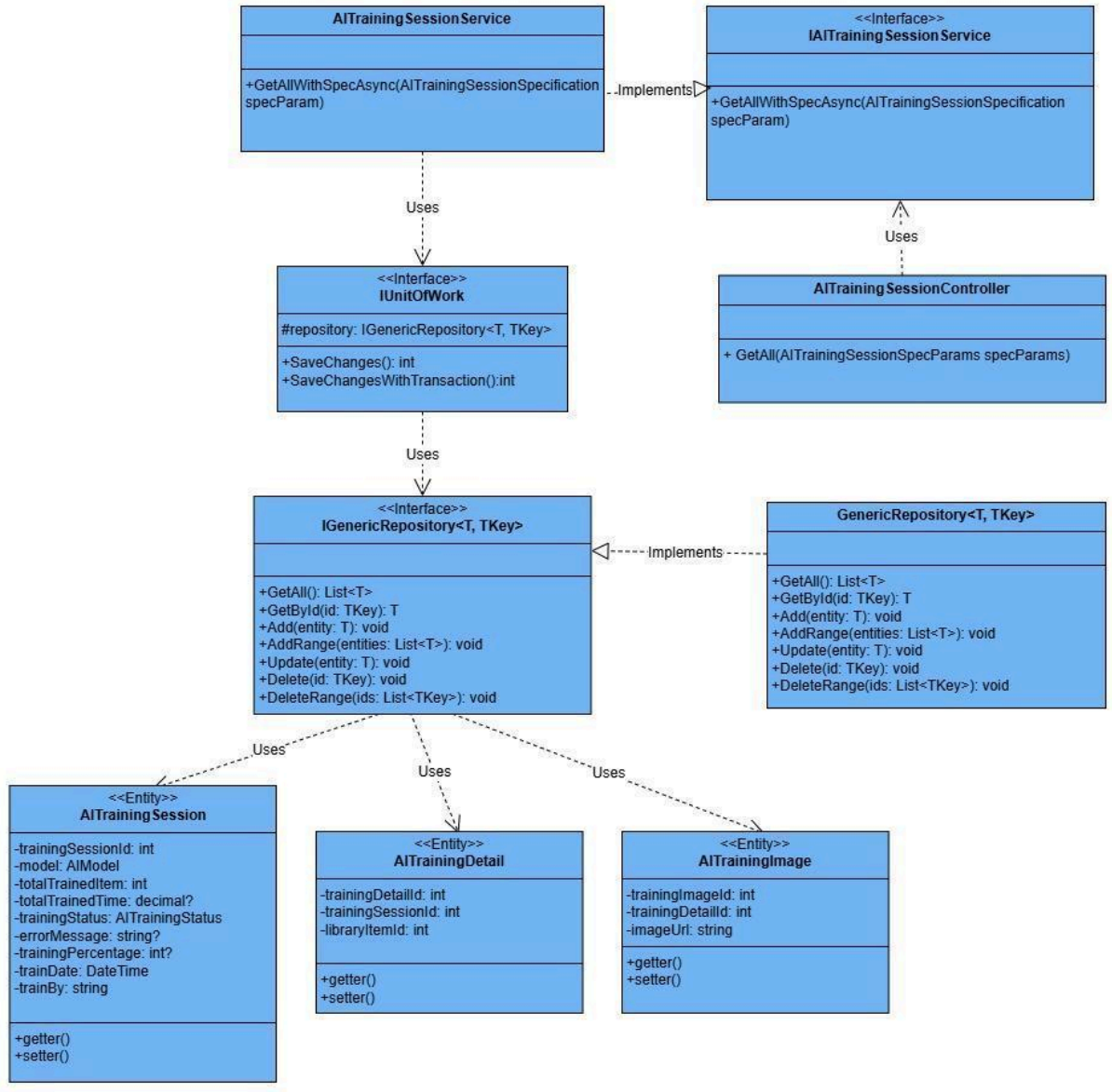


Figure 3.16.3.1 Class Diagram

3.16.3.2 Sequence Diagram

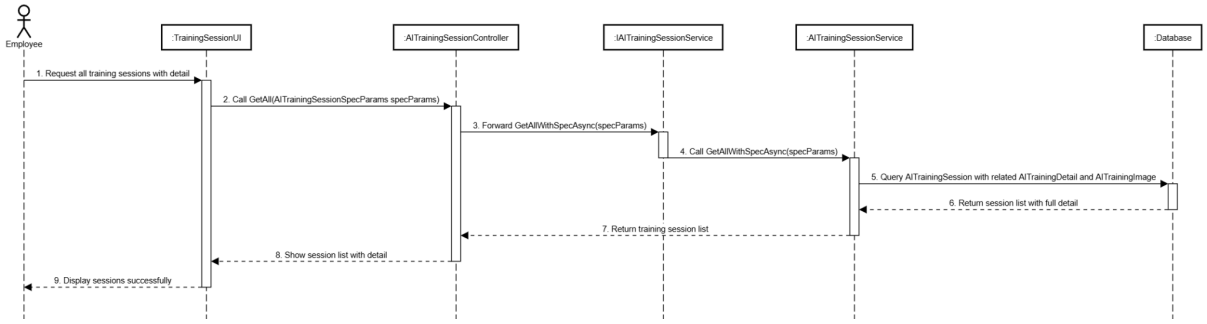


Figure 3.16.3.2 Sequence Diagram

3.16.3.3 Class Specification

3.16.3.3.a Custom Vision Controller

Table 3.16.3.3.a: Custom Vision Controller Description Table

No	Method	Description
1	GetAll(AITrainingSessionSpecParams specParams)	Get all AI training sessions with images that are used to train.

3.16.3.3.b AI Classification Service

Table 3.16.3.3.b: AI ClassificationService Description Table

No	Method	Description
1	GetAllWithSpec(AITrainingSessionSpecification specParam)	Get all AI training sessions with images that are used to train.

3.17. Manage Library Cards

3.17.1. Register Library Card

3.17.1.1 Class Diagram

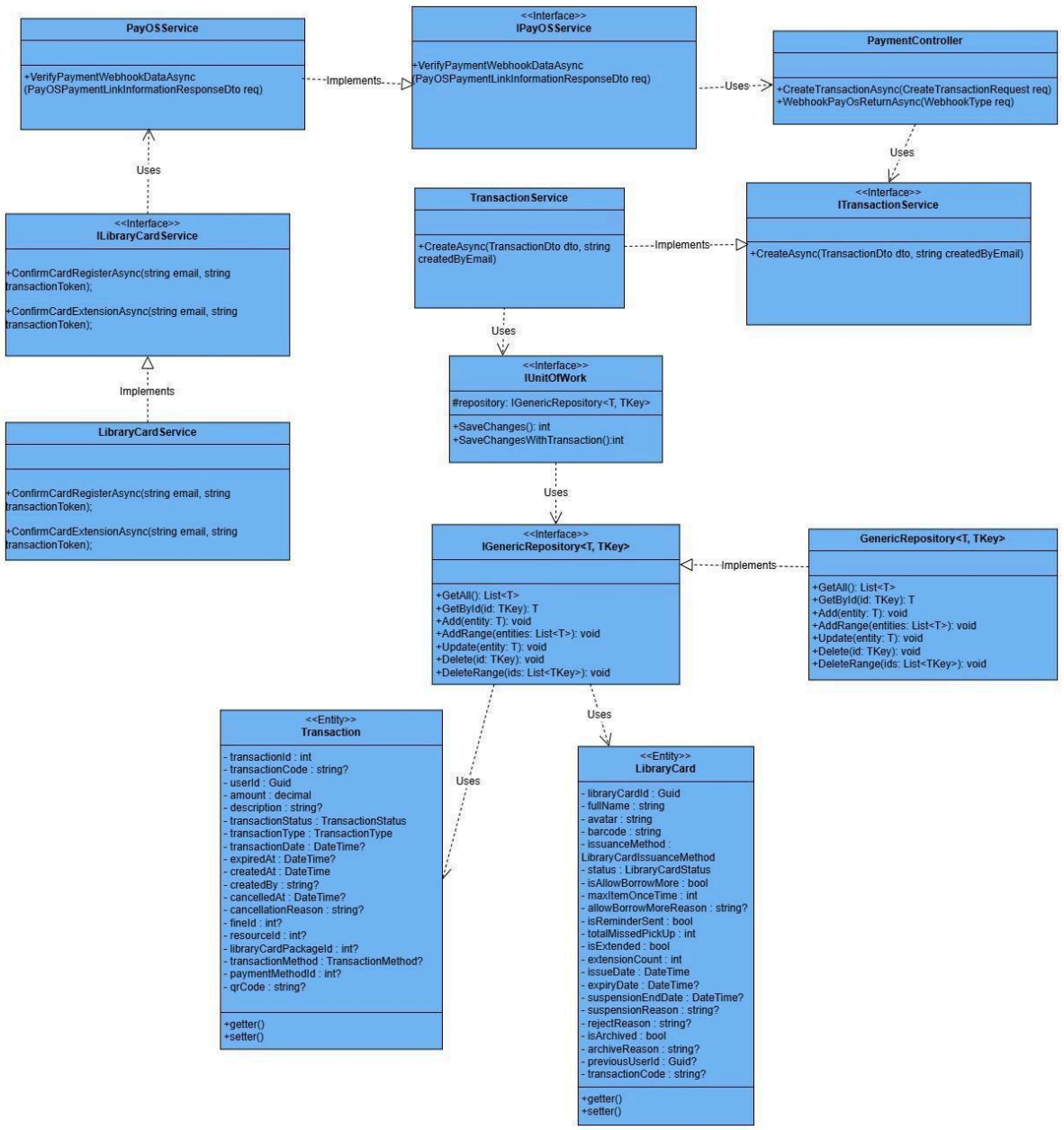


Figure 3.17.1.1 Class Diagram

3.17.1.2 Sequence Diagram

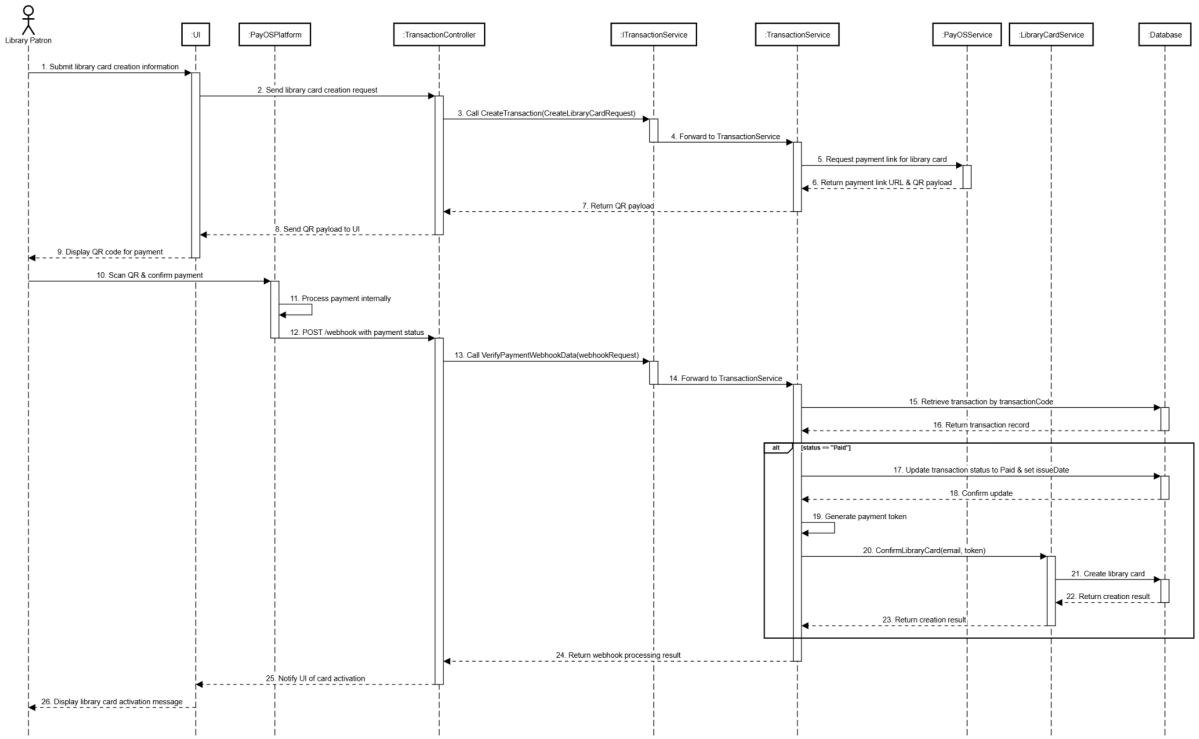


Figure 3.17.1.2 Sequence Diagram

3.17.1.3 Class Specification

3.17.1.3.a Library Card Controller

Table 3.17.1.3.a: Library Card Controller Description Table

No	Method	Description
1	CreateTransactionAsync(CreateTransactionRequest req)	Receives a request to initiate a payment transaction. Forwards the data to the service layer to create a payment link and record the transaction.
2	VerifyPayOsReturnAsync(WebhookType req)	Accepts webhook callback data from PayOS after a payment is completed.

3.17.1.3.b Library Card Service

Table 3.17.1.3.b: Library Card Description Table

No	Method	Description
1	CreateAsync(TransactionDto transactionDto, string createdByEmail)	Creates a PayOS payment link and stores a new transaction with pending status in the database, awaiting user payment.

2	VerifyPaymentWebhookDataAsync(PayOSPaymentLinkInformationResponse res)	Validates PayOS webhook data. If the data is correct, update the transaction status..

3.17.2. Confirm Card Creation

3.17.2.1 Class Diagram

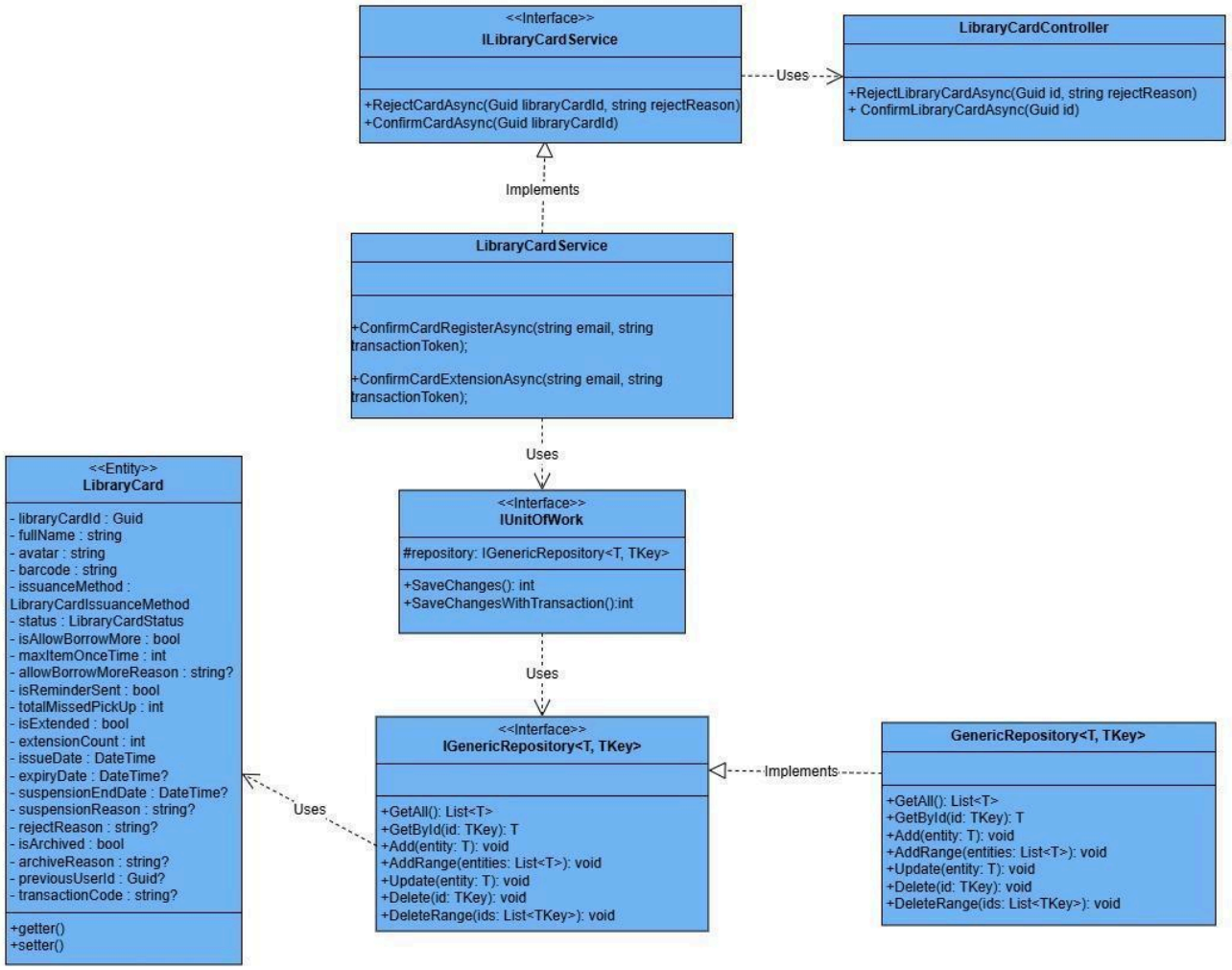


Figure 3.17.2.1 Class Diagram

3.17.2.2 Sequence Diagram

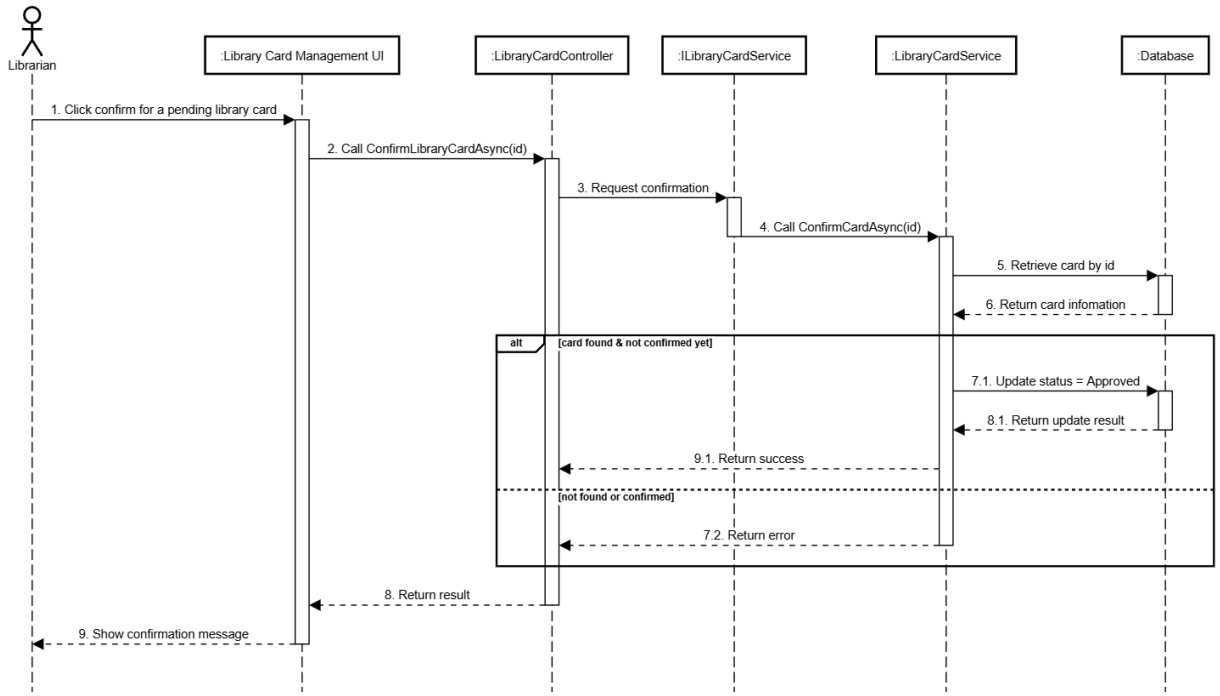


Figure 3.17.2.2 Sequence Diagram

3.17.2.3 Class Specification

3.17.2.3.a Library Card Controller

Table 3.17.2.3.a: Library Card Controller Description Table

No	Method	Description
1	ConfirmCardAsync(Guid libraryCardId)	Receives request to confirm a library card after payment.

3.17.2.3.b Library Card Service

Table 3.17.2.3.b: Library Card Description Table

No	Method	Description
1	ConfirmCardAsync(Guid libraryCardId)	Confirms and activates the library card with the specified ID.

3.17.3. Get All Library Card

3.17.3.1 Class Diagram

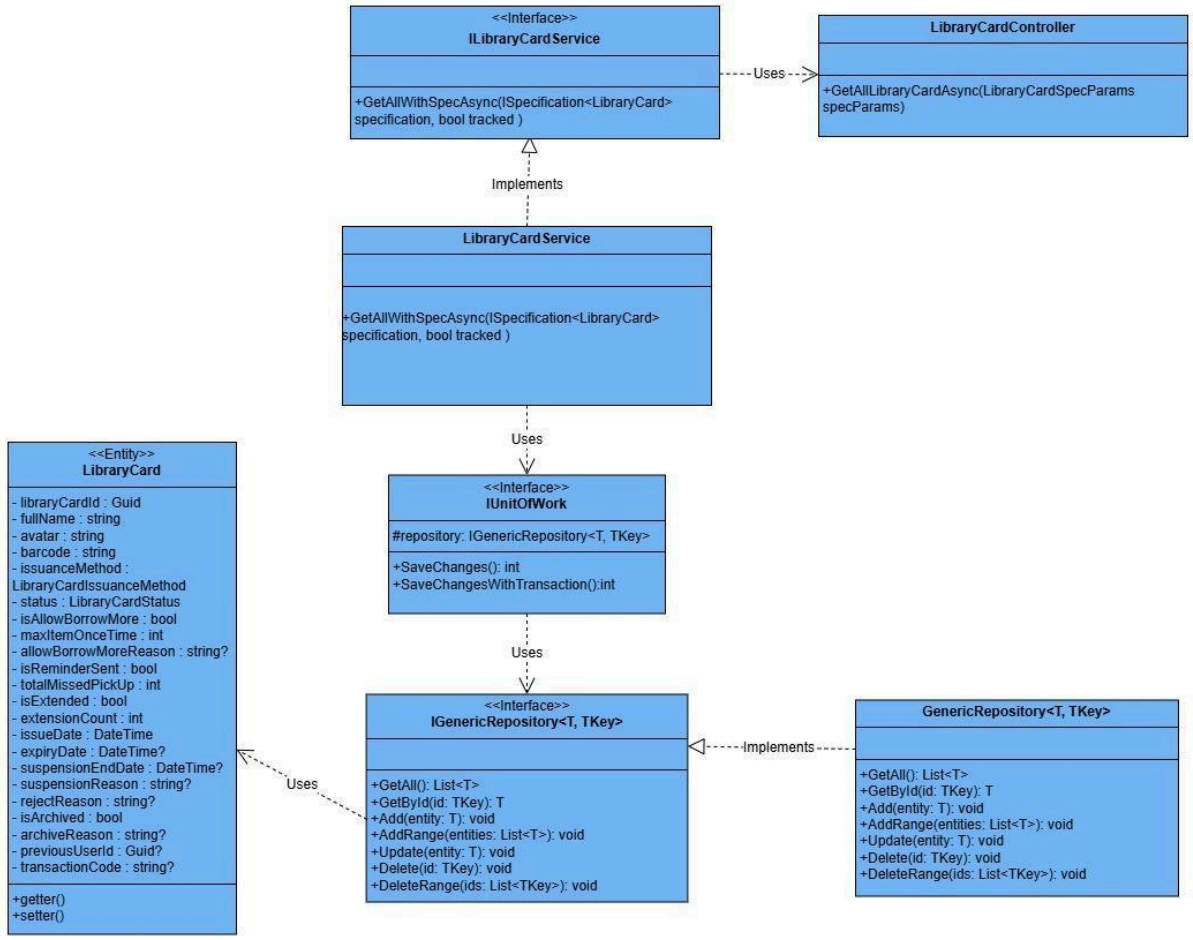


Figure 3.17.3.1 Class Diagram

3.17.3.2 Sequence Diagram

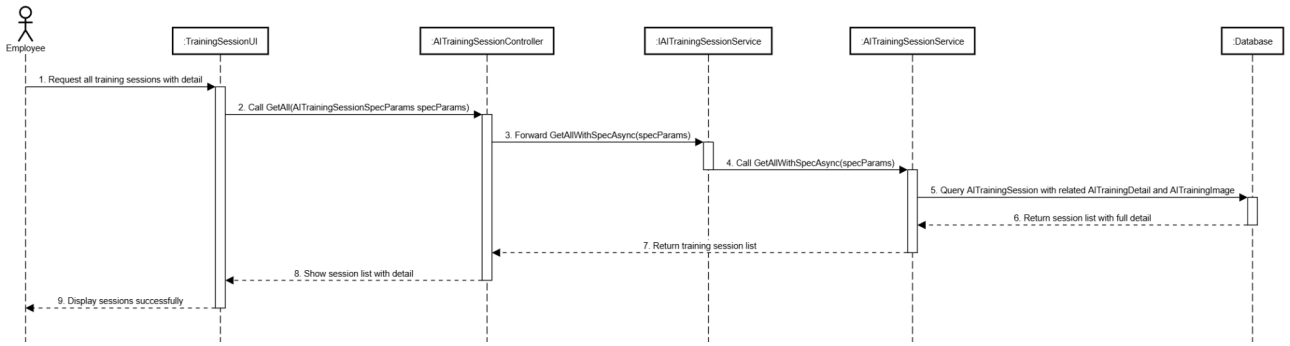


Figure 3.17.3.2 Sequence Diagram

3.17.3.3 Class Specification

3.17.3.3.a Library Card Controller

Table 3.17.3.3.a: Library Card Controller Description Table

No	Method	Description
1	GetAllLibraryCardAsync(LibraryCardSpecParams specParams)	Receives a request with filtering, sorting and pagination parameters.

3.17.3.3.b Library Card Service

Table 3.17.3.3.b: Library Card Description Table

No	Method	Description
1	GetAllWithSpec(LibraryCardSpecification spec)	Applies filtering, sorting, and pagination logic to retrieve all existing library cards.

3.18. Transactions

3.18.1. Create Transaction

3.18.1.1 Class Diagram

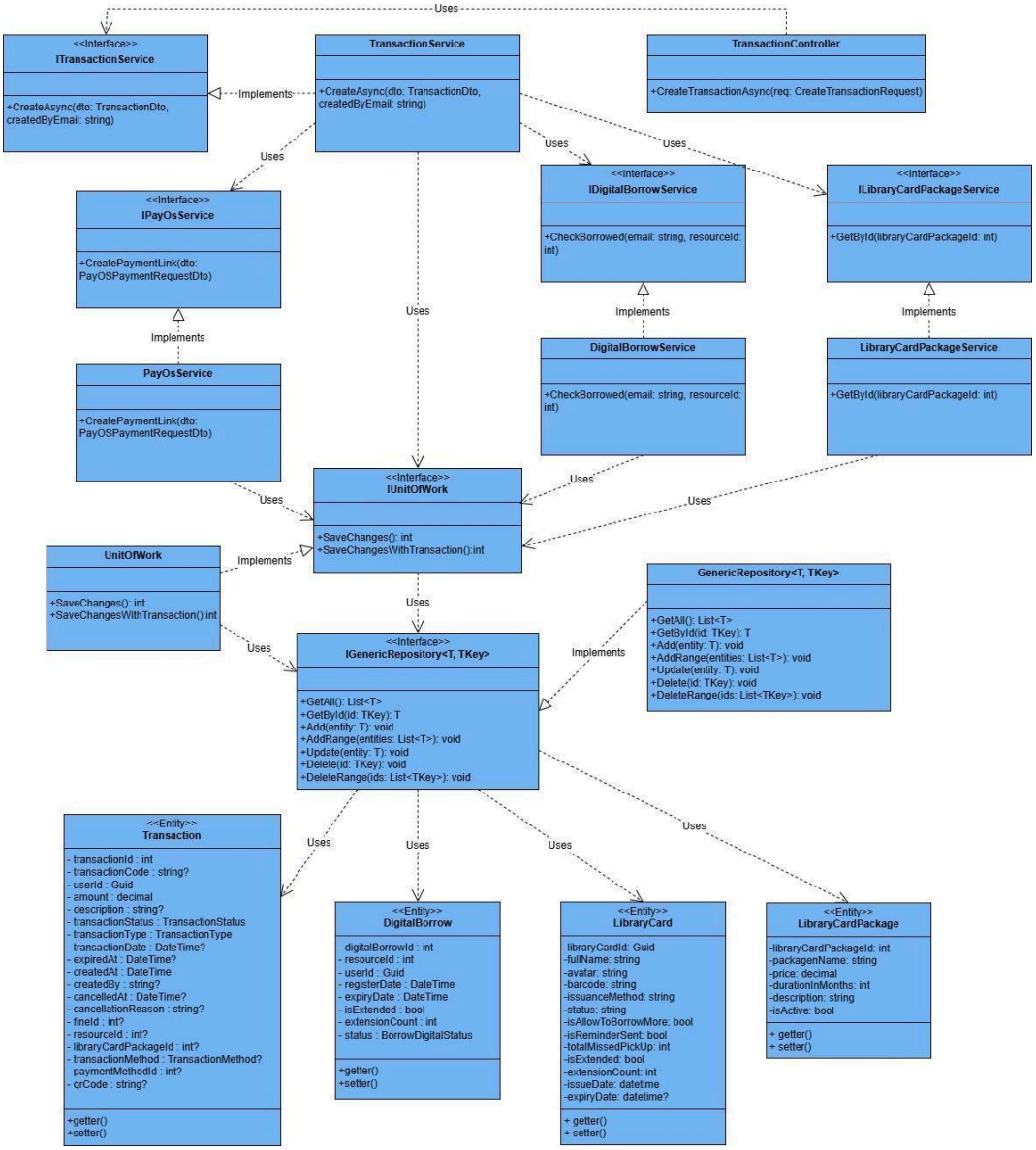


Figure 3.18.1.1 Class Diagram

3.18.1.2 Sequence Diagram

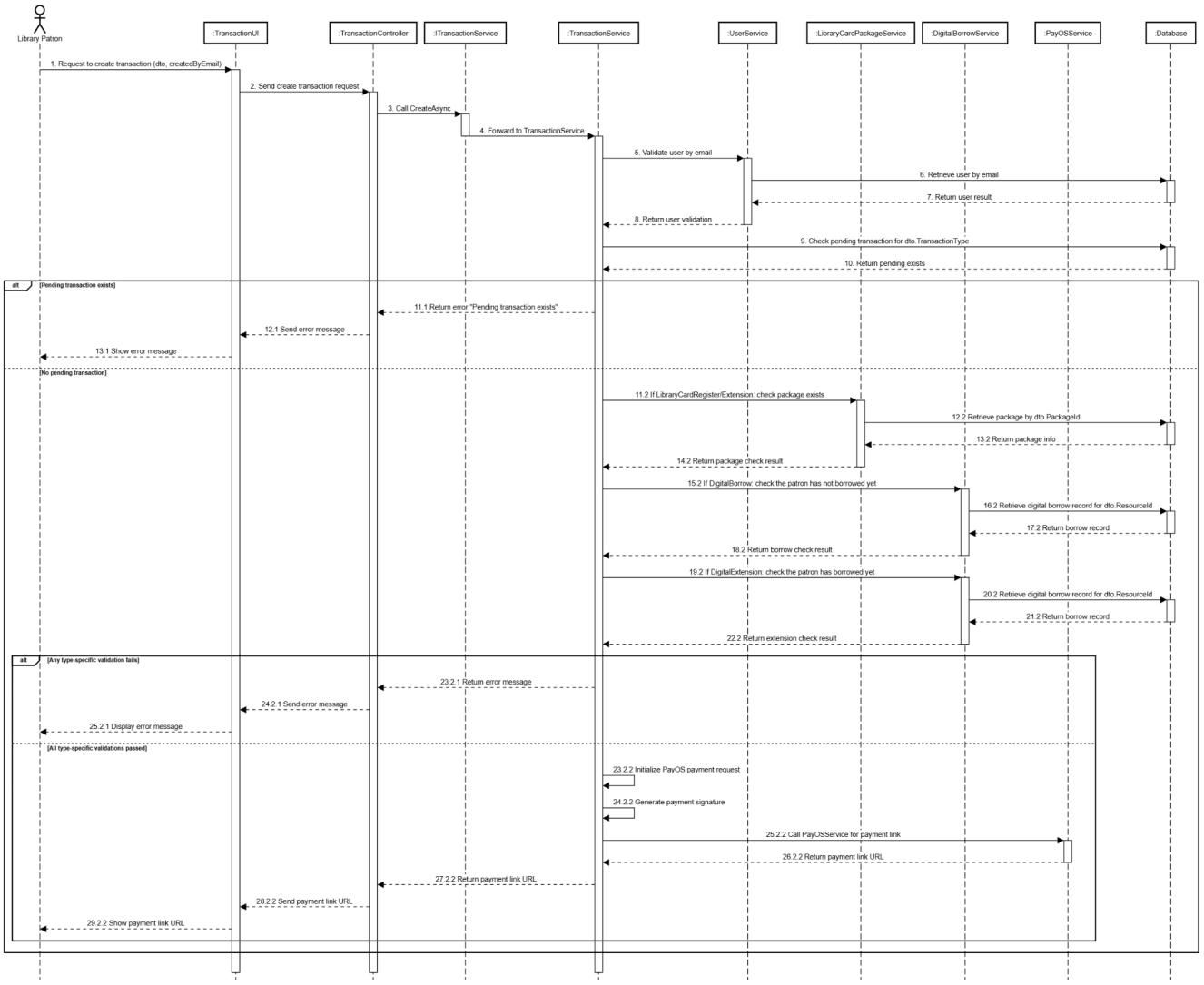


Figure 3.18.1.2 Sequence Diagram

3.18.1.3 Class Specification

3.18.1.3.a Transaction Controller

Table 3.18.1.3.a: Transaction Controller Description Table

No	Method	Description
1	CreateTransactionAsync(CreateTransactionRequest req)	Receives a request to initiate a payment transaction. Forwards the data to the service layer to create a payment link and record the transaction.

3.18.1.3.b Transaction Service

Table 3.18.1.3.b: Transaction Service Description Table

No	Method	Description
1	CreateAsync(TransactionDto transactionDto, string createdByEmail)	Creates a PayOS payment link and stores a new transaction with pending status in the database, awaiting user payment.

3.18.2. Create Fine Payment

3.18.2.1 Class Diagram

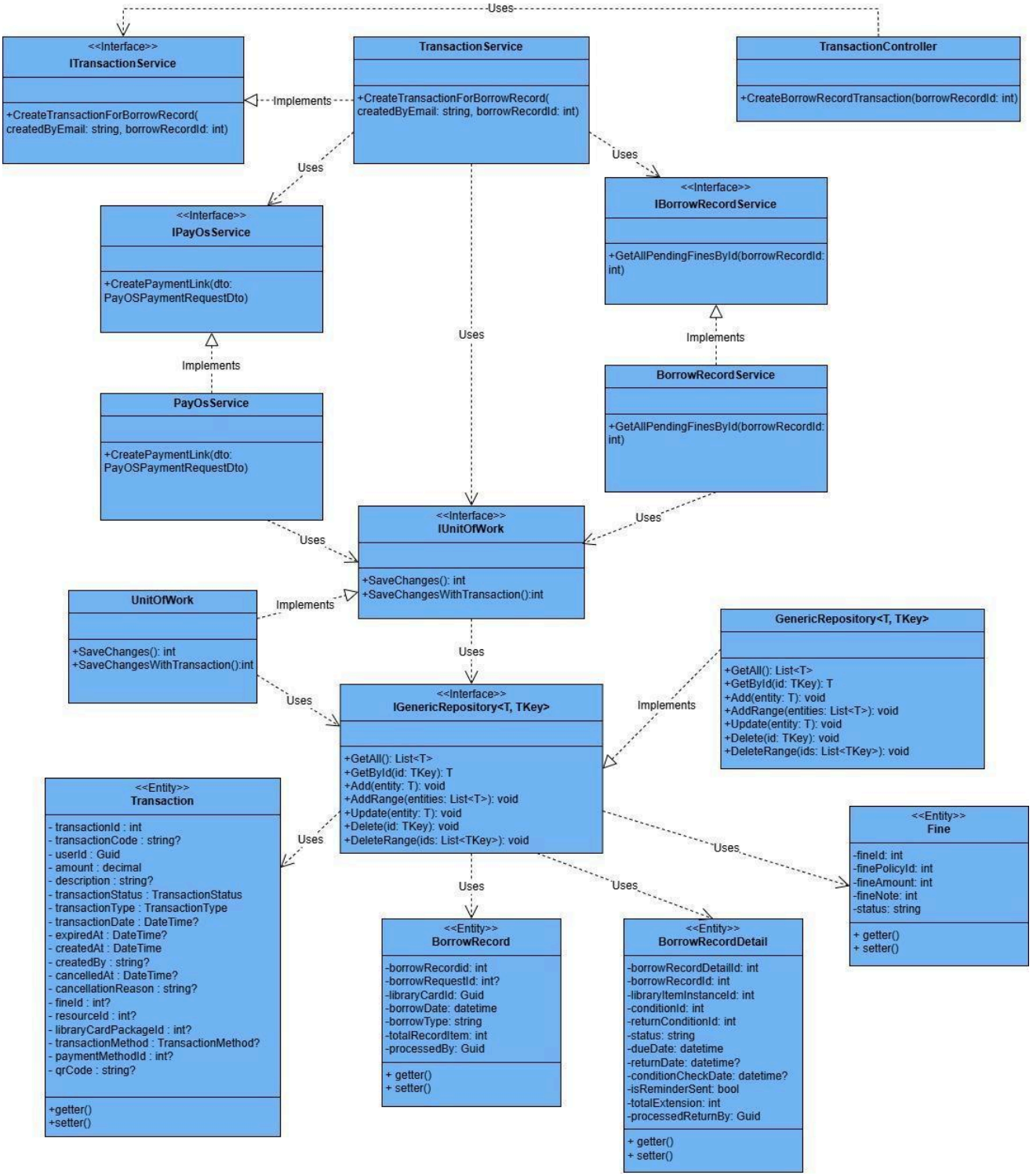


Figure 3.18.2.1 Class Diagram

3.18.2.2 Sequence Diagram

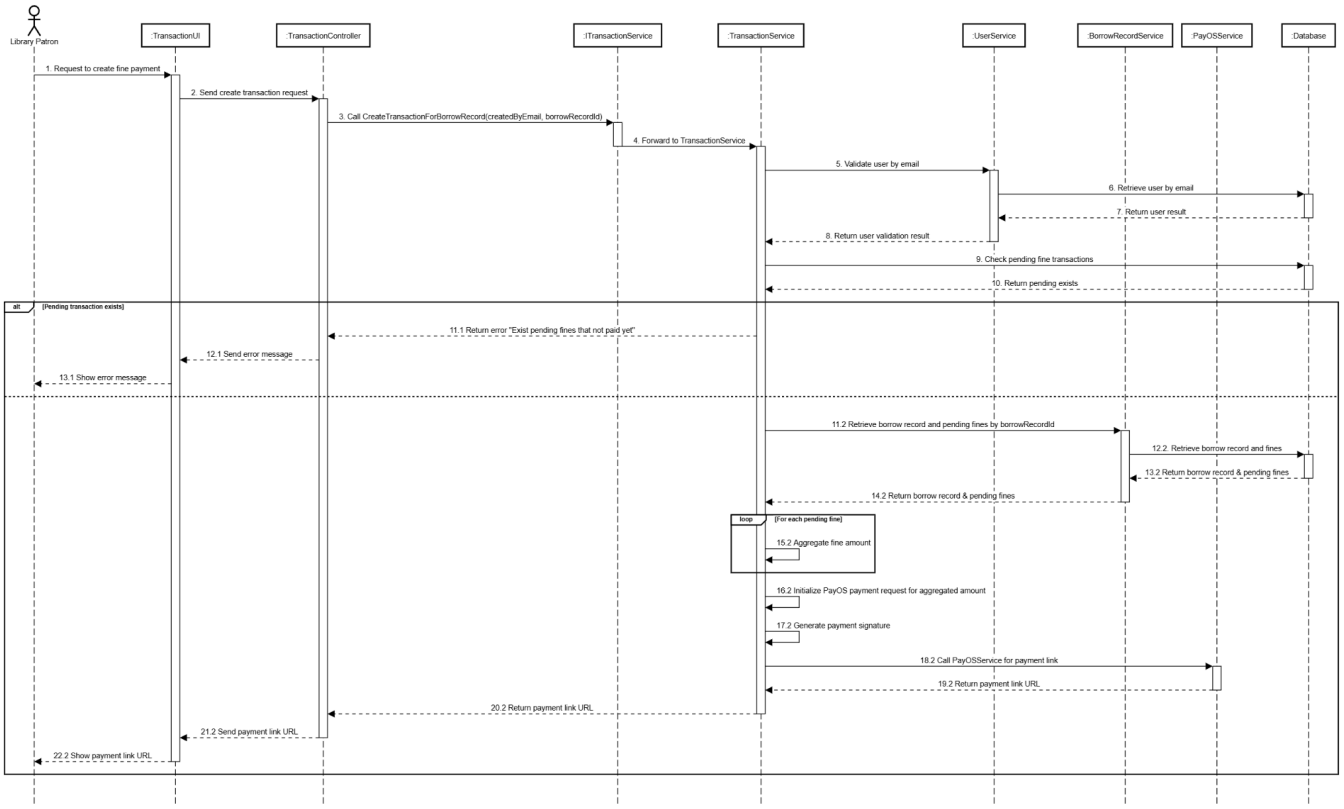


Figure 3.18.2.2 Sequence Diagram

3.18.2.3 Class Specification

3.18.2.3.a Transaction Controller

Table 3.18.2.3.a: Transaction Controller Description Table

No	Method	Description
1	CreateBorrowRecordTransaction(int borrowRecordId)	Receives a request to create a transaction related to borrow record's fines.

3.18.2.3.b Transaction Service

Table 3.18.2.3.b: Transaction Service Description Table

No	Method	Description
1	CreateTransactionForBorrowRecord(string createdByEmail, int borrowRecordId)	Creates a PayOS transaction linked to the specified borrow record. Generates a payment link and creates a new pending transaction.

3.18.3. Pay for Transaction

3.18.3.1 Class Diagram

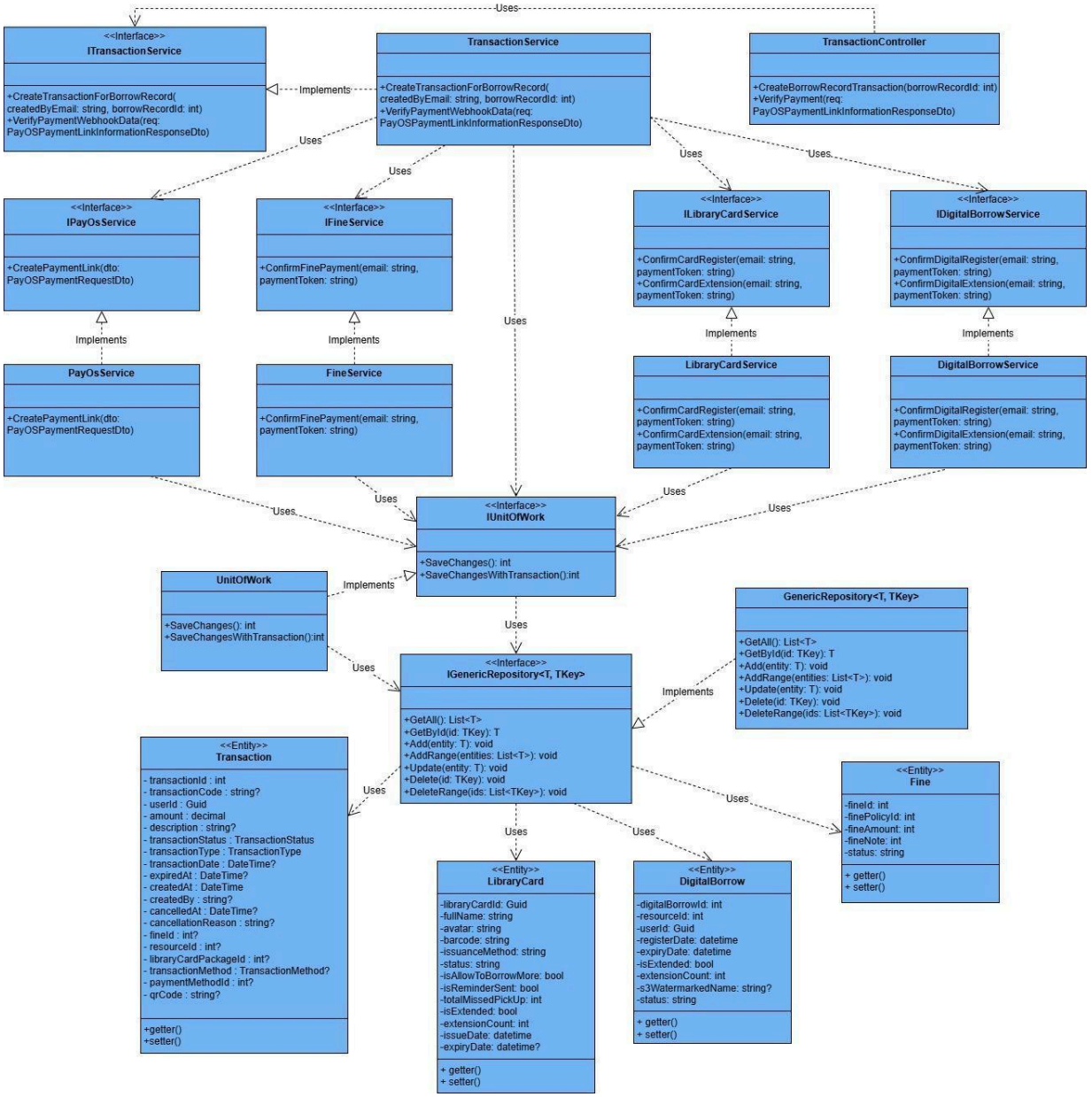


Figure 3.18.3.1 Class Diagram

3.18.3.2 Sequence Diagram

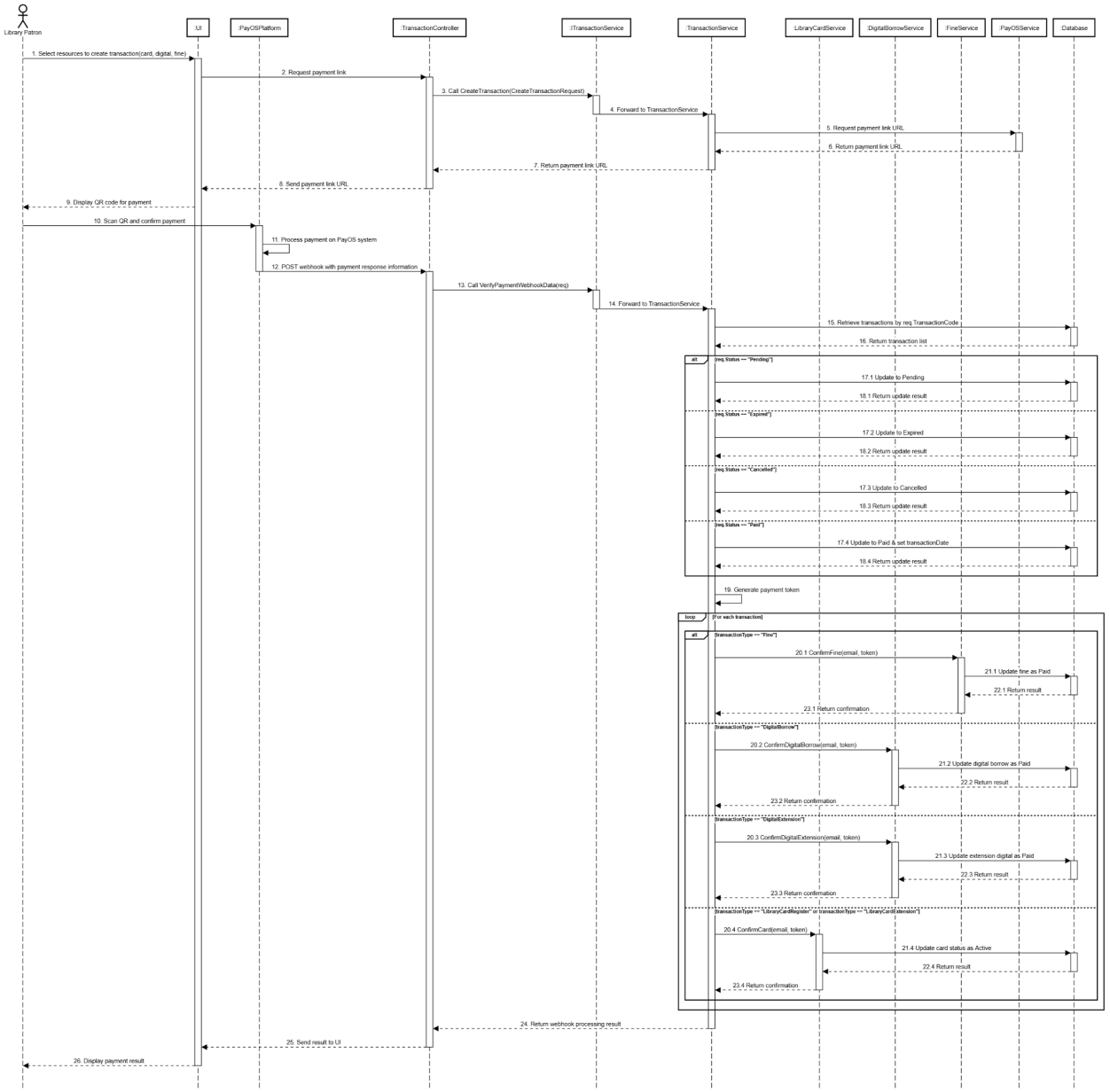


Figure 3.18.3.2 Sequence Diagram

3.18.3.3 Class Specification

3.18.3.3.a Transaction Controller

Table 3.18.3.3.a: Transaction Controller Description Table

No	Method	Description
1	CreateBorrowRecordTransaction(int borrowRecordId)	Receives a request to create a transaction related to borrow record's fines.
2	VerifyPayment(PayOSPaymentLinkInformationResponse req)	Accepts webhook callback data from PayOS after a payment is completed.

3.18.3.3.b Transaction Service

Table 3.18.3.3.b: Transaction Service Description Table

No	Method	Description
1	CreateTransactionForBorrowRecord(string createdByEmail, int borrowRecordId)	Creates a PayOS transaction linked to the specified borrow record. Generates a payment link and creates a new pending transaction.
2	VerifyPaymentWebhookData(PayOSPaymentLinkInformationResponse res)	Validates PayOS webhook data. If the data is correct, update the transaction status.

3.19. Manage Fine Policies

3.19.1. Create Fine Policy

3.19.1.1 Class Diagram

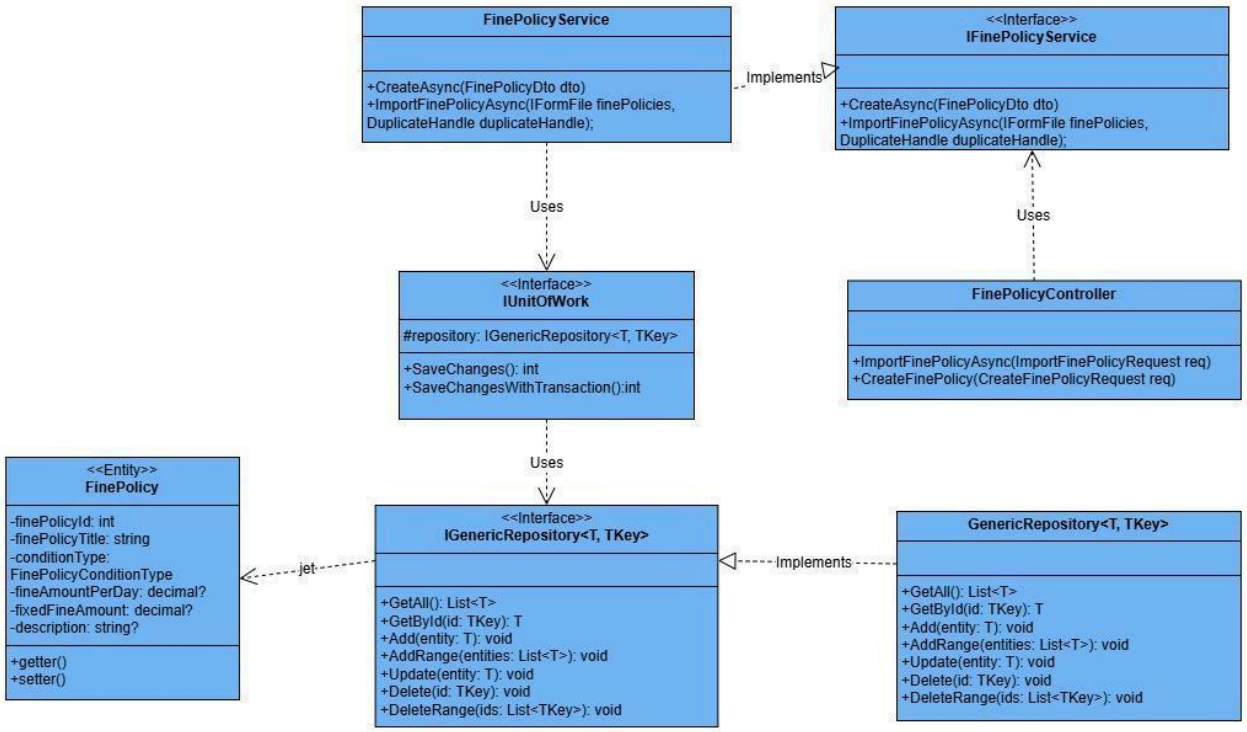


Figure 3.19.1.1 Class Diagram

3.19.1.2 Sequence Diagram

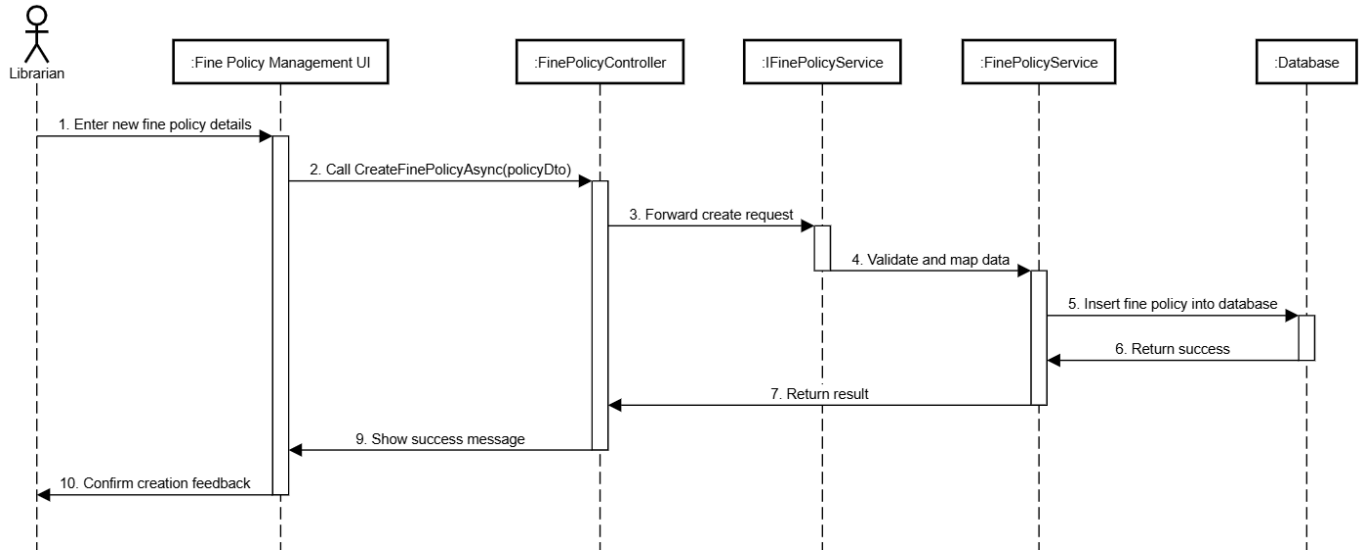


Figure 3.19.1.2 Sequence Diagram

3.19.1.3 Class Specification

3.19.1.3.a Fine Policy Controller

Table 3.19.1.3.a: Fine Policy Controller Description Table

No	Method	Description
1	CreateFinePolicy(CreateFinePolicyRequest req)	Create a new fine policy with required information.
2	ImportFinePolicy(ImportFine Policy Request req)	Create Fine Policies by importing an excel file that contains fine policy's information.

3.19.1.3.b Fine Policy Service

Table 3.19.1.3.b: Fine Policy Service Description Table

No	Method	Description
1	CreateFinePolicy(FinePolicyDto req)	Create a single fine policy with given information.
2	ImportFinePolicy(IFormFile file,DuplicateHandle duplicateHandle,string[]? ScanningField)	Handle import fine policies via excel file.

3.19.2. Delete Fine Policy

3.19.2.1 Class Diagram

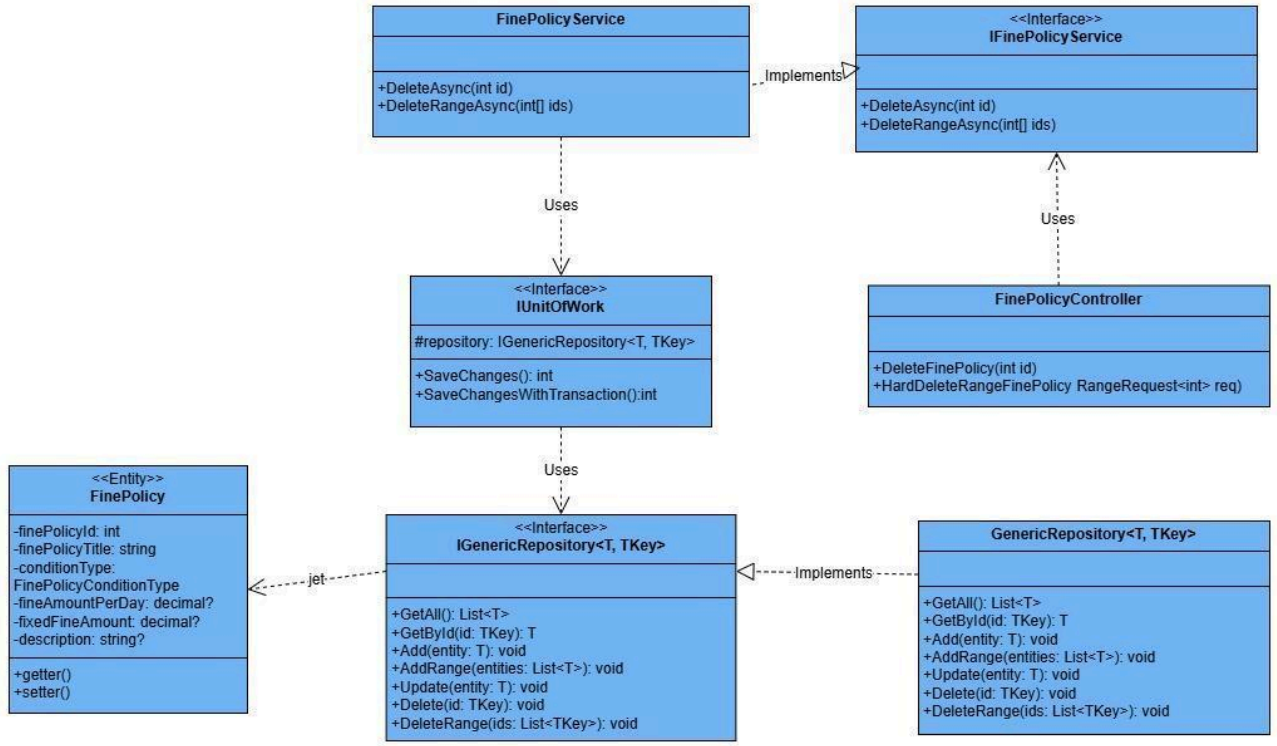


Figure 3.19.2.1 Class Diagram

3.19.2.2 Sequence Diagram

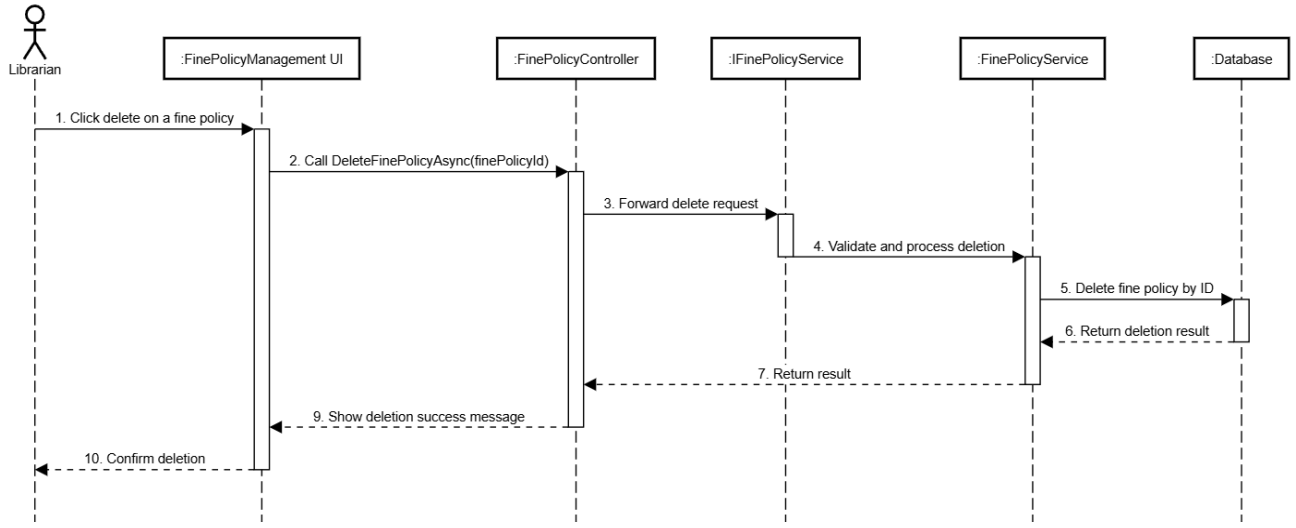


Figure 3.19.2.2 Sequence Diagram

3.19.2.3 Class Specification

3.19.2.3.a Fine Policy Controller

Table 3.19.2.3.a: Fine Policy Controller Description Table

No	Method	Description
1	DeleteFinePolicy(int id)	Choose a Fine Policy that is not used any more to delete.
2	HardDeleteRangeFinePolicy(RangeRequest <int> req)	Choose many Fine Policies that are not used to delete.

3.19.2.3.b Fine Policy Service

Table 3.19.2.3.b: Fine Policy Service Description Table

No	Method	Description
1	DeleteFinePolicy(int id)	Remove Fine Policy and it could not be recovered after deletion.
2	HardDeleteRangeFinePolicy(RangeRequest <int> req)	Remove Fine Policies from the database and could not be recovered after deleted

3.19.3. Get Fine Policy

3.19.3.1 Class Diagram

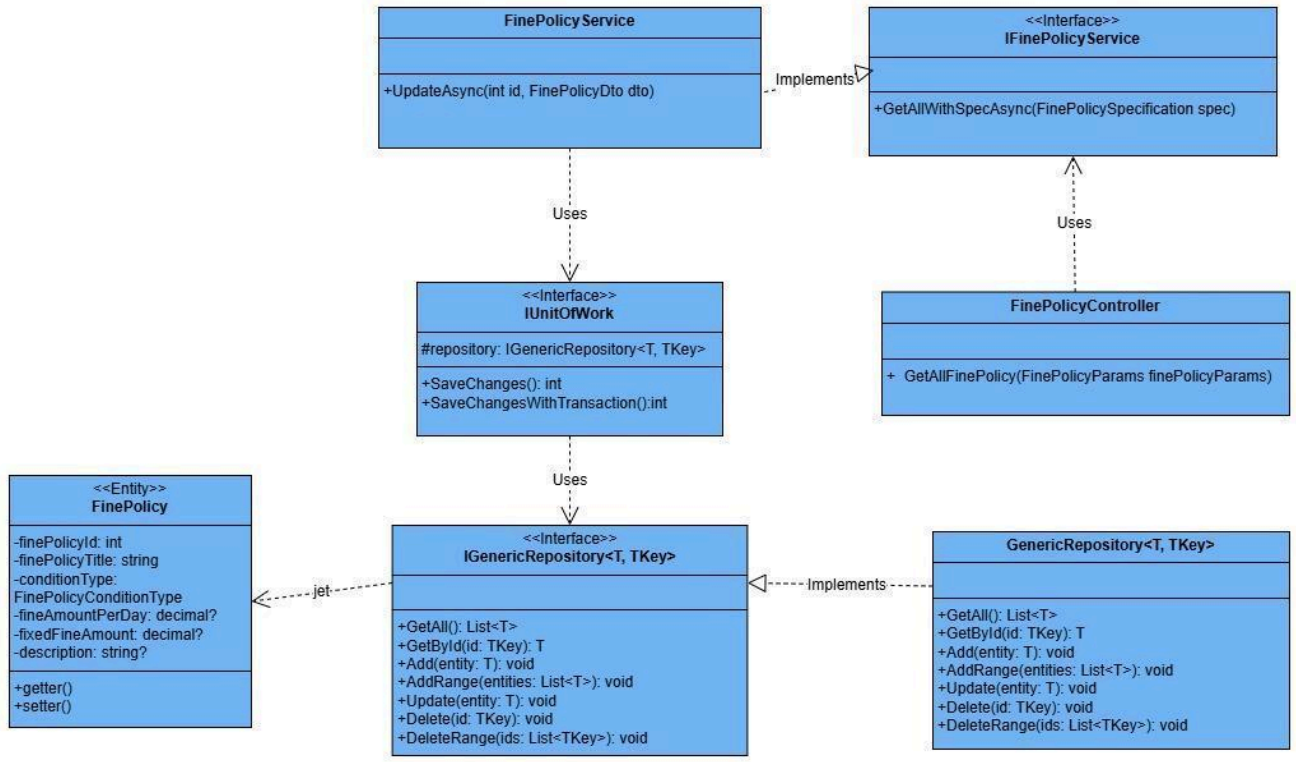


Figure 3.19.3.1 Class Diagram

3.19.3.2 Sequence Diagram

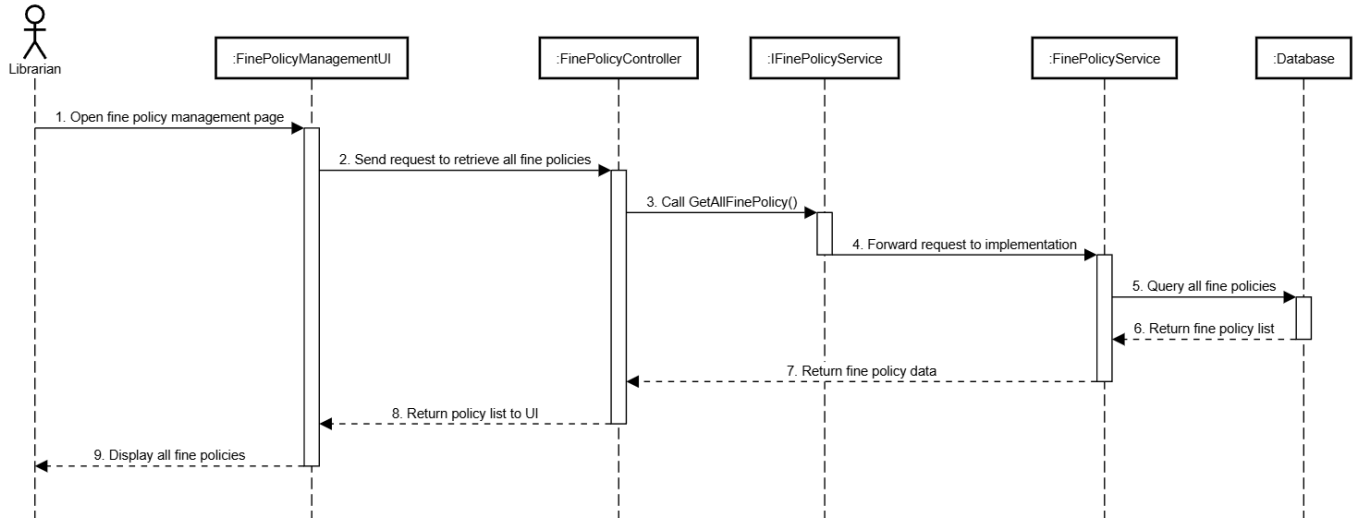


Figure 3.19.3.2 Sequence Diagram

3.16.3.3 Class Specification

3.16.3.3.a Fine Policy Controller

Table 3.16.3.3.a: Fine Policy Controller Description Table

No	Method	Description

1	GetAllFinePolicy(FinePolicyParams specParams)	Get all fine policies that are used in the system.

3.16.3.3.b Fine Policy Service

Table 3.16.3.3.b: Fine Policy Service Description Table

No	Method	Description
1	GetAllWithSpec(FinePolicySpecification specParam)	Get all fine policies that are used in the system.

3.19.4. Update Fine Policy

3.19.4.1 Class Diagram

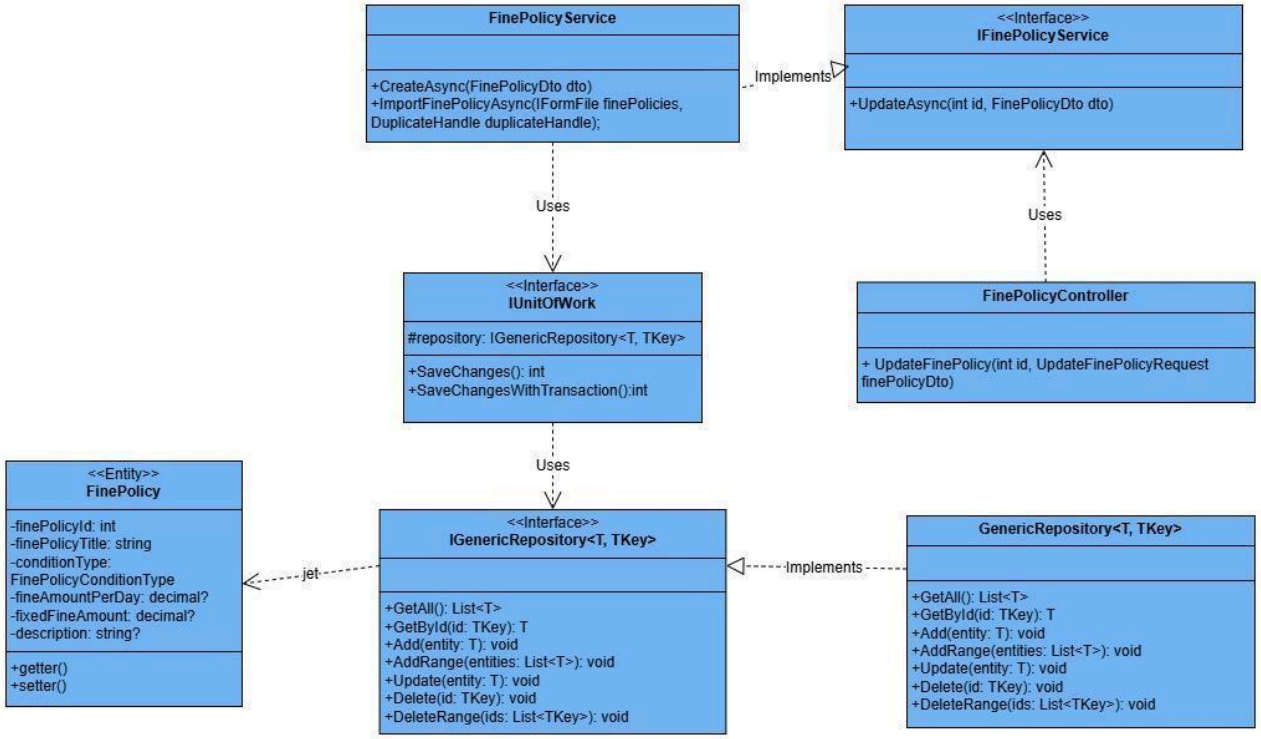


Figure 3.19.4.1 Class Diagram

3.19.4.2 Sequence Diagram

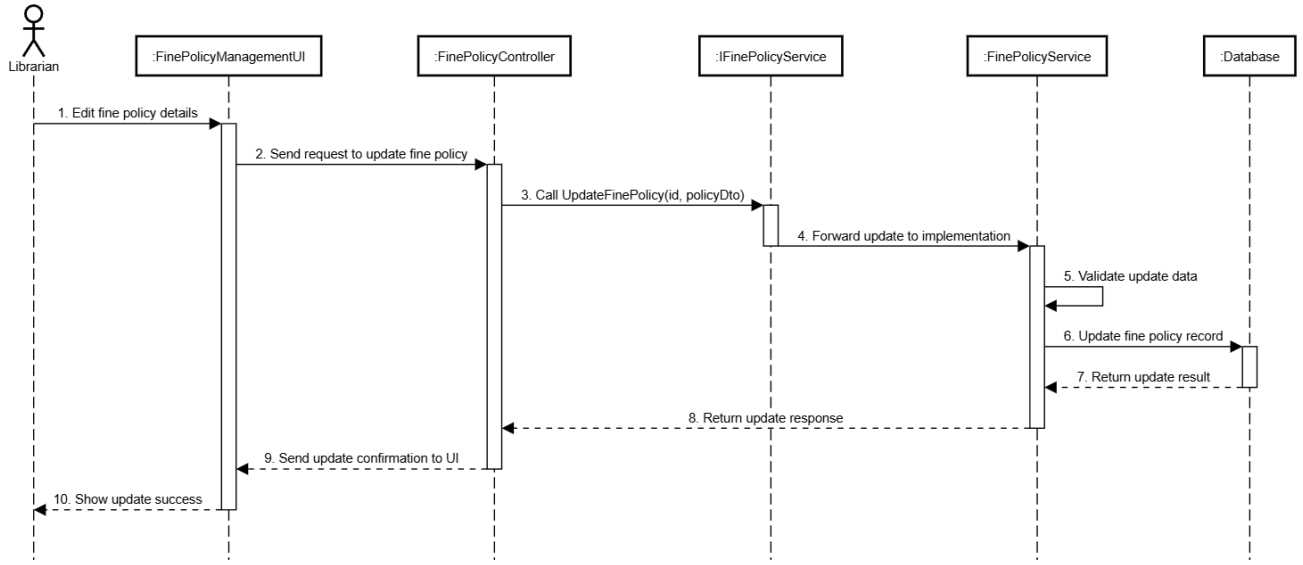


Figure 3.19.4.2 Sequence Diagram

3.19.4.3 Class Specification

3.19.4.3.a Fine Policy Controller

Table 3.19.4.3.a: Fine Policy Controller Description Table

No	Method	Description
1	UpdateFinePolicy(int finePolicyId, UpdateFinePolicyRequest updateRequest)	Update Fine Policy information that displays in the Fine Policy management UI.

3.19.4.3.b Fine Policy Service

Table 3.19.4.3.b:Fine Policy Service Description Table

No	Method	Description
1	Update(int FinePolicyId, FinePolicyDto finePolicyDto)	Validate and update the changed information of Fine Policy.

3.20. Borrow Requests

3.20.1. Create Borrow Request

3.20.1.1 Class Diagram

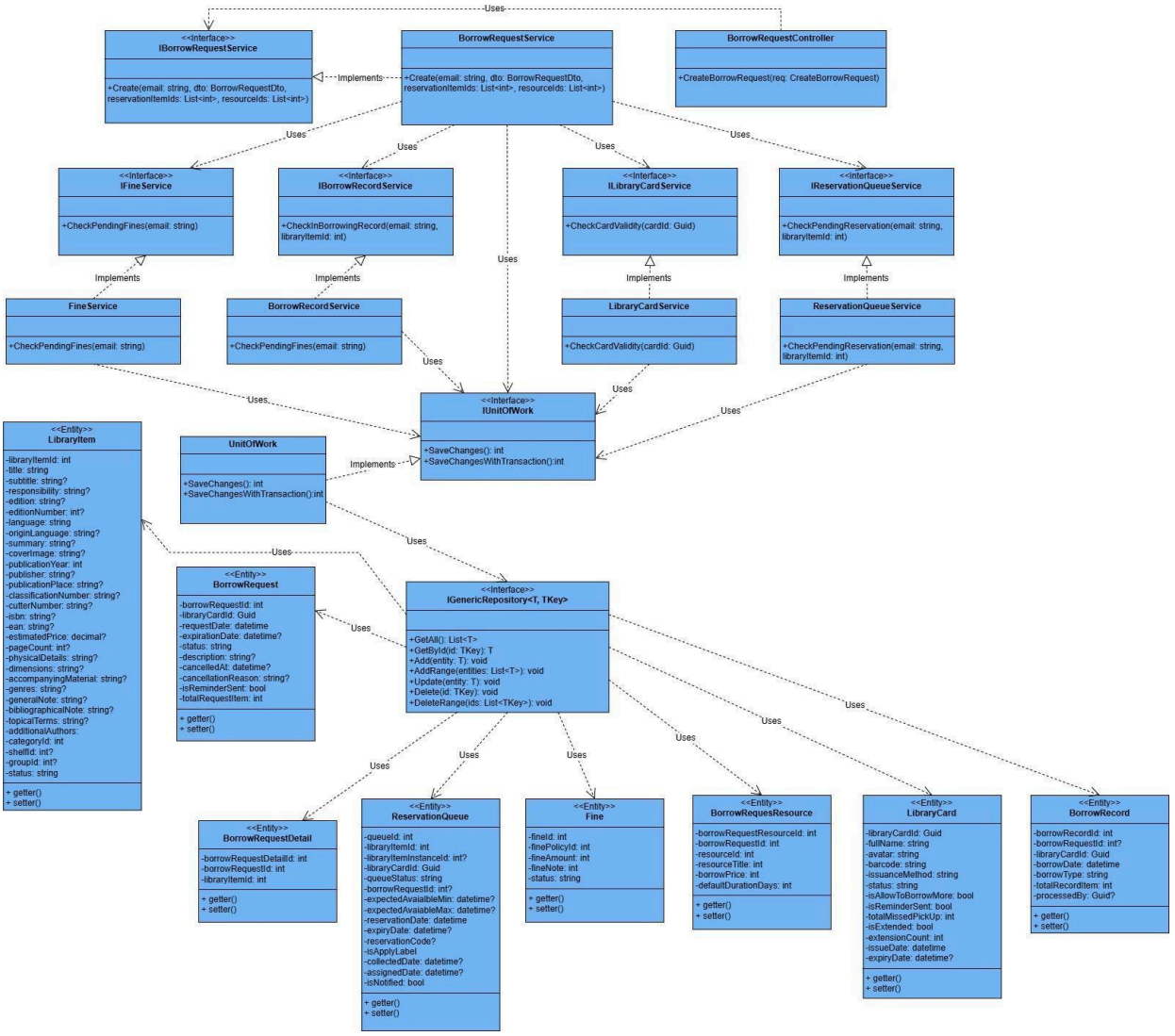


Figure 3.20.1.1 Class Diagram

3.20.1.2 Sequence Diagram

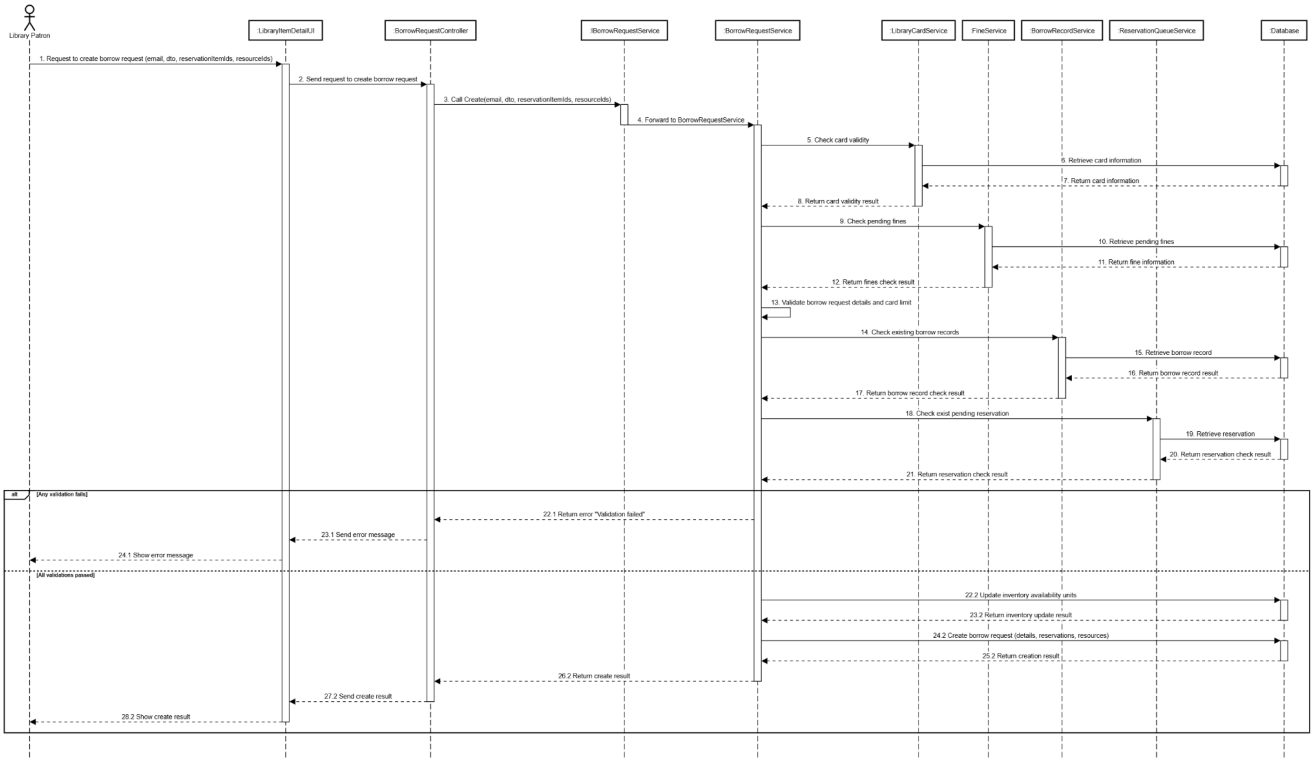


Figure 3.20.1.2 Sequence Diagram

3.20.1.3 Class Specification

3.20.1.3.a Borrow Request Controller

Table 3.20.1.3.a: Borrow Request Controller Description Table

No	Method	Description
1	CreateBorrowRequest(CreateBorrowRequest req)	Receives request containing the patron's borrow request payload (including item IDs, digital resource IDs, and reservation IDs), and delegates all processing to the service layer.

3.20.1.3.b Borrow Request Service

Table 3.20.1.3.b: Borrow Request Service Description Table

No	Method	Description
1	Create(string createdByEmail, BorrowRequestDto requestDto, List<int> reservationIds, List<int> resourceIds)	- Verifies that the library card of the requested email (patron) is active through CheckCardValidity. - Validates the borrow request details by checking the availability of physical items, digital resources existence and checking for reservation entries. If all validation passes, create a borrow request and reservations to the database.

3.20.2. Cancel Specific Item

3.20.2.1 Class Diagram

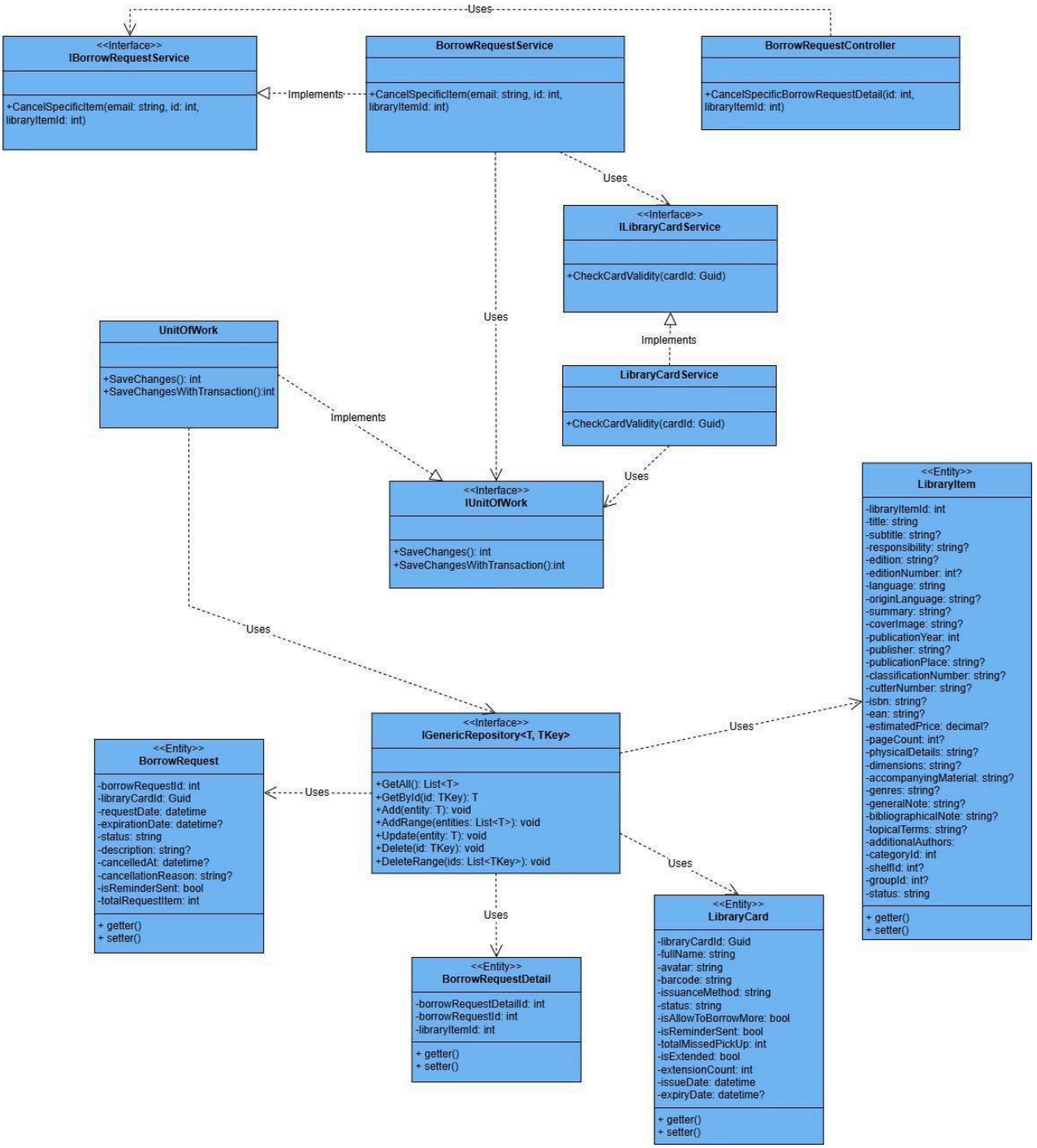


Figure 3.20.2.1 Class Diagram

3.20.2.2 Sequence Diagram

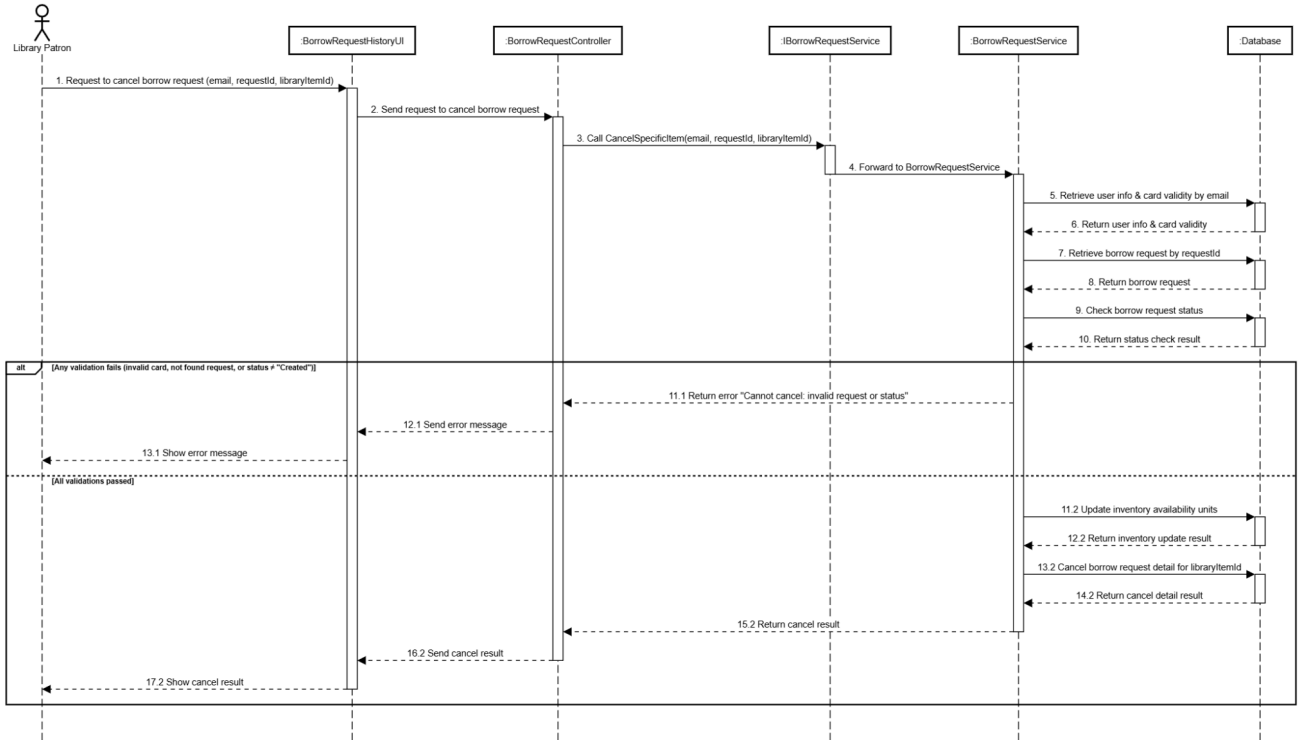


Figure 3.20.2.2 Sequence Diagram

3.20.2.3 Class Specification

3.20.2.3.a Borrow Request Controller

Table 3.20.2.3.a: Borrow Request Controller Description Table

No	Method	Description
1	CancelSpecificBorrowRequestDetail(int id, int libraryItemId)	Receives request containing the borrowed request ID and specified library item ID to cancel, delegates to service layer.

3.20.2.3.b Borrow Request Service

Table 3.20.2.3.b: Borrow Request Service Description Table

No	Method	Description
1	CancelSpecificItem(string email, int id, int libraryItemId)	- Verifies that the library card of the requested email (patron) is active through CheckCardValidity. - Verifies the borrow request ID and specified item exists, processes cancel specified borrow request detail, updates the borrow request details and inventory status.

3.20.3. Cancel Specific Digital Resource

3.20.3.1 Class Diagram

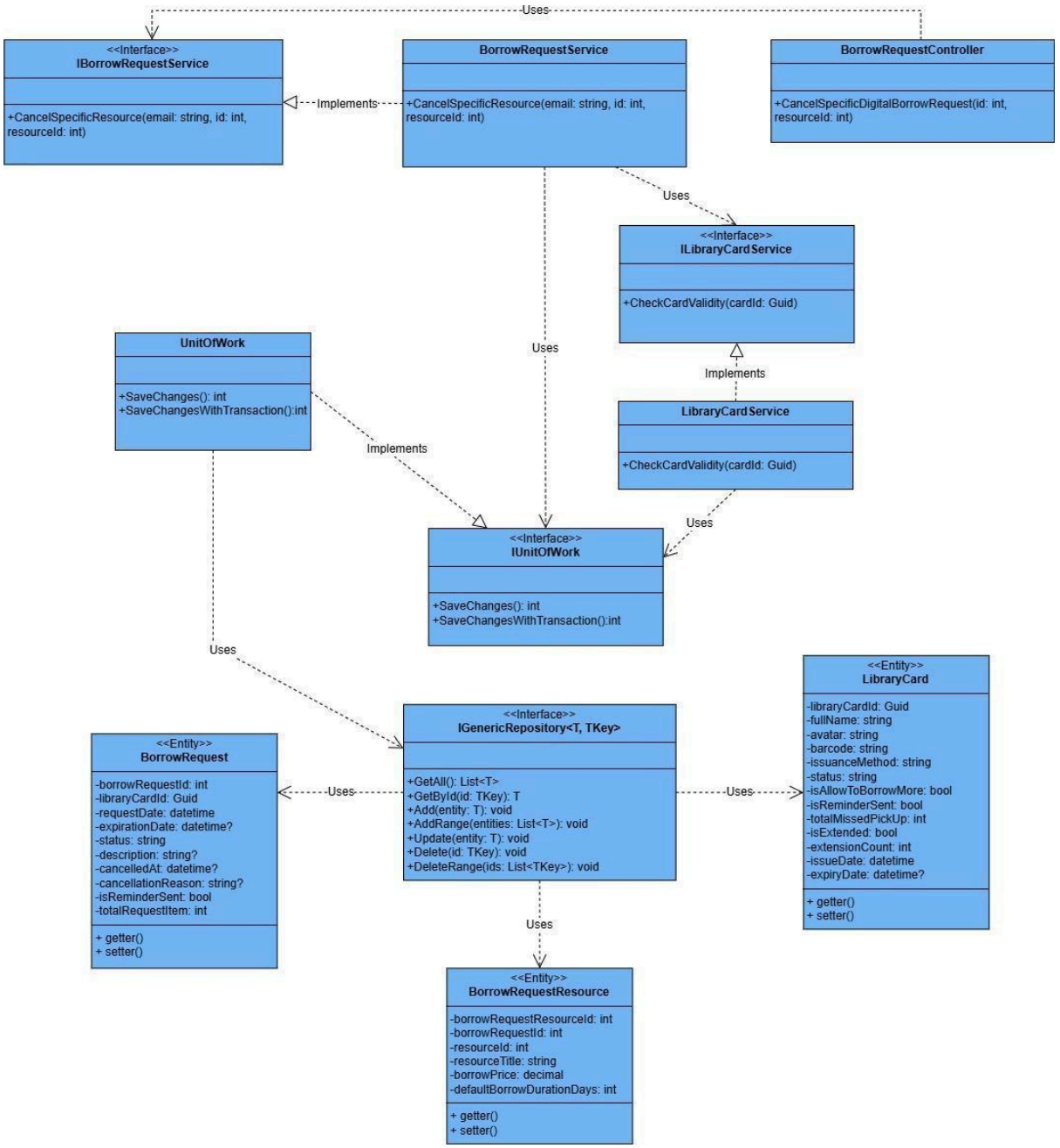


Figure 3.20.3.1 Class Diagram

3.20.3.2 Sequence Diagram

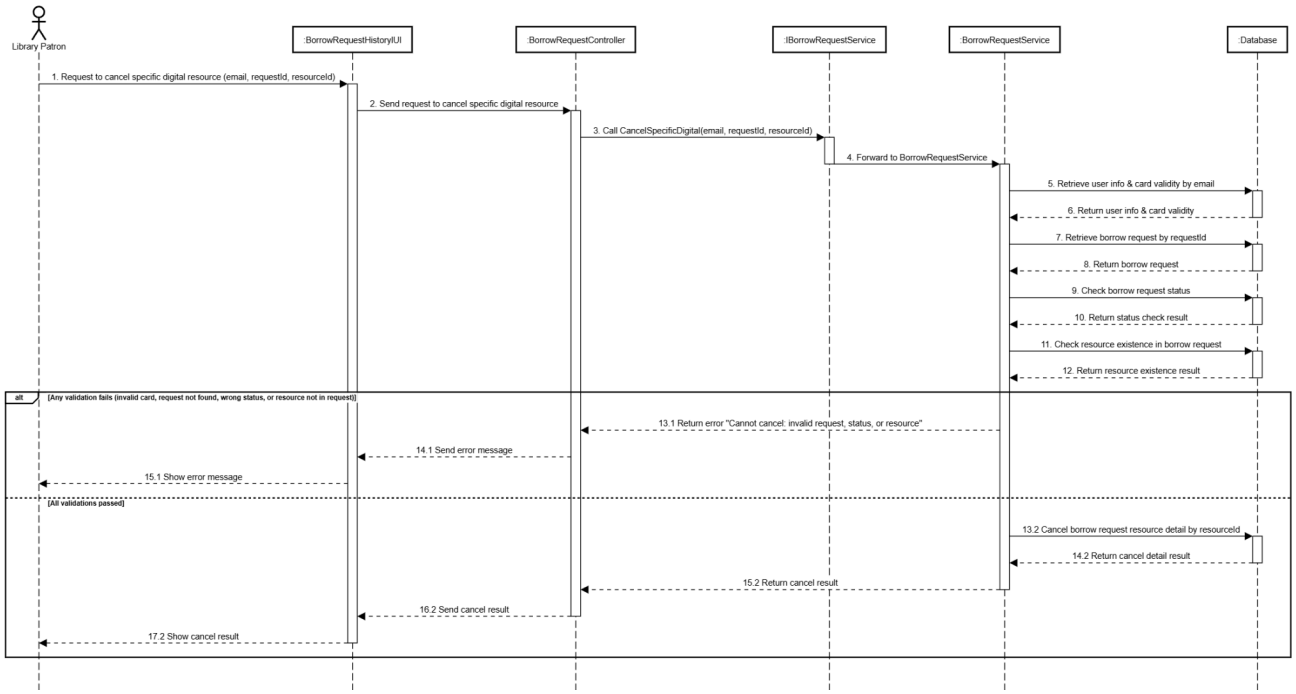


Figure 3.20.3.2 Sequence Diagram

3.20.3.3 Class Specification

3.20.3.3.a Borrow Request Controller

Table 3.20.3.3.a: Borrow Request Controller Description Table

No	Method	Description
1	CancelSpecificDigitalBorrowRequest(int id, int resourceId)	Receives request containing the borrow request ID and the digital resource ID to cancel, delegates to the service layer.

3.20.3.3.b Borrow Request Service

Table 3.20.3.3.b: Borrow Request Service Description Table

No	Method	Description
1	CancelSpecificDigital(int id, int resourceId)	- Verifies that the library card of the requested email (patron) is active through CheckCardValidity. - Verifies that the borrowed request ID and specified digital resource ID exists, processes cancel digital resource borrow and save changes to the database.

3.20.4. Cancel Specific Reservation

3.20.4.1 Class Diagram

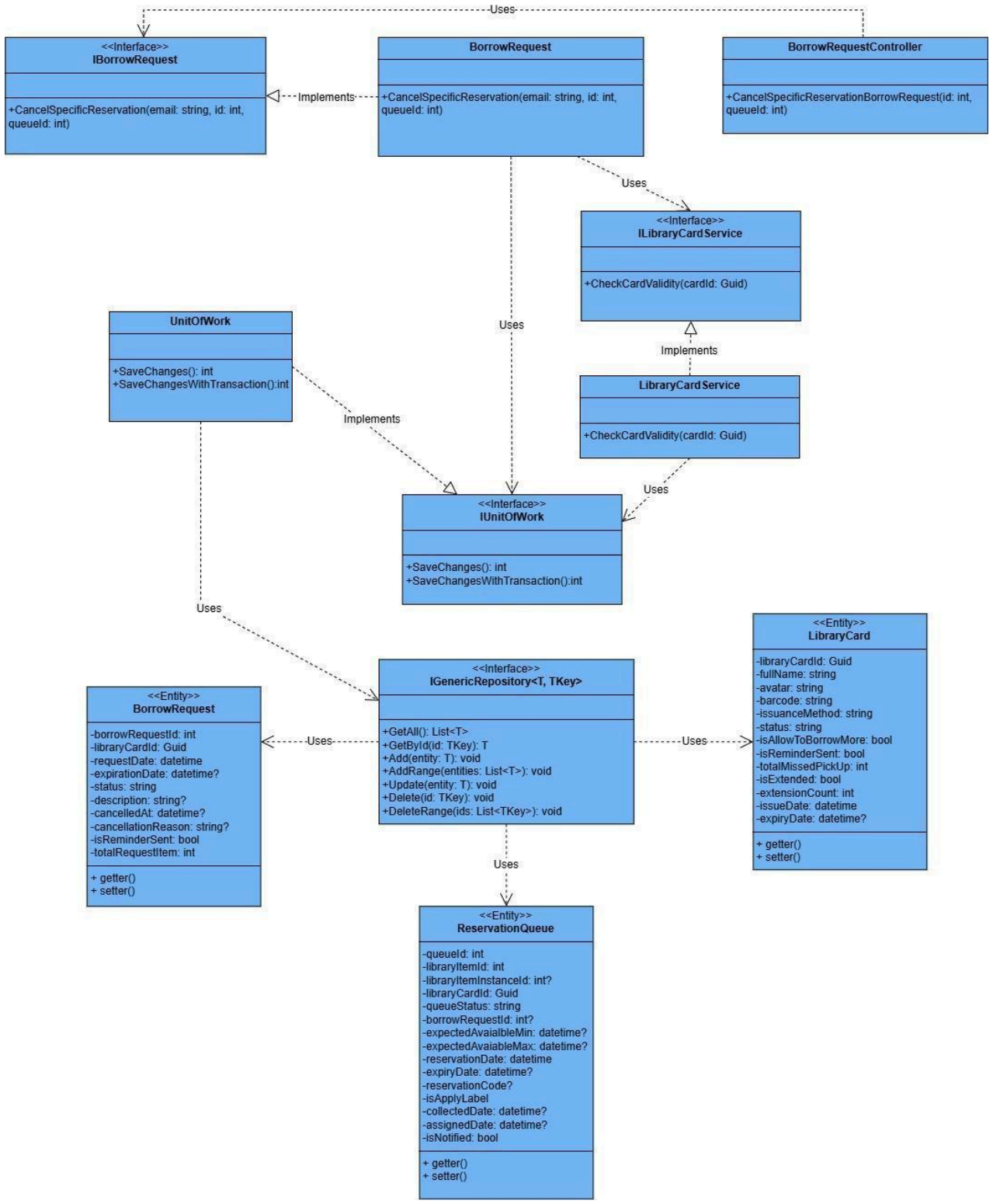


Figure 3.20.4.1 Class Diagram

3.20.4.2 Sequence Diagram

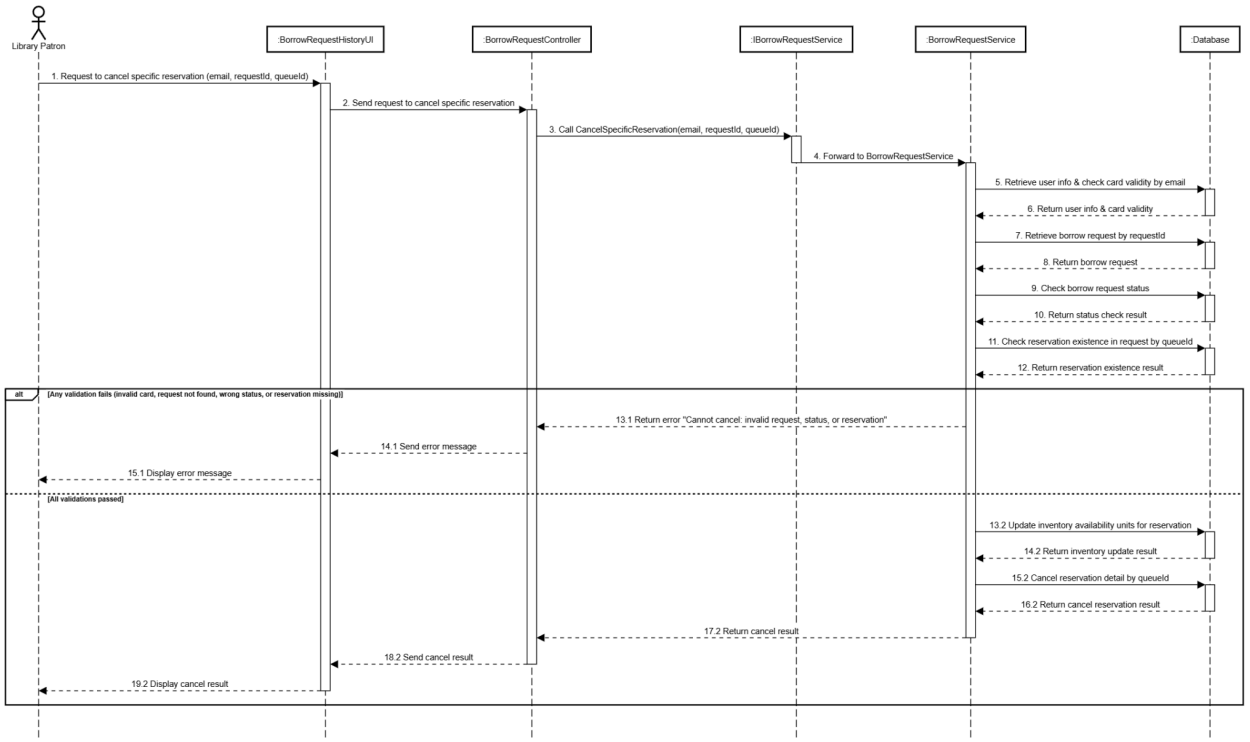


Figure 3.20.4.2 Sequence Diagram

3.20.4.3 Class Specification

3.20.4.3.a Borrow Request Controller

Table 3.20.4.3.a: Borrow Request Controller Description Table

No	Method	Description
1	CancelSpecificReservationBorrowRequest(int id, int queueId)	Receives request containing the borrow request ID and the reservation queue ID to cancel, delegates to the service layer.

3.20.4.3.b Borrow Request Service

Table 3.20.4.3.b: Borrow Request Service Description Table

No	Method	Description
1	CancelSpecificReservation(int id, int queueId)	- Verifies that the library card of the requested email (patron) is active through CheckCardValidity. - Verifies that the borrowed request ID and specified reservation queue ID exists, processes cancel reservation and save changes to the database.

3.20.5. Cancel Borrow Request

3.20.5.1 Class Diagram

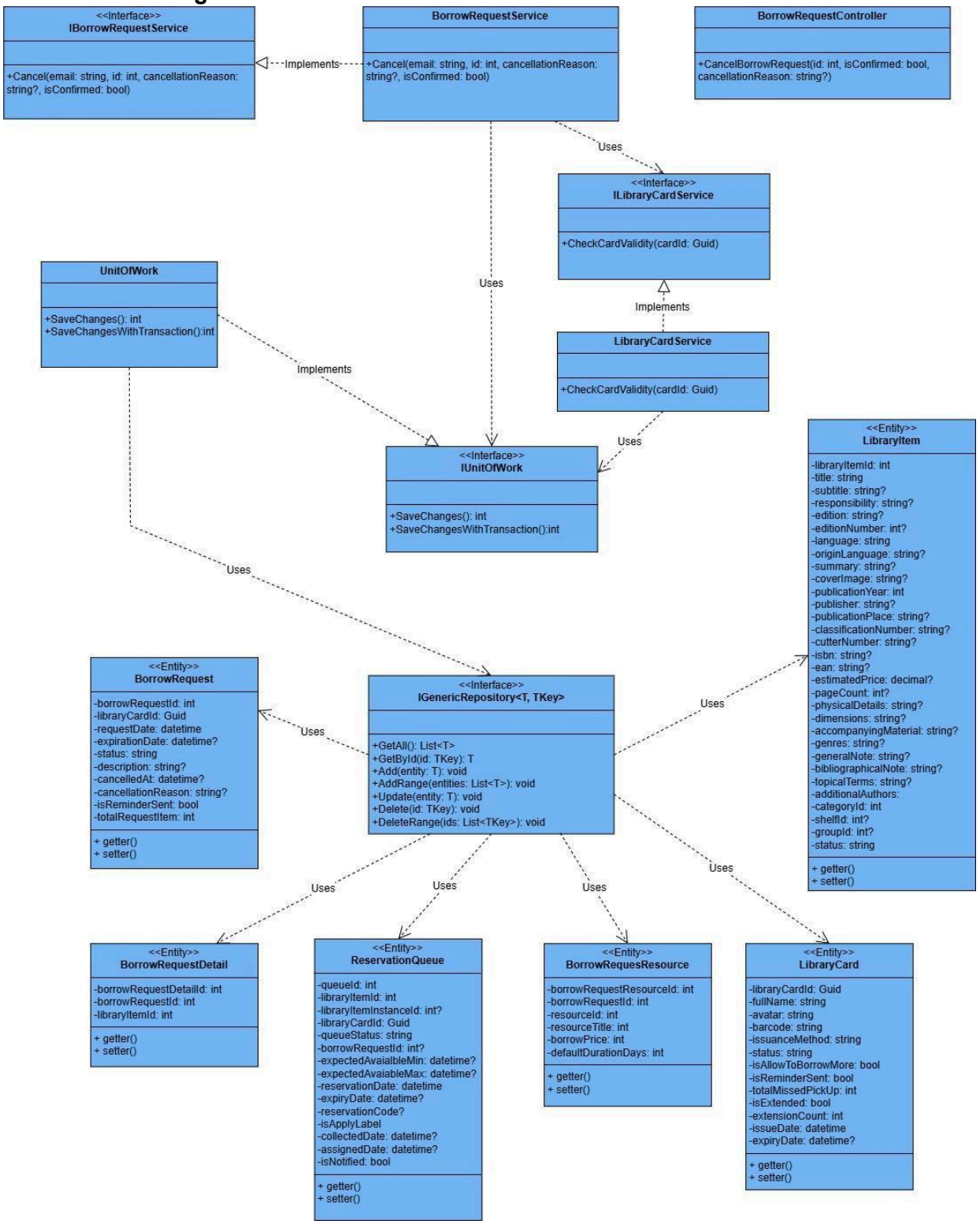


Figure 3.20.5.1 Class Diagram

3.20.5.2 Sequence Diagram

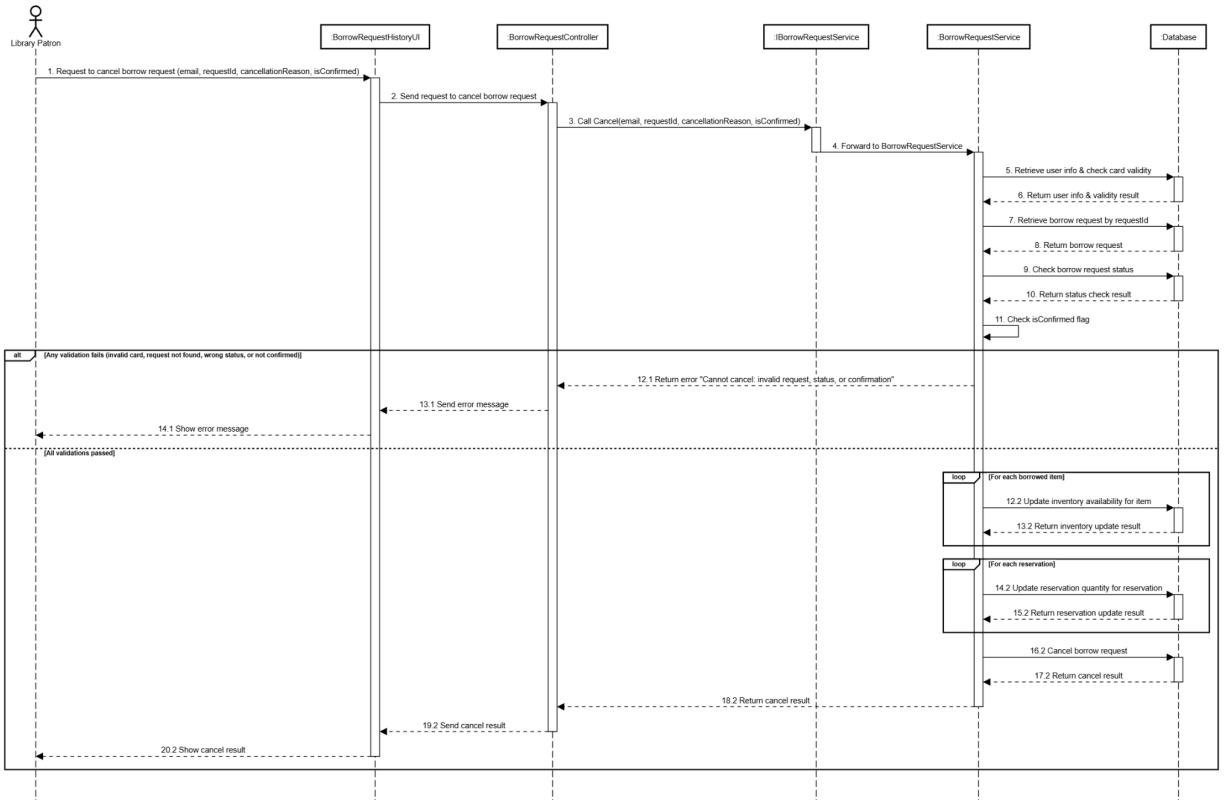


Figure 3.20.5.2 Sequence Diagram

3.20.5.3 Class Specification

3.20.5.3.a Borrow Request Controller

Table 3.20.5.3.a: Borrow Request Controller Description Table

No	Method	Description
1	CancelBorrowRequest(int id)	Receives a request containing the borrow request ID and delegate to the service layer.

3.20.5.3.b Borrow Request Service

Table 3.20.5.3.b: Borrow Request Service Description Table

No	Method	Description
2	Cancel(int id)	- Verifies that the library card of the requested email (patron) is active through CheckCardValidity. - Verifies the existence of the specified borrowed request ID, then cancels the borrowed request along with all its related details (physical items, digital resources, and reservations), and updates inventory.

3.20.6. Get All Borrow Request

3.20.6.1 Class Diagram

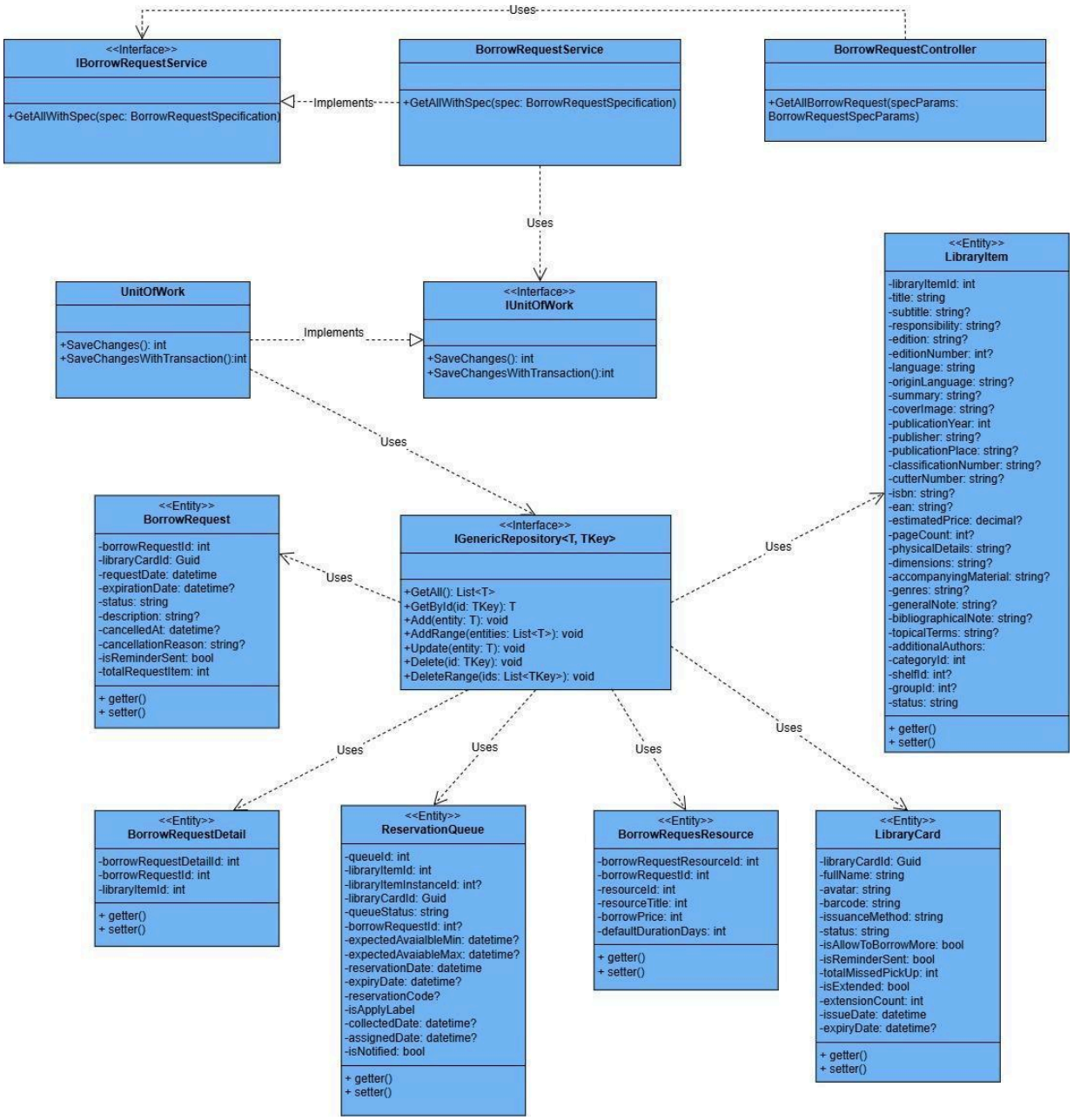


Figure 3.20.6.1 Class Diagram

3.20.6.2 Sequence Diagram

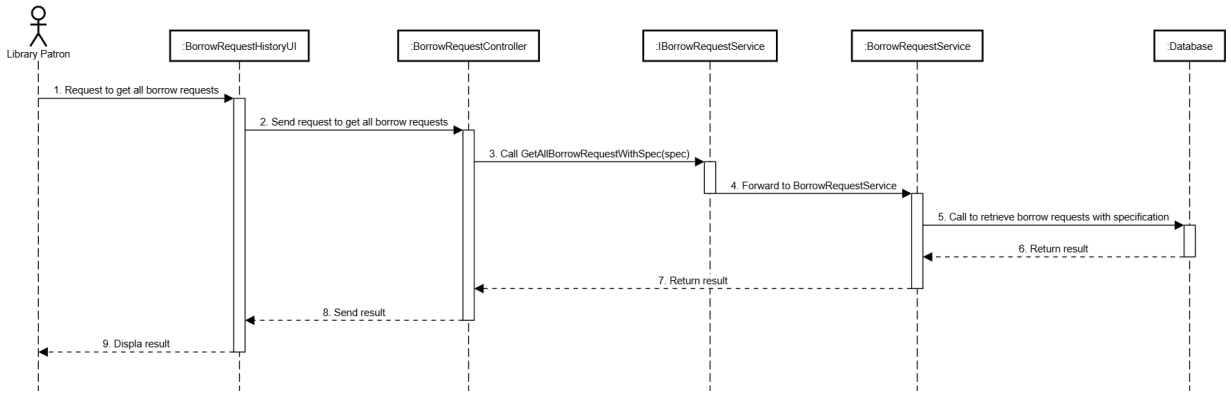


Figure 3.20.6.2 Sequence Diagram

3.20.6.3 Class Specification

3.20.6.3.a Borrow Request Controller

Table 3.20.6.3.a: Borrow Request Controller Description Table

No	Method	Description
1	GetAllBorrowRequest(BorrowRequestSpecParams specParams)	Receives request with filtering, sorting, and pagination parameters, delegates to the service layer.

3.20.6.3.b Borrow Request Service

Table 3.20.6.3.b: Borrow Request Service Description Table

No	Method	Description
1	GetAllWithSpec(BorrowRequestSpecification spec)	Applies filtering, sorting, and pagination logic to retrieve all existing borrow requests. Moreover, try to include borrow request details (physical items, digital resources and reservations).

3.21. Borrow Records

3.21.1. Create Borrow Record

3.21.1.1 Class Diagram

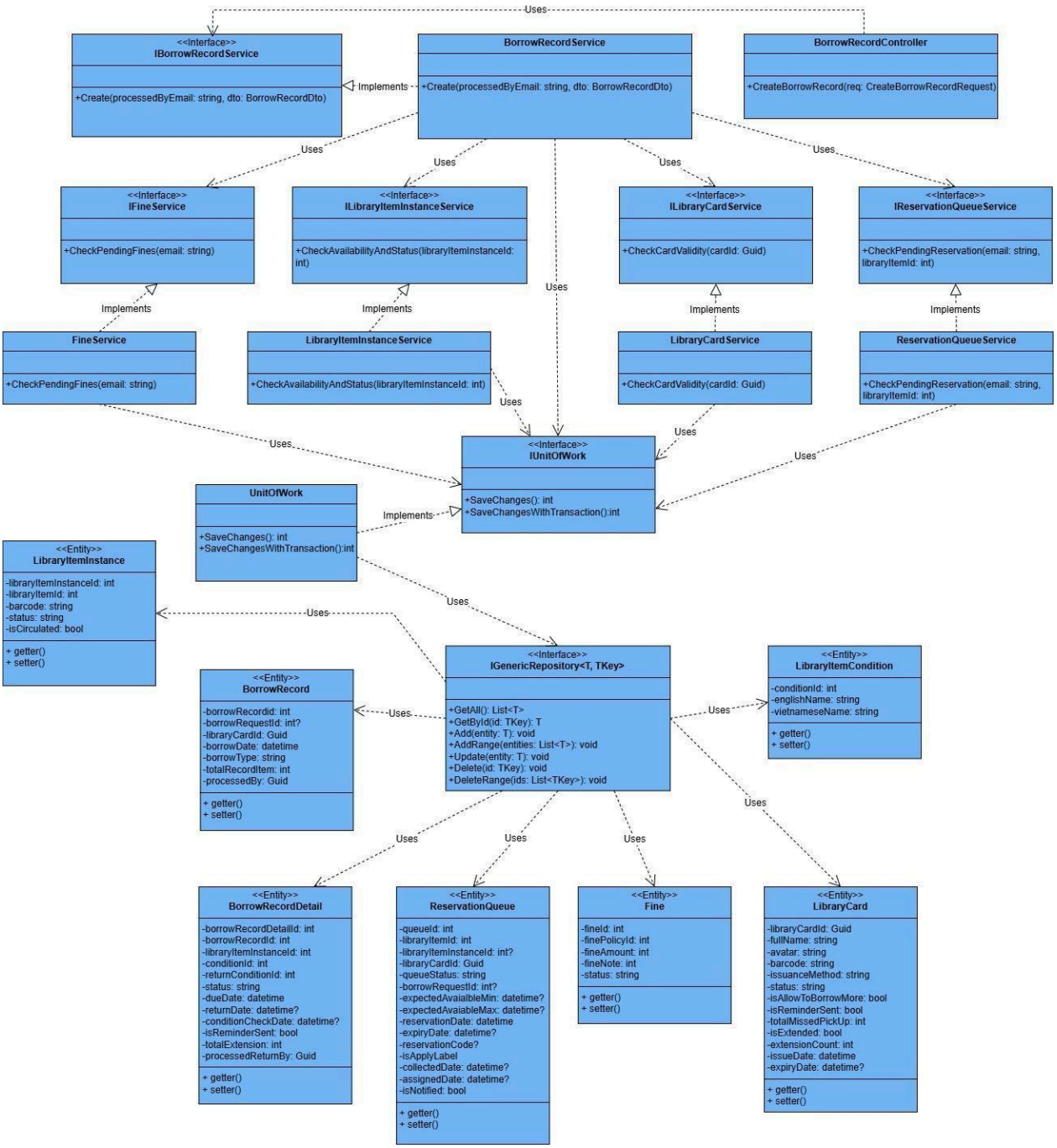


Figure 3.21.6.1 Class Diagram

3.21.6.2 Sequence Diagram

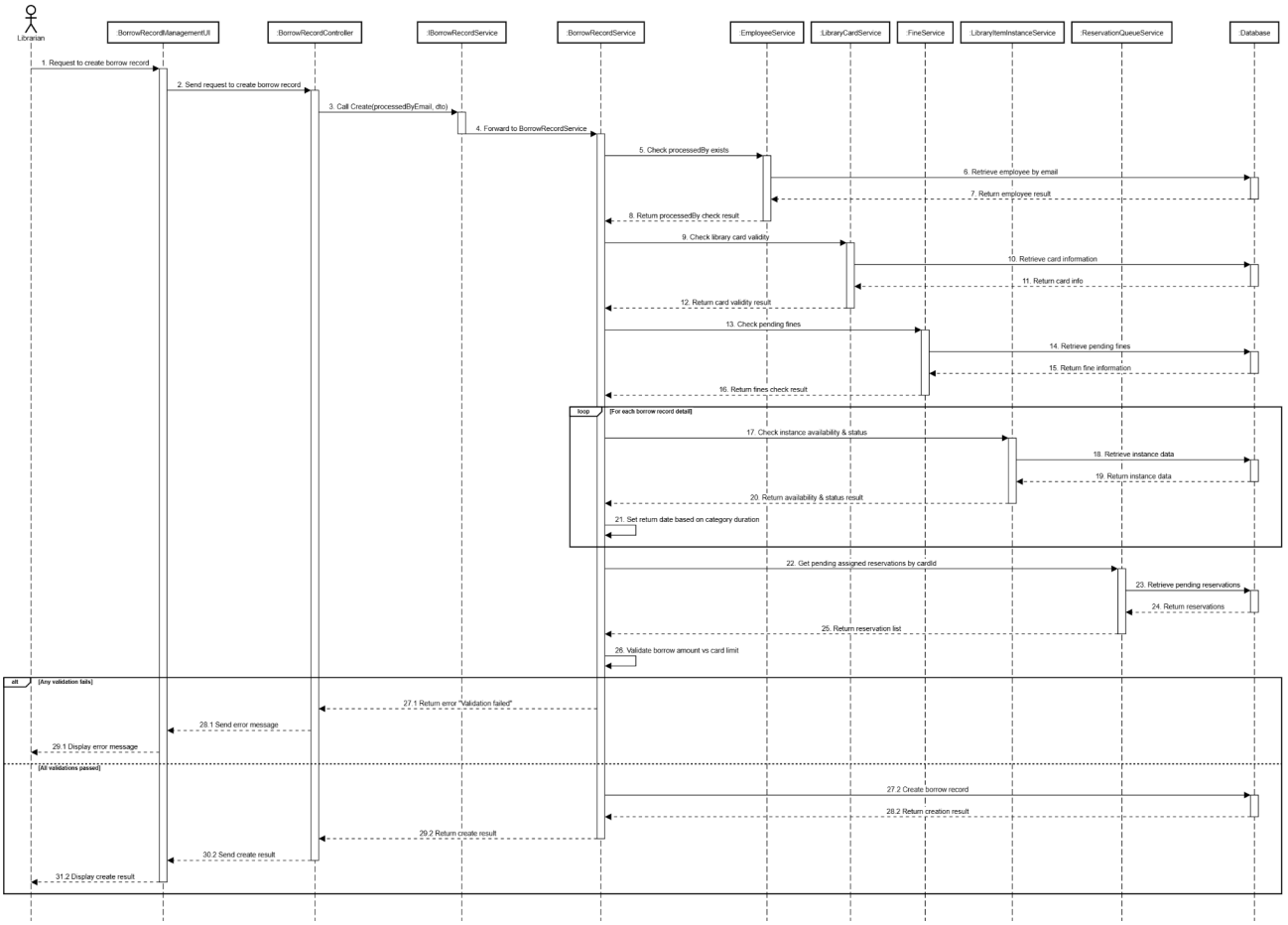


Figure 3.21.6.2 Sequence Diagram

3.21.6.3 Class Specification

3.21.6.3.a Borrow Record Controller

Table 3.21.6.3.a: Borrow Record Controller Description Table

No	Method	Description
1	CreateBorrowRecord(CreateBorrowRecordRequest req)	Receives request with the patron's borrow record payload. Extract the current user's email, calls to the service to create a borrow record.

3.21.6.3.b Borrow Record Service

Table 3.21.6.3.b: Borrow Record Service Description Table

No	Method	Description
1	Create(string processedByEmail, BorrowRecordDto recordDto)	- Verifies that the library card of the requested email (patron) is active through CheckCardValidity. - Ensure no pending fines through CheckPendingFines - Check each physical item's availability and status through CheckAvailabilityAndStatus

		- Verify there are no pending reservation through CheckPendingReservation - If all validations passed, create the borrow record and save changes to the database.

3.21.2. Extend Borrow Record

3.21.2.1 Class Diagram

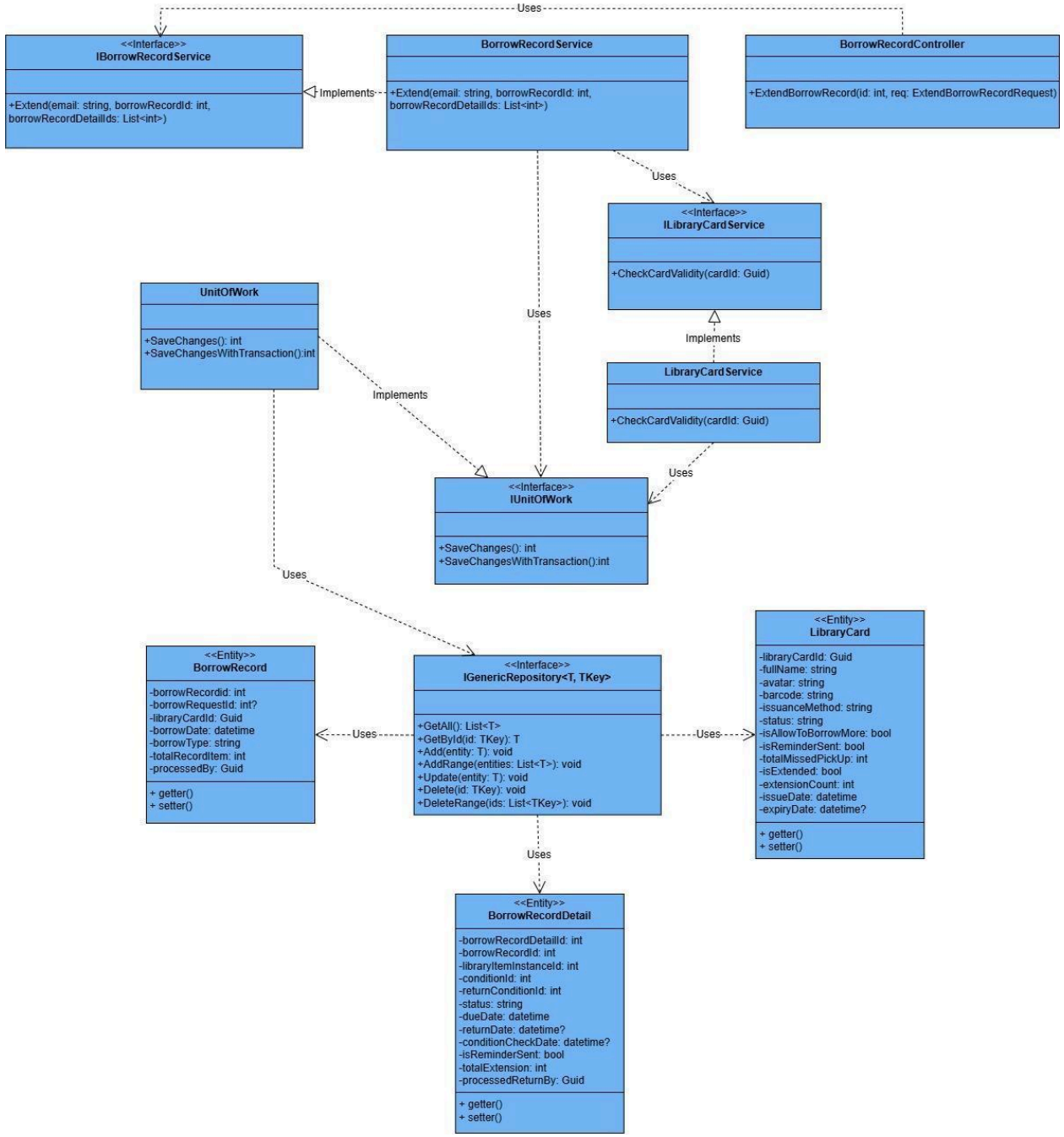


Figure 3.21.2.1 Class Diagram

3.21.2.2 Sequence Diagram

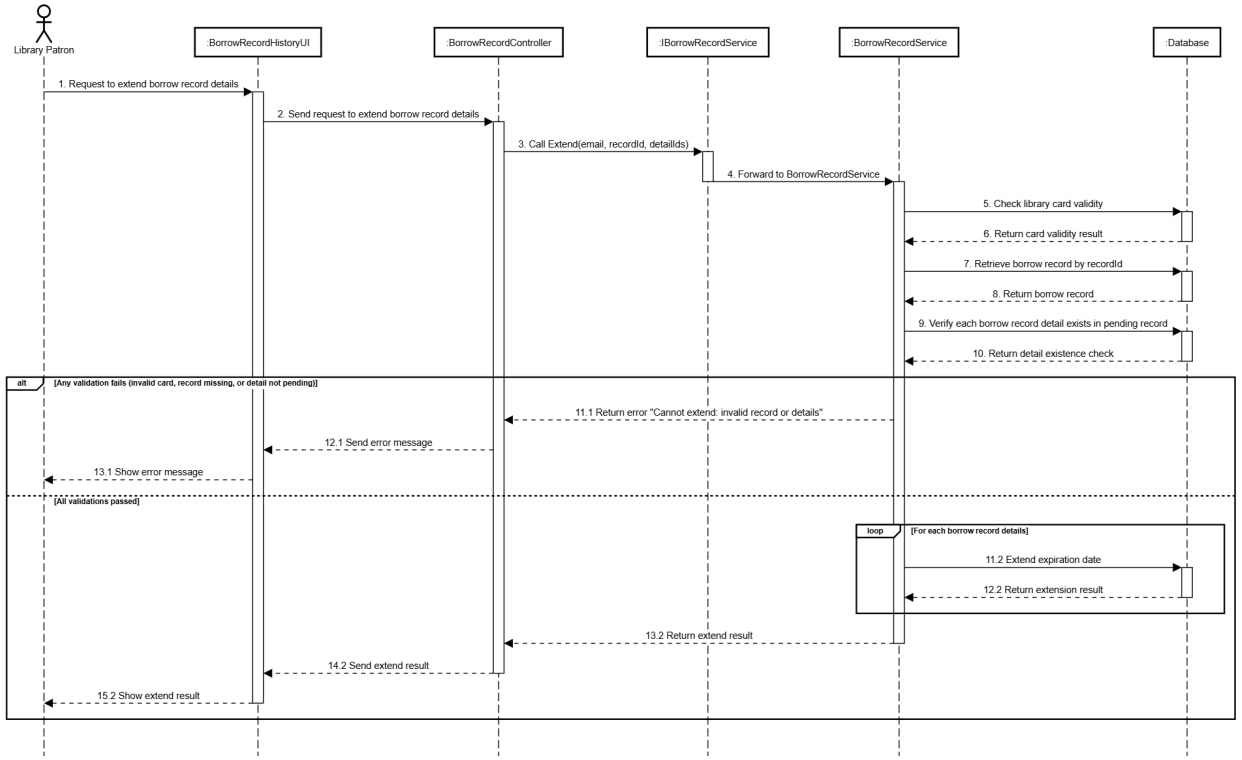


Figure 3.21.2.2 Sequence Diagram

3.21.2.3 Class Specification

3.21.2.3.a Borrow Record Controller

Table 3.21.2.3.a: Borrow Record Controller Description Table

No	Method	Description
1	ExtendBorrowRecord(int id, ExtendBorrowRecordRequest req)	Receives request with the borrow-record ID and a payload specifying which detailed entries to extend the due date.

3.21.2.3.b Borrow Record Service

Table 3.21.2.3.b: Borrow Record Service Description Table

No	Method	Description
1	Extend(string email, int borrowRecordId, List<int> borrowRecordDetailIds)	- Verifies that the library card of the requested email (patron) is active through CheckCardValidity. - Verifies each detailed entry is eligible for extension, updates their due date.

3.21.3. Process Return

3.21.3.1 Class Diagram

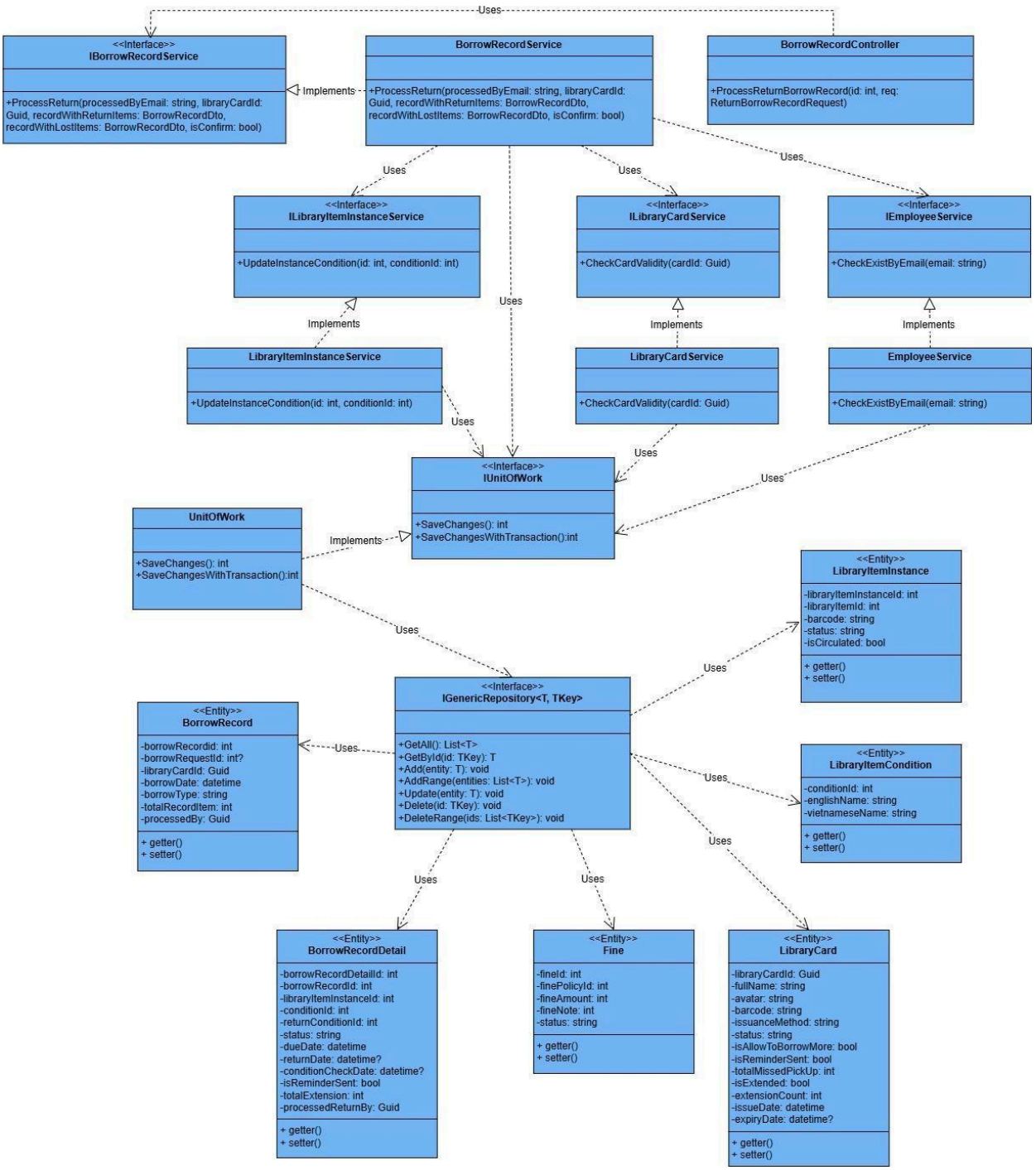


Figure 3.21.3.1 Class Diagram

3.21.3.2 Sequence Diagram

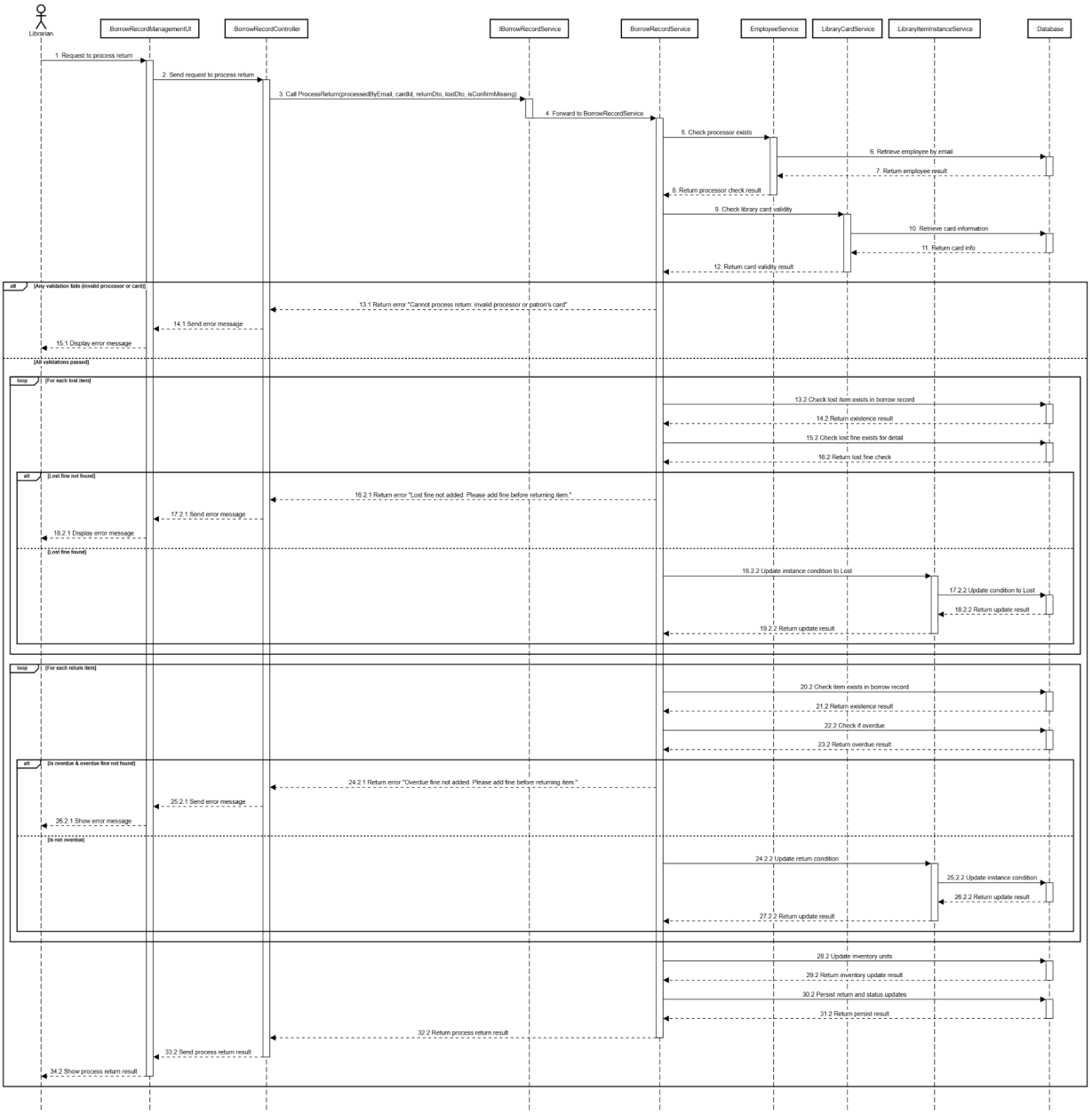


Figure 3.21.3.2 Sequence Diagram

3.21.3.3 Class Specification

3.21.3.3.a Borrow Record Controller

Table 3.21.3.3.a: Borrow Record Controller Description Table

No	Method	Description
1	ProcessReturnBorrowRecord(int id, ReturnBorrowRecordRequest req)	Receives request to process returned and lost items for specified borrow-record ID, extracts the current processed by email, delegates to the service layer.

3.21.3.3.b Borrow Record Service

Table 3.21.3.3.b: Borrow Record Service Description Table

No	Method	Description
1	ProcessReturn(string processedByEmail, Guid libraryCardId, BorrowRecordDto recordWithReturnItems, BorrowRecordDto recordWithLostItems, bool isConfirm)	- Verifies the processing user exists through CheckExistByEmail. - Verifies that the library card of the requested email (patron) is active through CheckCardValidity. - Update each item instance's condition through UpdateInstanceCondition. - Update borrow record status to return, set actual return date.

3.21.4. Get All Borrow Record

3.21.4.1 Class Diagram

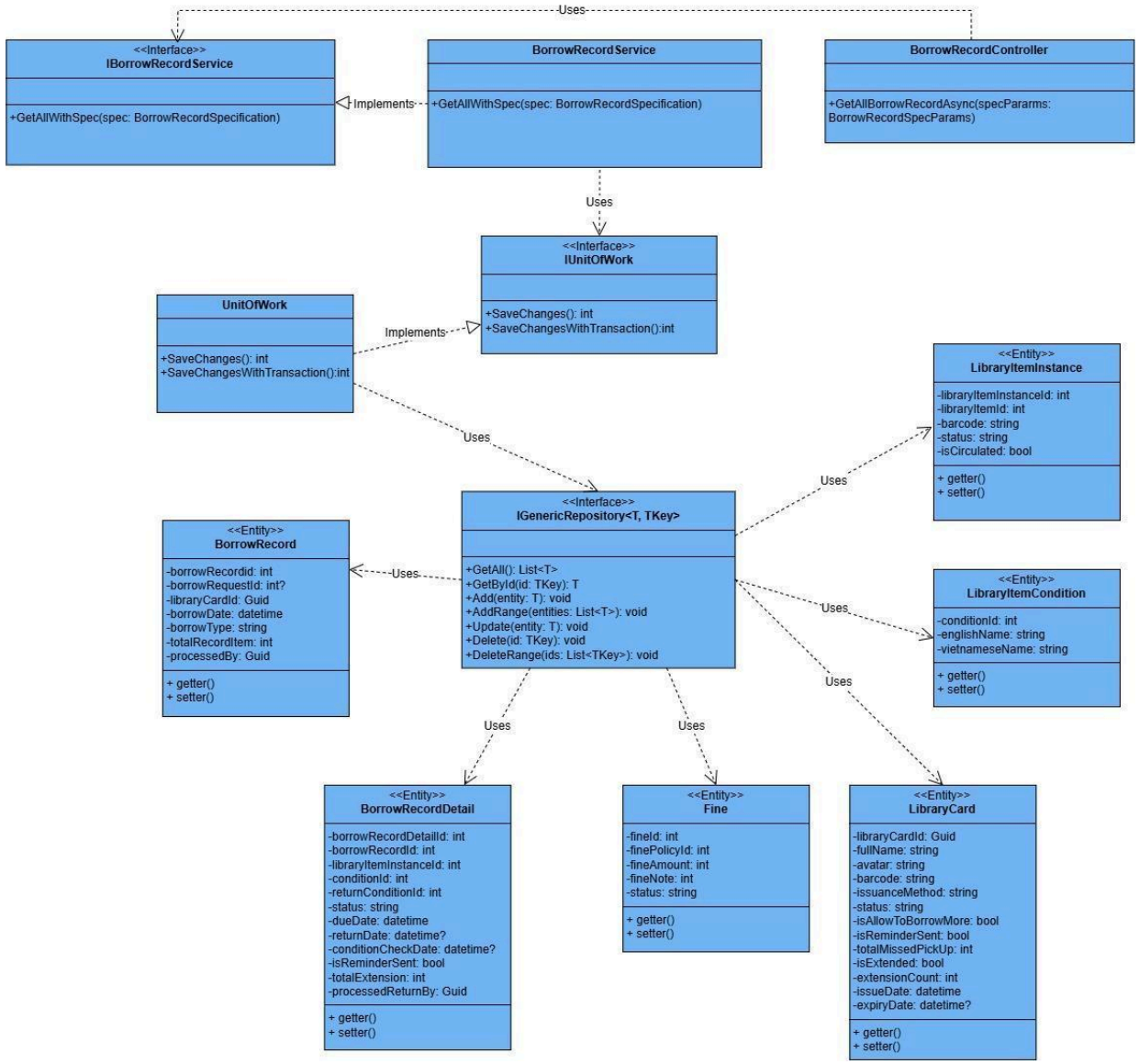


Figure 3.21.4.1 Class Diagram

3.21.4.2 Sequence Diagram

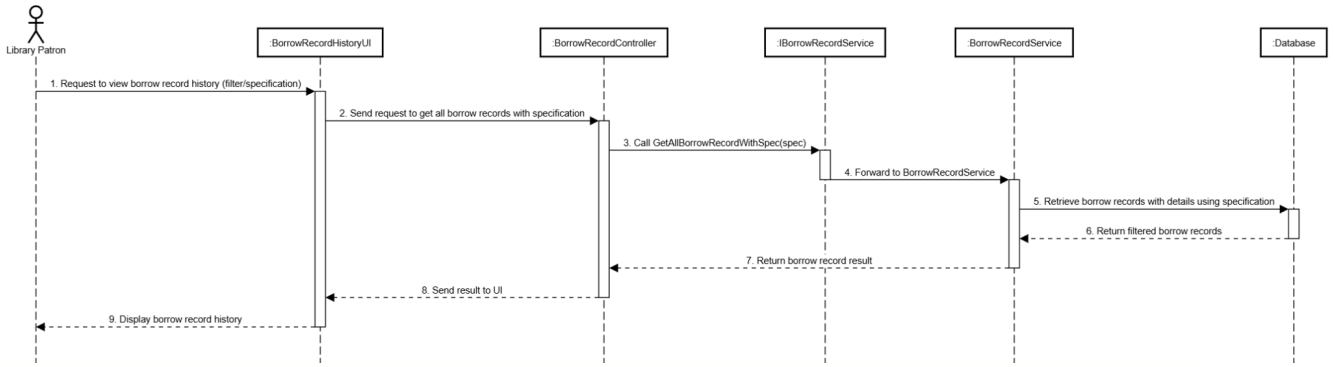


Figure 3.21.4.2 Sequence Diagram

3.21.4.3 Class Specification

3.21.4.3.a Borrow Record Controller

Table 3.21.4.3.a: Borrow Record Controller Description Table

No	Method	Description
1	GetAllBorrowRecord(BorrowRecordSpecParams specParams)	Receives request to retrieve a list of borrowed records. Accepts filtering, sorting, pagination parameters and delegates to the service layer.

3.21.4.3.b Borrow Record Service

Table 3.21.4.3.b: Borrow Record Service Description Table

No	Method	Description
1	GetAllWithSpec(BorrowRecordSpecification spec)	Applies filtering, sorting, and pagination logic to retrieve all existing borrow requests, including all borrow record details.

3.22. Manage Reservations

3.22.1. Get Assignable Item Instances by Reservation ID

3.22.1.1 Class Diagram

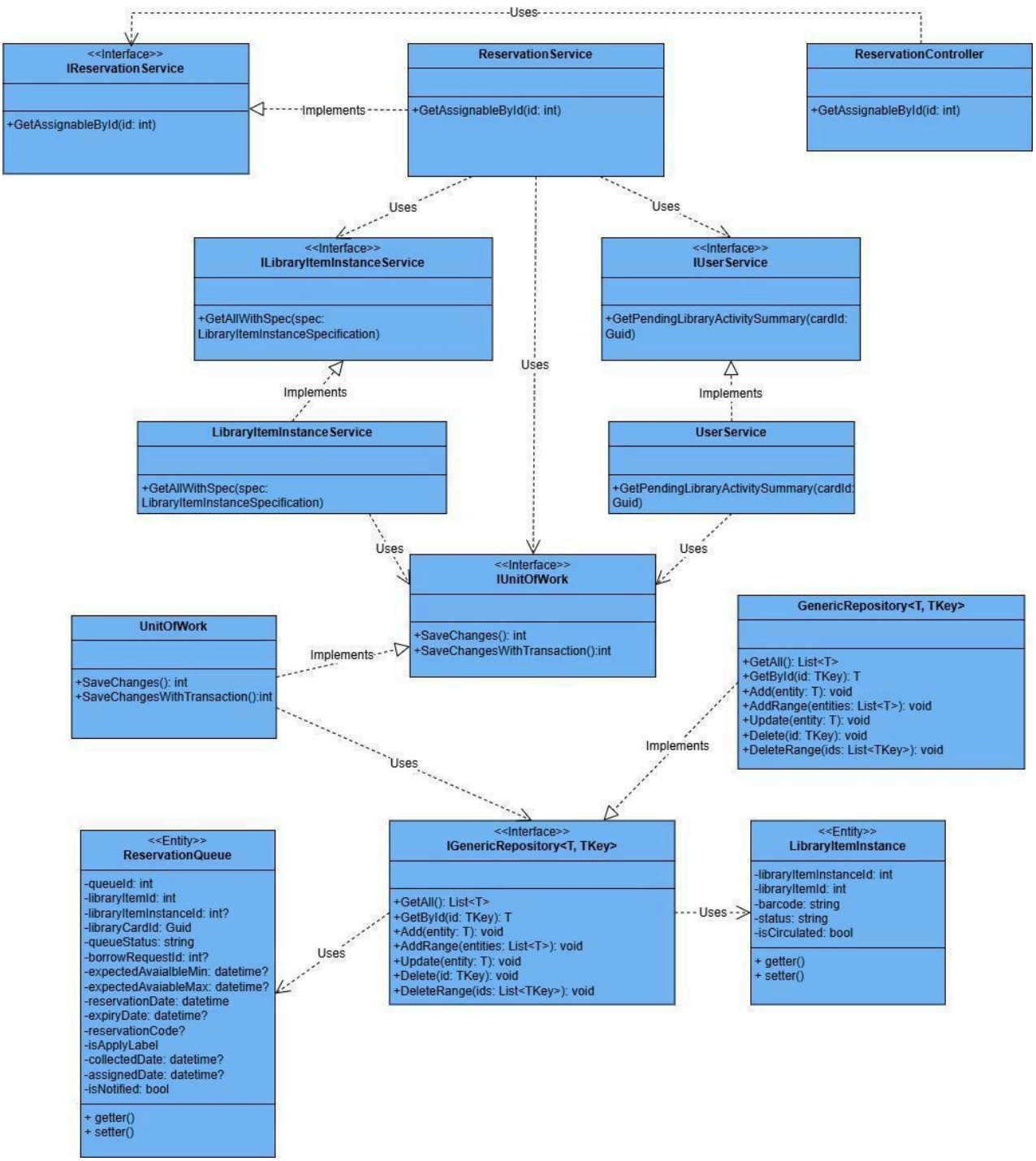


Figure 3.22.1.1 Class Diagram

3.22.1.2 Sequence Diagram

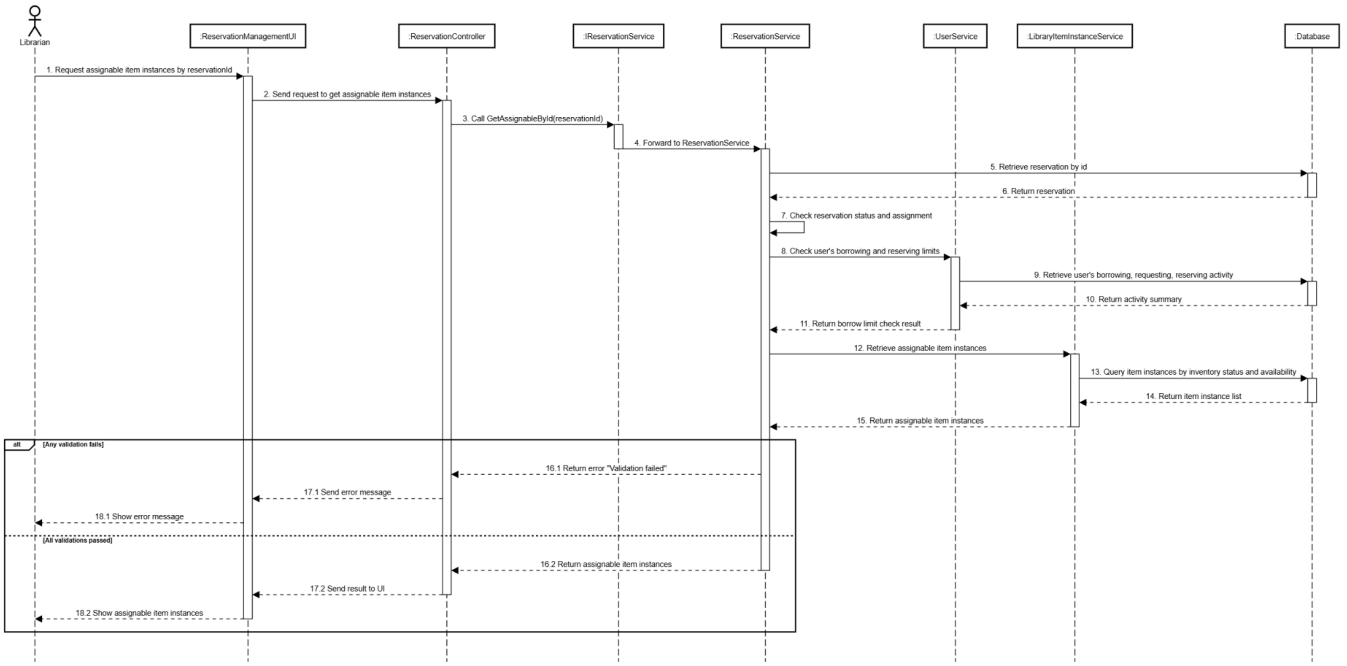


Figure 3.22.1.2 Sequence Diagram

3.22.1.3 Class Specification

3.22.1.3.a Reservation Controller

Table 3.22.1.3.a: Reservation Controller Description Table

No	Method	Description
1	GetAssignableById(int id)	Receives request to retrieve the list of items that can be assigned for the specified reservation ID, and delegates to the service layer.

3.22.1.3.b Reservation Service

Table 3.22.1.3.b: Reservation Service Description Table

No	Method	Description
1	GetAssignableById(int id)	- Retrieves the user's current borrowing summary to check for exceeding borrow limits through GetPendingLibraryActivitySummary. - Retrieves all assignable item instances through GetAllWithSpec. - Applies business rules based on borrow limits, then returns the list of item instances that can be assigned for the given reservation ID.

3.22.2. Assign Specific Item Instance

3.22.2.1 Class Diagram

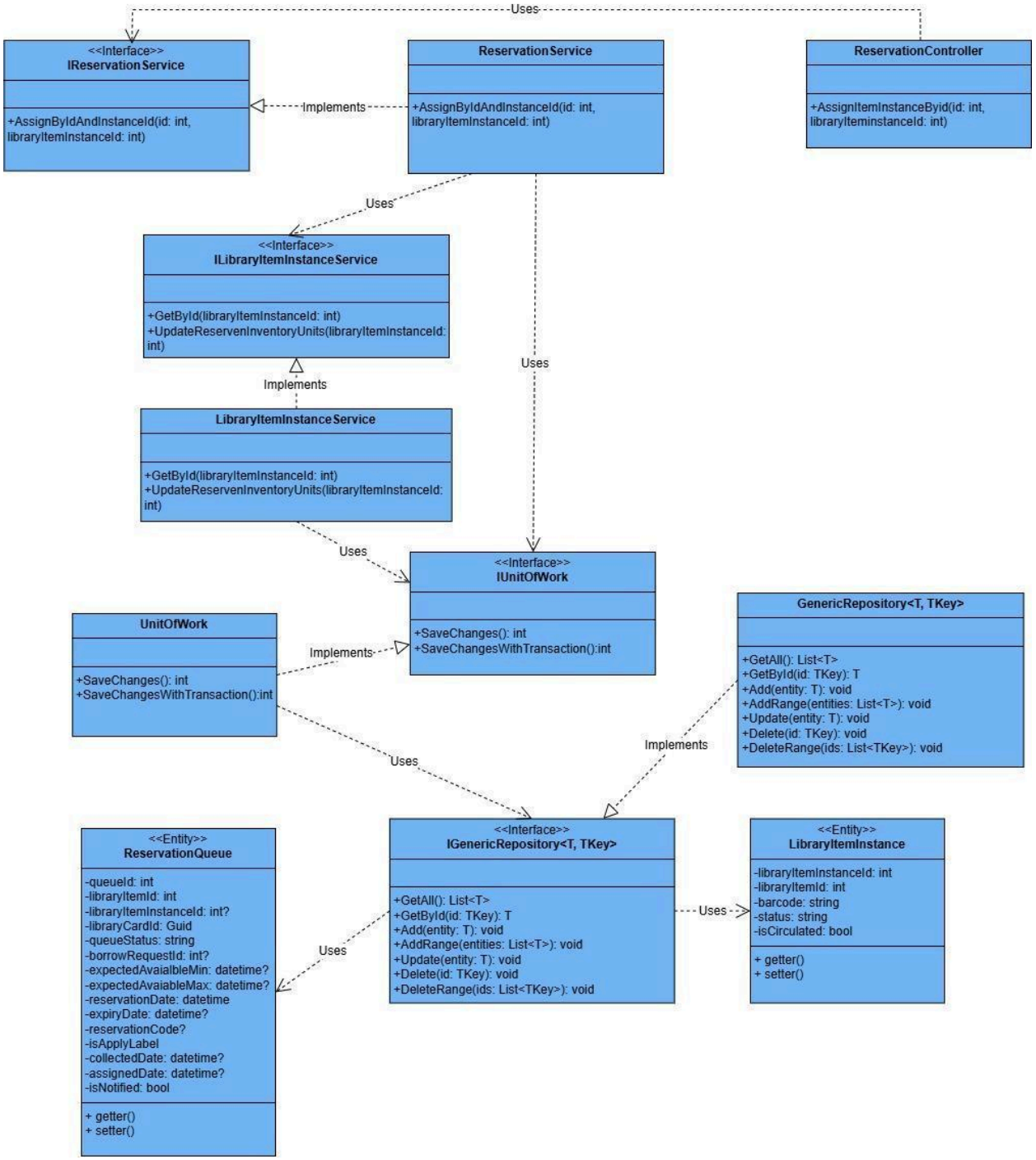


Figure 3.22.2.1 Class Diagram

3.22.2.2 Sequence Diagram

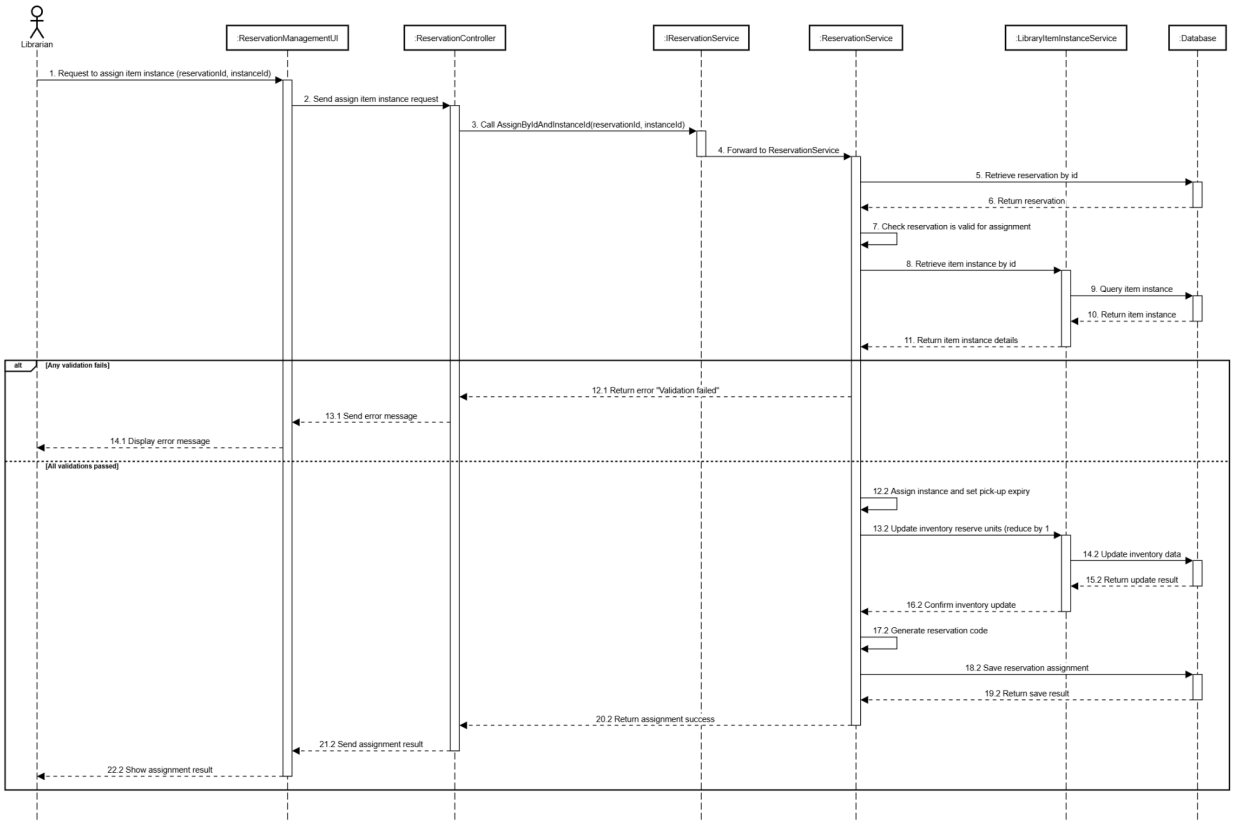


Figure 3.22.2.2 Sequence Diagram

3.22.2.3 Class Specification

3.22.2.3.a Reservation Controller

Table 3.22.2.3.a: Reservation Controller Description Table

No	Method	Description
1	AssignItemInstanceById(int id, int libraryItemInstanceId)	Receives request specifying the reservation ID and the library-item-instance ID. Delegates to the service layer to perform check exists, inventory updates, and the actual assignment.

3.22.2.3.b Reservation Service

Table 3.22.2.3.b: Reservation Service Description Table

No	Method	Description
1	AssignByAndInstanceId(int id, int libraryItemInstanceId)	- Check existing library-item-instance ID through GetById. - Update reserve inventory units through UpdateReserveInventoryUnits. - Process assign specified item instance ID to specified reservation ID.

3.22.3. Confirm Apply Label

3.22.3.1 Class Diagram

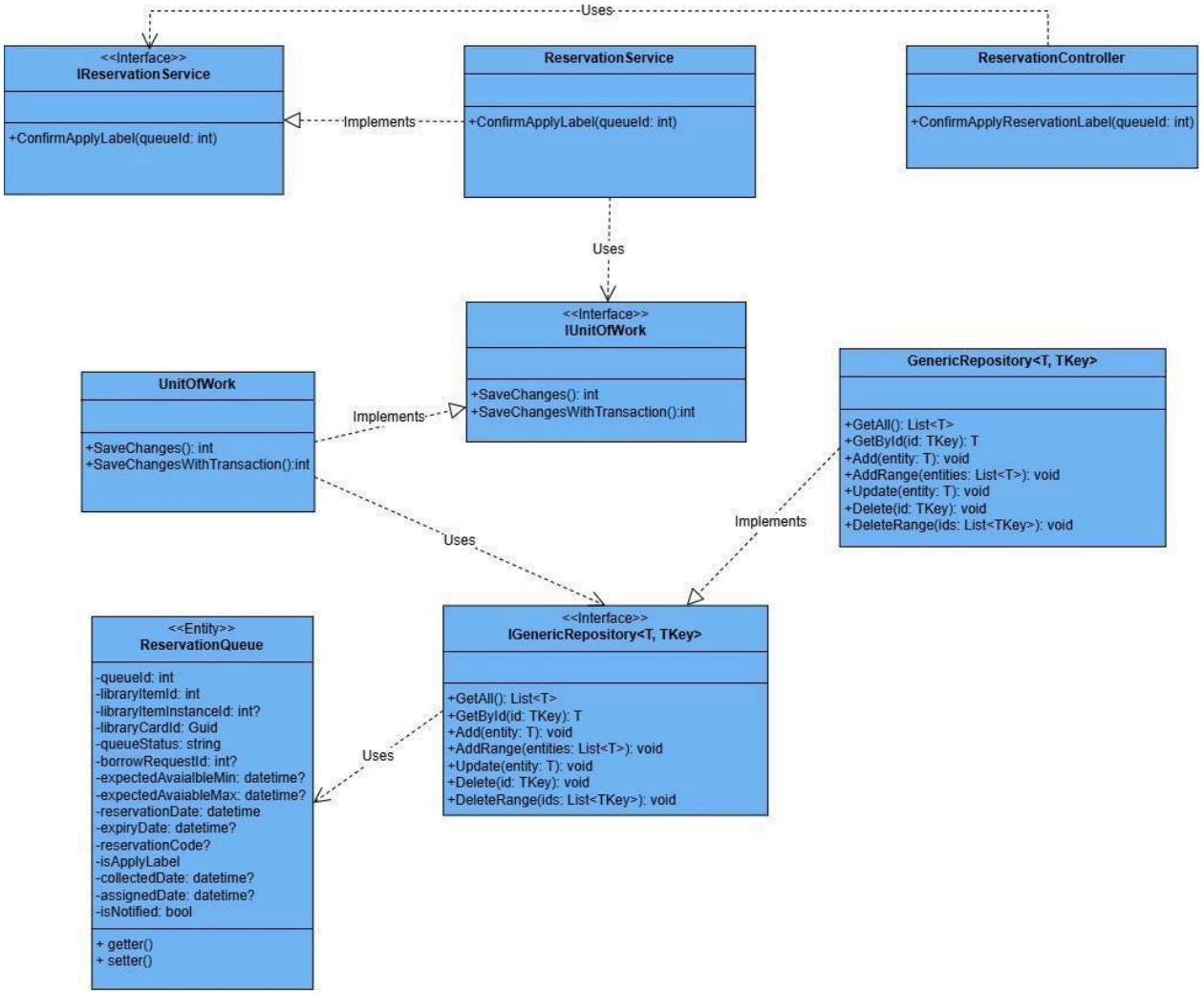


Figure 3.22.3.1 Class Diagram

3.22.3.2 Sequence Diagram

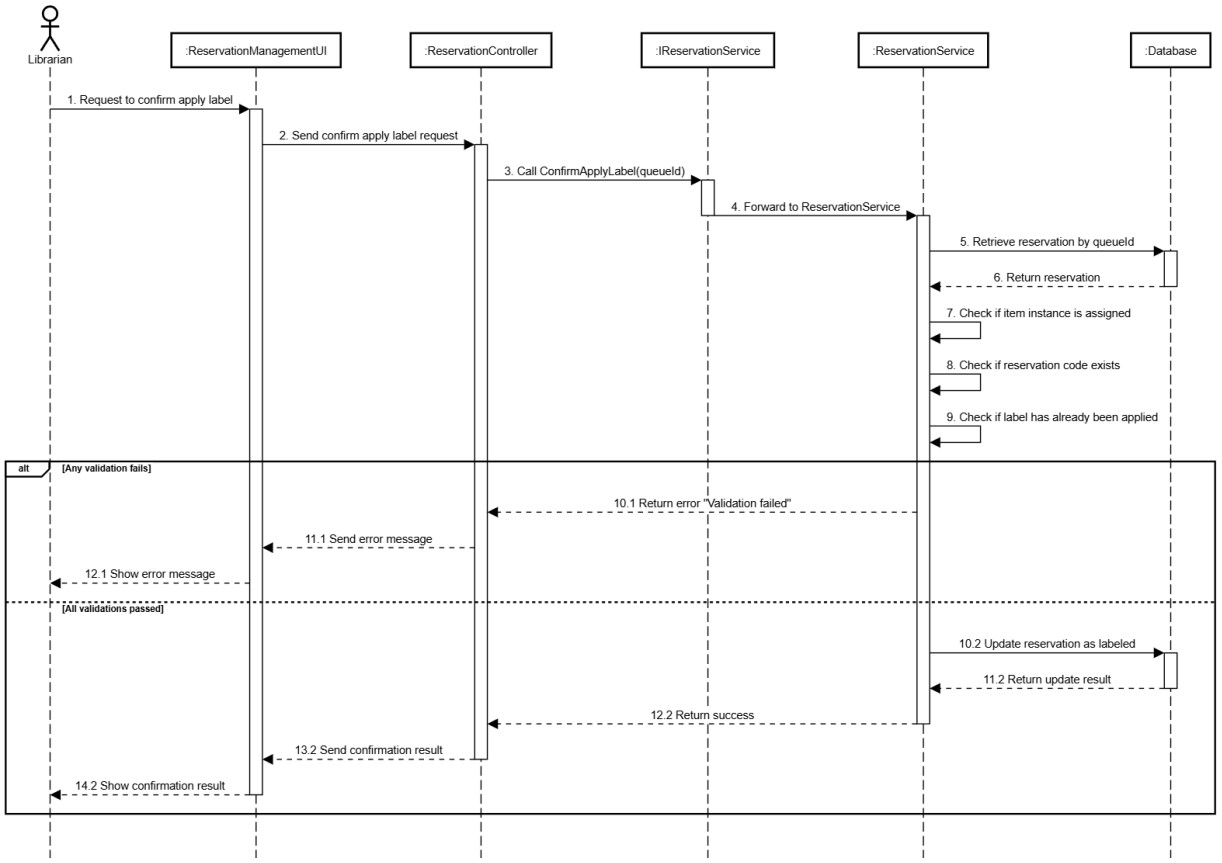


Figure 3.22.3.2 Sequence Diagram

3.22.3.3 Class Specification

3.22.3.3.a Reservation Controller

Table 3.22.3.3.a: Reservation Controller Description Table

No	Method	Description
1	ConfirmApplyReservationLabel(int queueId)	Receives request to mark a reservation's label as applied. Delegates to the service layer to update for specified queue entry.

3.22.3.3.b Reservation Service

Table 3.22.3.3.b: Reservation Service Description Table

No	Method	Description
1	ConfirmApplyLabel(int queueId)	Verifies the existence of the specific queue ID, and sets the applied label status to true.

3.23. Manage Digital Borrows

3.23.1. Pay For Digital Borrow

3.23.1.1 Class Diagram

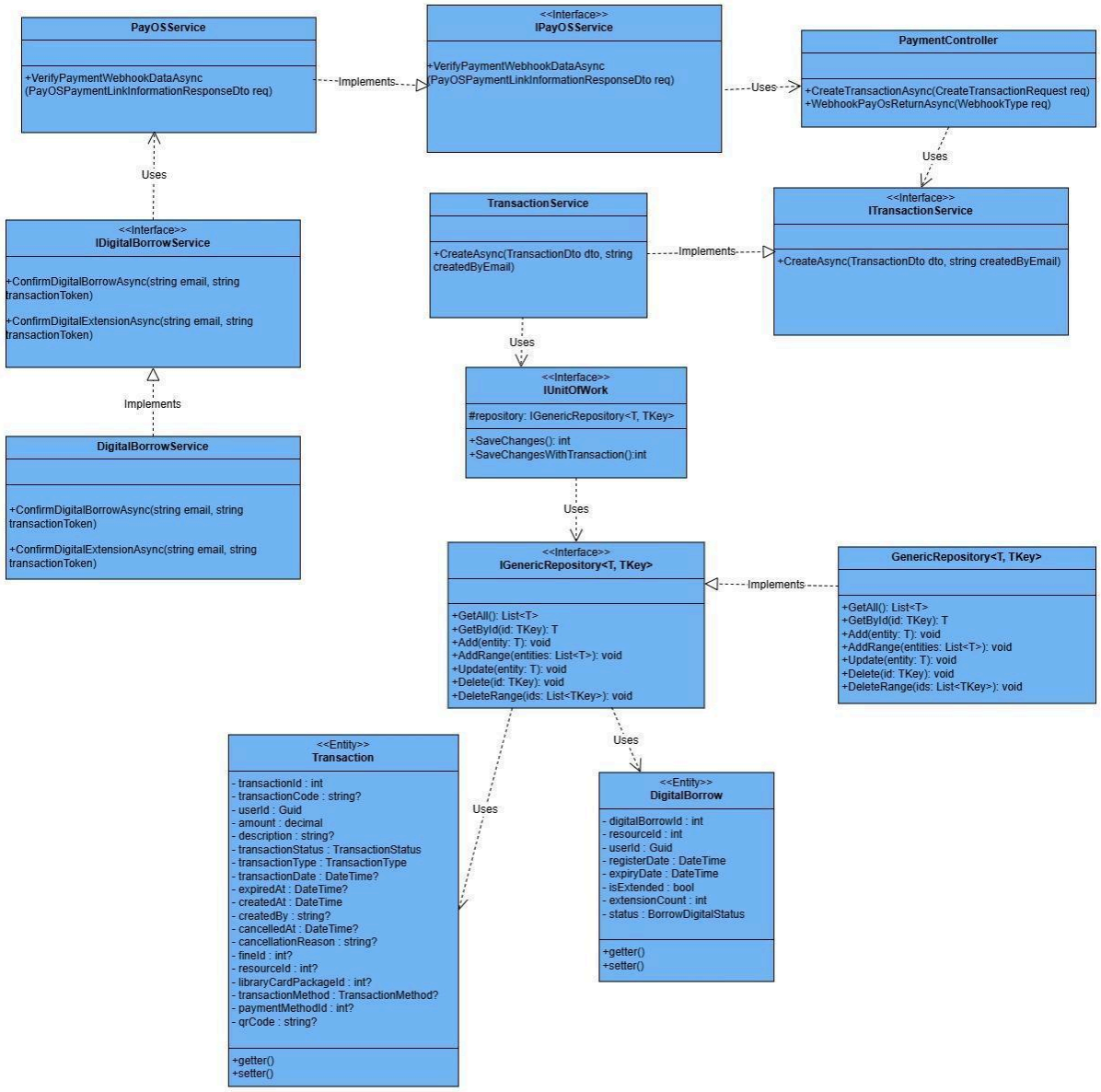


Figure 3.23.1.1 Class Diagram

3.23.1.2 Sequence Diagram

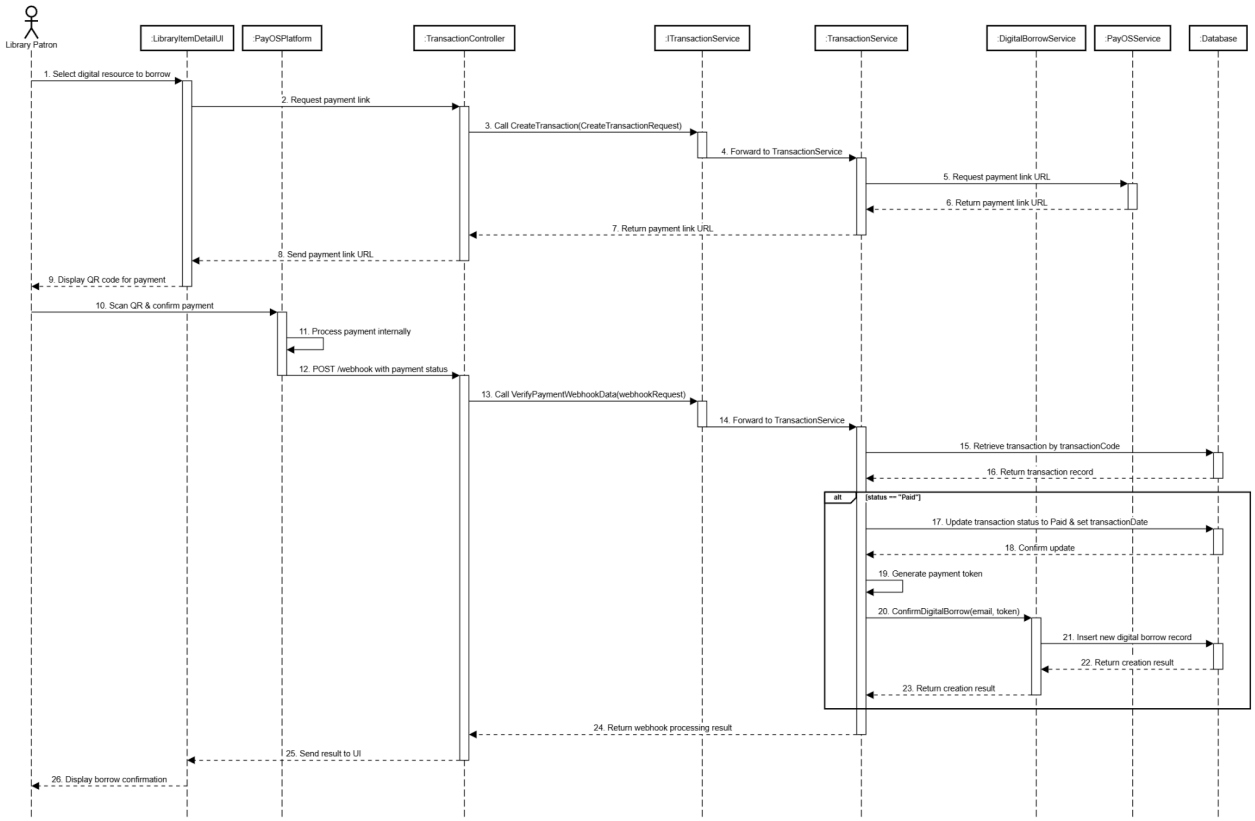


Figure 3.23.1.2 Sequence Diagram

3.23.1.3 Class Specification

3.23.1.3.a Payment Controller

Table 3.23.1.3.a: Payment Controller Description Table

No	Method	Description
1	CreateTransactionAsync(CreateTransactionRequest req)	Create transactions for users to pay to get complete borrowing digital resources process.
2	WebhookPayOsReturnAsync(WebhookType req)	Check status of the transaction to check borrowing digital resources process has finished or not

3.23.1.3.b Digital Borrow Service

Table 3.23.1.3.b: Digital Borrow Service Description Table

No	Method	Description

1	ConfirmDigitalBorrowAsync(string email, string transactionToken)	Receive transaction status to create a history of borrowing digital resources and give users permission to use them.

3.23.2. Get Digital Borrow

3.23.2.1 Class Diagram

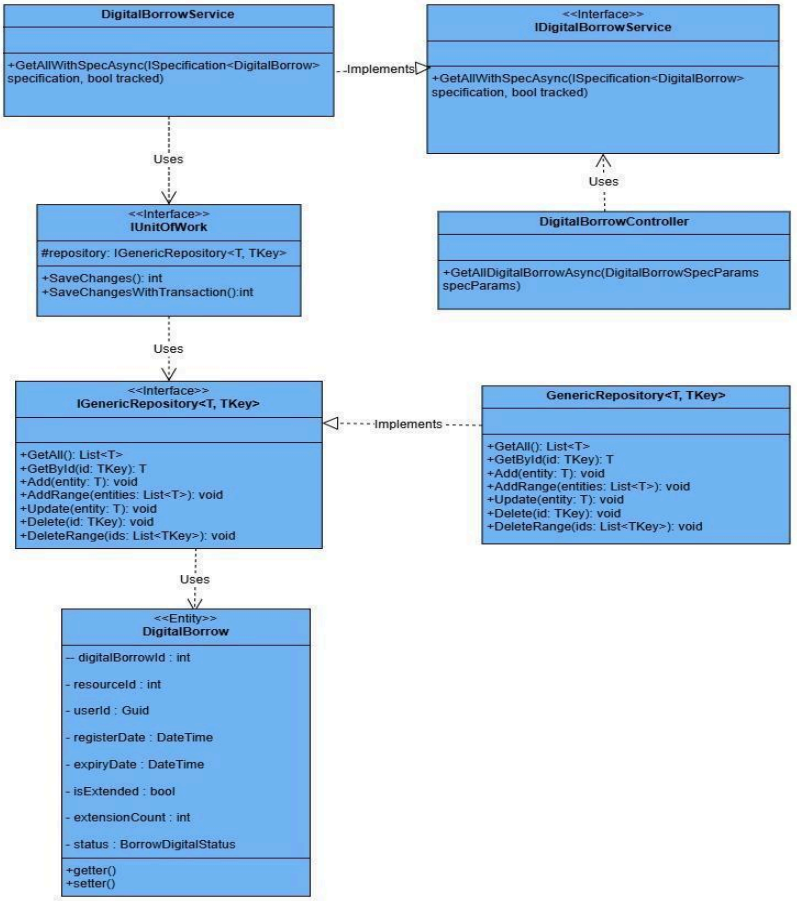


Figure 3.23.2.1 Class Diagram

3.23.2.2 Sequence Diagram

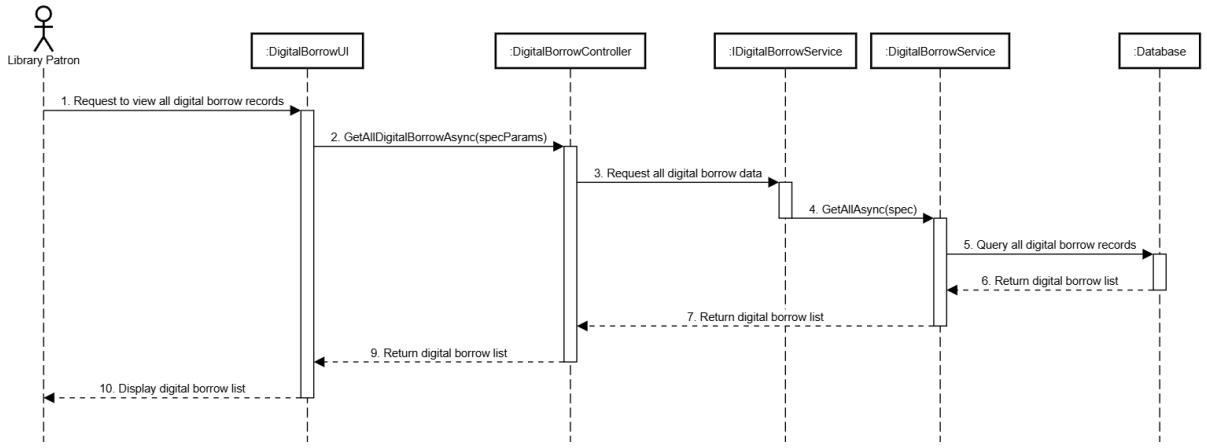


Figure 3.23.2.2 Sequence Diagram

3.23.2.3 Class Specification

3.23.2.3.a Digital Borrow Controller

Table 3.23.2.3.a: Digital Borrow Controller Description Table

No	Method	Description
1	GetAllDigitalBorrows(DigitalBorrowSpecParams specParams)	Get all digital borrow histories in the system and support searching and filtering.

3.23.2.3.b Digital Borrow Service

Table 3.23.2.3.b: Digital Borrow Service Description Table

No	Method	Description
1	GetAllWithSpec(ISpecification<DigitalBorrow>specification,bool tracked)	Get all digital borrow histories that suit conditions.

3.24. Manage Notification

3.24.1. Create Notification

3.24.1.1 Class Diagram

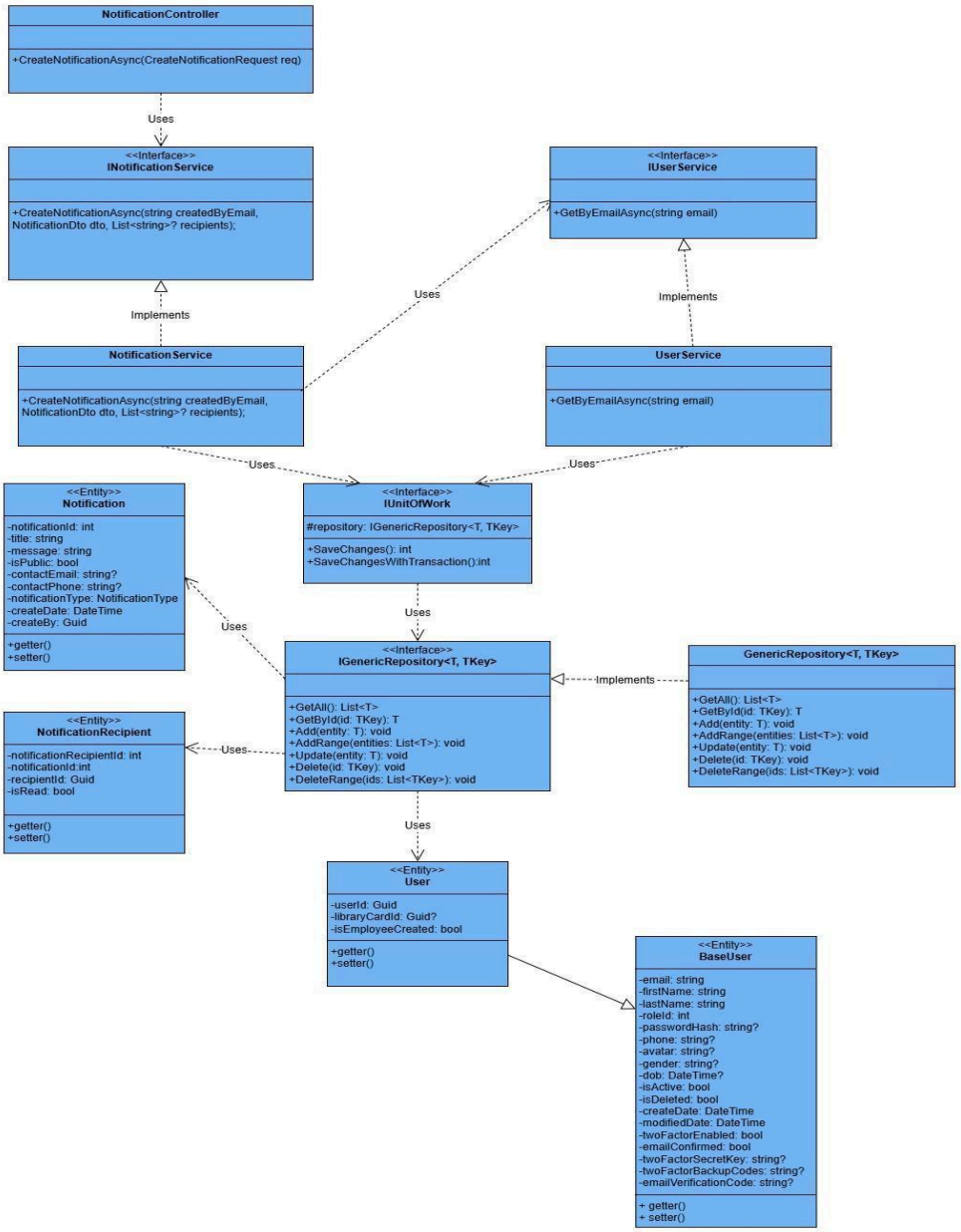


Figure 3.24.1.1 Class Diagram

3.24.1.2 Sequence Diagram

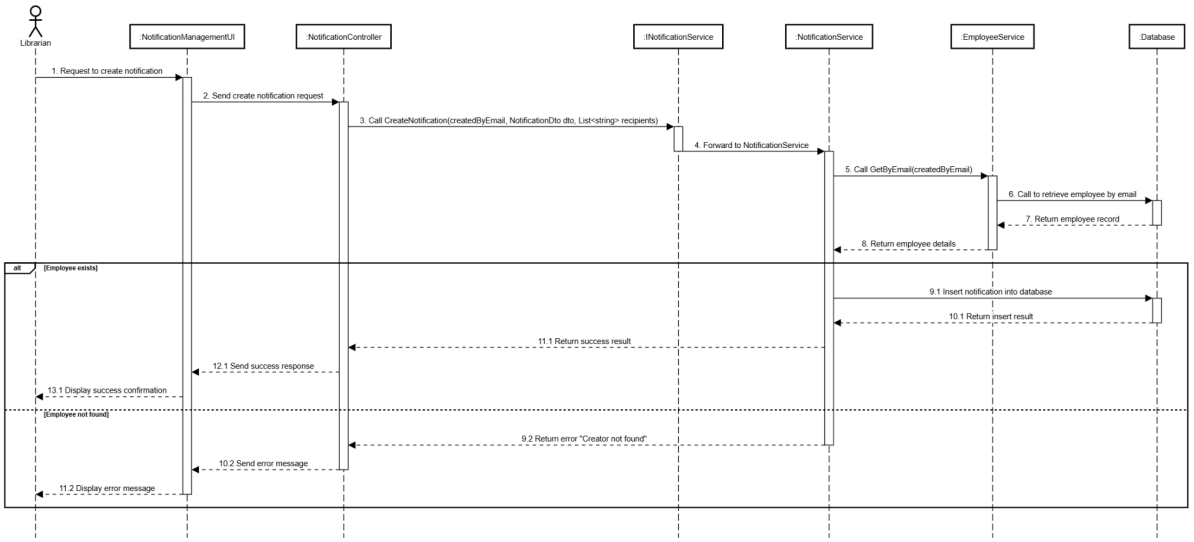


Figure 3.24.1.2 Sequence Diagram

3.24.1.3 Class Specification

3.24.1.3.a Notification Controller

Table 3.24.1.3.a: Notification Controller Description Table

No	Method	Description
1	CreateNotification(CreateNotificationRequest req)	Receives a request containing the new notification payload and delegates to the service layer.

3.24.1.3.b Notification Service

Table 3.24.1.3.b: Notification Service Description Table

No	Method	Description
1	CreateNotification(string createdByEmail, NotificationDto notificationDto, List<string>? recipients)	- Calls GetByEmail to confirm the creator exists. - Validate the notification and recipient list. - Process add new notification.

3.24.2. Get All Notification

3.24.2.1 Class Diagram

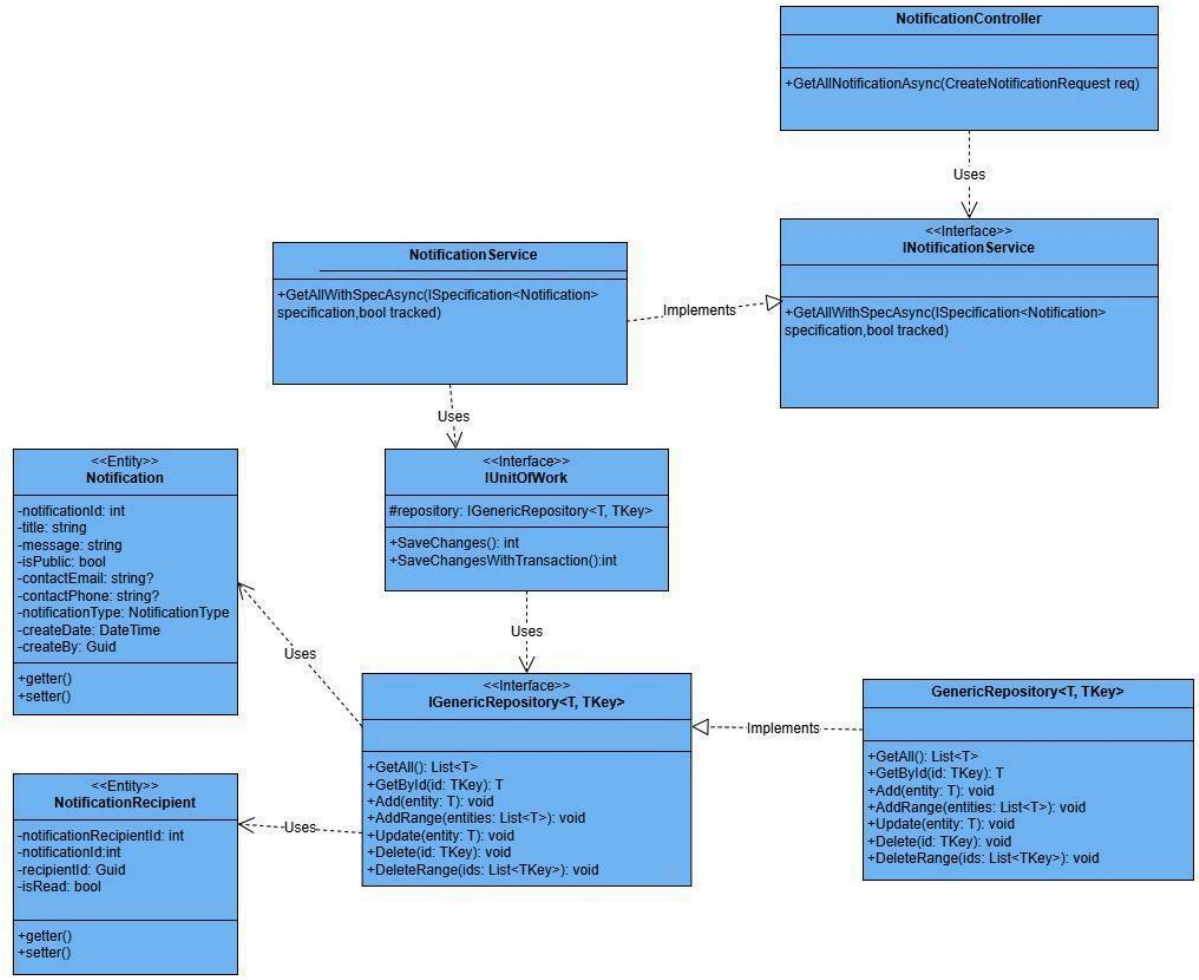


Figure 3.24.2.1 Class Diagram

3.24.2.2 Sequence Diagram

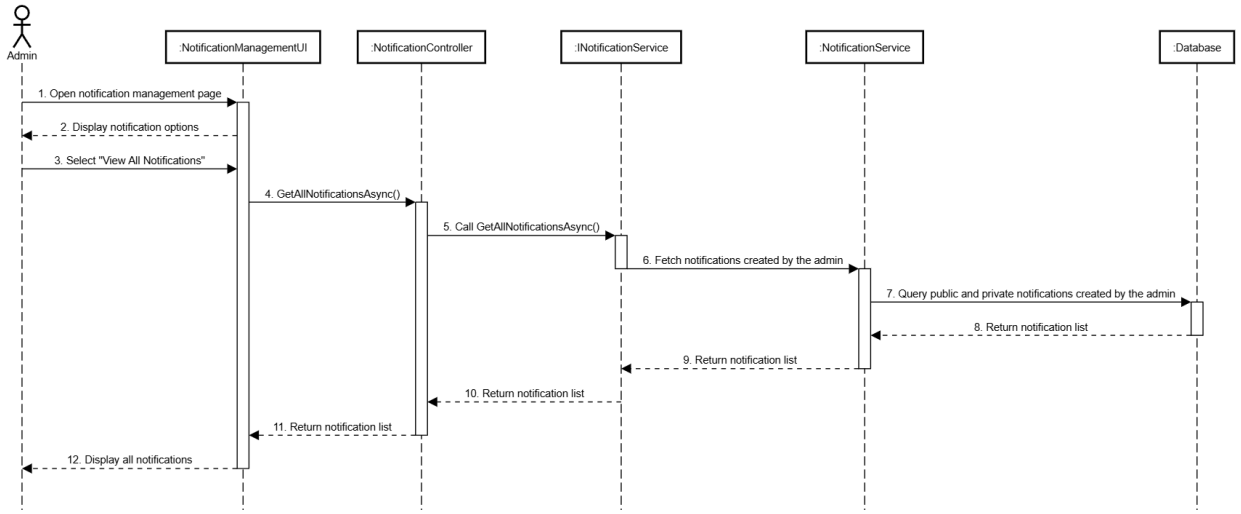


Figure 3.24.2.2 Sequence Diagram

3.24.2.3 Class Specification

3.24.2.3.a Notification Controller

Table 3.24.2.3.a: Notification Controller Description Table

No	Method	Description
1	GetAllNotification(NotificationSpecParams specParams)	Receives request to retrieve notifications. Accepts filtering, sorting, pagination parameters and delegates to the service layer.

3.24.2.3.b Notification Service

Table 3.24.2.3.b: Notification Service Description Table

No	Method	Description
1	GetAllWithSpec(NotificationSpecification spec)	Applies filtering, sorting, pagination to retrieve notifications.

3.24.3. Update Notification

3.24.3.1 Class Diagram

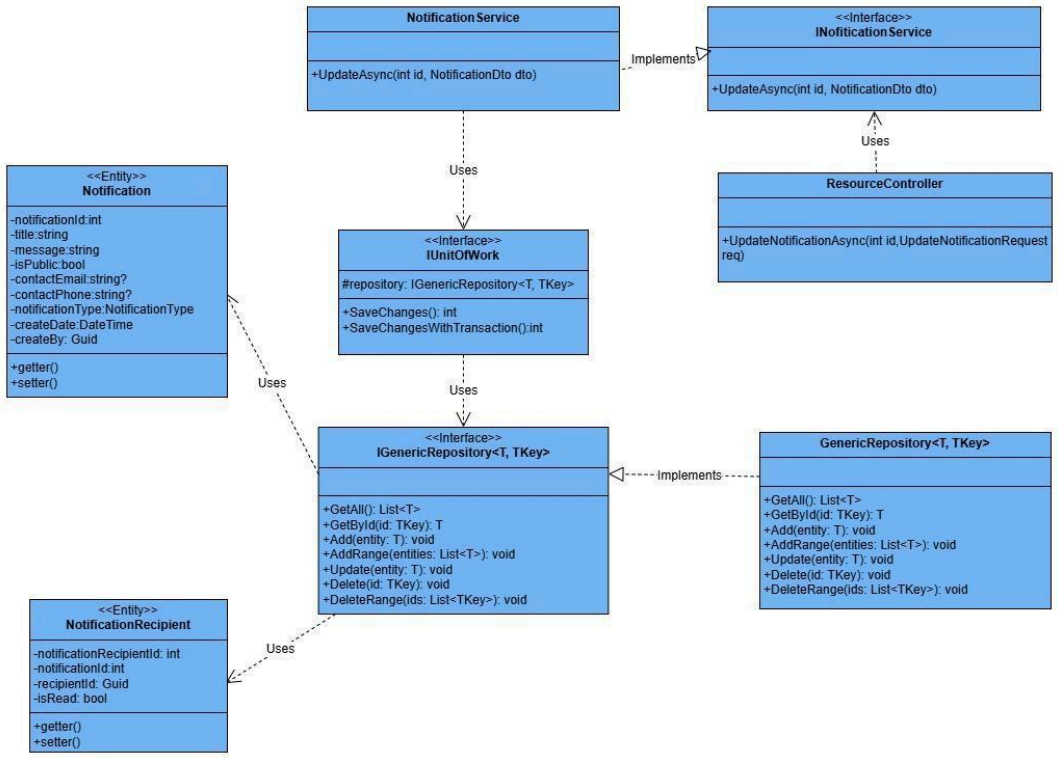


Figure 3.24.3.1 Class Diagram

3.24.3.2 Sequence Diagram

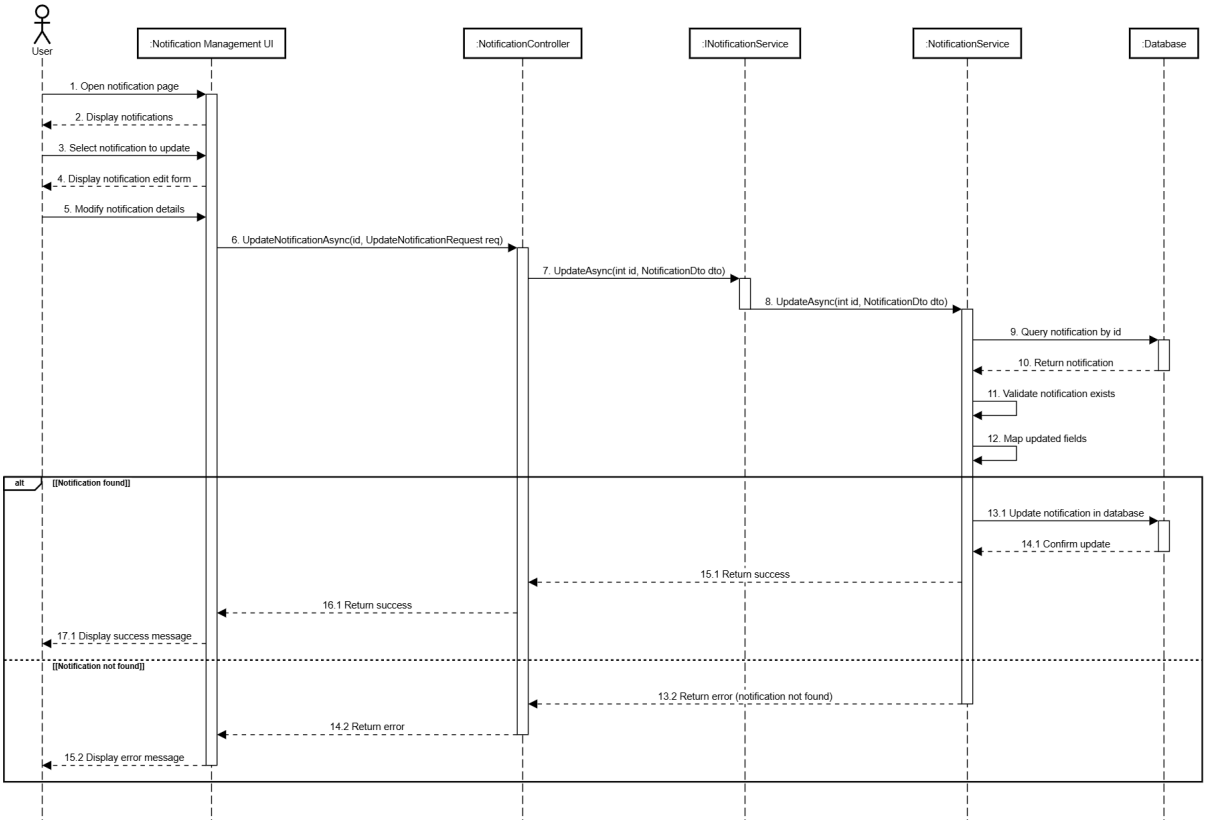


Figure 3.24.3.2 Sequence Diagram

3.24.3.3 Class Specification

3.24.3.3.a Notification Controller

Table 3.24.3.3.a: Notification Controller Description Table

No	Method	Description
1	UpdateNotification(int id, UpdateNotificationRequest req)	Receives request to update a nofition by its ID and delegates to the service layer.

3.24.3.3.b Notification Service

Table 3.24.3.3.b: Notification Service Description Table

No	Method	Description
1	Update(int id, NotificationDto notificationDto)	Validates that the notification with the given ID exists, applies the updated fields for existing notification.

V. Software Testing Documentation

1. Scope of Testing

Table 5.1 : Scope of Testing

Feature	Function	Role	Description
Authentication	Sign Up	Guest	Ensure the user can create a new account with valid information. The system shows an error for invalid inputs.
	Sign In with Password	All Roles	Check that the user can sign in with correct credentials.
	Sign In with OTP	All Roles	Ensure the one-time code is sent, accepted when correct, and rejected when expired or incorrect.
	Resend OTP	All Roles	Confirm the user can request a new OTP, and receive a different code.
	Forgot Password	All Roles	Test the password-reset flow: request link, receive email, follow link, set new password.
	Change Password	All Roles	Verify a registered user can update their password.
Profile Management	Update Profile	All Roles	Ensure all users can update their personal information.
	View Library Activity History	Library Patron	Ensure the library patron can access borrowing activities and interactions with the library system.

	View Transaction History	Library Patron	Ensure the library patron can review all transactions including borrows, returns, and payments with accurate date/time information.
	View Public/Private Notifications	All Roles	Ensure all users can access both public and their personal notifications.
Role Permission Management	Change Role Access Level	Admin	Ensure the admin can add or remove permissions for any role and that changes take effect immediately.
	Manage Role	Admin	Ensure the admin can create new roles, edit names/permissions, and delete roles.
User Management	Manage Accounts	Admin	Ensure the admin can create, edit, deactivate/reactivate user accounts, assign roles.
Dashboard	View Data Analytics Dashboard	Librarian, Head Librarian	Check that dashboard charts and tables load correctly and reflect the current data set and filters.
Supplier Management	Manage Suppliers	Head Librarian	Test CRUD of supplier records: add contact info, edit details.
Warehouse Tracking Management	Create Warehouse Stock-In	Head Librarian	Ensure the head librarian can create warehouse stock-in for new and additional library items with correct inventory counts.
	Create Warehouse Supplement Request	Head Librarian	Ensure the head library can submit for extra stock.

	Update Warehouse Tracking Status	Head Librarian	Check that the head librarian can update warehouse tracking status.
	Generate Warehouse Tracking Report	Head Librarian	Test that reports export correct data with expected fields and date ranges.
	View Warehouse Tracking Details	Librarian, Head Librarian	Ensure the librarian and the head librarian can view warehouse tracking information (quantities, dates, etc.)
	Apply Unique Barcode Registration	Librarian, Head Librarian	Ensure the librarian and head librarian can register unique barcodes to items.
Library Item Management	Manage Library Items	Librarian, Head Librarian	Ensure the librarian and head librarian can create, read, update, and delete item metadata (title, author, ISBN).
	Update Item Shelf Location	Librarian, Head Librarian	Ensure the librarian and head librarian can move items to new shelf locations.
	Train AI Image-based Detection	Librarian, Head Librarian	Ensure the librarian and head librarian can upload images, trigger an AI detection training, and view success or error messages.
	View AI Training History	Librarian, Head Librarian	Ensure the Librarian and Head Librarian can view history of AI detection training.
	Group Library Items	Librarian, Head Librarian	Ensure the librarian and head librarian can group items into collections.

	Get Personalized Library Item Recommendations	Library Patron	Ensure the library patron can view personalized item suggestions based on their borrowing history and preferences.
	Predict Library Items using AI Image-based detection	All Roles	Ensure all roles can upload an item cover to receive predicted item matches with confidence scores.
	Search Library Items with Quick, Basic, and Advanced Filtering	All Roles	Ensure all roles can perform quick, basic, and advanced searches, see correct results.
	Add Library Items to Favorite	Library Patron	Ensure the library patron can mark and unmark favorites and immediately see their updated favorites list.
	Review Library Items	Library Patron	Ensure the library patron can add, edit, and delete reviews or ratings.
Library Item Instance Management	Update Item Instance Shelf Status	Librarian, Head Librarian	Ensure the Librarian and Head Librarian can update an item instance's status (out-of-shelf, on-shelf, etc.).
	Mark Item Instance as Lost	Librarian, Head Librarian	Ensure the librarian and head librarian can mark instances as lost, preventing check-outs.
	Mark Lost Item Instance as Found	Librarian, Head Librarian	Ensure the librarian and head librarian can restore lost instances to available status.

Category Management	Manage Categories	Librarian, Head Librarian	Ensure the librarian and head librarian can create, edit, and delete categories.
Author Management	Manage Authors	Librarian, Head Librarian	Ensure the librarian and head librarian can add, edit, and delete authors.
Condition Management	Manage Conditions	Librarian, Head Librarian	Ensure the librarian and head librarian can manage item condition statuses (new, good, worn, etc.).
Fine Policy Management	Manage Fine Policies	Librarian, Head Librarian	Ensure the librarian and head librarian can create and edit fine rules.
Library Resource Management	Manage Resource In Cloud Storage	Librarian, Head Librarian	Ensure the librarian and head librarian can upload, retrieve, and delete digital resources in cloud storage.
	Preview Ebook, Audio	Guest, Library Patron	Ensure the guest and library patron can preview ebook and audio resources.
	View Watermarked Borrowed Resource	Library Patron	Ensure the library patron can open their borrowed digital resource in full, with a visible watermark on each page.
Library Card Package Management	Manage Library Card Packages	Librarian, HeadLibrarian	Ensure the librarian and head librarian can issue, view, update, and remove library cards.
Library Card Management	Manage Library Cards	Librarian, HeadLibrarian	Ensure librarians could use CRUD features to manage

			library cards, suspend or unsuspend cards.
	Extend Borrow Amount	Librarian, HeadLibrarian	Ensure the librarian and head librarian can increase a card's borrow limit.
	Confirm Card Register	Librarian, HeadLibrarian	Ensure the librarian and head librarian can review pending card requests, confirm registrations.
	Register Library Card	Library Patron	Ensure the library patron can apply for a new library card by submitting required details and receiving confirmation of approval.
	Extend Library Card Expiration	Library Patron	Ensure the library patron can request an extension of their card's expiration date.
Library Patron Management	Manage Library Patrons	Librarian, HeadLibrarian	Ensure librarians and head librarians can add, edit, or deactivate patron accounts.
	View Library Activity History	Librarian, HeadLibrarian	Ensure the librarian and head librarian can access comprehensive records of all patron activities.
Borrow Request Management	Create Borrow Requests	Library Patron	Ensure the library patron can create borrow requests for available items, digital resources or reserve for unavailable items.
	Cancel Specific Borrow Item	Library Patron, Librarian, Head Librarian	Ensure users can cancel pending borrow request items.

	Cancel Specific Digital Borrow	Library Patron, Librarian, Head Librarian	Ensure users can cancel pending borrow digital resources.
	Cancel Specific Reservation	Library Patron, Librarian, Head Librarian	Ensure users can cancel specific pending reservations.
	Manage Borrow Requests	Librarian, Head Librarian	Ensure the librarian and head librarian can view, sort all active borrow requests.
Borrow Record Management	Manage borrow Records	Librarian, Head Librarian	Ensure the librarian and head librarian can create, modify, add items to borrow records.
	Process Return Items	Librarian, Head Librarian	Ensure the librarian and head librarian can complete the return items process, update inventory, and verify item condition.
	Extend Borrow Record Due Date	Library Patron	Ensure the library patron can request extensions for current borrows.
	View All Borrow Policies	All Roles	Ensure all users can access comprehensive information about borrowing rules, limits.
Reservation Management	View All Assignable Reservations	Librarian, Head Librarian	Ensure the librarian and head librarian can view a complete list of pending reservations.
	Assign Item Instance to Pending Reservation	Librarian, Head Librarian	Ensure the librarian and head librarian can assign library item instances for pending reservations

	Assign Returned Item to Pending Reservation	Librarian, Head Librarian	Ensure the librarian and head librarian can assign returned item instances for pending reservations.
	Apply Reservation Hold Slip Label	Librarian, Head Librarian	Ensure the librarian and head librarian can generate and attach hold labels to reserved items.
Digital Borrow Management	Manage Digital Borrows	Librarian, Head Librarian	Ensure the librarian and head librarian can oversee all digital resource borrows.
	Extend Digital Borrow Expiration	Library Patron	Ensure the library patron can request extensions for digital resources.
Library Shelf Management	Manage Library Shelves	Librarian, Head Librarian	Ensure the librarian and head librarian can organize items in appropriate shelves based on classification number (DDC).
Notification Management	Manage Notifications	Librarian, Head Librarian	Ensure the librarian and head librarian can send private or public notifications for users.
	View Public Notifications	All Roles	Ensure all users can view all system public notifications.
	View Privacy Notifications	Library Patron	Ensure the library patron can view their privacy notifications.
Transaction Management	Impose Fine for Overdue Return Items	Librarian, Head Librarian	Ensure the librarian and head librarian are able to impose fines for overdue return items.

	Create Transaction for Card Register or Extension	Library Patron	Ensure the library patron can complete library card registration or extension transactions.
	Create Transaction for Digital Borrow or Extension	Library Patron	Ensure the library patron can complete digital resource borrow or extension transactions.
	View Transaction History	Library Patron	Ensure the library patron can view their transactions.
	Manage Transactions	Librarian, Head Librarian	Ensure the librarian and head librarian can oversee all system transactions.
Fine Management	Manage Fines	Librarian, Head Librarian	Ensure the librarian and head librarian can view all pending, paid, expired fines.
	Impose Fine for Overdue Return Items	Librarian, Head Librarian	Ensure librarians and head librarians are able to impose fines for overdue return items.
System Configuration Management	Manage system configuration	Admin	Ensure the admin could manage the system configuration (page size, borrowing duration, etc.) with CRUD based on the library business rule.
Library Location	View 2D indoor-map navigation	All Roles	Ensure all users can access and navigate interactive 2D indoor navigation maps.

2. Test Strategy

2.1 Testing Types

Table 5.2.1 : Testing Types

Objective	Technique	Completion criteria
Verify and validate each individual code units (functions, classes)	Unit testing	Most of the methods and classes covered by tests.
Testing application environment in real situations	End-to-end testing	- Database CRUD operations validated. - External integrations (payment, map, etc.) response properly.
Check correct interaction among modules	Integration testing	Data passed between modules matches expected schemas.
Testing how parts of a system communicate and exchange data.	Interface testing	Requests, response, etc.

2.2 Test Levels

Table 5.2.2 : Test Levels

Type of Tests	Test Level			
	Unit	Integration	System	Acceptance
Unit testing	X			
Interface testing	X	X		
Integration testing		X		
End-to-end testing				X

2.3 Supporting Tools

Table 5.2.3: Supporting Tools

Purpose	Tool	Vendor/In-house	Completion criteria
API testing	Postman/Swagger	Open Source	Latest stable

3. Test Plan

3.1 Human Resources

Table 5.3.1 Human Resources

Worker/Doer	Role	Specific Responsibilities/Comments
Lê Xuân Phước	Leader	Planning, assigning test functions to members and verifying test results.
Trần Trương Văn	Member	Execute test and report.
Đoàn Viết Thanh	Member	Execute test and report.
Nguyễn Vũ Quang Huy	Member	Execute test and report.

3.2 Test Environment

Table 5.3.2 : Test Environment

Purpose	Tool	Provider	Version
API testing	Window	Microsoft	Latest stable

3.3 Test Milestones

Table 5.3.3 : Test Milestones

Milestone Task	Start Date	End Date
Unit test		
Other Test Case (IT, ST, AT)		

4. Test Cases

- Unit Test Cases: *Report5_Unit Test.xls*

UNIT TEST REPORT

Project Name	*E-Library - Intelligent Library System*	**Creator**	PhuocLX
Project Code	*SP25SE126*	**Reviewer/Approver**	
Document Code	*SP25SE126_Test Report_v2.1*	**Issue Date**	*4/5/2025*
Notes			

No	Function code	Passed	Failed	Untested	N	A	B	Total Test
1	FO01	18	0	0	15	3	0	18
2	FO02	14	1	0	4	4	7	15
3	FO03	12	0	0	8	1	3	12
4	FO04	15	0	0	6	5	4	15
5	FO05	14	0	5	13	0	1	19
6	FO06	15	0	0	11	2	2	15
7	FO07	7	0	0	6	1	0	7
8	FO08	7	2	0	0	2	7	9
9	FO09	9	0	0	7	2	0	9
10	FO10	8	0	0	8	0	0	8
11	FO11	13	0	0	11	2	0	13
12	FO12	11	0	6	3	4	4	17
13	FO13	5	0	0	5	0	0	5
14	FO14	14	0	0	12	2	0	14
15	FO15	4	0	0	2	1	1	4
16	FO16	7	0	0	5	2	0	7
17	FO17	13	0	0	7	2	4	13
18	FO18	12	0	6	6	6	0	12
19	FO19	16	0	0	13	3	0	16
20	FO20	10	2	0	9	3	0	12
21	FO21	12	0	0	9	3	0	12
22	FO22	10	2	0	9	3	0	12
23	FO23	12	0	0	6	1	5	12
24	FO24	12	0	0	9	3	0	12
25	FO25	9	3	0	9	3	0	12
26	FO26	12	0	0	0	0	0	12
27	FO27	12	0	0	9	3	0	12
28	FO28	10	2	0	9	3	0	12
29	FO29	8	0	0	8	0	0	8
30	FO30	9	0	0	9	0	0	9
31	FO31	12	0	0	12	0	0	12
32	FO32	8	0	0	8	0	0	8
33	FO33	8	0	0	8	0	0	8
34	FO34	12	0	0	9	3	0	12
35	FO35	12	0	0	9	3	0	12
36	FO36	12	0	0	9	3	0	12
37	FO37	12	0	0	9	3	0	12
38	FO38	12	0	0	9	3	0	12
	Sub total	418	12	17	231	79	38	441

Test coverage	**97,51 %**
Test successful coverage	**94,78 %**
Normal case	**52,38 %**
Abnormal case	**17,91 %**
Boundary case	**8,62 %**

Figure 5.4.1 : Unit test case

- Test Cases and Test Reports: *Report5_Test Report*

TEST STATISTICS

Project Name	*E-Library - Intelligent Library System*	Creator		PhuocLX
Project Code	*SP25SE126*	Reviewer/Approver		
Document Code	*SP25SE126_Test Report_v2.1*	Issue Date		10/4/2025
Notes				

8	Module code	Passed	Failed	Pending	N/A	Number of test cases
1	Sign Up	9	0	0	0	9
2	Sign In with Password	7	0	0	0	7
3	Sign In with OTP	9	0	0	0	9
4	Create Stock In Warehouse Tracking	9	0	0	0	9
5	Create Supplement Warehouse Tracking	8	0	0	0	8
6	Add Item to Warehouse Tracking	8	0	1	0	9
7	Update Warehouse Tracking Status	7	0	0	0	7
8	Update Barcode Registration	9	0	0	0	9
9	Create Library Item	11	2	0	0	13
10	Get All Library Item	12	2	0	0	14
11	Update Library Item Status	8	0	0	0	8
12	Check Library Item with OCR	3	0	0	0	3
13	Search Library Item with Elastic	9	0	1	0	10
14	Get Full PDF File with Watermark	6	0	0	0	6
15	Get Full Audio File with Watermark	7	0	0	0	7
16	Add Resource to Library Item	9	0	0	0	9
17	Update Item Shelf Location	7	0	0	0	7
18	Update In Shelf	7	0	0	0	7
19	Mark Item Instance as Lost	7	1	0	0	8
20	Get All Suitable Items for Grouping	7	1	0	0	8
21	Train AI Image-based Detection	9	2	0	0	11
22	Predict Library Item by Image	5	2	0	0	7
23	Create Library Patron	10	0	0	0	10
24	Get All Library Patron	7	0	0	0	7
25	Register Library Card	7	0	0	0	7
26	Get All Library Card	8	0	0	0	8
27	Confirm Library Card Registration	6	0	0	0	6
28	Extend Library Card Expiration	7	0	0	0	7
29	Create Transaction	8	0	0	0	8
30	Get All Transaction	7	0	0	0	7
31	Create Borrow Request	8	0	0	0	8
32	Get All Borrow Request	8	0	0	0	8
33	Create Borrow Record	13	0	0	0	13
34	Get All Borrow Record	10	0	0	0	10
35	Process Return	9	0	0	0	9
36	Get All Assignable Reservation	7	0	0	0	7
37	Assign Item Instance to Reservation	8	0	0	0	8
38	Get Dashboard Circulation And Borrowing Analytics	7	0	0	0	7
39	Get Dashboard Financial And Transaction Analytics	7	0	0	0	6
40	Get Top Circulation Items	8	0	0	0	8
	Sub total	318	10	2	0	329

Test coverage	99,70 %
Test successful coverage	96,66 %

Figure 5.4.2 : Test Statistic

VI. Release Package & User Guides

1. Deliverable Package

No.	Deliverable Item	Description
1	Schedule/Task Tracking	We use Jira, Google Sheet, Google meet to track our progress, Messenger and Notepad to store schedules, notes and Google Doc to write the documents: • **Jira** • **Google meet** • **Messenger**

		• **Notepad** • **Google sheet** • **Google Doc**
2	Project Backlog	The project backlog is managed using Jira for task tracking and prioritisation.
3	Source Codes	We have multiple private repositories for different parts of the application: • Frontend website: [Github ELibrary Web] • Mobile: [Github ELibrary Mobile] • Backend: [Github ELibrary Server]
4	Database Script(s)	The database scripts are stored in the project repository. The database used is **SQL Server** for managing all data.
5	Final Report Document	The final report is prepared using Google Docs and is stored in Google Drive for sharing and version control.
6	Test Cases Document	The test case documentation is managed using Google Docs and Google Sheets. It includes all test cases for functionality validation.
7	Defects List	The defect list is maintained in Jira for tracking and resolution.
8	Issues List	The issues list is managed in GitHub (via Issues tab) and Jira for task management and team coordination.
9	Slide	Presentation slides are designed using Canva and stored in Google Drive.

2. Installation Guides

2.1 System Requirements

2.1.1 Hardware requirements

Table 6.2.1.1 : Hardware requirements

	Minimum	Recommended
Internet connection	LAN Cable, Wi-Fi (4 Mbps)	LAN Cable, Wi-Fi (4 Mbps)
Processor	1.8GHz	Multi-core

Memory	4GB	8GB or higher
Storage	128GB	256GB or higher

2.1.2 Software requirements

Table 6.2.1.2 : Software requirements

	Name/Version	Description
Operating System	Windows 10 or higher	OS for PC
Browser	Latest version	Browser for web application
.NET SDK	.NET 8 SDK	.NET Standard Development Kit
Nodejs Runtime	Version 18 or higher	
SQL Server	Version 19 or higher	
IDEs	Visual Studio Code, Visual Studio 2022, Jetbrains Rider 2024	
Version Control	GitHub	
Deployment Servers	Azure Container Registry, AWS Amplify	

2.2 Installation Instruction

2.2.1 Front-end

- Step 1: Download & Install Visual Studio Code

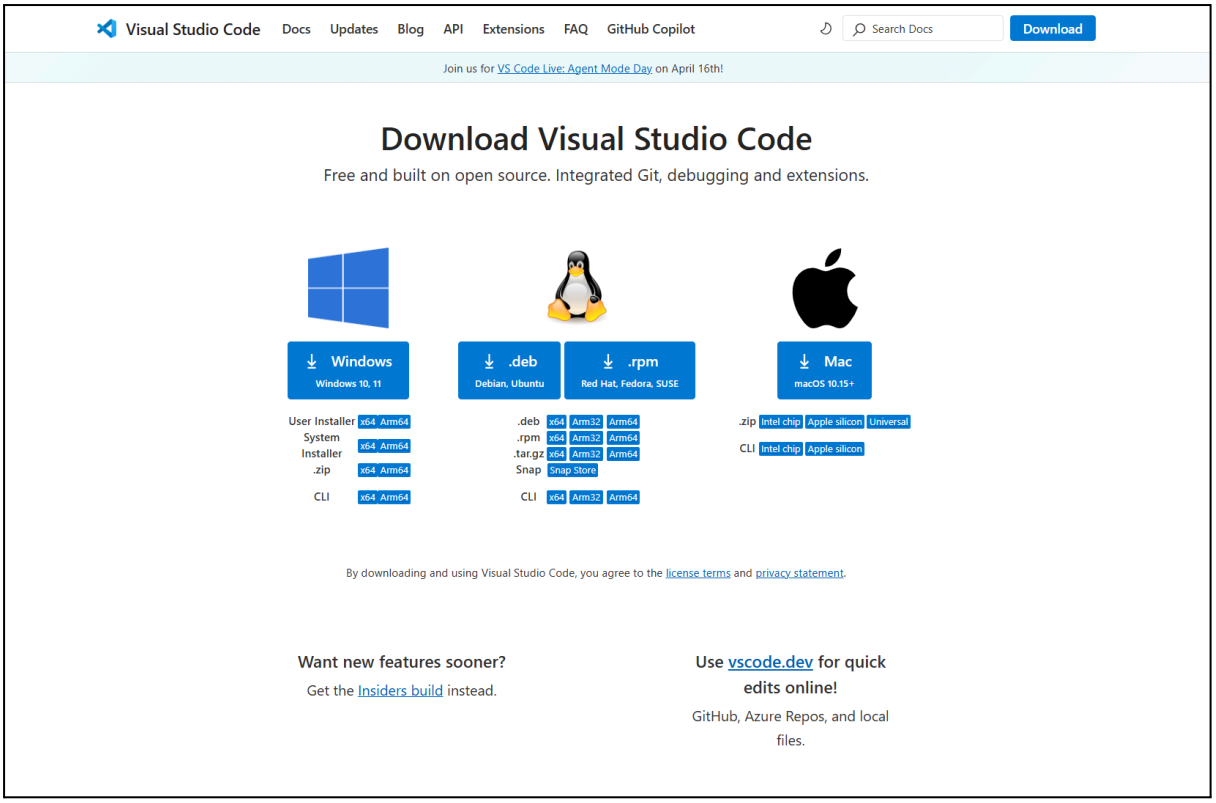


Figure 6.2.2.1.1: Front-end

- Step 2: [Download and install git]

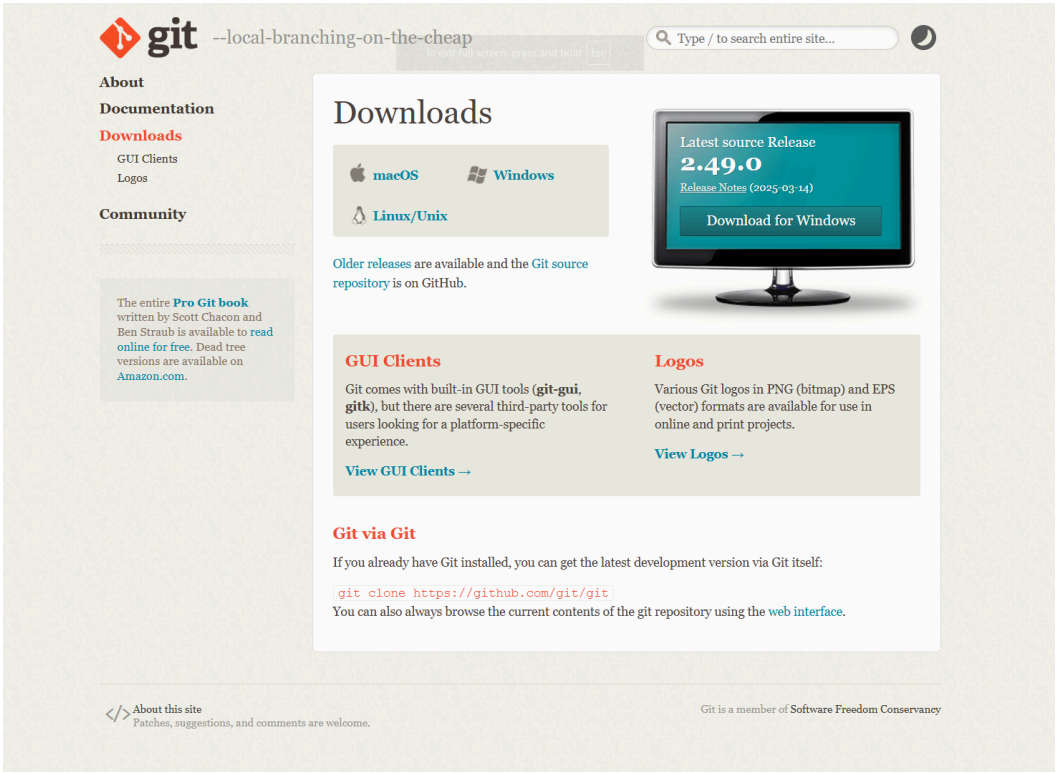


Figure 6.2.2.1.2: Front-end

- Step 3: Move to the specific folder, then open the terminal

  - Right click to the specific folder, then open the terminal and enter the command line:

  “git clone git@github.com:KingChen711/FPT_Library_Web.git”

  - Type “code .” in the terminal window to open the source code.

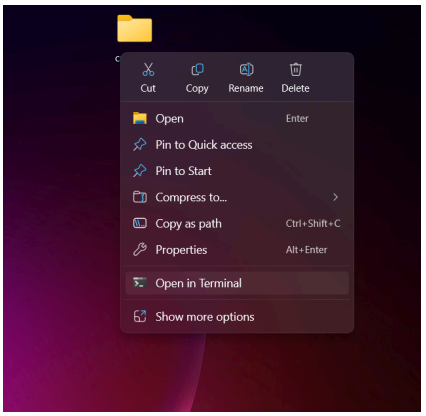


Figure 6.2.2.1.3: Front-end

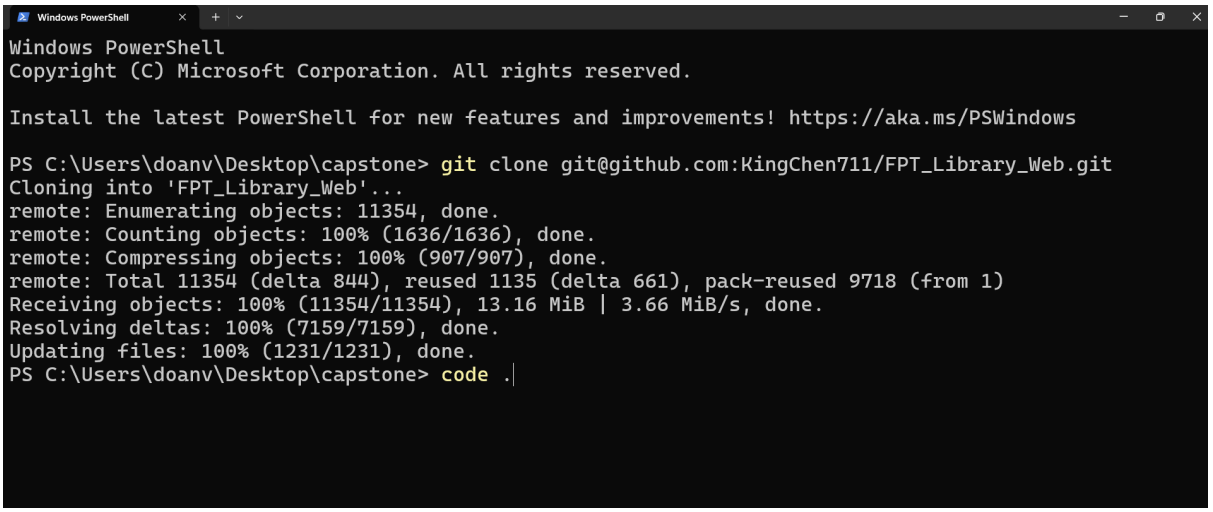


Figure 6.2.2.1.4: Front-end

- Step 4: To install the library packages (node_modules), enter the command line “*npm install --legacy-peer-deps*”

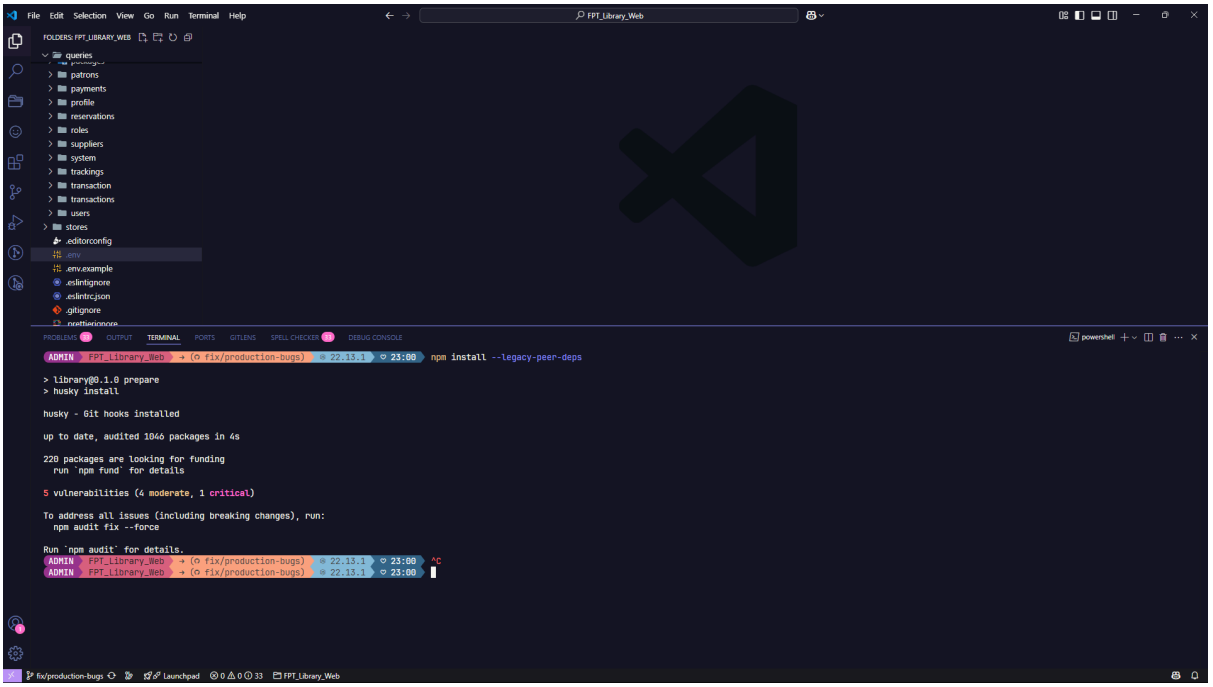


Figure 6.2.2.1.5: Front-end

- Step 5: Create the .env file to verify some keys

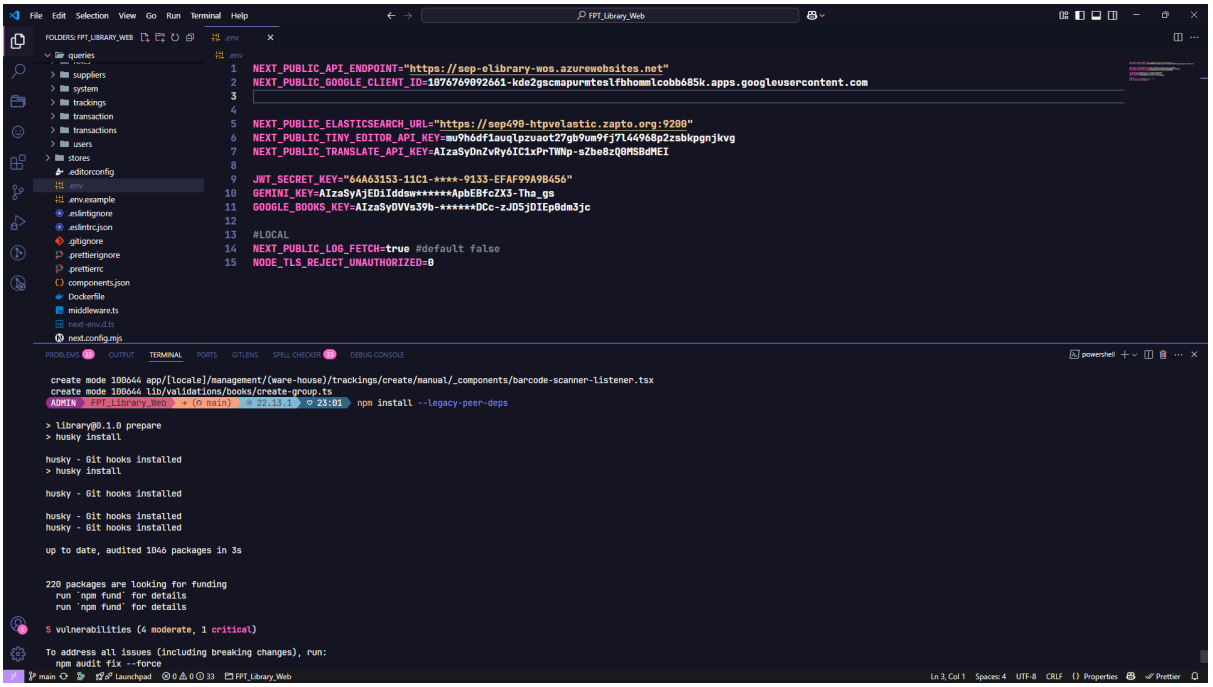


Figure 6.2.2.1.6: Front-end

- Step 6: Run command line "*npm run dev*" to run the frontend source code at the http://localhost:3000/.

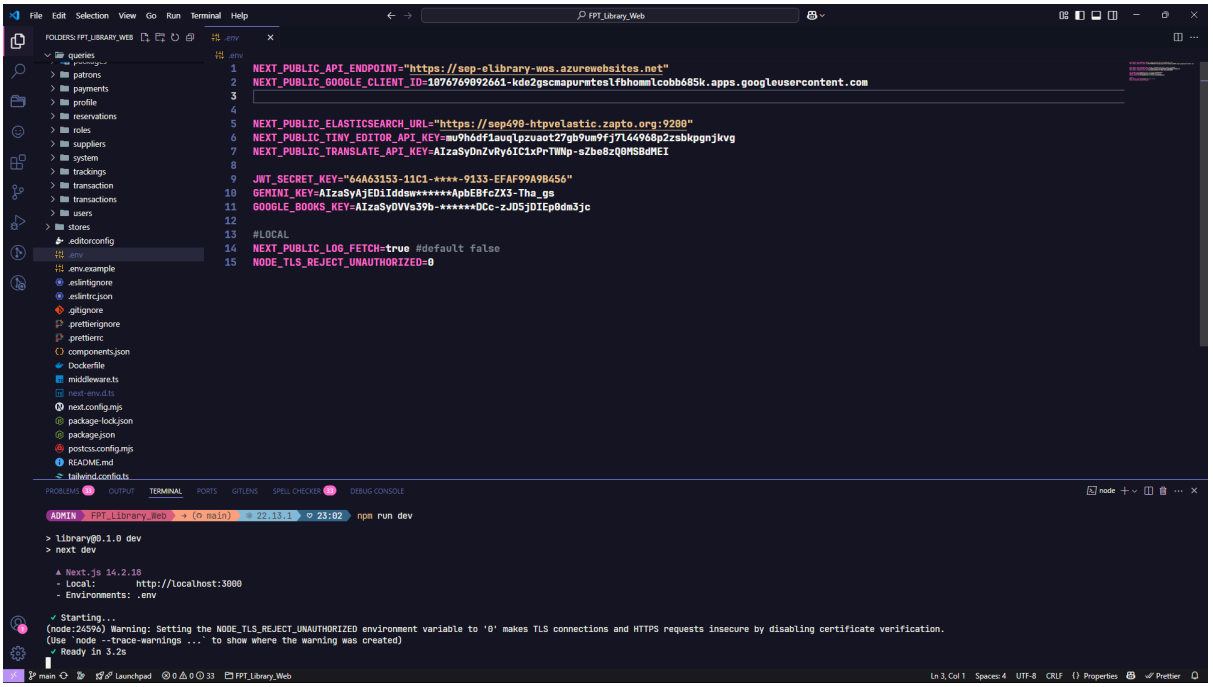


Figure 6.2.2.1.7: Front-end

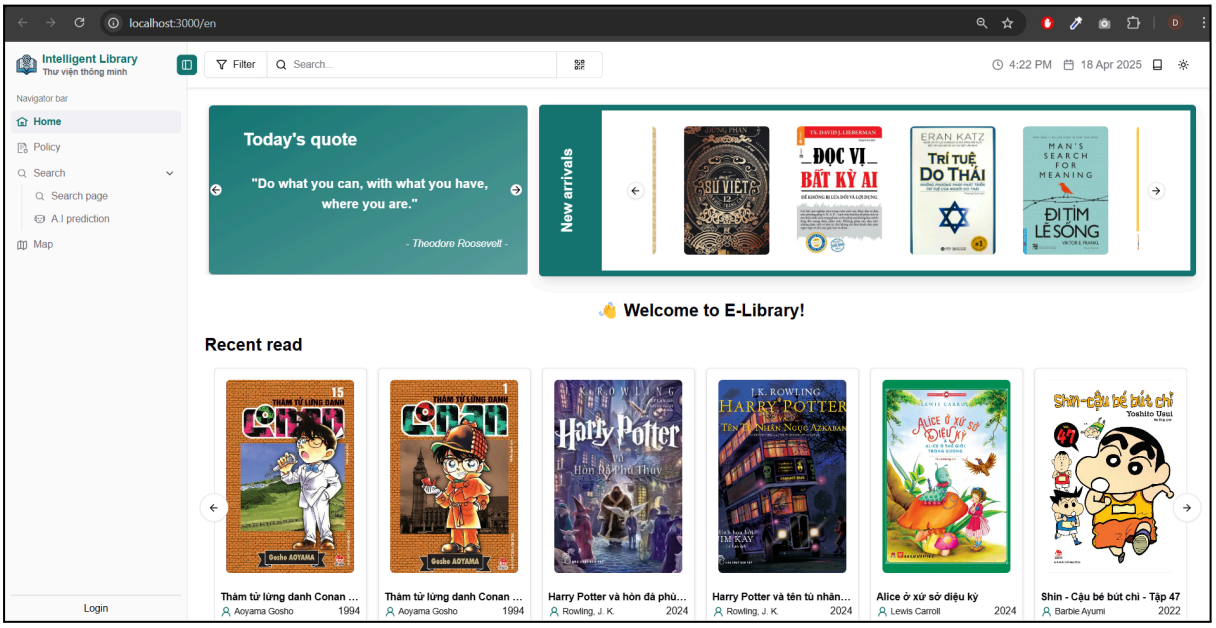


Figure 6.2.2.1.8: Front-end

2.2.2 Mobile

- Step 1: Download & Install Visual Studio Code
- Step 2: Download and install git
- Step 3: There are 2 ways to setup the device to run the Mobile project (Download Android Studio or Download Expo Go on ChPlay)

- **Android Studio**

+ [Download Android Studio]

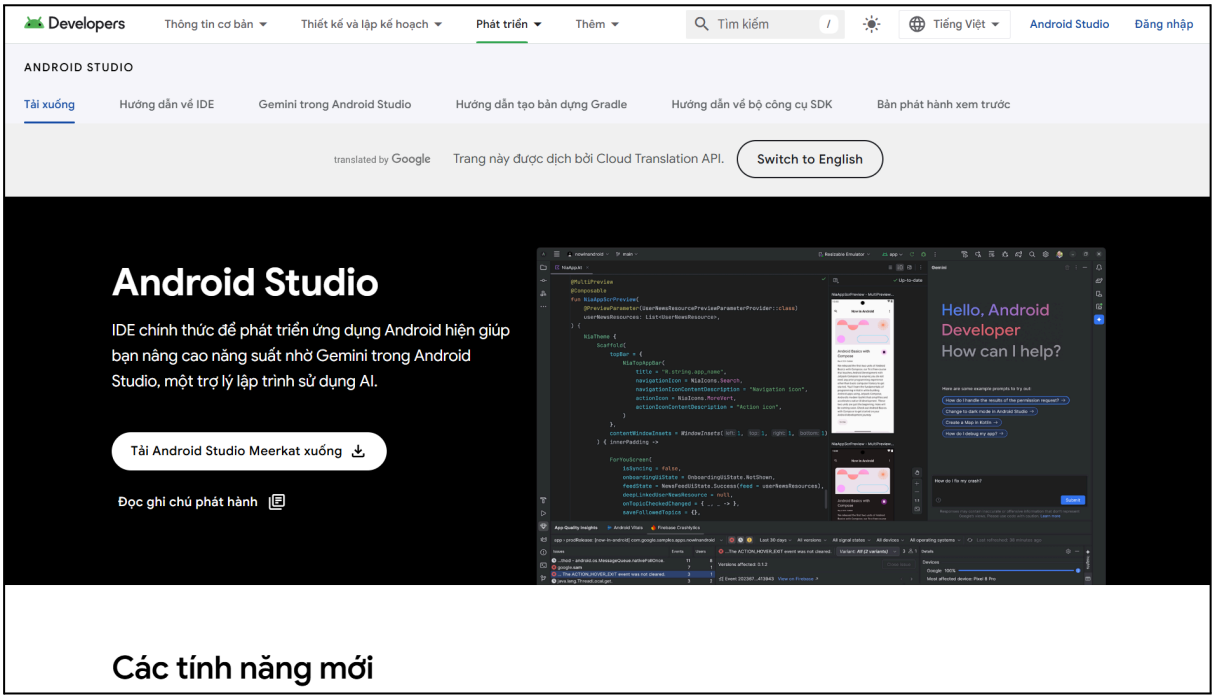


Figure 6.2.2.2.1: Mobile

+ Click a “**Virtual Device Manager**”, then click a “**Create virtual device**” button to install the suitable device.

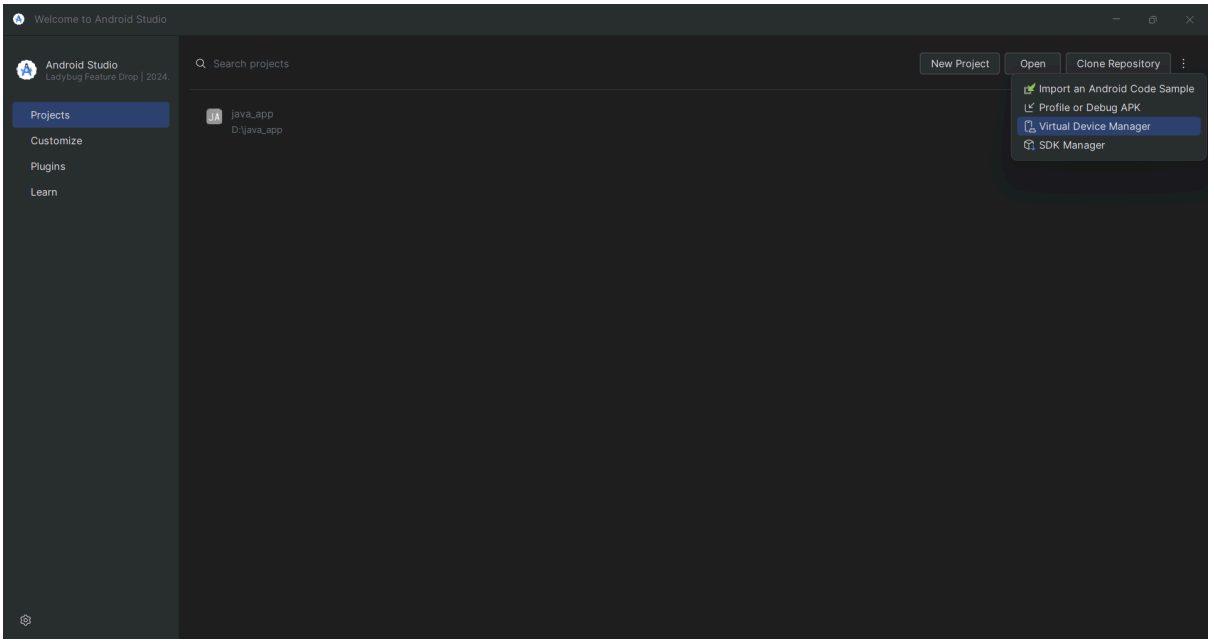


Figure 6.2.2.2.2: Mobile

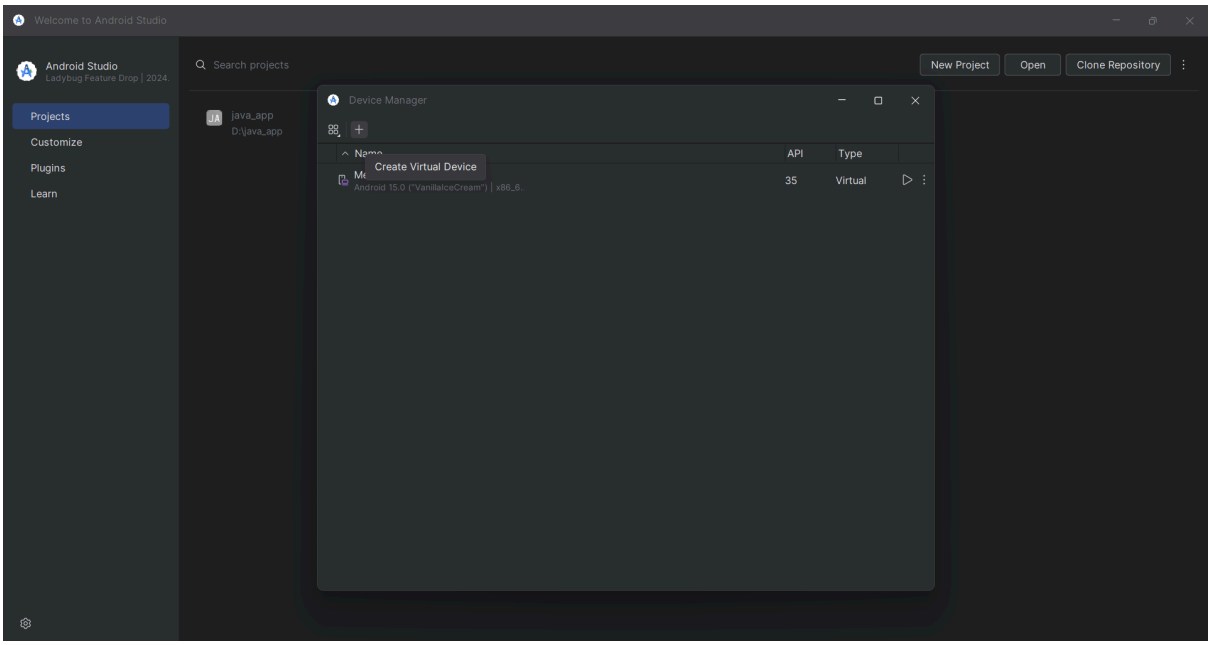


Figure 6.2.2.2.3: Mobile

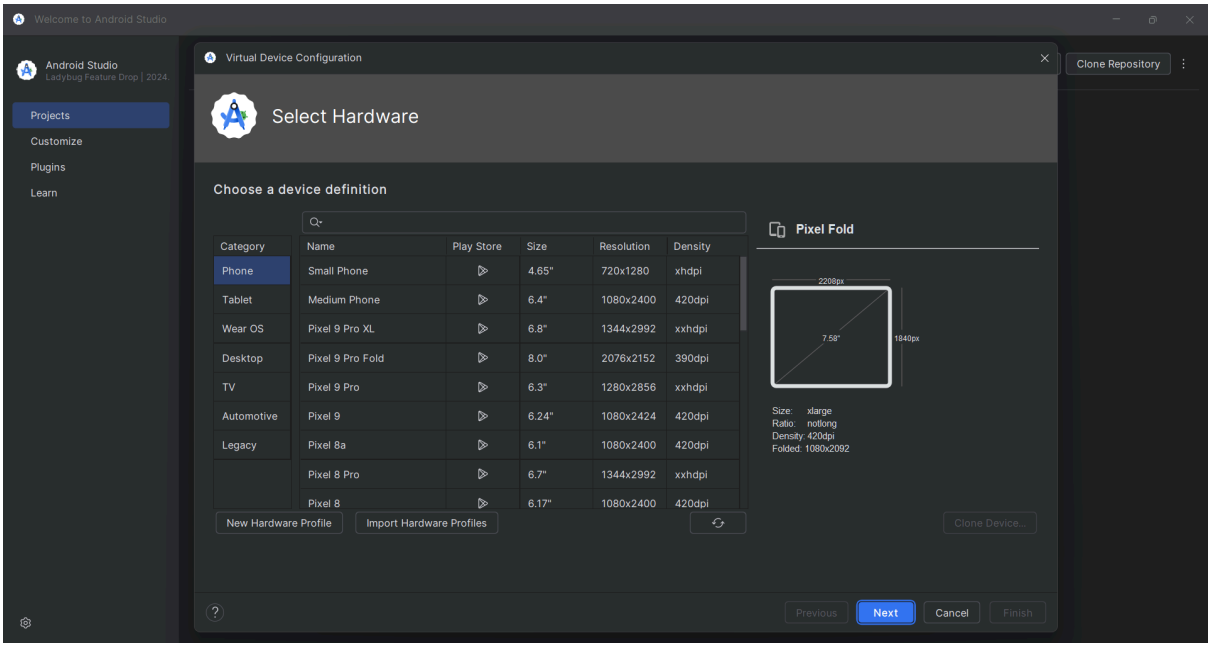


Figure 6.2.2.2.4: Mobile

+ Click a “**SDK Manage**r” button, then install Android SDK Platform (Android 15.0 _ API 35), and SDK Tools

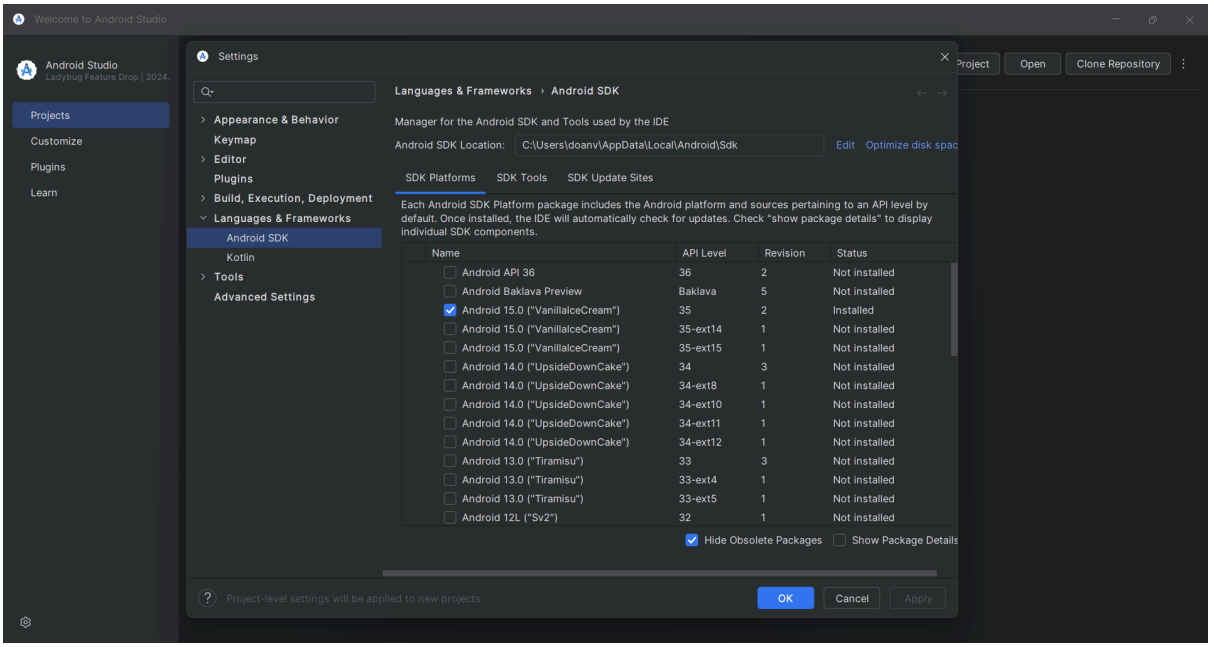


Figure 6.2.2.2.5: Mobile

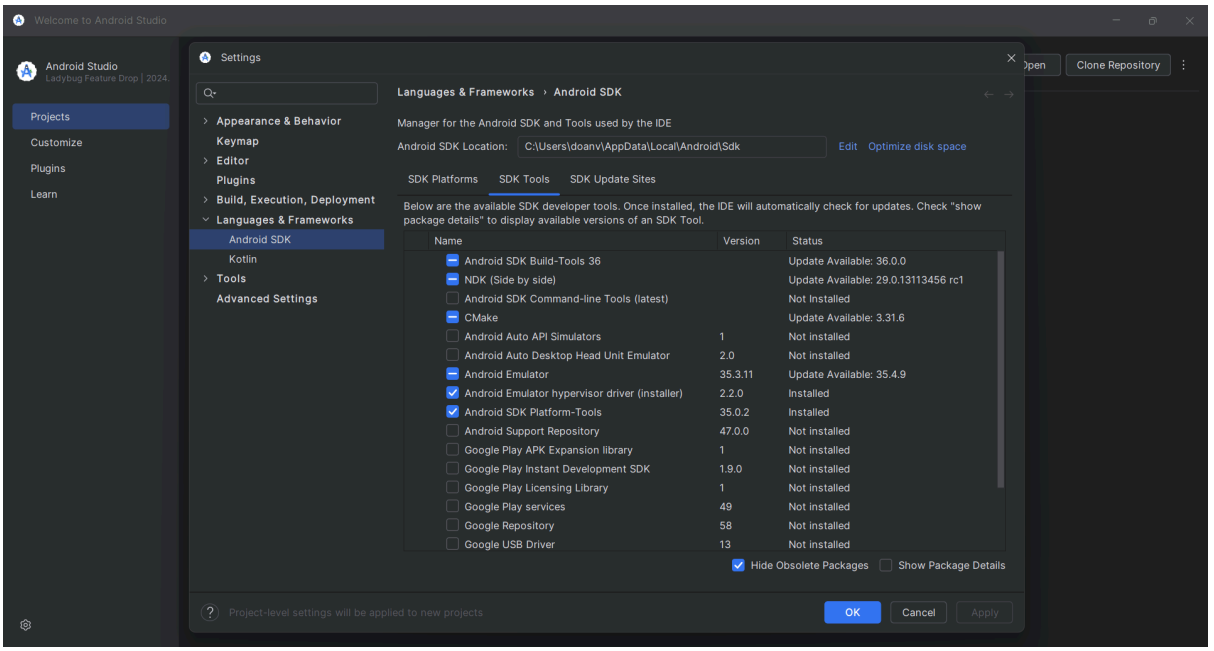


Figure 6.2.2.2.6: Mobile

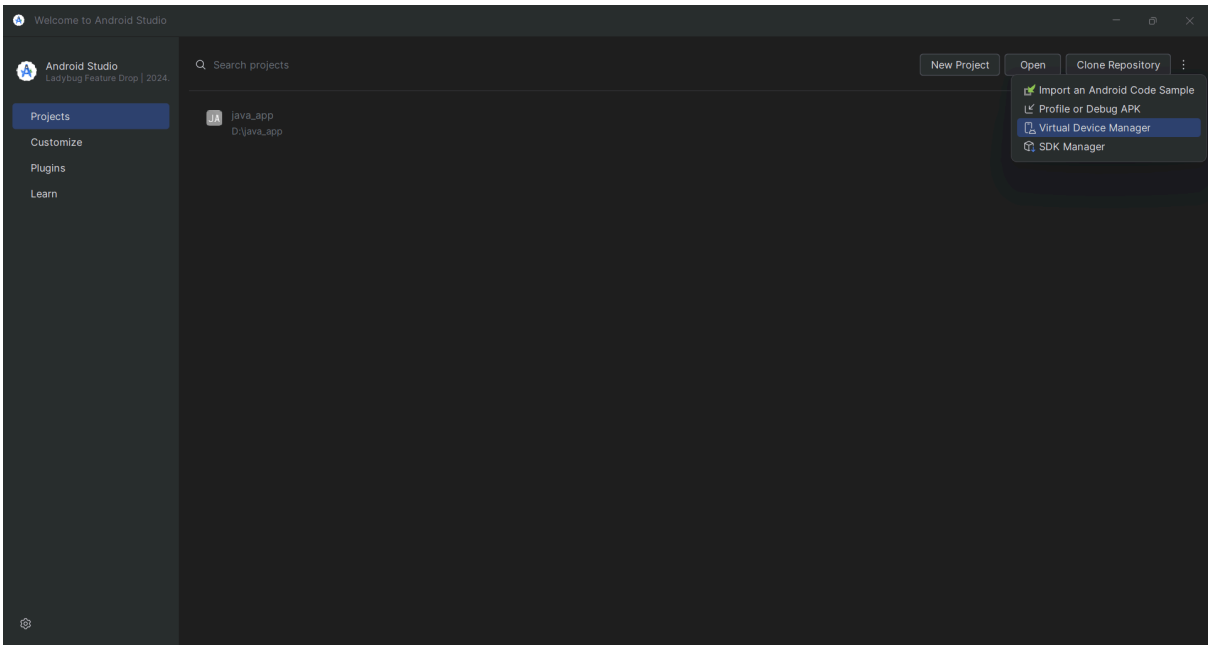


Figure 6.2.2.2.7: Mobile

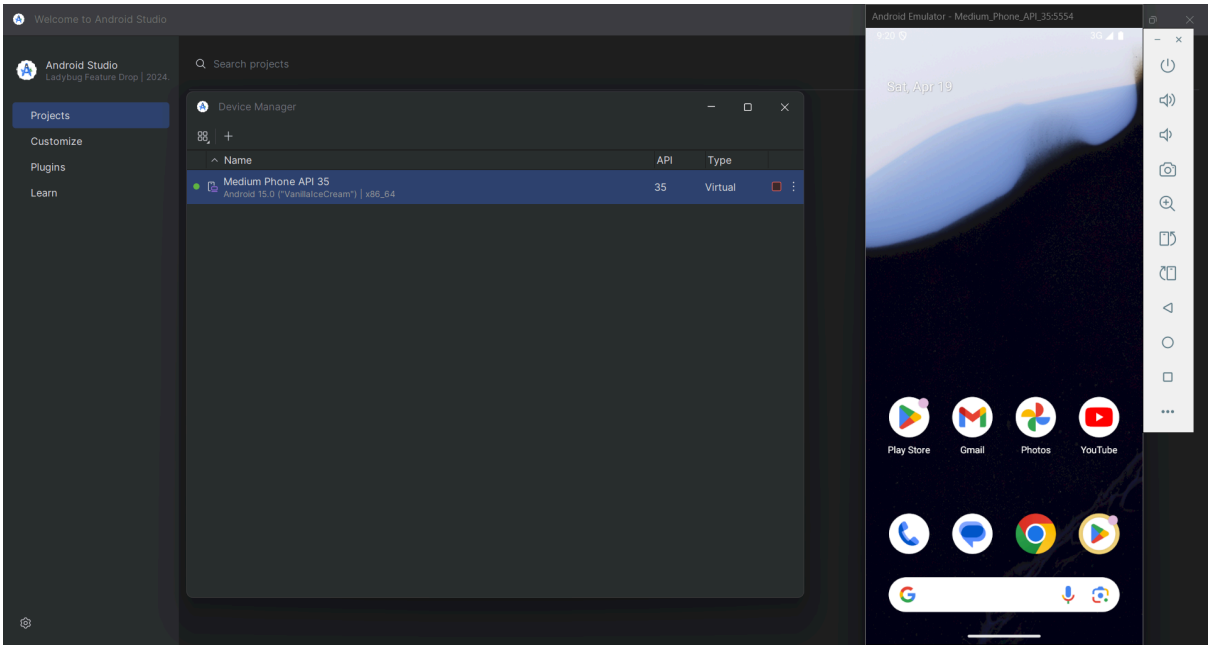


Figure 6.2.2.2.8: Mobile

- Install **Expo Go** (Android device) via CH Play

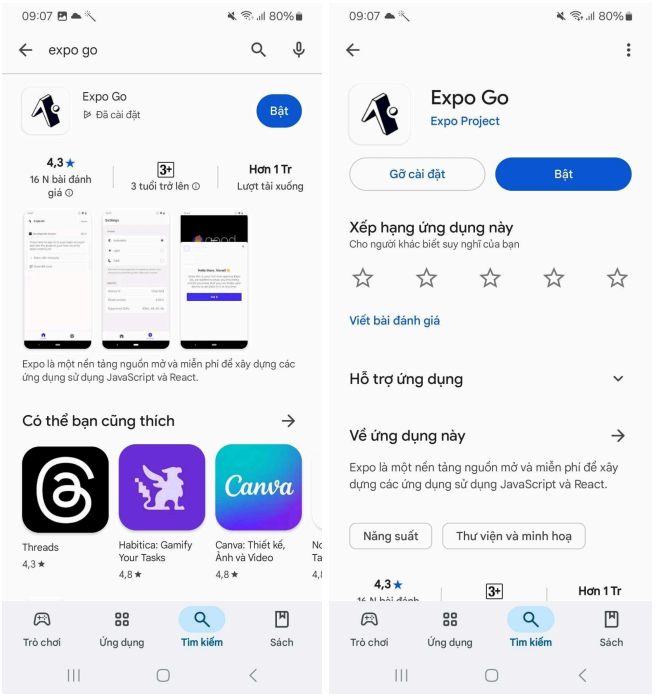


Figure 6.2.2.2.9: Mobile

- Step 4: Install **NodeJS** and **Expo CLI**

  - Download NodeJS

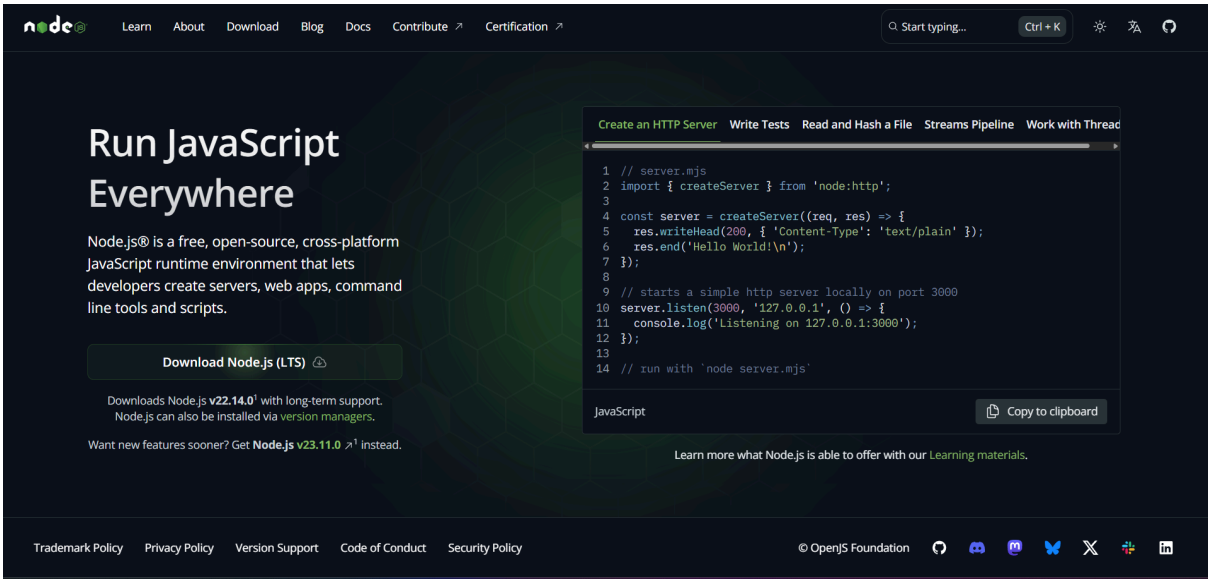


Figure 6.2.2.2.10: Mobile

  - Download Expo CLI

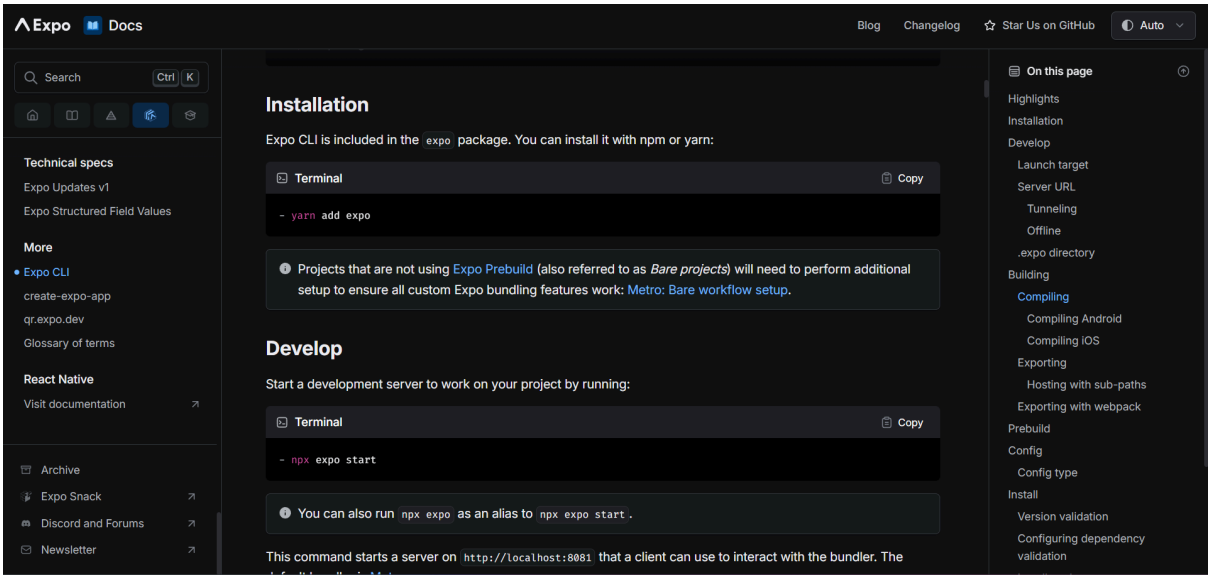


Figure 6.2.2.2.11: Mobile

- Step 5: Open terminal

- Type “git clone git@github.com:DoanVietThanh/FPT_Library_Mobile.git”, then type “code .” to open source code in Visual Studio Code

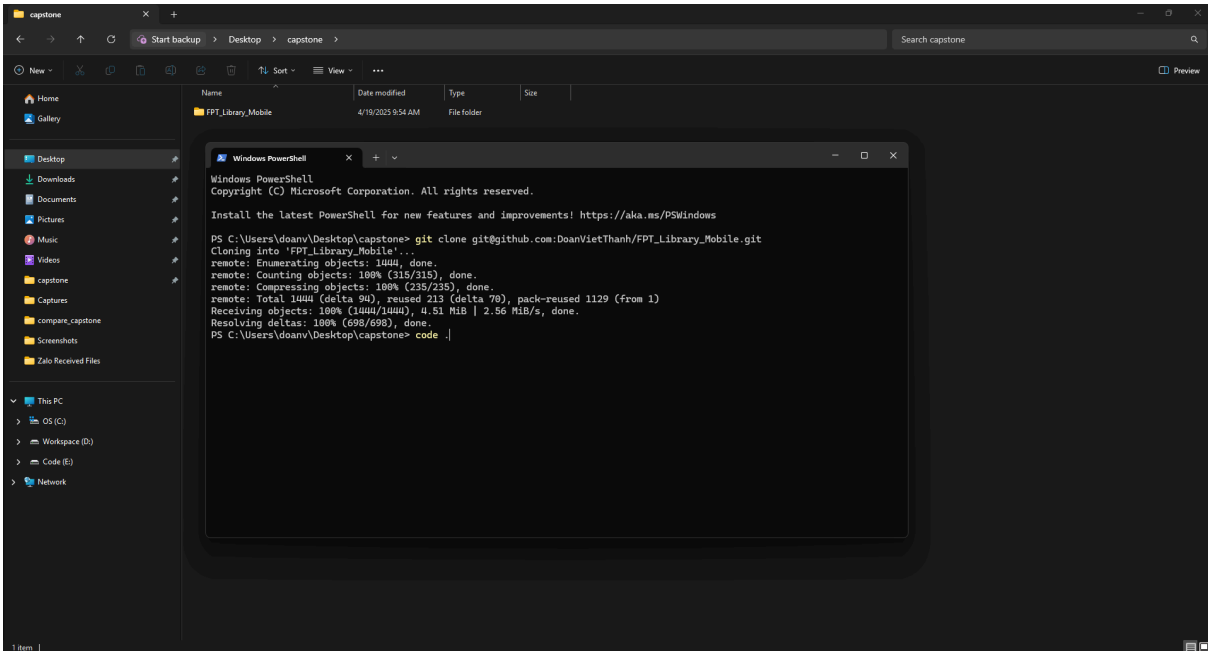


Figure 6.2.2.2.12: Mobile

- Create a “.env” file and run “npm i” in a command line to install node_modules

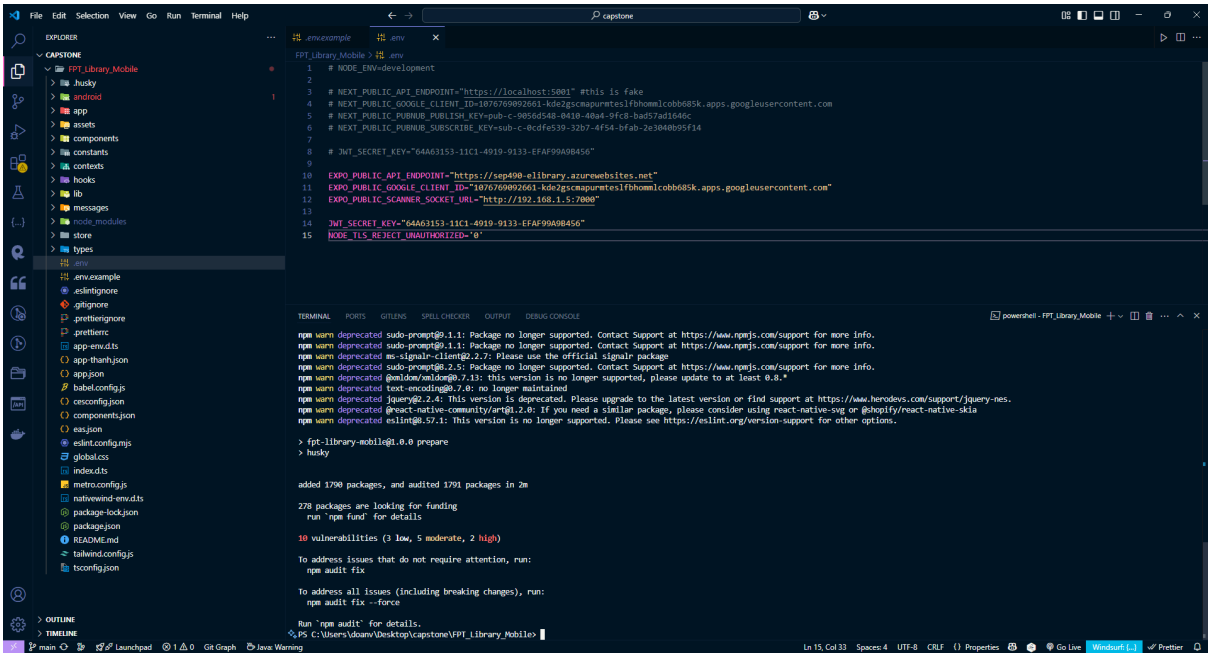


Figure 6.2.2.2.13: Mobile

- Run “npm run start”

Figure 6.2.2.2.14: Mobile

- Press “s” to switch to Expo Go or Press “a” to open mobile project on android device directly (make sure that it is connected with a cable)

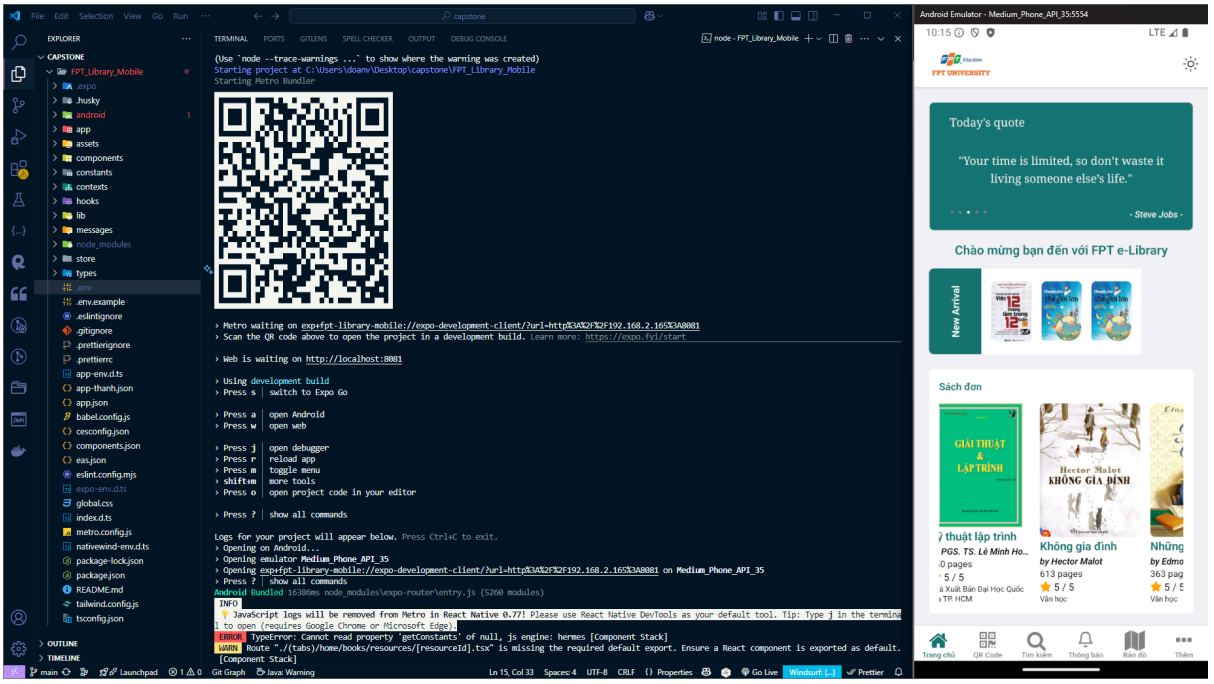


Figure 6.2.2.2.15: Mobile

2.2.3 Back-end

- Step 1: Download and install git
- Step 2: Connect to git repository with GitHub account
- Step 3: Clone git repository: https://github.com/SP25-Captone-ELibrary/SP25_SEP490_ELibrary.git
- Step 4: Download and install Docker
- Step 5: Navigate to folder **SP25_SEP490_ELibrary**

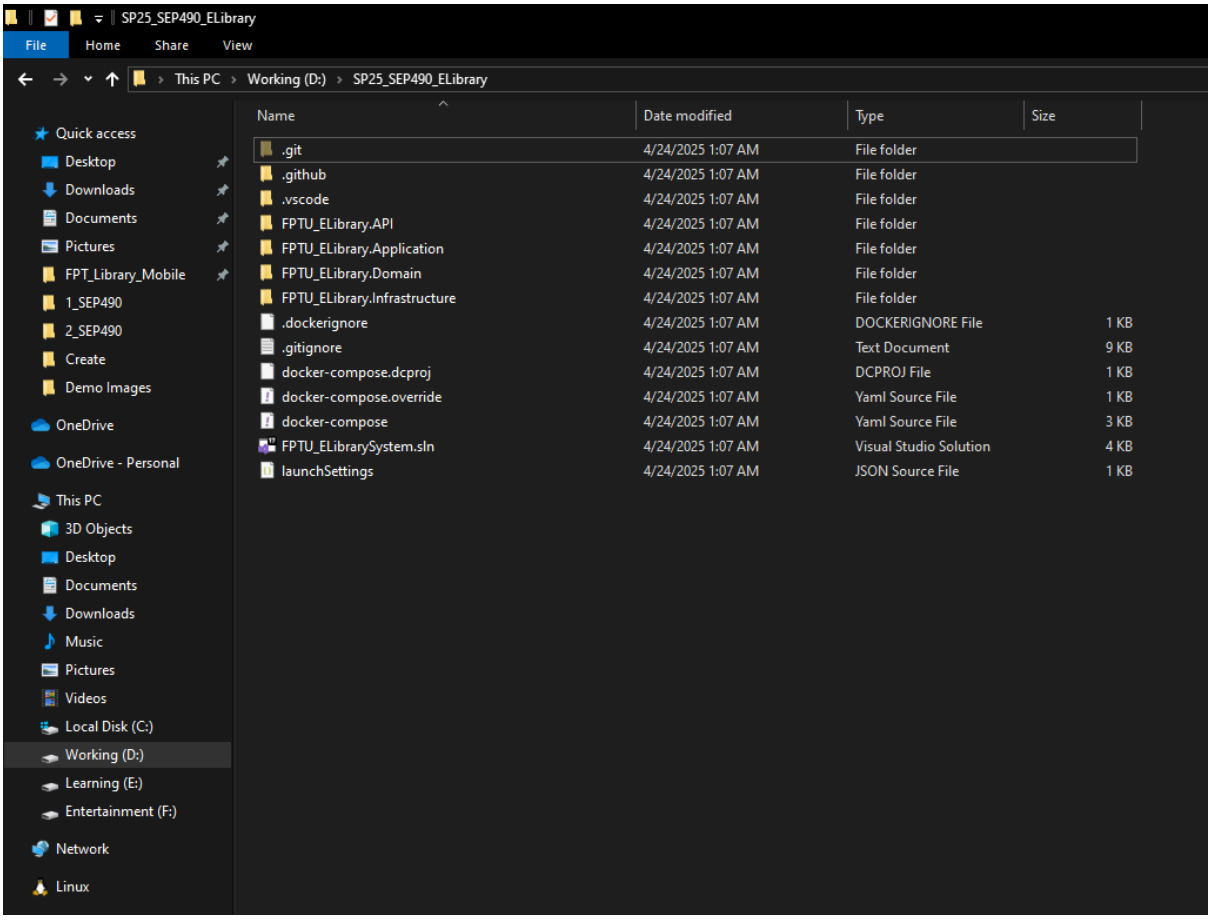


Figure 6.2.2.3.1: Project Folder

- Step 6: Type **cmd** in the address bar and hit Enter:

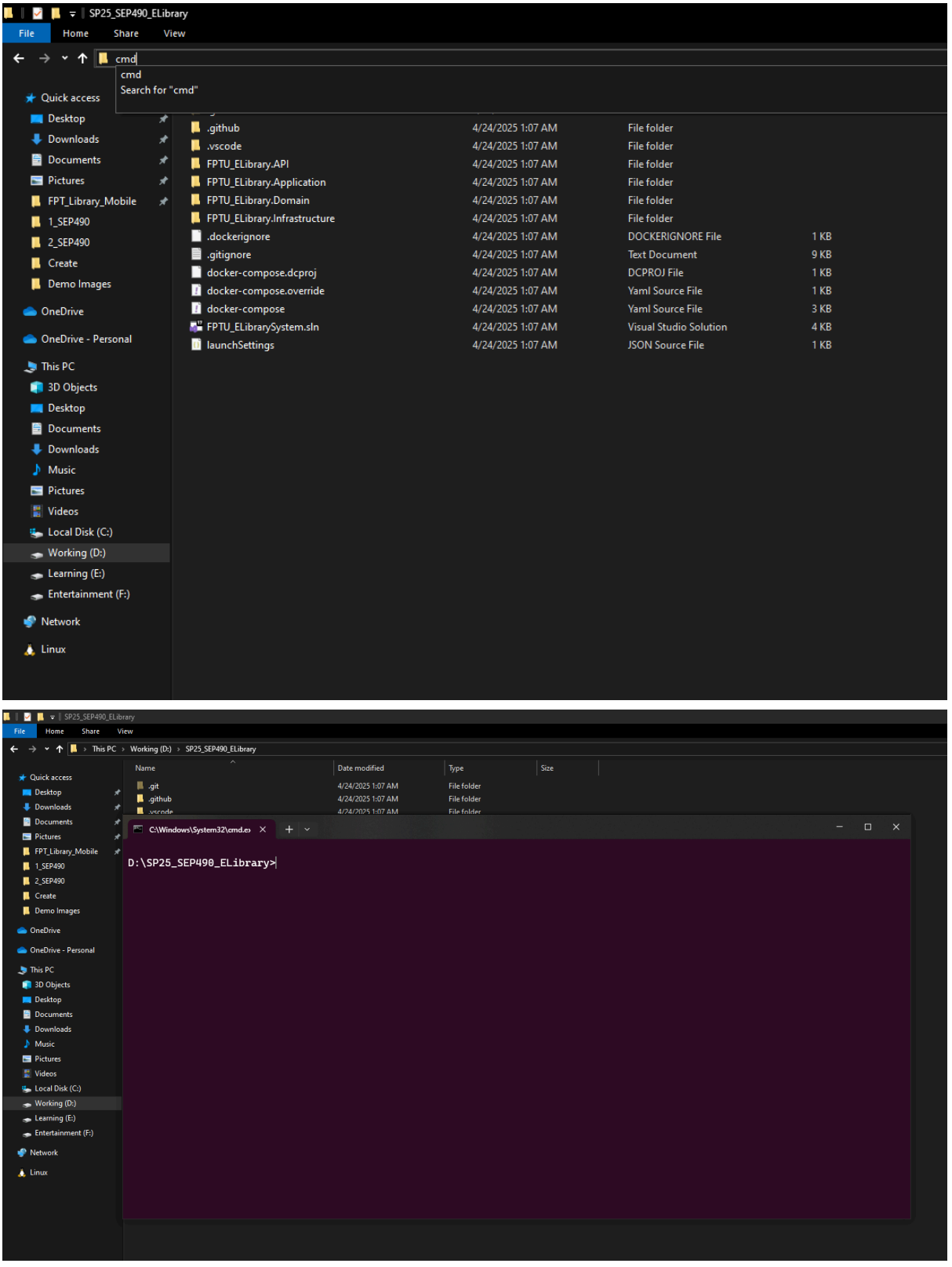
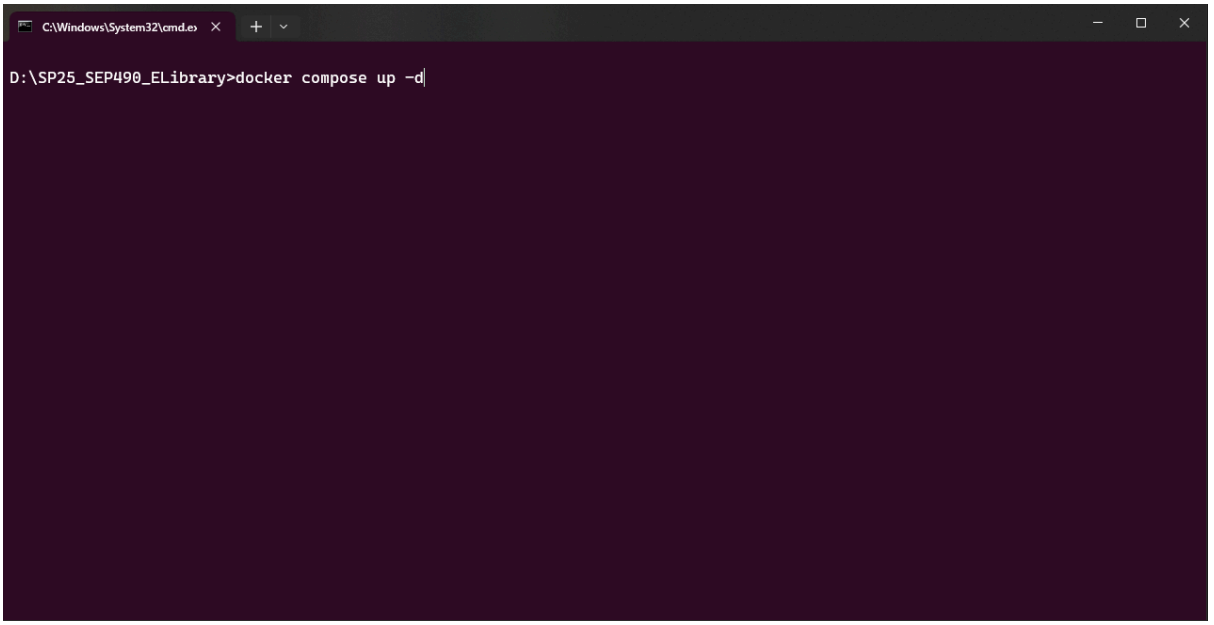
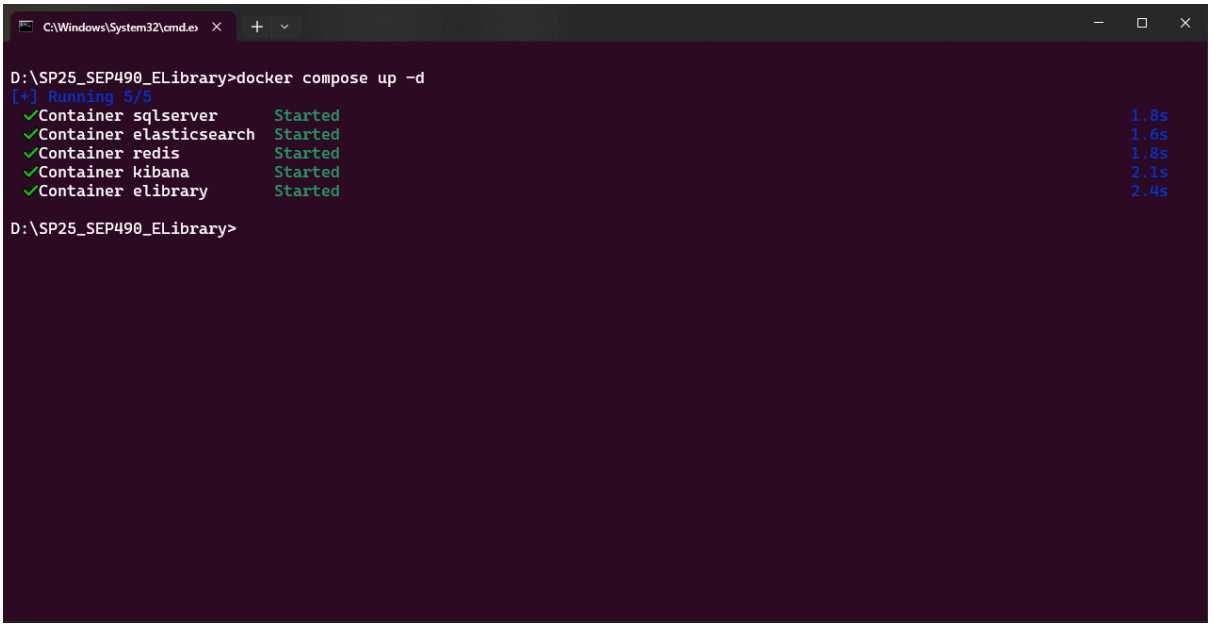


Figure 6.2.2.3.2: Open cmd screen

- Step 7: Use **docker-compose up -d** command to run the app


C:\Windows\System32\cmd.e
D:\SP25_SEP490_ELibrary>docker compose up -d


C:\Windows\System32\cmd.e
D:\SP25_SEP490_ELibrary>docker compose up -d
[+] Running 5/5
✓Container sqlserver Started 1.8s
✓Container elasticsearch Started 1.6s
✓Container redis Started 1.8s
✓Container kibana Started 2.1s
✓Container elibrary Started 2.4s
D:\SP25_SEP490_ELibrary>

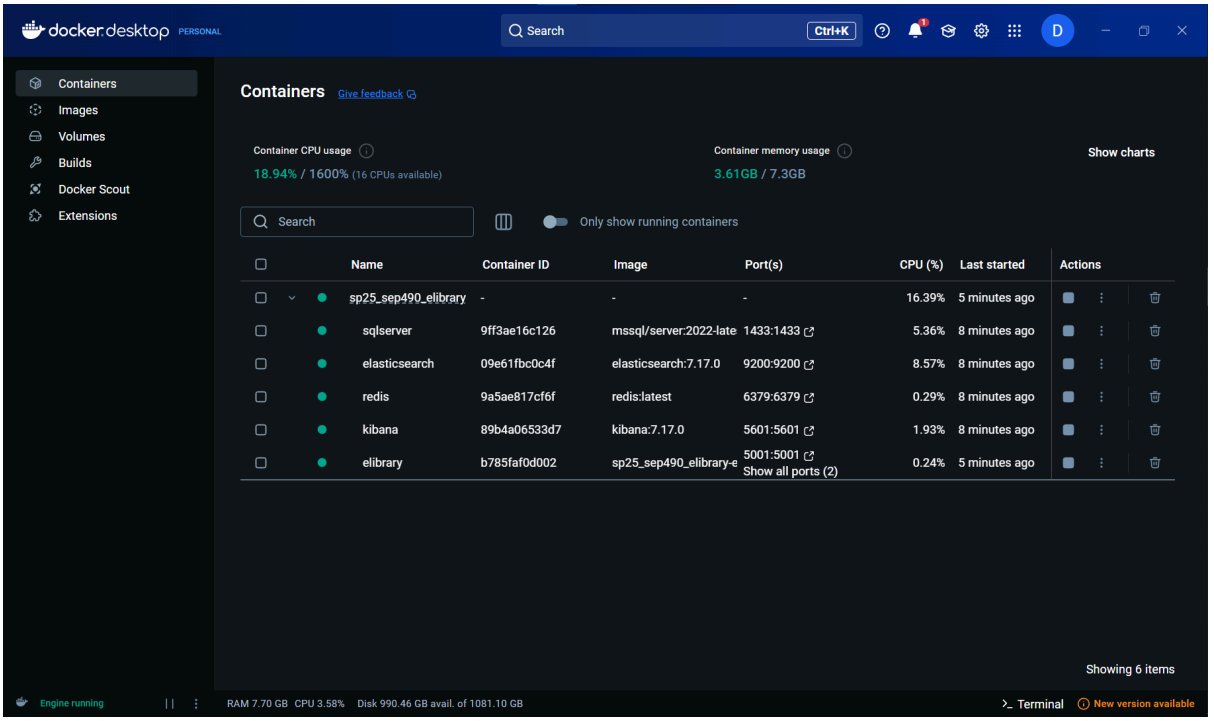


Figure 6.2.2.3.3: Run the command screen

- Step 8: Go to http://localhost:5000/swagger/index.html to see the website.

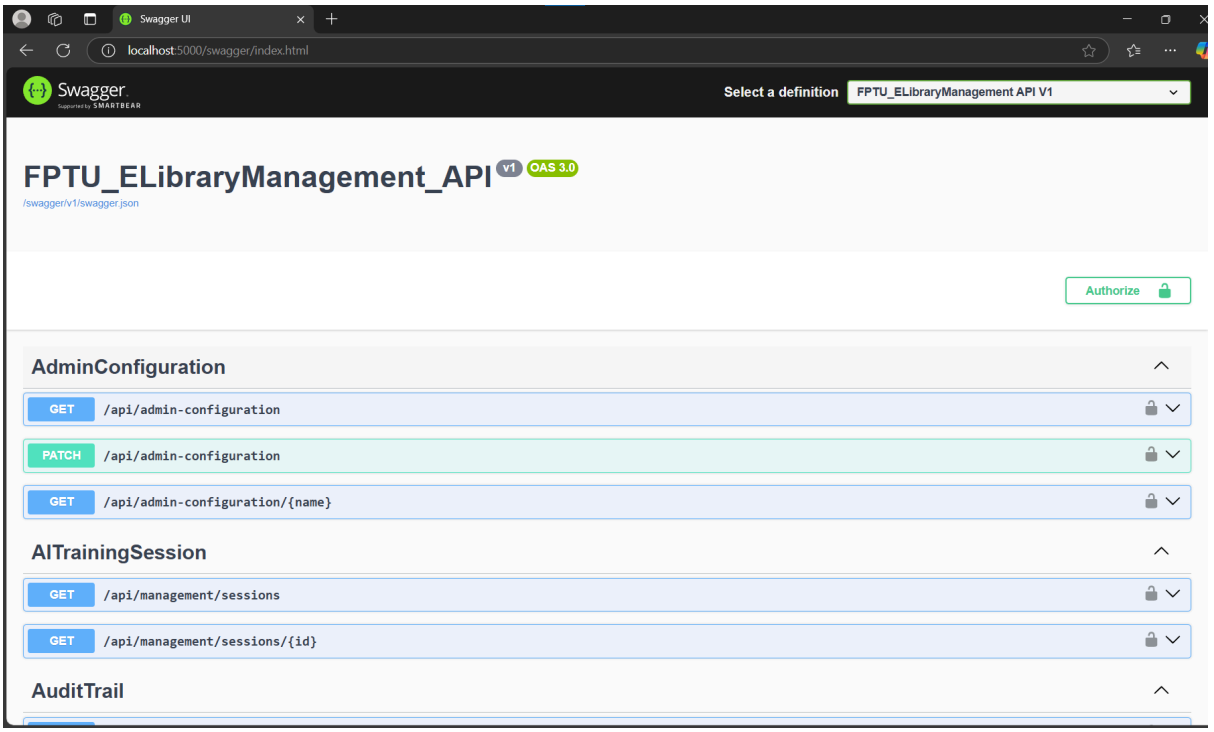


Figure 6.2.2.3.4 : API screen

3. User Manual

3.1 Overview

- **Workflow 1: New Arrival Cataloging and Notification**
    1. Head librarian login to the management page, then create a warehouse stock-in file.
    2. Librarian login to the management page, then create a catalog for the library item.
    3. Librarian registers the unique barcode and attaches it to the library item.
    4. Librarian adds the digital resources to the library item.
    5. Librarian pushes the library item to the reasonable shelf.
    6. Librarian imports the images and starts to train AI detection.
    7. Librarian publishes the library items after training AI detection successfully.
    8. Librarian sends the notification to the library patrons.

- **Workflow 2: Borrowing and Circulation**
    1. Librarian logins to the management page, then fills the personal patron information, selects the payment method to create a new library card.
    2. After the patron pays for the library card, the patron checks email to get the login account (email, and password).
    3. To borrow the library item patron wants, the patron can login and use the search feature (quick search, basic search, advanced search, and AI-prediction).
    4. The patron adds the library items, reserved items, and digital resources to the borrow list, and creates the borrow request.
    5. The patron can check the borrow request detail in the profile page, then pay for it.
    6. The patron can login by mobile, locate the library items with a visual 2D indoor map, and bring them to the librarian.
    7. The Librarian scans the QR code of each library item to create the borrowed record.

- **Workflow 3: Return books and assign to pending reservations**
    1. Librarian scans the patron's library card to identify the patron.
    2. Librarian scans the Barcode of library items to check the condition of returned library items, checks overdue status and can take note of fines.
    3. Librarian assigns return the library items to the pending reservations, and then attach to hold slip.
    4. The patron can view the pending fines and pay them.

- **Workflow 4: Analytics and Reporting management**
    1. The Librarian can view the data analytics on the Dashboard page and circulation analytics.
    2. The librarian selects the understocked library items, then uses the potential book purchase recommendations.
    3. The librarian selects high-rated purchase options.
    4. The librarian creates the book supplement request.

3.2 Workflow 1

- Head librarian login to Management Page.

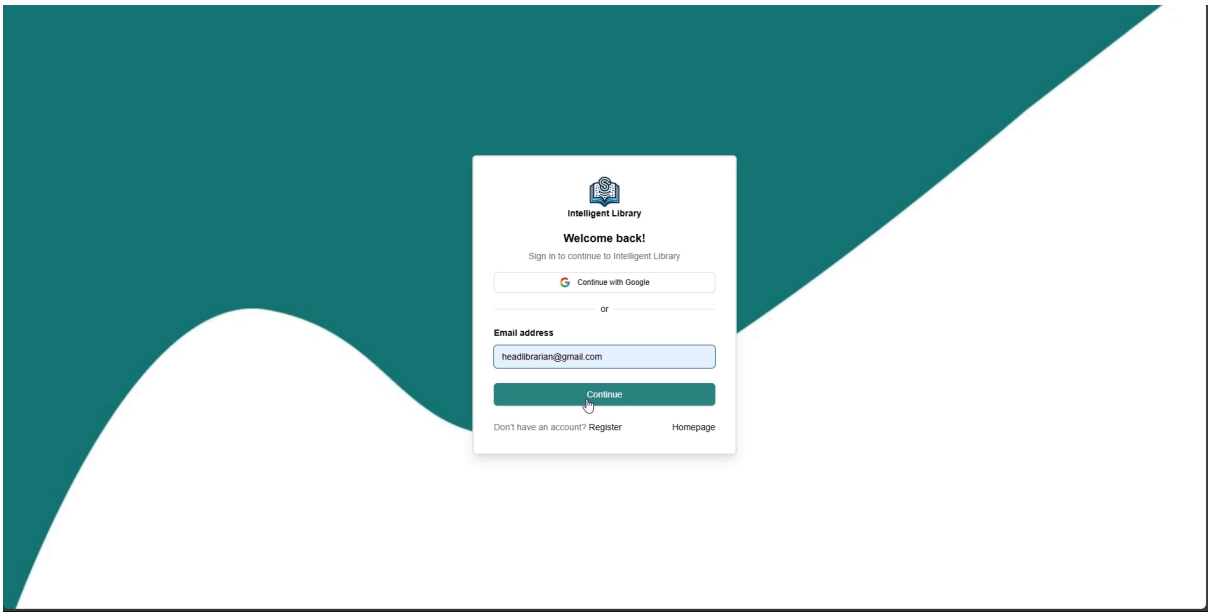


Figure 6.3.2.1: Login screen

- Head librarian creates new warehouse tracking.

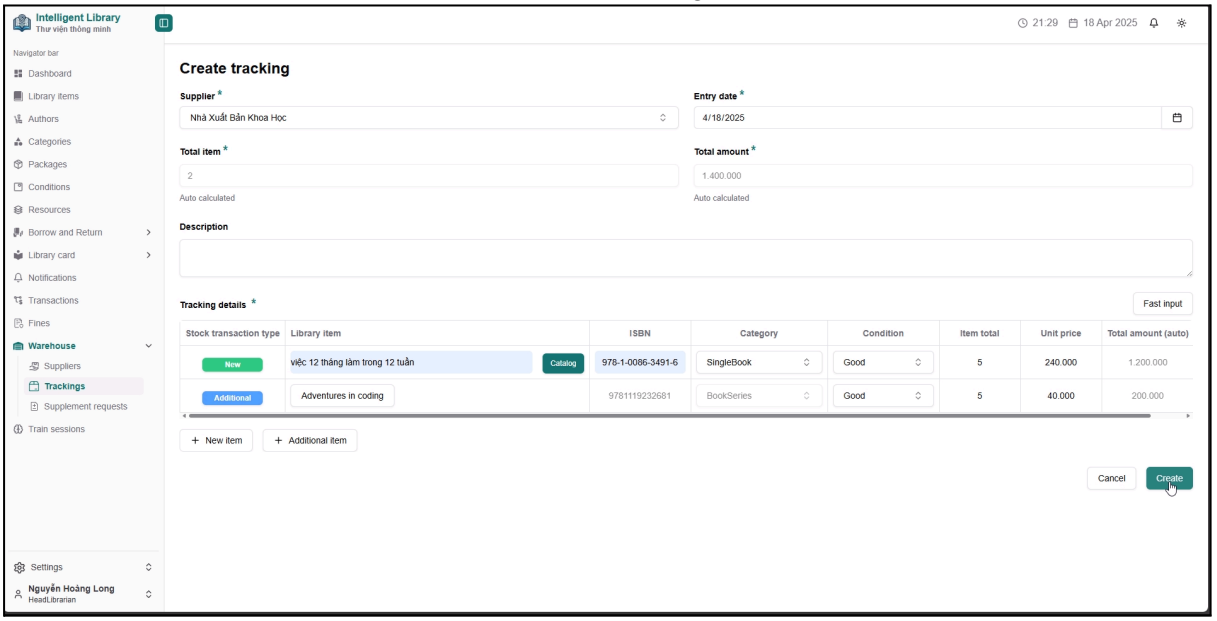


Figure 6.3.2.2: Create warehouse tracking screen

- Librarian logins to the Library Web

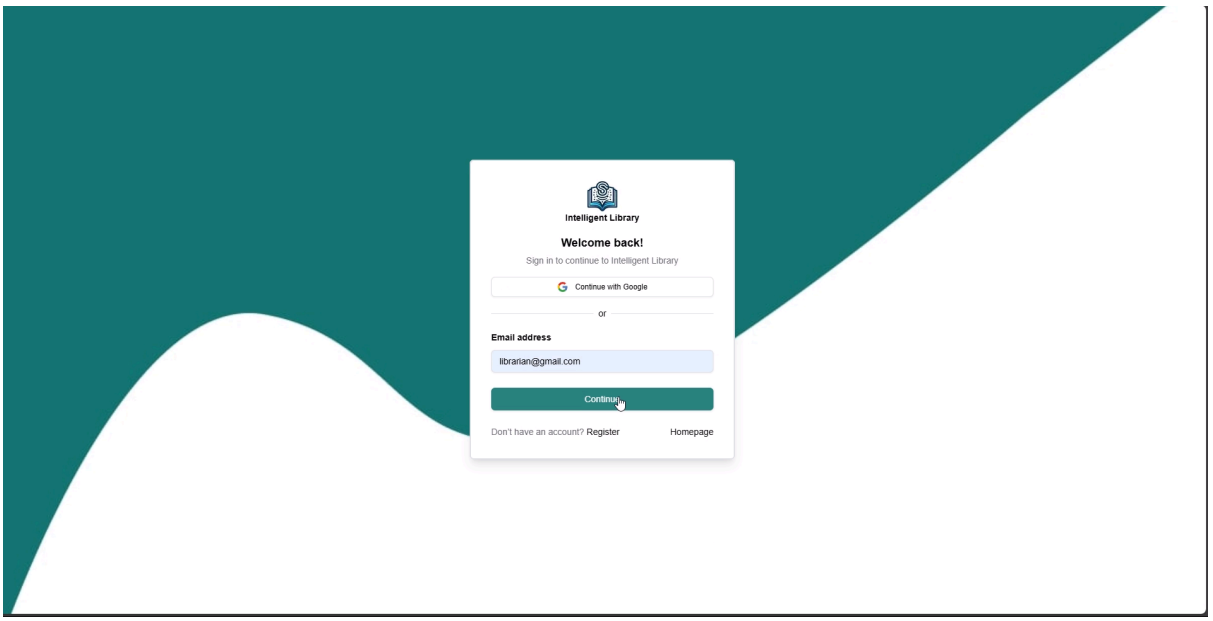


Figure 6.3.2.3: Login screen

- Librarian views the tracking details and navigate to the ‘Catalog Page’ from the entry of the ‘Uncatalog’ library item.

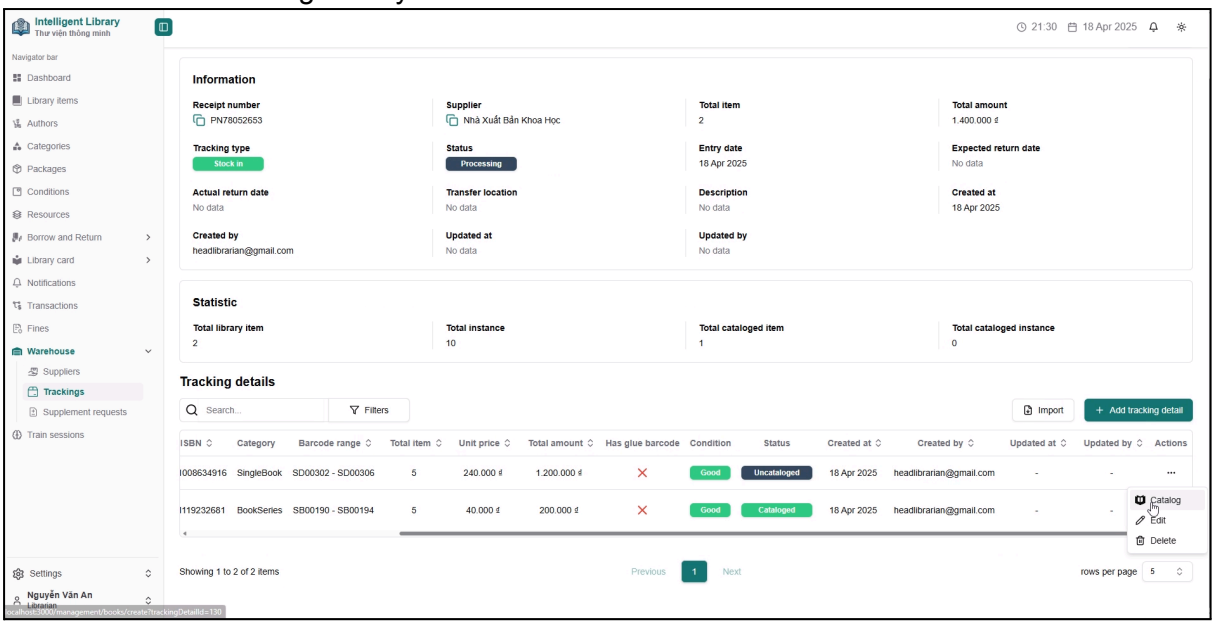


Figure 6.3.2.4: Warehouse tracking detail screen

- Librarian checks the pre-selected library item category from the warehouse tracking creation step, then clicks continue.

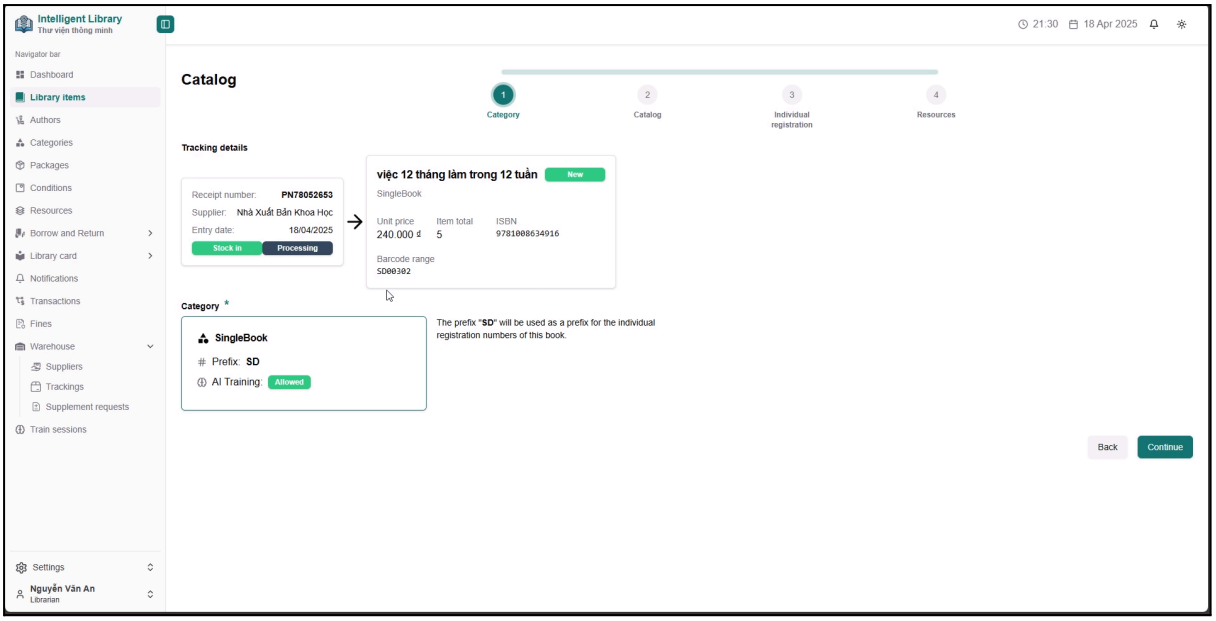


Figure 6.3.2.5: Catalog screen

- Librarian proceeds to catalog the library item (either by manual cataloging or by copy cataloging using MARC21).

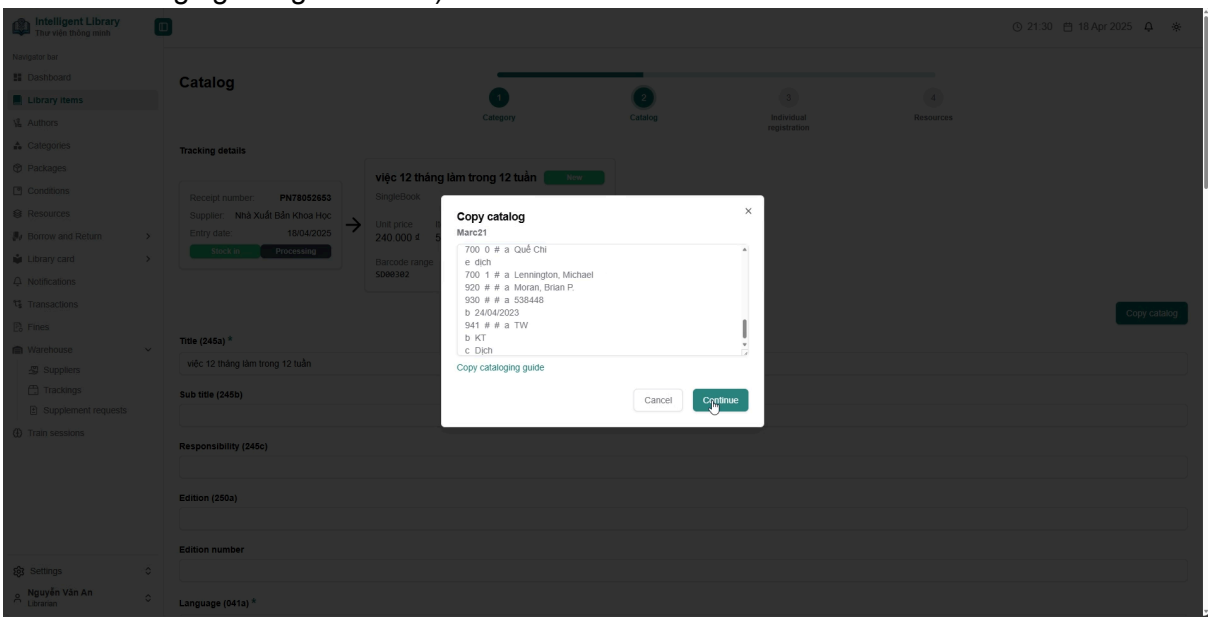


Figure 6.3.2.6: Catalog screen

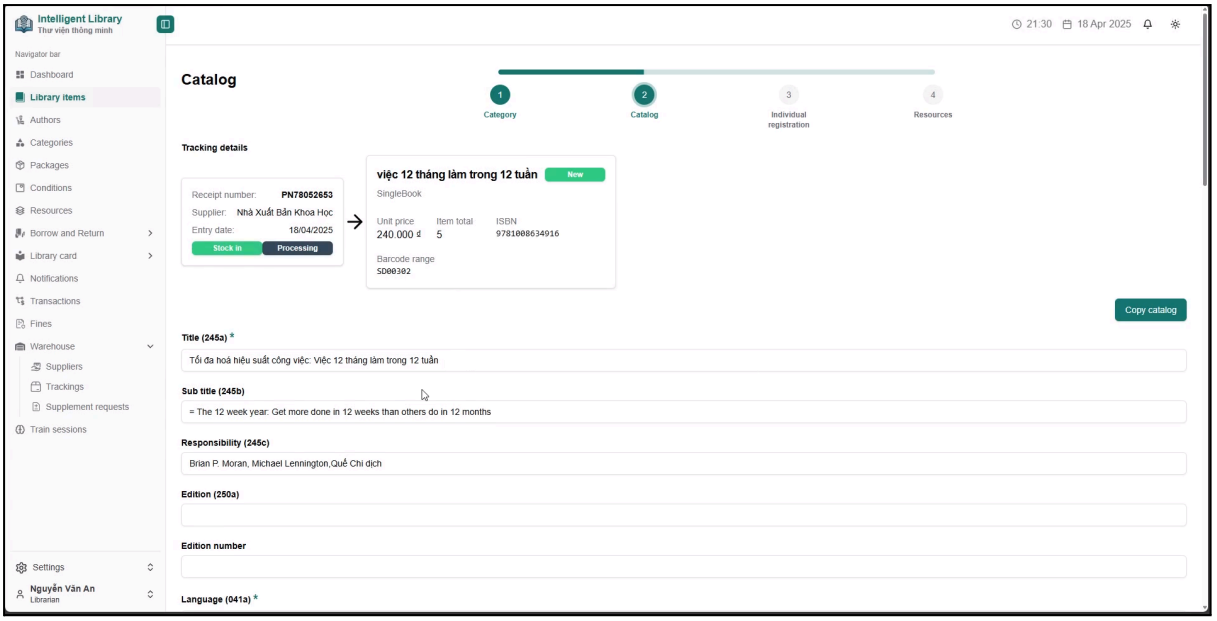


Figure 6.3.2.7: Catalog screen

- Librarian proceeds to print and glue the unique registration barcodes and classification barcodes for the instances of the library item.

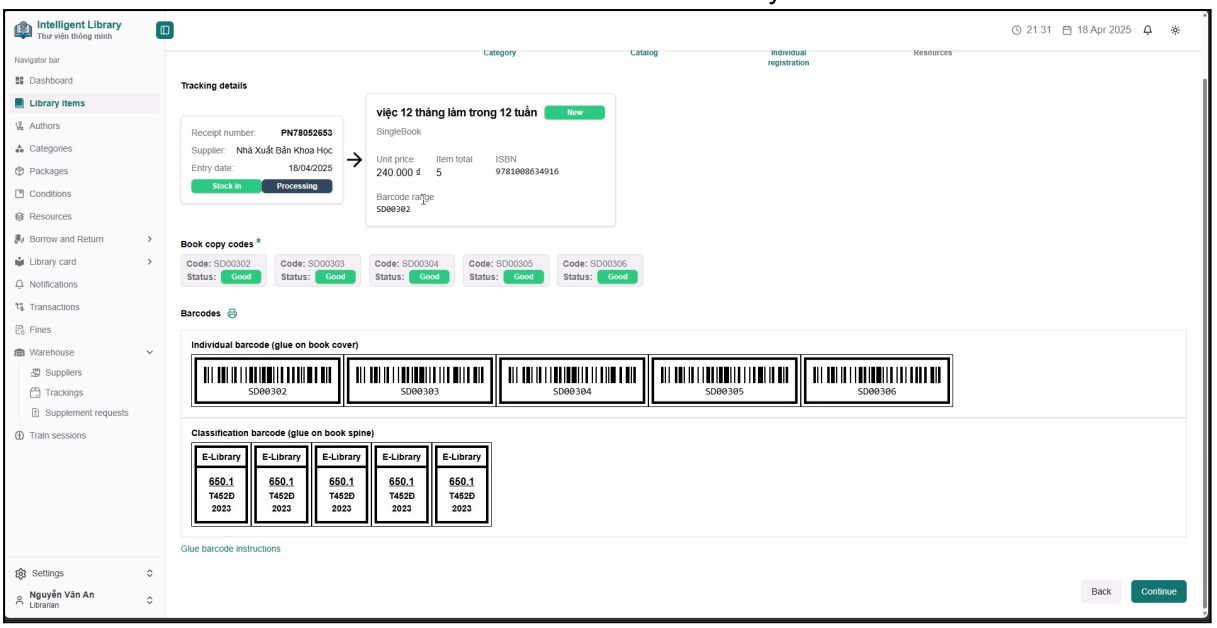


Figure 6.3.2.8: Catalog screen

Figure 6.3.2.9: Printer screen

- Librarian adds ebooks and audiobooks to the library item(optional), then clicks “Continue” to complete the cataloging process.

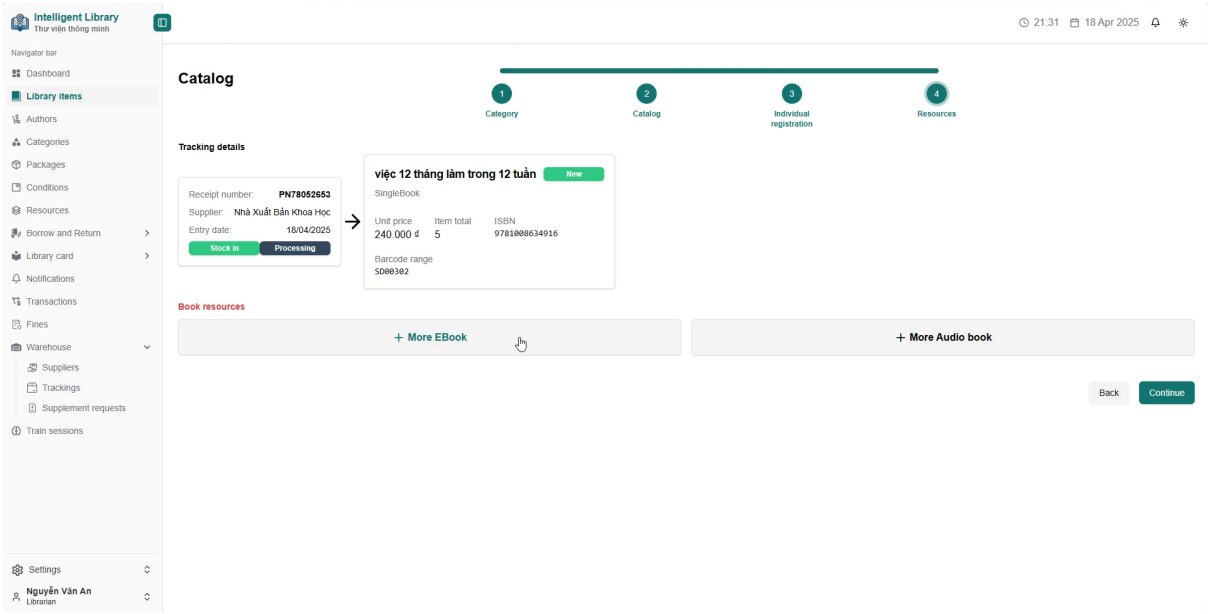


Figure 6.3.2.10: Catalog screen

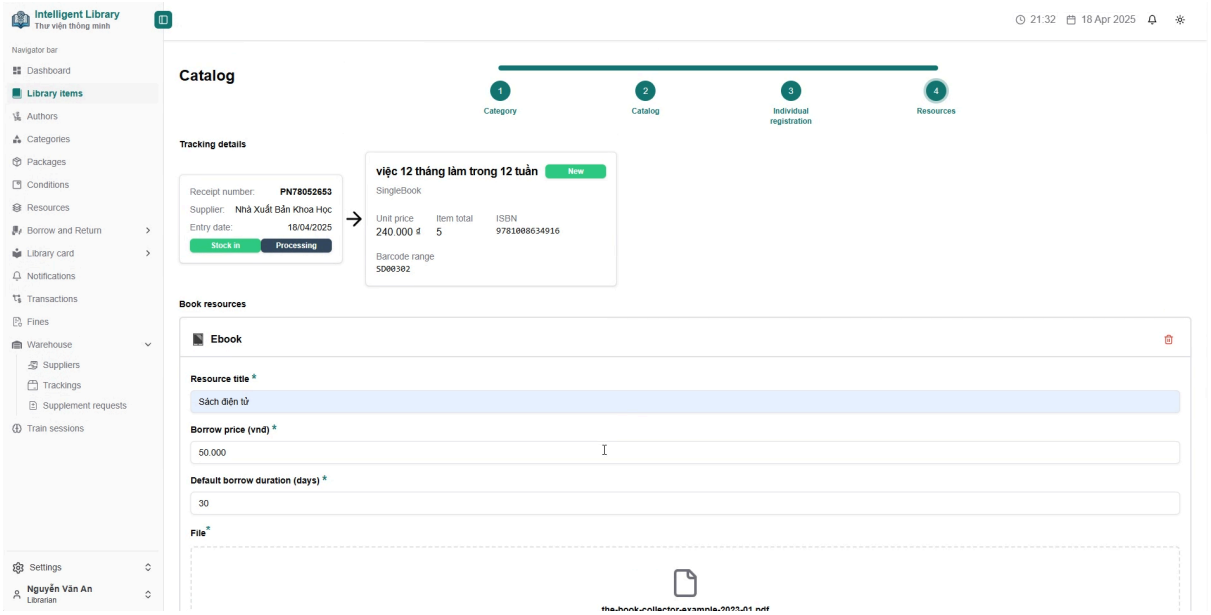


Figure 6.3.2.11: Catalog screen

- Librarian classifies the shelf based on the DDC format.

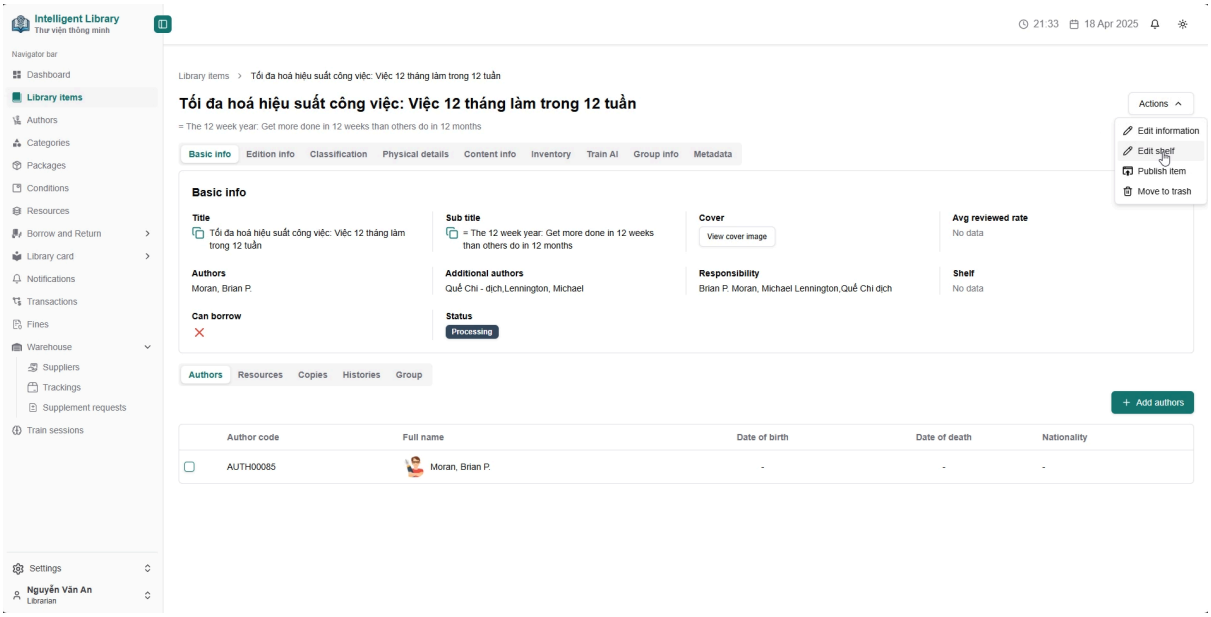


Figure 6.3.2.12: Library item management detail screen

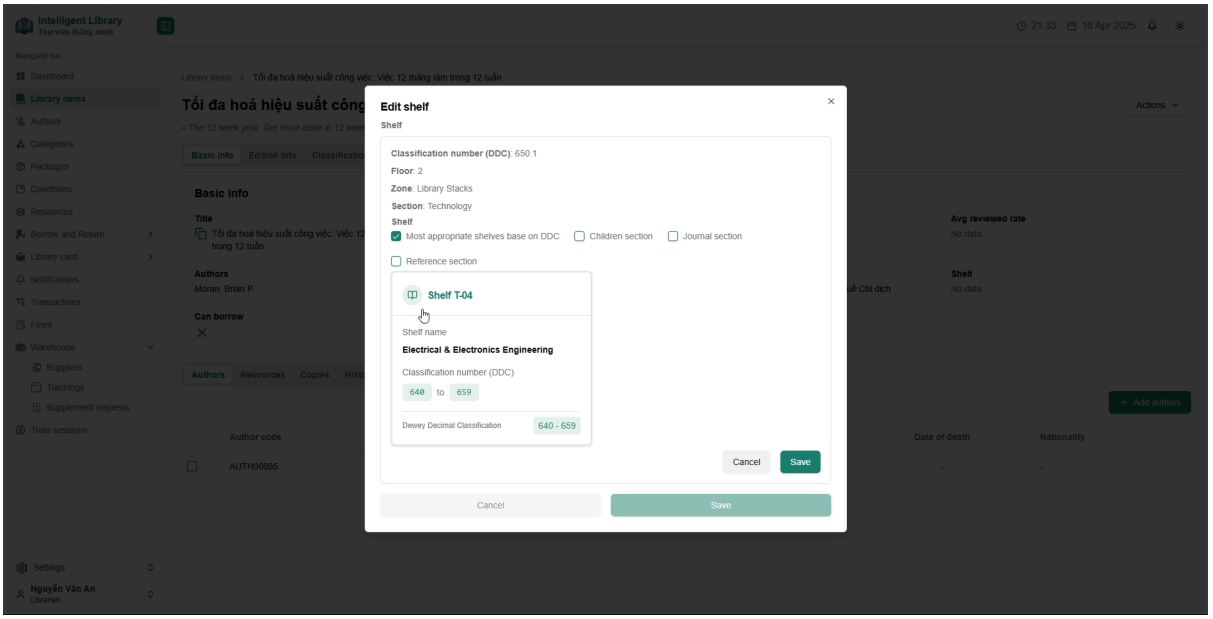
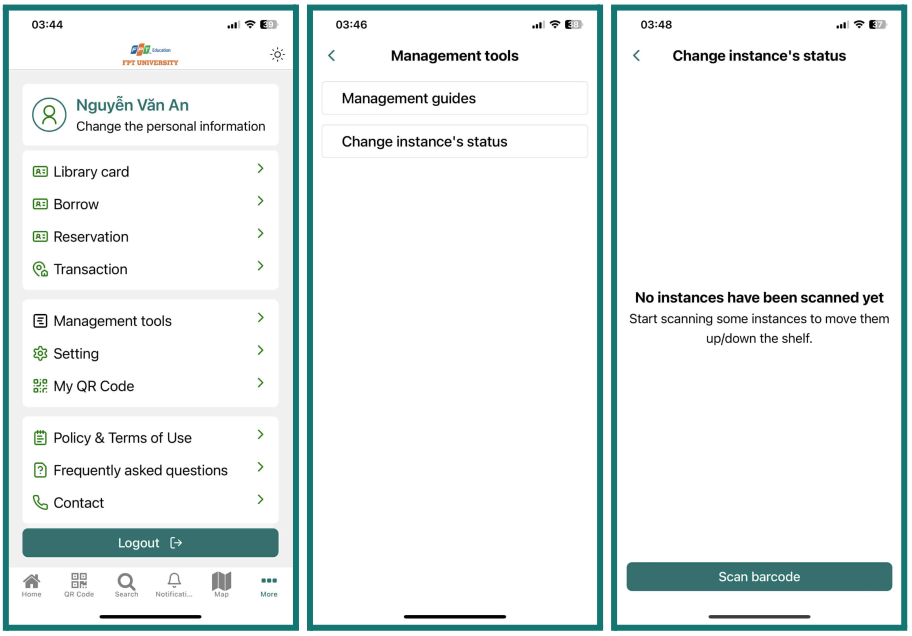


Figure 6.3.2.13: Library item management detail screen

- Librarian takes the instances of library item whose status needs to be changed to "In Shelf," goes to the shelf, and uses the "Change instance's status" tool on the system via their phone to scan and update the status of those instances



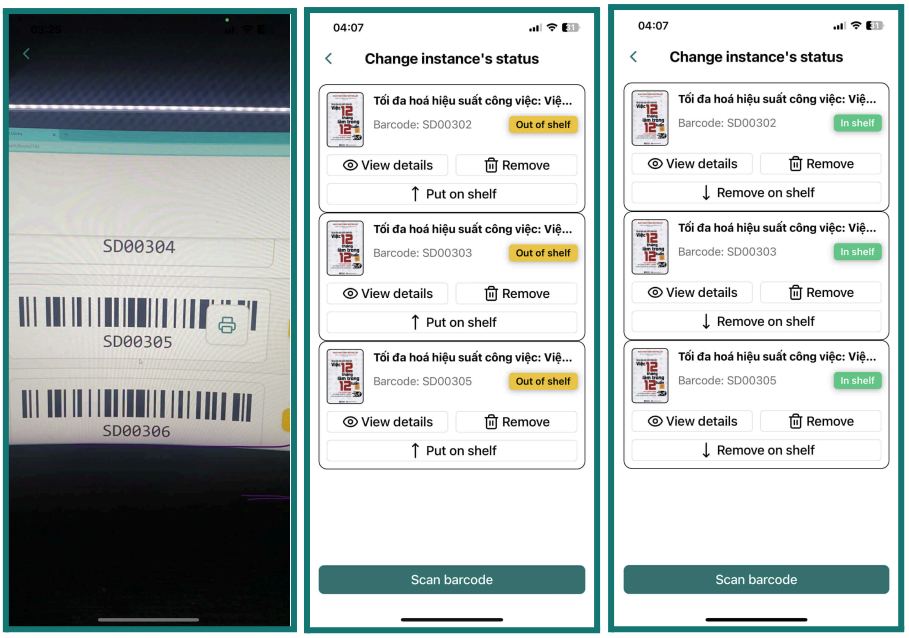


Figure 6.3.2.14: Change instance's status screen (Mobile app)

- Librarian changes the status of the library item to "Published" so that patrons can search for and borrow it.

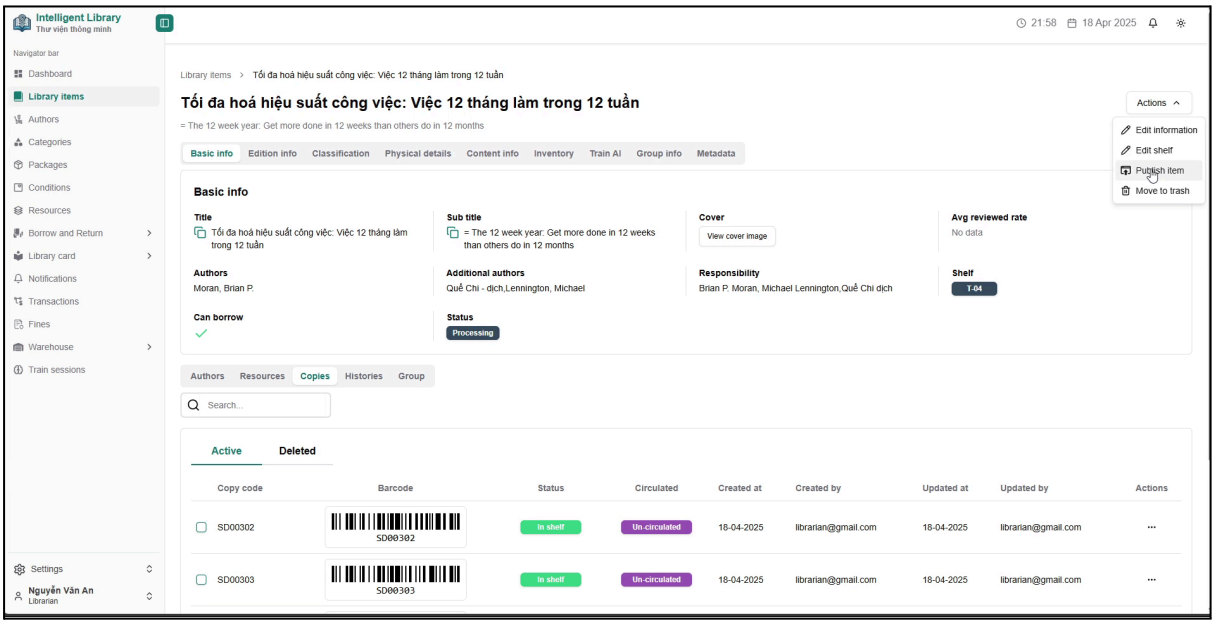


Figure 6.3.2.15: Library item management detail screen

- Librarian trains the AI on the new library item so that patrons can search for it using the "AI Predict" tool.

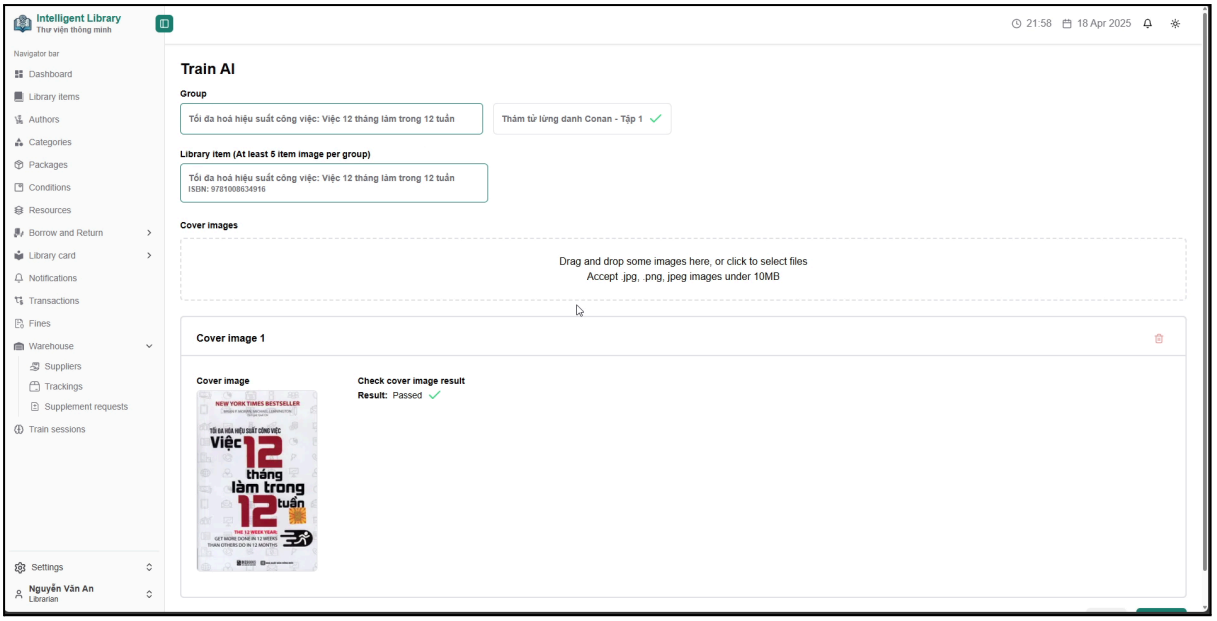


Figure 6.3.2.16: Train AI screen

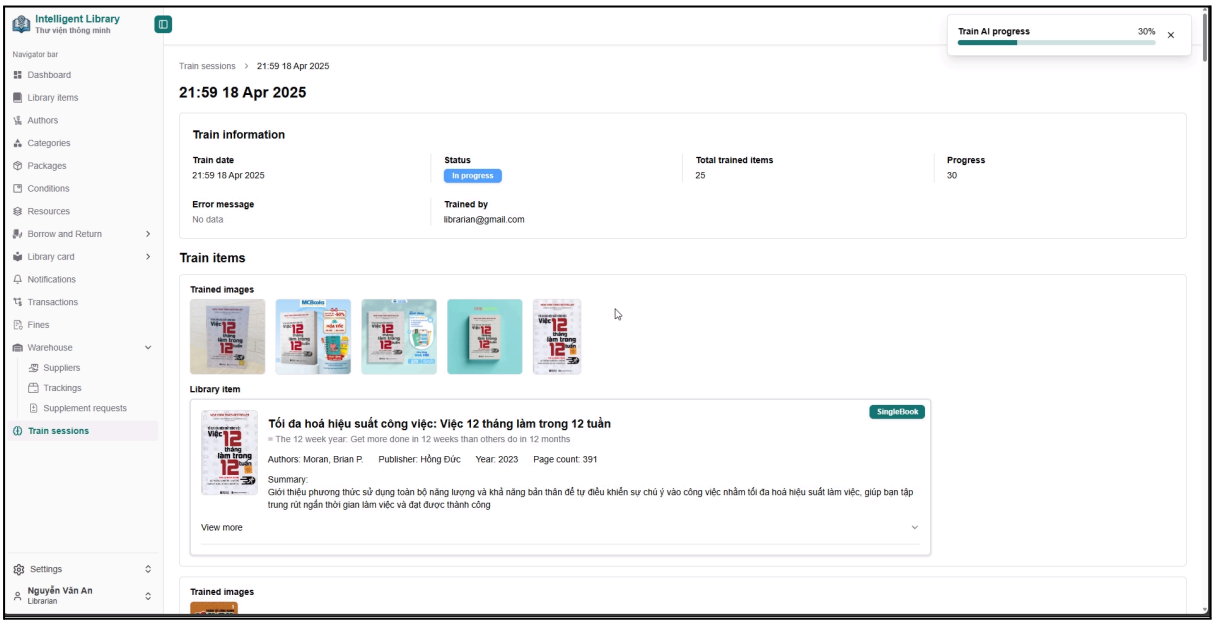


Figure 6.3.2.17: Train session detail screen

- Librarian creates a public notification to the patrons after creating the library item successfully.

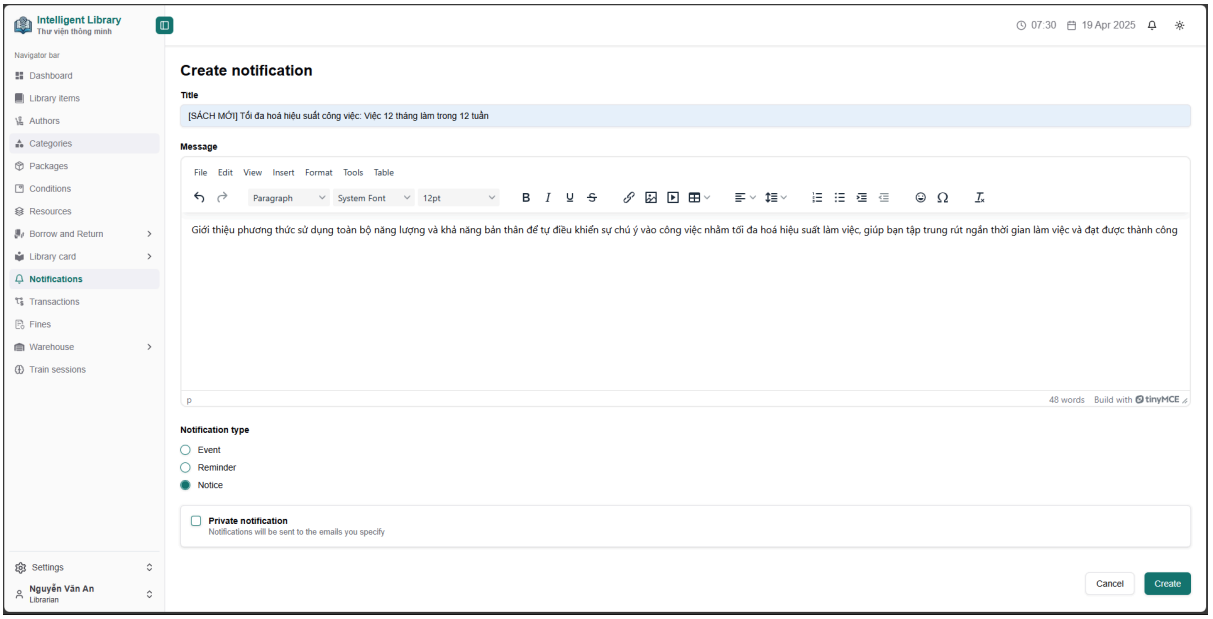


Figure 6.3.2.18: Create notification screen

3.3 Workflow 2

- Librarian logins to the Management Page.

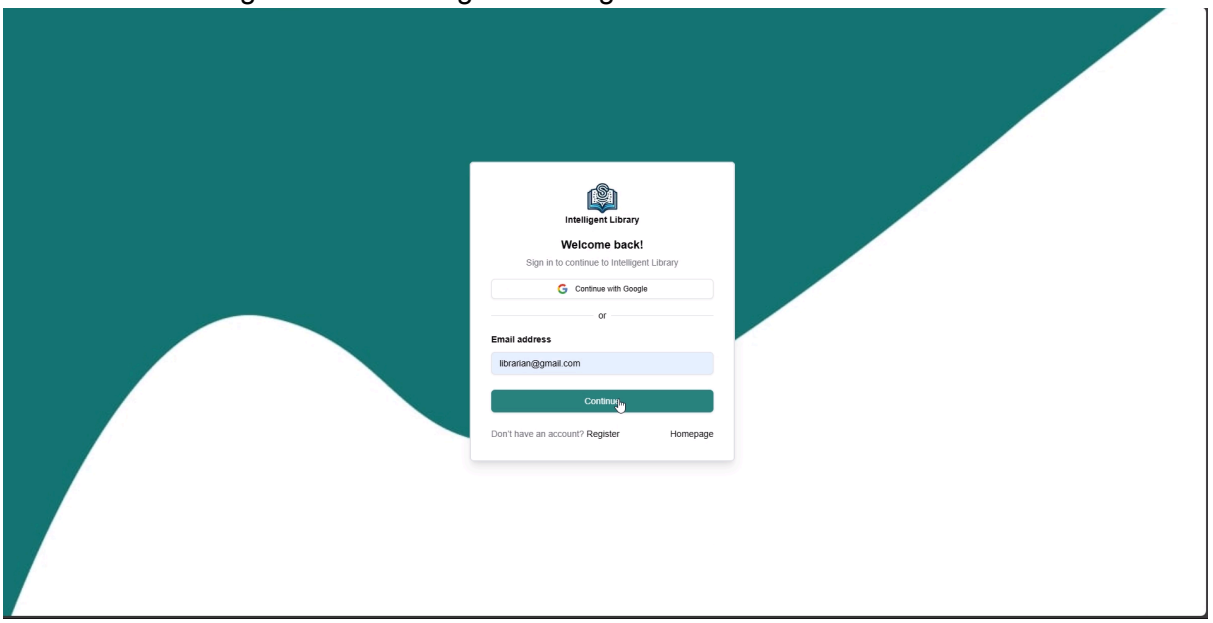


Figure 6.3.3.1: Login screen

- Librarian creates a new patron by selecting a library package and filling in the required fields.

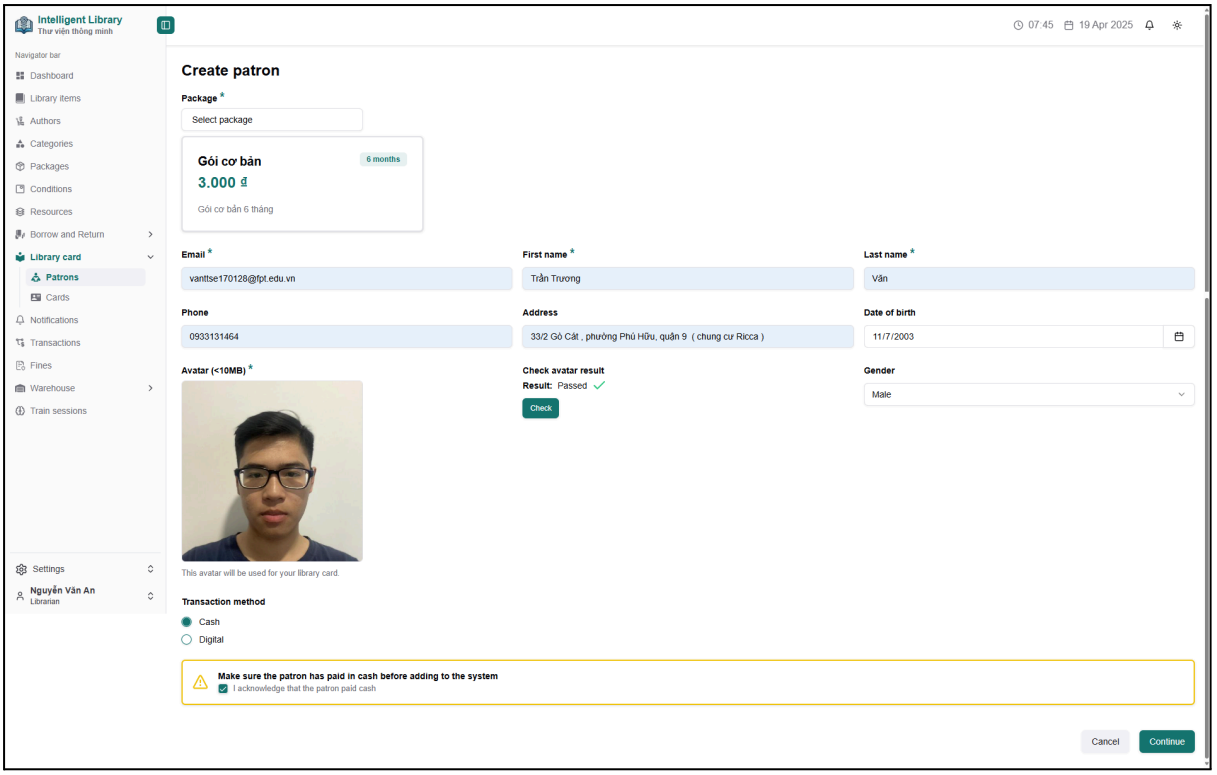


Figure 6.3.3.2: Create patron screen

- After creating the patron successfully on the system, the patron can check the email to view the details of account, library card, and transaction.

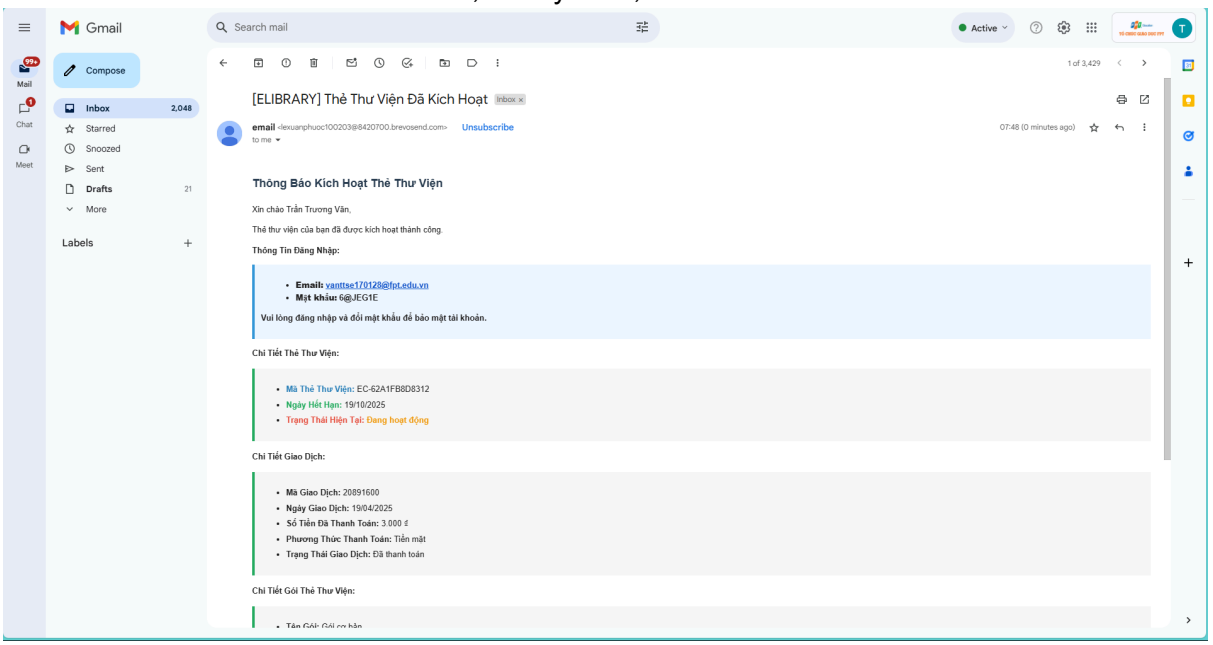


Figure 6.3.3.3: Patron's Gmail Inbox

- Patron logins to the Library Web

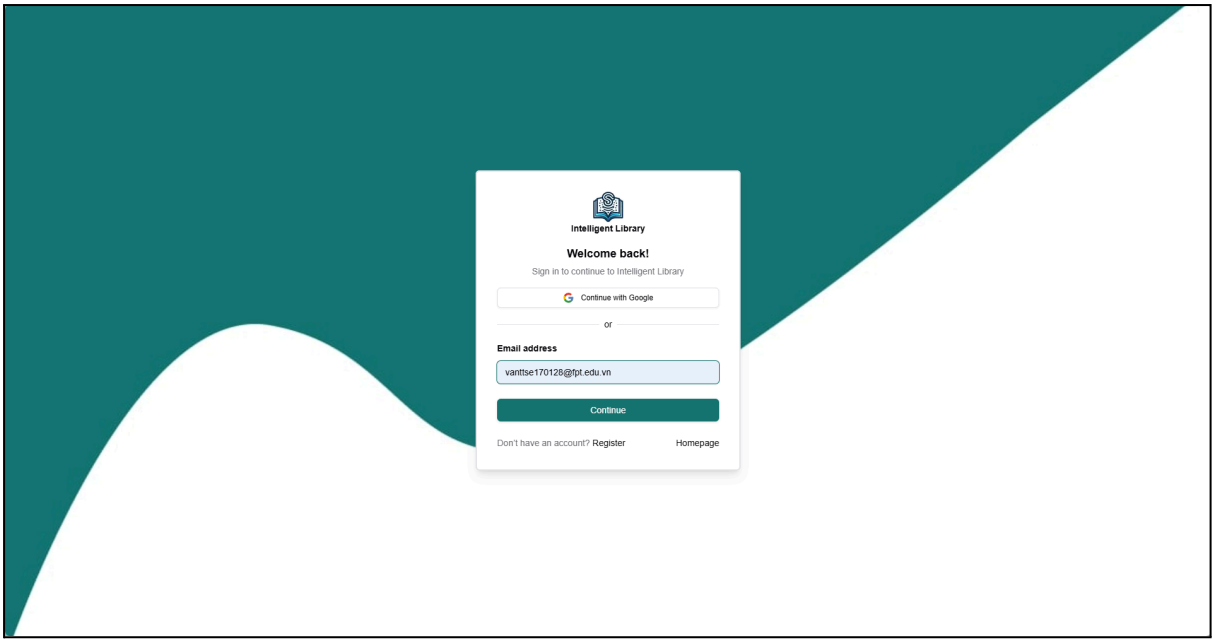


Figure 6.3.3.4: Login screen

- Patron can search for library items using the search tool.

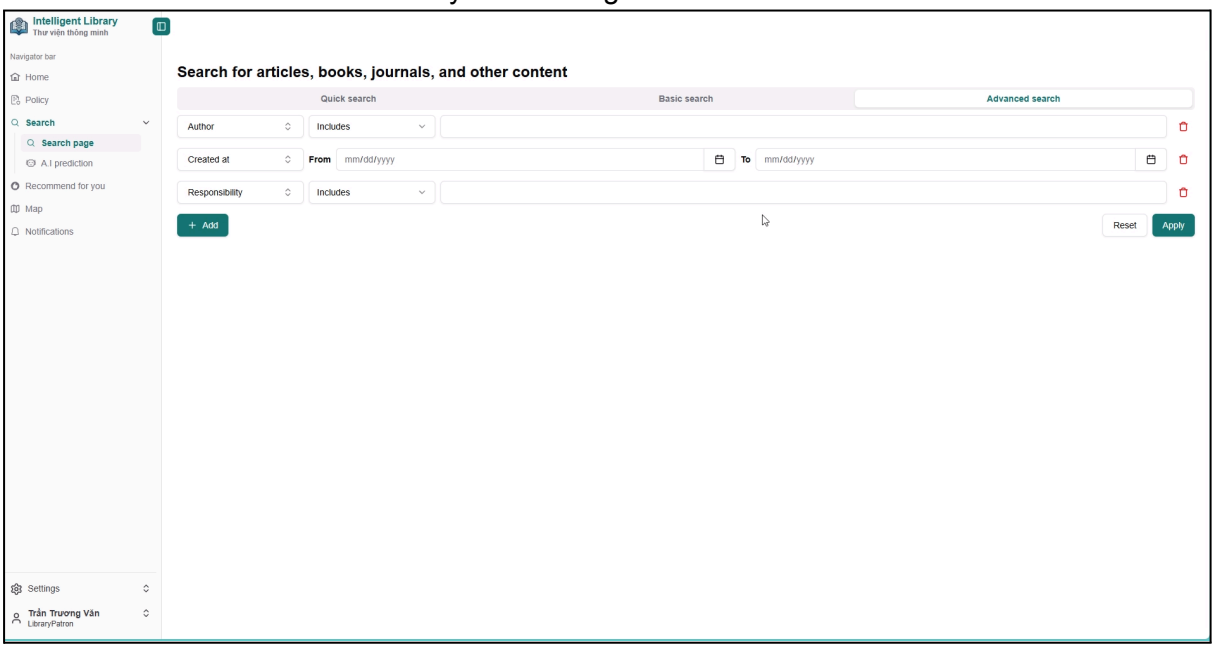


Figure 6.3.3.5: Search screen

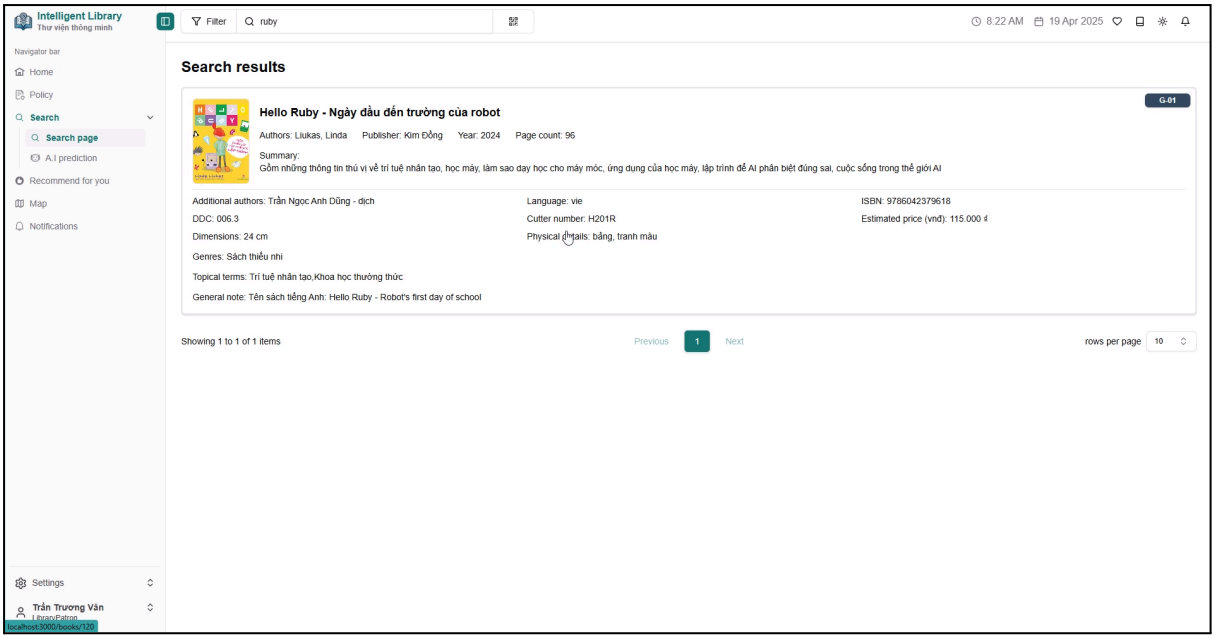


Figure 6.3.3.6: Search results screen

- Patron can also search using the "AI Predict" tool.

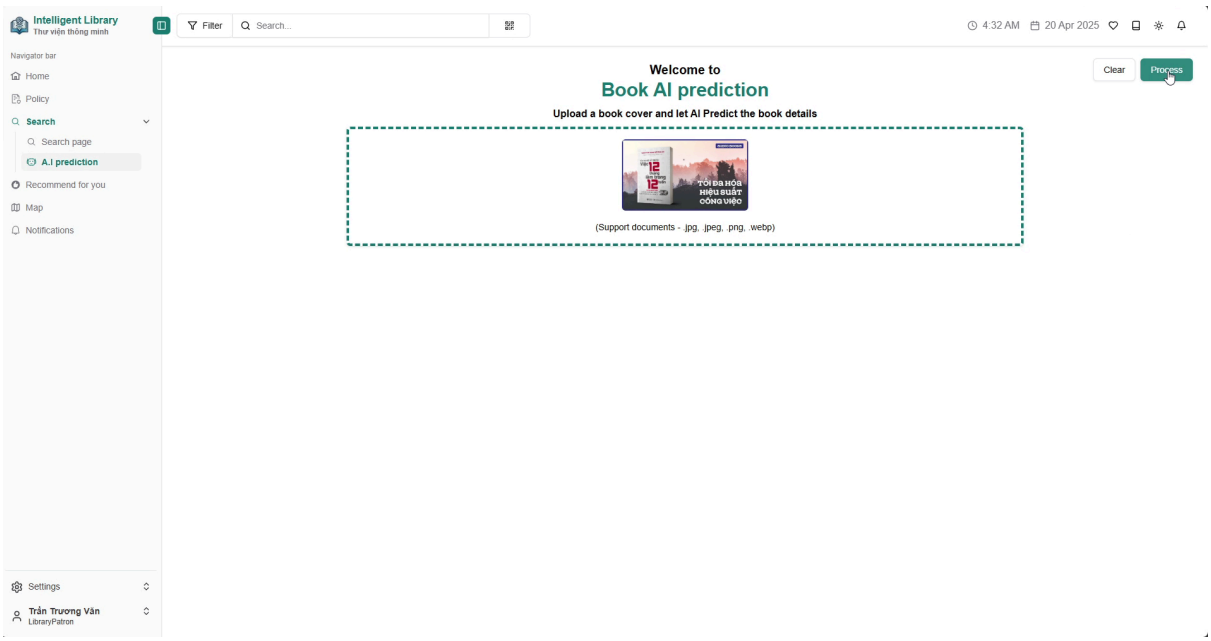


Figure 6.3.3.7: AI prediction screen

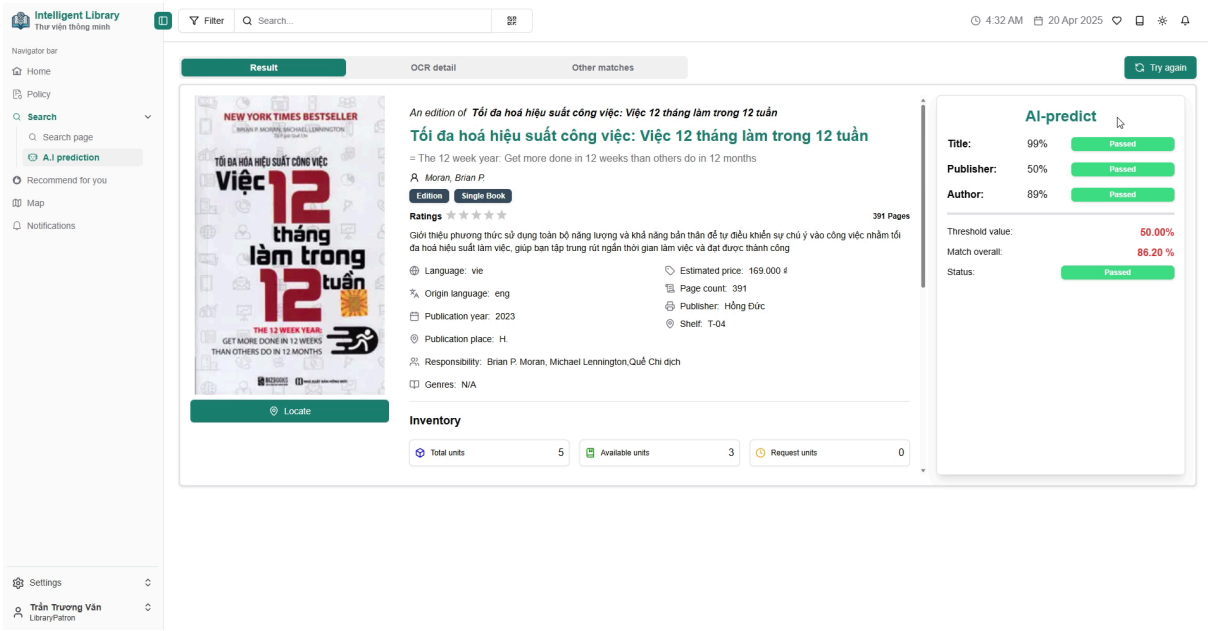


Figure 6.3.3.8: AI prediction screen

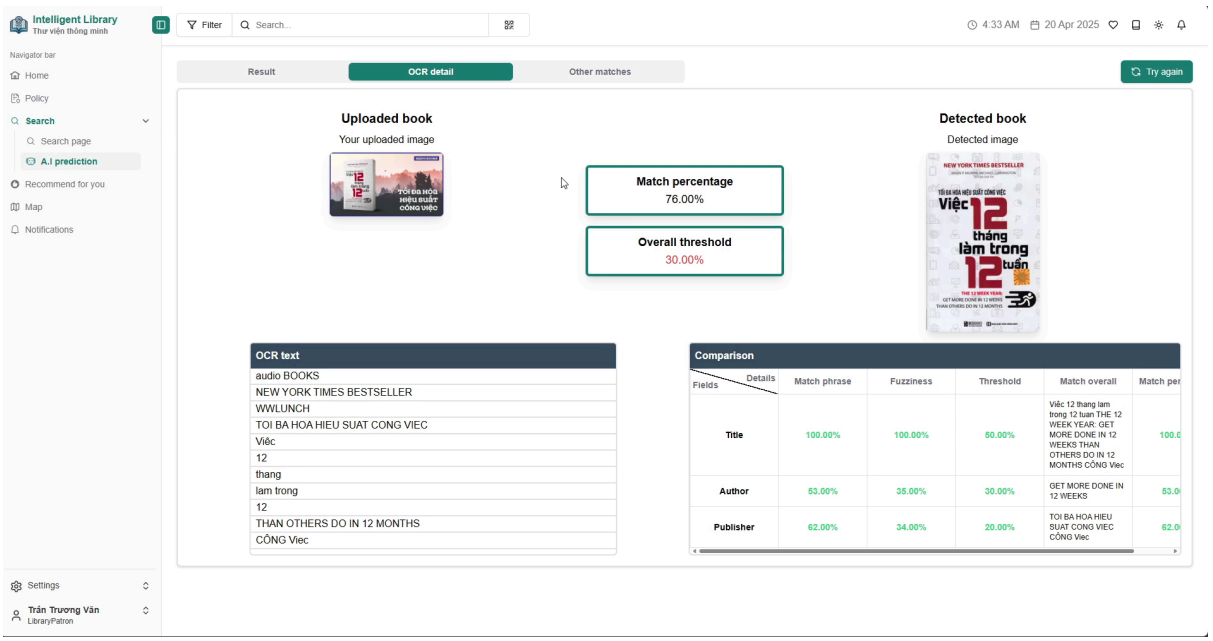


Figure 6.3.3.9: AI prediction screen

- Patron adds library items and digital resources he wishes to borrow to his borrowing list.

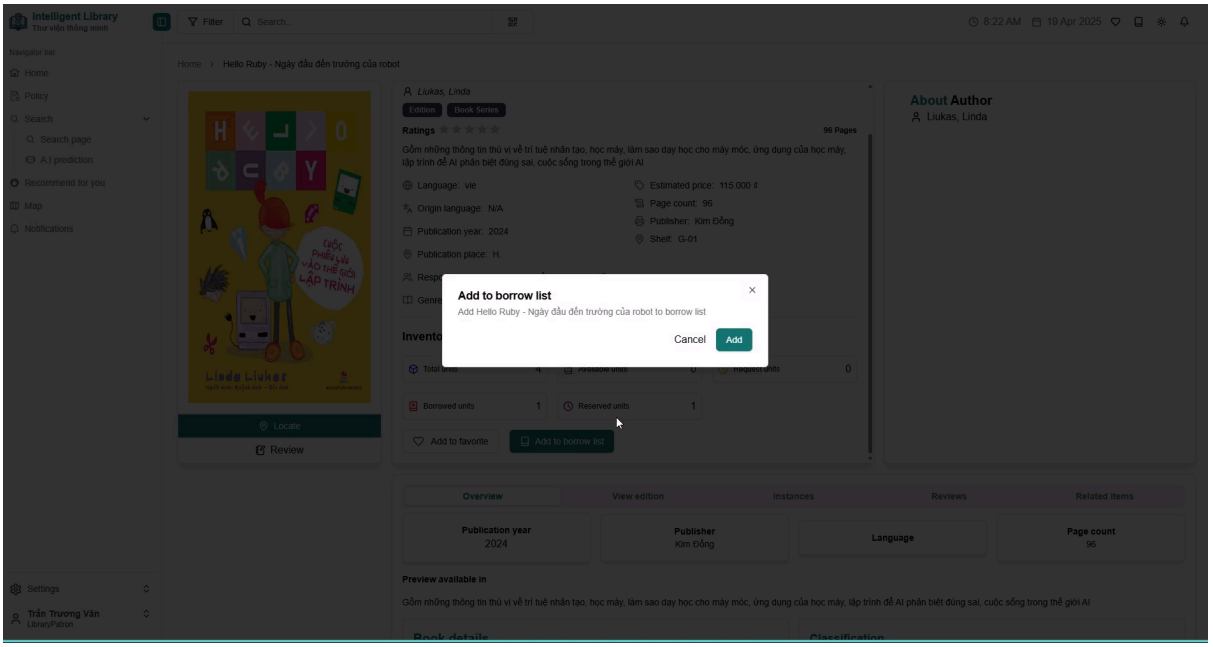


Figure 6.3.3.10: Library item detail screen

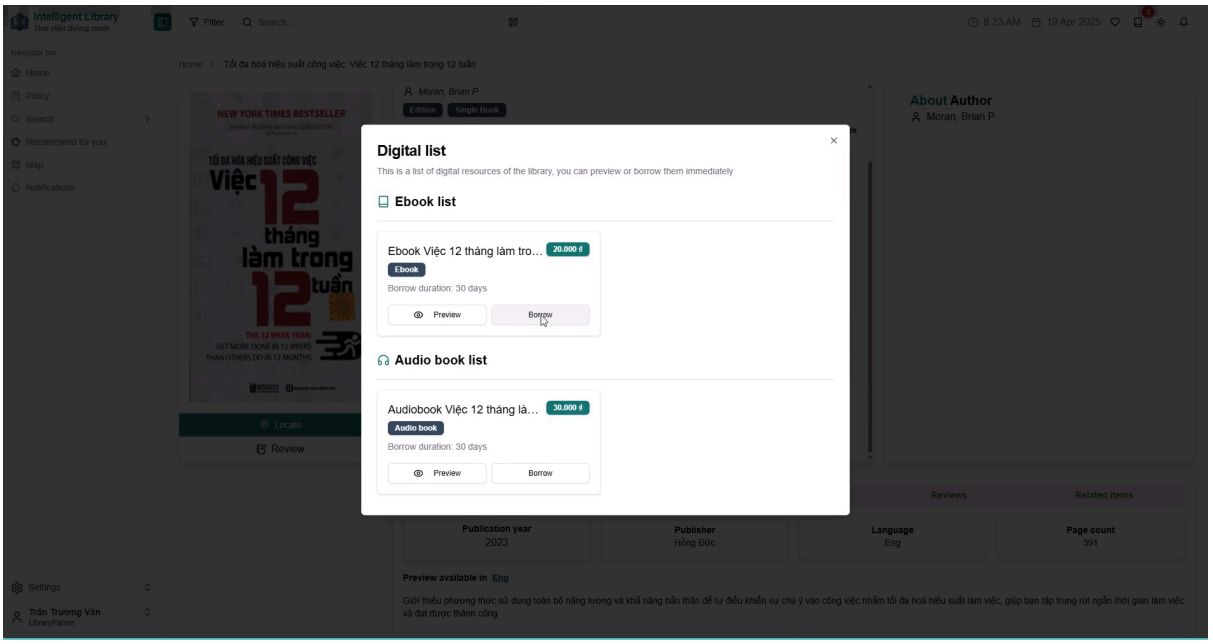


Figure 6.3.3.11: Library item detail screen

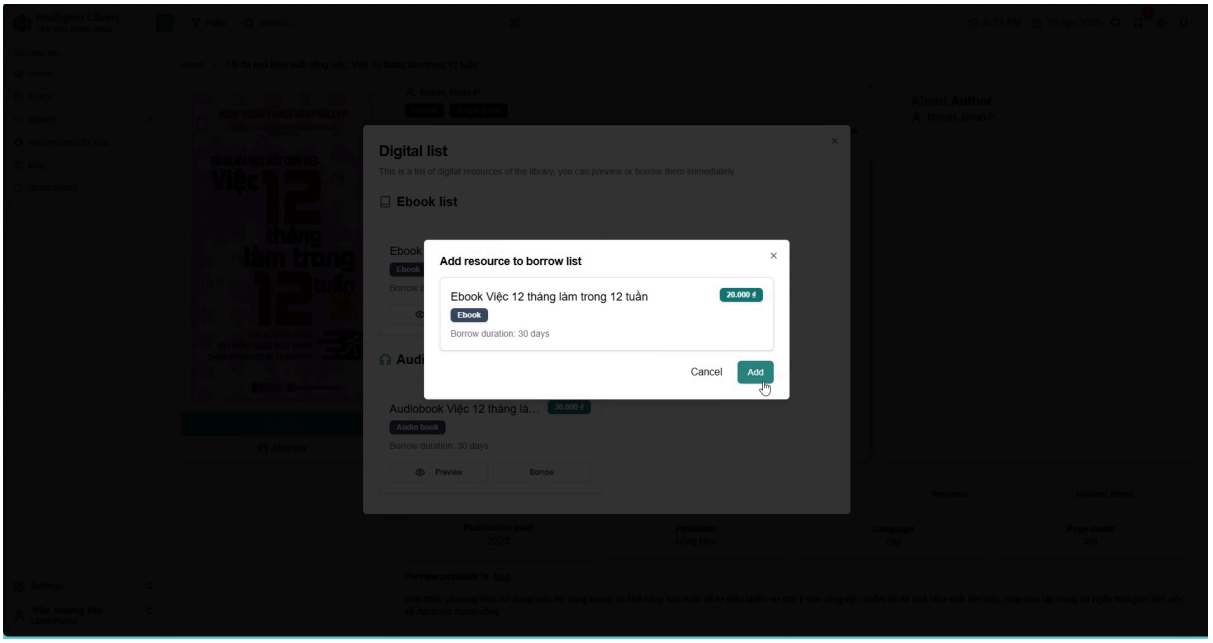


Figure 6.3.3.12: Library item detail screen

- Patron reviews his borrowing list and clicks "Continue" to submit a borrow request.

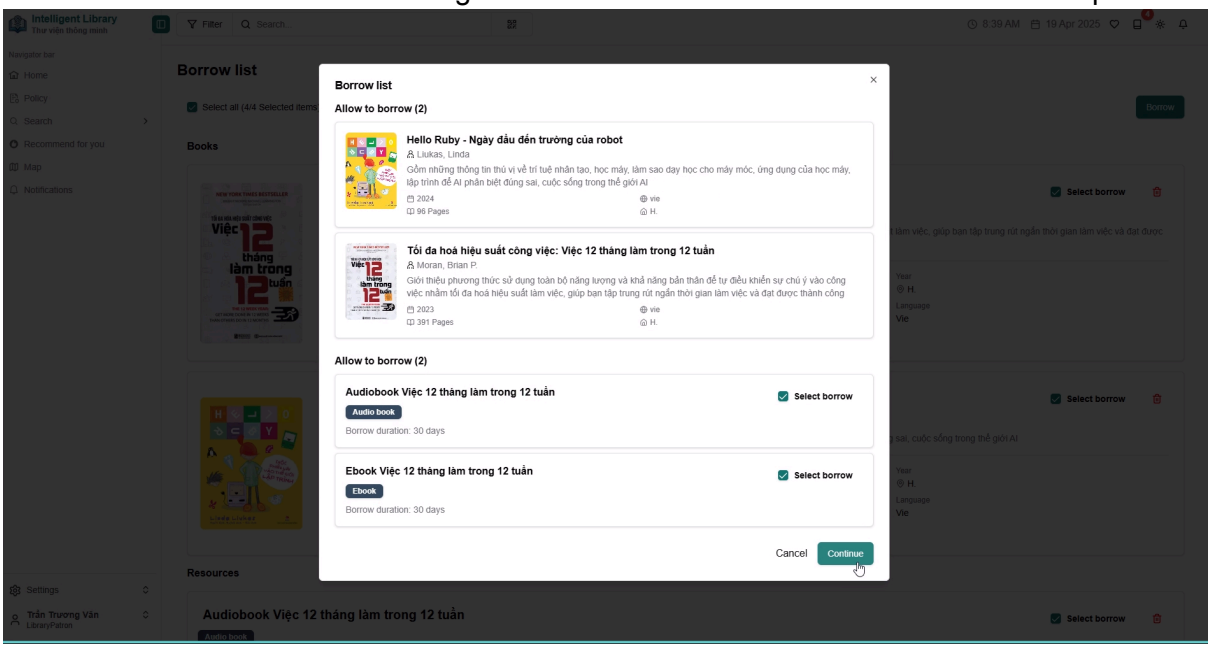


Figure 6.3.3.13: Borrow list screen

- Patron can view the borrow request detail on the “Borrow” tab in his account and make payment for the digital resources.

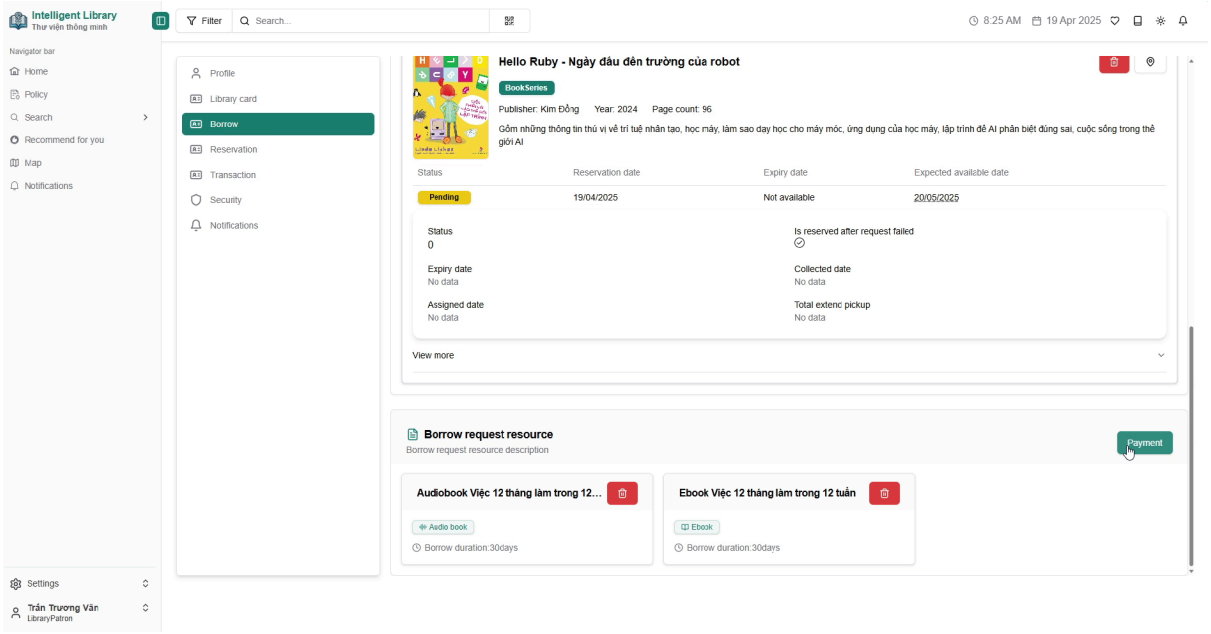


Figure 6.3.3.14: Borrow request detail screen

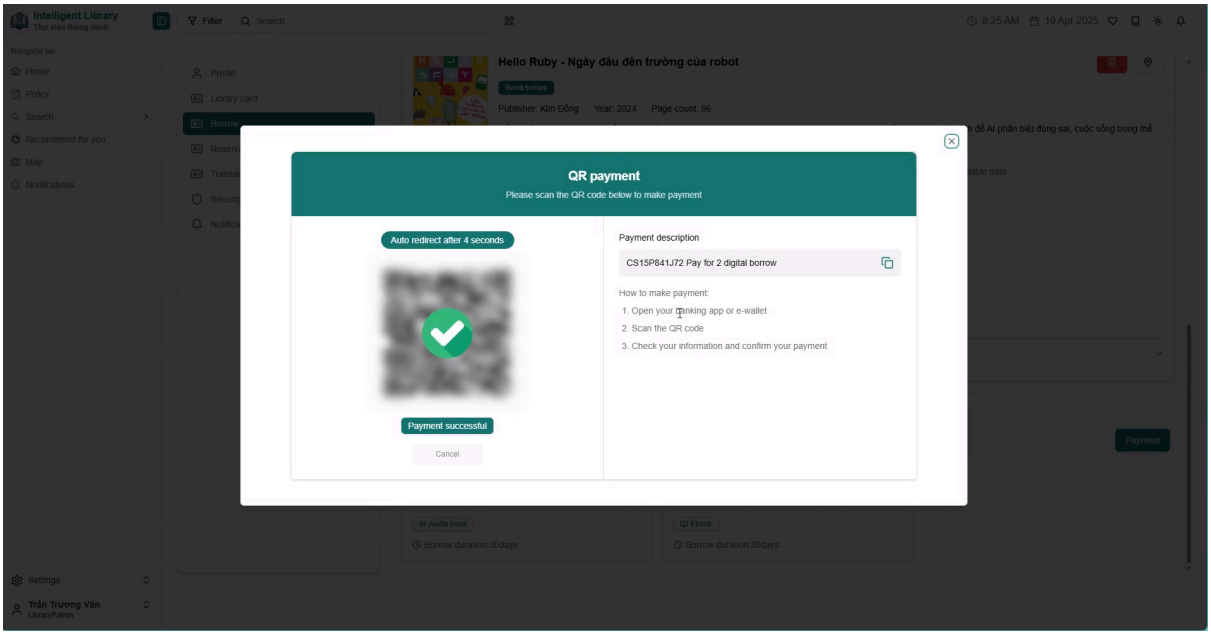


Figure 6.3.3.15: Borrow request detail screen

- After paying for the digital resources, the patron can view them in the “Borrow digital” tab as watermarked by his email.

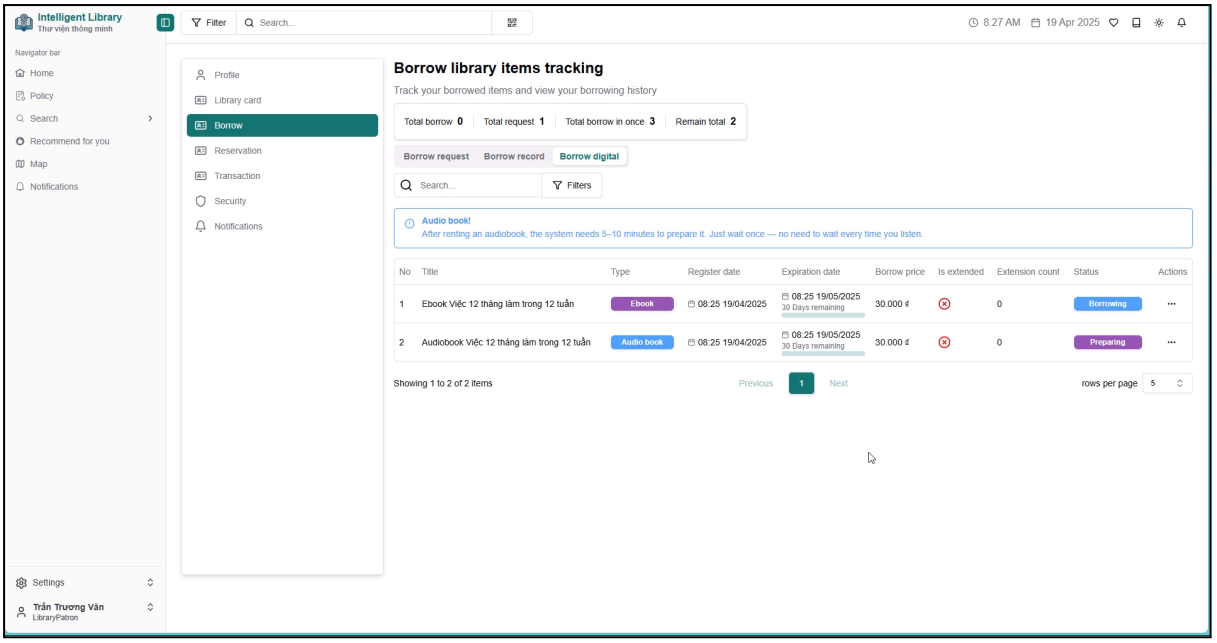


Figure 6.3.3.16: Borrow digitals screen

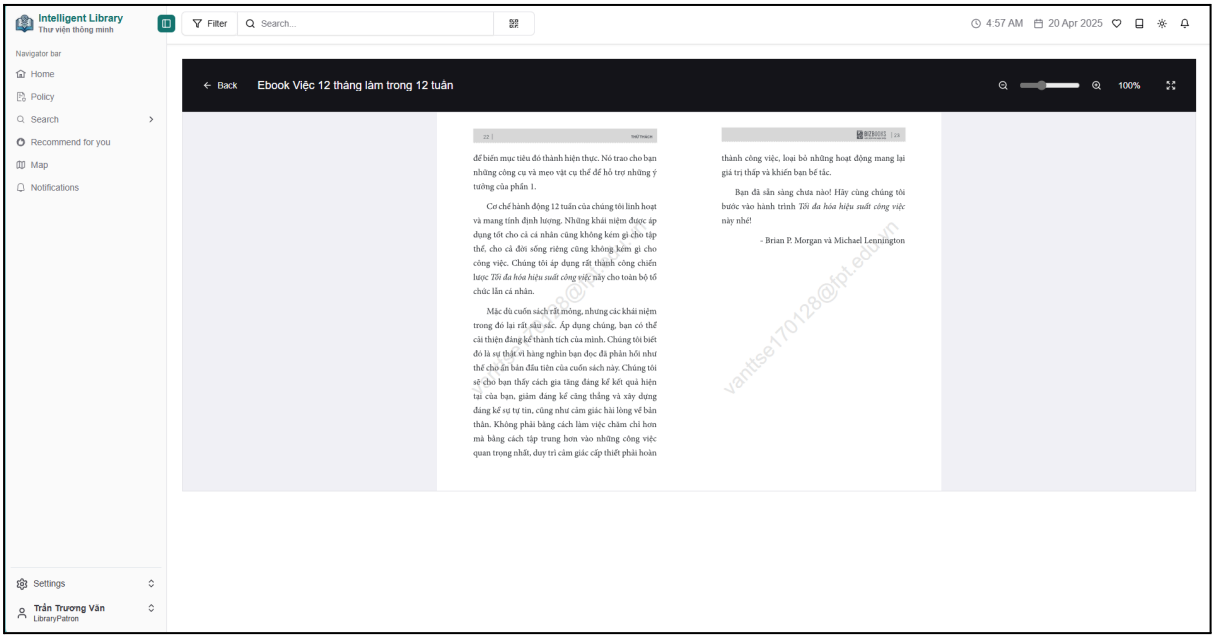


Figure 6.3.3.17: Borrow resource detail screen (ebook)

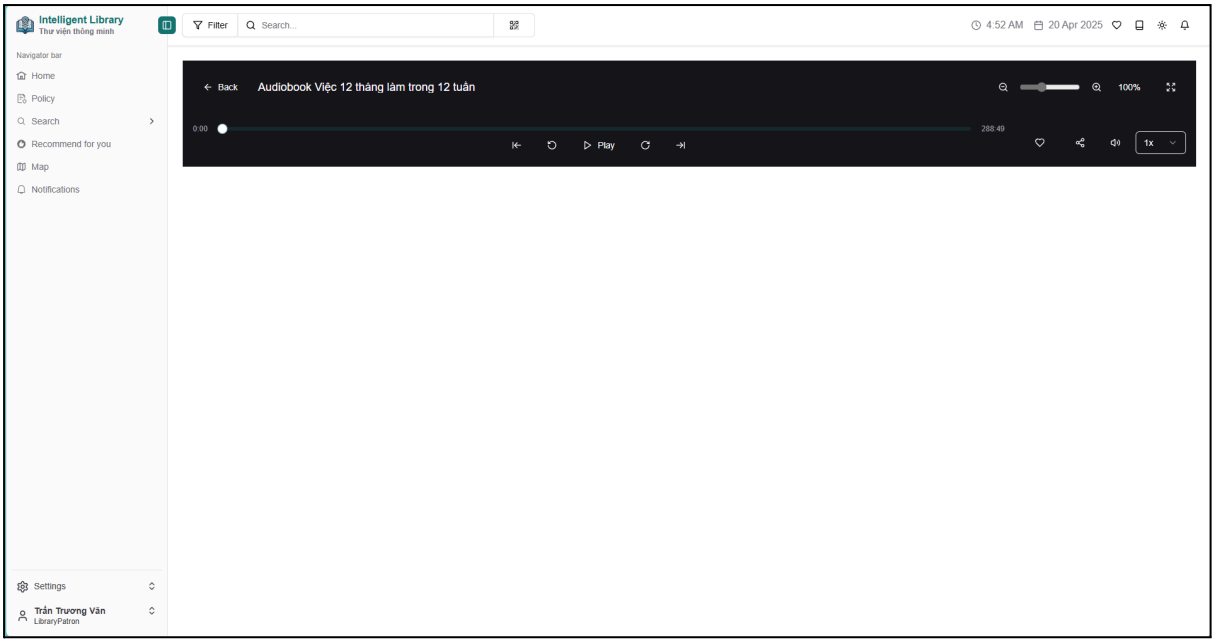


Figure 6.3.3.18: Borrow resource detail screen (audiobook)

- Patron accesses the Mobile App to locate the library items position, then bring them to the librarian.

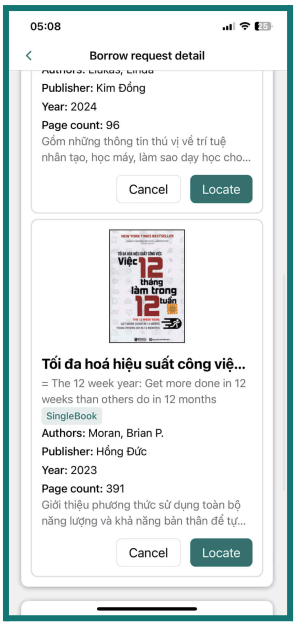


Figure 6.3.3.19: Borrow request detail screen (Mobile App)

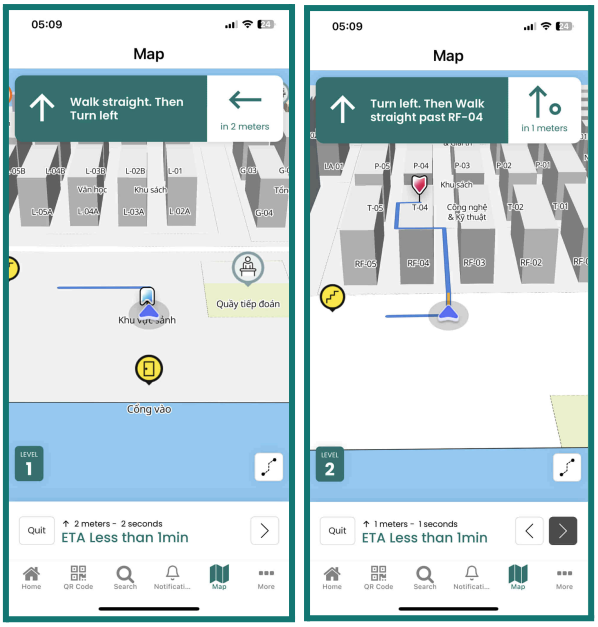


Figure 6.3.3.20: Map screen (Mobile App)

- Librarian logins to the Management Page

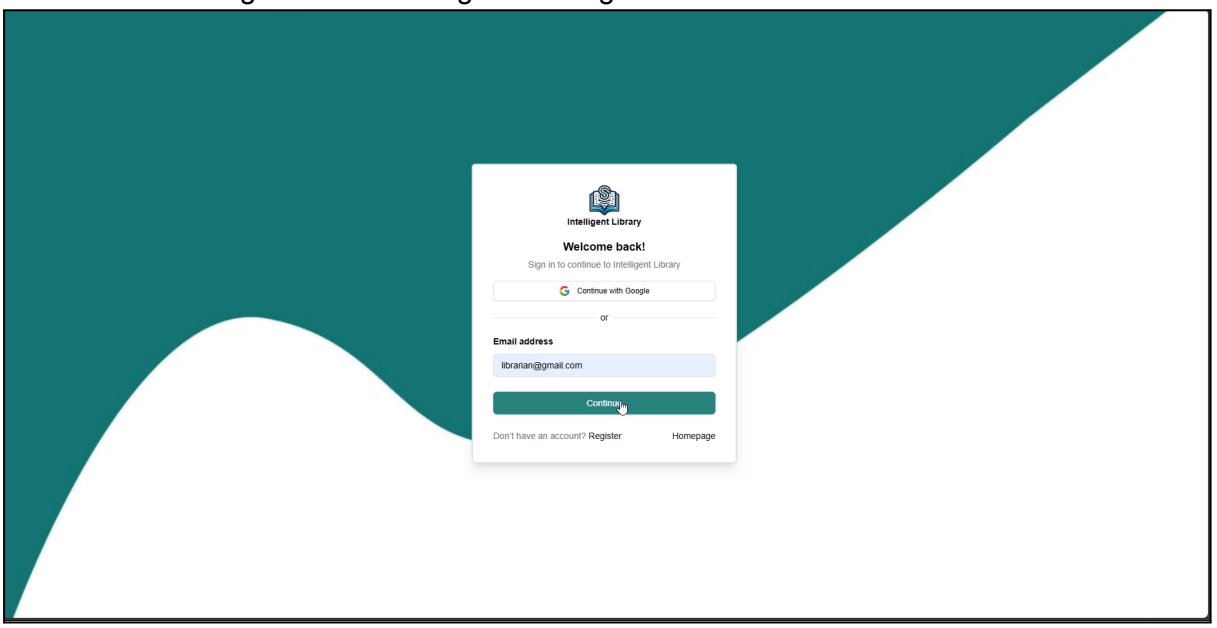


Figure 6.3.3.21: Login screen

- Librarian scan the barcode of the patron's card

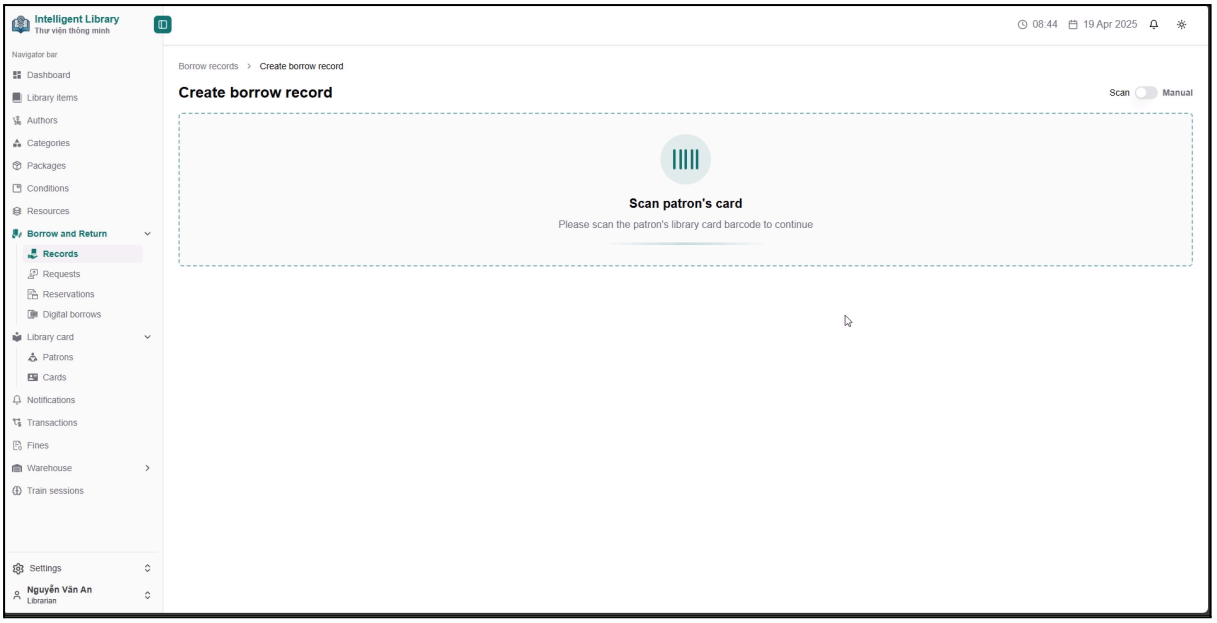


Figure 6.3.3.22: Create borrow record screen

- Librarian scans all the library items that the patron brings, which were previously collected from the shelves, then clicks "Continue" to create the borrowed record.

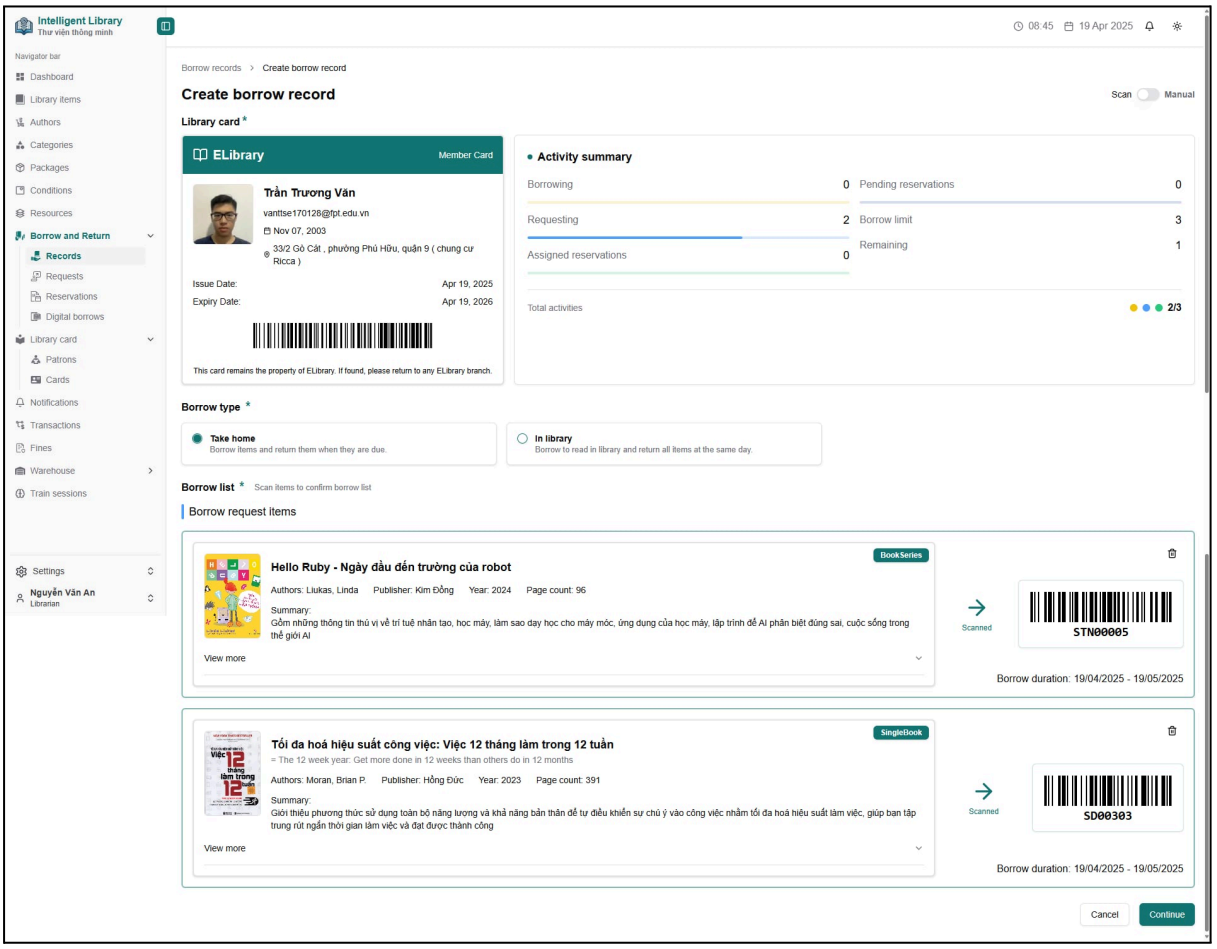


Figure 6.3.3.23: Create borrow record screen

3.4 Workflow 3

- Librarian logins to the Management Page.

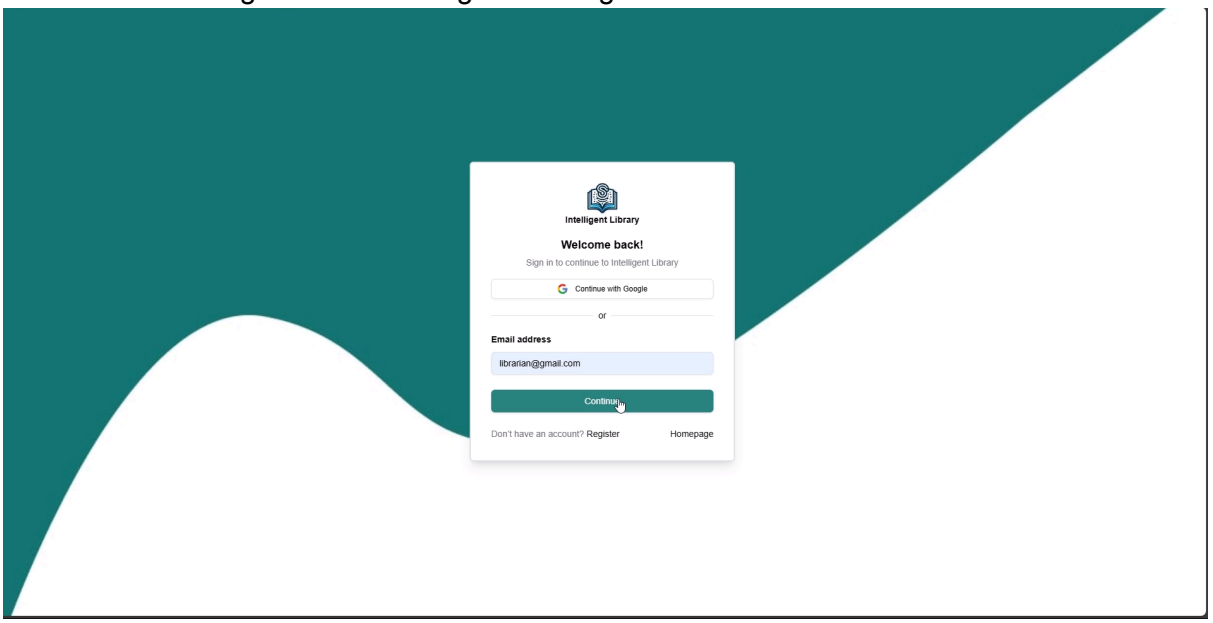


Figure 6.3.4.1: Login screen

- Librarian scan the barcode of the patron's card

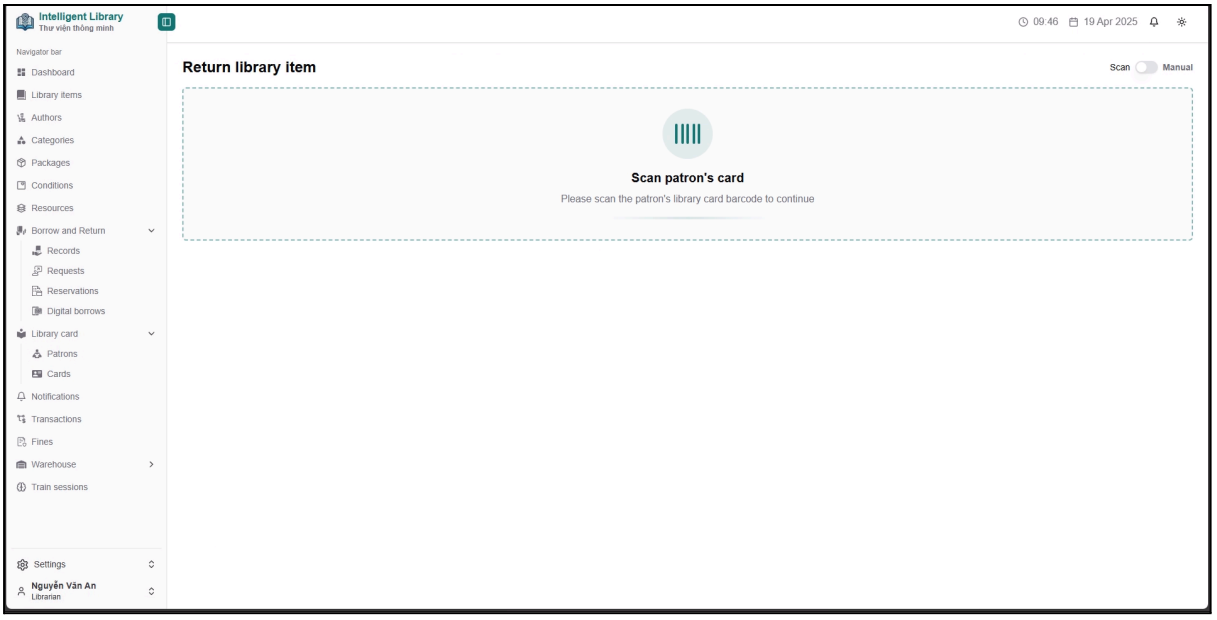


Figure 6.3.4.2: Return library items screen

- Librarian scans the library items that the patron returns.

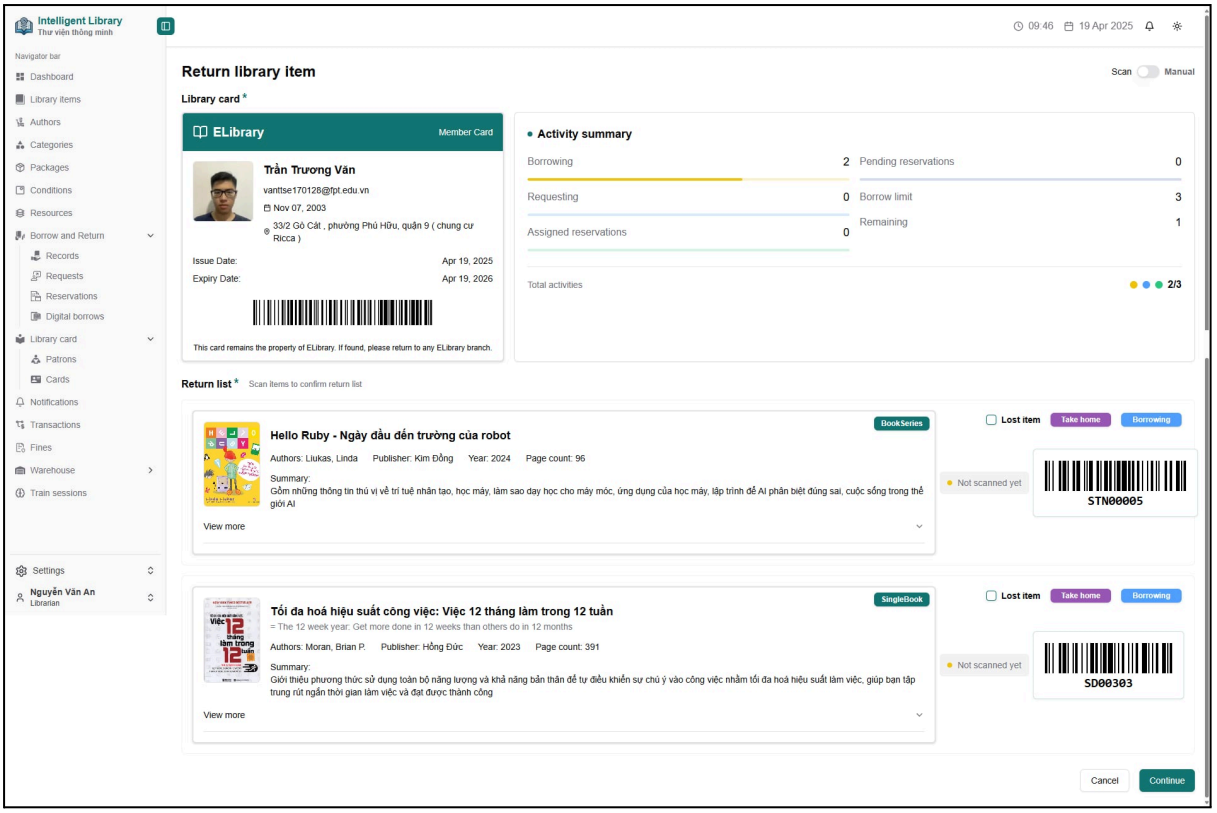


Figure 6.3.4.3: Return library items screen

- Librarian adds the fines to the specific library item (optional), then clicks “Continue” to complete the return items process.

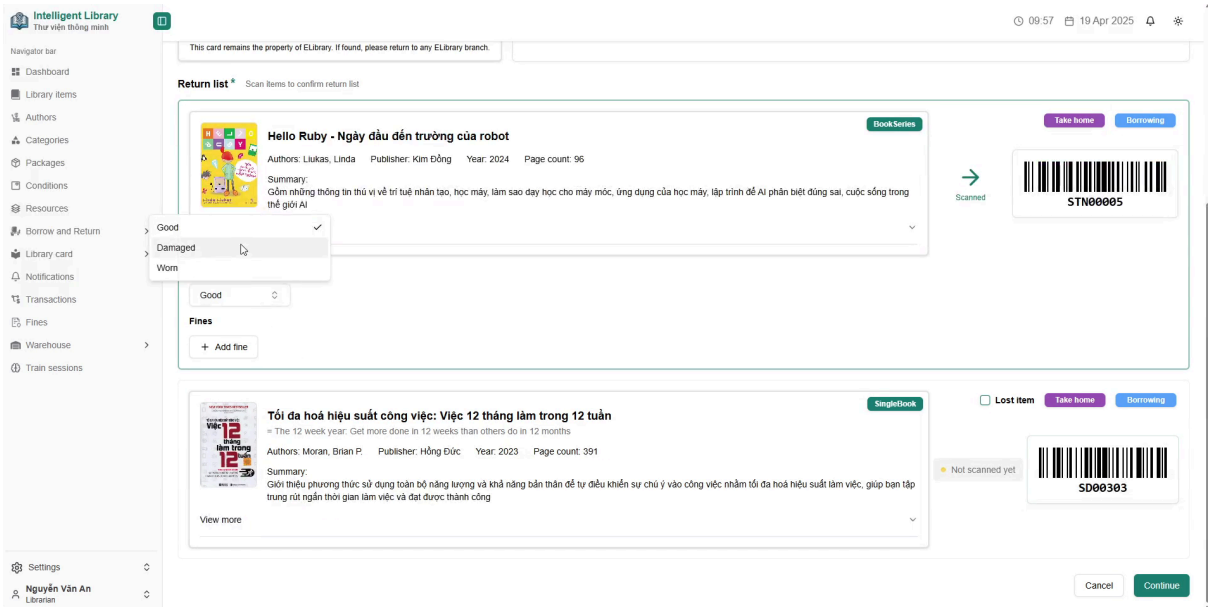


Figure 6.3.4.4: Return library items screen

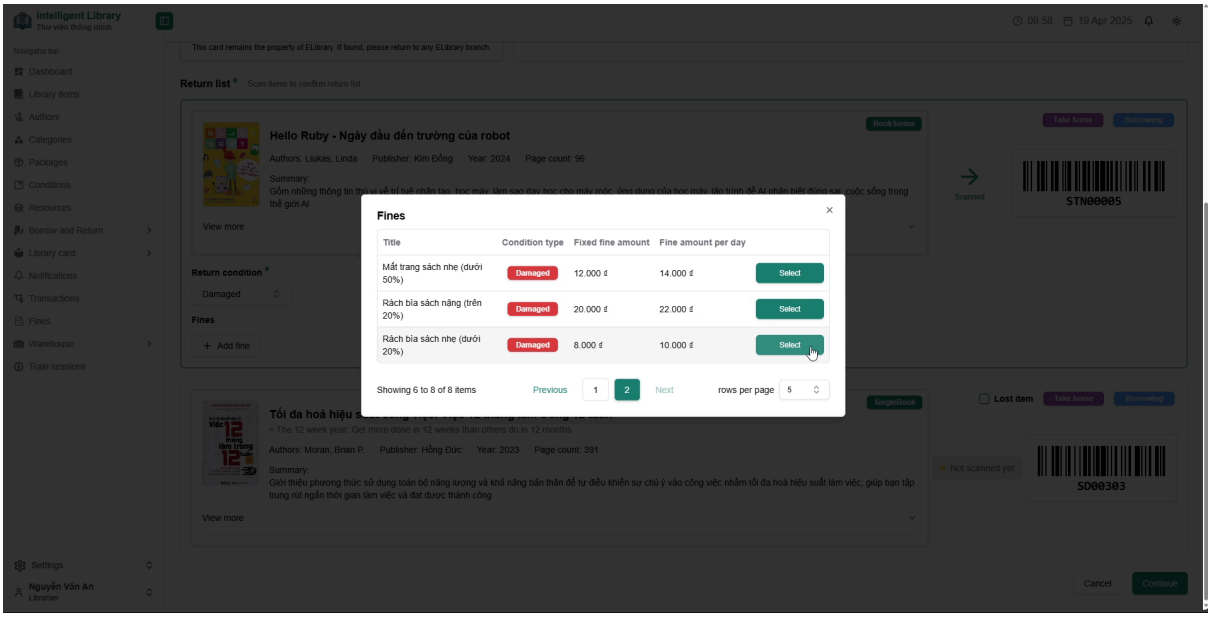


Figure 6.3.4.5: Return library items screen

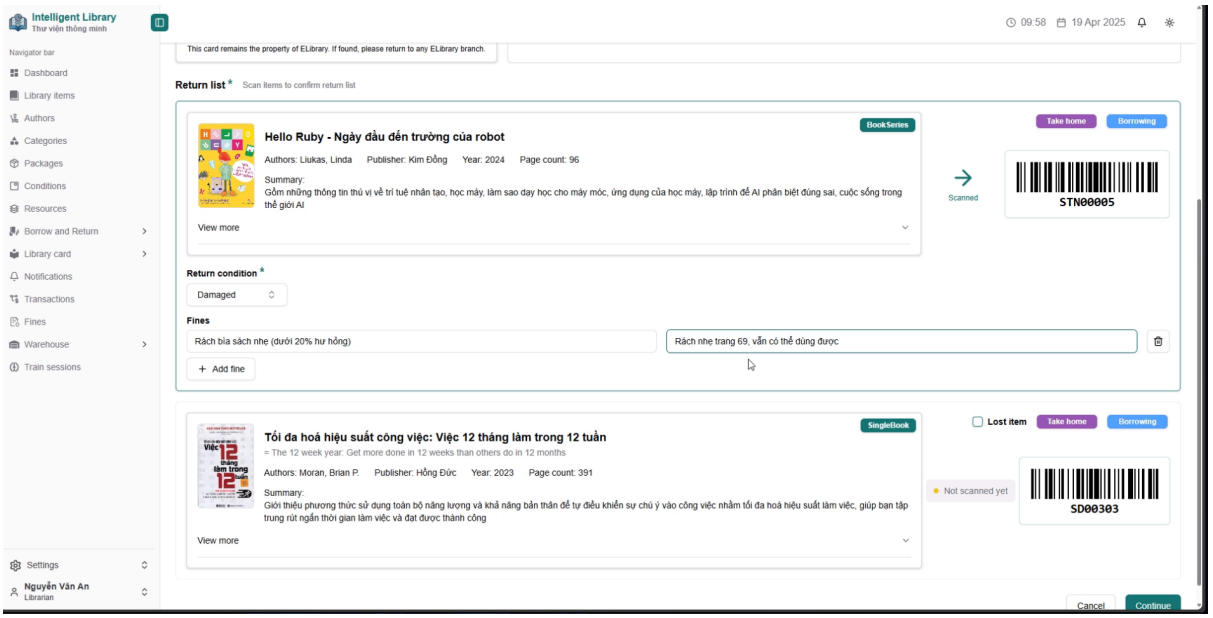


Figure 6.3.4.6: Return library items screen

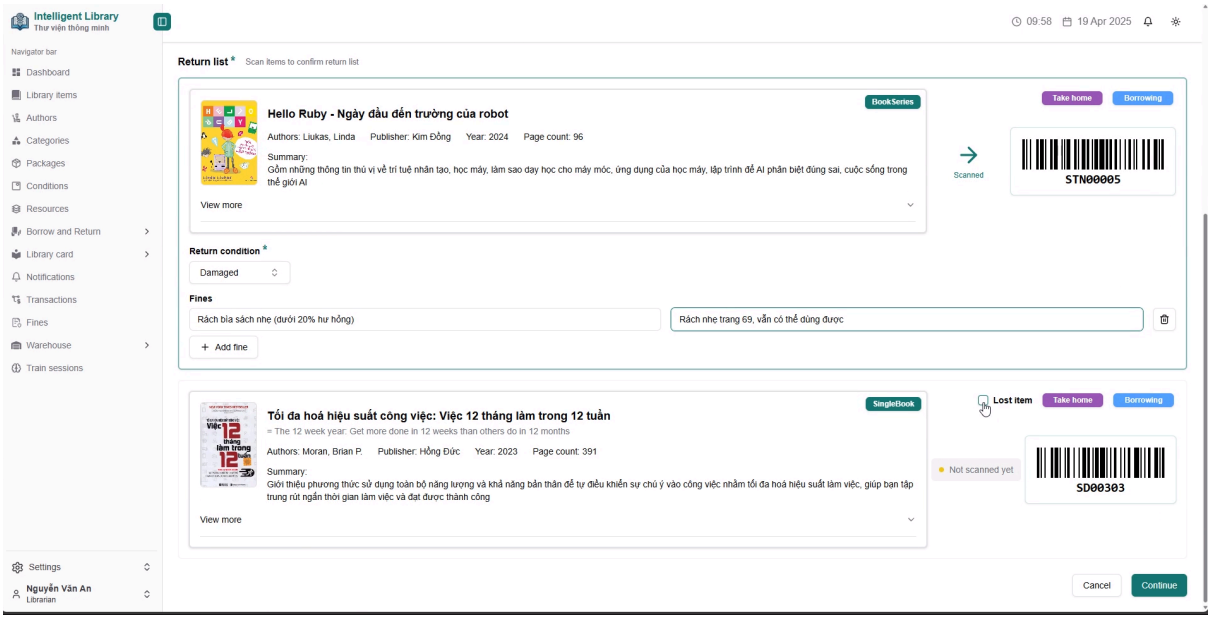


Figure 6.3.4.7: Return library items screen

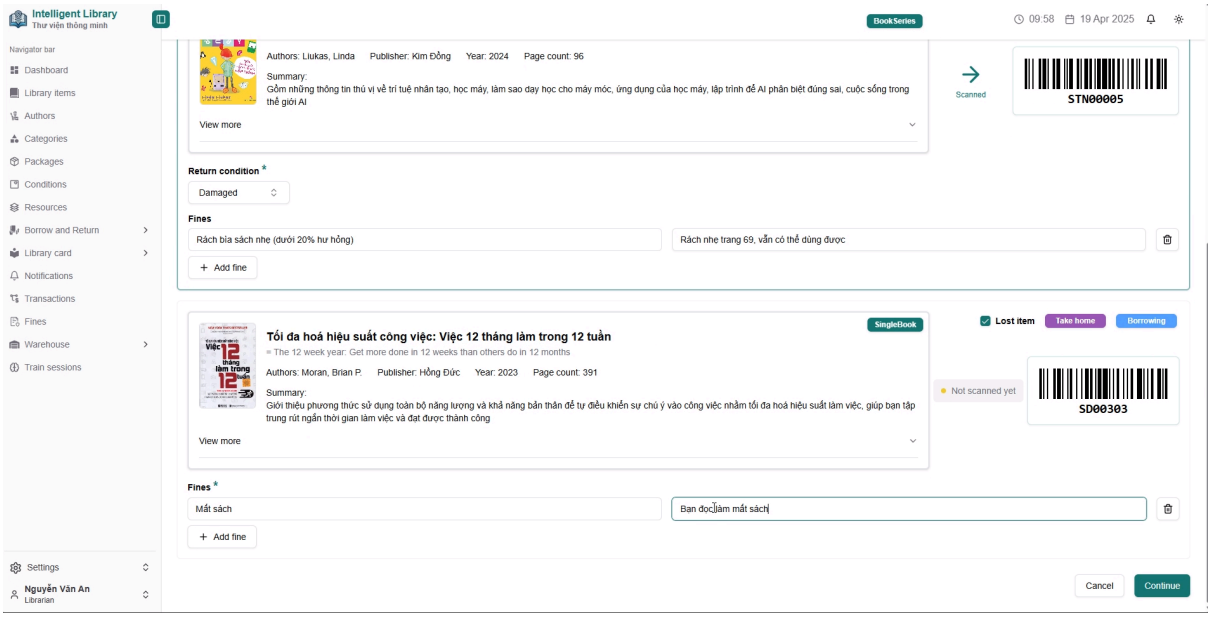


Figure 6.3.4.8: Return library items screen

- Librarian confirms that the returned library items can be assigned to existing reservations, then clicks “Continue”.

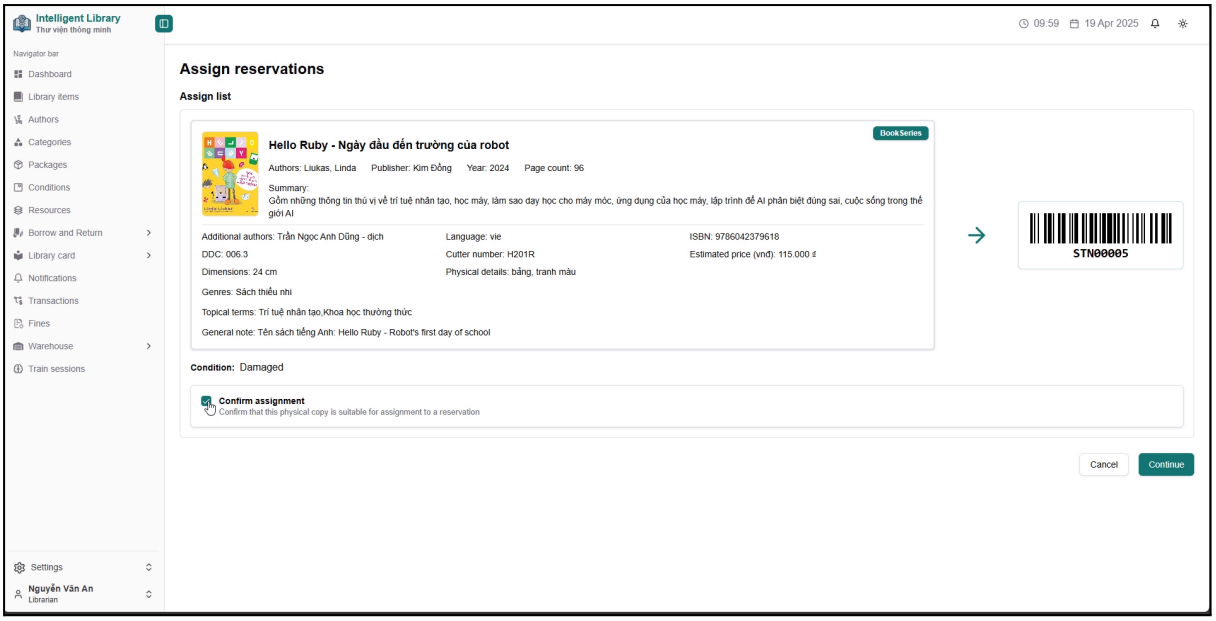


Figure 6.3.4.9: Assign reservations screen

- Librarian prints the “hold slips” and marks the library items that have been successfully assigned to reservations, so that other patrons can come and pick them up.

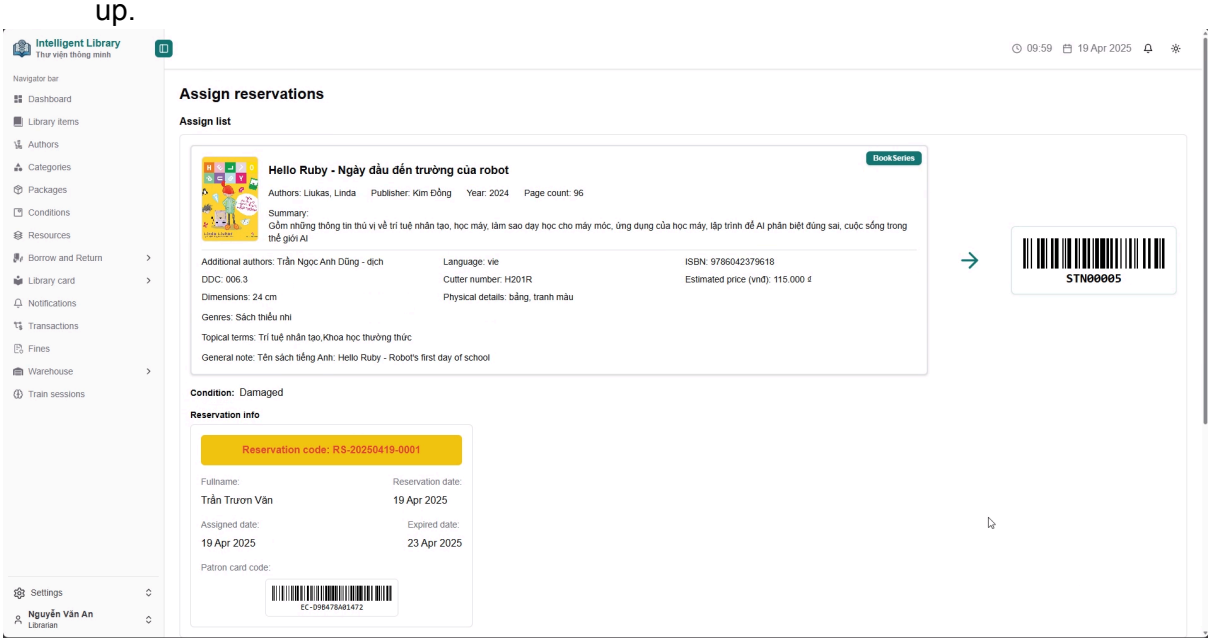


Figure 6.3.4.10: Assign reservations screen

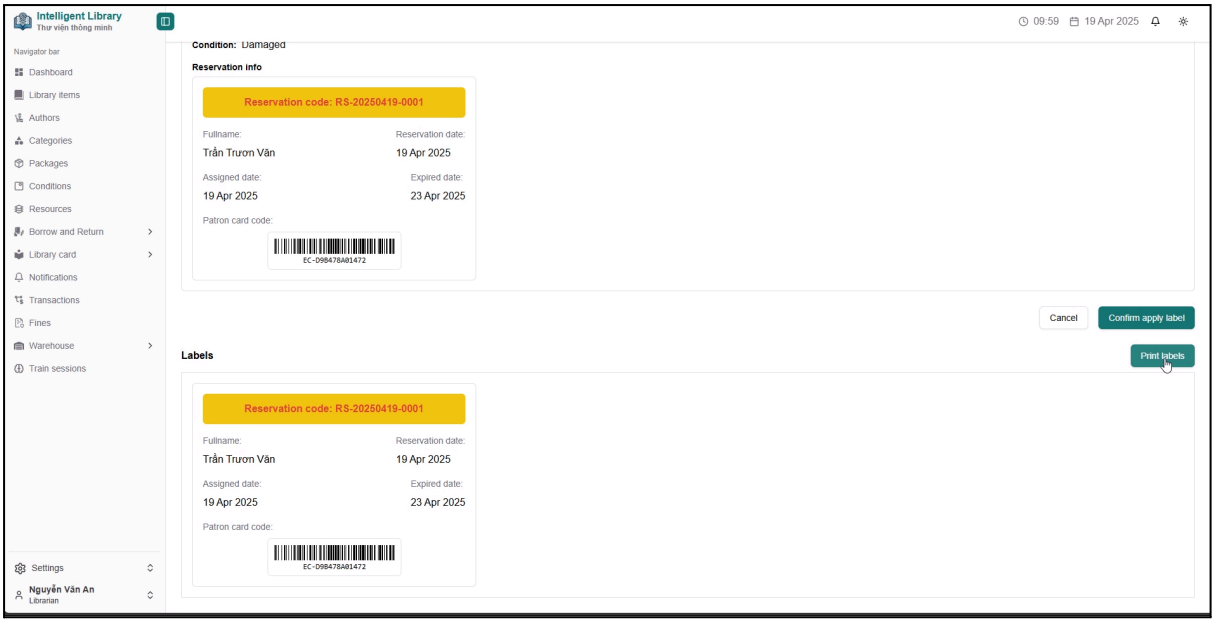


Figure 6.3.4.11: Assign reservations screen

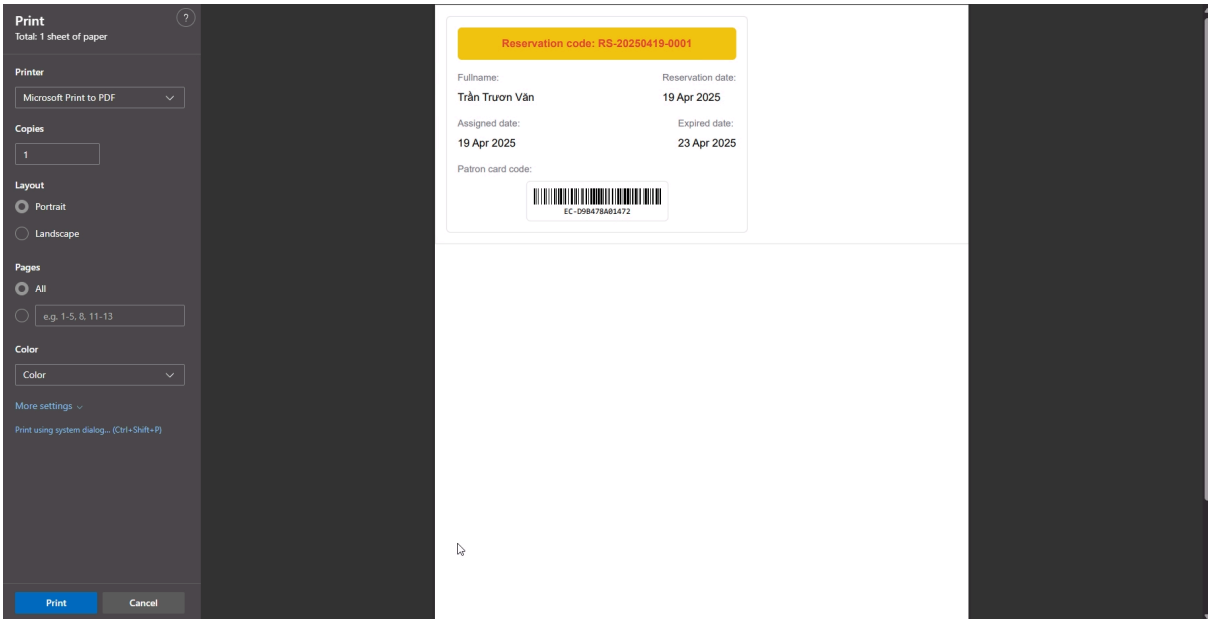


Figure 6.3.4.12: Printer screen

- Librarian needs to confirm that the “hold slips” have been attached by entering the number of slips applied.

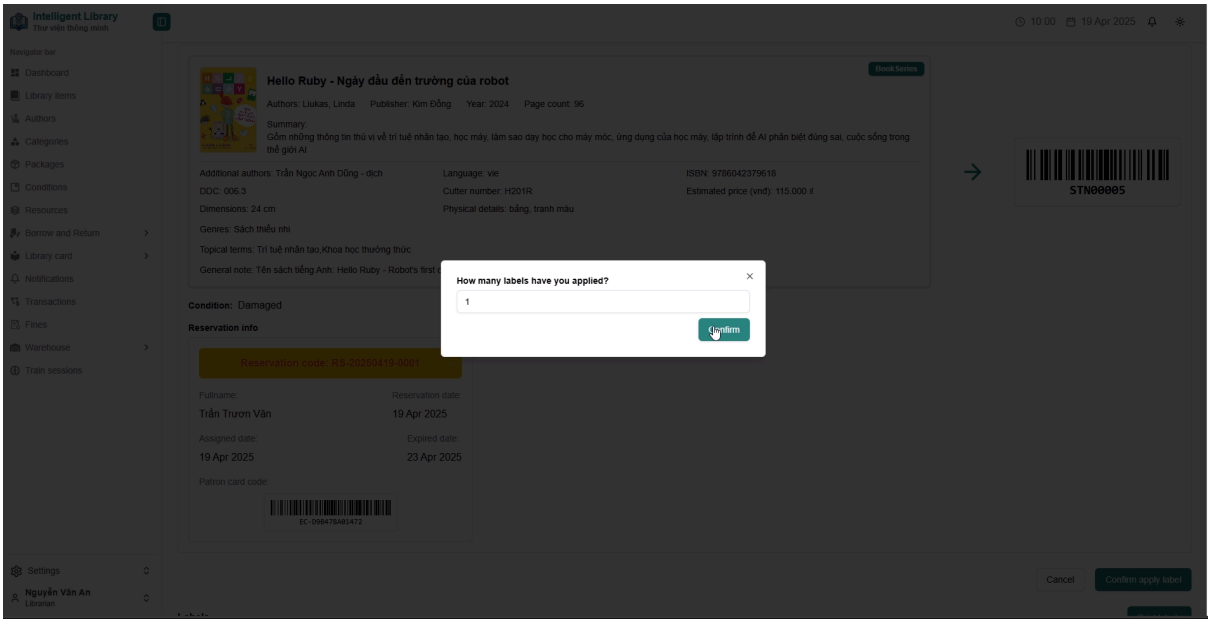


Figure 6.3.4.13: Assign reservations screen

- Patron logins to the Library Web

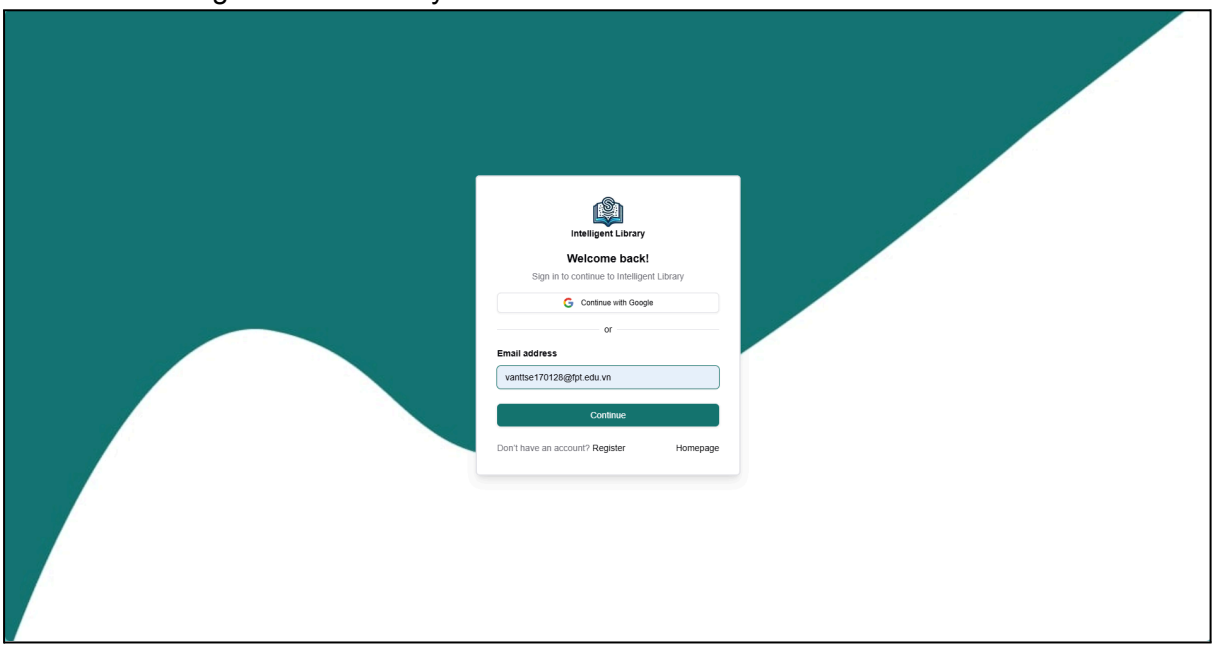


Figure 6.3.4.14: Login screen

- Patron navigates to the “Borrow record” tab in his profile to pay the fines if it exists.

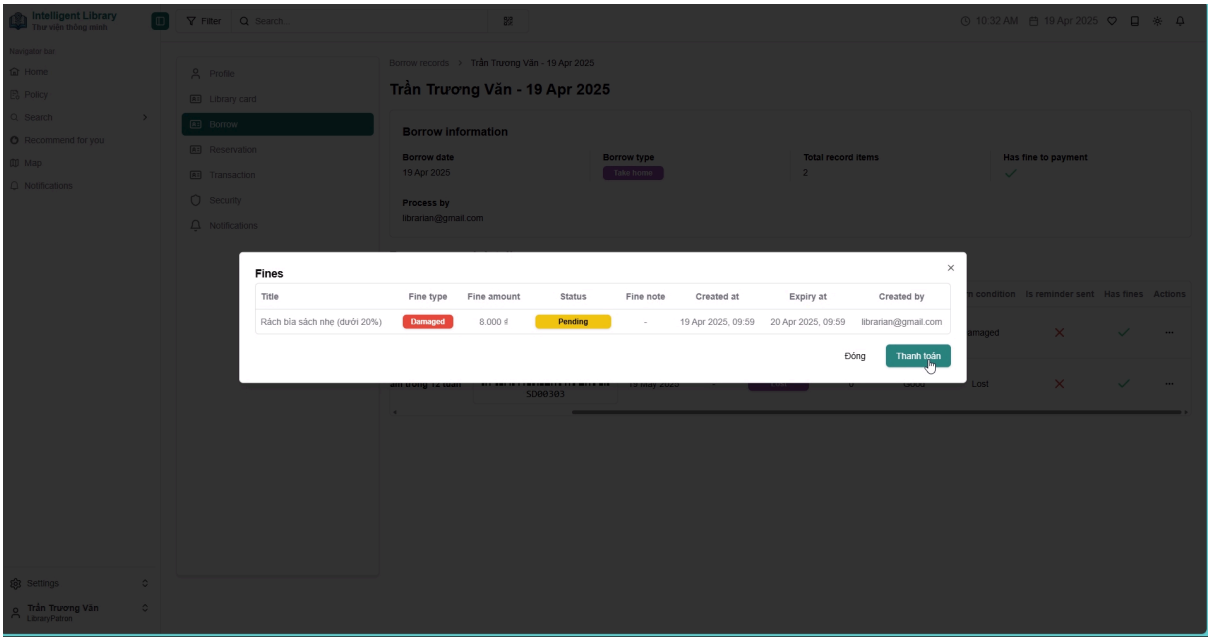


Figure 6.3.4.15: Borrow record detail screen

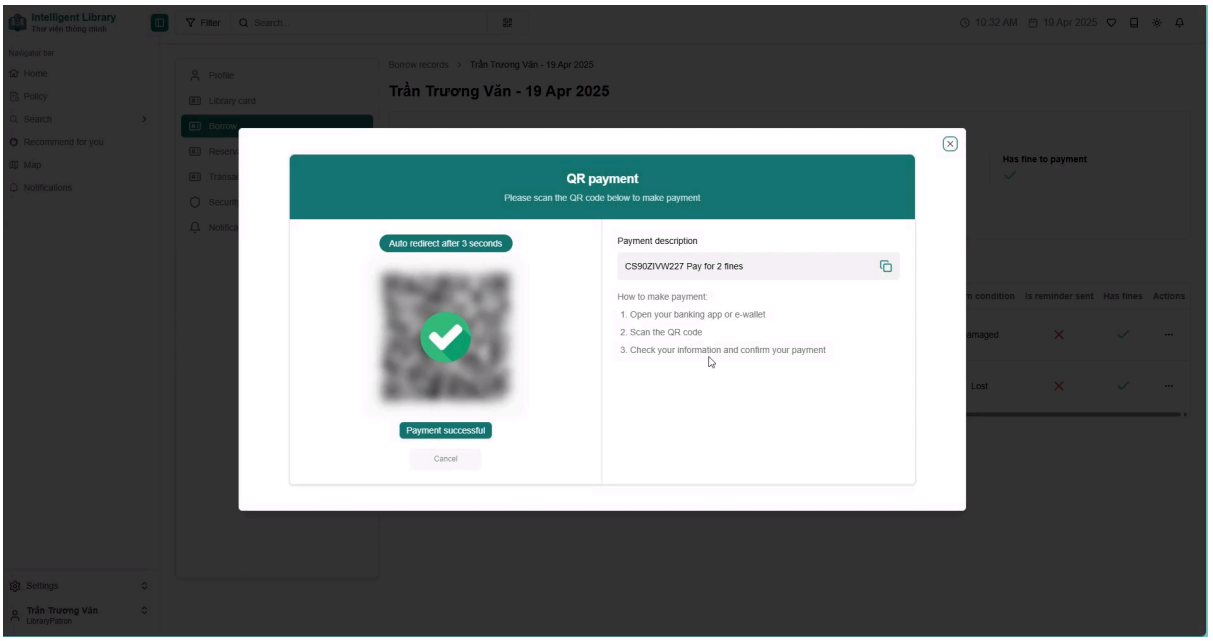


Figure 6.3.4.16: Borrow record detail screen

- Additionally, patron can check the "Recommendation for you" page and see that library items similar to those they have borrowed are being suggested.

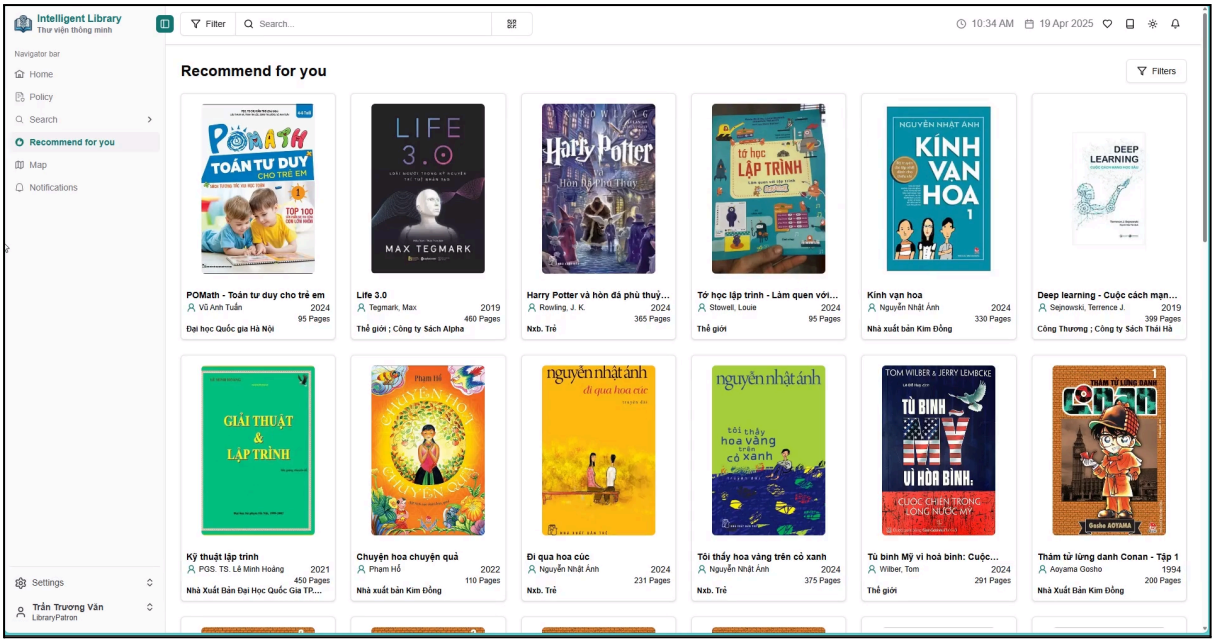


Figure 6.3.4.17: Recommend for you screen

- The patron marked the item *"Shin-Cậu bé bút chì"* as a favorite.

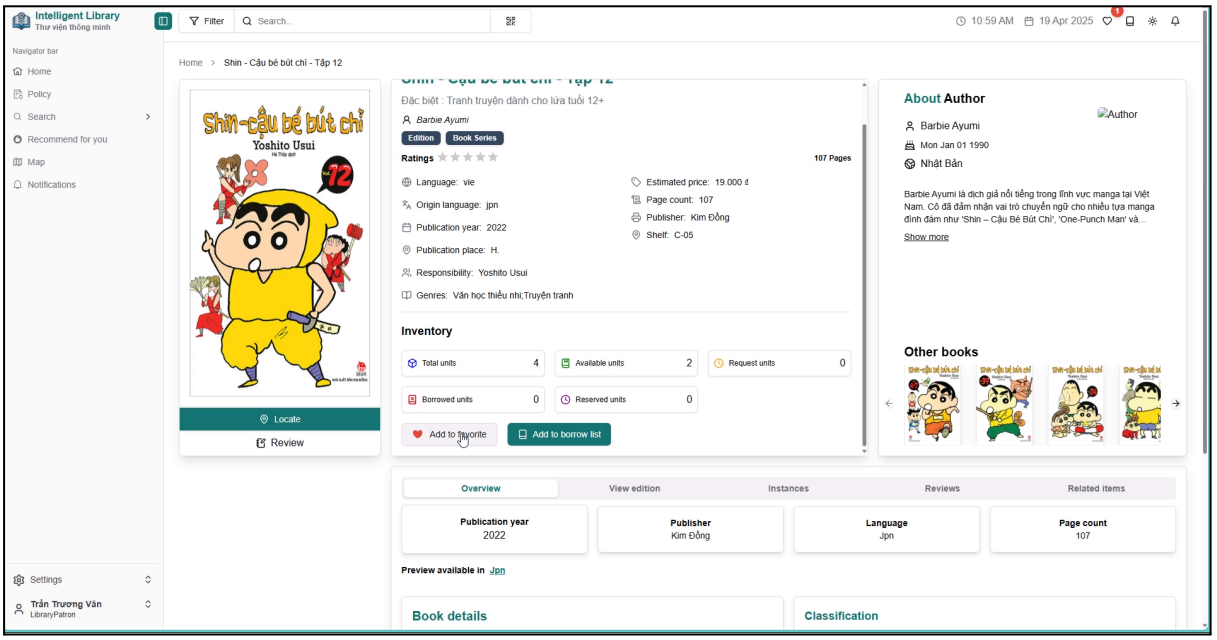


Figure 6.3.4.18: Library item detail screen

- The patron adds a review for the item *"Conan-Thám tử lừng danh"*.

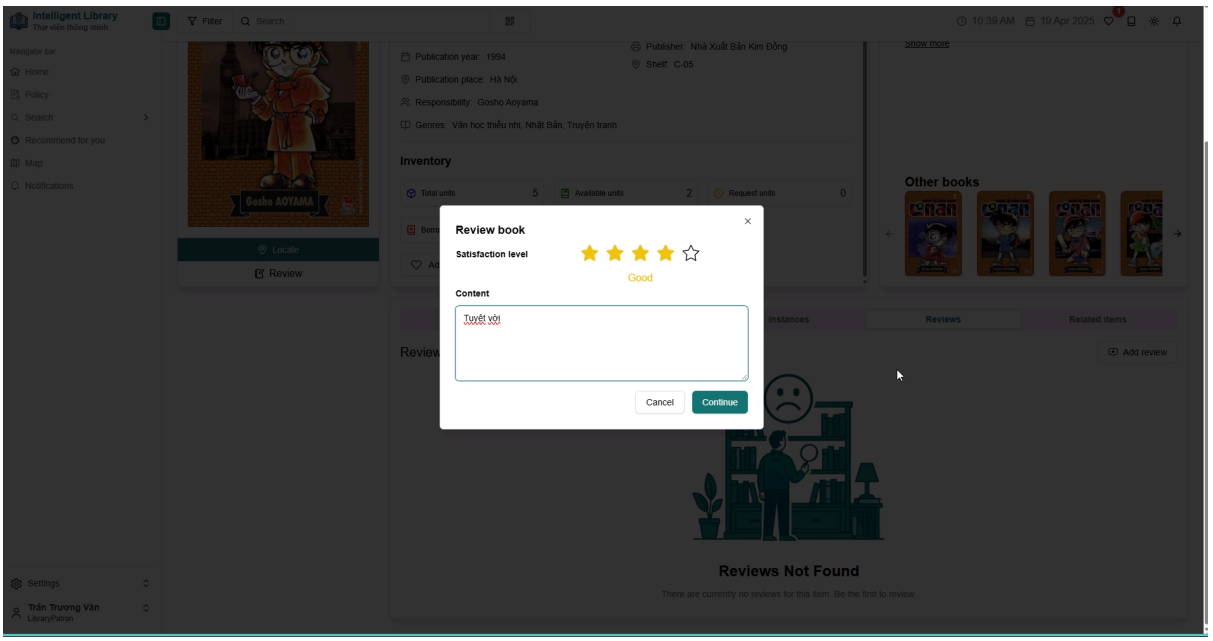


Figure 6.3.4.19: Library item detail screen

- The patron returns to the "Recommendation for you" page and sees that the results have been updated.

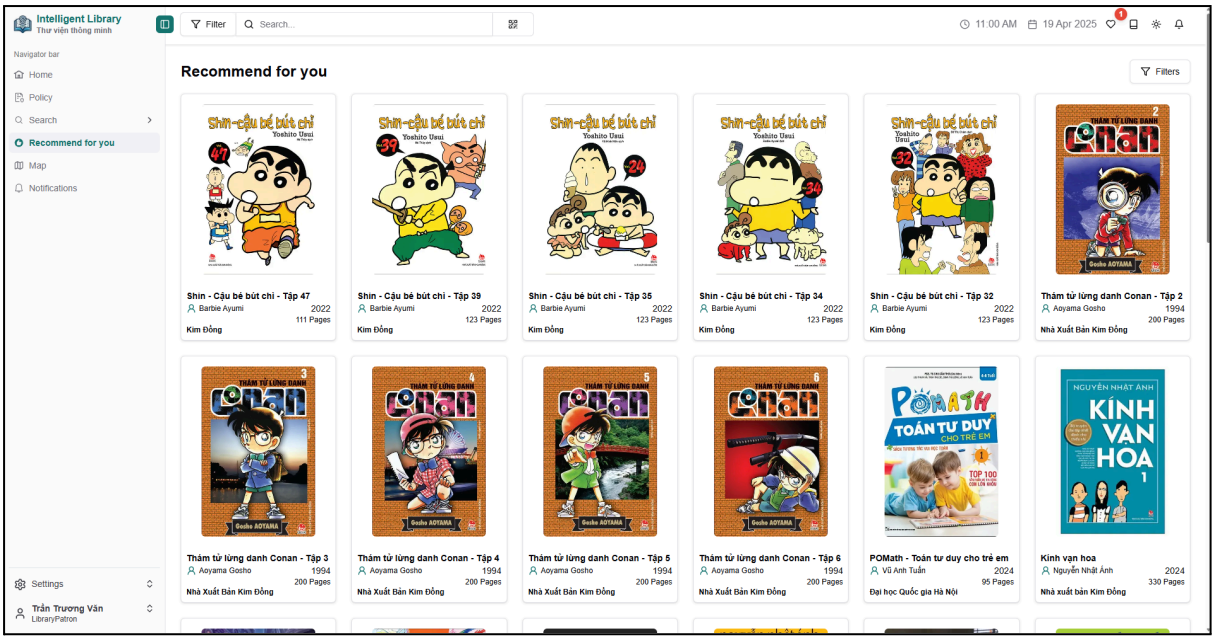


Figure 6.3.4.20: Recommend for you screen

3.5 Workflow 4

- Librarian views an overview of the library items inventory in the system.

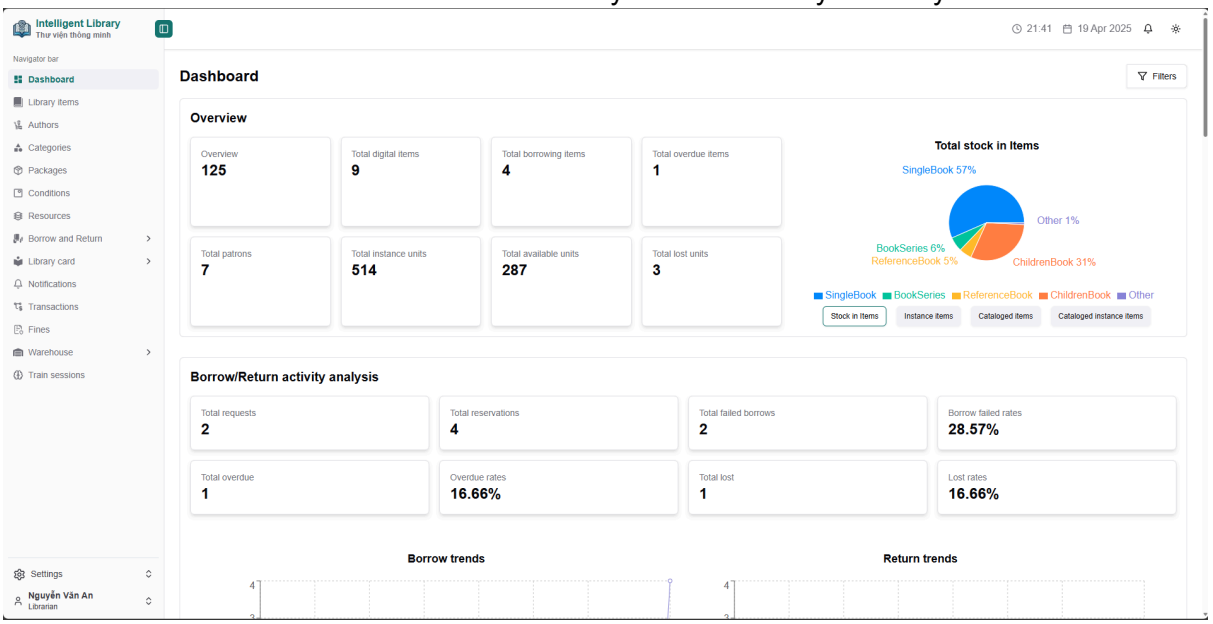


Figure 6.3.5.1: Dashboard screen

- Librarian reviews analytics on the library's borrowing and returning activities.

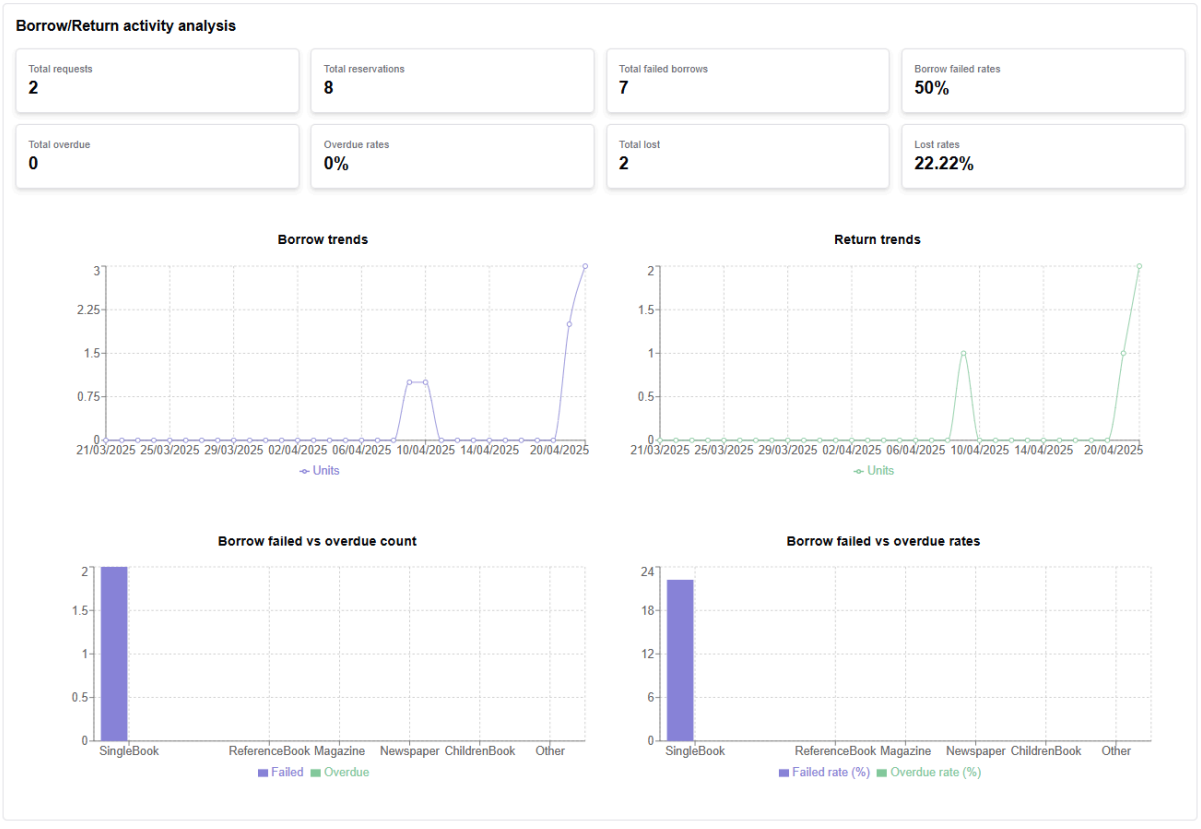


Figure 6.3.5.2: Dashboard screen

- Librarian reviews statistics on digital resources and the most popular digital resources being borrowed

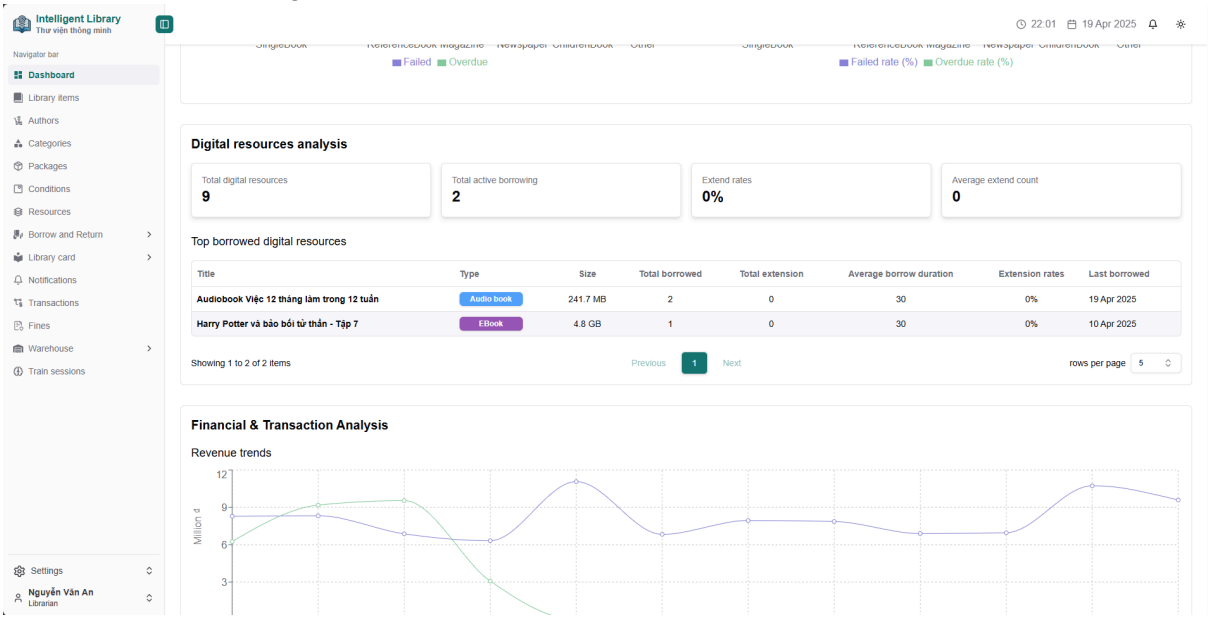


Figure 6.3.5.3: Dashboard screen

- Librarian views a chart comparing revenue between last year and this year, along with recent transactions.

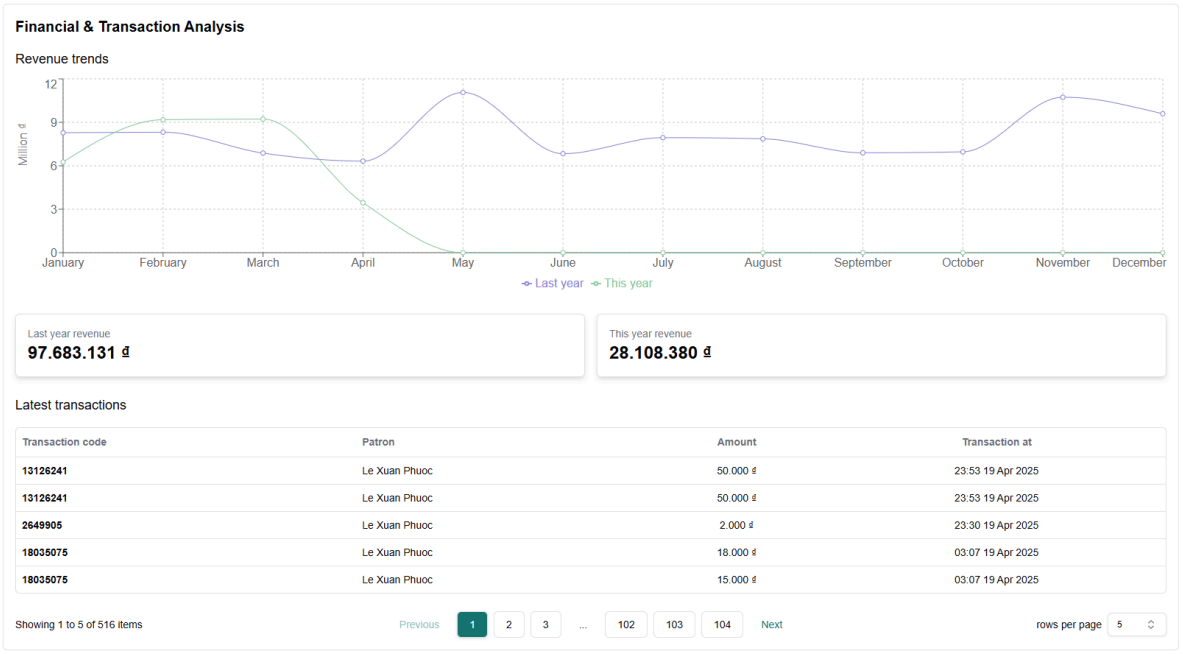


Figure 6.3.5.4: Dashboard screen

- Librarian views the overdue borrow record and can send reminders to the patrons.

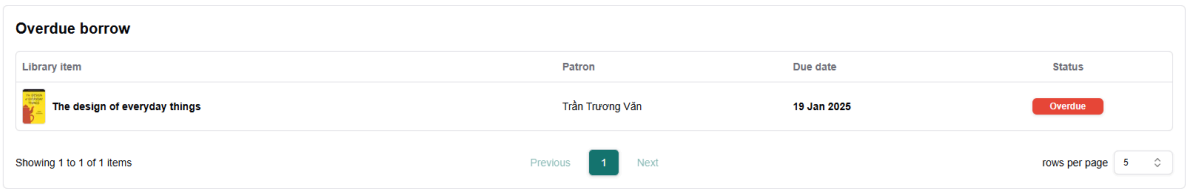


Figure 6.3.5.5: Dashboard screen

- Librarian views the assignable reservations and proceeds to assign library items to them.

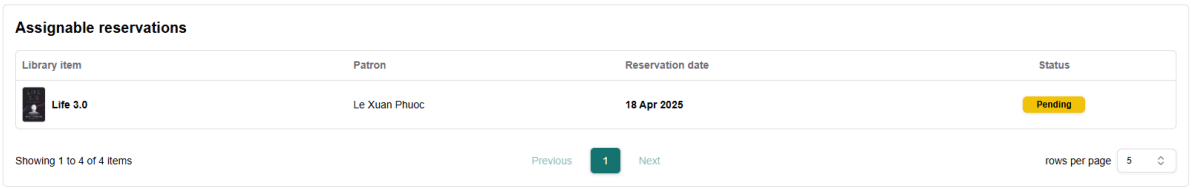


Figure 6.3.5.6: Dashboard screen

- The librarian views the recent active borrow records.

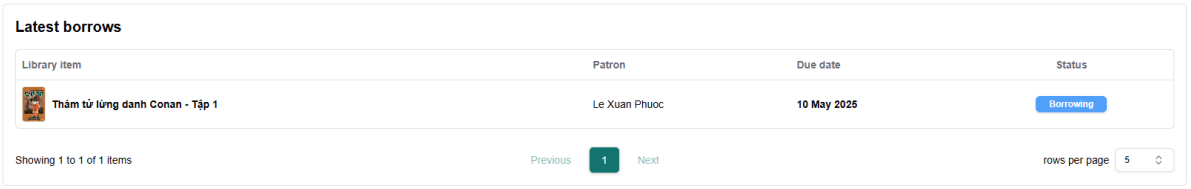


Figure 6.3.5.7: Dashboard screen

- Librarian reviews the analysis of the top circulated library items and navigate to the page for creating a new supplement request.

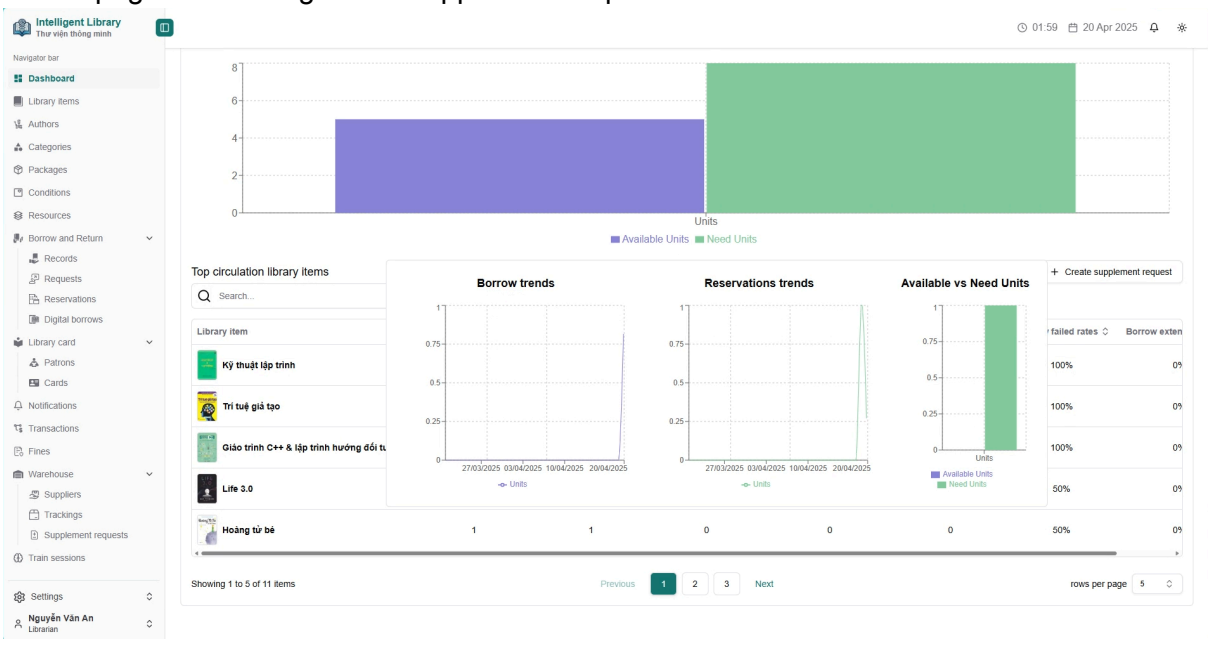


Figure 6.3.5.8: Dashboard screen

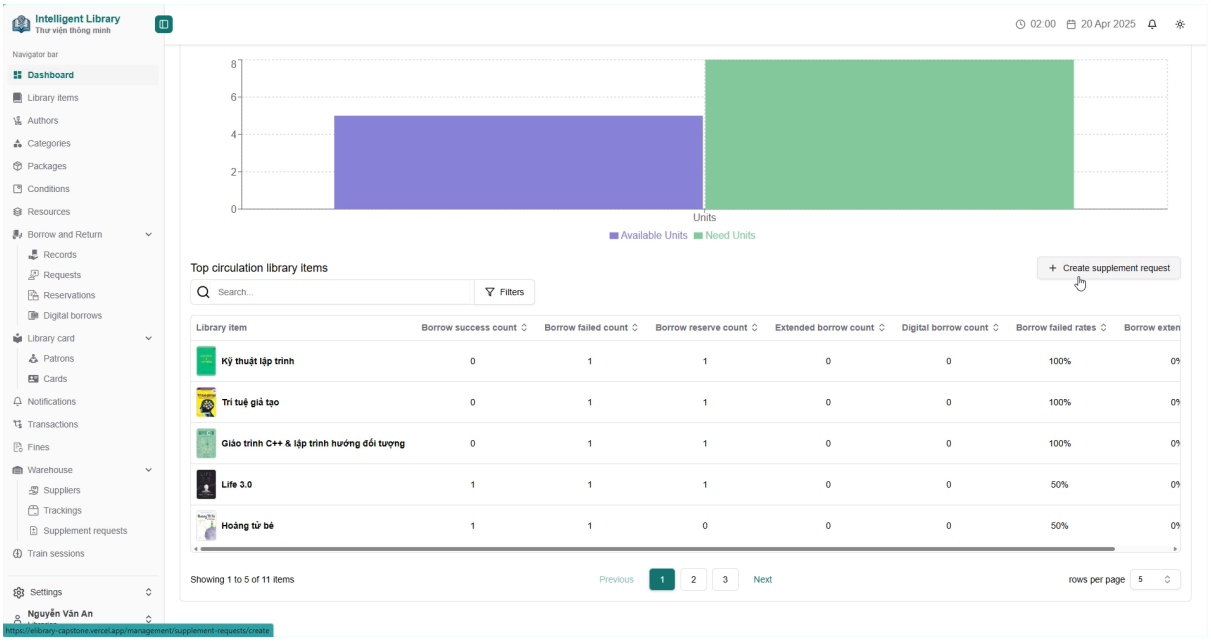


Figure 6.3.5.9: Dashboard screen

- The librarian adds the library items they wish to acquire and can also request suggestions for library items that are not yet available in the system.

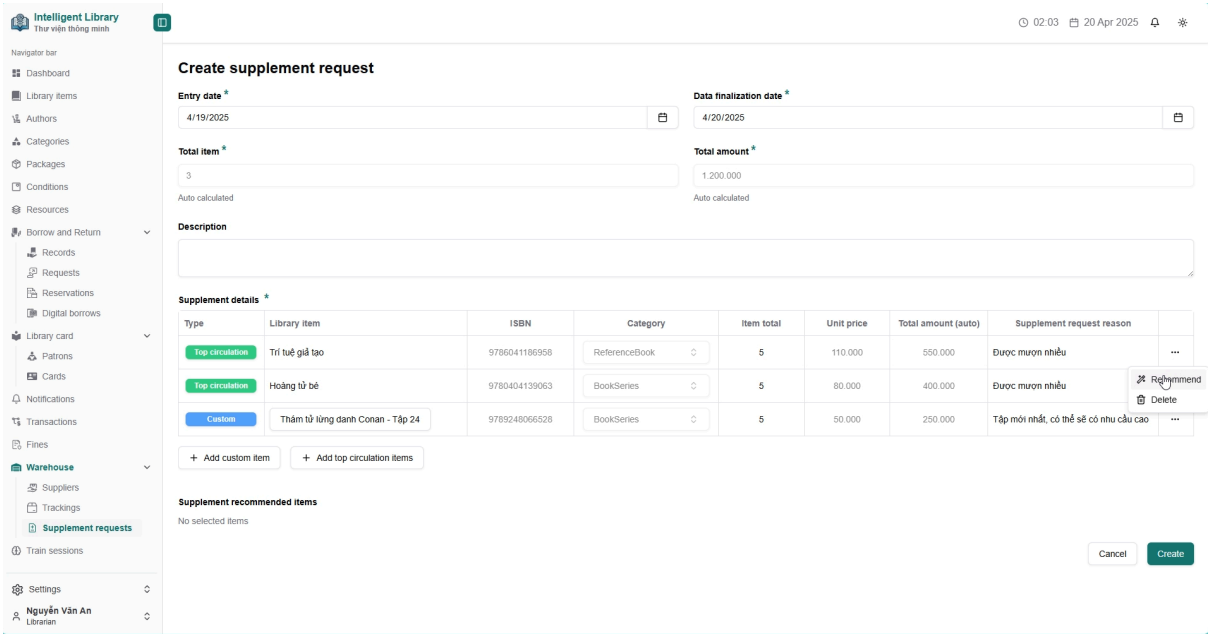


Figure 6.3.5.10: Create supplement request screen

- The librarian selects the library items they wish to recommend for addition and then clicks "Continue" to complete the creating supplement request process.

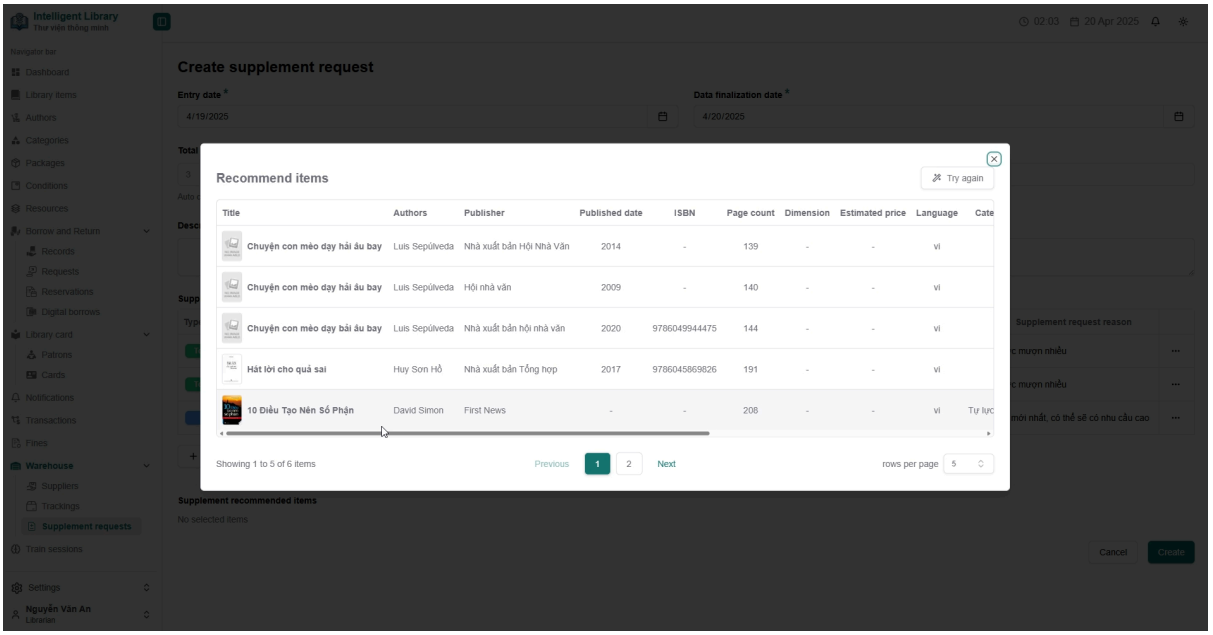


Figure 6.3.5.11: Create supplement request screen

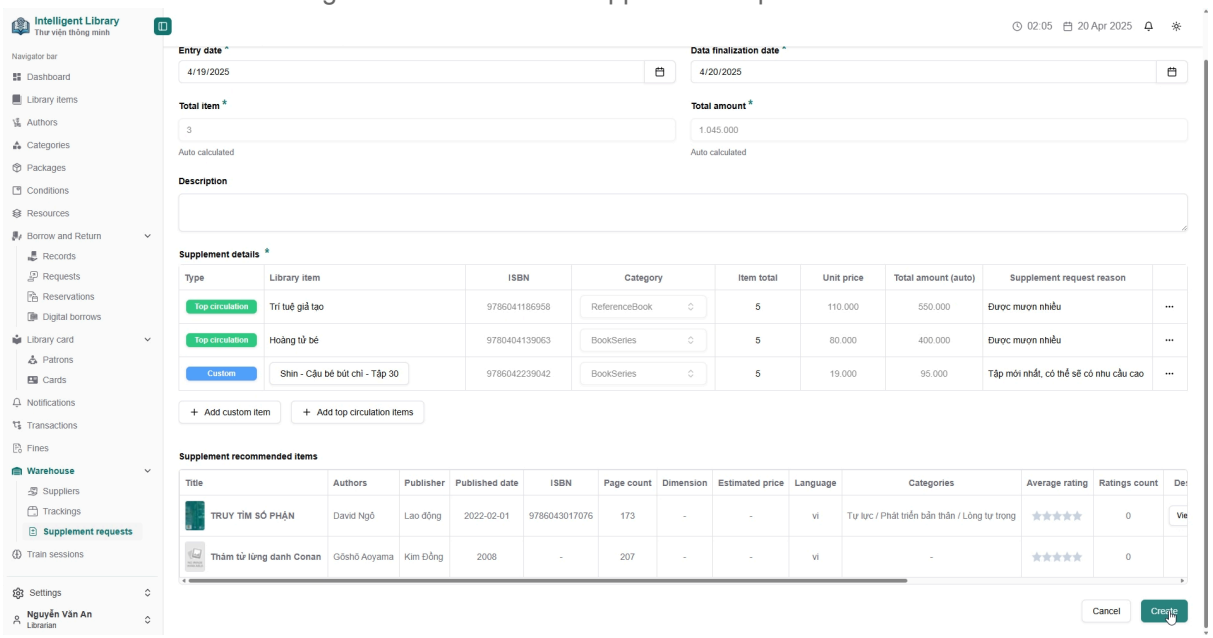


Figure 6.3.5.12: Supplement request detail screen

- The librarian reviews the supplement request that has just been created.

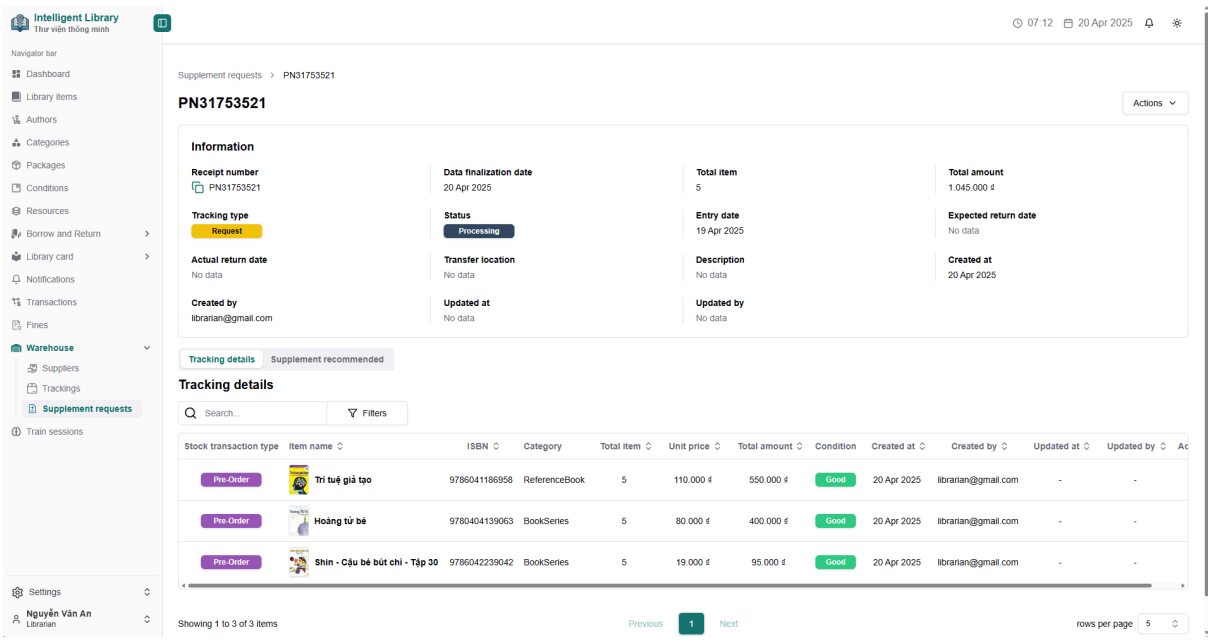


Figure 6.3.5.13: Supplement request detail screen

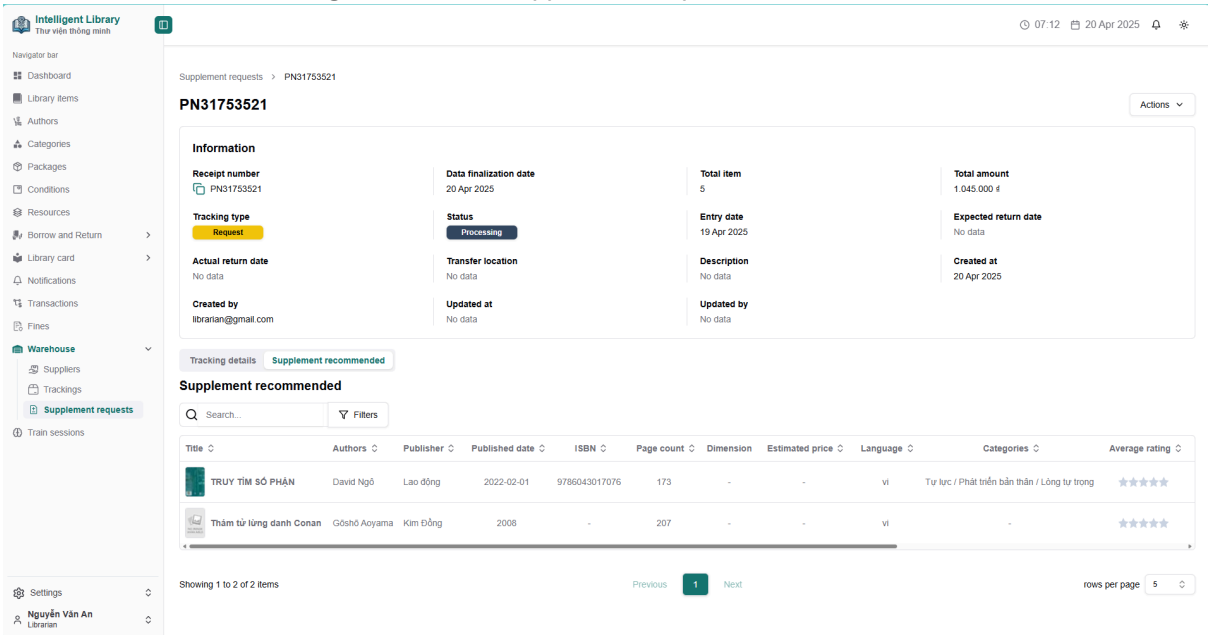


Figure 6.3.5.14: Supplement request detail screen

- Head librarian login to Management Page.

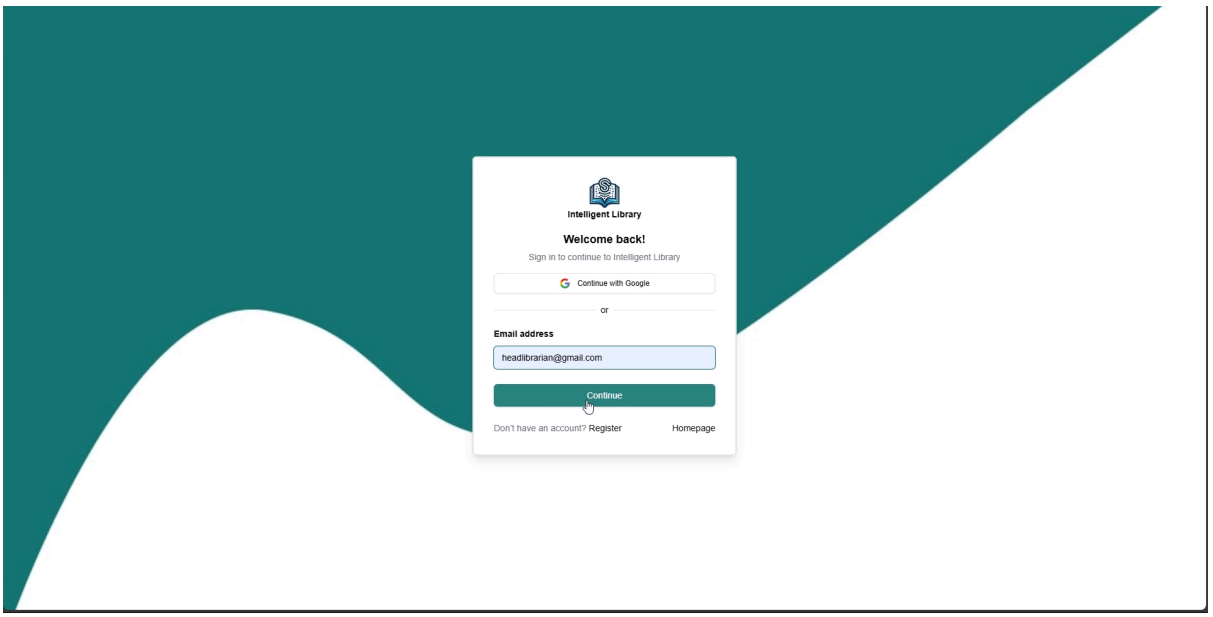


Figure 6.3.5.15: Login screen

- Head librarian checks and marks the warehouse tracking as completed.

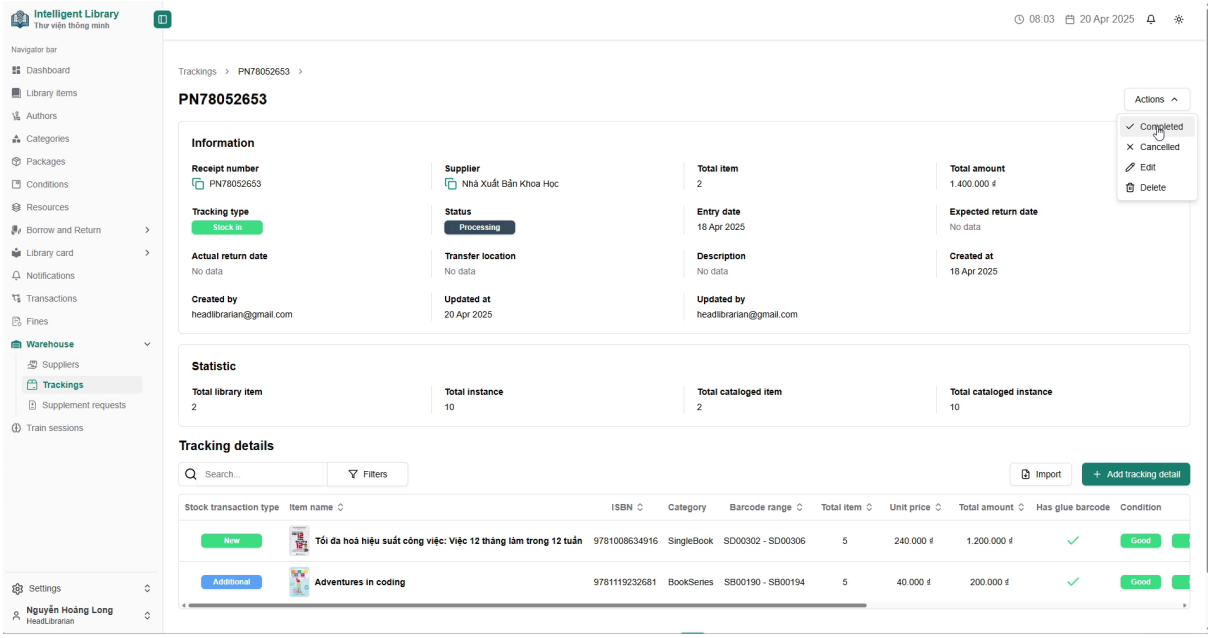
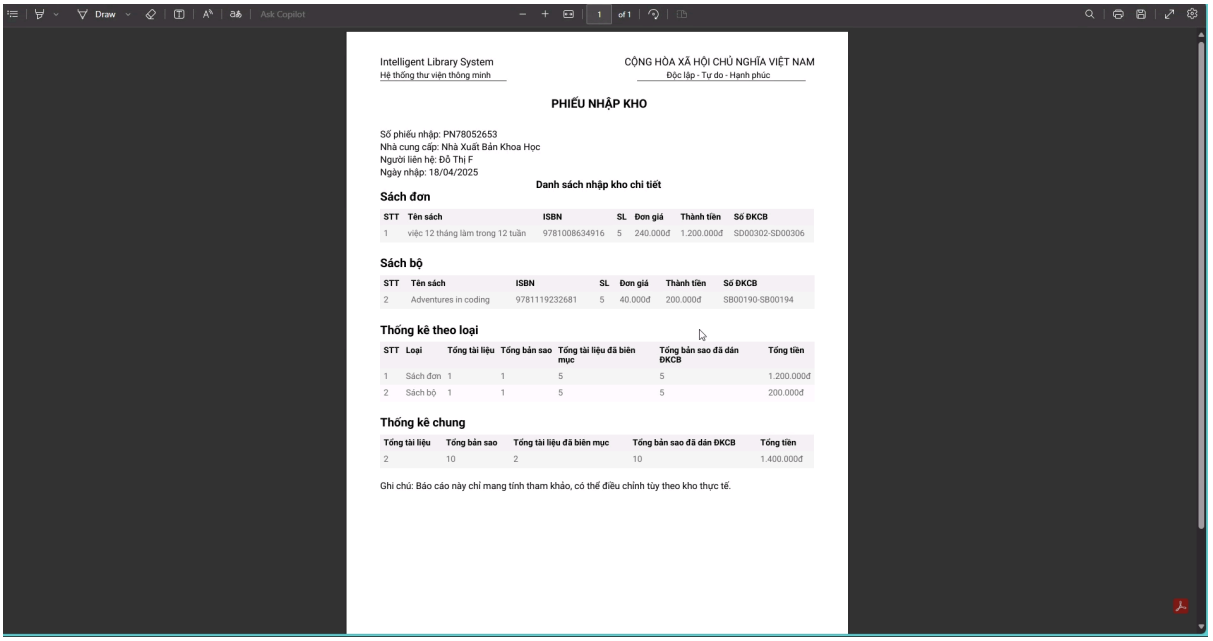


Figure 6.3.5.16: Warehouse tracking detail screen

- Head librarian generates the warehouse tracking pdf.



Intelligent Library System
Hệ thống thư viện thông minh

CỘNG HÒA XÃ HỘI CHỦ NGHĨA VIỆT NAM
Độc lập - Tự do - Hạnh phúc

PHIẾU NHẬP KHO

Số phiếu nhập: PN78052653
Nhà cung cấp: Nhà Xuất Bản Khoa Học
Người liên hệ: Đỗ Thị F
Ngày nhập: 18/04/2025

Danh sách nhập kho chi tiết

Sách đơn

STT	Tên sách	ISBN	SL	Đơn giá	Thành tiền	Số ĐKCB
1	việc 12 tháng làm trong 12 tuần	9781008634916	5	240.000đ	1.200.000đ	SD00302-SD00306

Sách bộ

STT	Tên sách	ISBN	SL	Đơn giá	Thành tiền	Số ĐKCB
2	Adventures in coding	9781119232681	5	40.000đ	200.000đ	SB00190-SB00194

Thống kê theo loại

STT	Loại	Tổng tài liệu	Tổng bản sao	Tổng tài liệu đã biên mục	Tổng bản sao đã dán ĐKCB	Tổng tiền
1	Sách đơn	1	1	5	5	1.200.000đ
2	Sách bộ	1	1	5	5	200.000đ

Thống kê chung

Tổng tài liệu	Tổng bản sao	Tổng tài liệu đã biên mục	Tổng bản sao đã dán ĐKCB	Tổng tiền
2	10	2	10	1.400.000đ

Ghi chú: Báo cáo này chỉ mang tính tham khảo, có thể điều chỉnh tùy theo kho thực tế.

Figure 6.3.5.17: View PDF screen